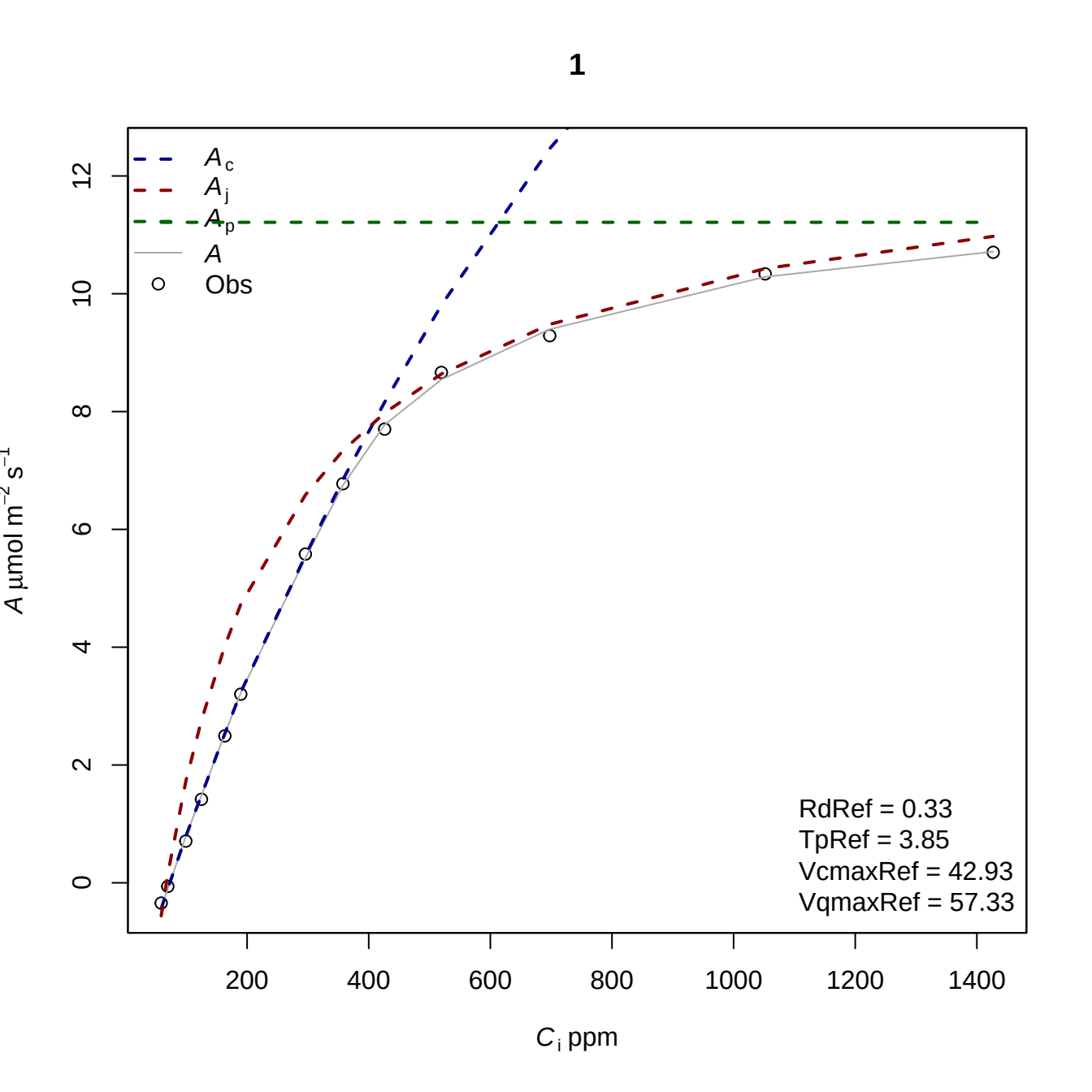
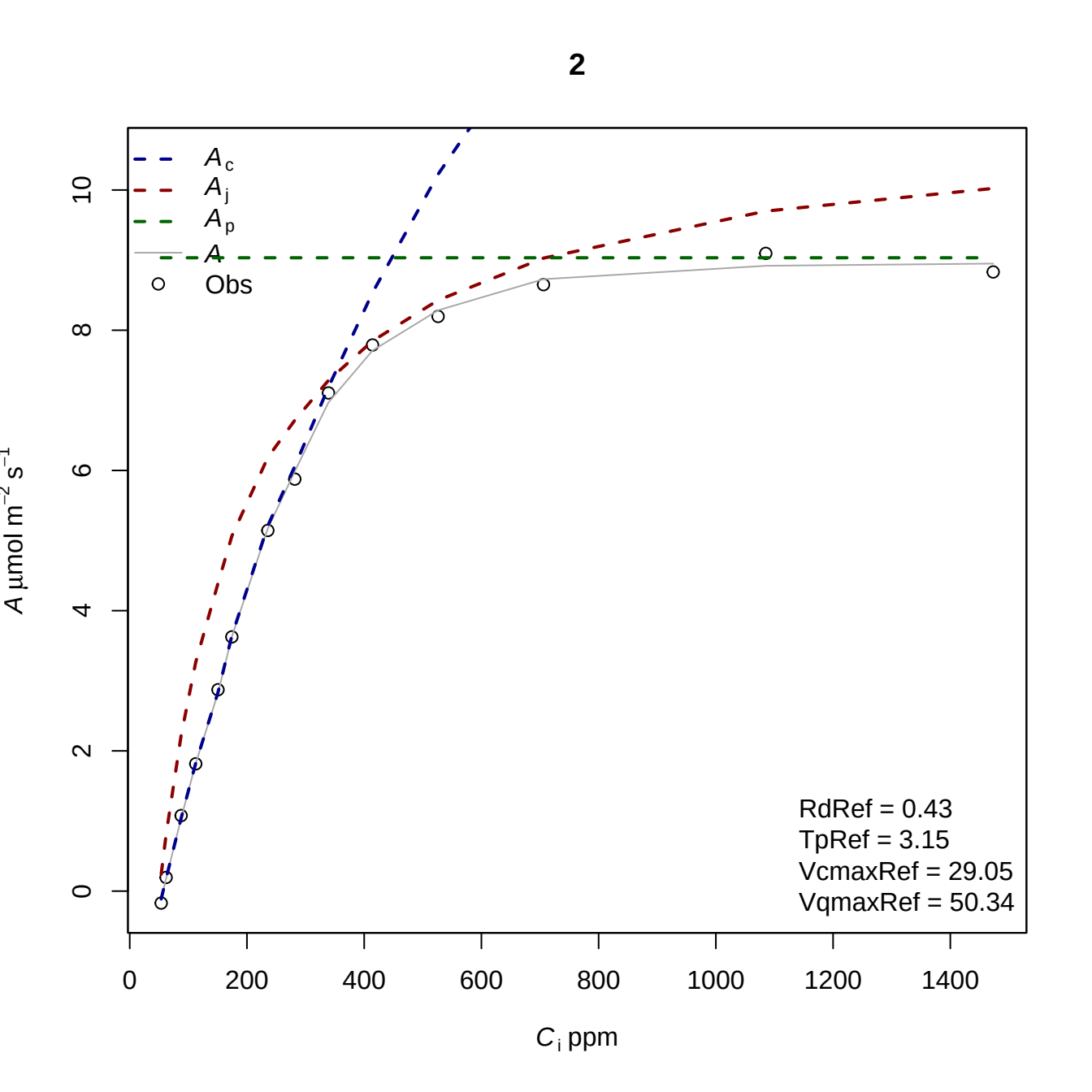
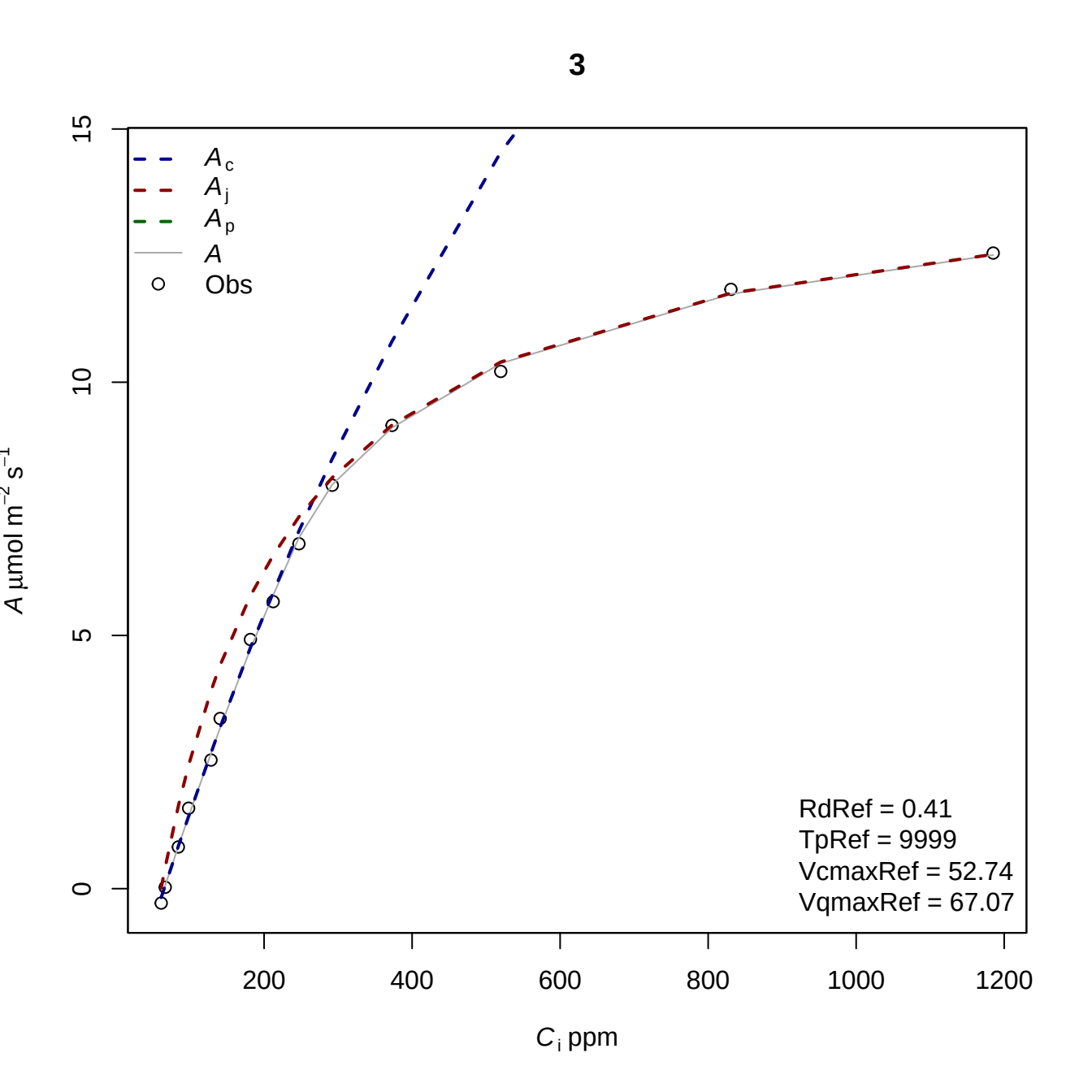
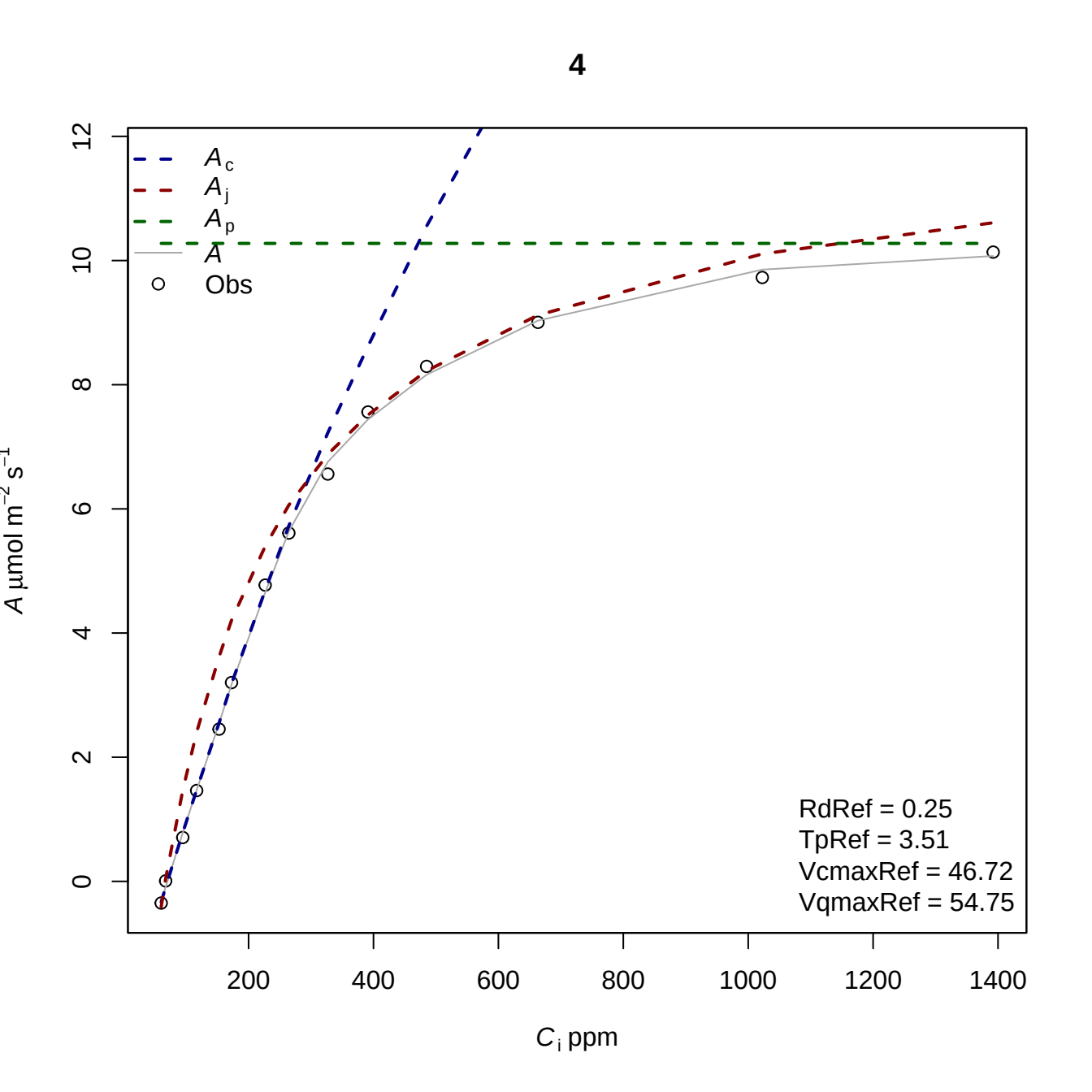
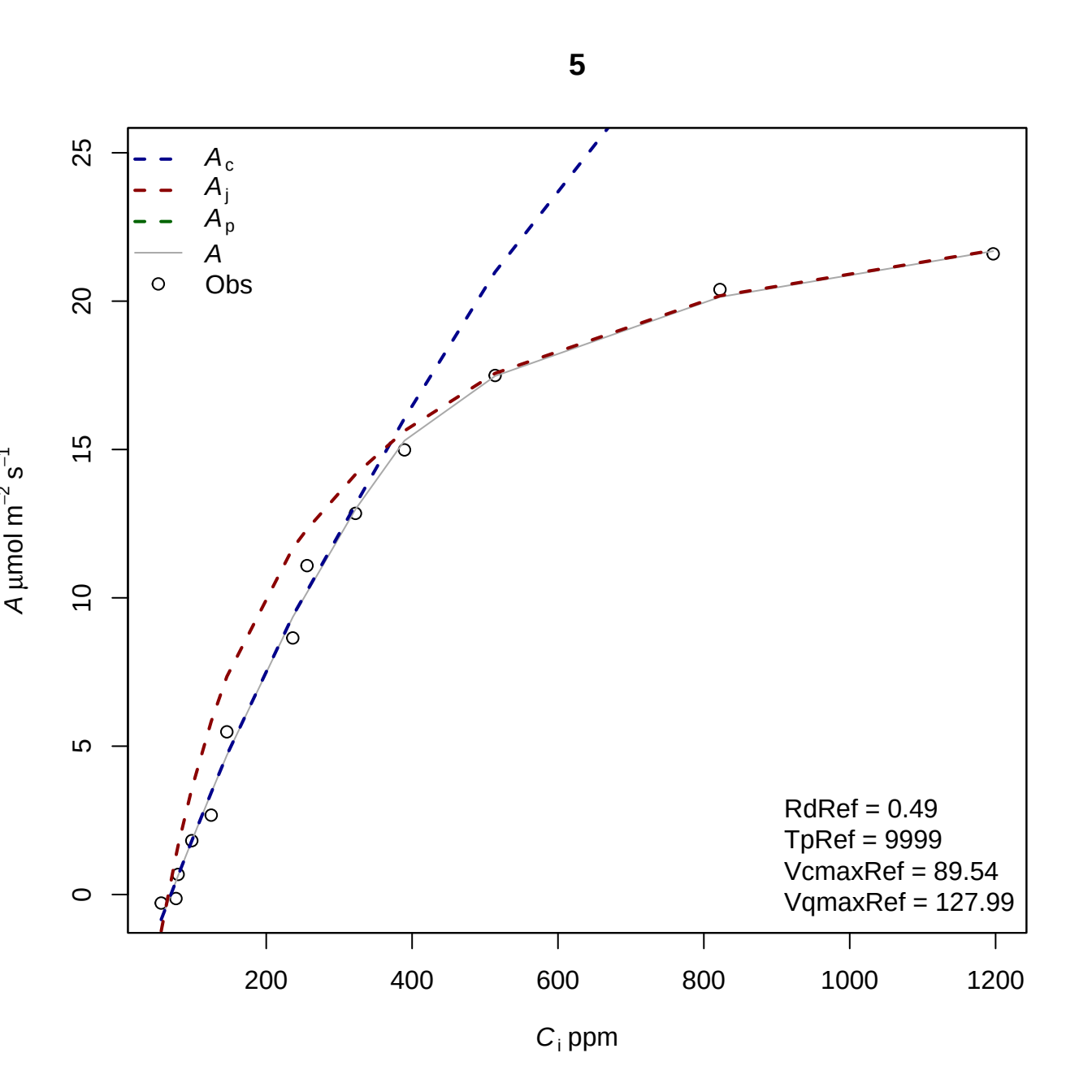
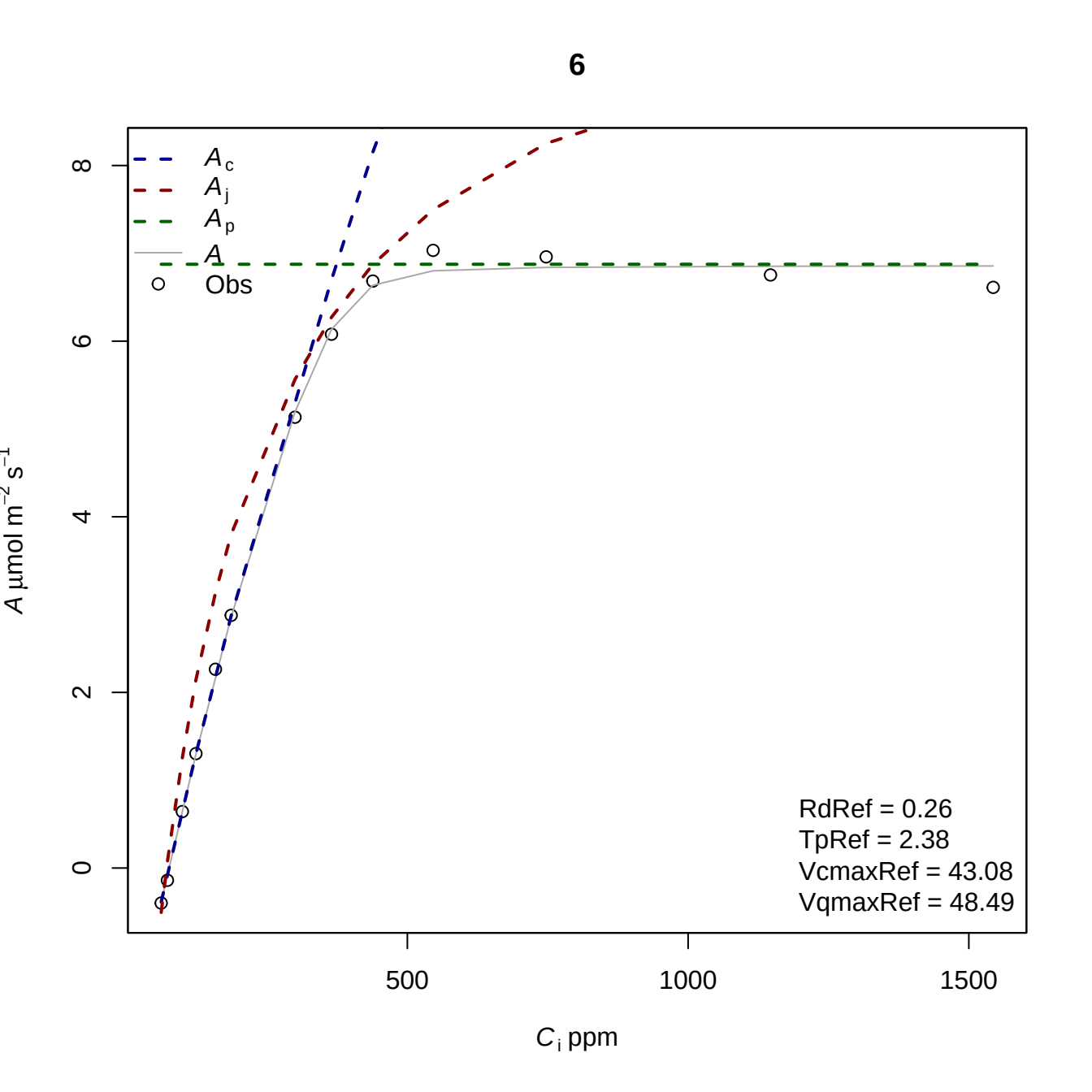

1
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.33
TpRef = 3.85
VcmaxRef = 42.93
VqmaxRef = 57.33
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

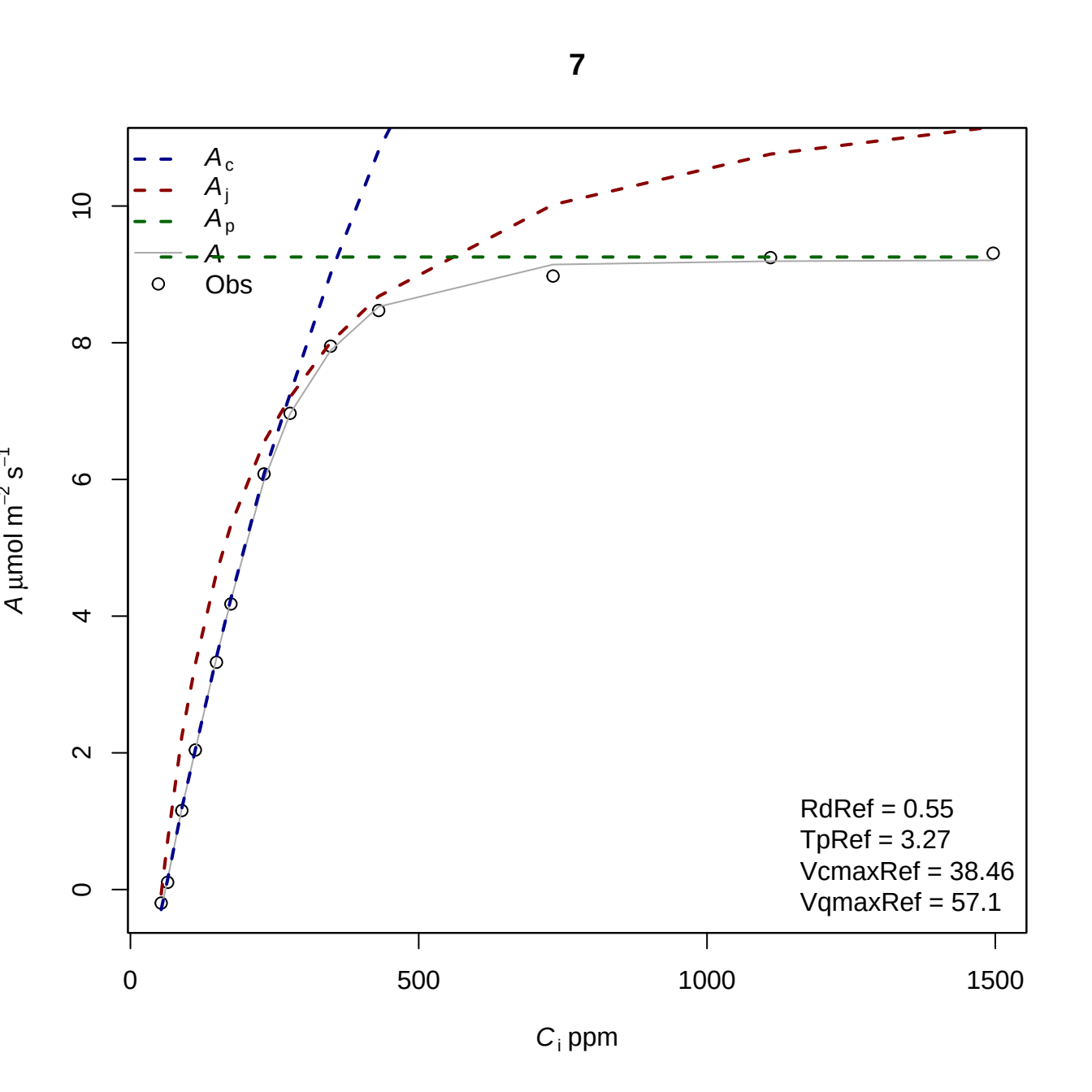

2
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.43
TpRef = 3.15
VcmaxRef = 29.05
VqmaxRef = 50.34
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

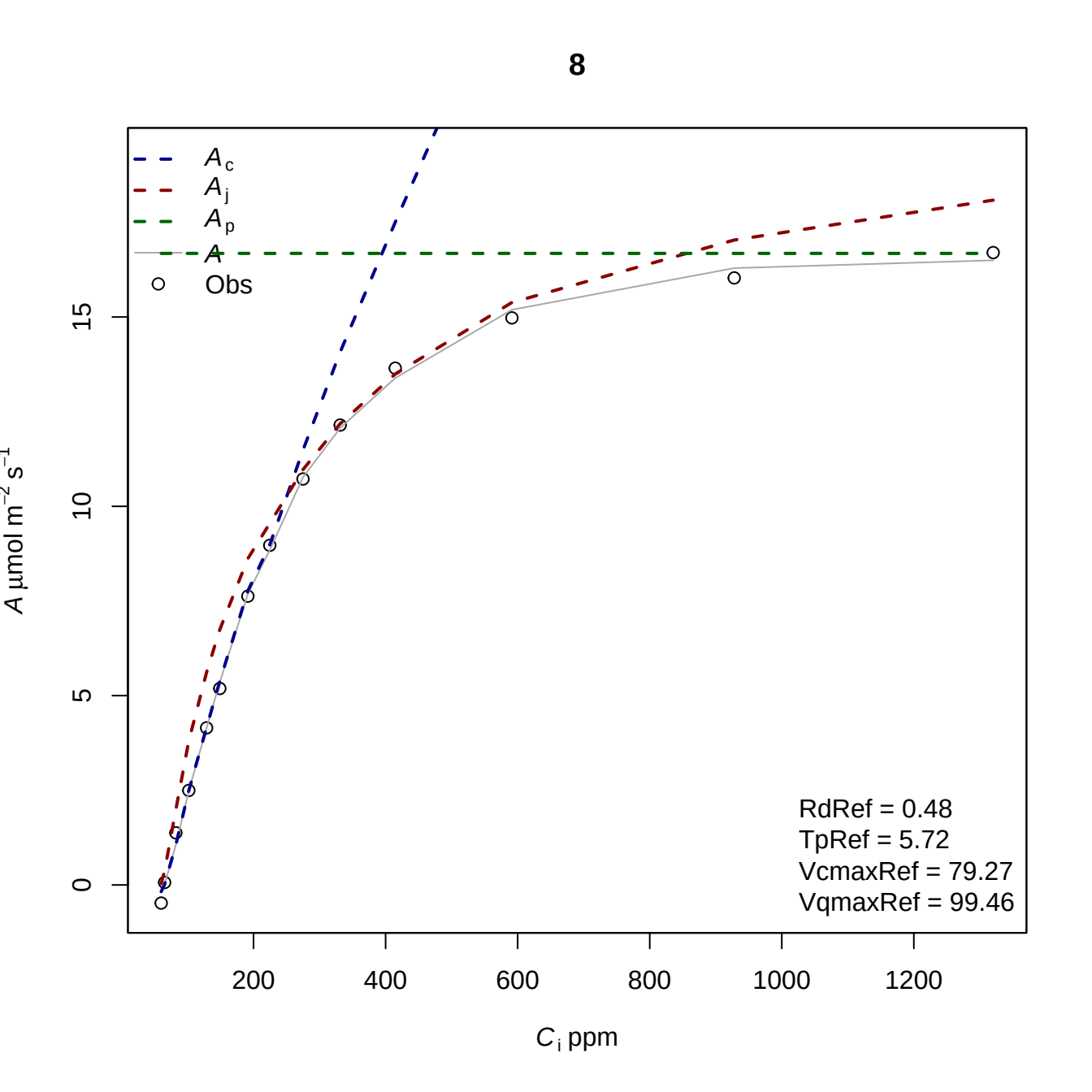

3
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.41
TpRef = 9999
VcmaxRef = 52.74
VqmaxRef = 67.07
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200

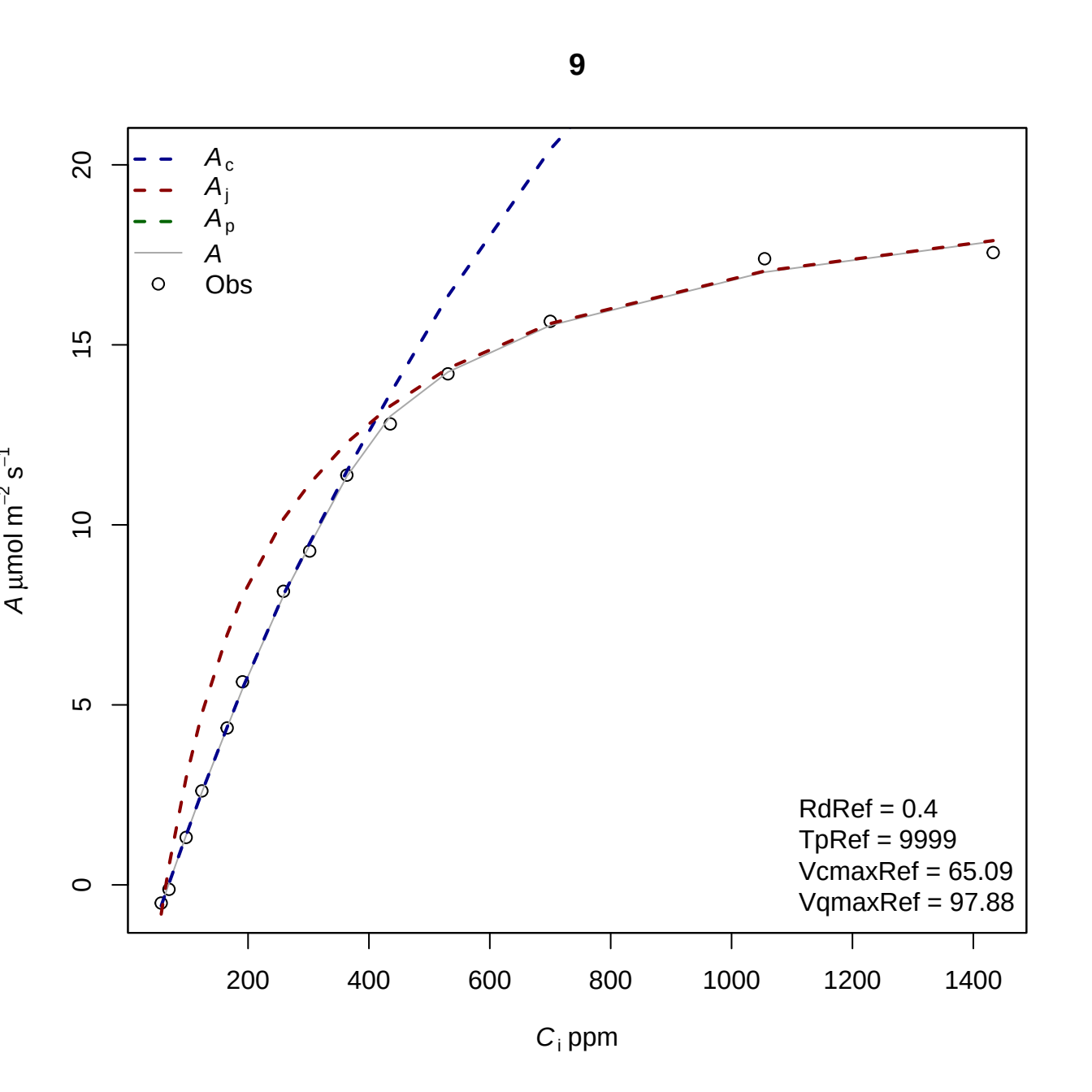

4
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.25
TpRef = 3.51
VcmaxRef = 46.72
VqmaxRef = 54.75
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

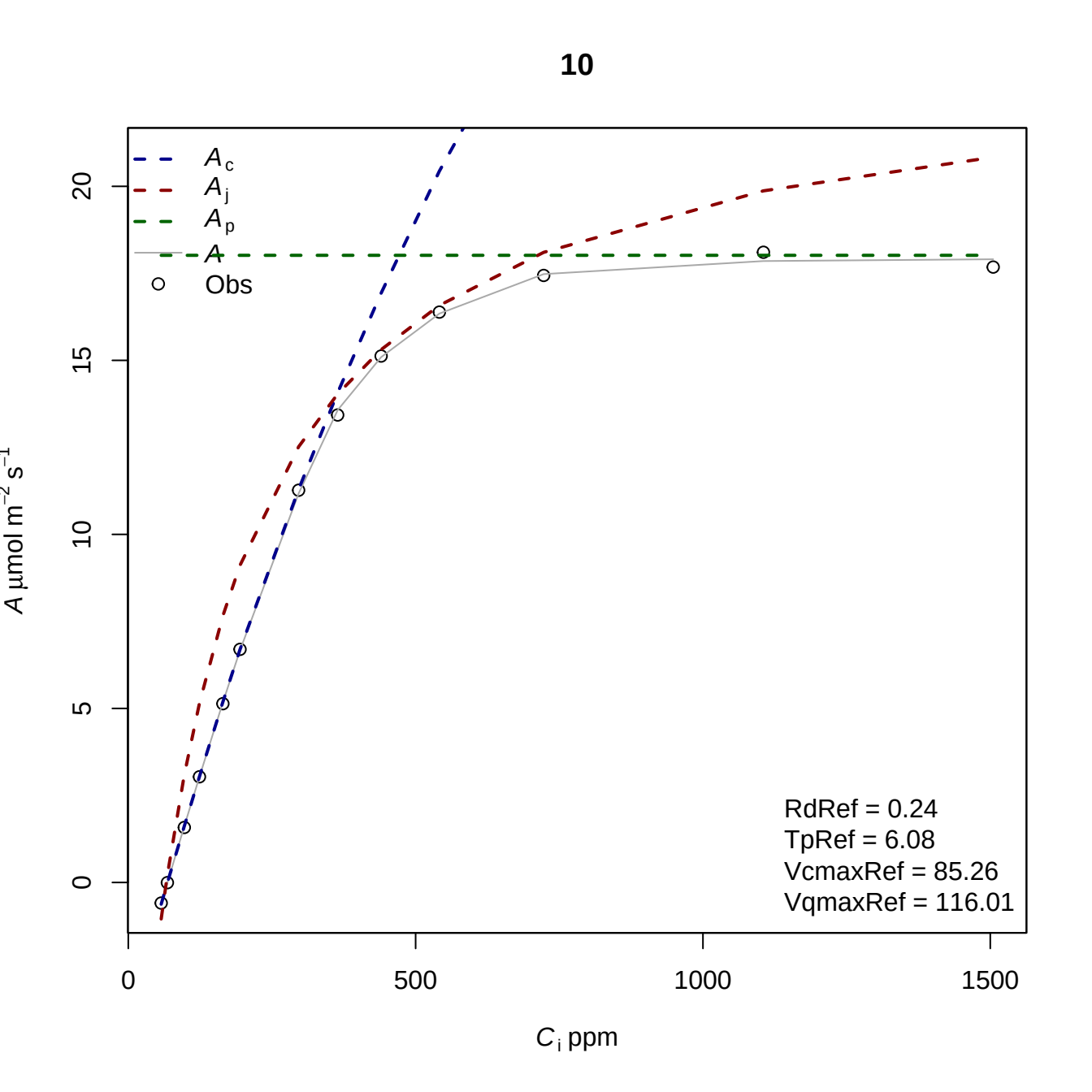

5
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.49
TpRef = 9999
VcmaxRef = 89.54
VqmaxRef = 127.99
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

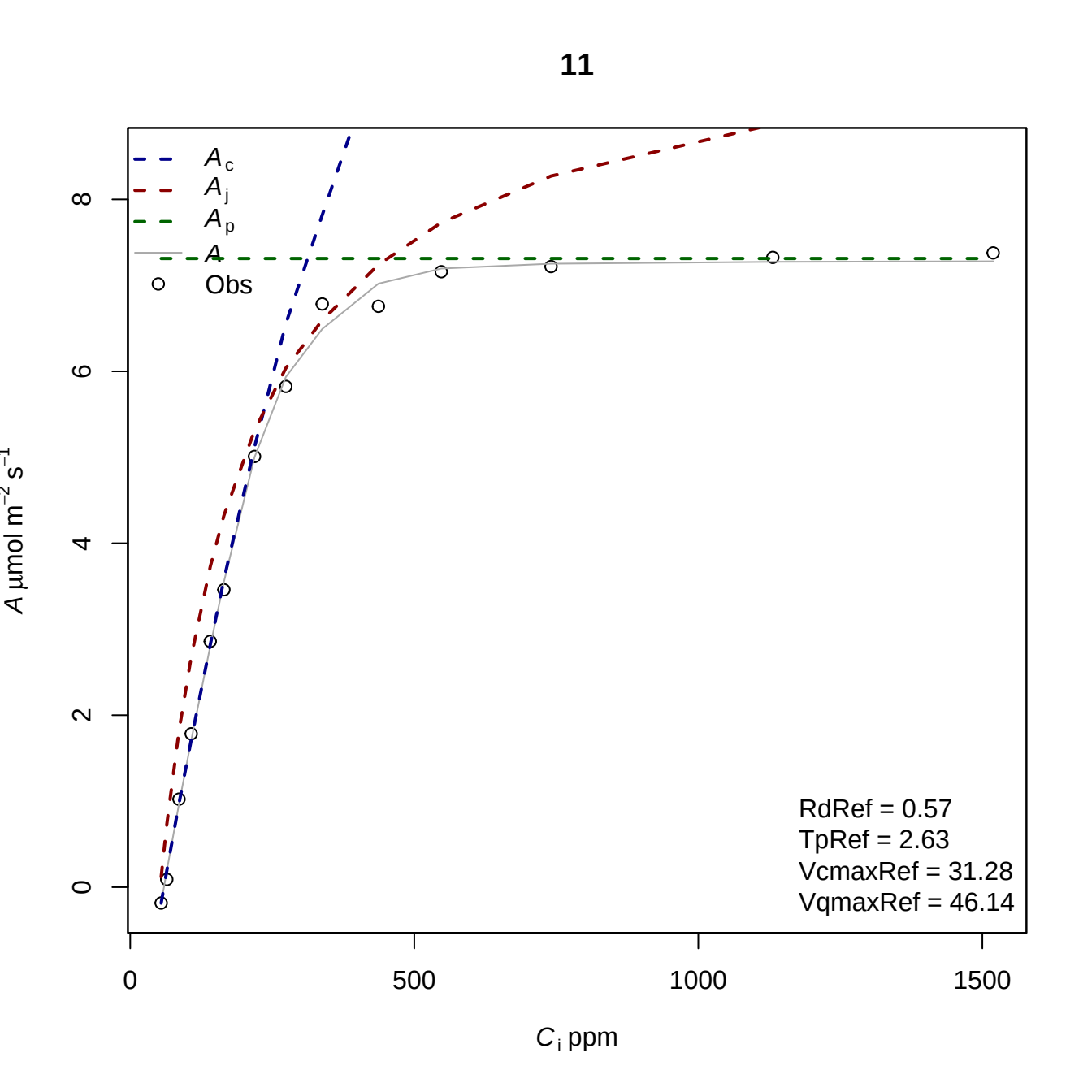

6
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.26
TpRef = 2.38
VcmaxRef = 43.08
VqmaxRef = 48.49
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
500
1000
1500

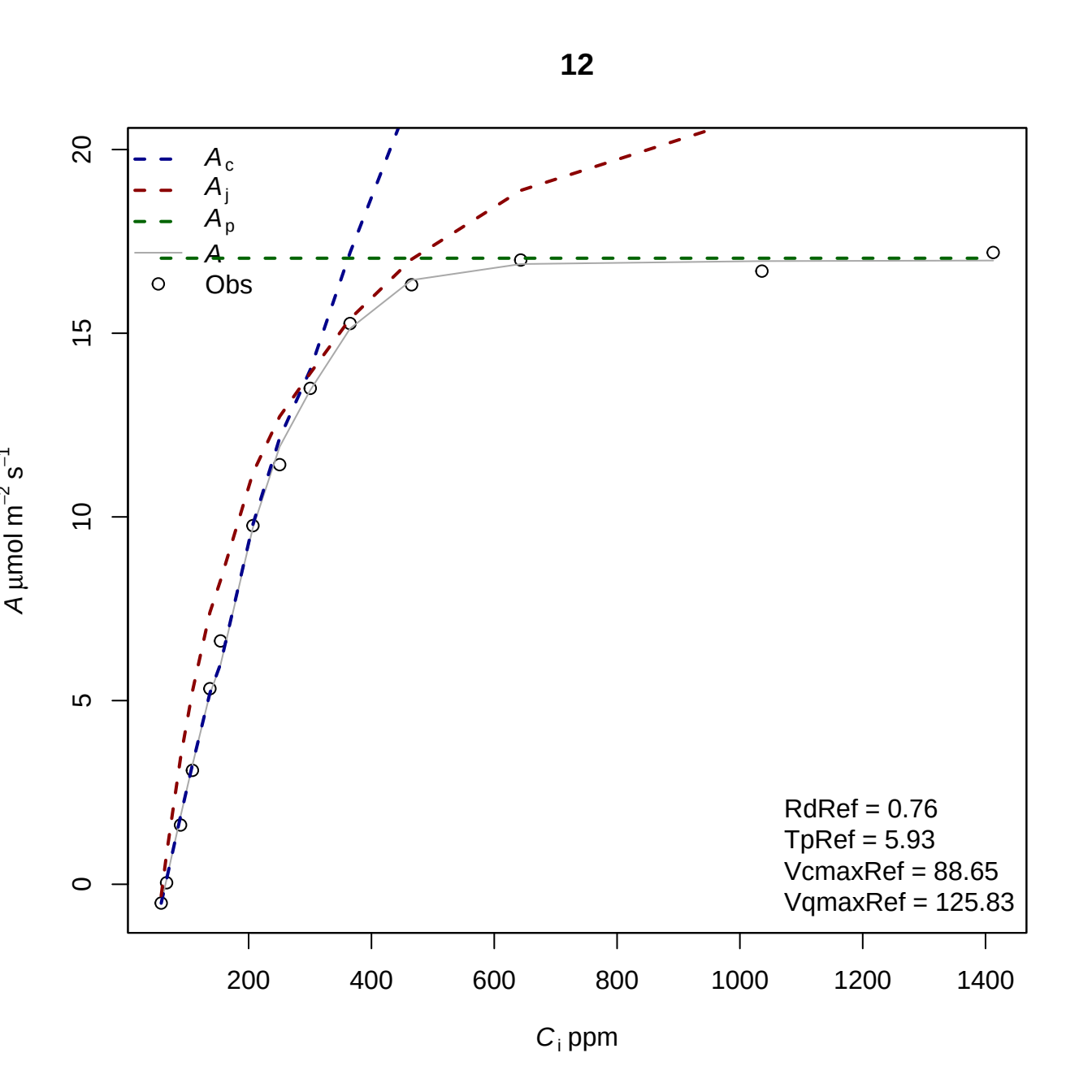

7
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.55
TpRef = 3.27
VcmaxRef = 38.46
VqmaxRef = 57.1
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

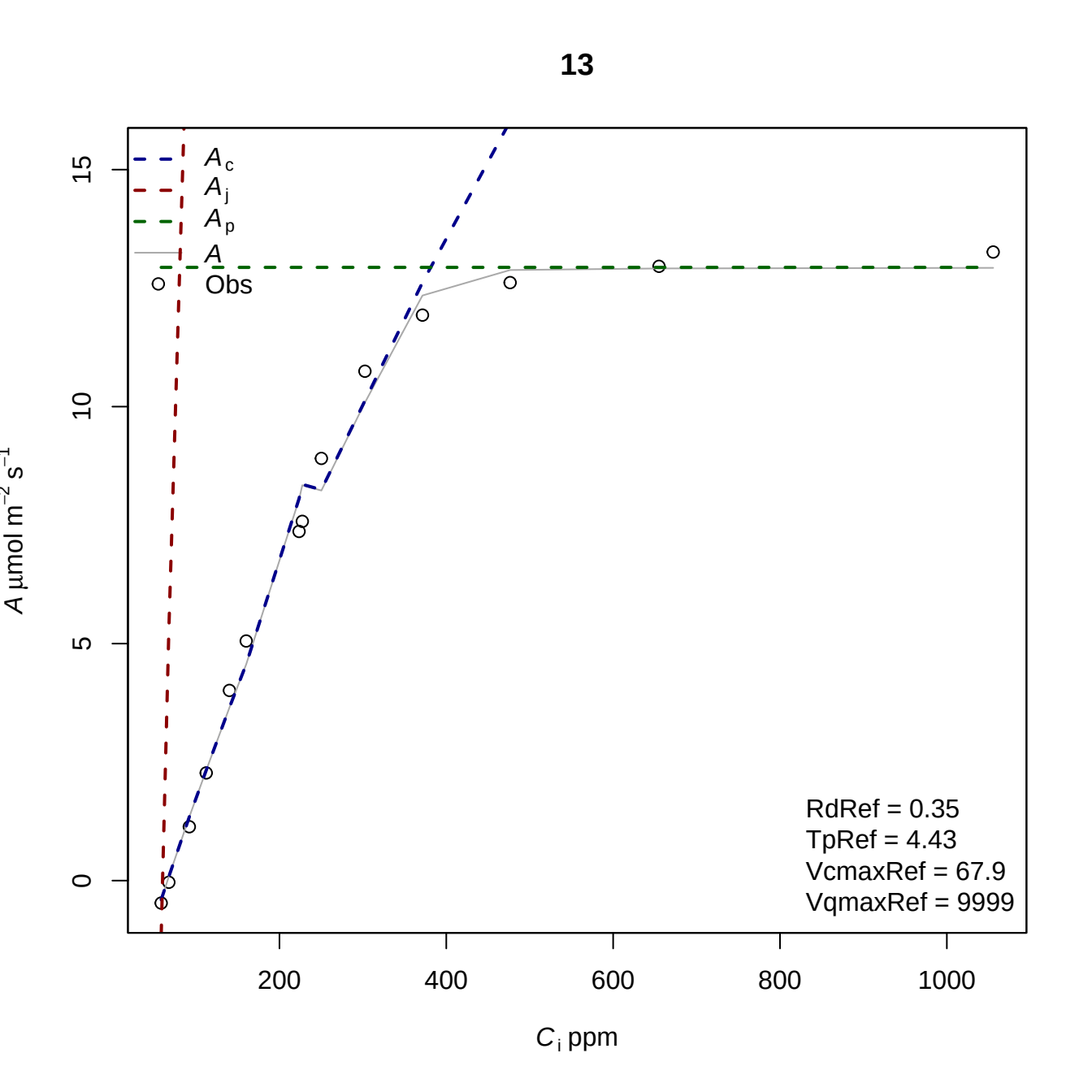

8
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.48
TpRef = 5.72
VcmaxRef = 79.27
VqmaxRef = 99.46
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200

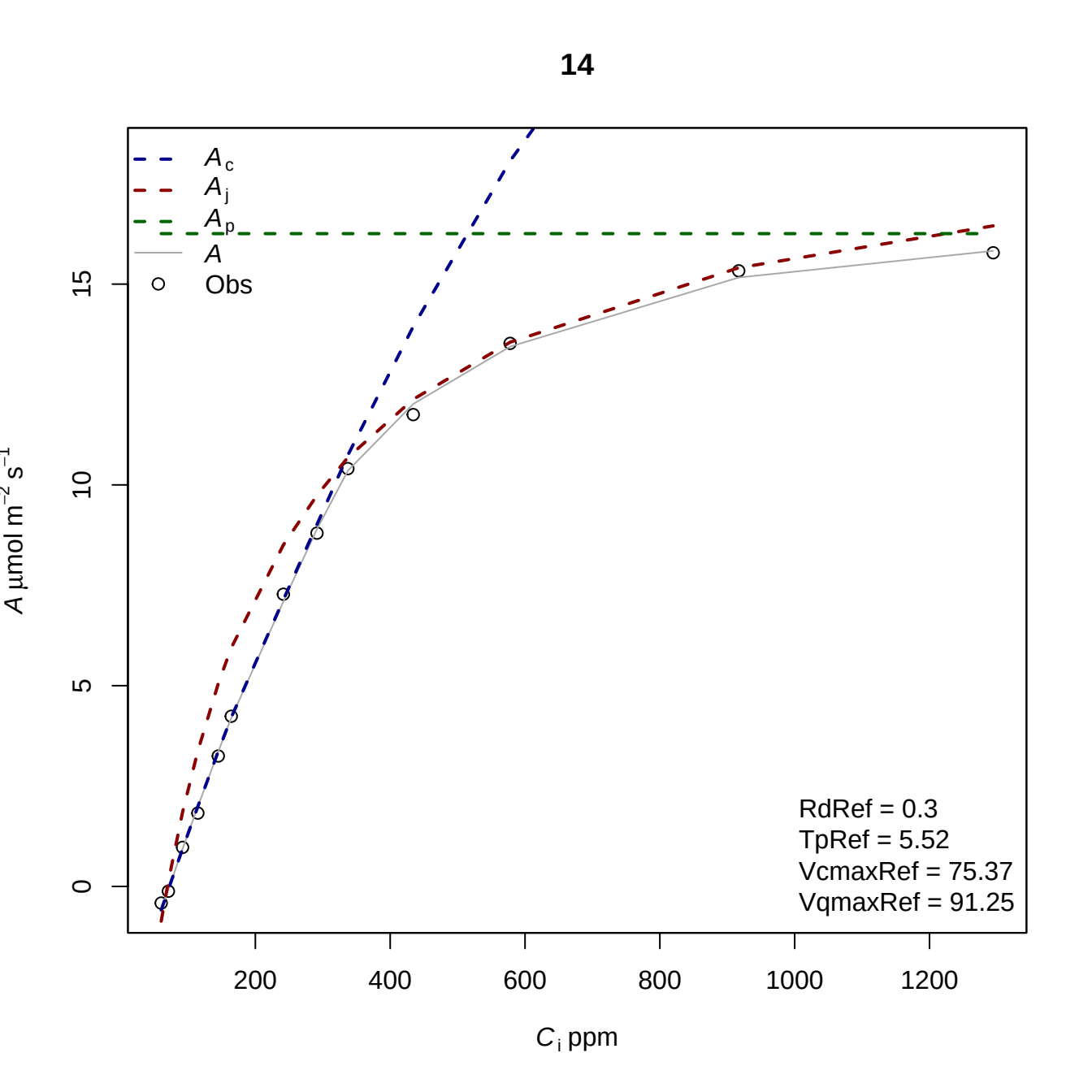

9
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.4
TpRef = 9999
VcmaxRef = 65.09
VqmaxRef = 97.88
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

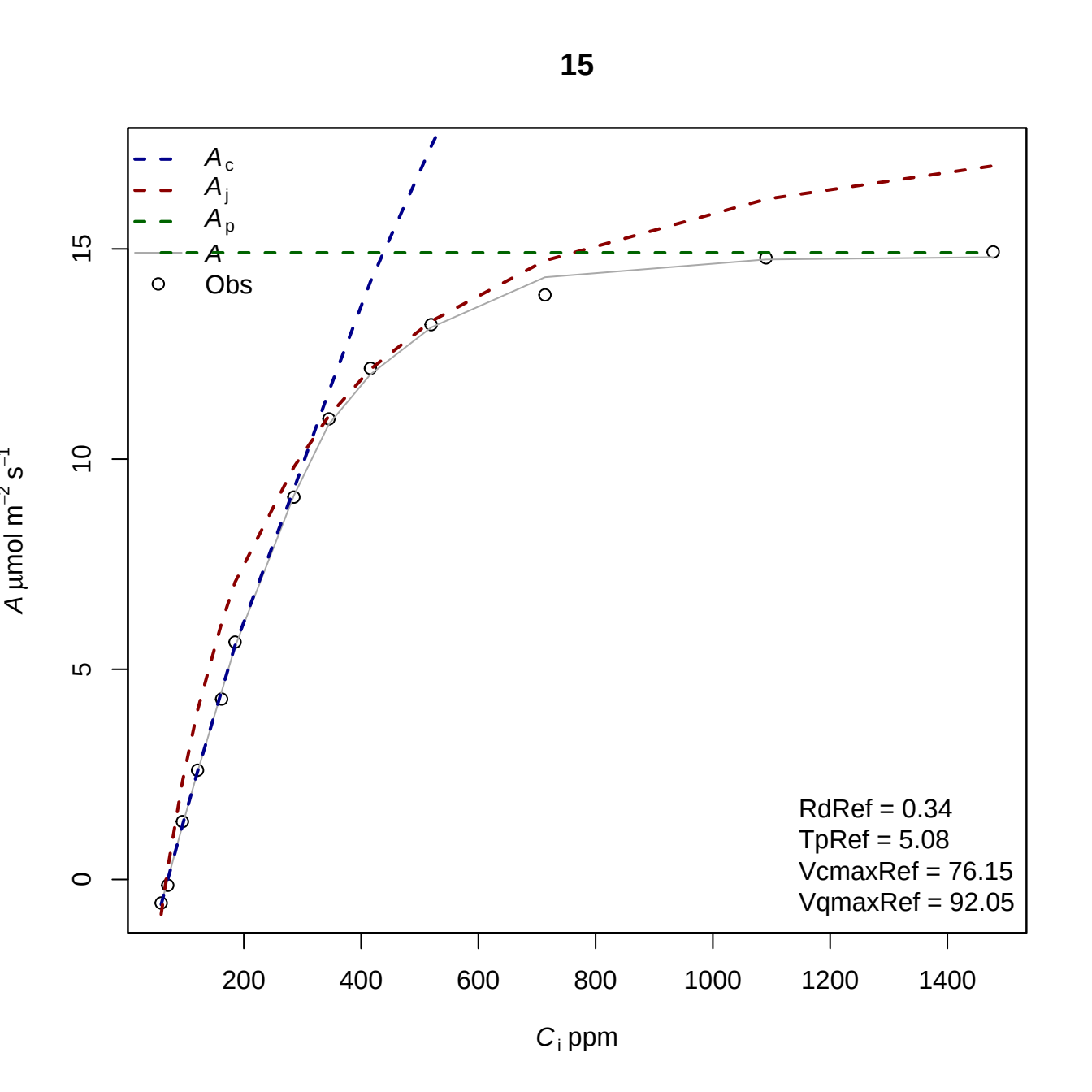

10
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.24
TpRef = 6.08
VcmaxRef = 85.26
VqmaxRef = 116.01
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

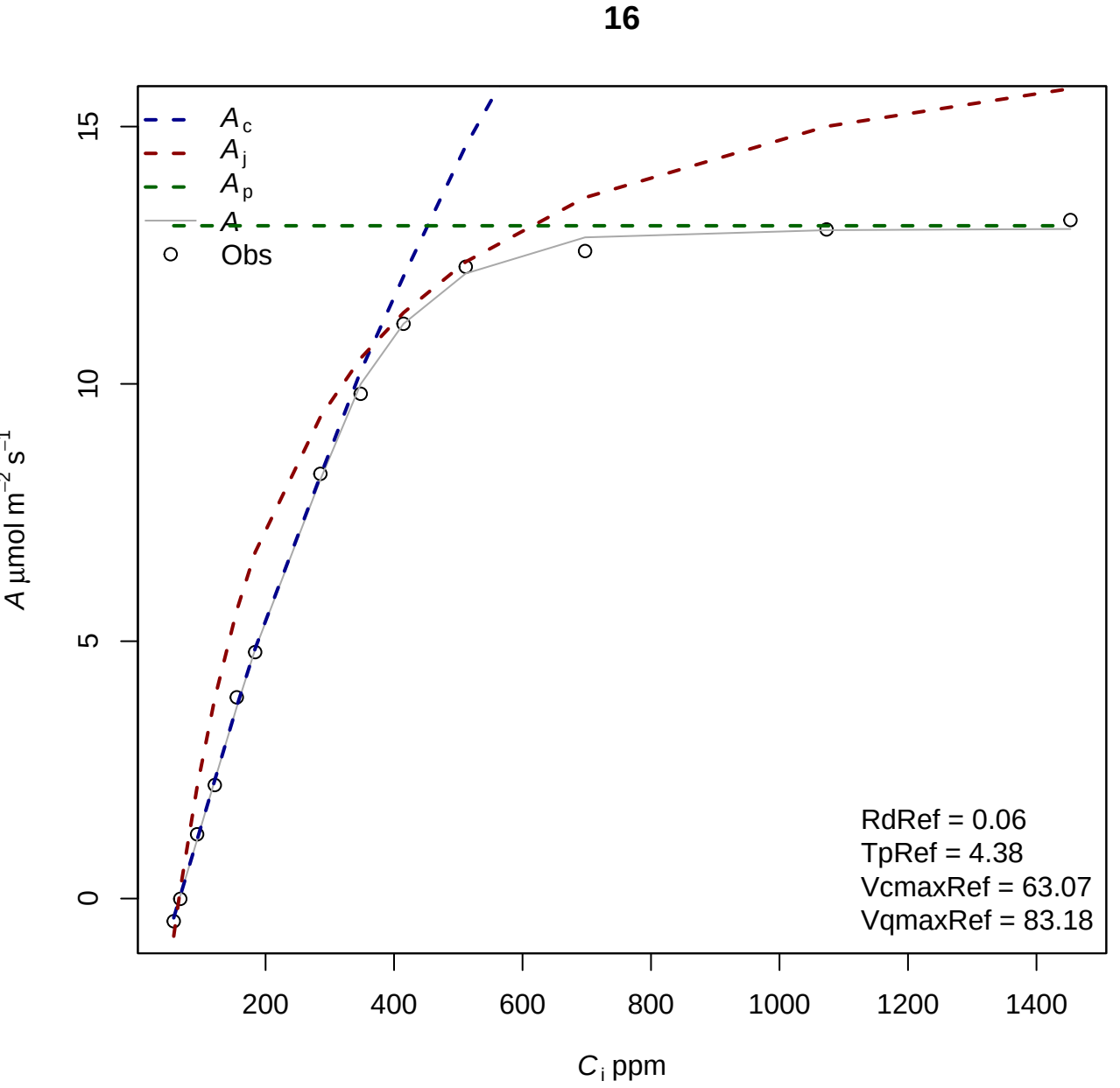

11
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.57
TpRef = 2.63
VcmaxRef = 31.28
VqmaxRef = 46.14
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
0
500
1000
1500

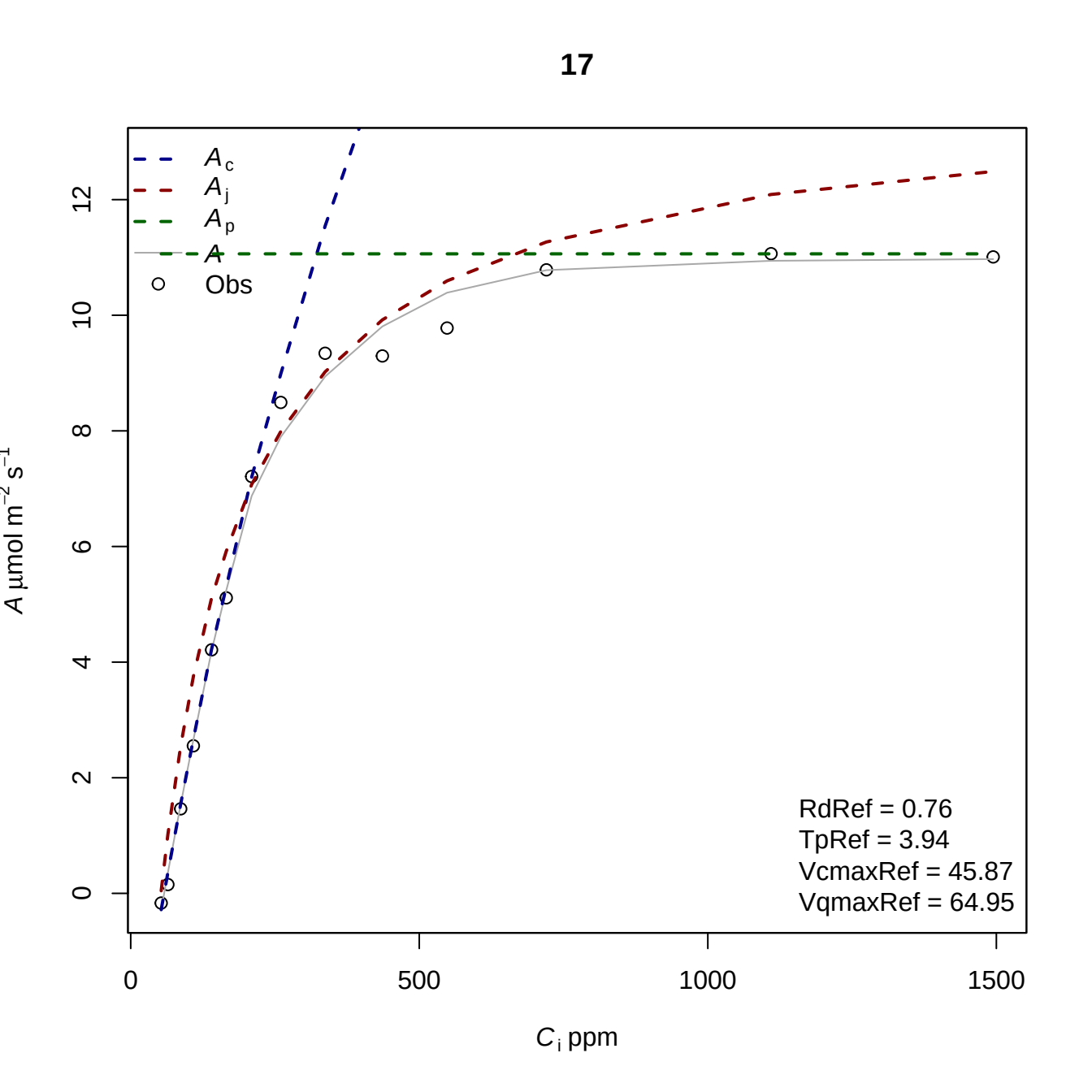

12
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.76
TpRef = 5.93
VcmaxRef = 88.65
VqmaxRef = 125.83
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

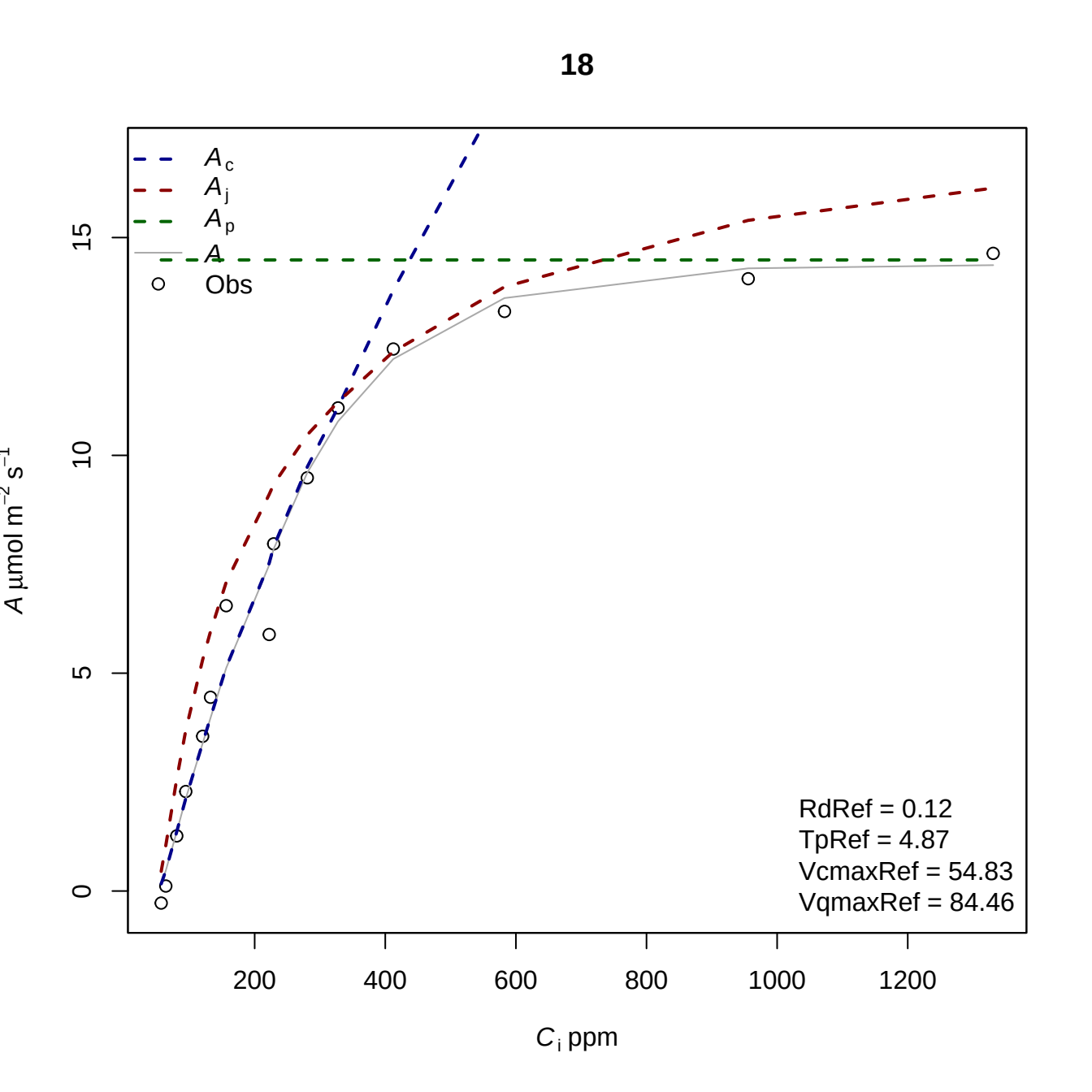

13
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.35
TpRef = 4.43
VcmaxRef = 67.9
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

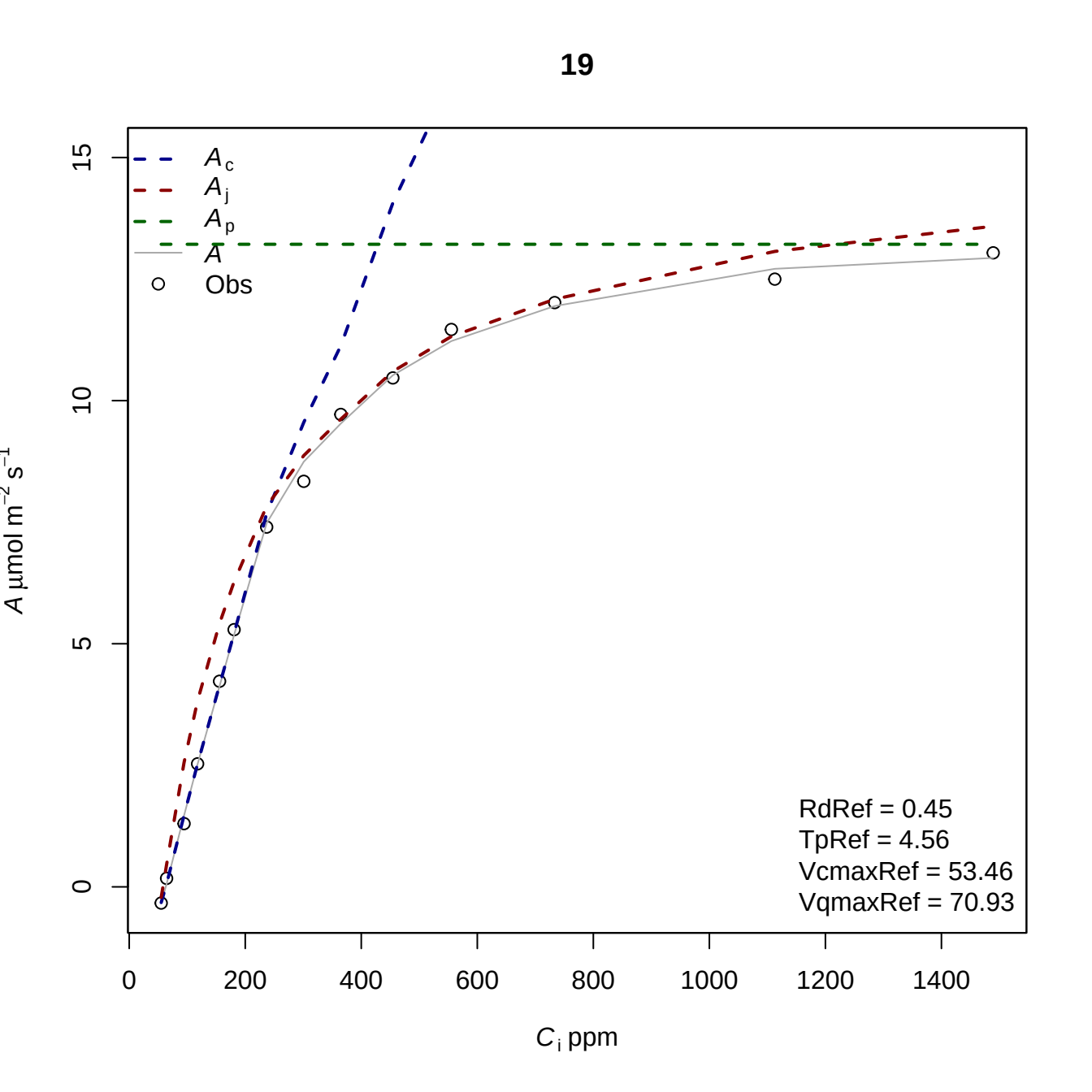

14
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.3
TpRef = 5.52
VcmaxRef = 75.37
VqmaxRef = 91.25
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200

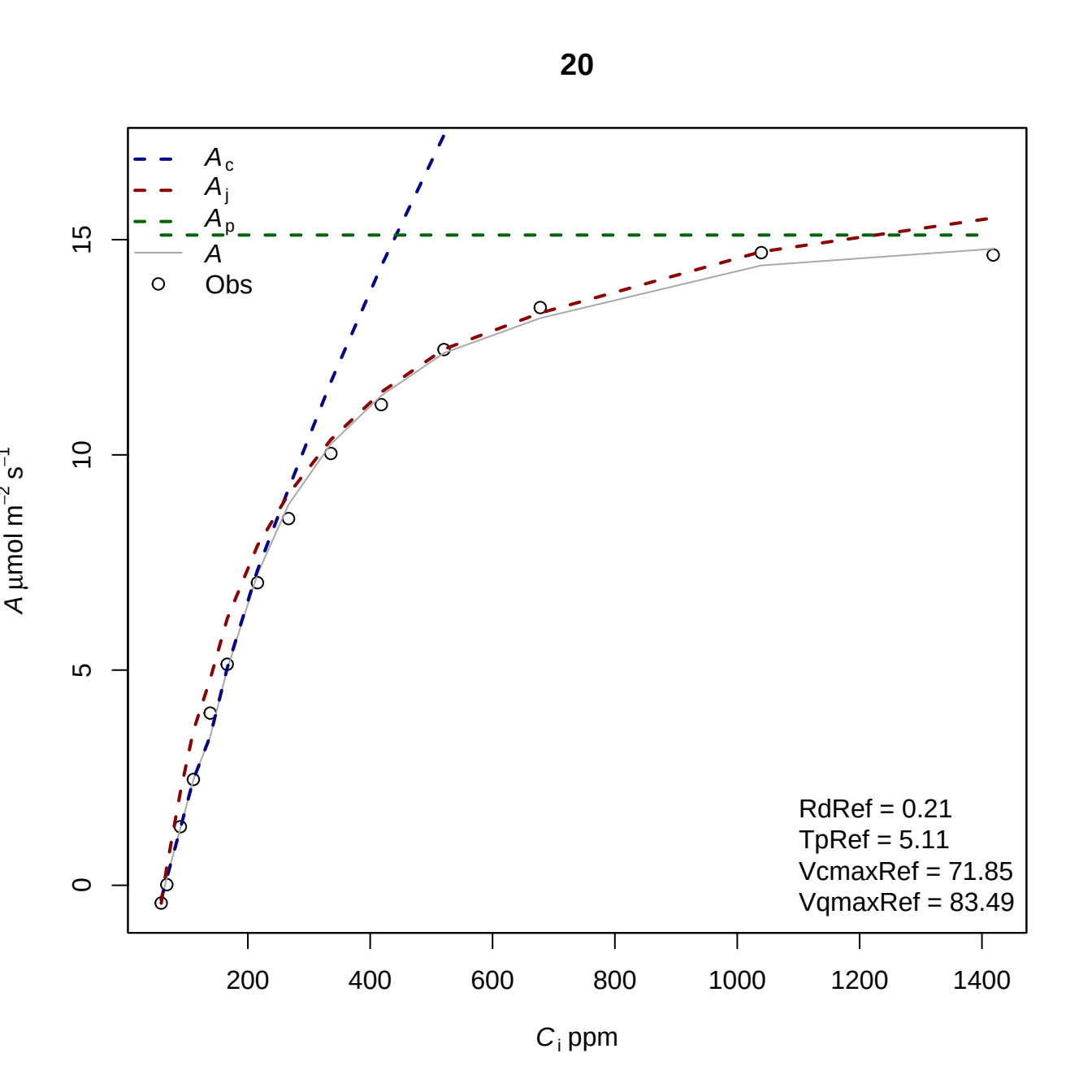

15
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.34
TpRef = 5.08
VcmaxRef = 76.15
VqmaxRef = 92.05
0
5
10
15
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

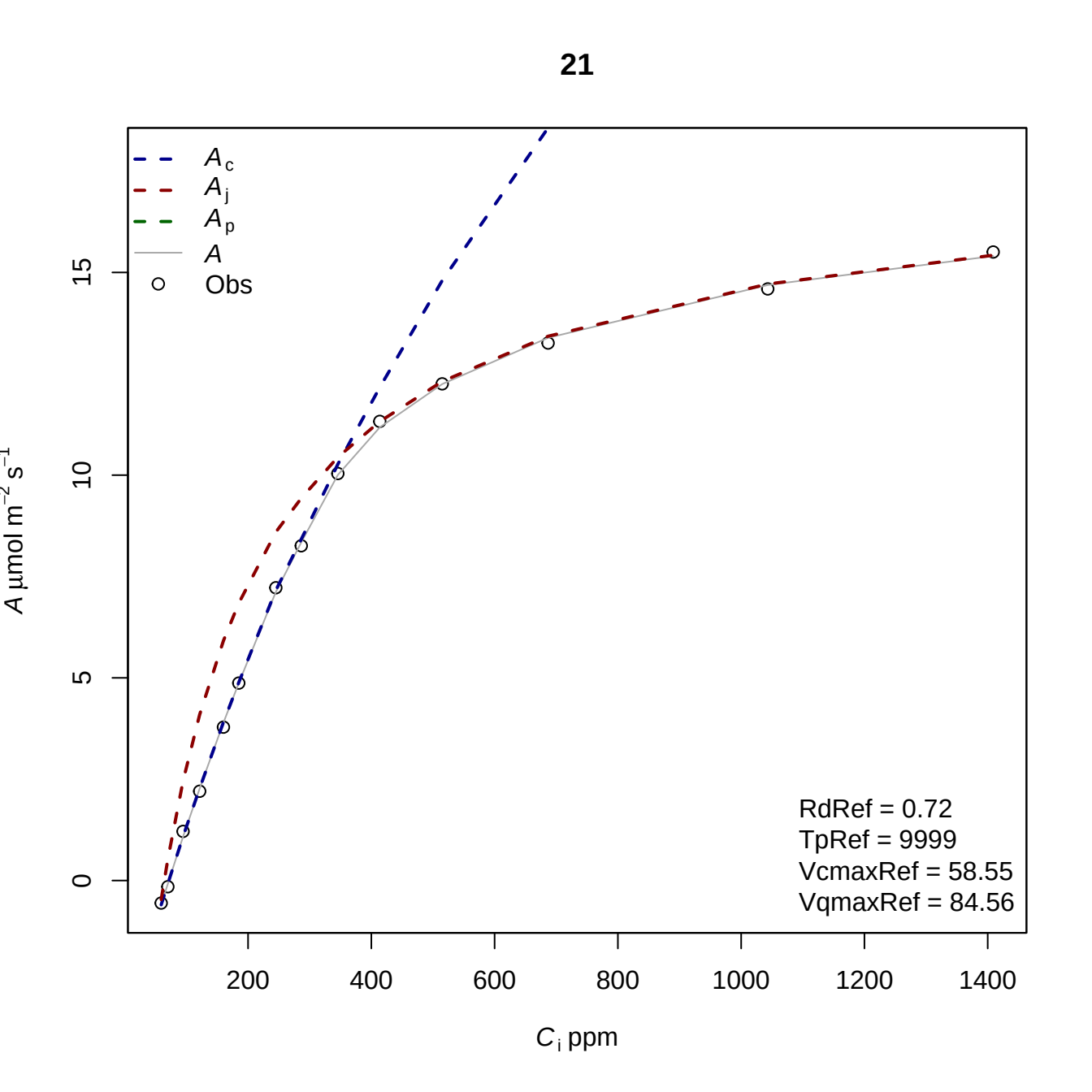

16
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.06
TpRef = 4.38
VcmaxRef = 63.07
VqmaxRef = 83.18
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

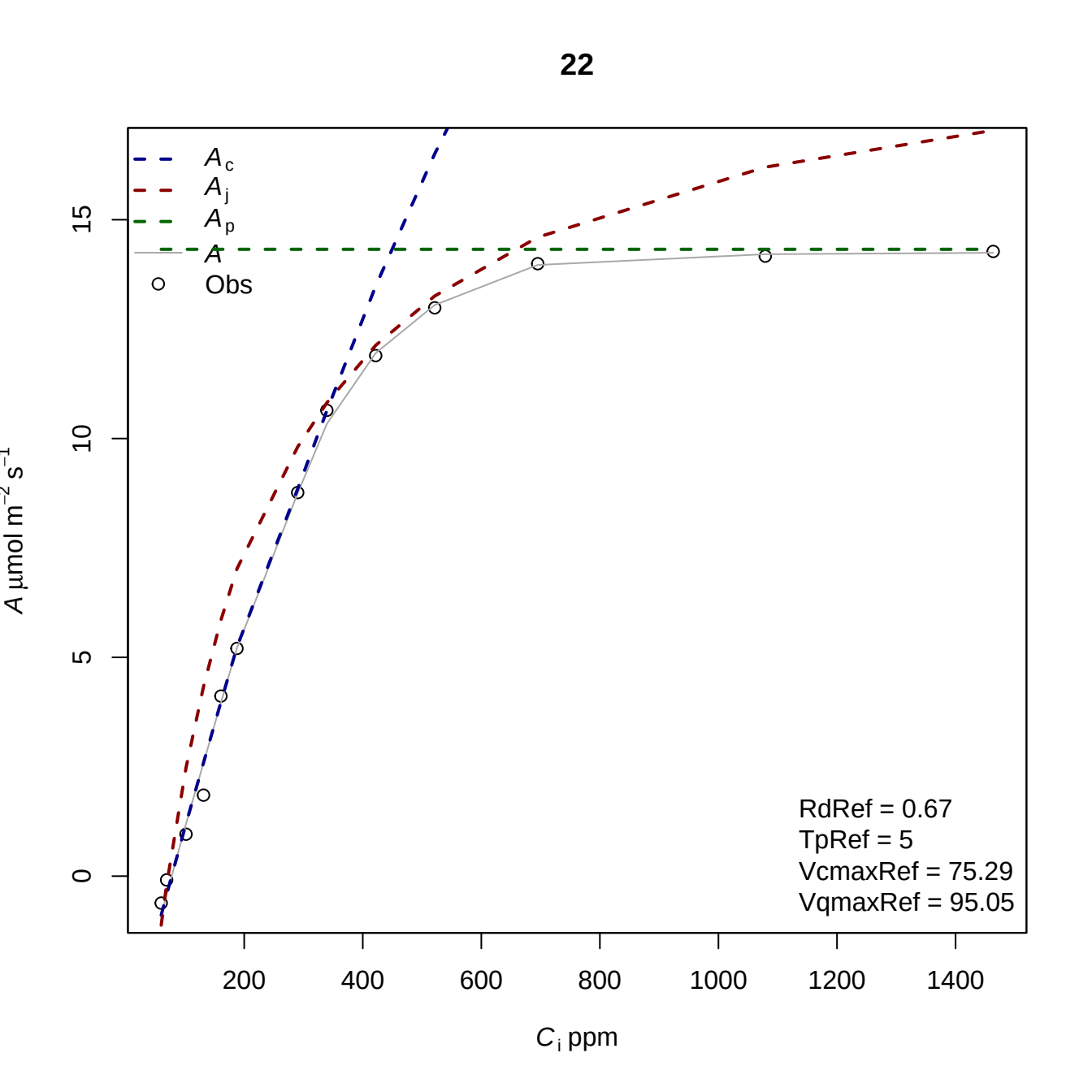

17
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.76
TpRef = 3.94
VcmaxRef = 45.87
VqmaxRef = 64.95
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

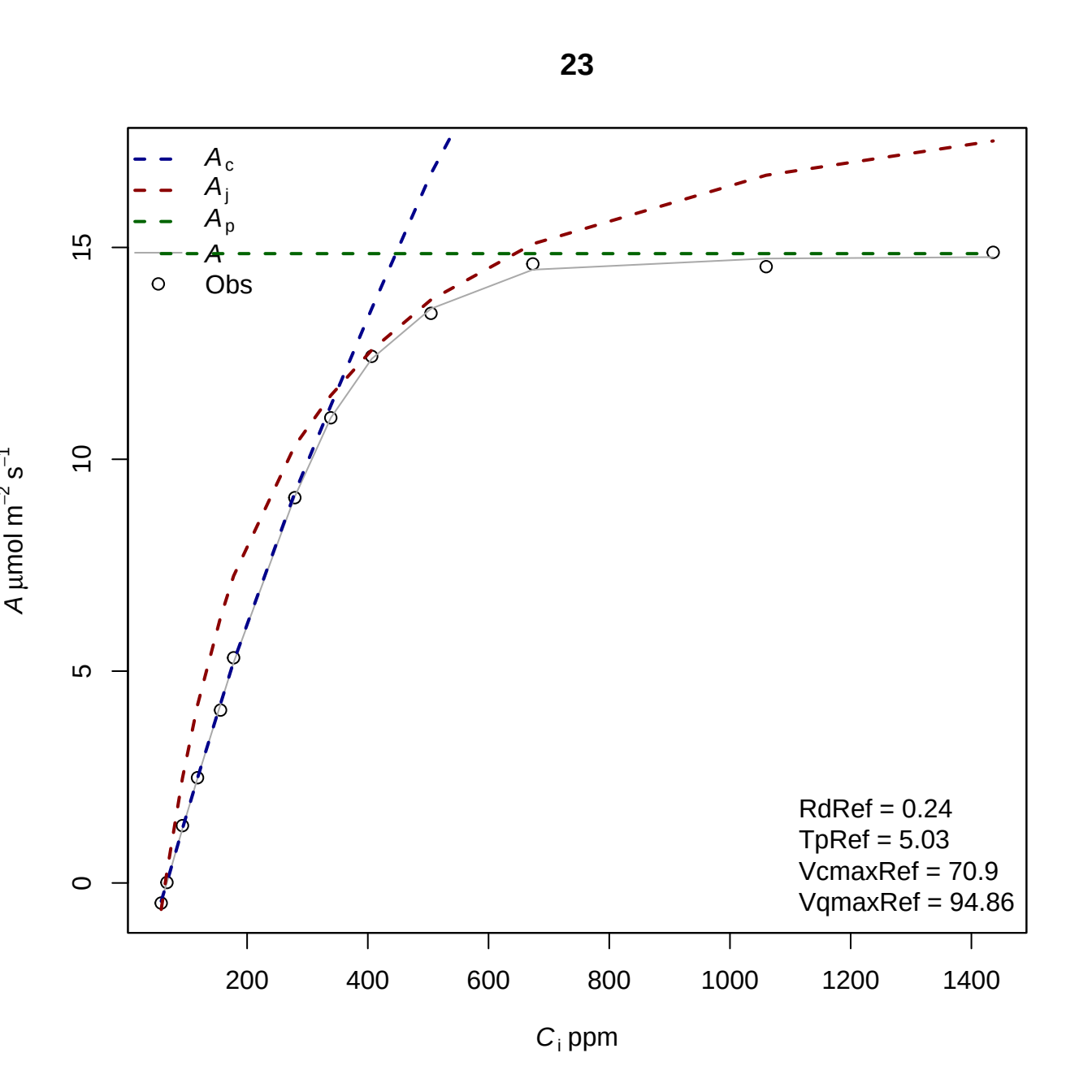

18
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.12
TpRef = 4.87
VcmaxRef = 54.83
VqmaxRef = 84.46
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200

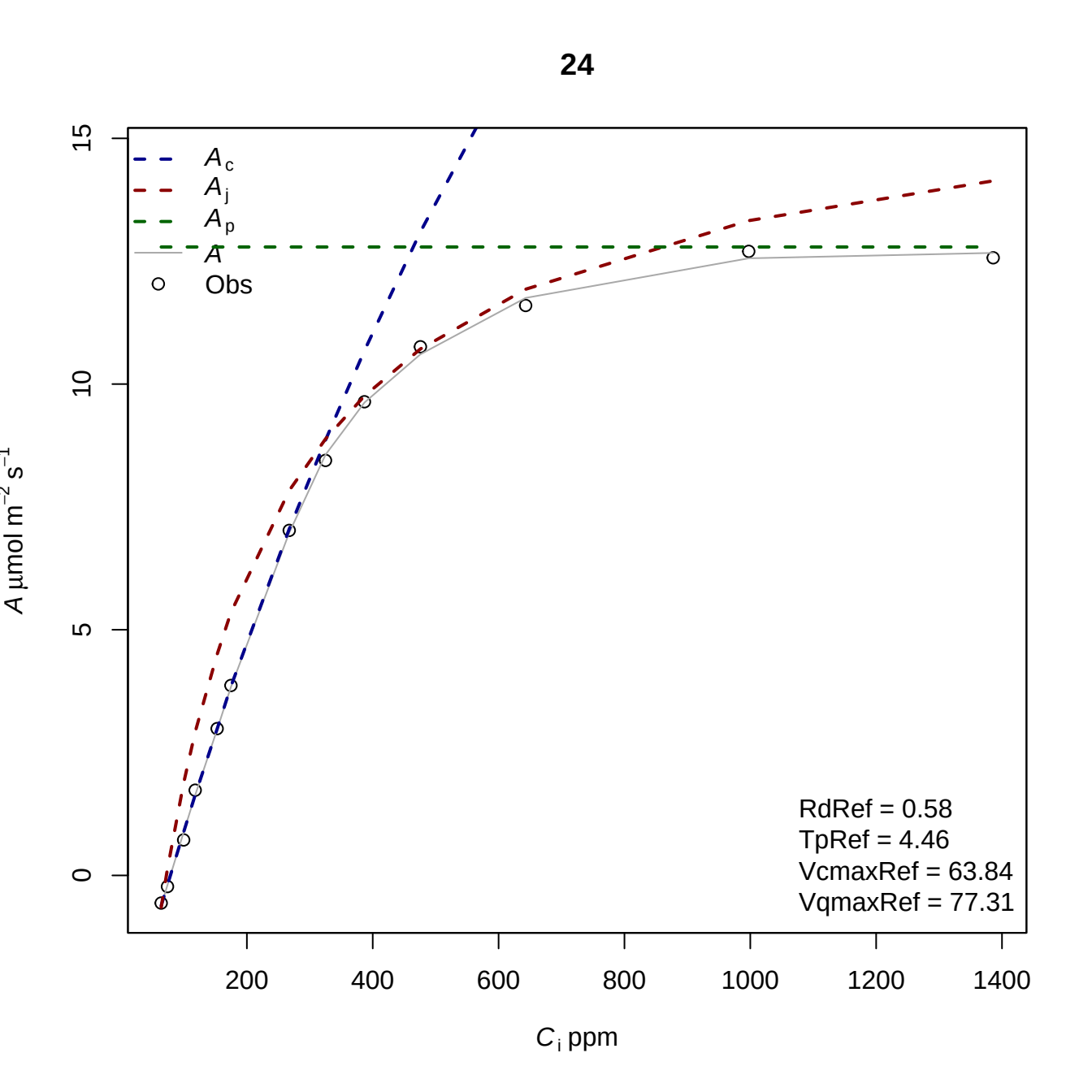

19
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.45
TpRef = 4.56
VcmaxRef = 53.46
VqmaxRef = 70.93
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

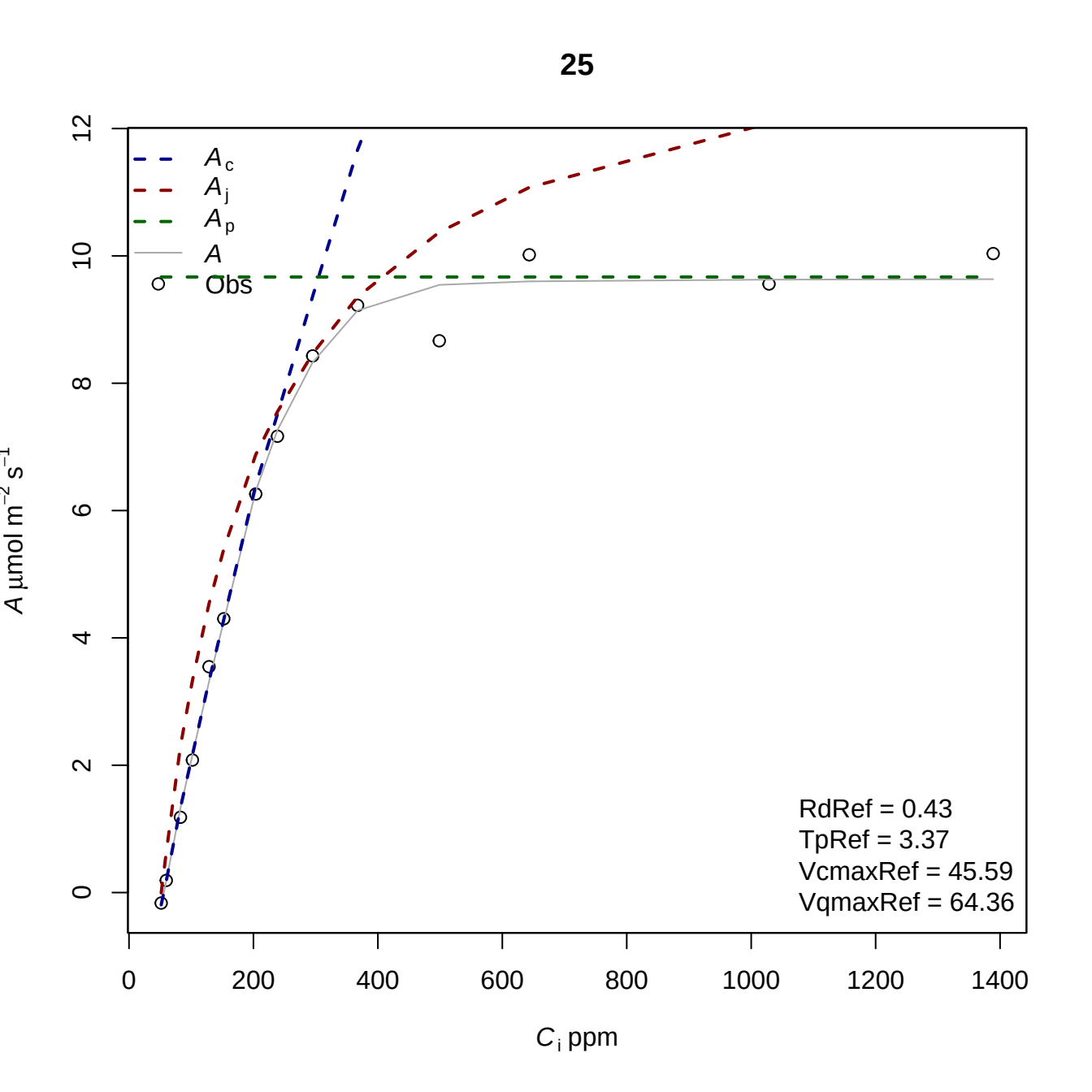

20
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.21
TpRef = 5.11
VcmaxRef = 71.85
VqmaxRef = 83.49
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

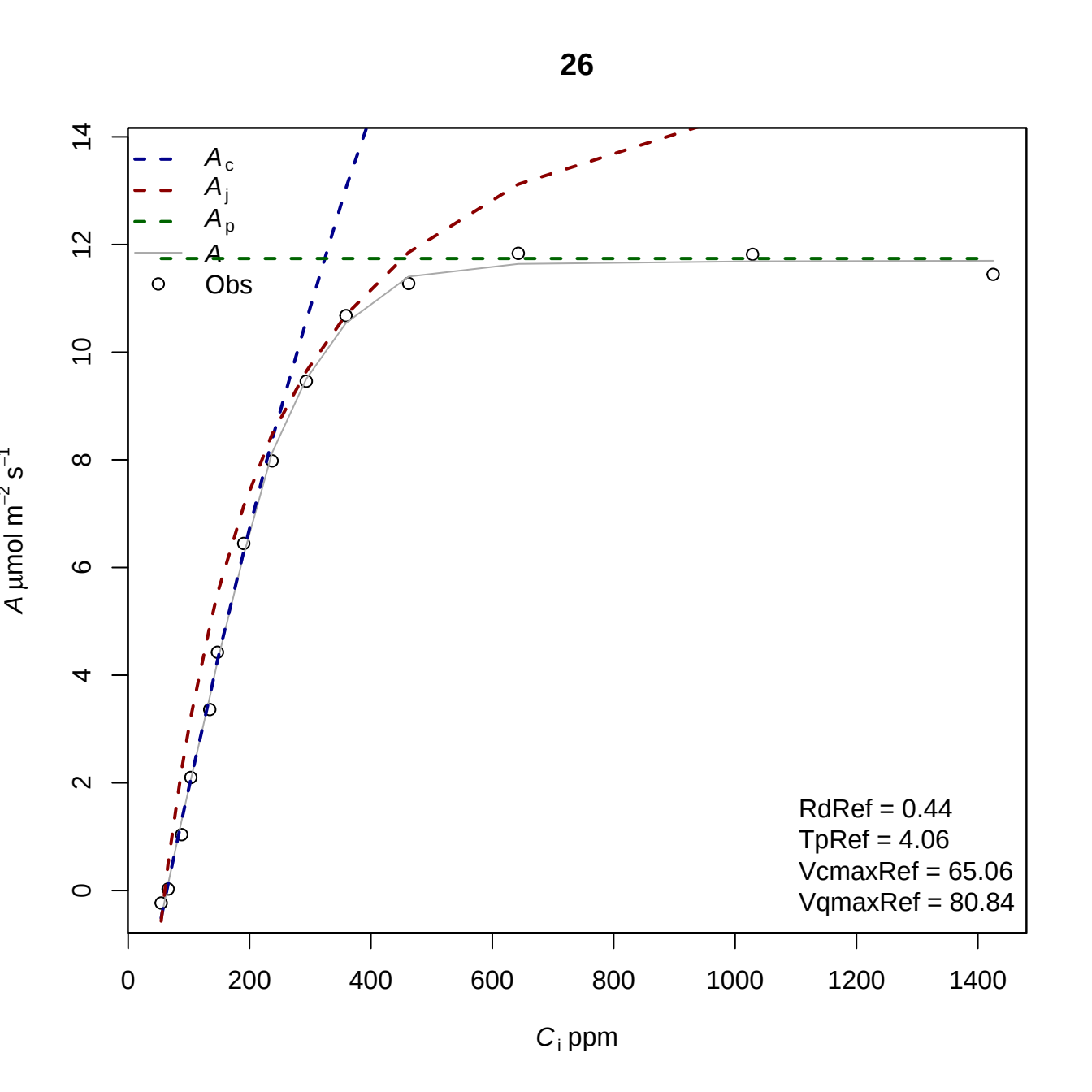

21
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.72
TpRef = 9999
VcmaxRef = 58.55
VqmaxRef = 84.56
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

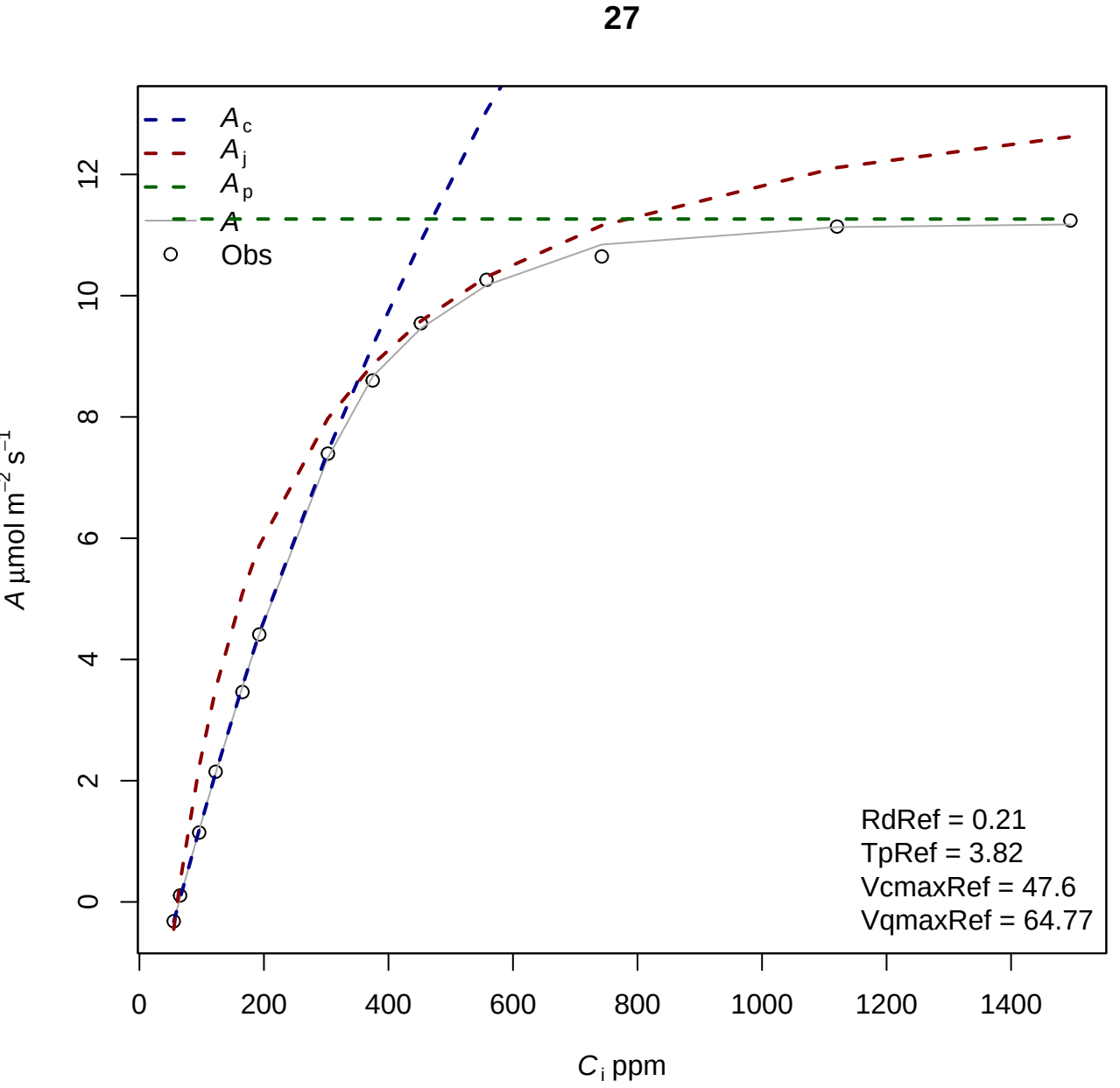

22
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.67
TpRef = 5
VcmaxRef = 75.29
VqmaxRef = 95.05
0
5
10
15
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

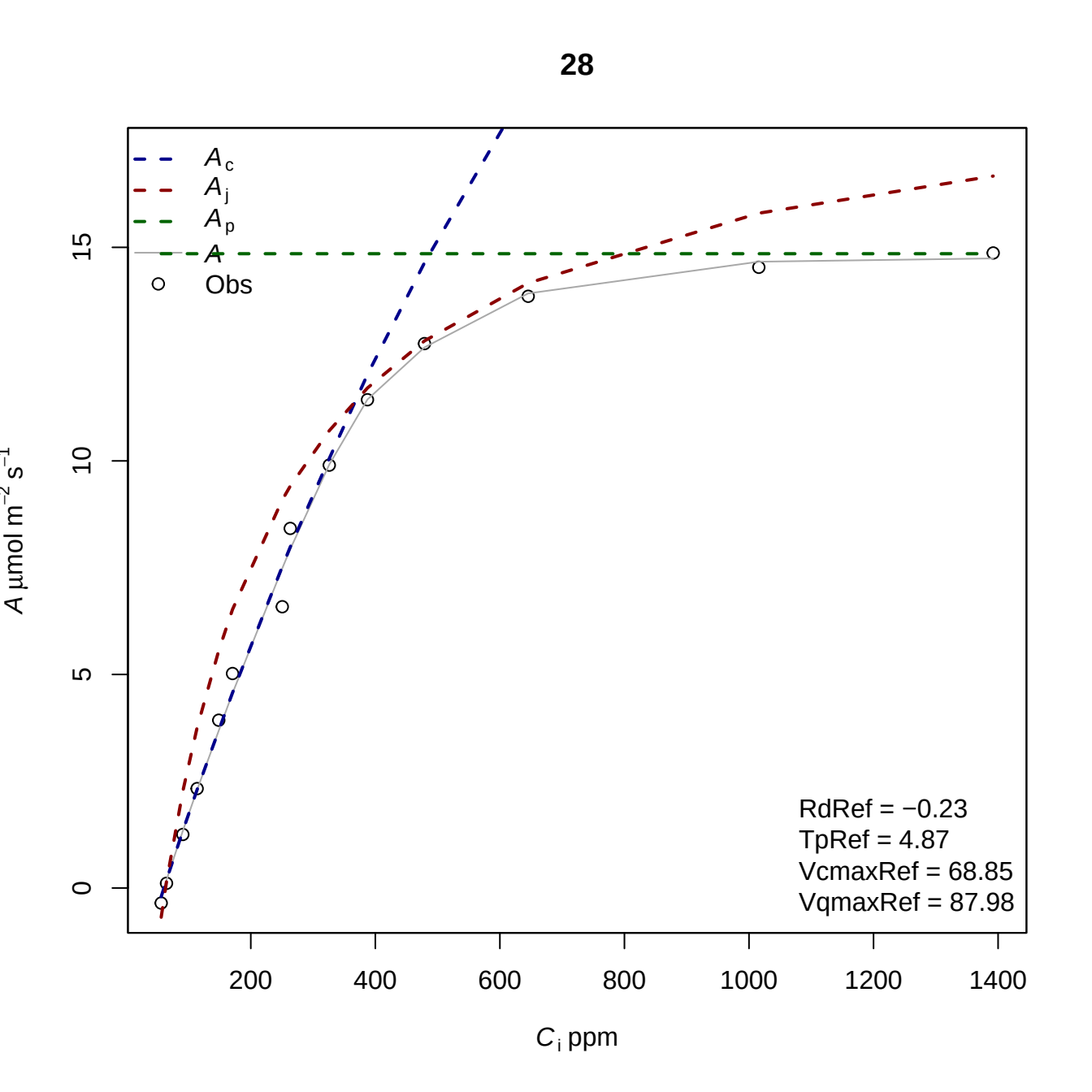

23
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.24
TpRef = 5.03
VcmaxRef = 70.9
VqmaxRef = 94.86
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

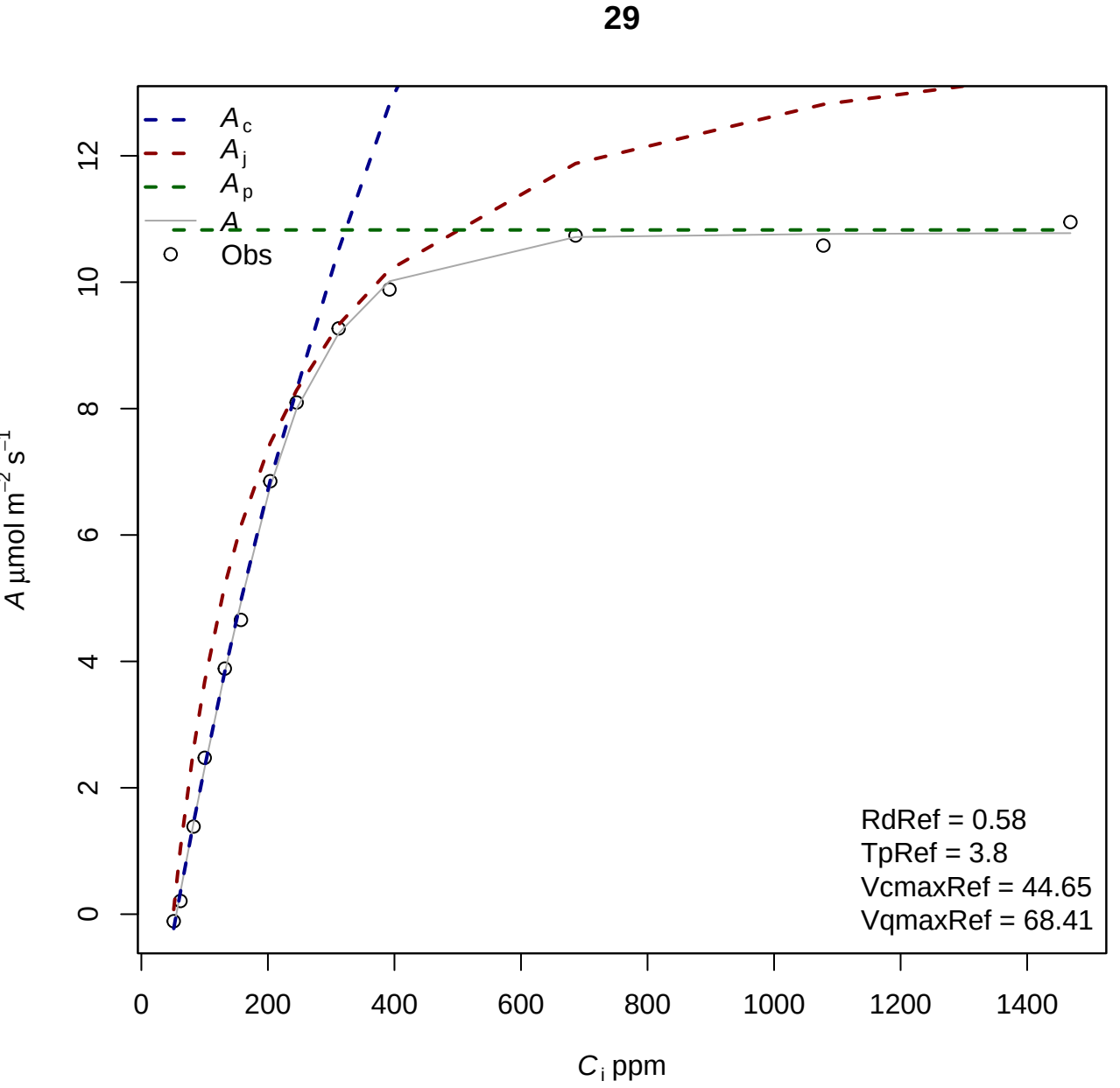

24
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.58
TpRef = 4.46
VcmaxRef = 63.84
VqmaxRef = 77.31
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

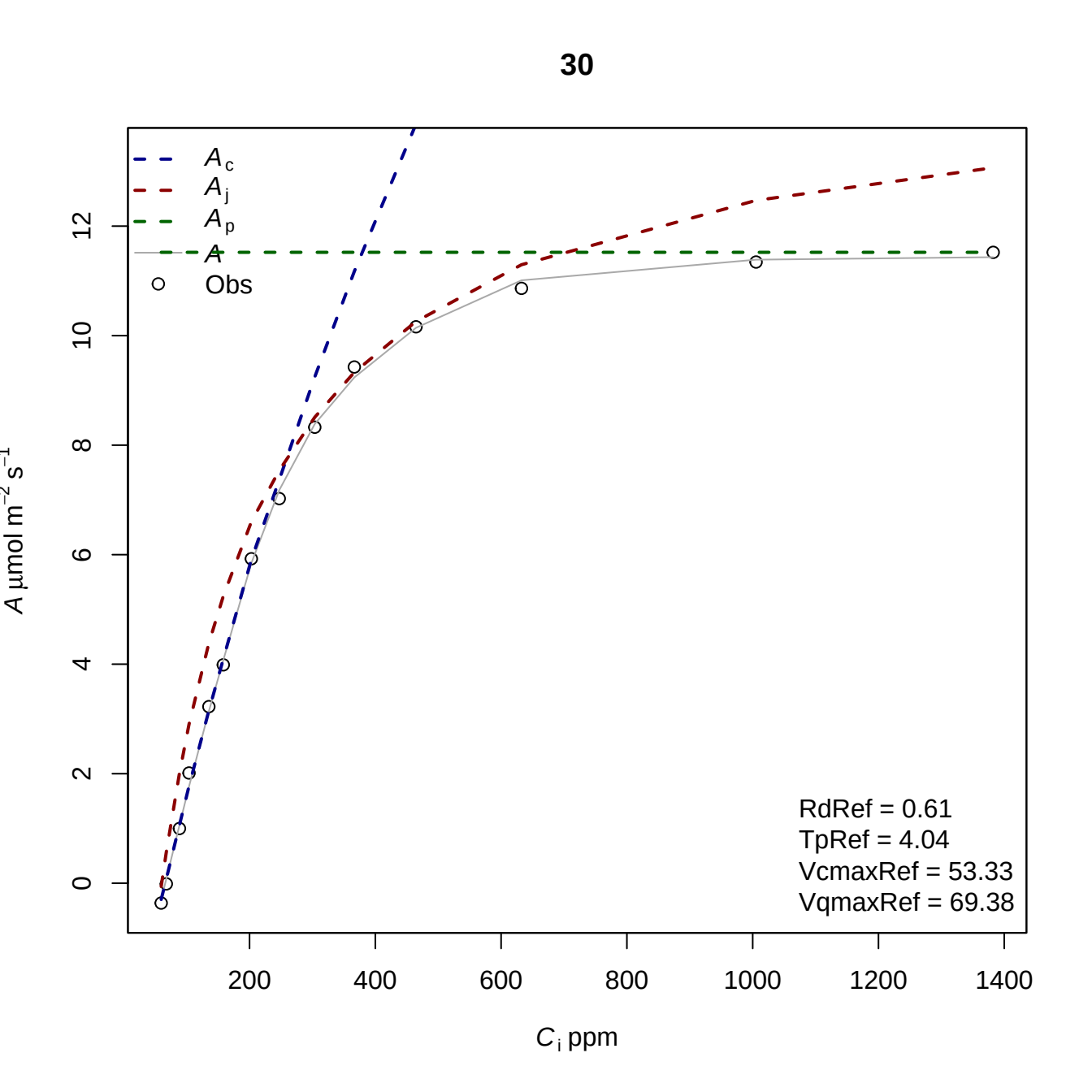

25
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.43
TpRef = 3.37
VcmaxRef = 45.59
VqmaxRef = 64.36
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

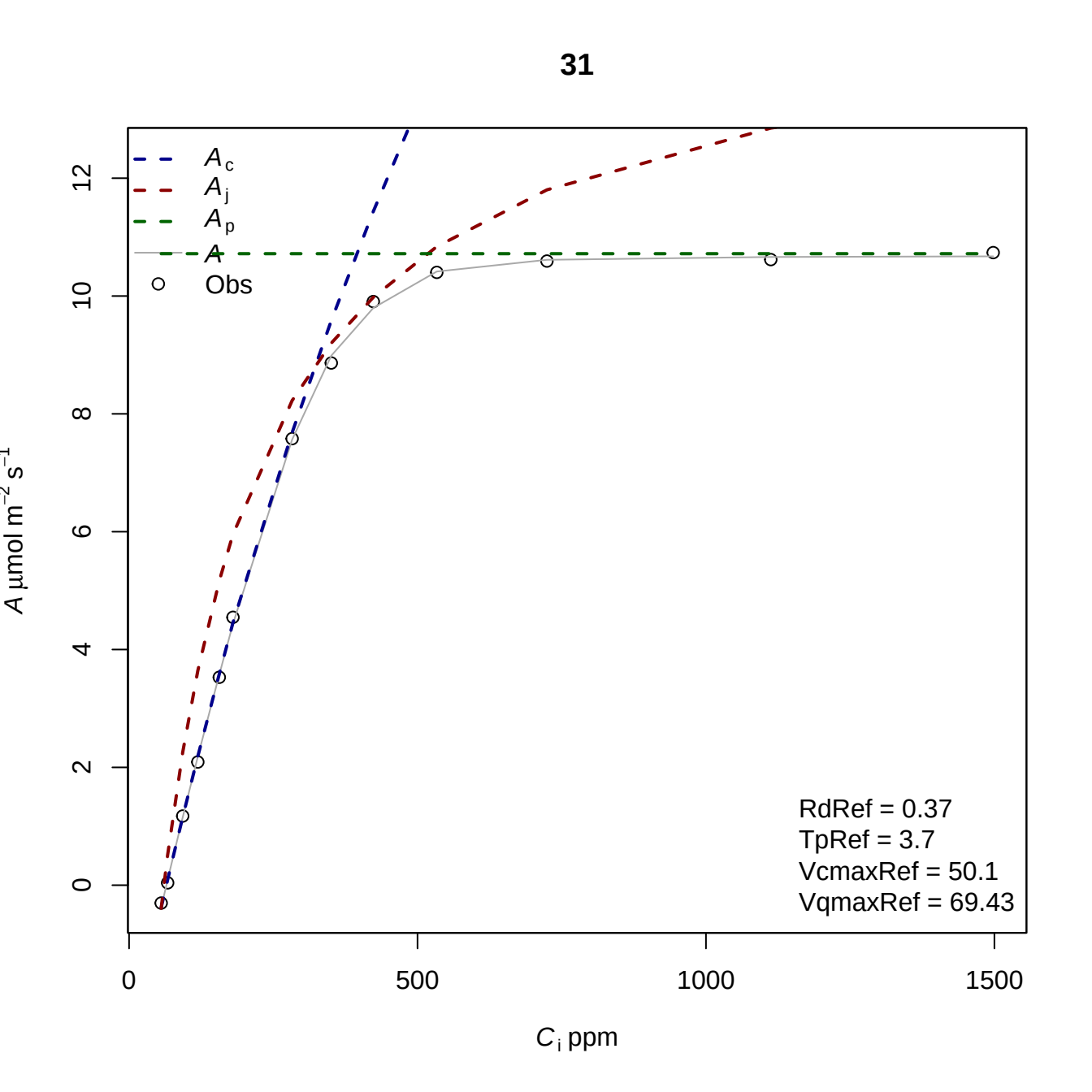

26
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.44
TpRef = 4.06
VcmaxRef = 65.06
VqmaxRef = 80.84
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
1400

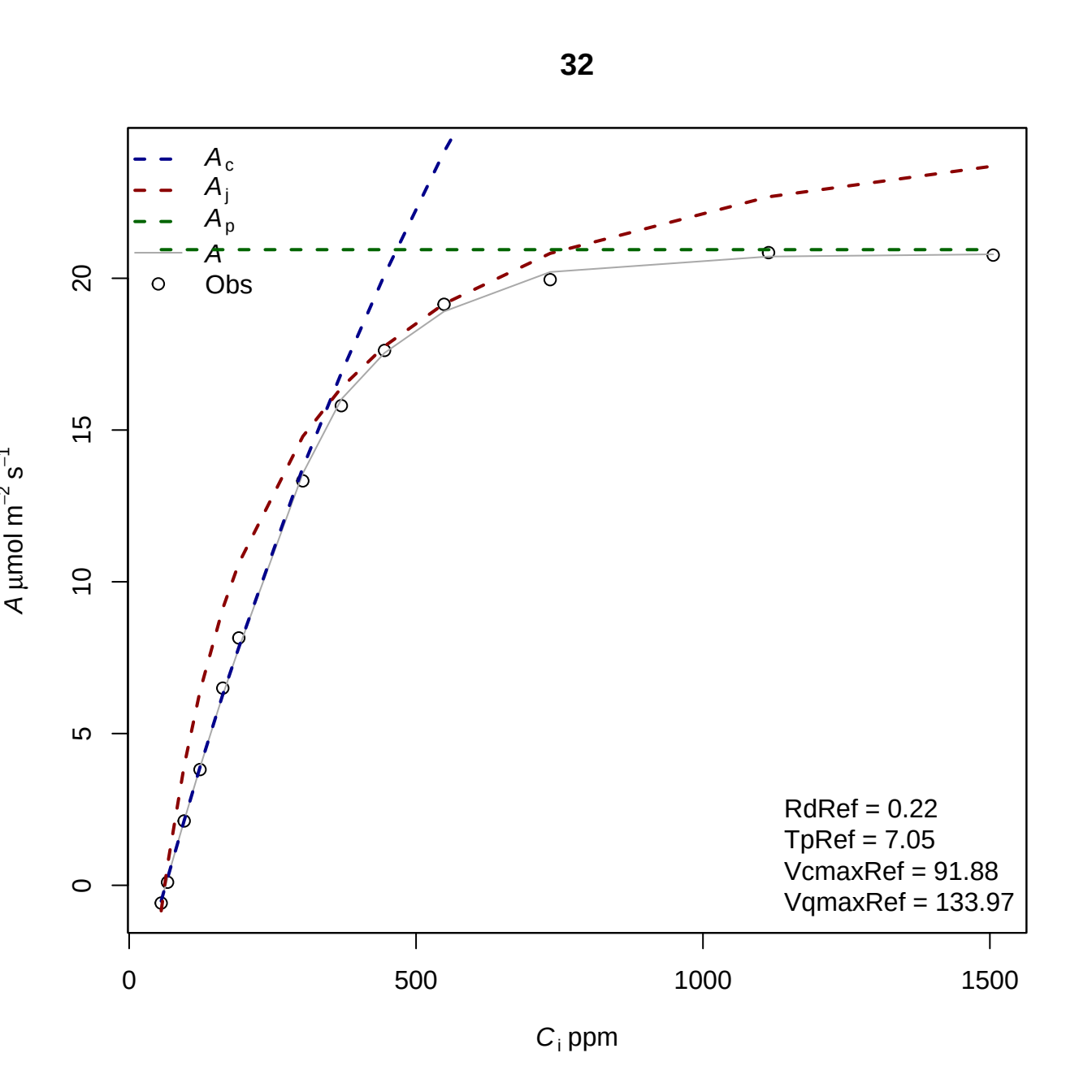

27
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.21
TpRef = 3.82
VcmaxRef = 47.6
VqmaxRef = 64.77
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

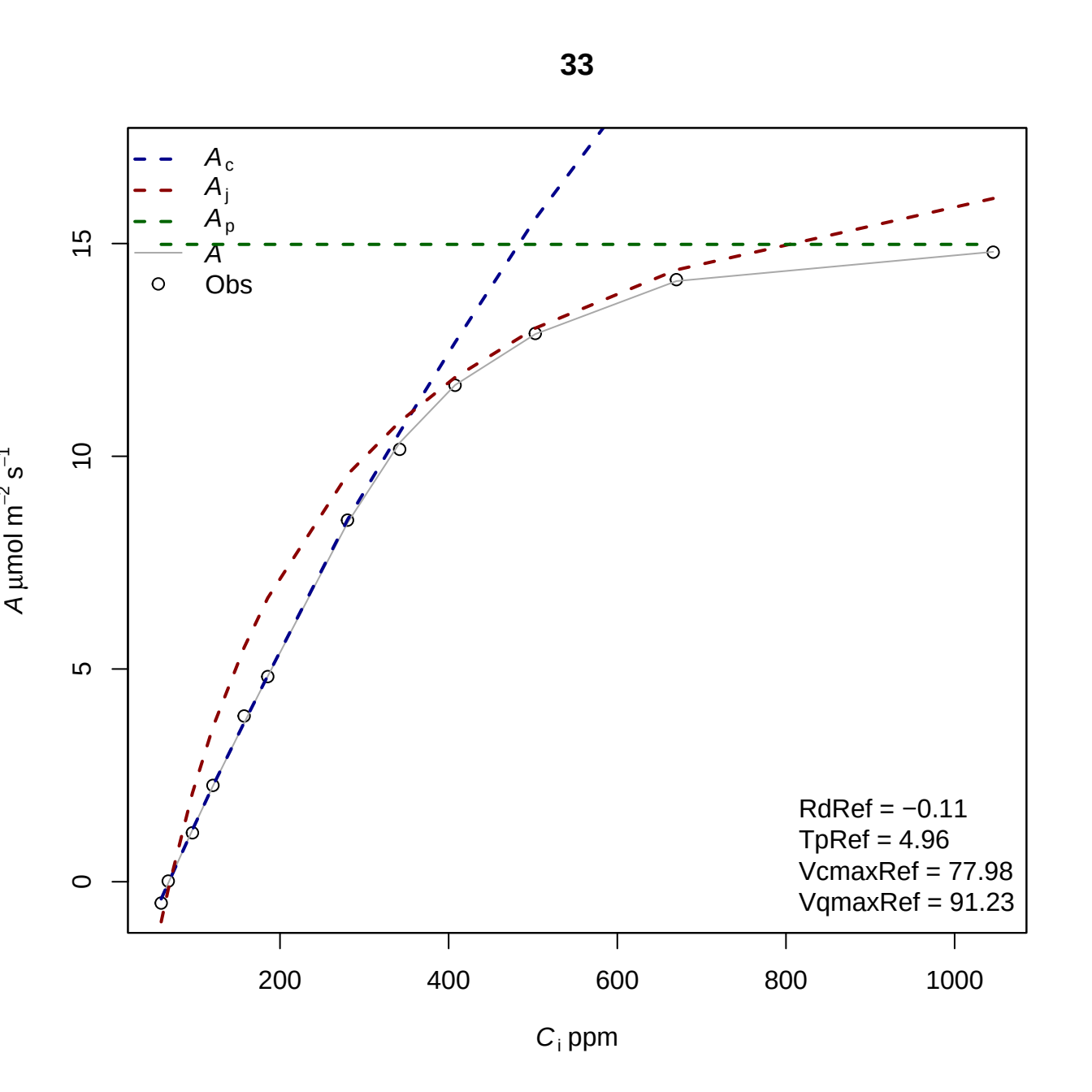

28
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.23
TpRef = 4.87
VcmaxRef = 68.85
VqmaxRef = 87.98
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

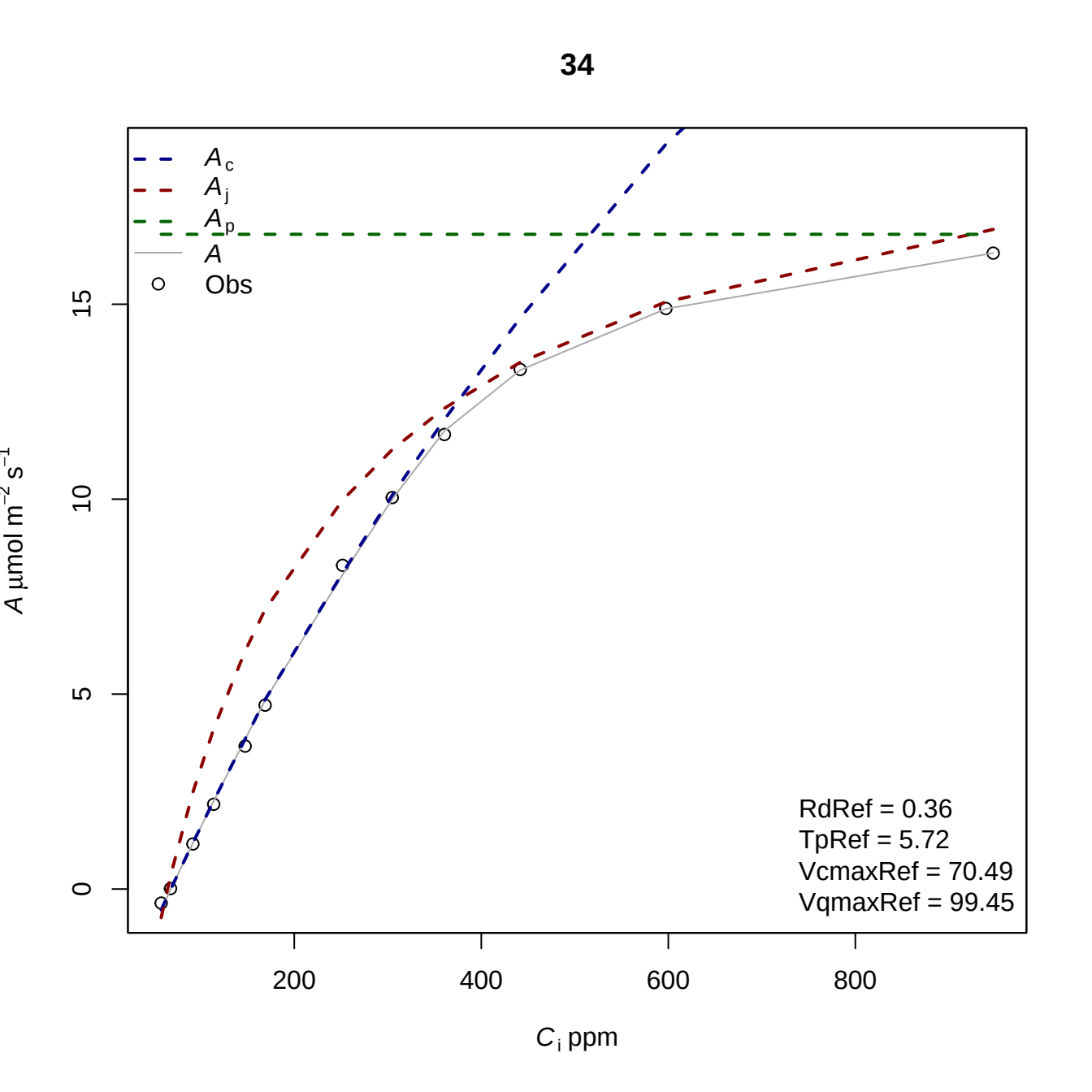

29
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.58
TpRef = 3.8
VcmaxRef = 44.65
VqmaxRef = 68.41
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

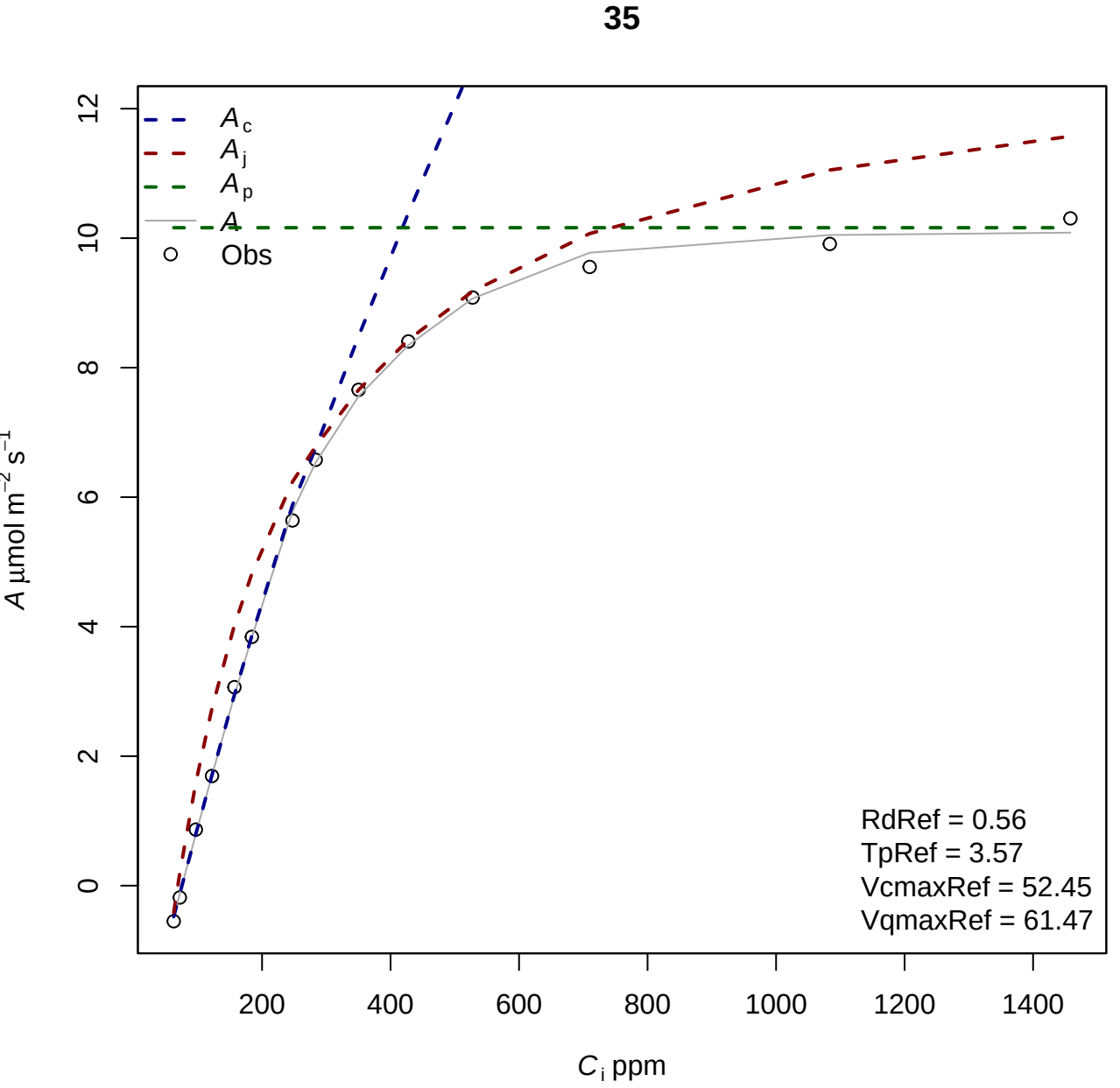

30
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.61
TpRef = 4.04
VcmaxRef = 53.33
VqmaxRef = 69.38
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

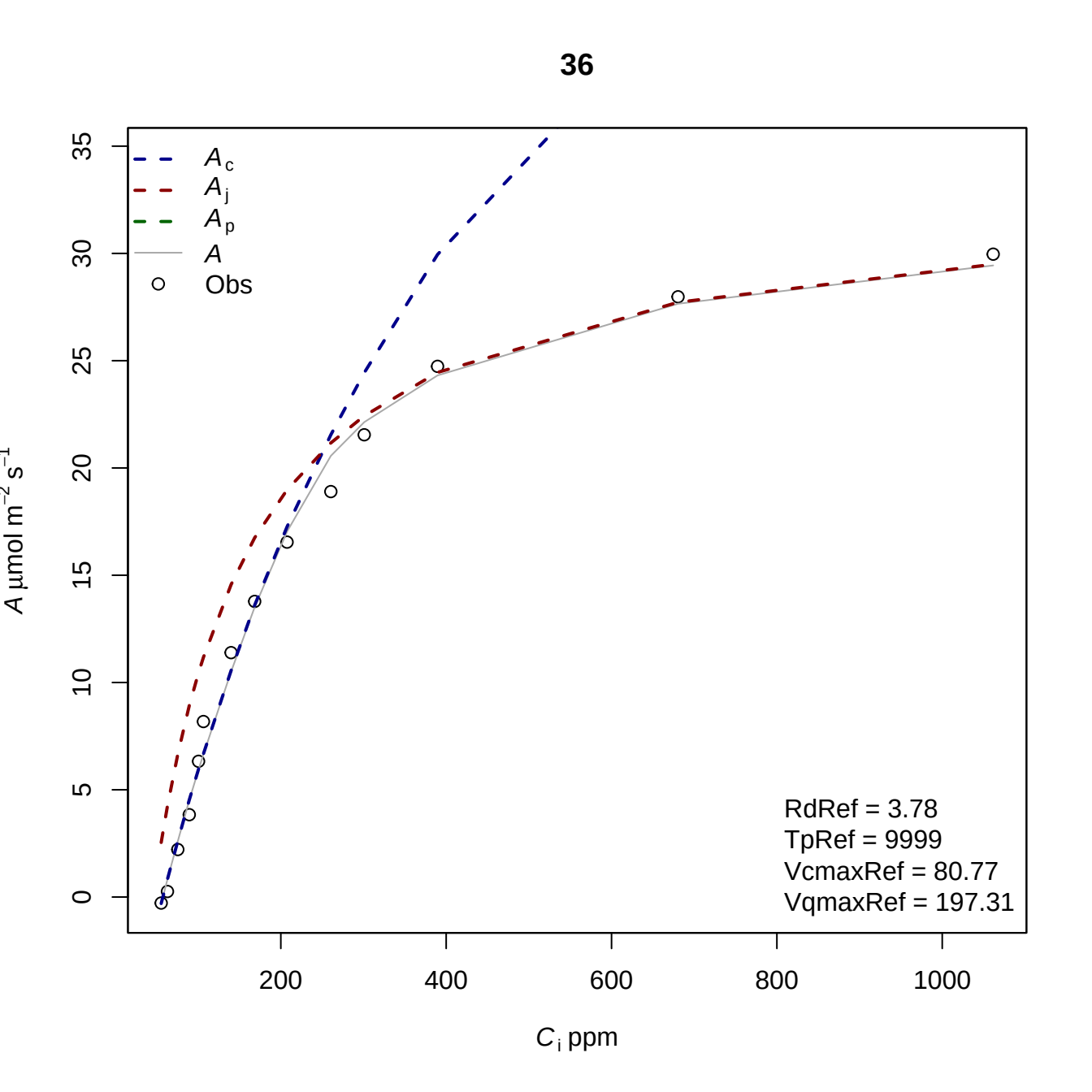

31
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.37
TpRef = 3.7
VcmaxRef = 50.1
VqmaxRef = 69.43
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

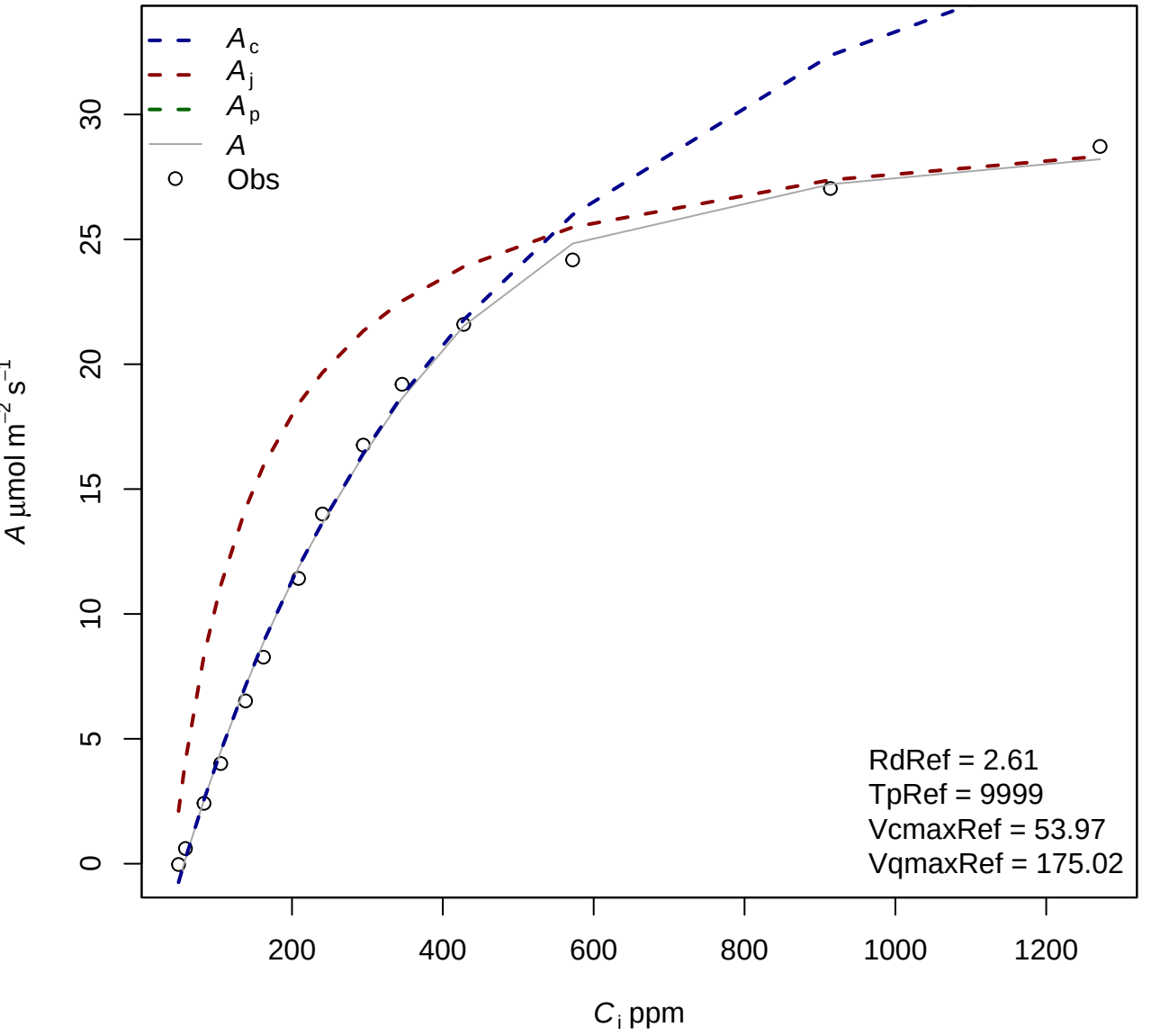

32
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.22
TpRef = 7.05
VcmaxRef = 91.88
VqmaxRef = 133.97
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

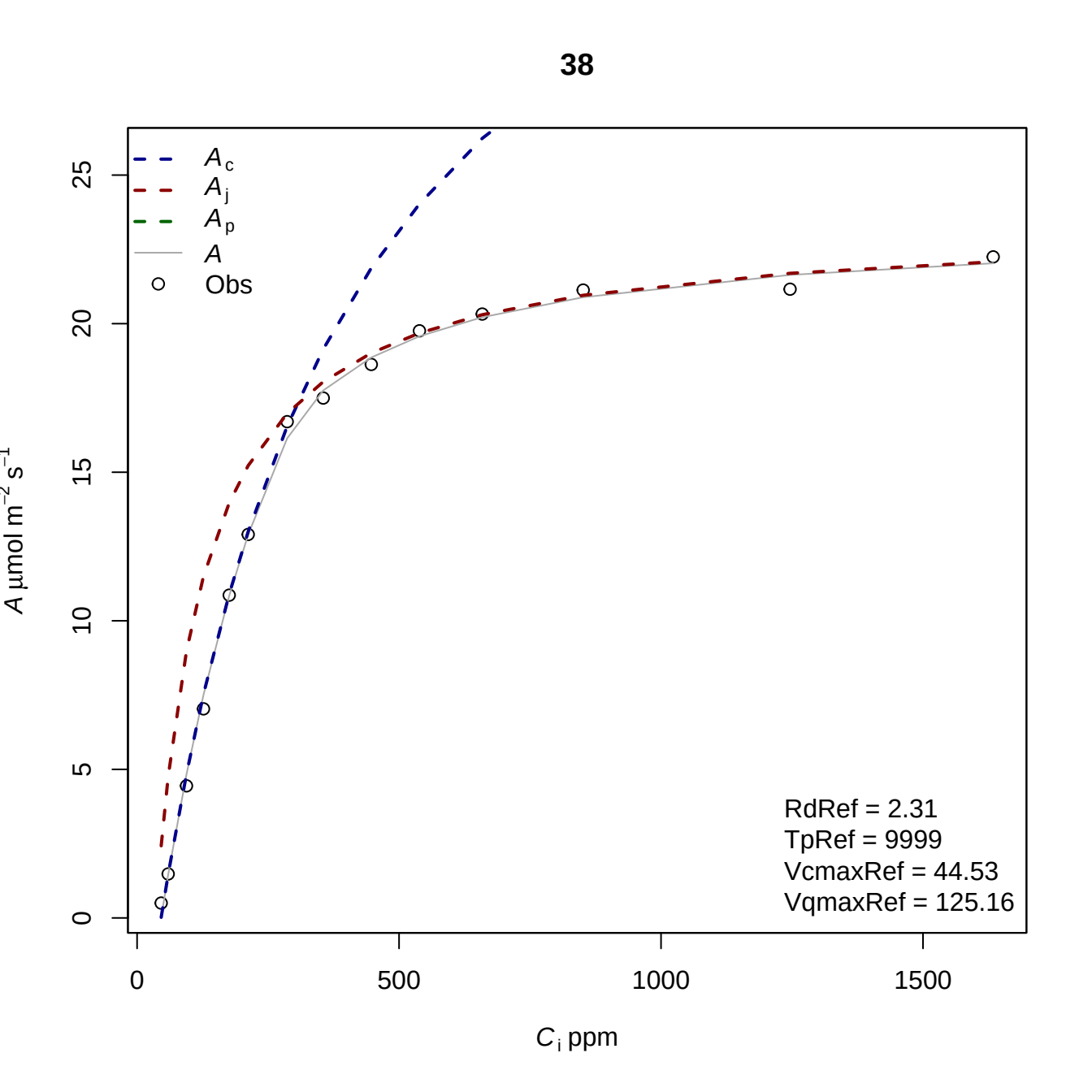

33
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.11
TpRef = 4.96
VcmaxRef = 77.98
VqmaxRef = 91.23
0
5
10
15
200
400
600
800
1000
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

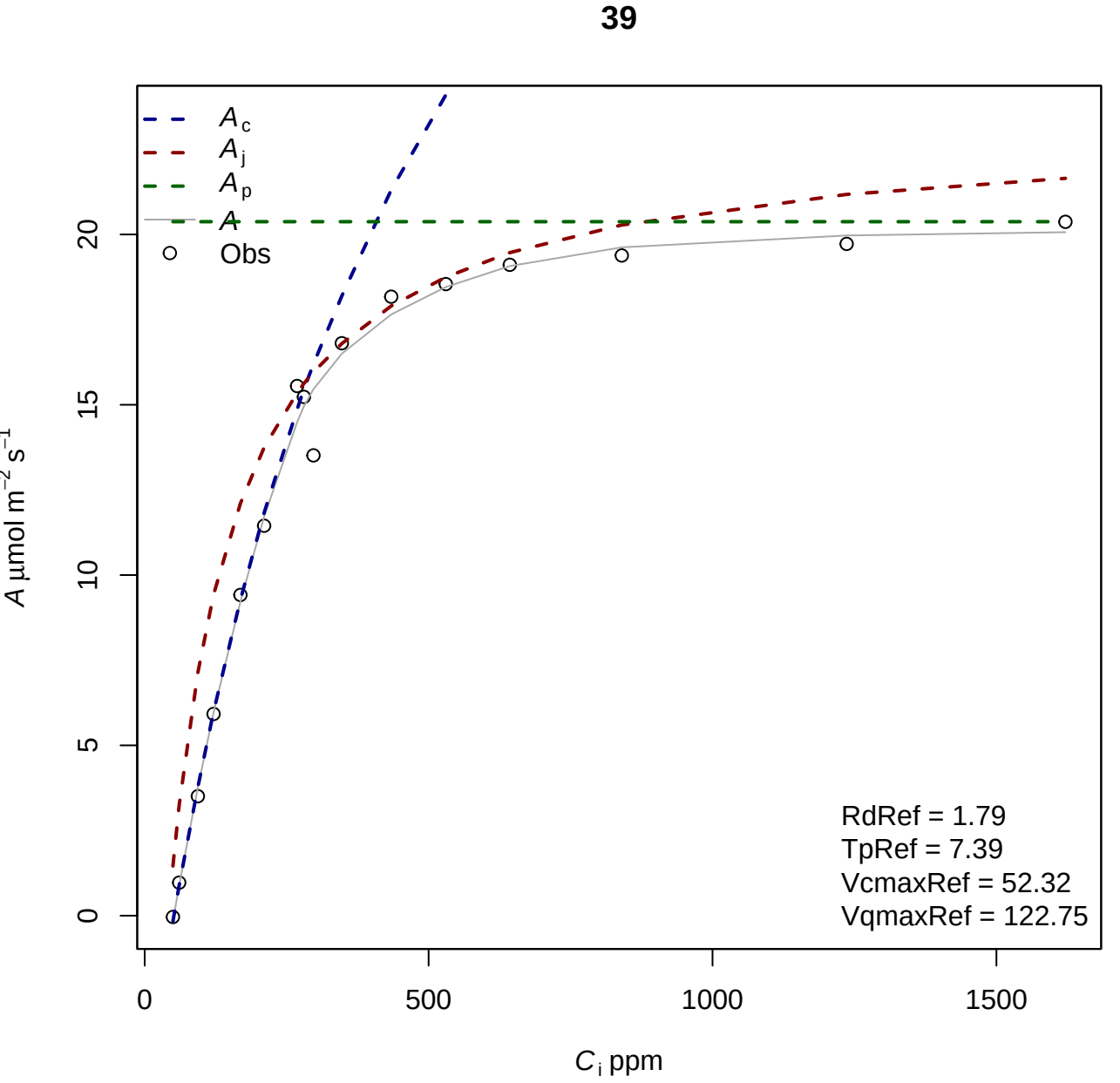

34
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.36
TpRef = 5.72
VcmaxRef = 70.49
VqmaxRef = 99.45
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800

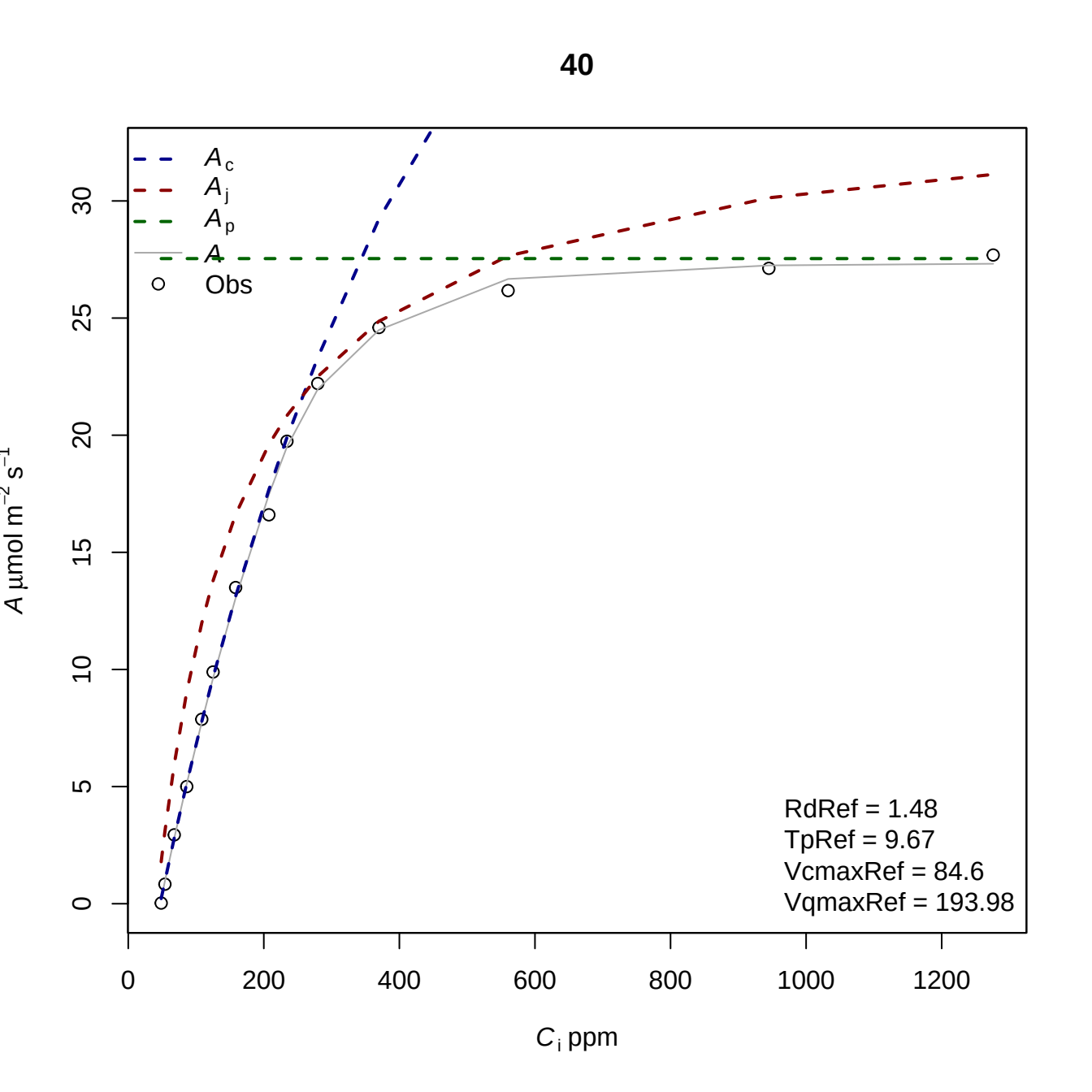

35
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.56
TpRef = 3.57
VcmaxRef = 52.45
VqmaxRef = 61.47
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

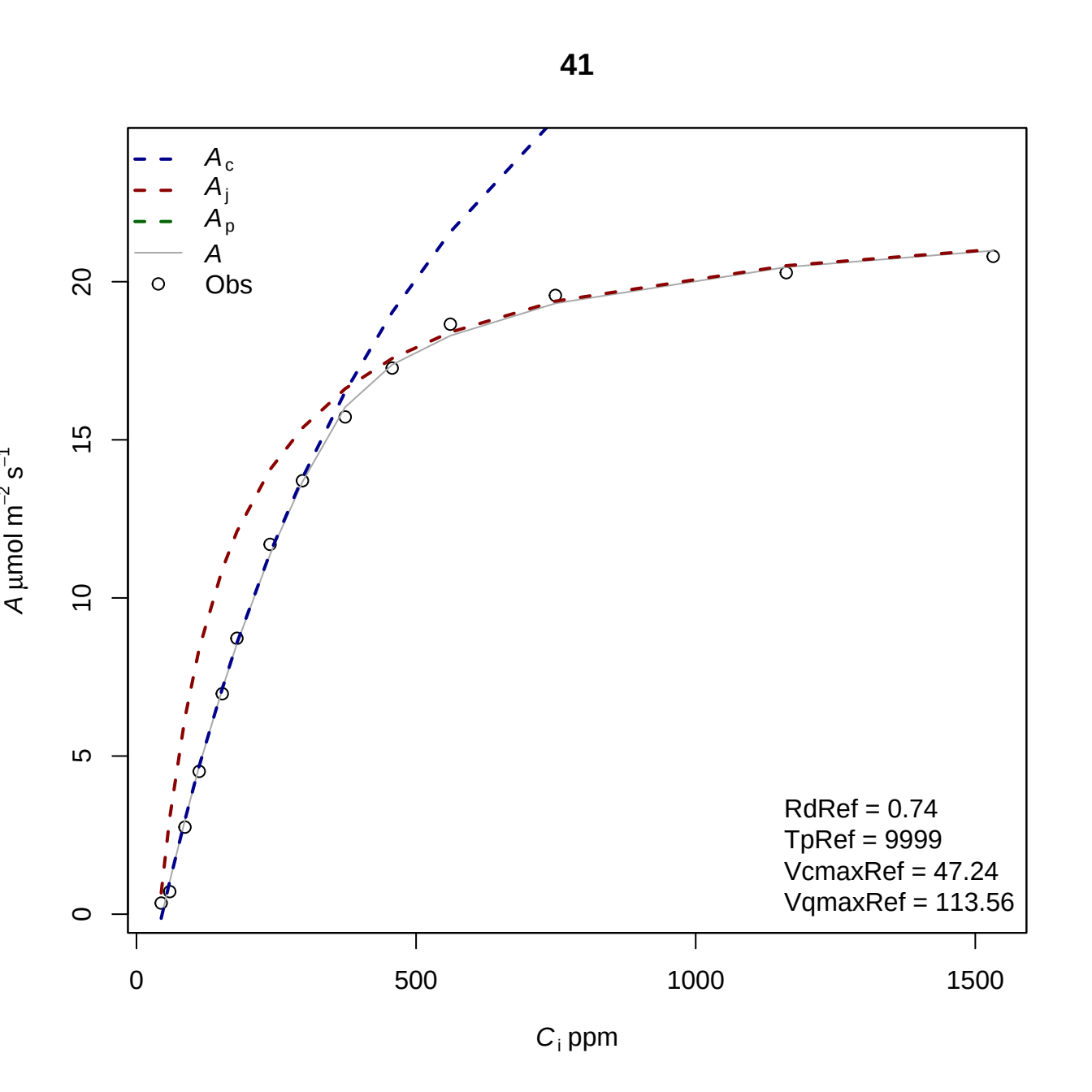

36
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.78
TpRef = 9999
VcmaxRef = 80.77
VqmaxRef = 197.31
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
200
400
600
800
1000

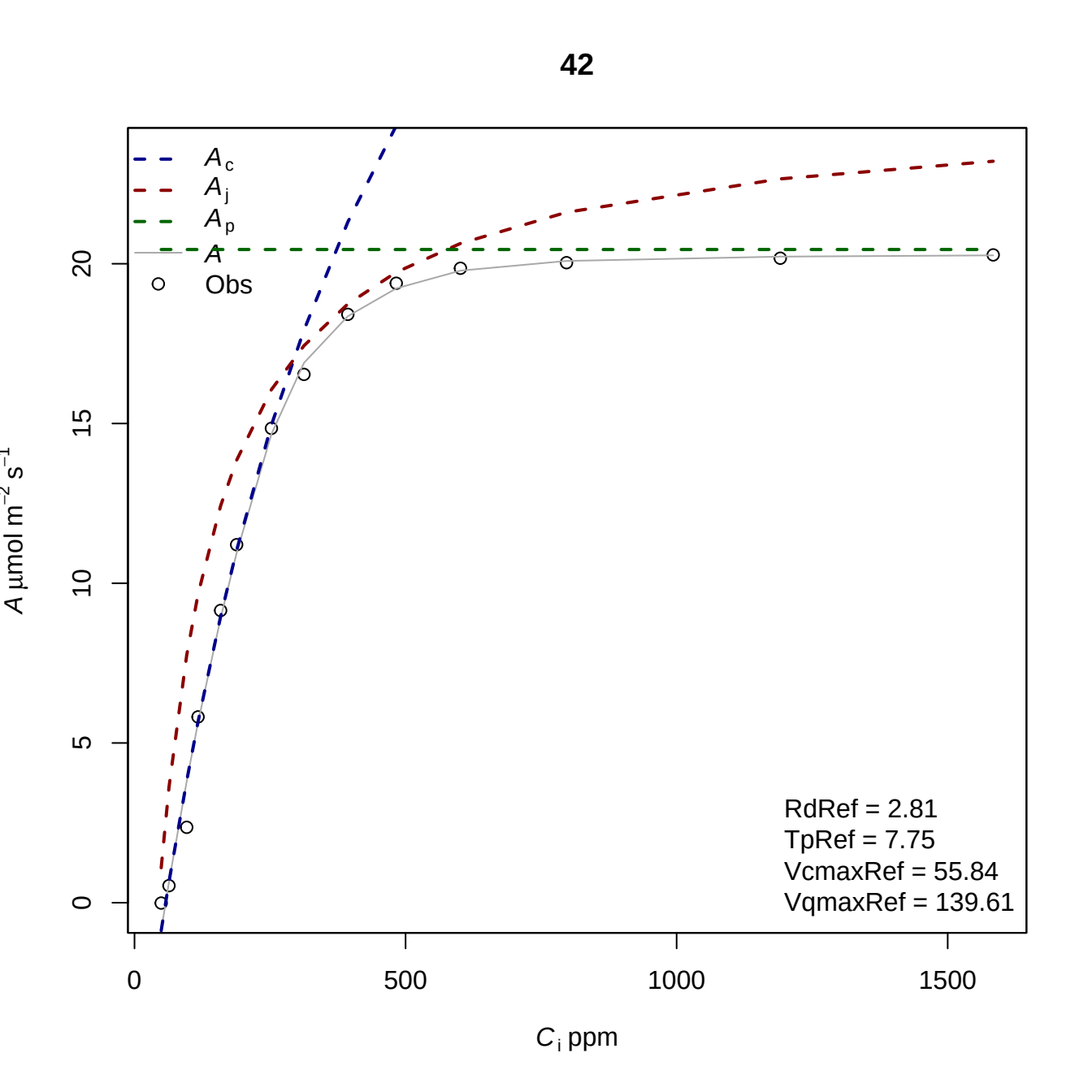

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.61
TpRef = 9999
VcmaxRef = 53.97
VqmaxRef = 175.02
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

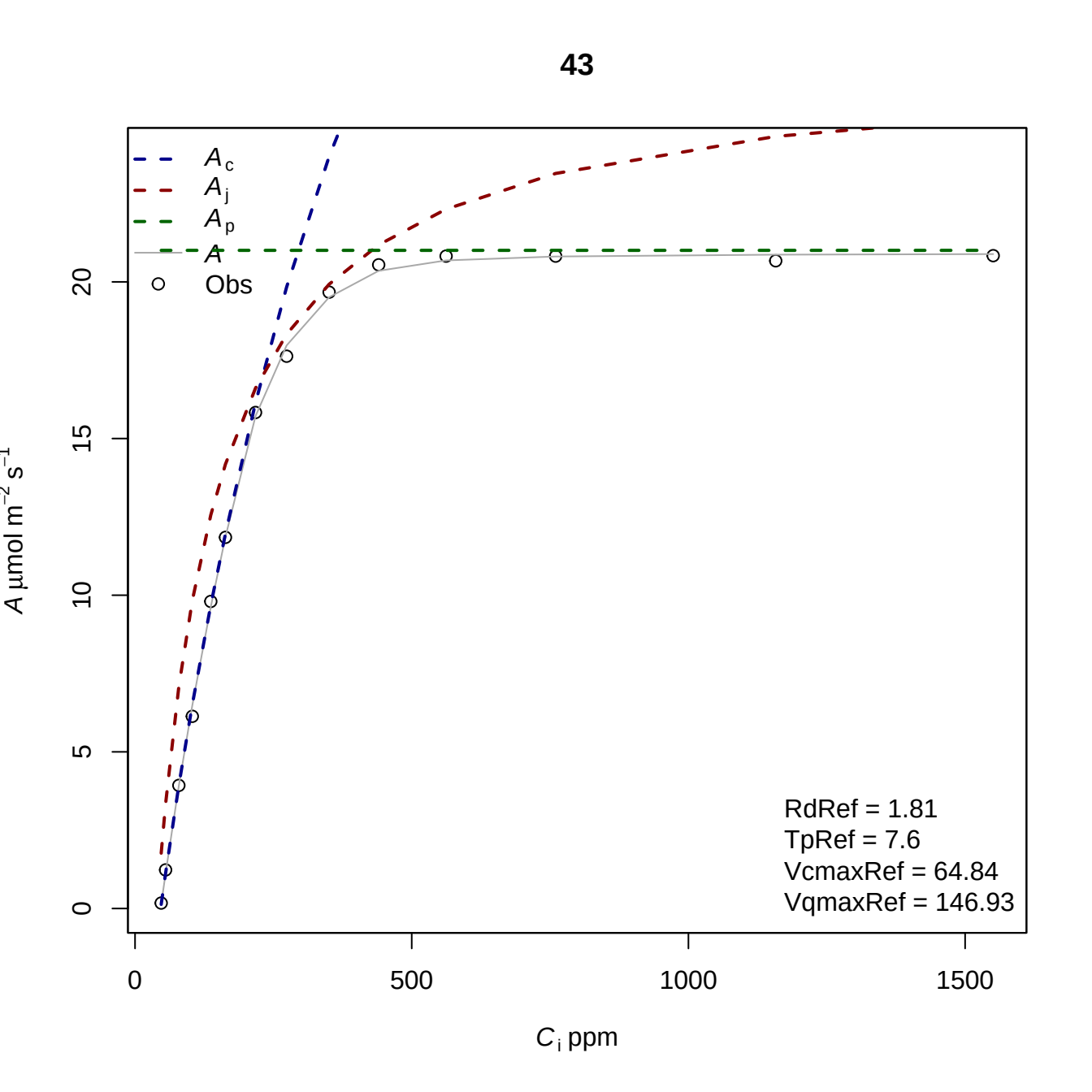

38
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.31
TpRef = 9999
VcmaxRef = 44.53
VqmaxRef = 125.16
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

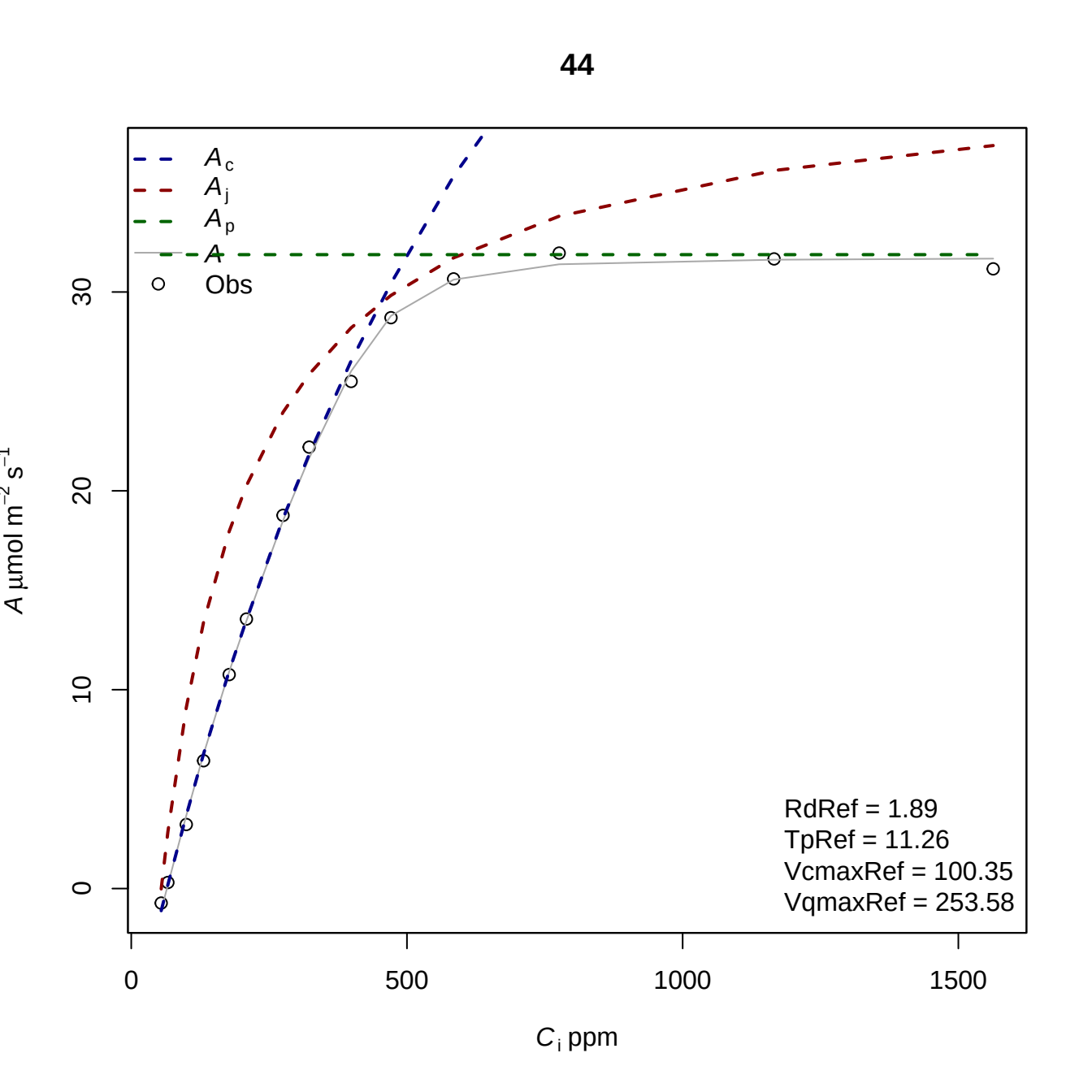

39
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.79
TpRef = 7.39
VcmaxRef = 52.32
VqmaxRef = 122.75
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

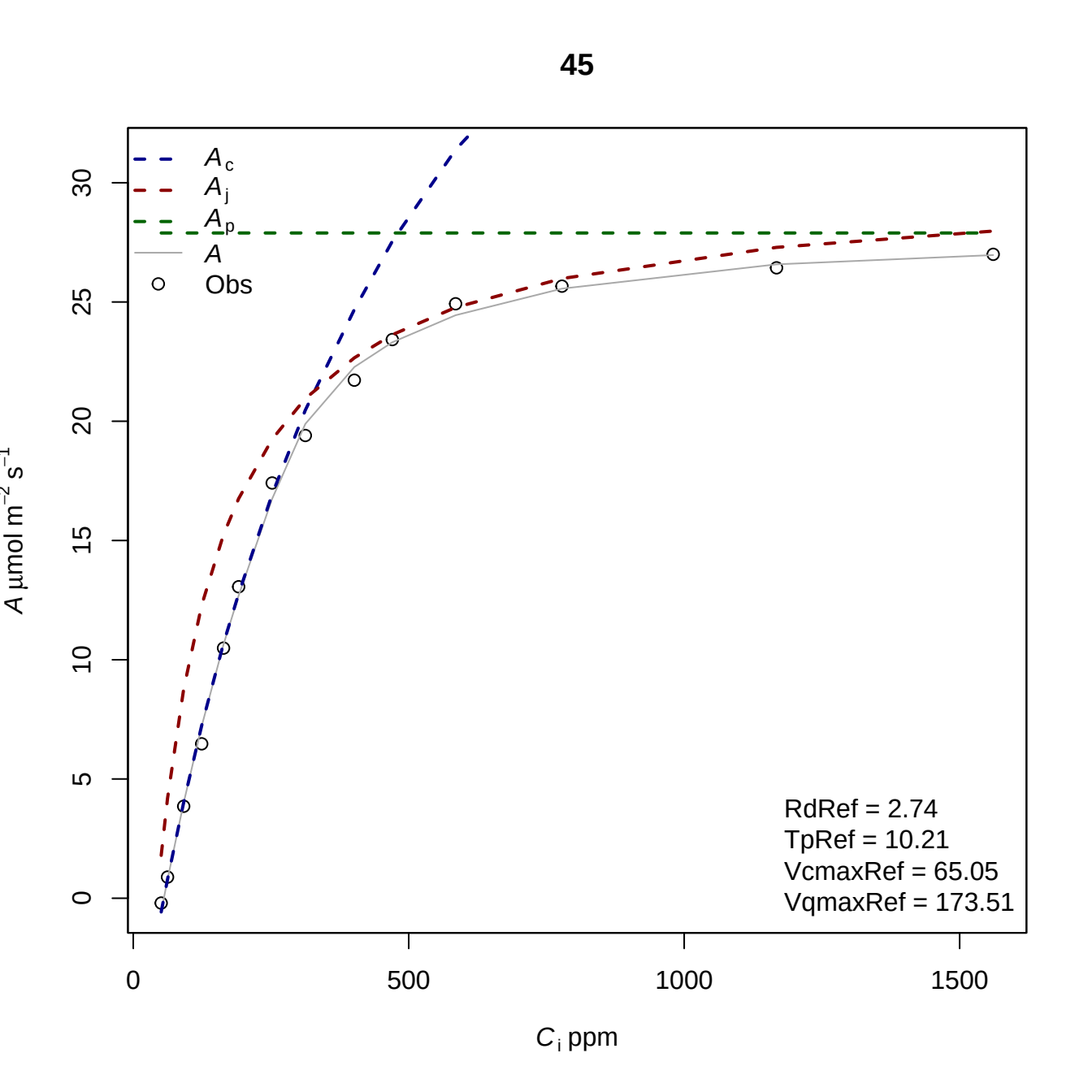

40
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.48
TpRef = 9.67
VcmaxRef = 84.6
VqmaxRef = 193.98
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

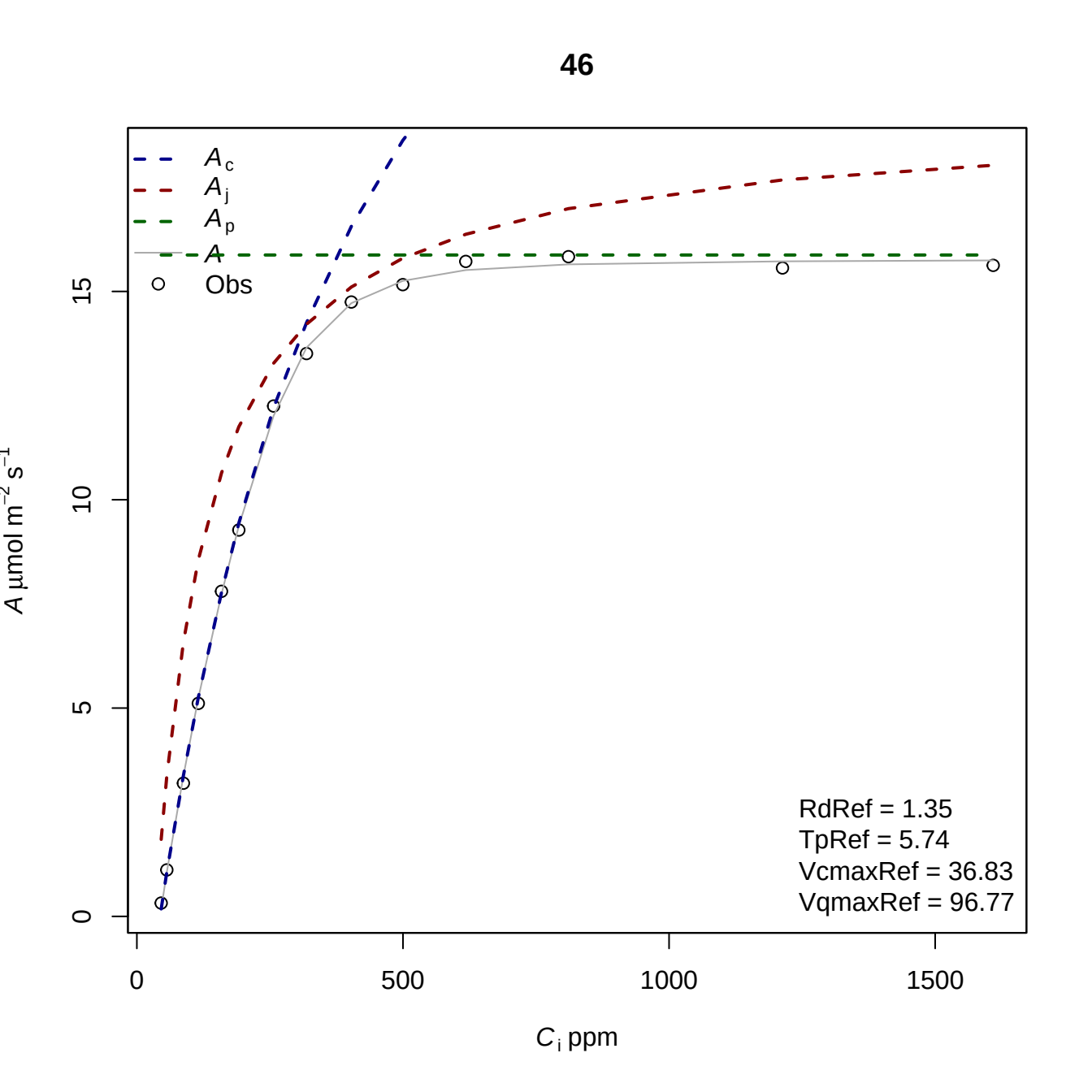

41
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.74
TpRef = 9999
VcmaxRef = 47.24
VqmaxRef = 113.56
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

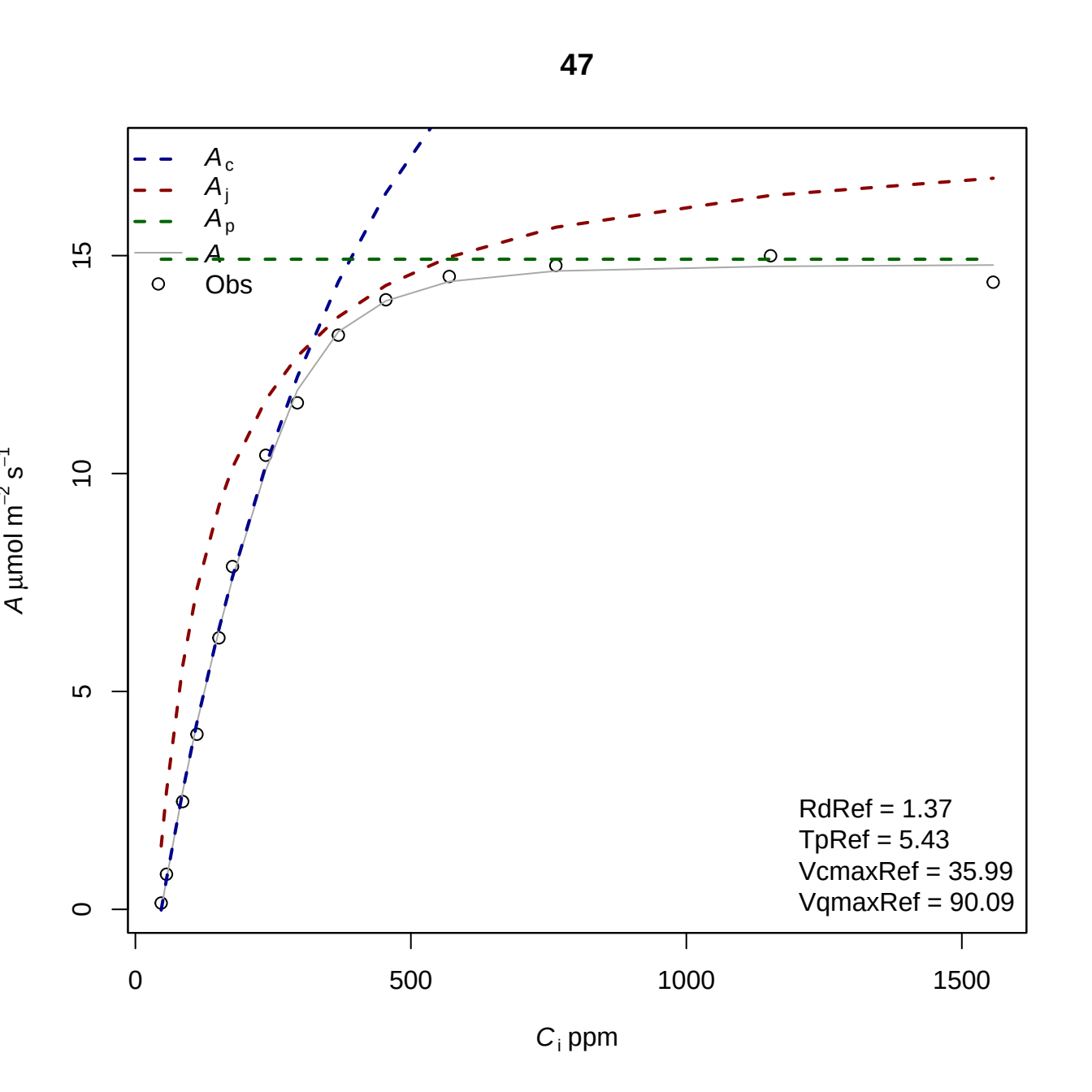

42
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.81
TpRef = 7.75
VcmaxRef = 55.84
VqmaxRef = 139.61
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

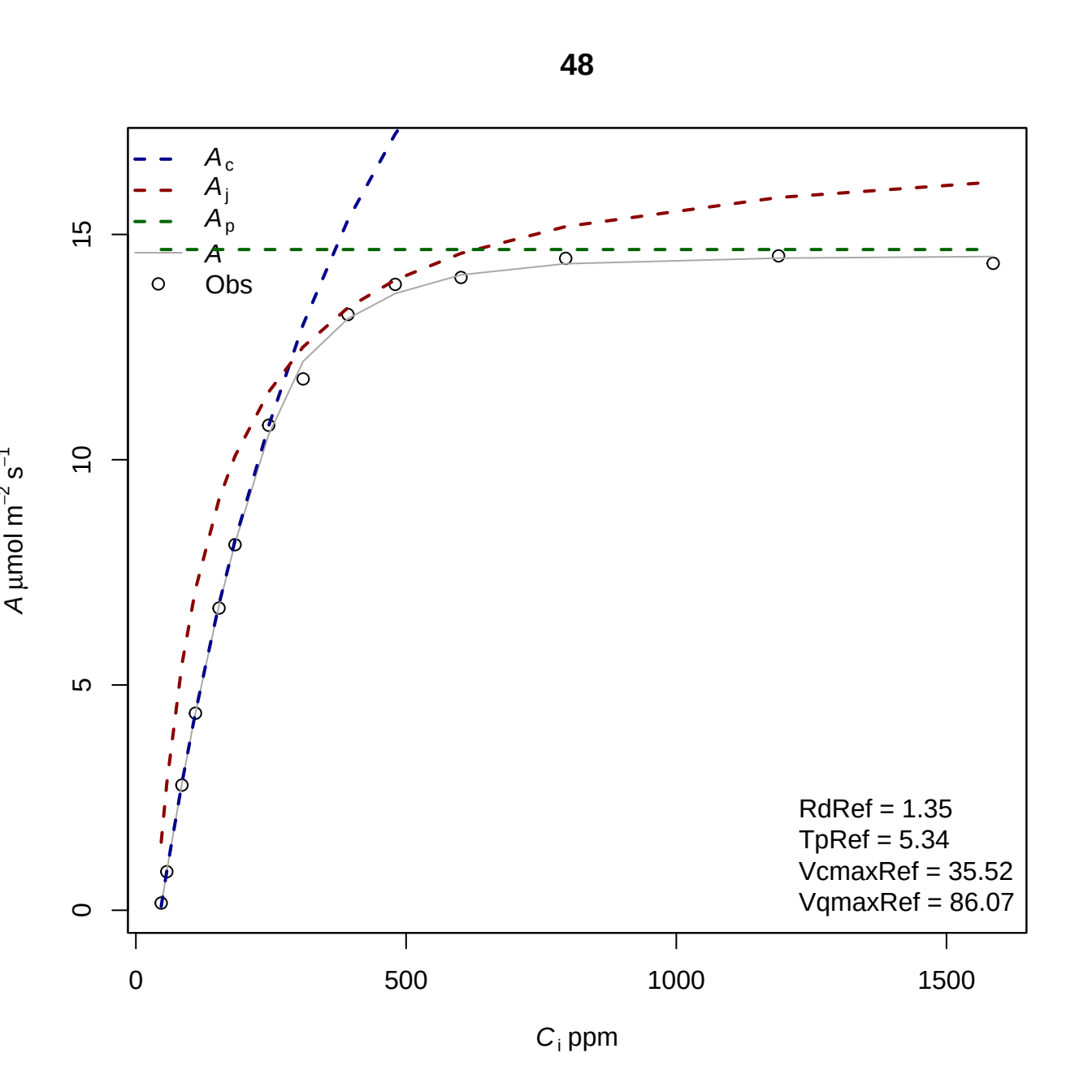

43
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.81
TpRef = 7.6
VcmaxRef = 64.84
VqmaxRef = 146.93
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

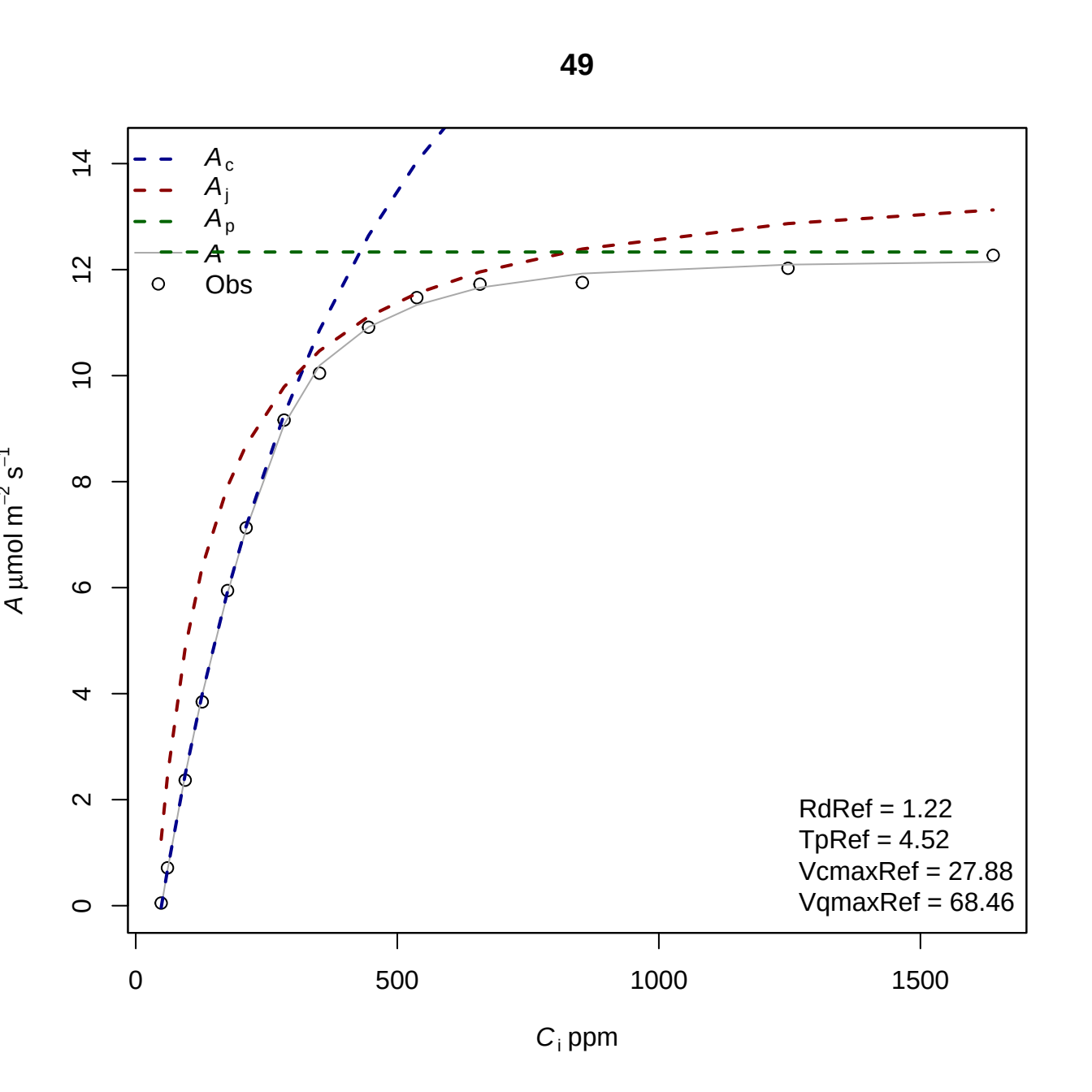

44
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.89
TpRef = 11.26
VcmaxRef = 100.35
VqmaxRef = 253.58
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
0
500
1000
1500

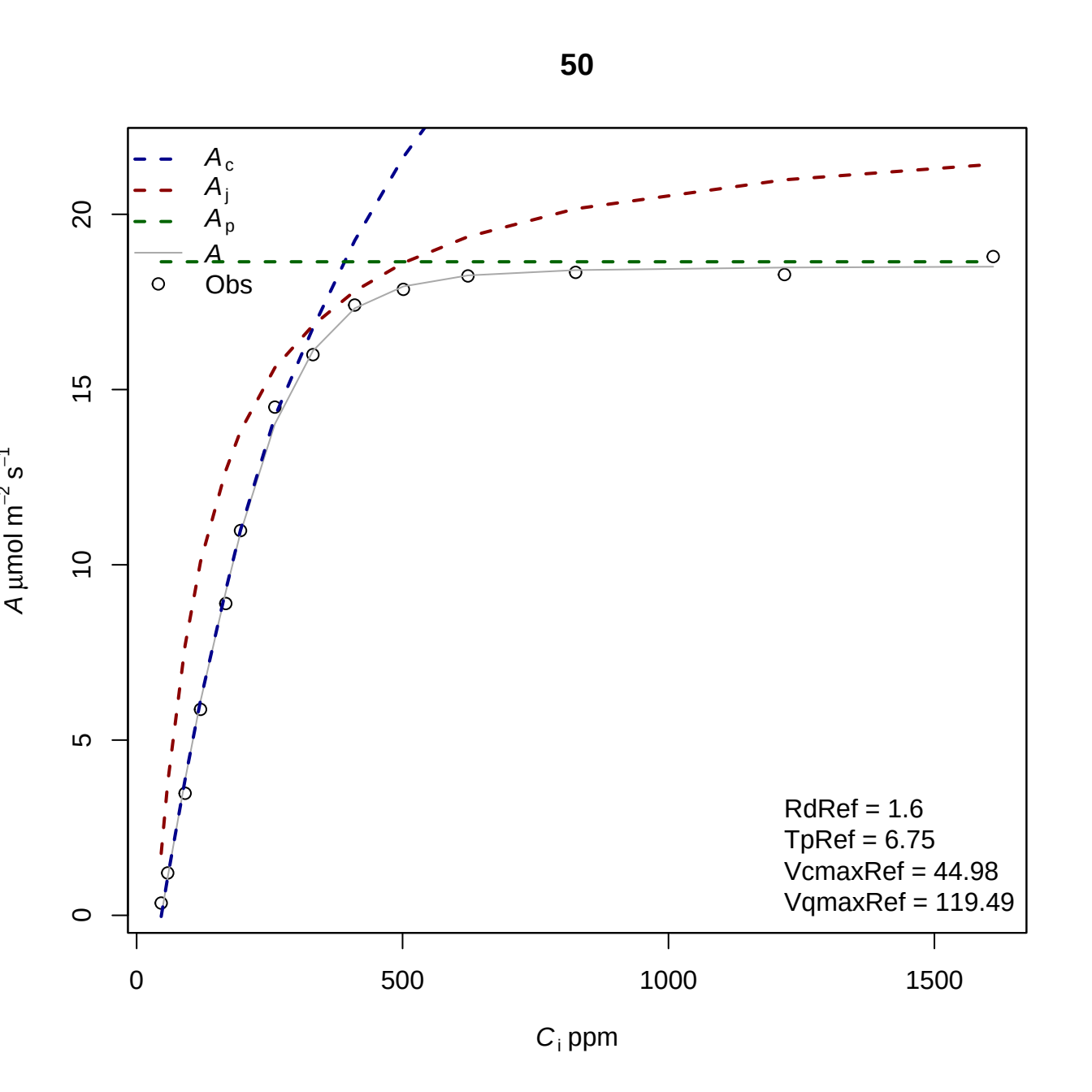

45
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.74
TpRef = 10.21
VcmaxRef = 65.05
VqmaxRef = 173.51
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

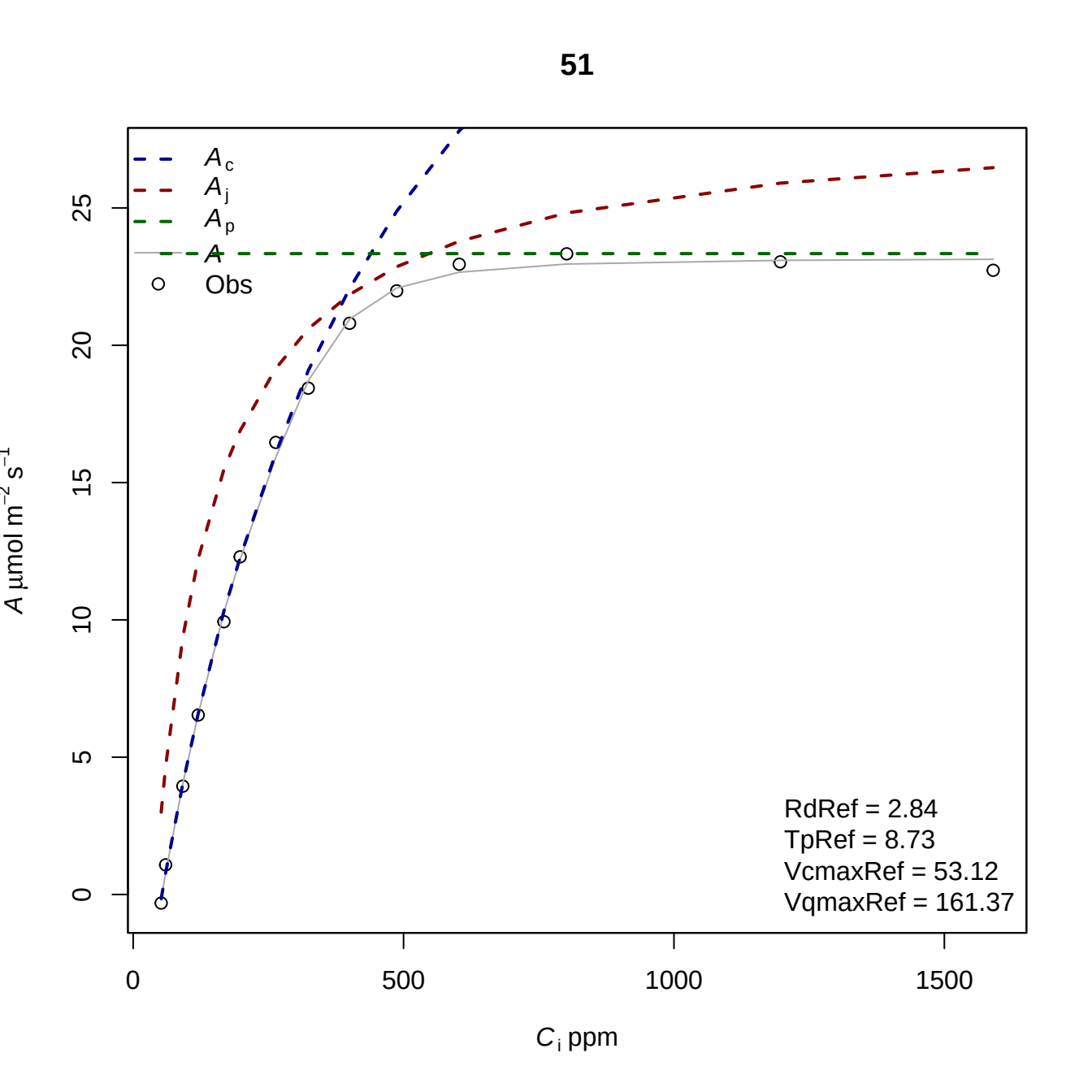

46
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.35
TpRef = 5.74
VcmaxRef = 36.83
VqmaxRef = 96.77
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

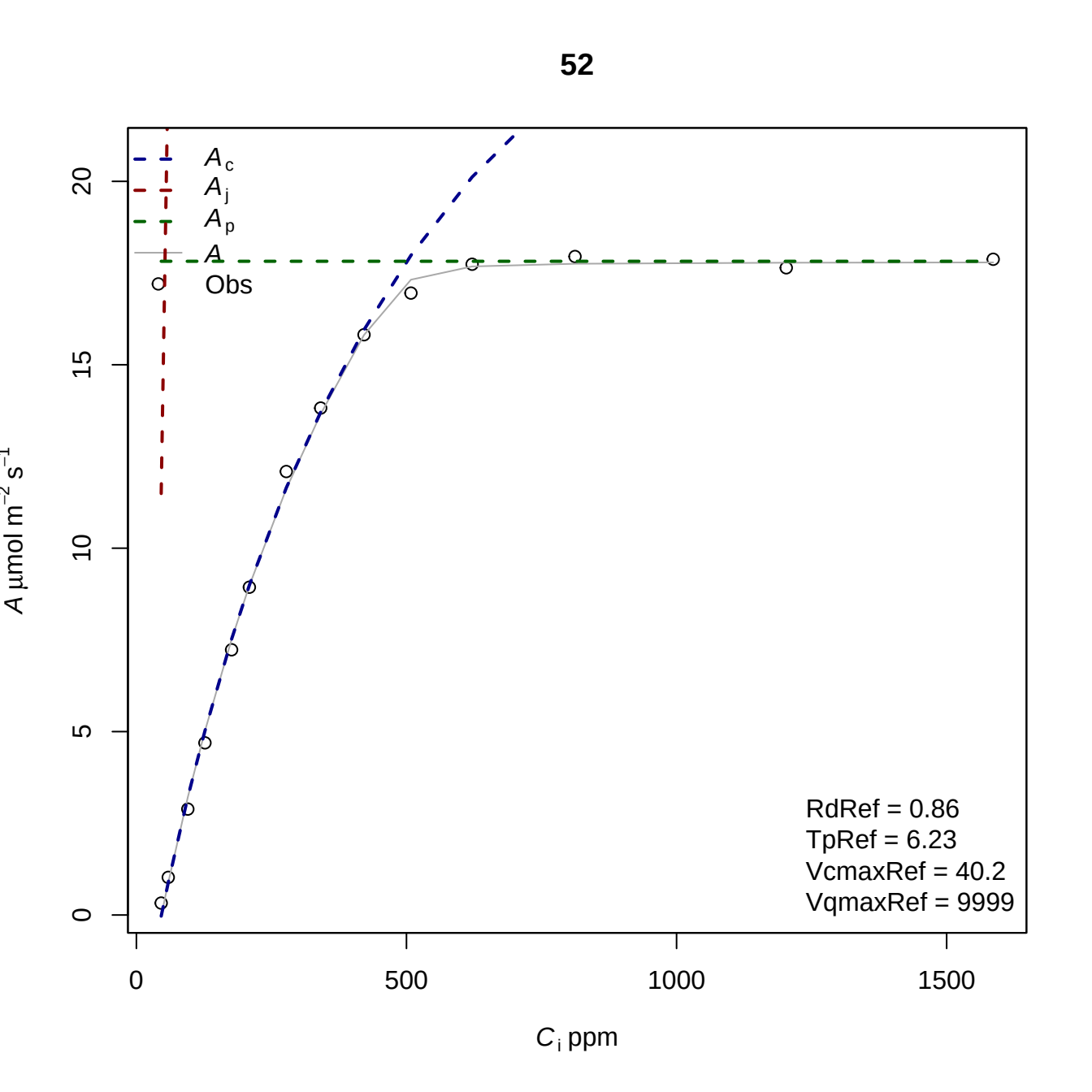

47
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.37
TpRef = 5.43
VcmaxRef = 35.99
VqmaxRef = 90.09
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

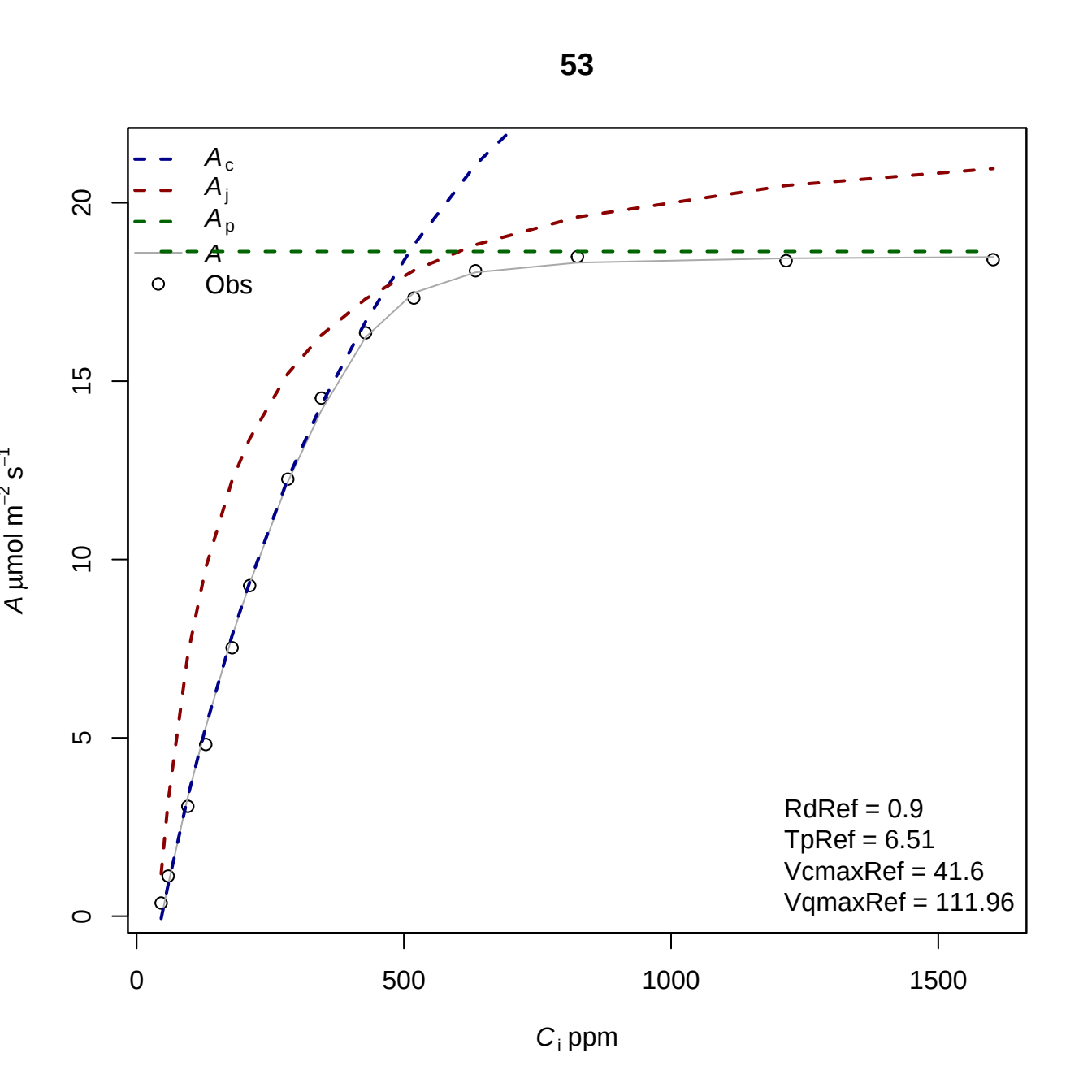

48
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.35
TpRef = 5.34
VcmaxRef = 35.52
VqmaxRef = 86.07
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

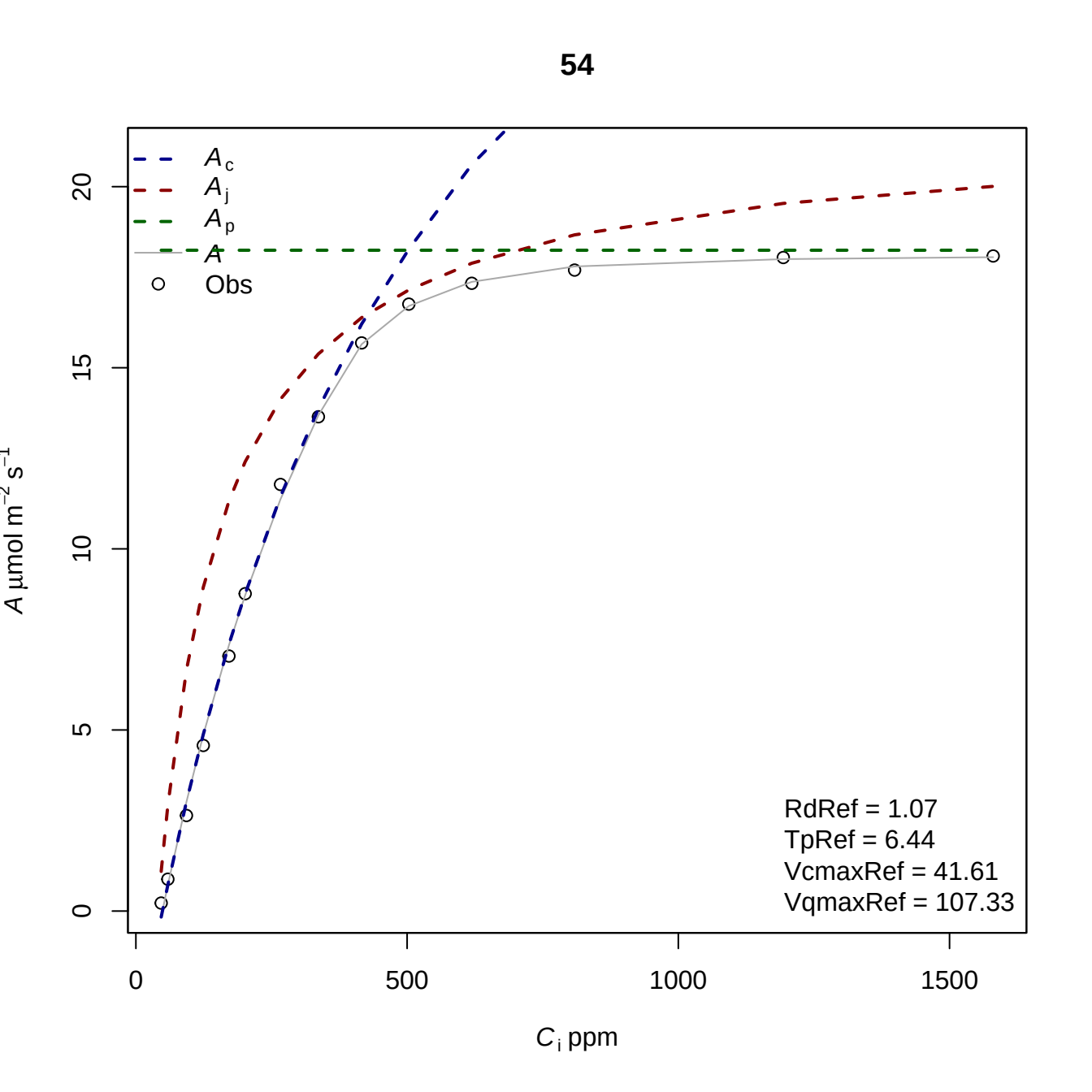

49
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.22
TpRef = 4.52
VcmaxRef = 27.88
VqmaxRef = 68.46
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
0
500
1000
1500

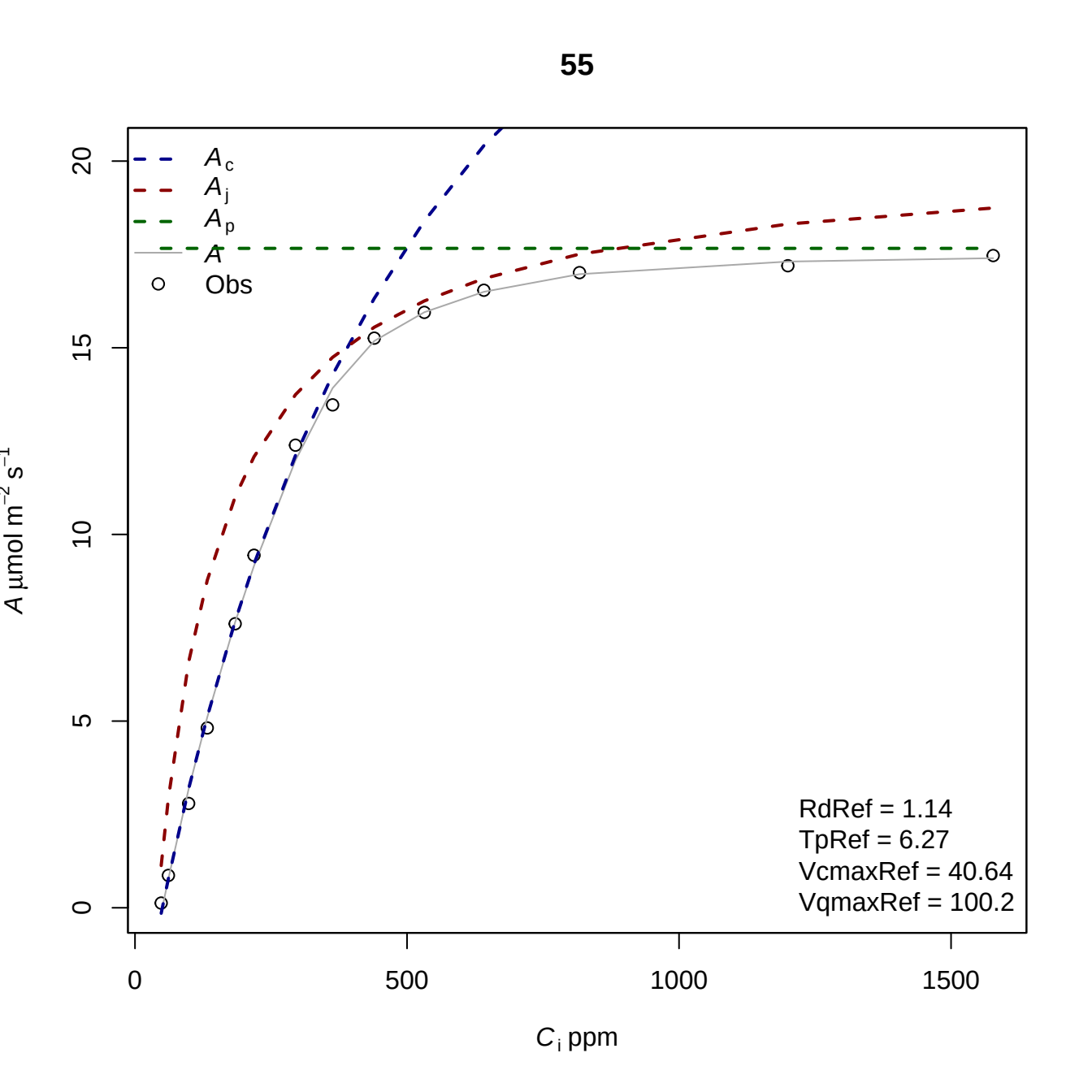

50
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.6
TpRef = 6.75
VcmaxRef = 44.98
VqmaxRef = 119.49
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

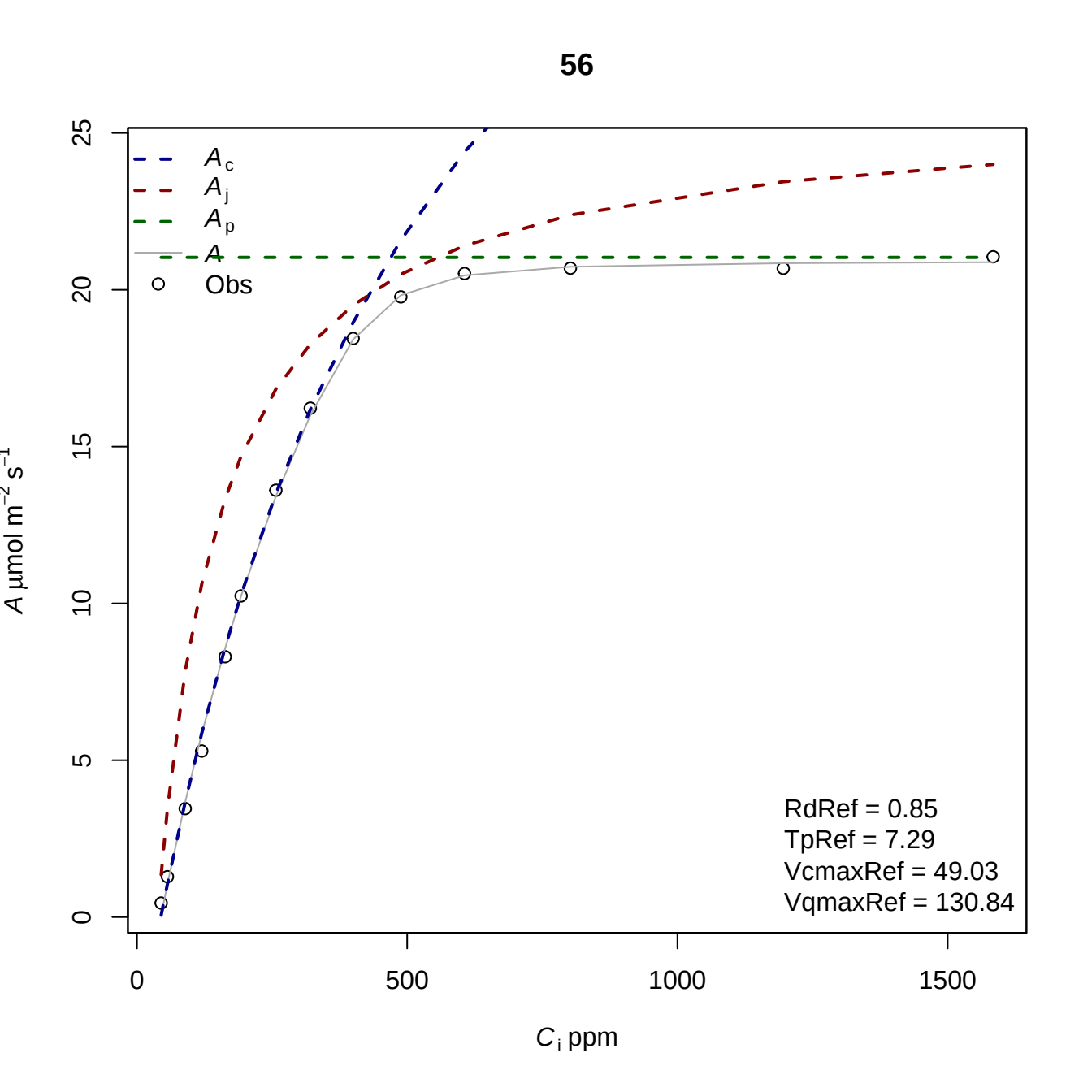

51
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.84
TpRef = 8.73
VcmaxRef = 53.12
VqmaxRef = 161.37
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

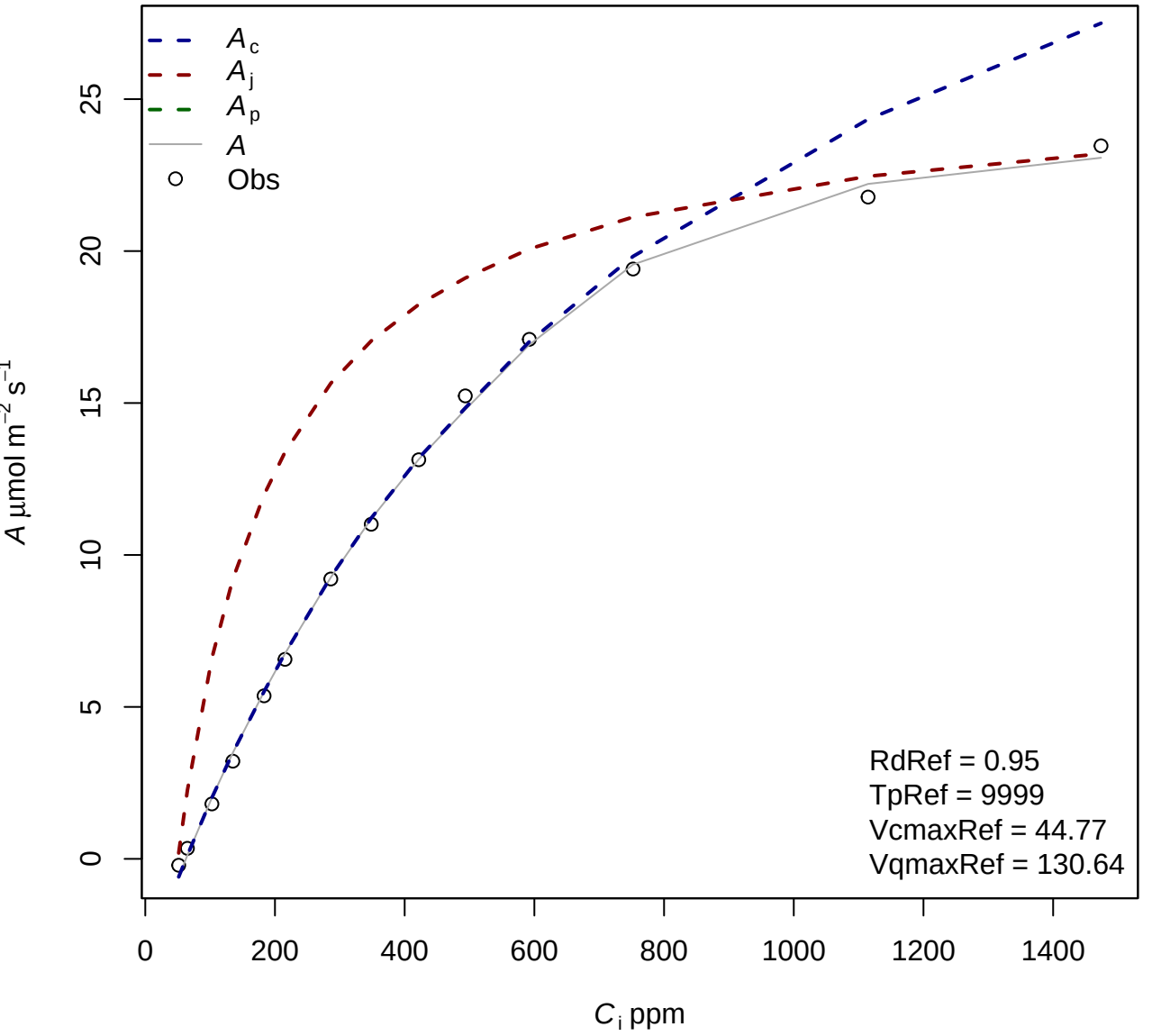

52
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.86
TpRef = 6.23
VcmaxRef = 40.2
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

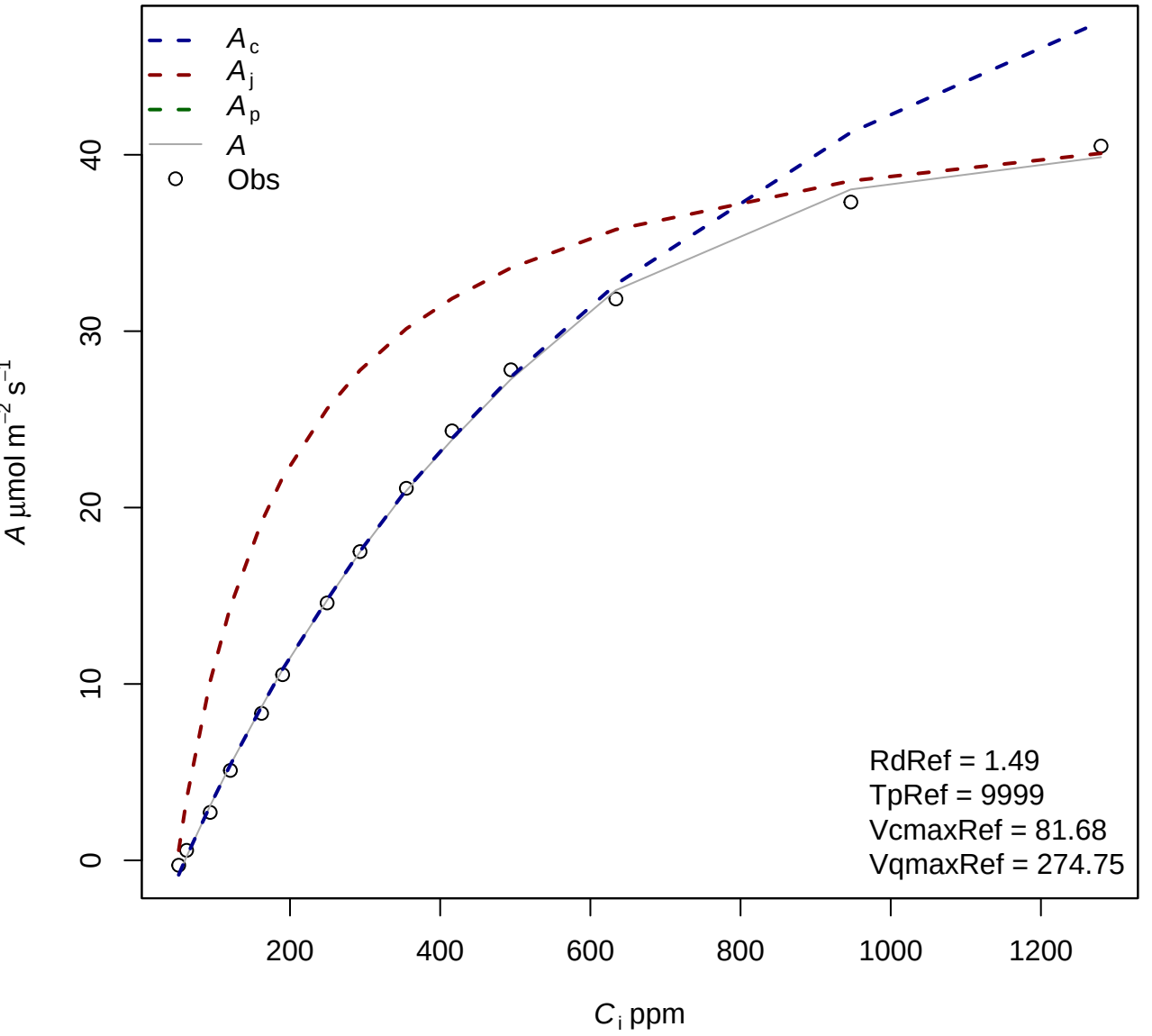

53
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.9
TpRef = 6.51
VcmaxRef = 41.6
VqmaxRef = 111.96
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

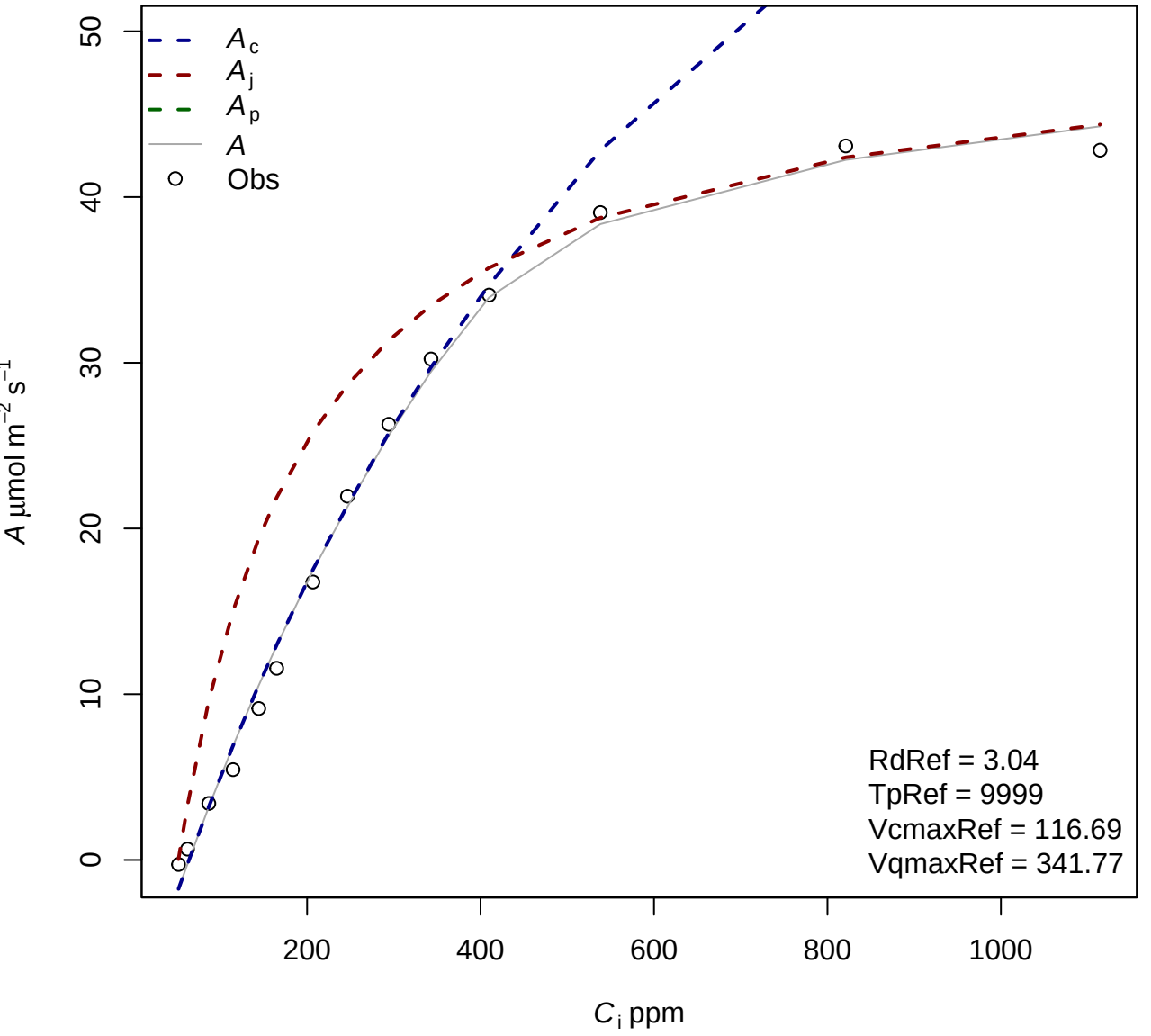

54
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.07
TpRef = 6.44
VcmaxRef = 41.61
VqmaxRef = 107.33
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

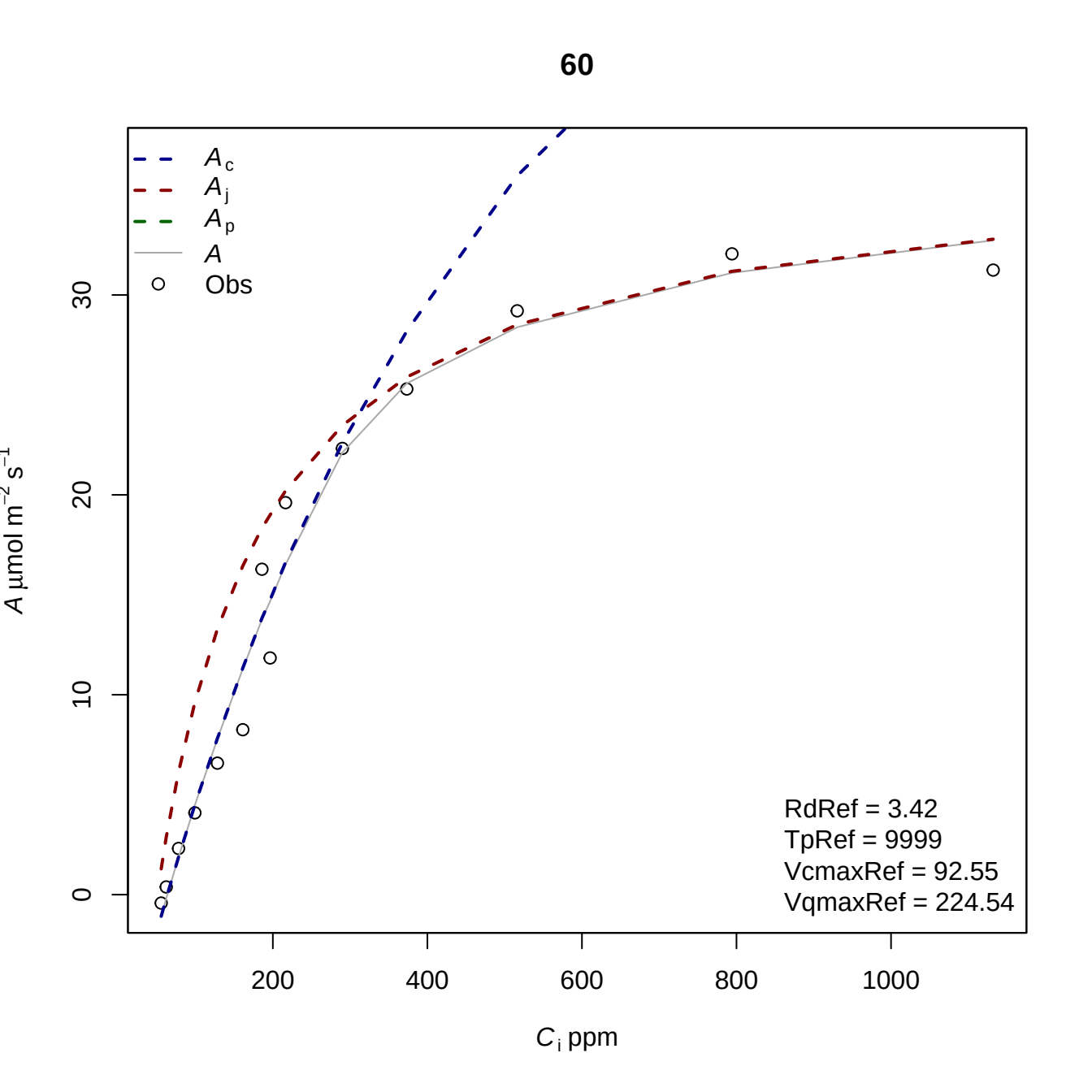

55
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.14
TpRef = 6.27
VcmaxRef = 40.64
VqmaxRef = 100.2
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

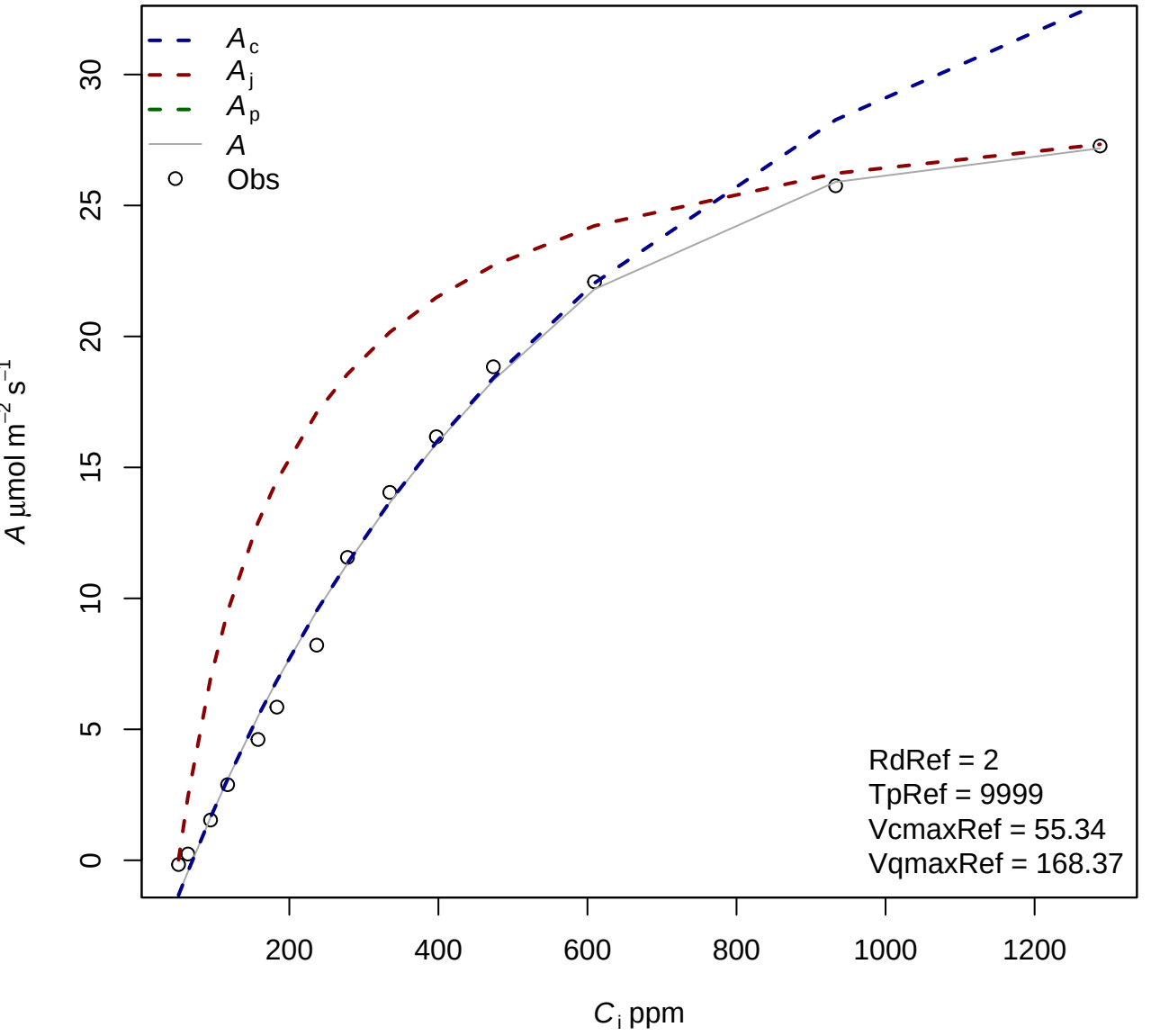

56
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.85
TpRef = 7.29
VcmaxRef = 49.03
VqmaxRef = 130.84
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

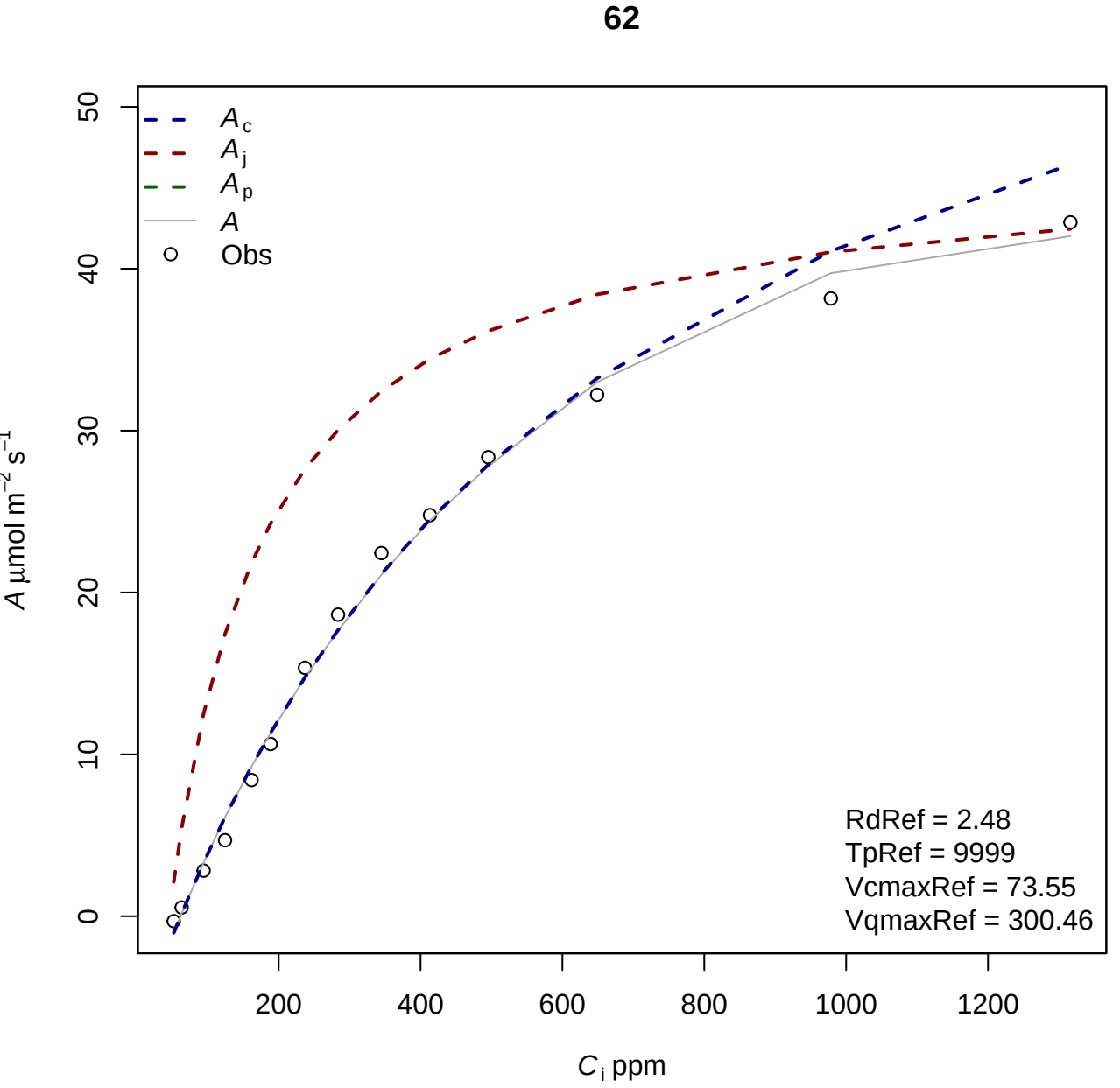

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.95
TpRef = 9999
VcmaxRef = 44.77
VqmaxRef = 130.64
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400

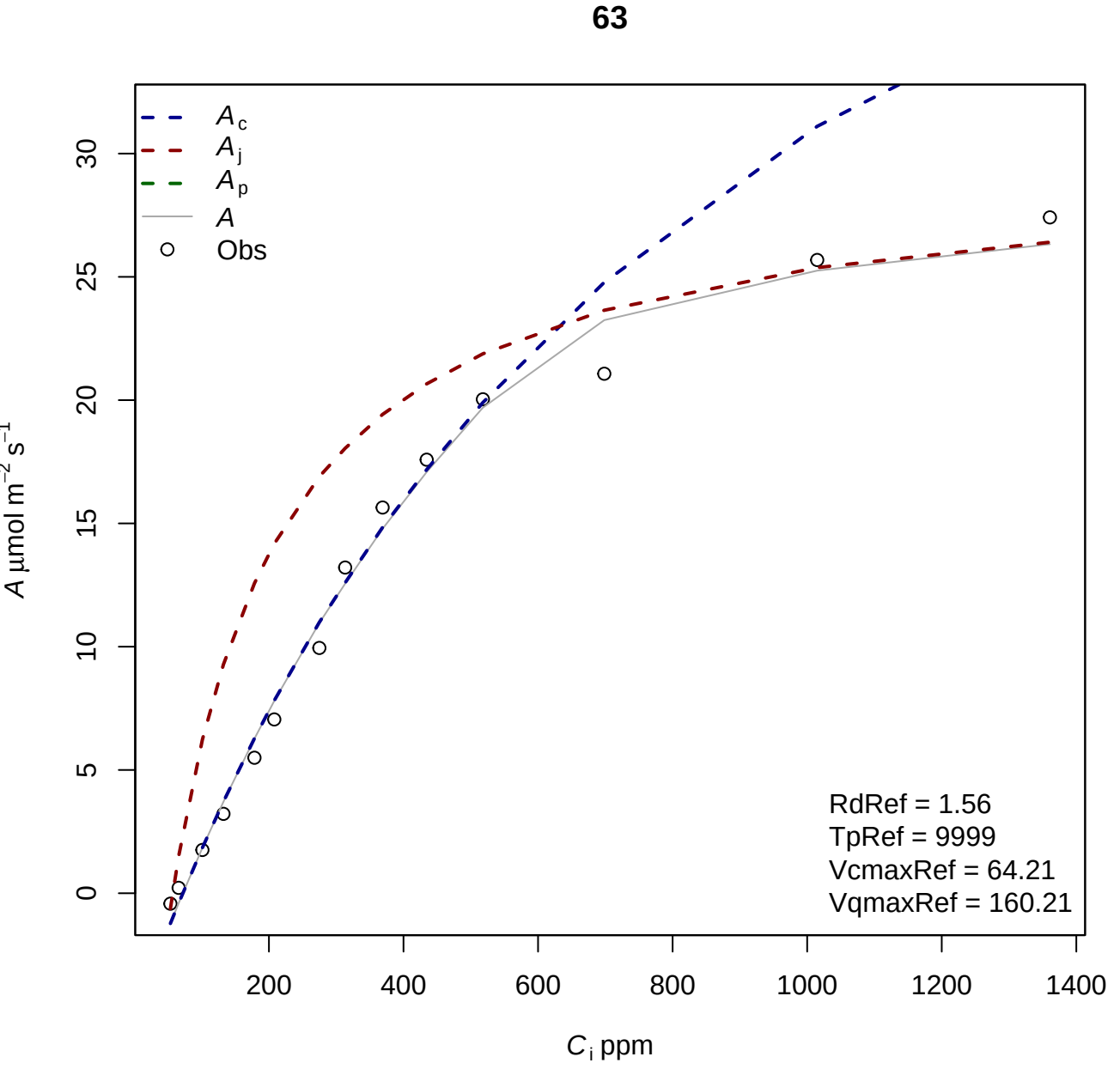

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.49
TpRef = 9999
VcmaxRef = 81.68
VqmaxRef = 274.75
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

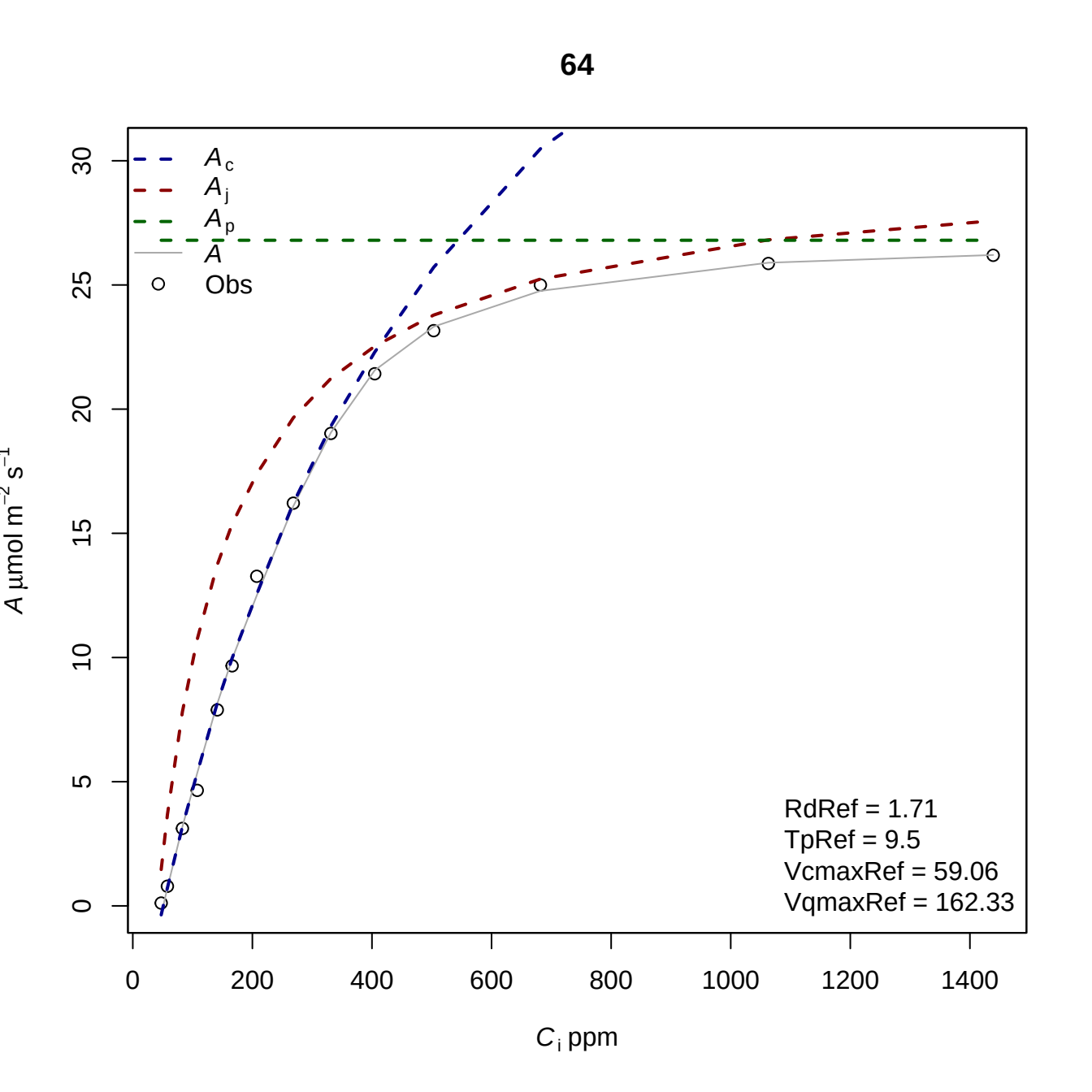

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.04
TpRef = 9999
VcmaxRef = 116.69
VqmaxRef = 341.77
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000

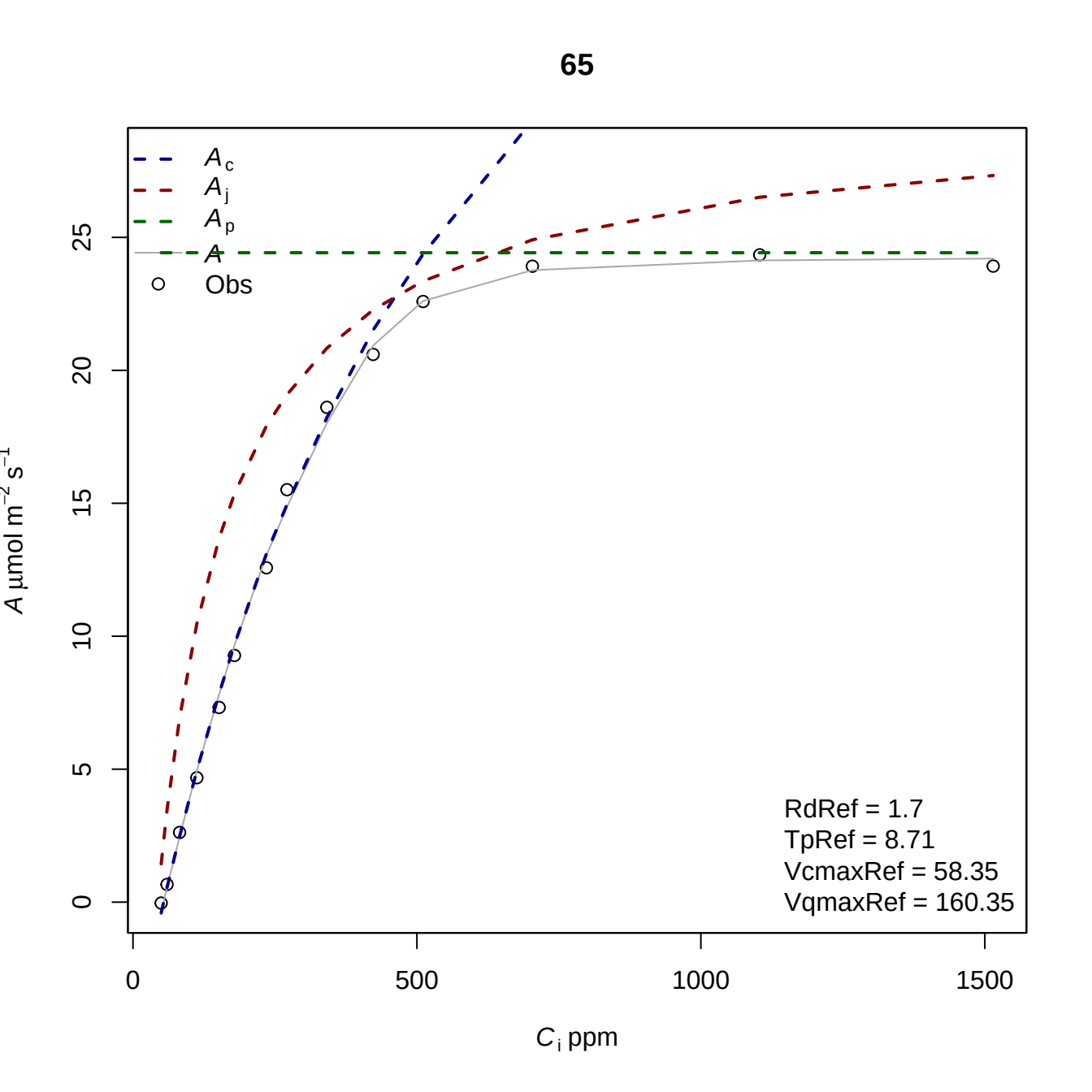

60
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.42
TpRef = 9999
VcmaxRef = 92.55
VqmaxRef = 224.54
0
10
20
30
200
400
600
800
1000
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

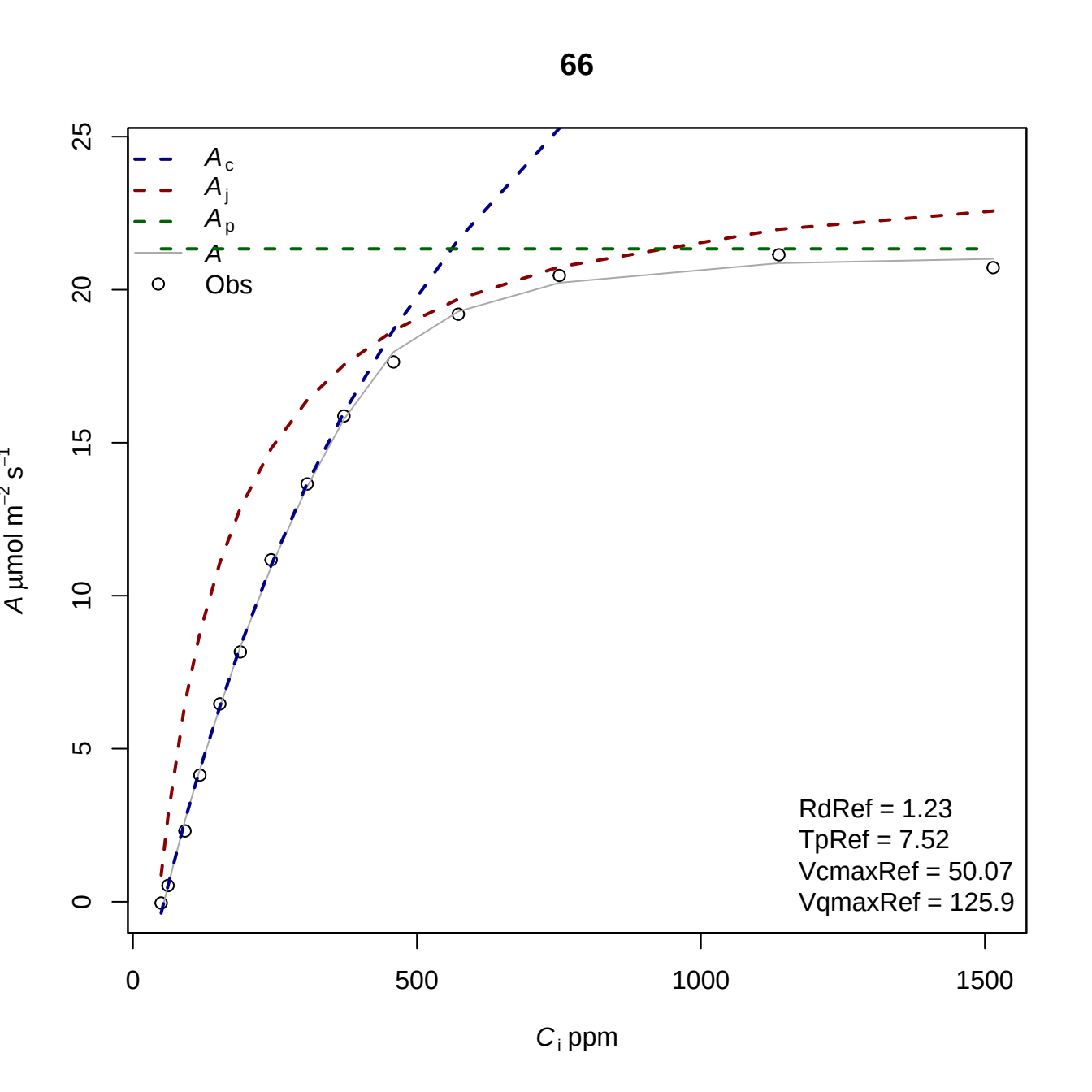

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2
TpRef = 9999
VcmaxRef = 55.34
VqmaxRef = 168.37
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

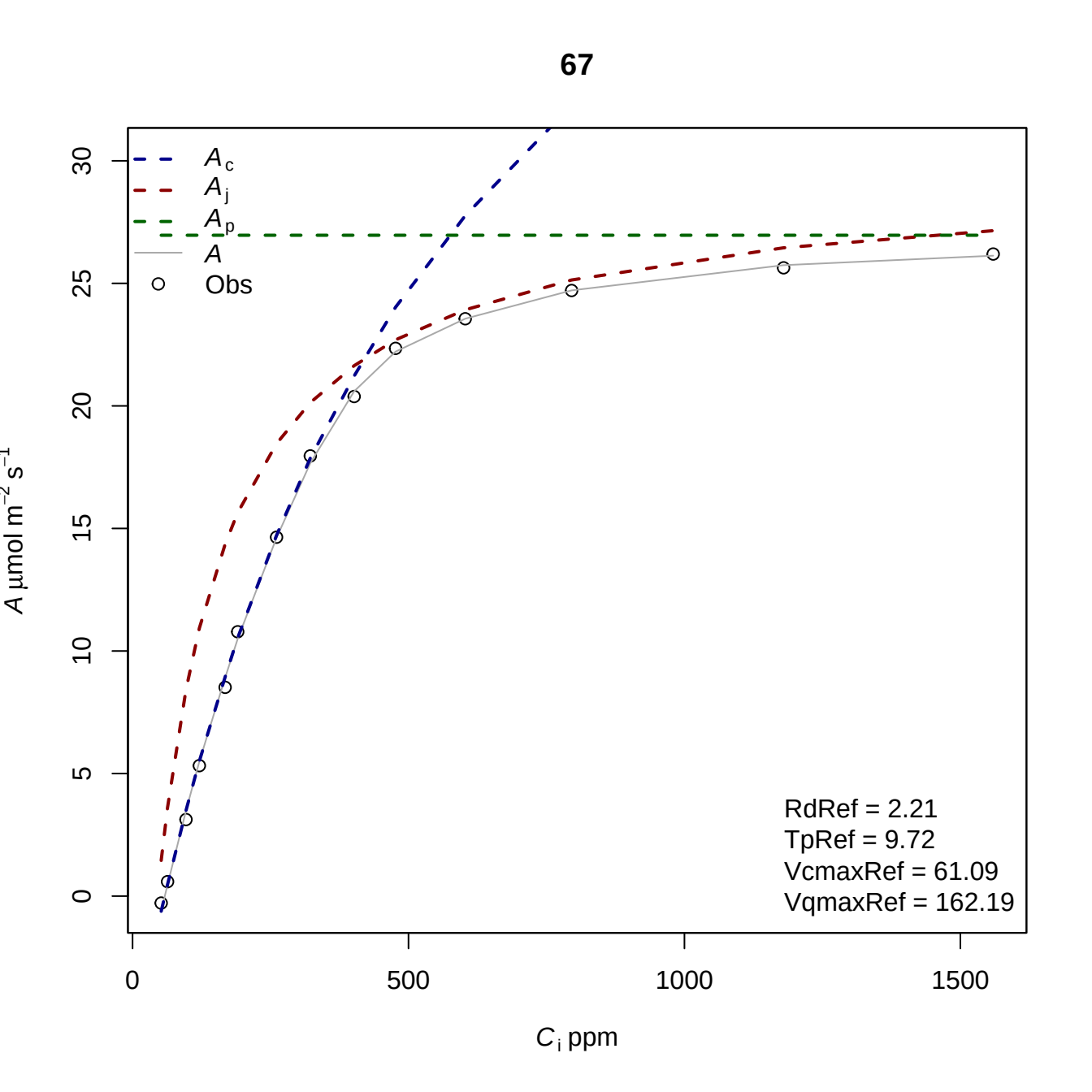

62
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.48
TpRef = 9999
VcmaxRef = 73.55
VqmaxRef = 300.46
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

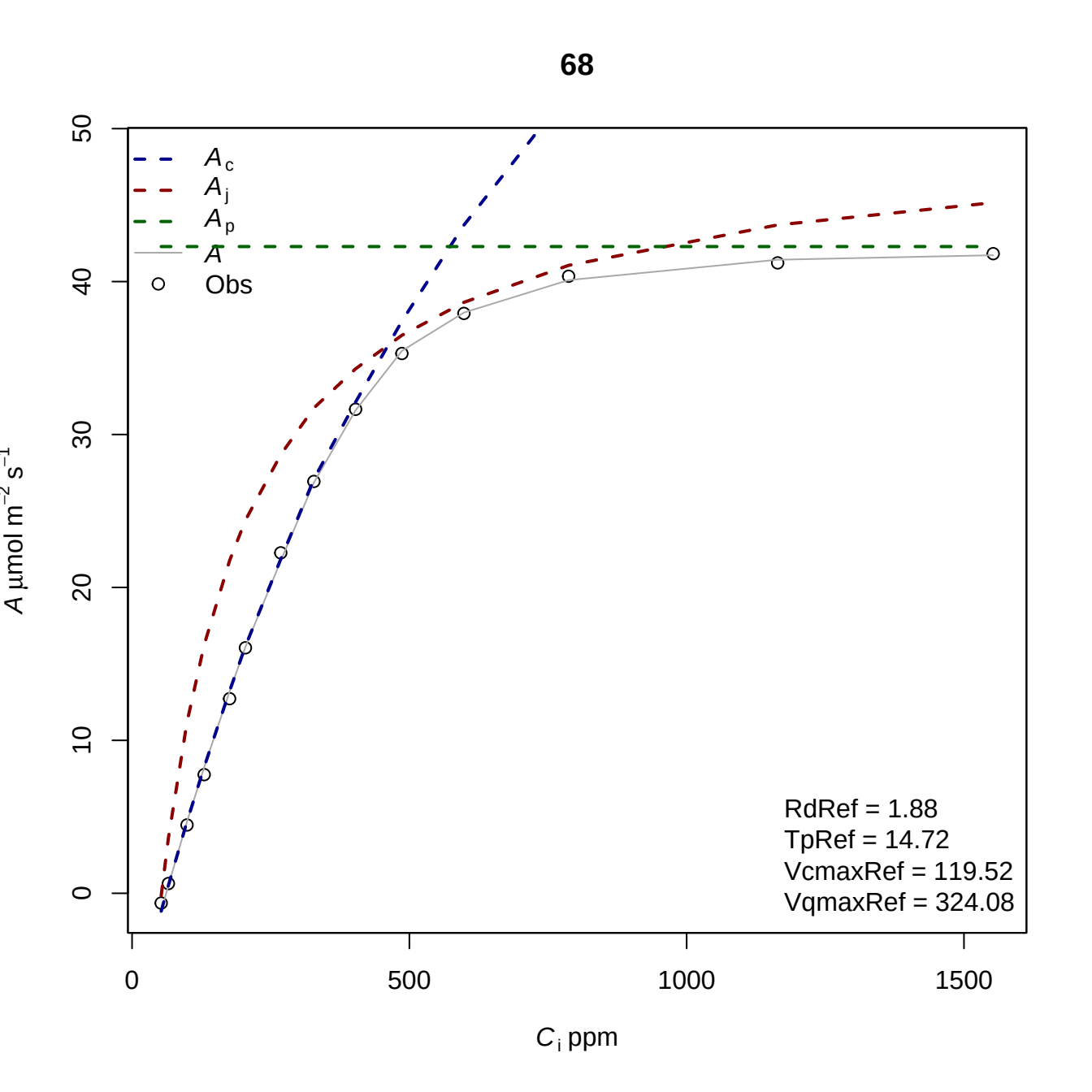

63
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.56
TpRef = 9999
VcmaxRef = 64.21
VqmaxRef = 160.21
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200
1400

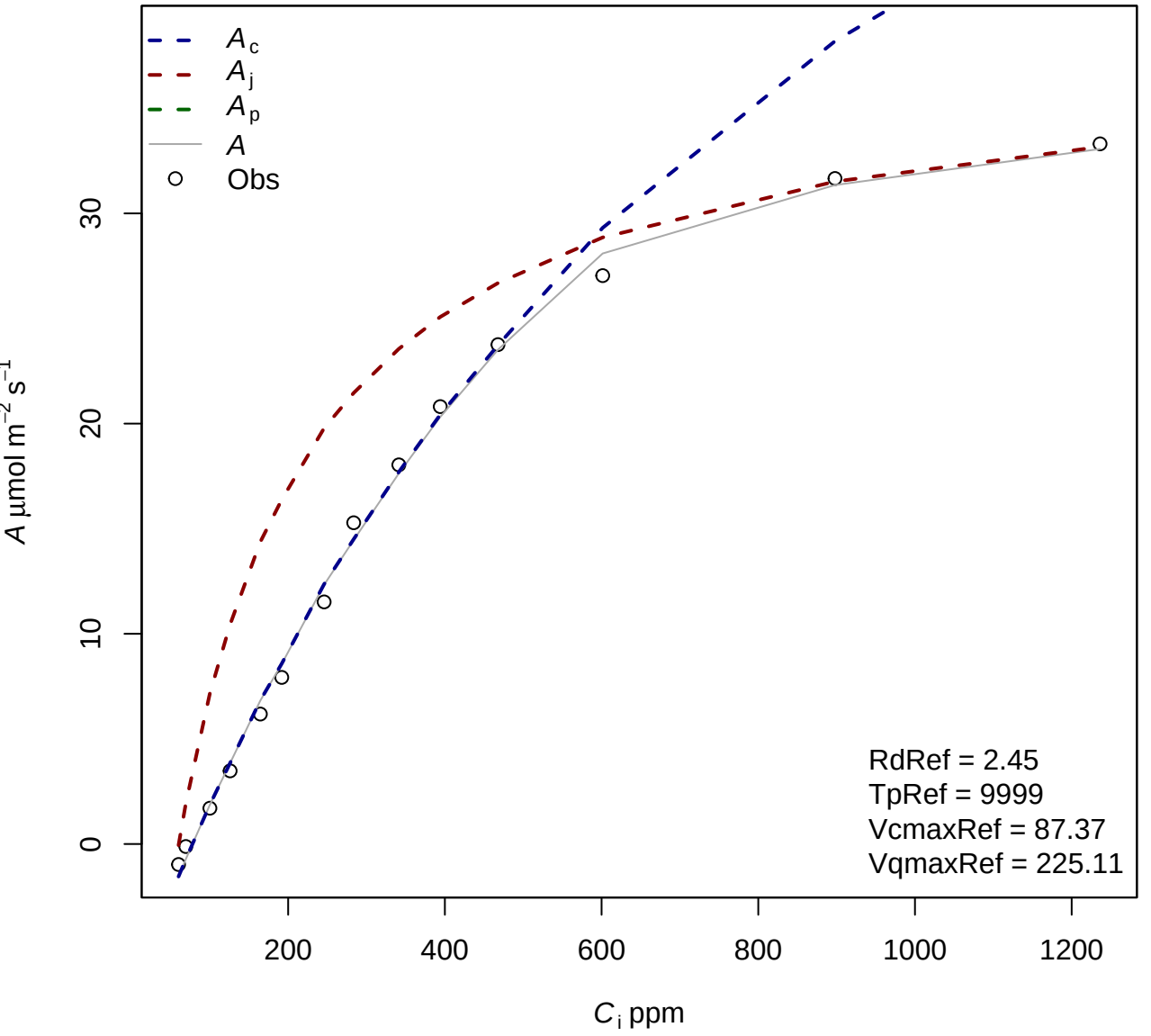

64
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.71
TpRef = 9.5
VcmaxRef = 59.06
VqmaxRef = 162.33
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
1400

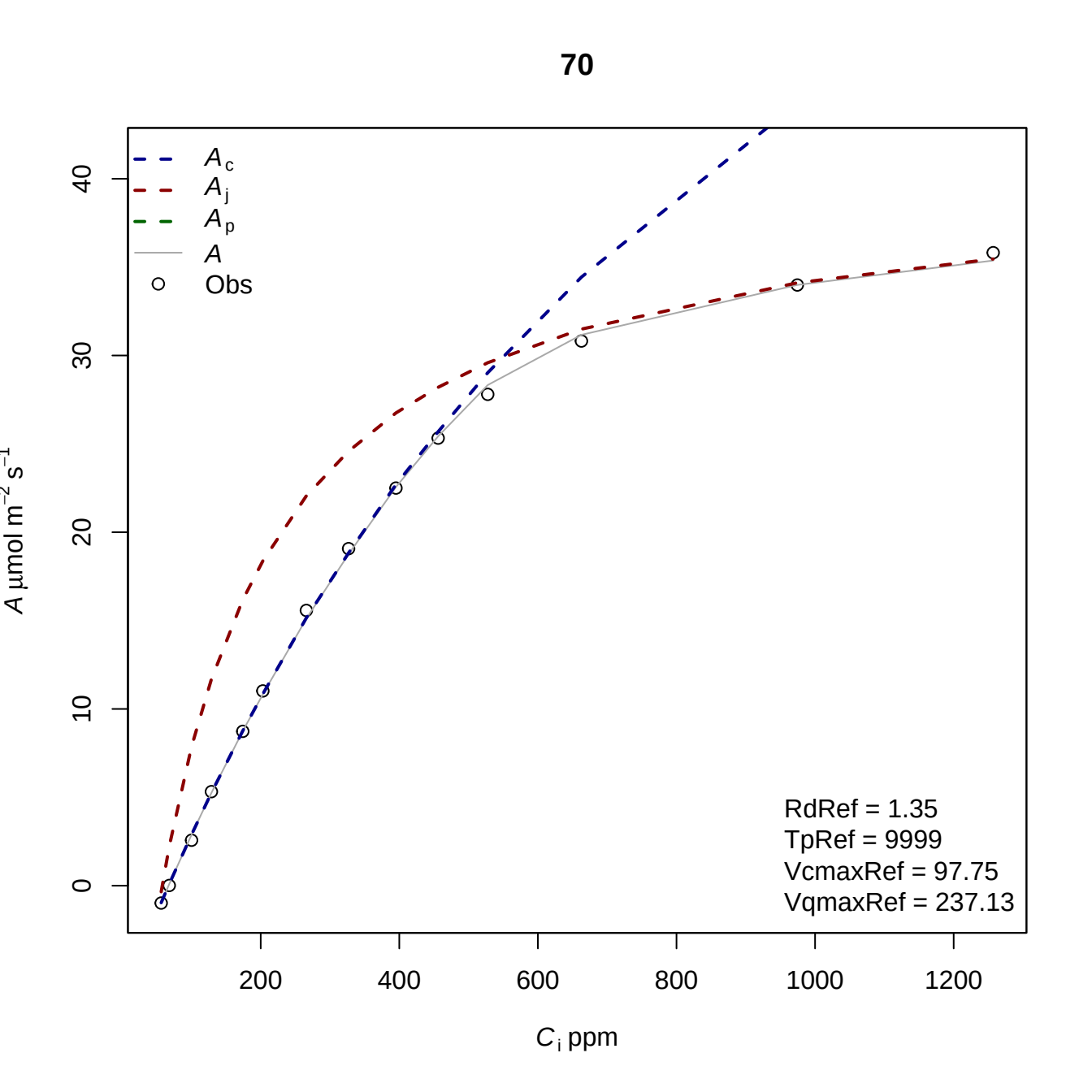

65
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.7
TpRef = 8.71
VcmaxRef = 58.35
VqmaxRef = 160.35
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

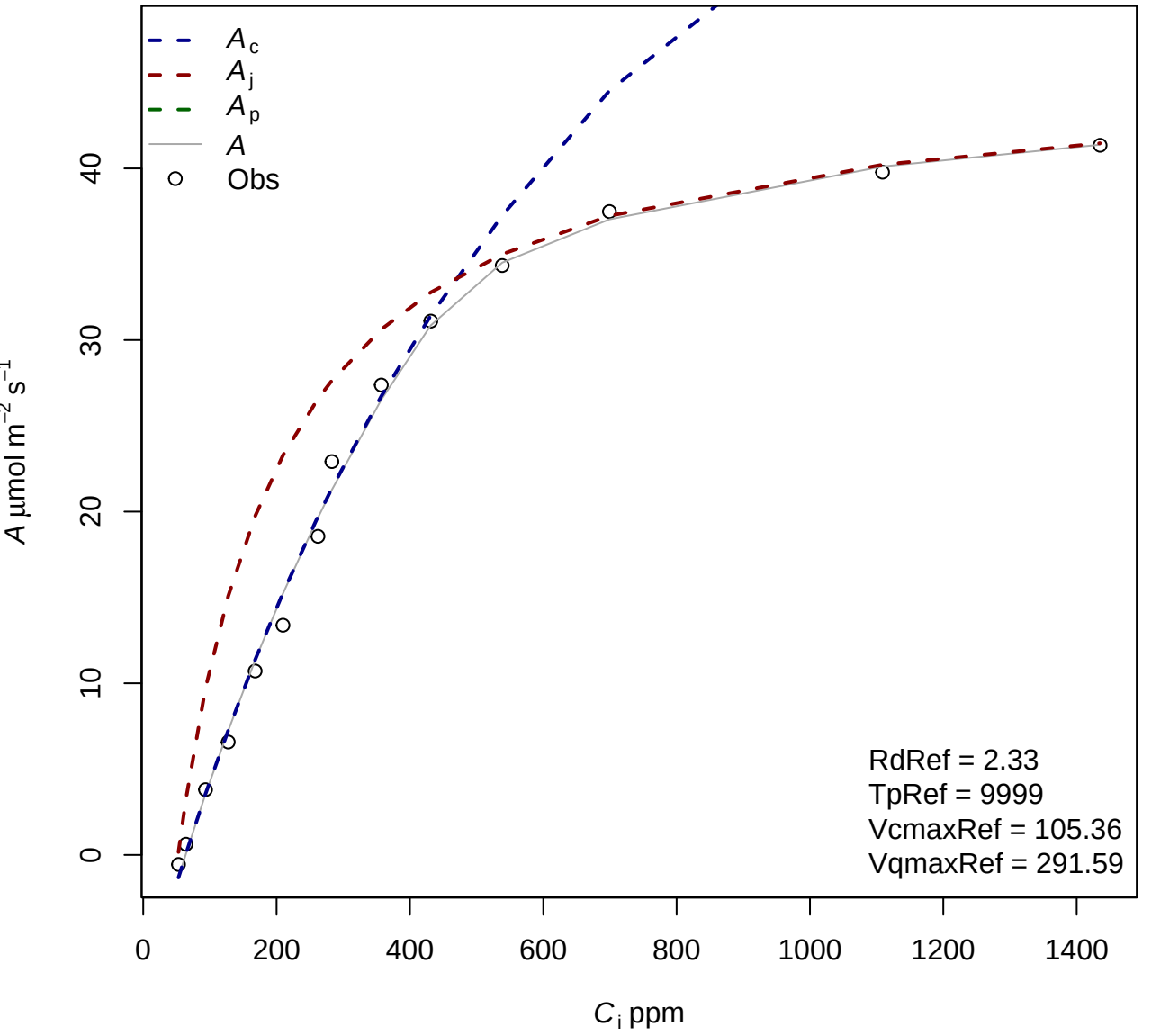

66
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.23
TpRef = 7.52
VcmaxRef = 50.07
VqmaxRef = 125.9
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

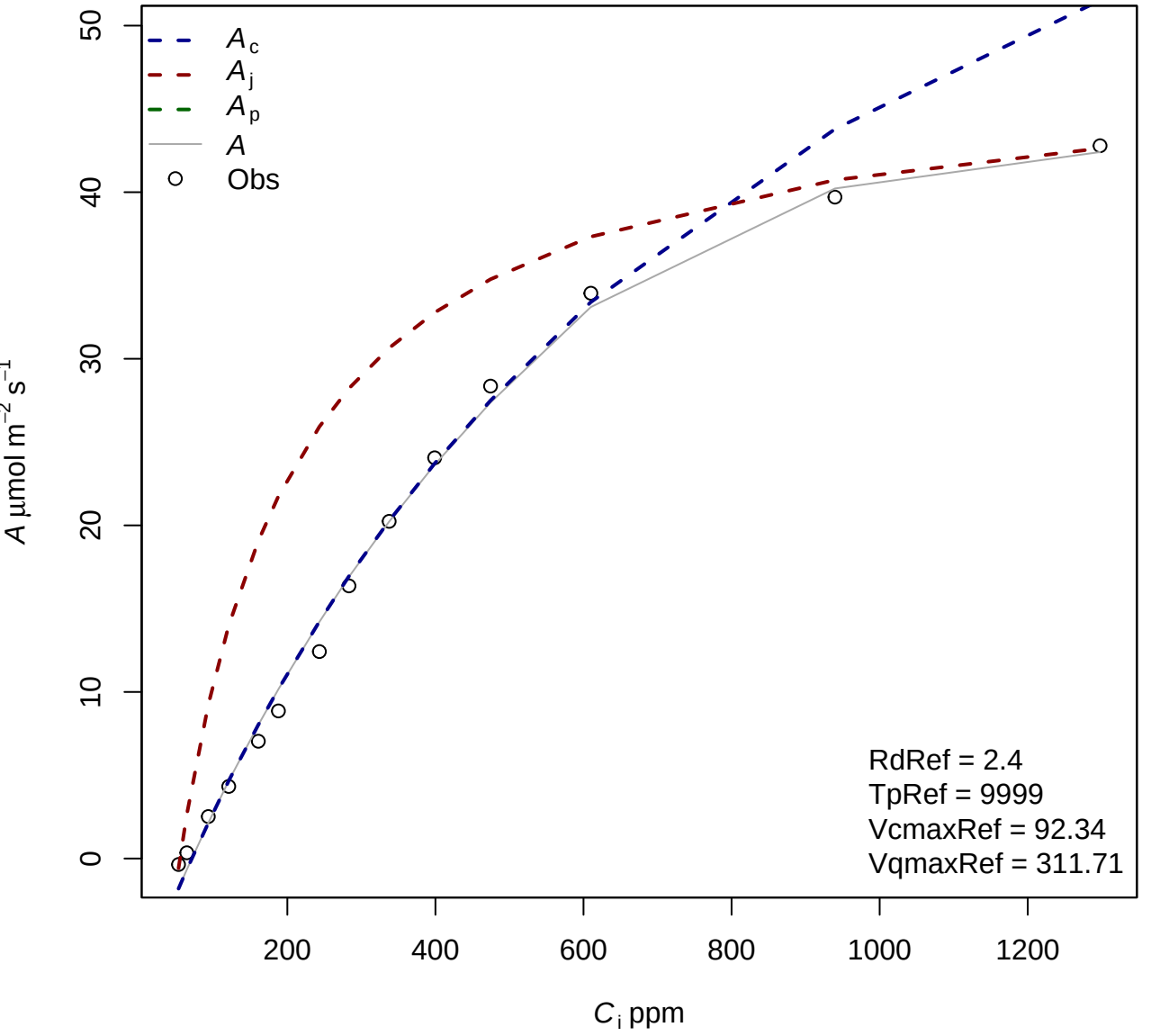

67
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.21
TpRef = 9.72
VcmaxRef = 61.09
VqmaxRef = 162.19
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

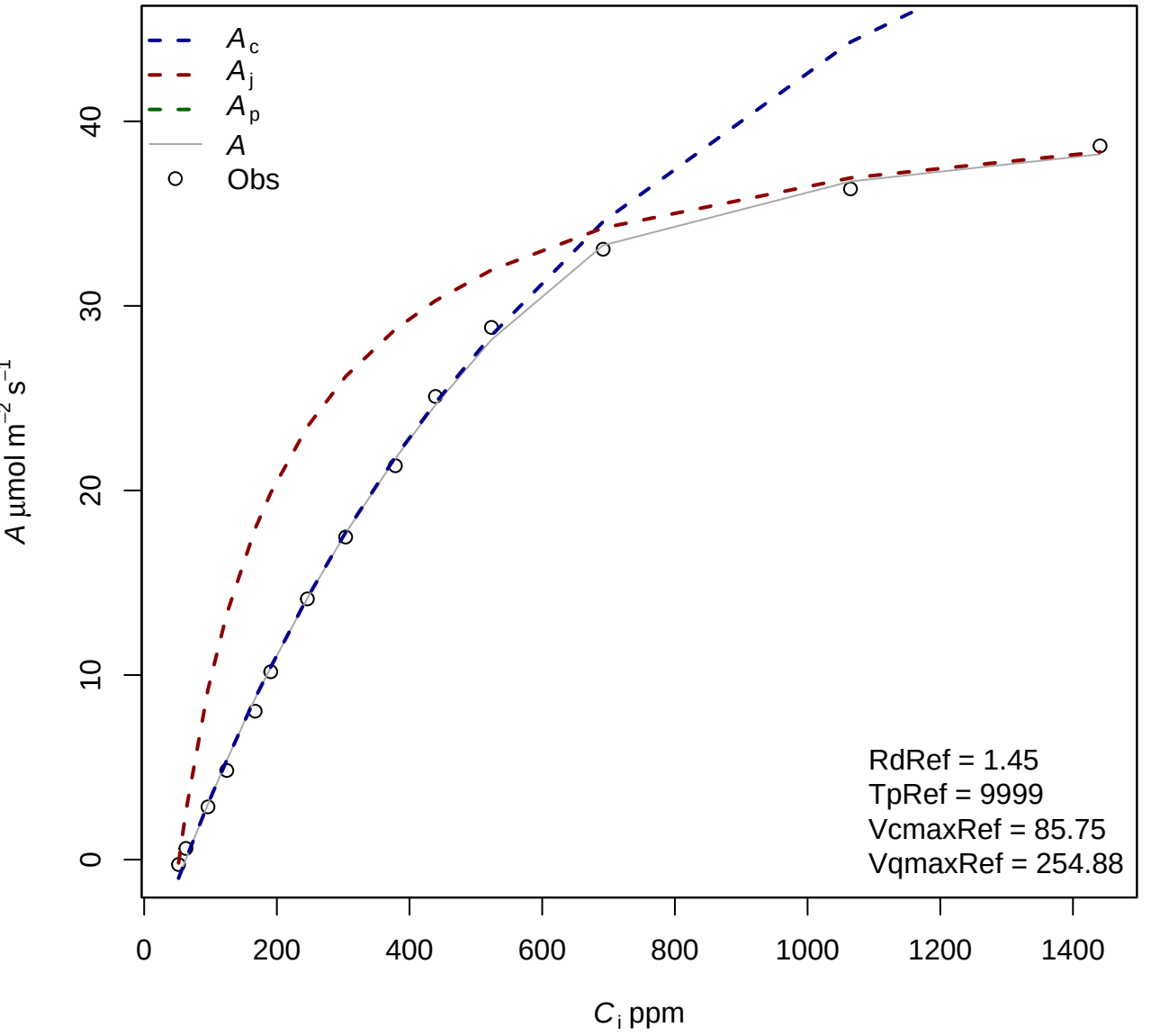

68
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.88
TpRef = 14.72
VcmaxRef = 119.52
VqmaxRef = 324.08
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
0
500
1000
1500

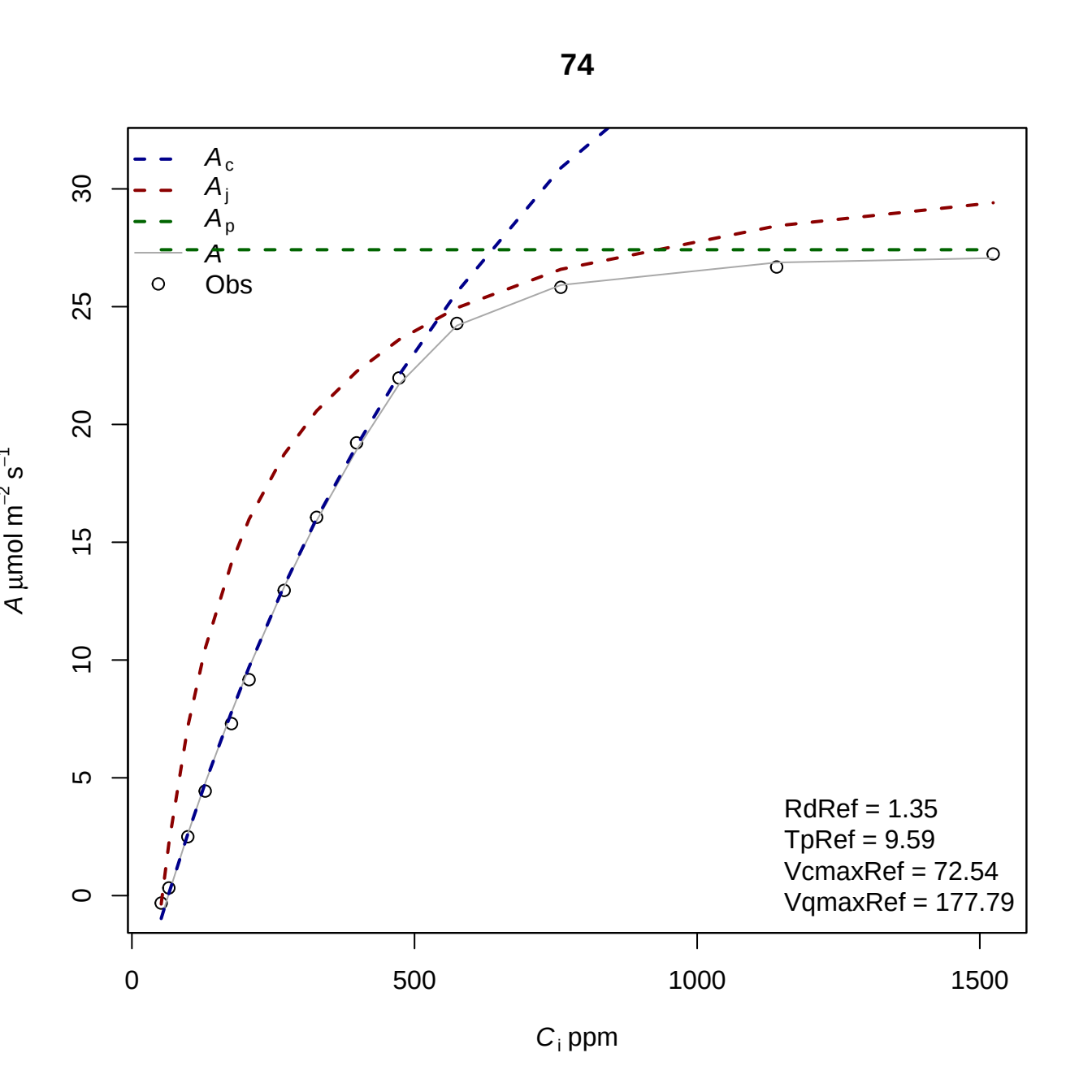

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.45
TpRef = 9999
VcmaxRef = 87.37
VqmaxRef = 225.11
0
10
20
30
200
400
600
800
1000
1200
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

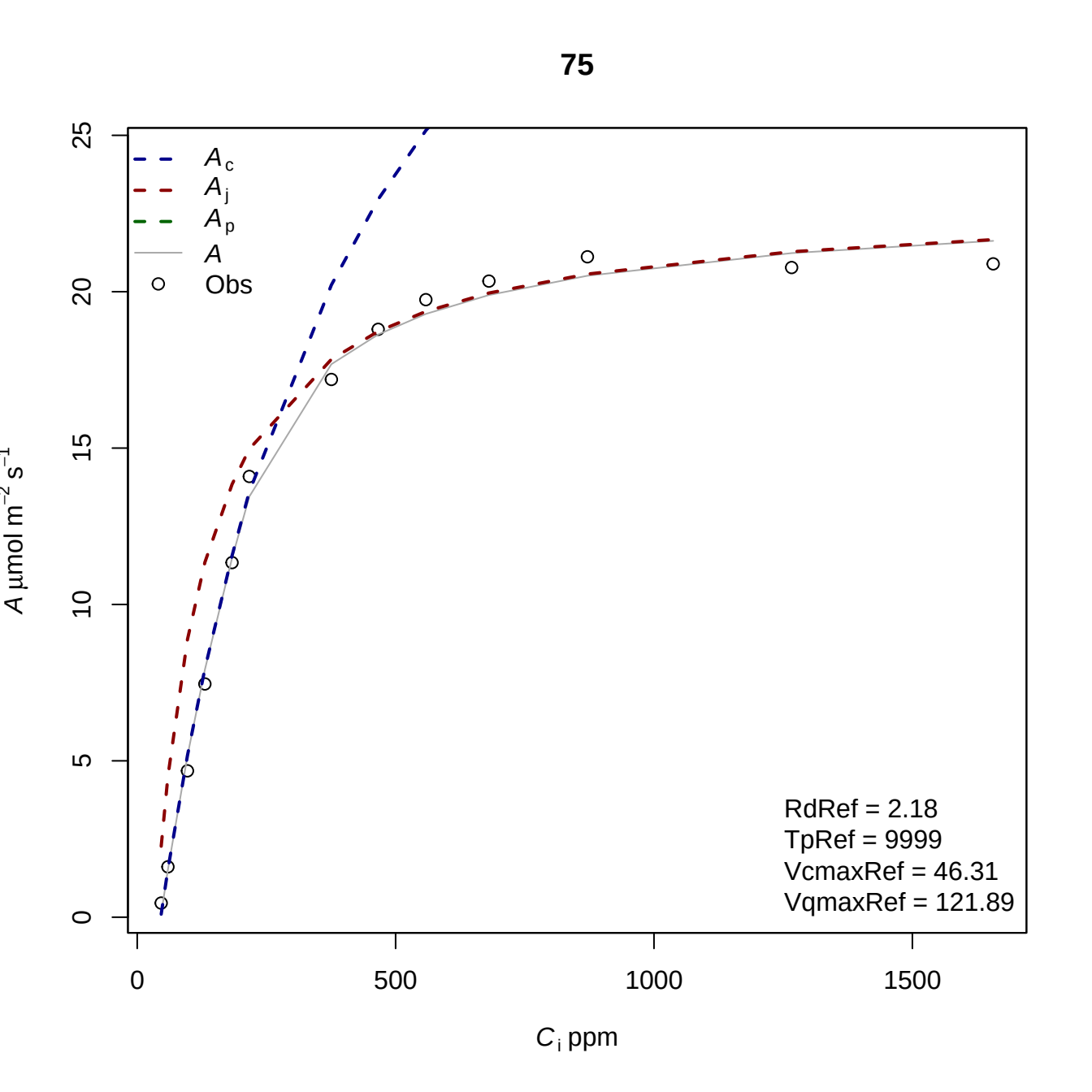

70
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.35
TpRef = 9999
VcmaxRef = 97.75
VqmaxRef = 237.13
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

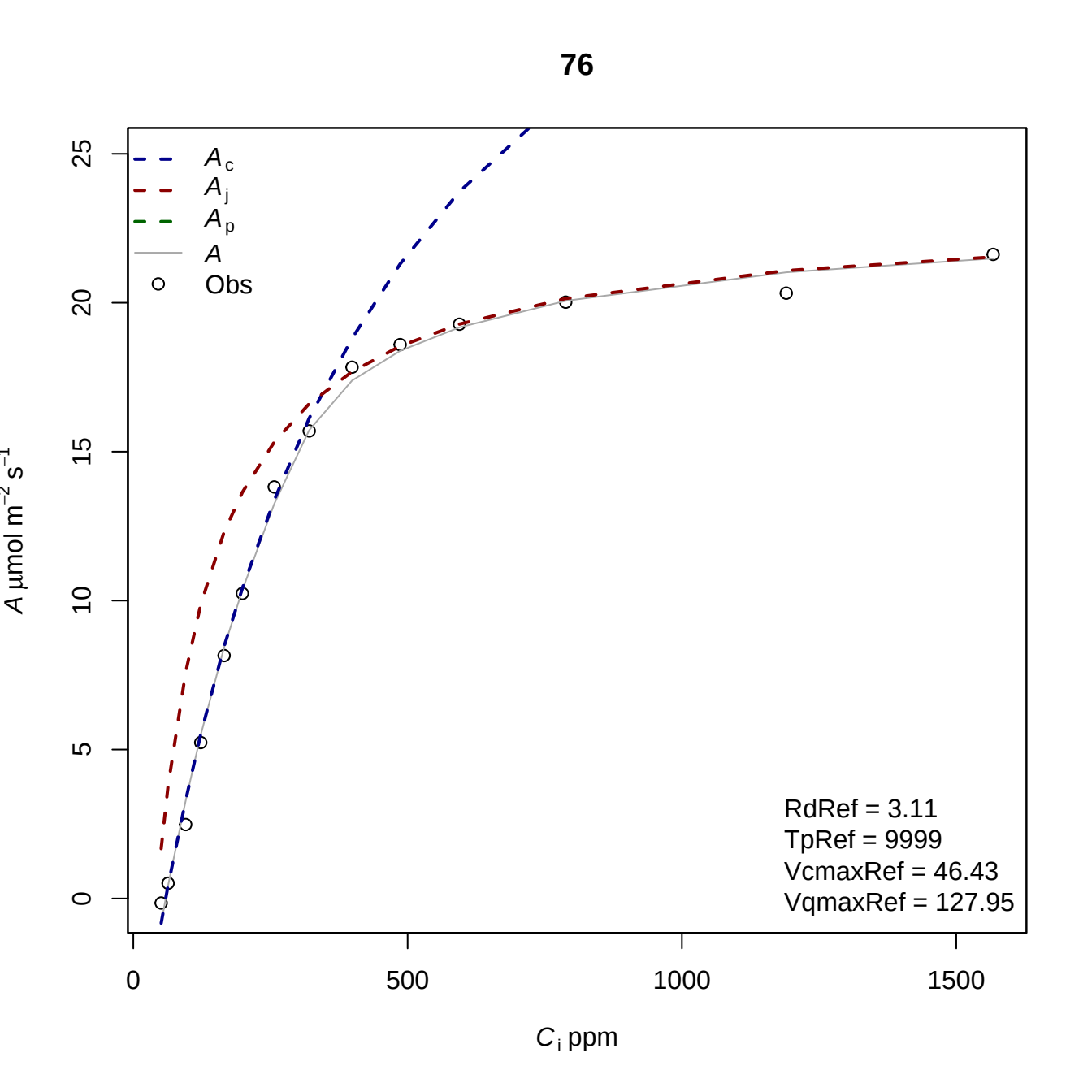

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.33
TpRef = 9999
VcmaxRef = 105.36
VqmaxRef = 291.59
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200
1400

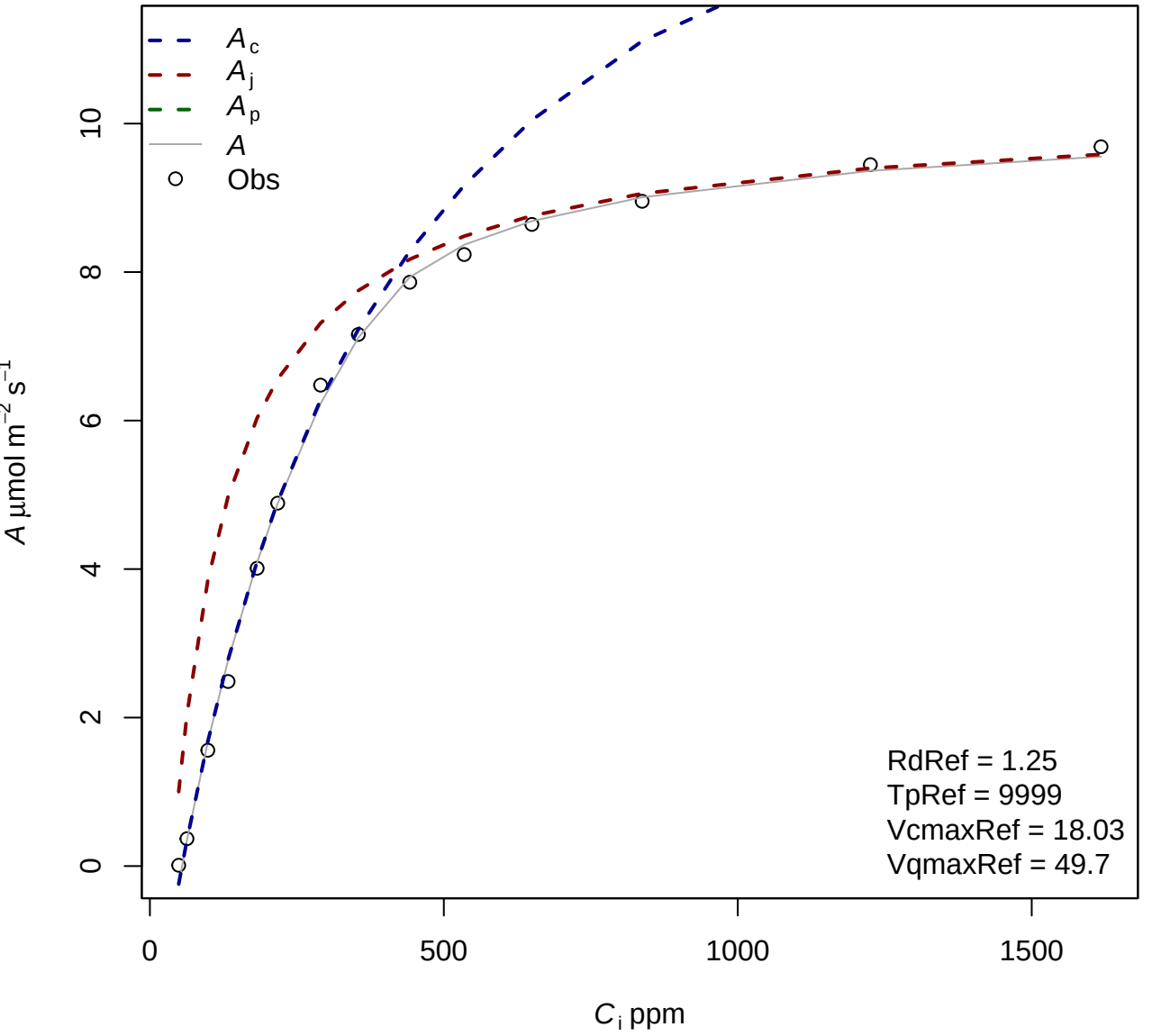

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.4
TpRef = 9999
VcmaxRef = 92.34
VqmaxRef = 311.71
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

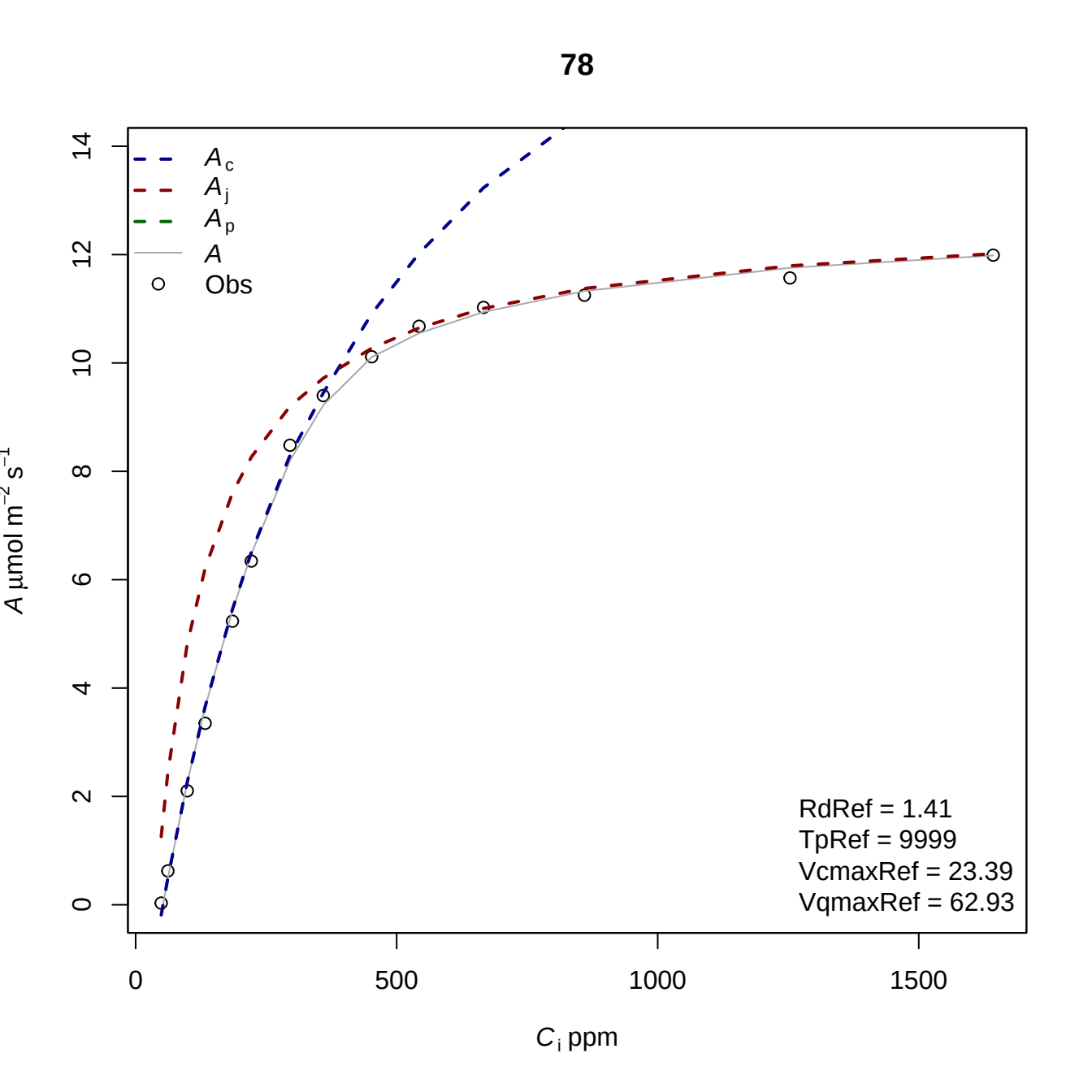

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.45
TpRef = 9999
VcmaxRef = 85.75
VqmaxRef = 254.88
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200
1400

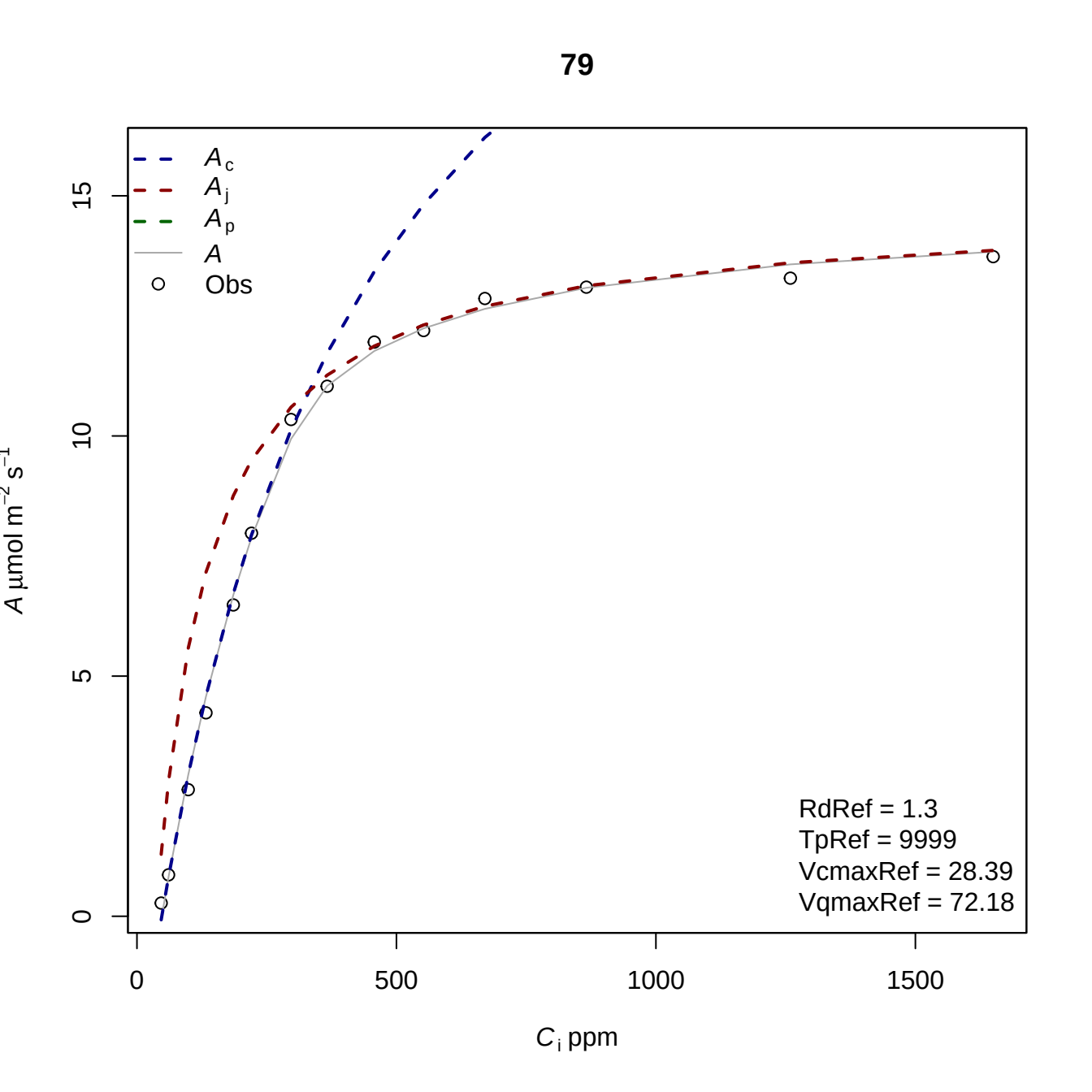

74
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.35
TpRef = 9.59
VcmaxRef = 72.54
VqmaxRef = 177.79
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

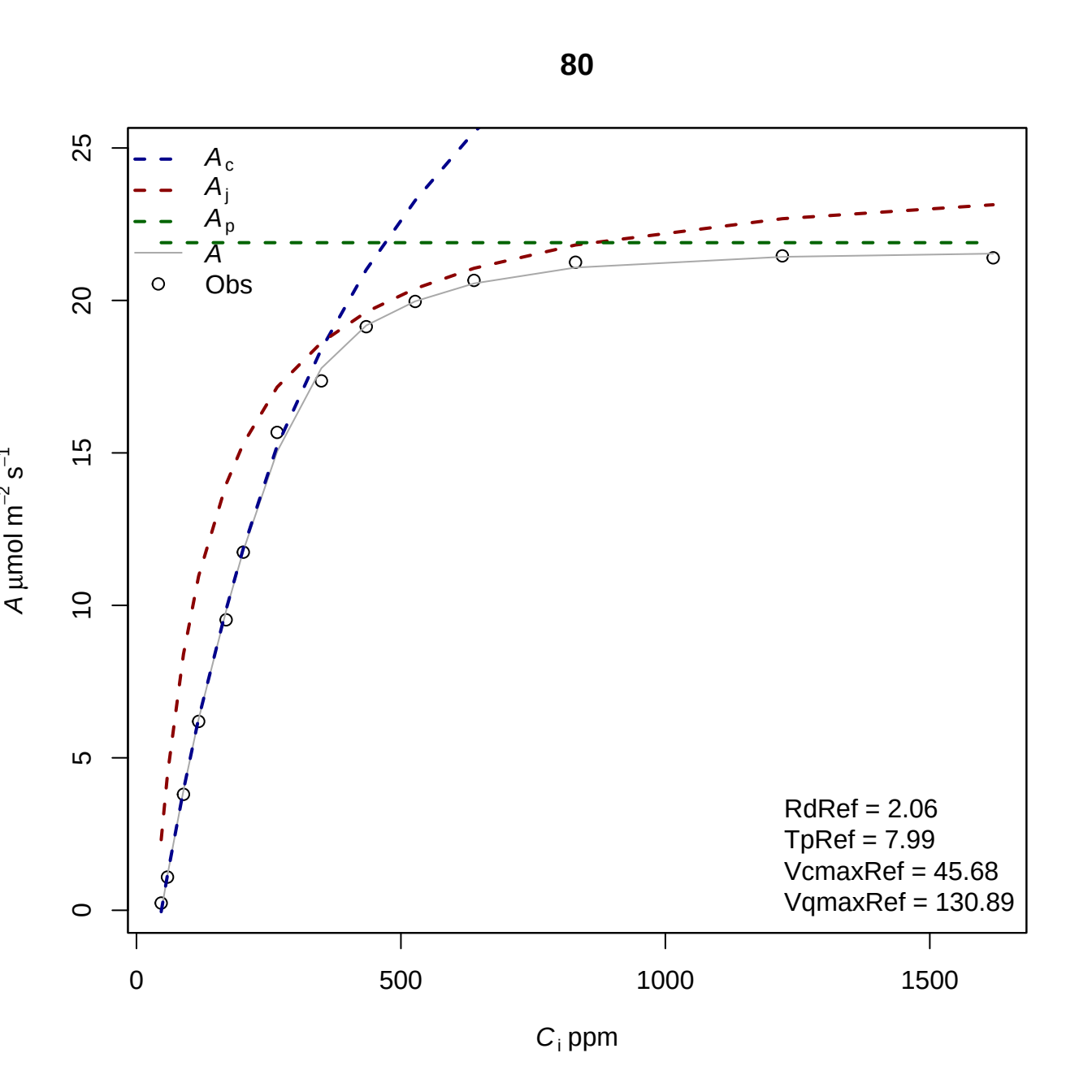

75
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.18
TpRef = 9999
VcmaxRef = 46.31
VqmaxRef = 121.89
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

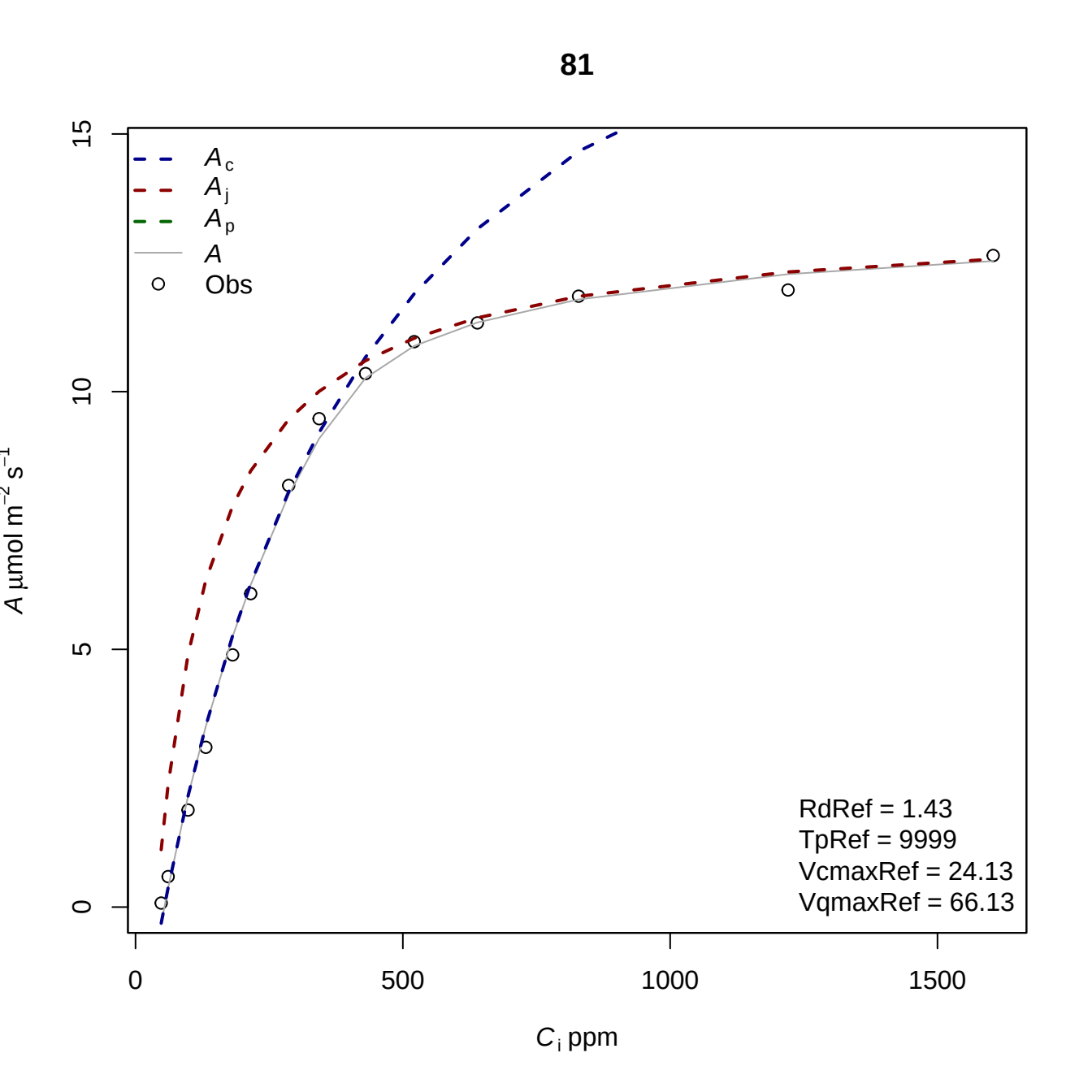

76
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.11
TpRef = 9999
VcmaxRef = 46.43
VqmaxRef = 127.95
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

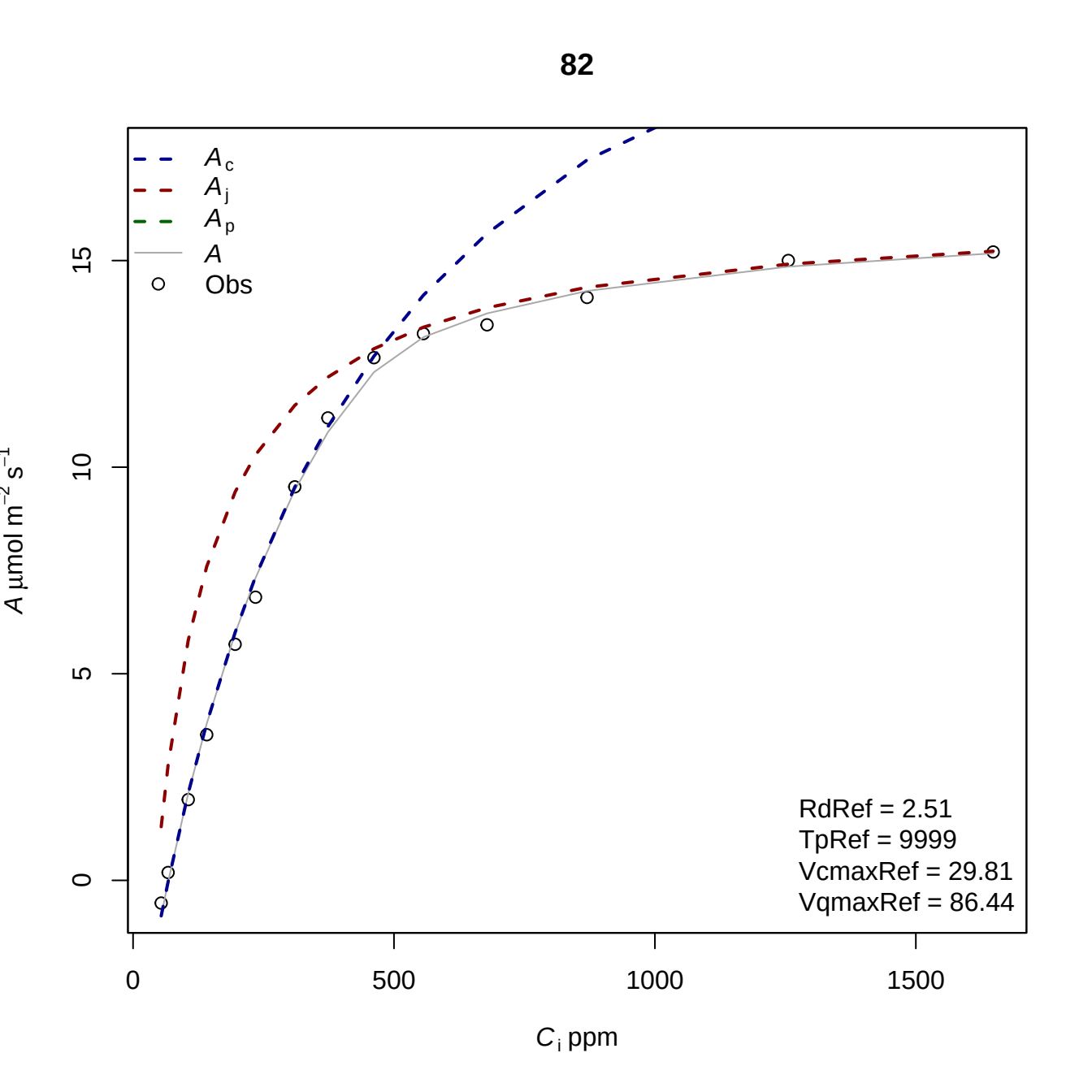

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.25
TpRef = 9999
VcmaxRef = 18.03
VqmaxRef = 49.7
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

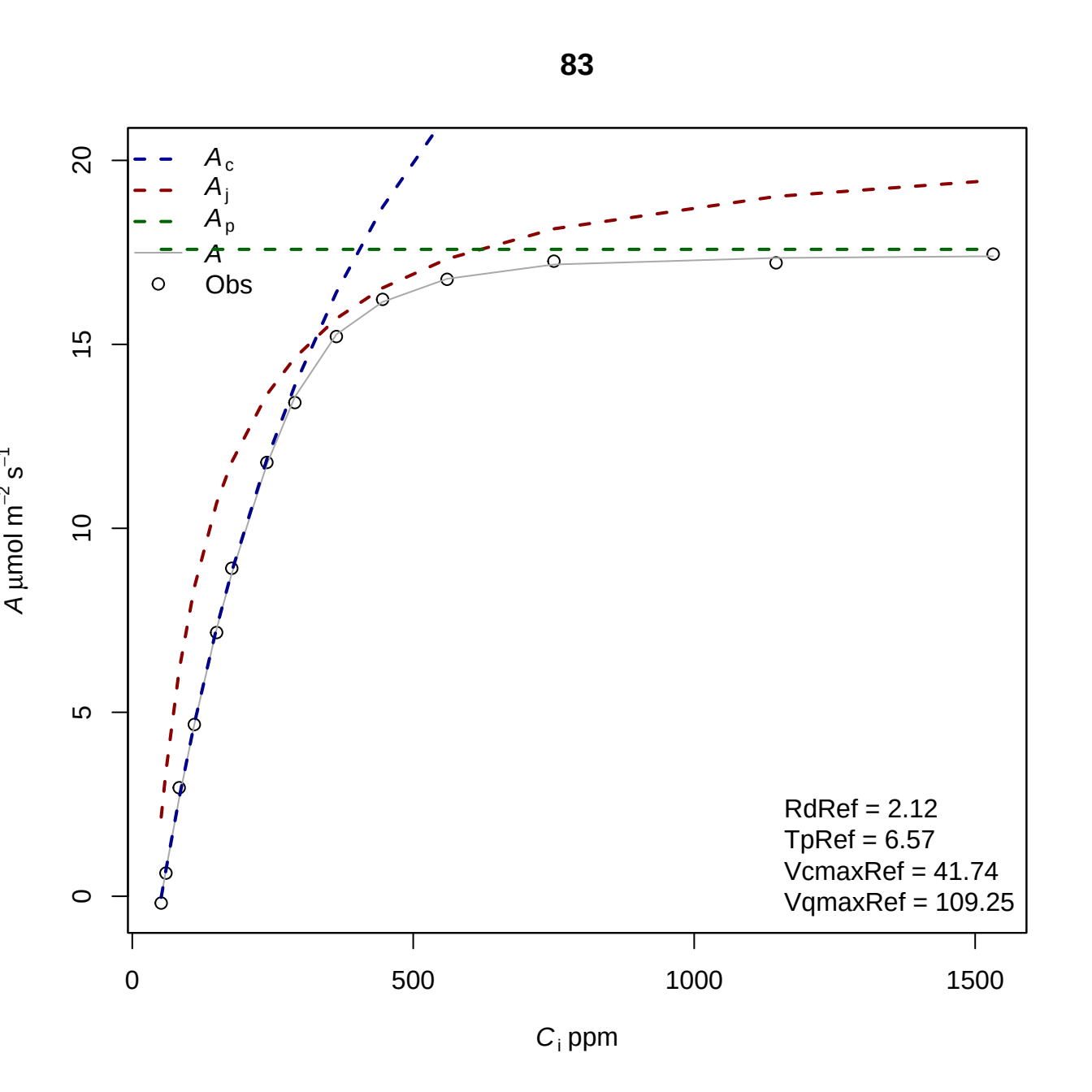

78
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.41
TpRef = 9999
VcmaxRef = 23.39
VqmaxRef = 62.93
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
0
500
1000
1500

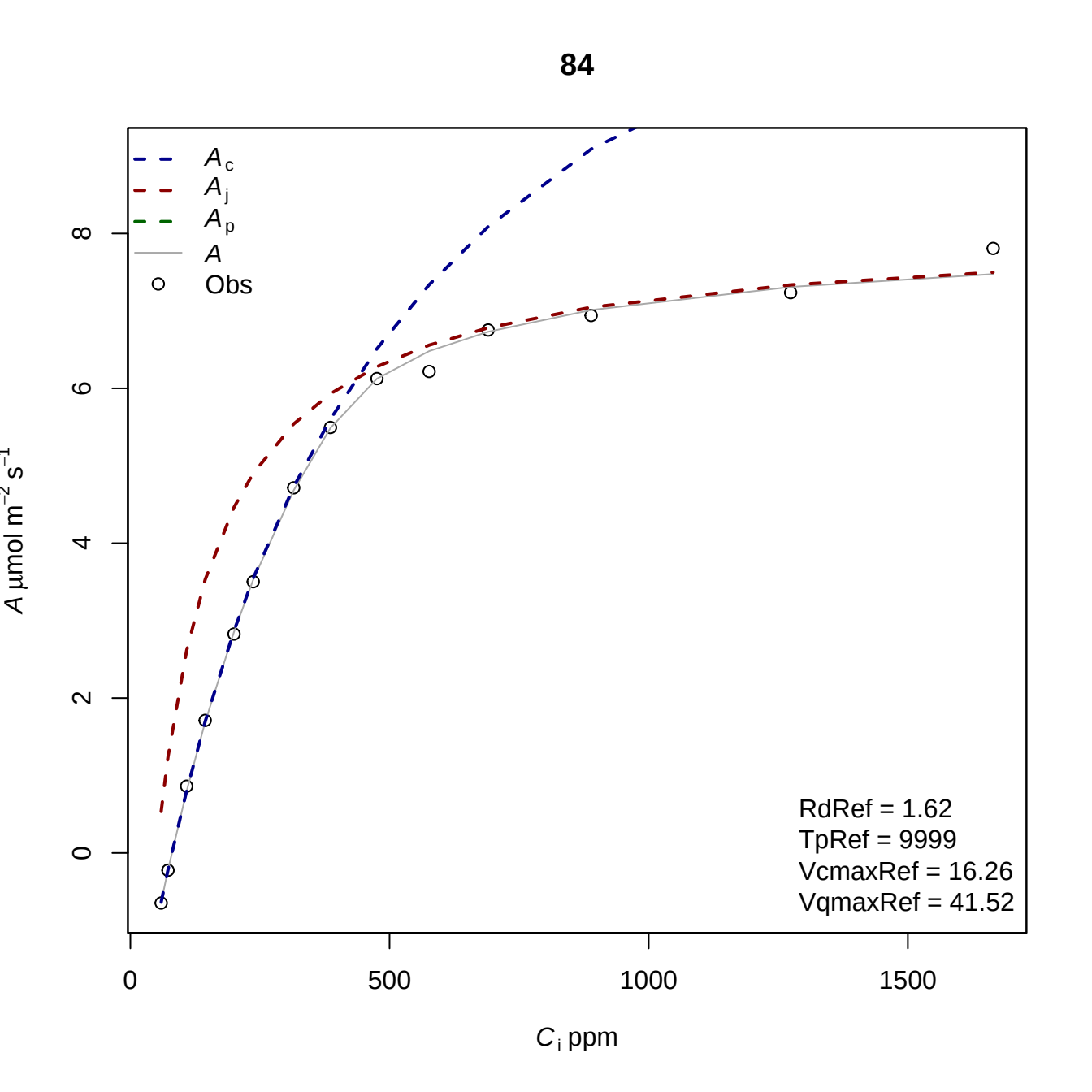

79
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.3
TpRef = 9999
VcmaxRef = 28.39
VqmaxRef = 72.18
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

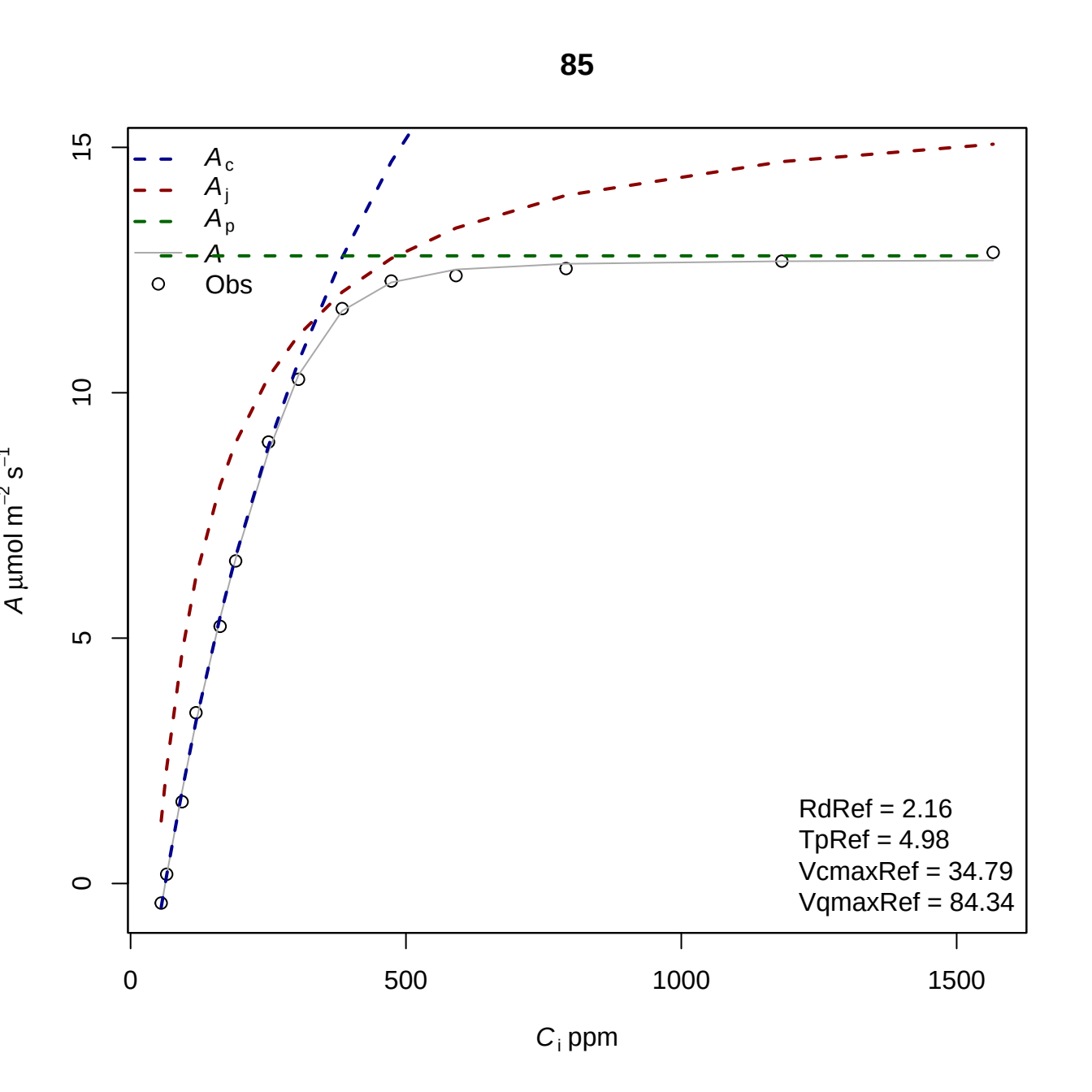

80
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.06
TpRef = 7.99
VcmaxRef = 45.68
VqmaxRef = 130.89
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

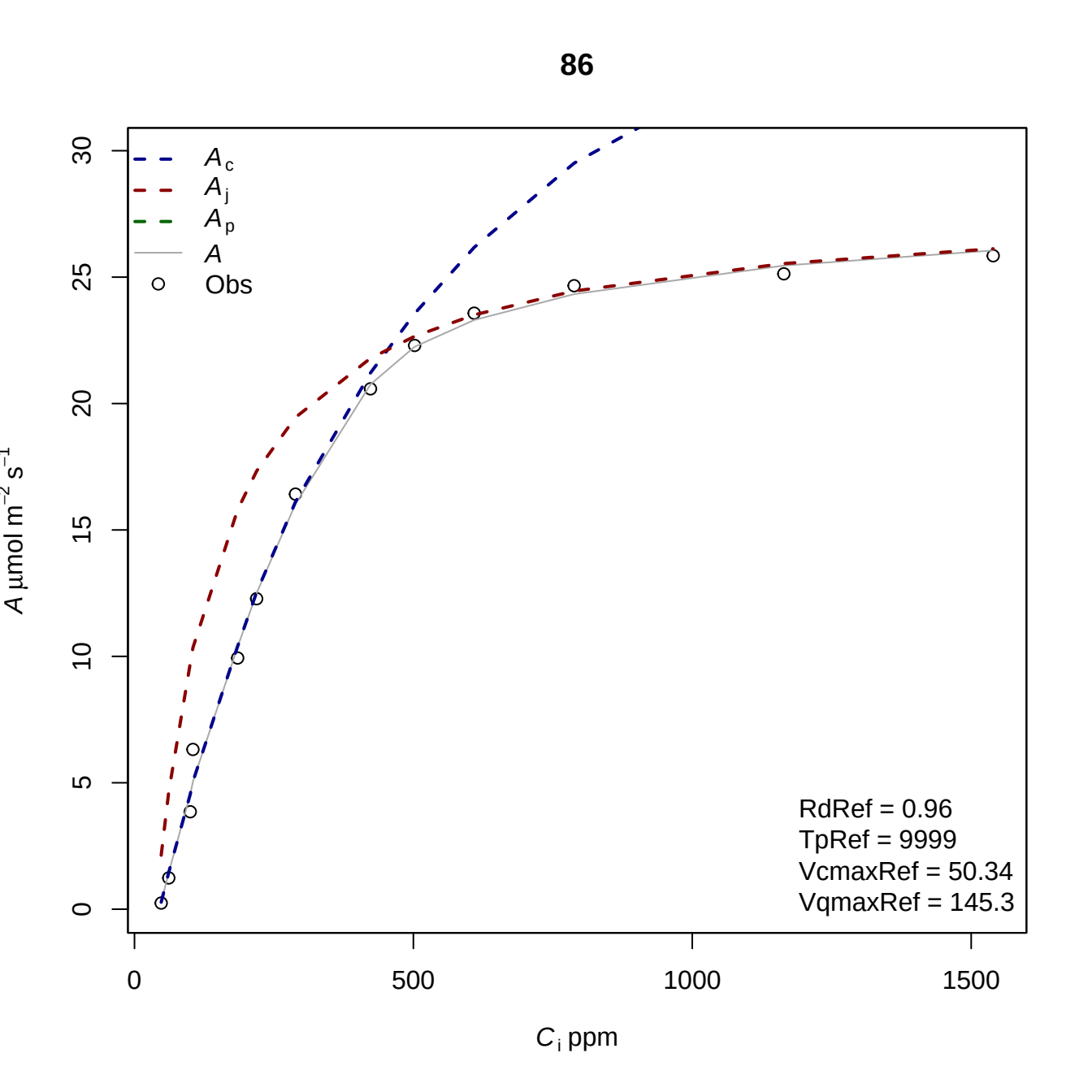

81
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.43
TpRef = 9999
VcmaxRef = 24.13
VqmaxRef = 66.13
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

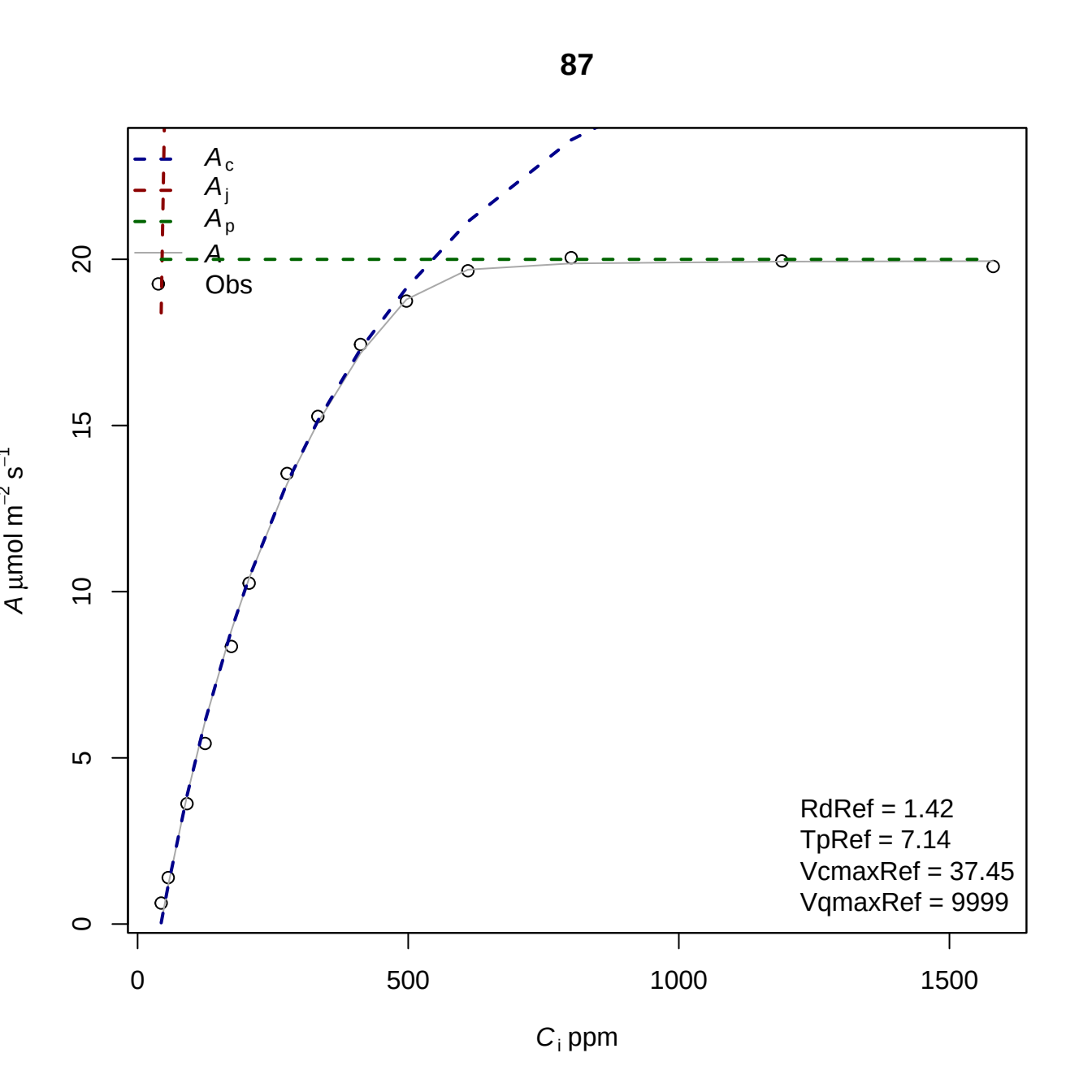

82
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.51
TpRef = 9999
VcmaxRef = 29.81
VqmaxRef = 86.44
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

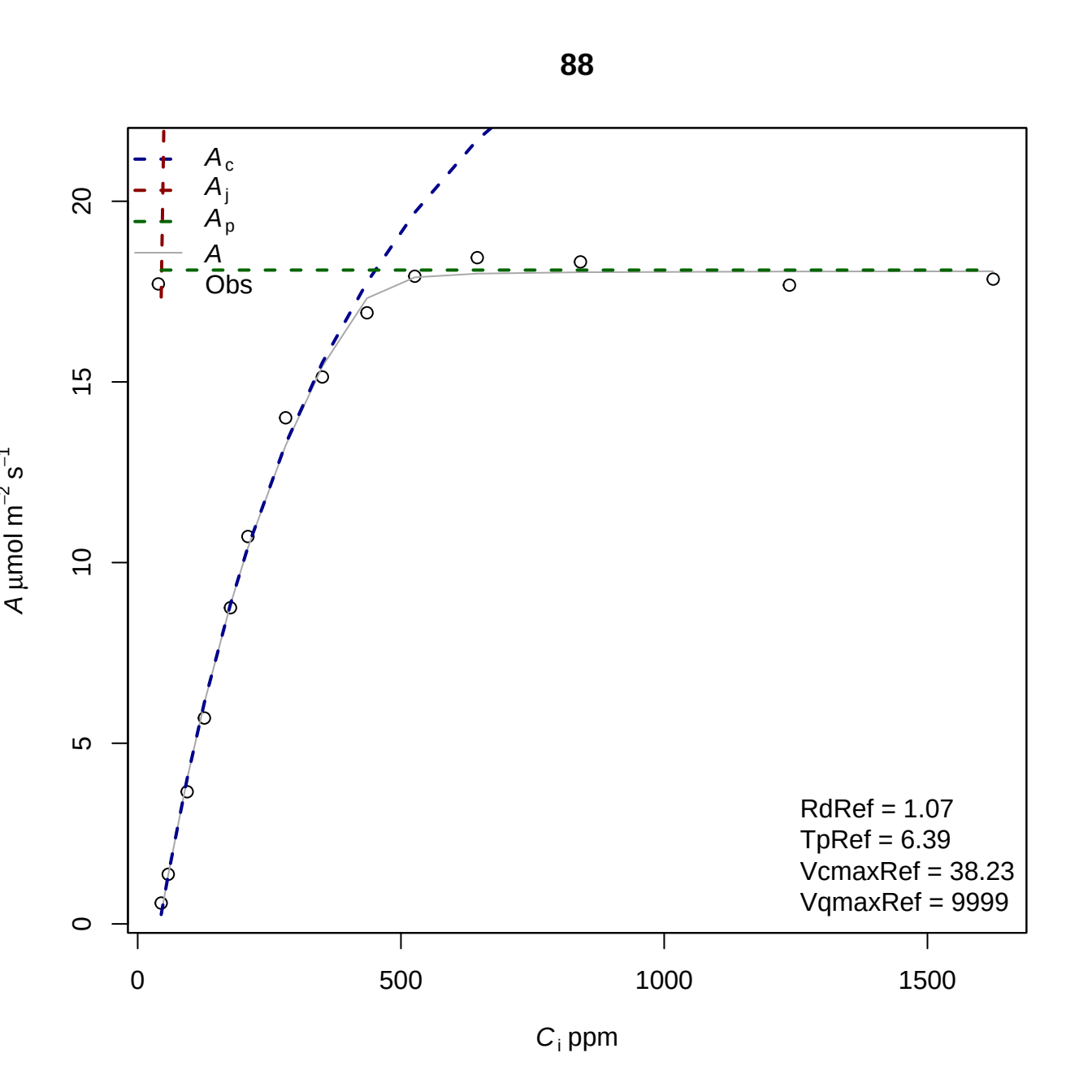

83
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.12
TpRef = 6.57
VcmaxRef = 41.74
VqmaxRef = 109.25
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

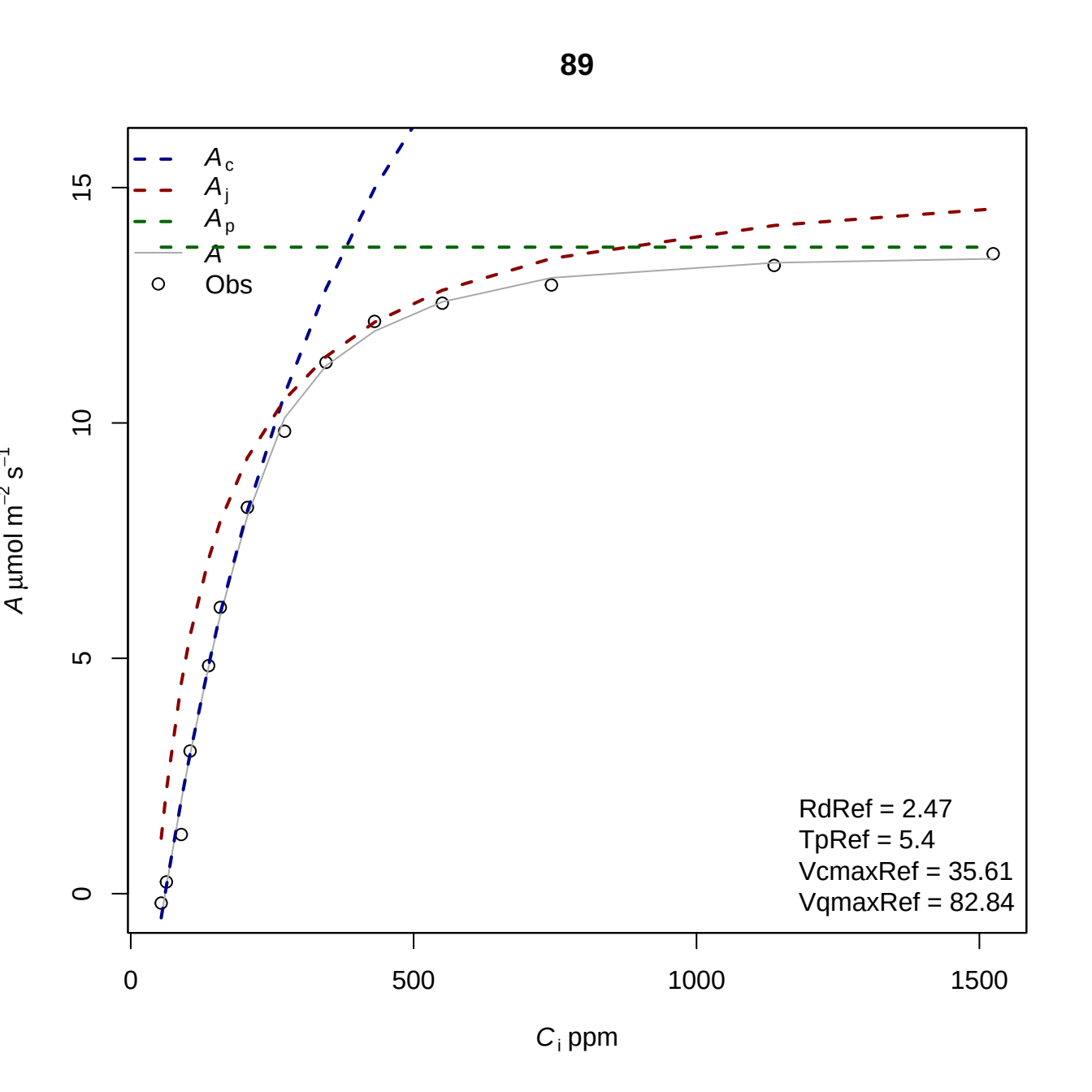

84
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.62
TpRef = 9999
VcmaxRef = 16.26
VqmaxRef = 41.52
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
0
500
1000
1500

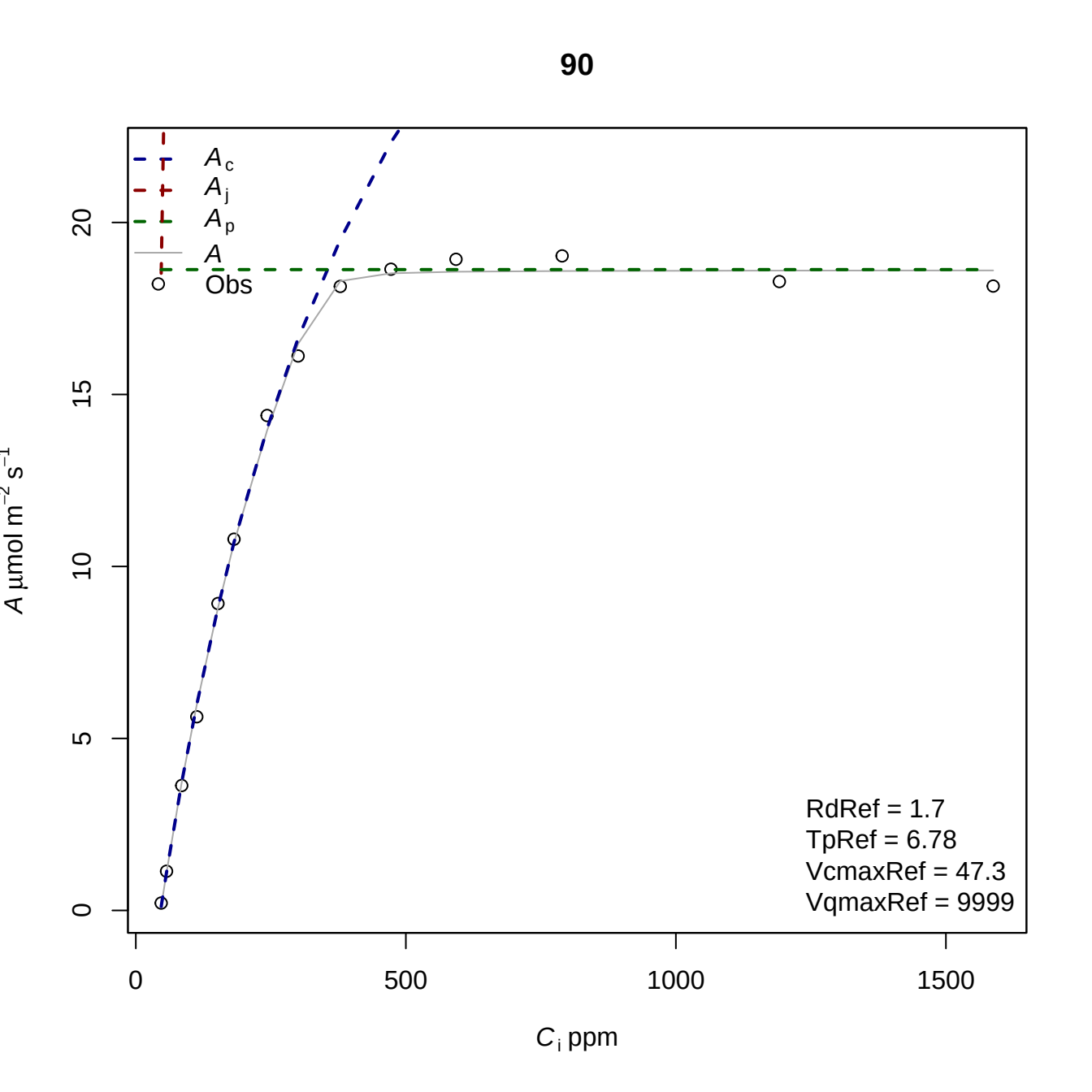

85
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.16
TpRef = 4.98
VcmaxRef = 34.79
VqmaxRef = 84.34
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

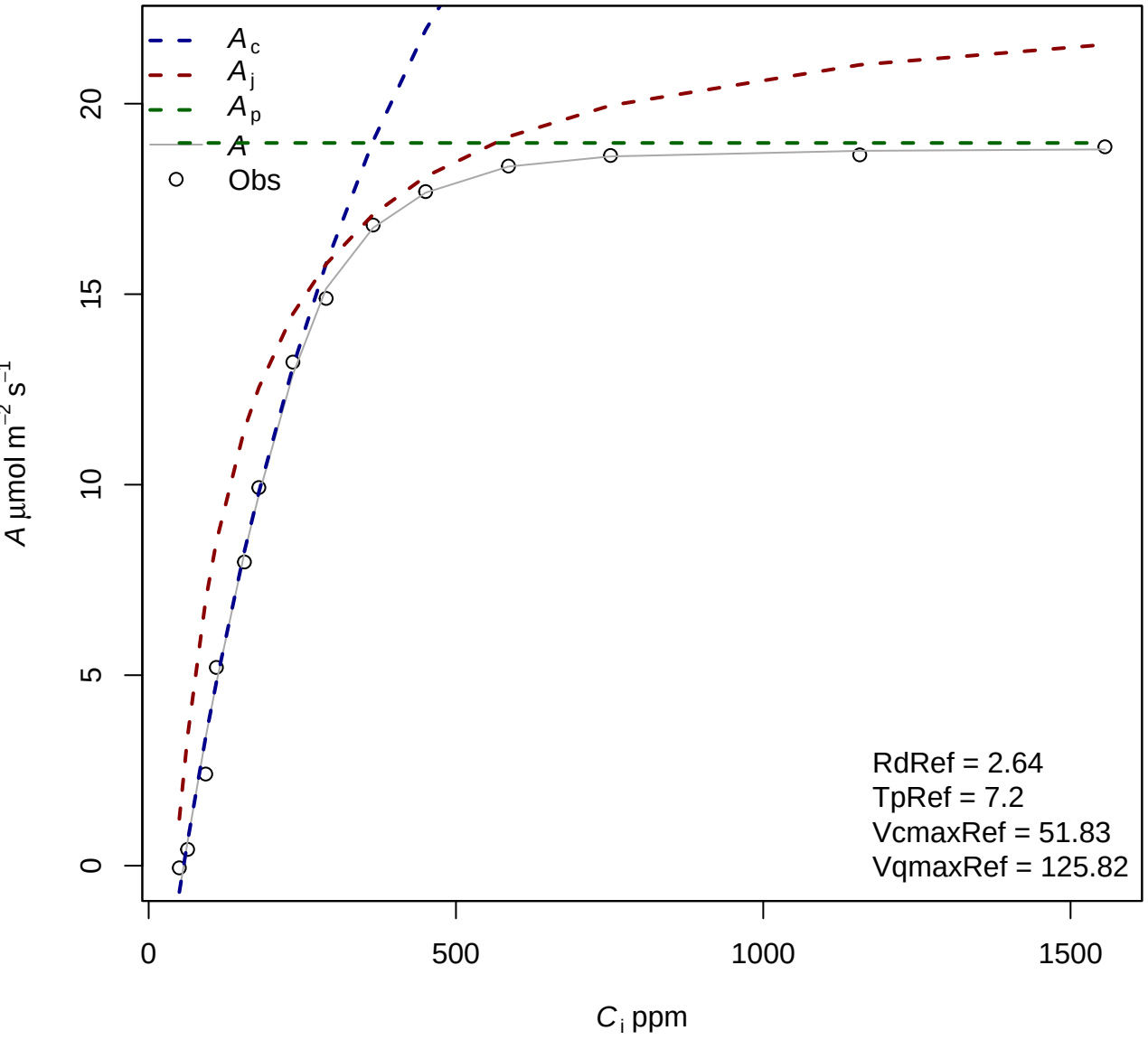

86
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.96
TpRef = 9999
VcmaxRef = 50.34
VqmaxRef = 145.3
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

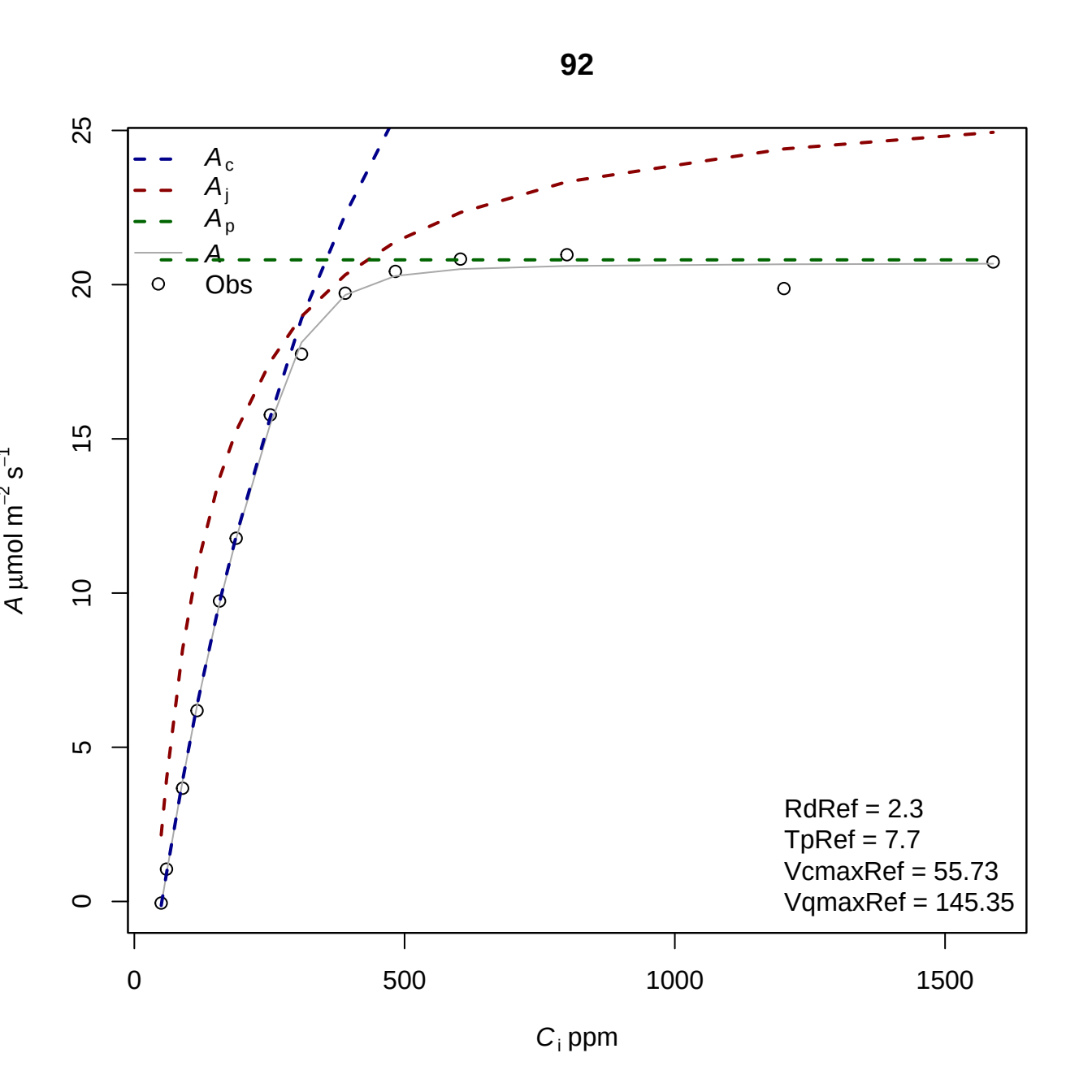

87
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.42
TpRef = 7.14
VcmaxRef = 37.45
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

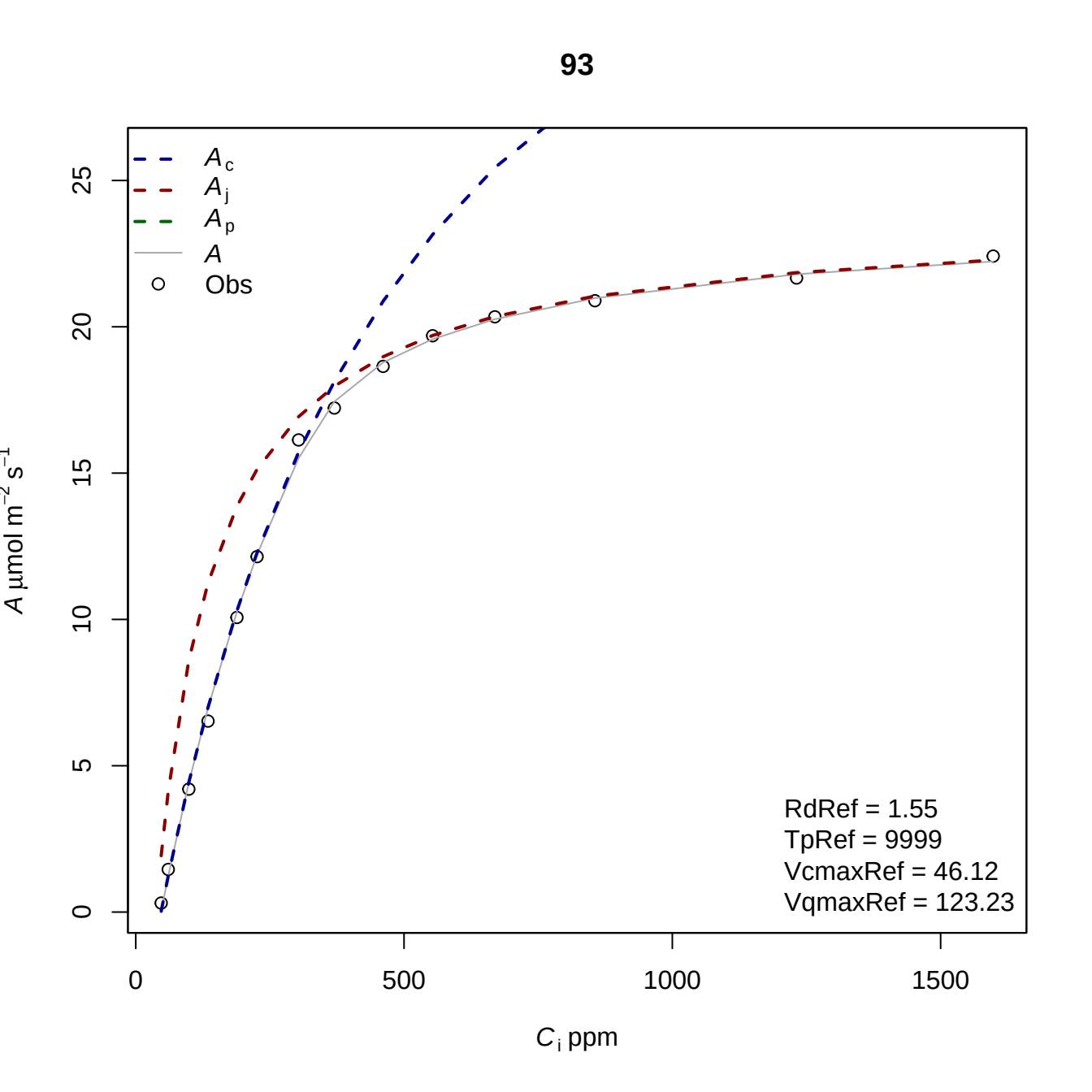

88
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.07
TpRef = 6.39
VcmaxRef = 38.23
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

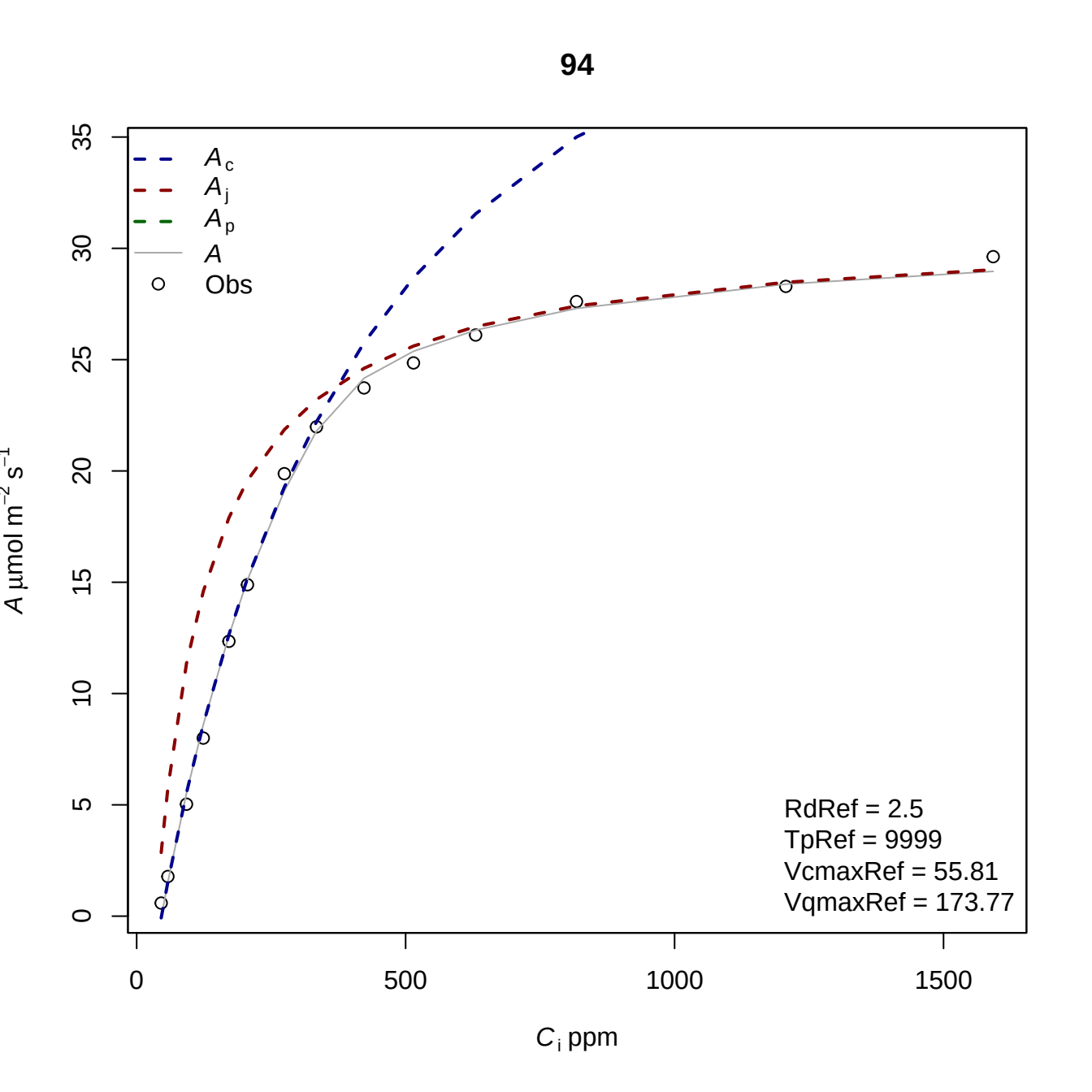

89
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.47
TpRef = 5.4
VcmaxRef = 35.61
VqmaxRef = 82.84
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

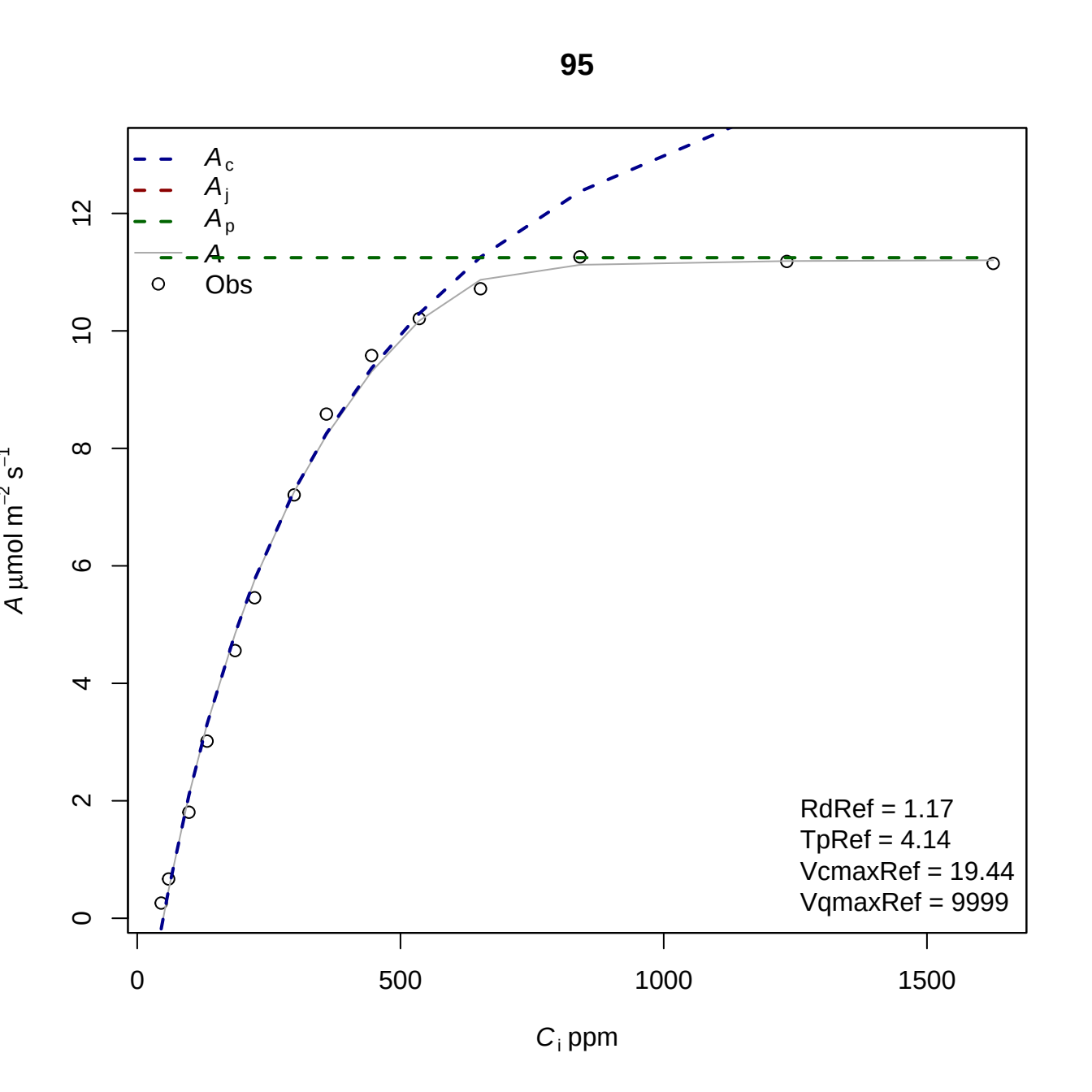

90
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.7
TpRef = 6.78
VcmaxRef = 47.3
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

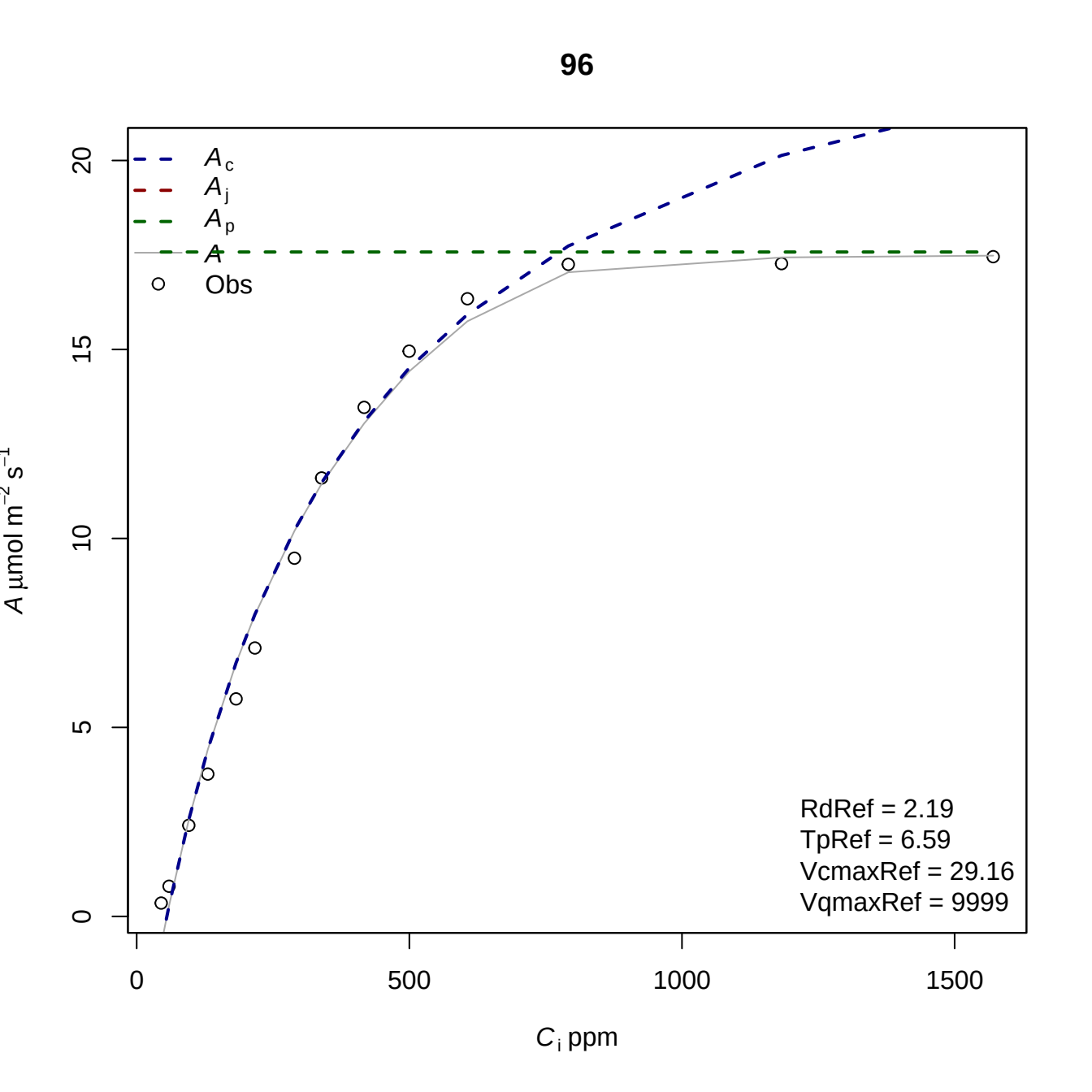

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.64
TpRef = 7.2
VcmaxRef = 51.83
VqmaxRef = 125.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

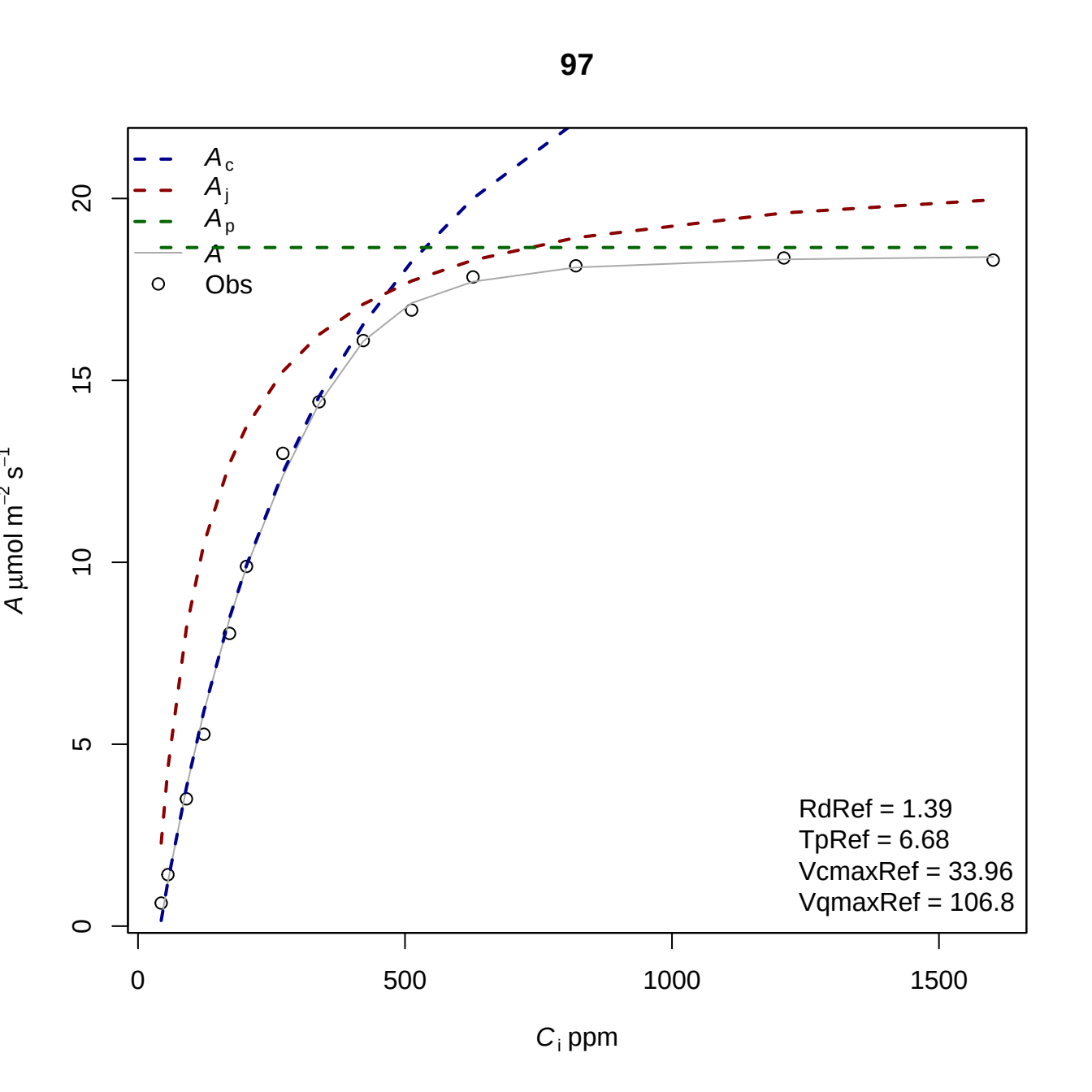

92
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.3
TpRef = 7.7
VcmaxRef = 55.73
VqmaxRef = 145.35
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

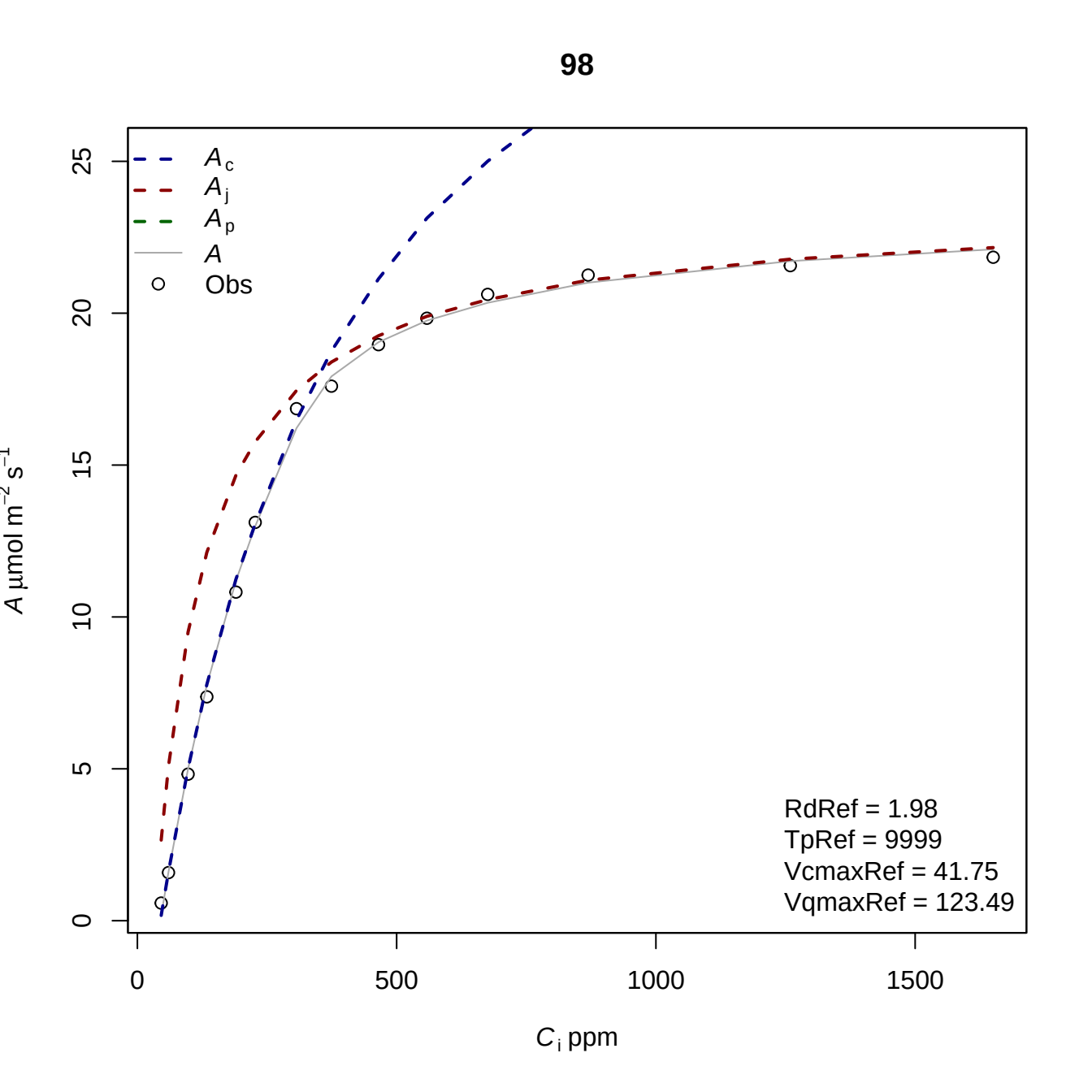

93
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.55
TpRef = 9999
VcmaxRef = 46.12
VqmaxRef = 123.23
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

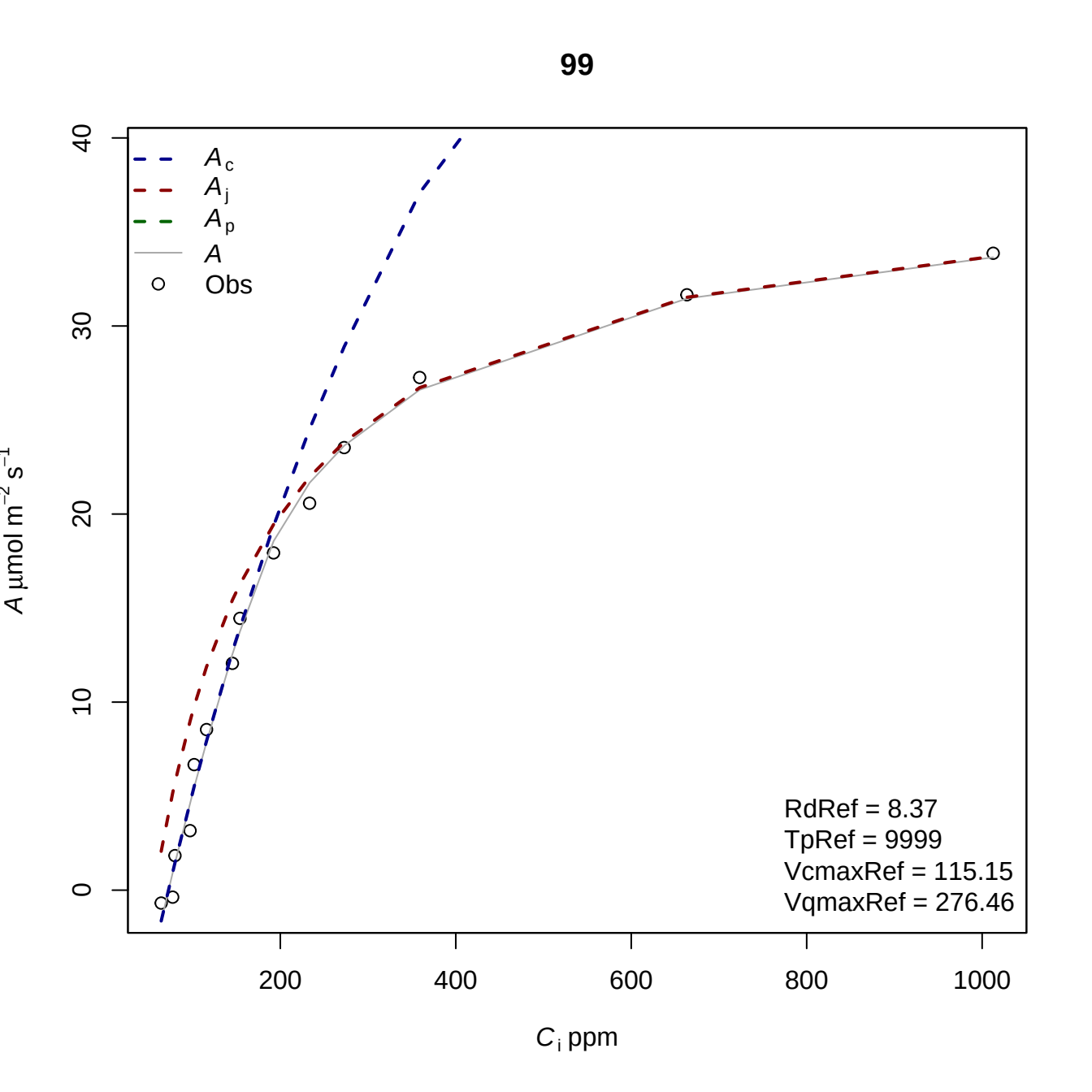

94
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.5
TpRef = 9999
VcmaxRef = 55.81
VqmaxRef = 173.77
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
500
1000
1500

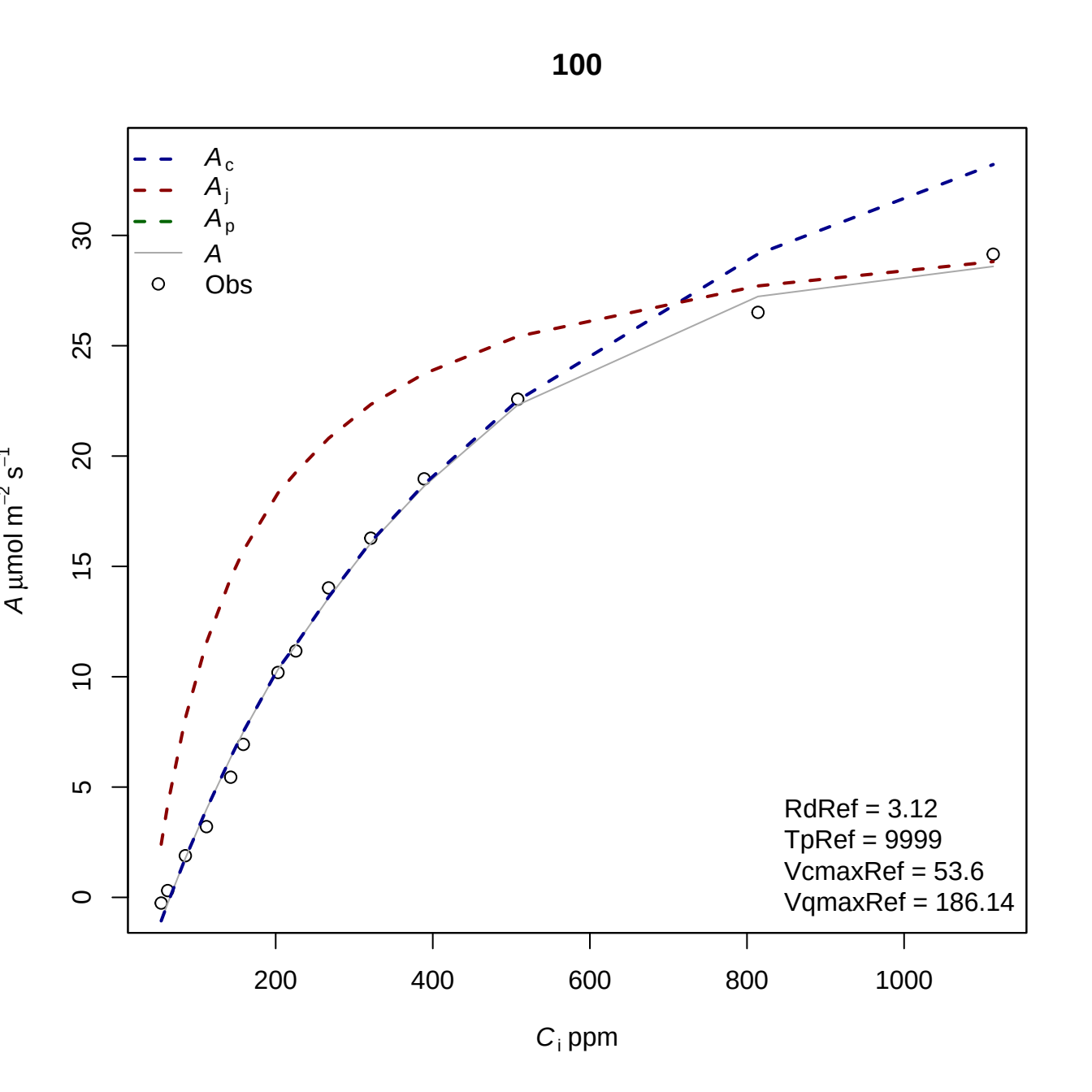

95
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.17
TpRef = 4.14
VcmaxRef = 19.44
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

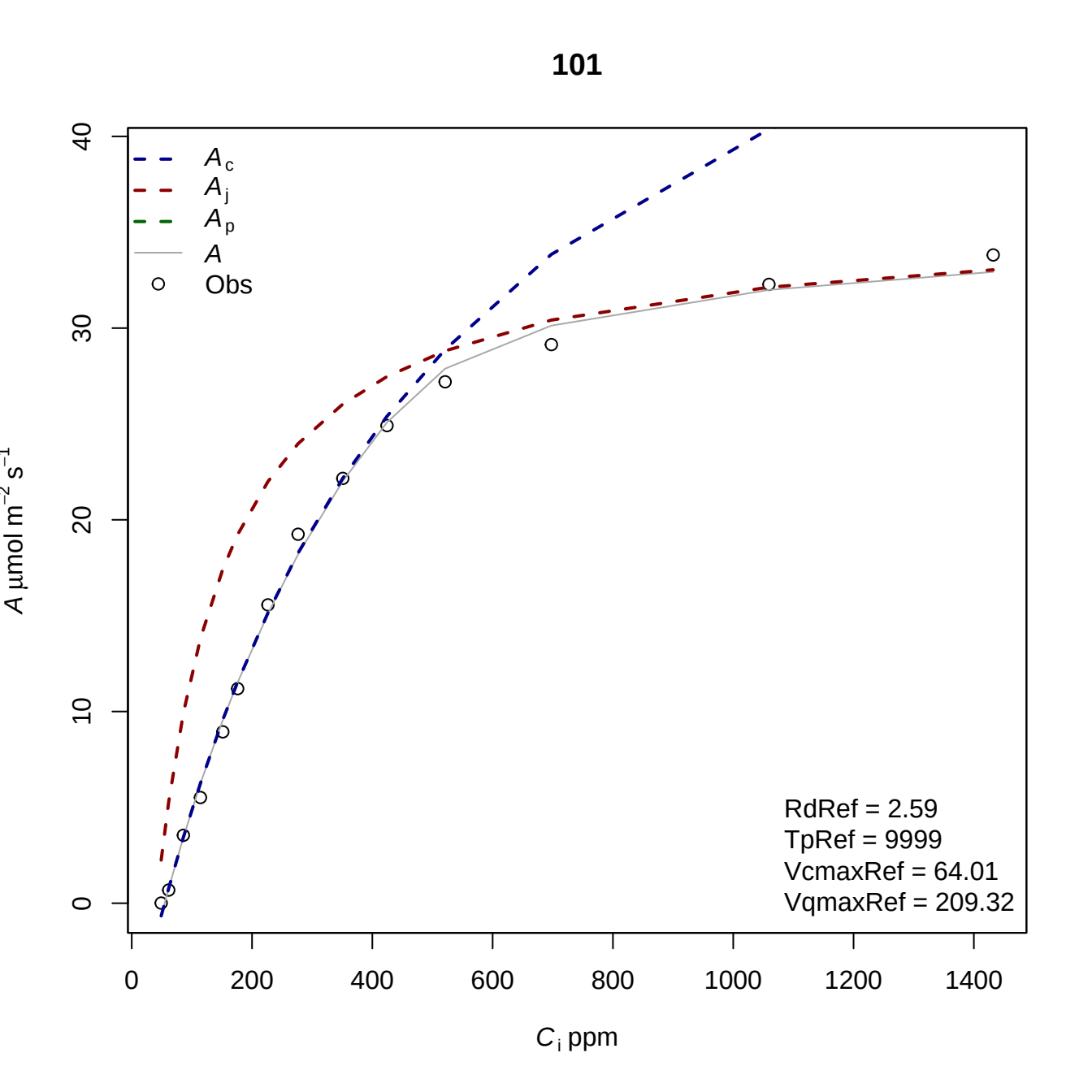

96
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.19
TpRef = 6.59
VcmaxRef = 29.16
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

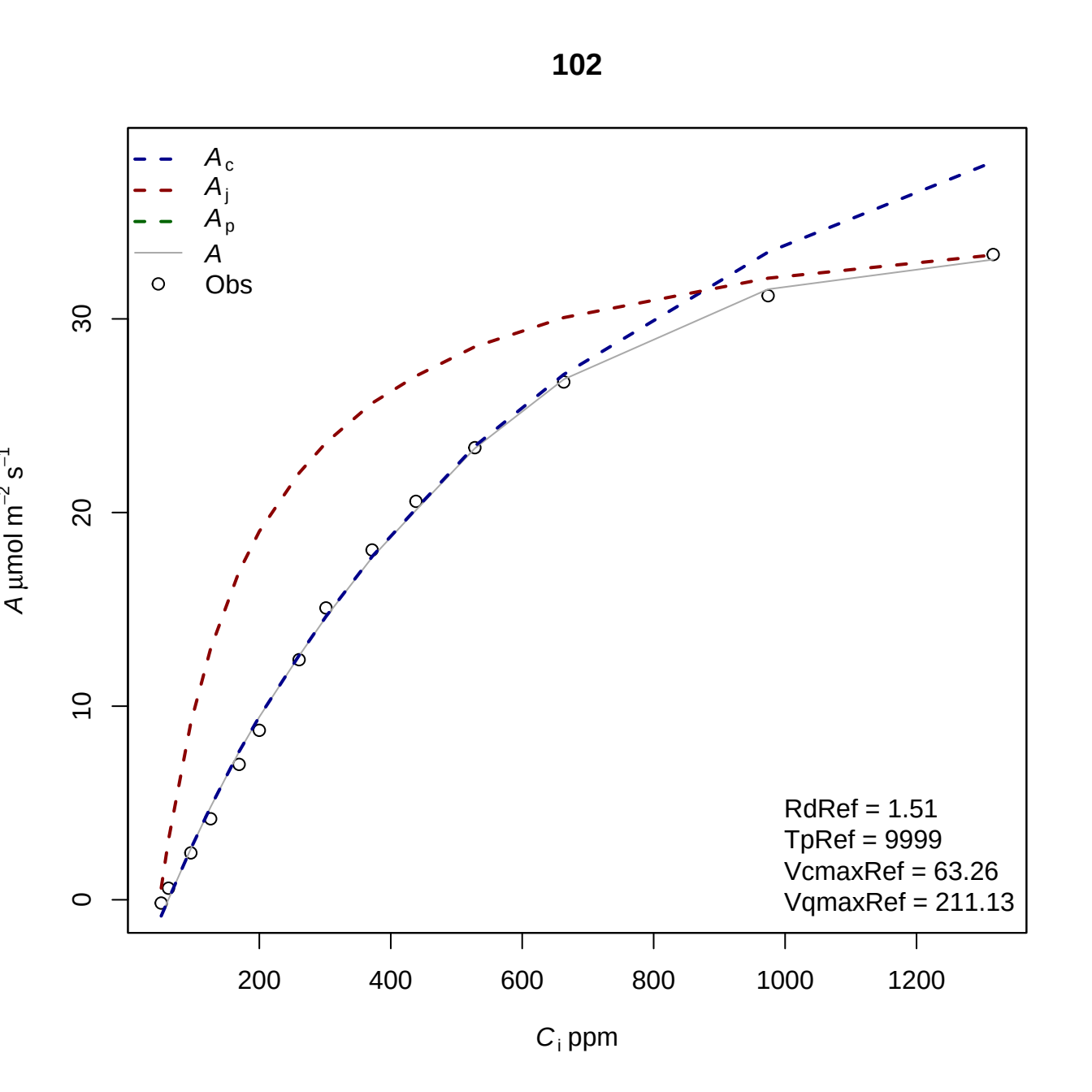

97
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.39
TpRef = 6.68
VcmaxRef = 33.96
VqmaxRef = 106.8
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

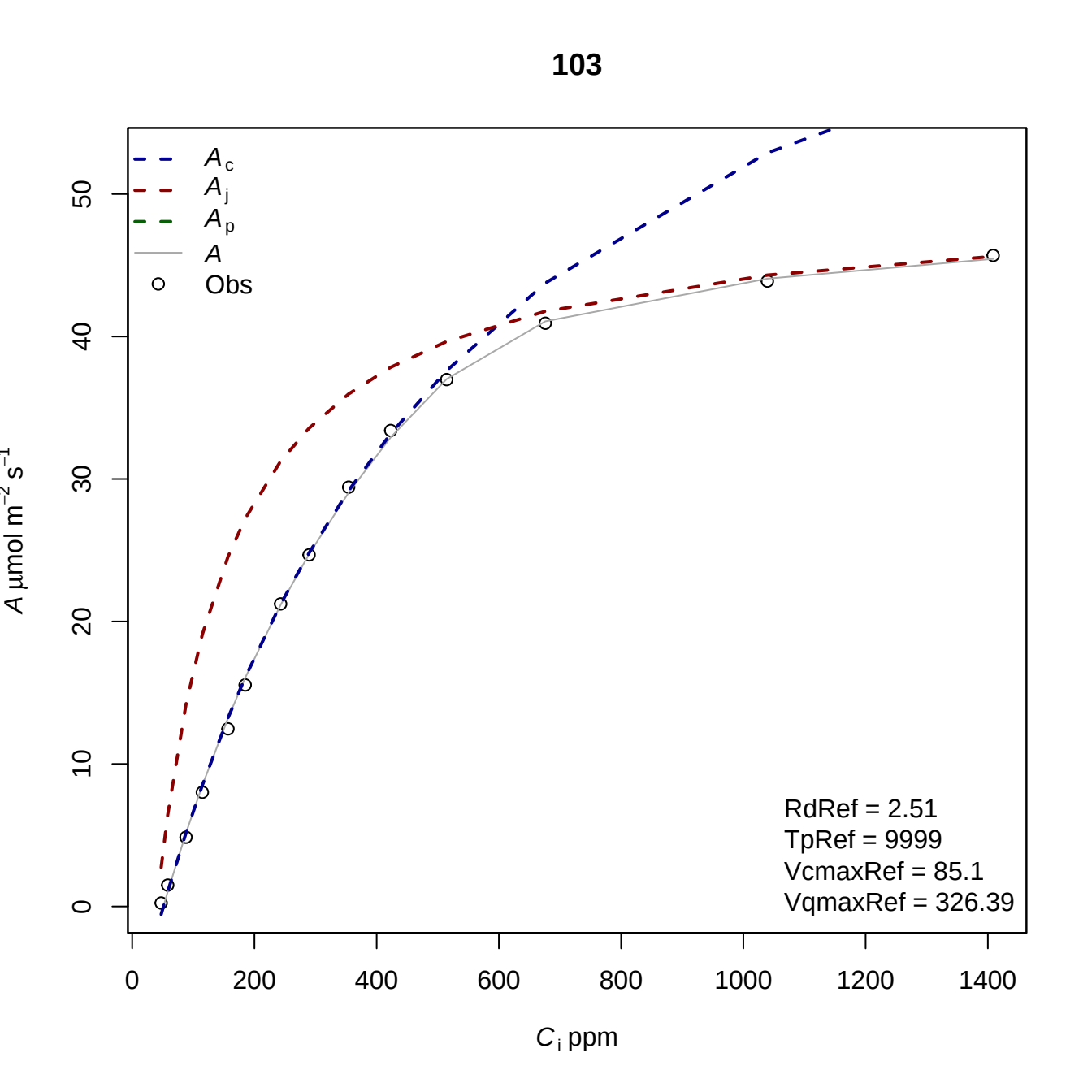

98
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.98
TpRef = 9999
VcmaxRef = 41.75
VqmaxRef = 123.49
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

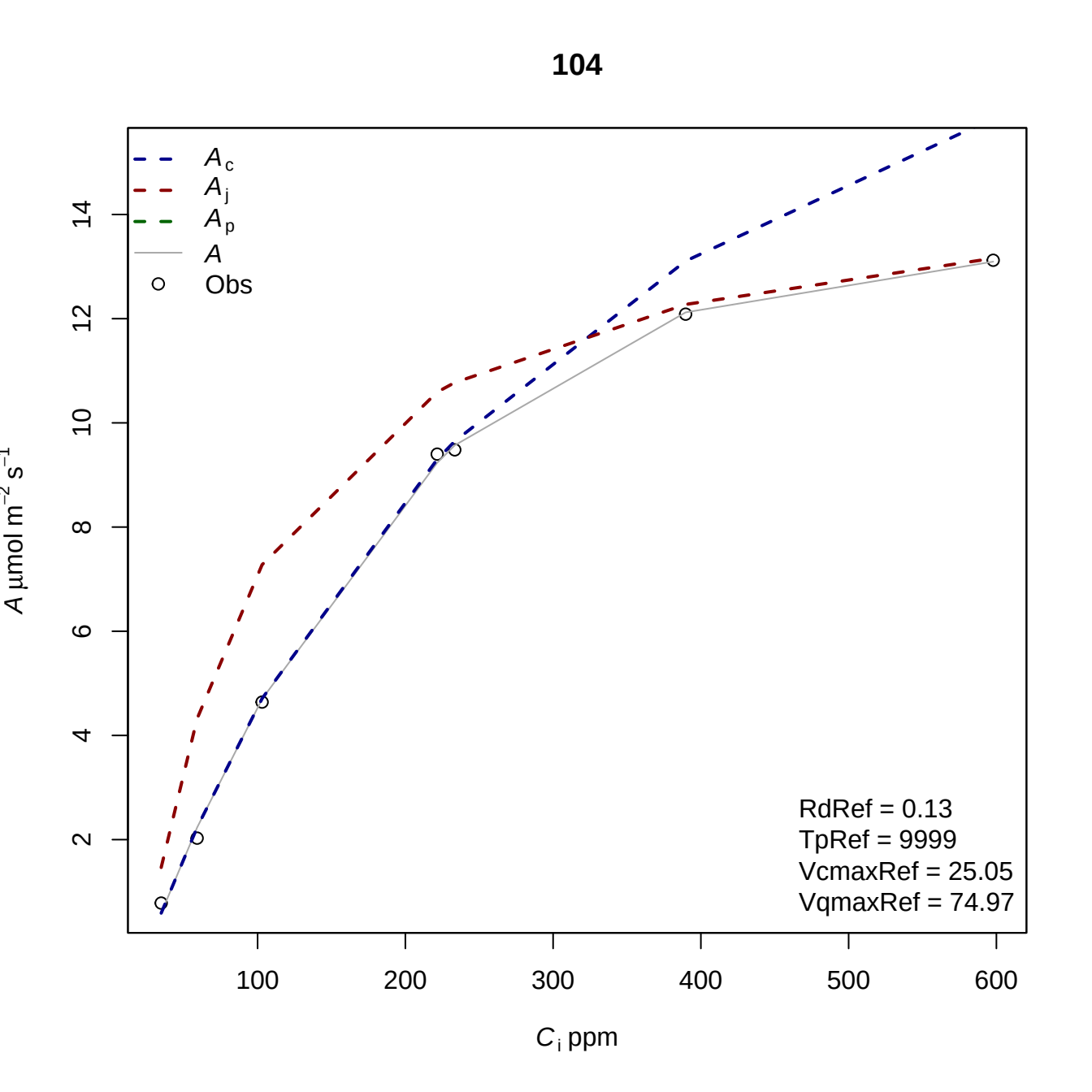

99
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 8.37
TpRef = 9999
VcmaxRef = 115.15
VqmaxRef = 276.46
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000

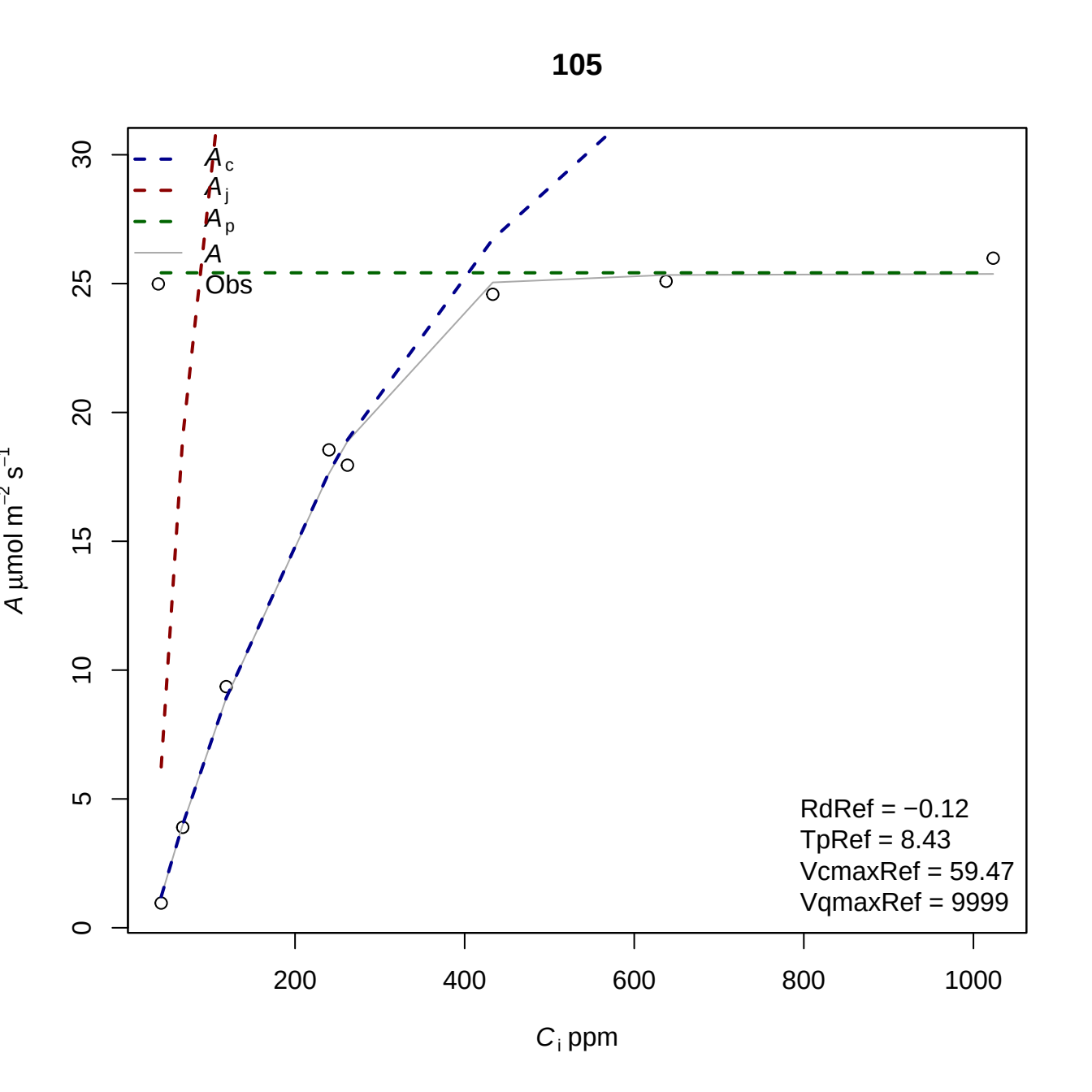

100
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.12
TpRef = 9999
VcmaxRef = 53.6
VqmaxRef = 186.14
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000

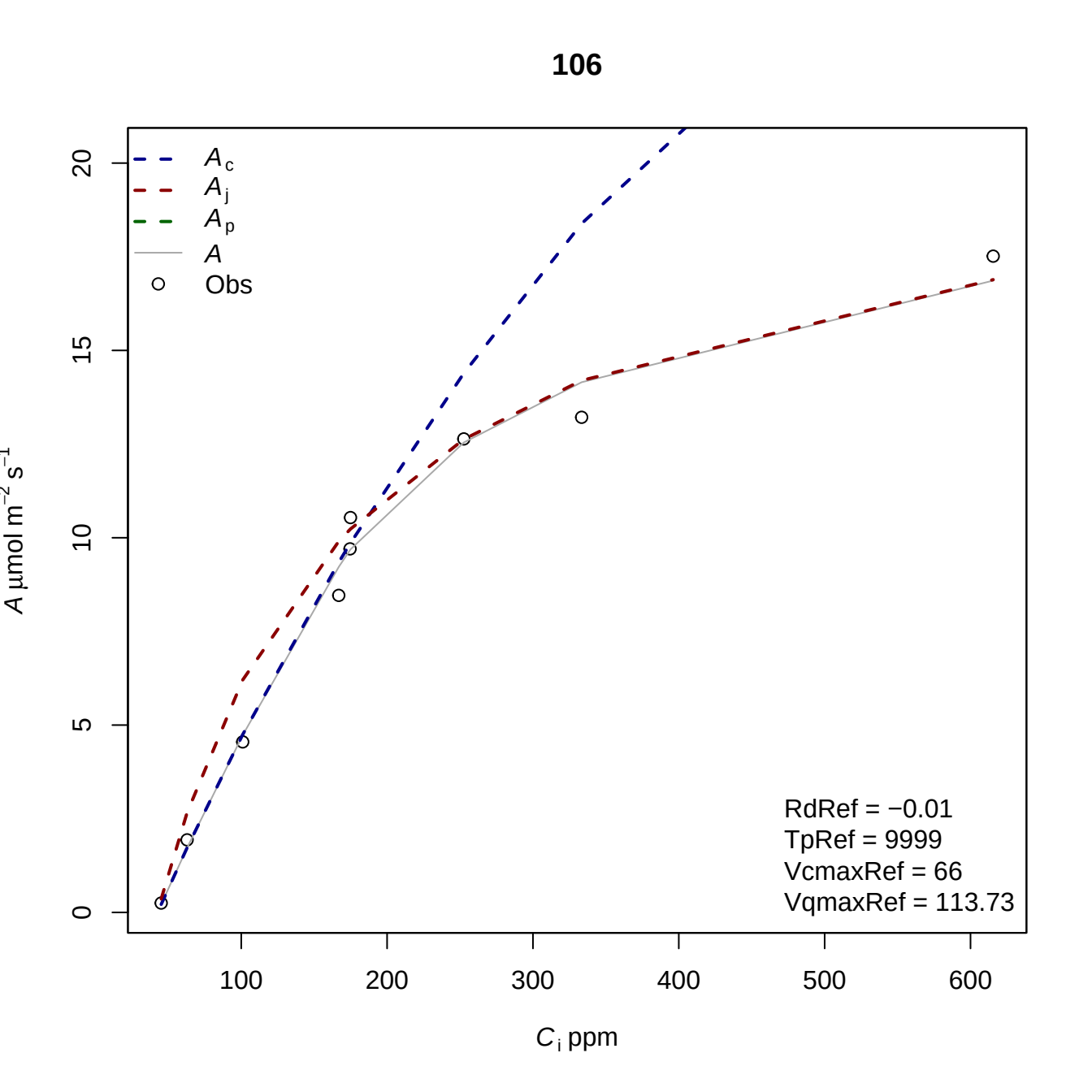

101
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.59
TpRef = 9999
VcmaxRef = 64.01
VqmaxRef = 209.32
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200
1400

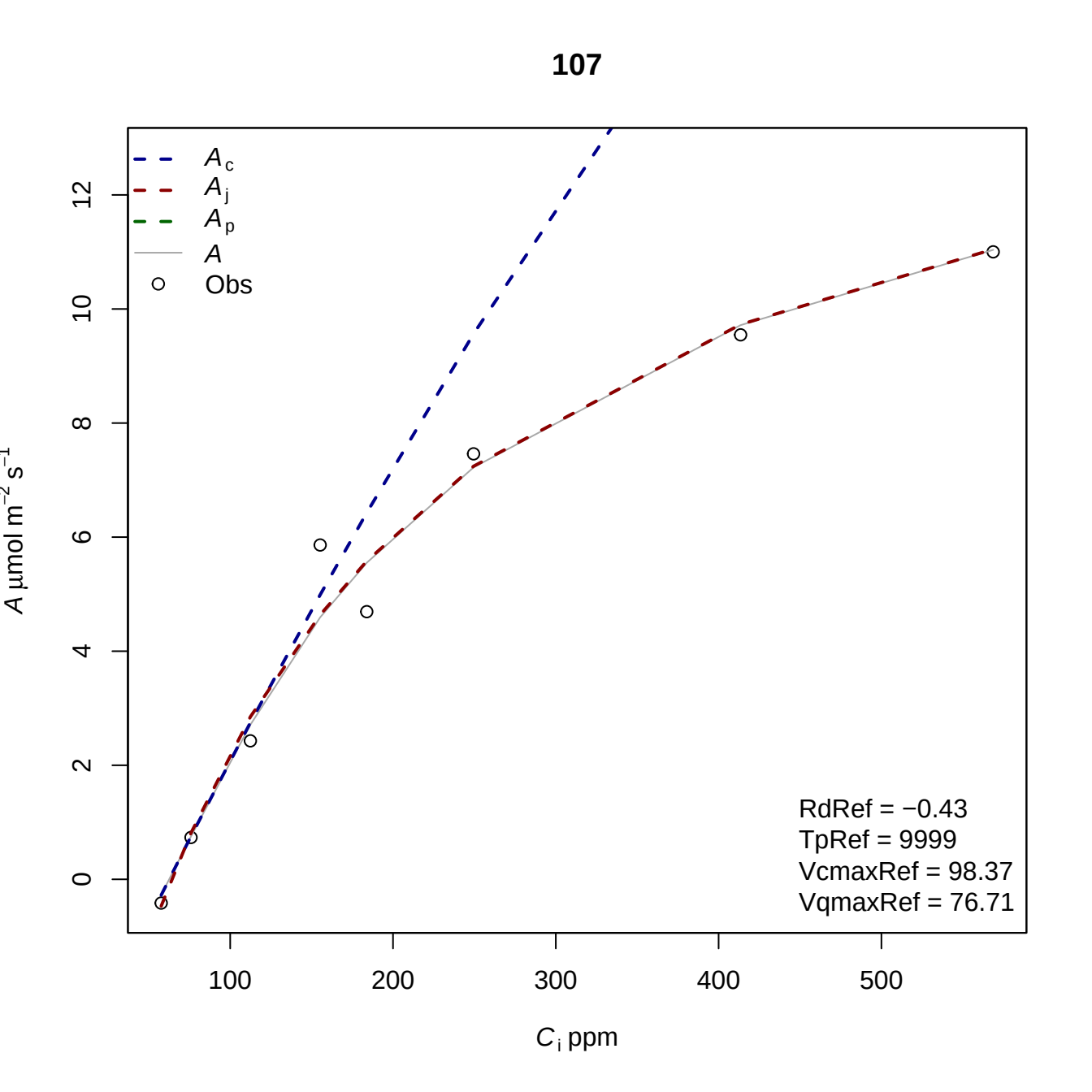

102
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.51
TpRef = 9999
VcmaxRef = 63.26
VqmaxRef = 211.13
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
200
400
600
800
1000
1200

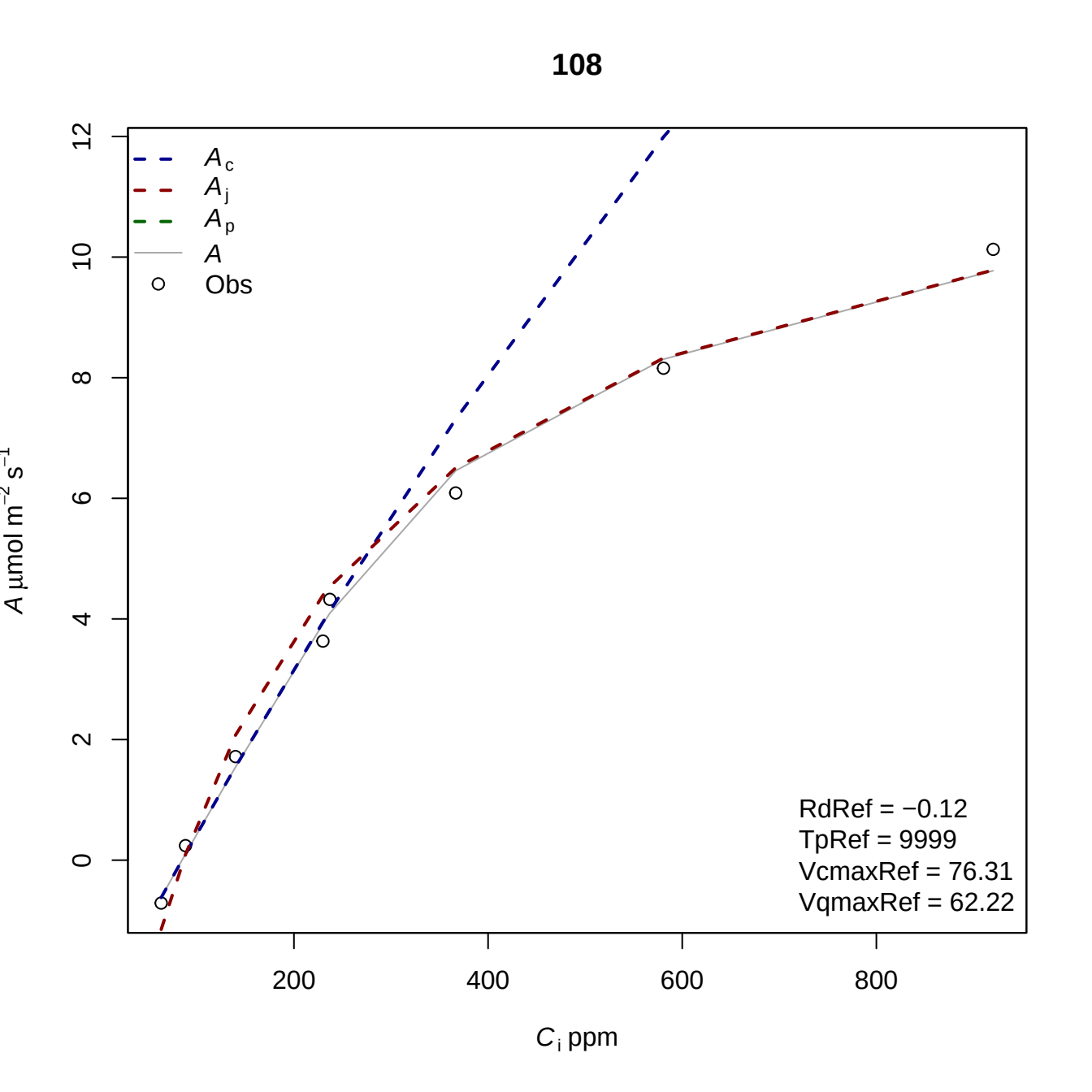

103
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.51
TpRef = 9999
VcmaxRef = 85.1
VqmaxRef = 326.39
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
0
200
400
600
800
1000
1200
1400

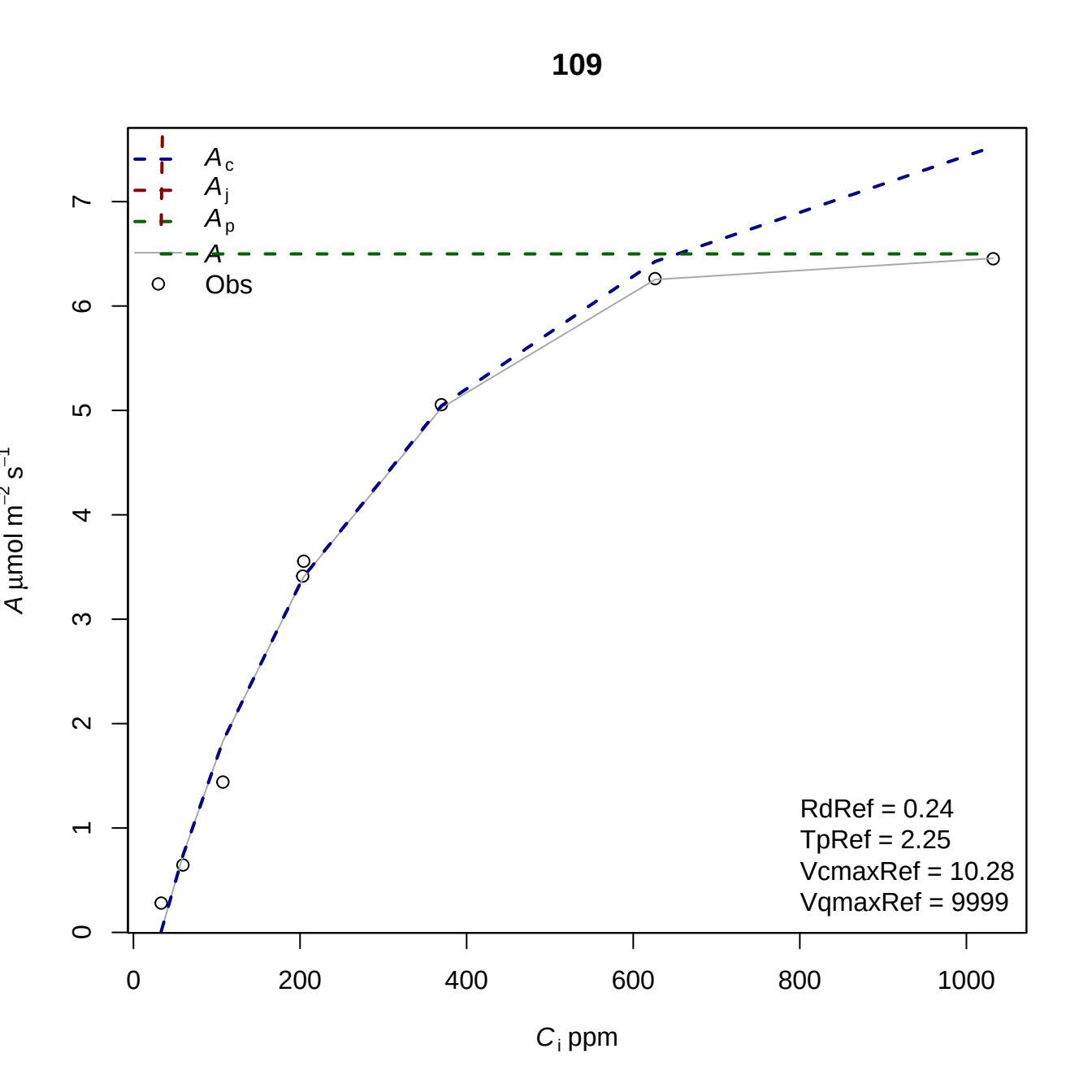

104
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.13
TpRef = 9999
VcmaxRef = 25.05
VqmaxRef = 74.97
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
2
4
6
8
10
12
14
100
200
300
400
500
600

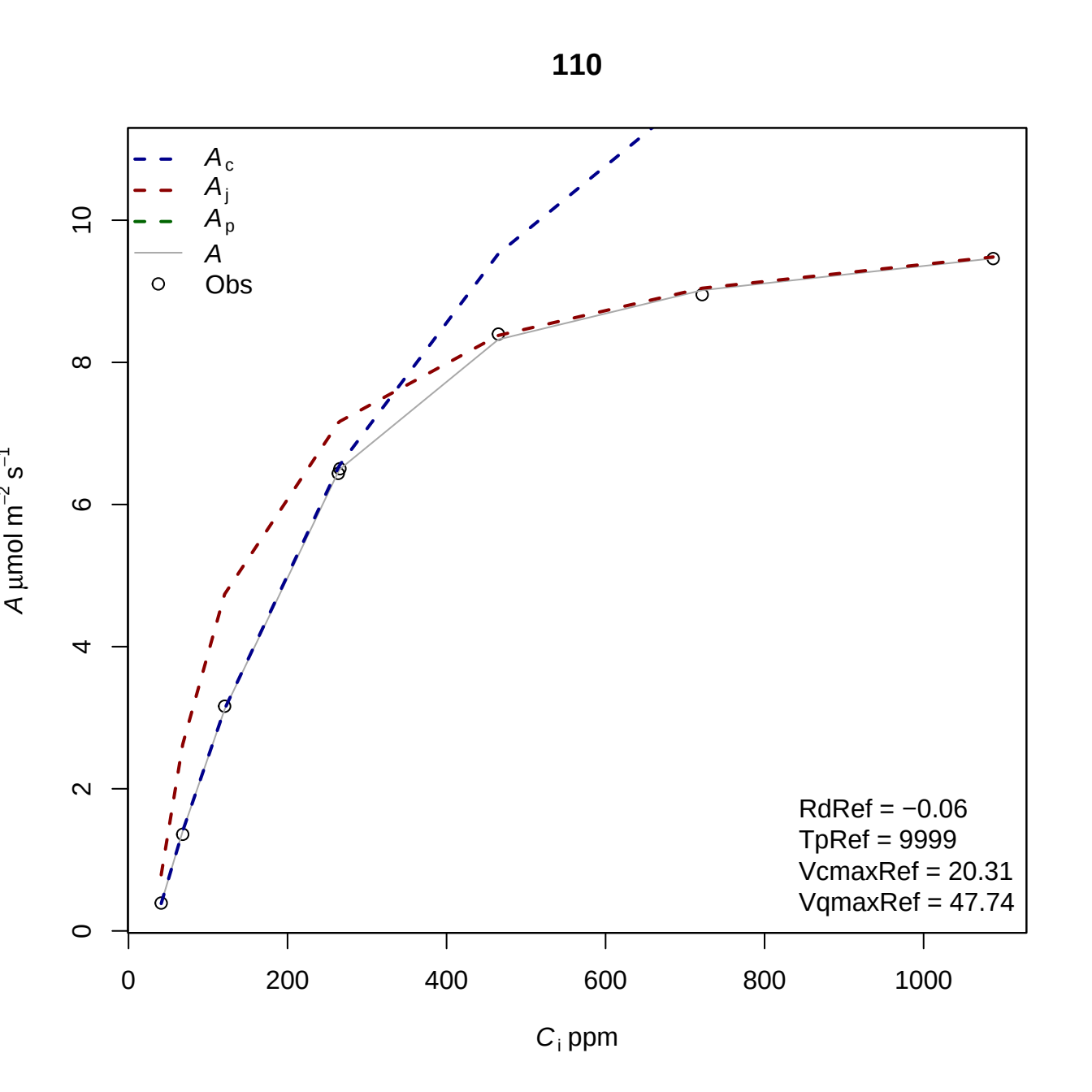

105
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.12
TpRef = 8.43
VcmaxRef = 59.47
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000

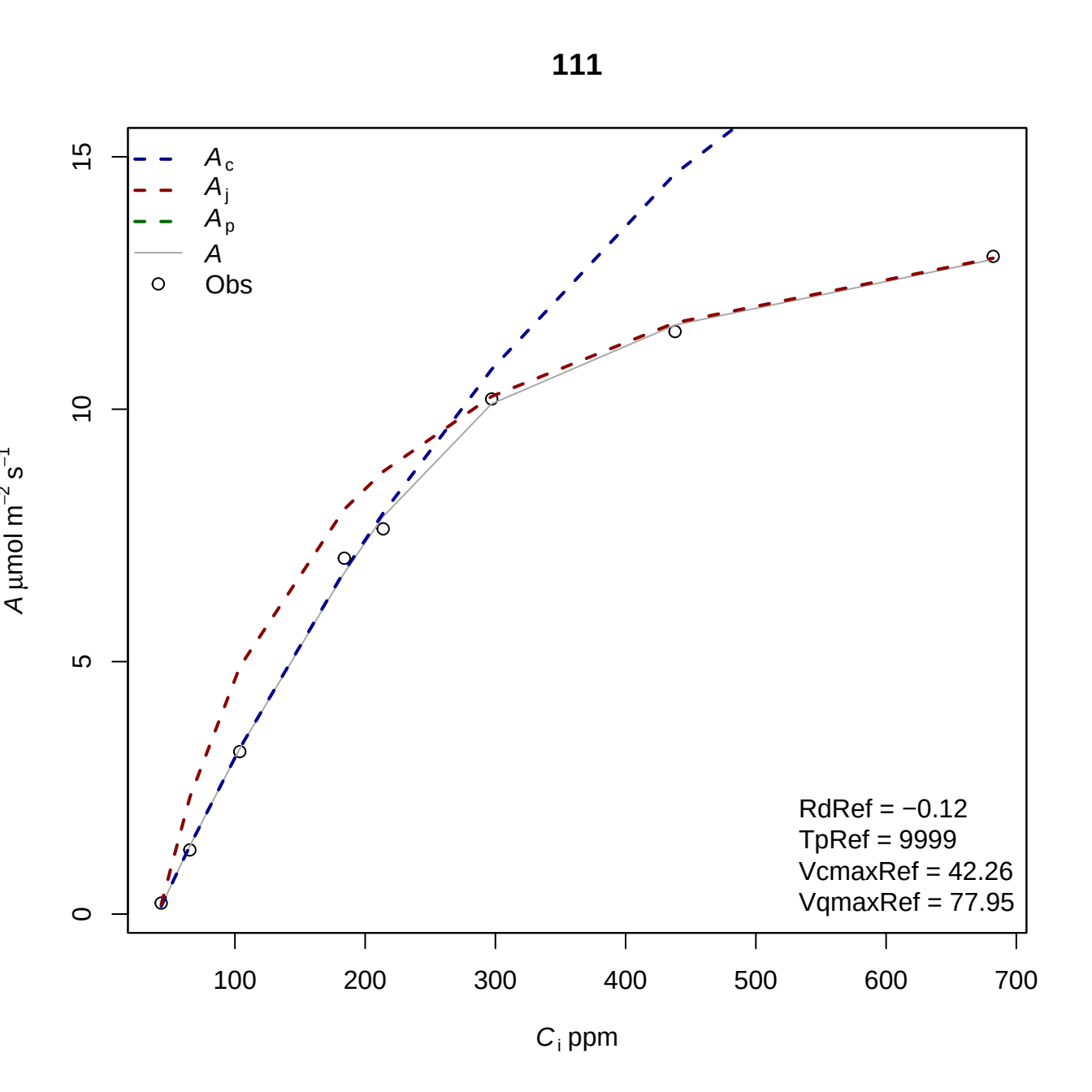

106
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.01
TpRef = 9999
VcmaxRef = 66
VqmaxRef = 113.73
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
100
200
300
400
500
600

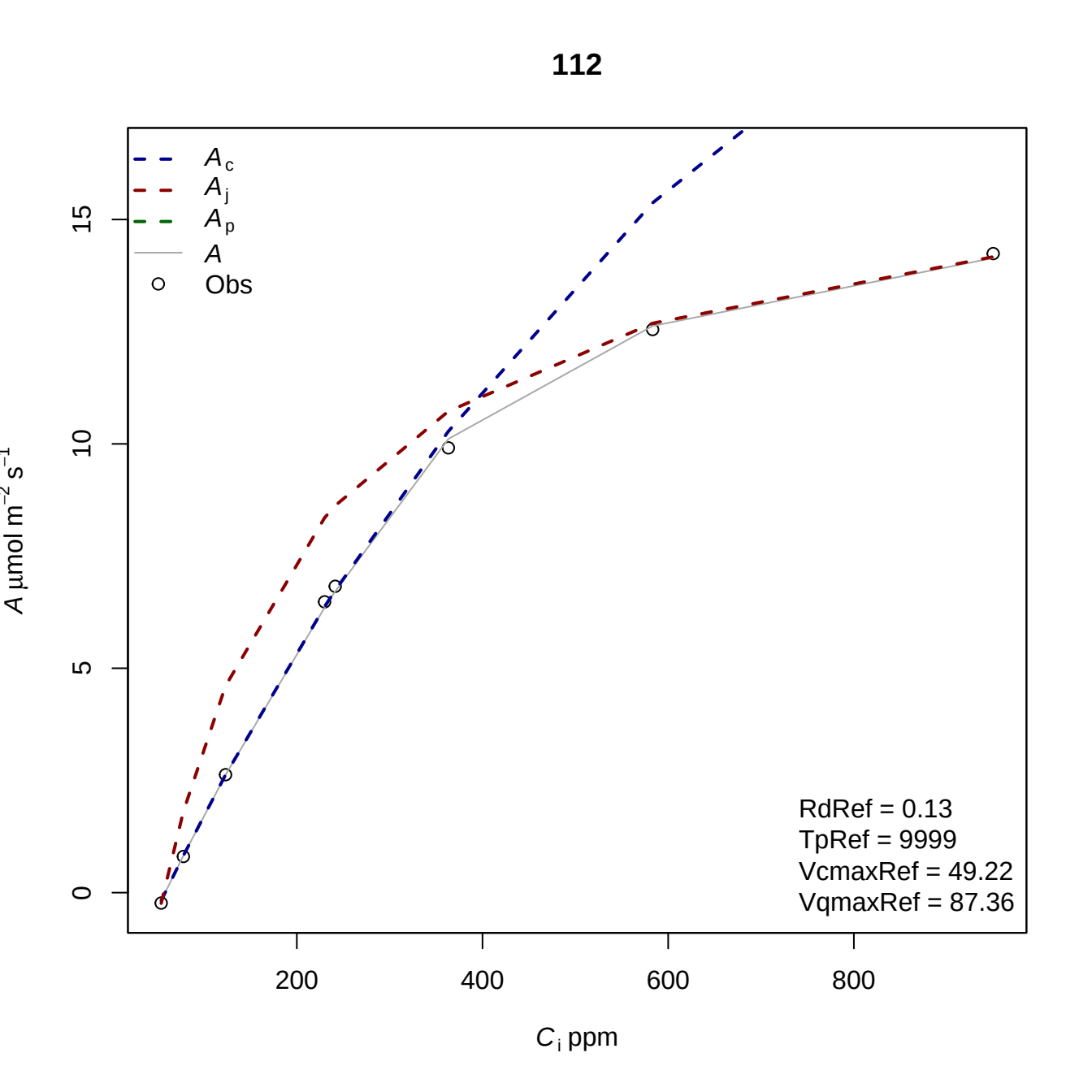

107
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.43
TpRef = 9999
VcmaxRef = 98.37
VqmaxRef = 76.71
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
100
200
300
400
500

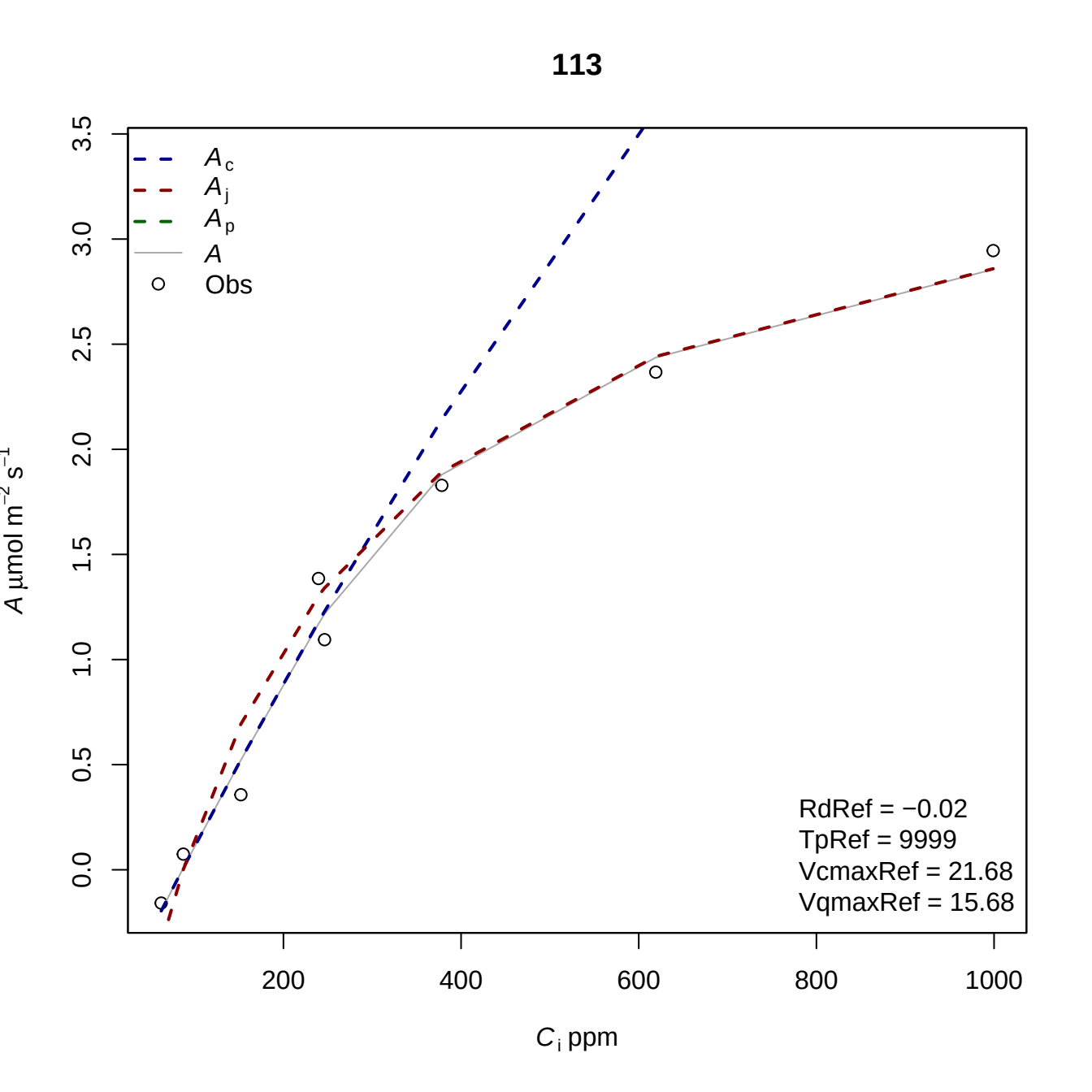

108
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.12
TpRef = 9999
VcmaxRef = 76.31
VqmaxRef = 62.22
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800

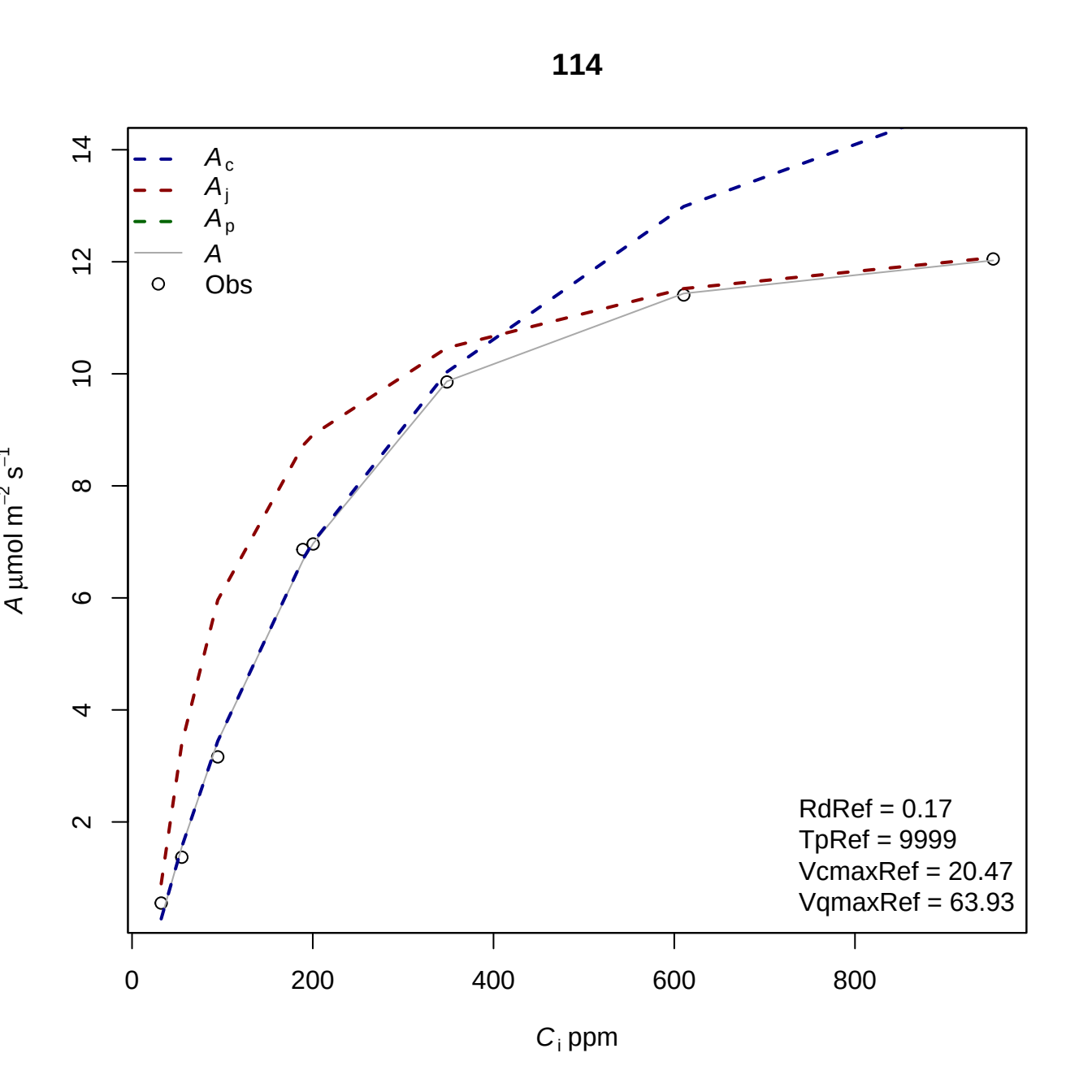

109
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.24
TpRef = 2.25
VcmaxRef = 10.28
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
7
0
200
400
600
800
1000

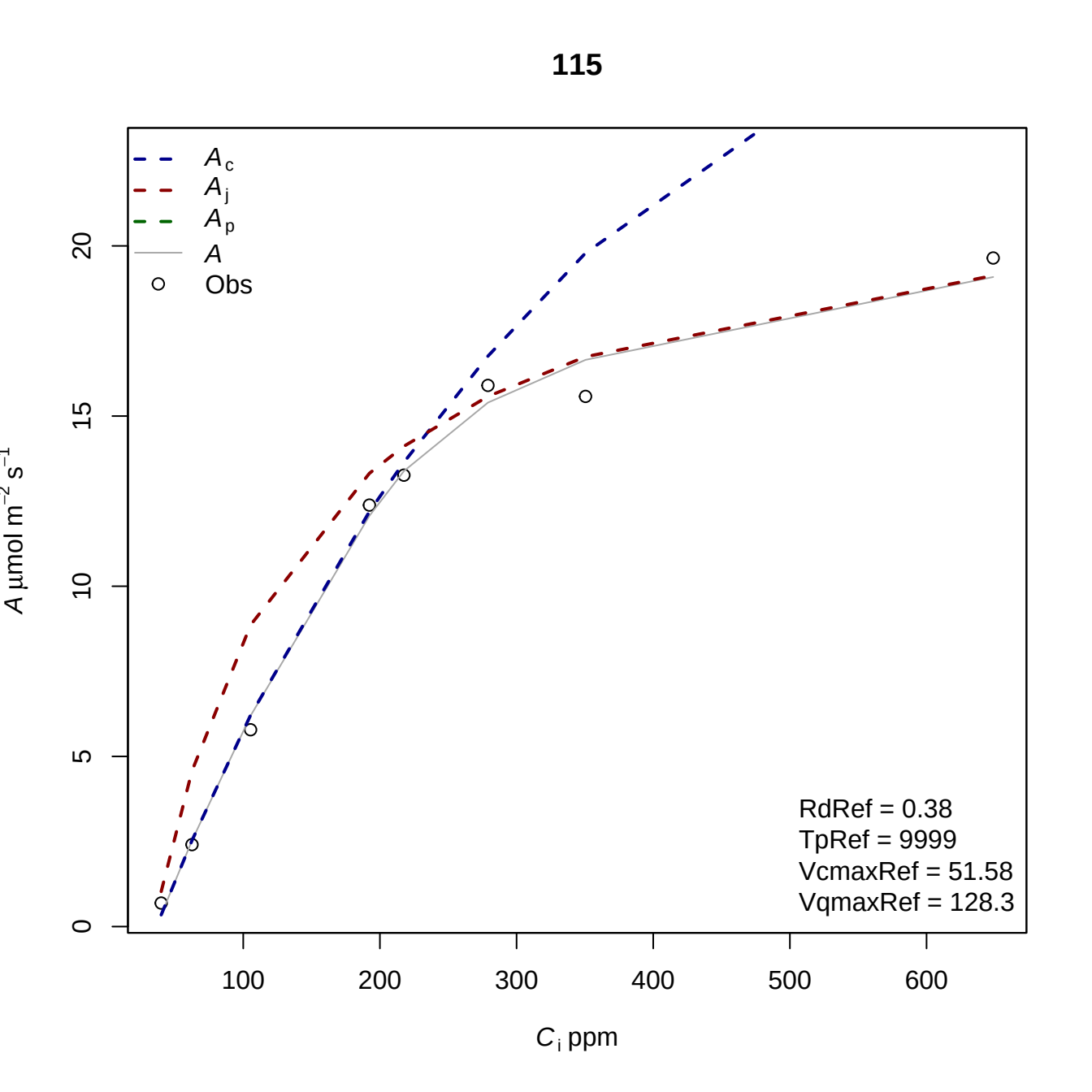

110
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.06
TpRef = 9999
VcmaxRef = 20.31
VqmaxRef = 47.74
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
200
400
600
800
1000

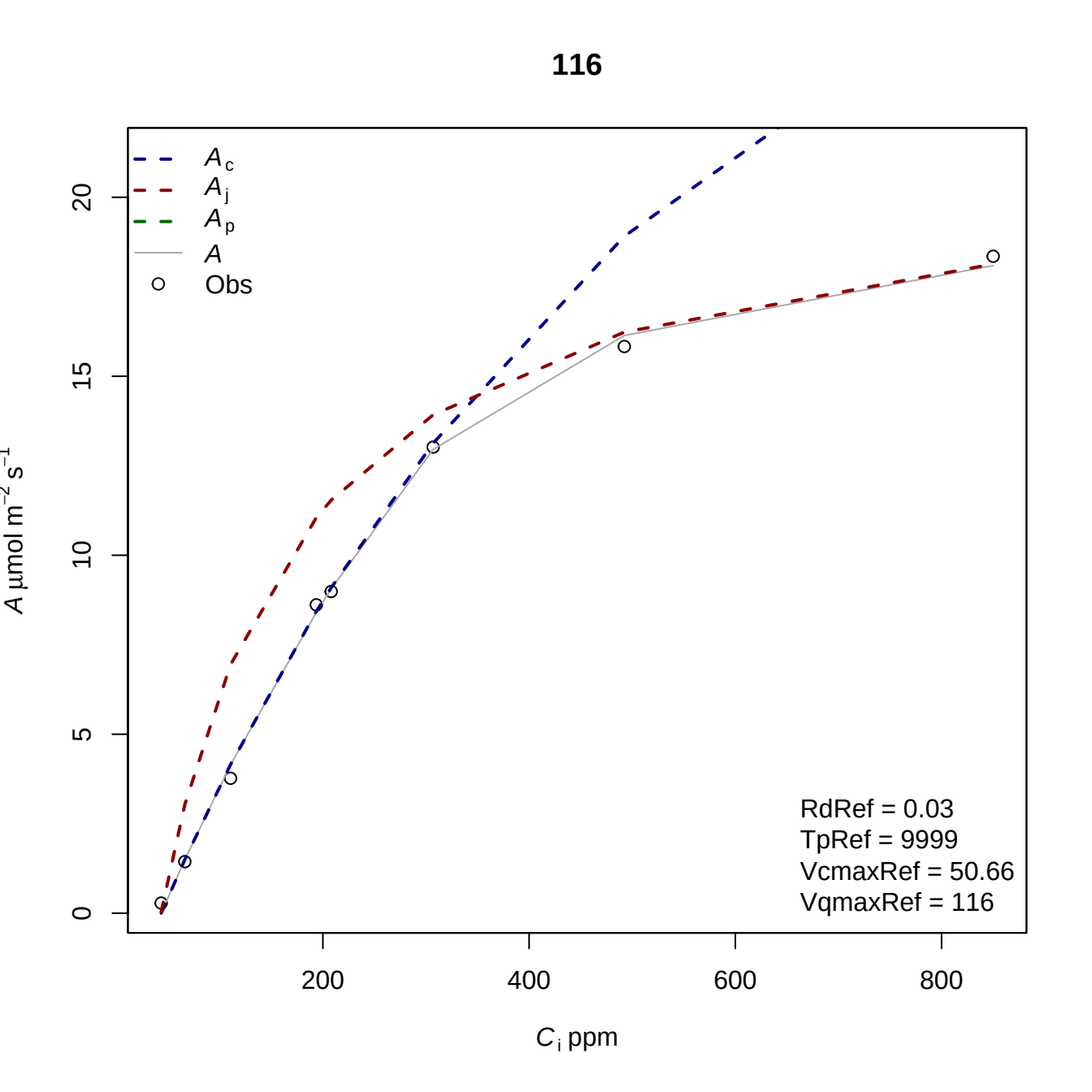

111
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.12
TpRef = 9999
VcmaxRef = 42.26
VqmaxRef = 77.95
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500
600
700

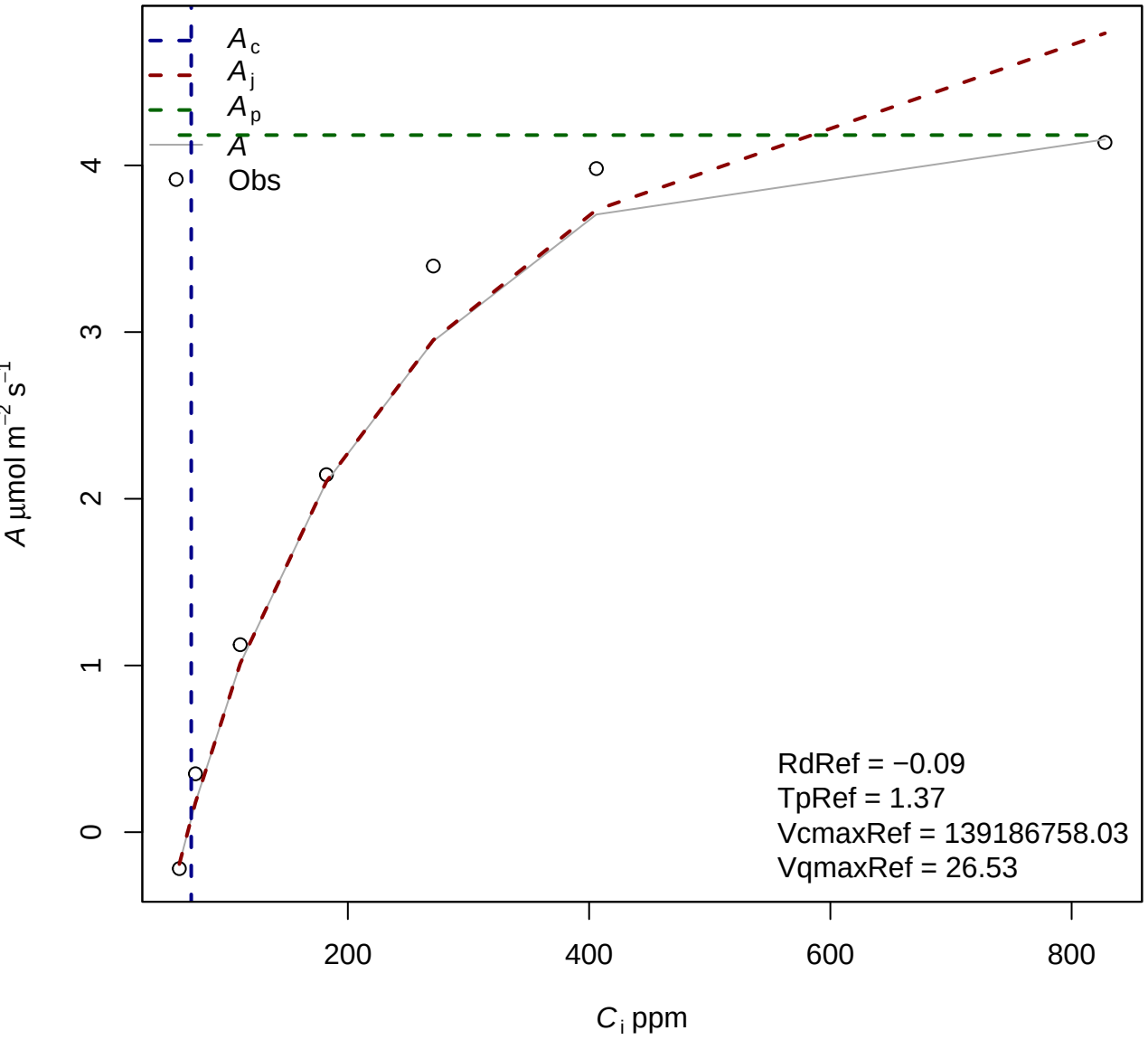

112
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.13
TpRef = 9999
VcmaxRef = 49.22
VqmaxRef = 87.36
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800

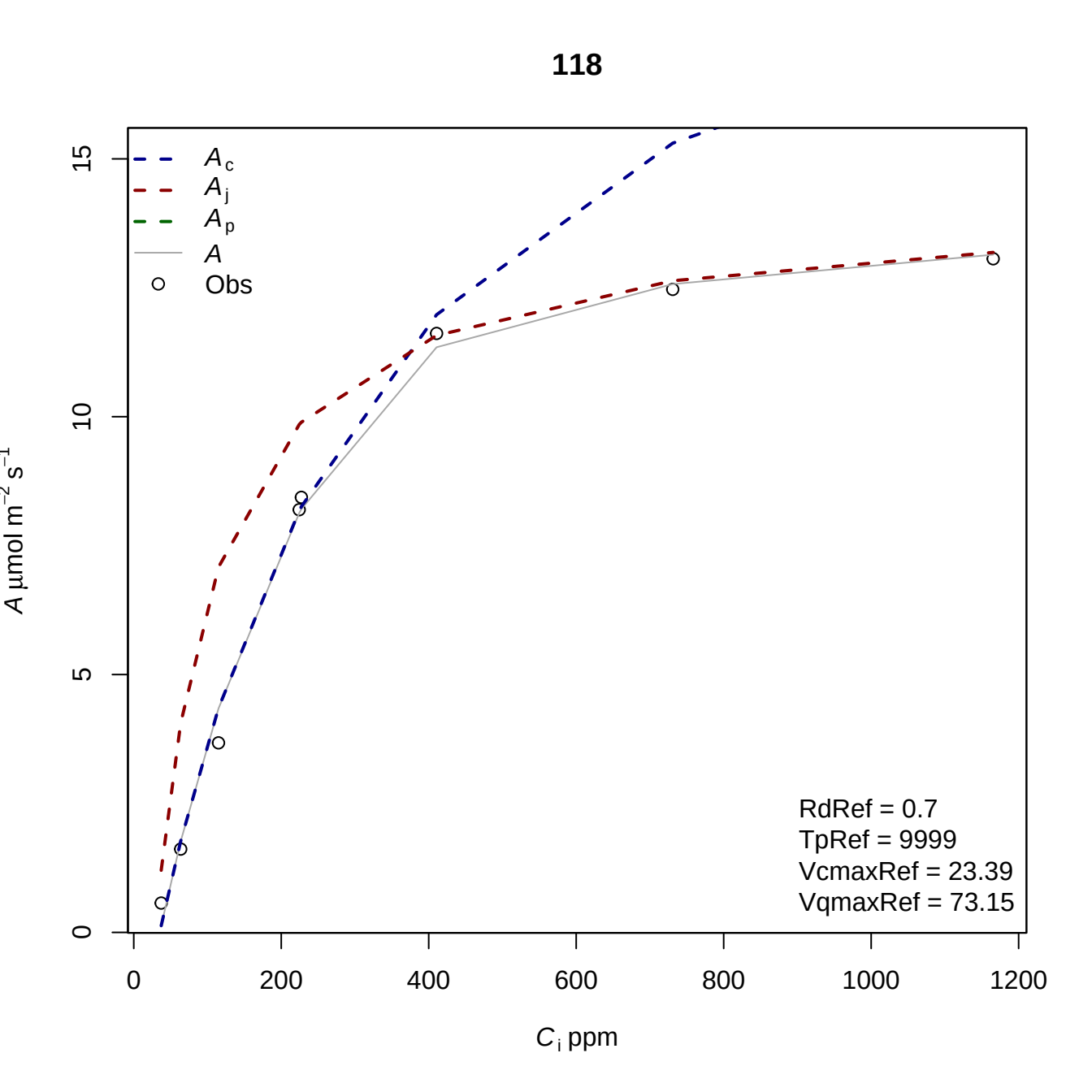

113
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.02
TpRef = 9999
VcmaxRef = 21.68
VqmaxRef = 15.68
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
200
400
600
800
1000

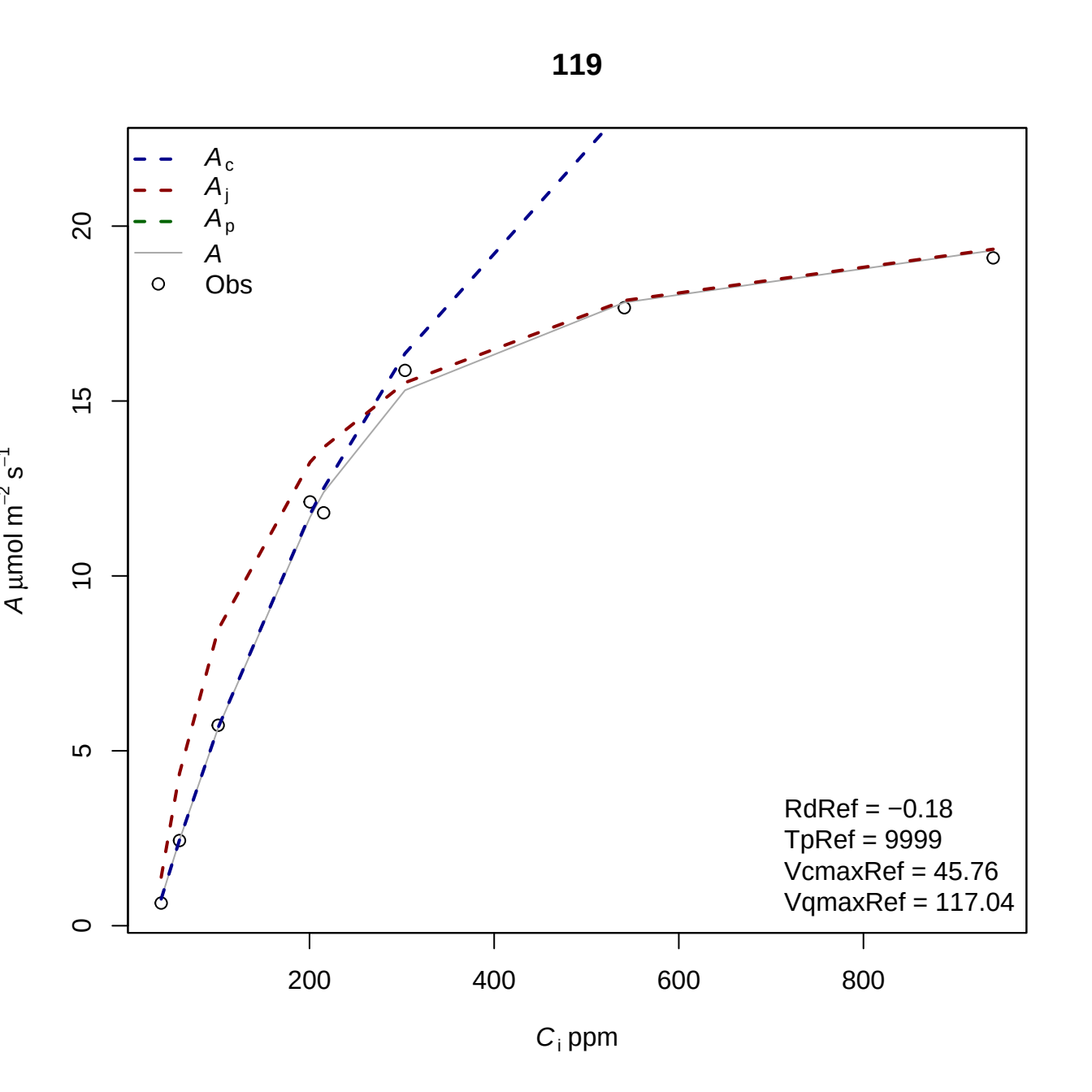

114
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.17
TpRef = 9999
VcmaxRef = 20.47
VqmaxRef = 63.93
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
2
4
6
8
10
12
14

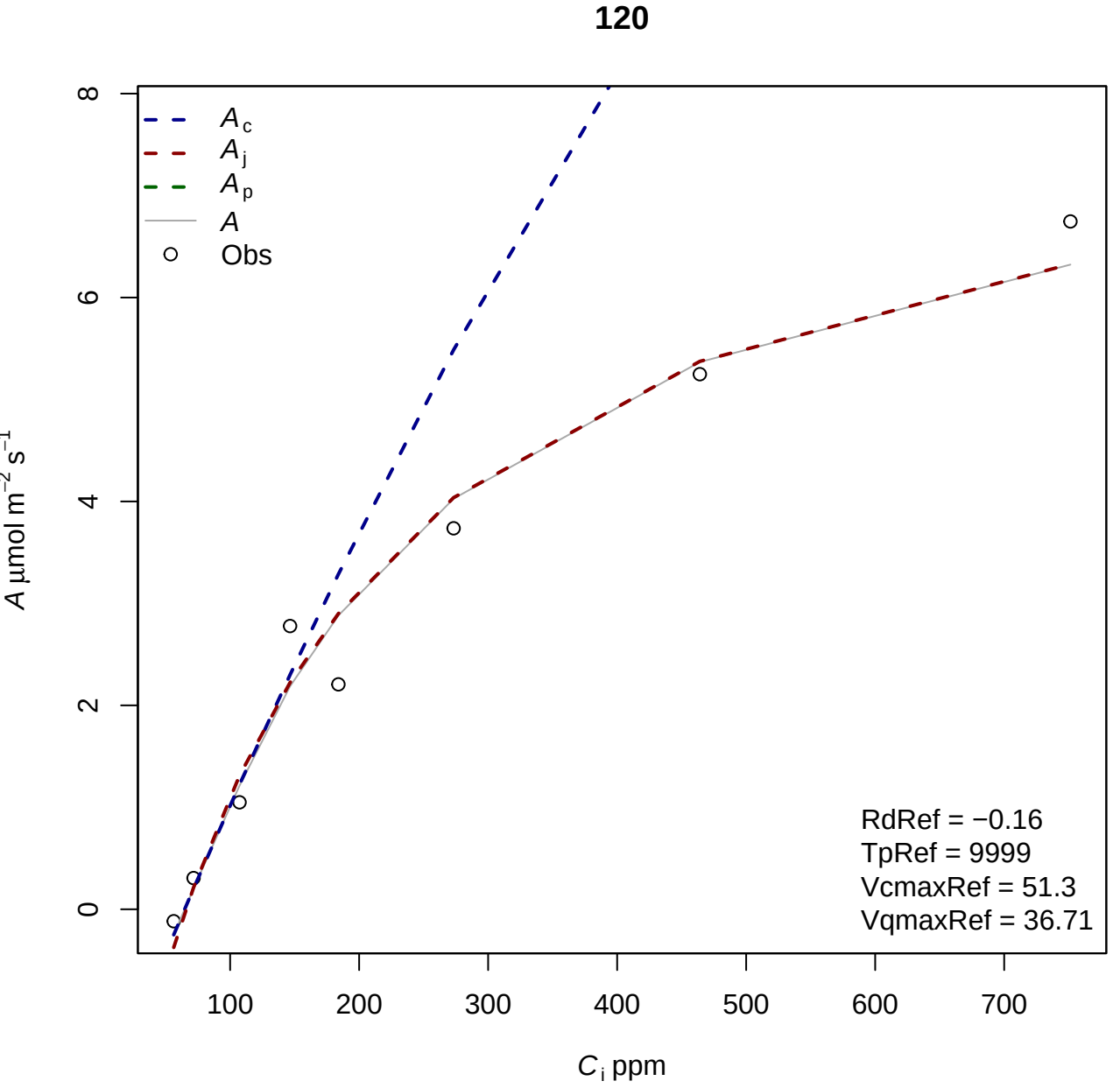

115
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.38
TpRef = 9999
VcmaxRef = 51.58
VqmaxRef = 128.3
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
100
200
300
400
500
600

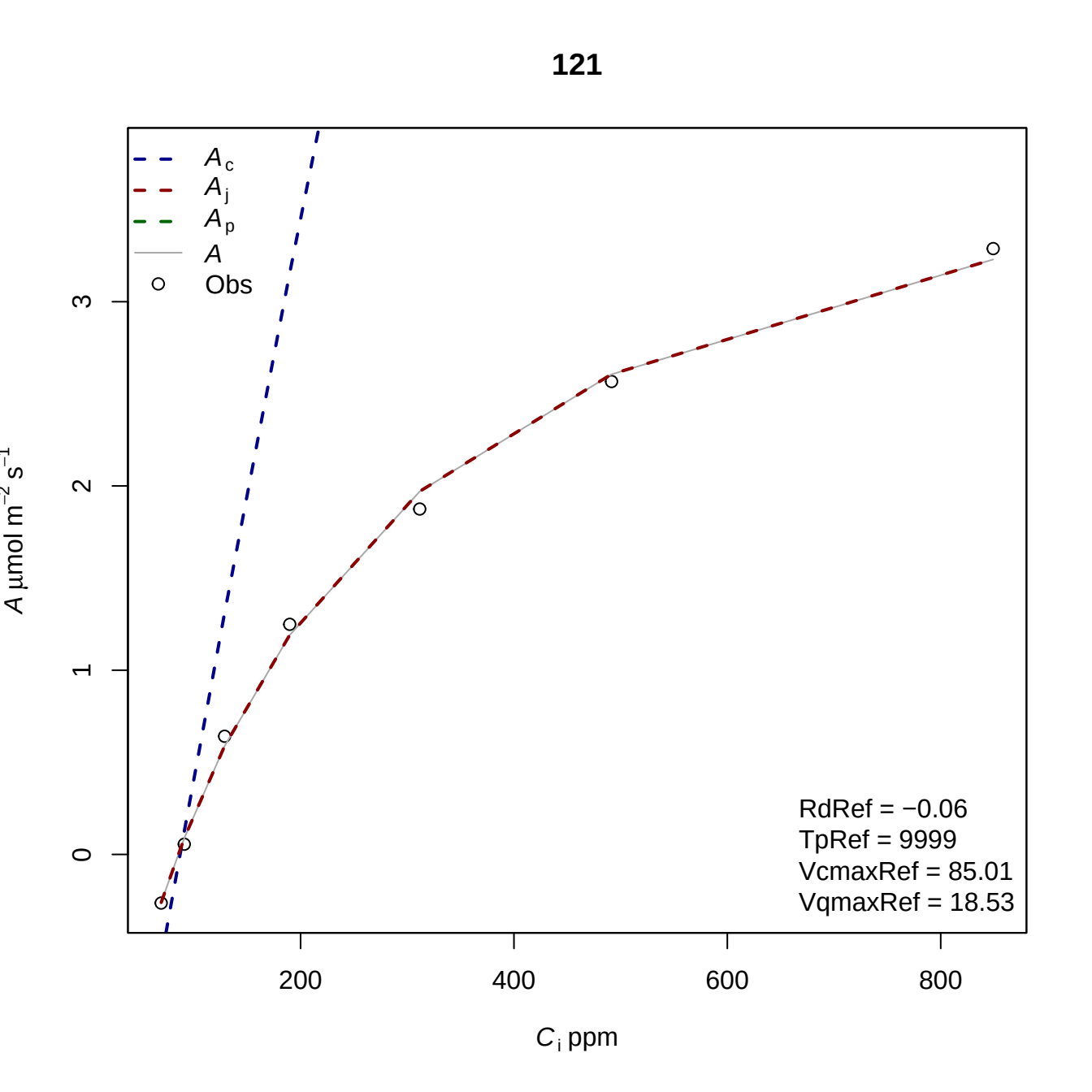

116
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.03
TpRef = 9999
VcmaxRef = 50.66
VqmaxRef = 116
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800

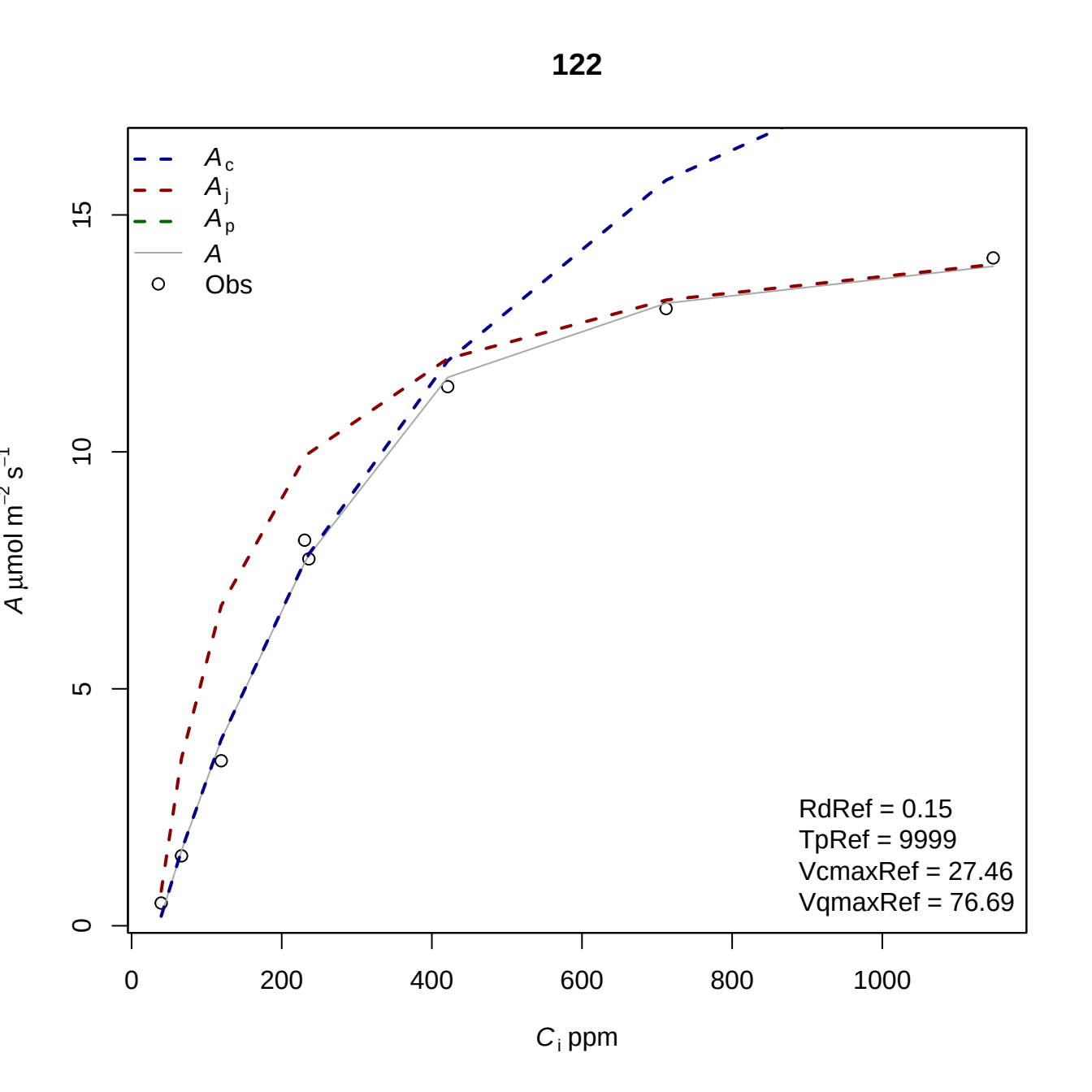

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.09
TpRef = 1.37
VcmaxRef = 139186758.03
VqmaxRef = 26.53
0
1
2
3
4
200
400
600
800
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

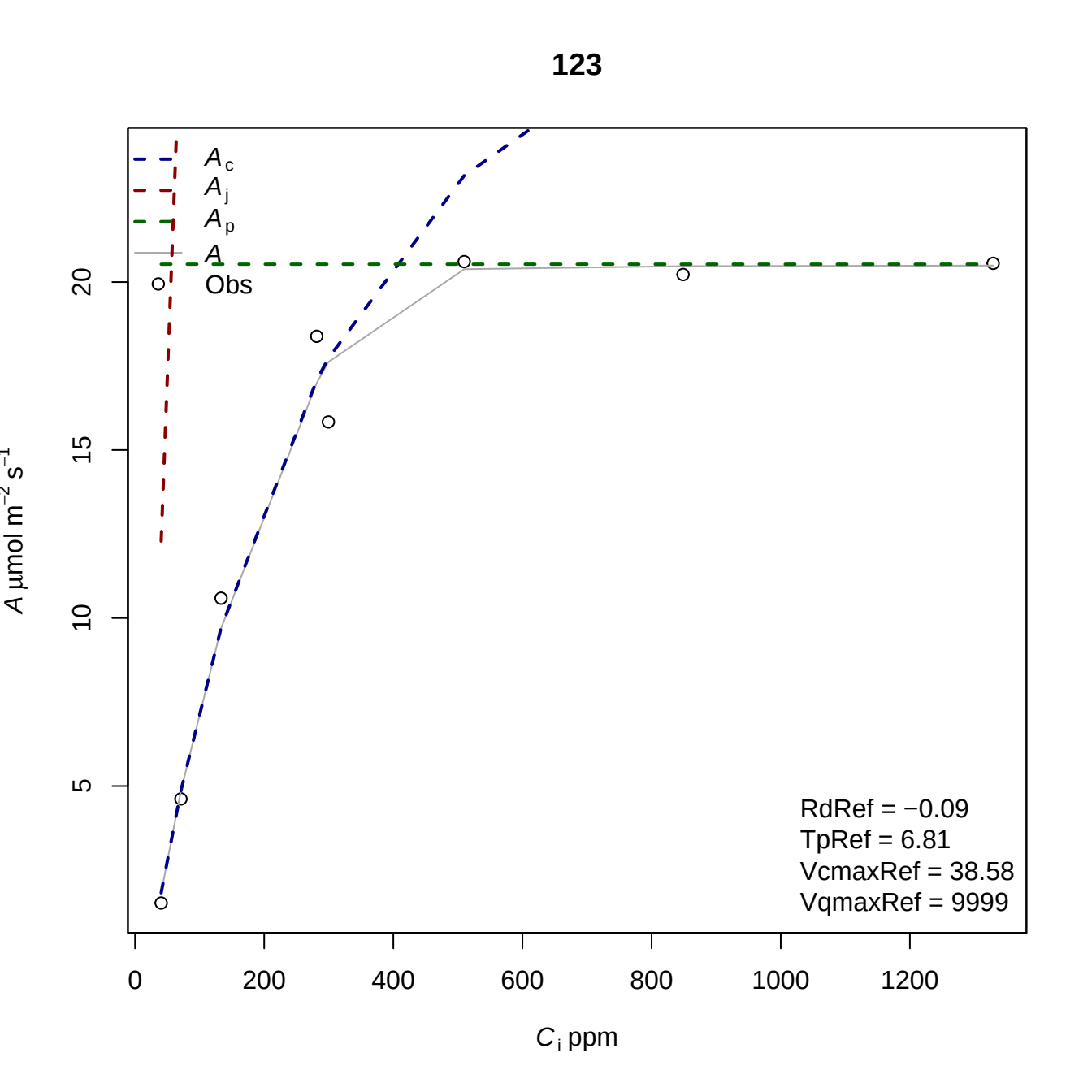

118
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.7
TpRef = 9999
VcmaxRef = 23.39
VqmaxRef = 73.15
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

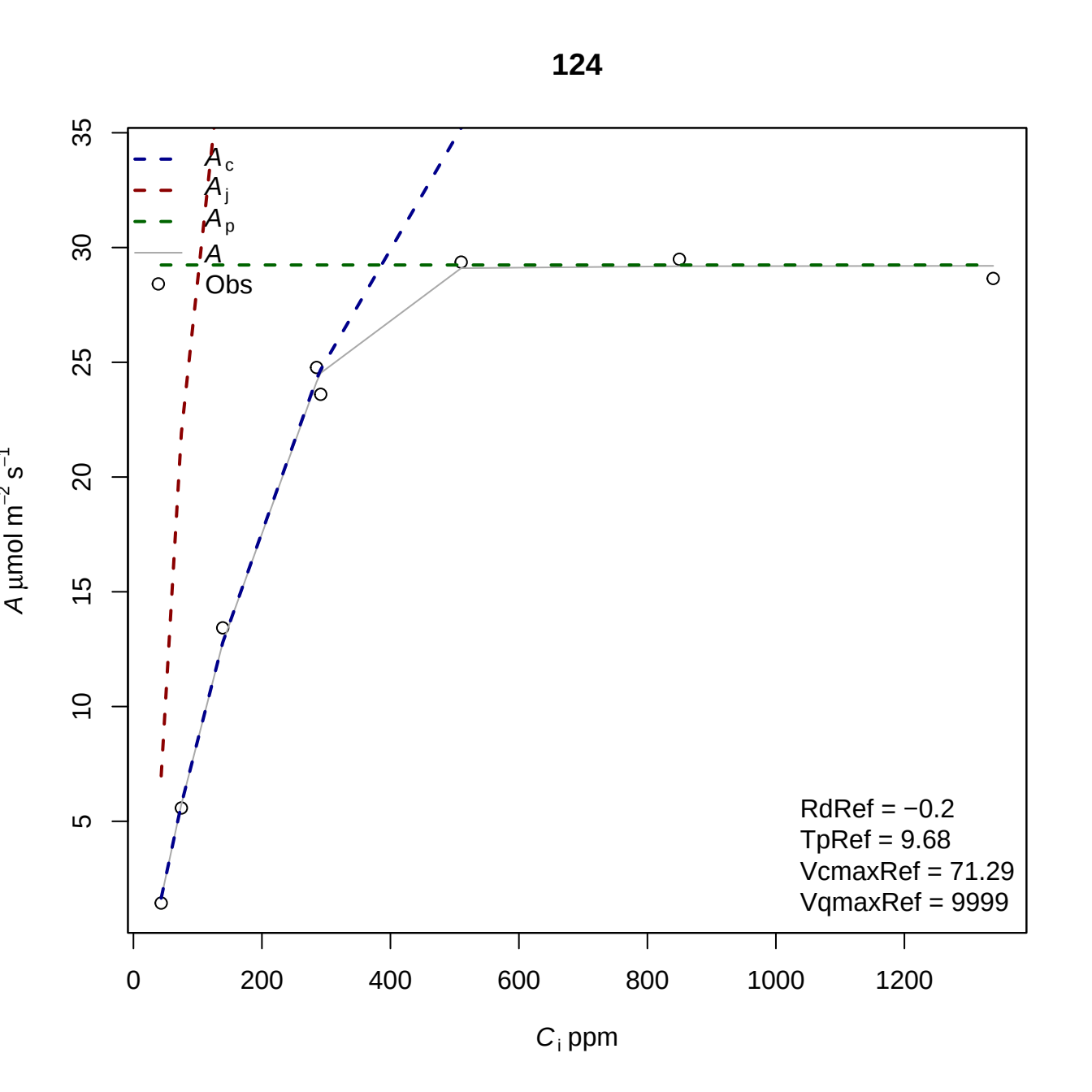

119
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.18
TpRef = 9999
VcmaxRef = 45.76
VqmaxRef = 117.04
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800

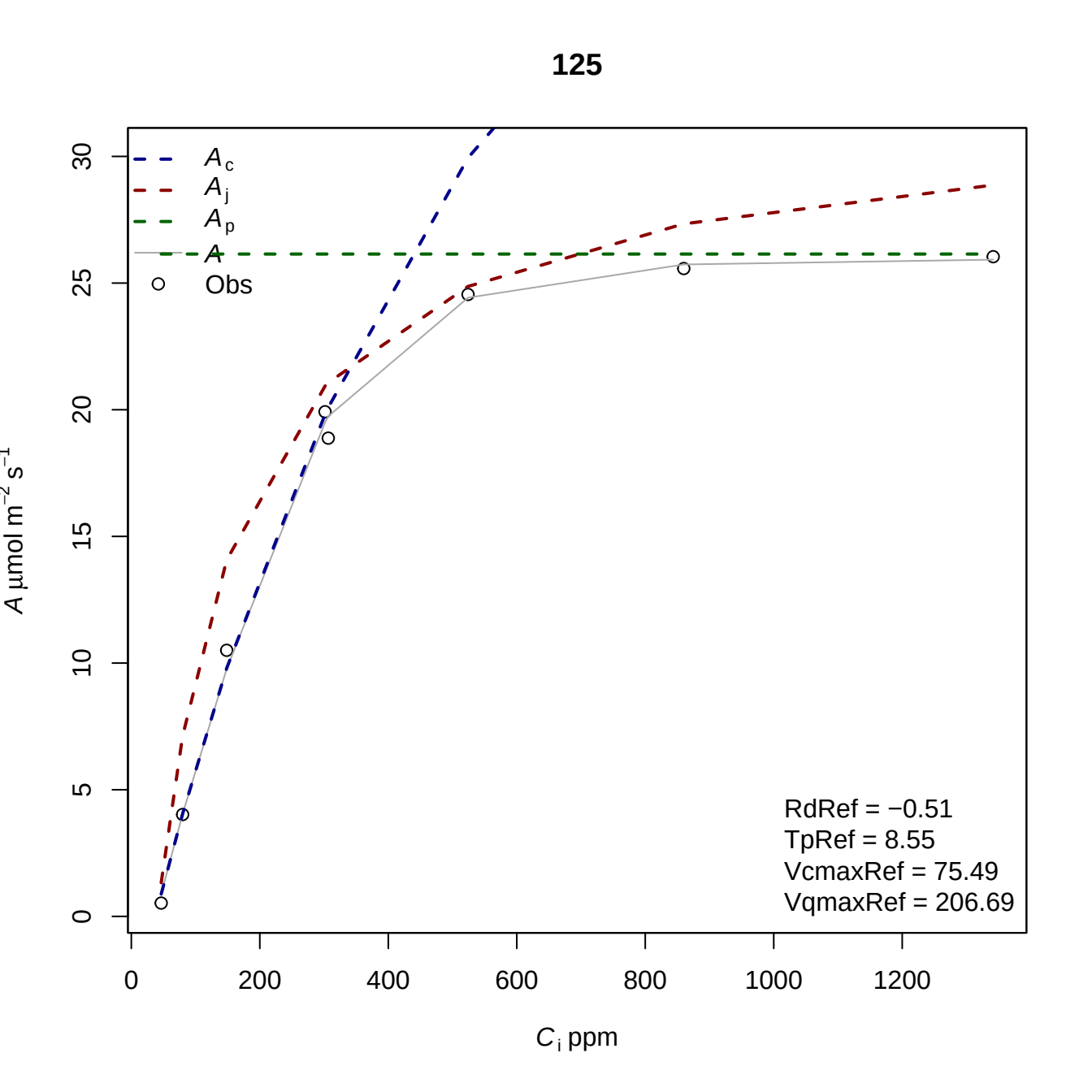

120
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.16
TpRef = 9999
VcmaxRef = 51.3
VqmaxRef = 36.71
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
100
200
300
400
500
600
700

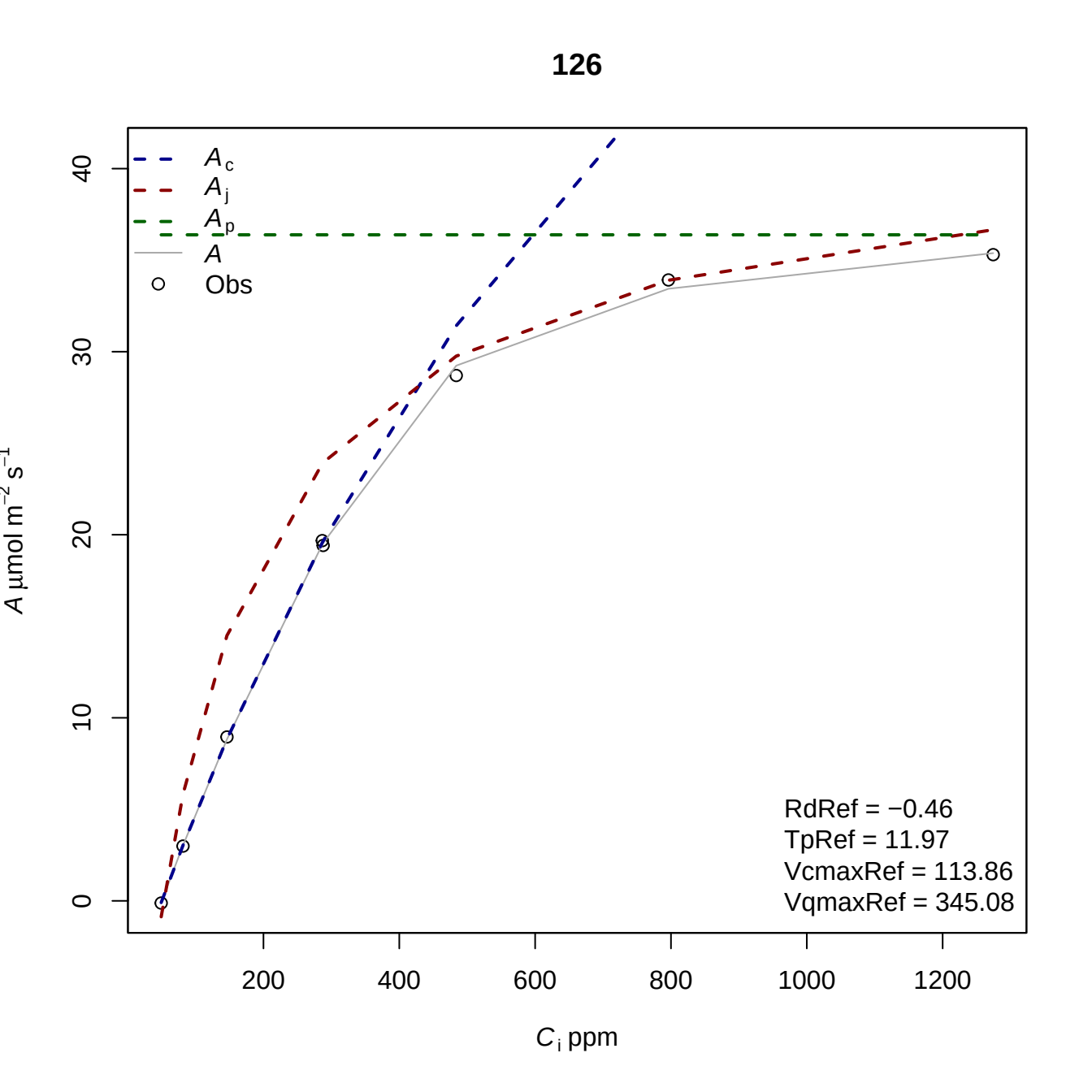

121
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.06
TpRef = 9999
VcmaxRef = 85.01
VqmaxRef = 18.53
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
200
400
600
800

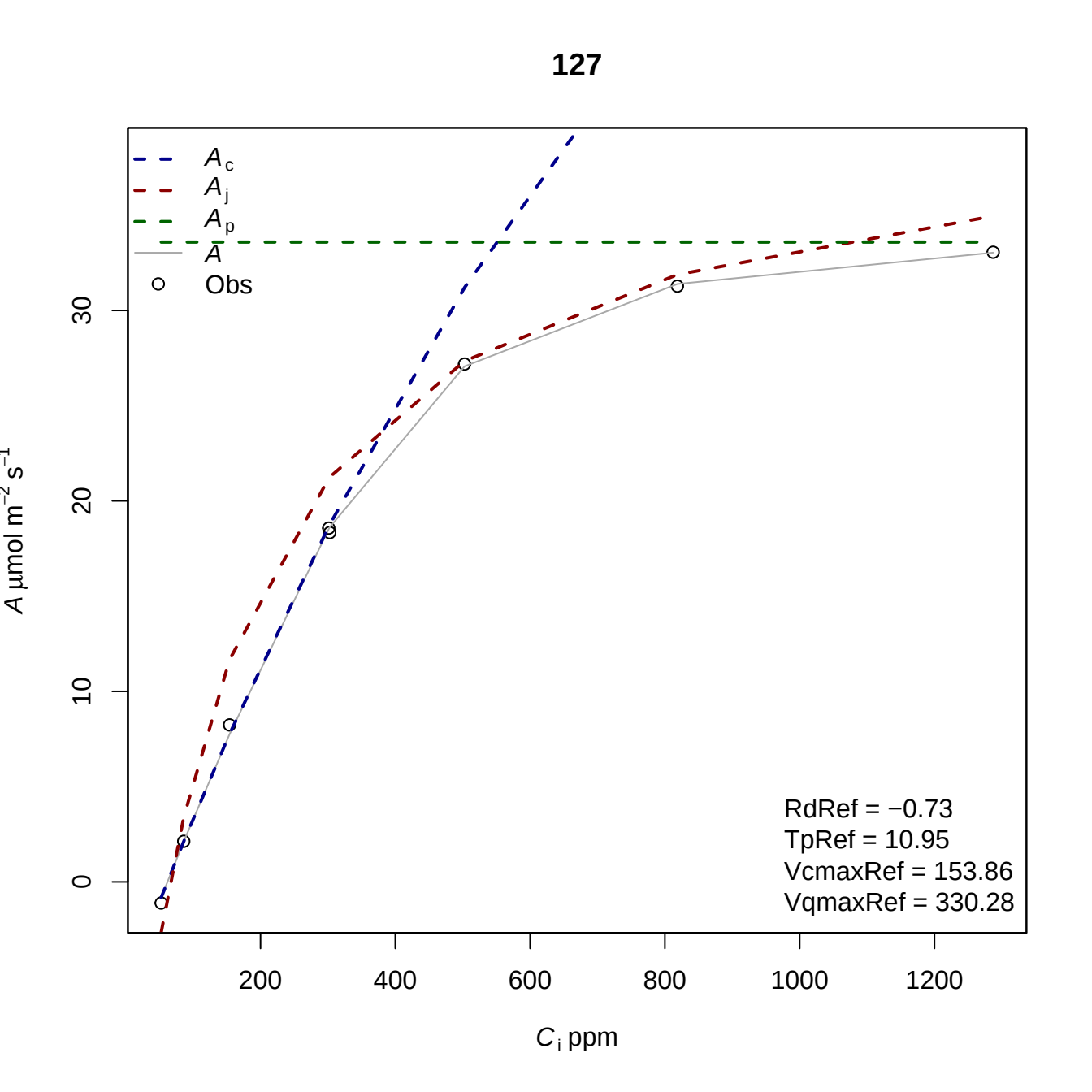

122
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.15
TpRef = 9999
VcmaxRef = 27.46
VqmaxRef = 76.69
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000

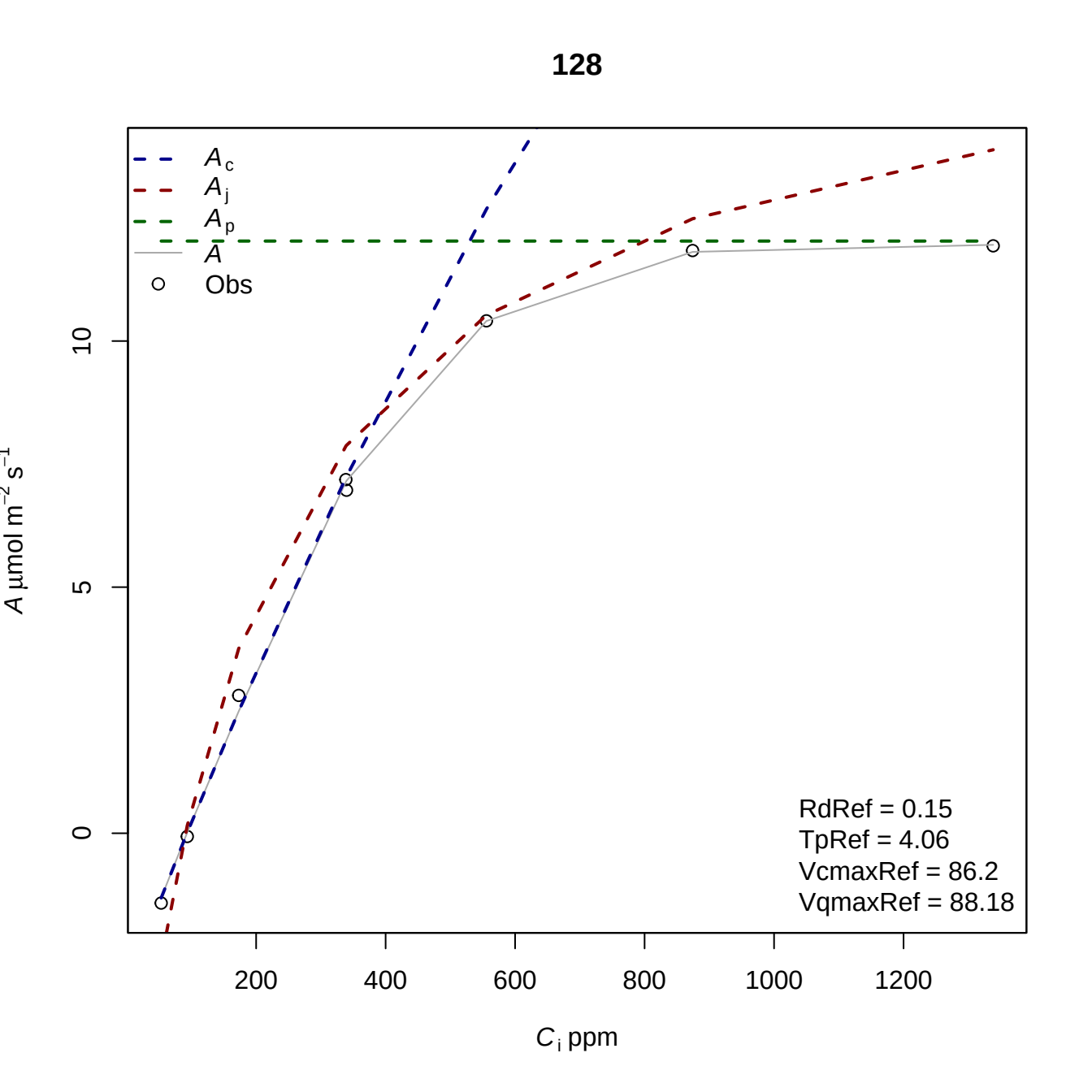

123
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.09
TpRef = 6.81
VcmaxRef = 38.58
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20

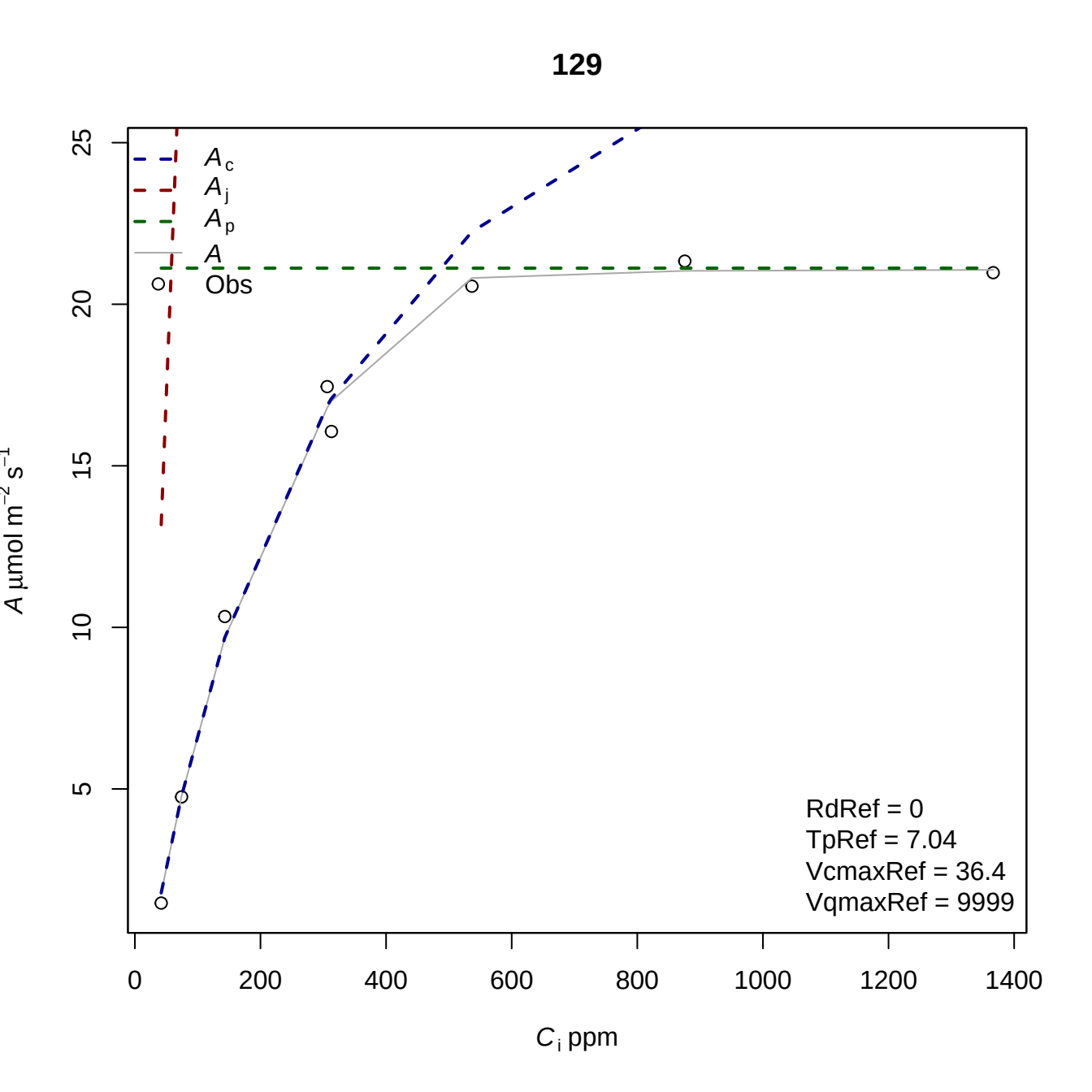

124
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.2
TpRef = 9.68
VcmaxRef = 71.29
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35

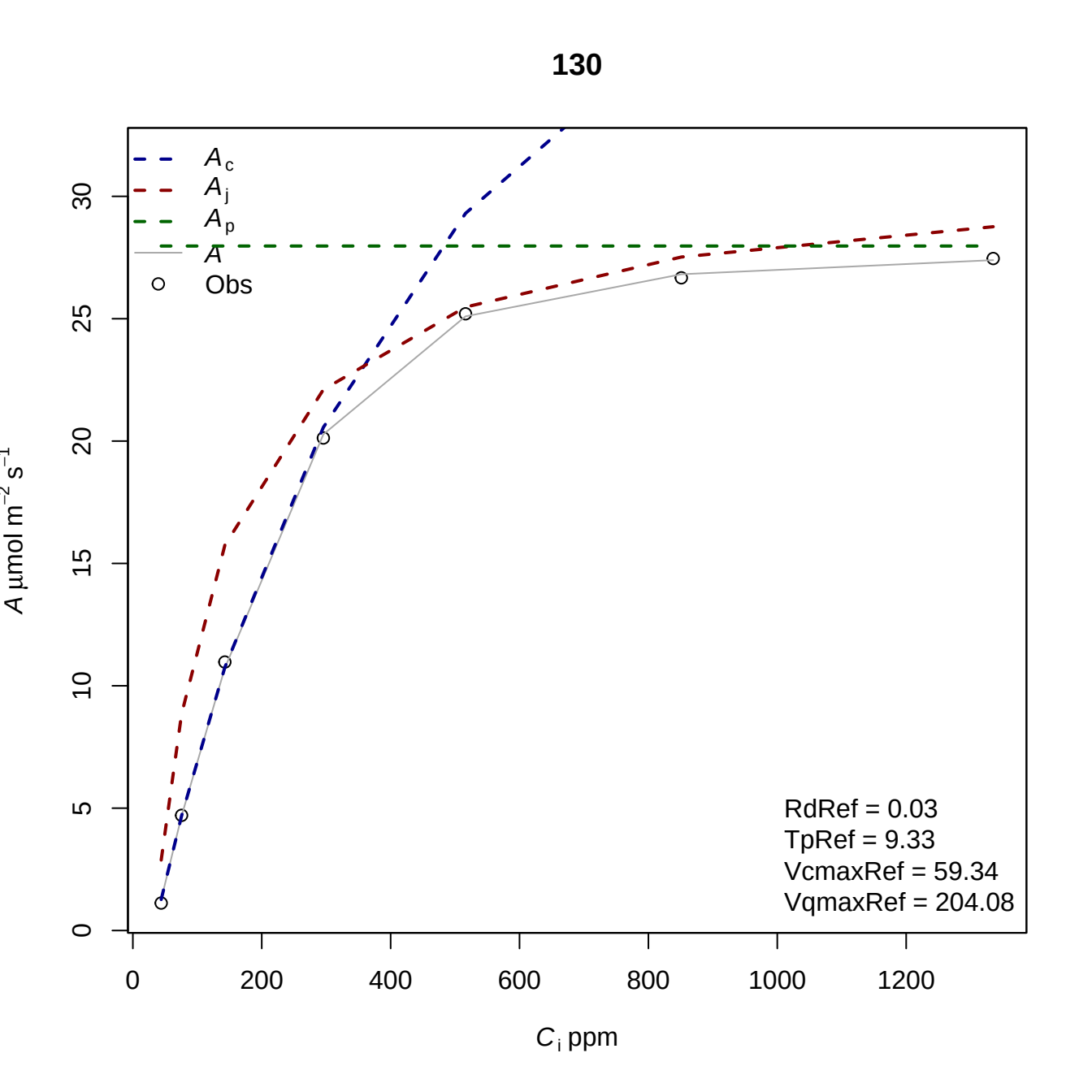

125
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.51
TpRef = 8.55
VcmaxRef = 75.49
VqmaxRef = 206.69
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

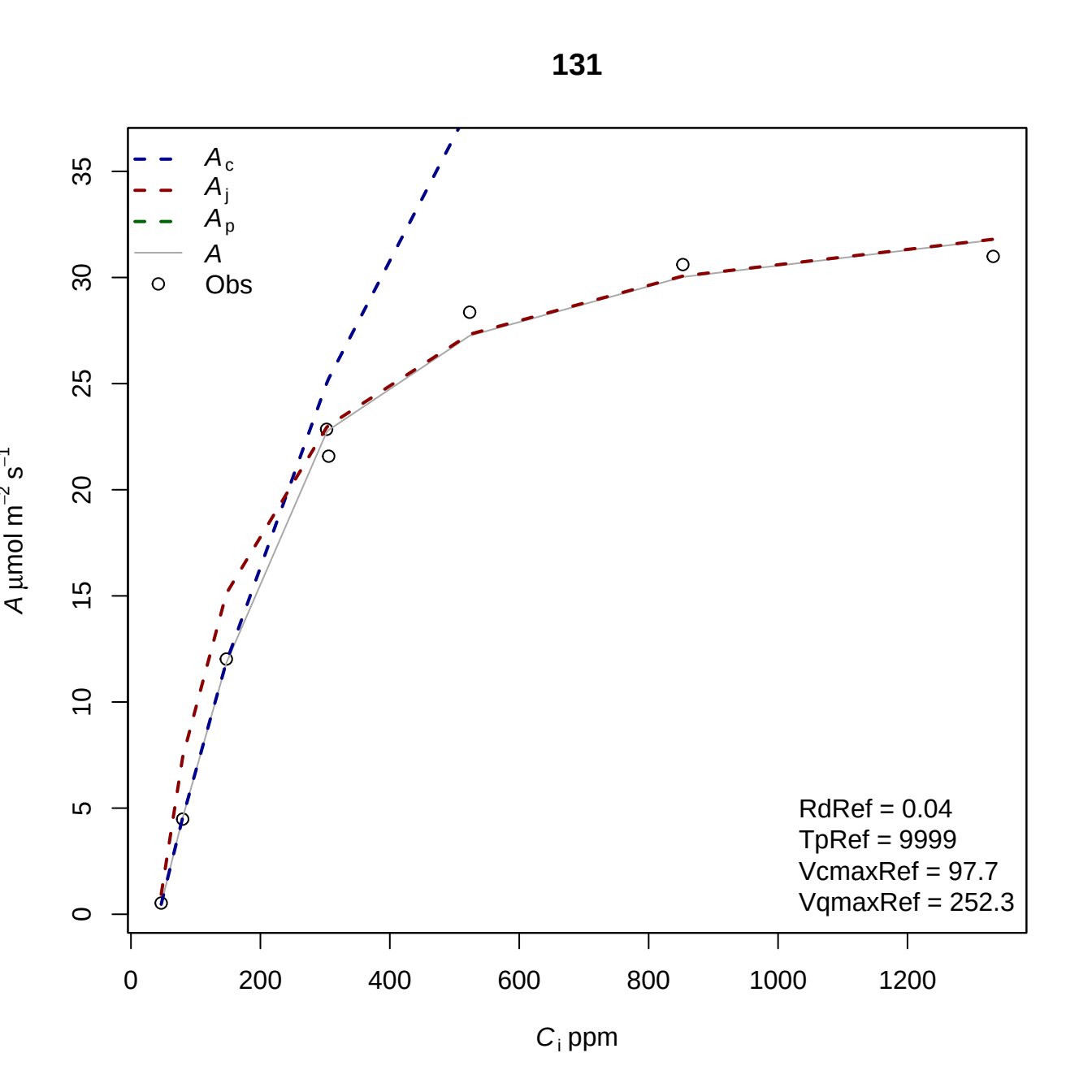

126
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.46
TpRef = 11.97
VcmaxRef = 113.86
VqmaxRef = 345.08
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

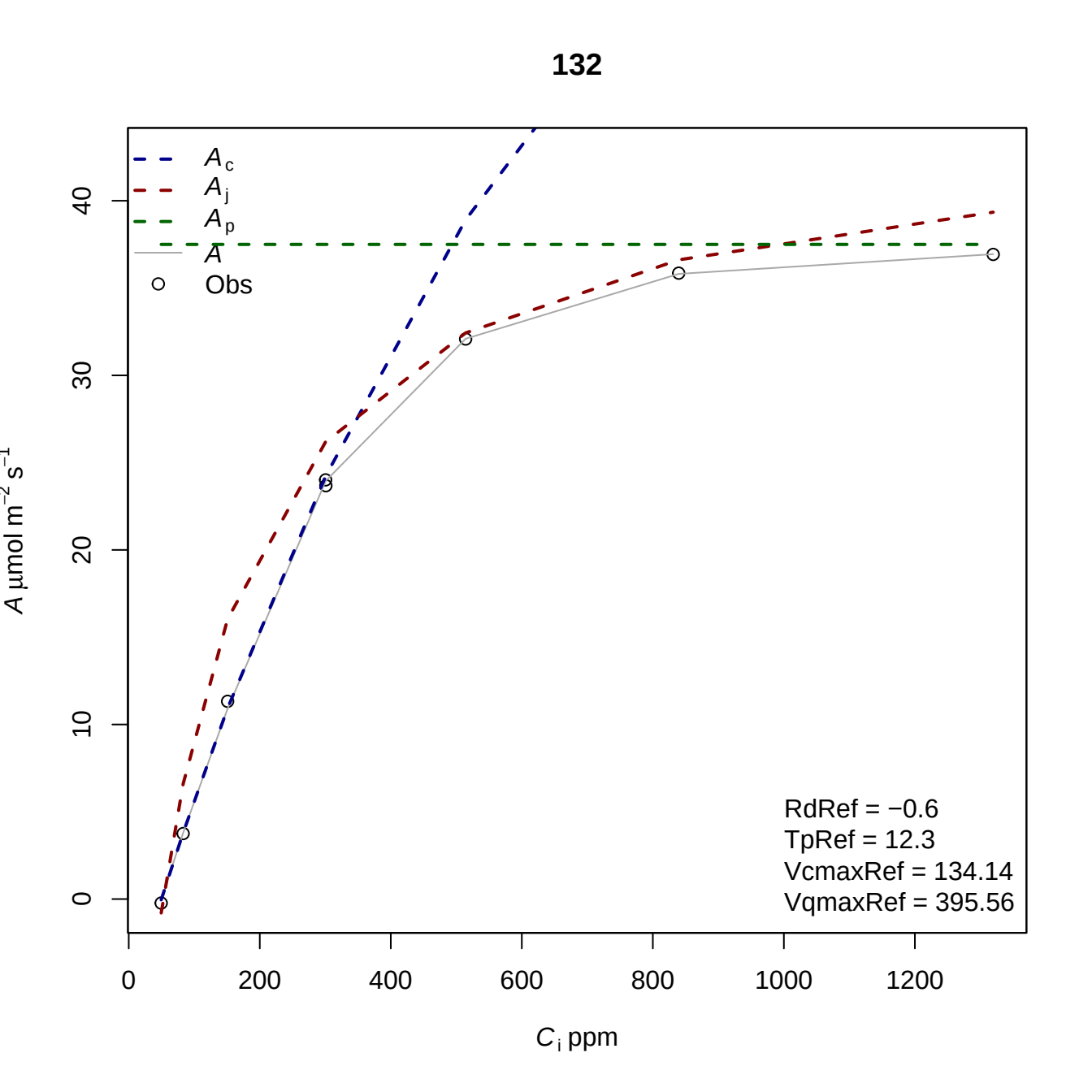

127
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.73
TpRef = 10.95
VcmaxRef = 153.86
VqmaxRef = 330.28
0
10
20
30
200
400
600
800
1000
1200
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

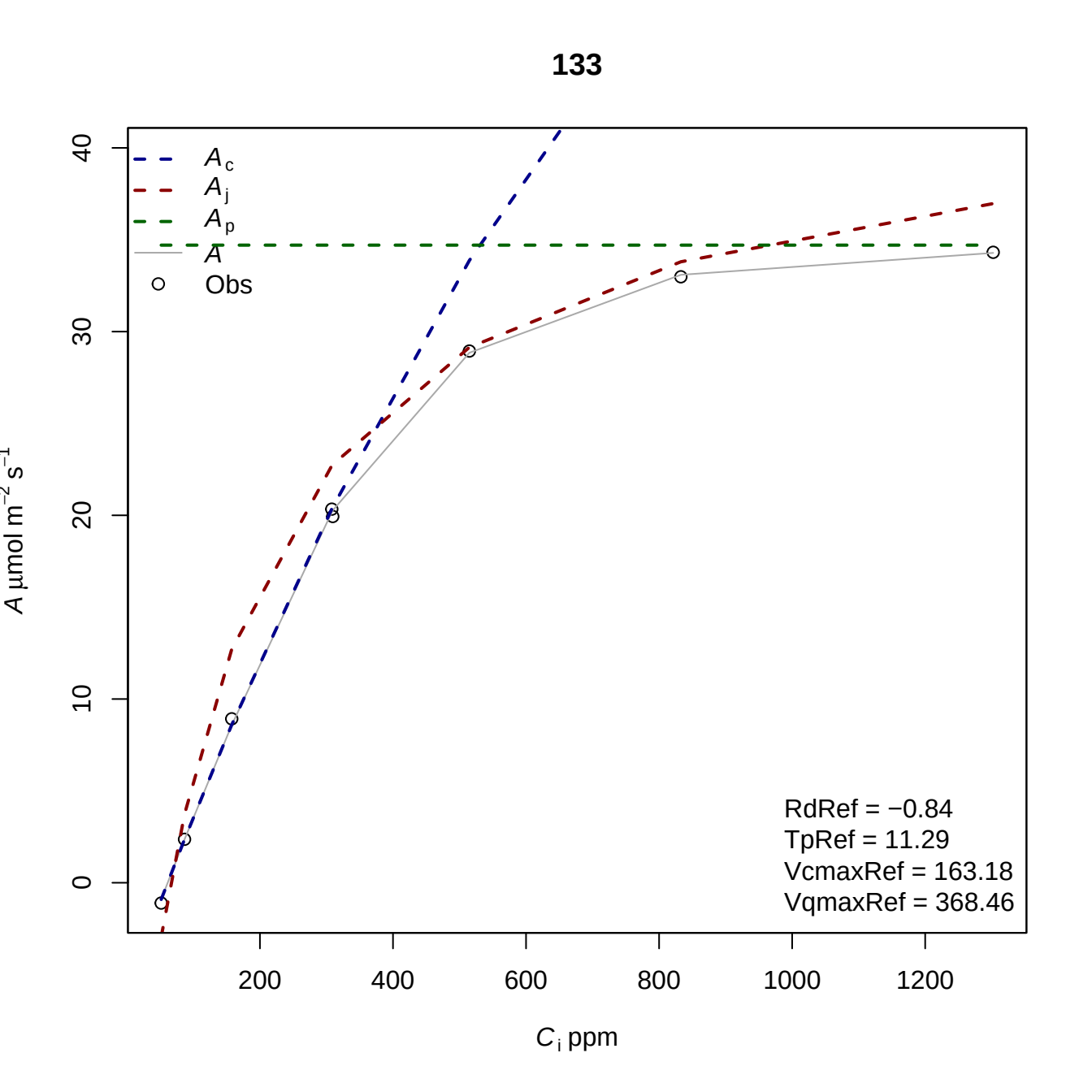

128
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.15
TpRef = 4.06
VcmaxRef = 86.2
VqmaxRef = 88.18
0
5
10
200
400
600
800
1000
1200
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

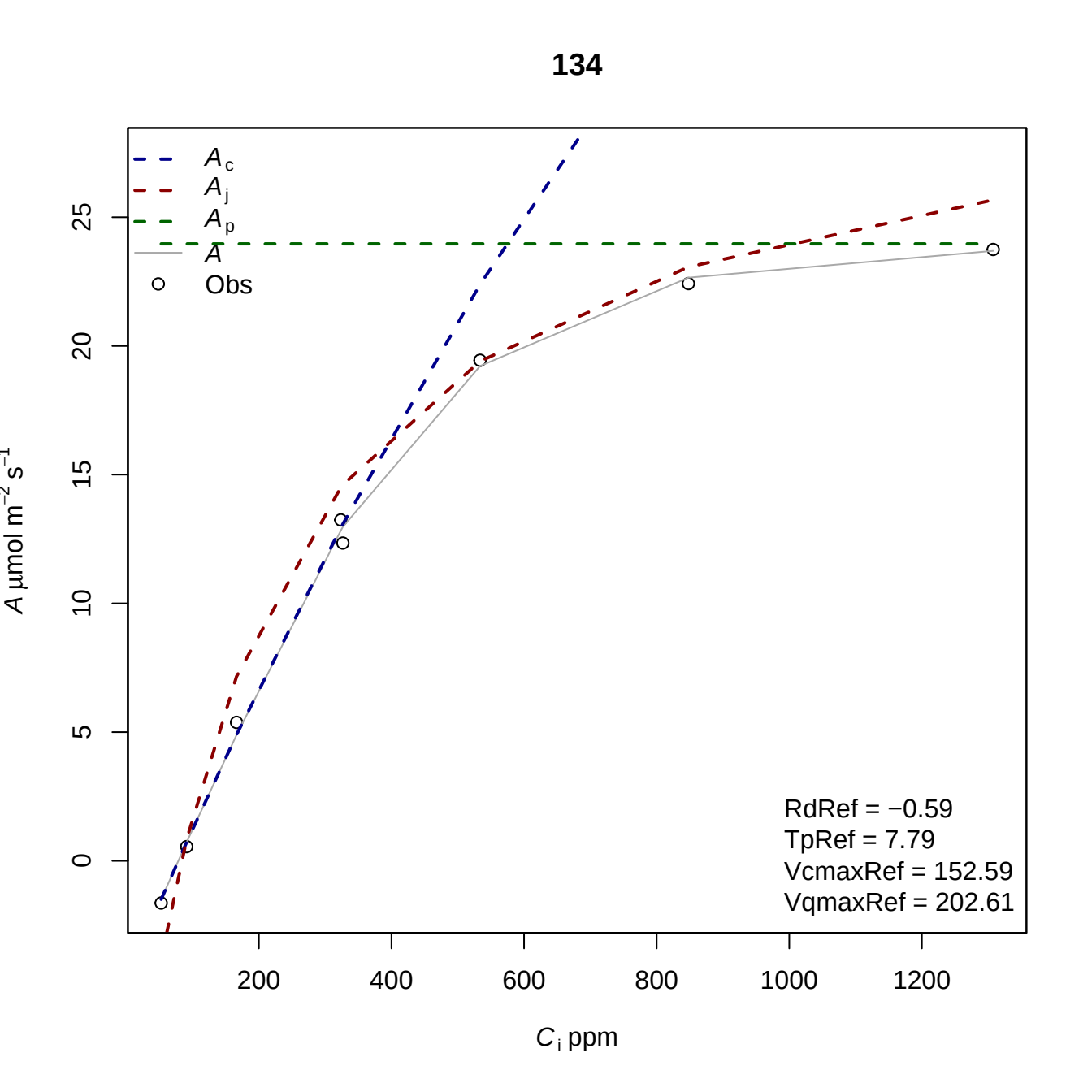

129
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0
TpRef = 7.04
VcmaxRef = 36.4
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
1400
5
10
15
20
25

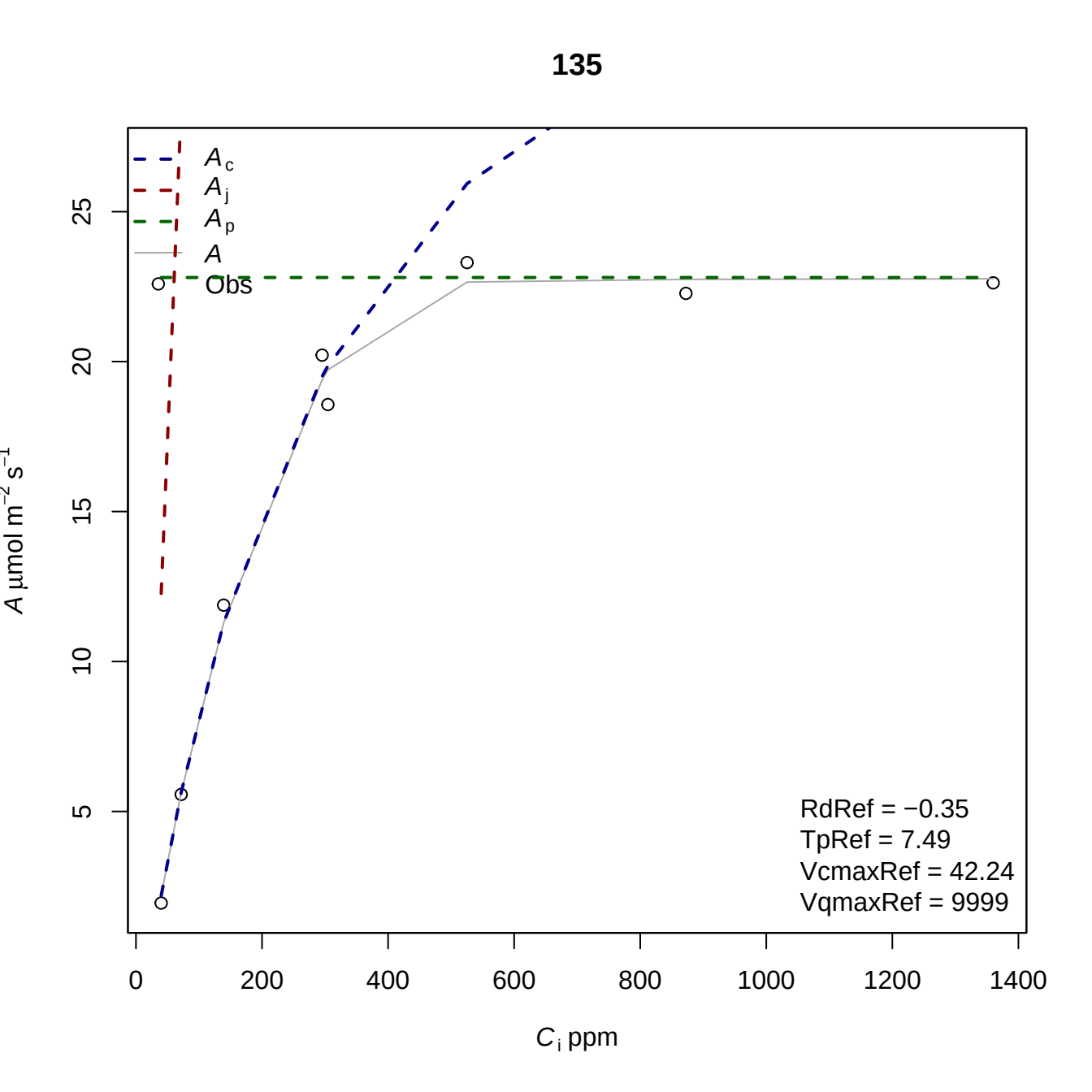

130
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.03
TpRef = 9.33
VcmaxRef = 59.34
VqmaxRef = 204.08
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

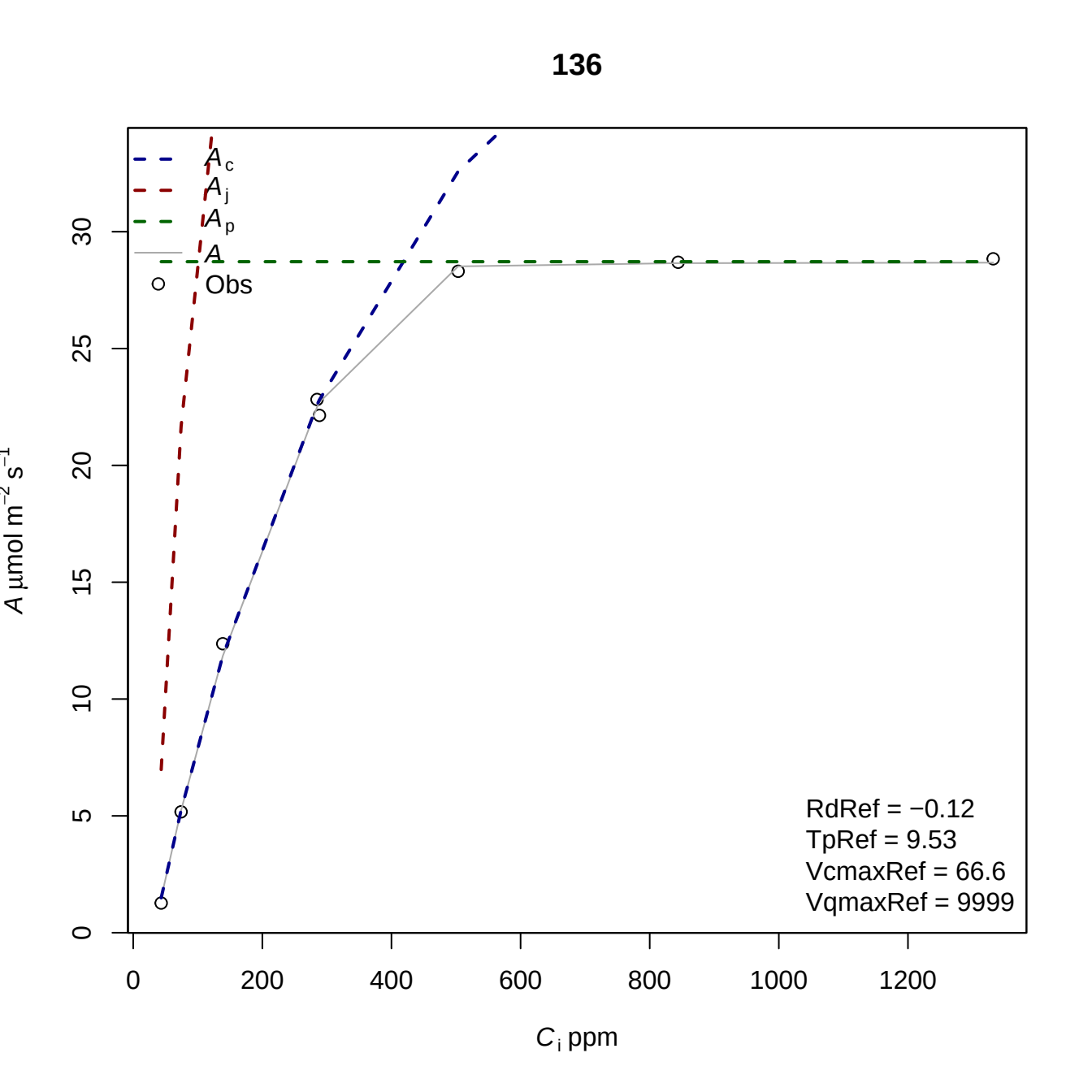

131
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.04
TpRef = 9999
VcmaxRef = 97.7
VqmaxRef = 252.3
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

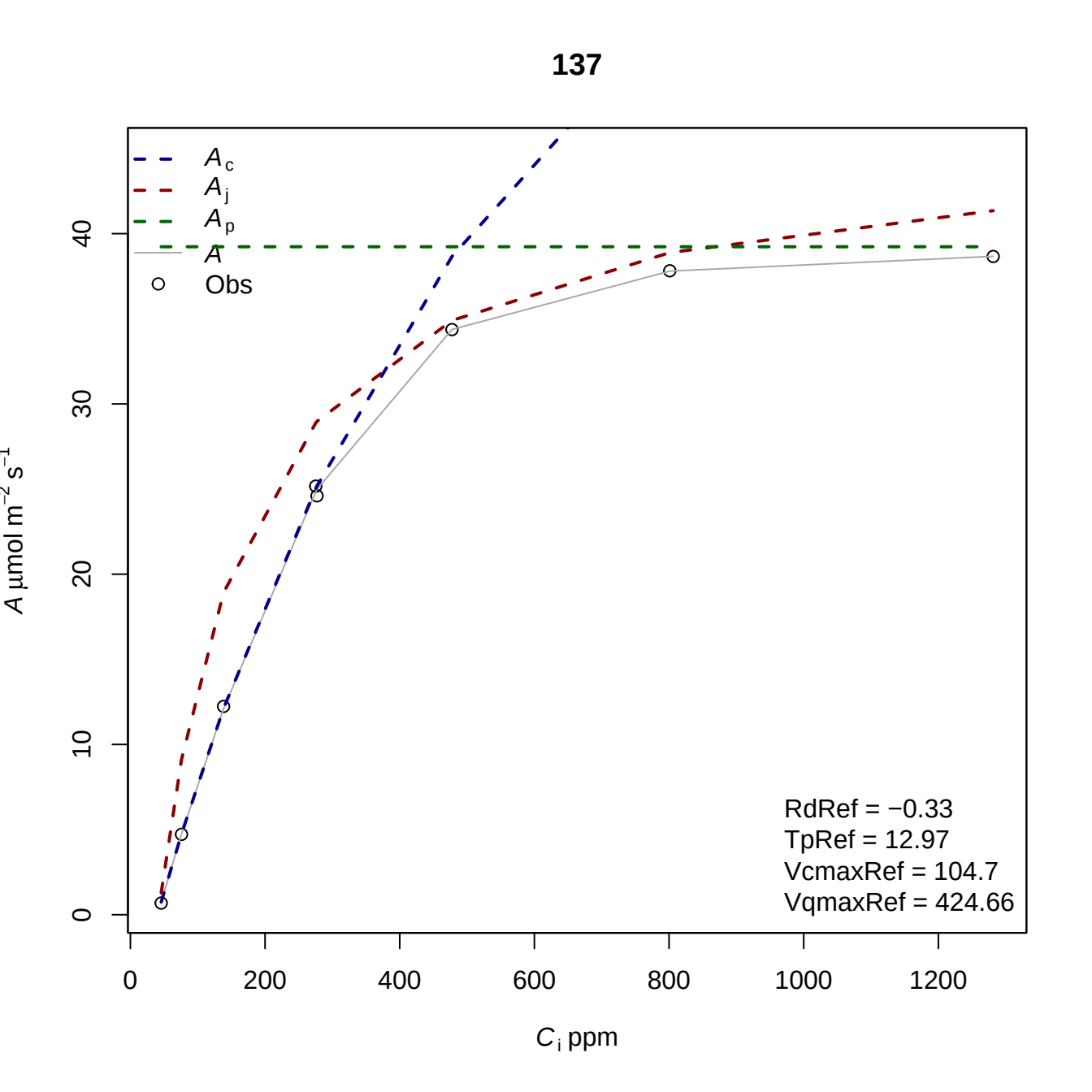

132
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.6
TpRef = 12.3
VcmaxRef = 134.14
VqmaxRef = 395.56
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

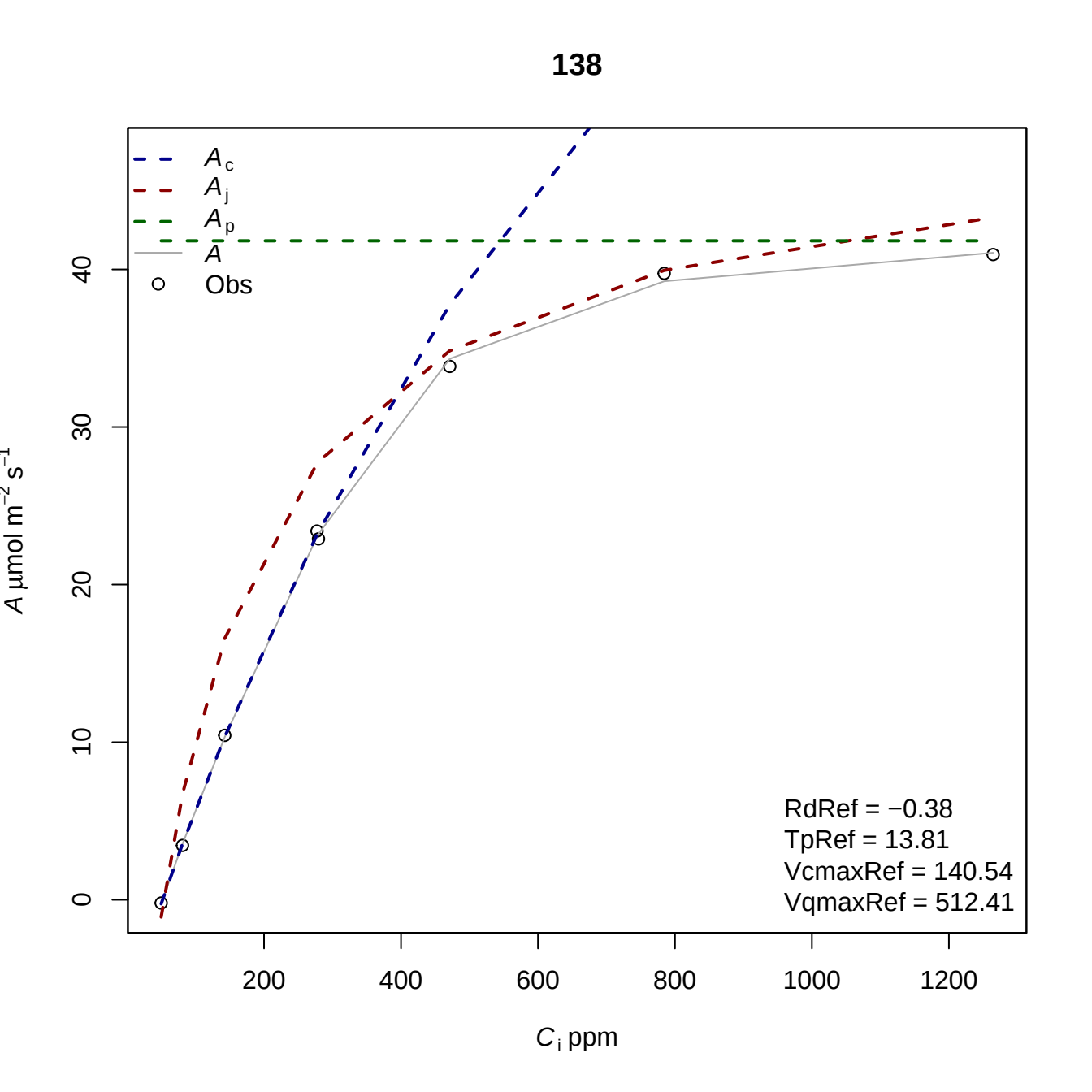

133
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.84
TpRef = 11.29
VcmaxRef = 163.18
VqmaxRef = 368.46
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

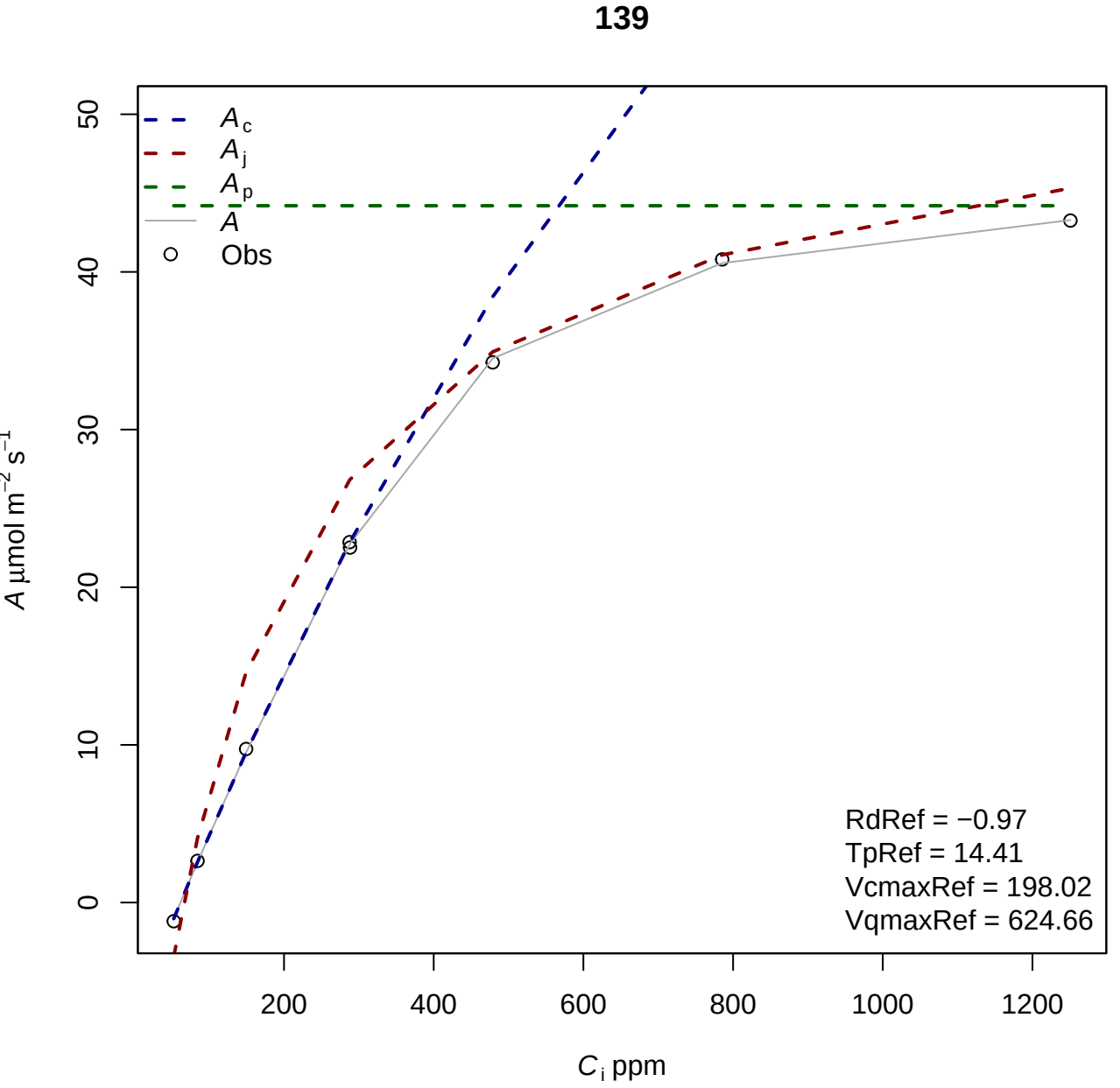

134
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.59
TpRef = 7.79
VcmaxRef = 152.59
VqmaxRef = 202.61
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

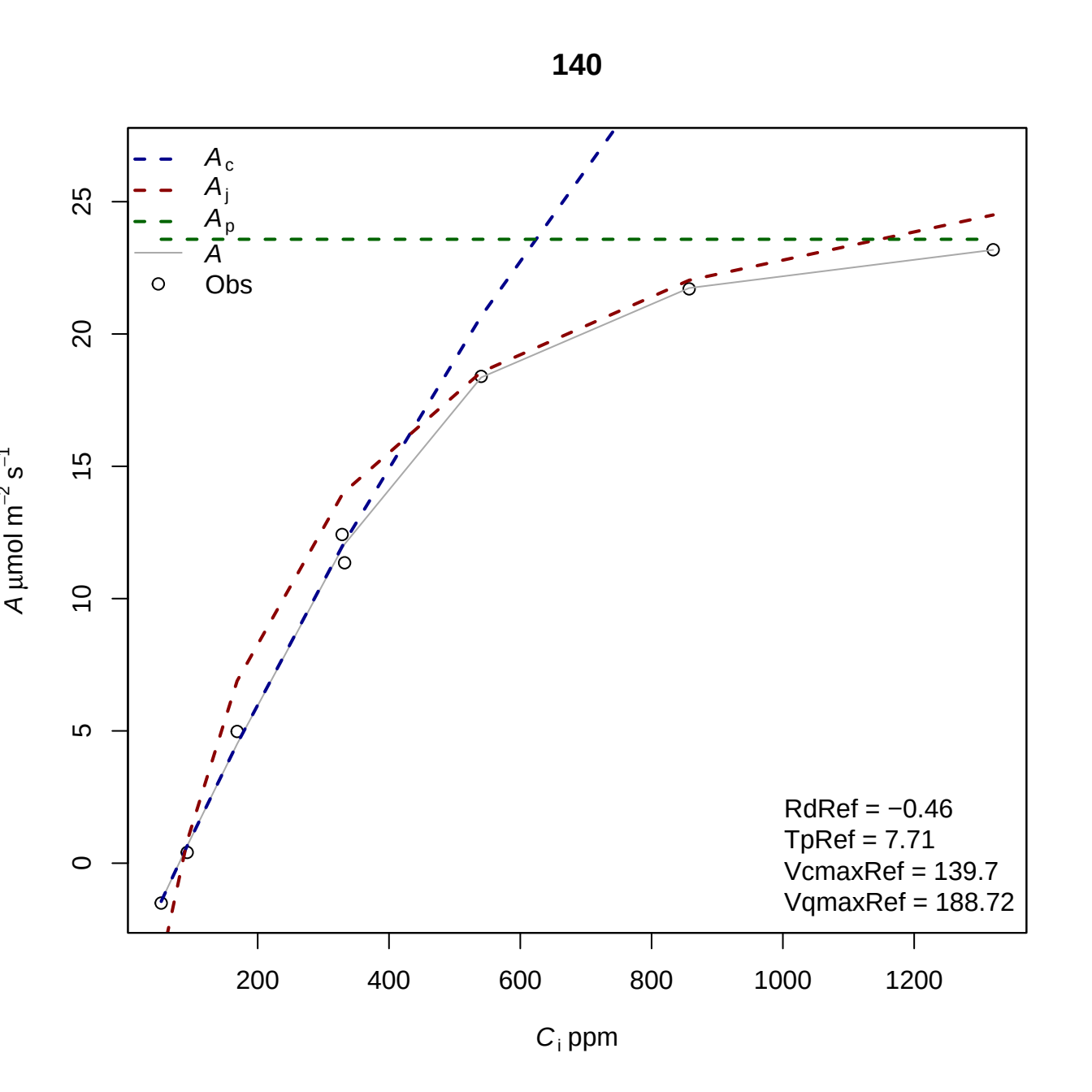

135
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.35
TpRef = 7.49
VcmaxRef = 42.24
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
1400
5
10
15
20
25

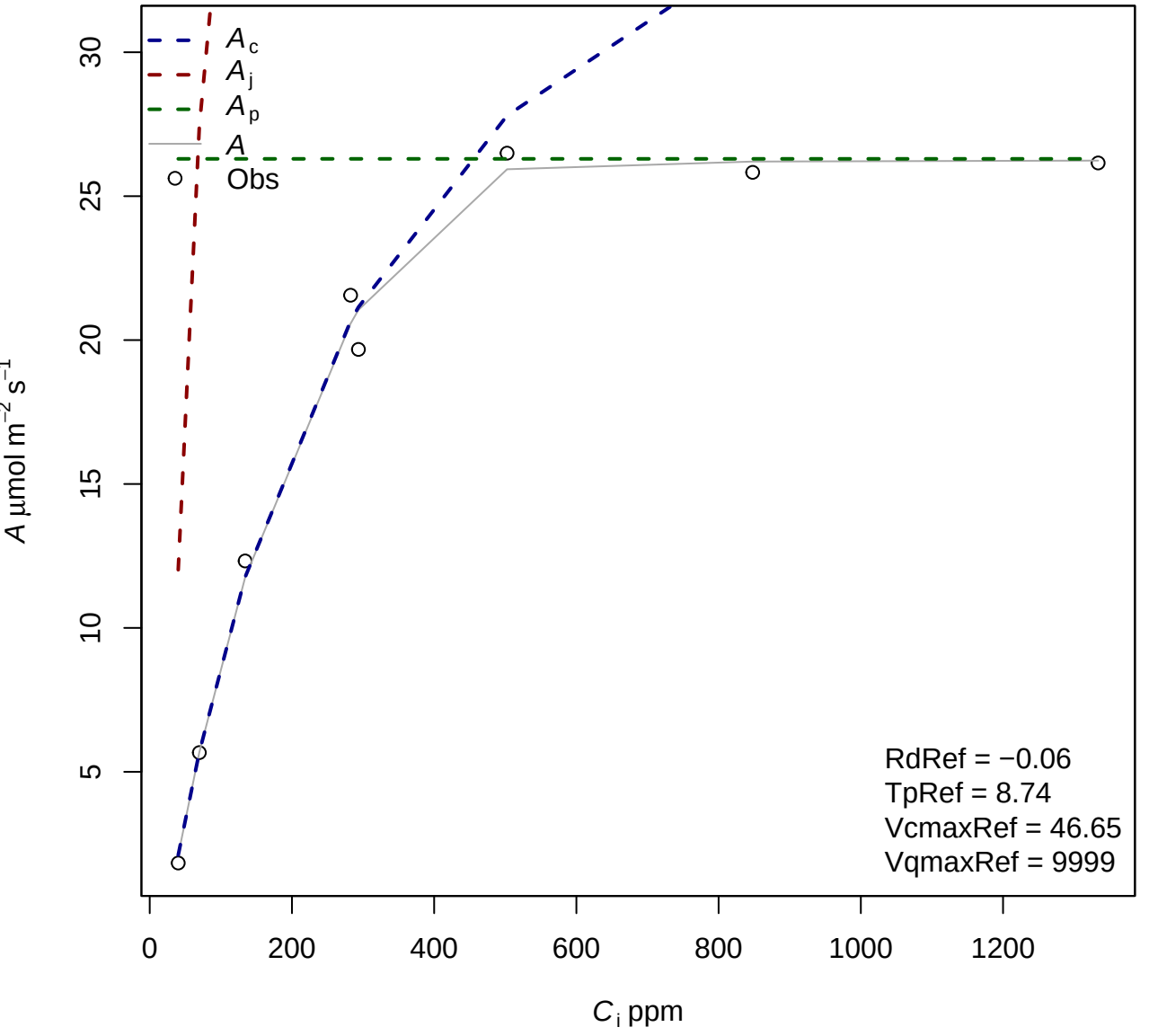

136
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.12
TpRef = 9.53
VcmaxRef = 66.6
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

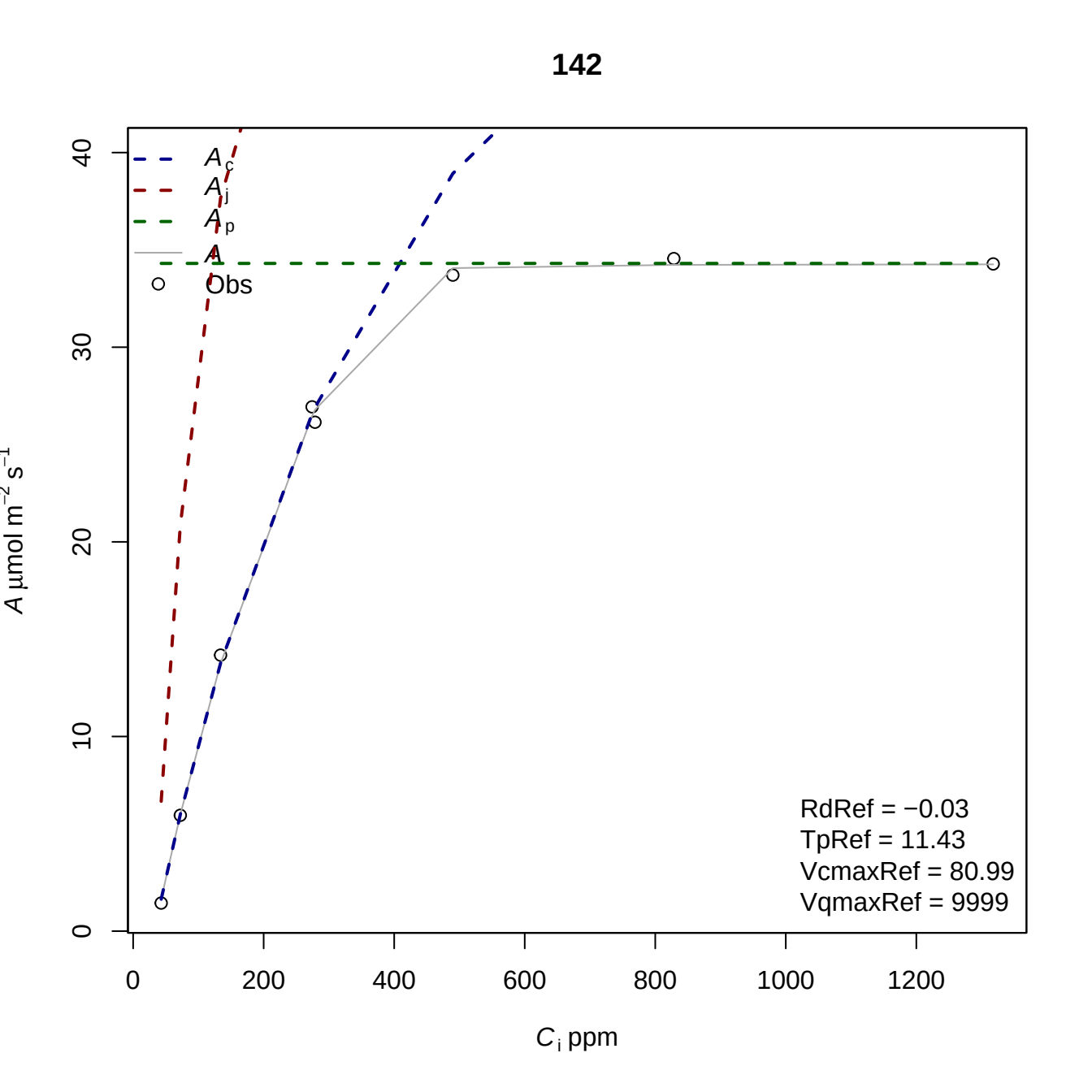

137
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.33
TpRef = 12.97
VcmaxRef = 104.7
VqmaxRef = 424.66
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

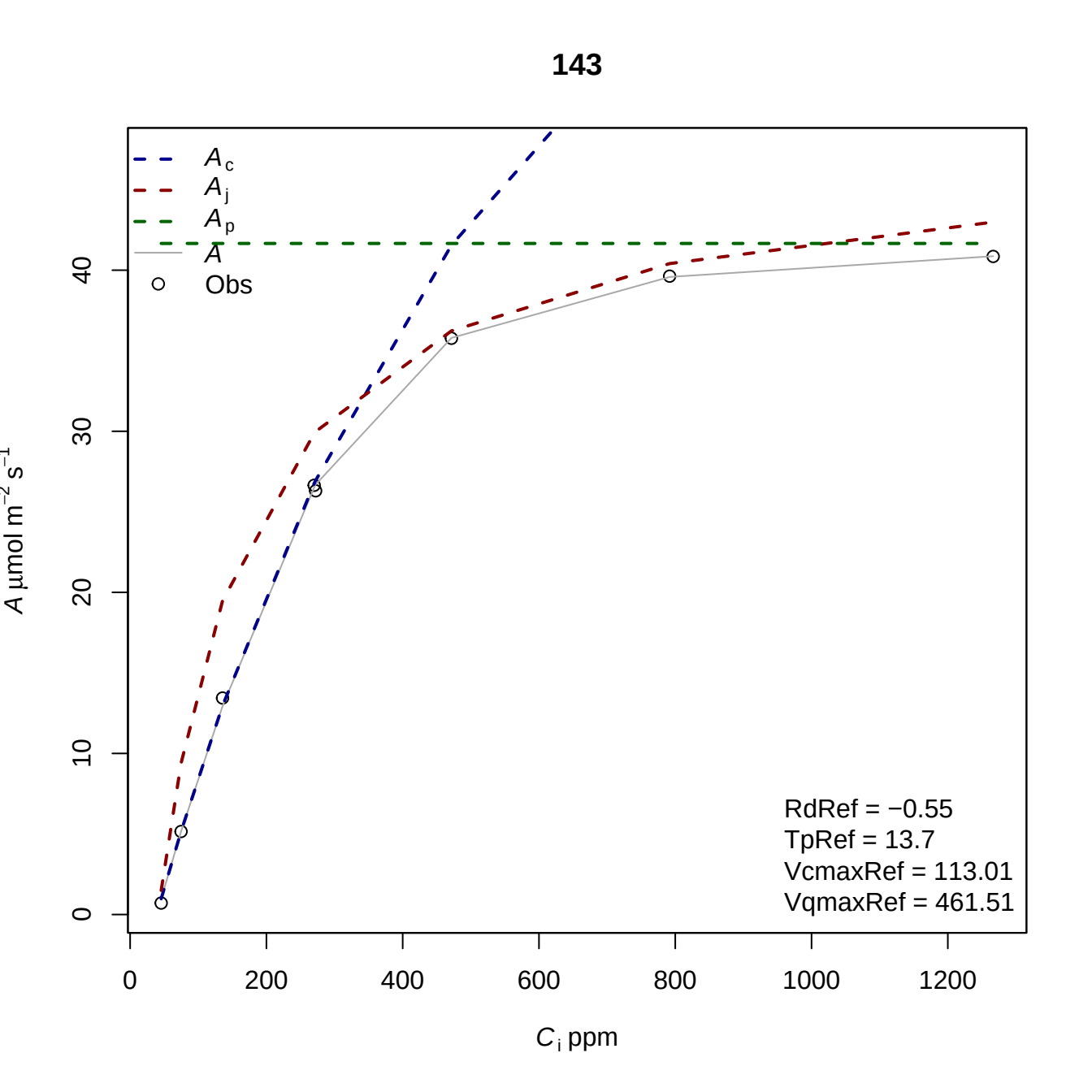

138
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.38
TpRef = 13.81
VcmaxRef = 140.54
VqmaxRef = 512.41
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

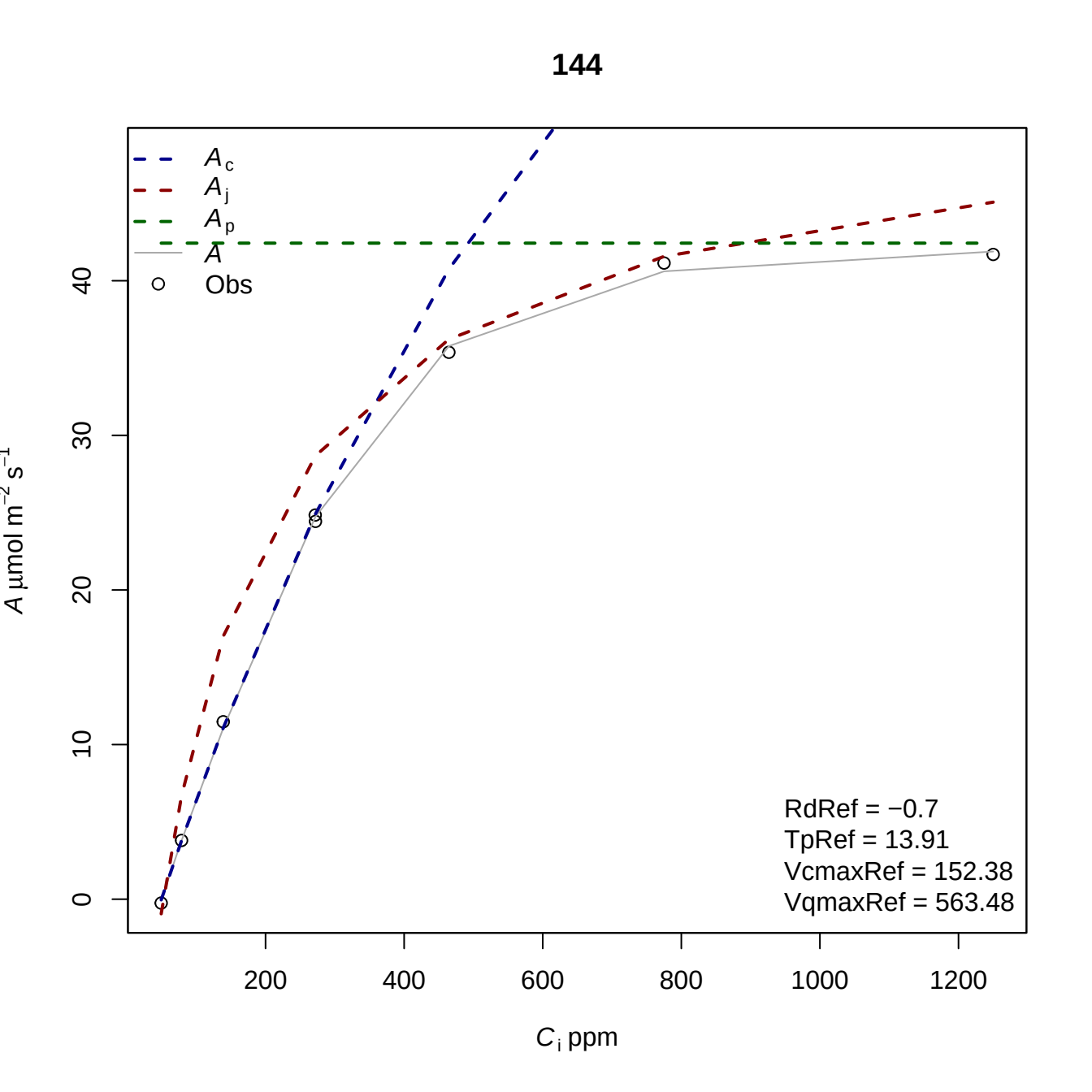

139
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.97
TpRef = 14.41
VcmaxRef = 198.02
VqmaxRef = 624.66
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

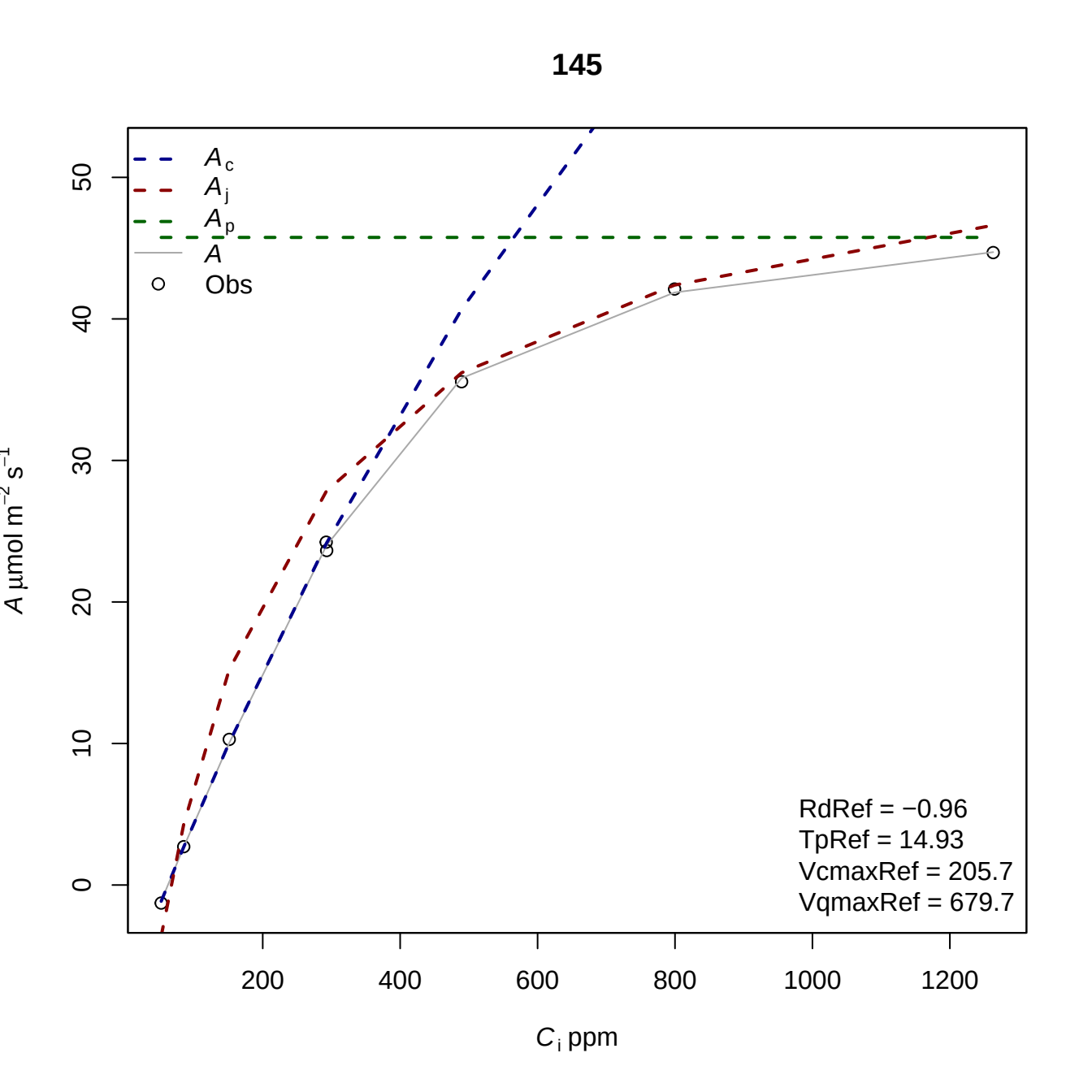

140
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.46
TpRef = 7.71
VcmaxRef = 139.7
VqmaxRef = 188.72
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

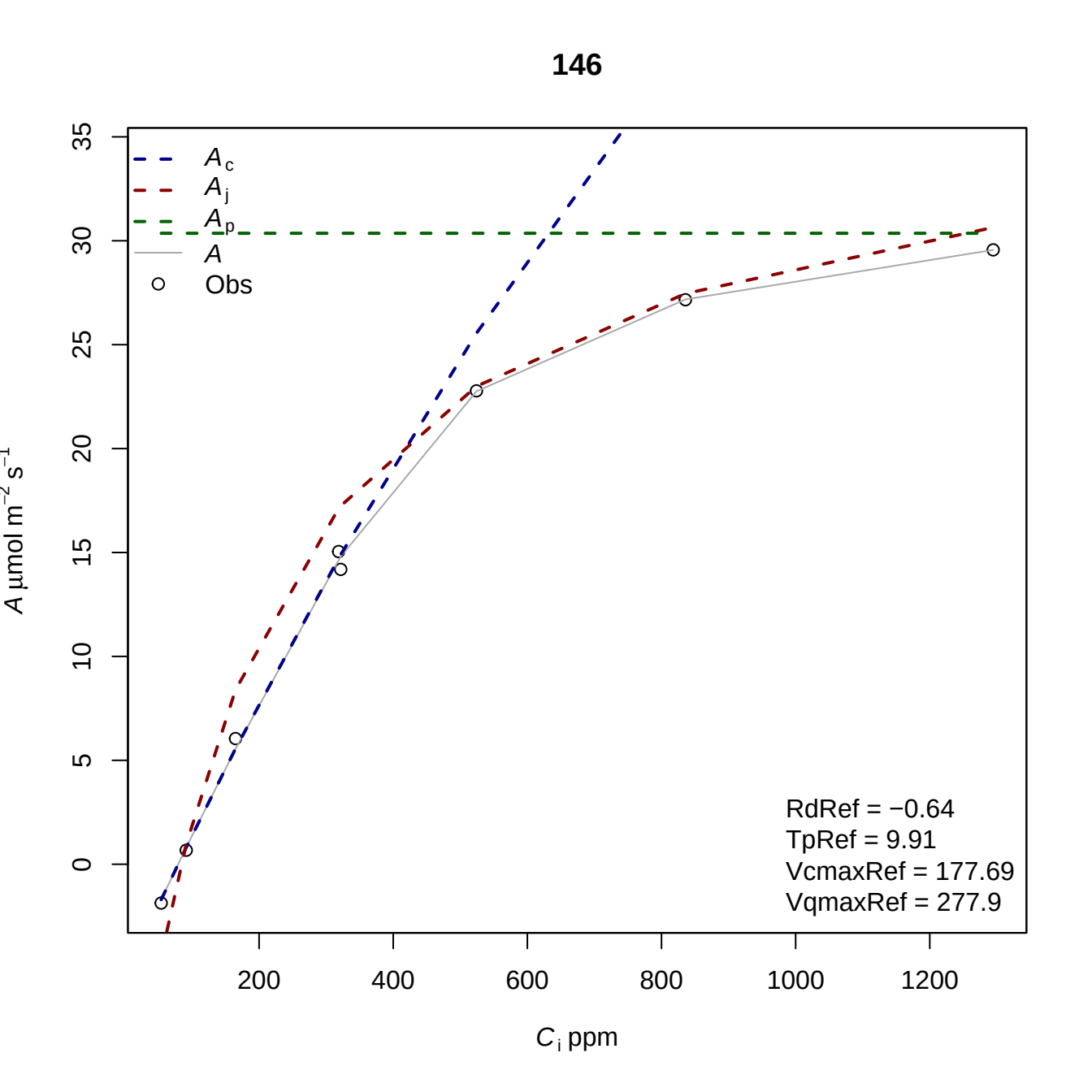

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.06
TpRef = 8.74
VcmaxRef = 46.65
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25
30

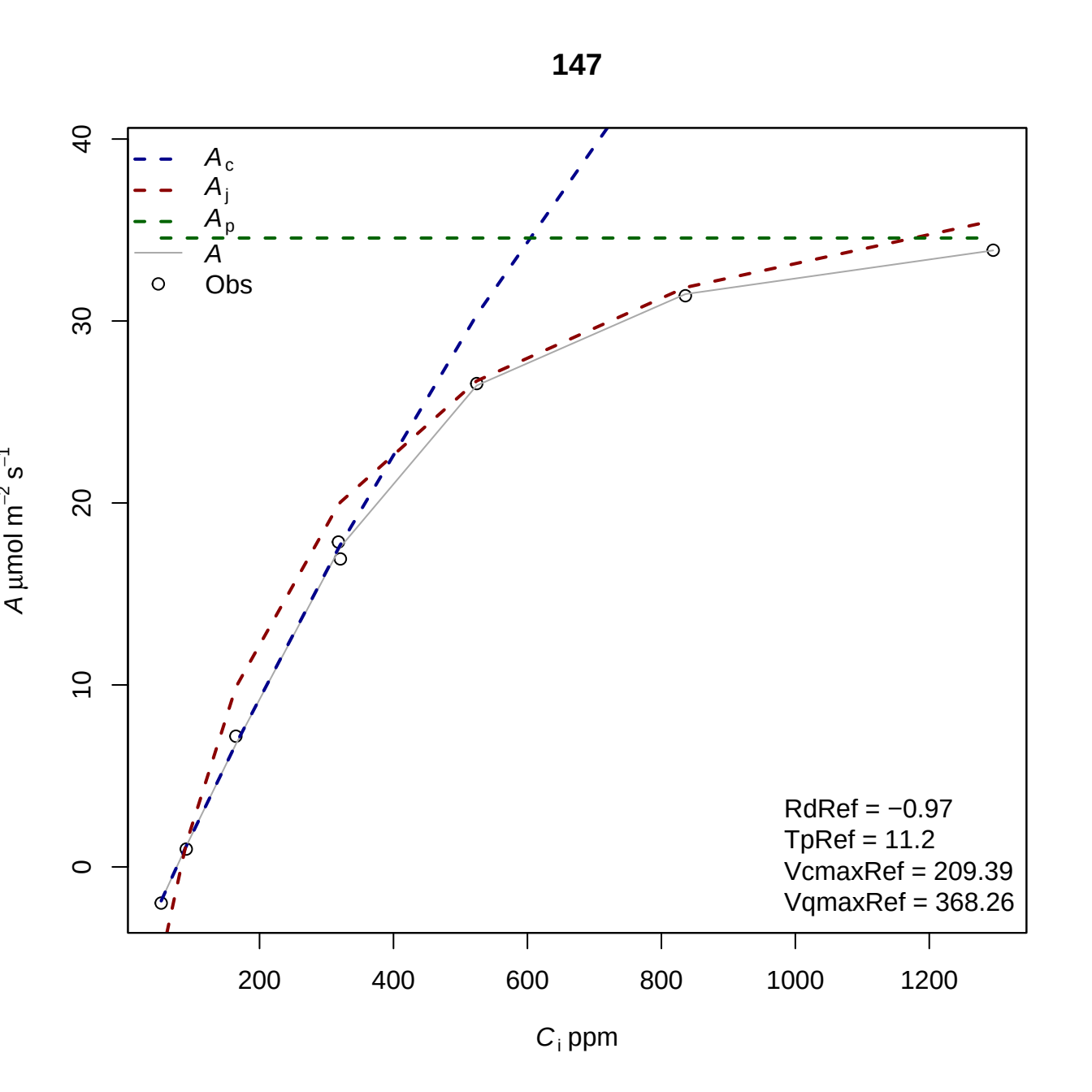

142
$A_c$
$A_j$
$A_p$
A
Obs
RdRef = −0.03
TpRef = 11.43
VcmaxRef = 80.99
VqmaxRef = 9999
A µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

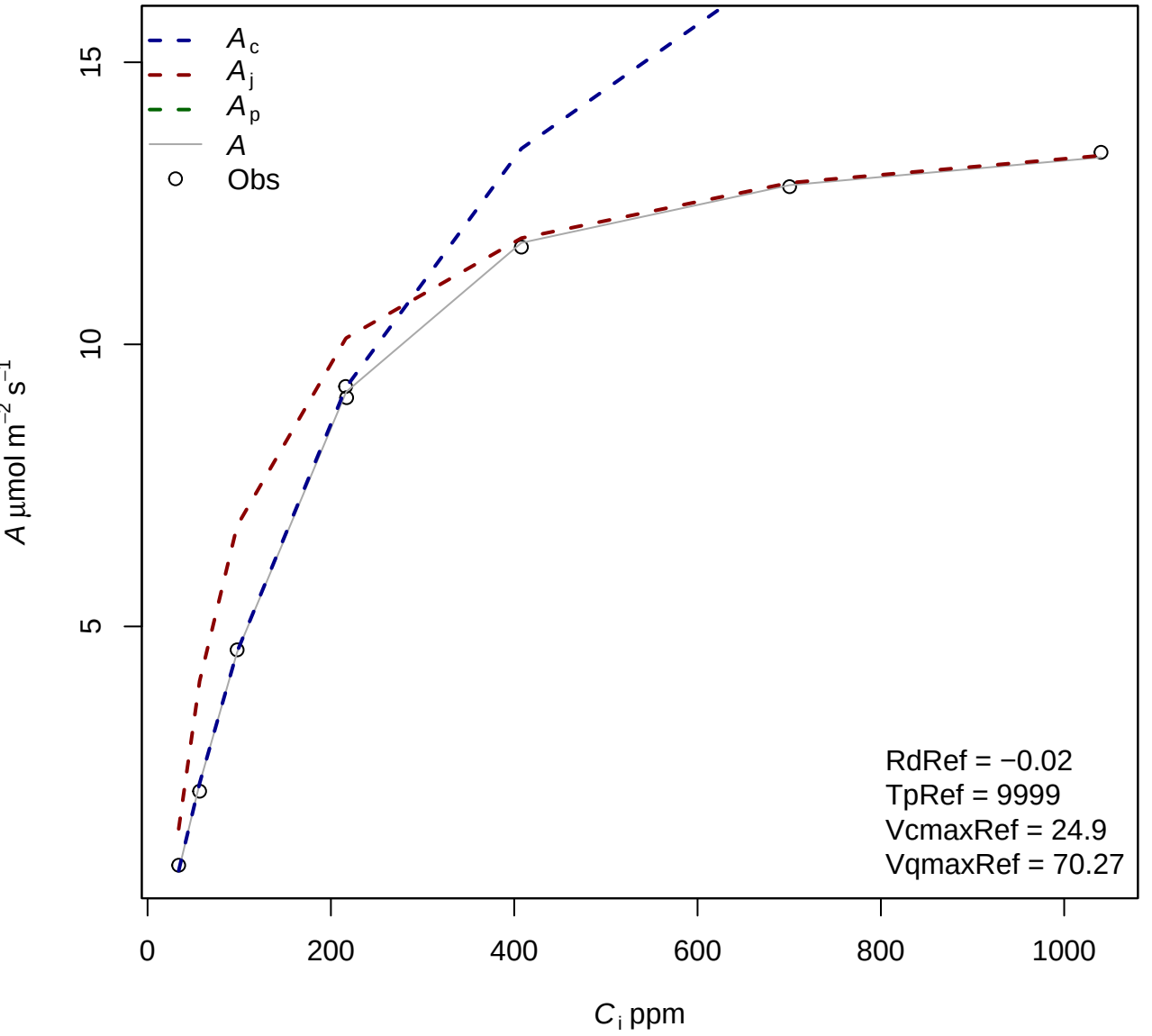

143
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.55
TpRef = 13.7
VcmaxRef = 113.01
VqmaxRef = 461.51
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

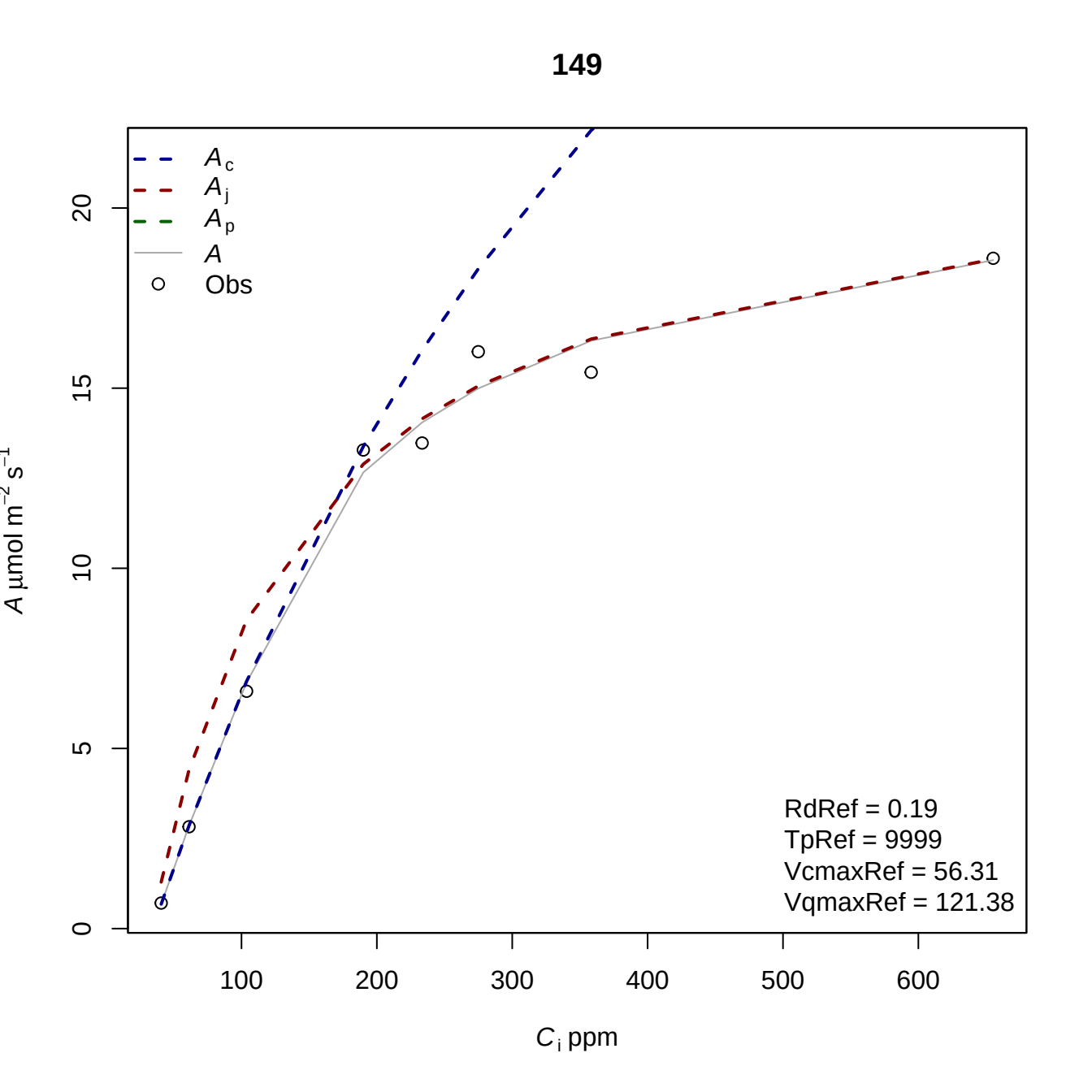

144
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.7
TpRef = 13.91
VcmaxRef = 152.38
VqmaxRef = 563.48
0
10
20
30
40
200
400
600
800
1000
1200
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

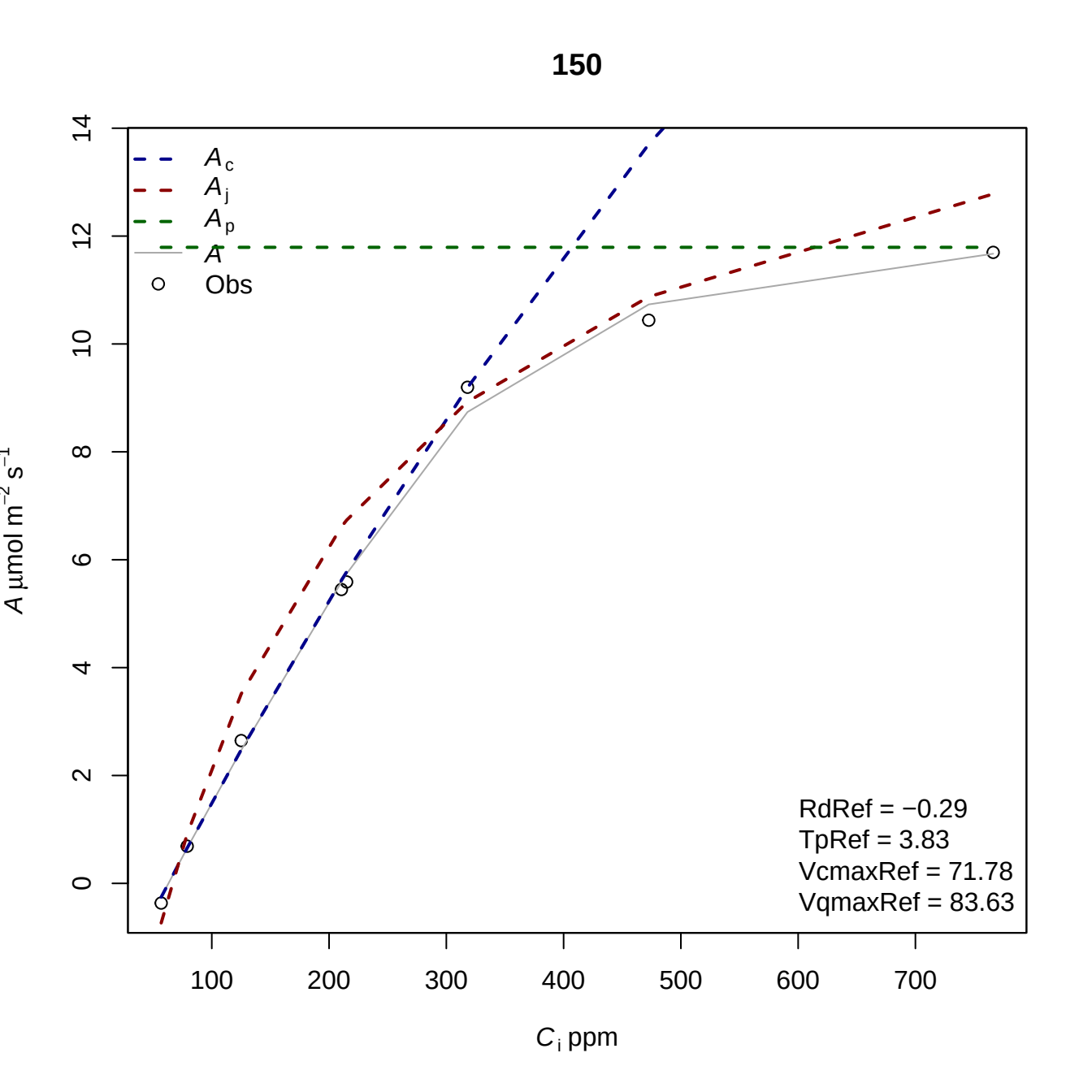

145
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.96
TpRef = 14.93
VcmaxRef = 205.7
VqmaxRef = 679.7
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

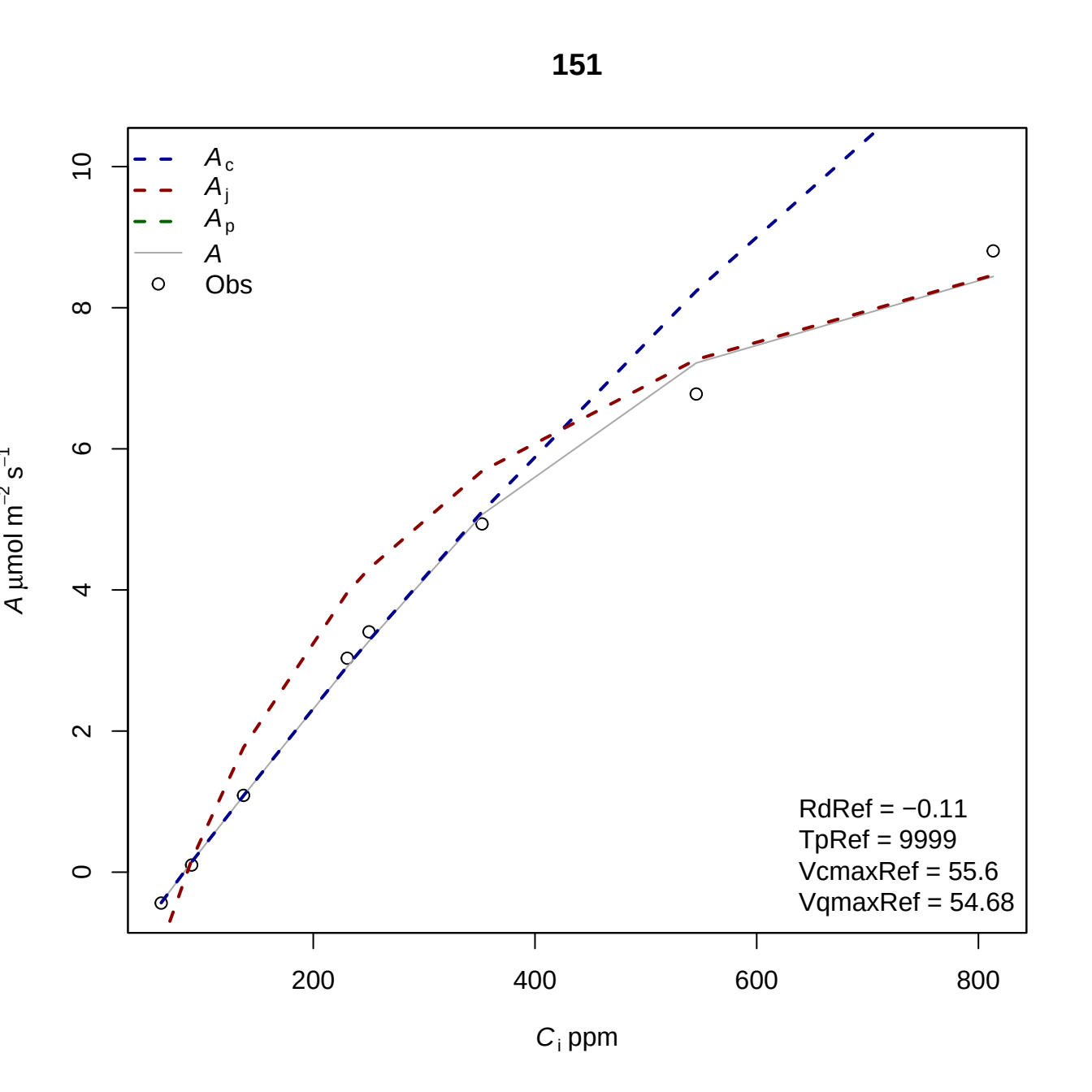

146
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.64
TpRef = 9.91
VcmaxRef = 177.69
VqmaxRef = 277.9
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
200
400
600
800
1000
1200

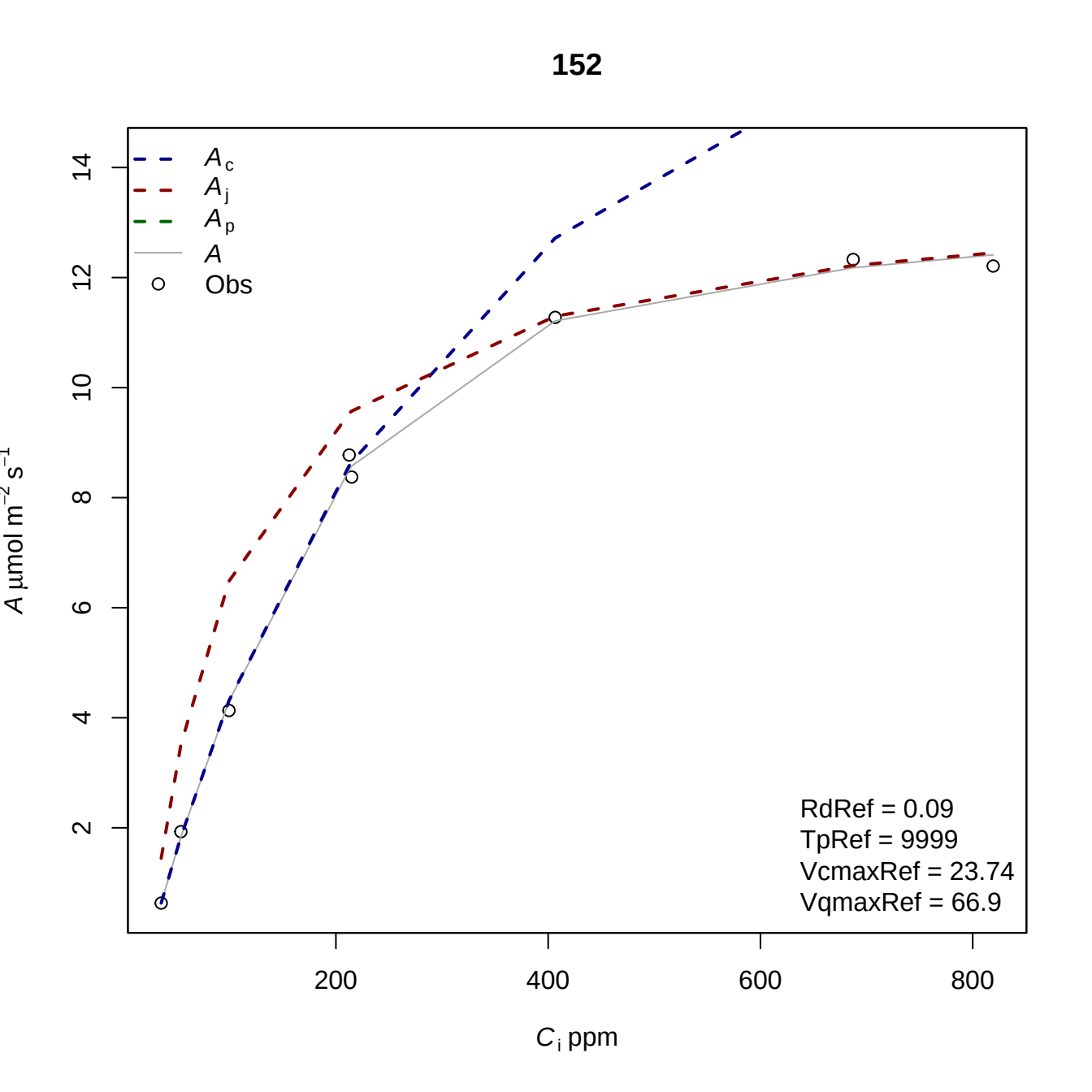

147
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.97
TpRef = 11.2
VcmaxRef = 209.39
VqmaxRef = 368.26
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

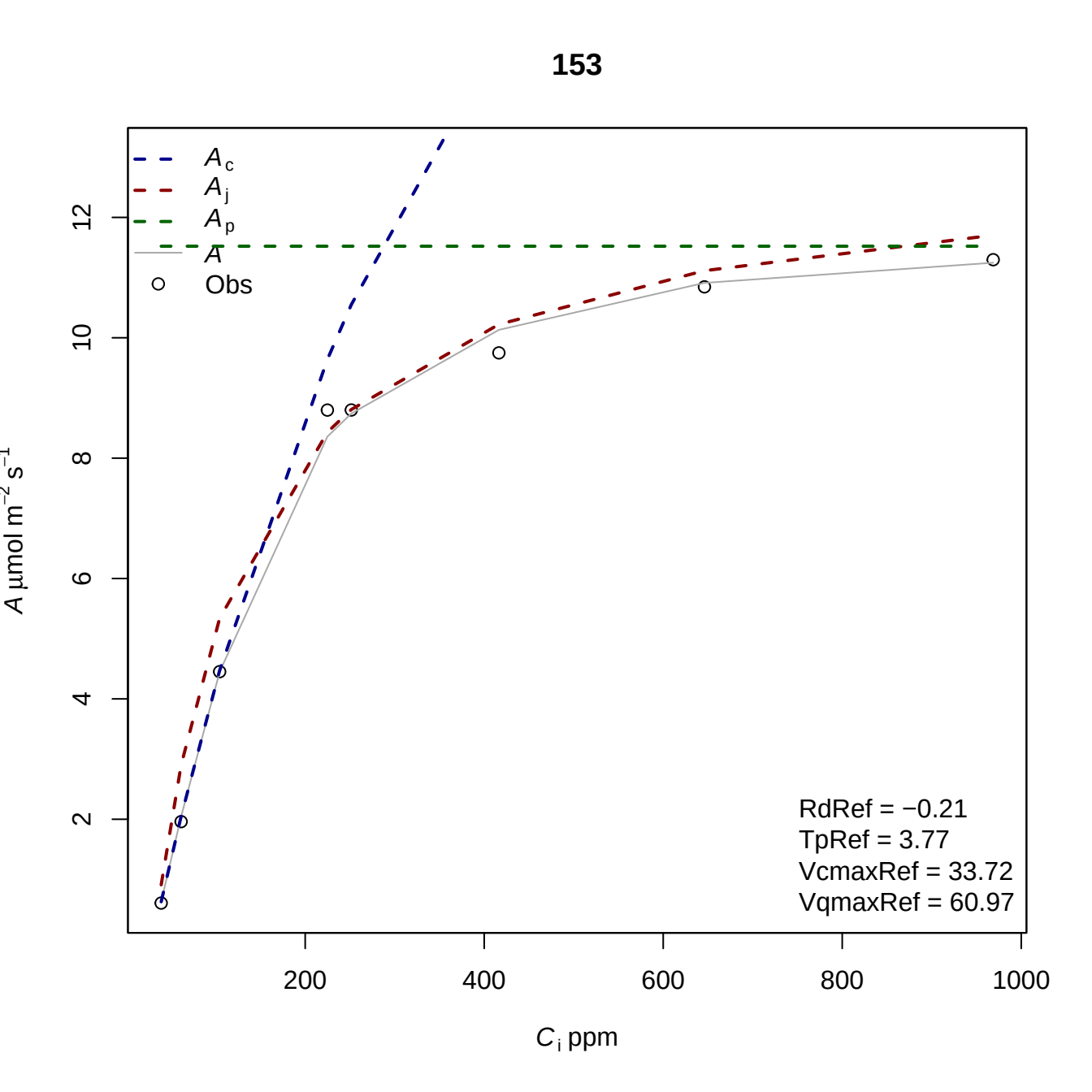

$A_c$
$A_j$
$A_p$
$A$
Obs
15
10
5
$A$ µmol m$^{-2}$ s$^{-1}$
0
200
400
600
800
1000
$C_i$ ppm
RdRef = −0.02
TpRef = 9999
VcmaxRef = 24.9
VqmaxRef = 70.27

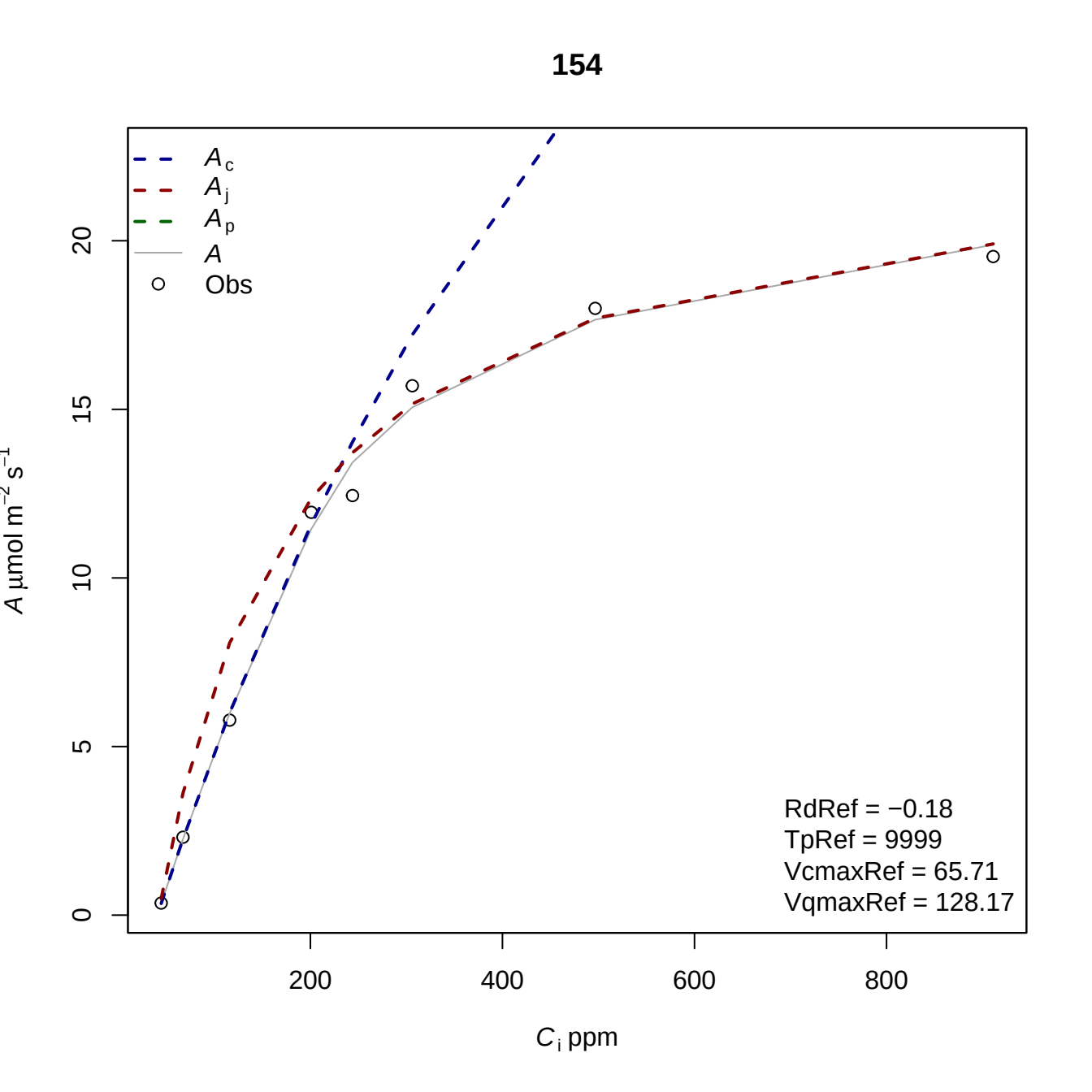

149
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.19
TpRef = 9999
VcmaxRef = 56.31
VqmaxRef = 121.38
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
100
200
300
400
500
600

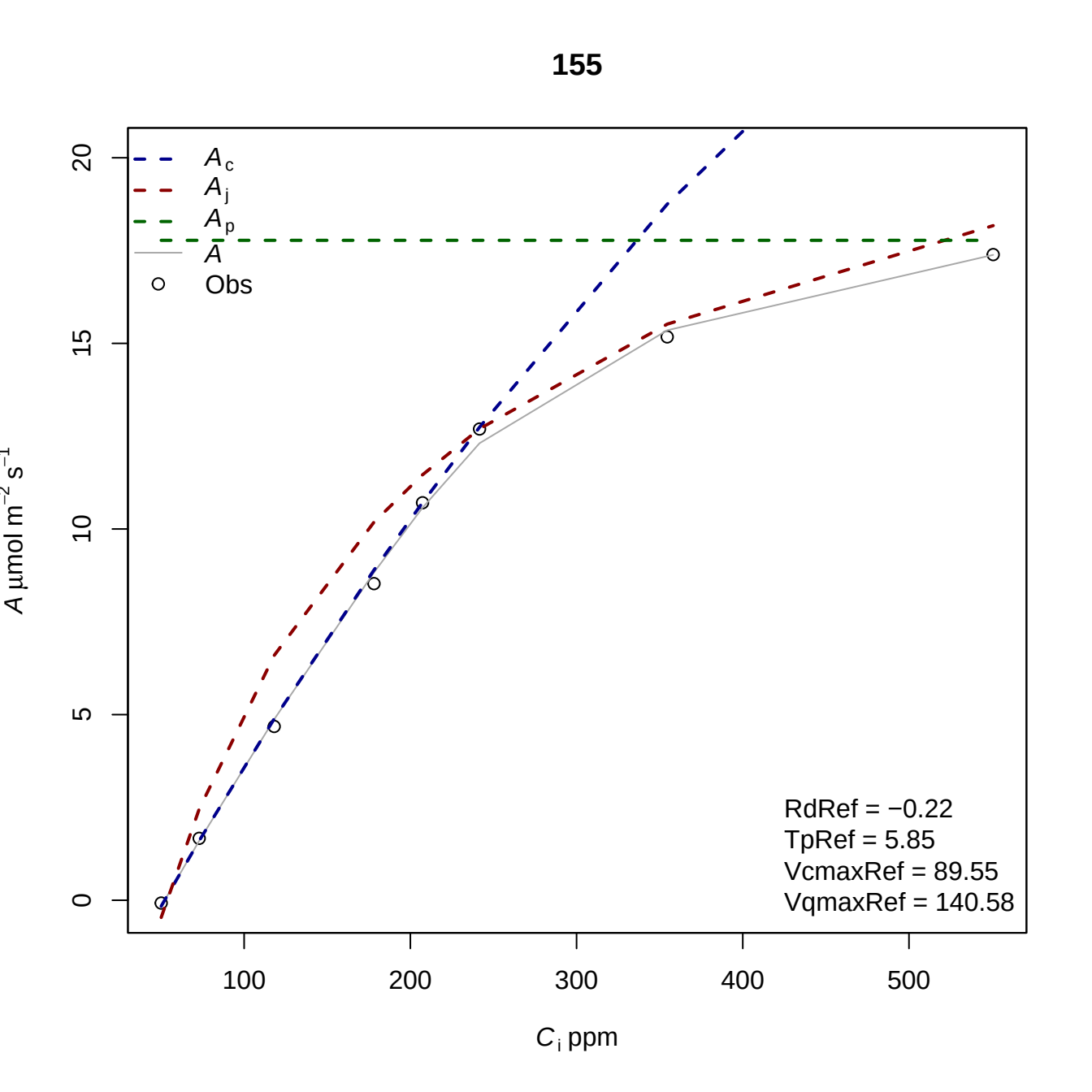

150
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.29
TpRef = 3.83
VcmaxRef = 71.78
VqmaxRef = 83.63
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
100
200
300
400
500
600
700

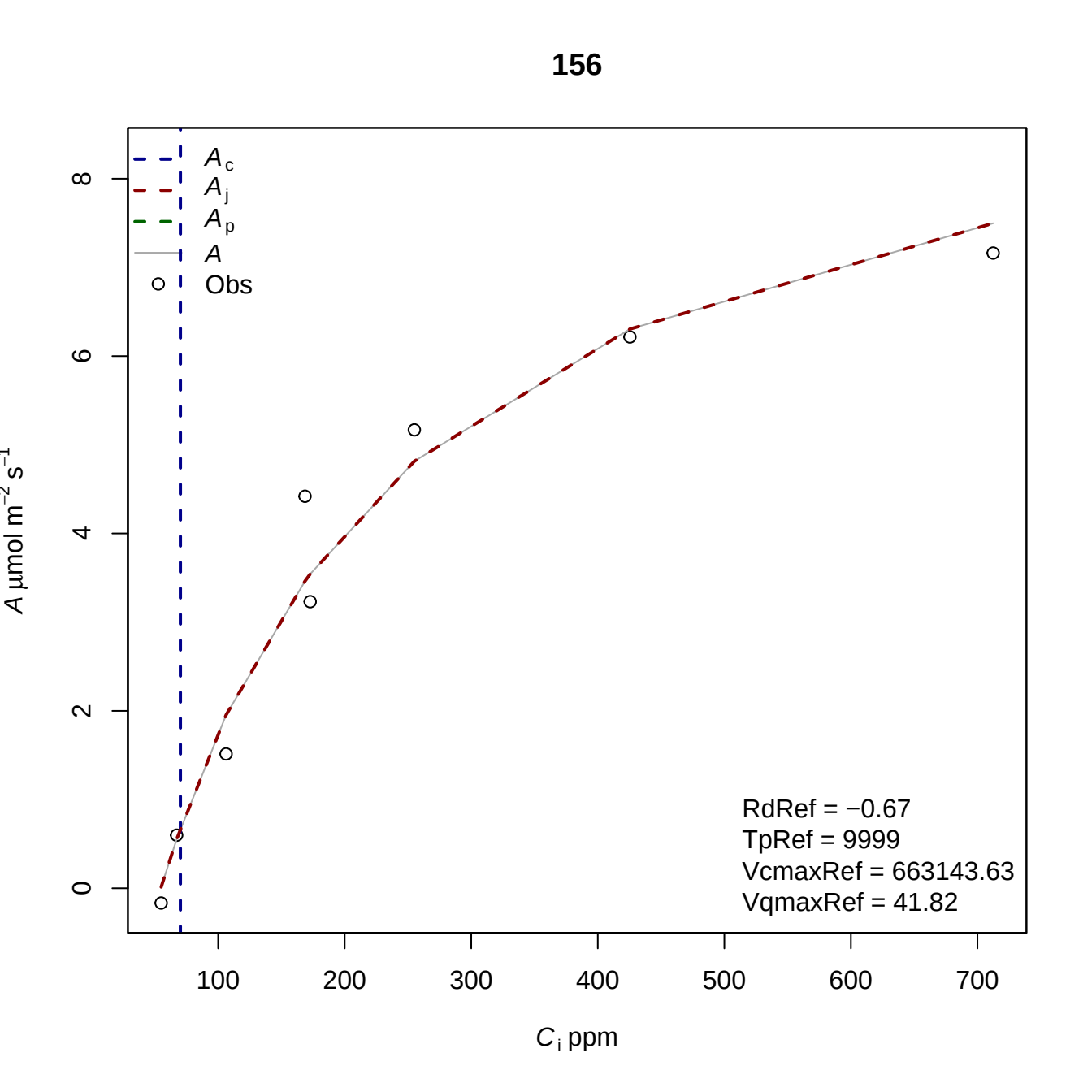

151
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.11
TpRef = 9999
VcmaxRef = 55.6
VqmaxRef = 54.68
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
200
400
600
800

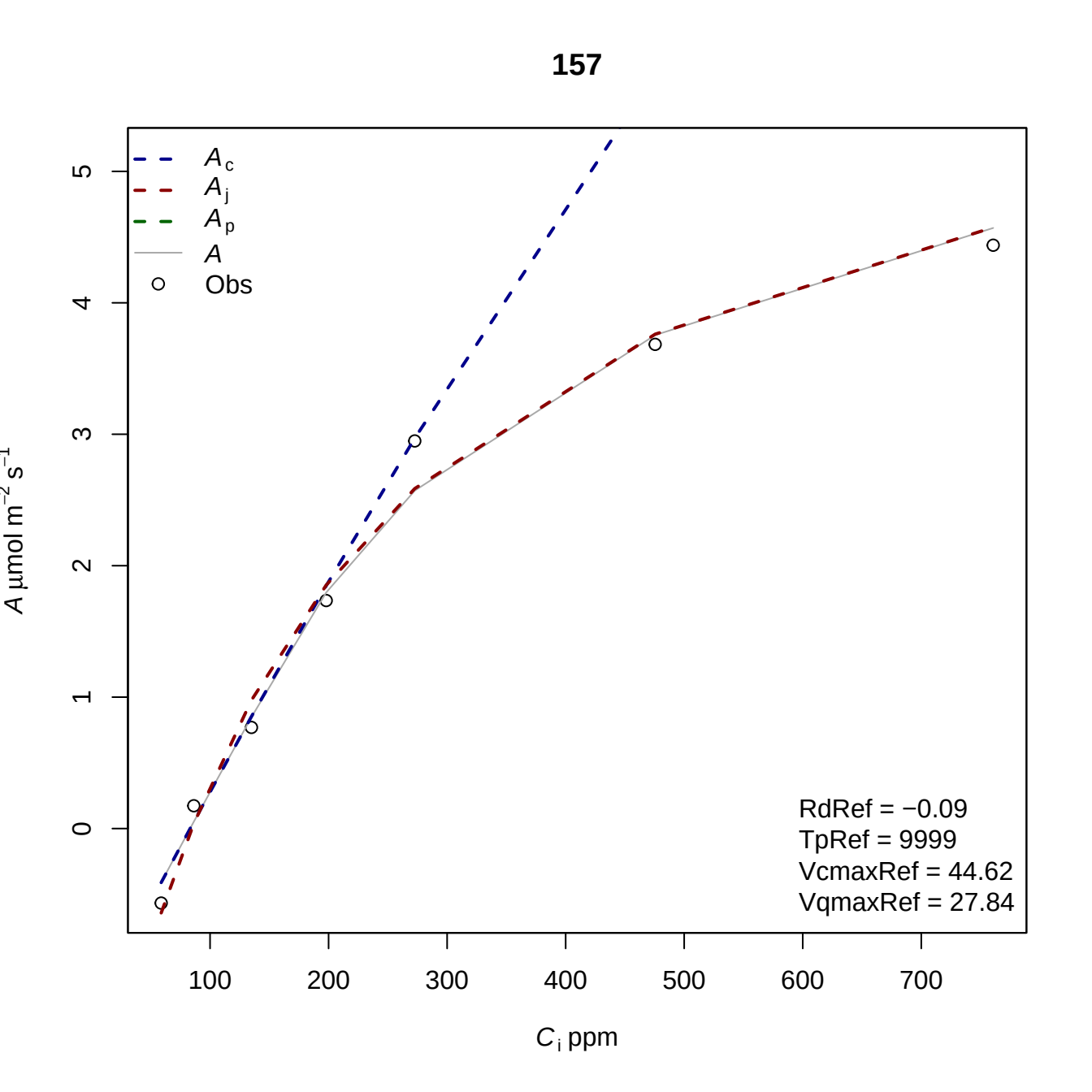

152
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.09
TpRef = 9999
VcmaxRef = 23.74
VqmaxRef = 66.9
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
2
4
6
8
10
12
14
200
400
600
800

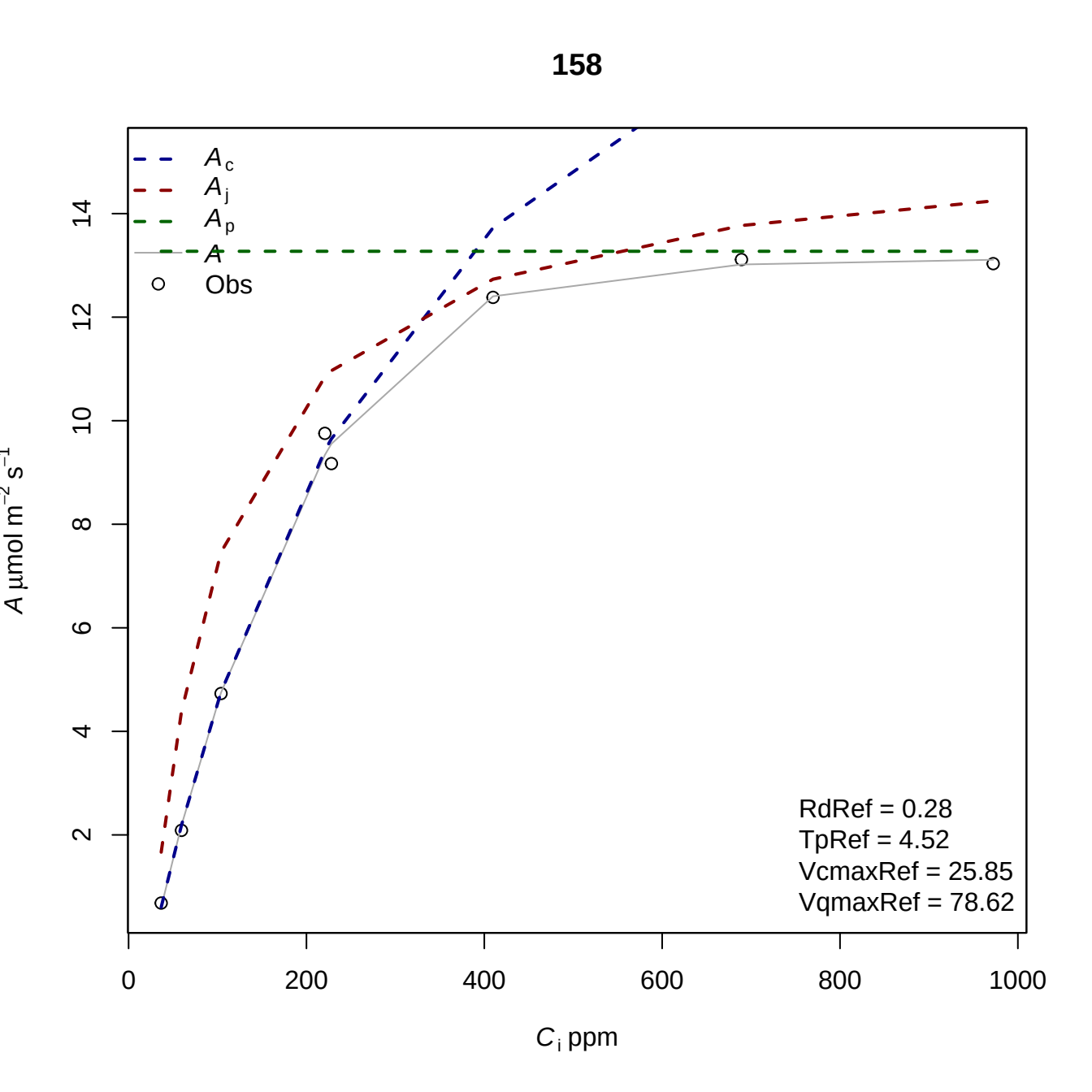

153
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.21
TpRef = 3.77
VcmaxRef = 33.72
VqmaxRef = 60.97
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
2
4
6
8
10
12
200
400
600
800
1000

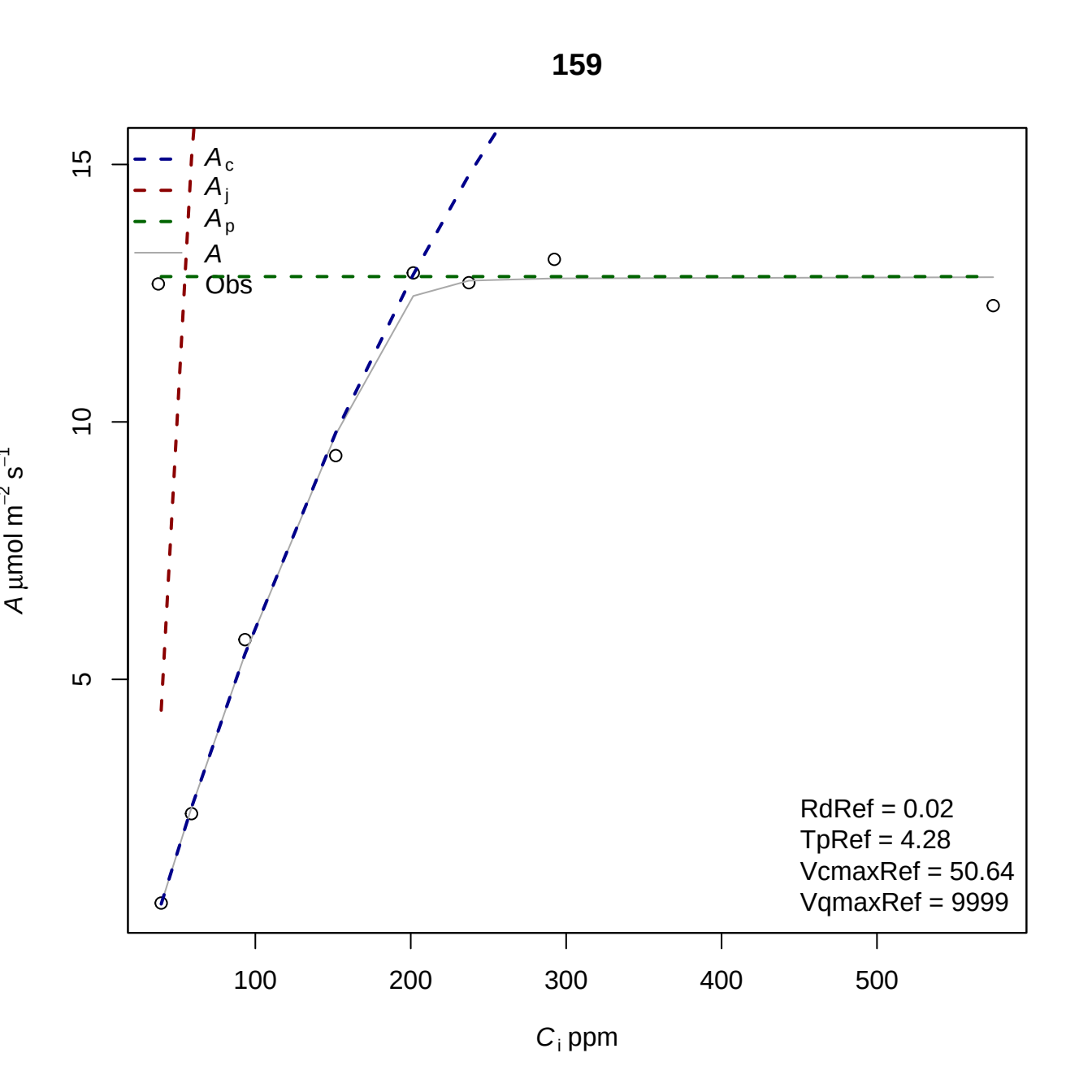

154
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.18
TpRef = 9999
VcmaxRef = 65.71
VqmaxRef = 128.17
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800

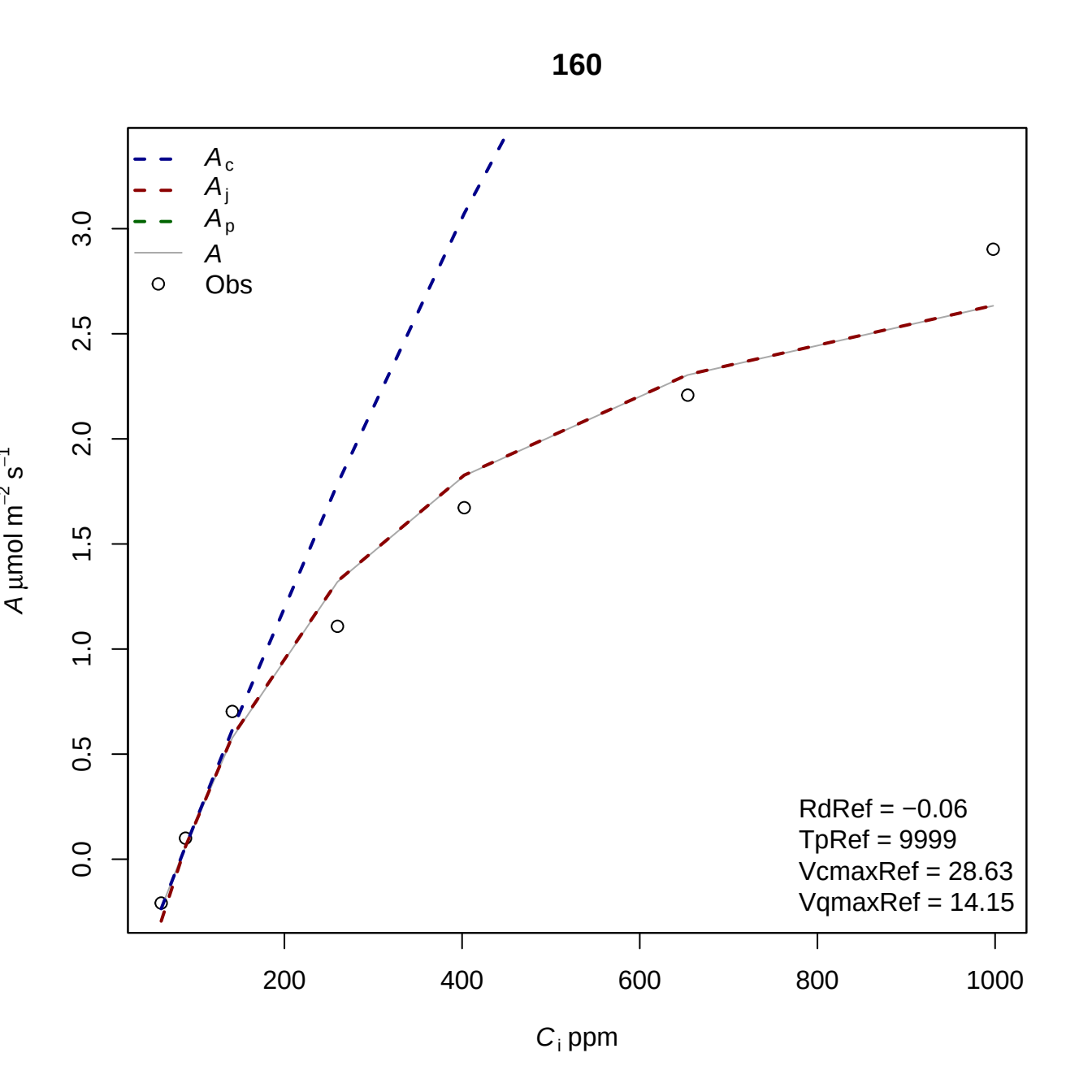

155
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.22
TpRef = 5.85
VcmaxRef = 89.55
VqmaxRef = 140.58
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
100
200
300
400
500

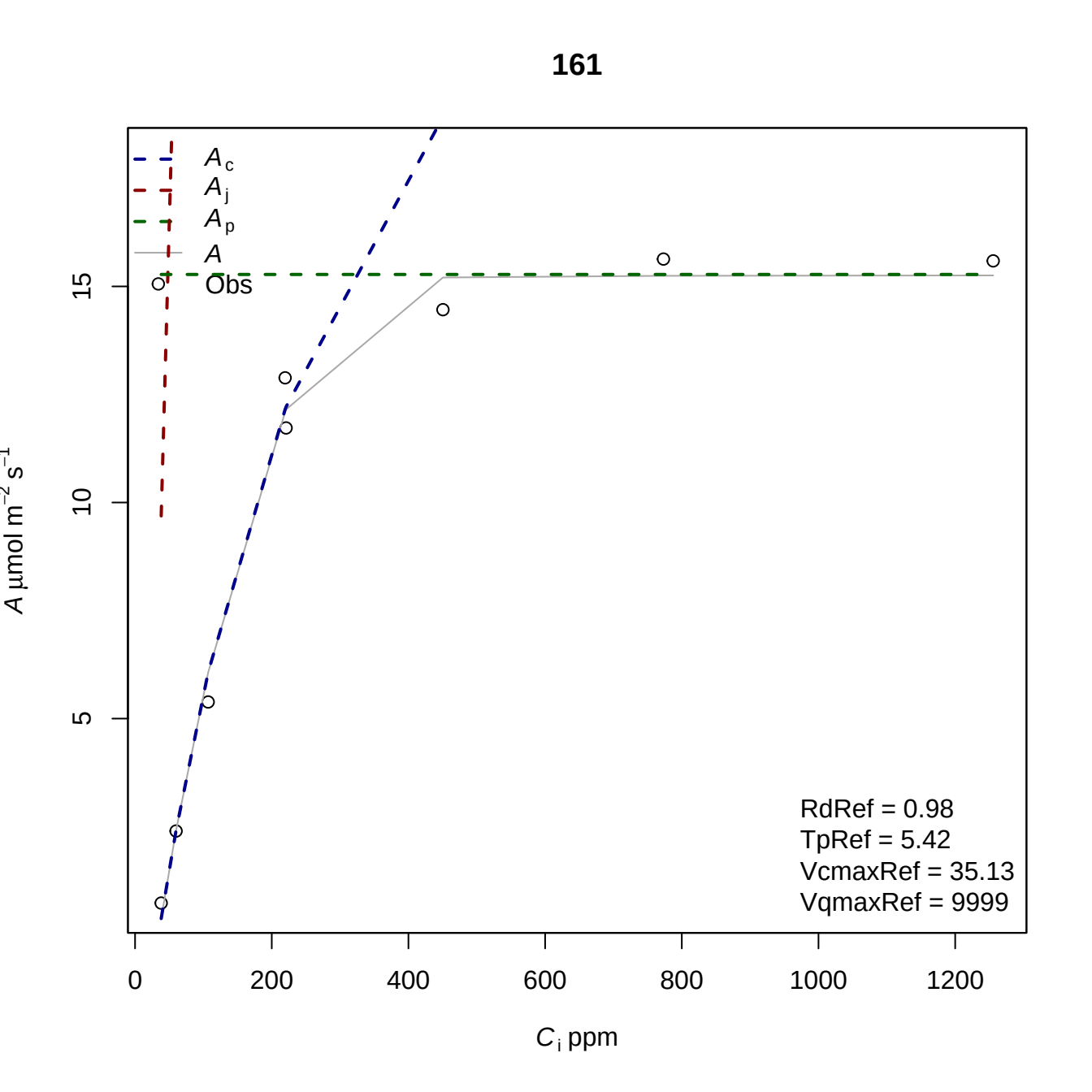

156
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.67
TpRef = 9999
VcmaxRef = 663143.63
VqmaxRef = 41.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
100
200
300
400
500
600
700

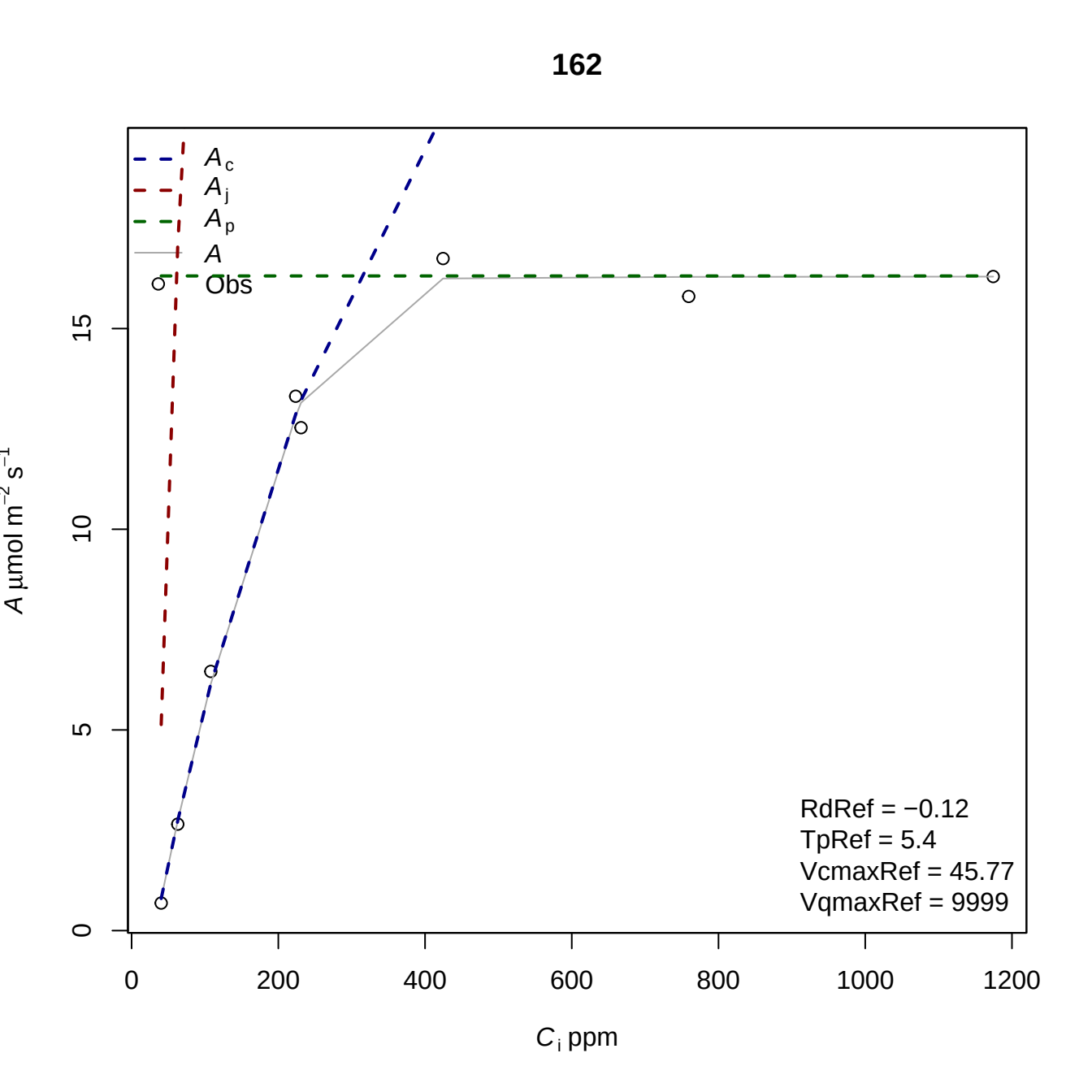

157
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.09
TpRef = 9999
VcmaxRef = 44.62
VqmaxRef = 27.84
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
100
200
300
400
500
600
700

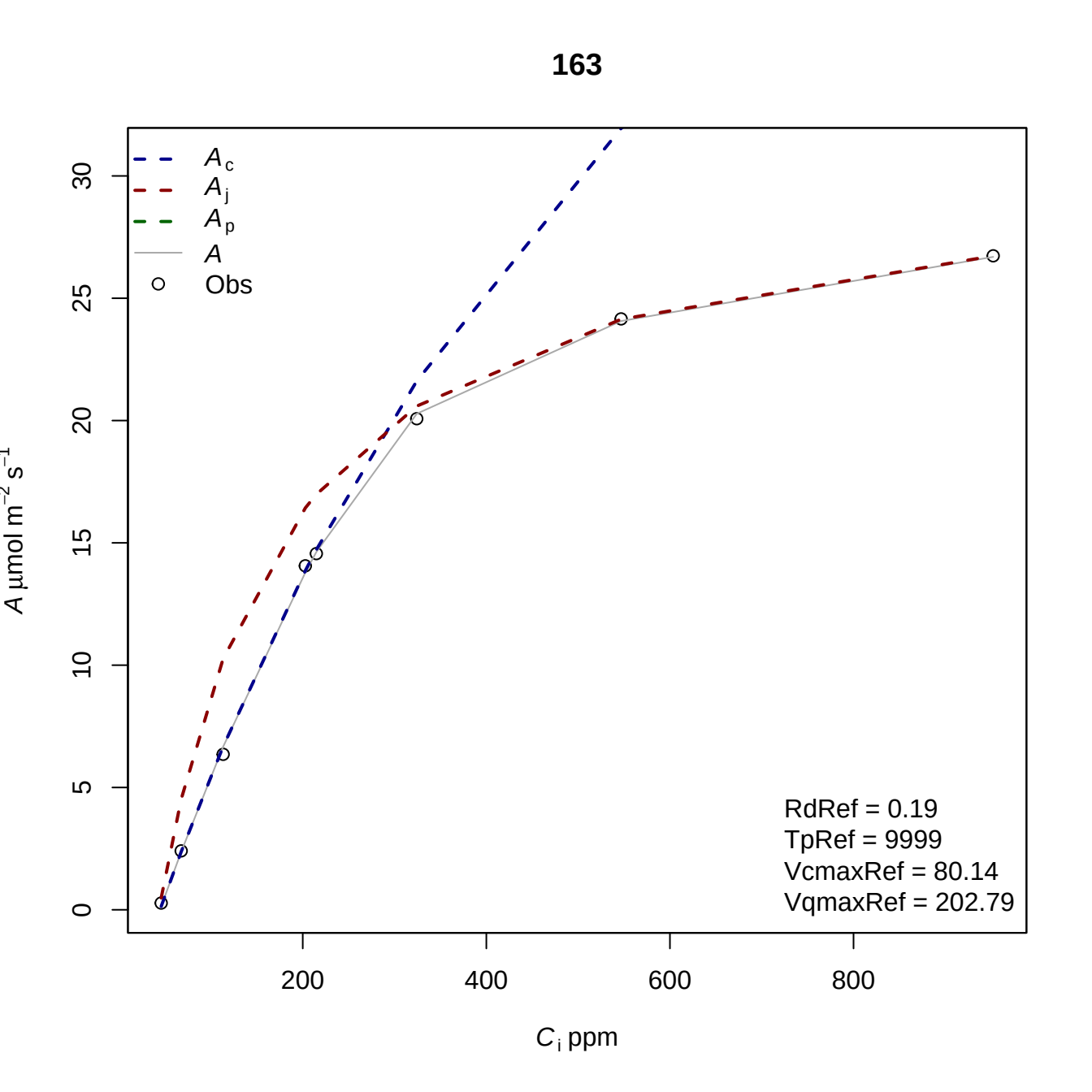

158
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.28
TpRef = 4.52
VcmaxRef = 25.85
VqmaxRef = 78.62
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
2
4
6
8
10
12
14

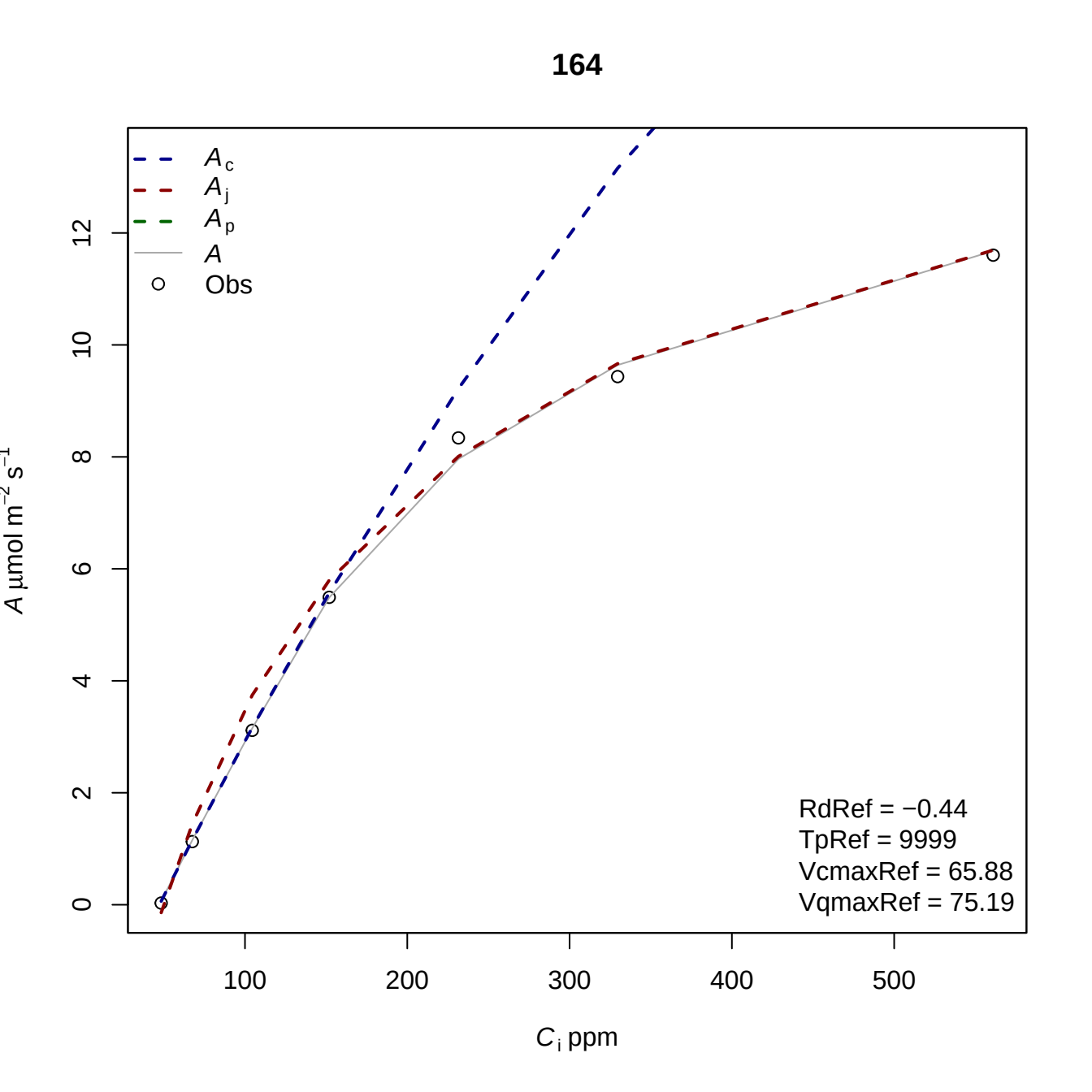

159
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.02
TpRef = 4.28
VcmaxRef = 50.64
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
100
200
300
400
500

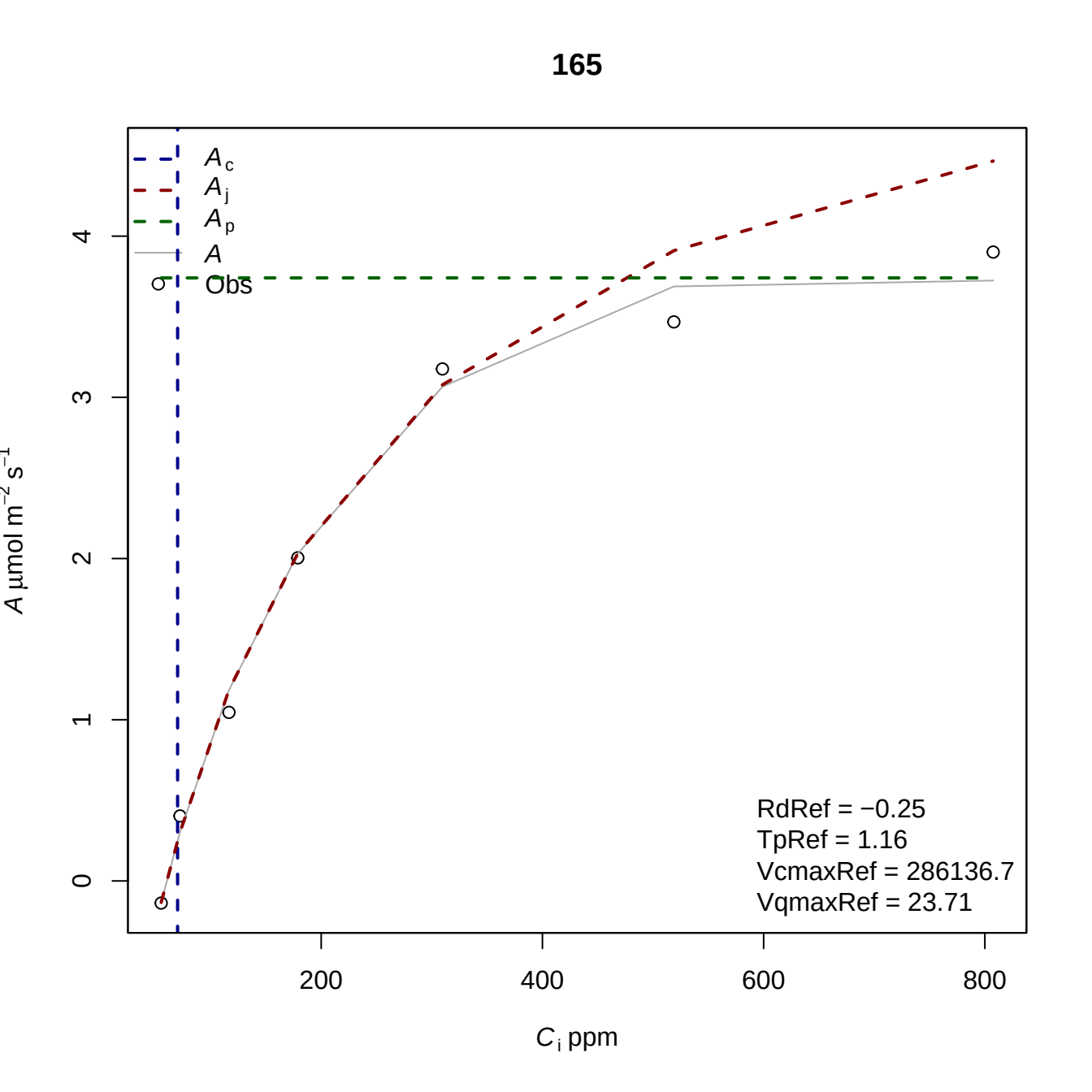

160
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.06
TpRef = 9999
VcmaxRef = 28.63
VqmaxRef = 14.15
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
200
400
600
800
1000

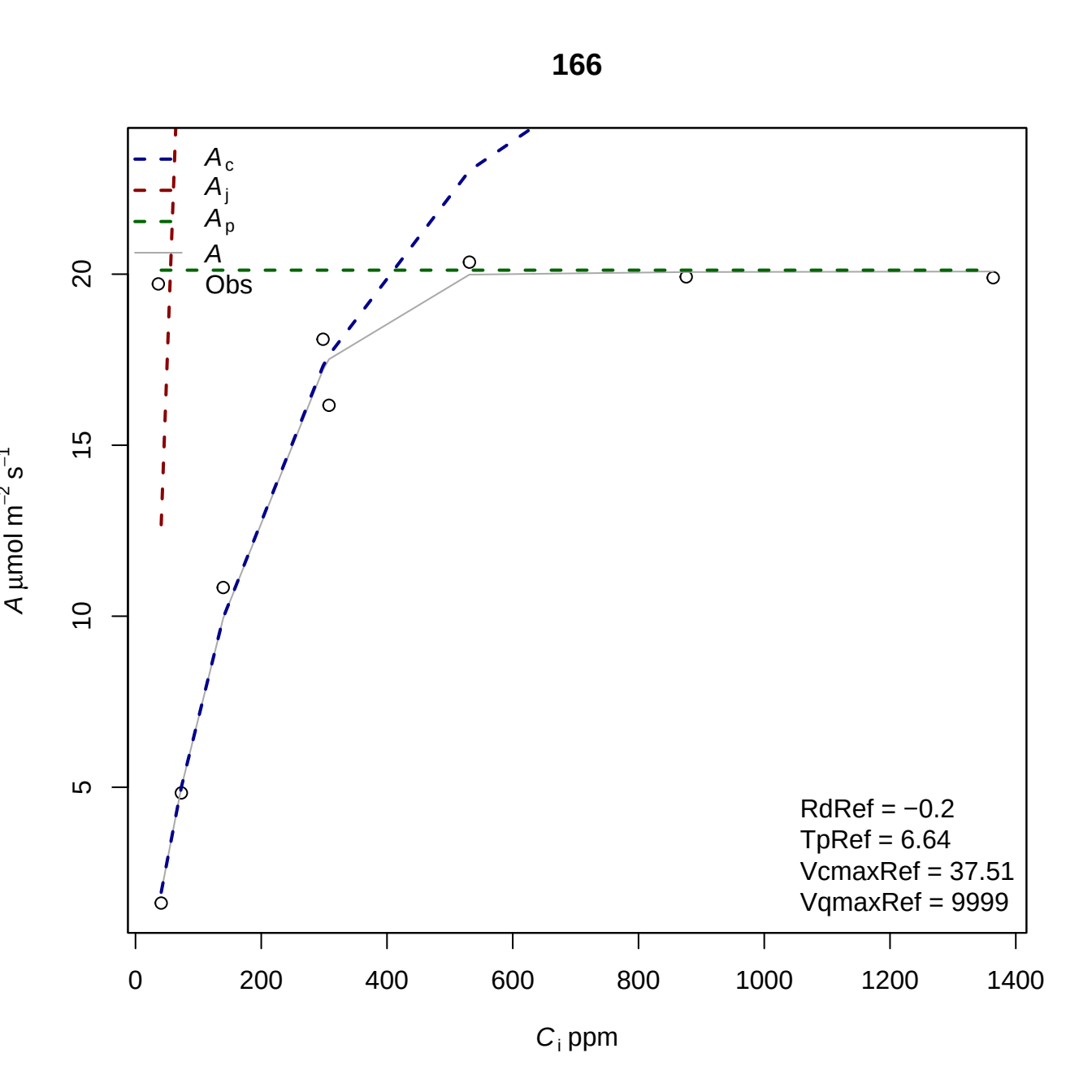

161
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.98
TpRef = 5.42
VcmaxRef = 35.13
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15

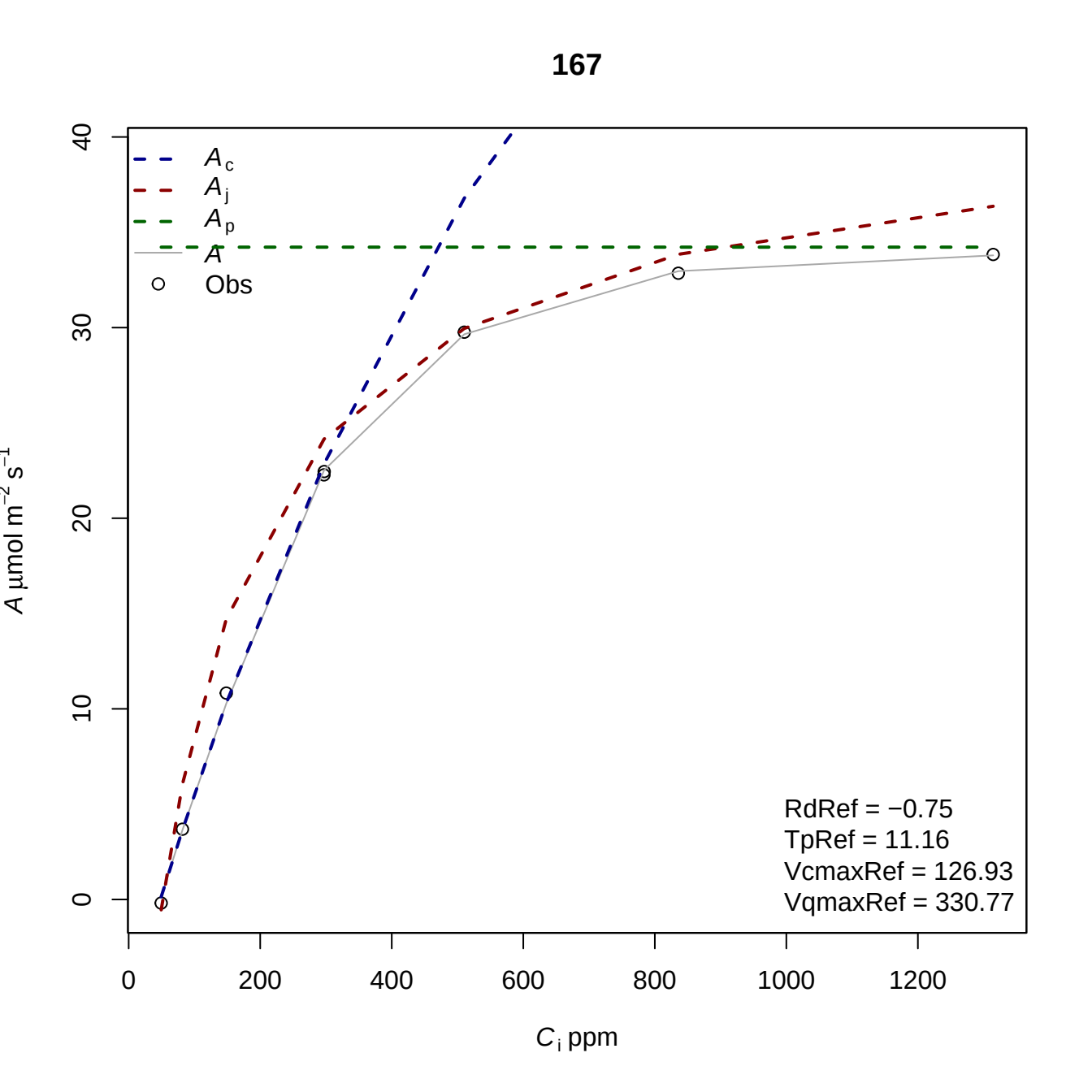

162
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.12
TpRef = 5.4
VcmaxRef = 45.77
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

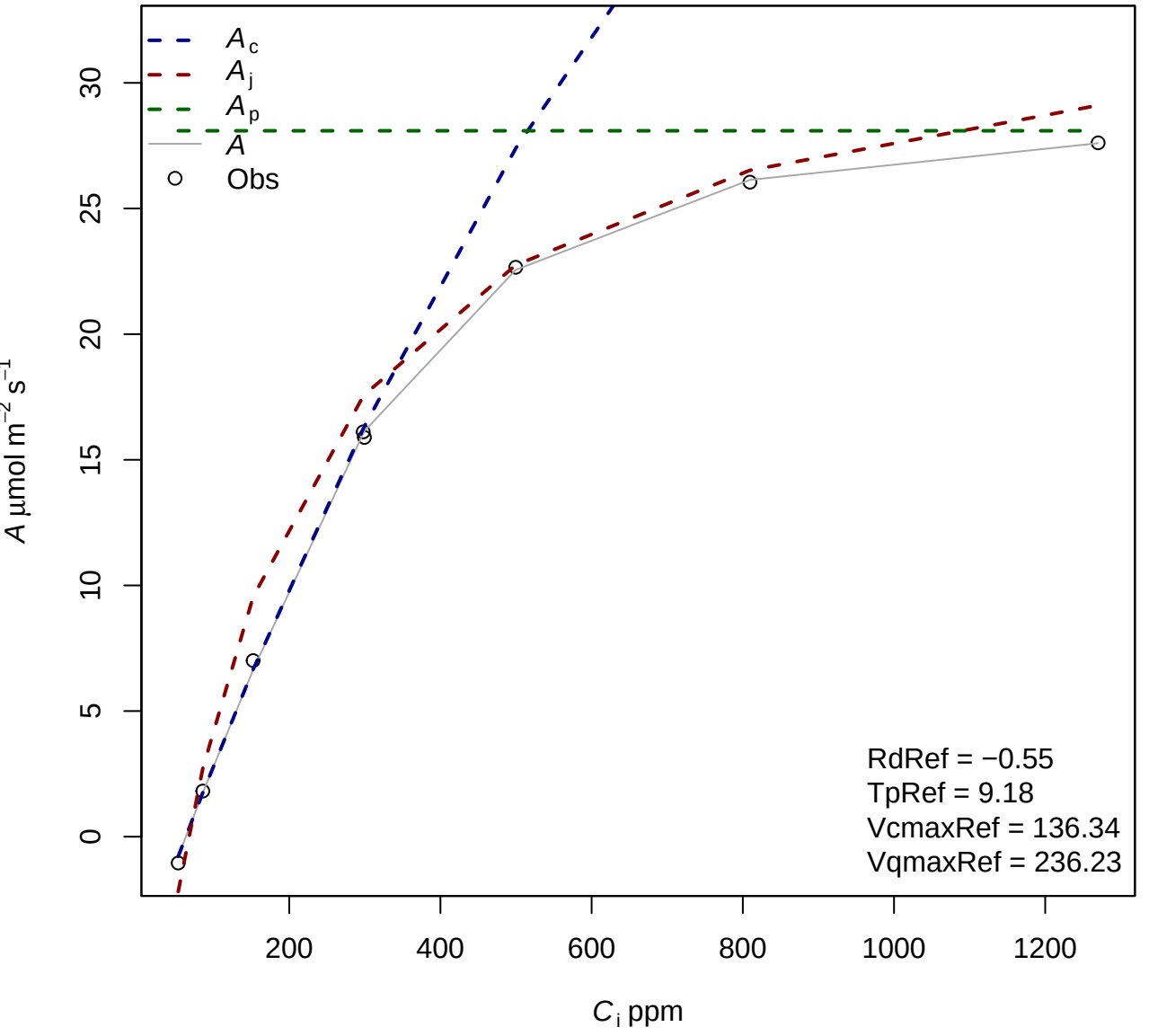

163
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.19
TpRef = 9999
VcmaxRef = 80.14
VqmaxRef = 202.79
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800

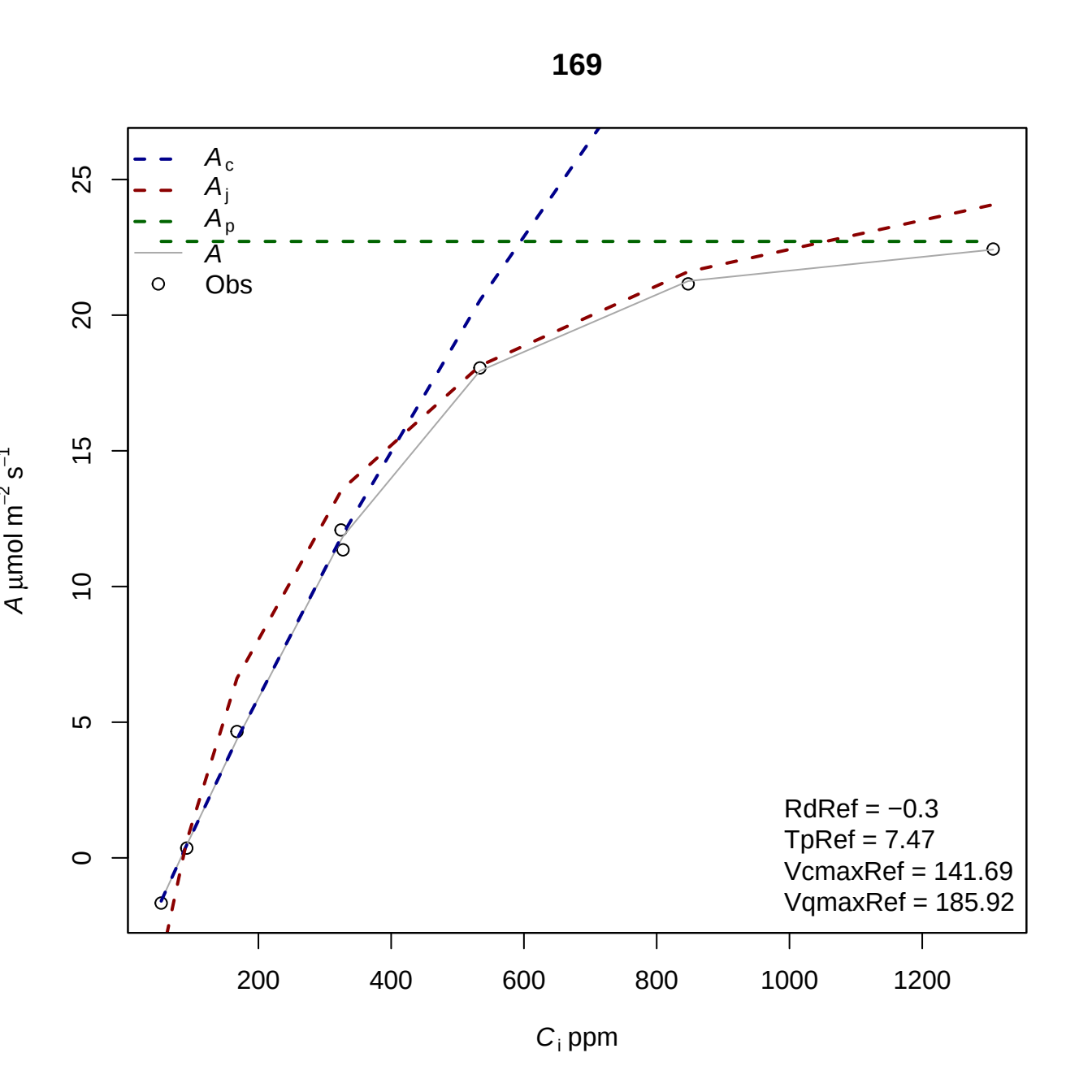

164
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.44
TpRef = 9999
VcmaxRef = 65.88
VqmaxRef = 75.19
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
100
200
300
400
500

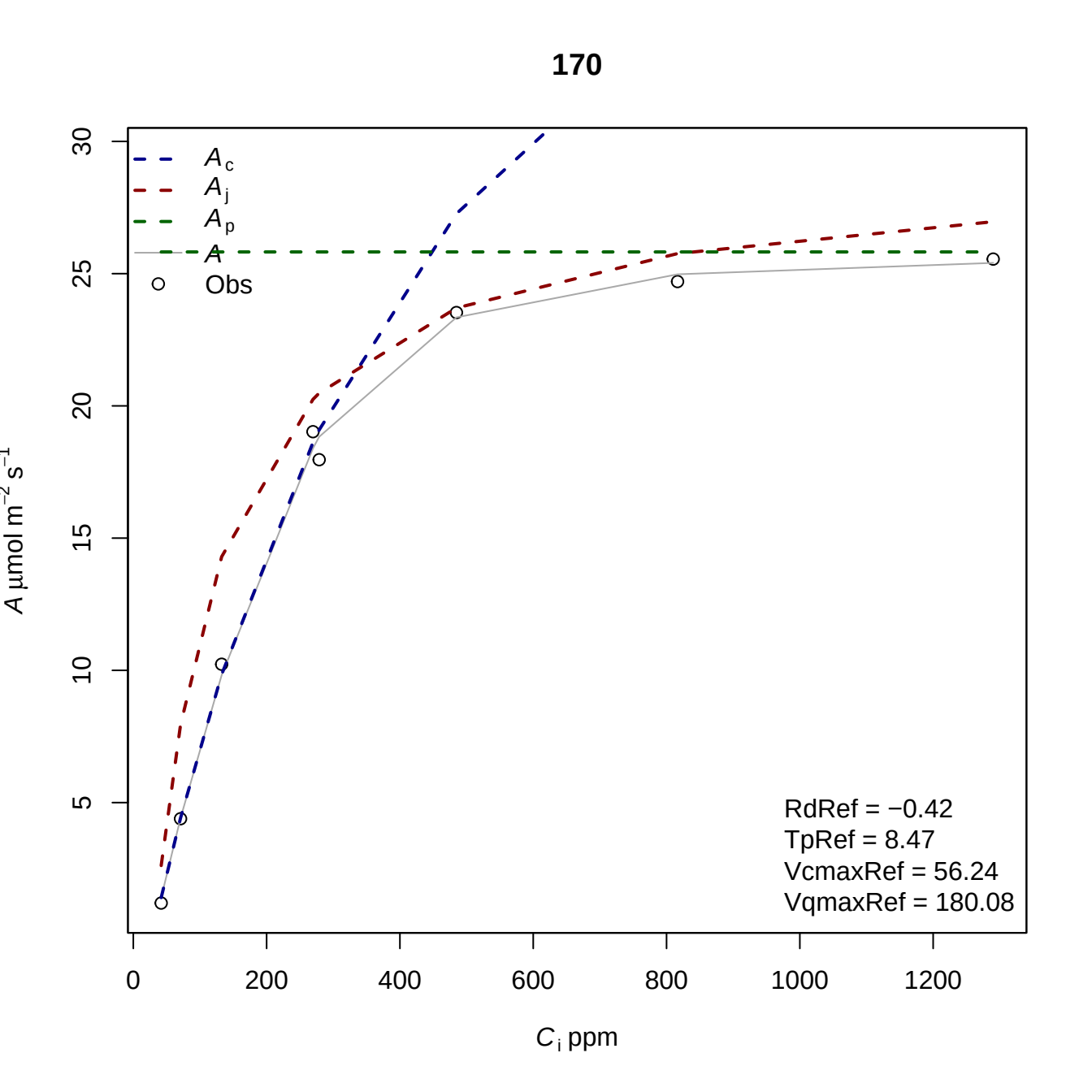

165
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.25
TpRef = 1.16
VcmaxRef = 286136.7
VqmaxRef = 23.71
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
200
400
600
800

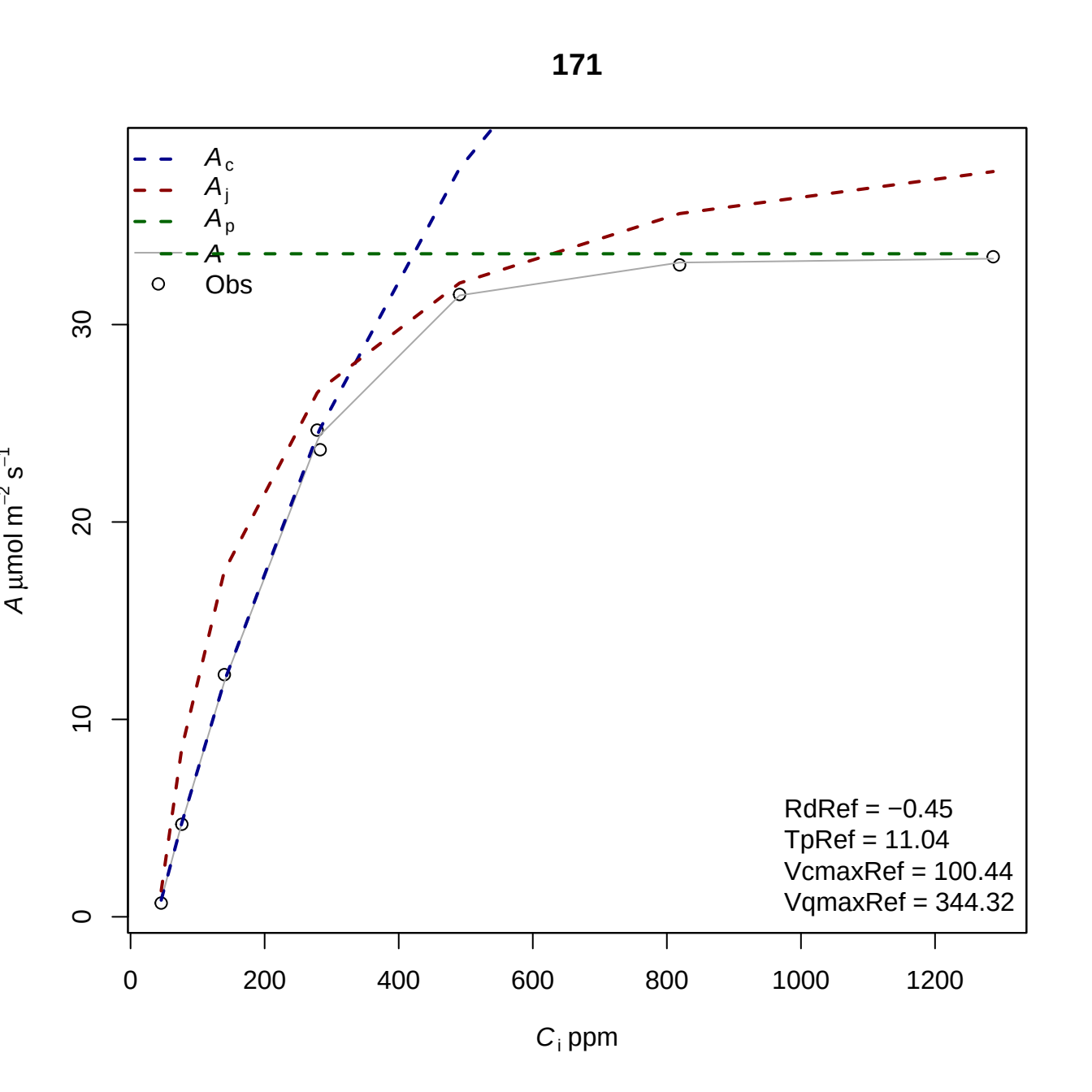

166
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.2
TpRef = 6.64
VcmaxRef = 37.51
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
1400
5
10
15
20

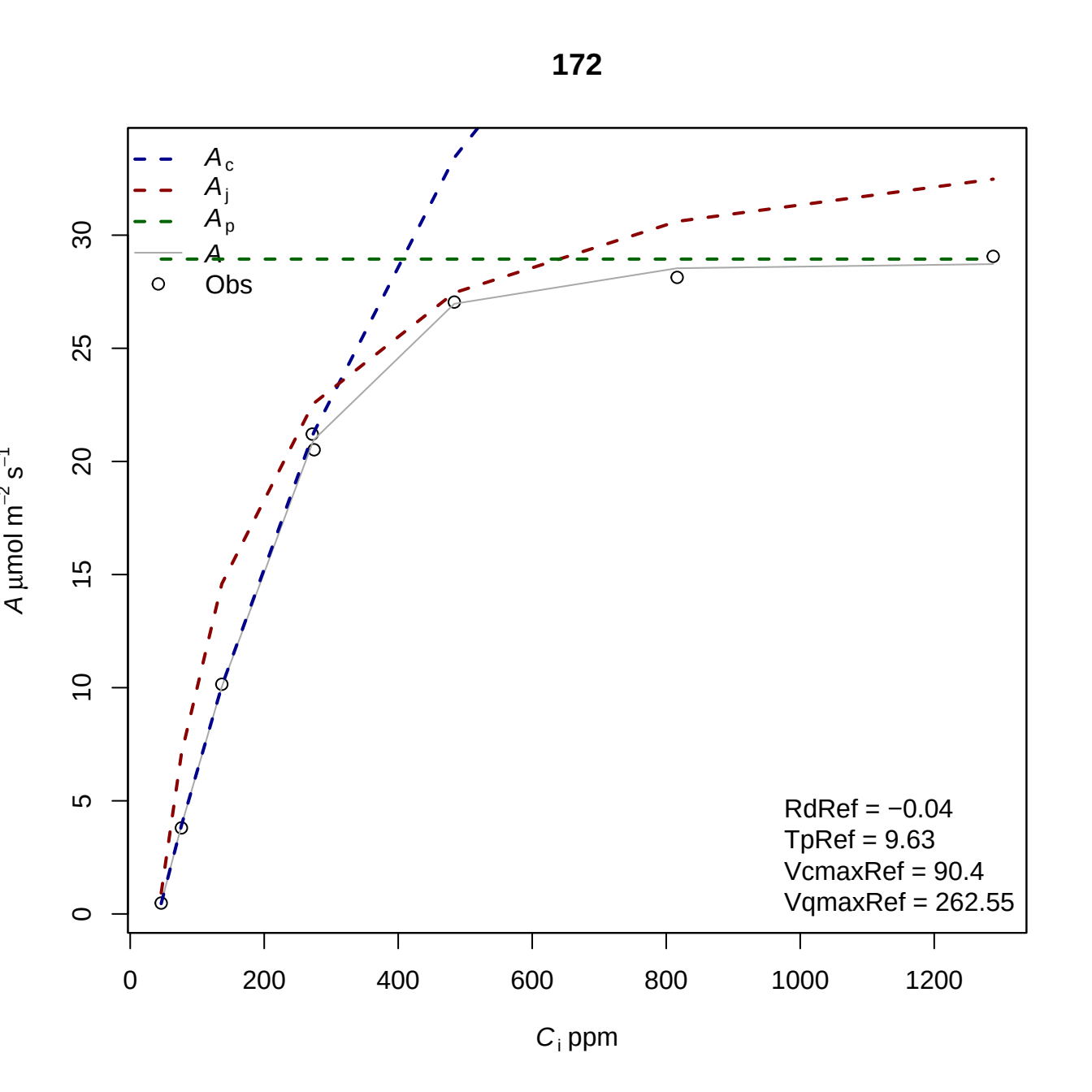

167
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.75
TpRef = 11.16
VcmaxRef = 126.93
VqmaxRef = 330.77
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

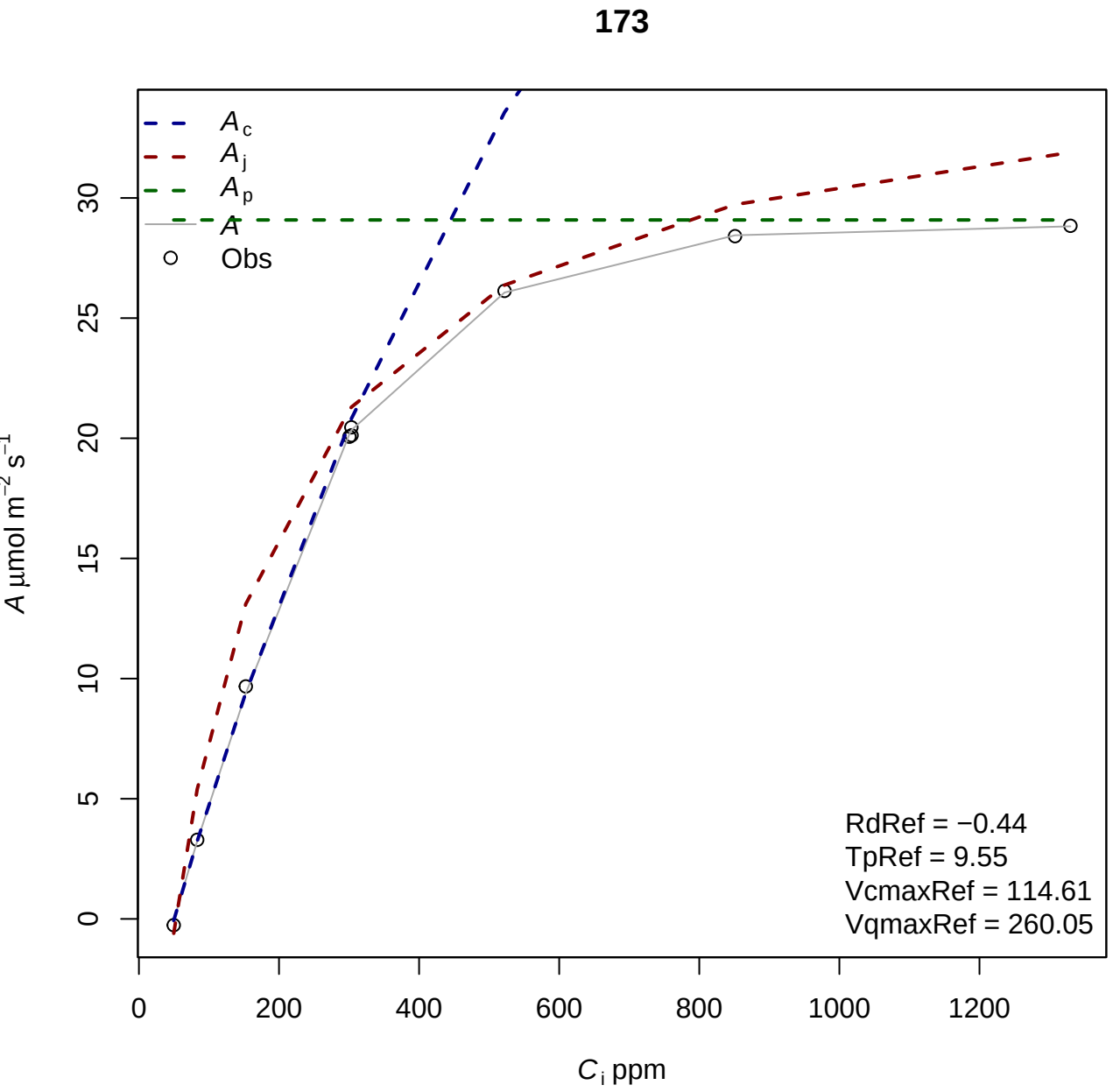

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.55
TpRef = 9.18
VcmaxRef = 136.34
VqmaxRef = 236.23
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

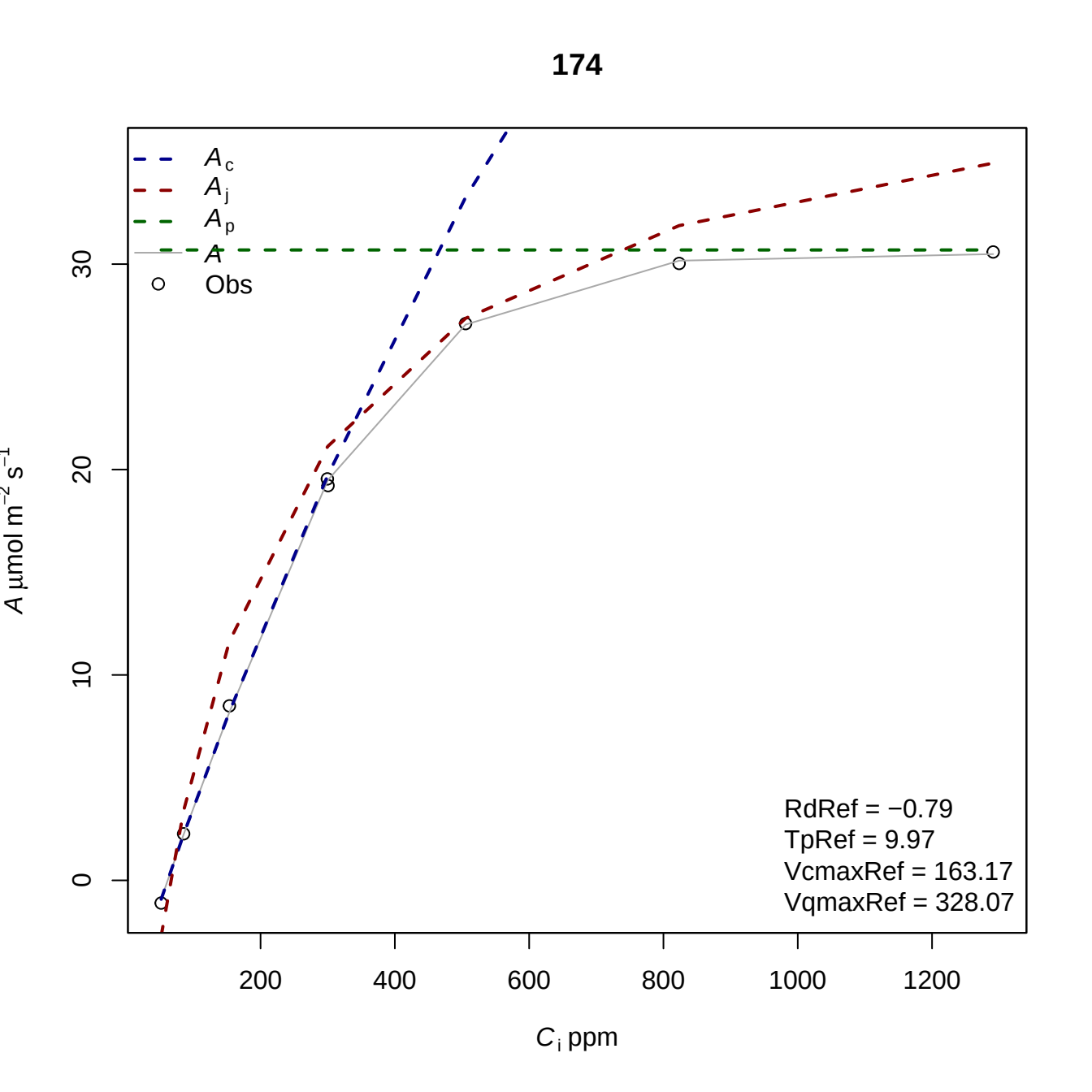

169
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.3
TpRef = 7.47
VcmaxRef = 141.69
VqmaxRef = 185.92
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

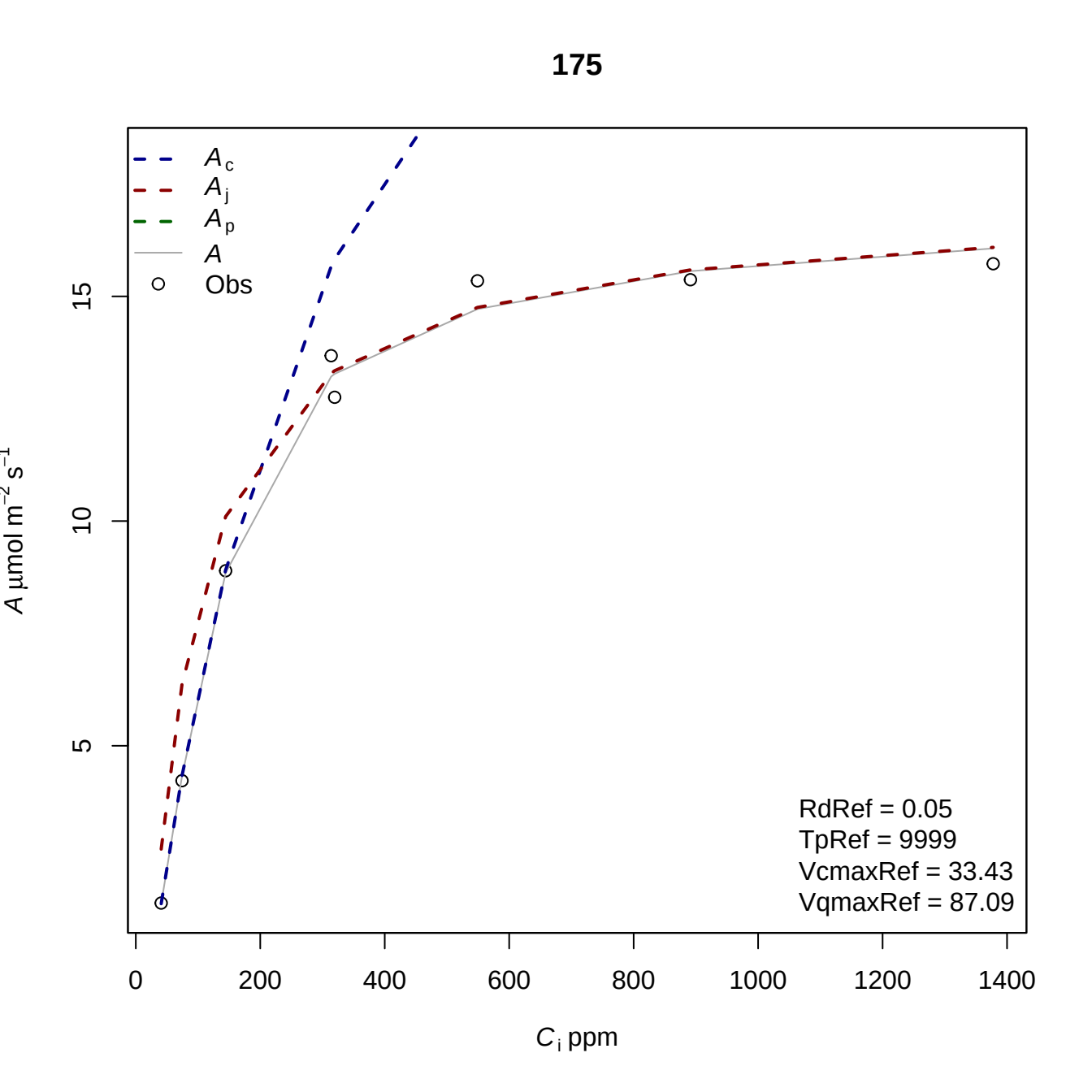

170
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.42
TpRef = 8.47
VcmaxRef = 56.24
VqmaxRef = 180.08
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25
30

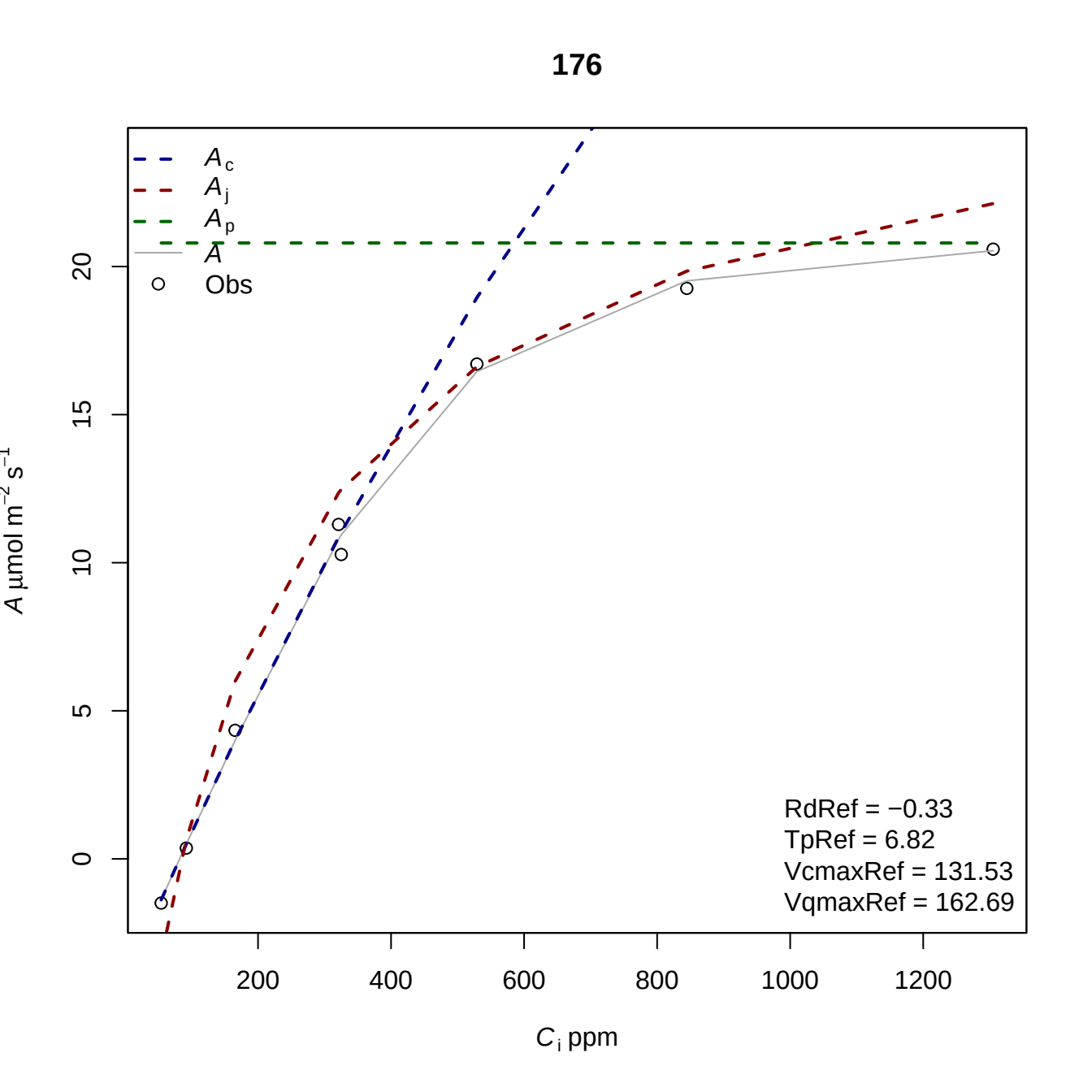

171
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.45
TpRef = 11.04
VcmaxRef = 100.44
VqmaxRef = 344.32
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
0
200
400
600
800
1000
1200

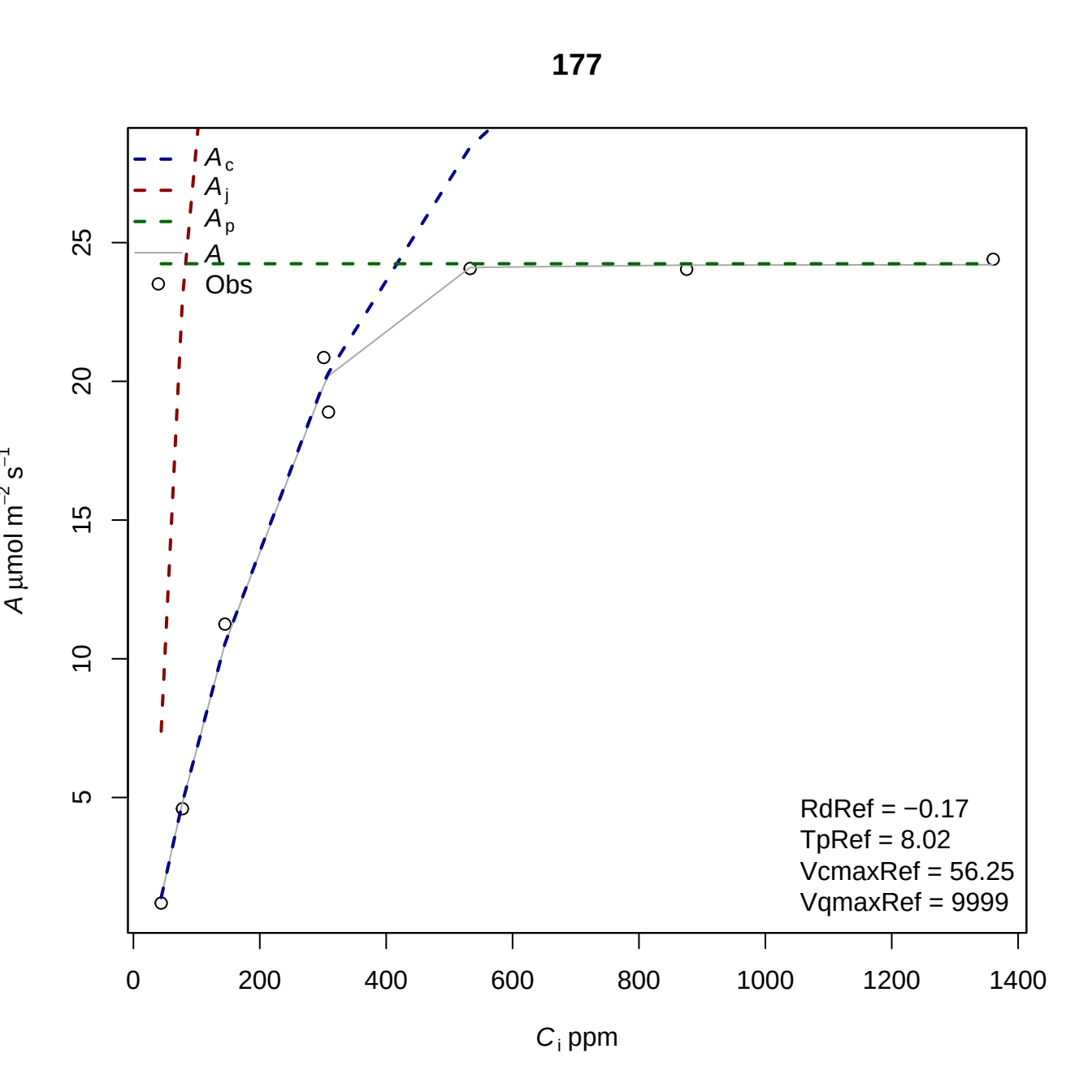

172
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.04
TpRef = 9.63
VcmaxRef = 90.4
VqmaxRef = 262.55
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

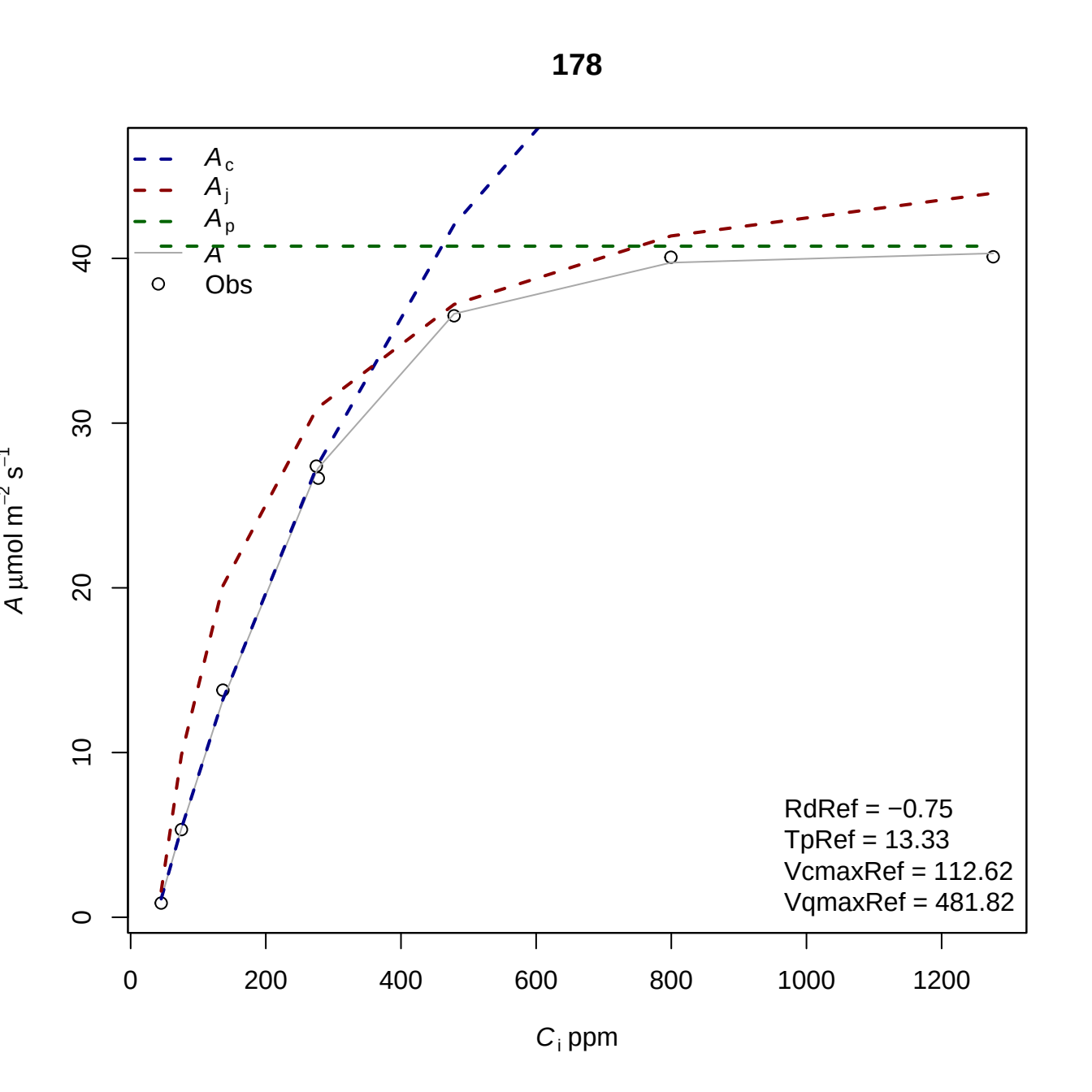

173
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.44
TpRef = 9.55
VcmaxRef = 114.61
VqmaxRef = 260.05
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

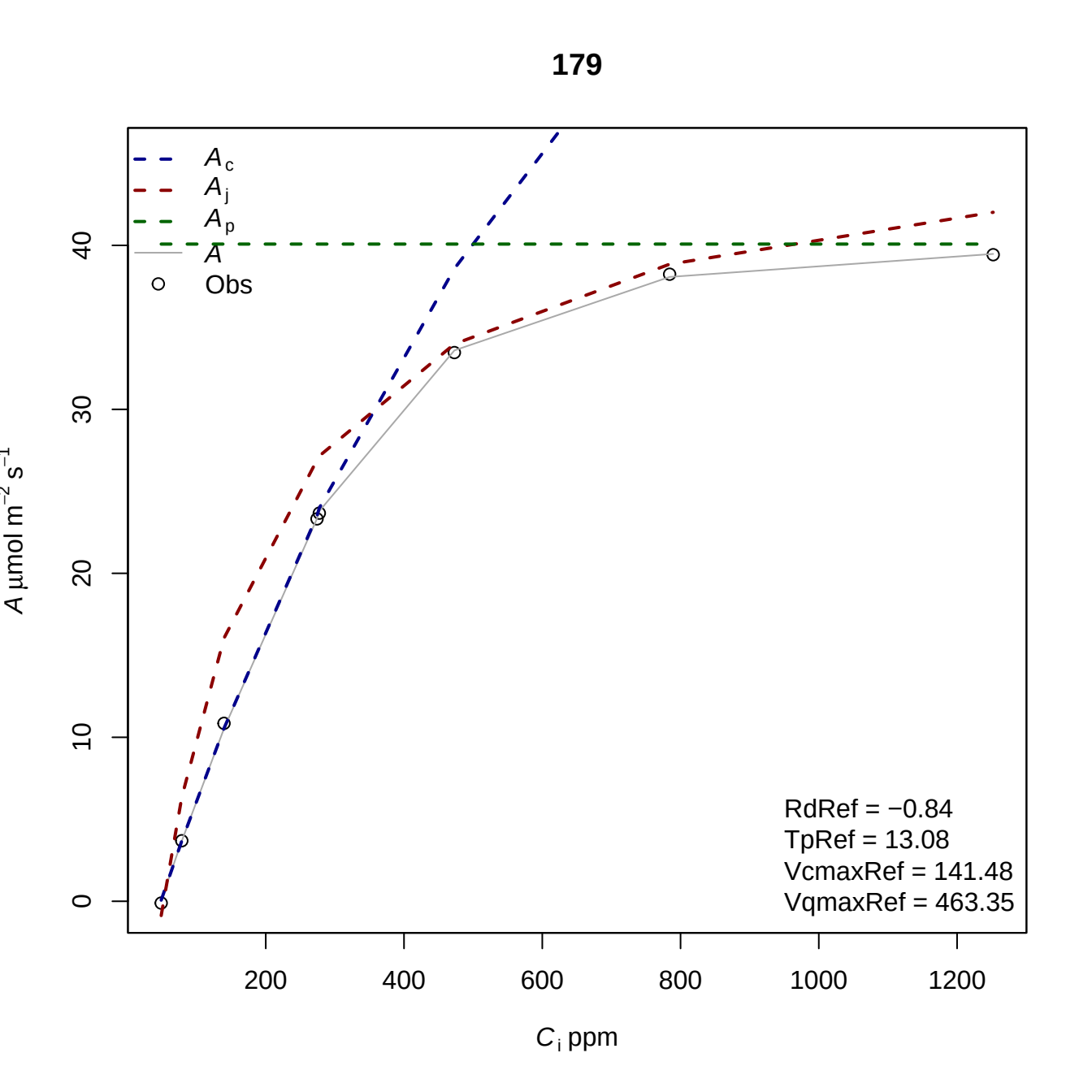

174
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.79
TpRef = 9.97
VcmaxRef = 163.17
VqmaxRef = 328.07
0
10
20
30
200
400
600
800
1000
1200
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

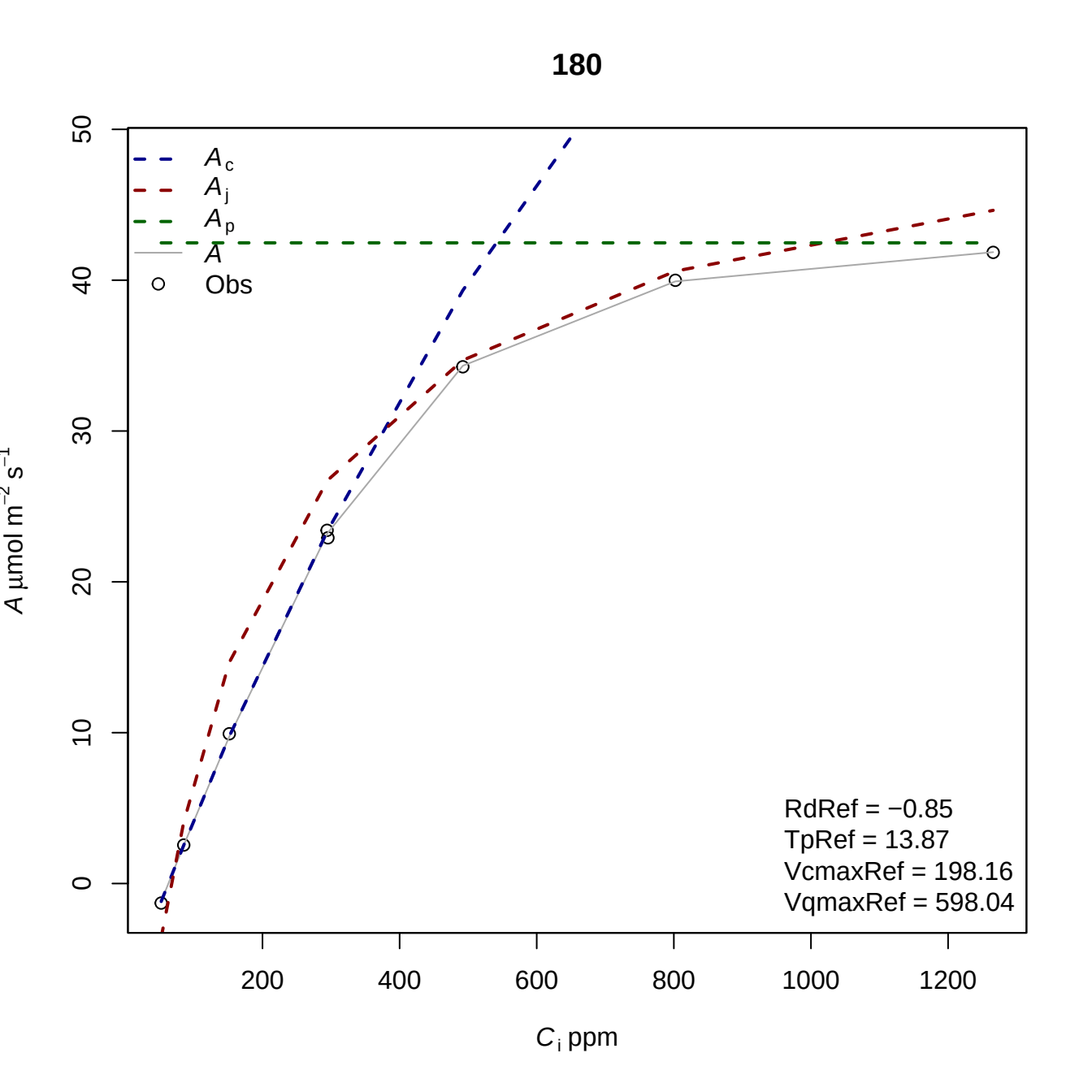

175
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.05
TpRef = 9999
VcmaxRef = 33.43
VqmaxRef = 87.09
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
1400
5
10
15

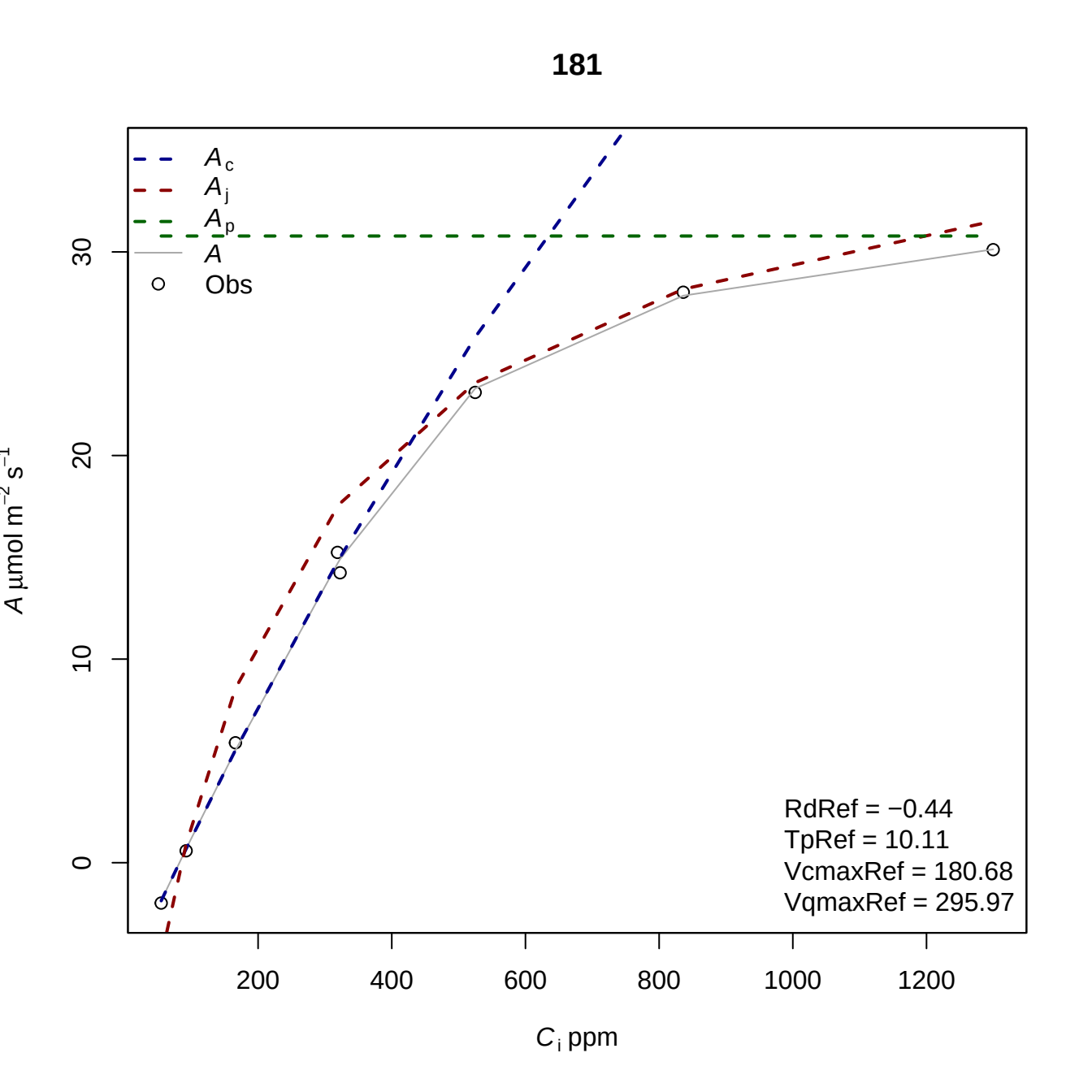

176
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.33
TpRef = 6.82
VcmaxRef = 131.53
VqmaxRef = 162.69
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200

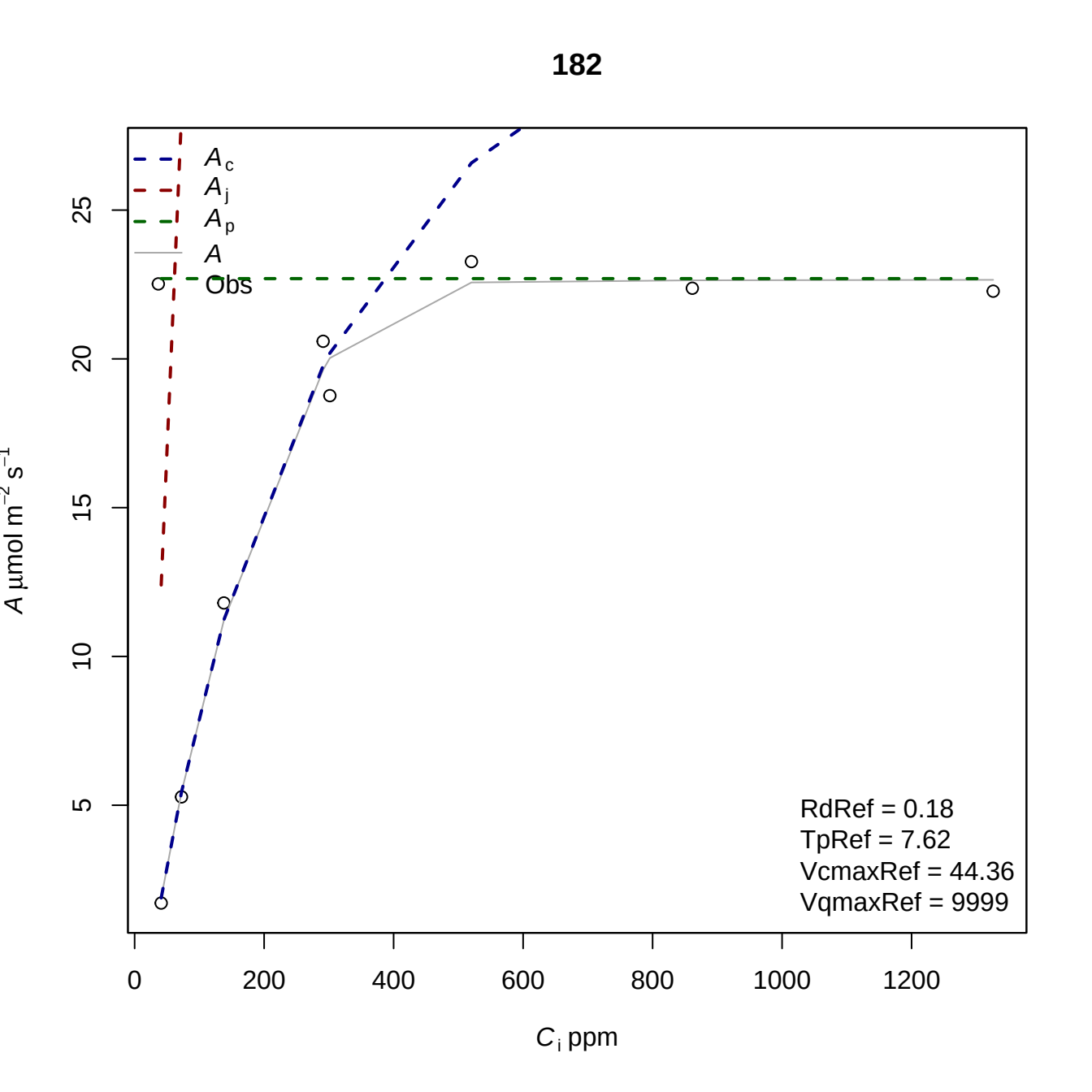

177
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.17
TpRef = 8.02
VcmaxRef = 56.25
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
1400
5
10
15
20
25

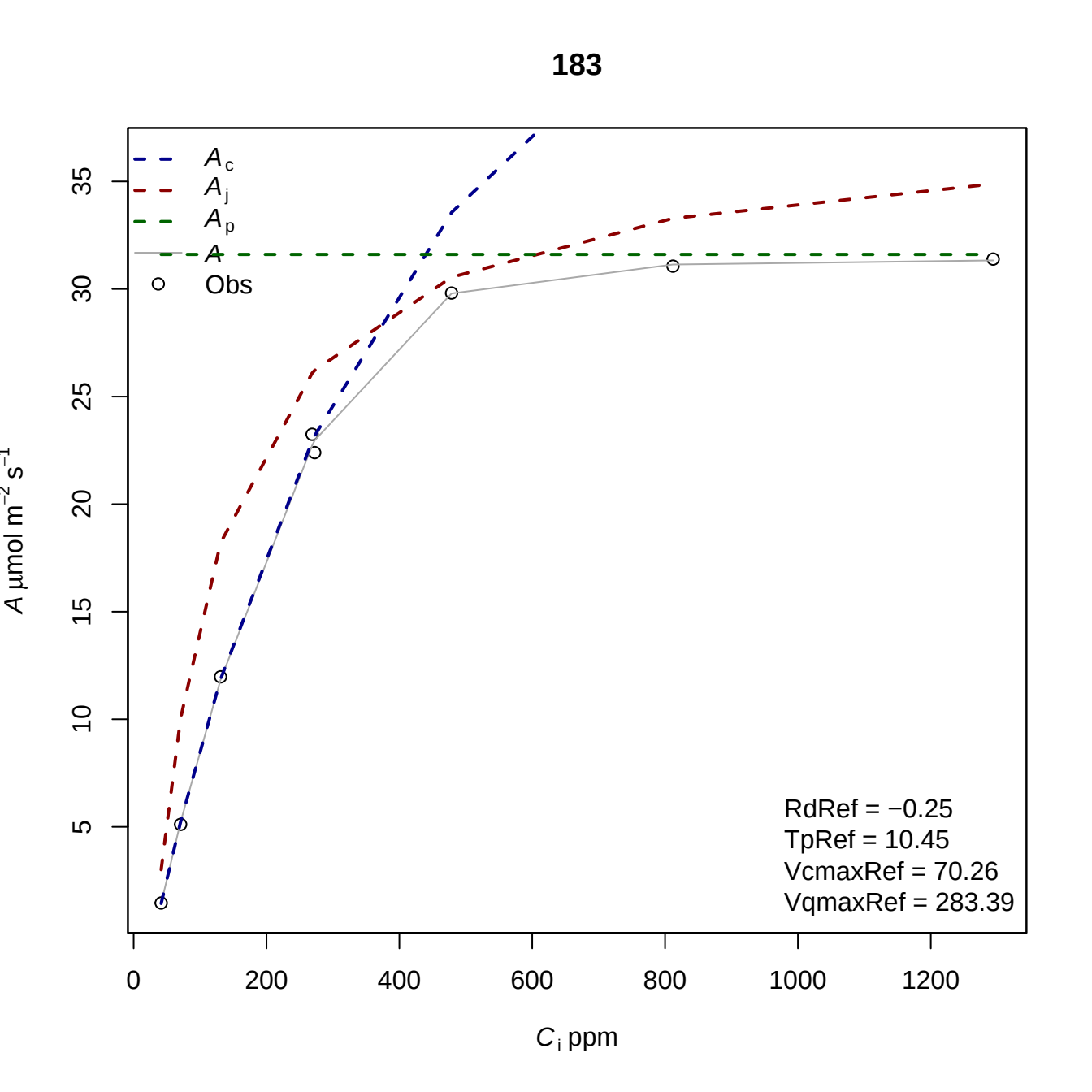

178
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.75
TpRef = 13.33
VcmaxRef = 112.62
VqmaxRef = 481.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

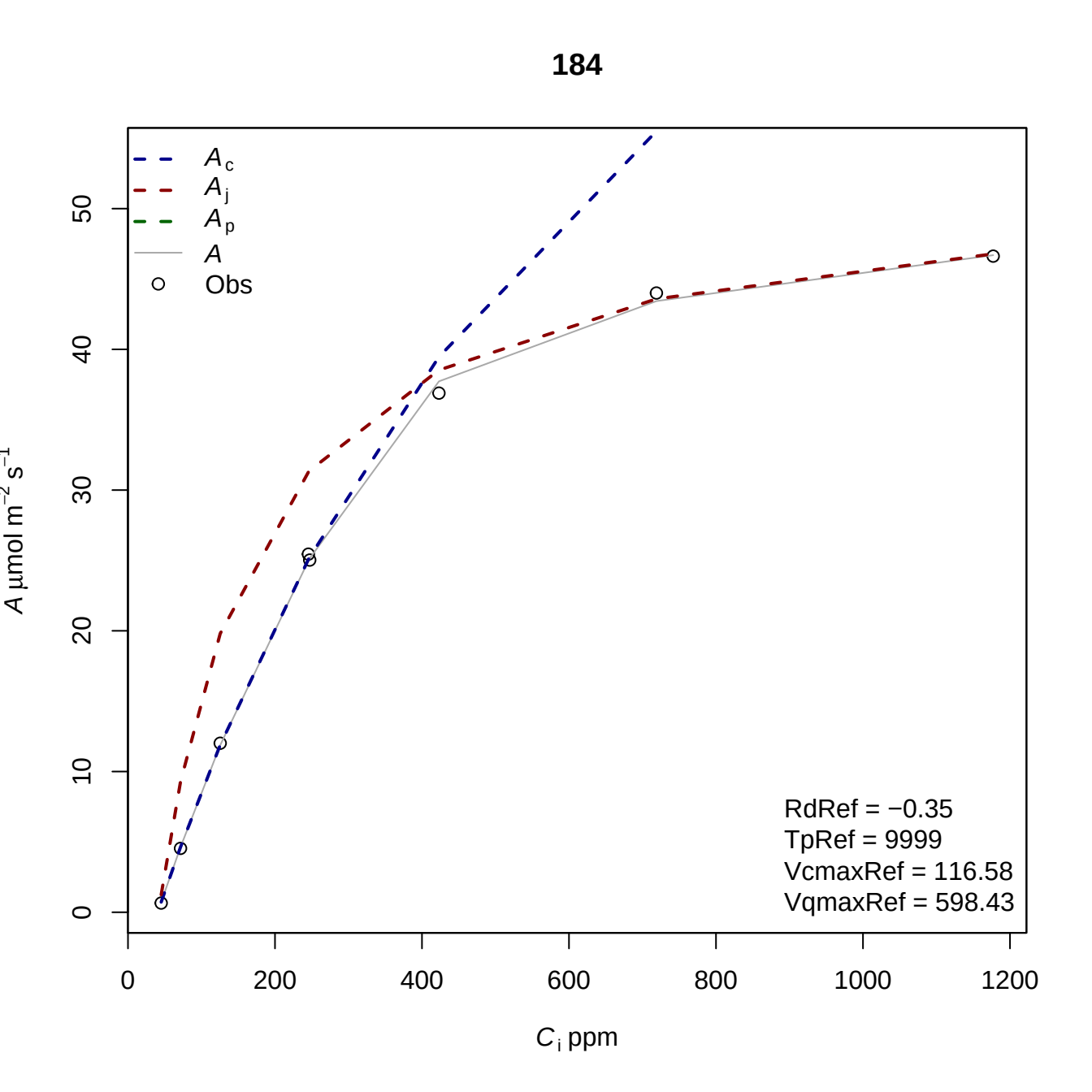

179
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.84
TpRef = 13.08
VcmaxRef = 141.48
VqmaxRef = 463.35
0
10
20
30
40
200
400
600
800
1000
1200
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

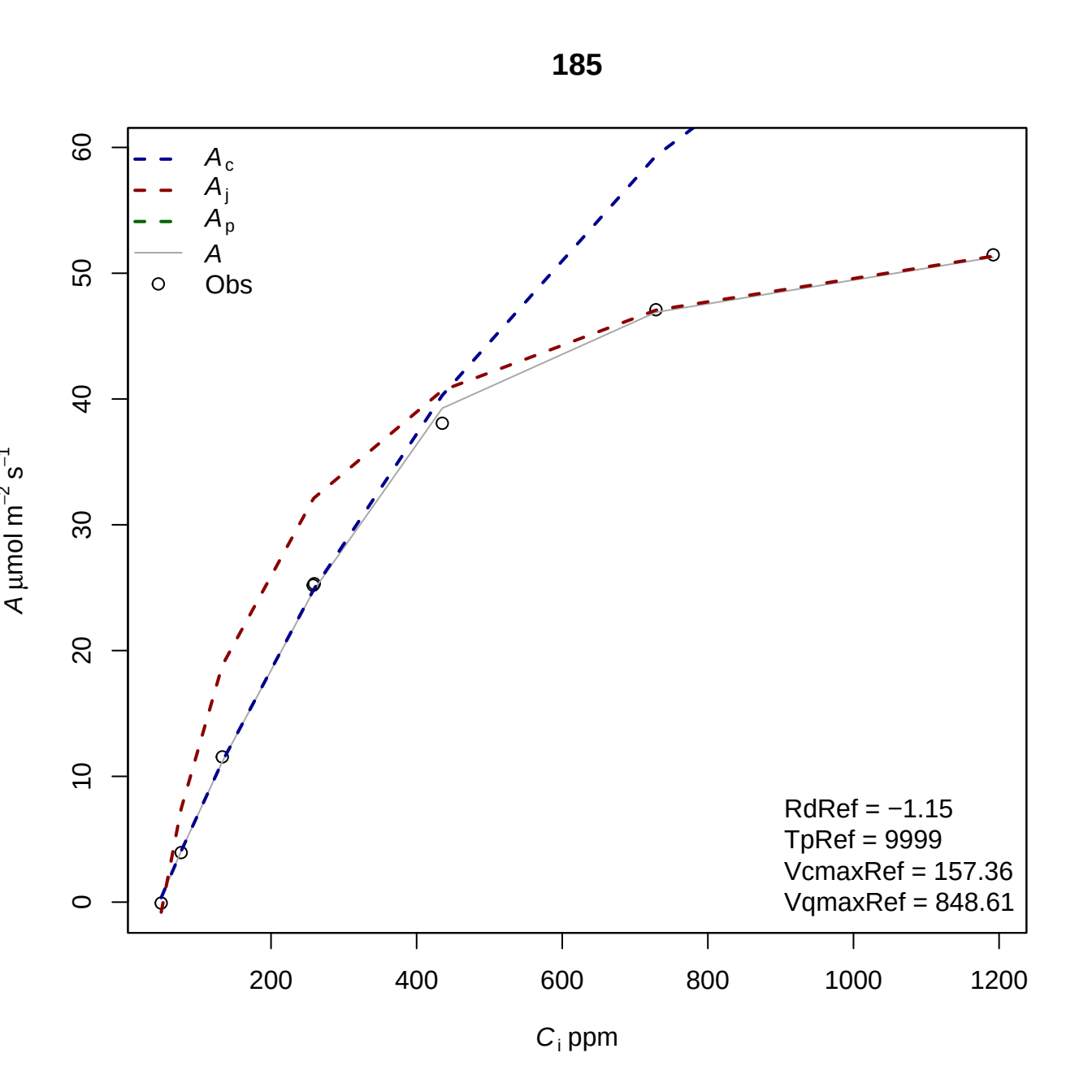

180
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.85
TpRef = 13.87
VcmaxRef = 198.16
VqmaxRef = 598.04
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

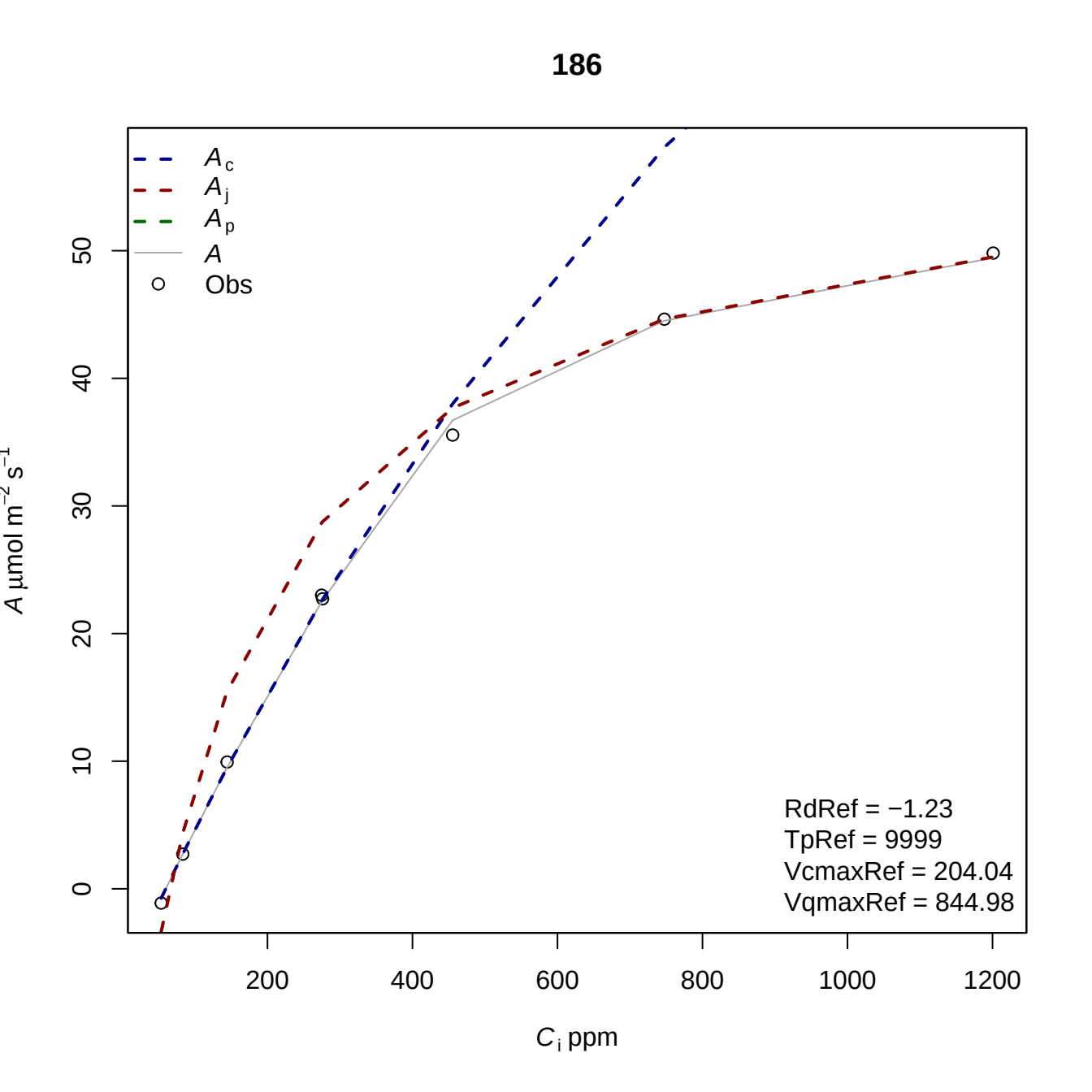

181
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.44
TpRef = 10.11
VcmaxRef = 180.68
VqmaxRef = 295.97
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
200
400
600
800
1000
1200

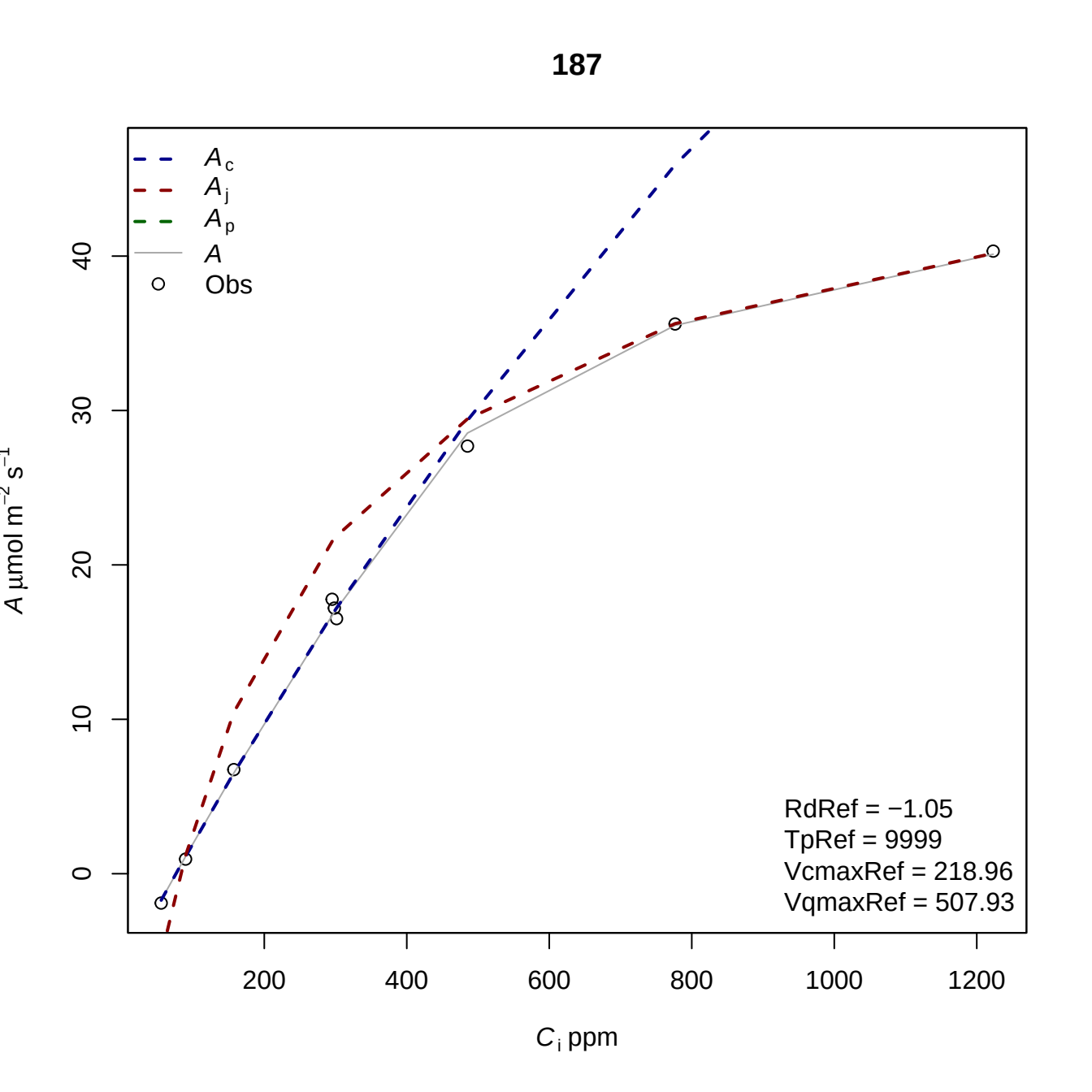

182
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.18
TpRef = 7.62
VcmaxRef = 44.36
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

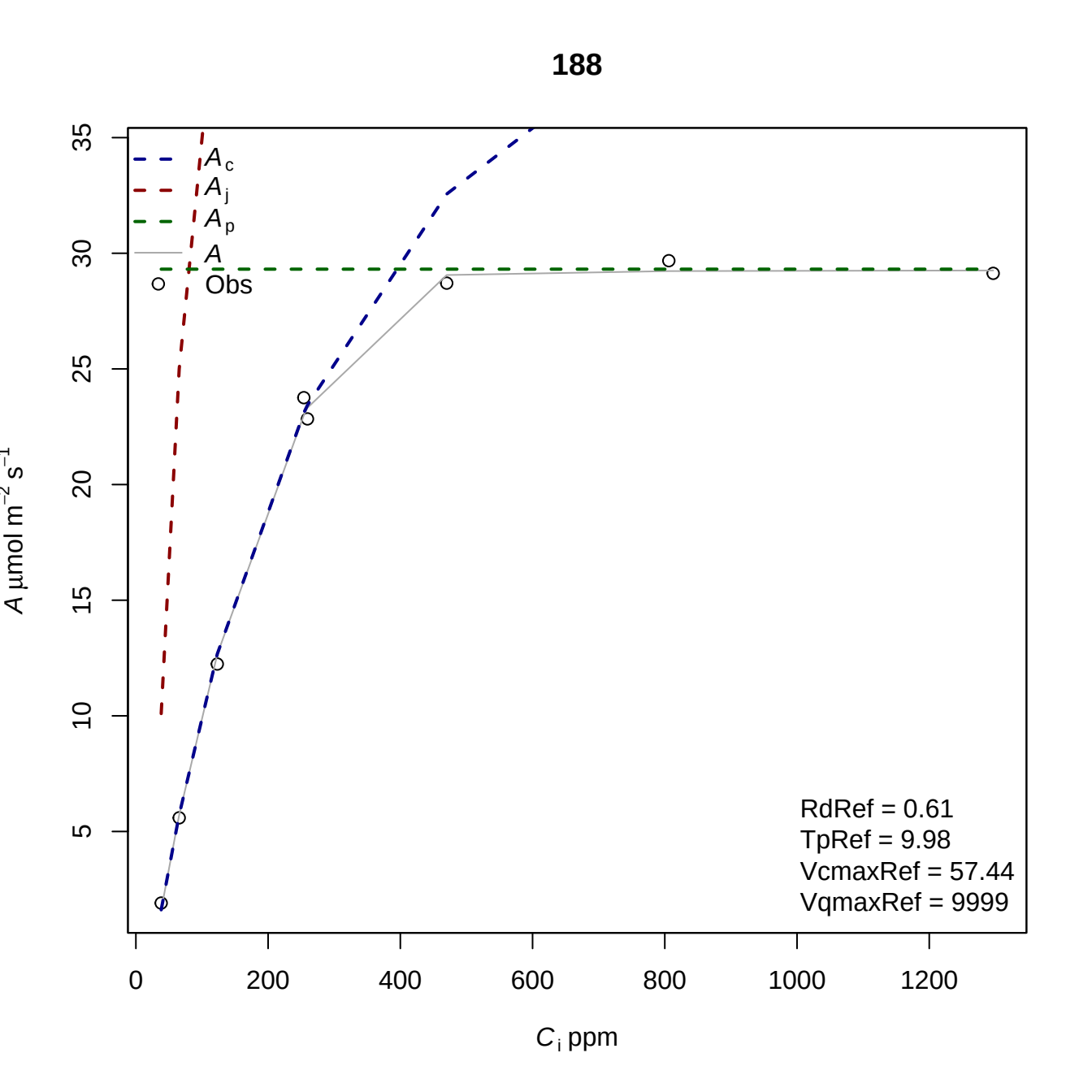

183
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.25
TpRef = 10.45
VcmaxRef = 70.26
VqmaxRef = 283.39
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35

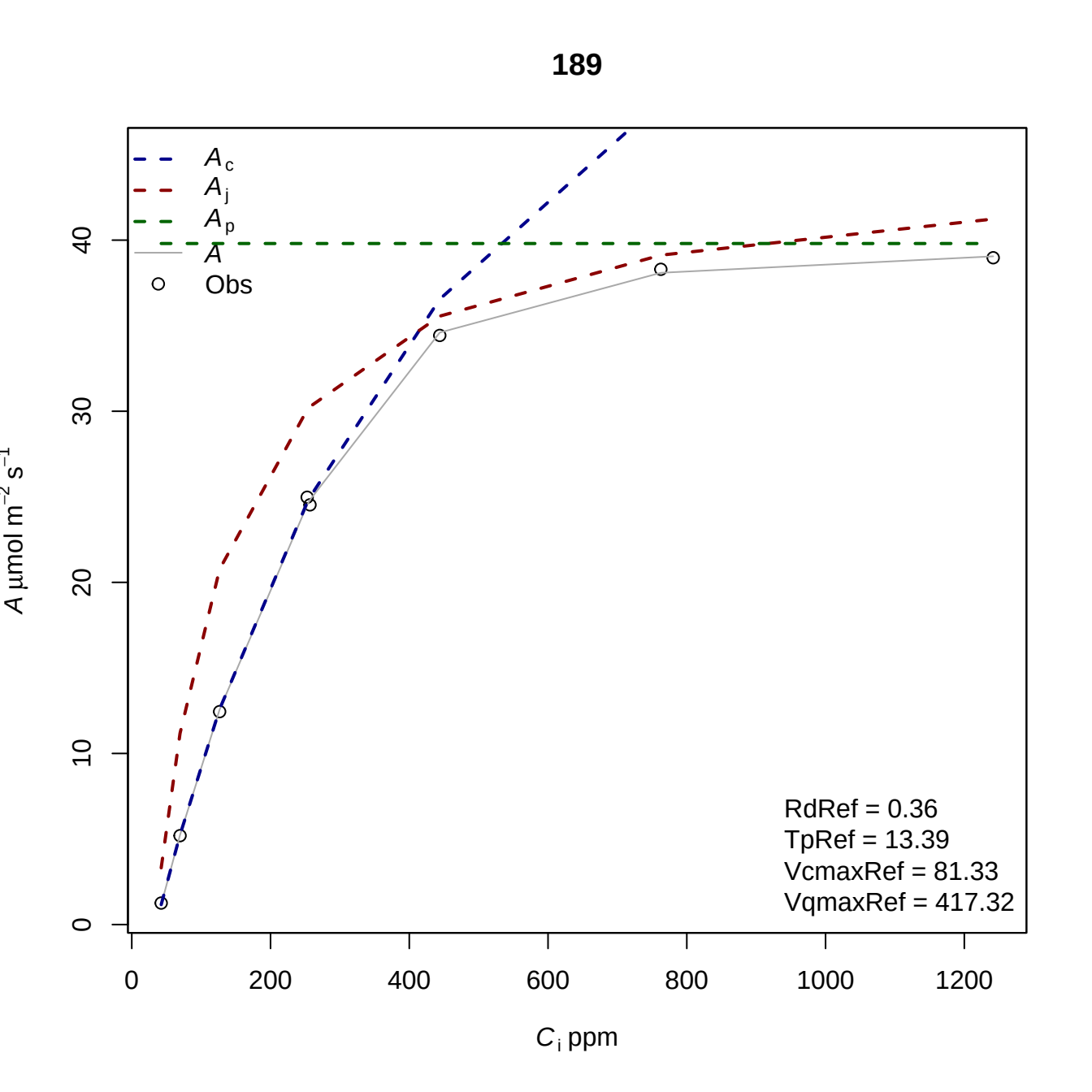

184
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.35
TpRef = 9999
VcmaxRef = 116.58
VqmaxRef = 598.43
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
0
200
400
600
800
1000
1200

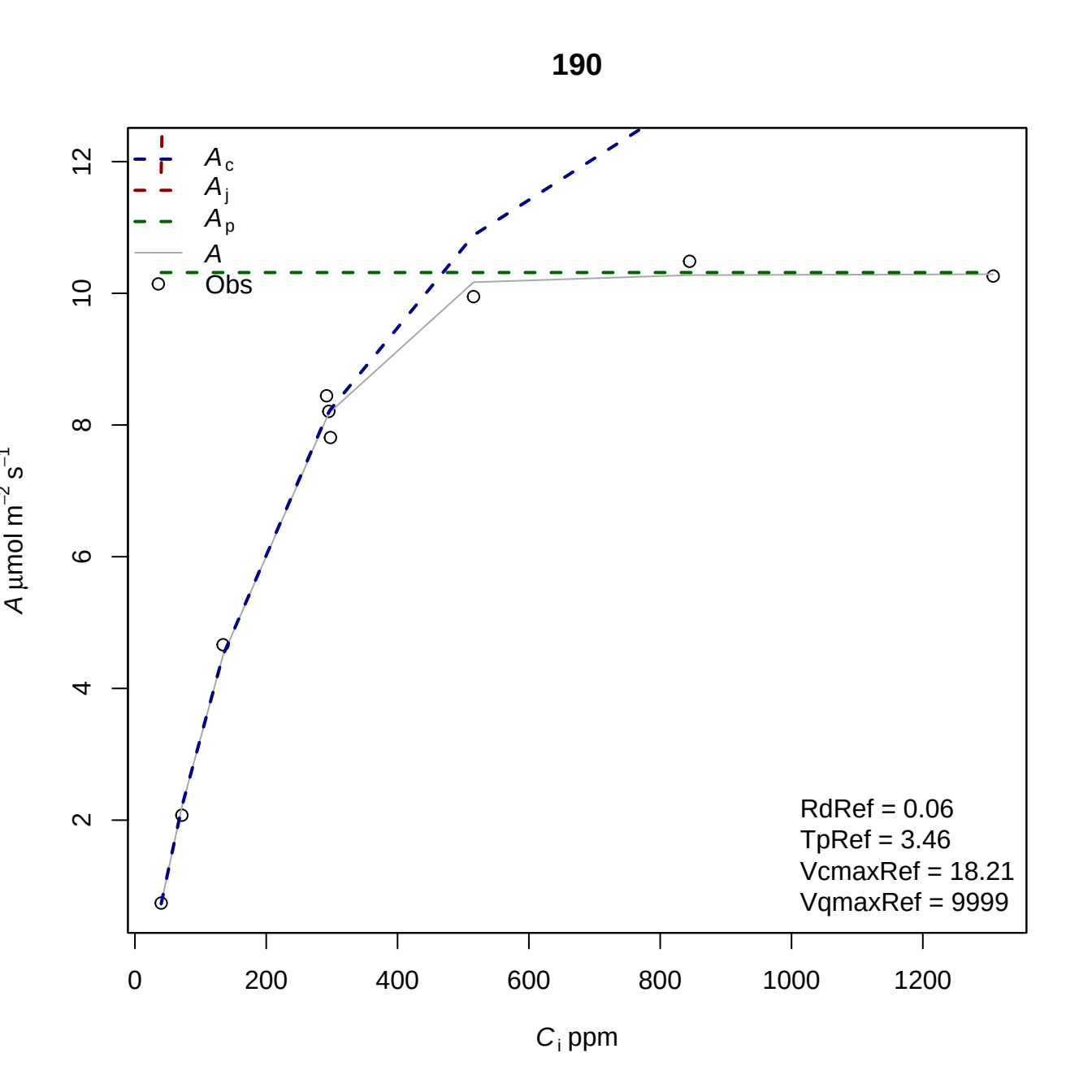

185
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −1.15
TpRef = 9999
VcmaxRef = 157.36
VqmaxRef = 848.61
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
60
200
400
600
800
1000
1200

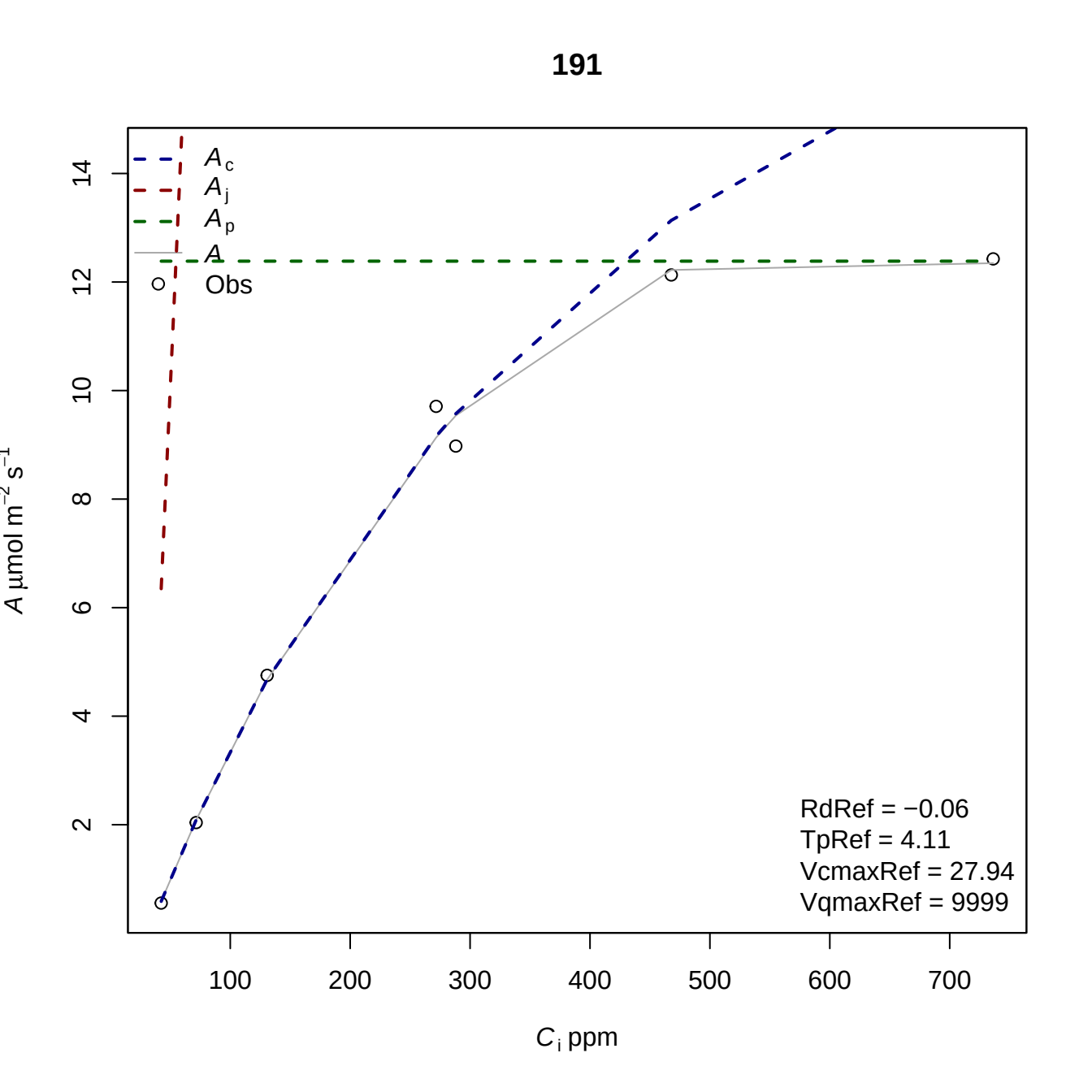

186
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −1.23
TpRef = 9999
VcmaxRef = 204.04
VqmaxRef = 844.98
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

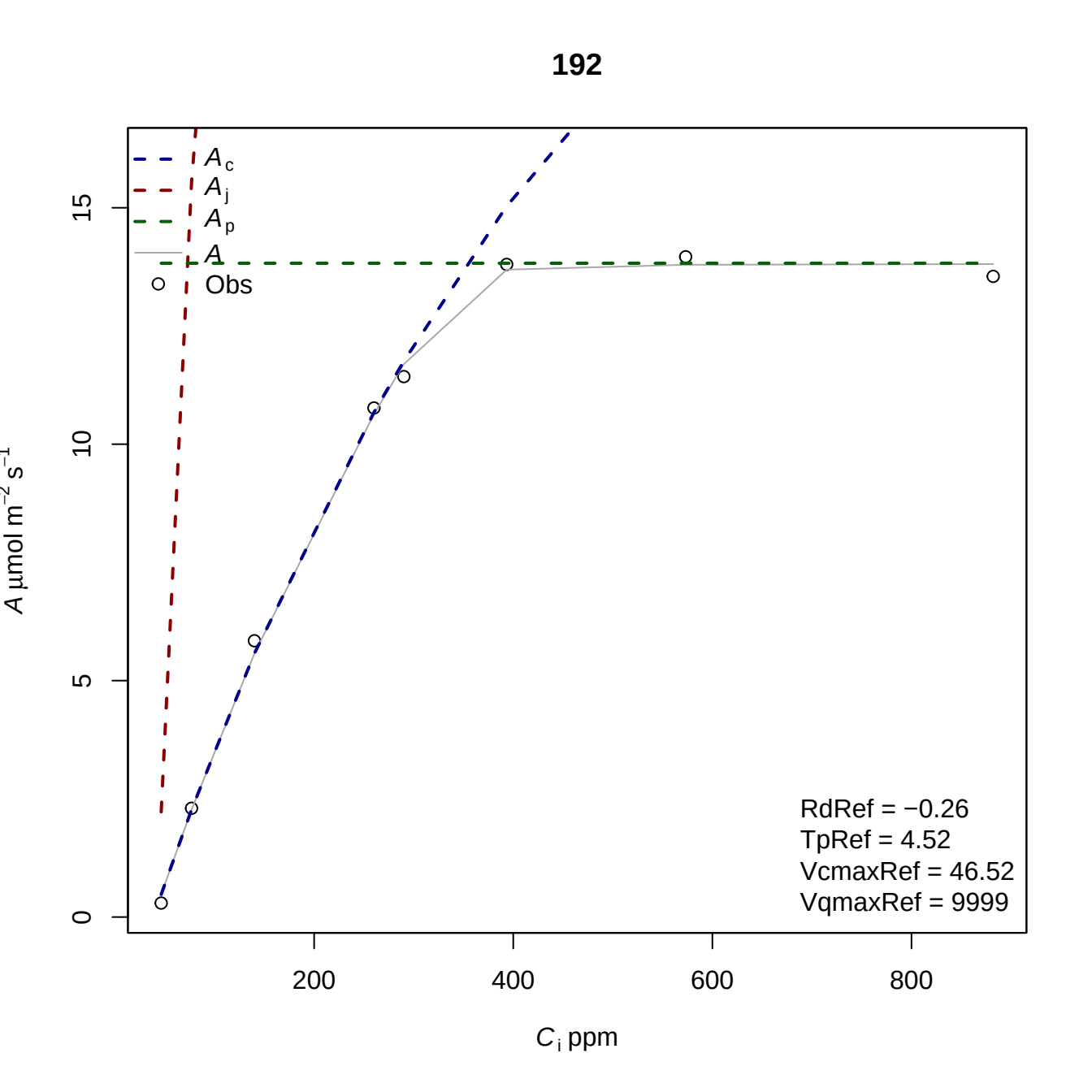

187
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −1.05
TpRef = 9999
VcmaxRef = 218.96
VqmaxRef = 507.93
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

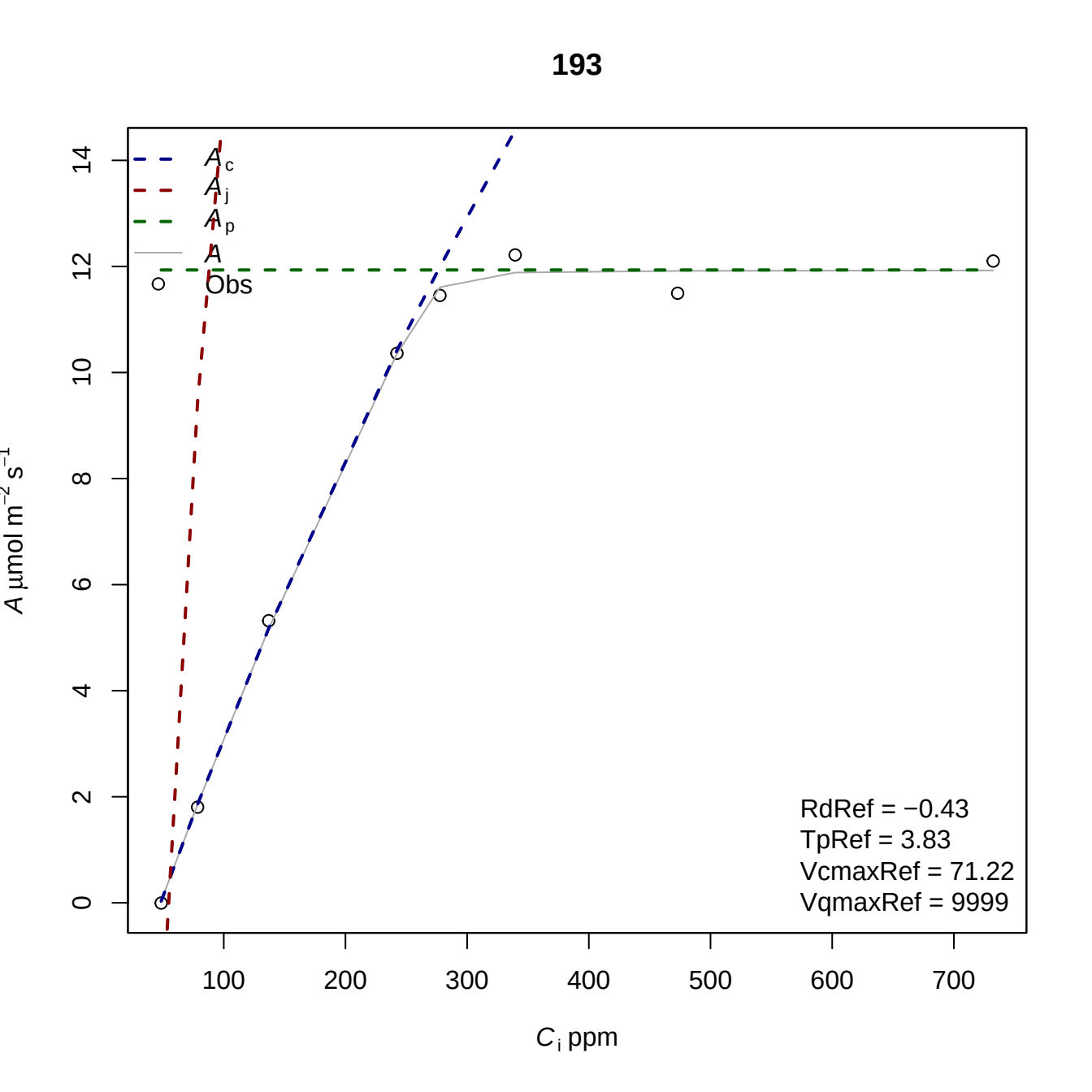

188
A_c
A_j
A_p
A
Obs
RdRef = 0.61
TpRef = 9.98
VcmaxRef = 57.44
VqmaxRef = 9999
A µmol m−2 s−1
C_i ppm
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35

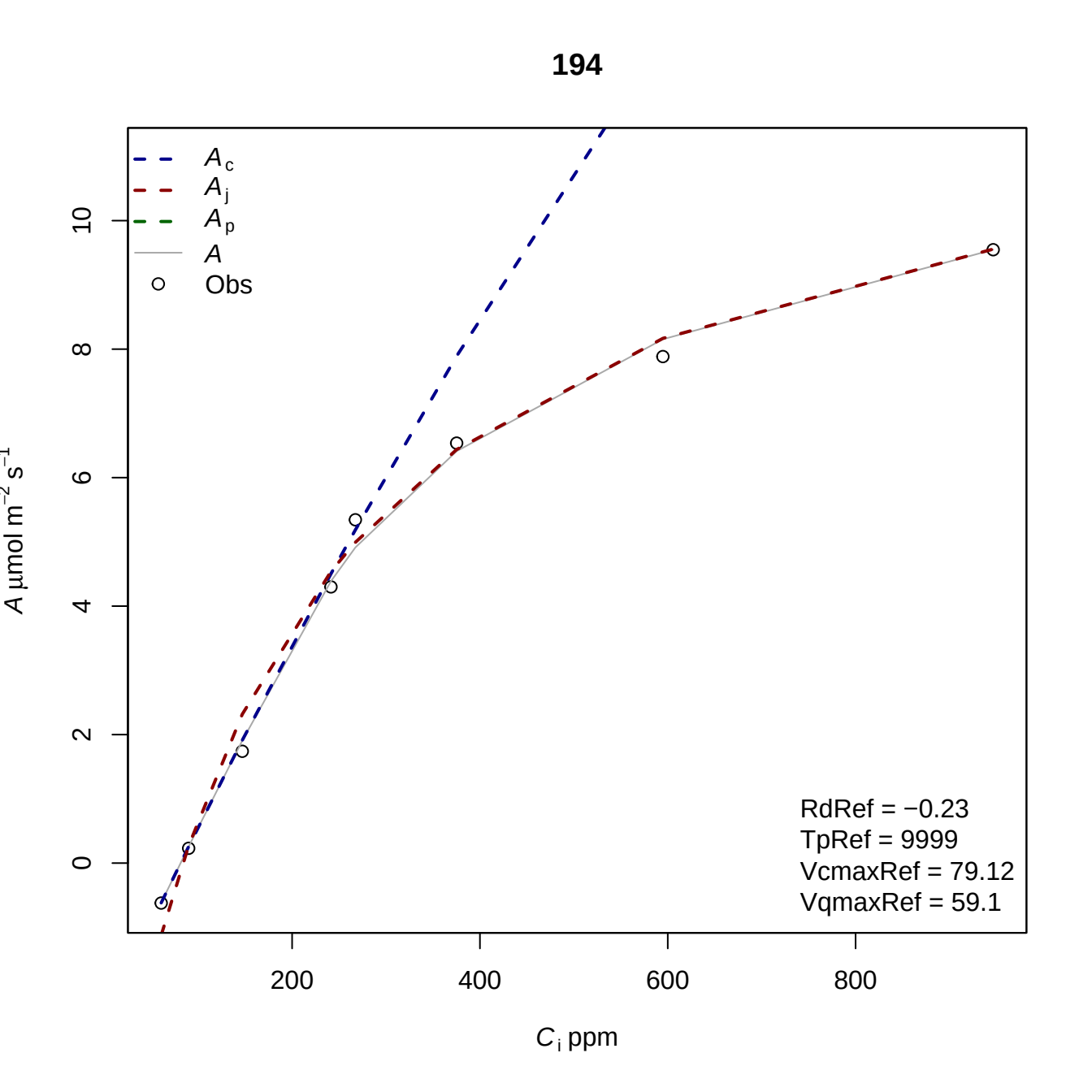

189
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.36
TpRef = 13.39
VcmaxRef = 81.33
VqmaxRef = 417.32
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

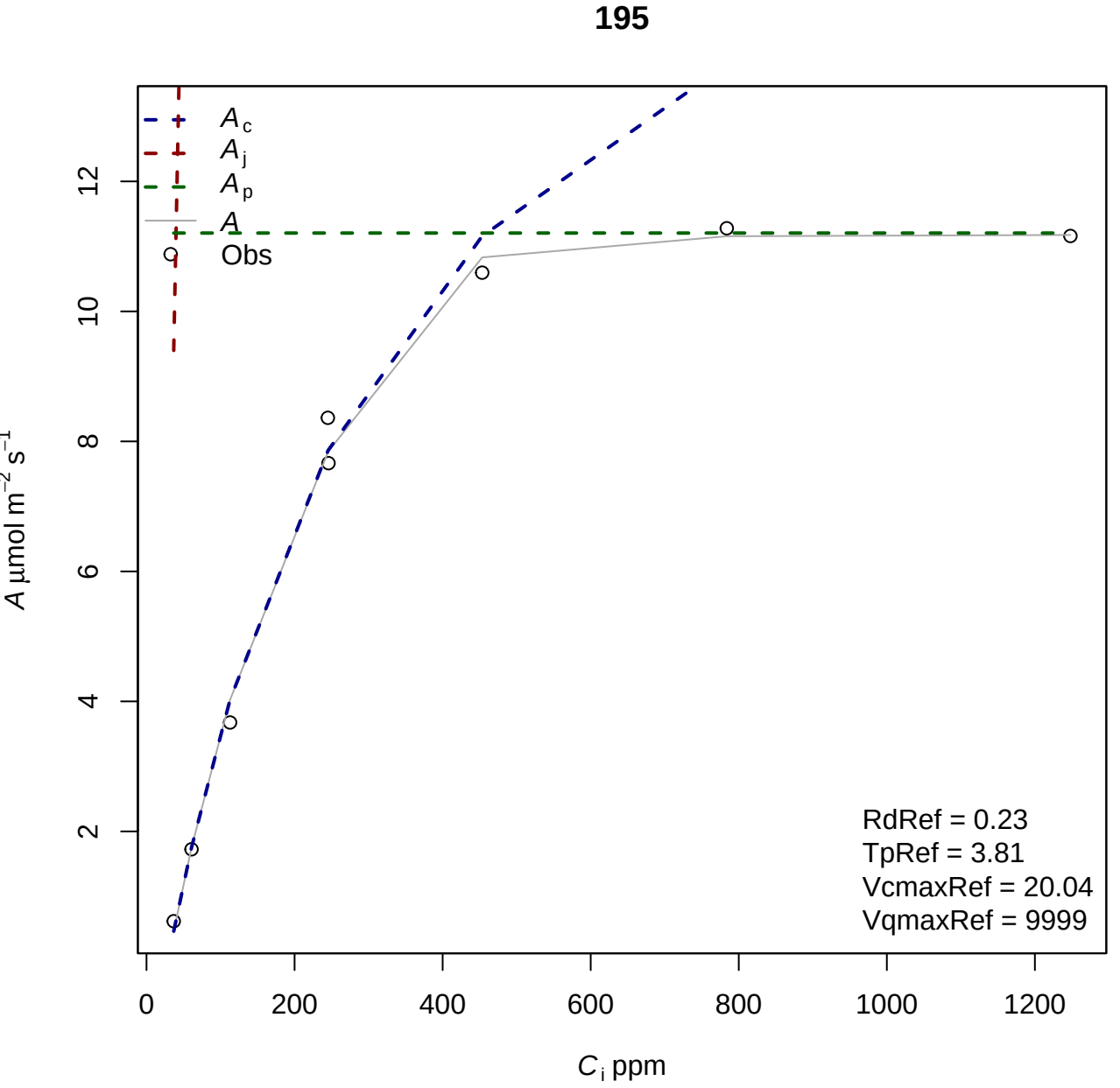

190
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.06
TpRef = 3.46
VcmaxRef = 18.21
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
2
4
6
8
10
12
0
200
400
600
800
1000
1200

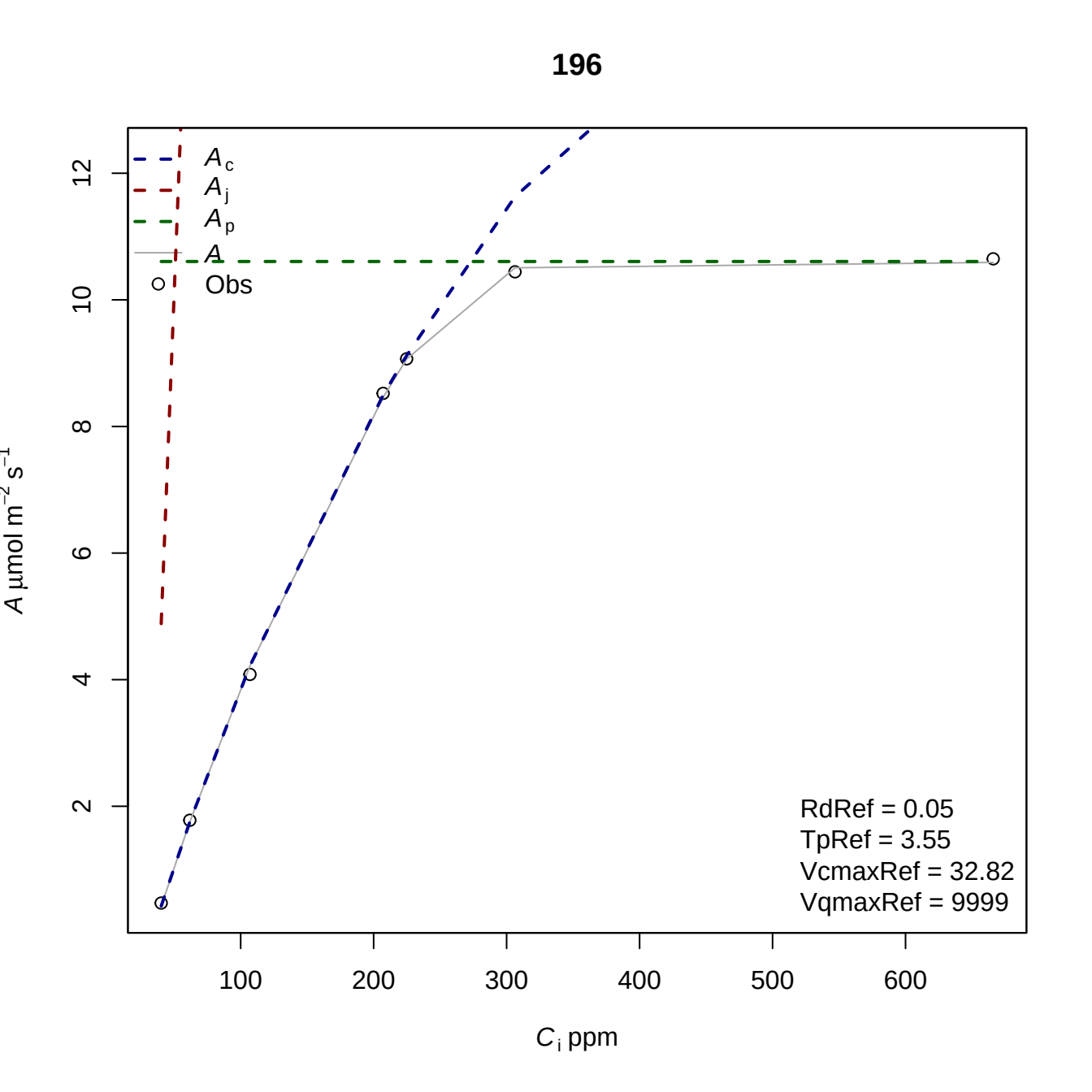

191
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.06
TpRef = 4.11
VcmaxRef = 27.94
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
2
4
6
8
10
12
14
100
200
300
400
500
600
700

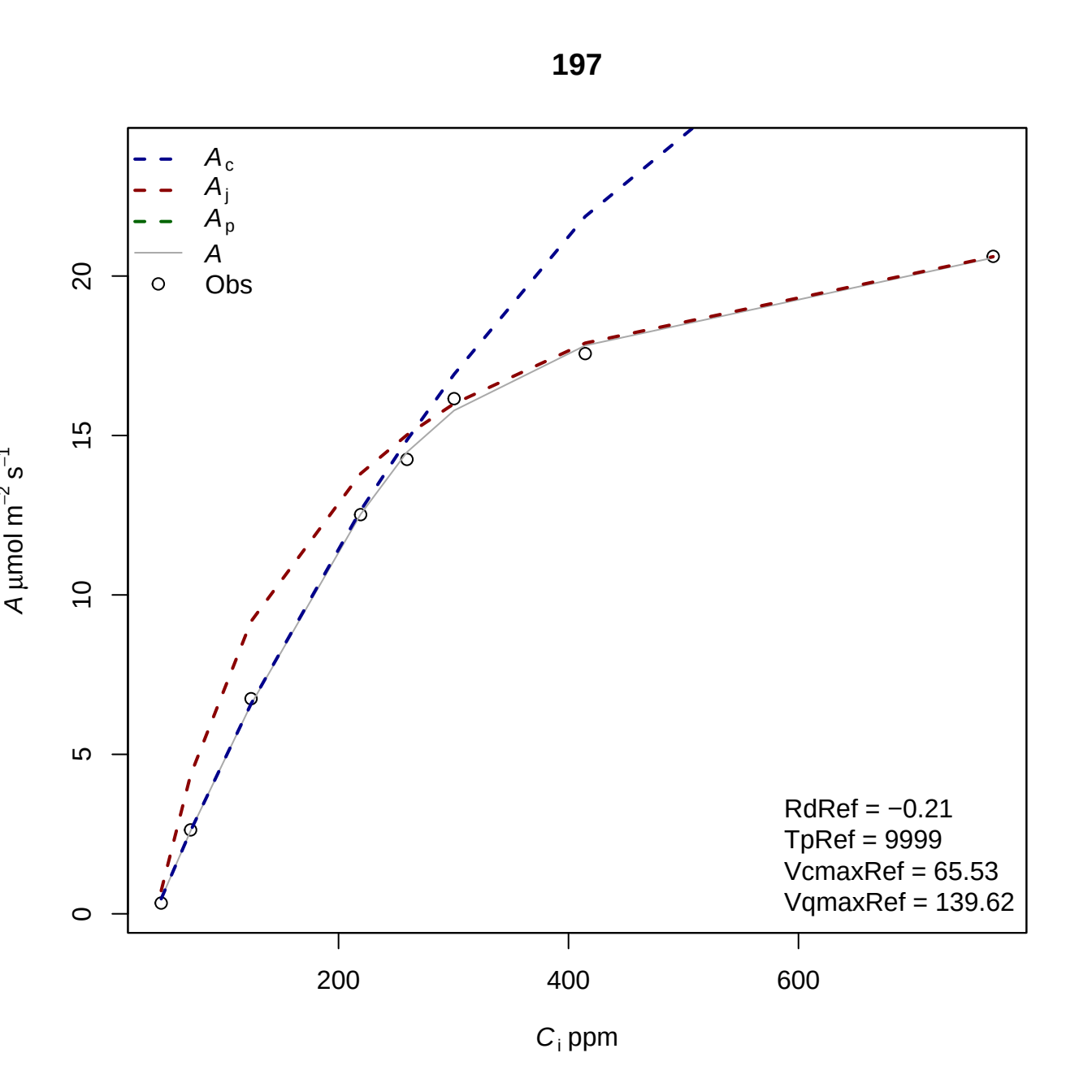

192
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.26
TpRef = 4.52
VcmaxRef = 46.52
VqmaxRef = 9999
0
5
10
15
200
400
600
800
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

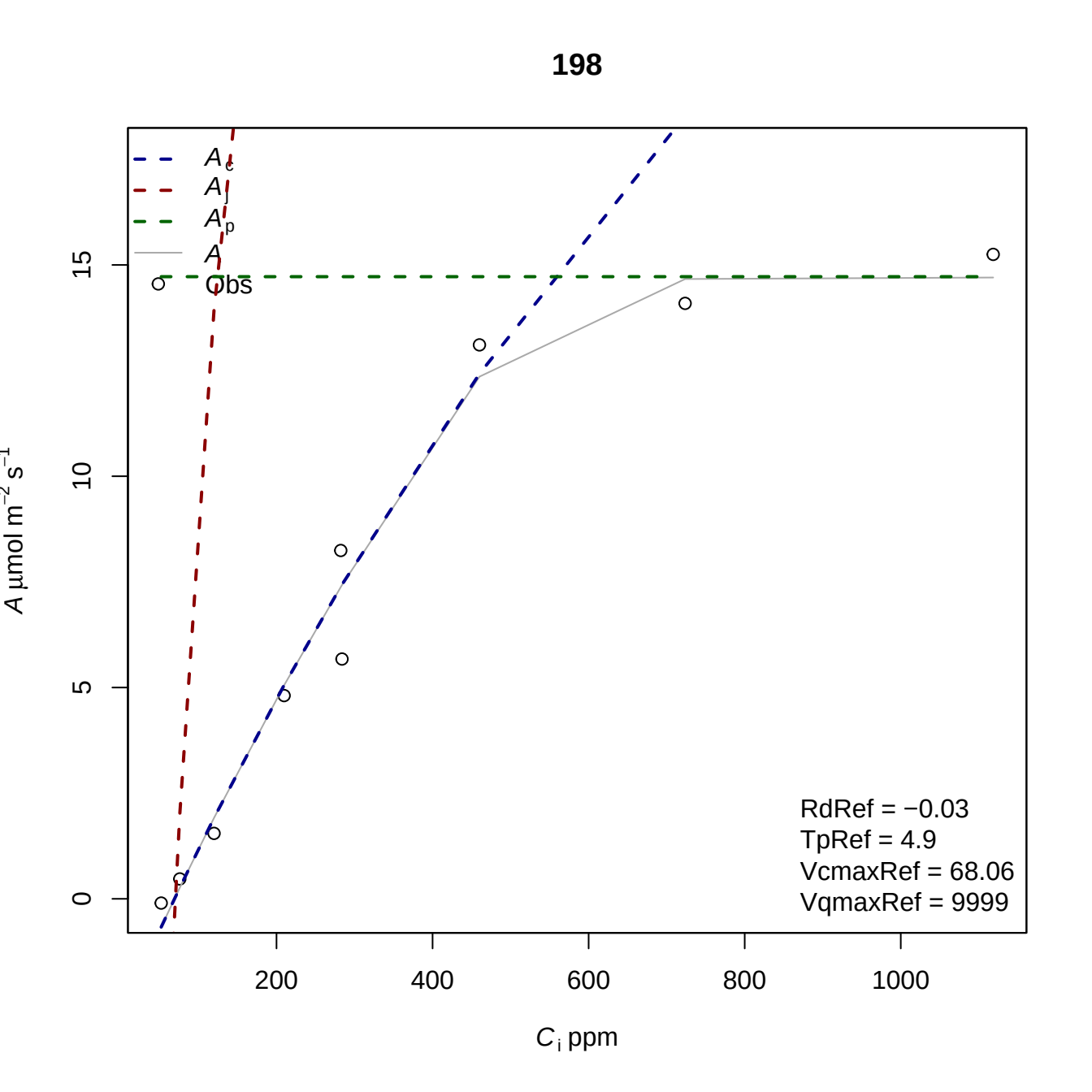

193
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.43
TpRef = 3.83
VcmaxRef = 71.22
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
100
200
300
400
500
600
700

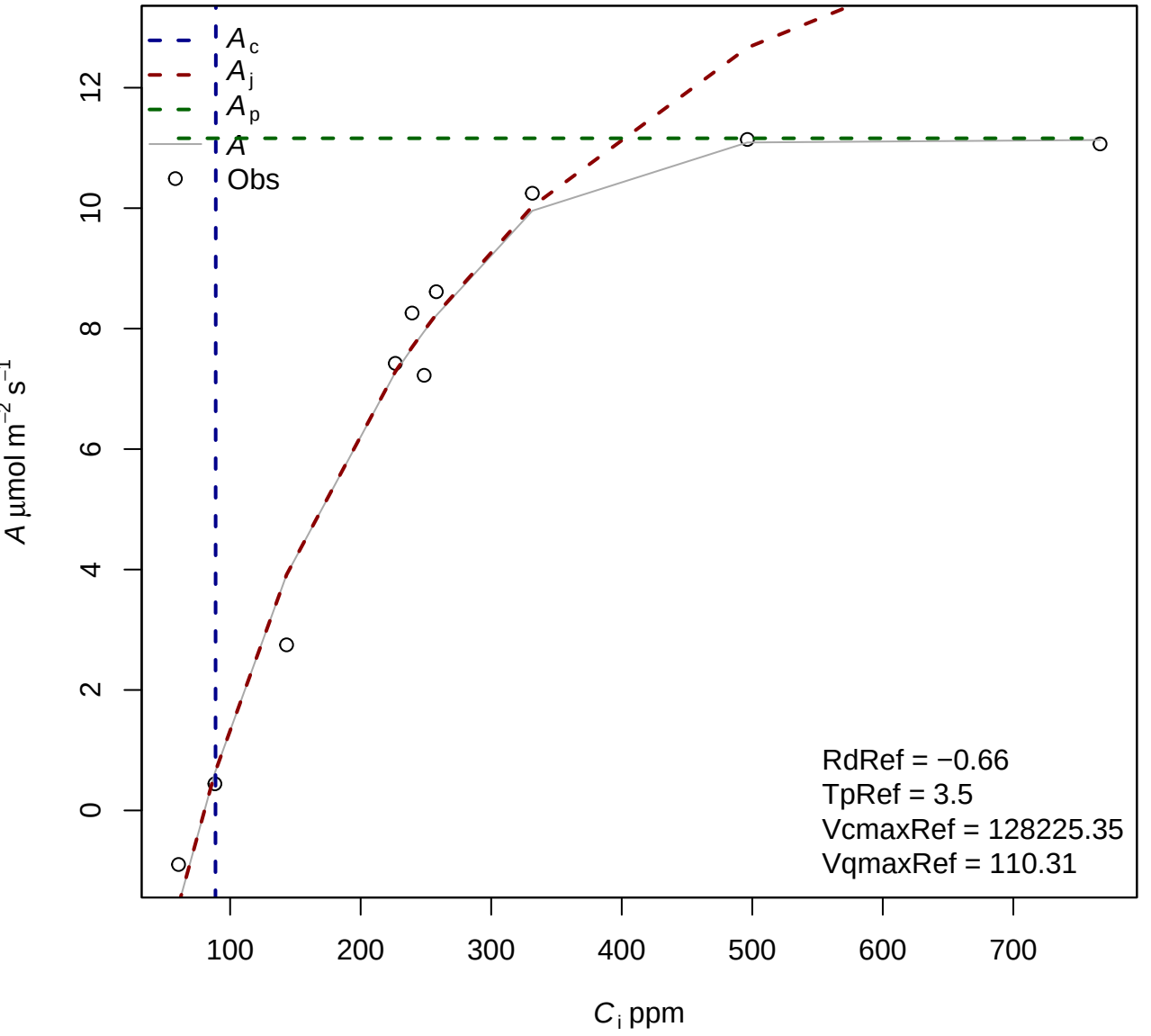

194
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.23
TpRef = 9999
VcmaxRef = 79.12
VqmaxRef = 59.1
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
200
400
600
800

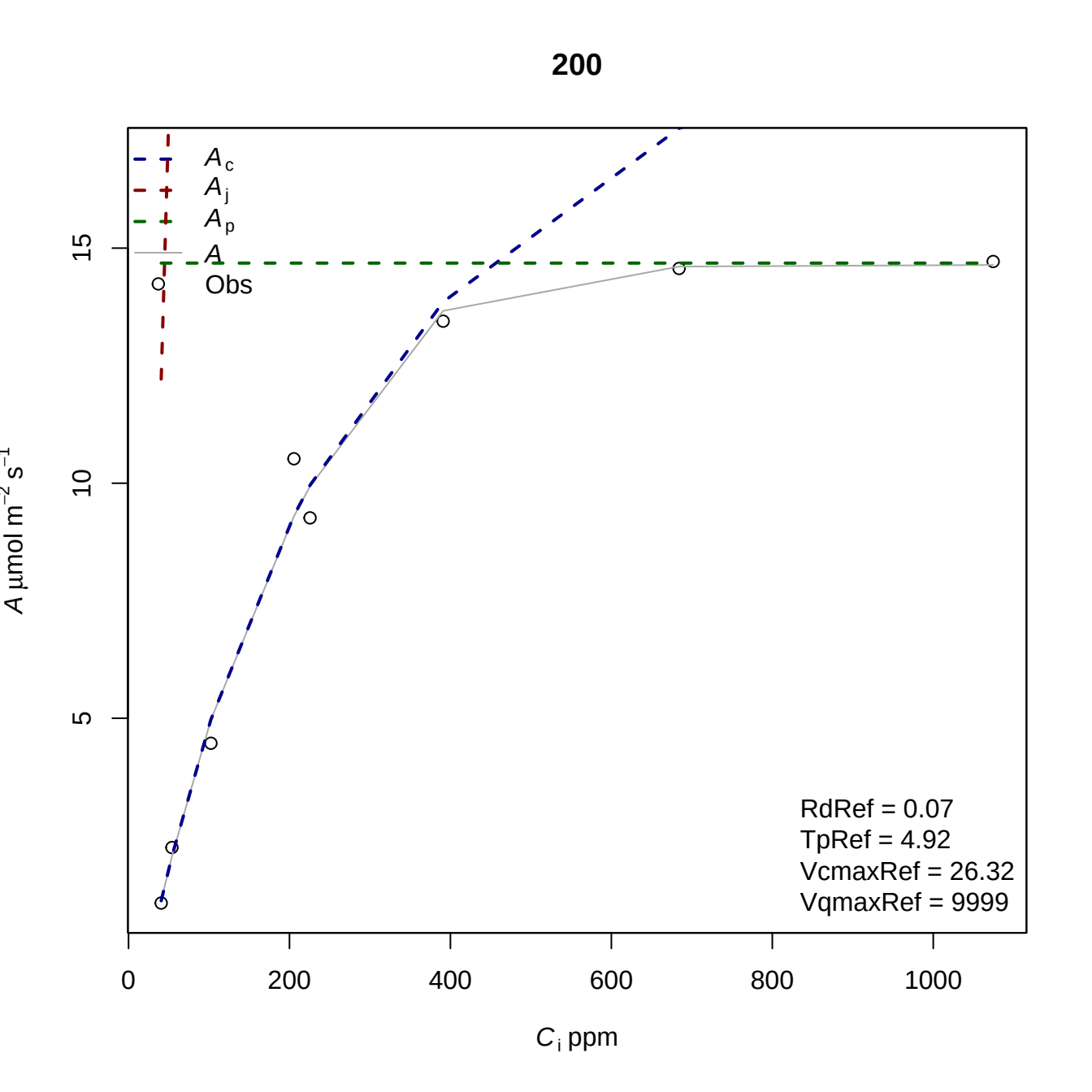

195
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.23
TpRef = 3.81
VcmaxRef = 20.04
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
2
4
6
8
10
12

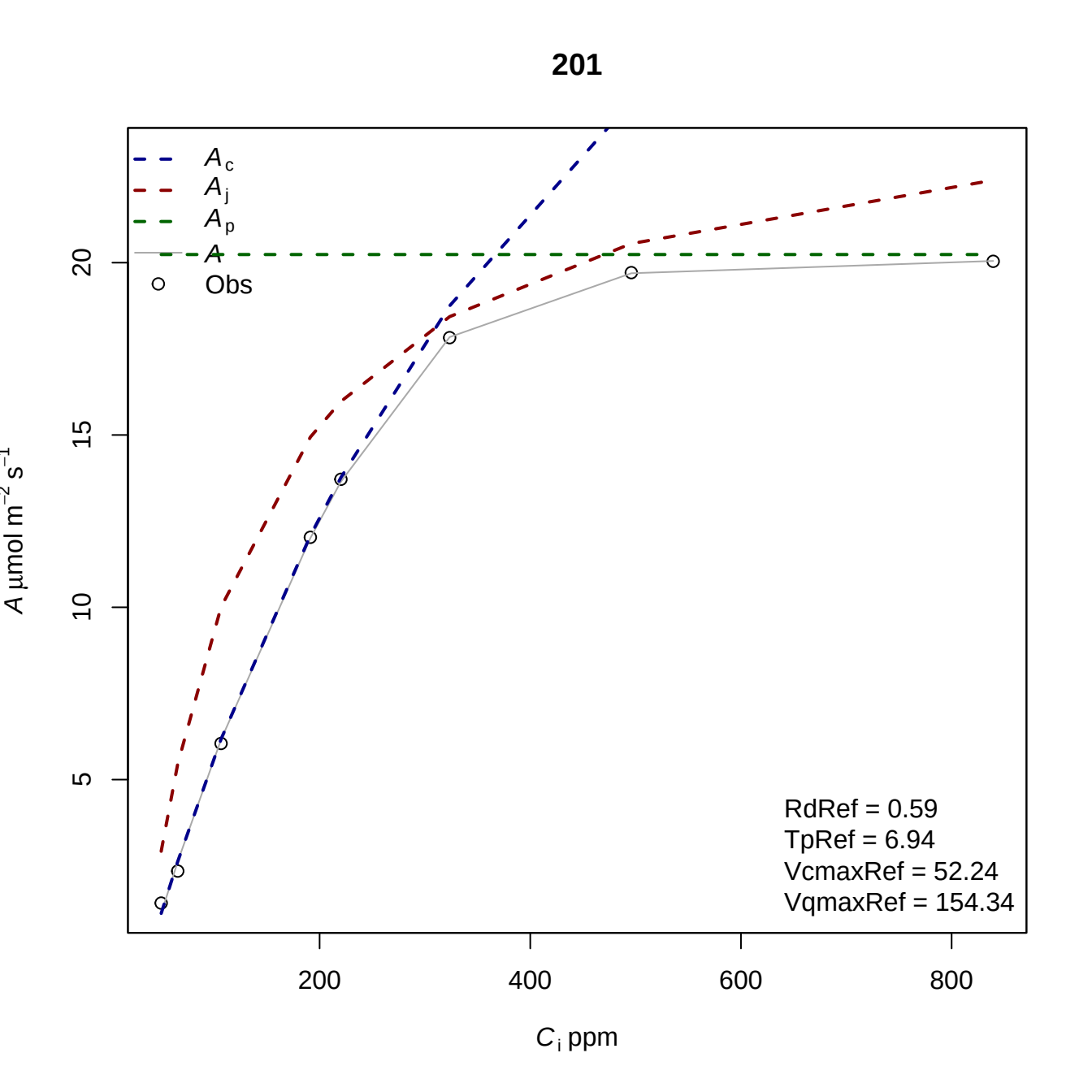

196
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.05
TpRef = 3.55
VcmaxRef = 32.82
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
2
4
6
8
10
12
100
200
300
400
500
600

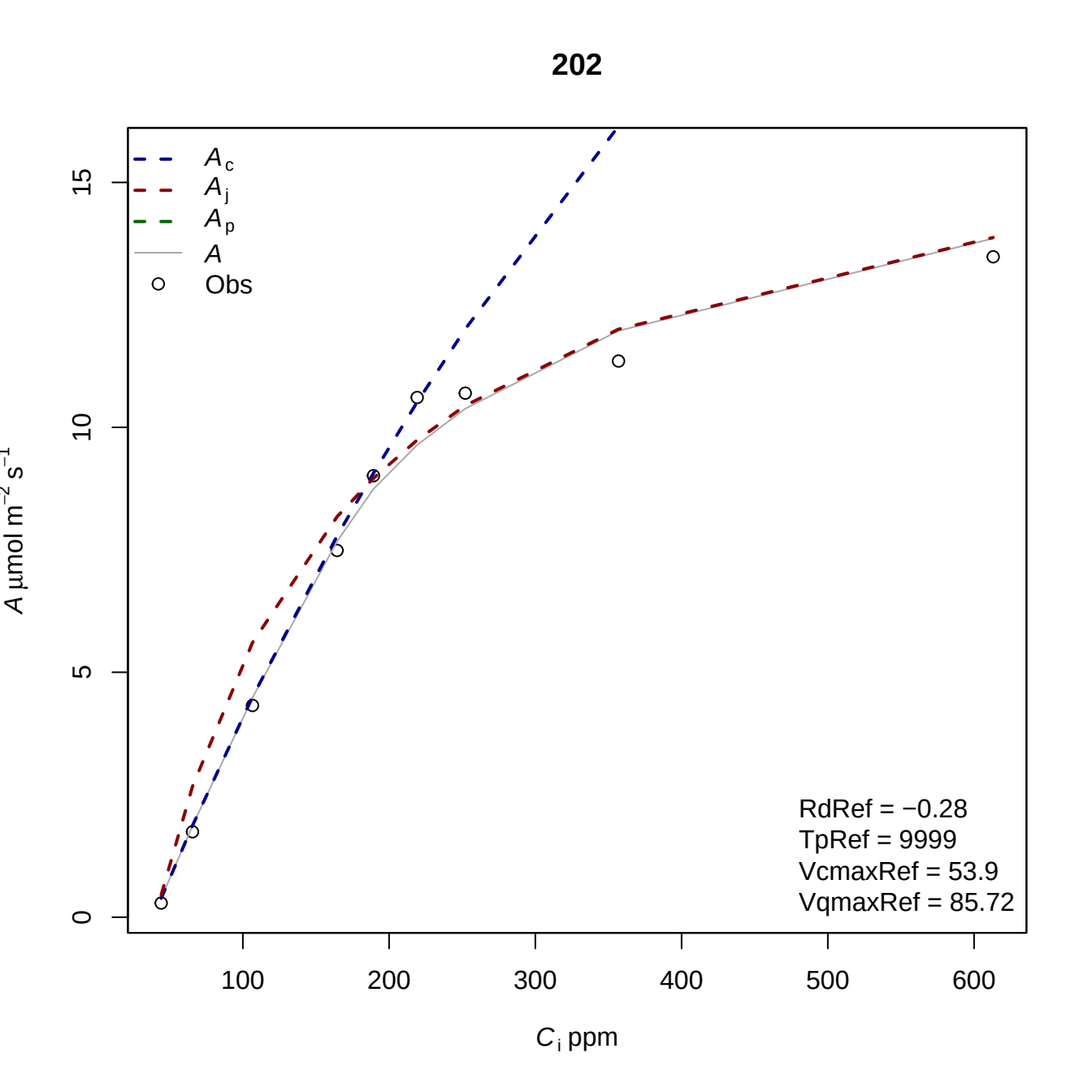

197
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.21
TpRef = 9999
VcmaxRef = 65.53
VqmaxRef = 139.62
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600

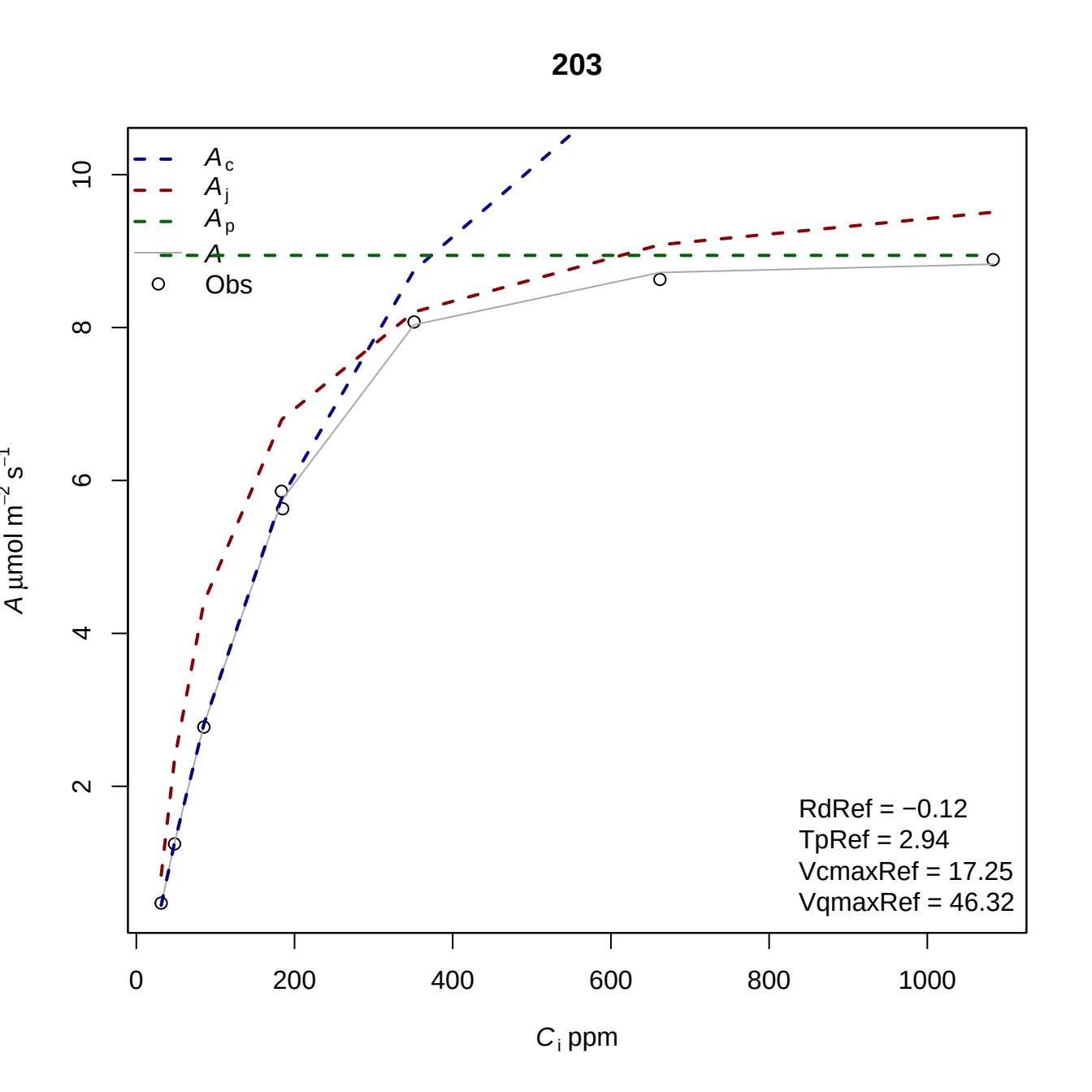

198
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.03
TpRef = 4.9
VcmaxRef = 68.06
VqmaxRef = 9999
0
5
10
15
200
400
600
800
1000
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

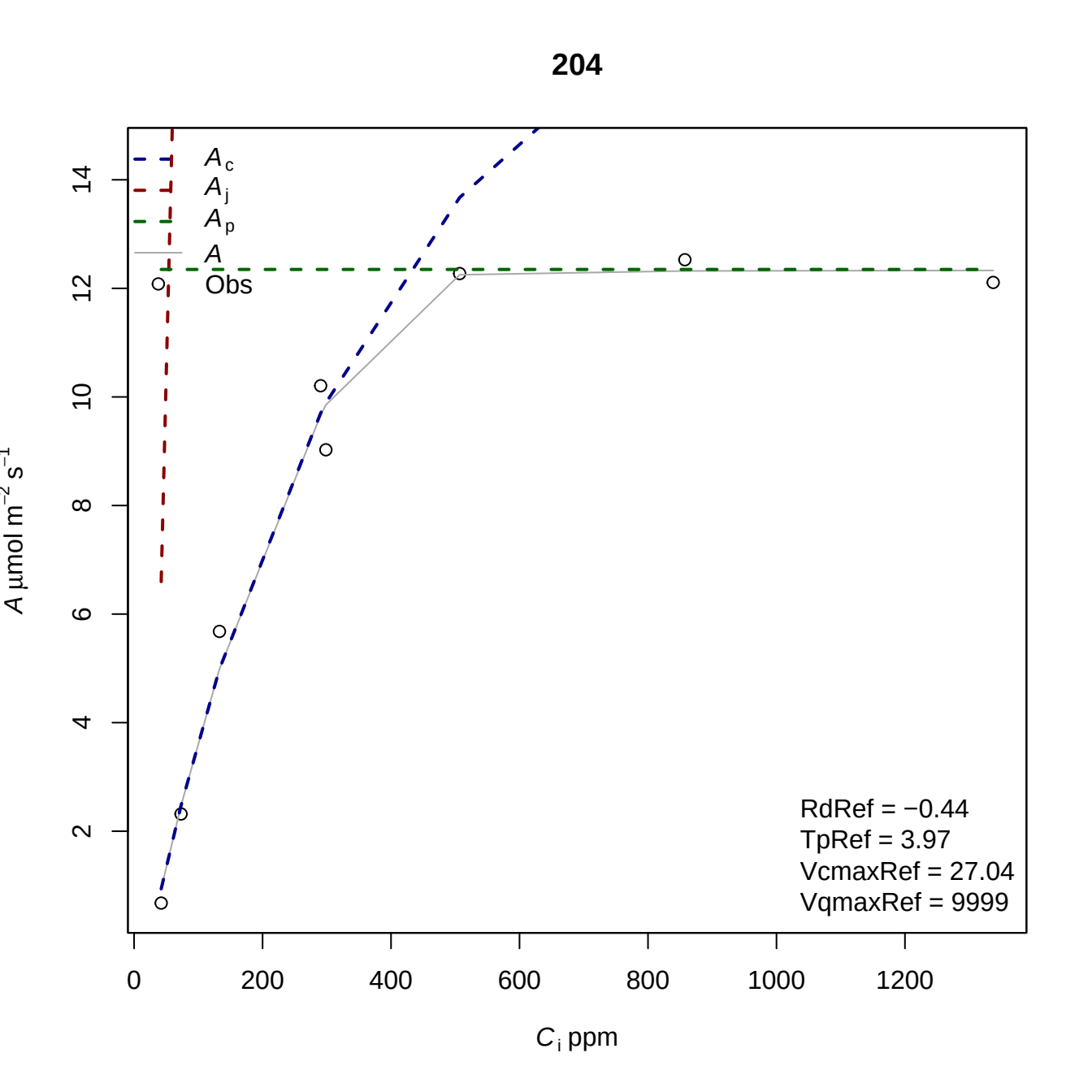

$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.66
TpRef = 3.5
VcmaxRef = 128225.35
VqmaxRef = 110.31
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
100
200
300
400
500
600
700

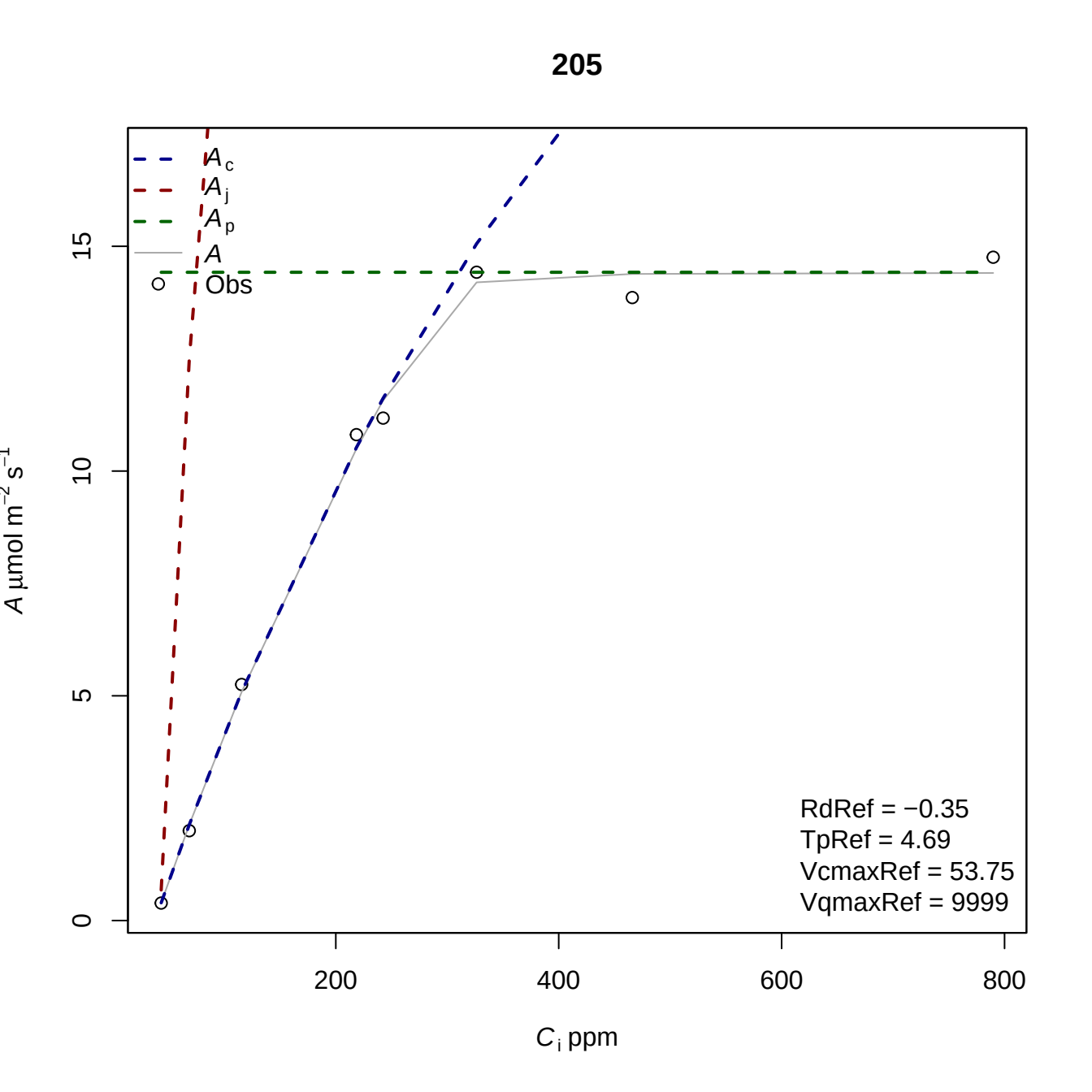

200
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.07
TpRef = 4.92
VcmaxRef = 26.32
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
5
10
15

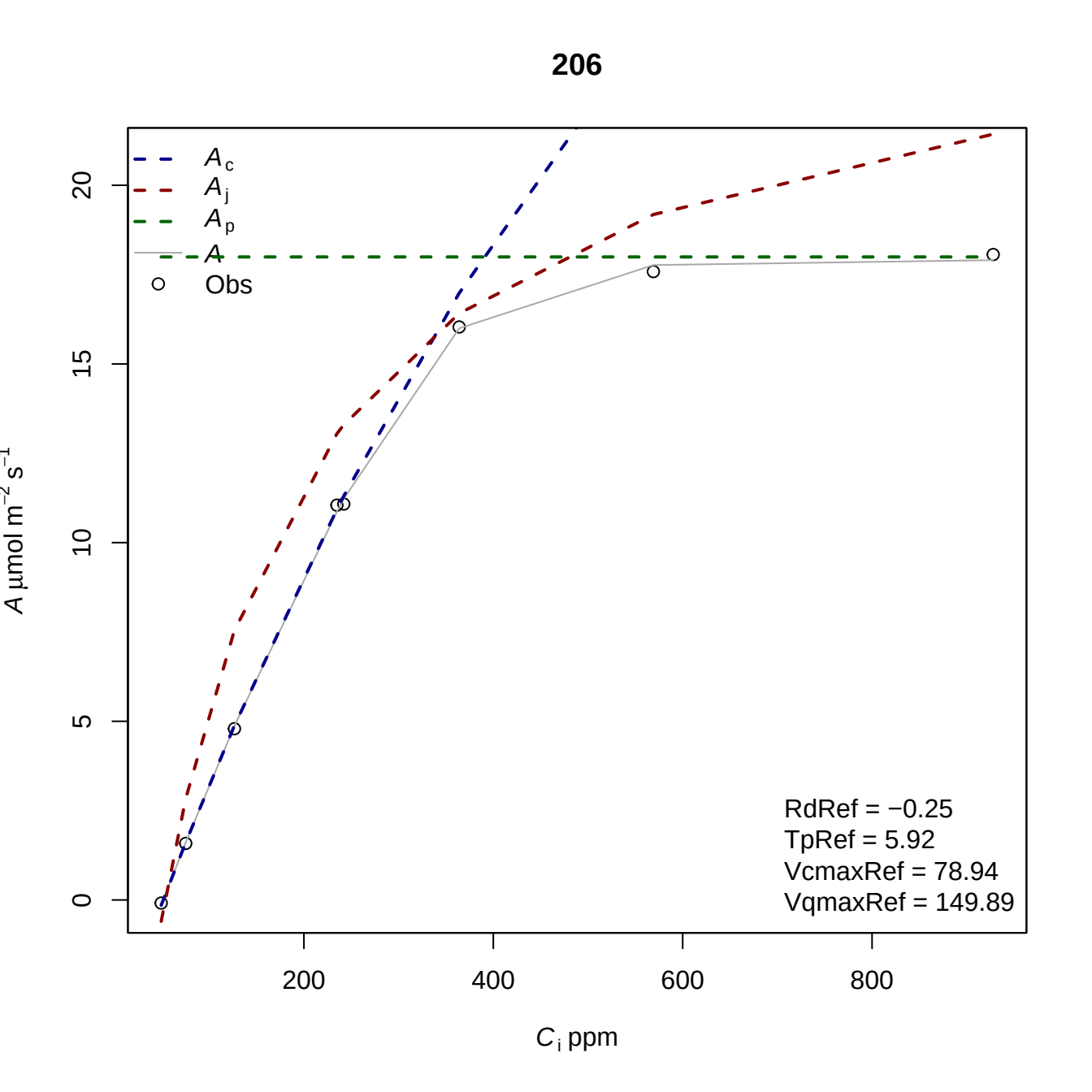

201
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.59
TpRef = 6.94
VcmaxRef = 52.24
VqmaxRef = 154.34
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
20
200
400
600
800

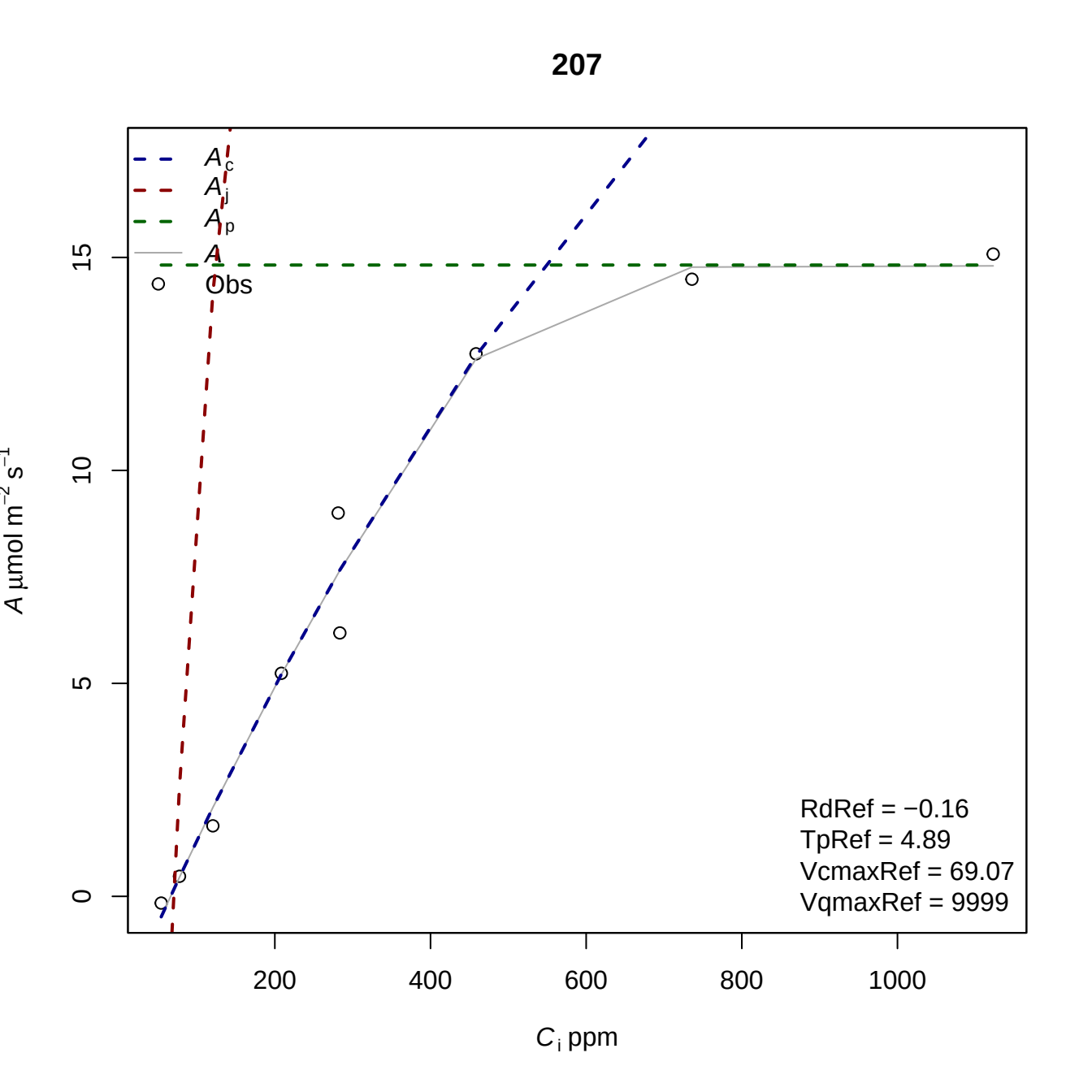

202
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.28
TpRef = 9999
VcmaxRef = 53.9
VqmaxRef = 85.72
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500
600

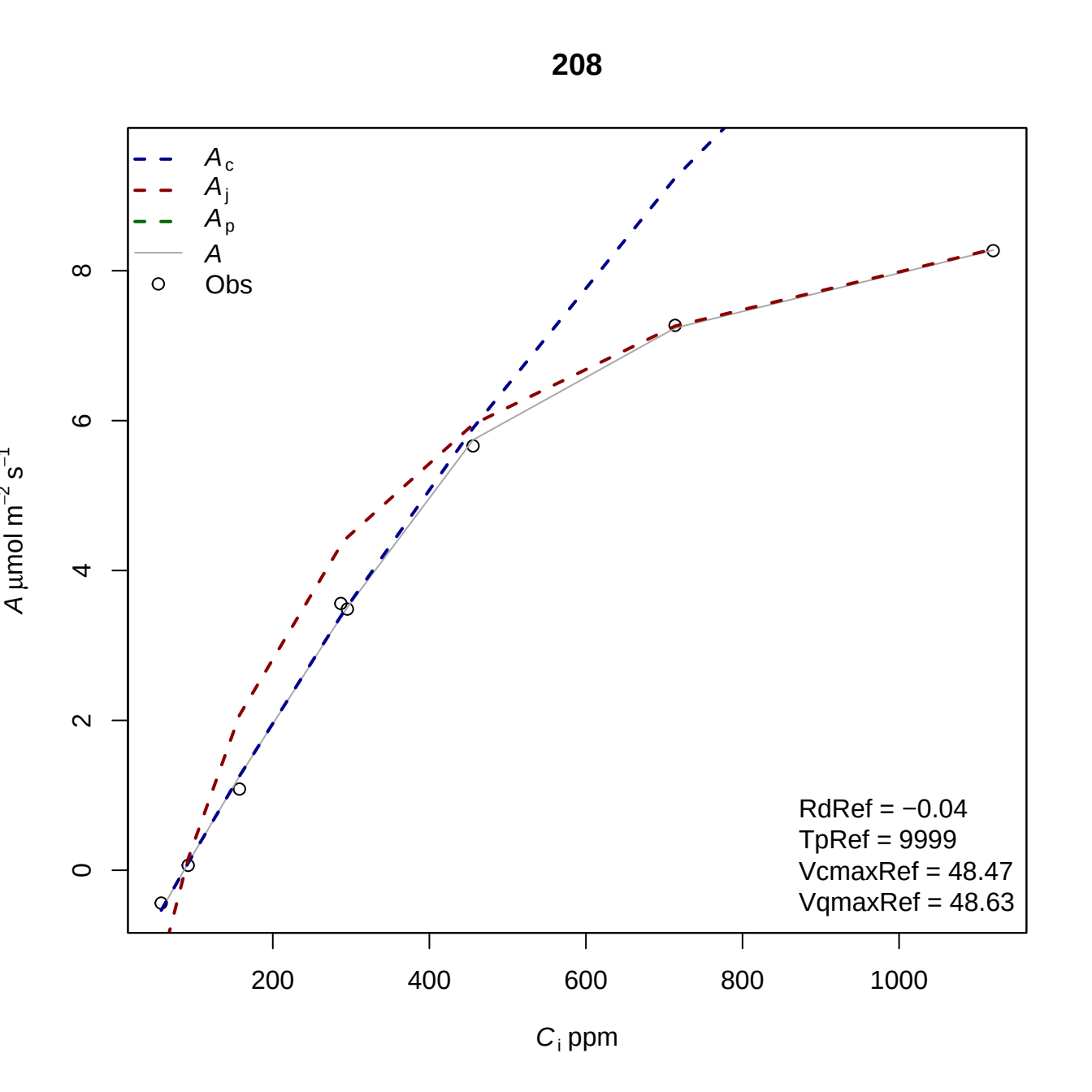

203
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.12
TpRef = 2.94
VcmaxRef = 17.25
VqmaxRef = 46.32
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
2
4
6
8
10

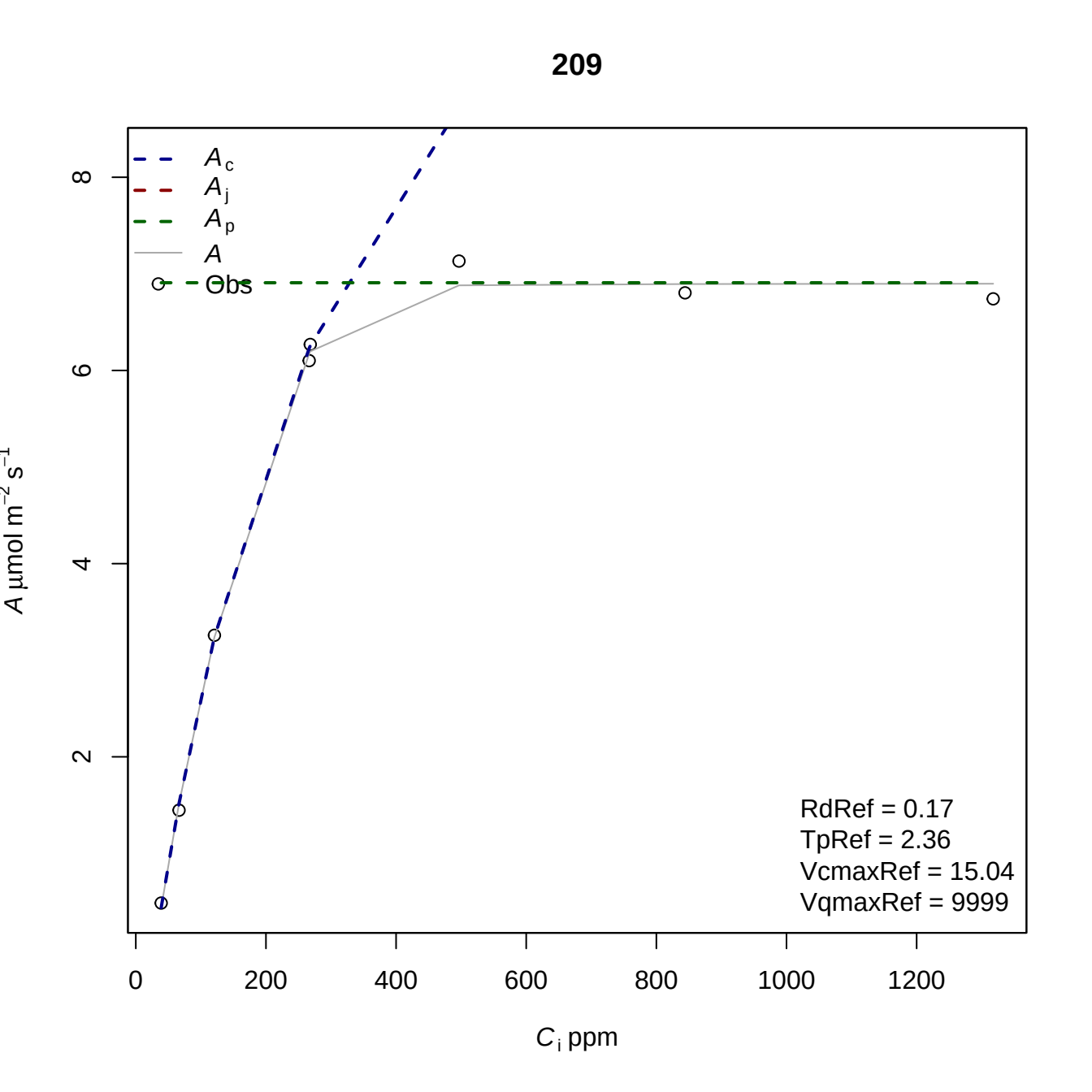

204
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.44
TpRef = 3.97
VcmaxRef = 27.04
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14

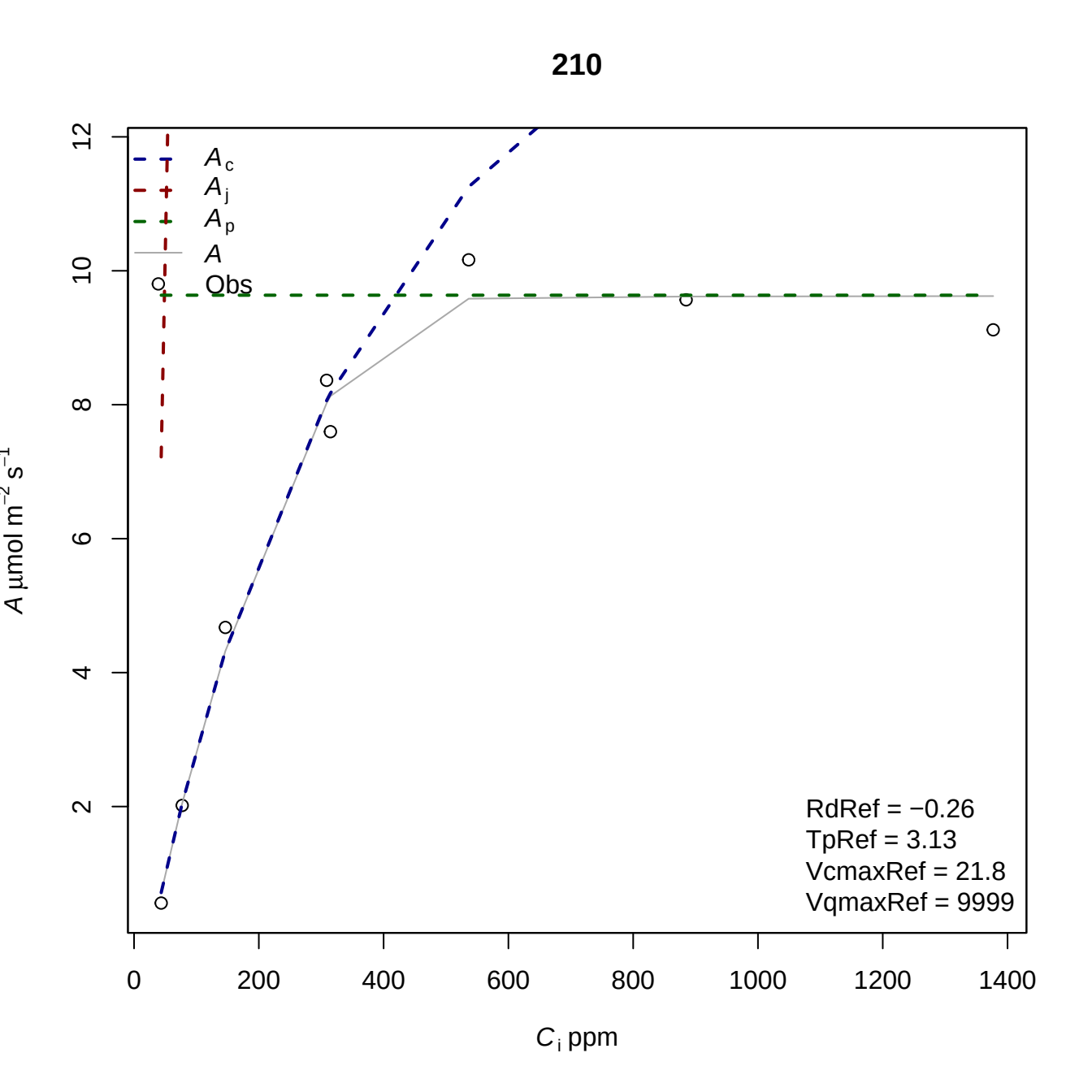

205
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.35
TpRef = 4.69
VcmaxRef = 53.75
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800

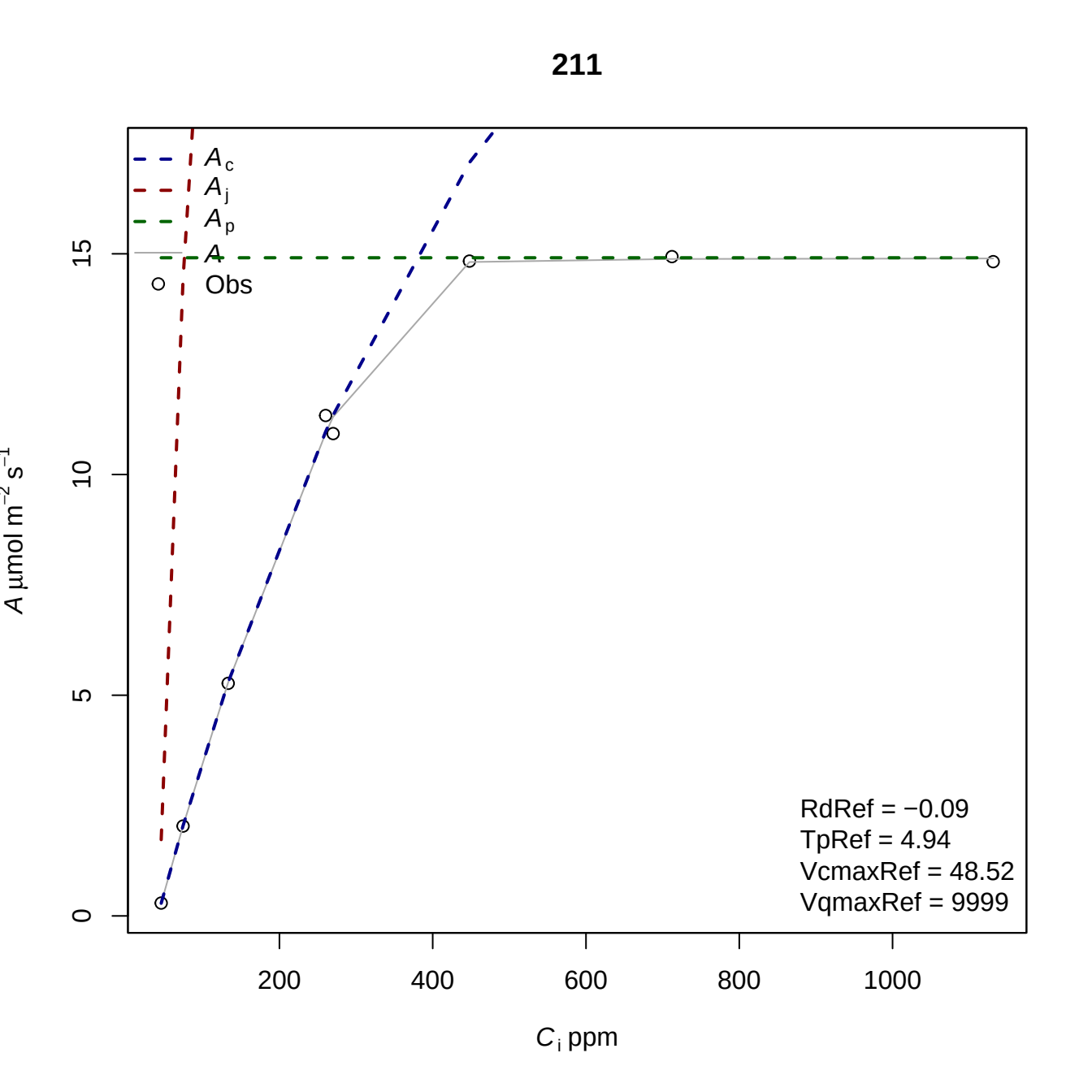

206
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.25
TpRef = 5.92
VcmaxRef = 78.94
VqmaxRef = 149.89
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800

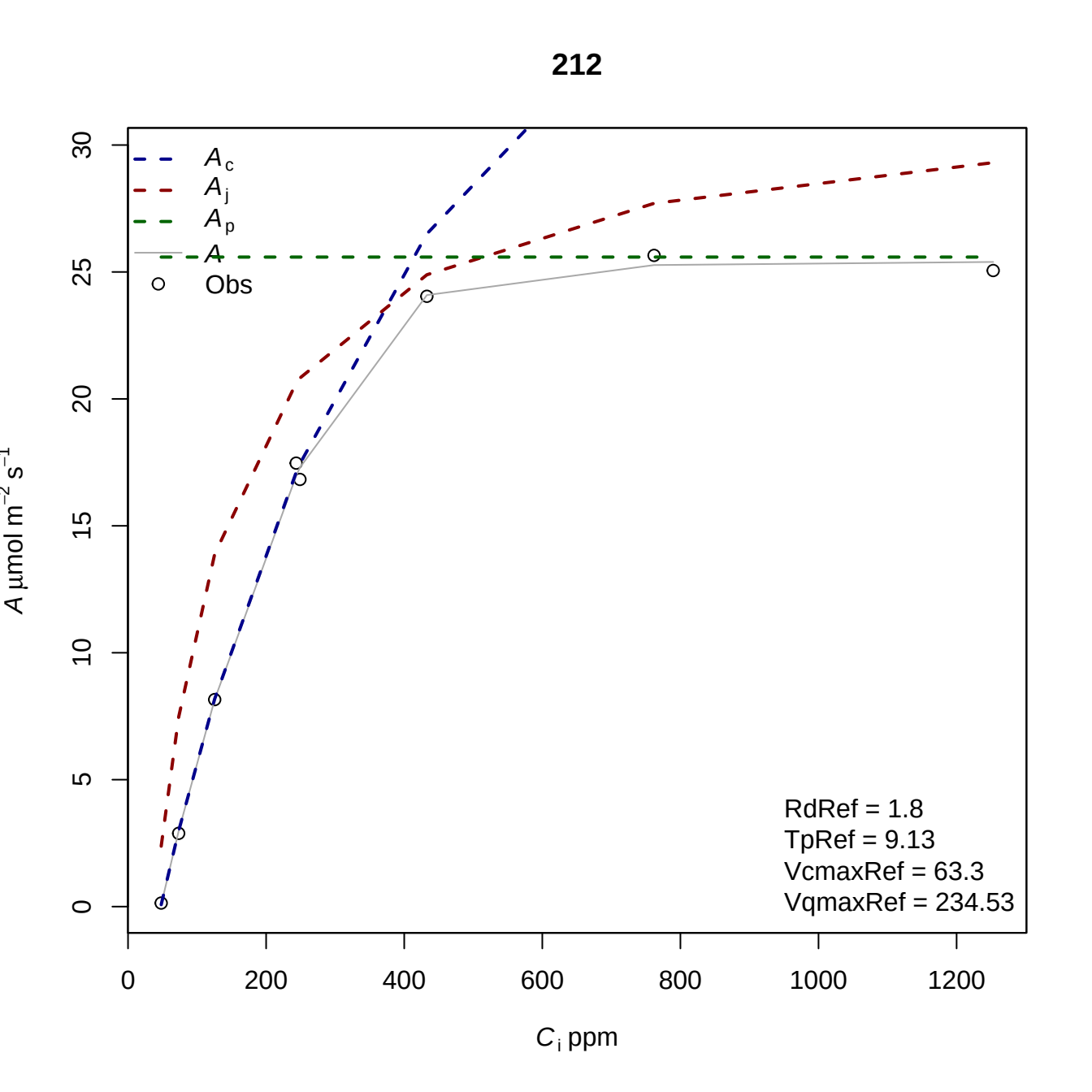

207
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.16
TpRef = 4.89
VcmaxRef = 69.07
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

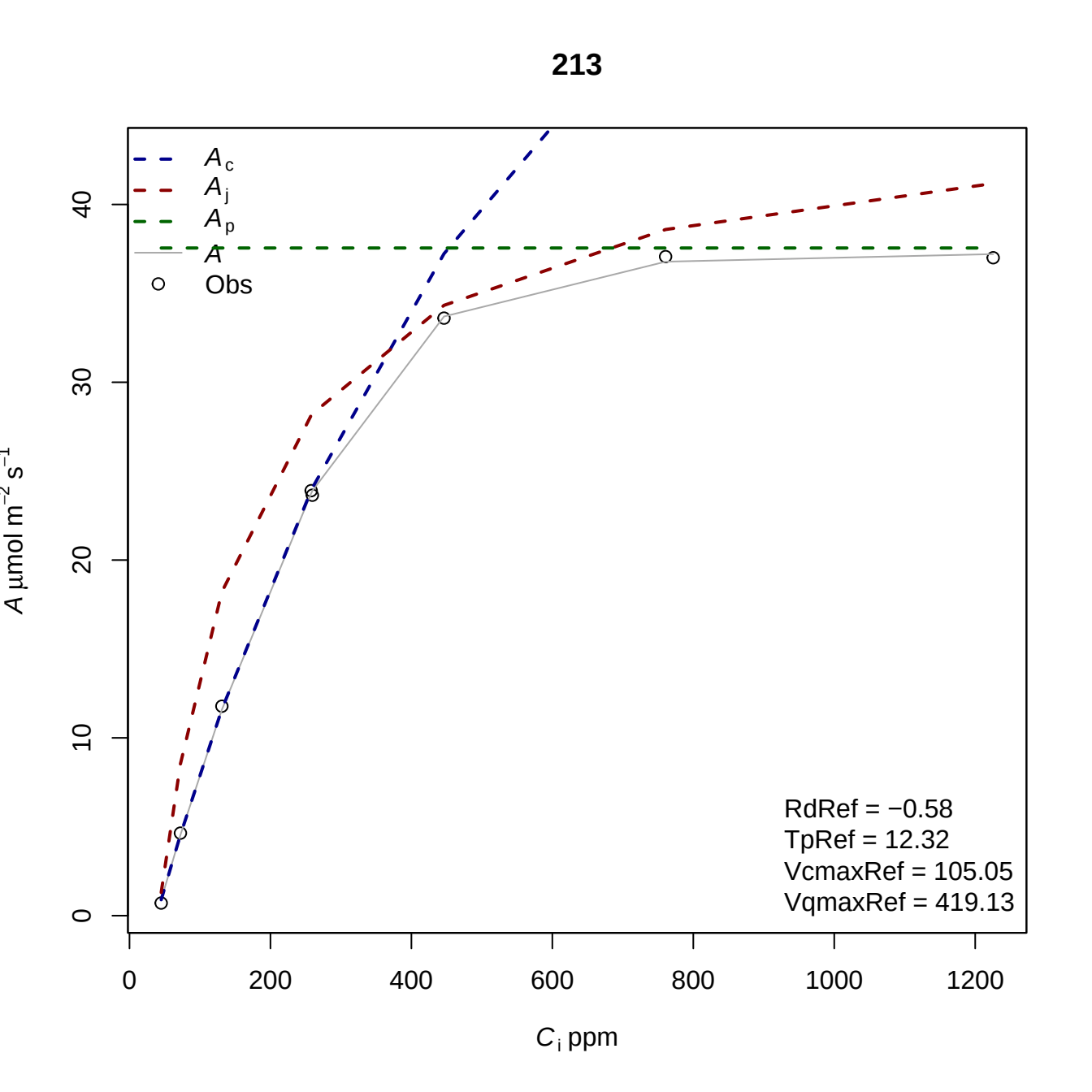

208
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.04
TpRef = 9999
VcmaxRef = 48.47
VqmaxRef = 48.63
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
200
400
600
800
1000

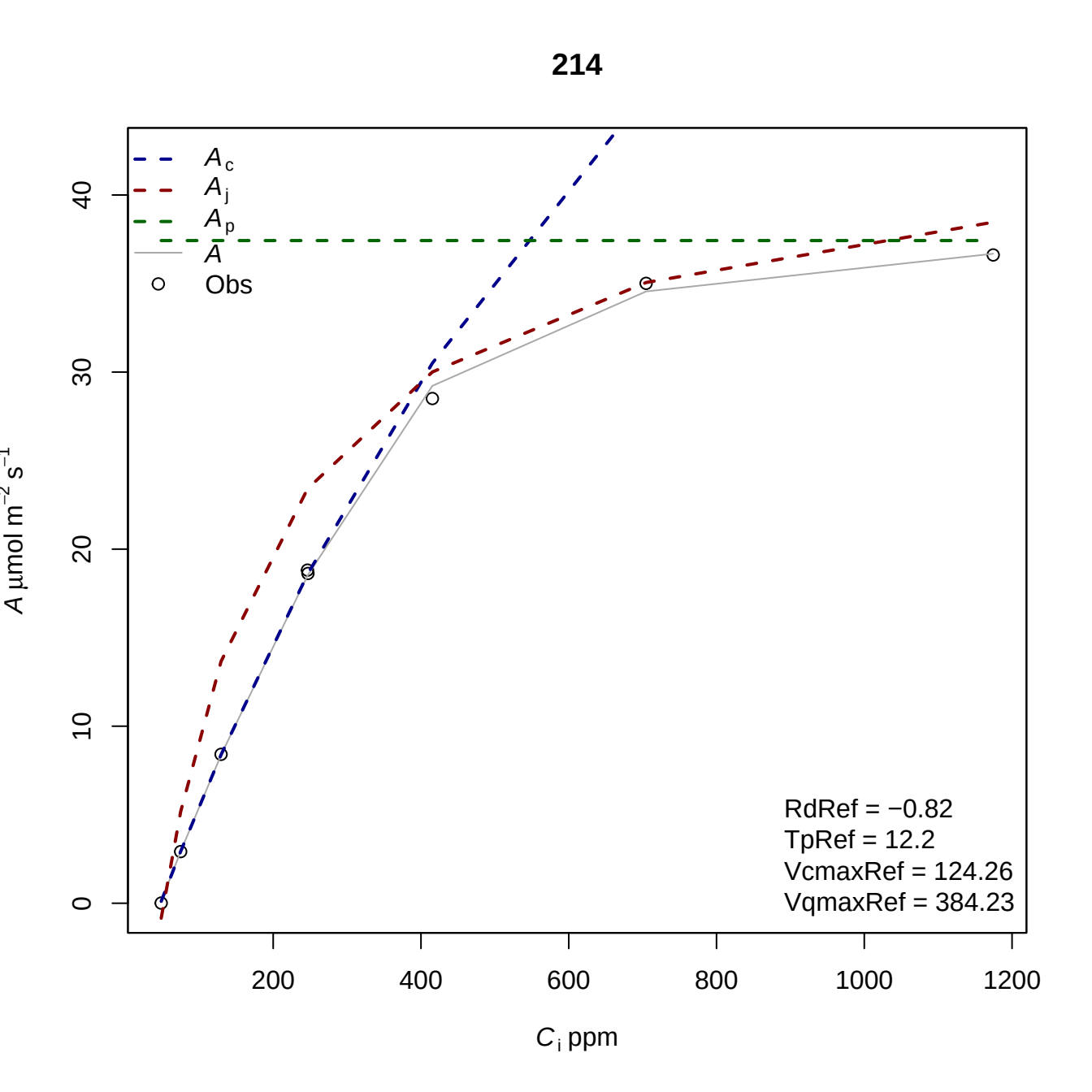

209
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.17
TpRef = 2.36
VcmaxRef = 15.04
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
2
4
6
8

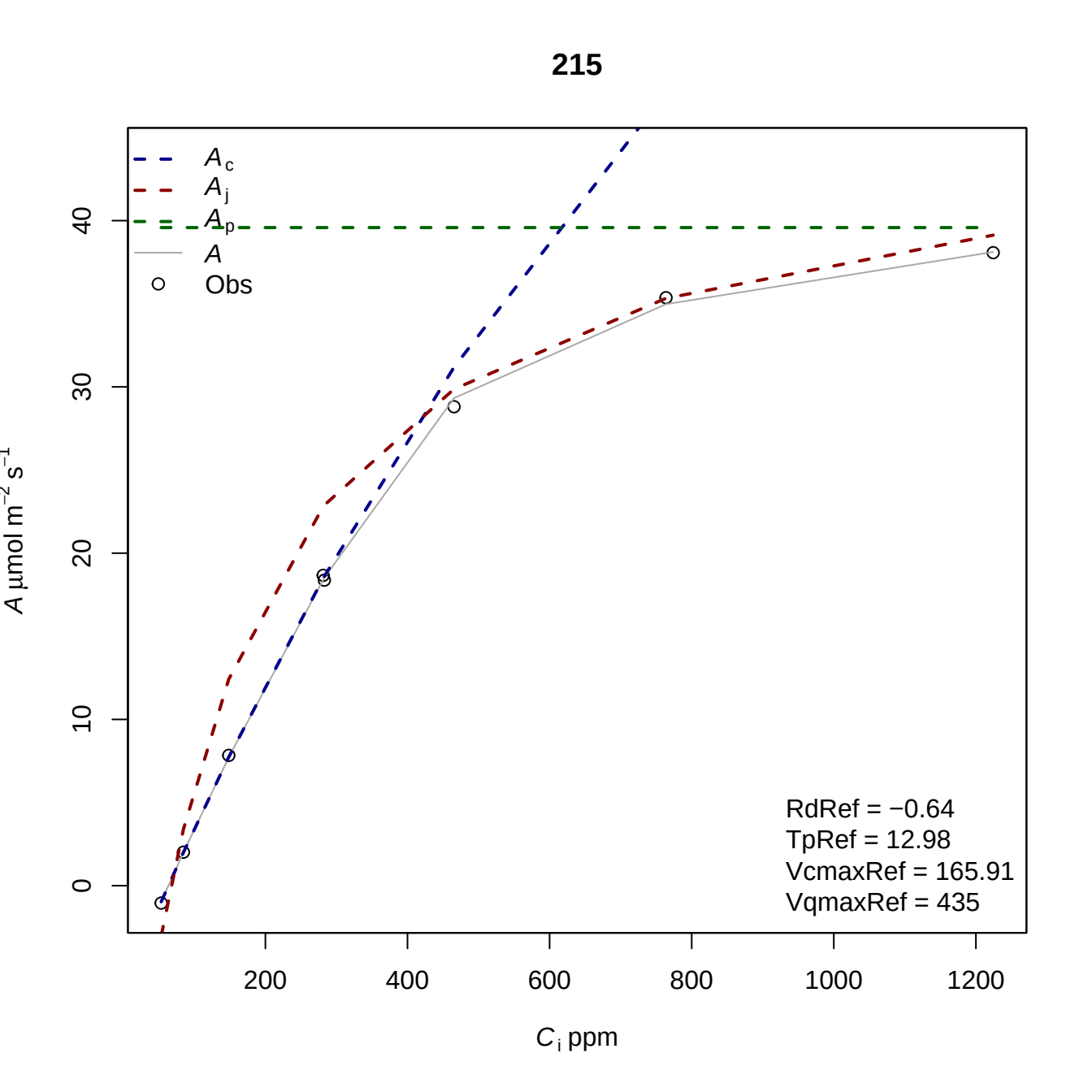

210
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.26
TpRef = 3.13
VcmaxRef = 21.8
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
1400
2
4
6
8
10
12

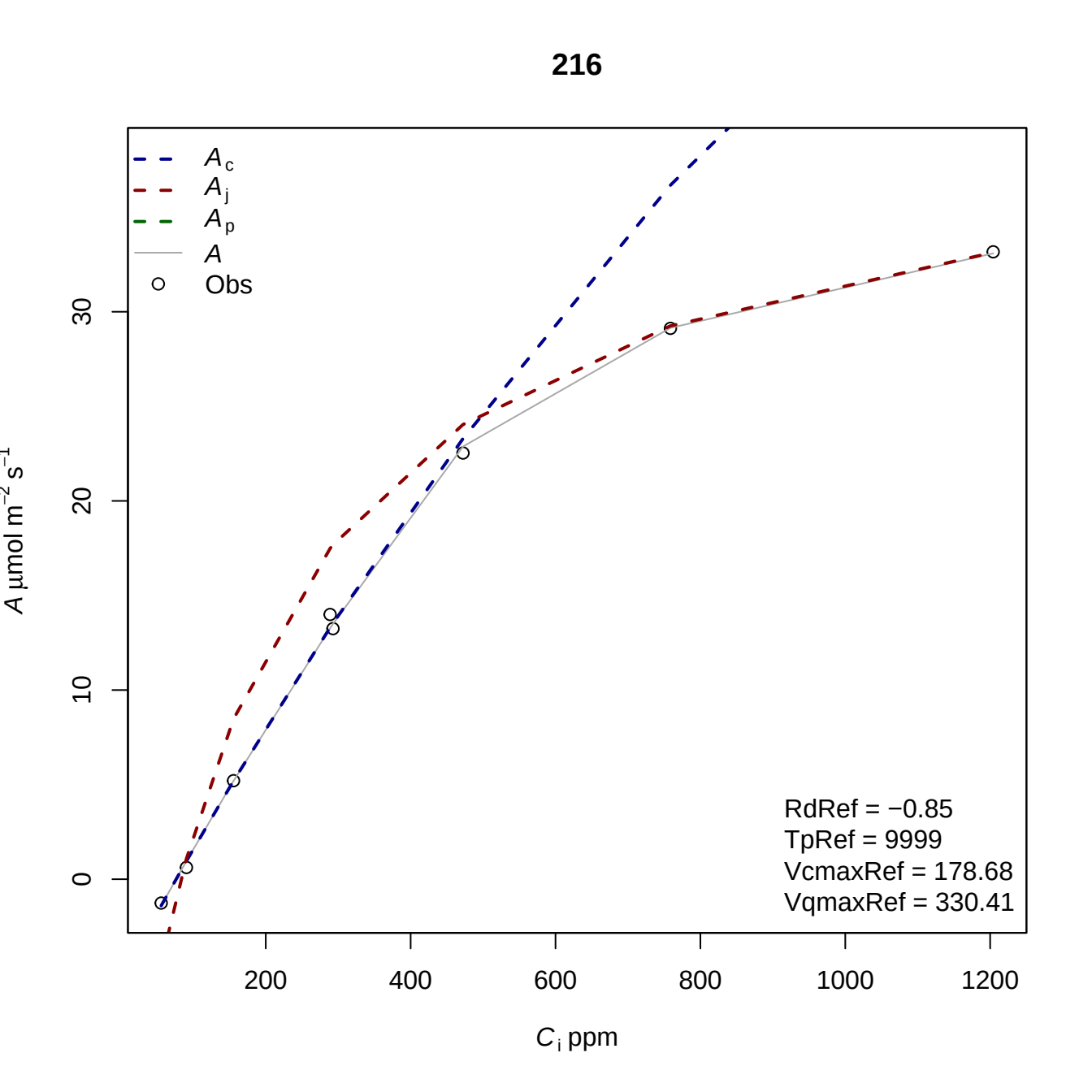

211
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.09
TpRef = 4.94
VcmaxRef = 48.52
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

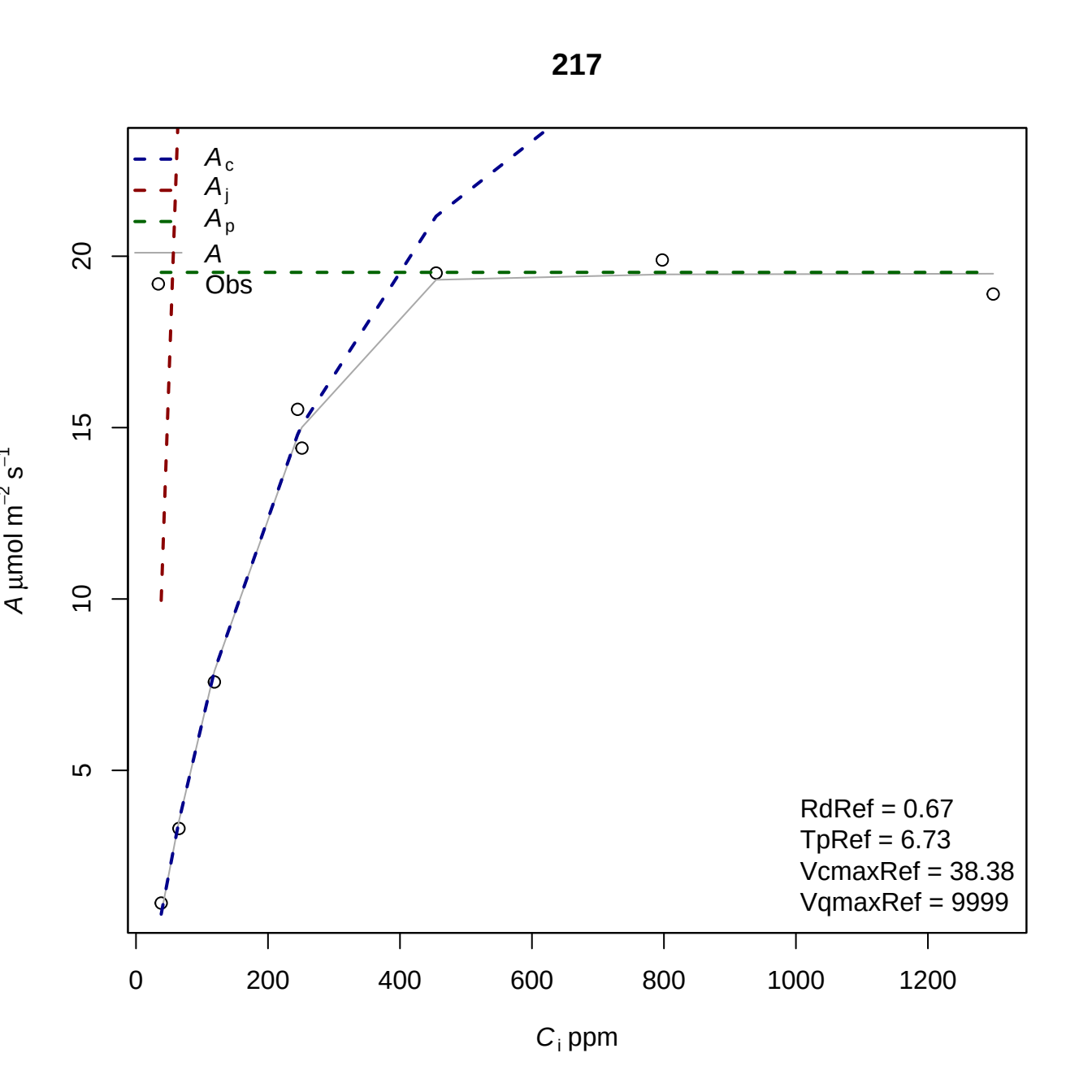

212
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.8
TpRef = 9.13
VcmaxRef = 63.3
VqmaxRef = 234.53
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

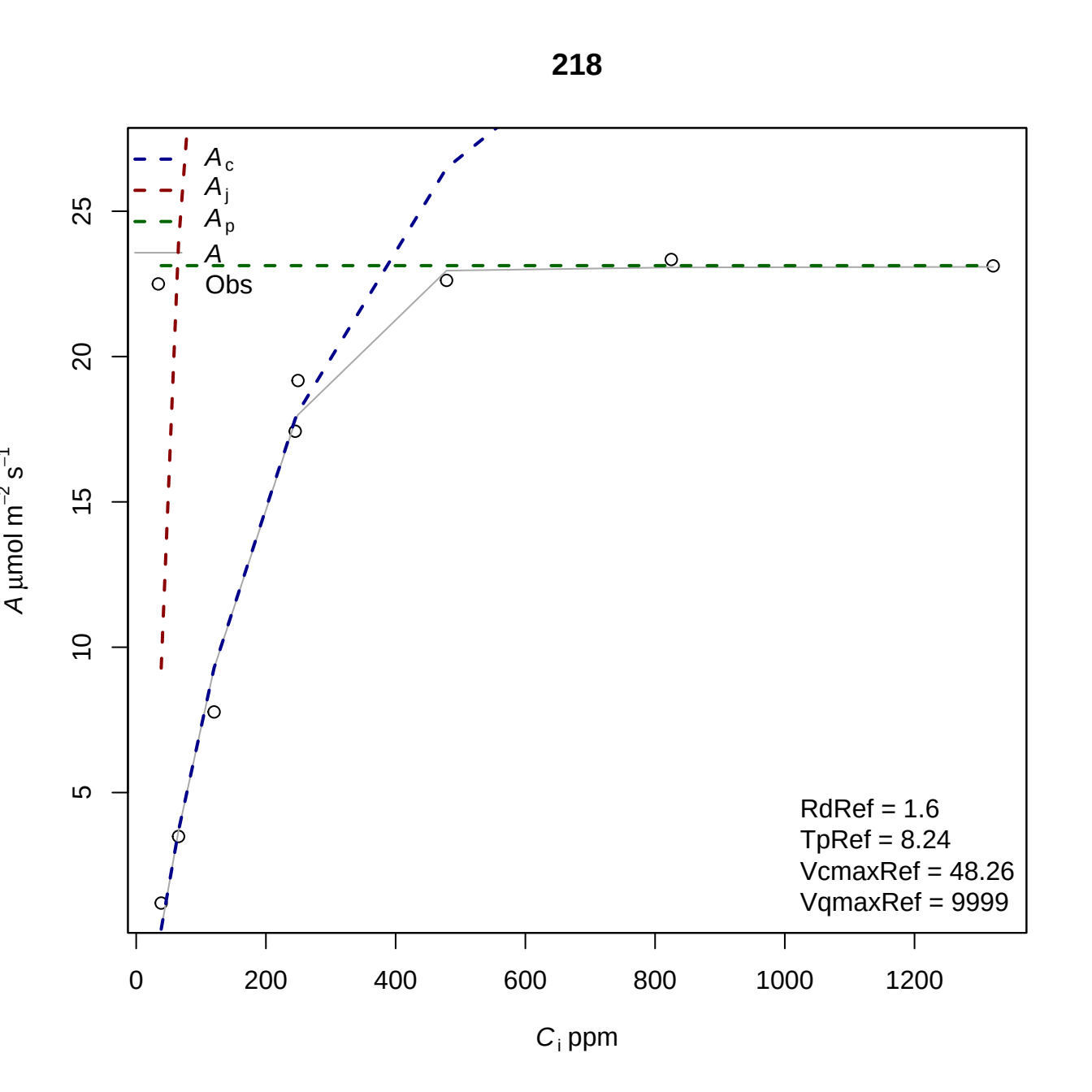

213
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.58
TpRef = 12.32
VcmaxRef = 105.05
VqmaxRef = 419.13
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

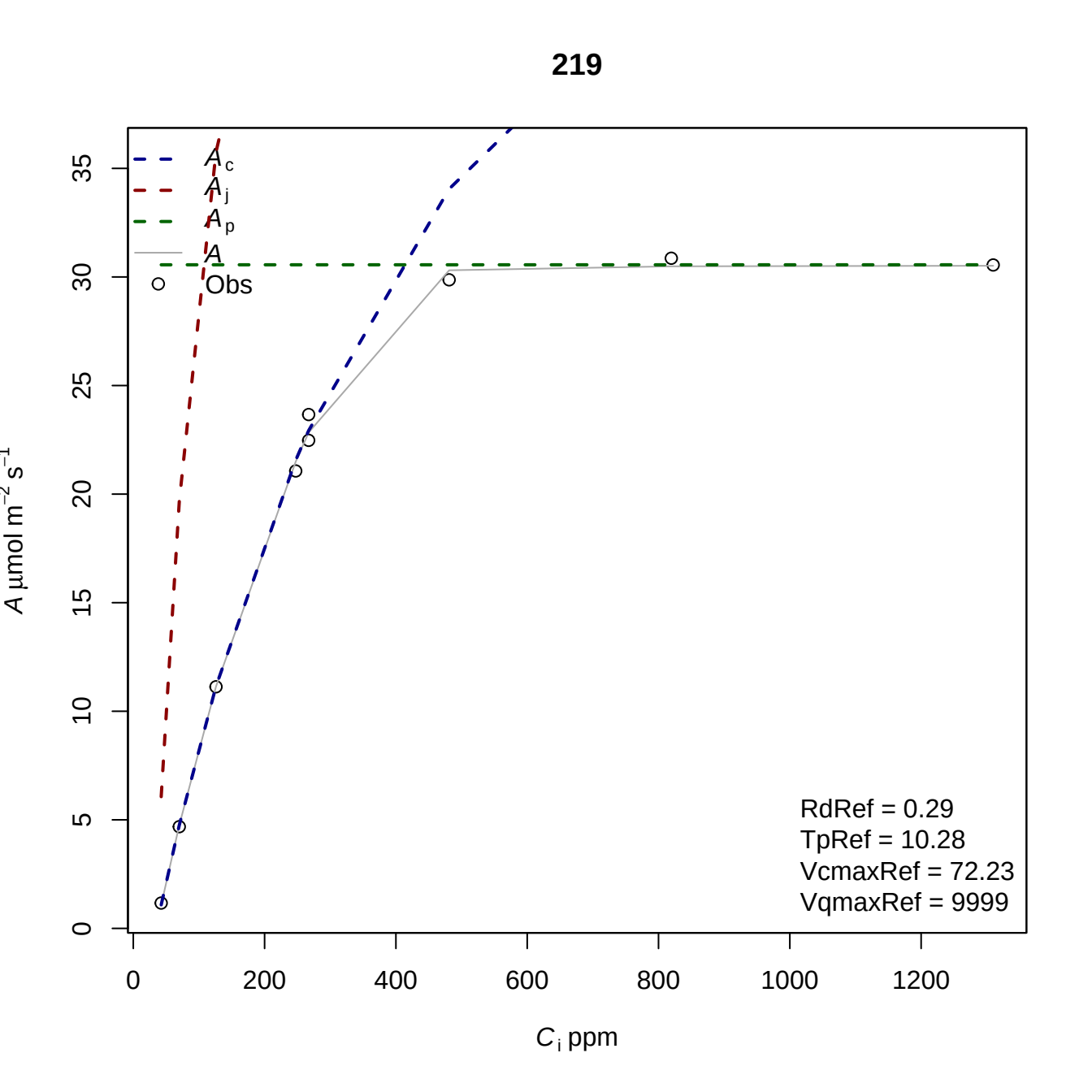

214
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.82
TpRef = 12.2
VcmaxRef = 124.26
VqmaxRef = 384.23
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

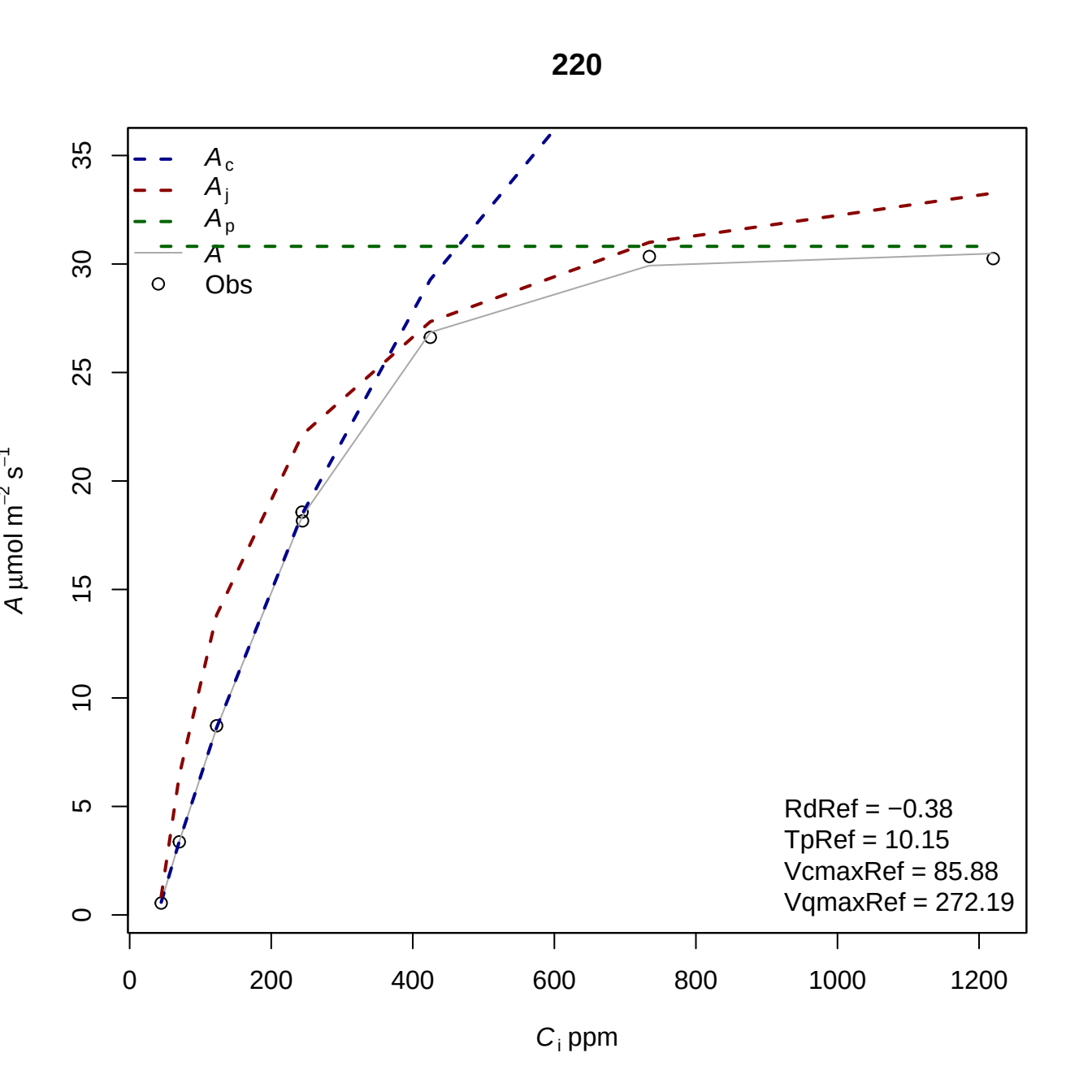

215
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.64
TpRef = 12.98
VcmaxRef = 165.91
VqmaxRef = 435
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

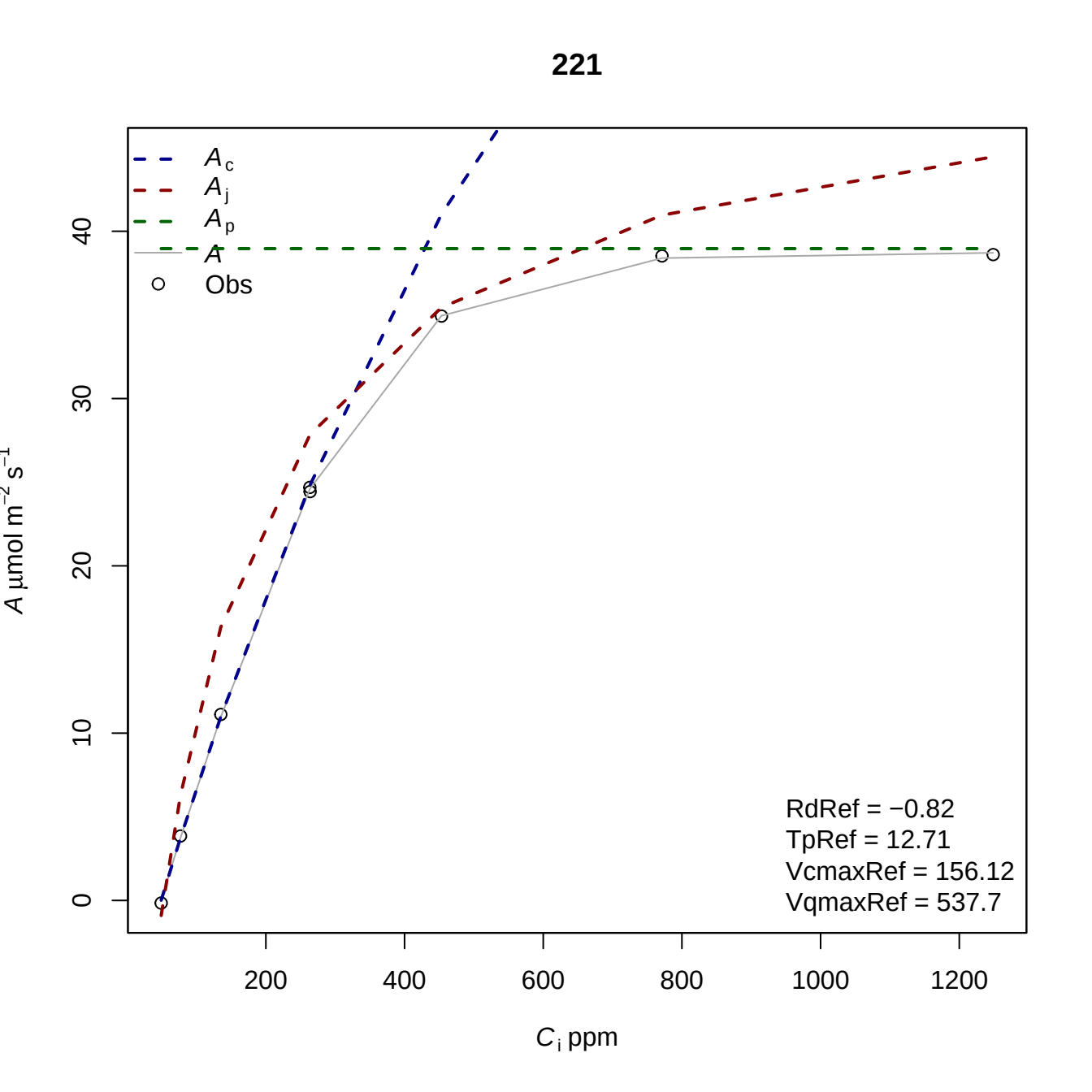

216
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.85
TpRef = 9999
VcmaxRef = 178.68
VqmaxRef = 330.41
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
200
400
600
800
1000
1200

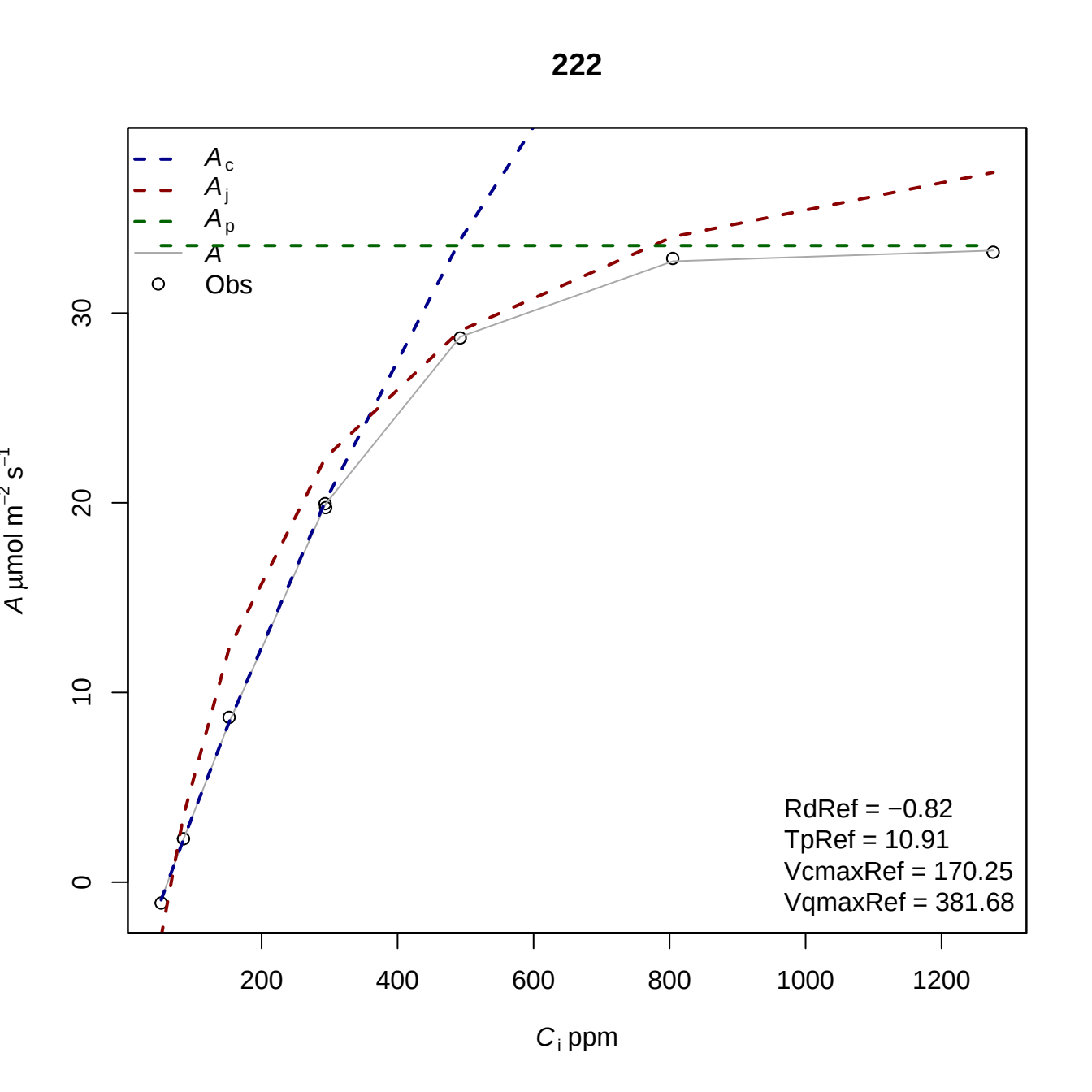

217
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.67
TpRef = 6.73
VcmaxRef = 38.38
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20

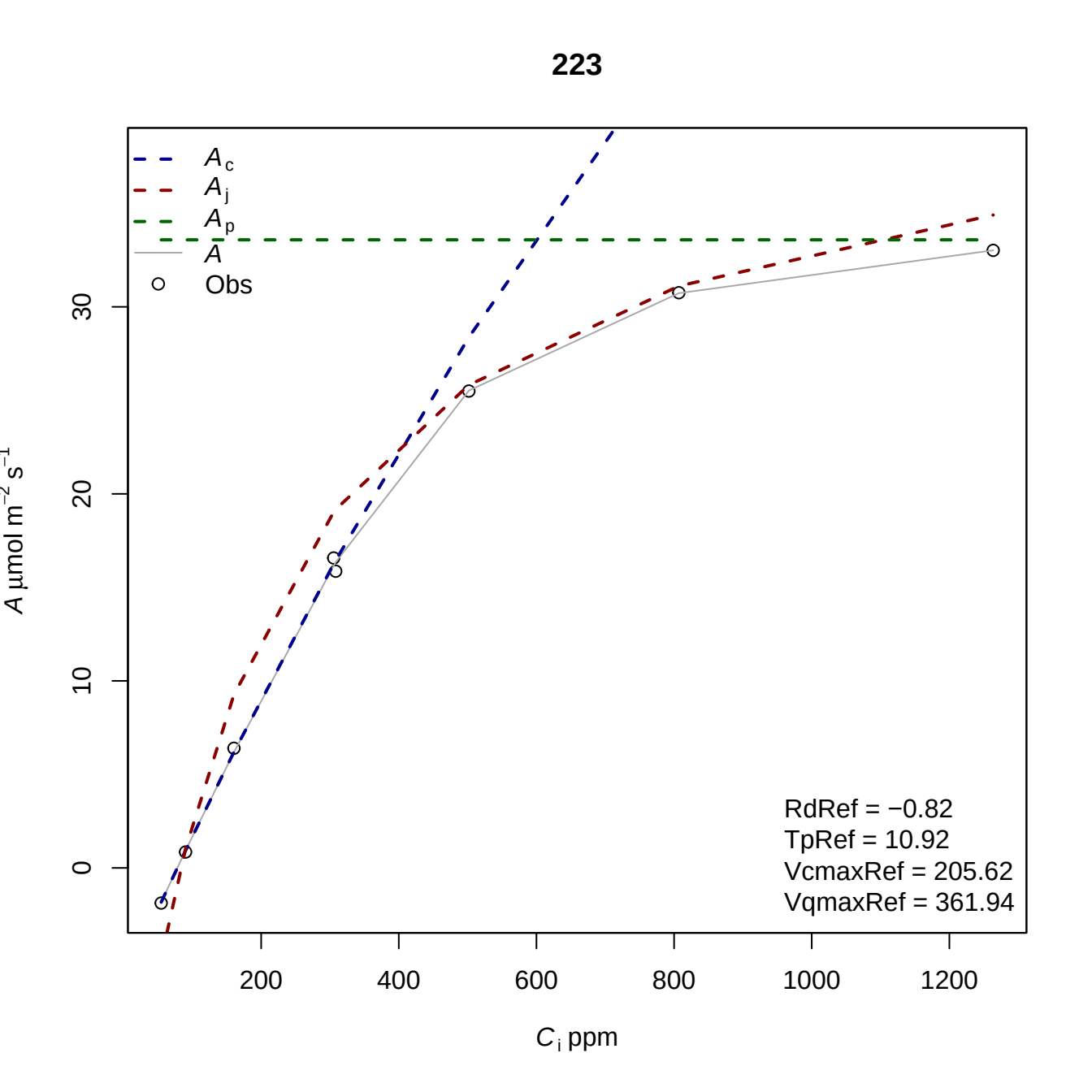

218
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.6
TpRef = 8.24
VcmaxRef = 48.26
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

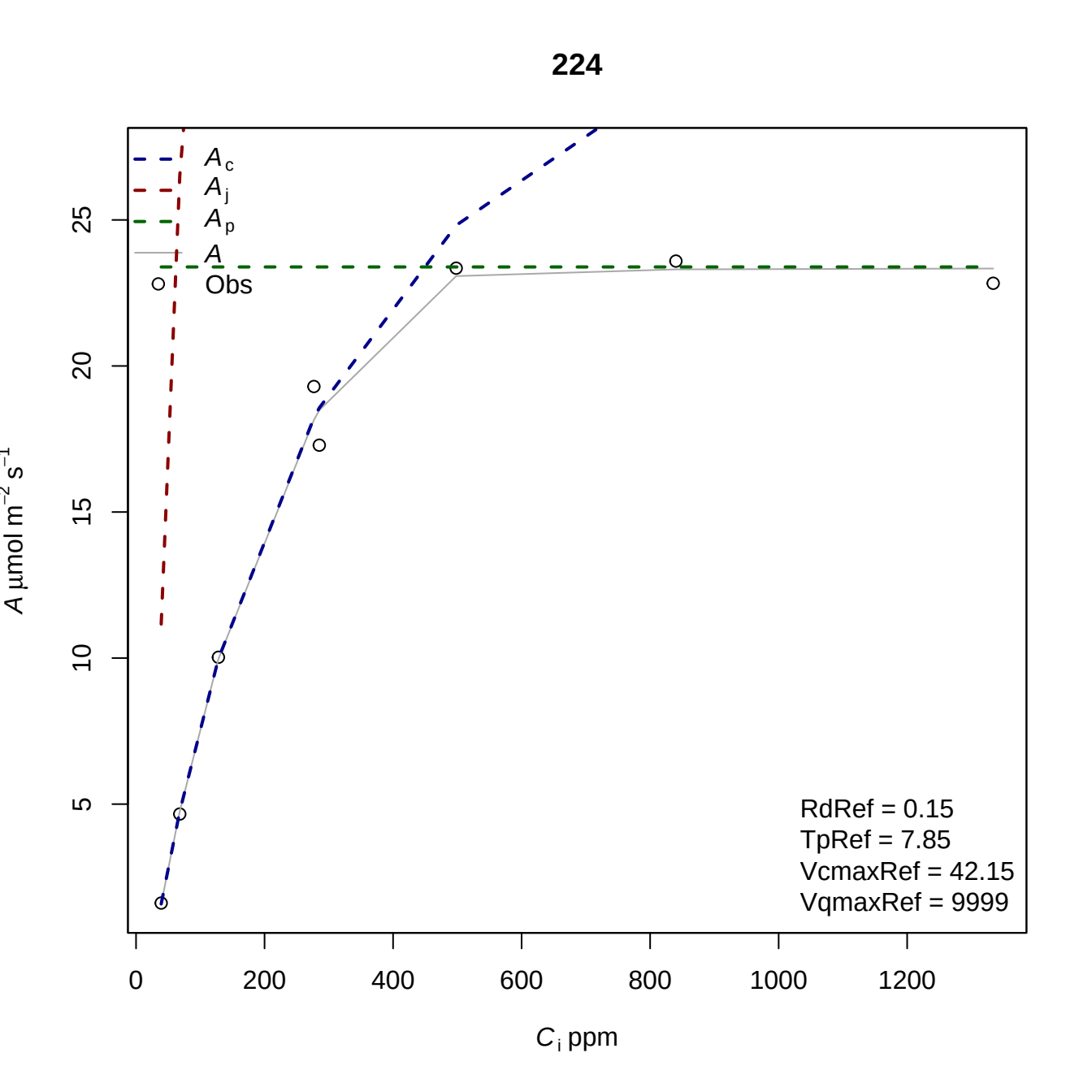

219
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.29
TpRef = 10.28
VcmaxRef = 72.23
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

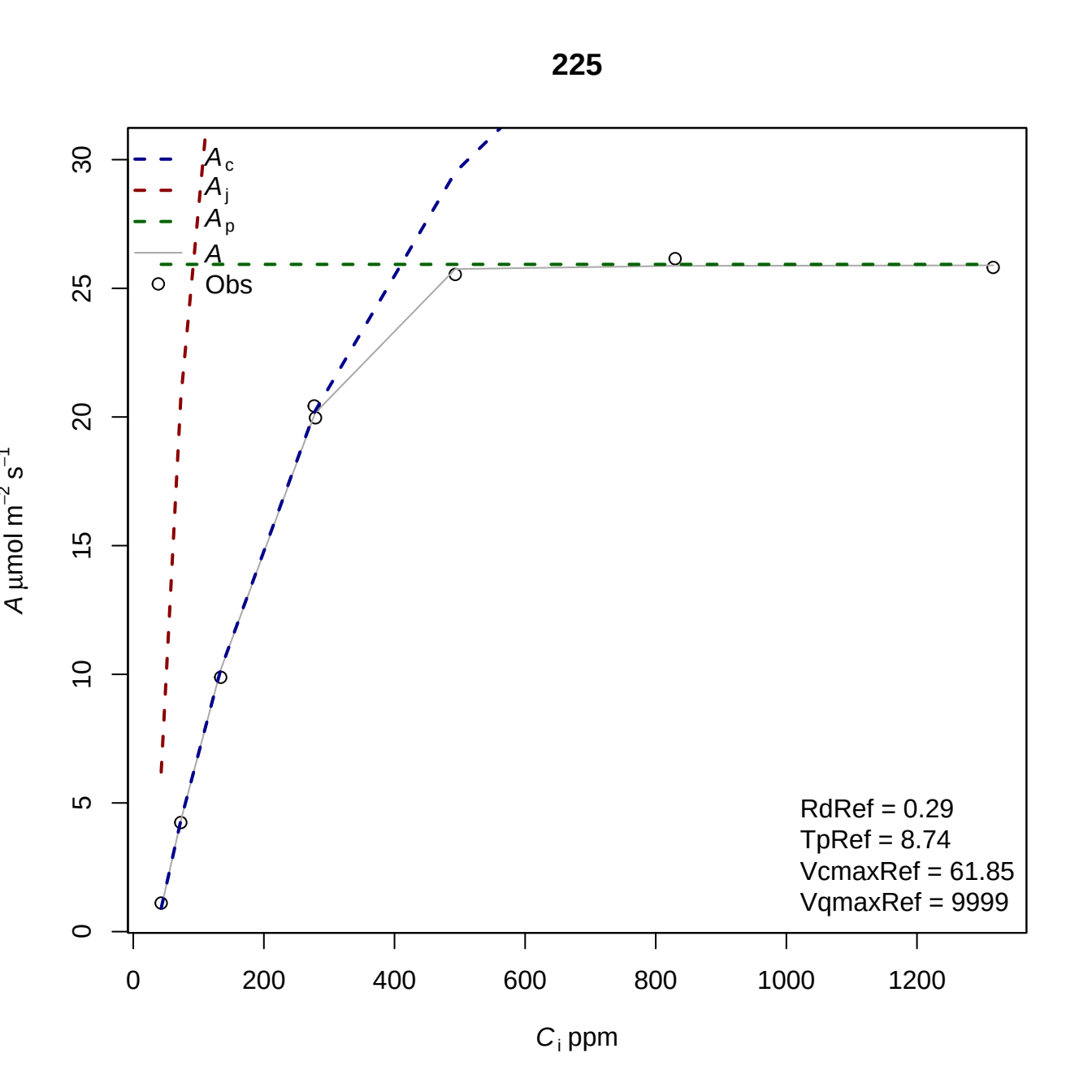

220
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.38
TpRef = 10.15
VcmaxRef = 85.88
VqmaxRef = 272.19
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

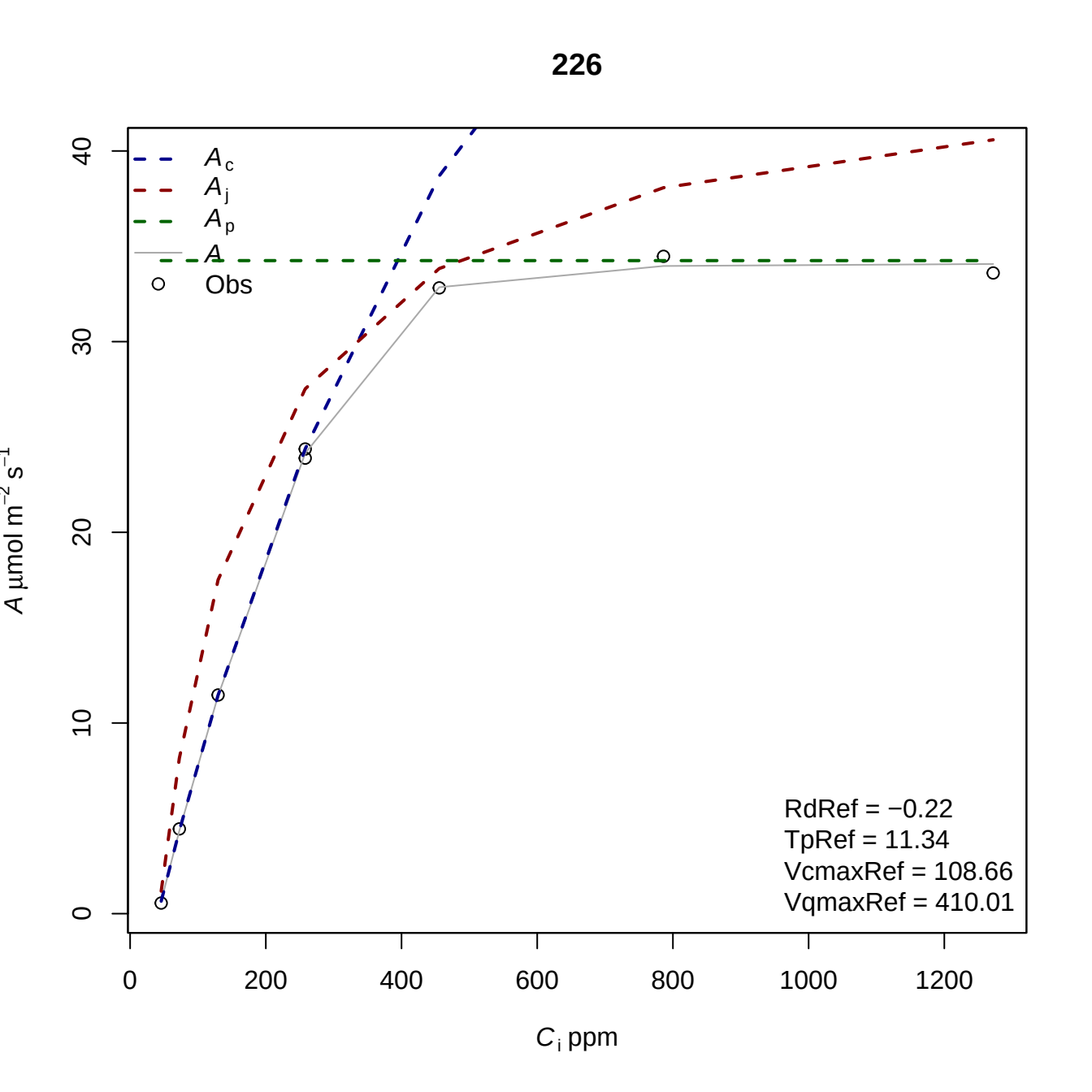

221
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.82
TpRef = 12.71
VcmaxRef = 156.12
VqmaxRef = 537.7
0
10
20
30
40
200
400
600
800
1000
1200
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

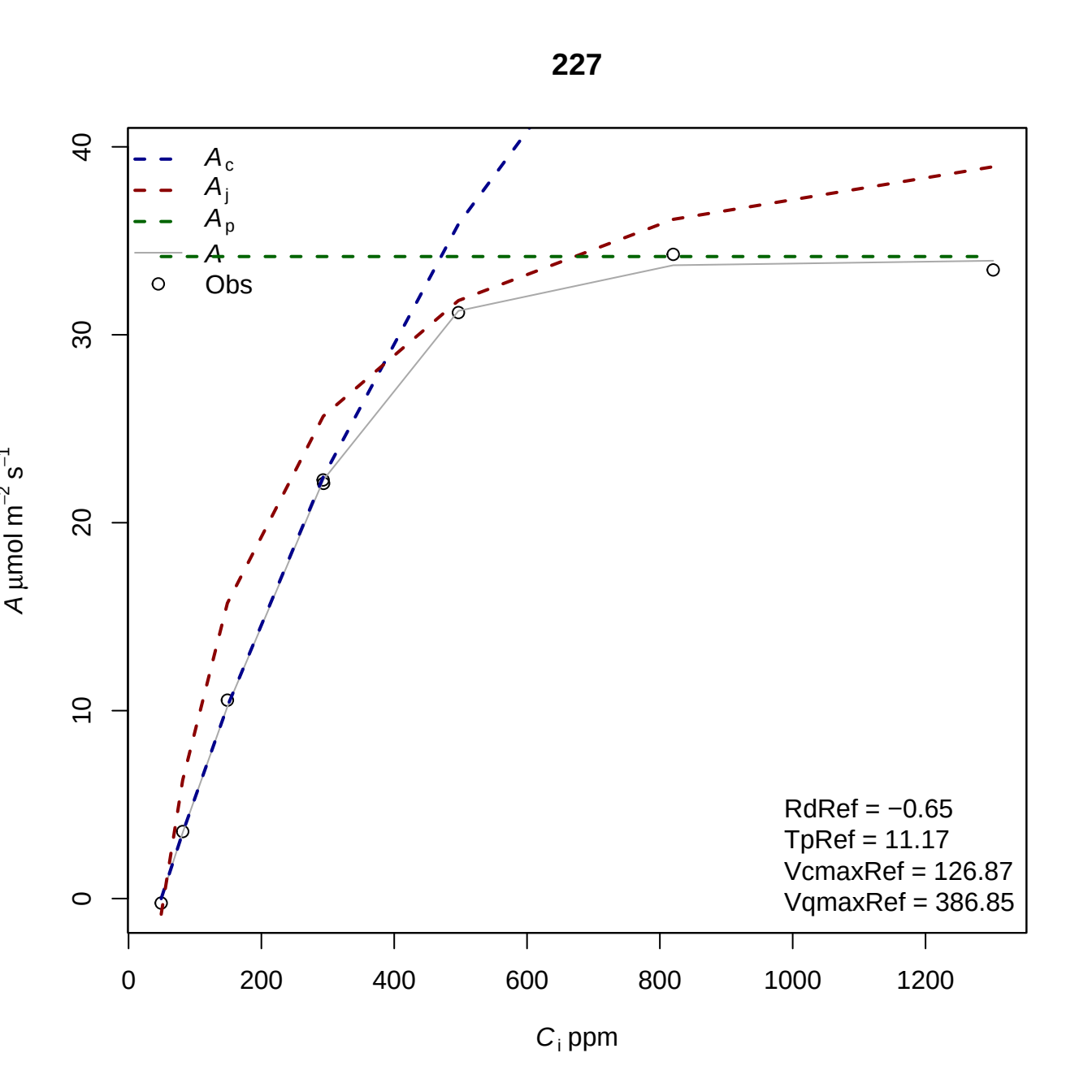

222
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.82
TpRef = 10.91
VcmaxRef = 170.25
VqmaxRef = 381.68
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
200
400
600
800
1000
1200

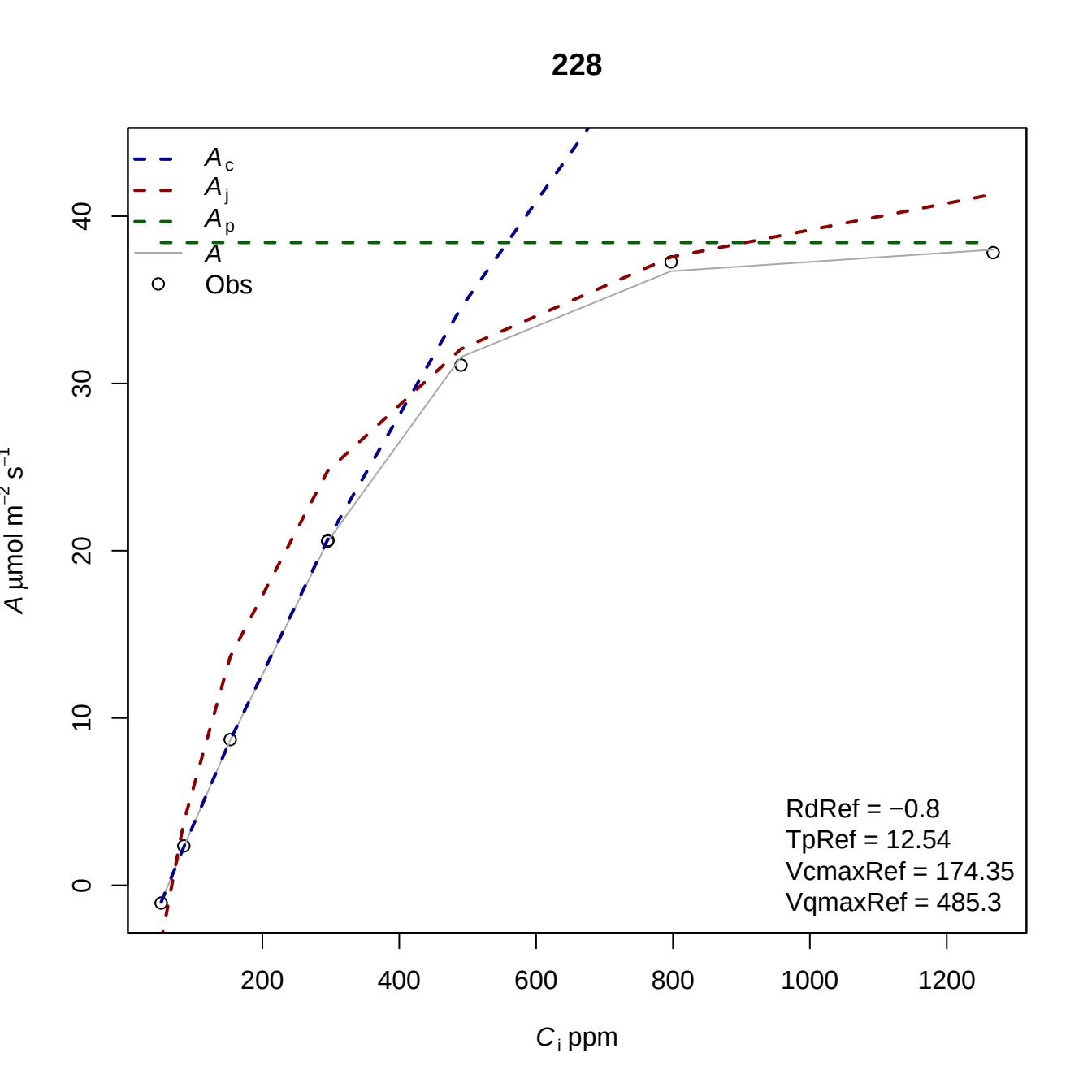

223
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.82
TpRef = 10.92
VcmaxRef = 205.62
VqmaxRef = 361.94
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
200
400
600
800
1000
1200

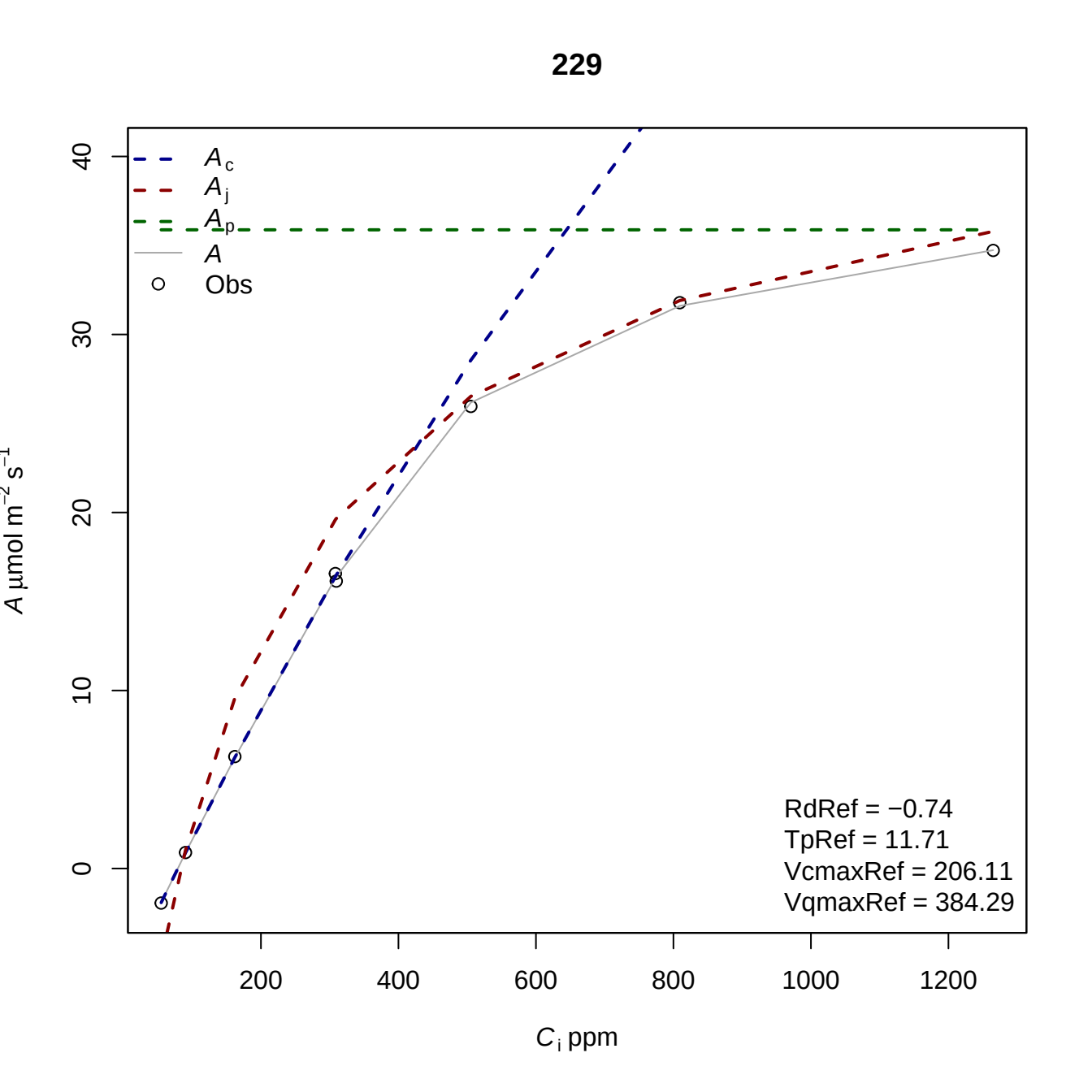

224
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.15
TpRef = 7.85
VcmaxRef = 42.15
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

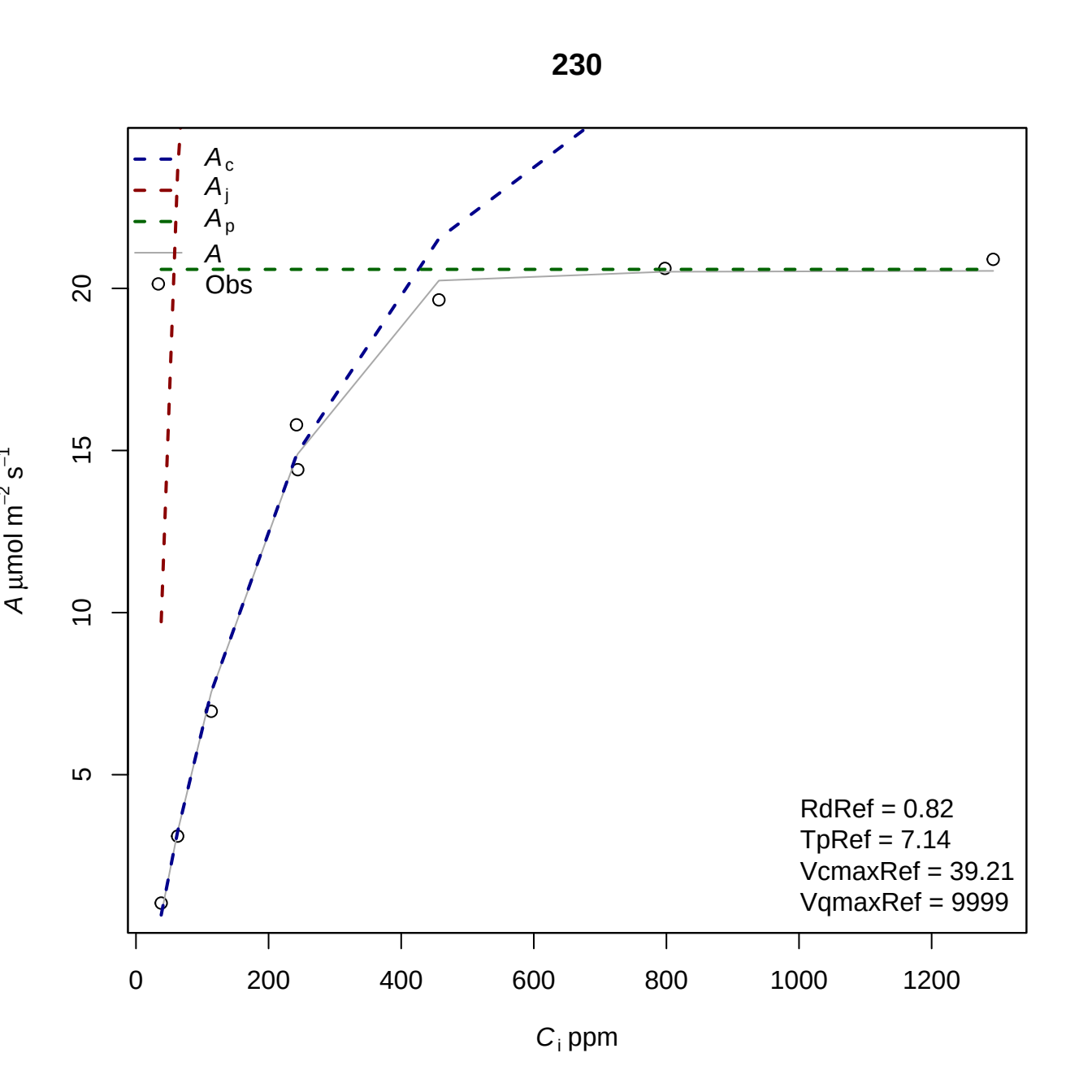

225
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.29
TpRef = 8.74
VcmaxRef = 61.85
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

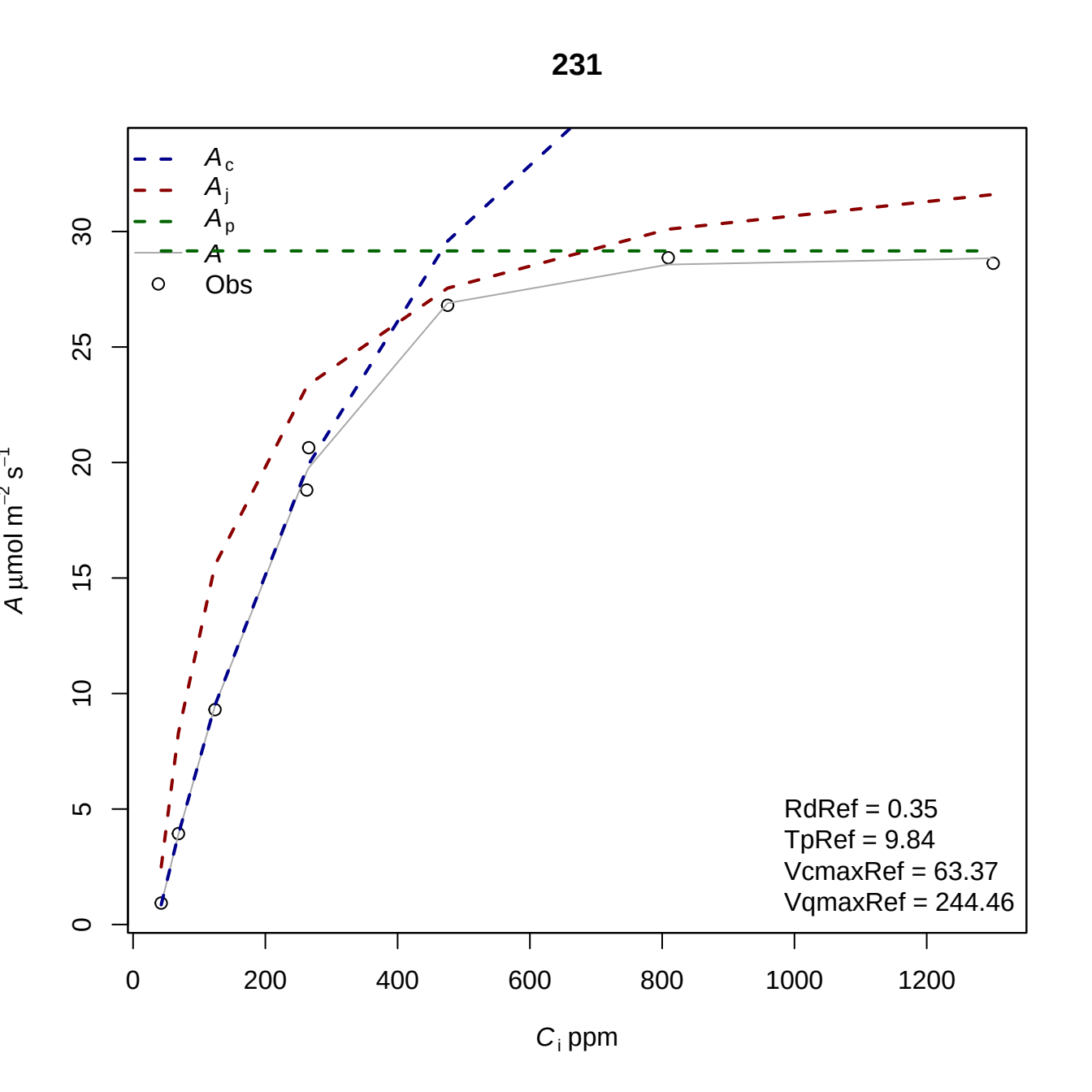

226
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.22
TpRef = 11.34
VcmaxRef = 108.66
VqmaxRef = 410.01
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

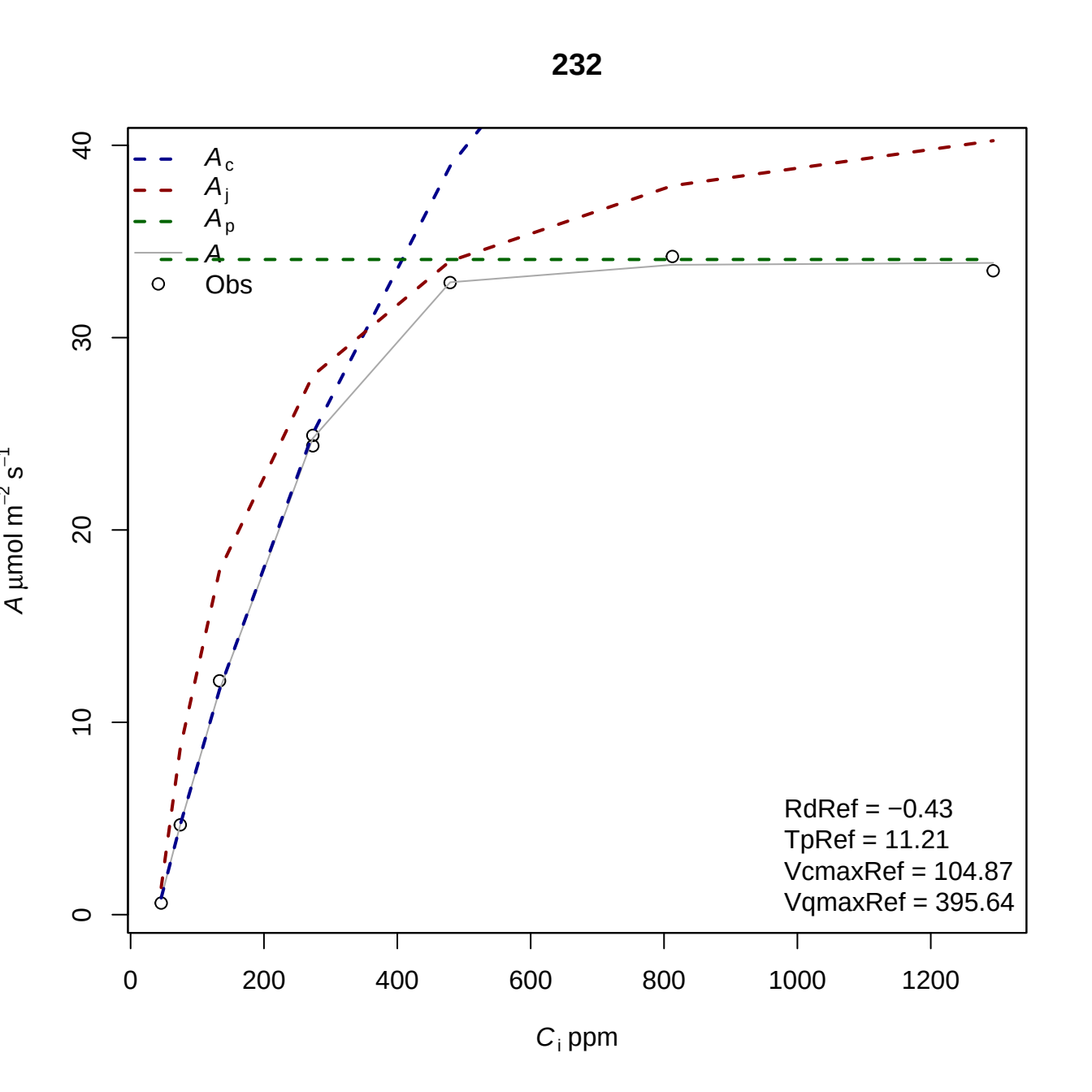

227
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.65
TpRef = 11.17
VcmaxRef = 126.87
VqmaxRef = 386.85
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

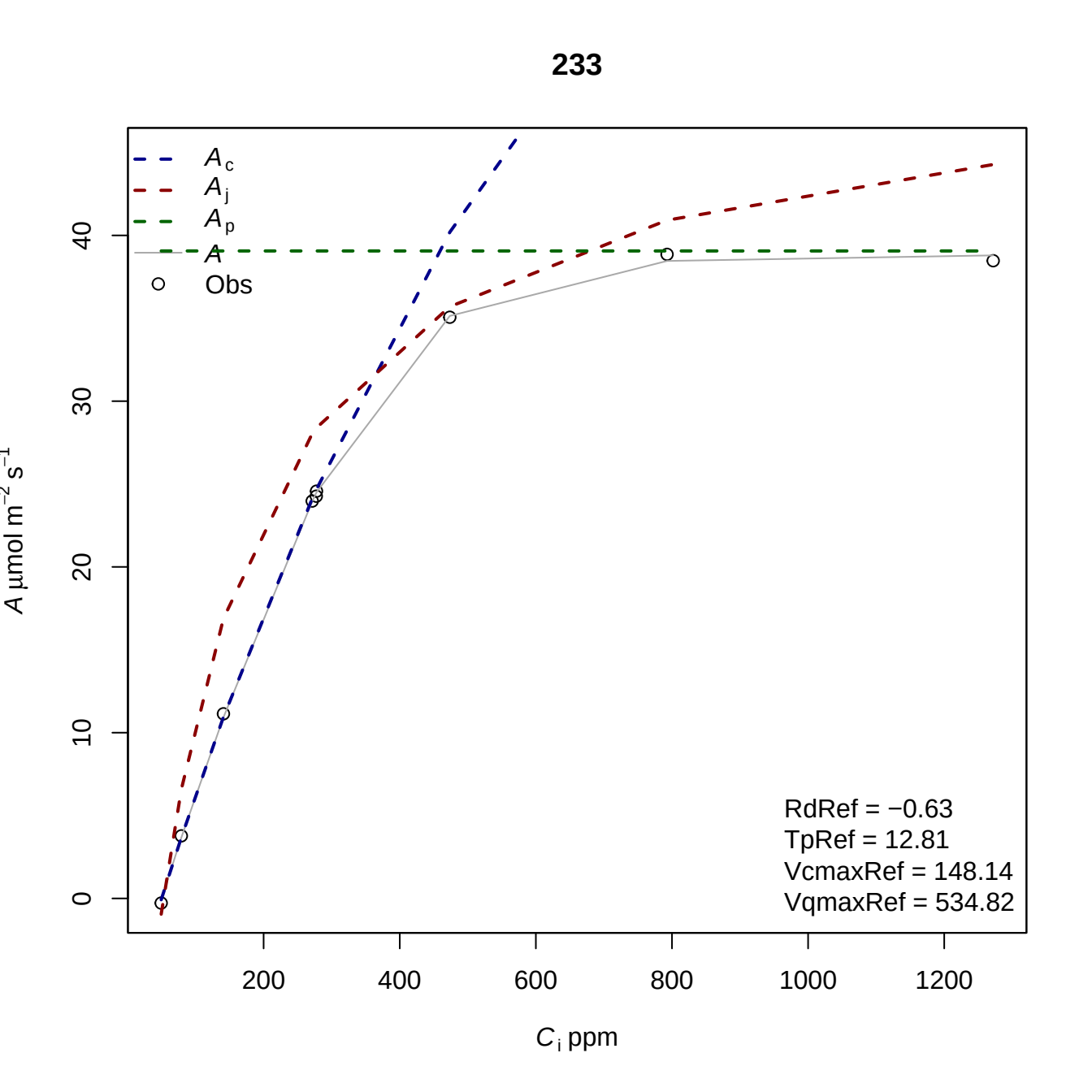

228
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.8
TpRef = 12.54
VcmaxRef = 174.35
VqmaxRef = 485.3
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

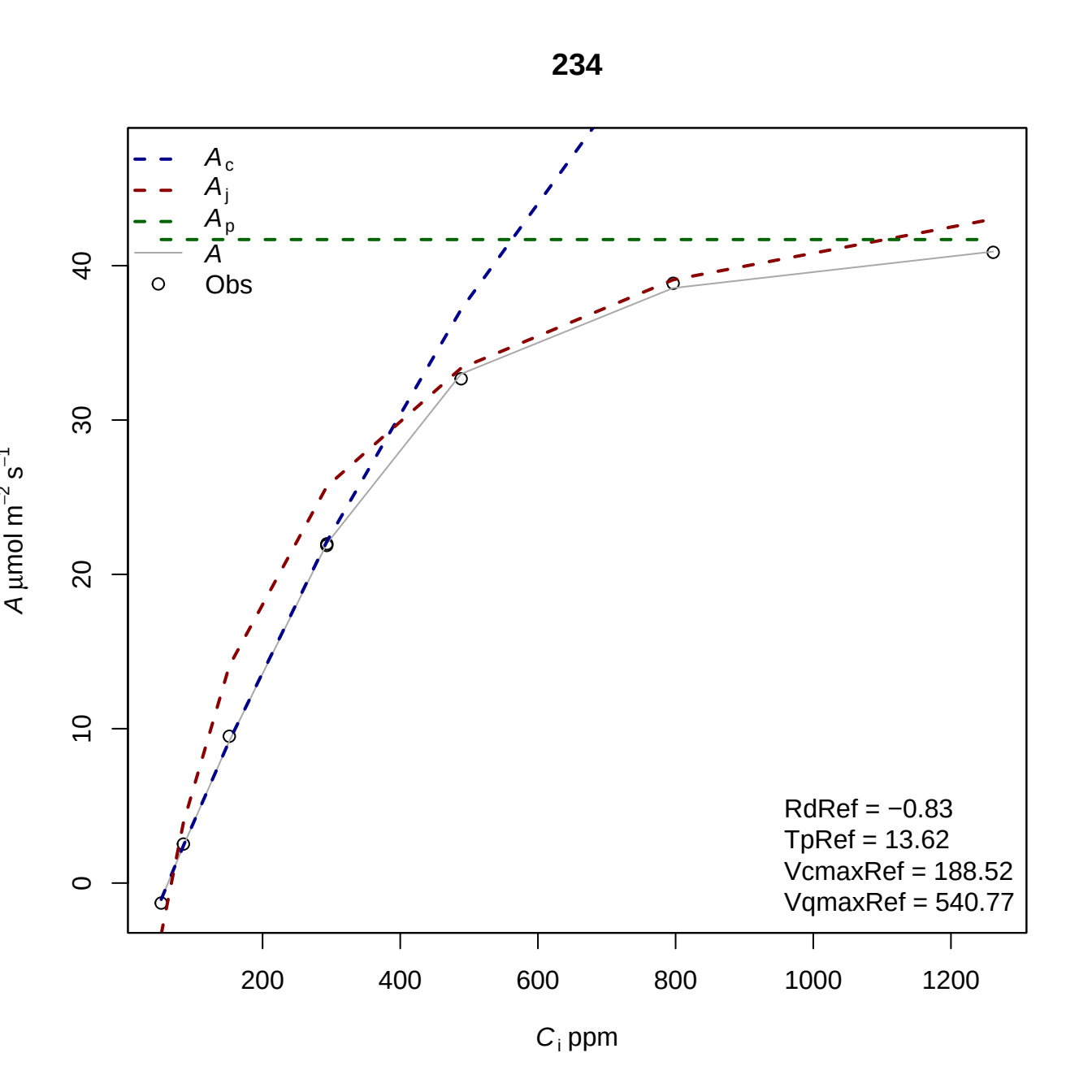

229
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.74
TpRef = 11.71
VcmaxRef = 206.11
VqmaxRef = 384.29
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

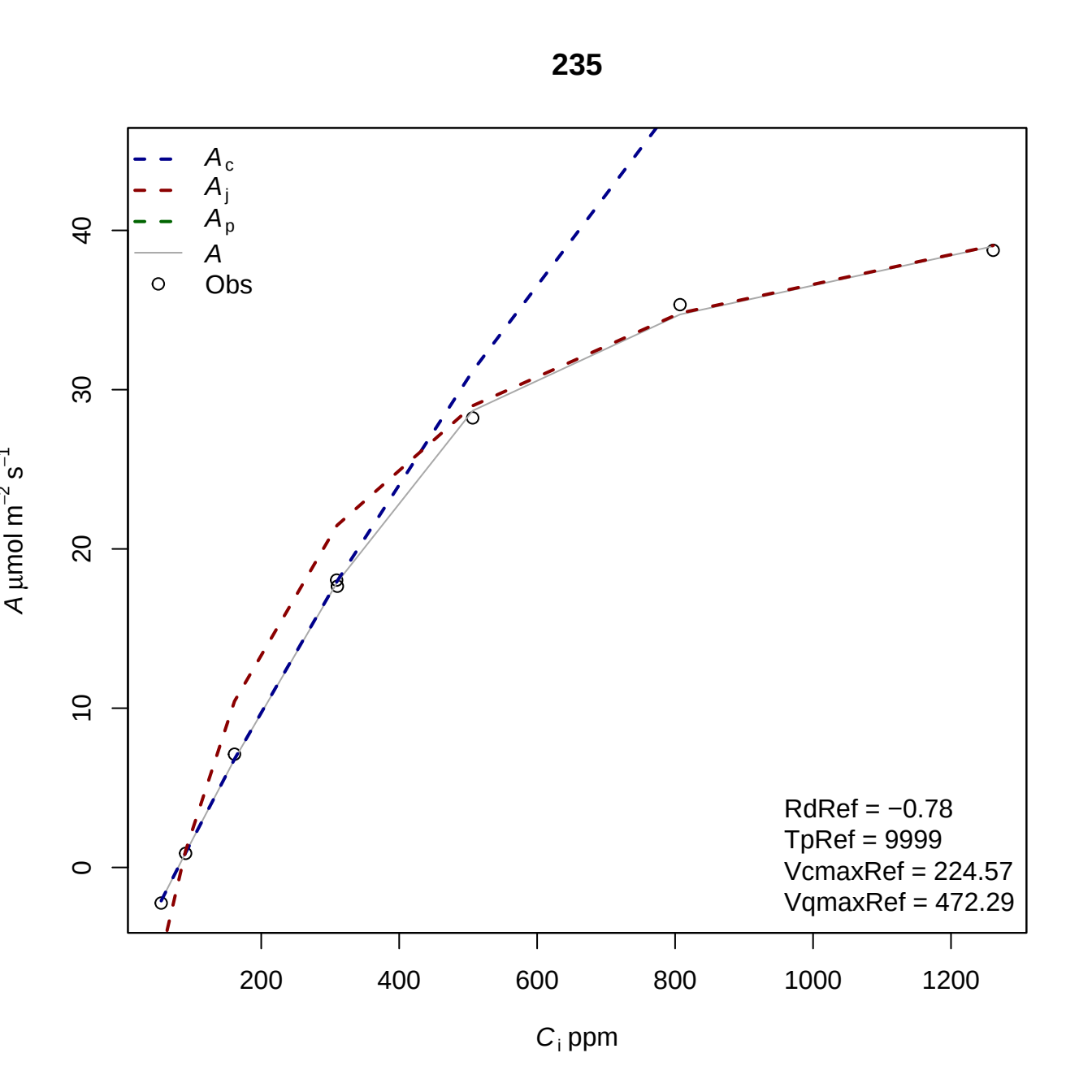

230
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.82
TpRef = 7.14
VcmaxRef = 39.21
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20

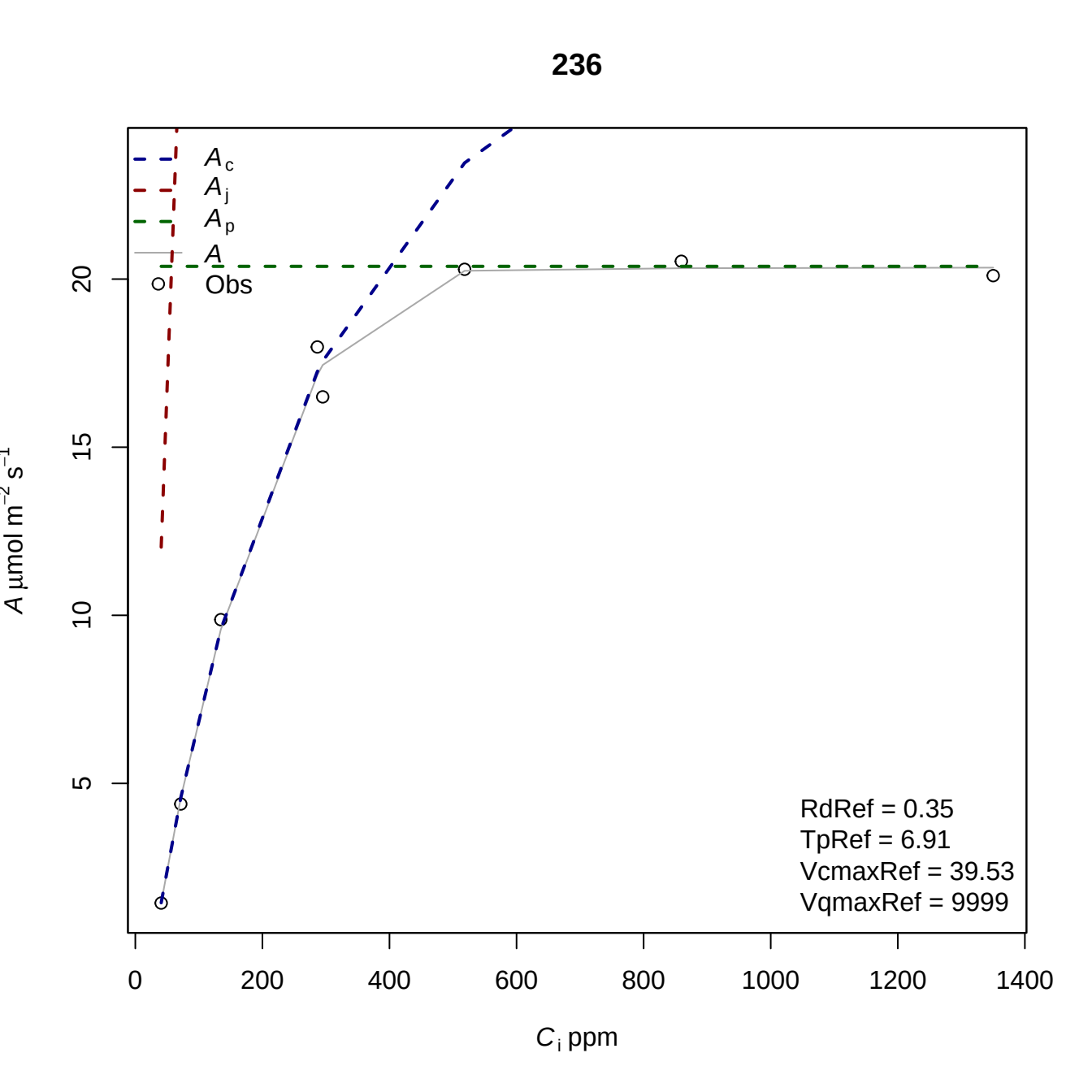

231
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.35
TpRef = 9.84
VcmaxRef = 63.37
VqmaxRef = 244.46
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

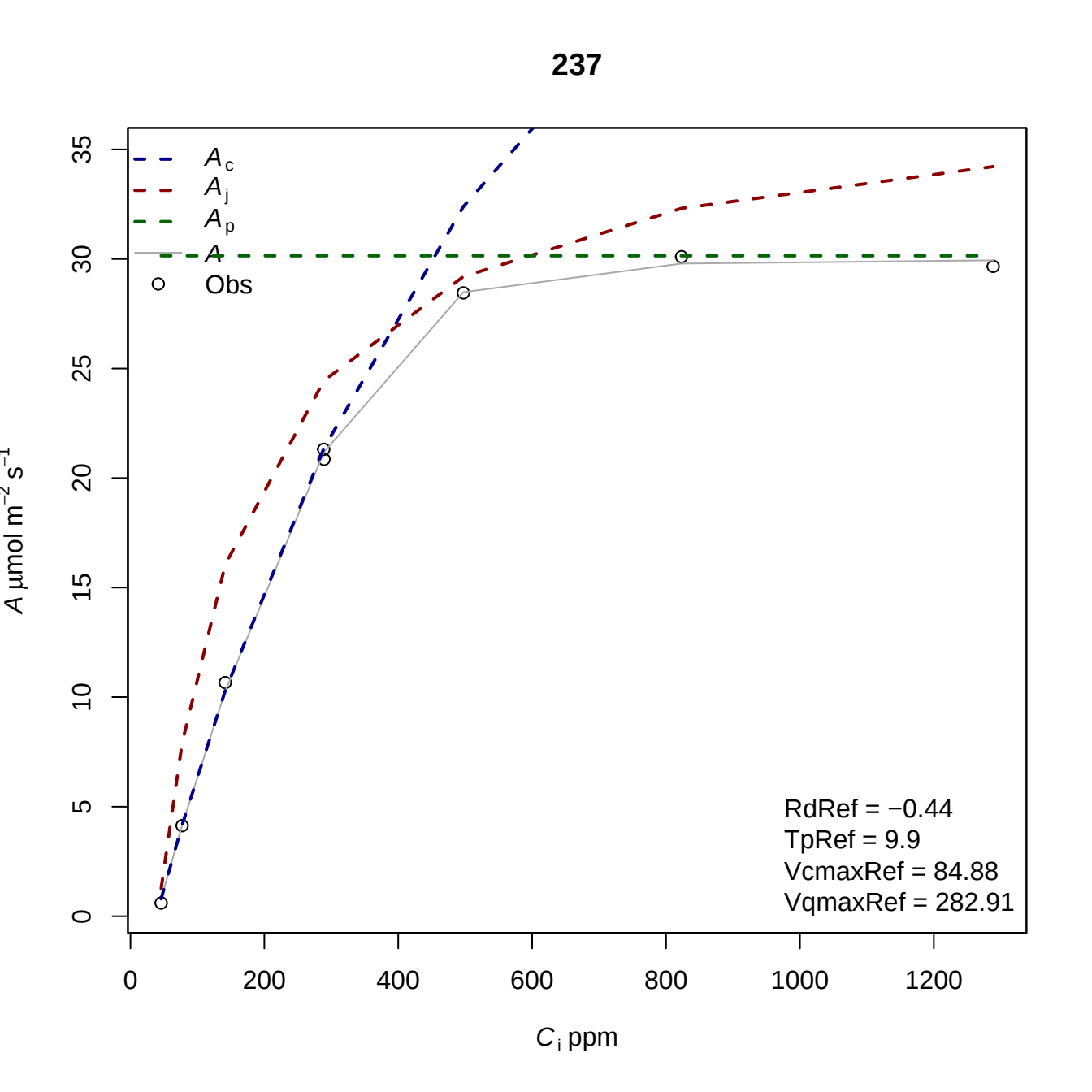

232
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.43
TpRef = 11.21
VcmaxRef = 104.87
VqmaxRef = 395.64
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

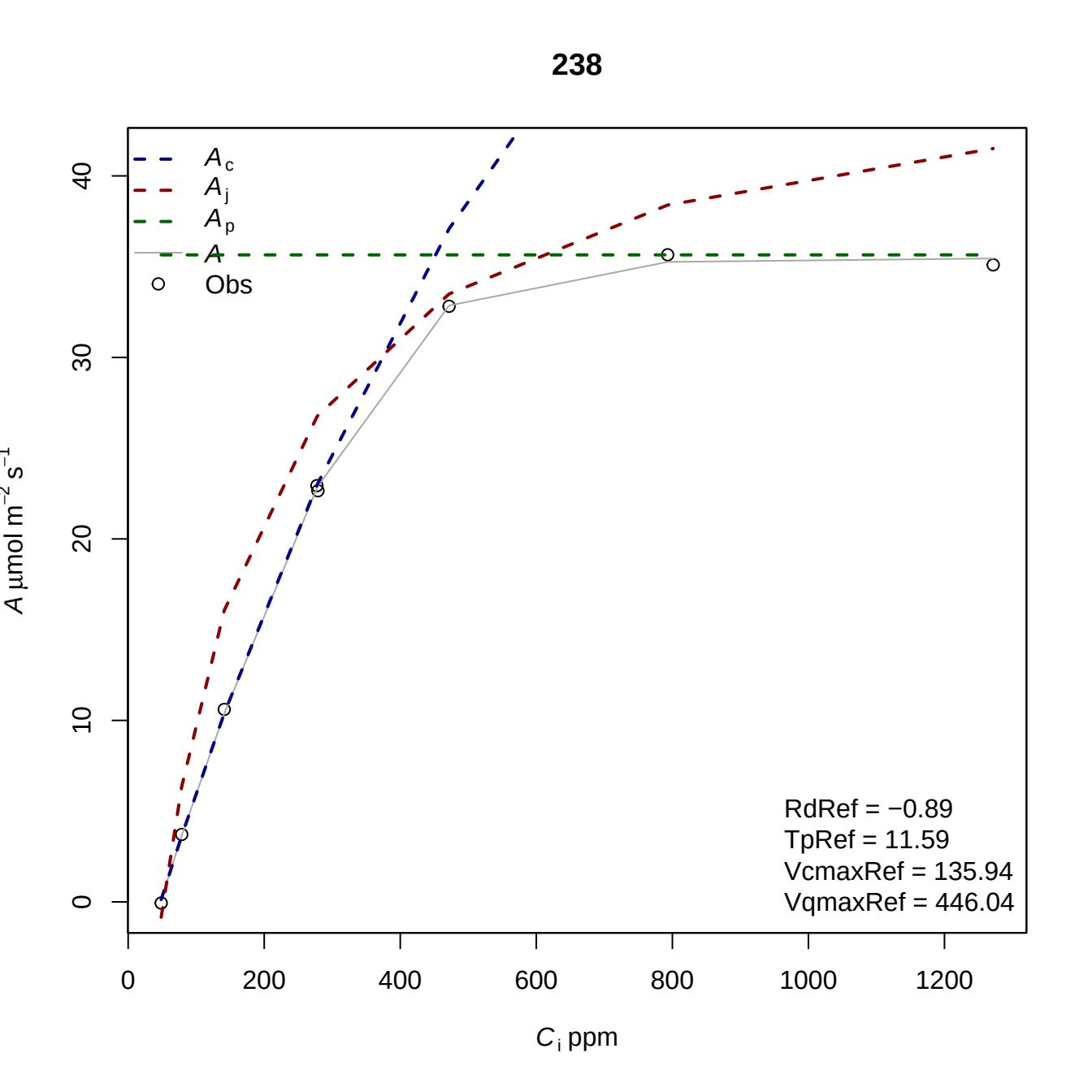

233
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.63
TpRef = 12.81
VcmaxRef = 148.14
VqmaxRef = 534.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

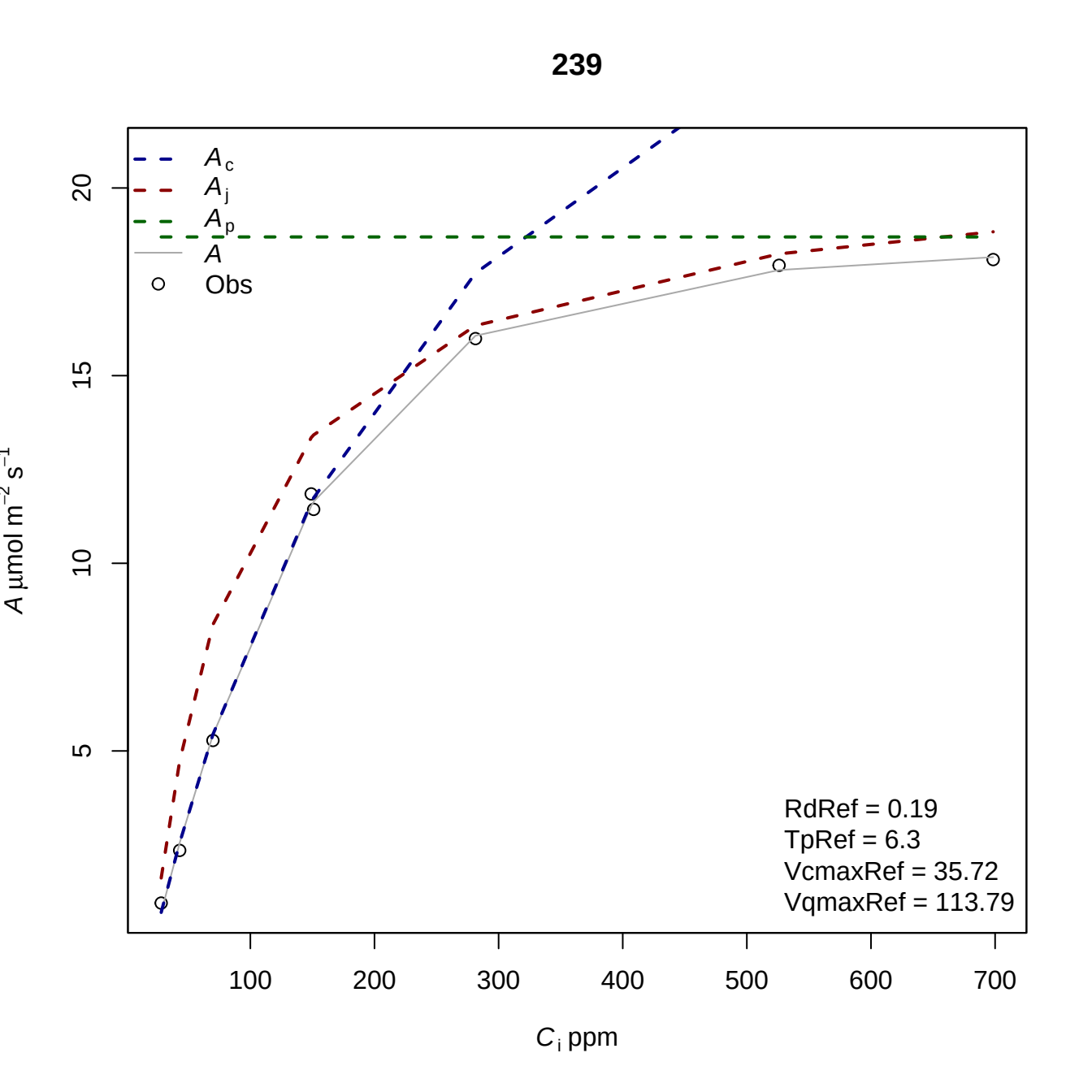

234
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.83
TpRef = 13.62
VcmaxRef = 188.52
VqmaxRef = 540.77
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

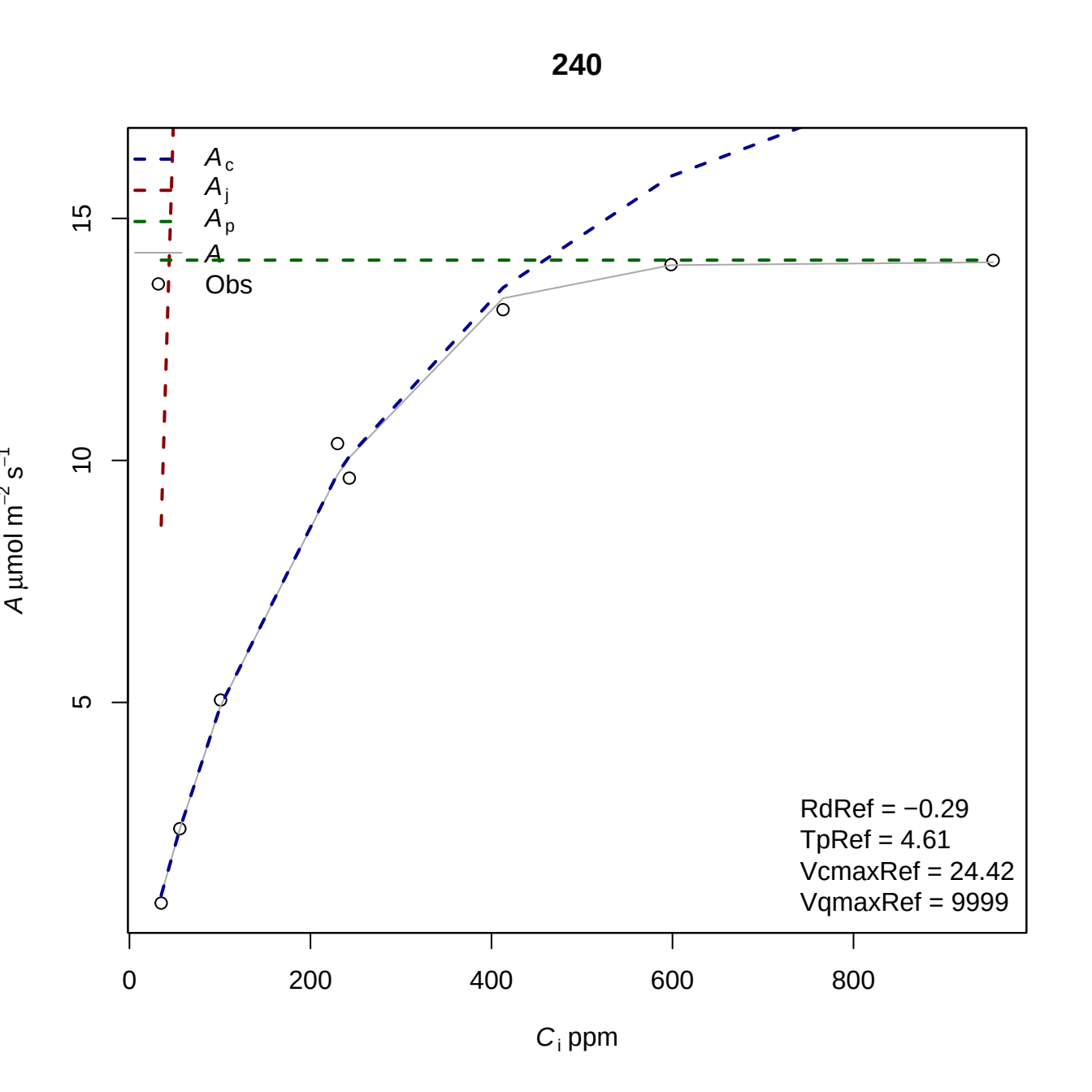

235
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.78
TpRef = 9999
VcmaxRef = 224.57
VqmaxRef = 472.29
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

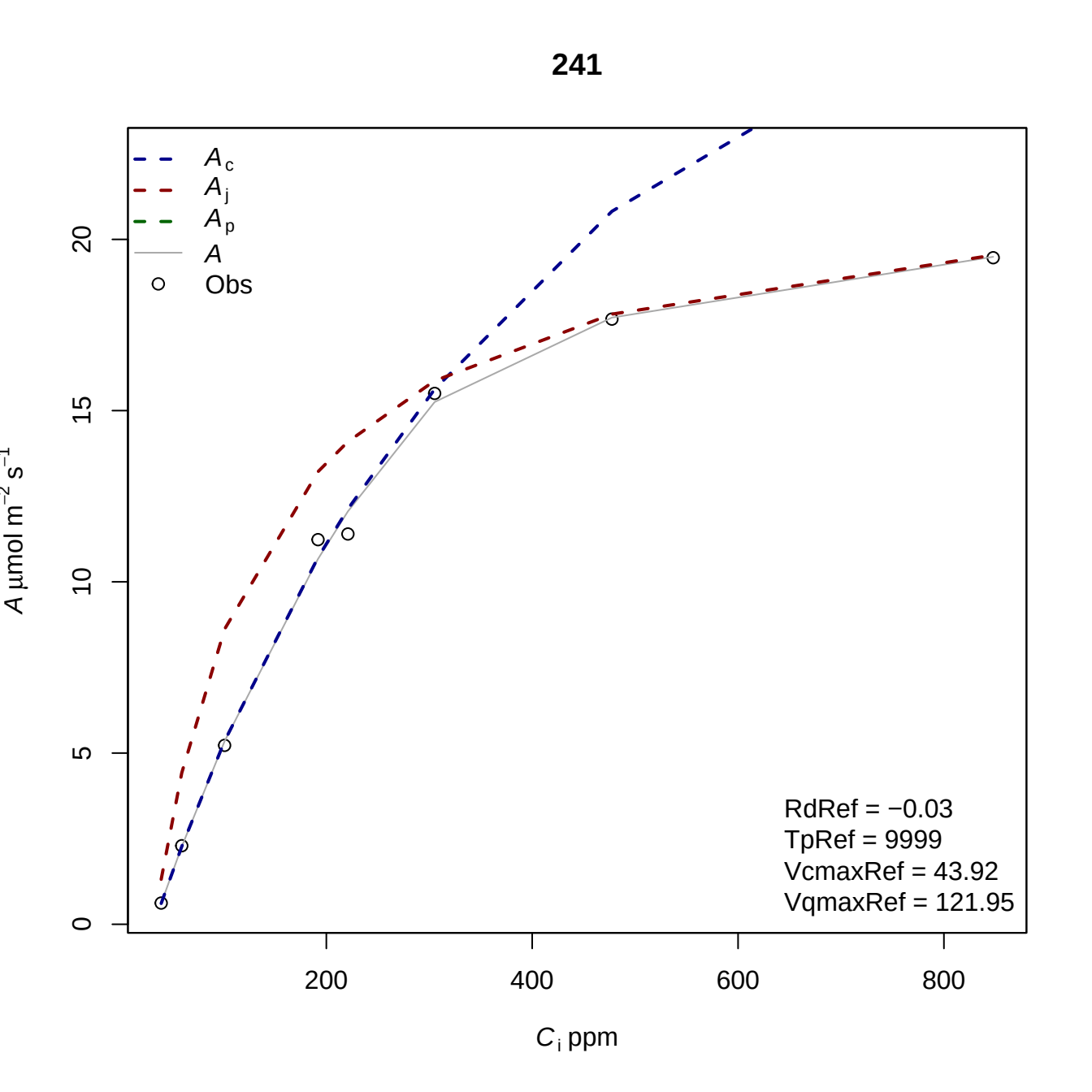

236
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.35
TpRef = 6.91
VcmaxRef = 39.53
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
1400
5
10
15
20

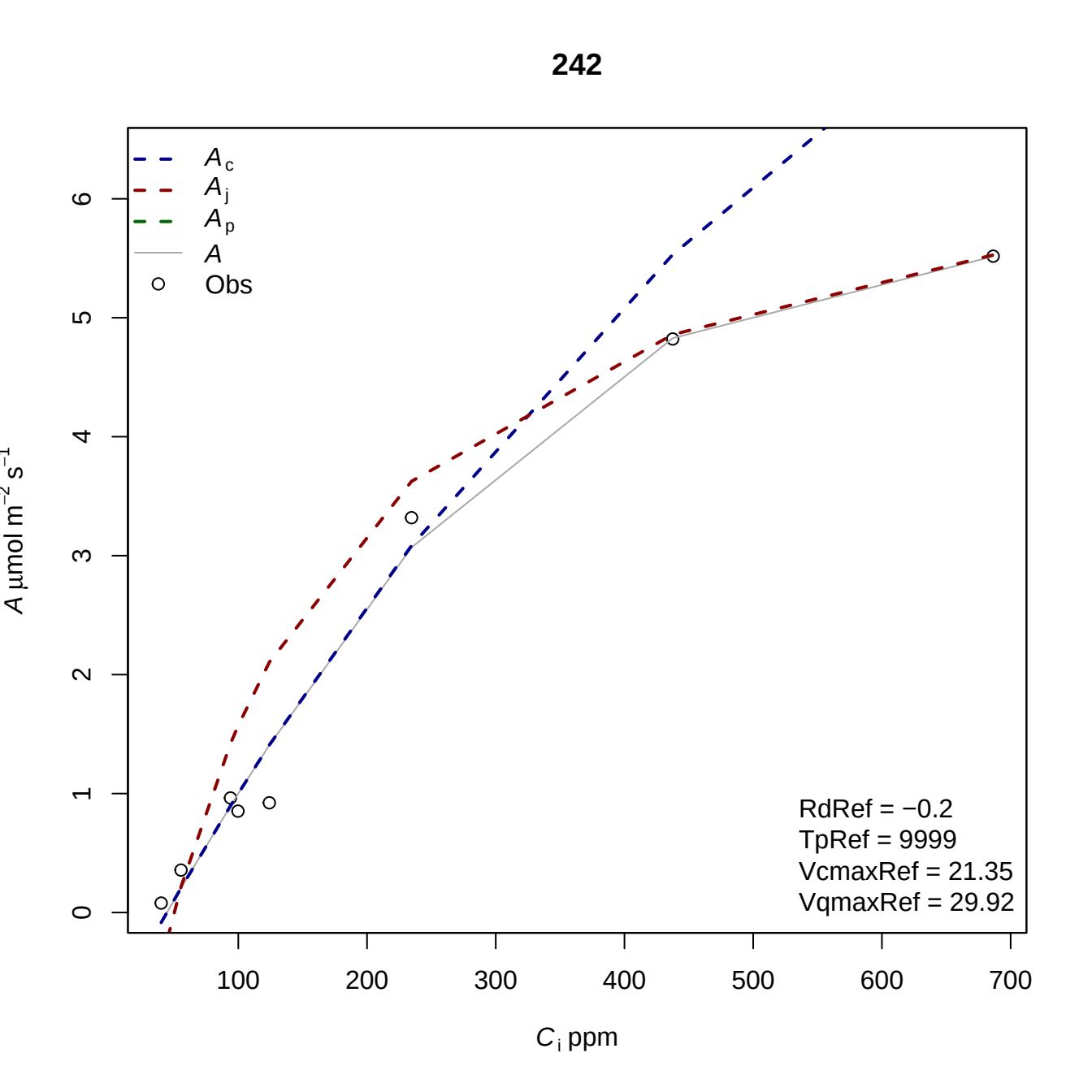

237
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.44
TpRef = 9.9
VcmaxRef = 84.88
VqmaxRef = 282.91
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

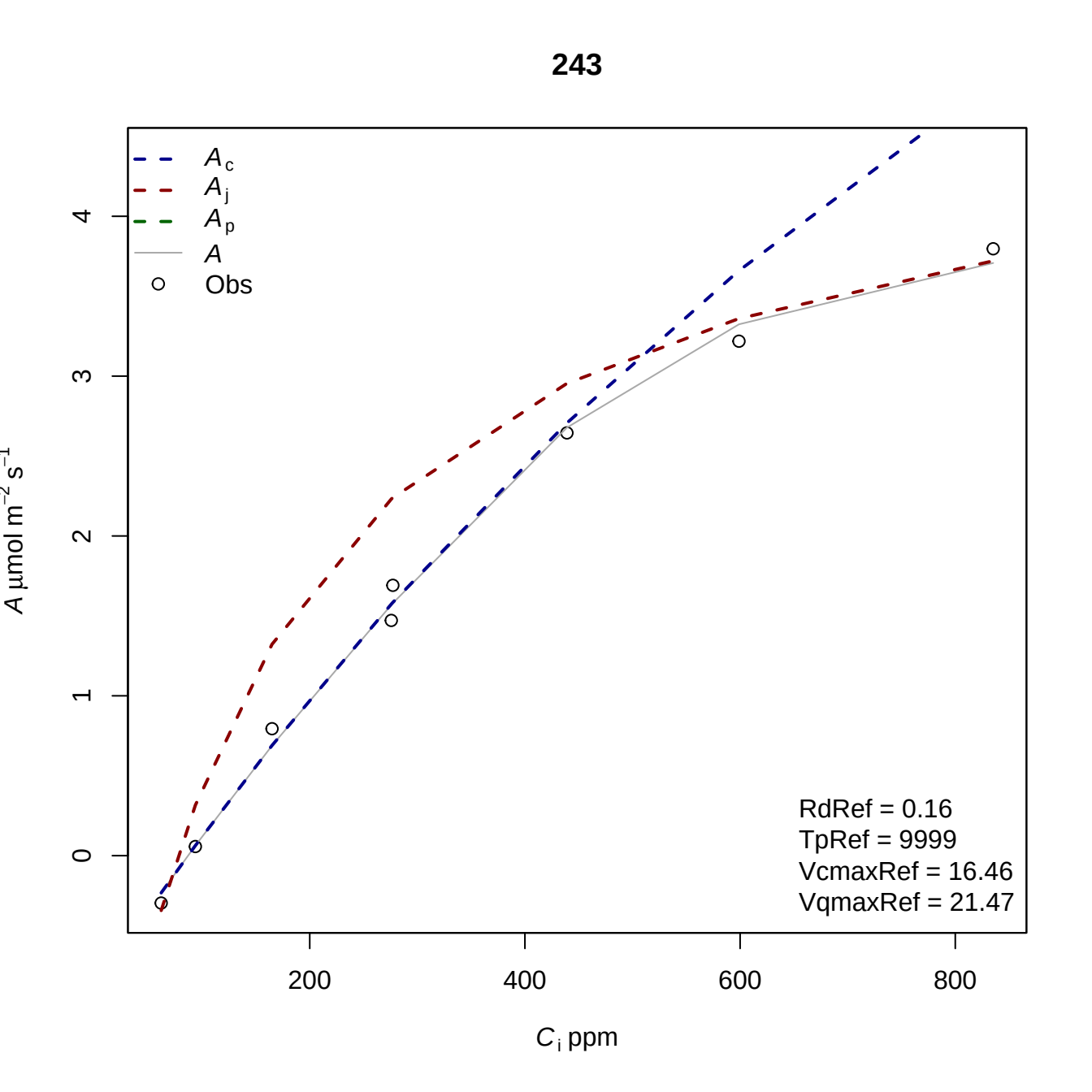

238
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.89
TpRef = 11.59
VcmaxRef = 135.94
VqmaxRef = 446.04
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

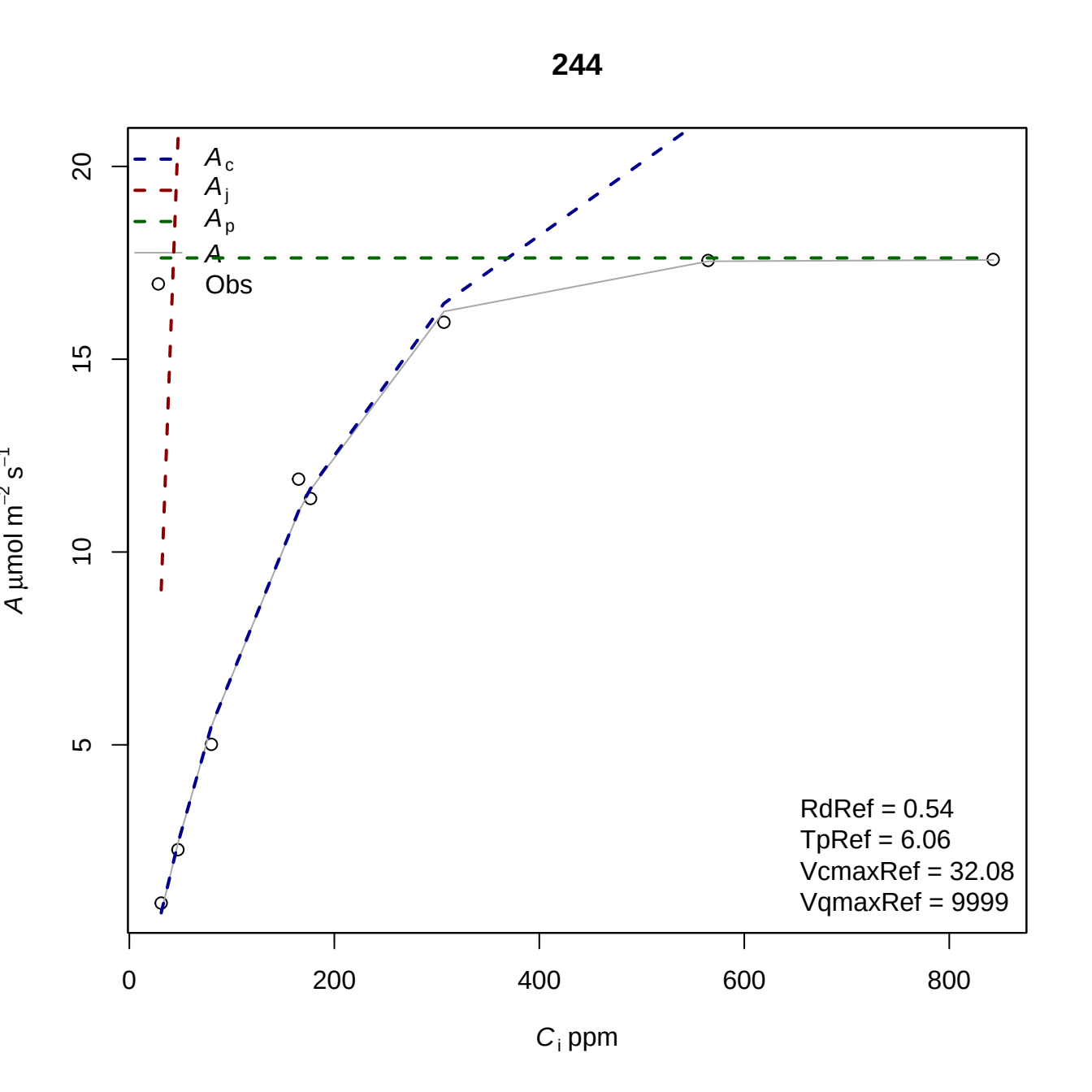

239
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.19
TpRef = 6.3
VcmaxRef = 35.72
VqmaxRef = 113.79
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
20
100
200
300
400
500
600
700

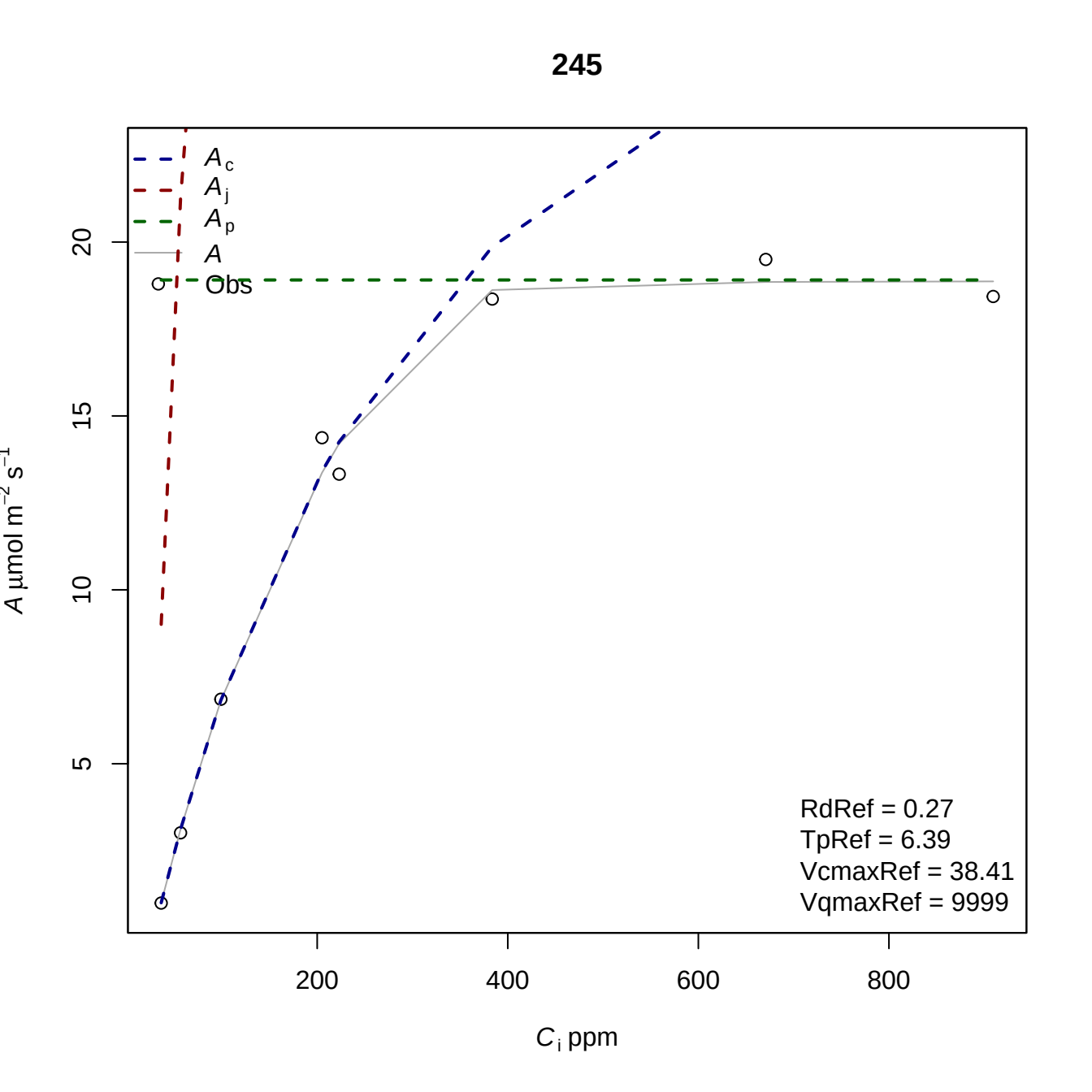

240
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.29
TpRef = 4.61
VcmaxRef = 24.42
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
5
10
15

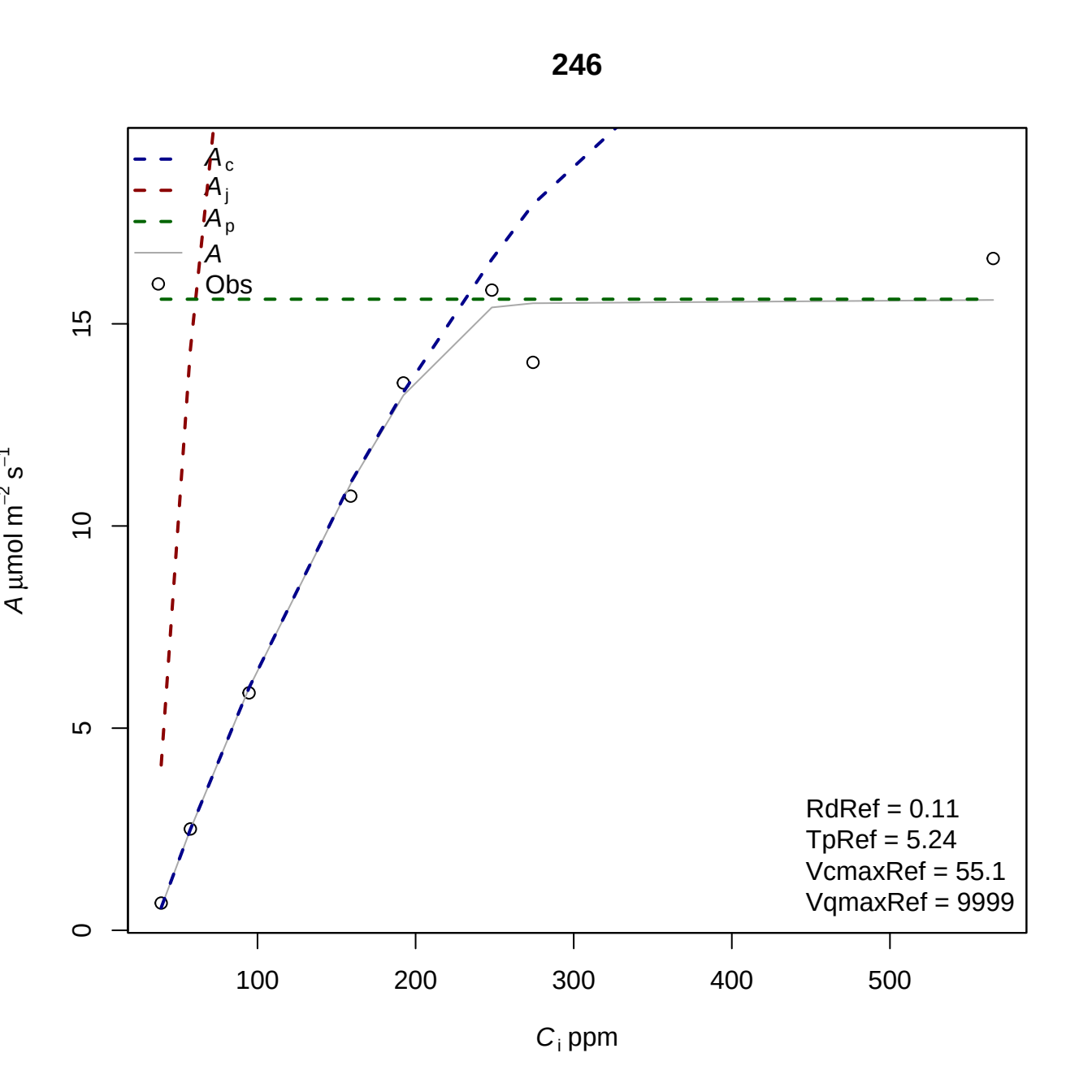

241
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.03
TpRef = 9999
VcmaxRef = 43.92
VqmaxRef = 121.95
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800

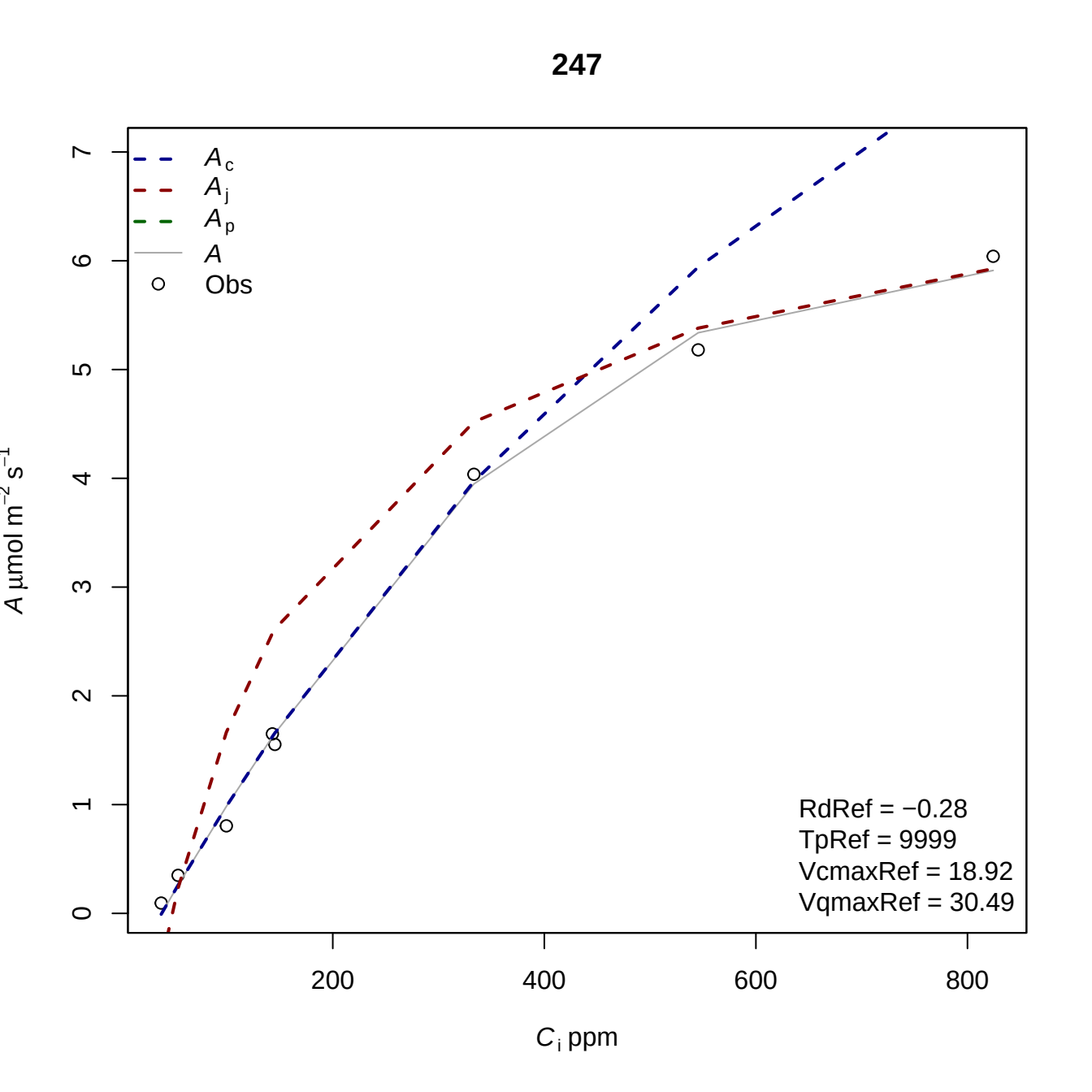

242
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.2
TpRef = 9999
VcmaxRef = 21.35
VqmaxRef = 29.92
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
100
200
300
400
500
600
700

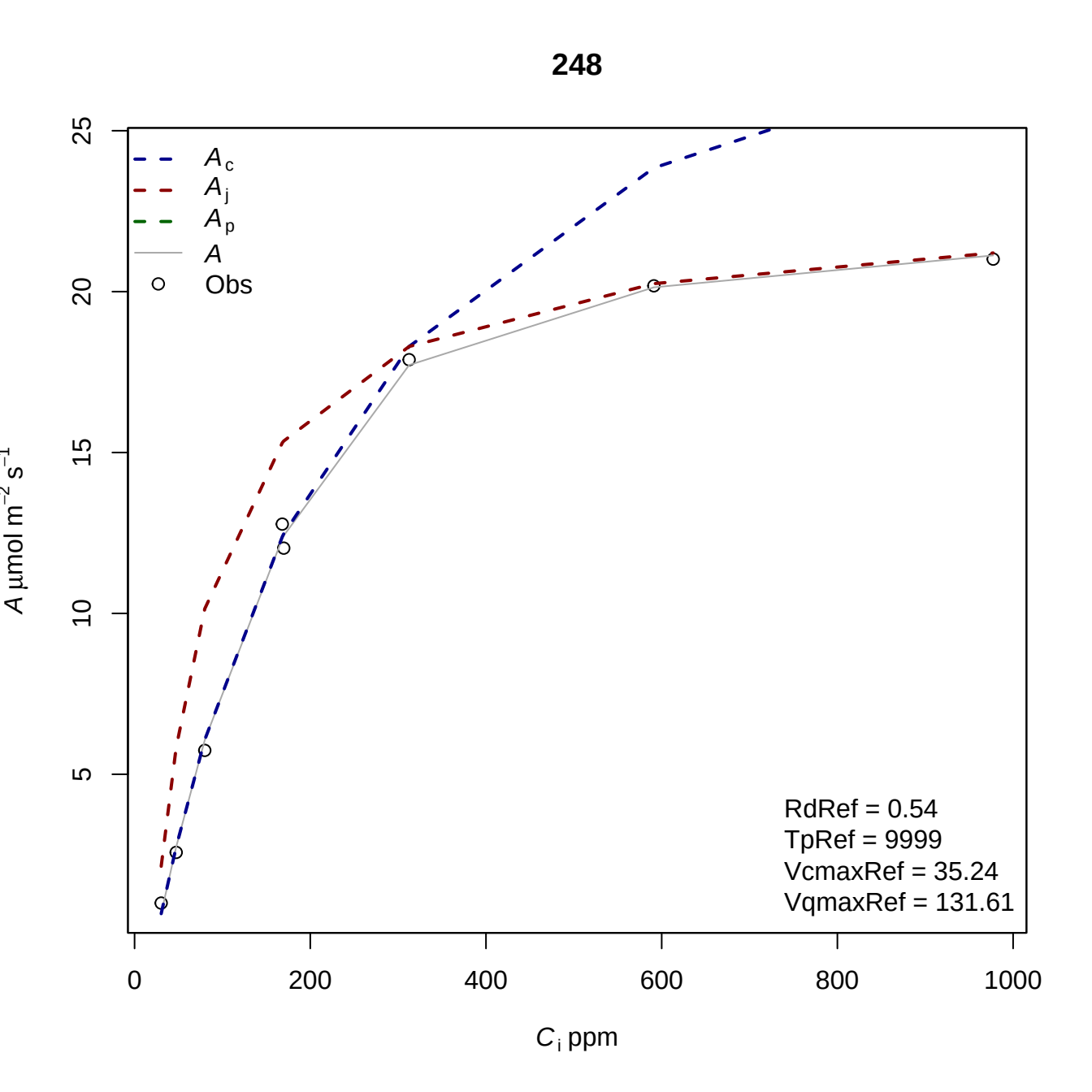

243
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.16
TpRef = 9999
VcmaxRef = 16.46
VqmaxRef = 21.47
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
200
400
600
800

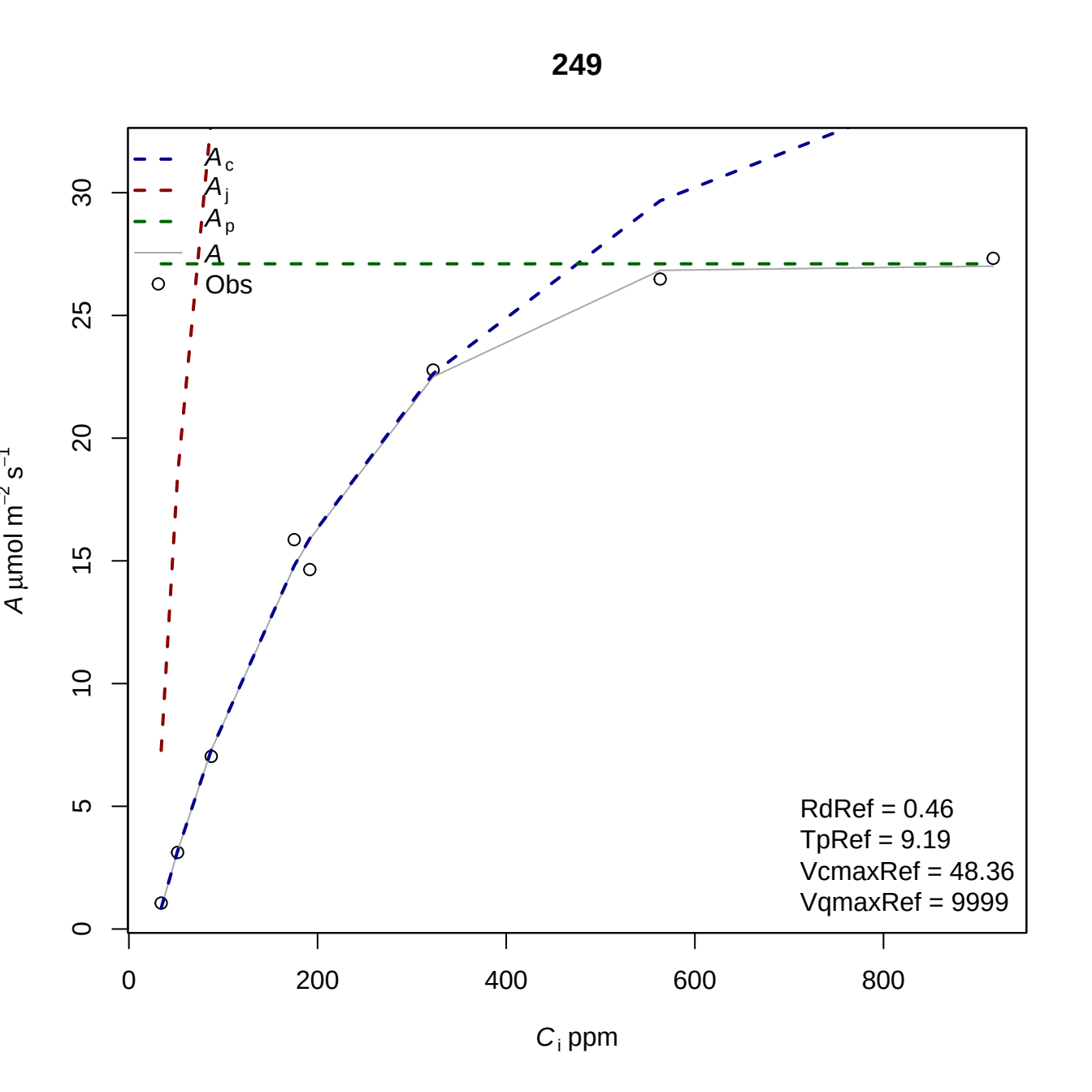

244
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.54
TpRef = 6.06
VcmaxRef = 32.08
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
5
10
15
20

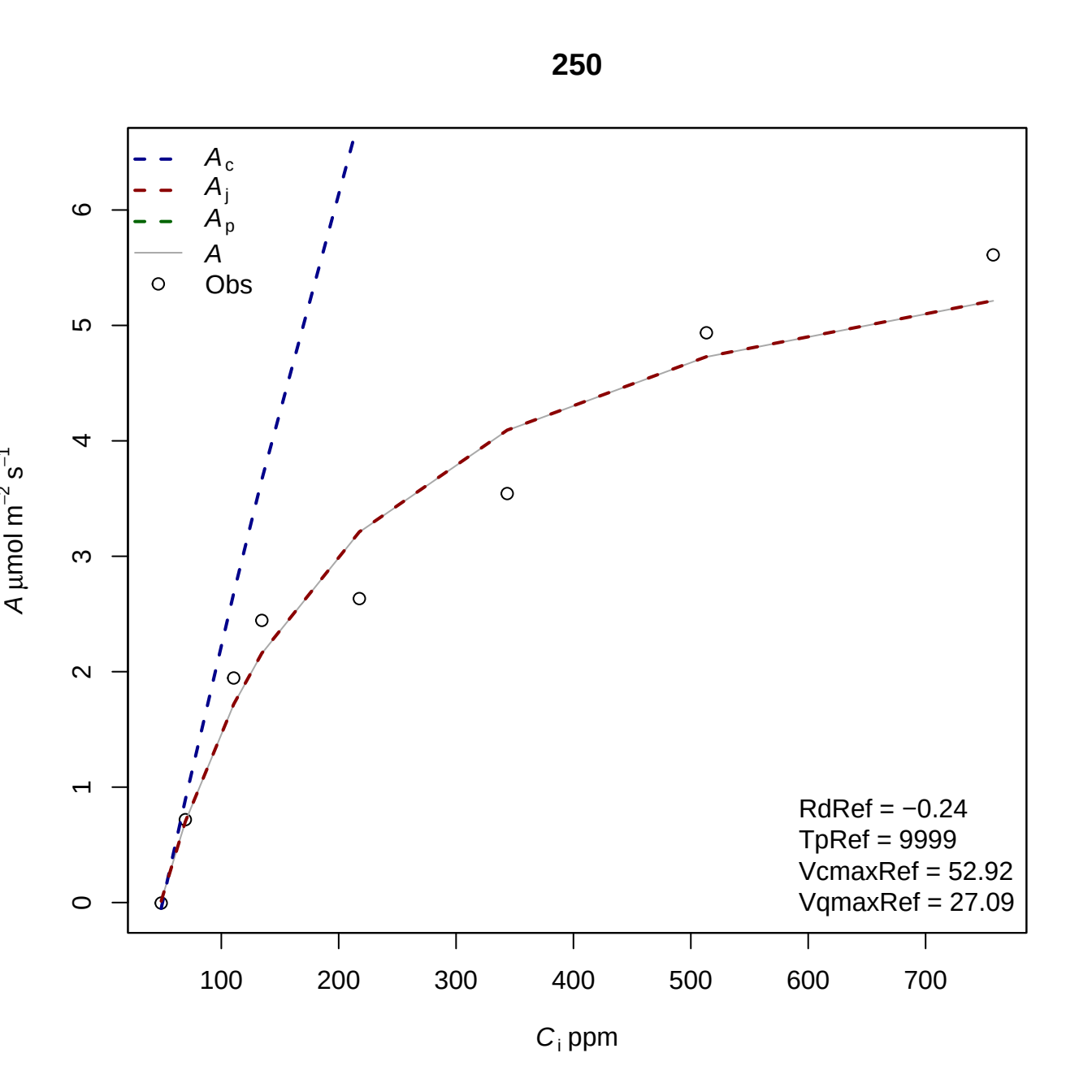

245
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.27
TpRef = 6.39
VcmaxRef = 38.41
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
20
200
400
600
800

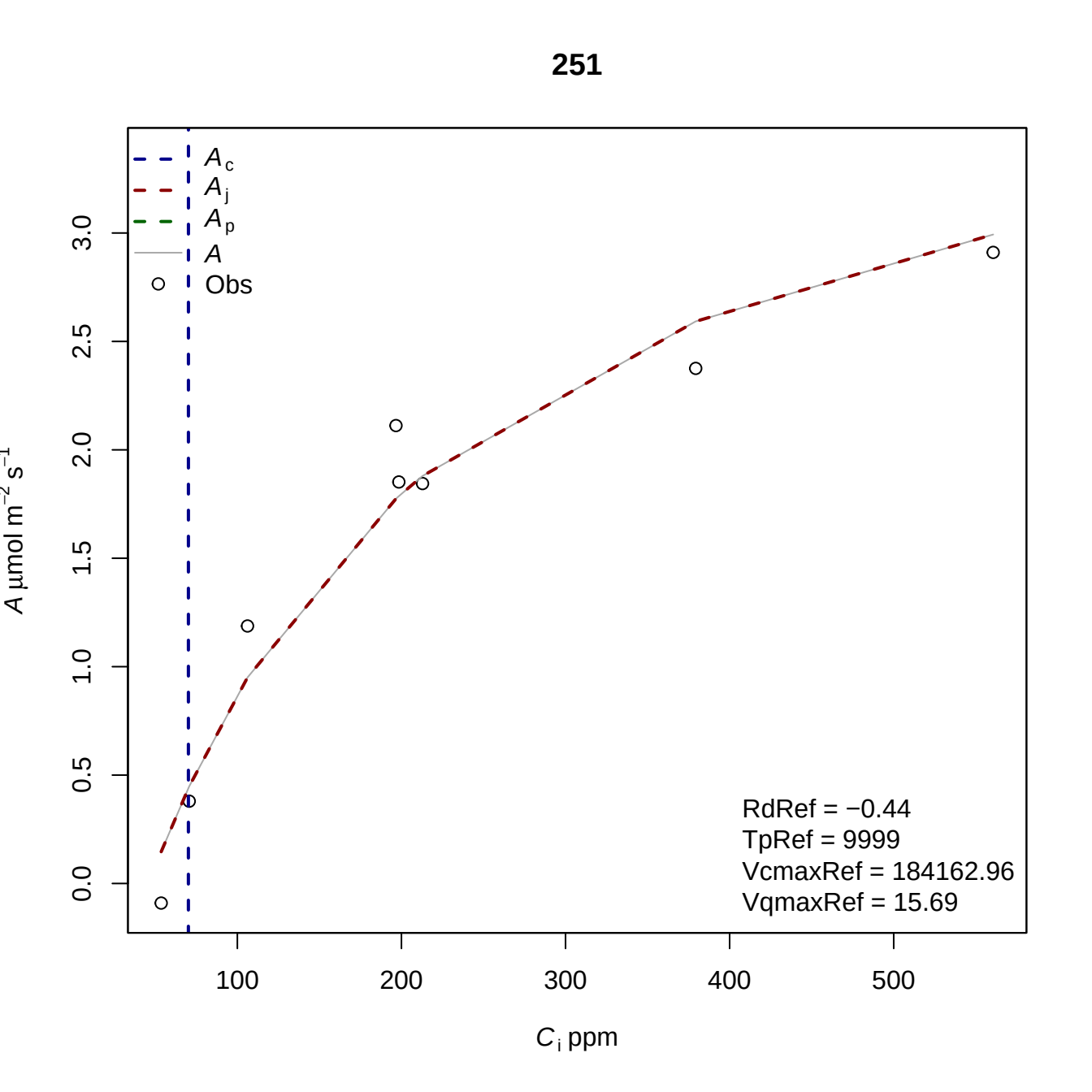

246
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.11
TpRef = 5.24
VcmaxRef = 55.1
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500

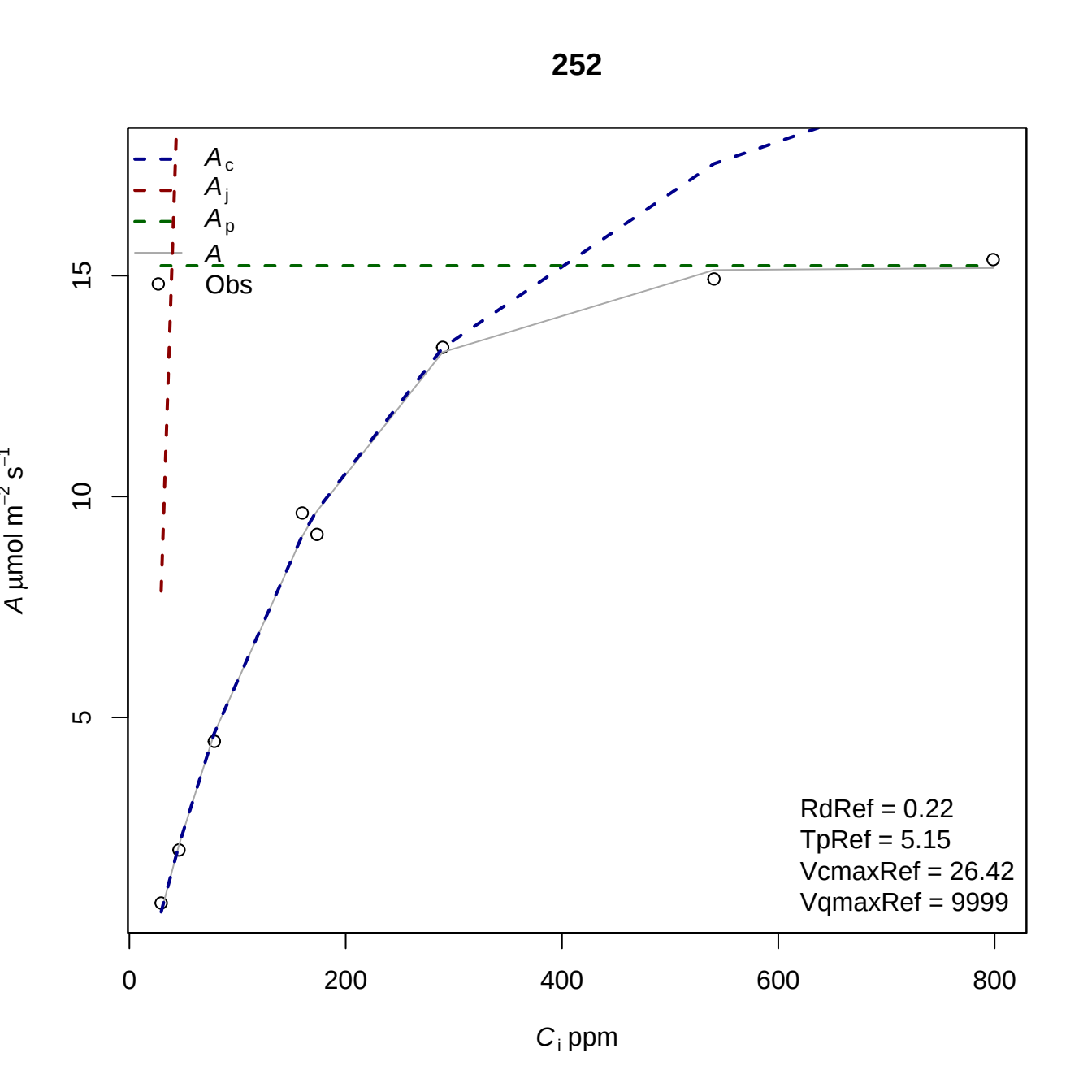

247
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.28
TpRef = 9999
VcmaxRef = 18.92
VqmaxRef = 30.49
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
7
200
400
600
800

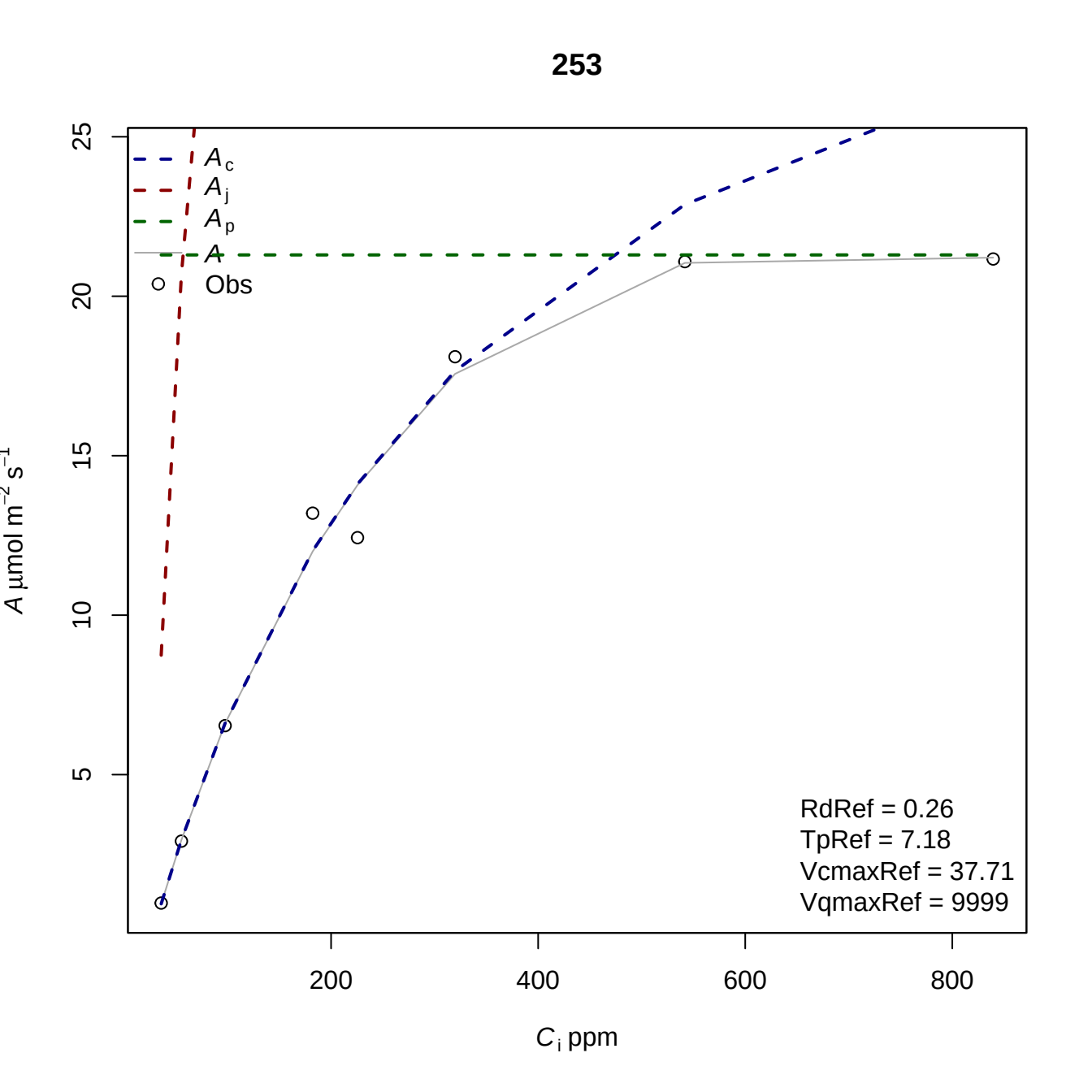

248
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.54
TpRef = 9999
VcmaxRef = 35.24
VqmaxRef = 131.61
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
5
10
15
20
25

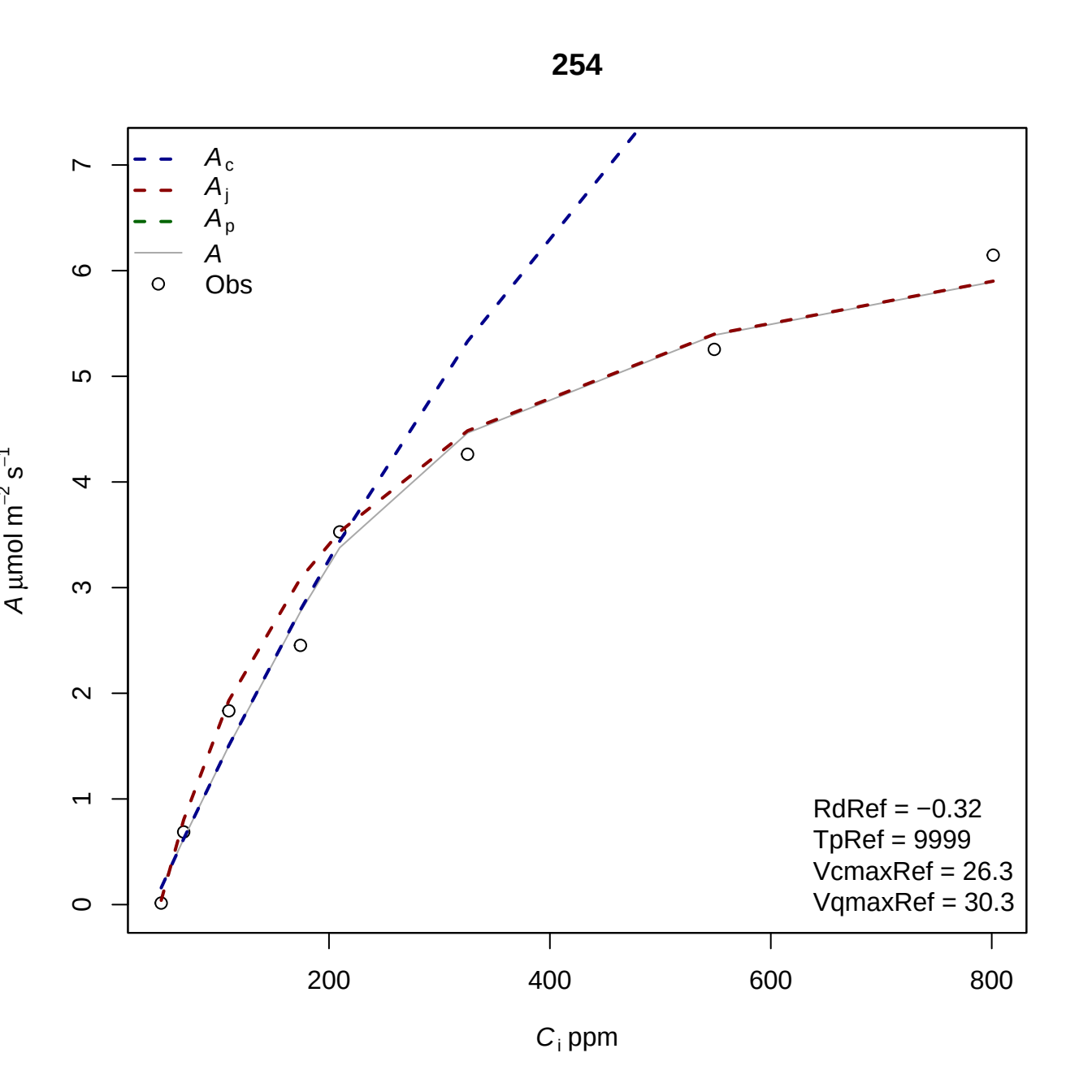

249
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.46
TpRef = 9.19
VcmaxRef = 48.36
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800

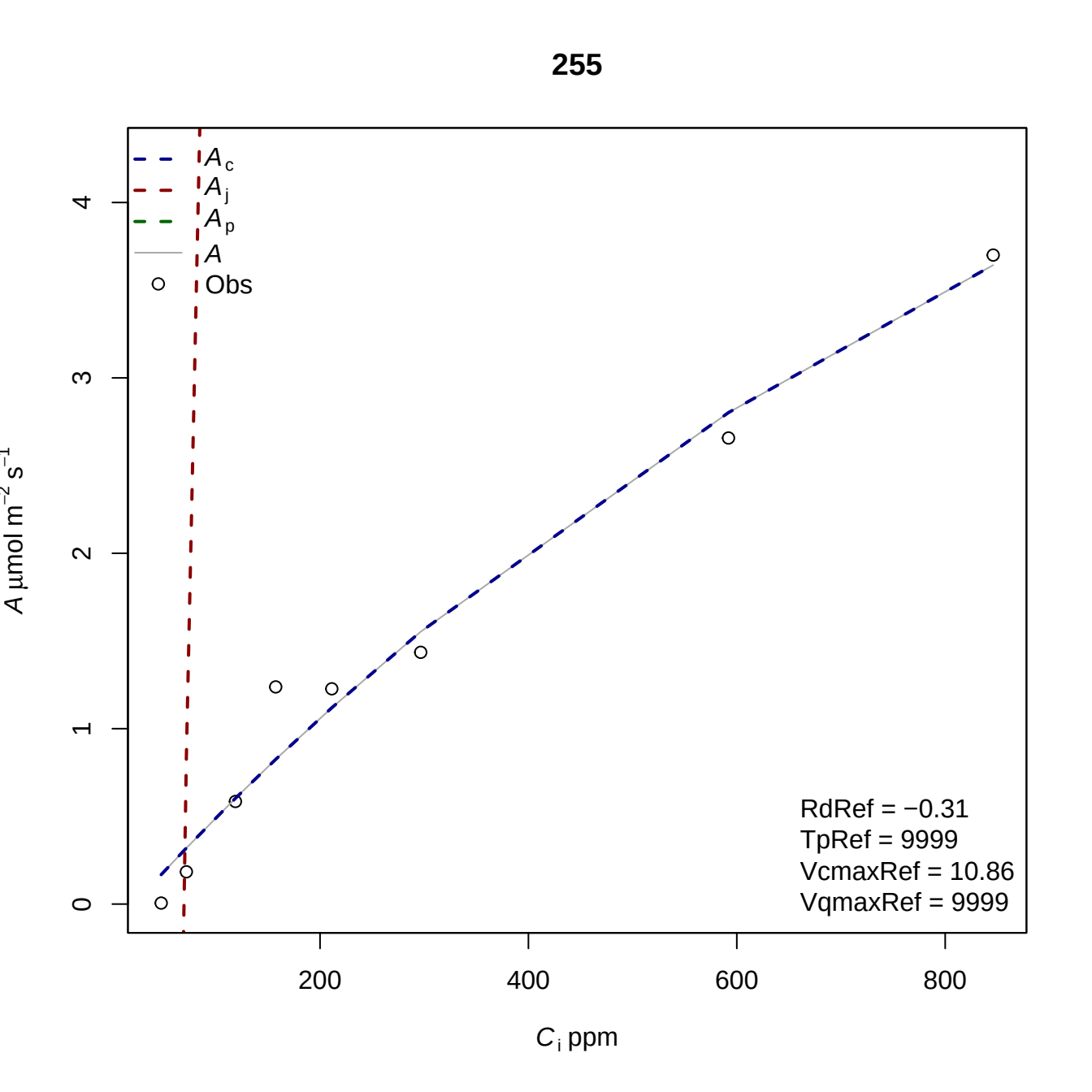

250
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.24
TpRef = 9999
VcmaxRef = 52.92
VqmaxRef = 27.09
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
100
200
300
400
500
600
700

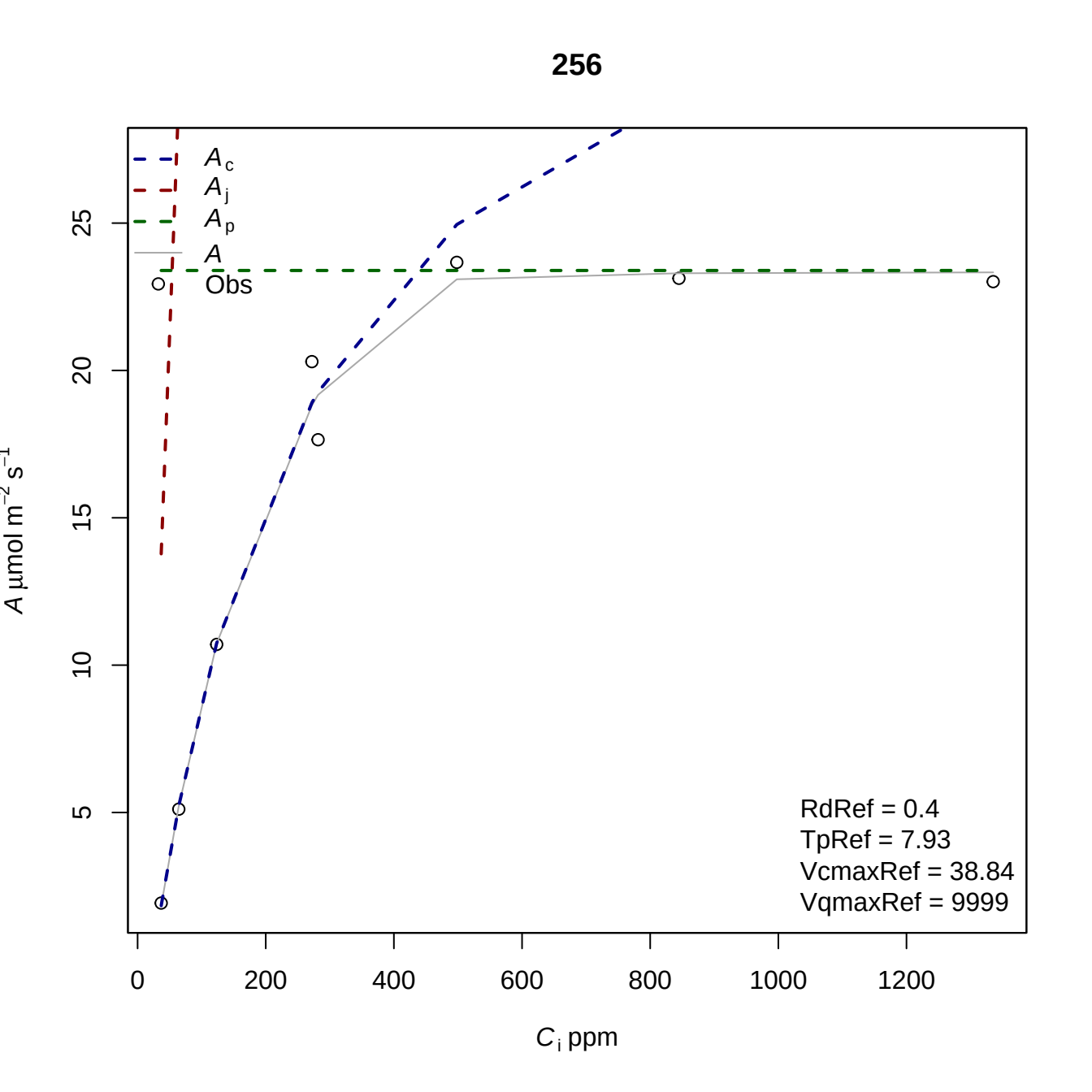

251
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.44
TpRef = 9999
VcmaxRef = 184162.96
VqmaxRef = 15.69
$A$ µmol m$^{-2}$ s$^{-1}$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
100
200
300
400
500
$C_i$ ppm

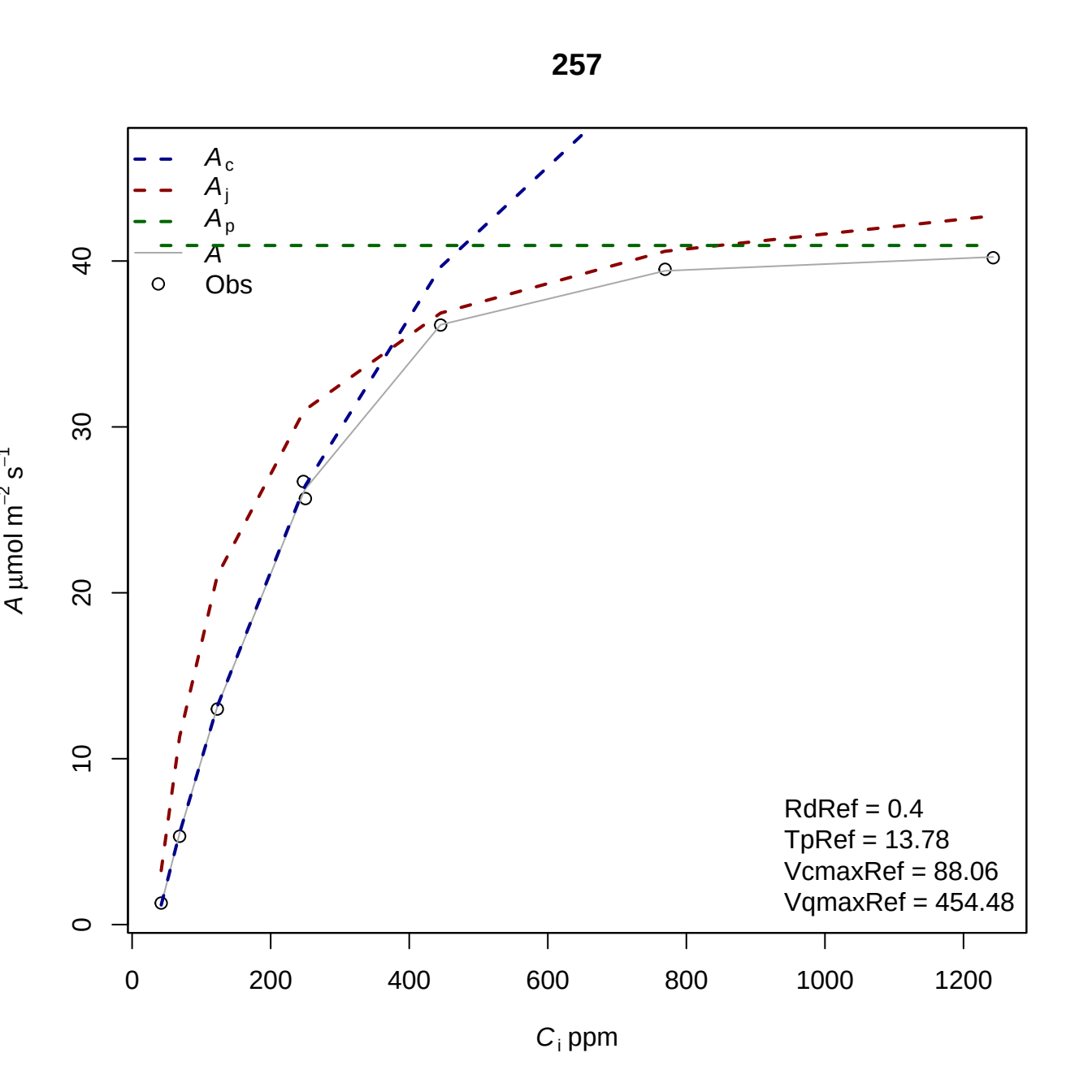

252
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.22
TpRef = 5.15
VcmaxRef = 26.42
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
0
200
400
600
800

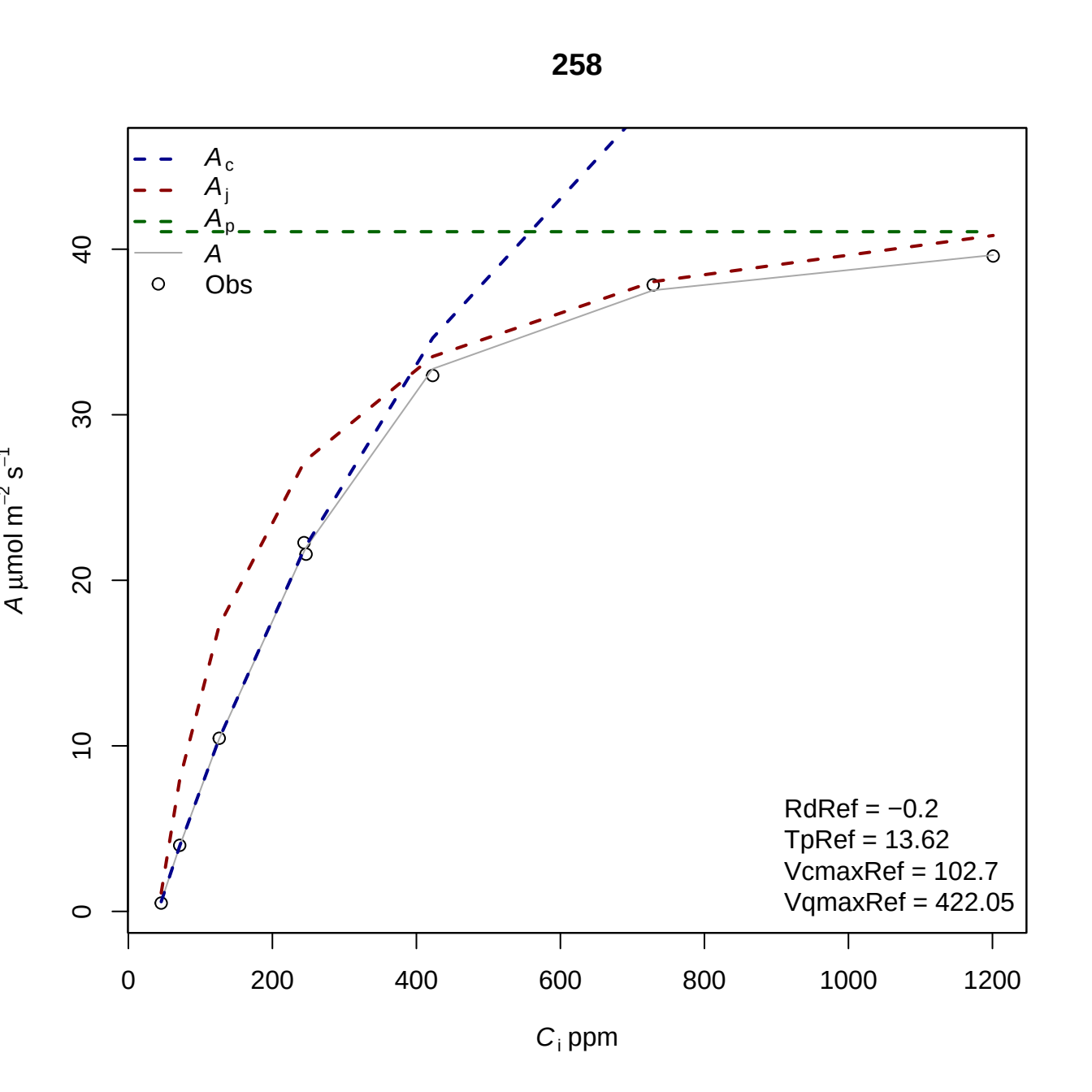

253
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.26
TpRef = 7.18
VcmaxRef = 37.71
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
20
25
200
400
600
800

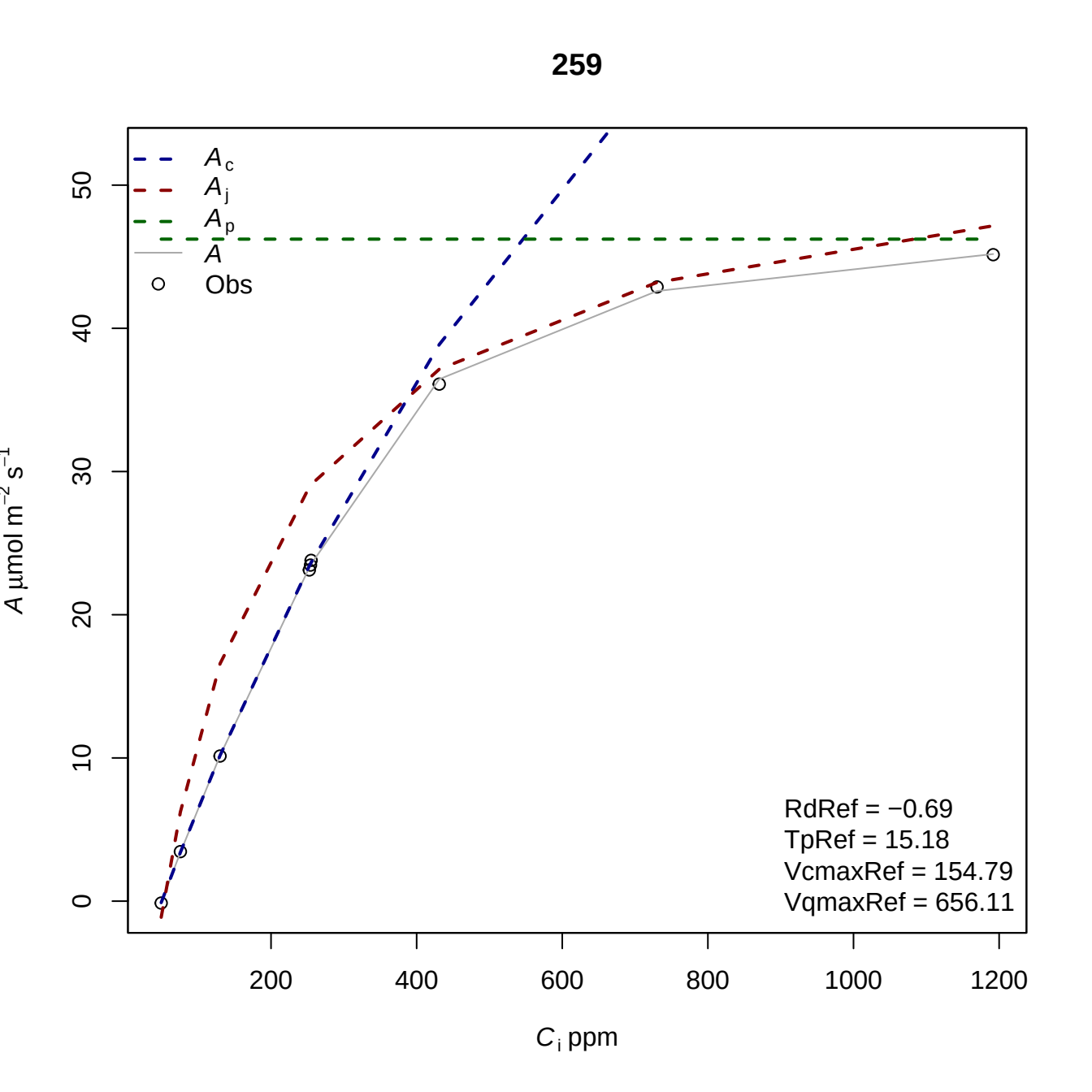

254
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.32
TpRef = 9999
VcmaxRef = 26.3
VqmaxRef = 30.3
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
7
200
400
600
800

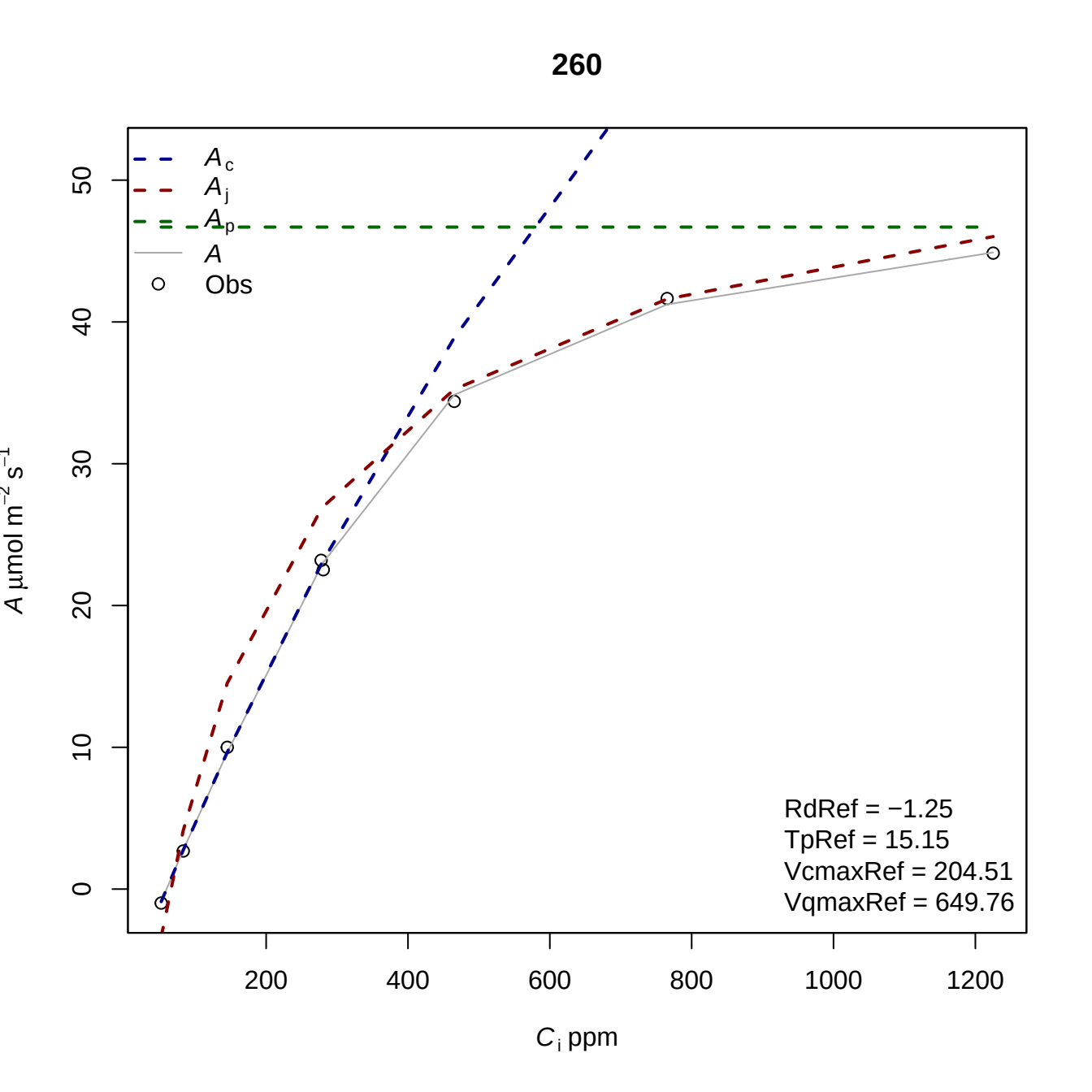

255
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.31
TpRef = 9999
VcmaxRef = 10.86
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
200
400
600
800

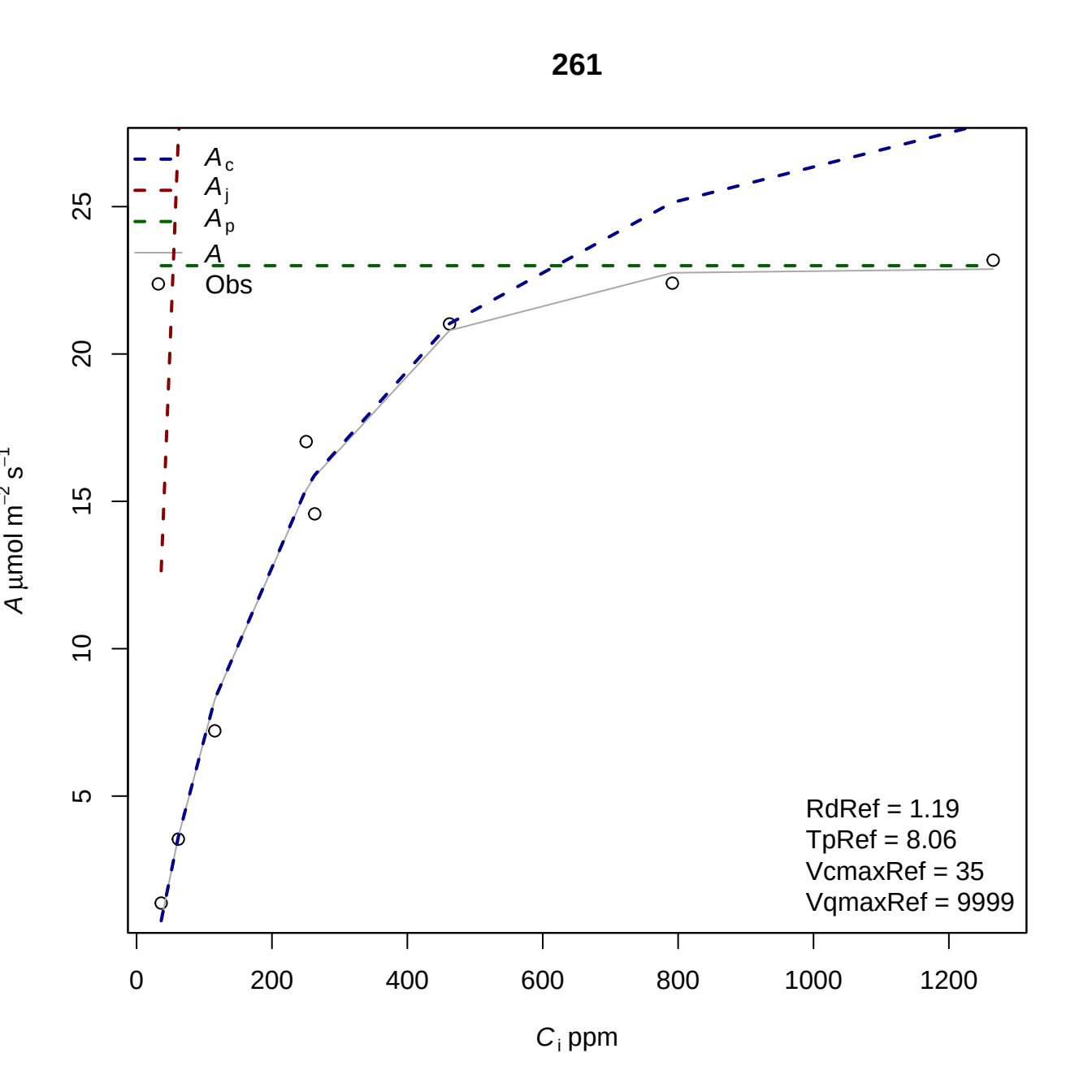

256
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.4
TpRef = 7.93
VcmaxRef = 38.84
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

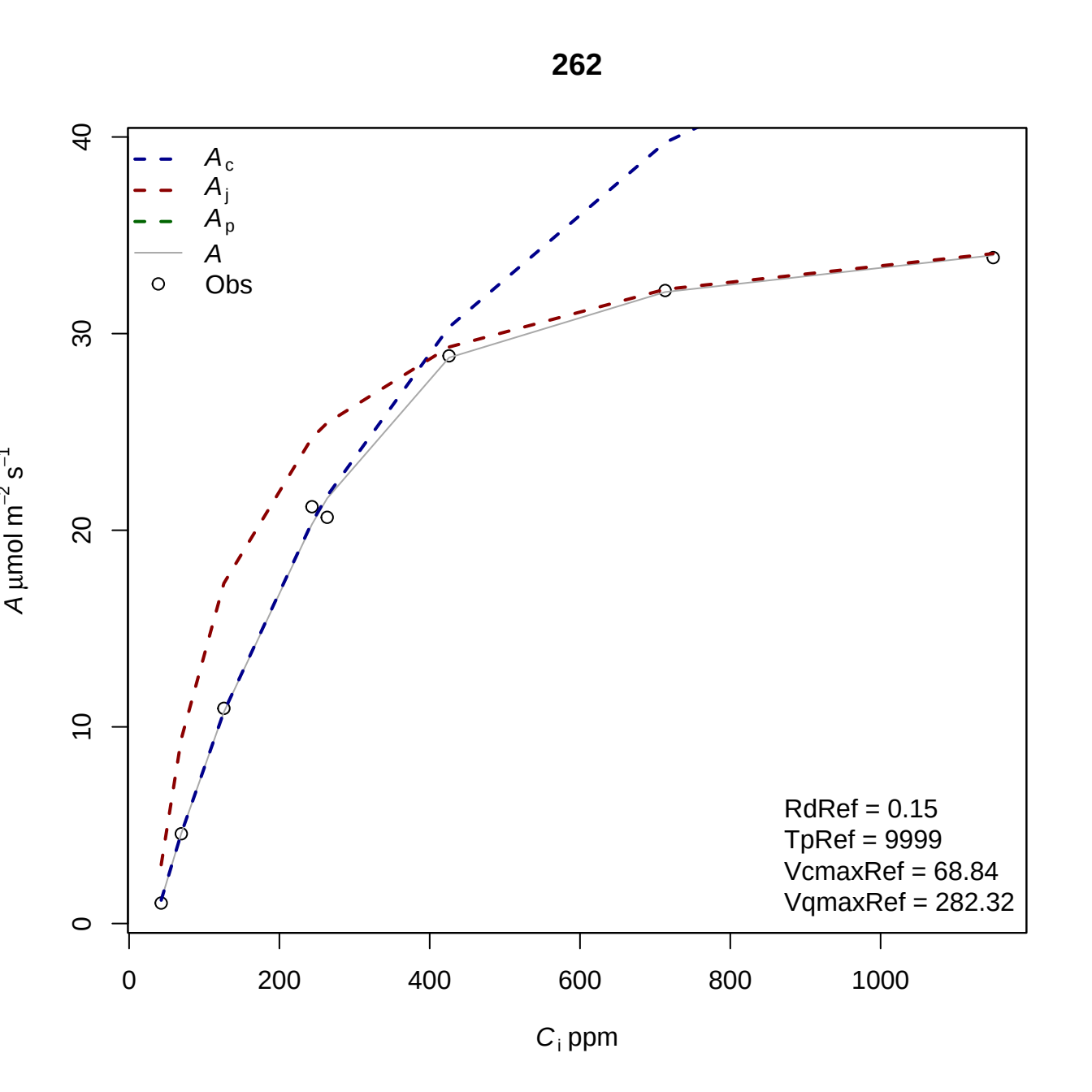

257
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.4
TpRef = 13.78
VcmaxRef = 88.06
VqmaxRef = 454.48
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

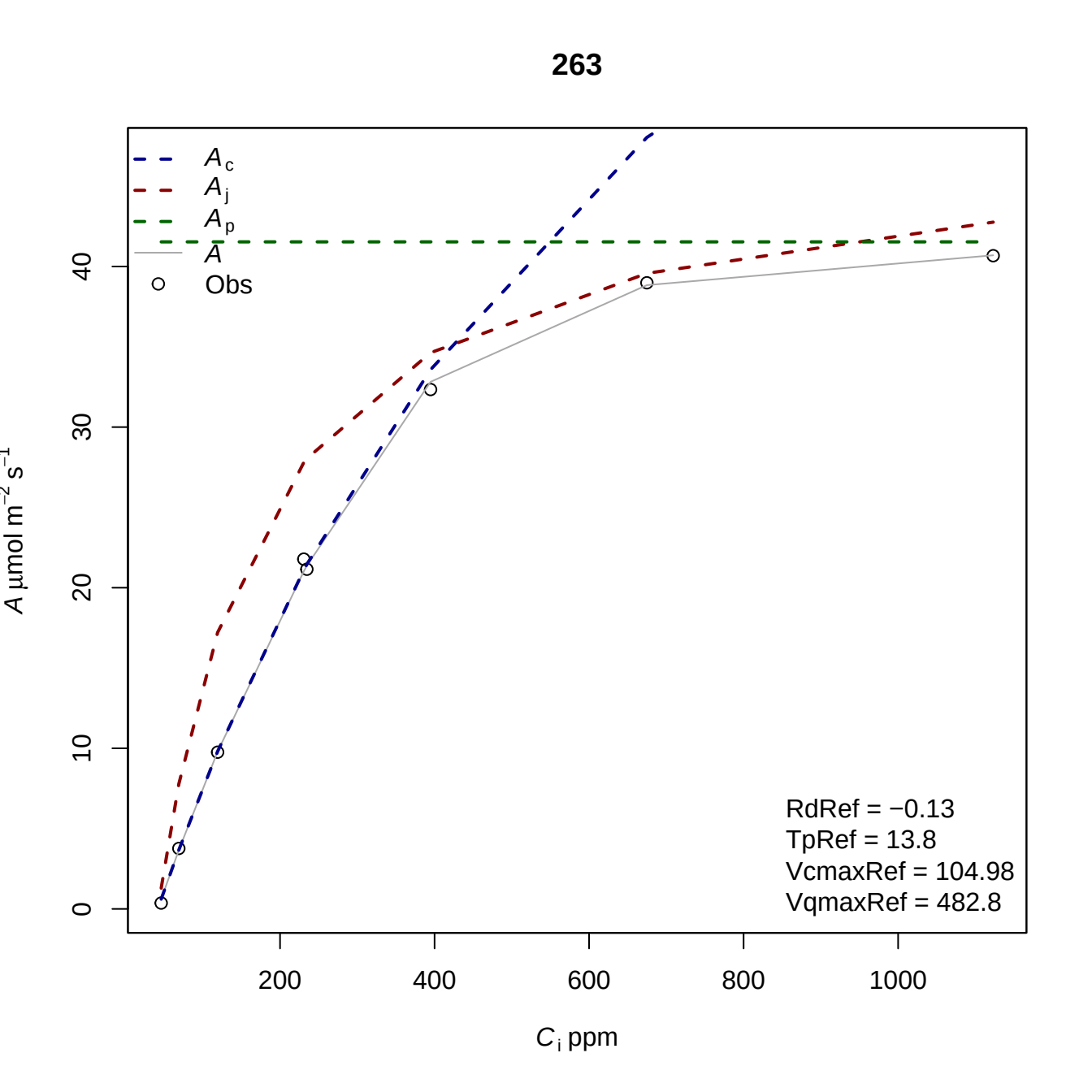

258
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.2
TpRef = 13.62
VcmaxRef = 102.7
VqmaxRef = 422.05
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

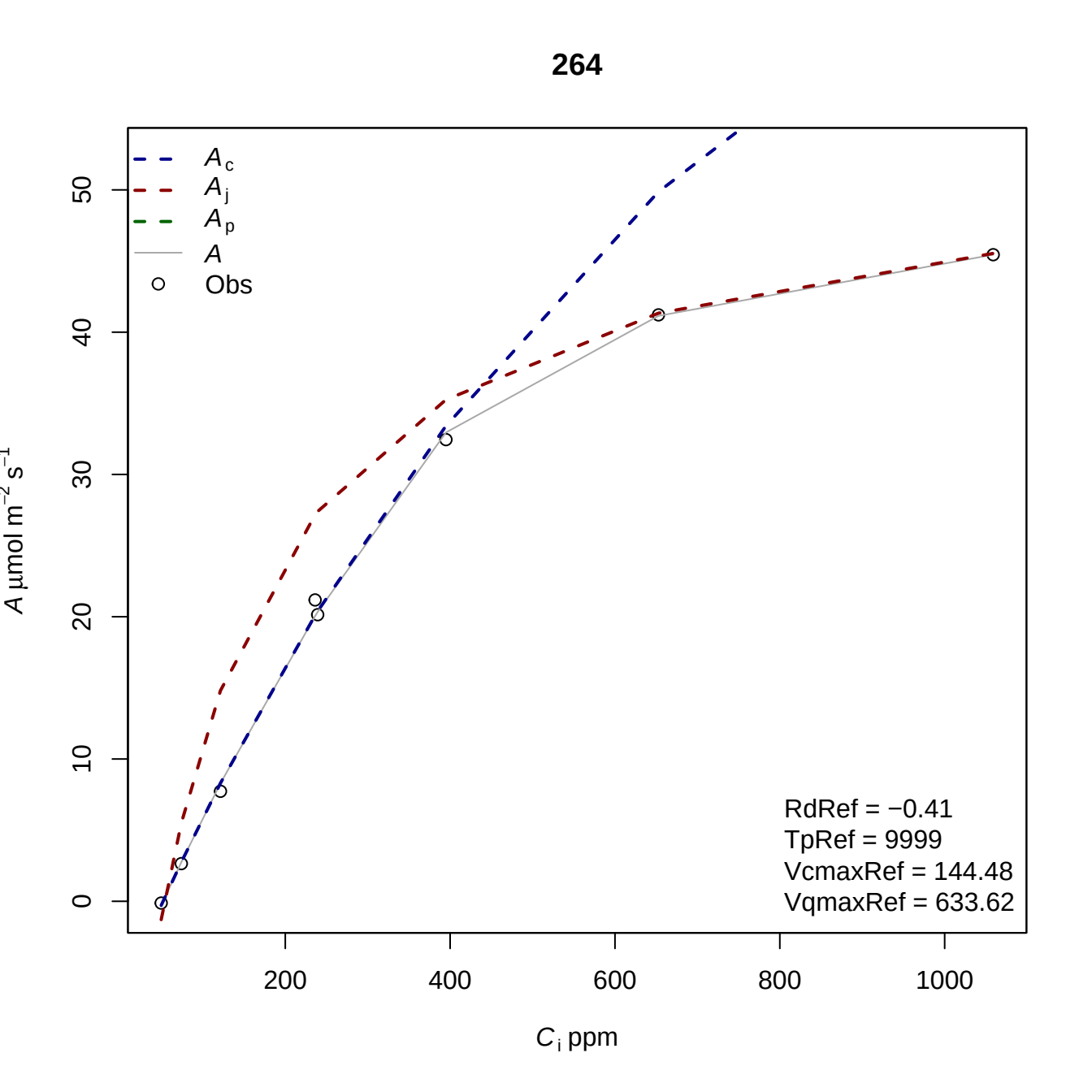

259
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.69
TpRef = 15.18
VcmaxRef = 154.79
VqmaxRef = 656.11
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

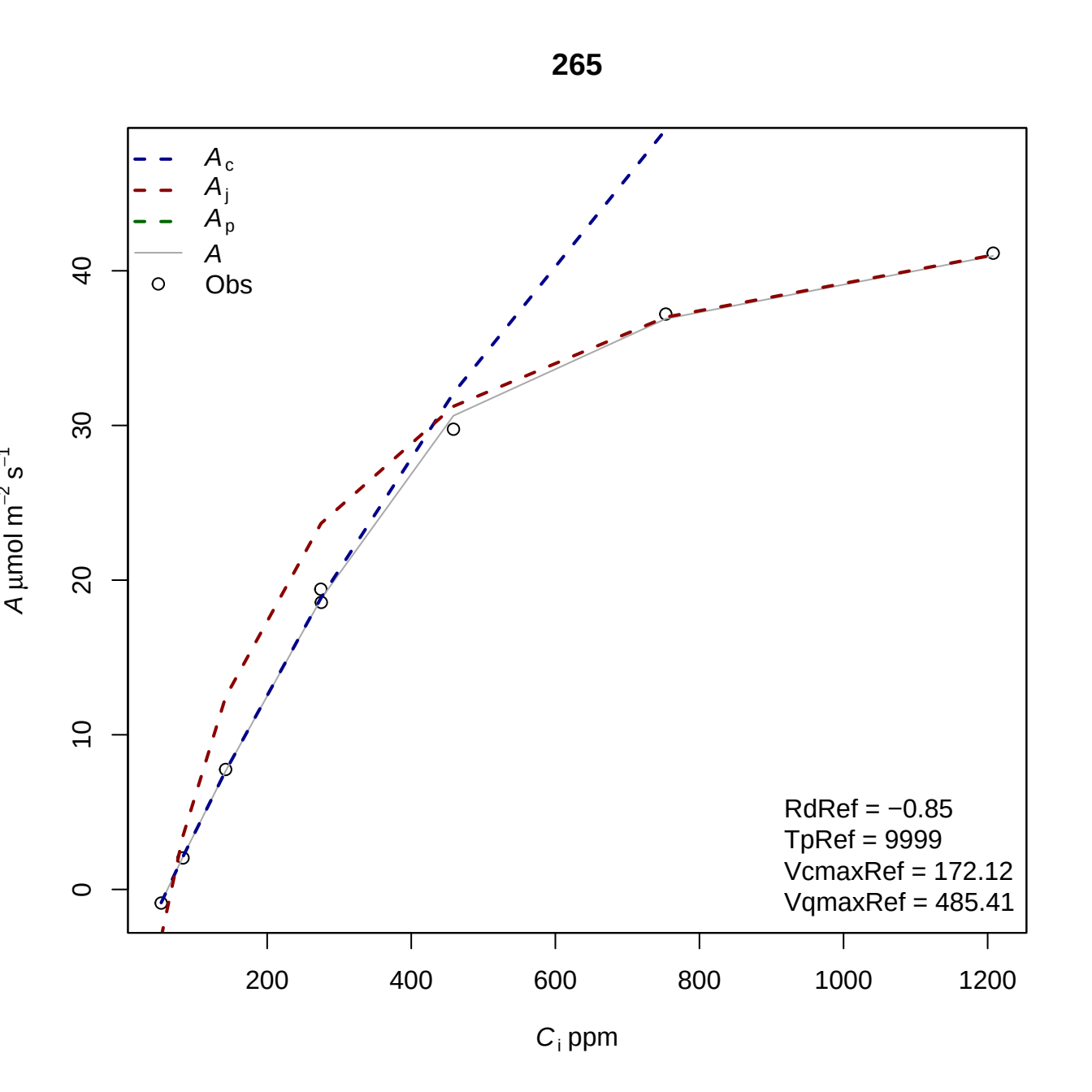

260
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −1.25
TpRef = 15.15
VcmaxRef = 204.51
VqmaxRef = 649.76
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

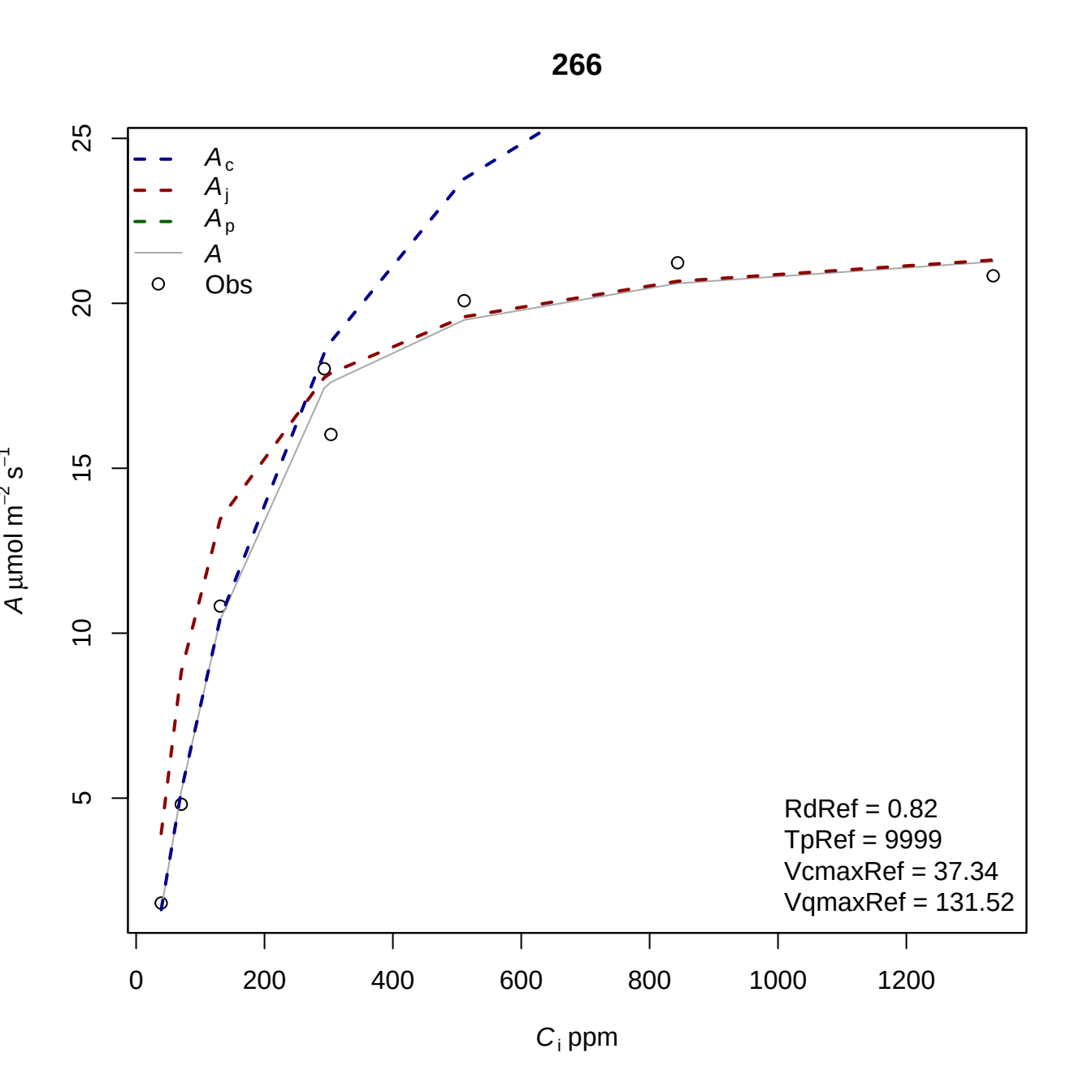

261
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.19
TpRef = 8.06
VcmaxRef = 35
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

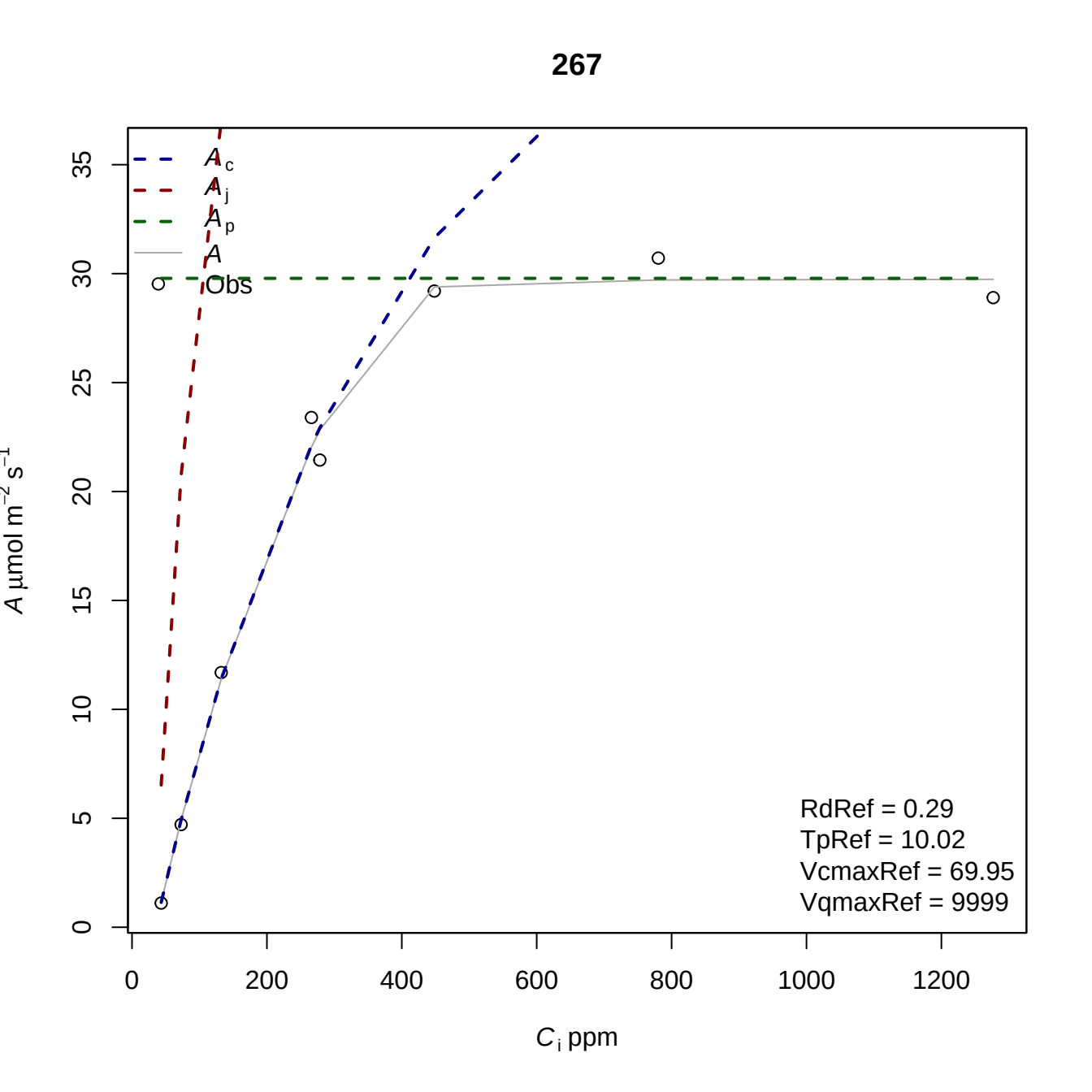

262
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.15
TpRef = 9999
VcmaxRef = 68.84
VqmaxRef = 282.32
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000

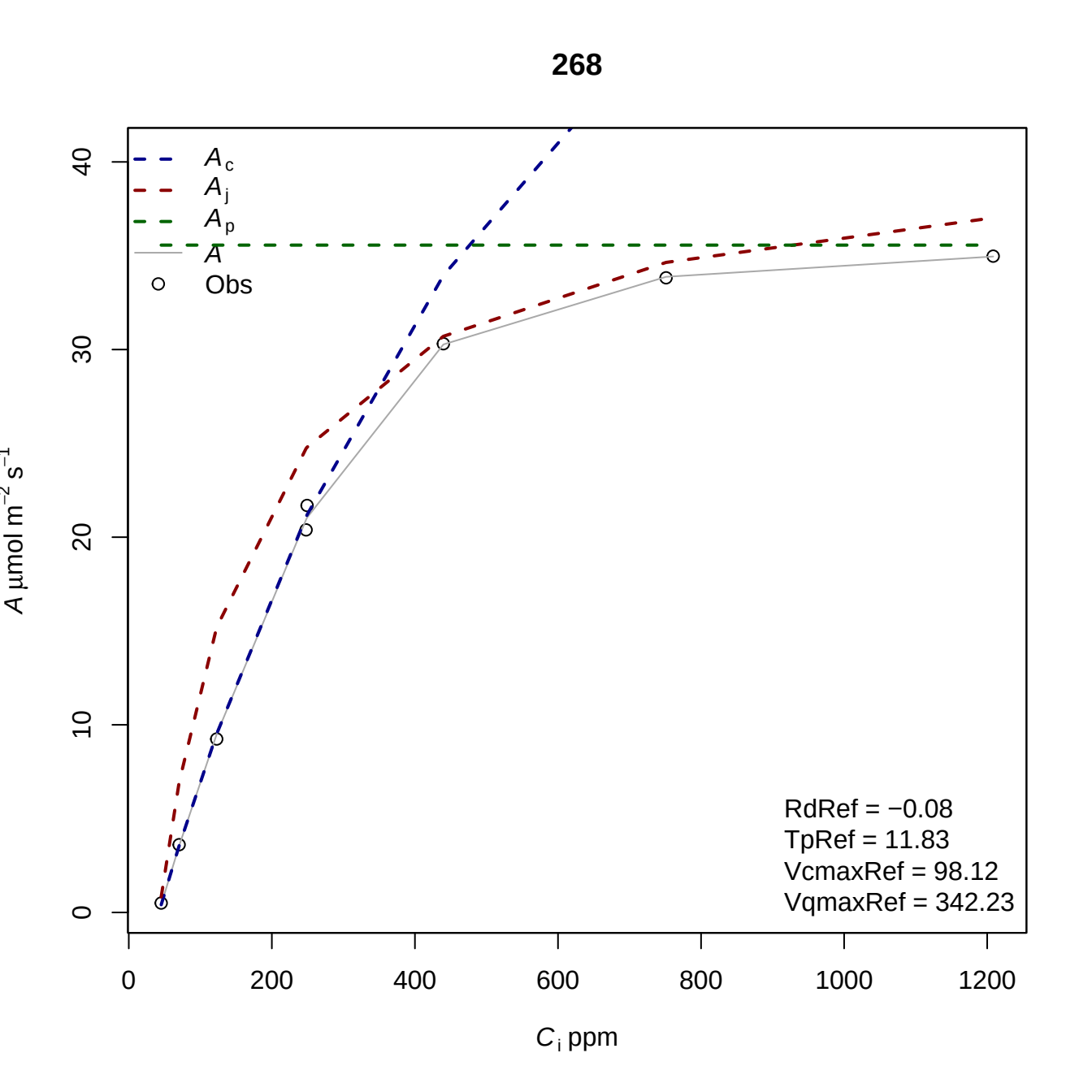

263
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.13
TpRef = 13.8
VcmaxRef = 104.98
VqmaxRef = 482.8
0
10
20
30
40
200
400
600
800
1000
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

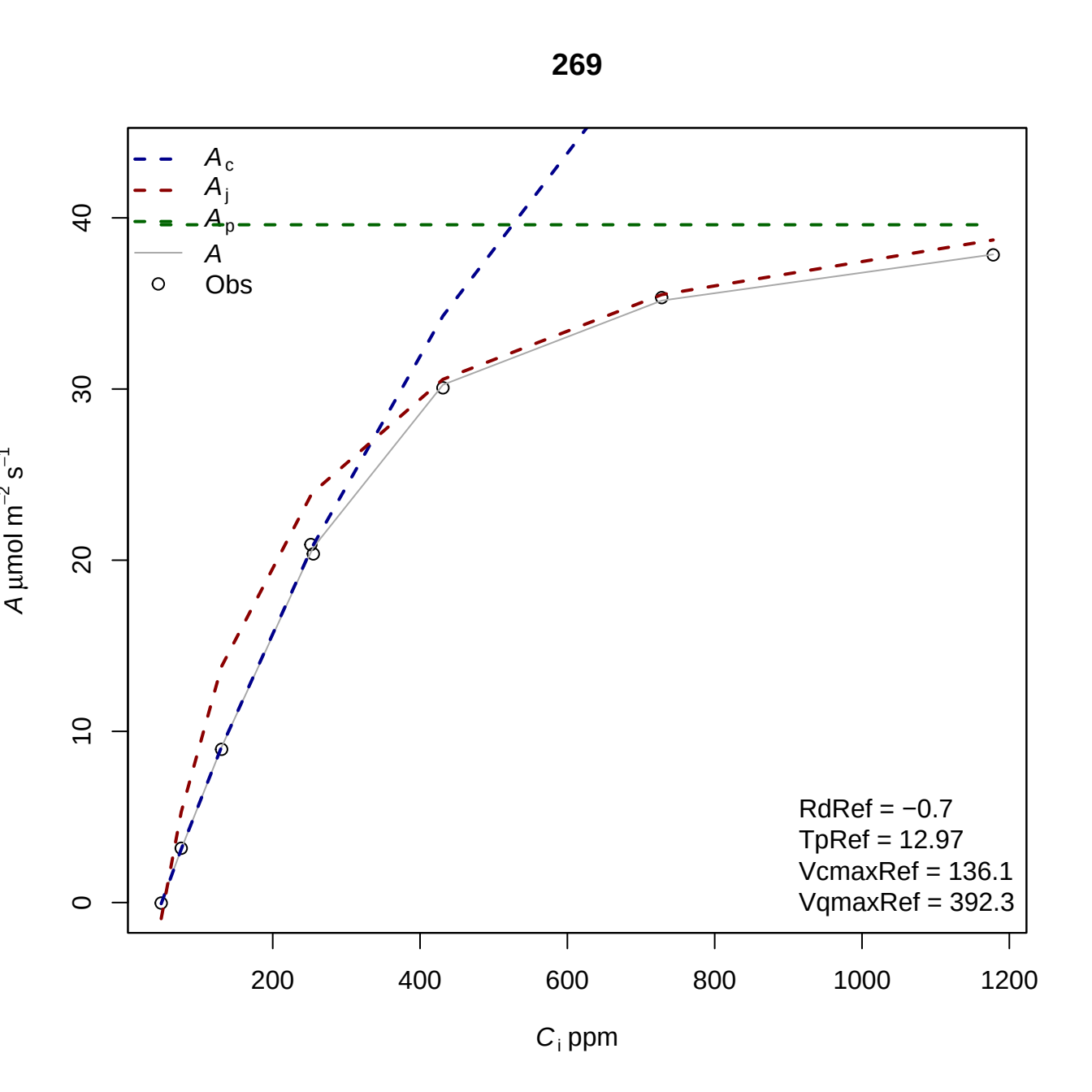

264
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.41
TpRef = 9999
VcmaxRef = 144.48
VqmaxRef = 633.62
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000

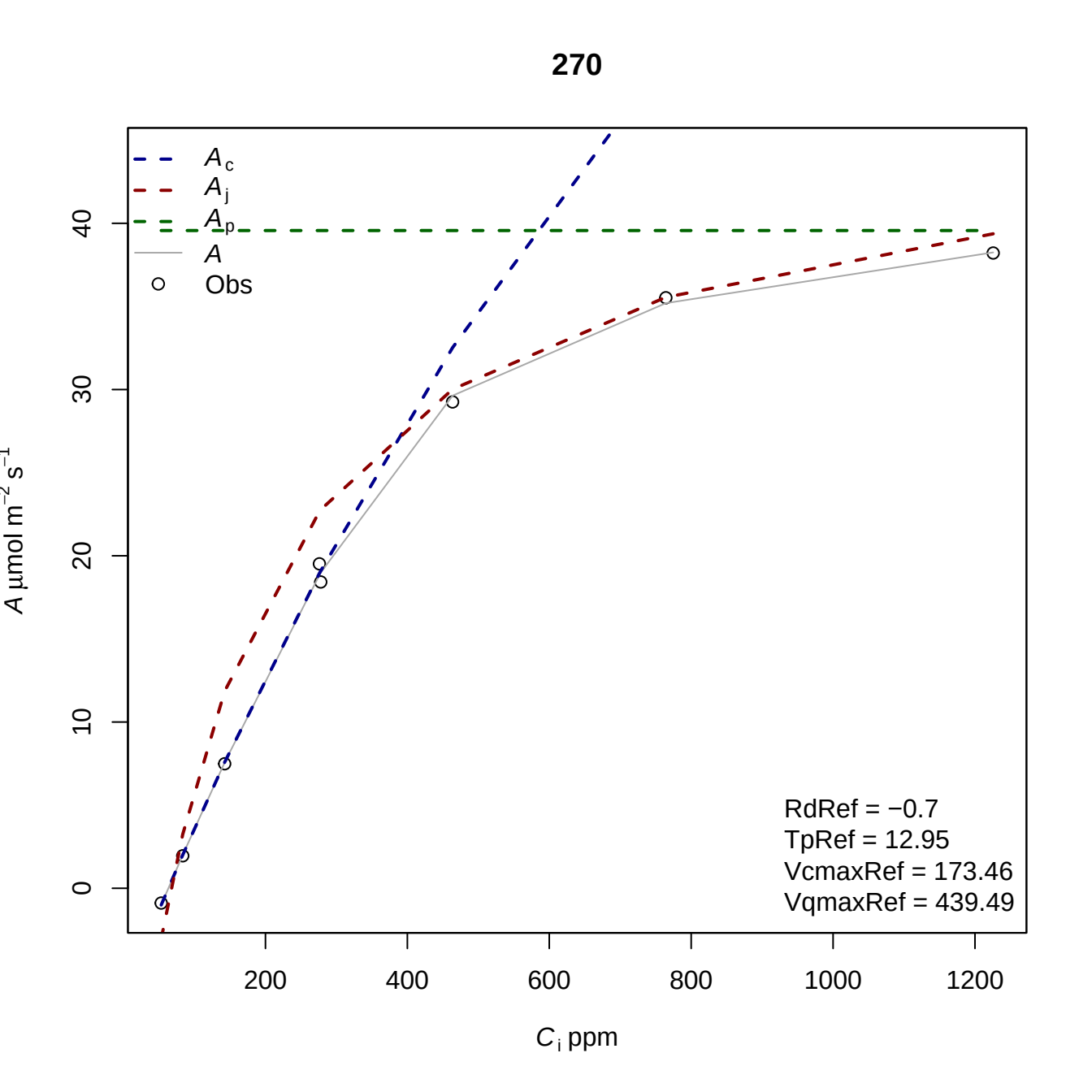

265
$A_c$
$A_j$
$A_p$
$A$
Obs
$A$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
200
400
600
800
1000
1200
$C_i$ ppm
RdRef = −0.85
TpRef = 9999
VcmaxRef = 172.12
VqmaxRef = 485.41

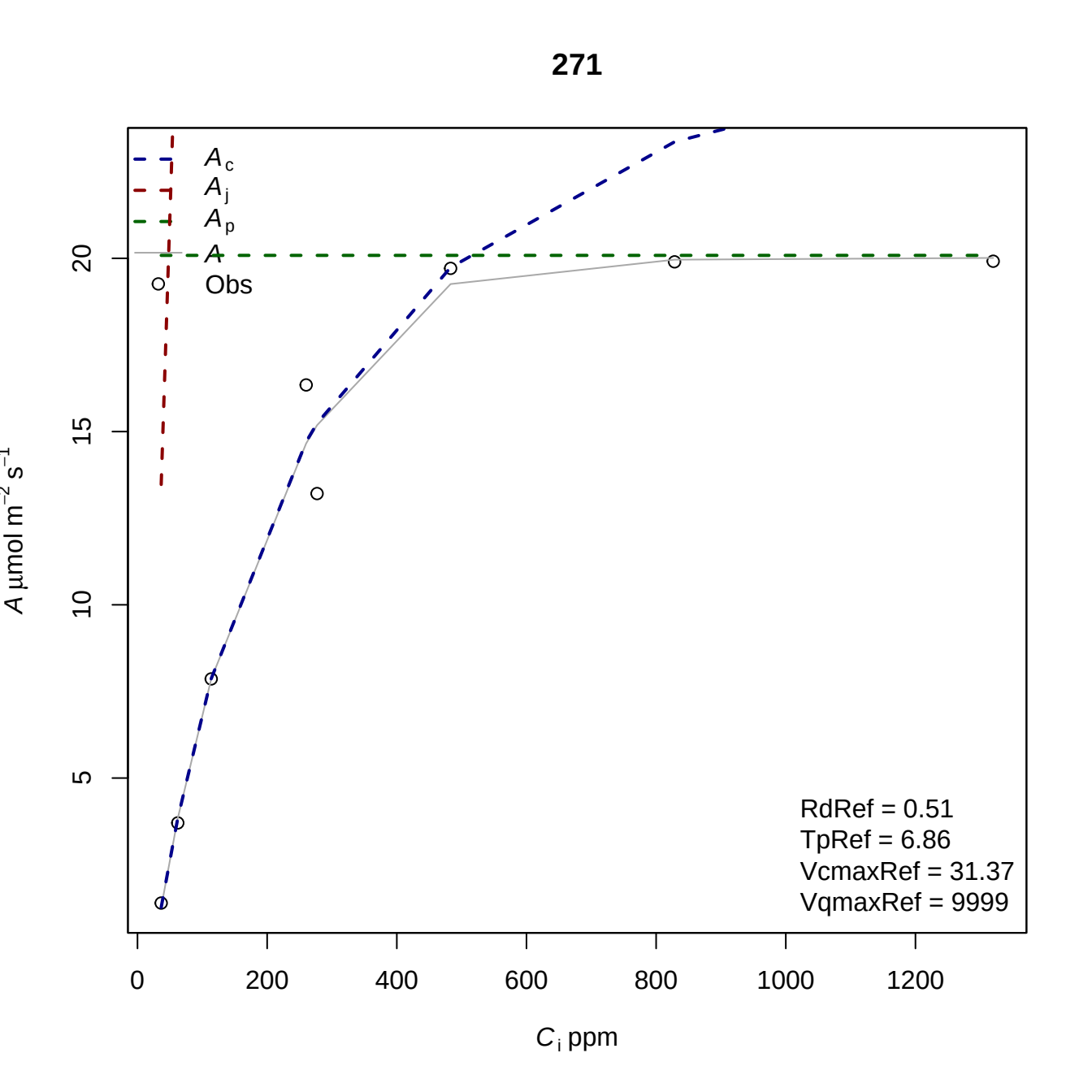

266
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.82
TpRef = 9999
VcmaxRef = 37.34
VqmaxRef = 131.52
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

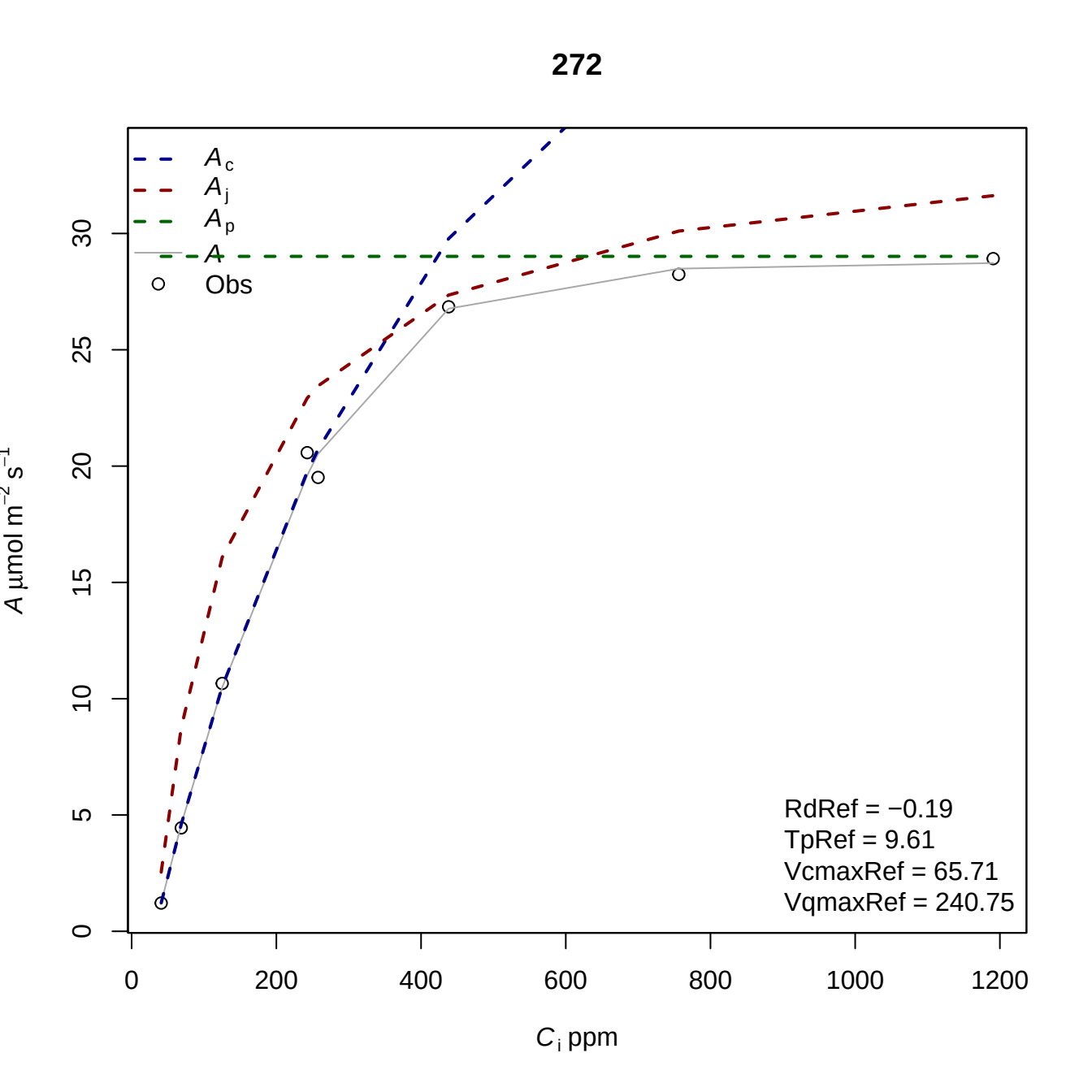

267
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.29
TpRef = 10.02
VcmaxRef = 69.95
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

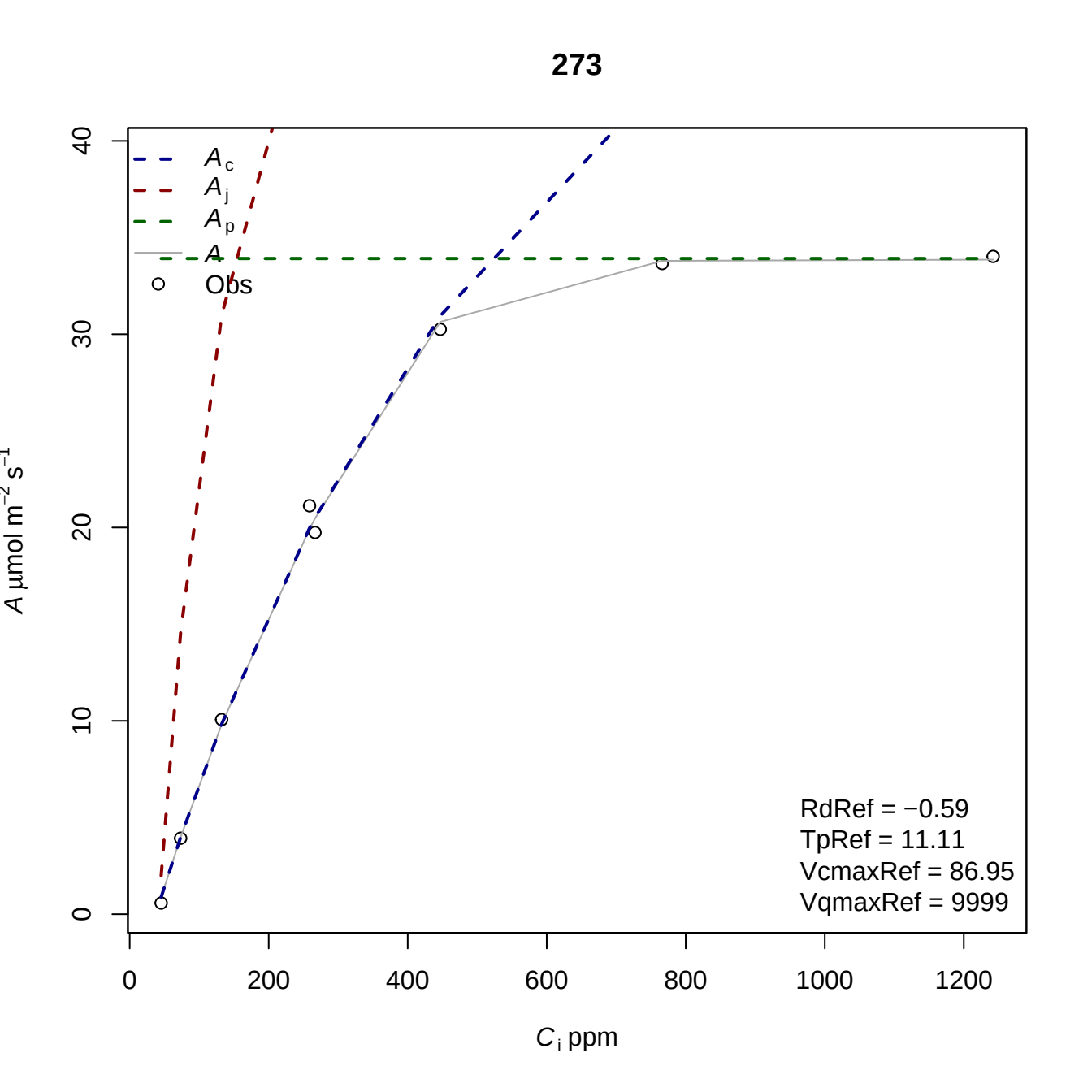

268
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.08
TpRef = 11.83
VcmaxRef = 98.12
VqmaxRef = 342.23
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

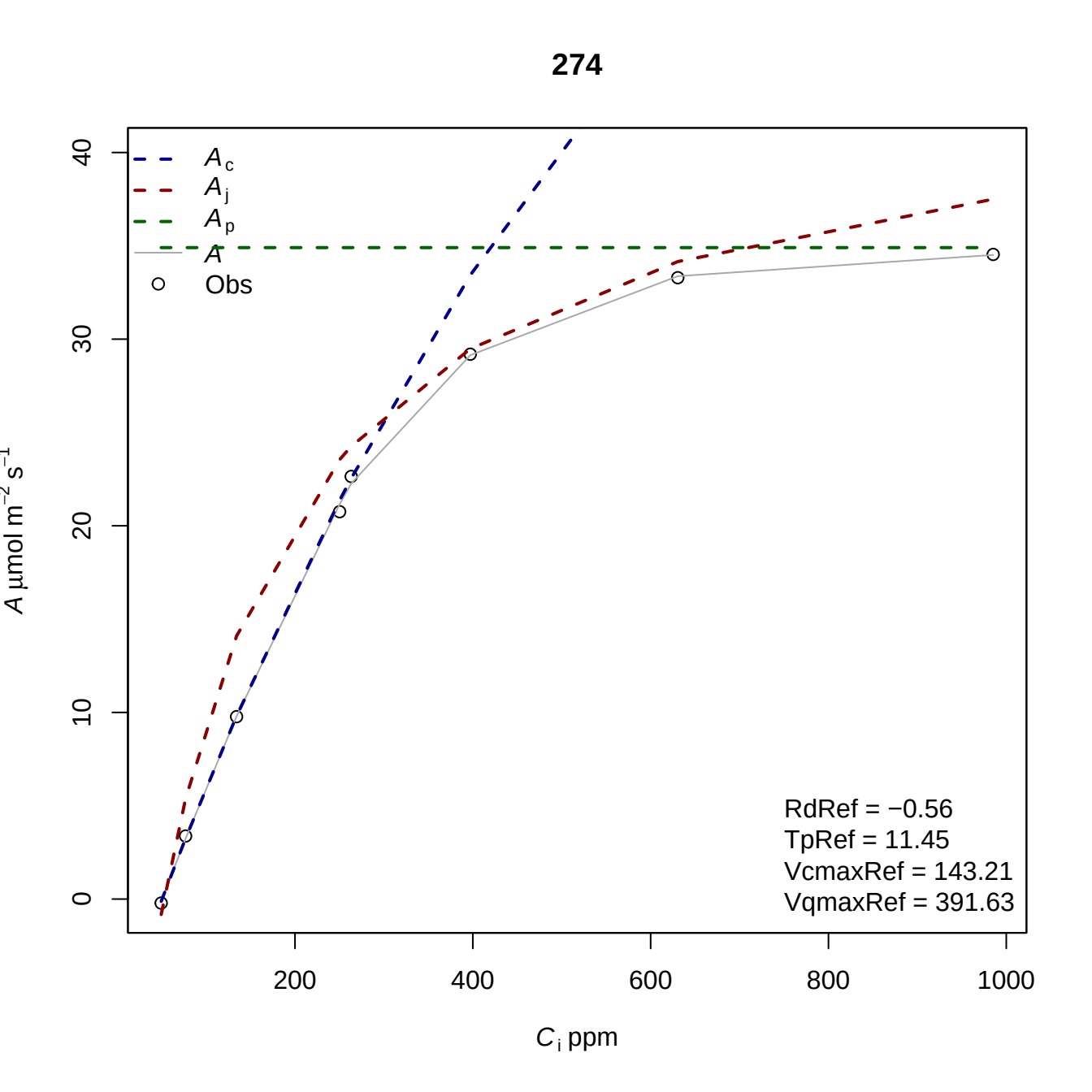

269
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.7
TpRef = 12.97
VcmaxRef = 136.1
VqmaxRef = 392.3
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

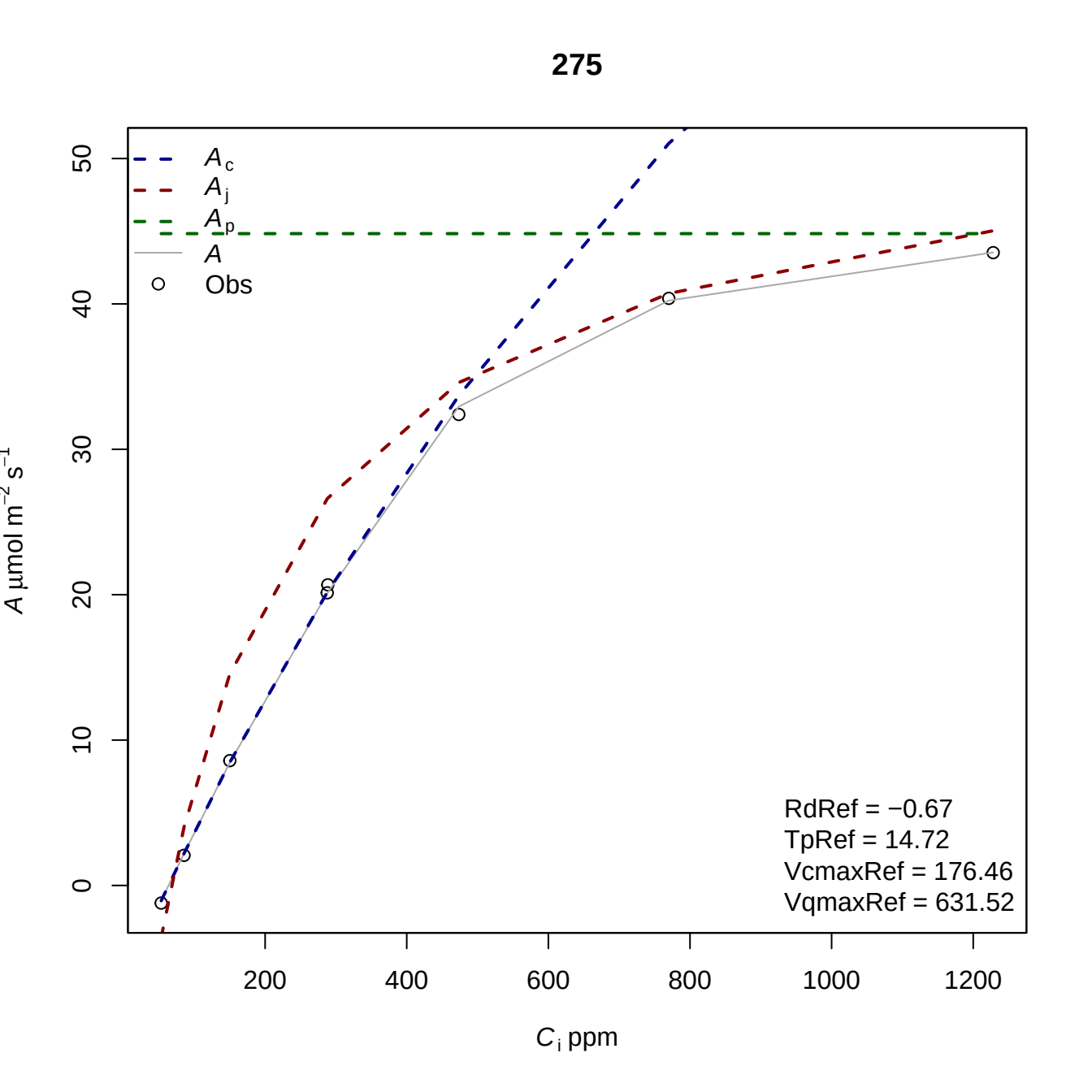

270
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.7
TpRef = 12.95
VcmaxRef = 173.46
VqmaxRef = 439.49
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

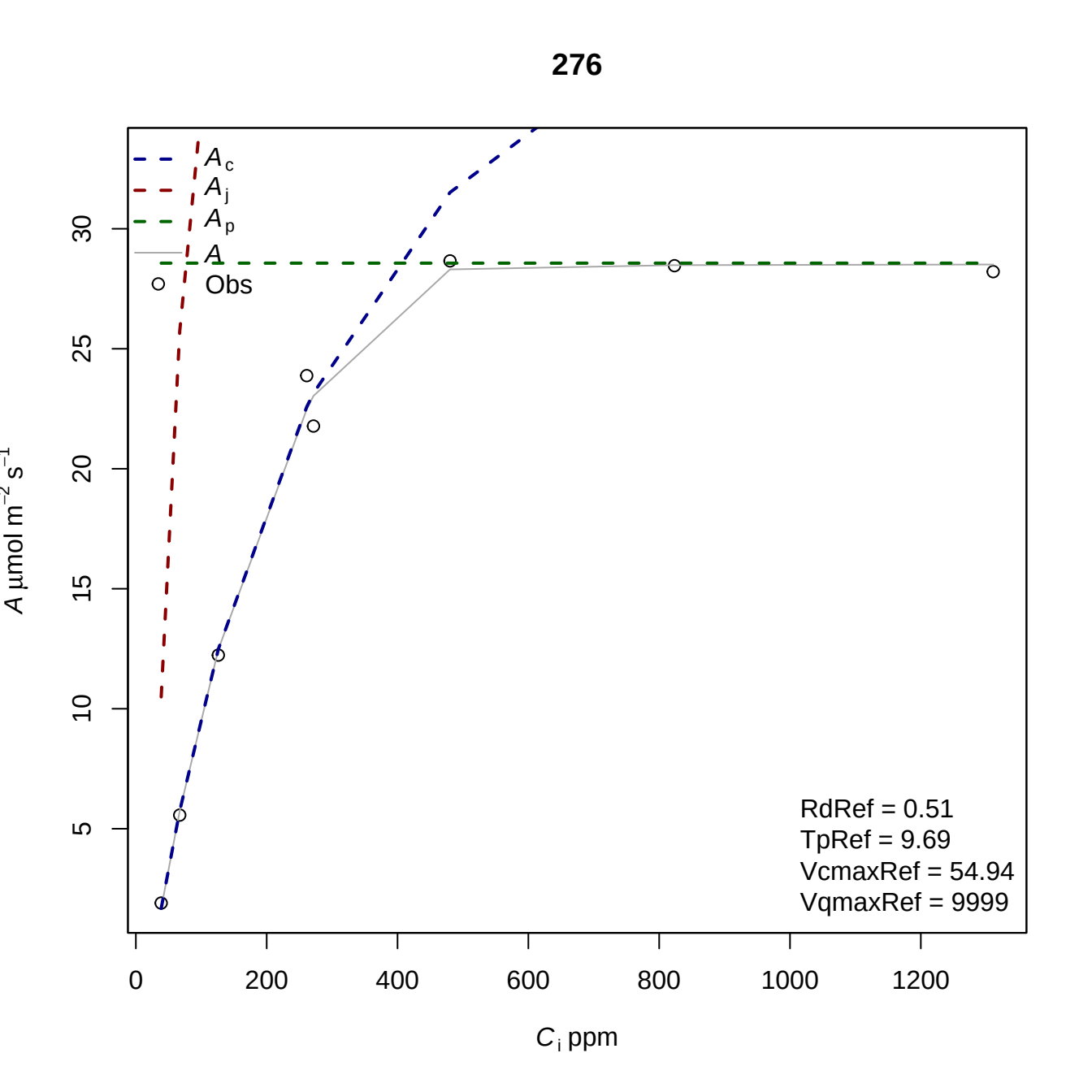

271
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.51
TpRef = 6.86
VcmaxRef = 31.37
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20

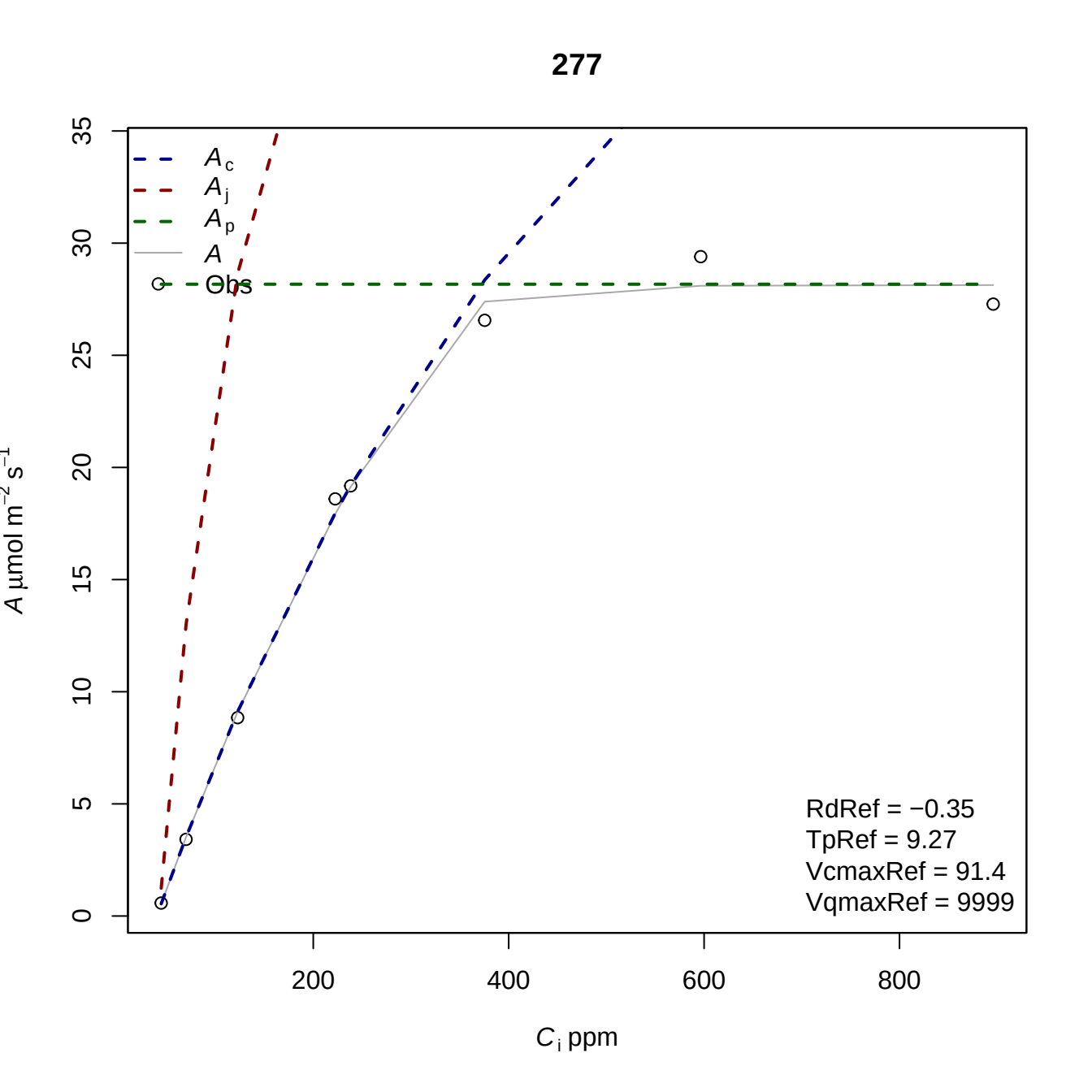

272
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.19
TpRef = 9.61
VcmaxRef = 65.71
VqmaxRef = 240.75
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

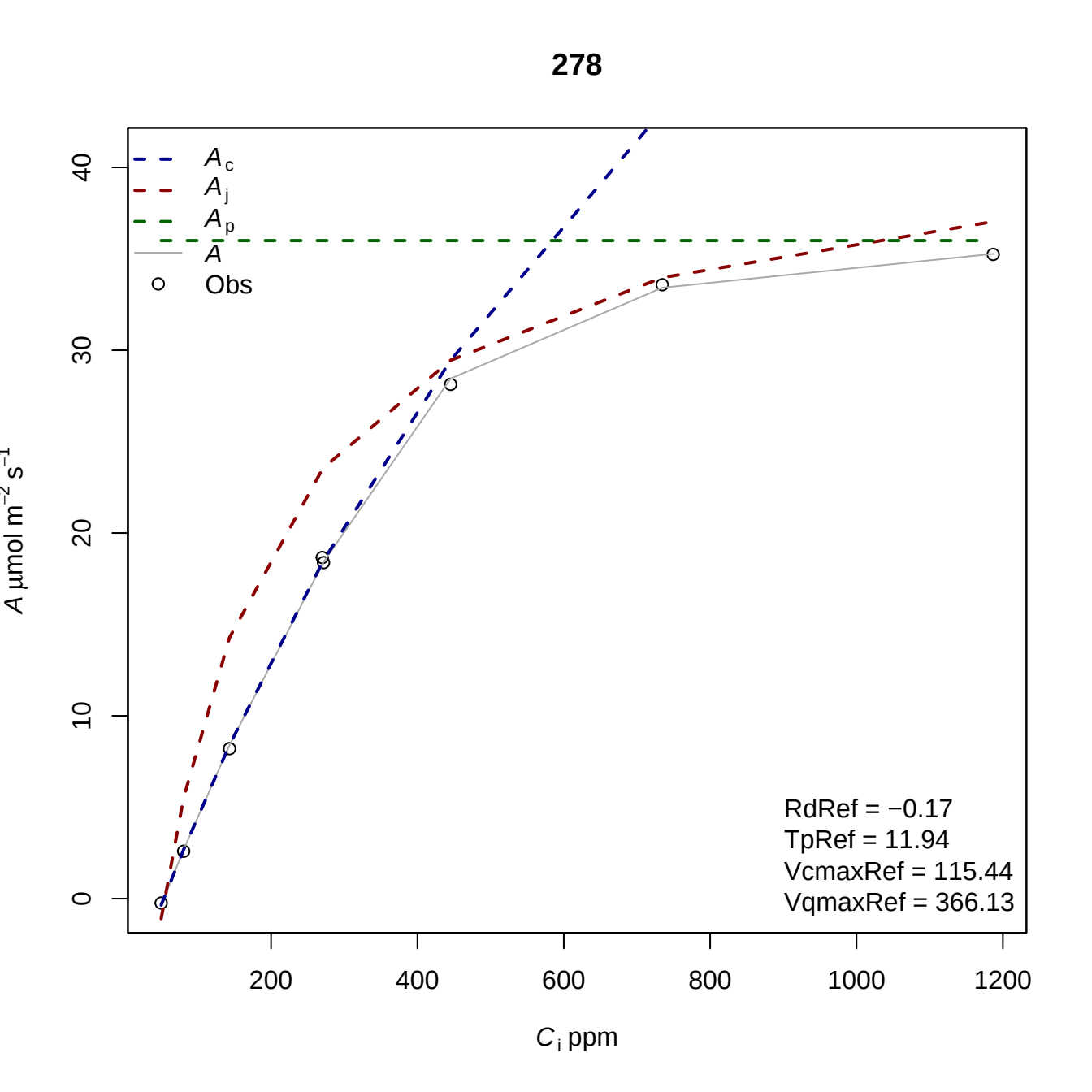

273
$A_c$
$A_j$
$A_p$
A
Obs
RdRef = −0.59
TpRef = 11.11
VcmaxRef = 86.95
VqmaxRef = 9999
A µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

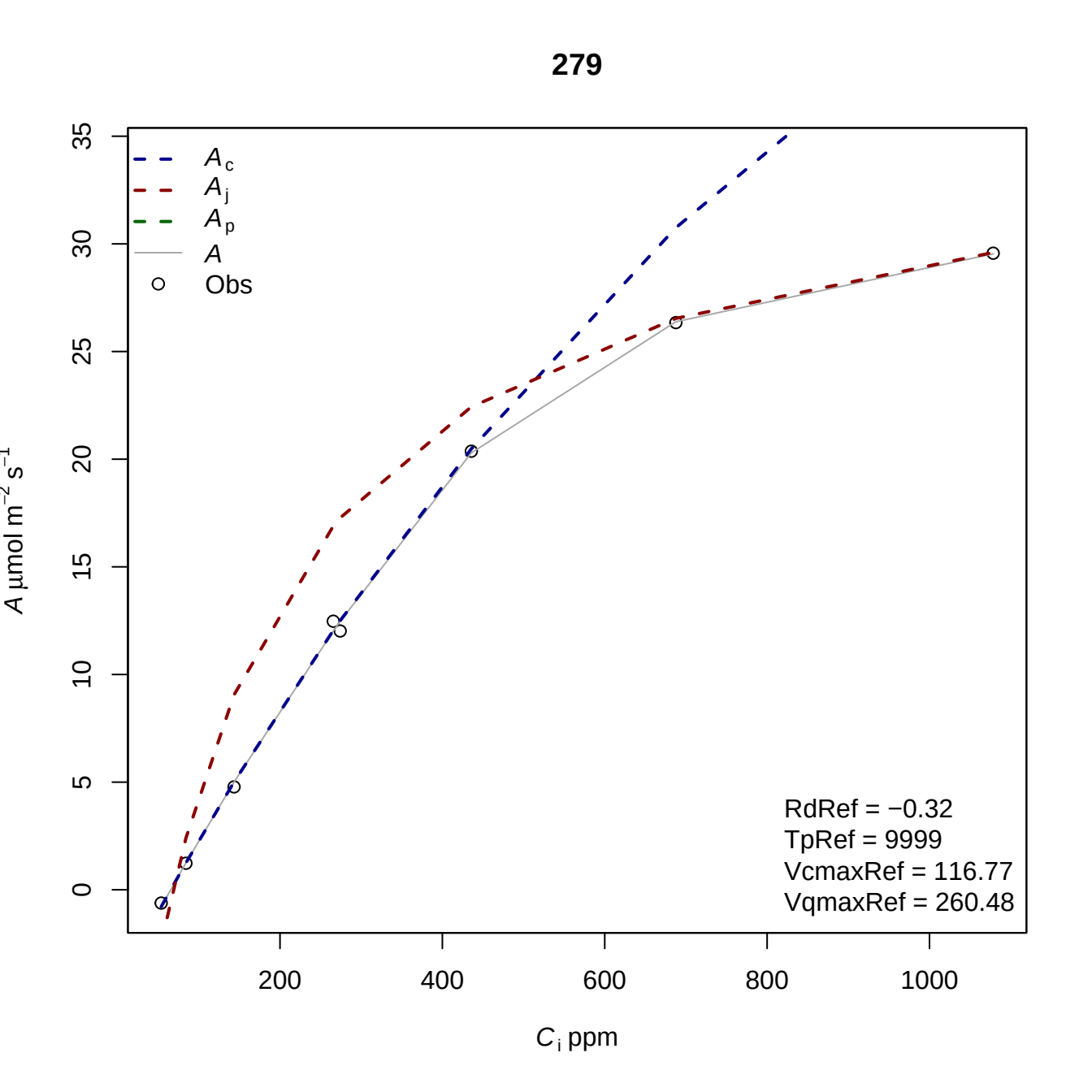

274
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.56
TpRef = 11.45
VcmaxRef = 143.21
VqmaxRef = 391.63
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000

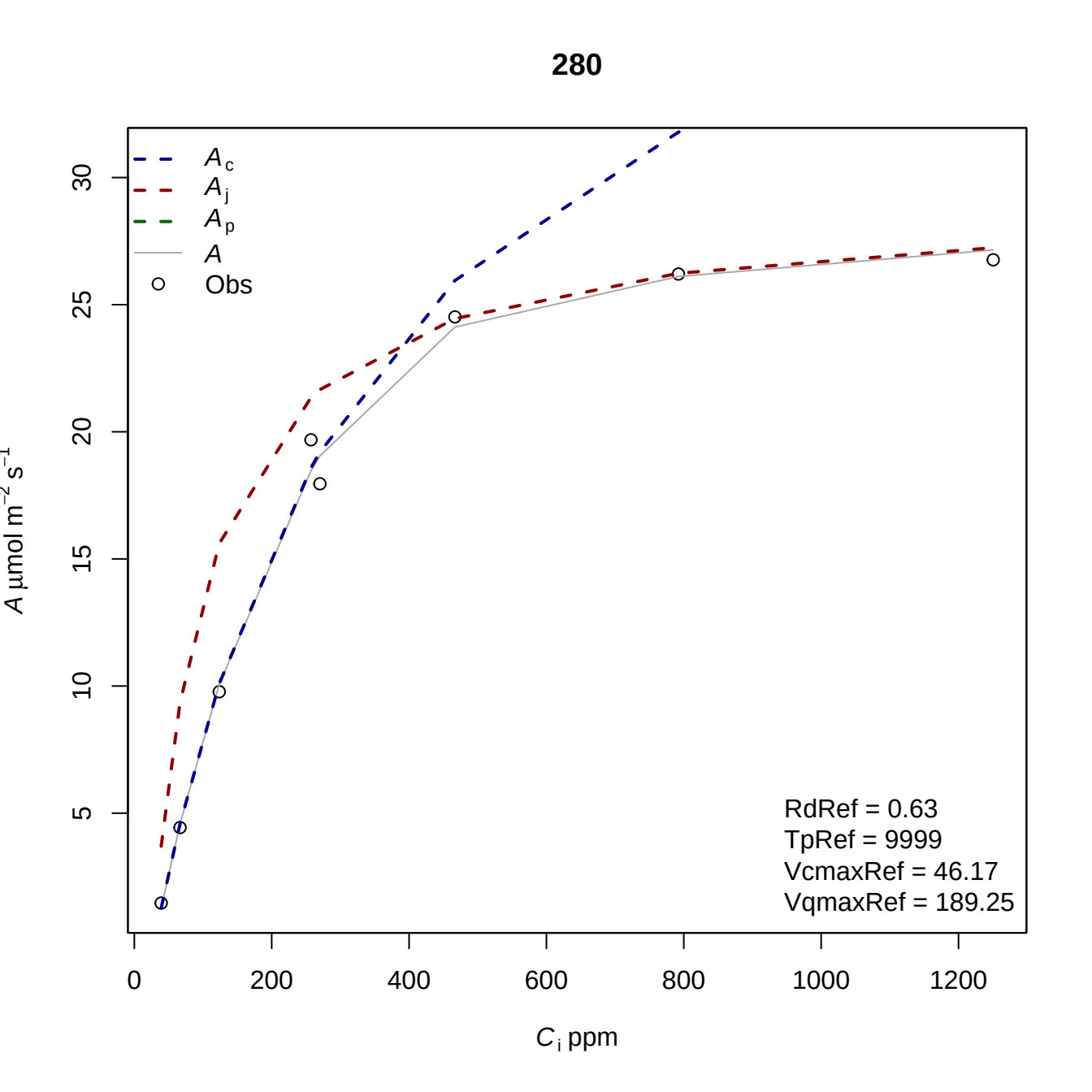

275
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.67
TpRef = 14.72
VcmaxRef = 176.46
VqmaxRef = 631.52
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

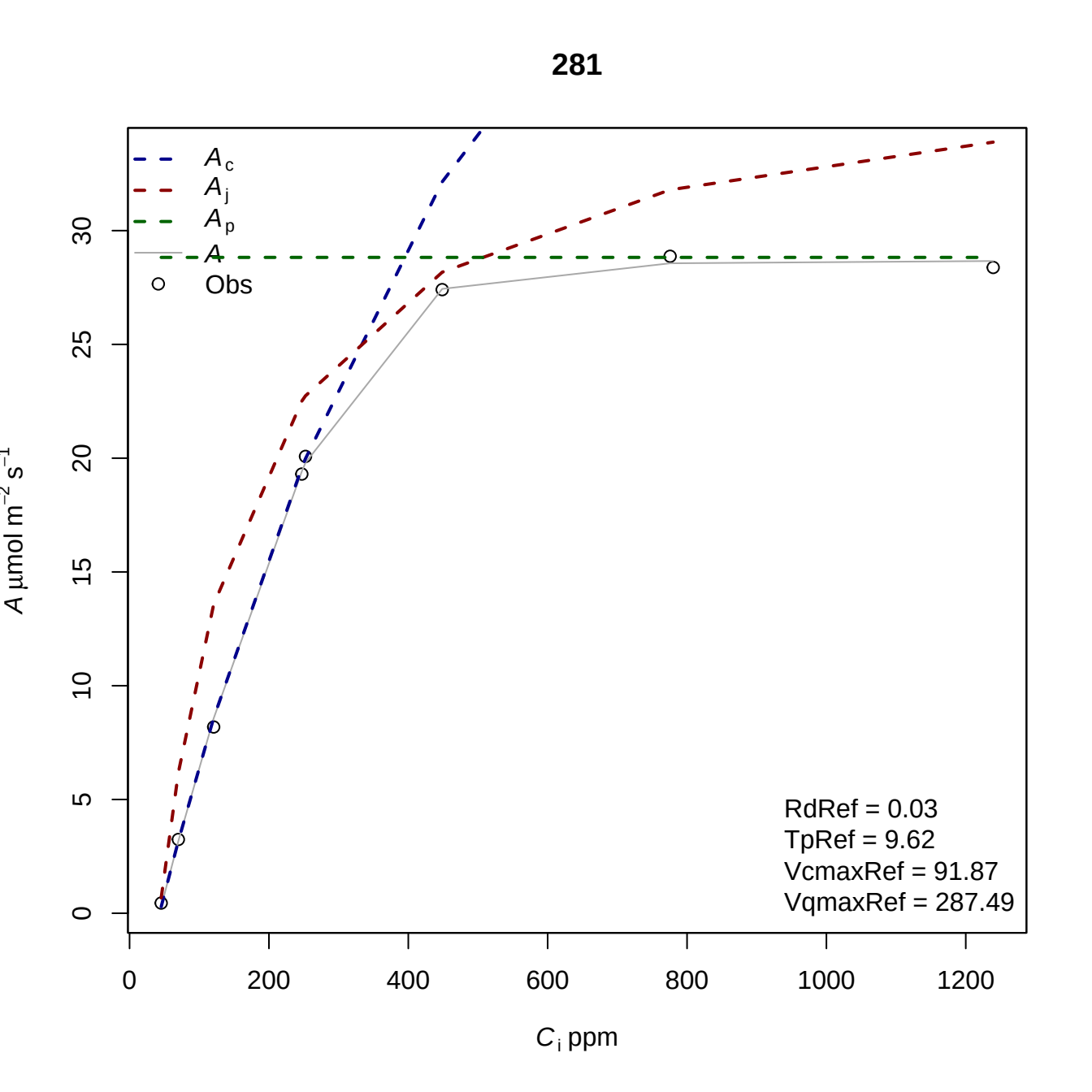

276
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.51
TpRef = 9.69
VcmaxRef = 54.94
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25
30

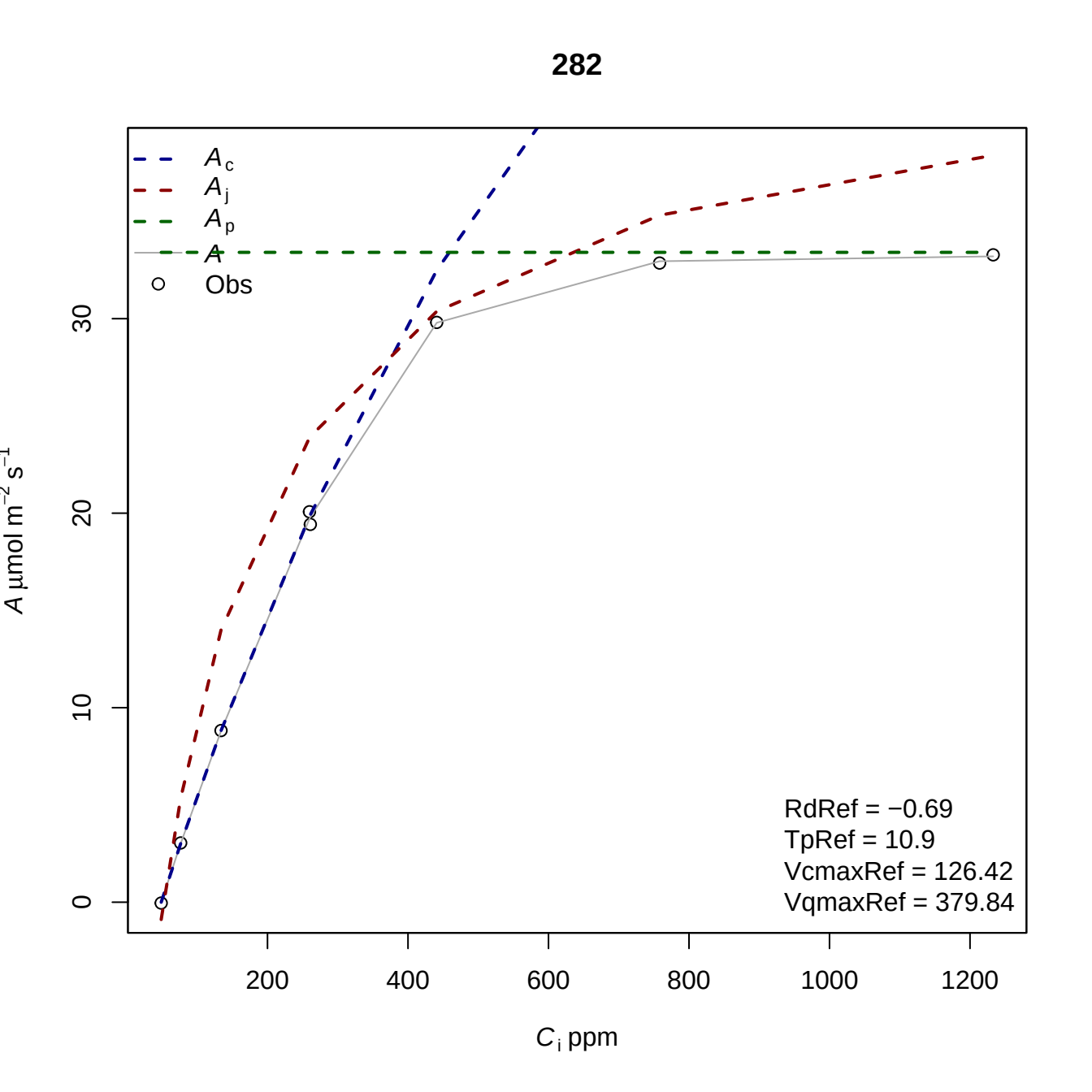

277
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.35
TpRef = 9.27
VcmaxRef = 91.4
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
200
400
600
800

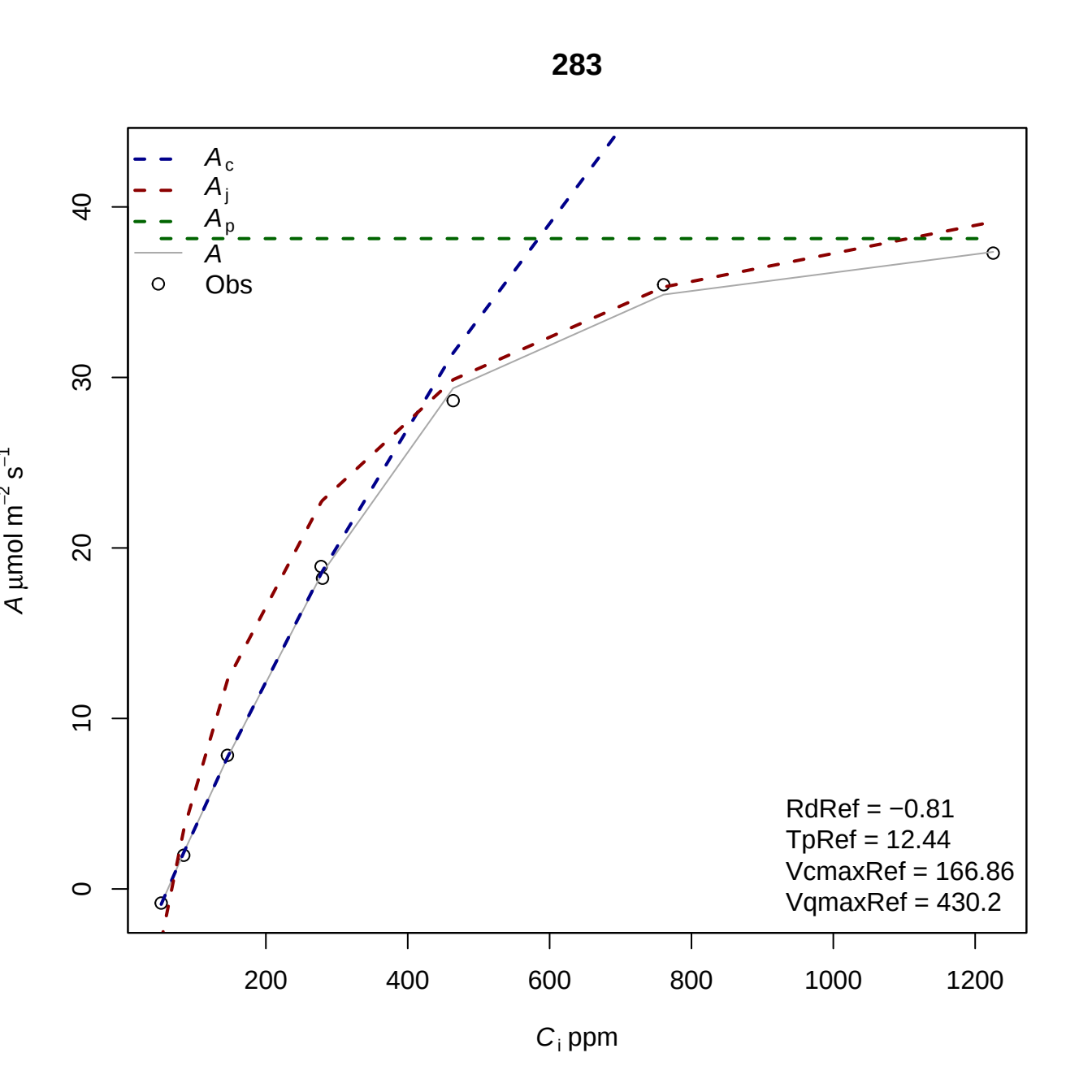

278
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.17
TpRef = 11.94
VcmaxRef = 115.44
VqmaxRef = 366.13
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

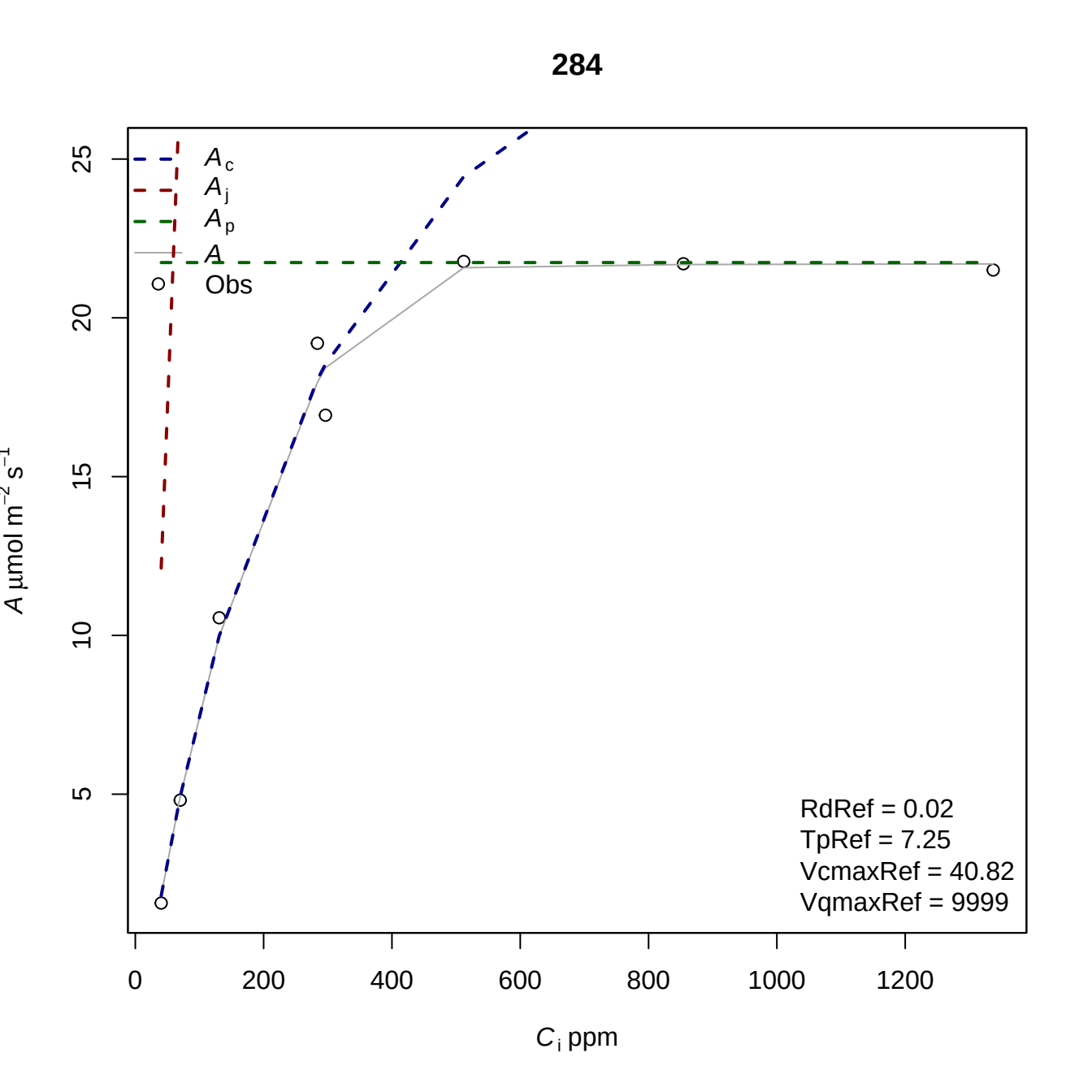

279
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.32
TpRef = 9999
VcmaxRef = 116.77
VqmaxRef = 260.48
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
200
400
600
800
1000

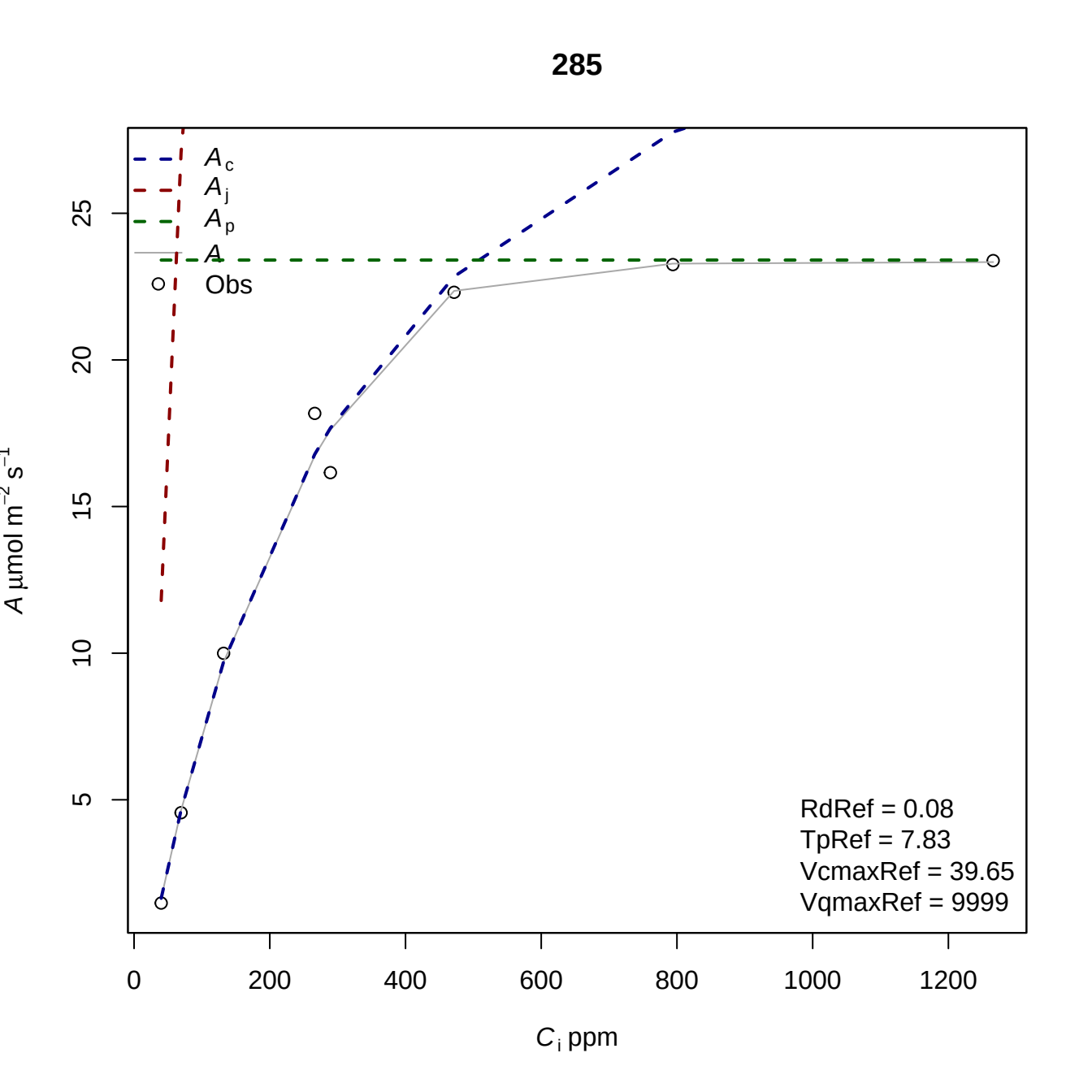

280
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.63
TpRef = 9999
VcmaxRef = 46.17
VqmaxRef = 189.25
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25
30

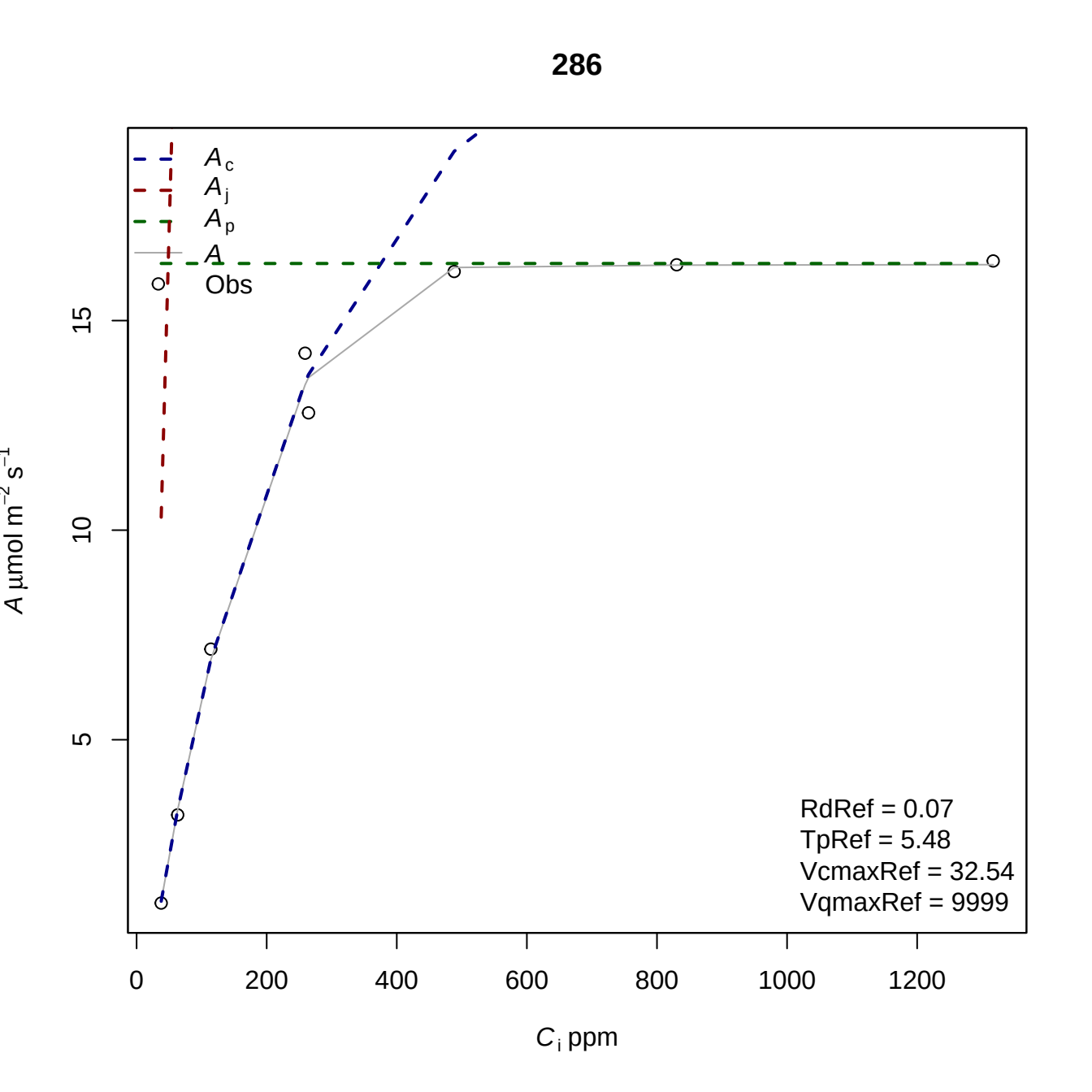

281
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.03
TpRef = 9.62
VcmaxRef = 91.87
VqmaxRef = 287.49
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

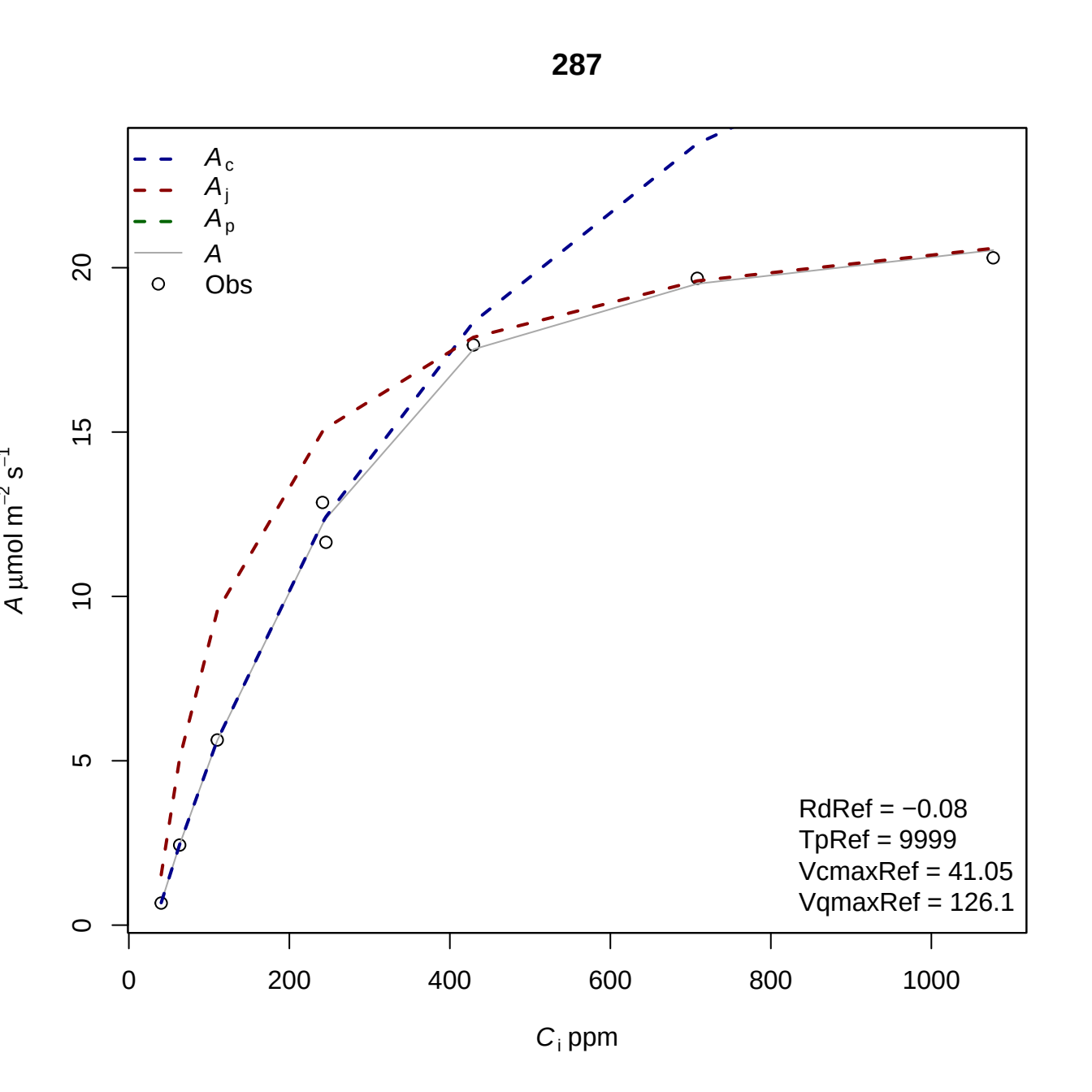

282
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.69
TpRef = 10.9
VcmaxRef = 126.42
VqmaxRef = 379.84
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
200
400
600
800
1000
1200

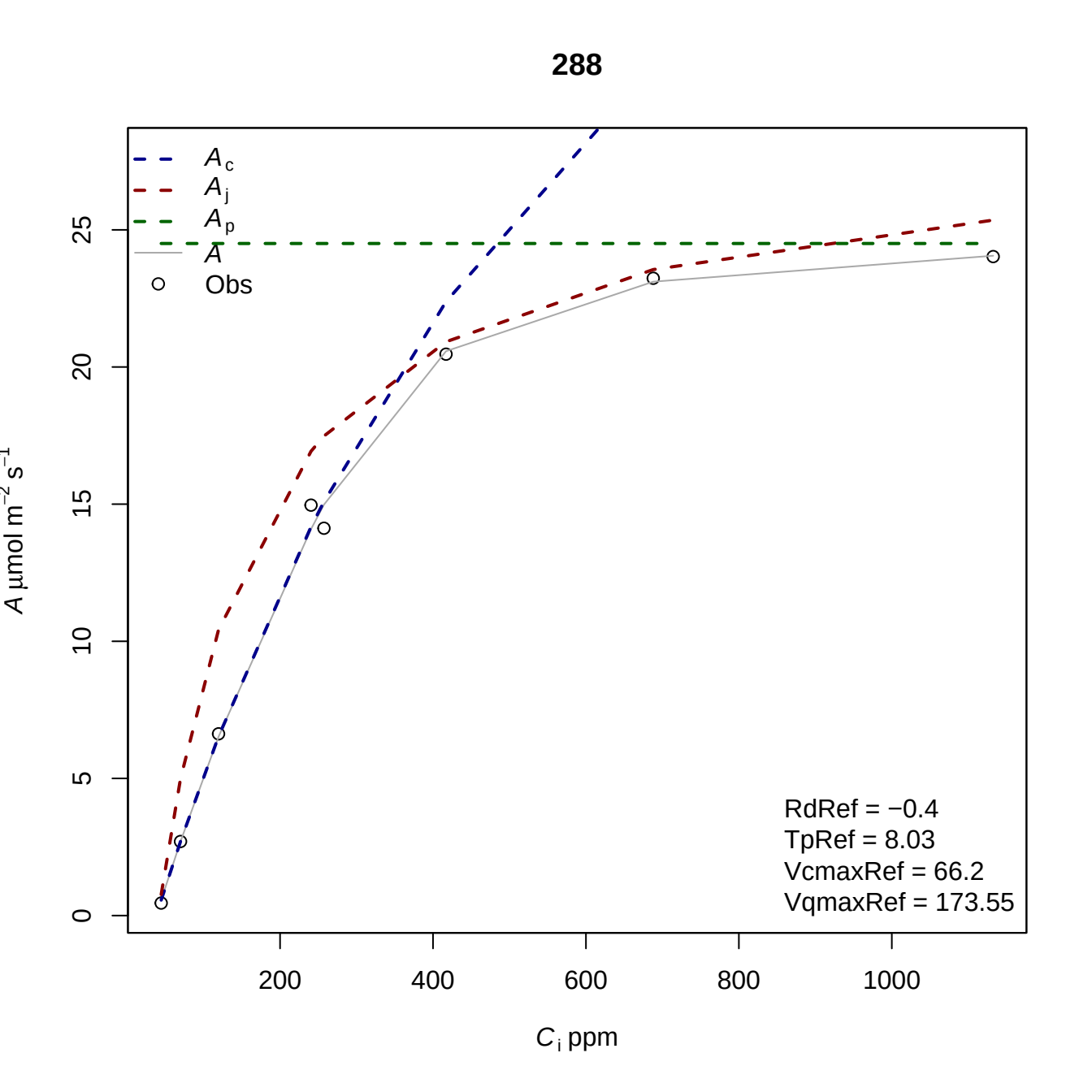

283
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.81
TpRef = 12.44
VcmaxRef = 166.86
VqmaxRef = 430.2
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

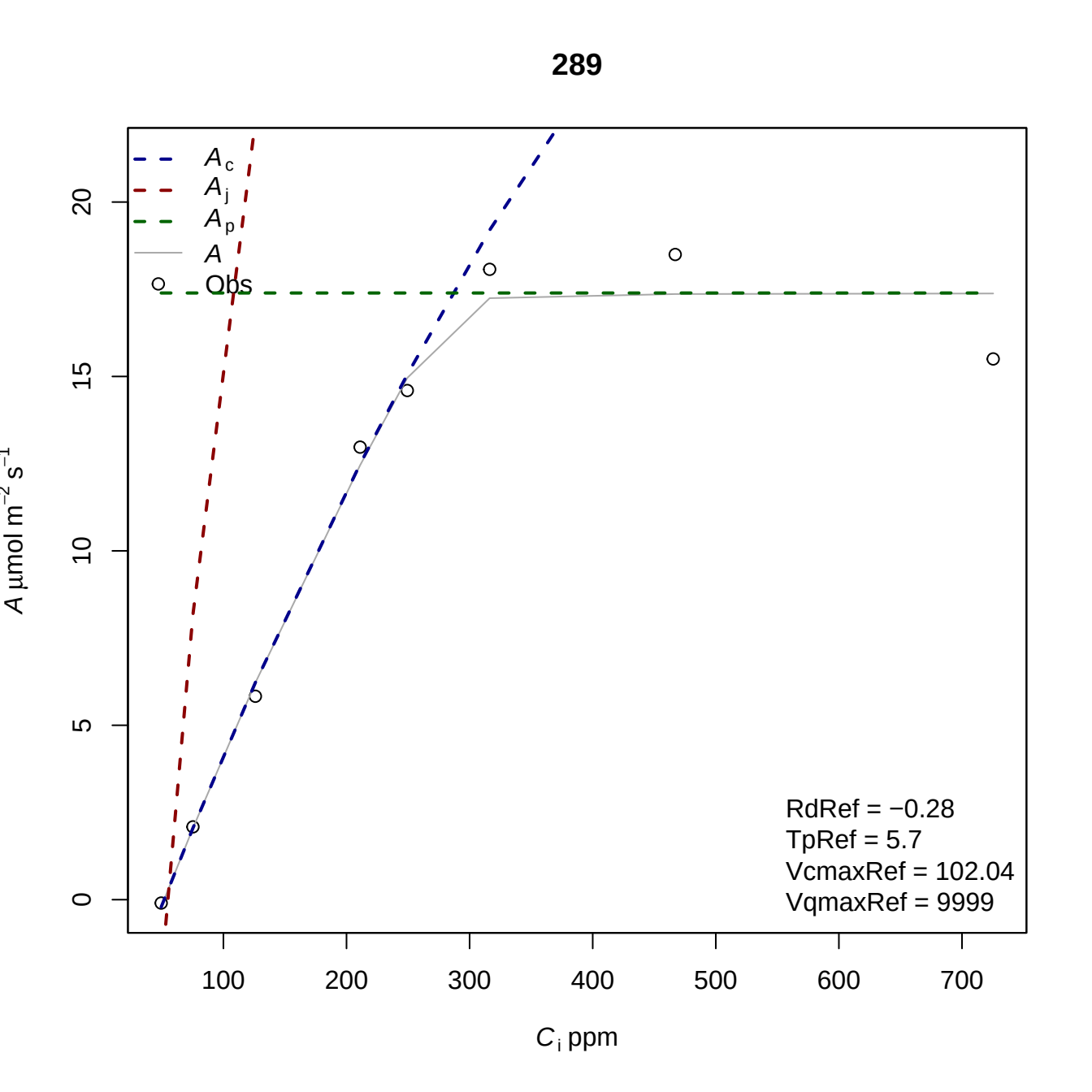

284
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.02
TpRef = 7.25
VcmaxRef = 40.82
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

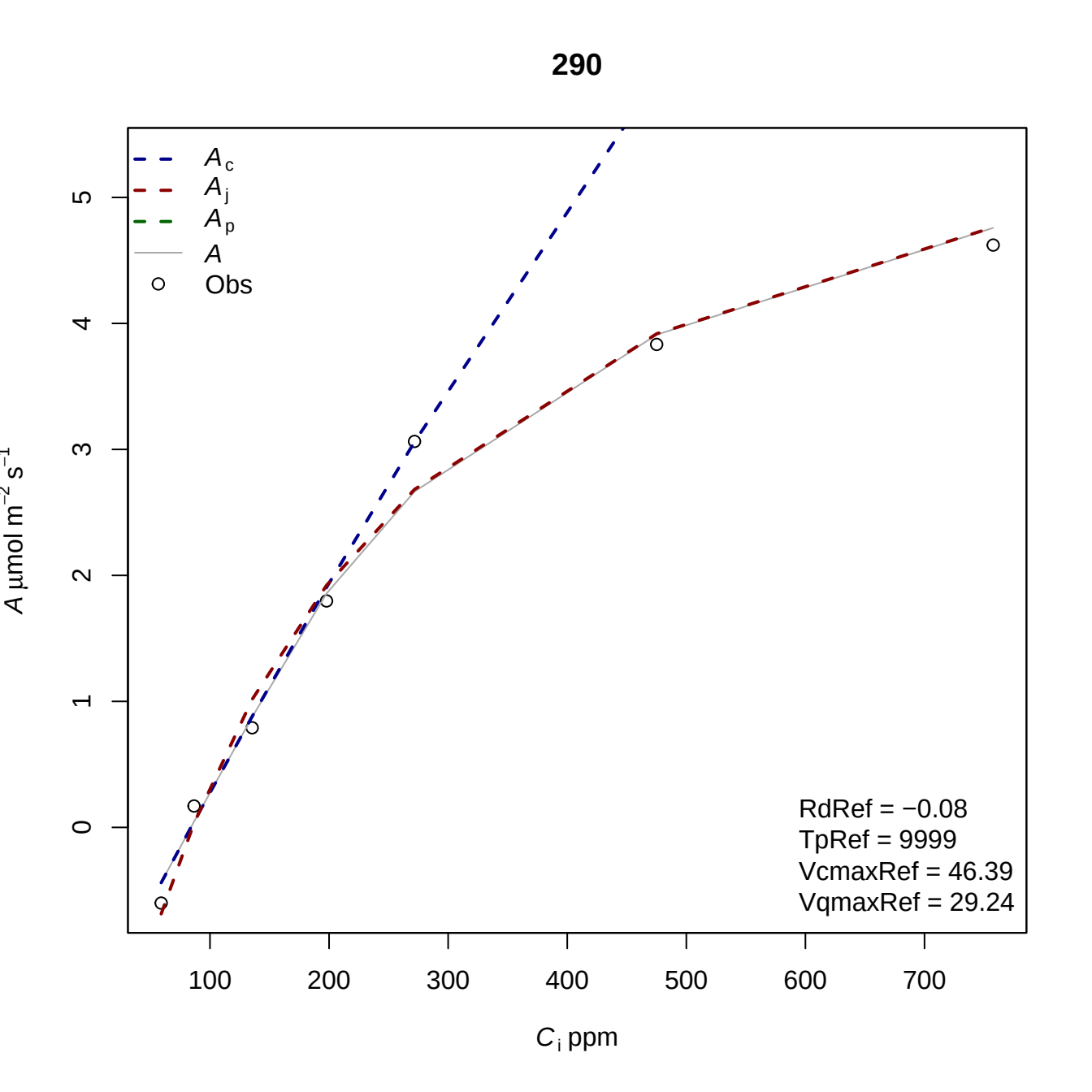

285
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.08
TpRef = 7.83
VcmaxRef = 39.65
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

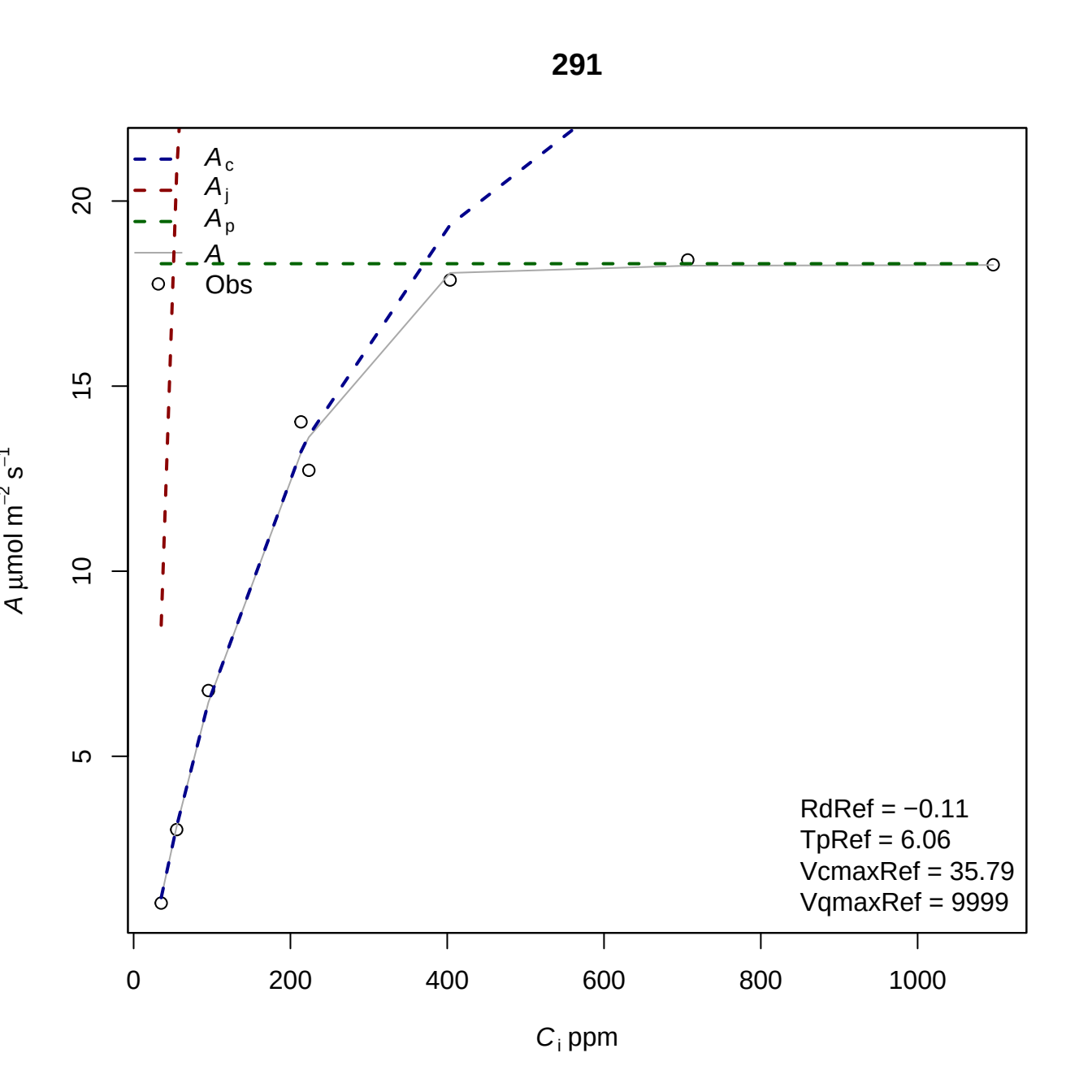

286
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.07
TpRef = 5.48
VcmaxRef = 32.54
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15

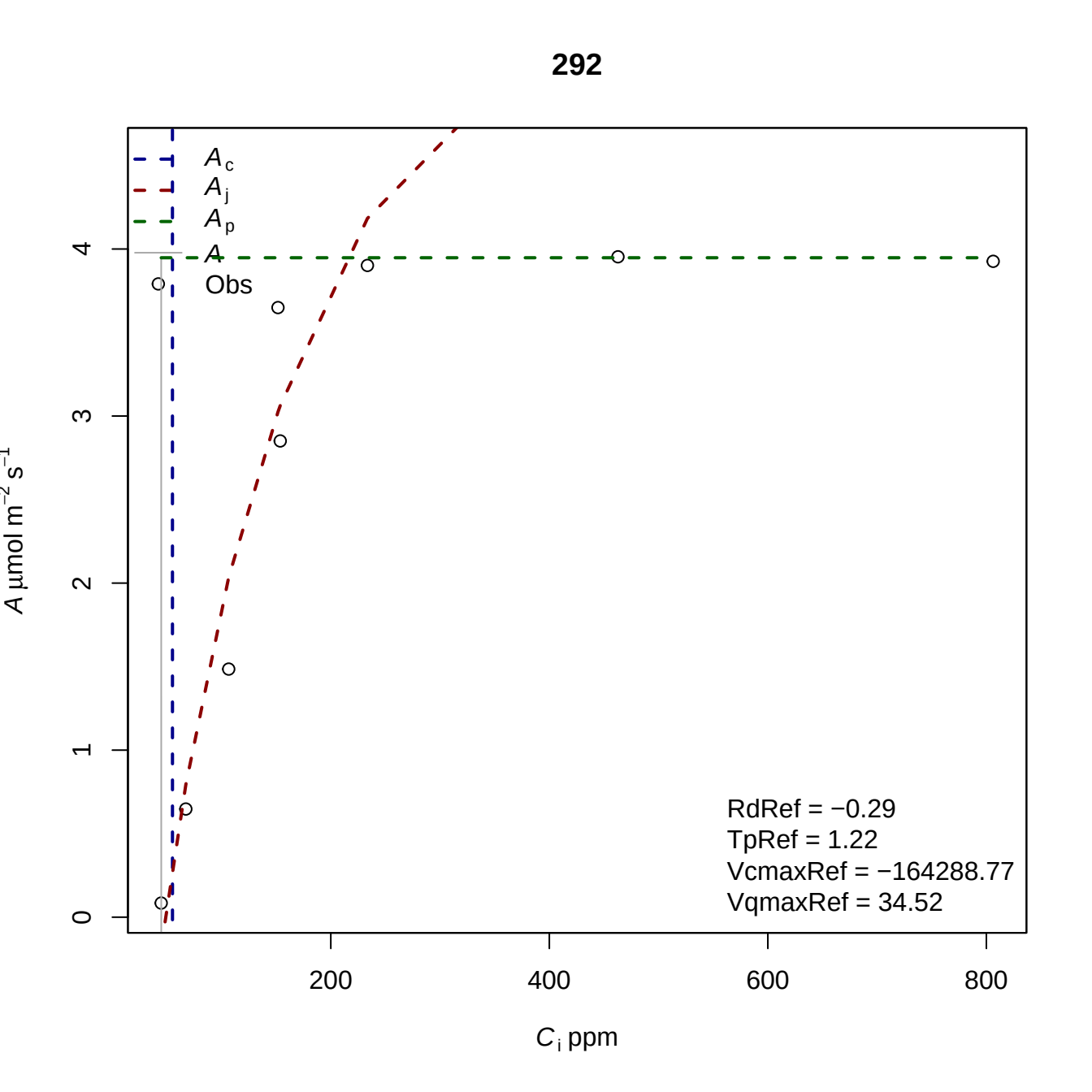

287
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.08
TpRef = 9999
VcmaxRef = 41.05
VqmaxRef = 126.1
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000

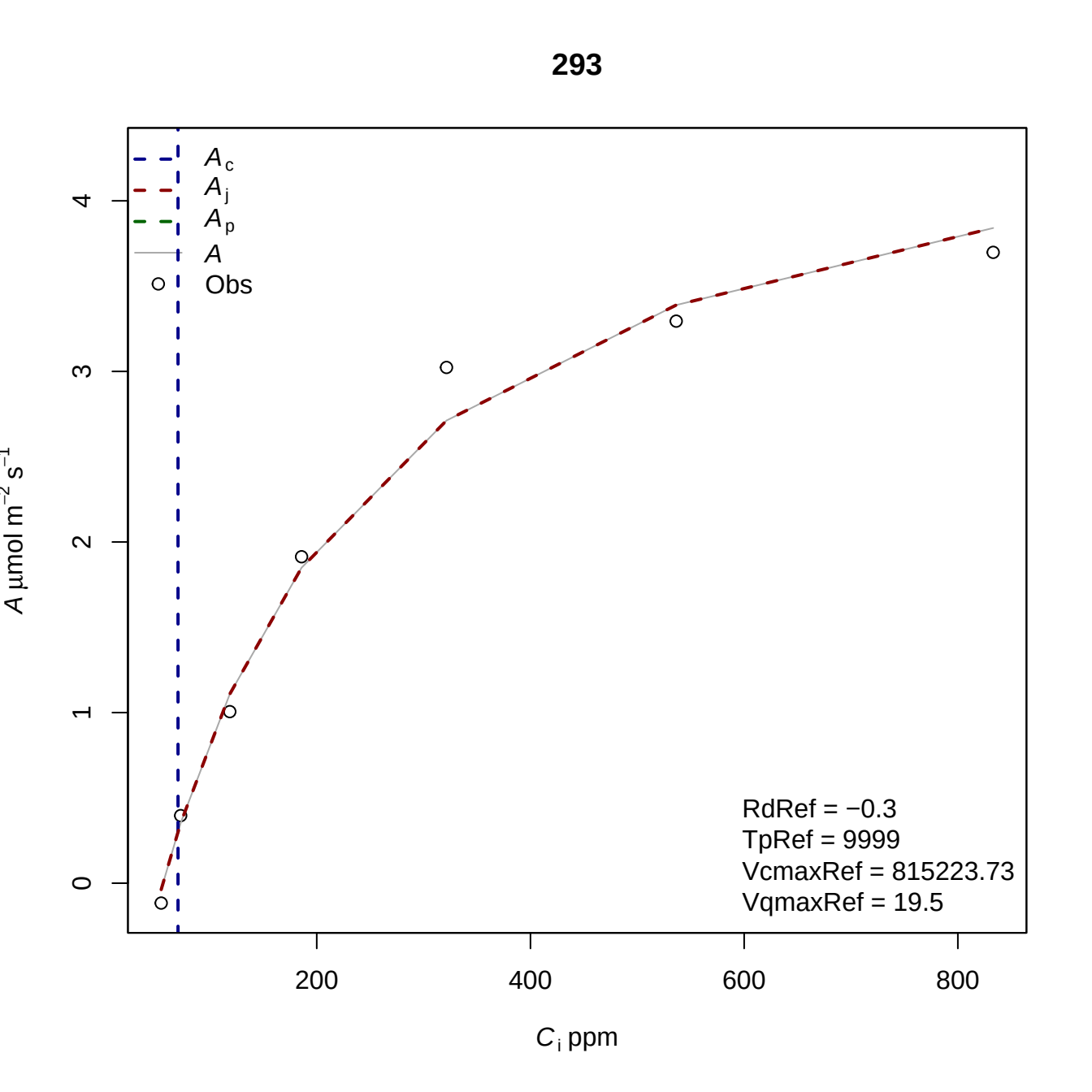

288
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.4
TpRef = 8.03
VcmaxRef = 66.2
VqmaxRef = 173.55
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000

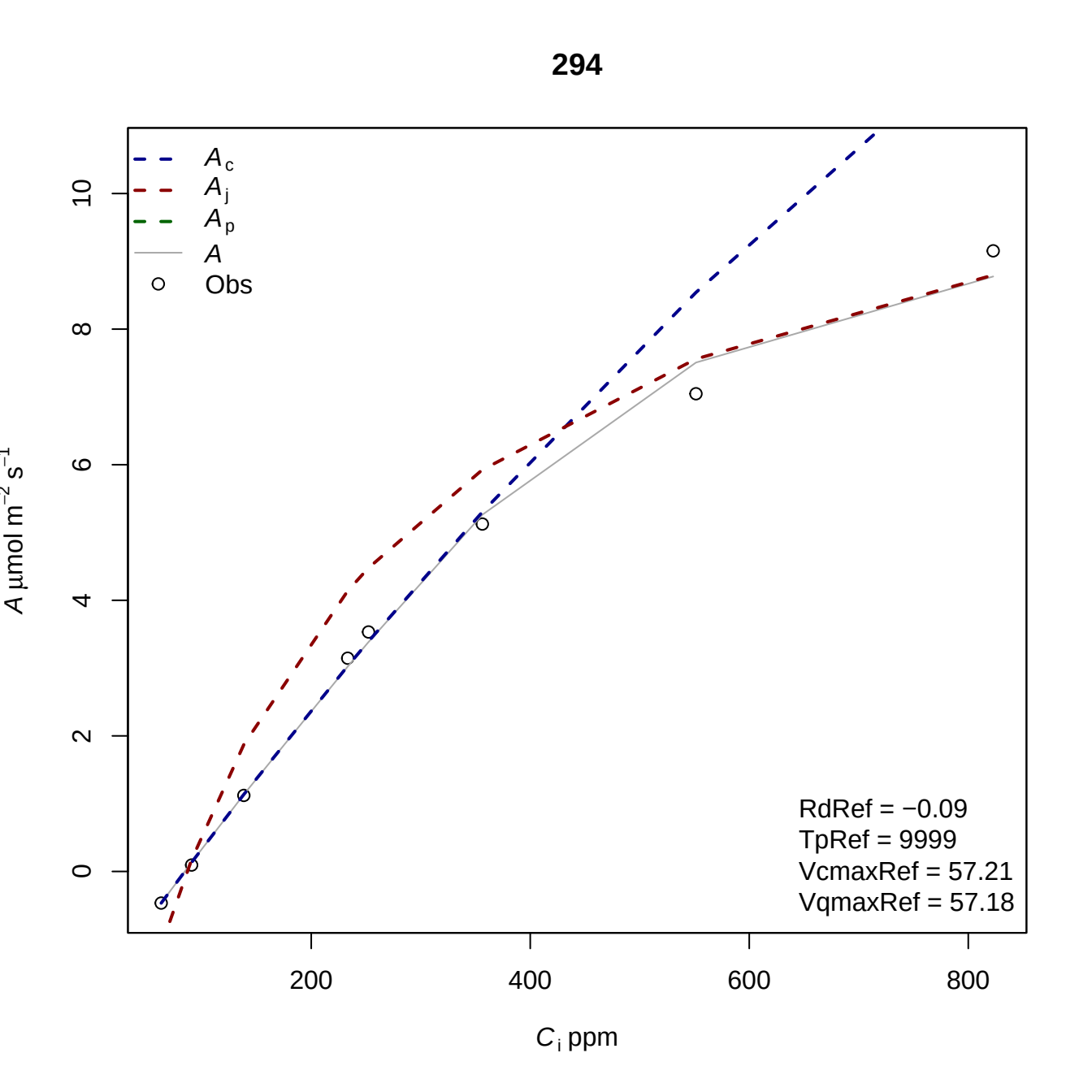

289
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.28
TpRef = 5.7
VcmaxRef = 102.04
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
100
200
300
400
500
600
700

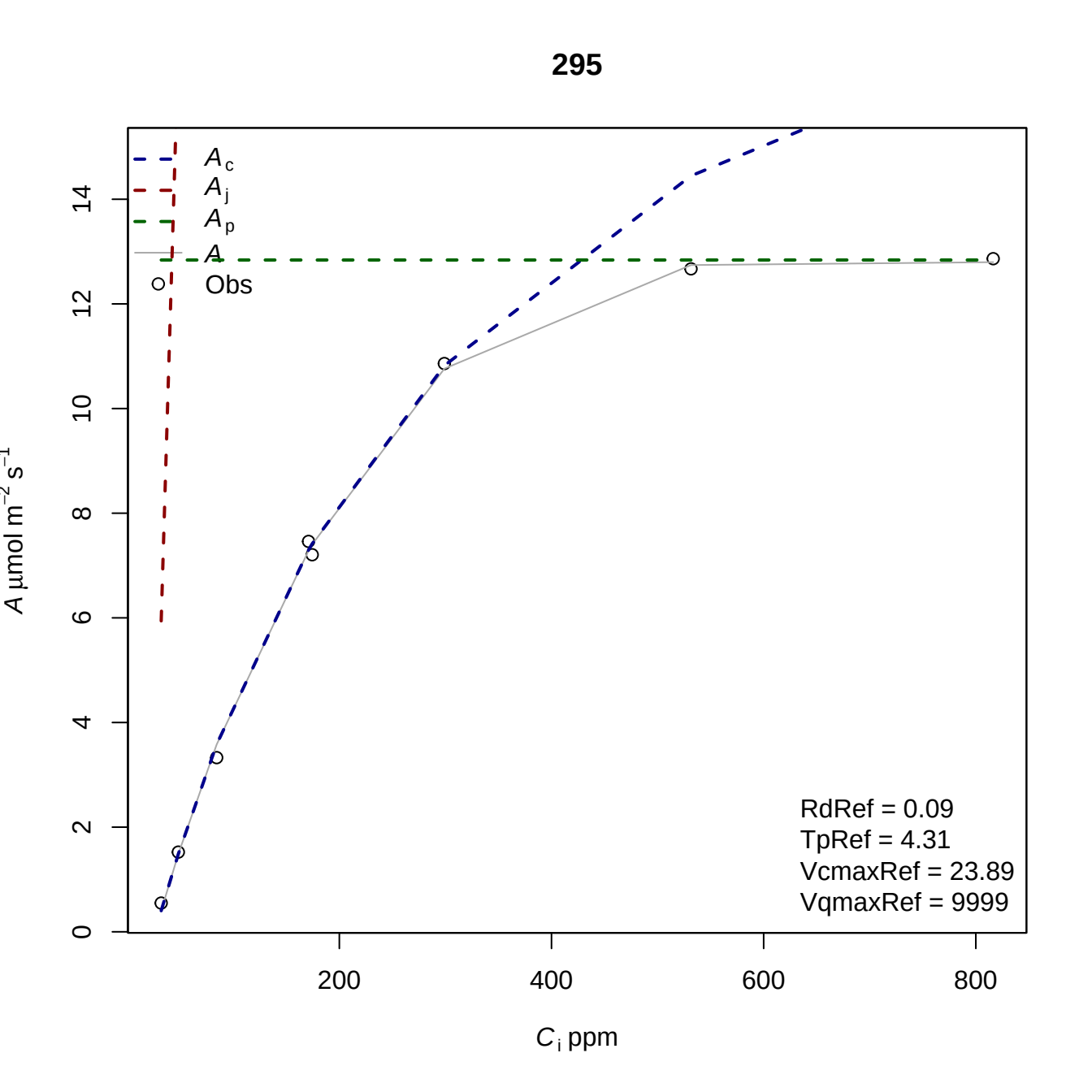

290
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.08
TpRef = 9999
VcmaxRef = 46.39
VqmaxRef = 29.24
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
100
200
300
400
500
600
700

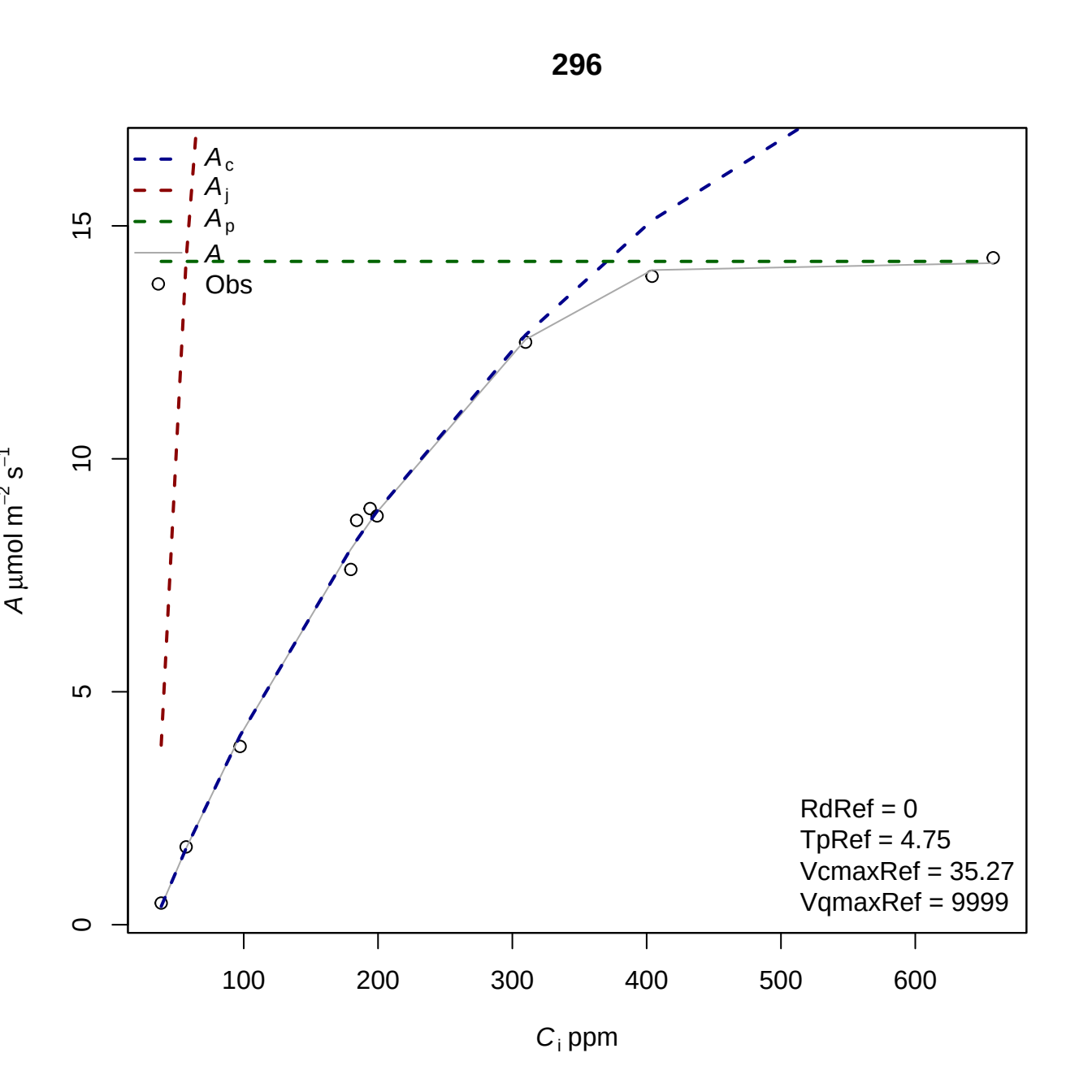

291
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.11
TpRef = 6.06
VcmaxRef = 35.79
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
5
10
15
20

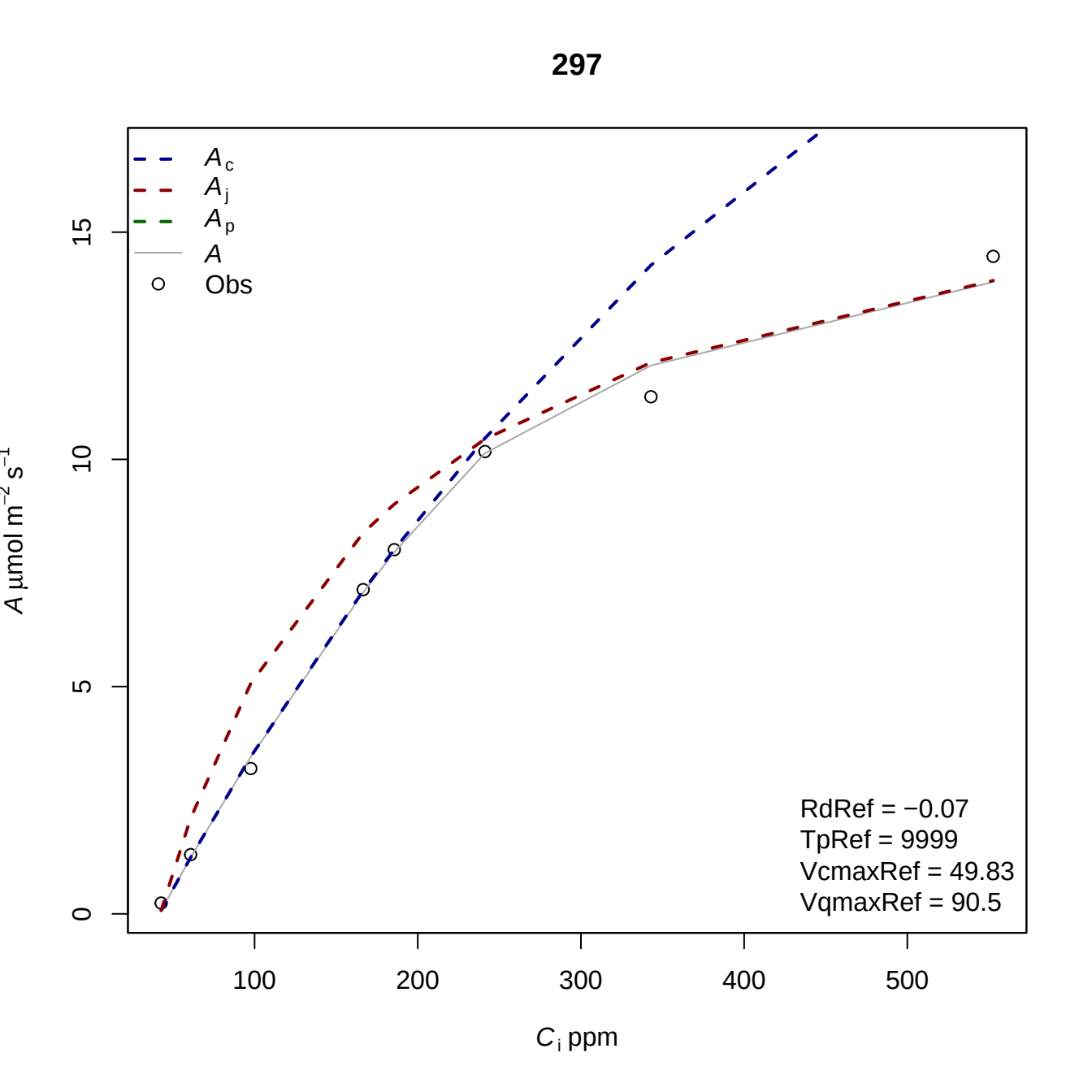

292
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.29
TpRef = 1.22
VcmaxRef = −164288.77
VqmaxRef = 34.52
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
200
400
600
800

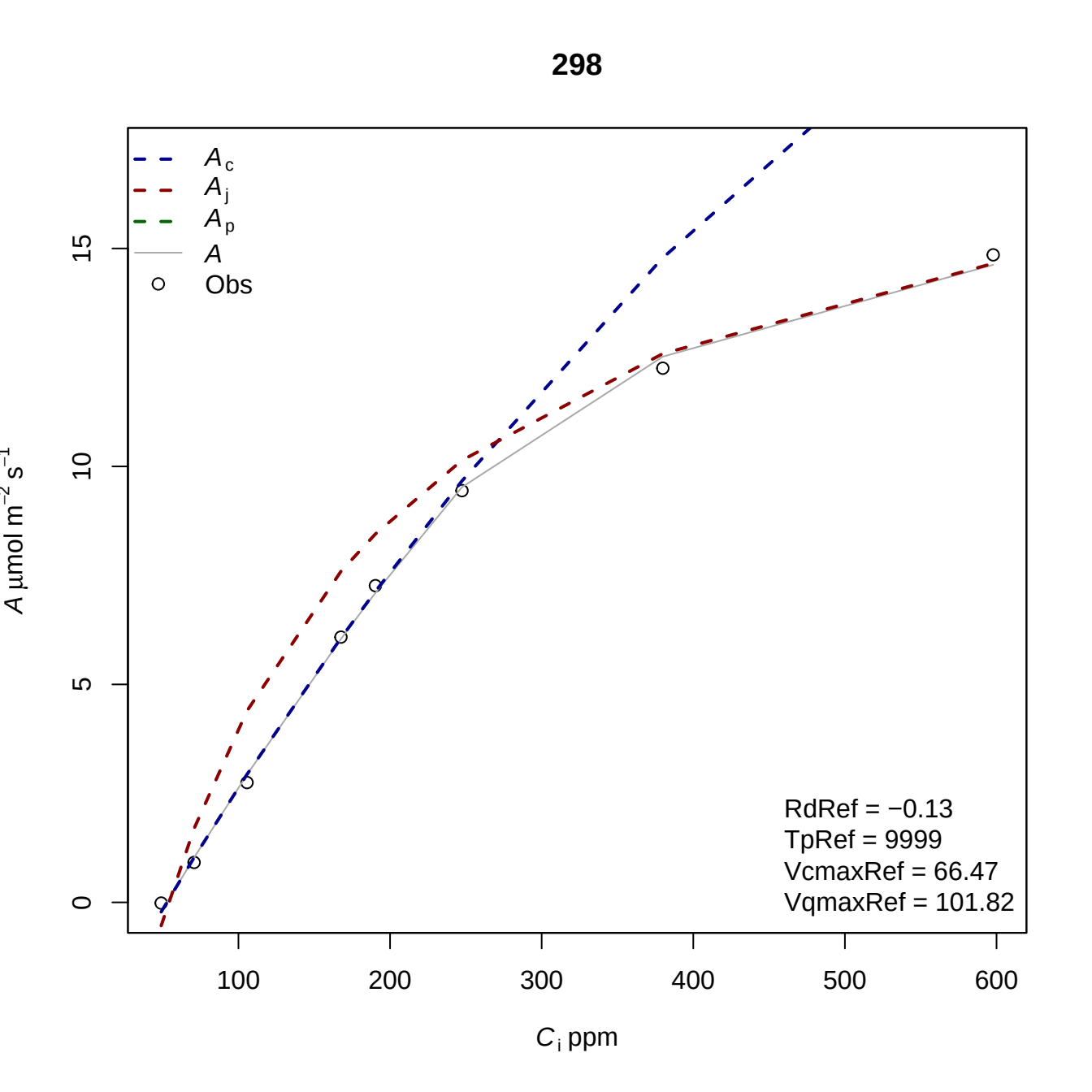

293
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.3
TpRef = 9999
VcmaxRef = 815223.73
VqmaxRef = 19.5
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
200
400
600
800

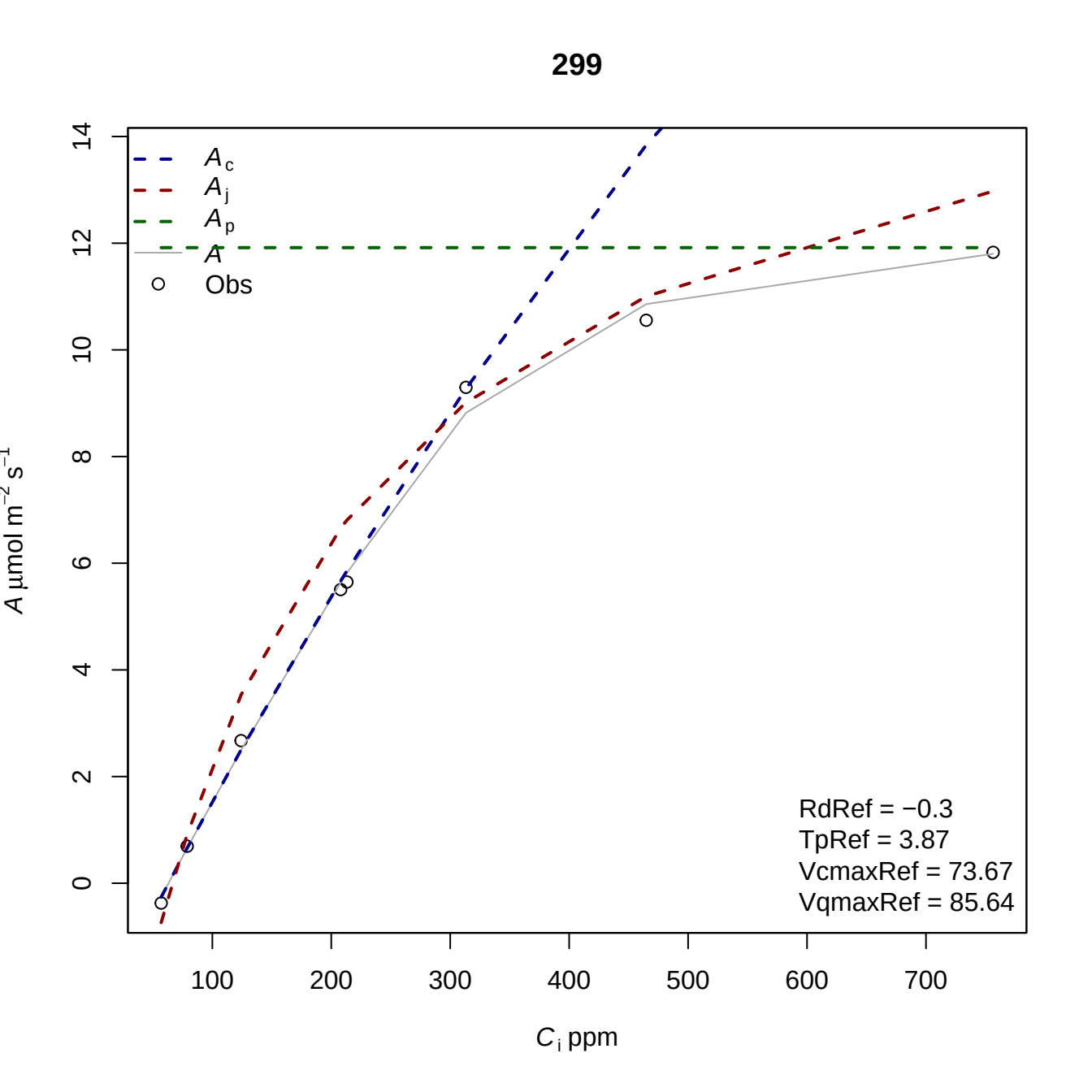

294
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.09
TpRef = 9999
VcmaxRef = 57.21
VqmaxRef = 57.18
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
200
400
600
800

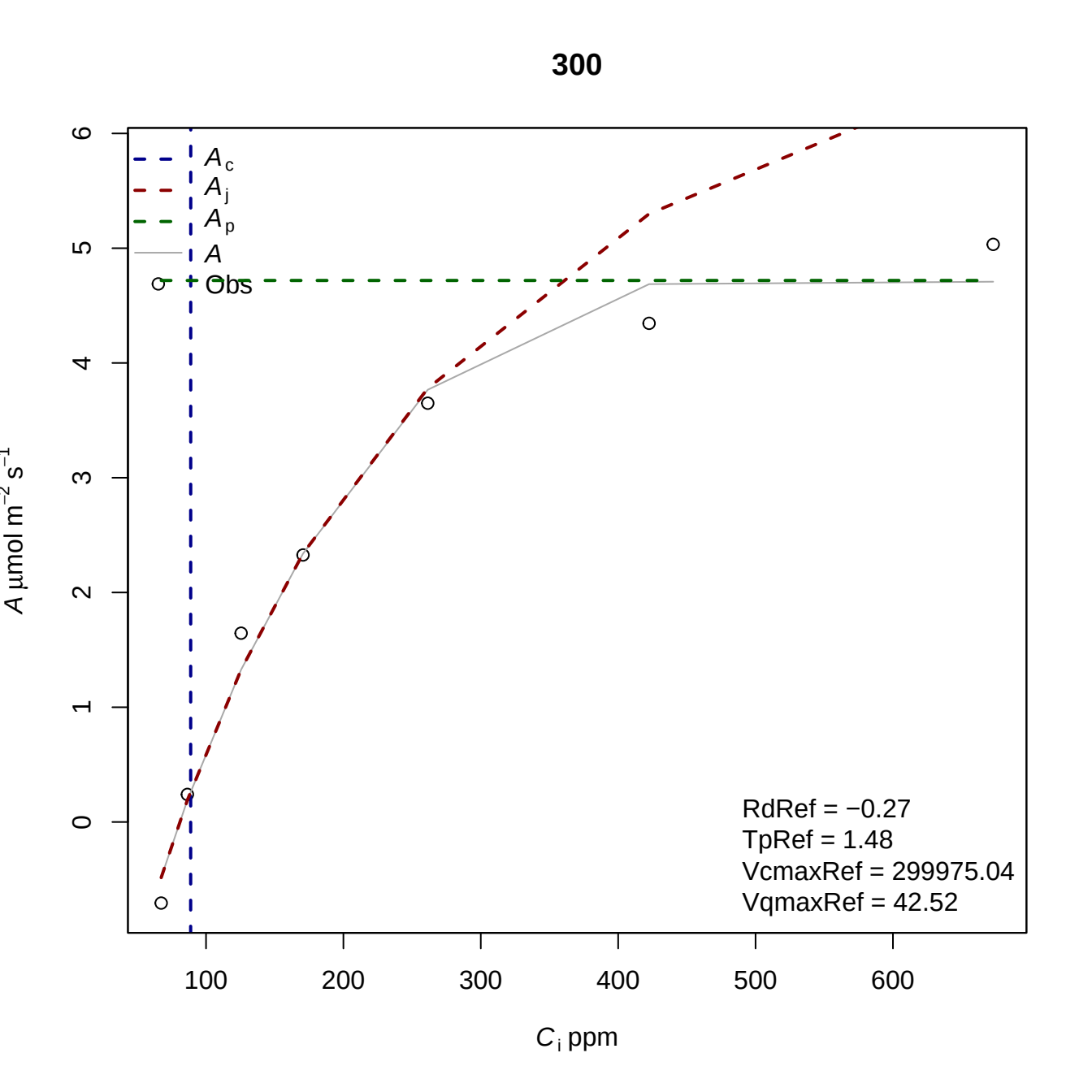

295
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.09
TpRef = 4.31
VcmaxRef = 23.89
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
200
400
600
800

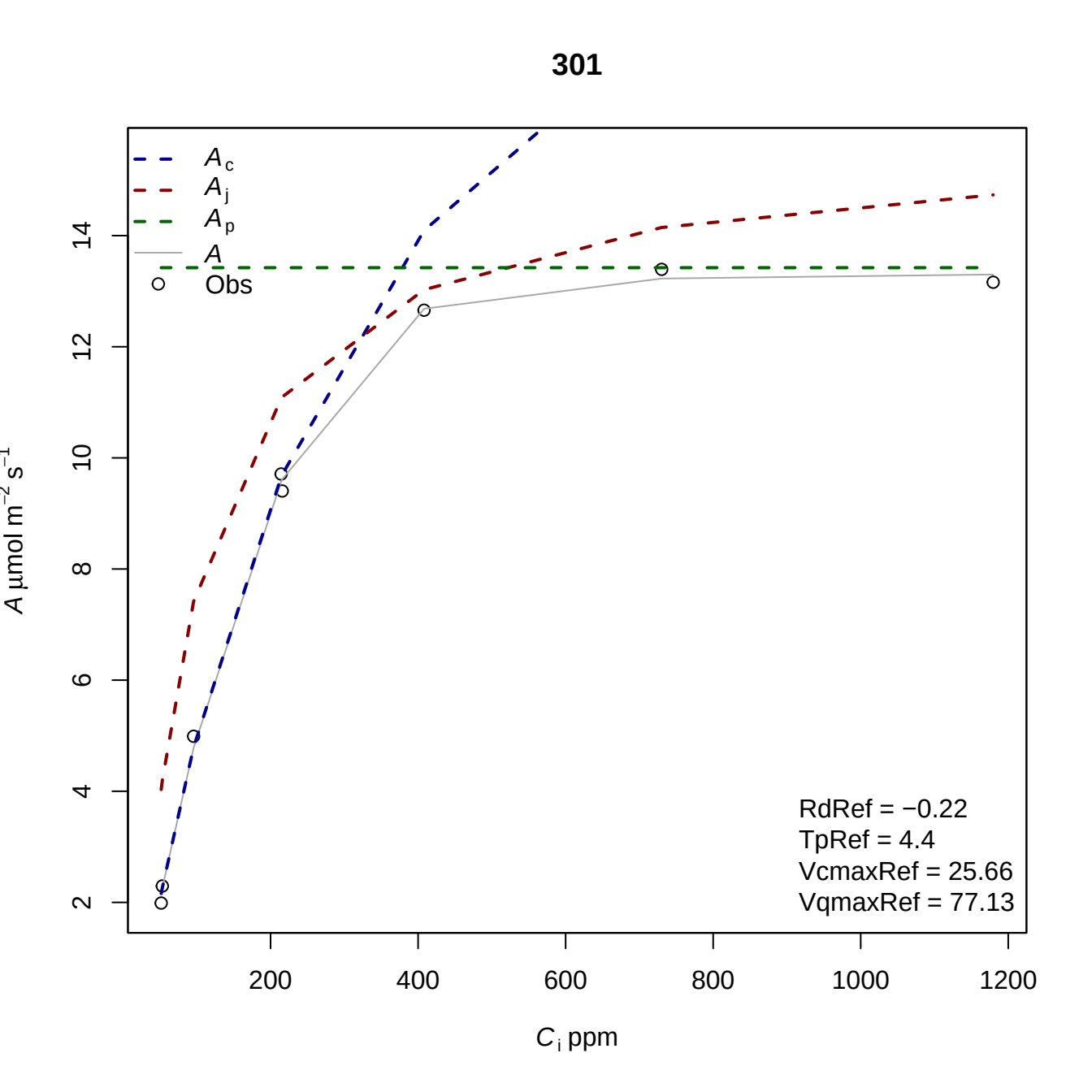

296
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0
TpRef = 4.75
VcmaxRef = 35.27
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500
600

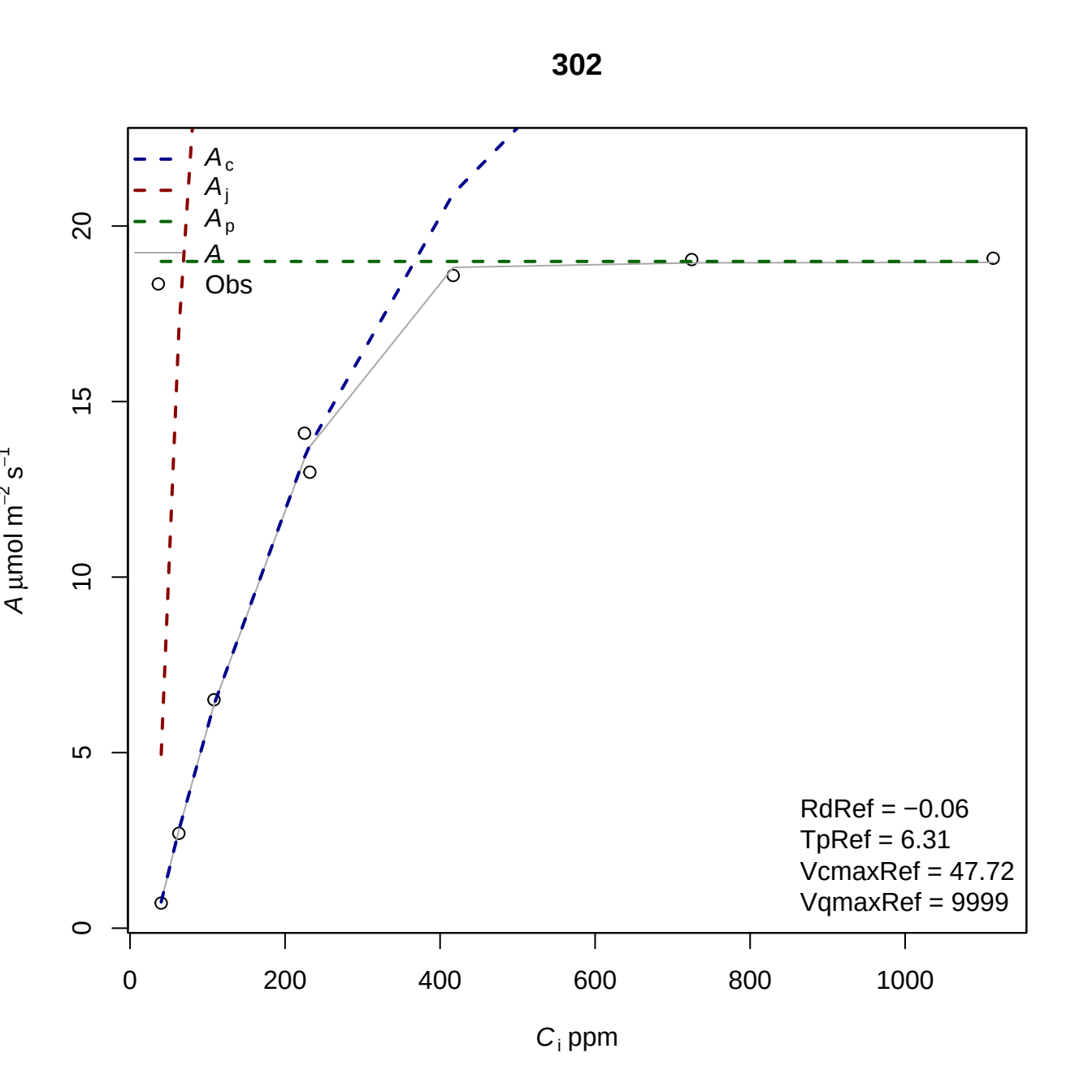

297
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.07
TpRef = 9999
VcmaxRef = 49.83
VqmaxRef = 90.5
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500

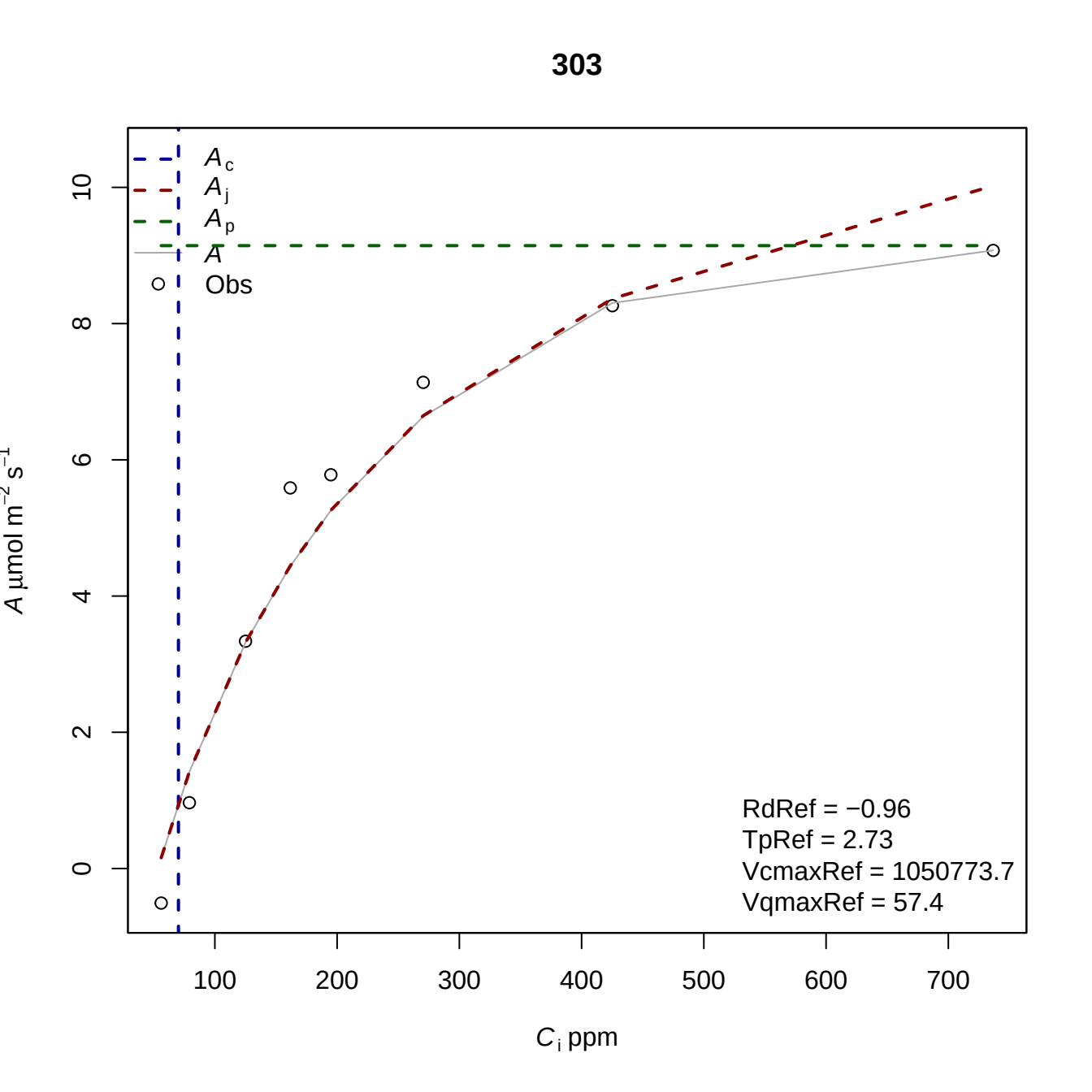

298
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.13
TpRef = 9999
VcmaxRef = 66.47
VqmaxRef = 101.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500
600

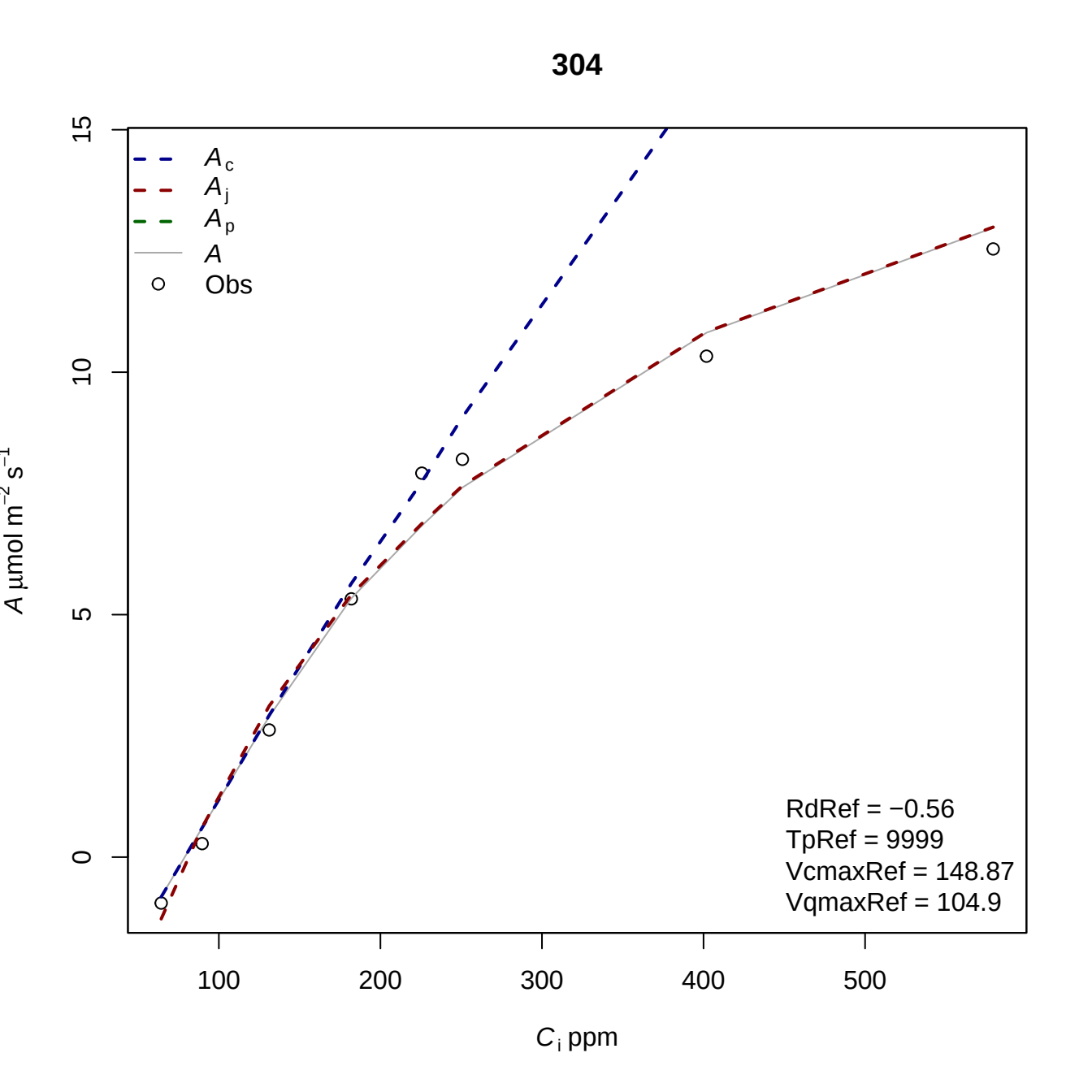

299
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.3
TpRef = 3.87
VcmaxRef = 73.67
VqmaxRef = 85.64
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
100
200
300
400
500
600
700

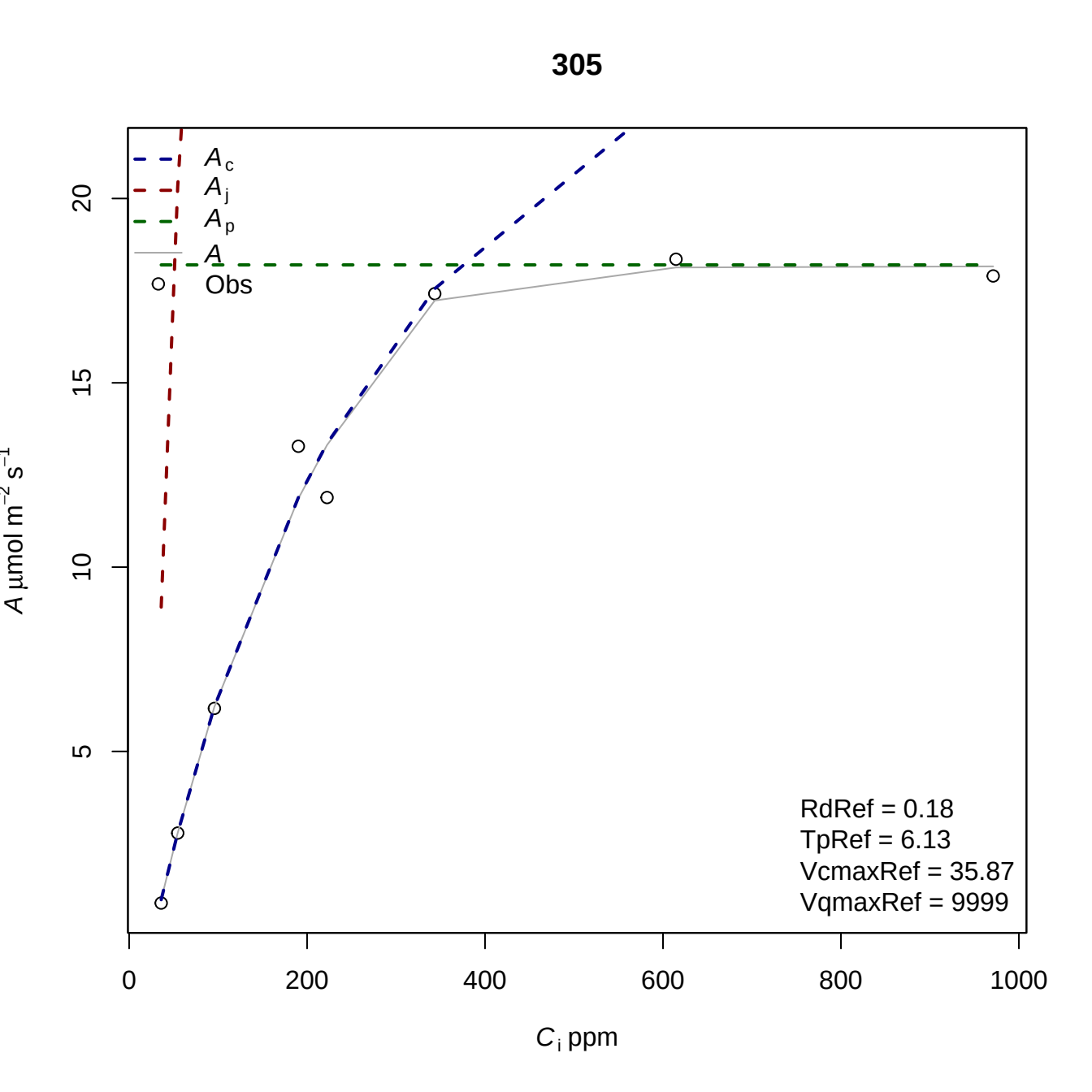

300
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.27
TpRef = 1.48
VcmaxRef = 299975.04
VqmaxRef = 42.52
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
100
200
300
400
500
600

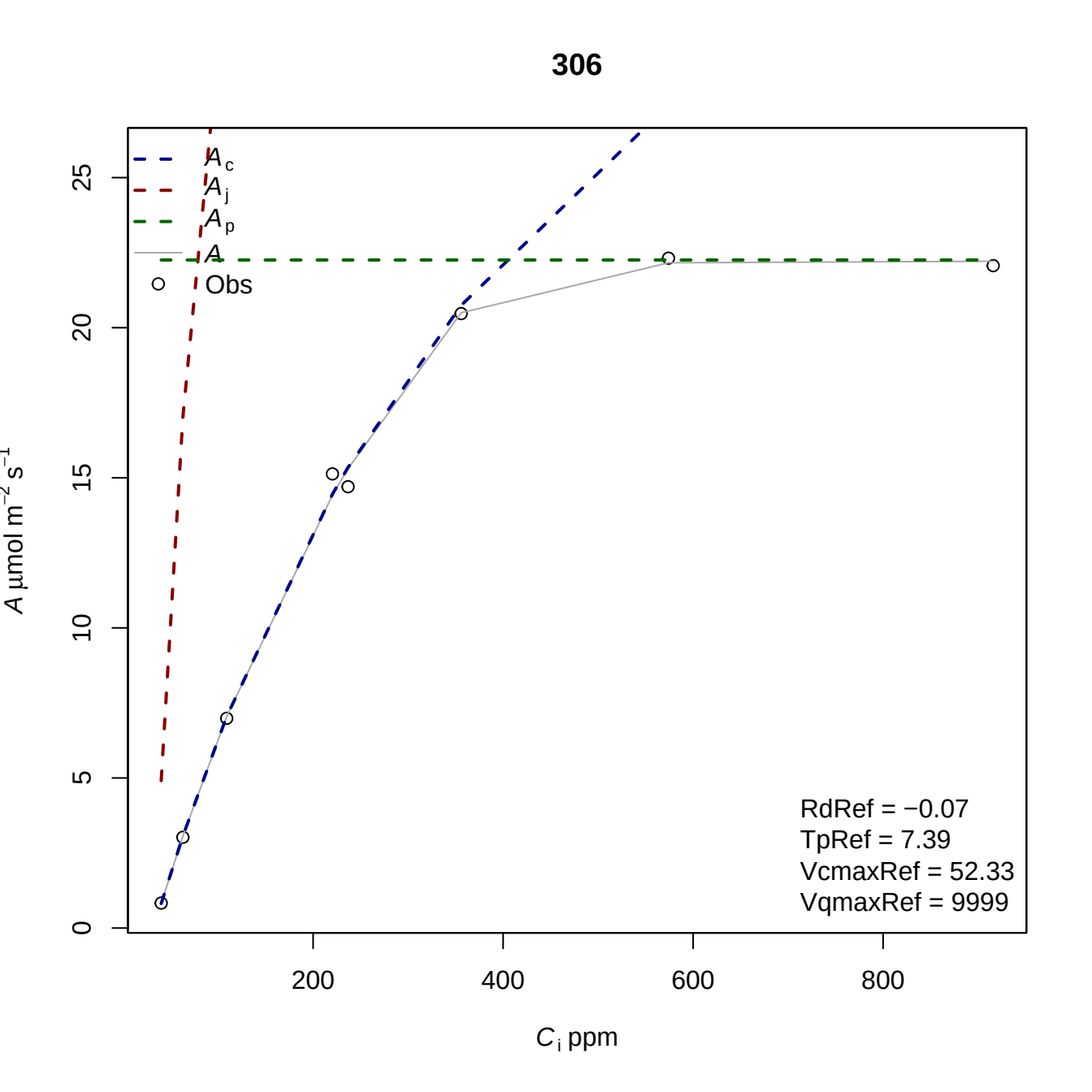

301
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.22
TpRef = 4.4
VcmaxRef = 25.66
VqmaxRef = 77.13
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
2
4
6
8
10
12
14
200
400
600
800
1000
1200

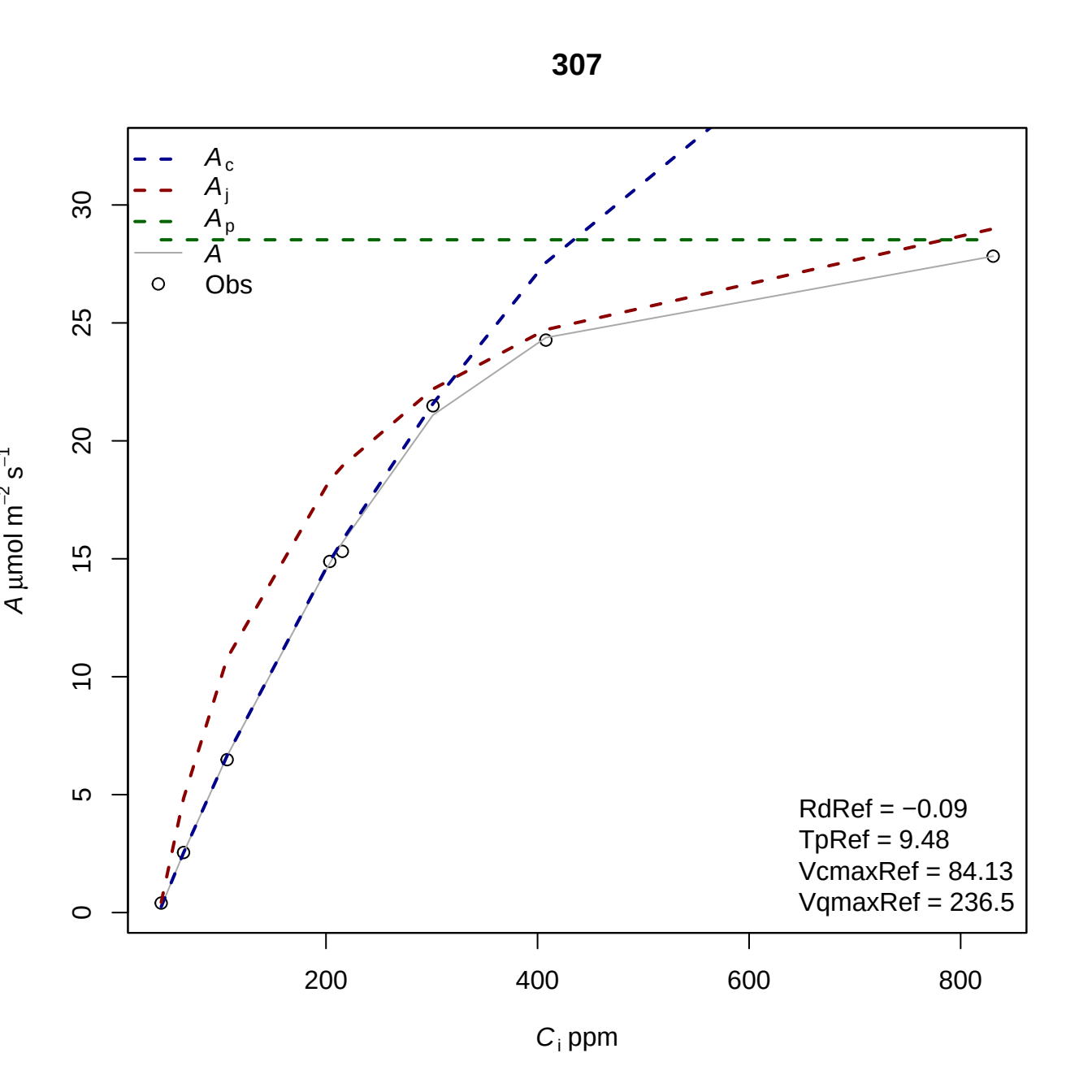

302
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.06
TpRef = 6.31
VcmaxRef = 47.72
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000

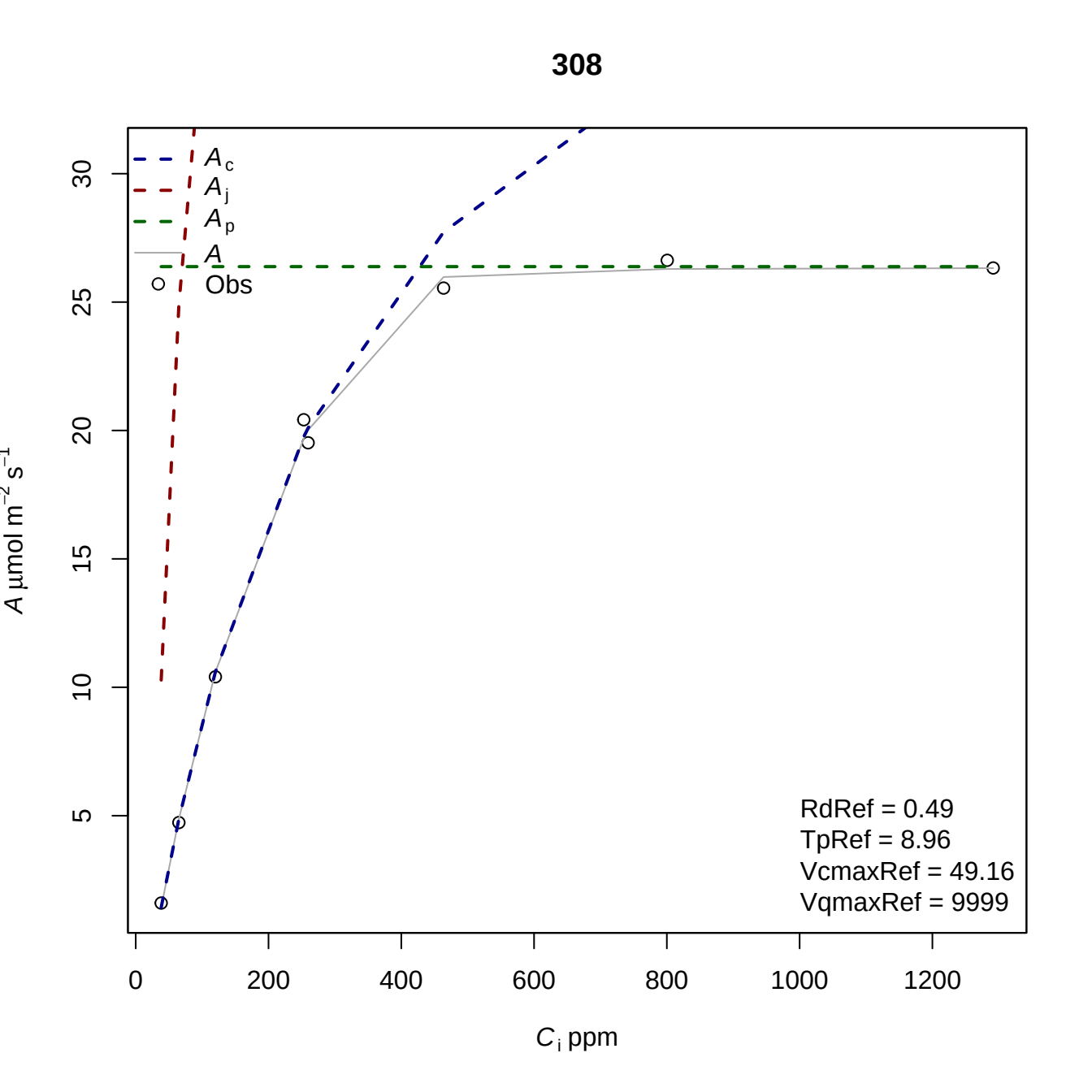

303
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.96
TpRef = 2.73
VcmaxRef = 1050773.7
VqmaxRef = 57.4
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
100
200
300
400
500
600
700

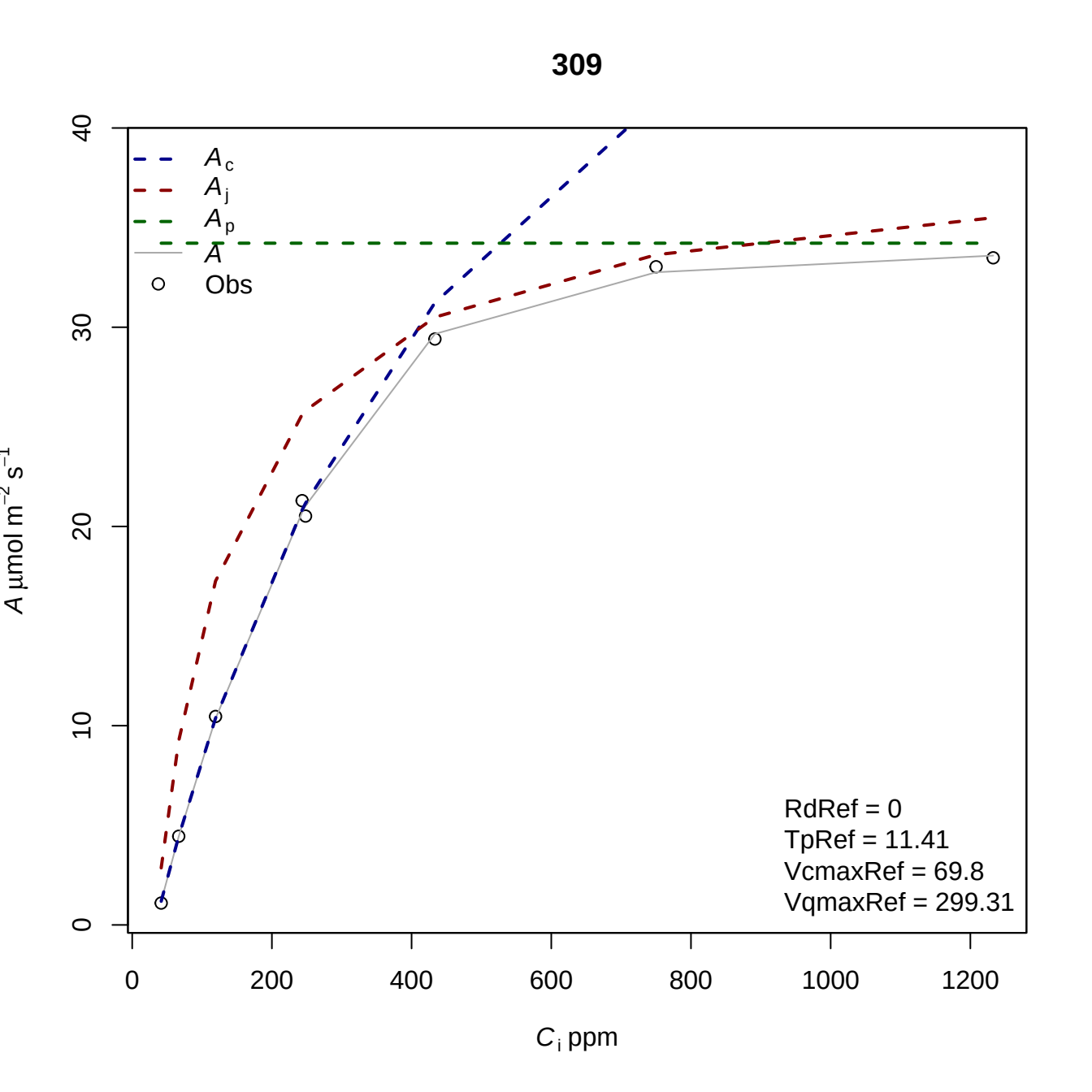

304
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.56
TpRef = 9999
VcmaxRef = 148.87
VqmaxRef = 104.9
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500

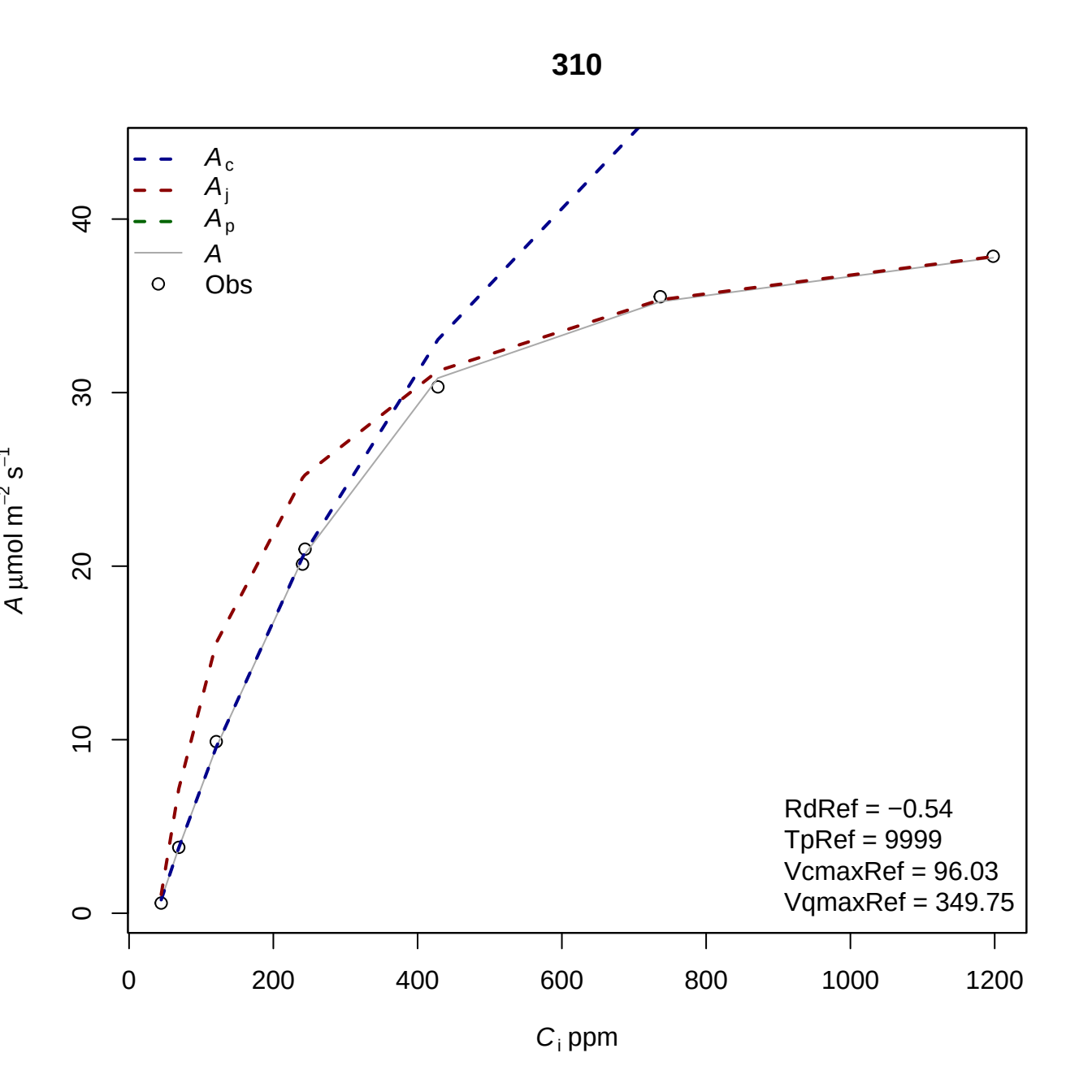

305
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.18
TpRef = 6.13
VcmaxRef = 35.87
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
5
10
15
20

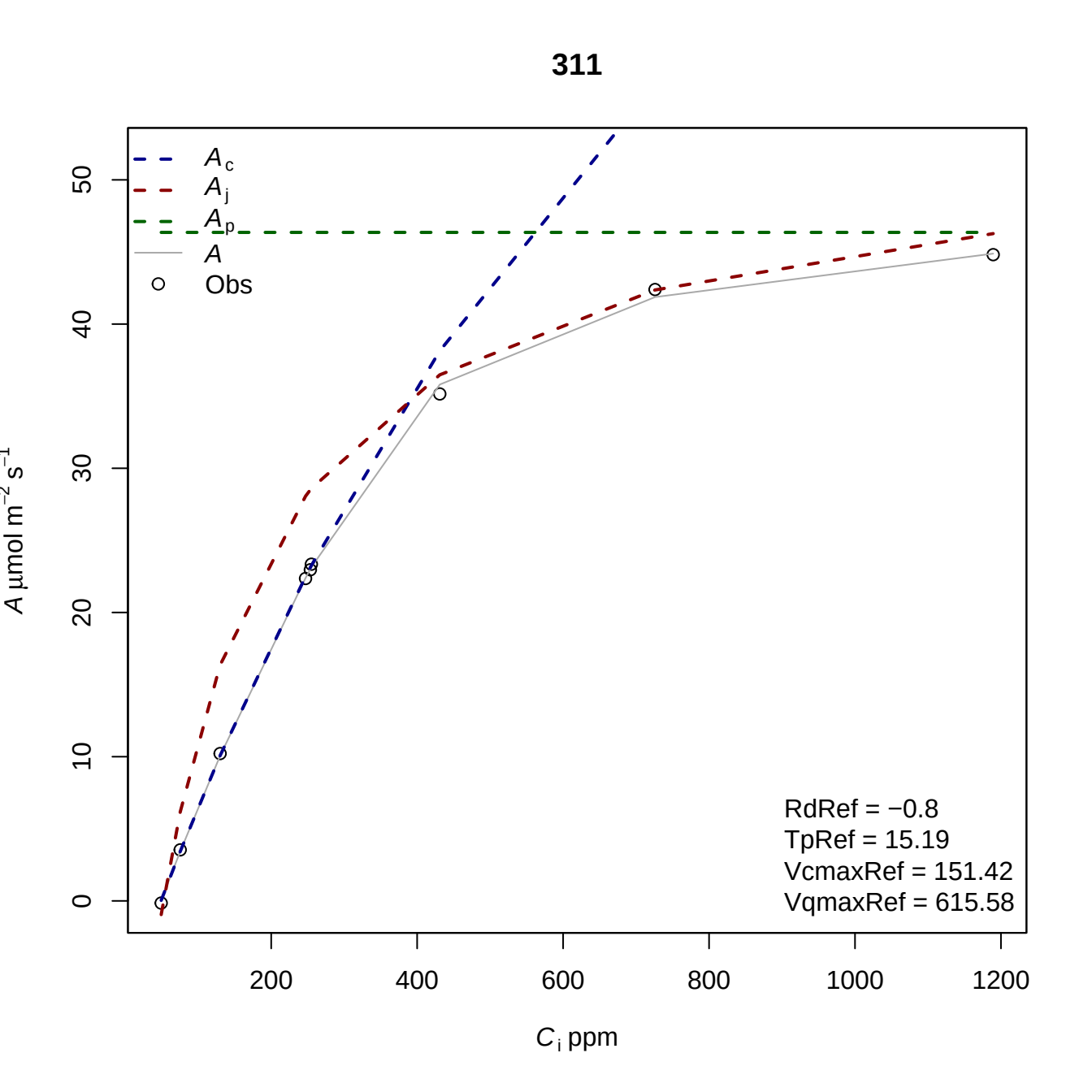

306
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.07
TpRef = 7.39
VcmaxRef = 52.33
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800

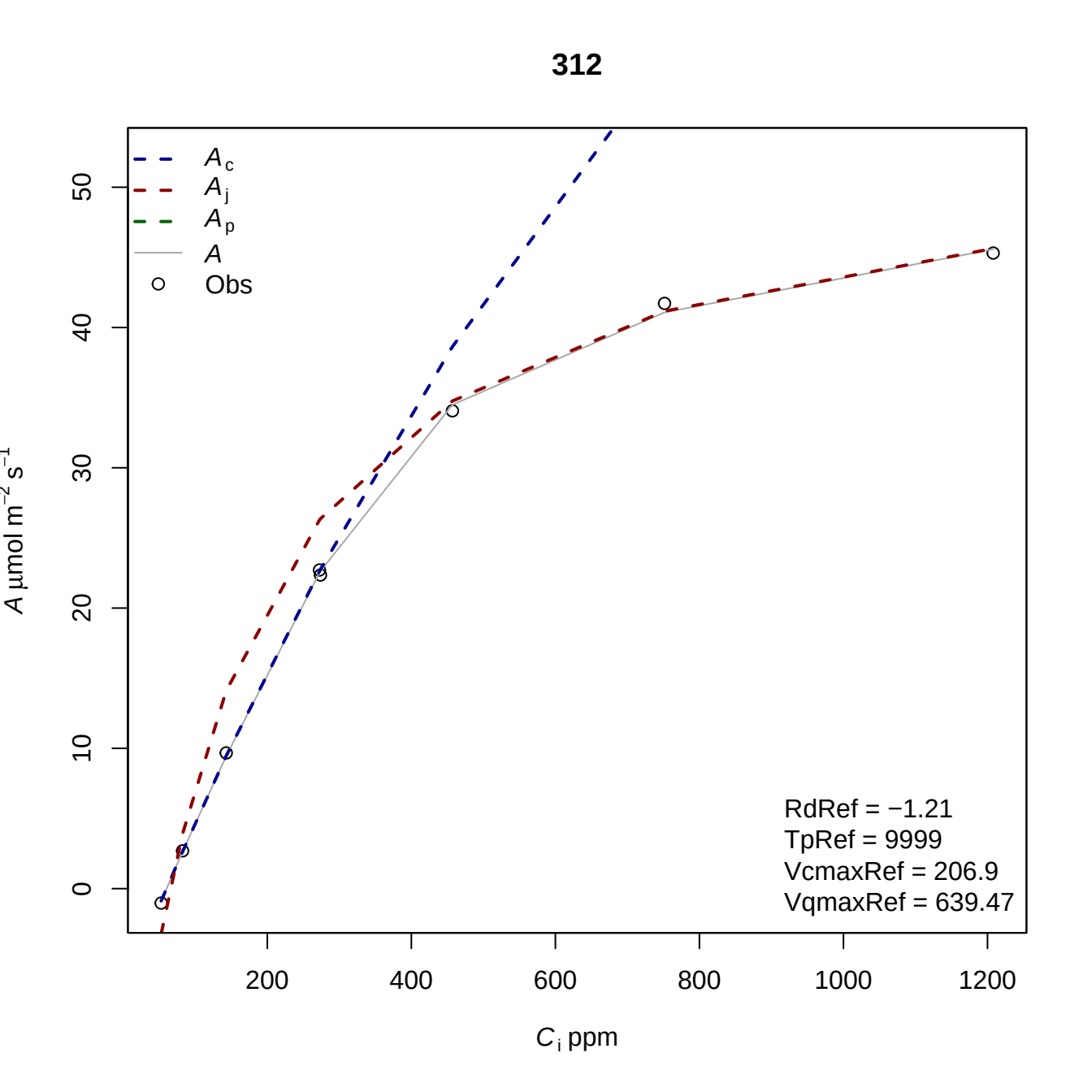

307
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.09
TpRef = 9.48
VcmaxRef = 84.13
VqmaxRef = 236.5
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800

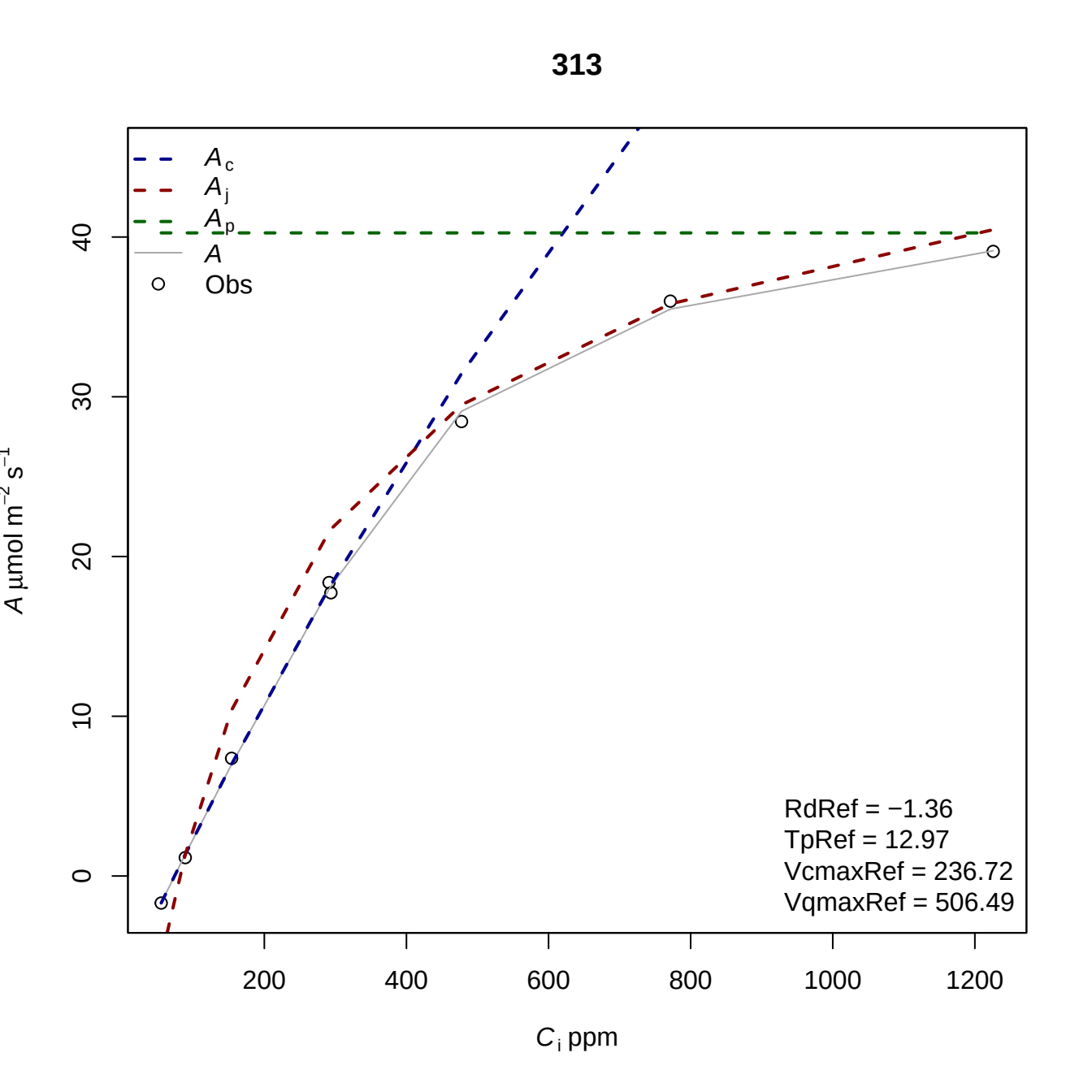

308
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.49
TpRef = 8.96
VcmaxRef = 49.16
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25
30

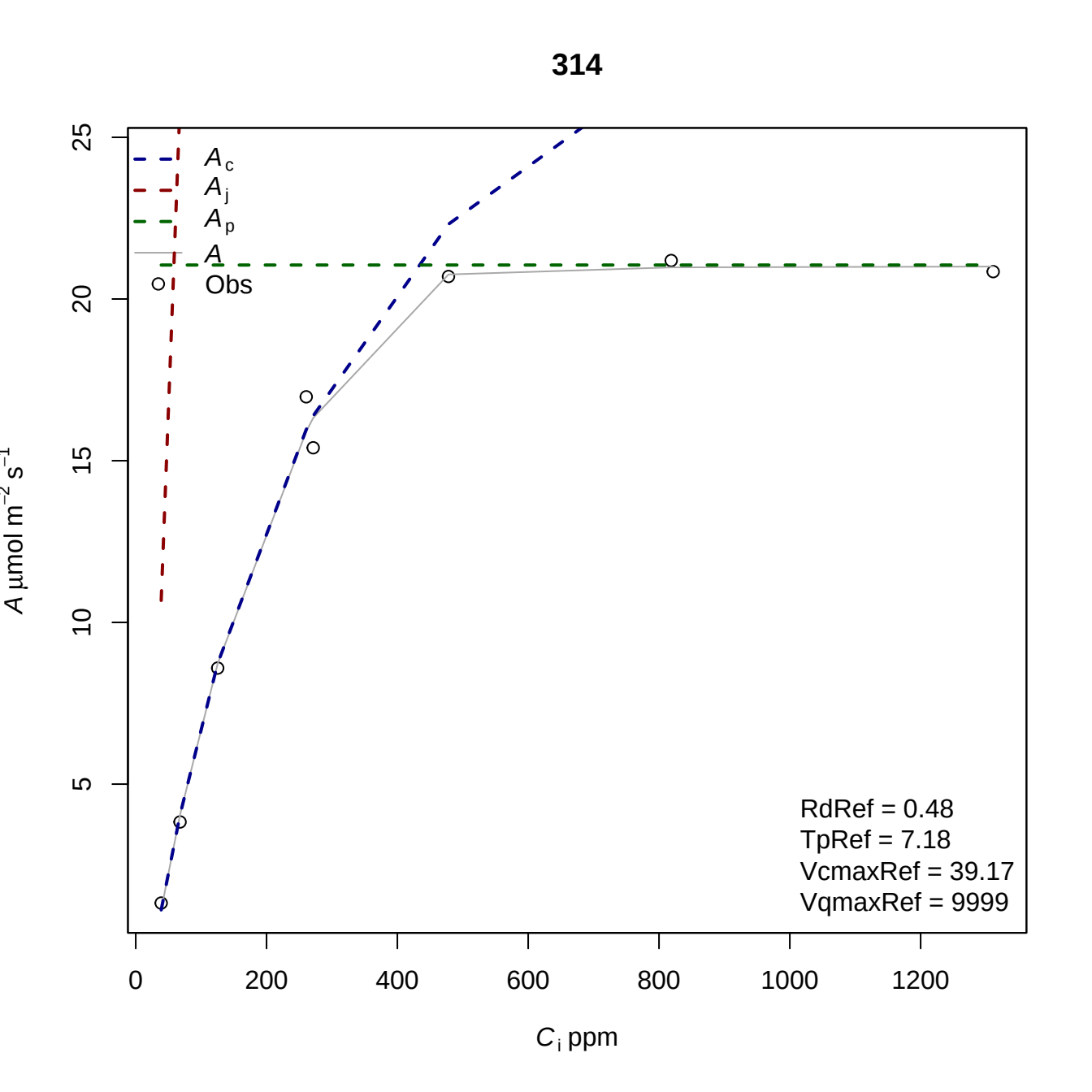

309
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0
TpRef = 11.41
VcmaxRef = 69.8
VqmaxRef = 299.31
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

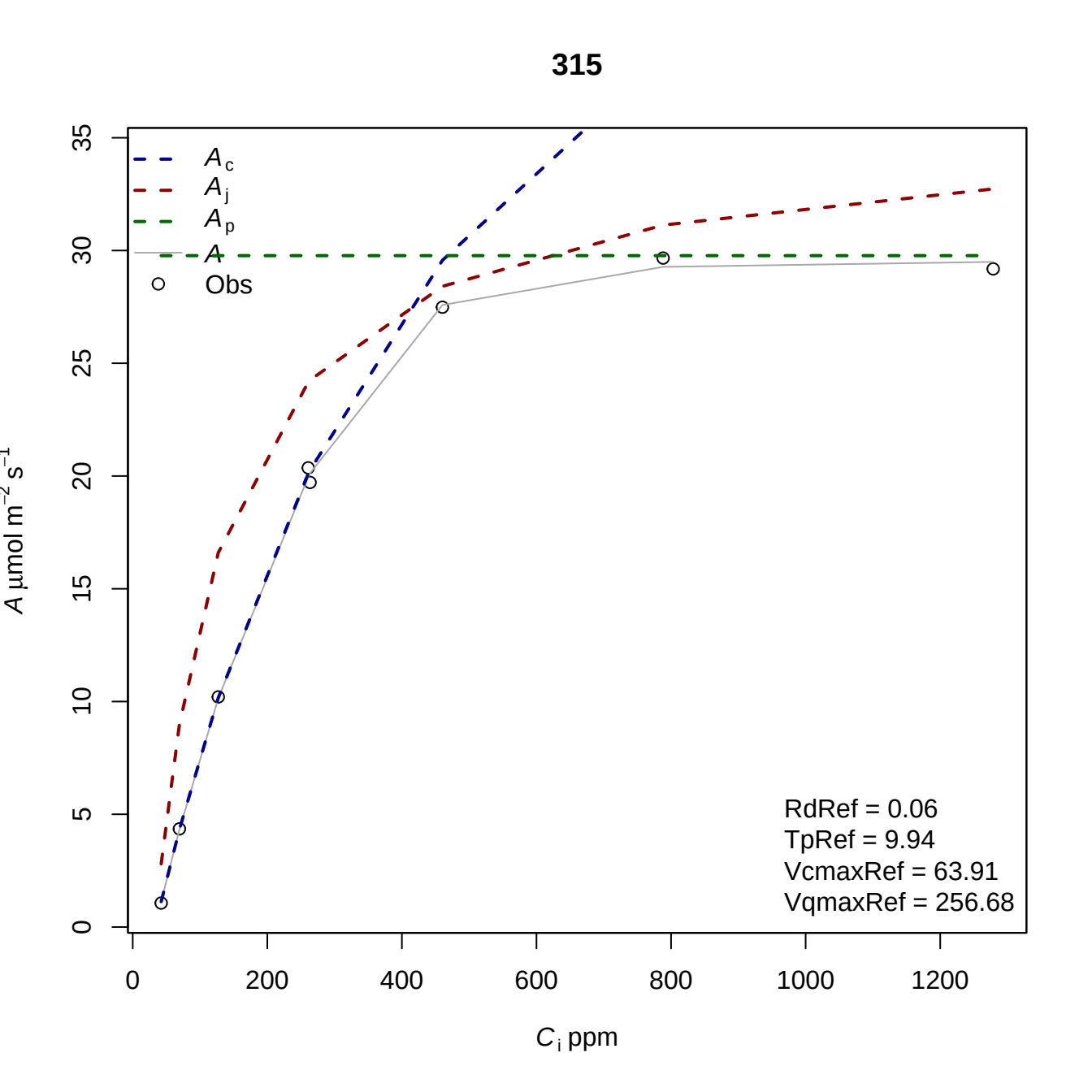

310
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.54
TpRef = 9999
VcmaxRef = 96.03
VqmaxRef = 349.75
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

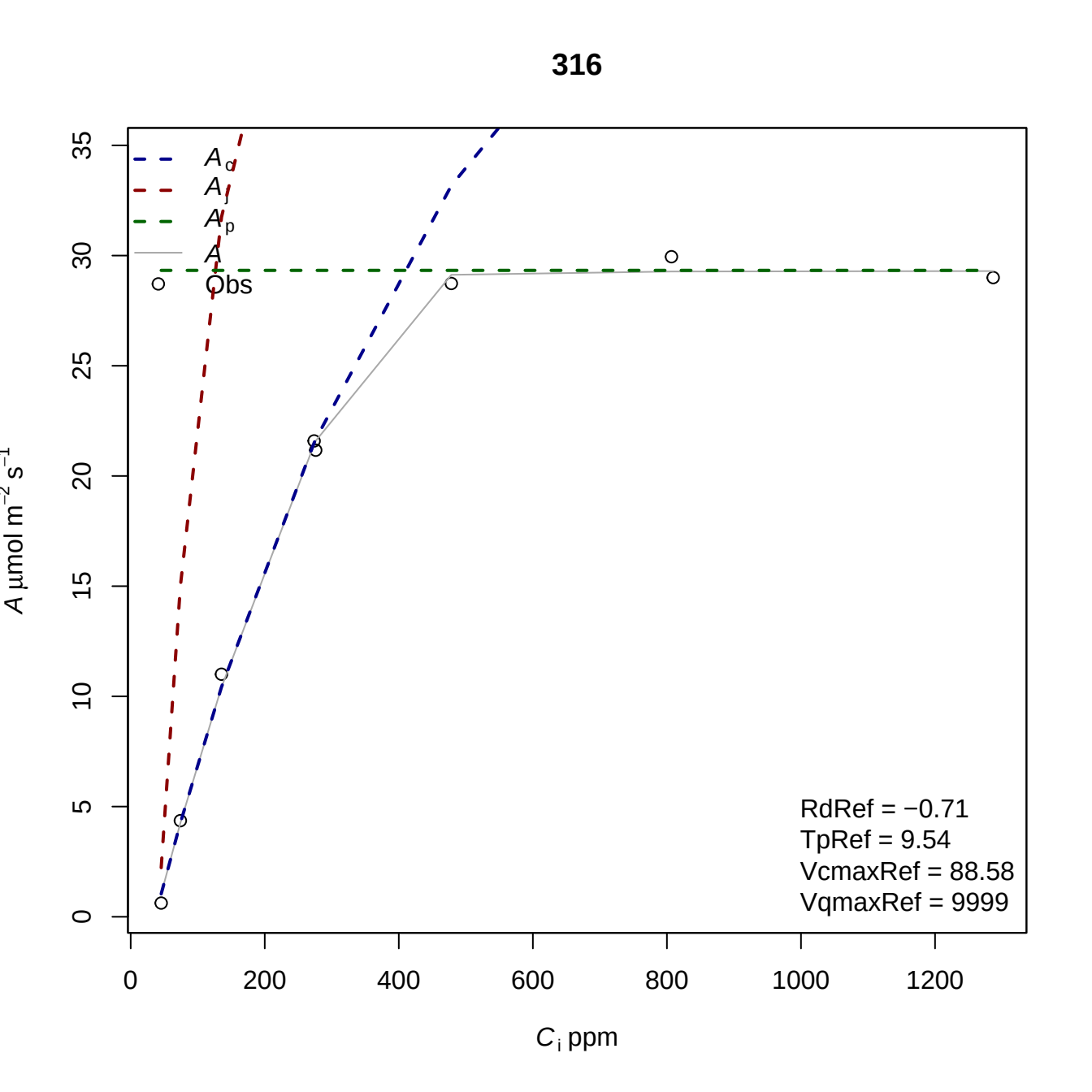

311
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.8
TpRef = 15.19
VcmaxRef = 151.42
VqmaxRef = 615.58
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

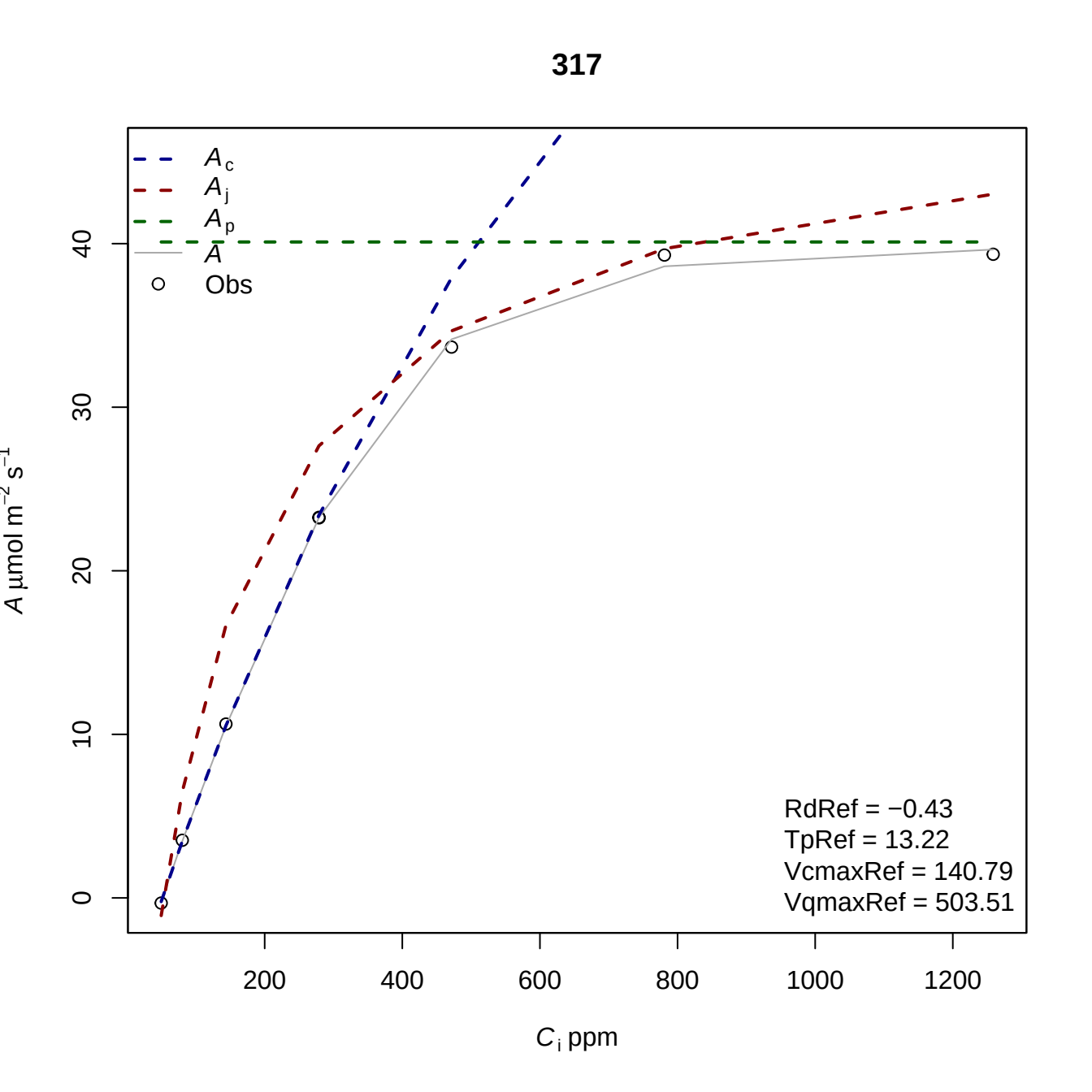

312
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −1.21
TpRef = 9999
VcmaxRef = 206.9
VqmaxRef = 639.47
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

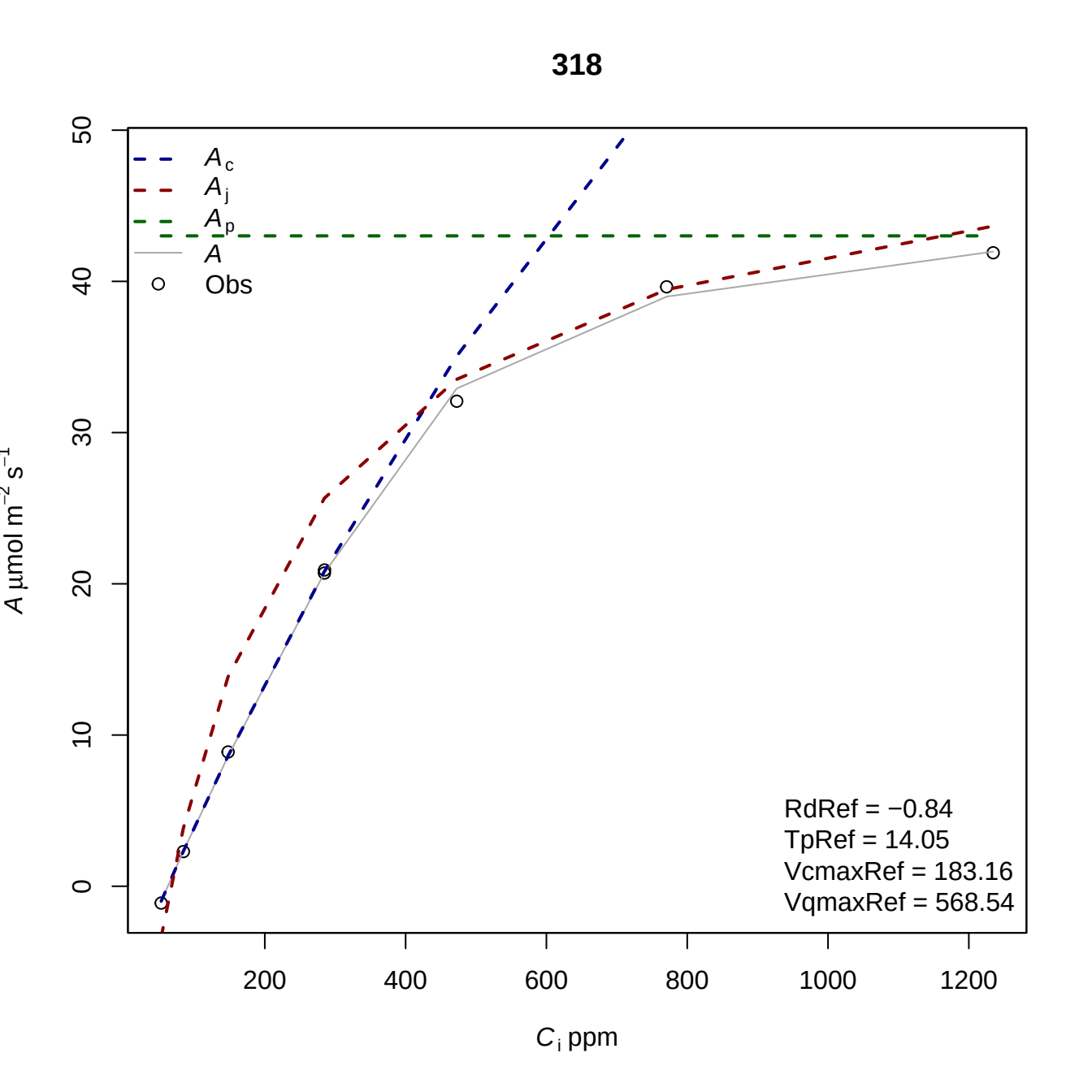

313
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −1.36
TpRef = 12.97
VcmaxRef = 236.72
VqmaxRef = 506.49
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

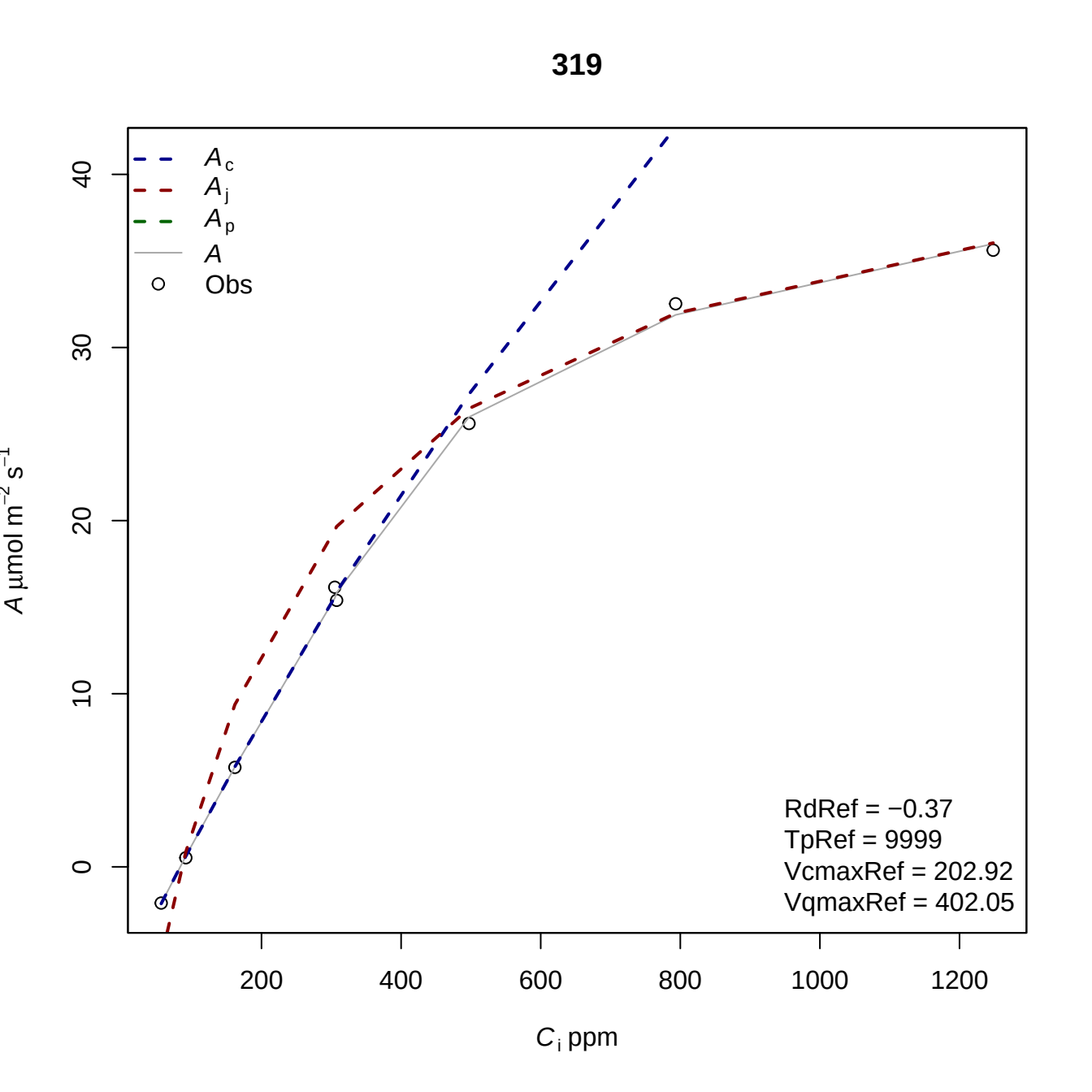

314
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.48
TpRef = 7.18
VcmaxRef = 39.17
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

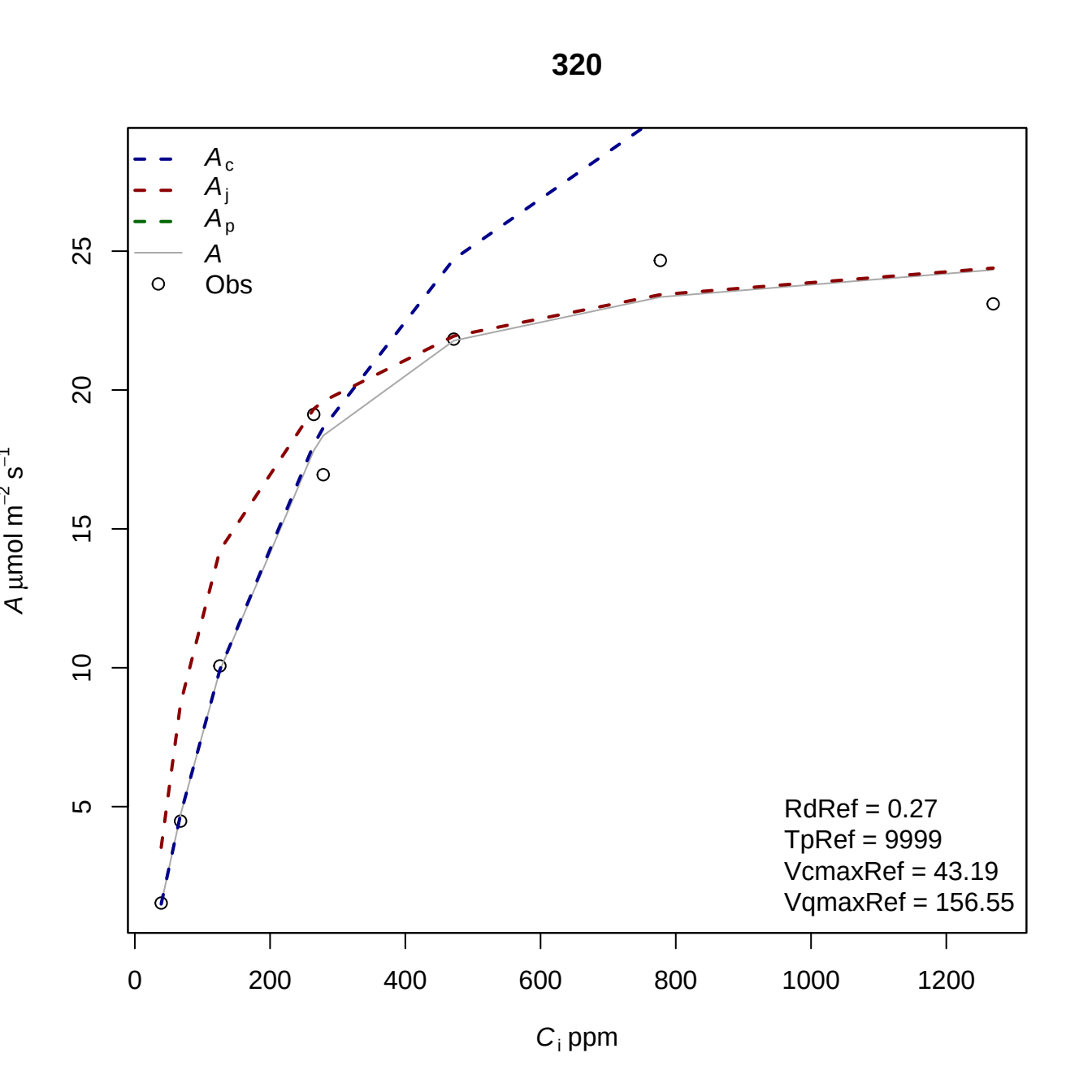

315
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.06
TpRef = 9.94
VcmaxRef = 63.91
VqmaxRef = 256.68
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

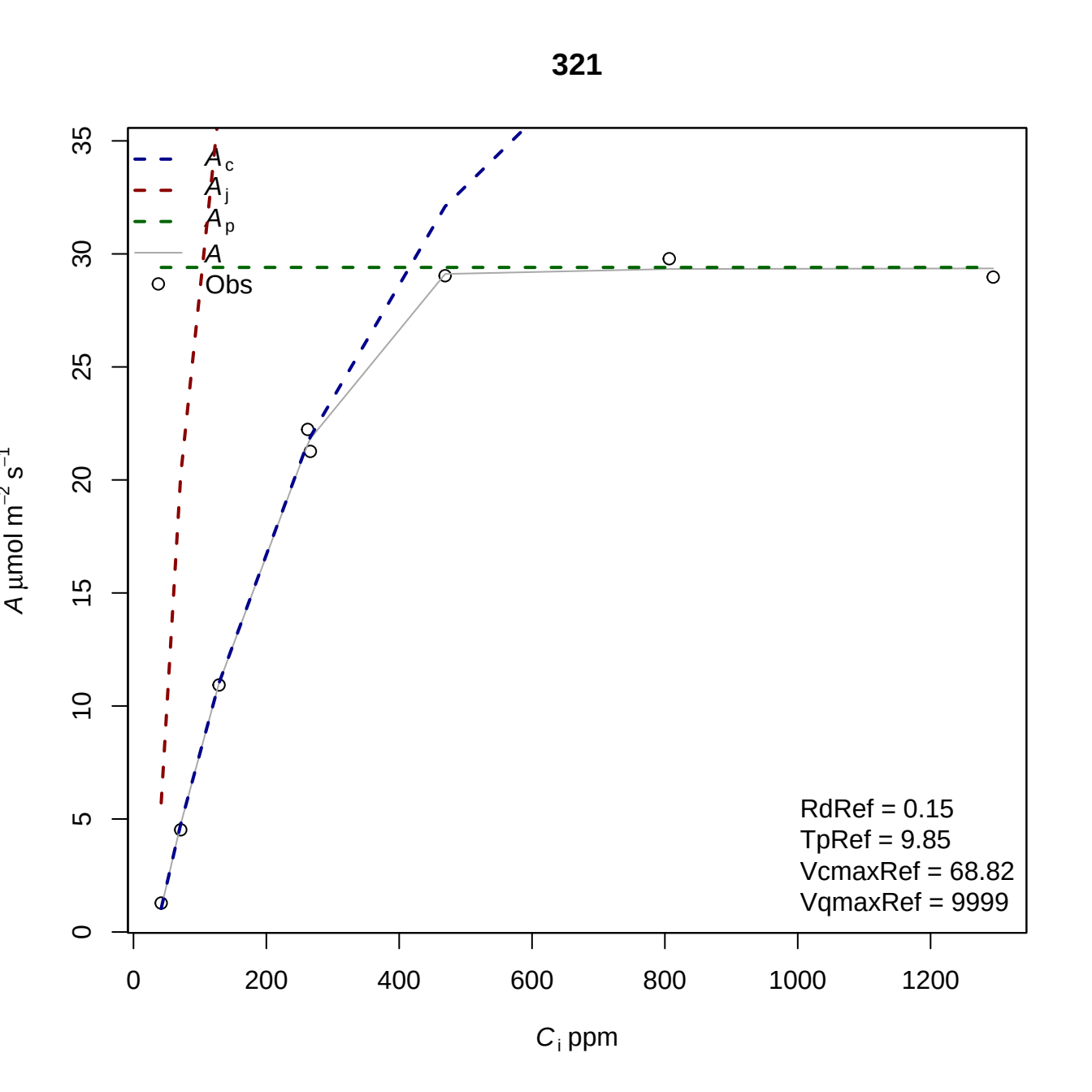

316
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.71
TpRef = 9.54
VcmaxRef = 88.58
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

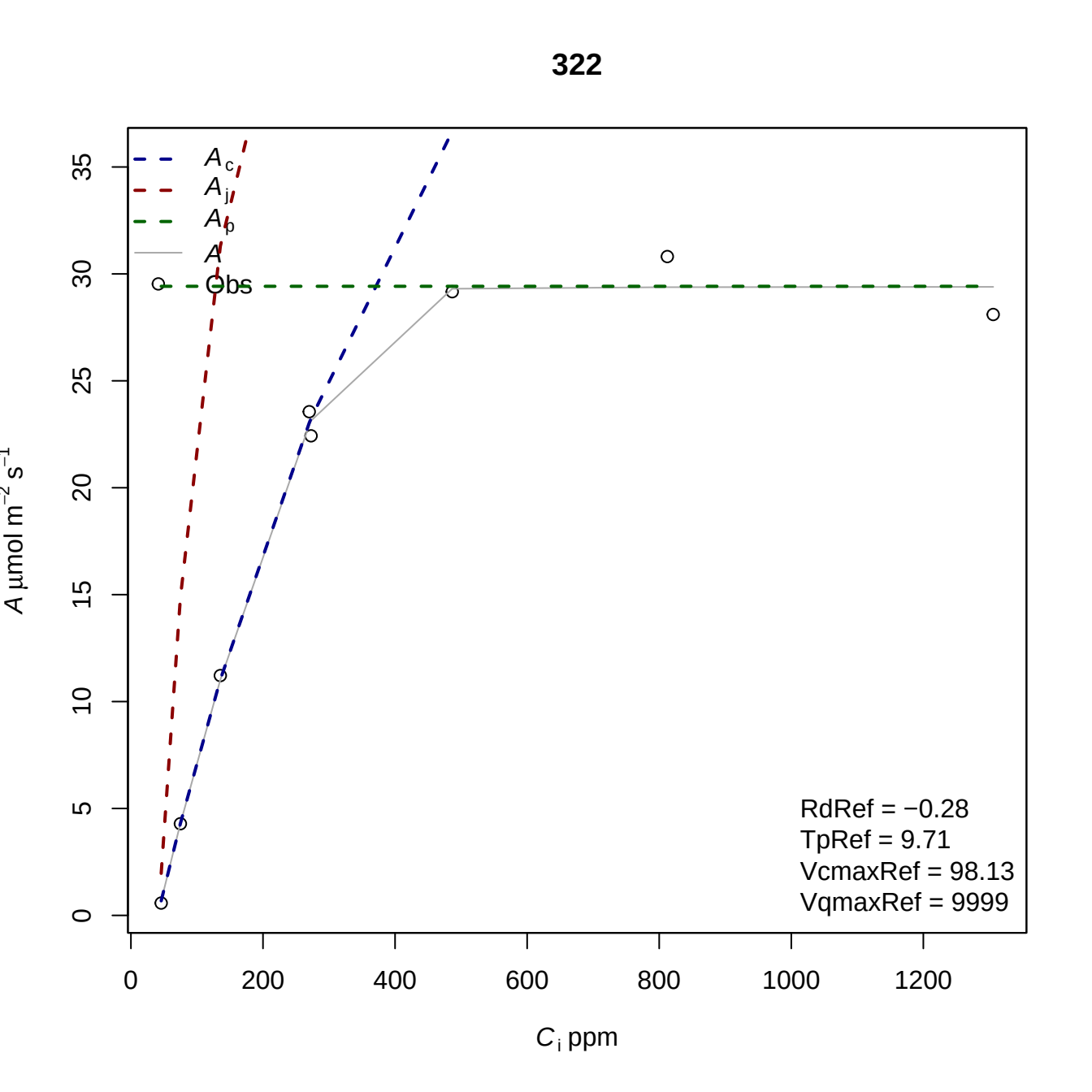

317
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.43
TpRef = 13.22
VcmaxRef = 140.79
VqmaxRef = 503.51
$A$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
200
400
600
800
1000
1200
$C_i$ ppm

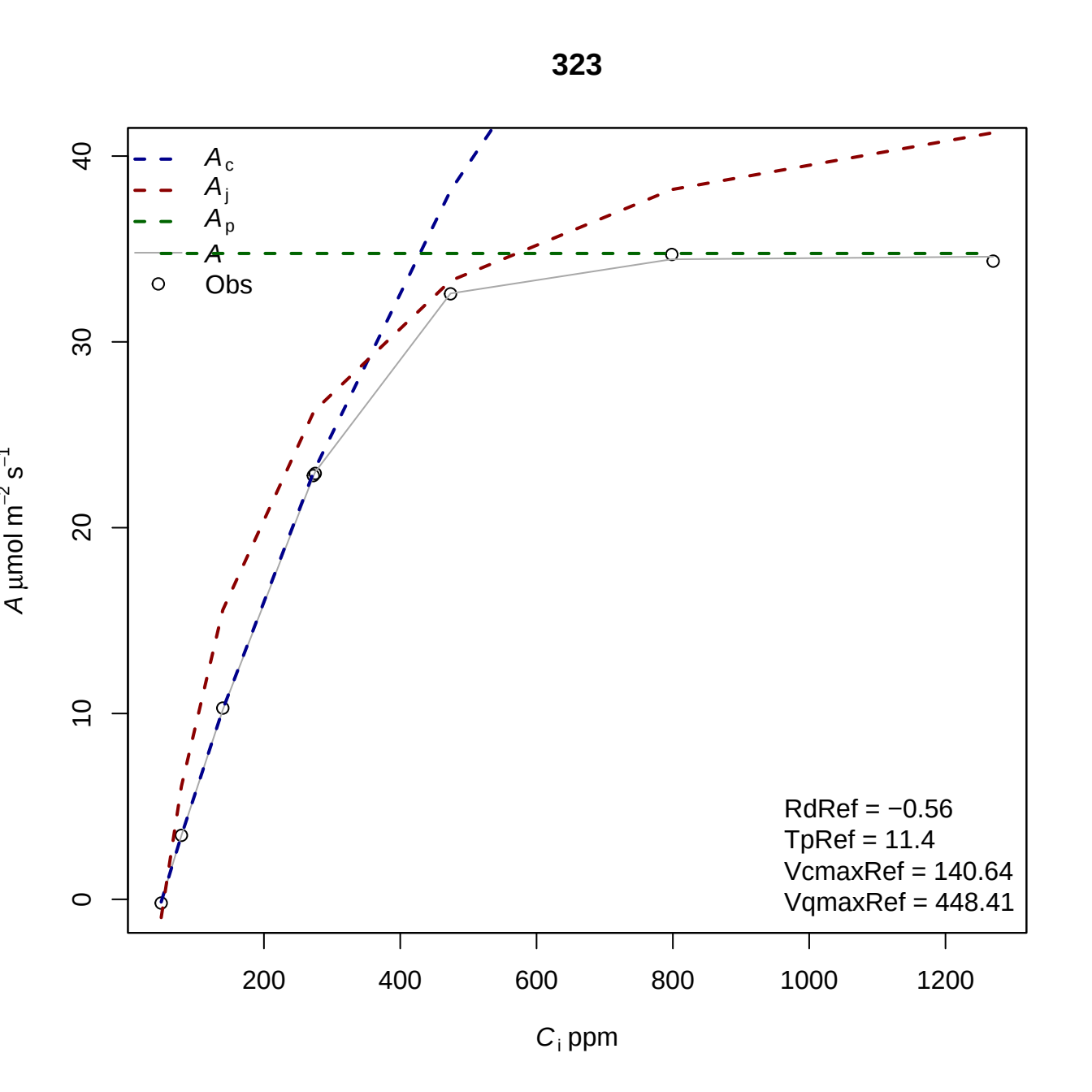

318
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.84
TpRef = 14.05
VcmaxRef = 183.16
VqmaxRef = 568.54
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

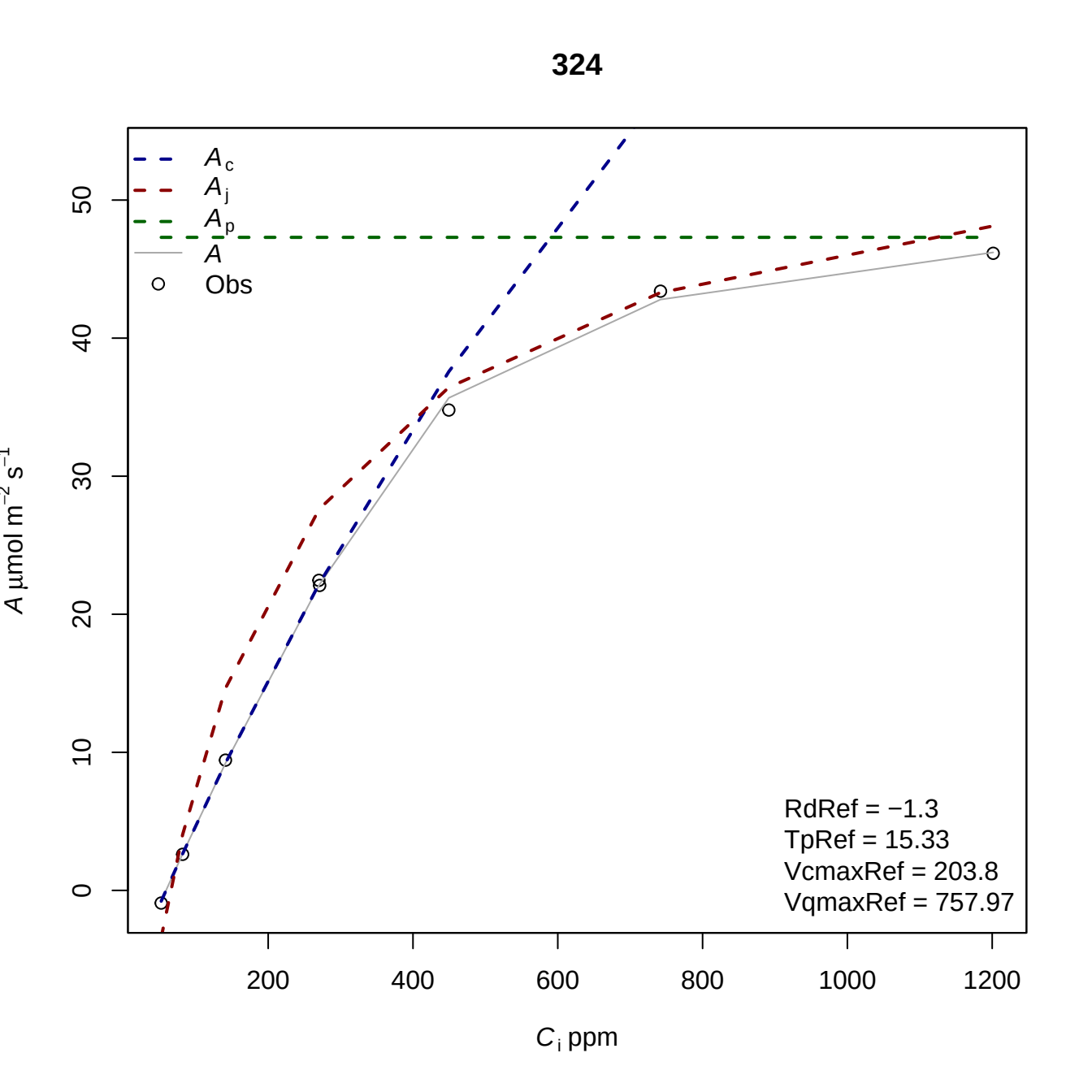

319
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.37
TpRef = 9999
VcmaxRef = 202.92
VqmaxRef = 402.05
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

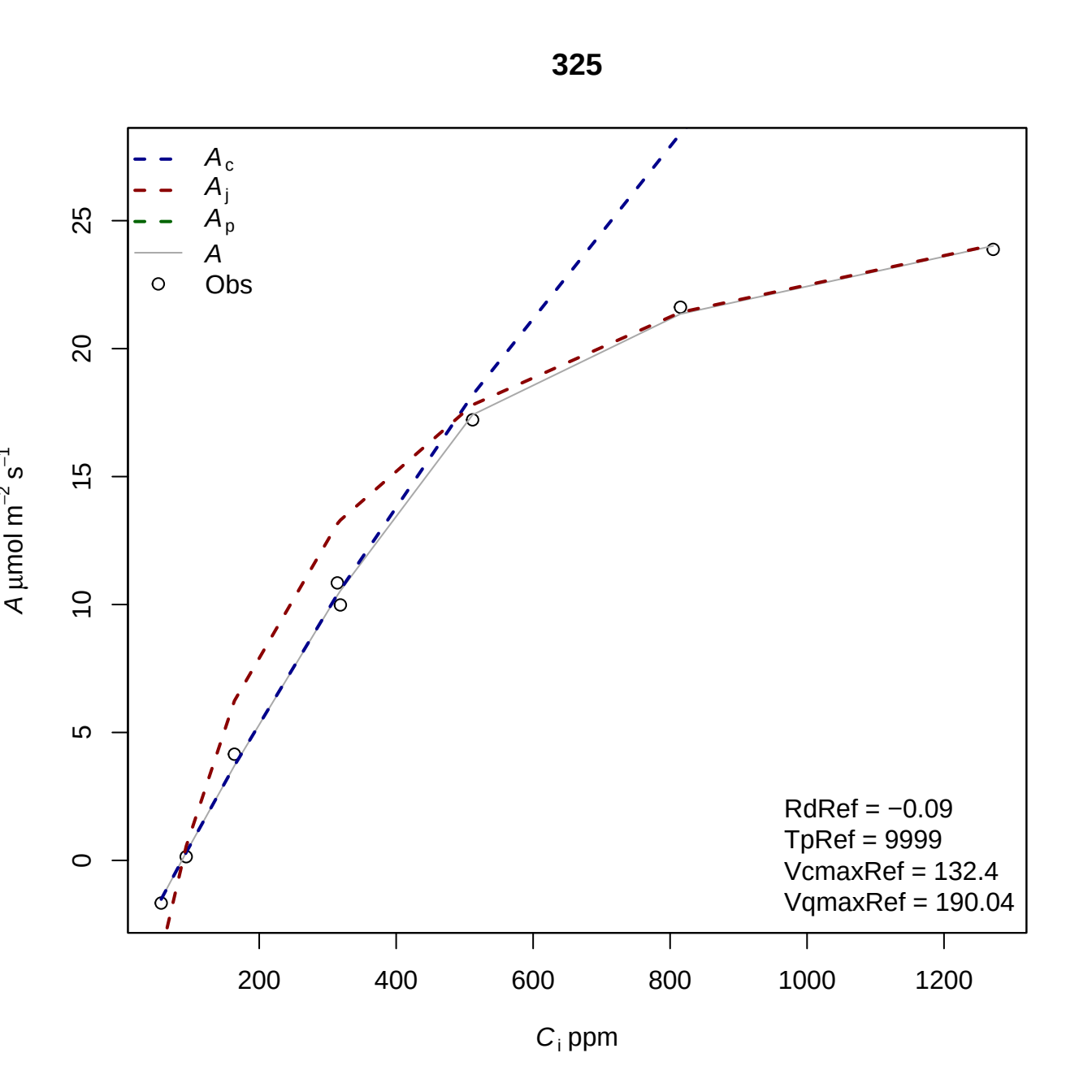

320
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.27
TpRef = 9999
VcmaxRef = 43.19
VqmaxRef = 156.55
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

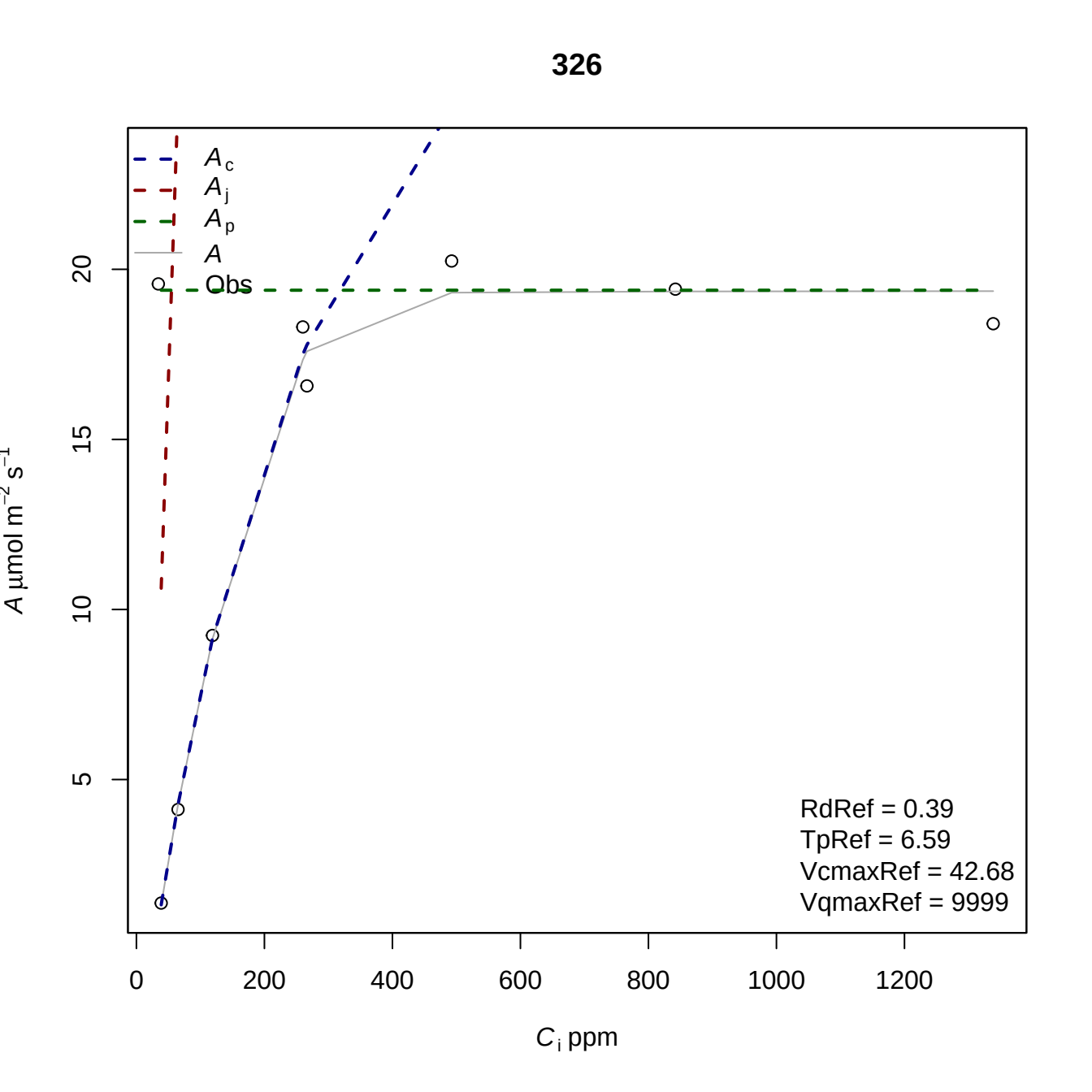

321
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.15
TpRef = 9.85
VcmaxRef = 68.82
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

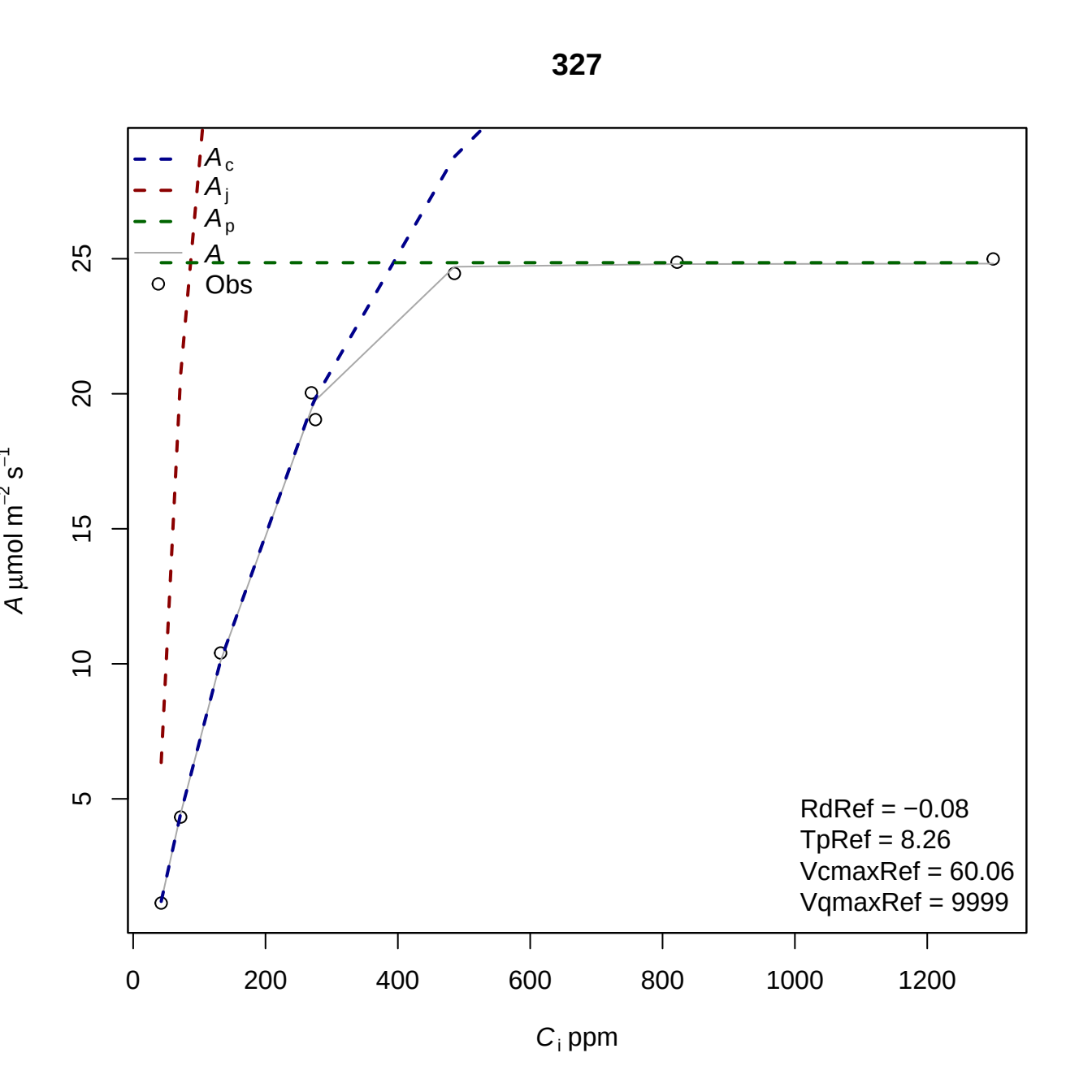

322
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.28
TpRef = 9.71
VcmaxRef = 98.13
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

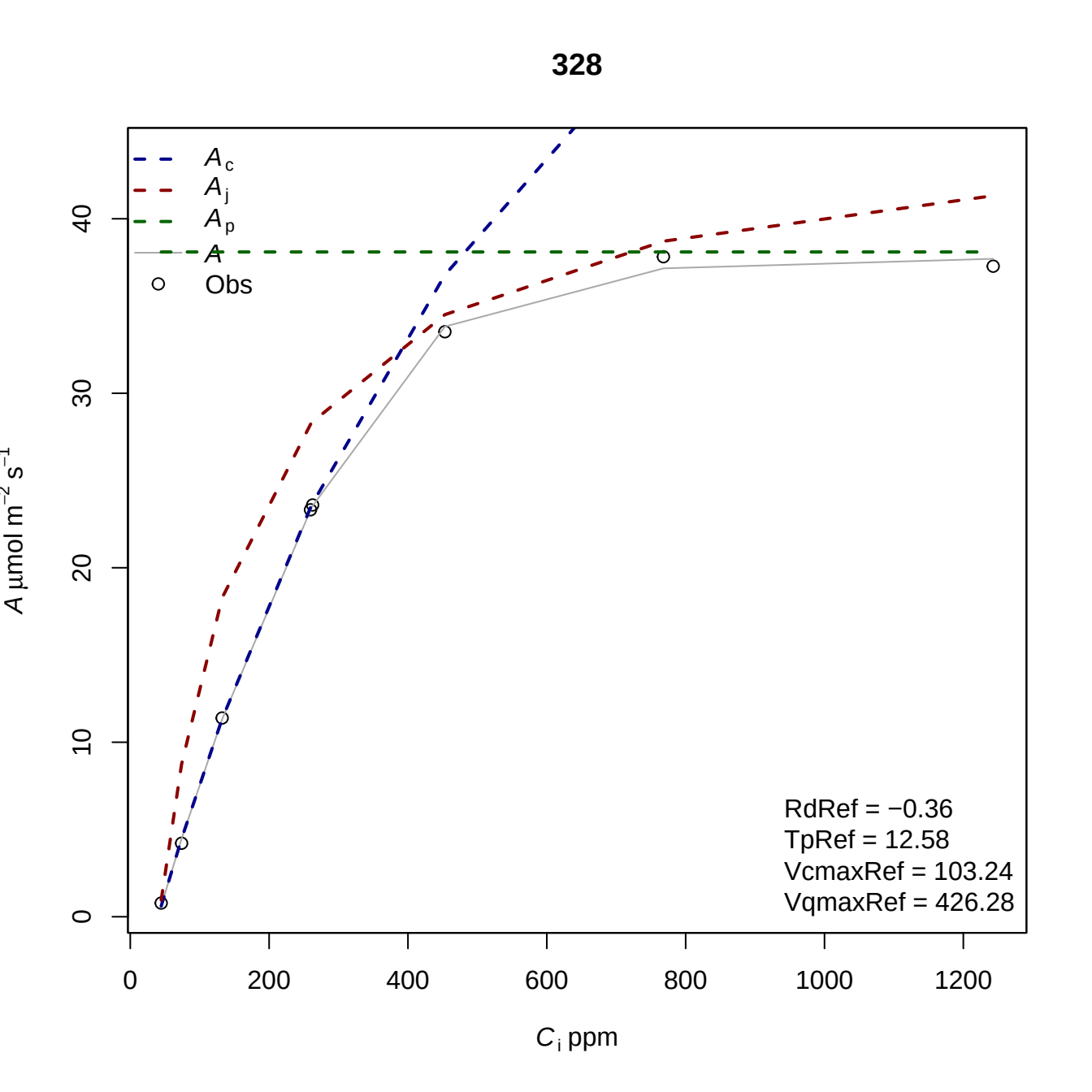

323
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.56
TpRef = 11.4
VcmaxRef = 140.64
VqmaxRef = 448.41
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

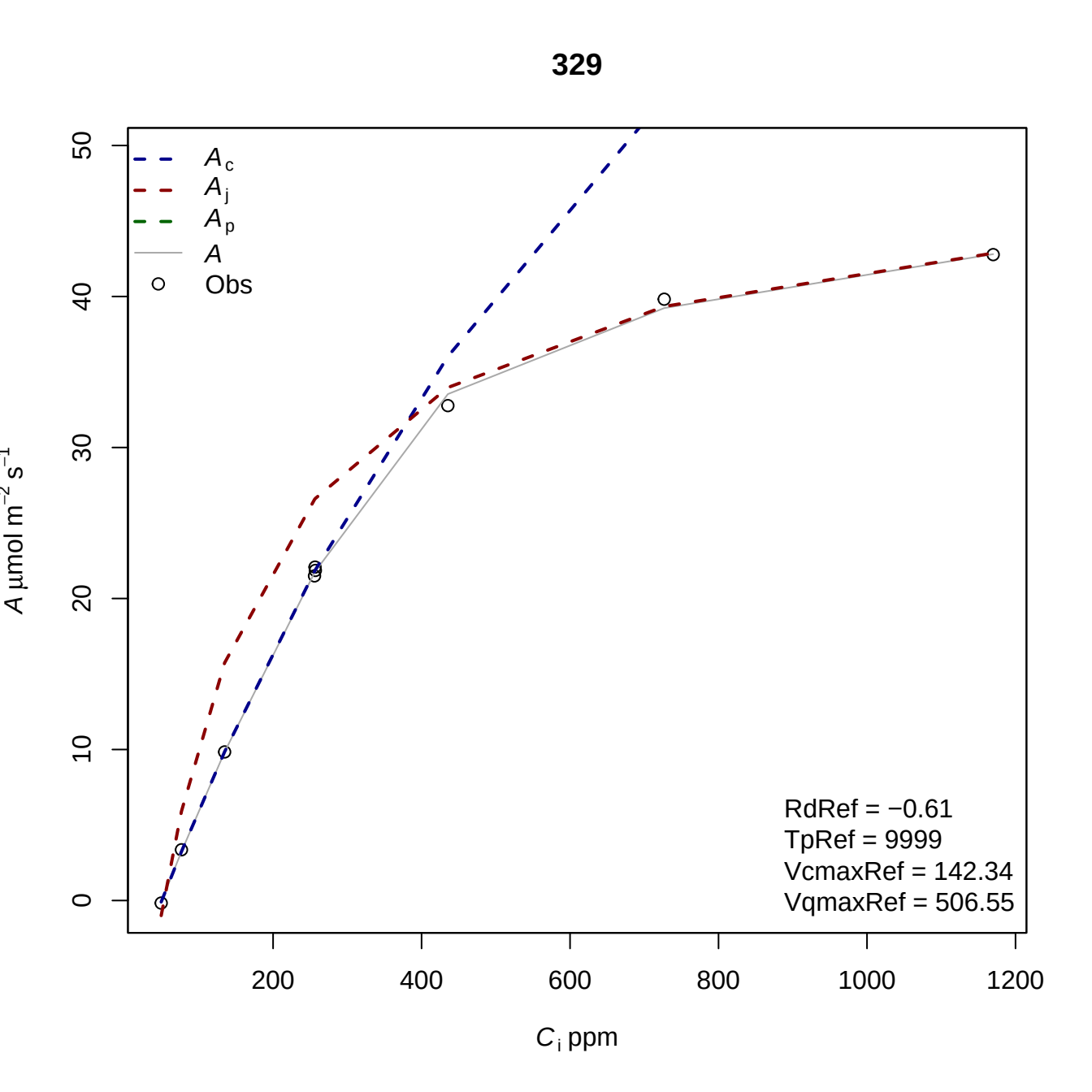

324
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −1.3
TpRef = 15.33
VcmaxRef = 203.8
VqmaxRef = 757.97
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

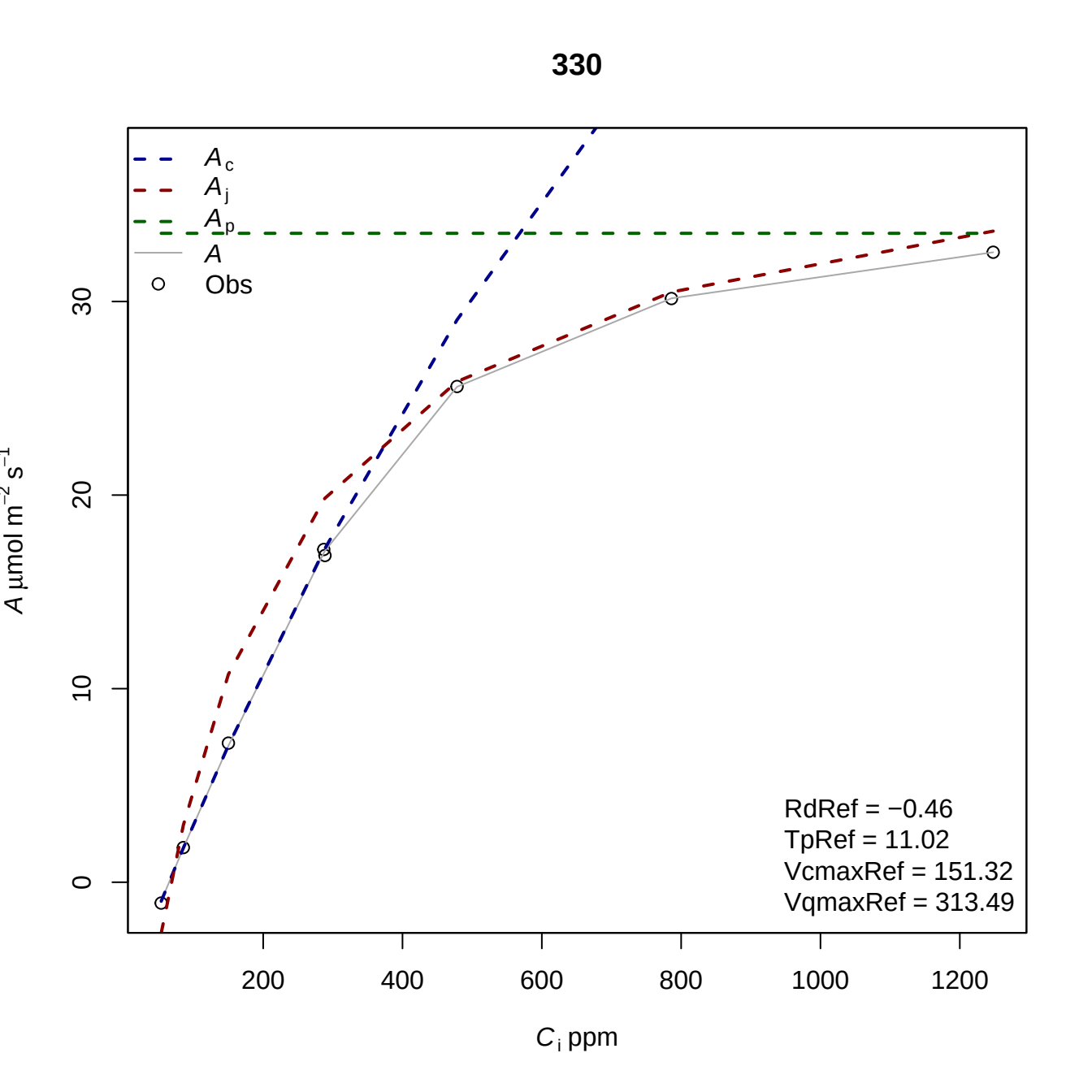

325
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.09
TpRef = 9999
VcmaxRef = 132.4
VqmaxRef = 190.04
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

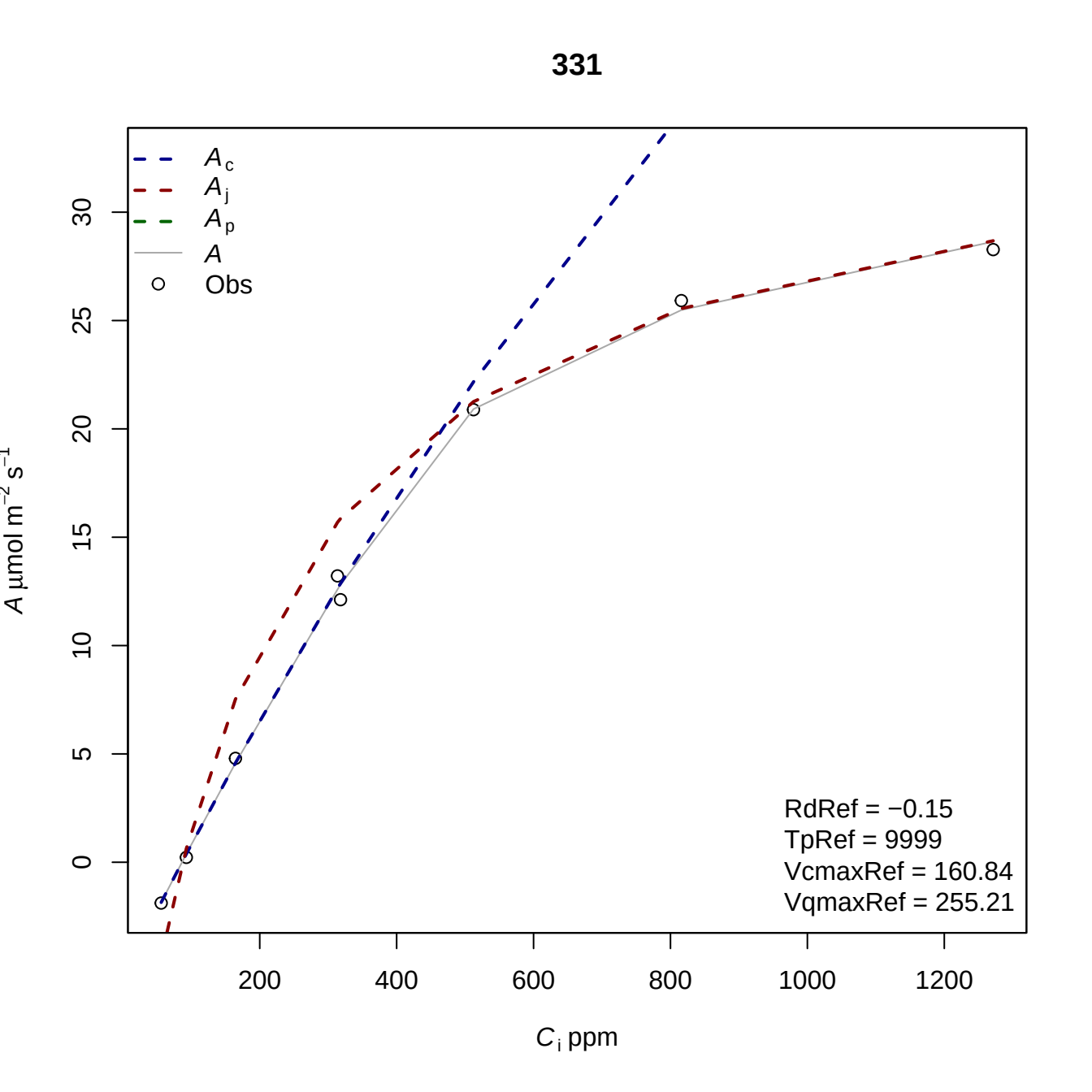

326
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.39
TpRef = 6.59
VcmaxRef = 42.68
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20

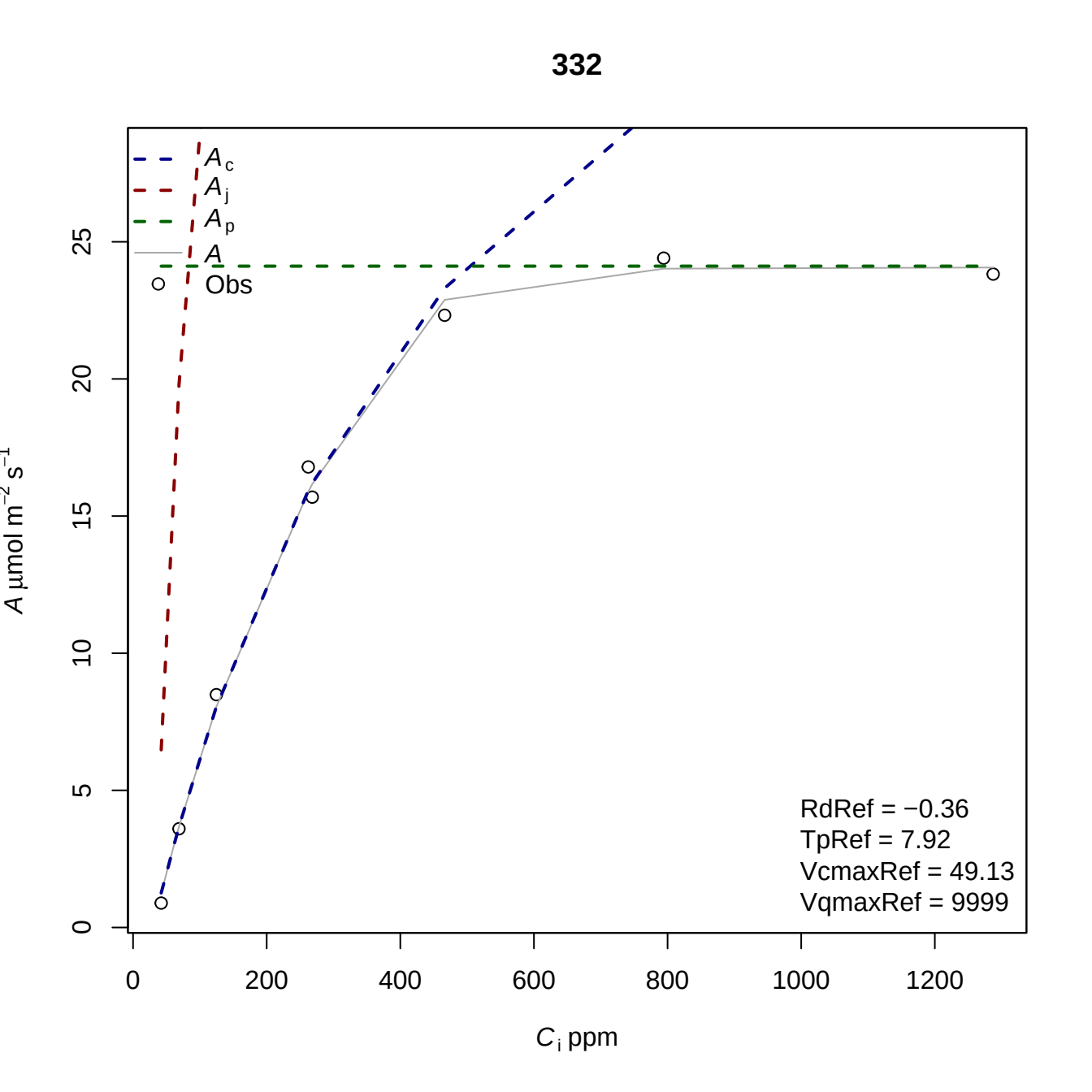

327
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.08
TpRef = 8.26
VcmaxRef = 60.06
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
200
400
600
800
1000
1200
5
10
15
20
25

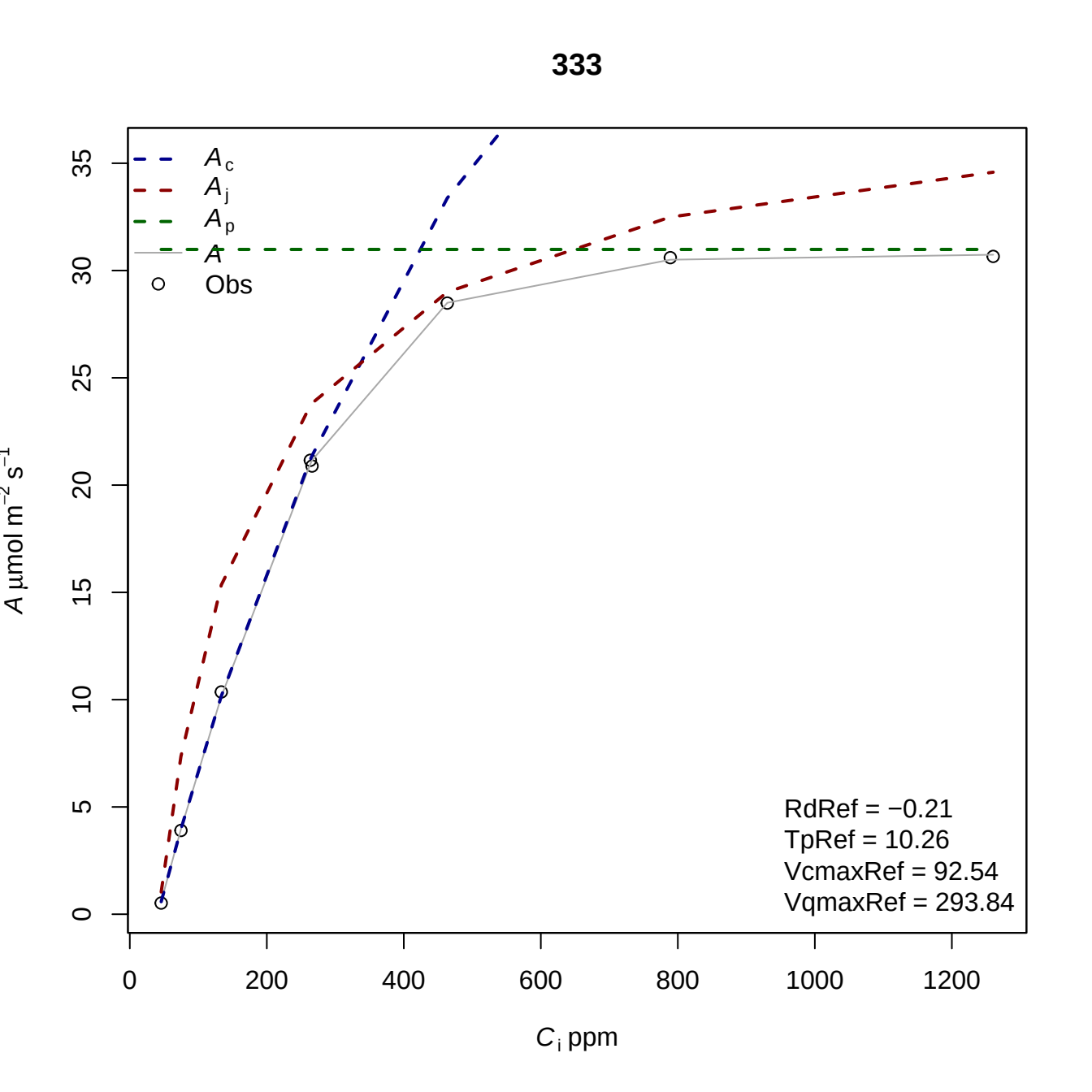

328
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.36
TpRef = 12.58
VcmaxRef = 103.24
VqmaxRef = 426.28
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

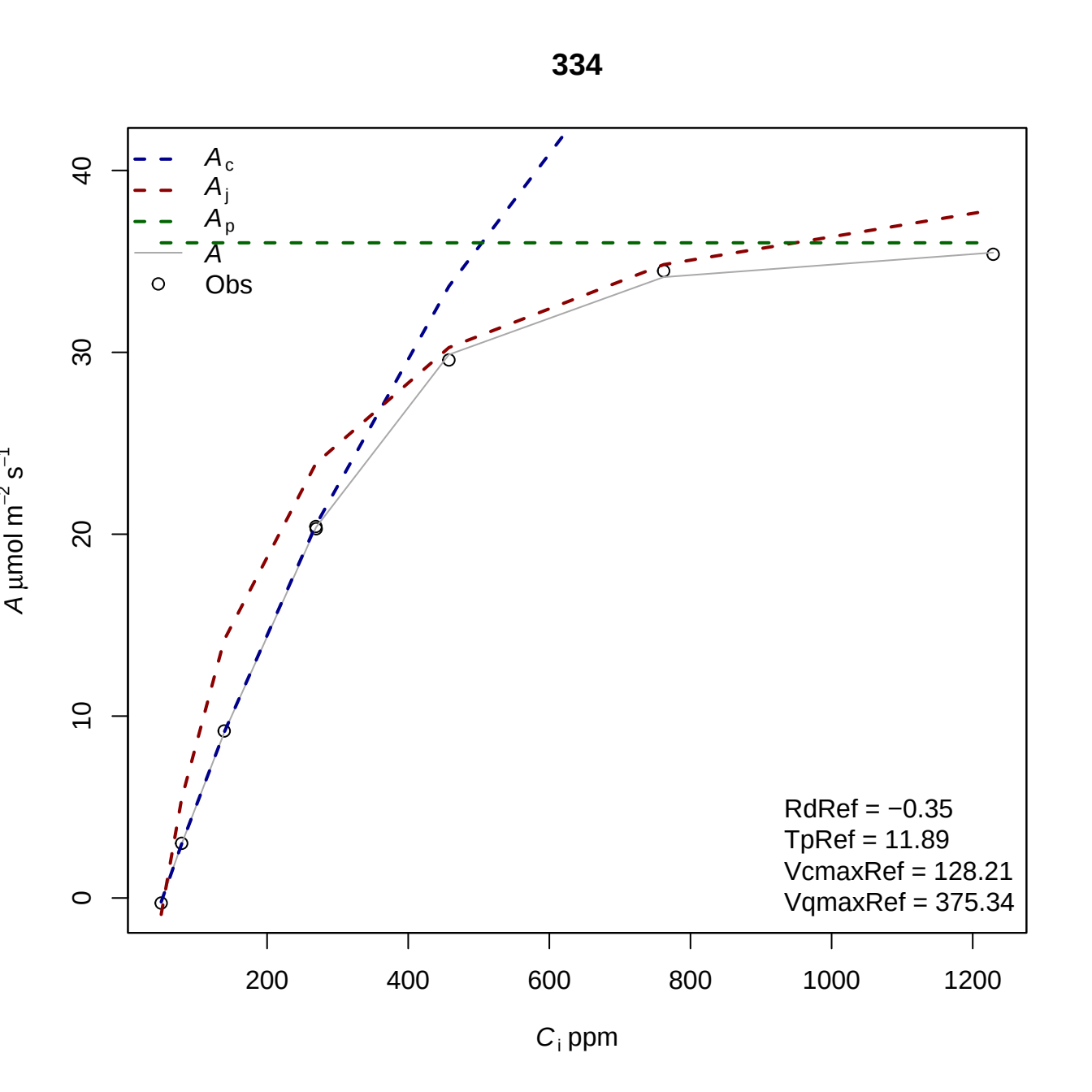

329
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.61
TpRef = 9999
VcmaxRef = 142.34
VqmaxRef = 506.55
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

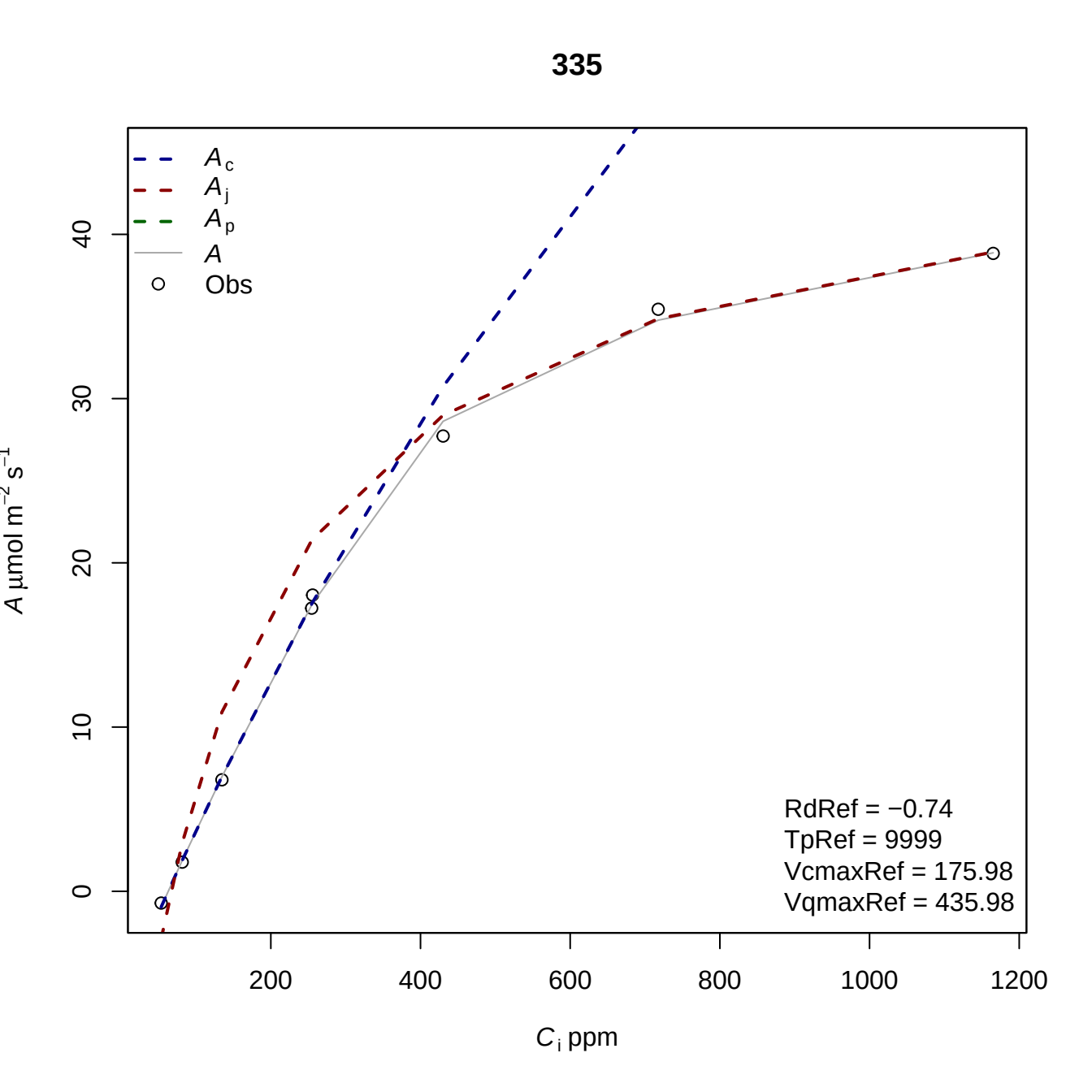

330
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.46
TpRef = 11.02
VcmaxRef = 151.32
VqmaxRef = 313.49
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
200
400
600
800
1000
1200

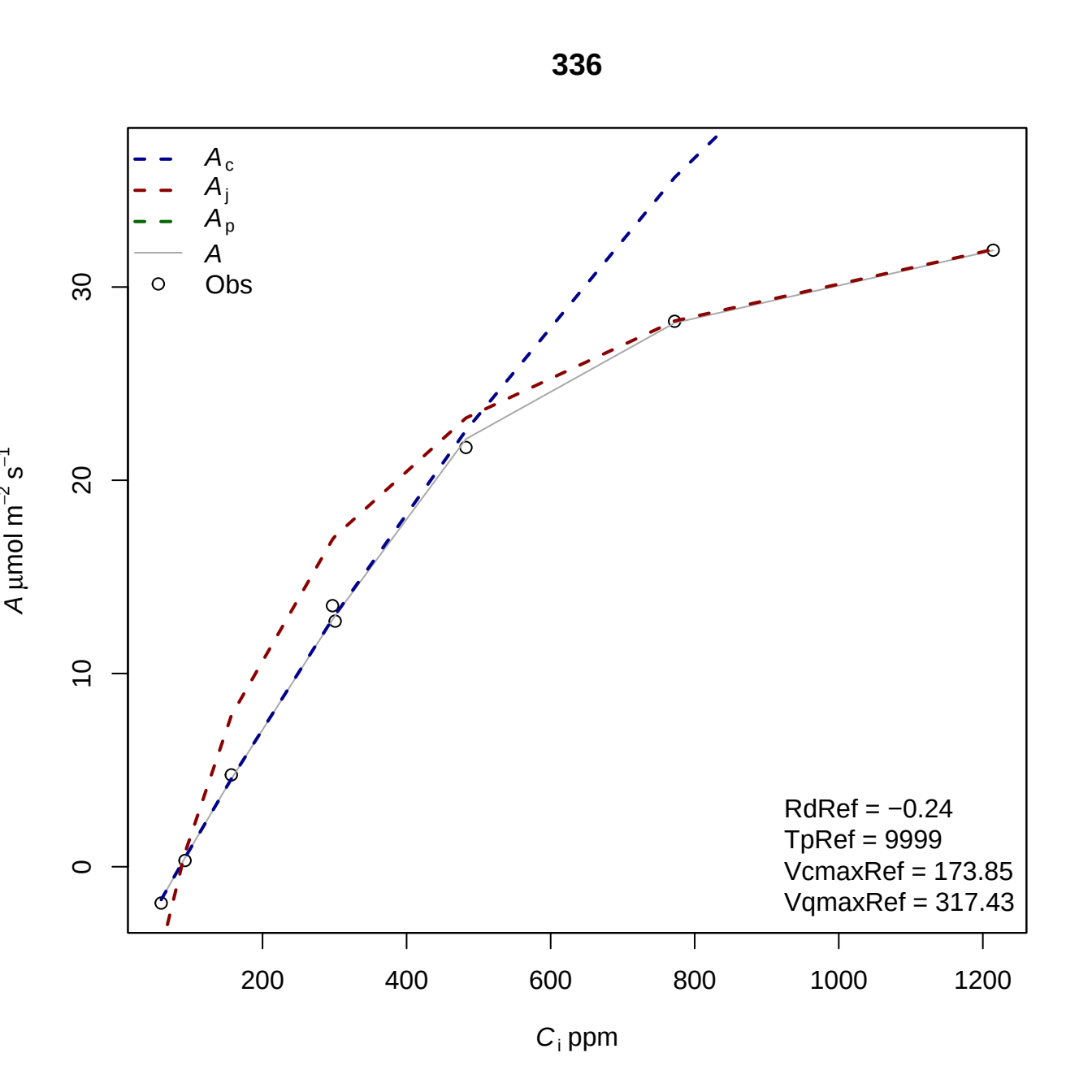

331
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.15
TpRef = 9999
VcmaxRef = 160.84
VqmaxRef = 255.21
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

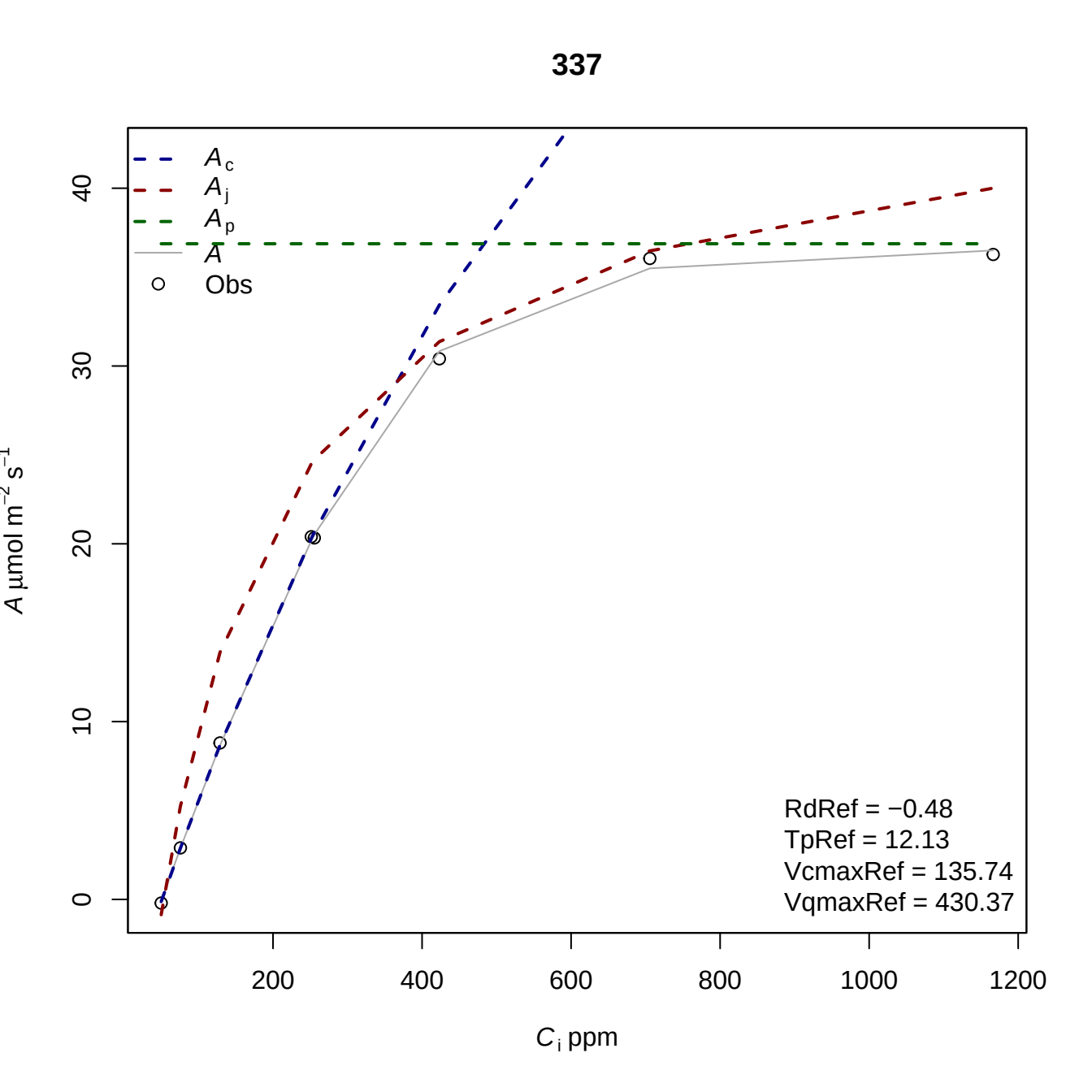

332
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.36
TpRef = 7.92
VcmaxRef = 49.13
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

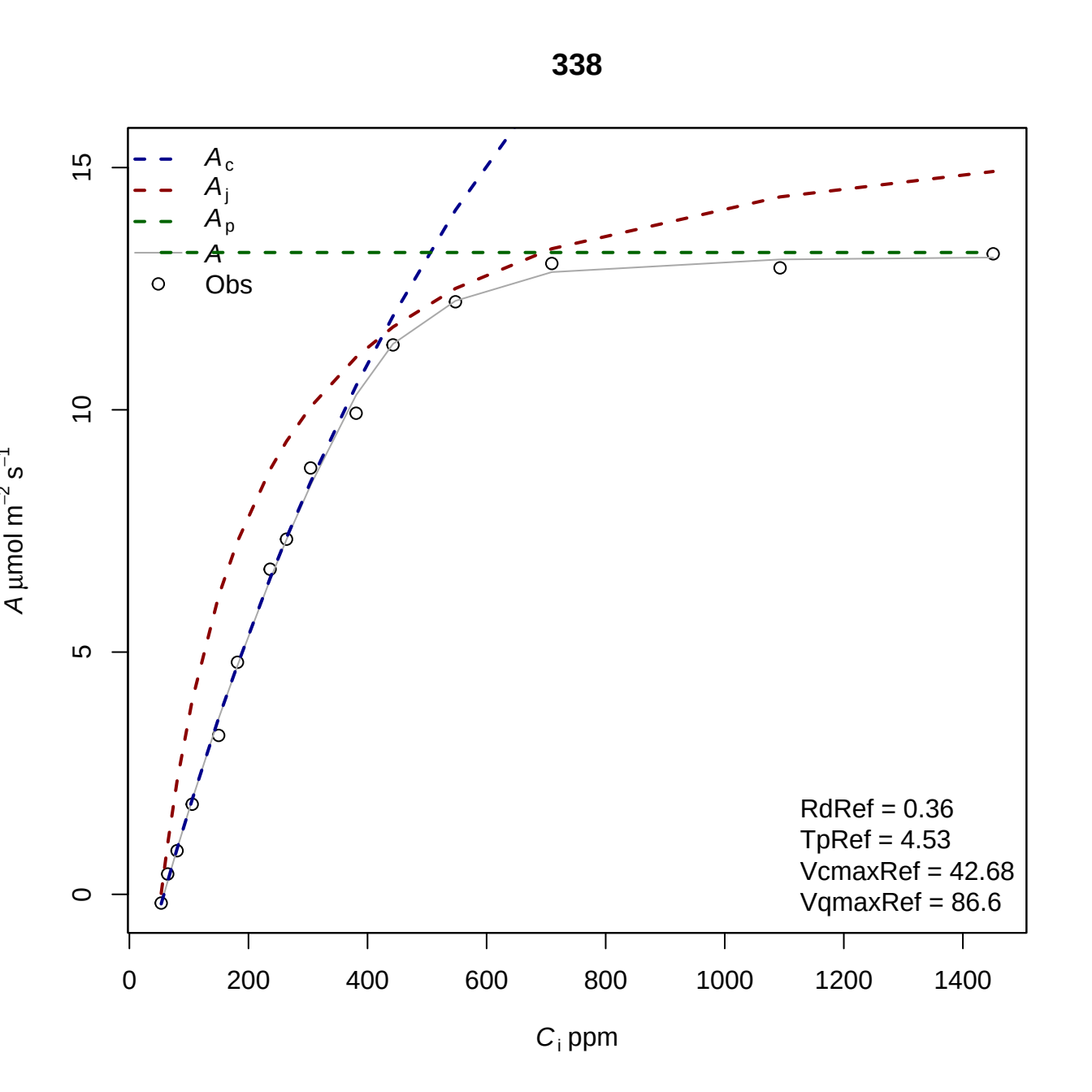

333
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.21
TpRef = 10.26
VcmaxRef = 92.54
VqmaxRef = 293.84
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

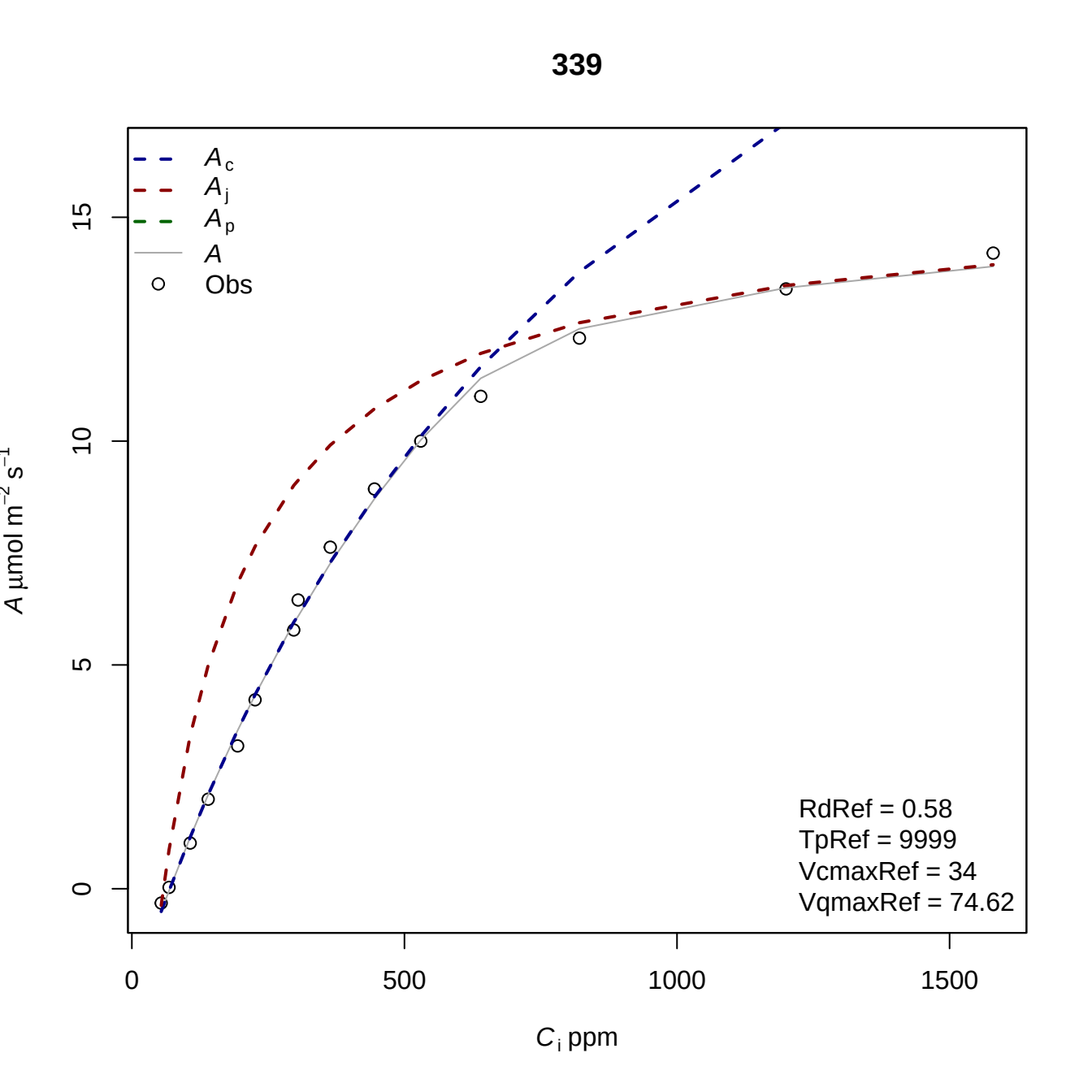

334
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.35
TpRef = 11.89
VcmaxRef = 128.21
VqmaxRef = 375.34
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

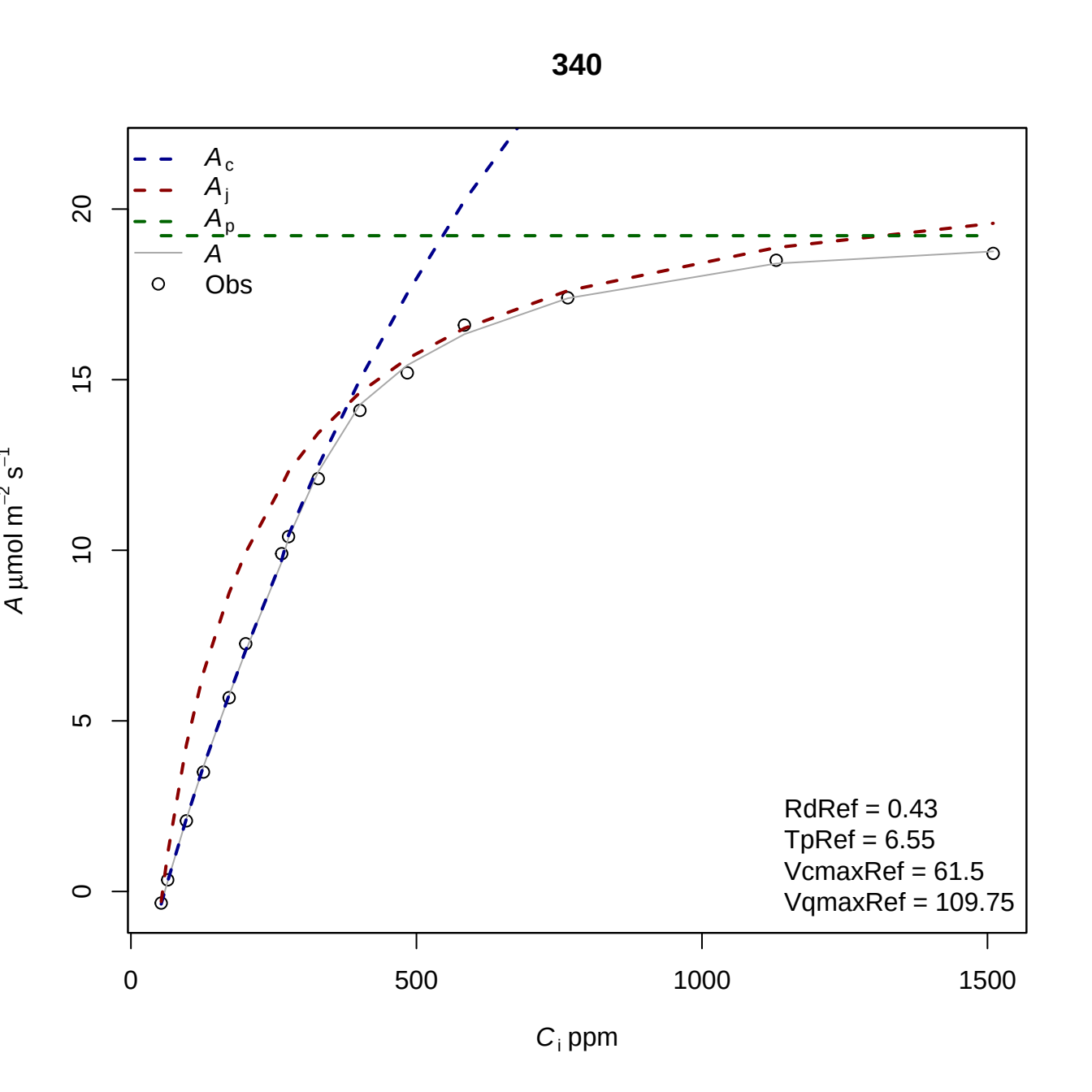

335
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.74
TpRef = 9999
VcmaxRef = 175.98
VqmaxRef = 435.98
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

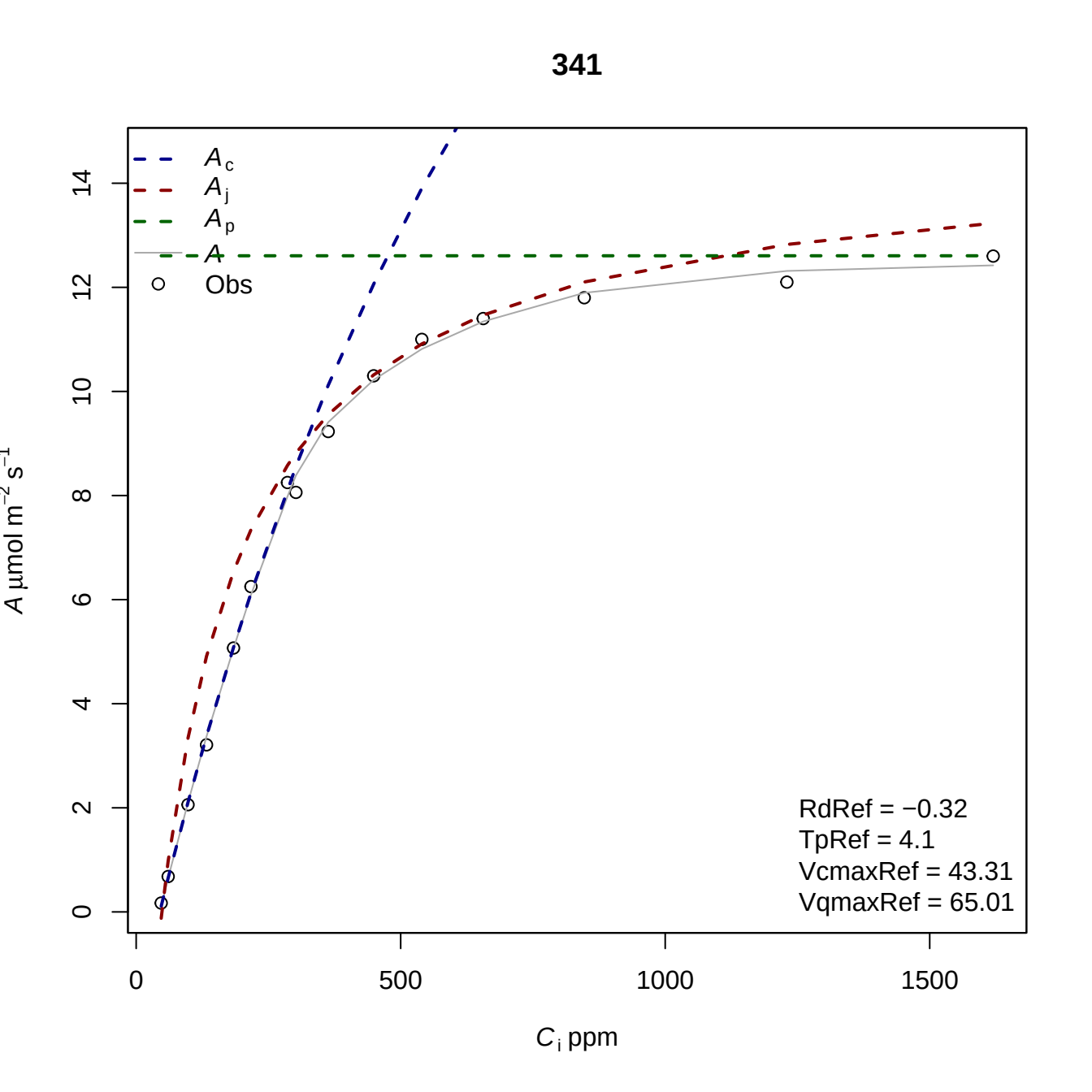

336
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.24
TpRef = 9999
VcmaxRef = 173.85
VqmaxRef = 317.43
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
200
400
600
800
1000
1200

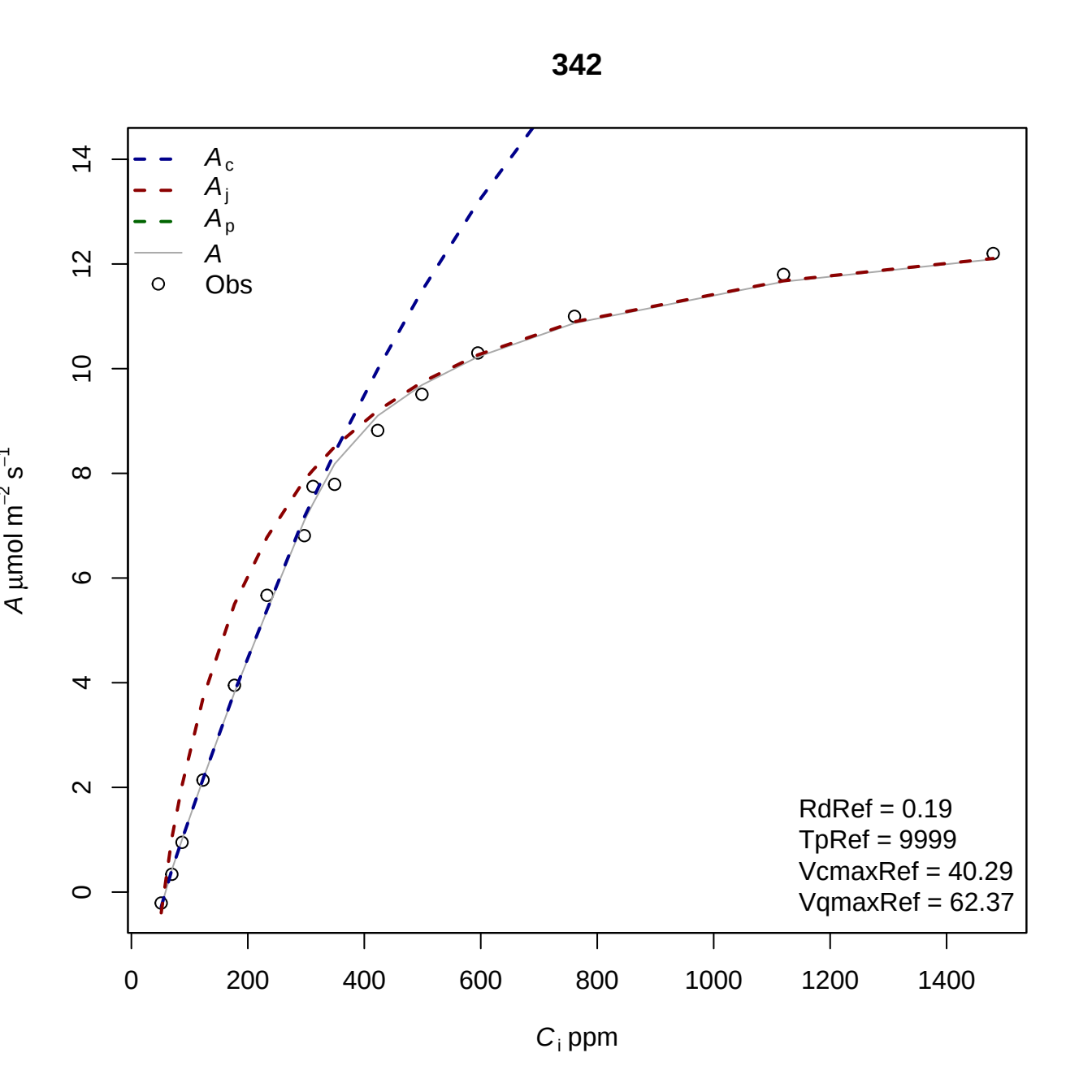

337
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.48
TpRef = 12.13
VcmaxRef = 135.74
VqmaxRef = 430.37
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

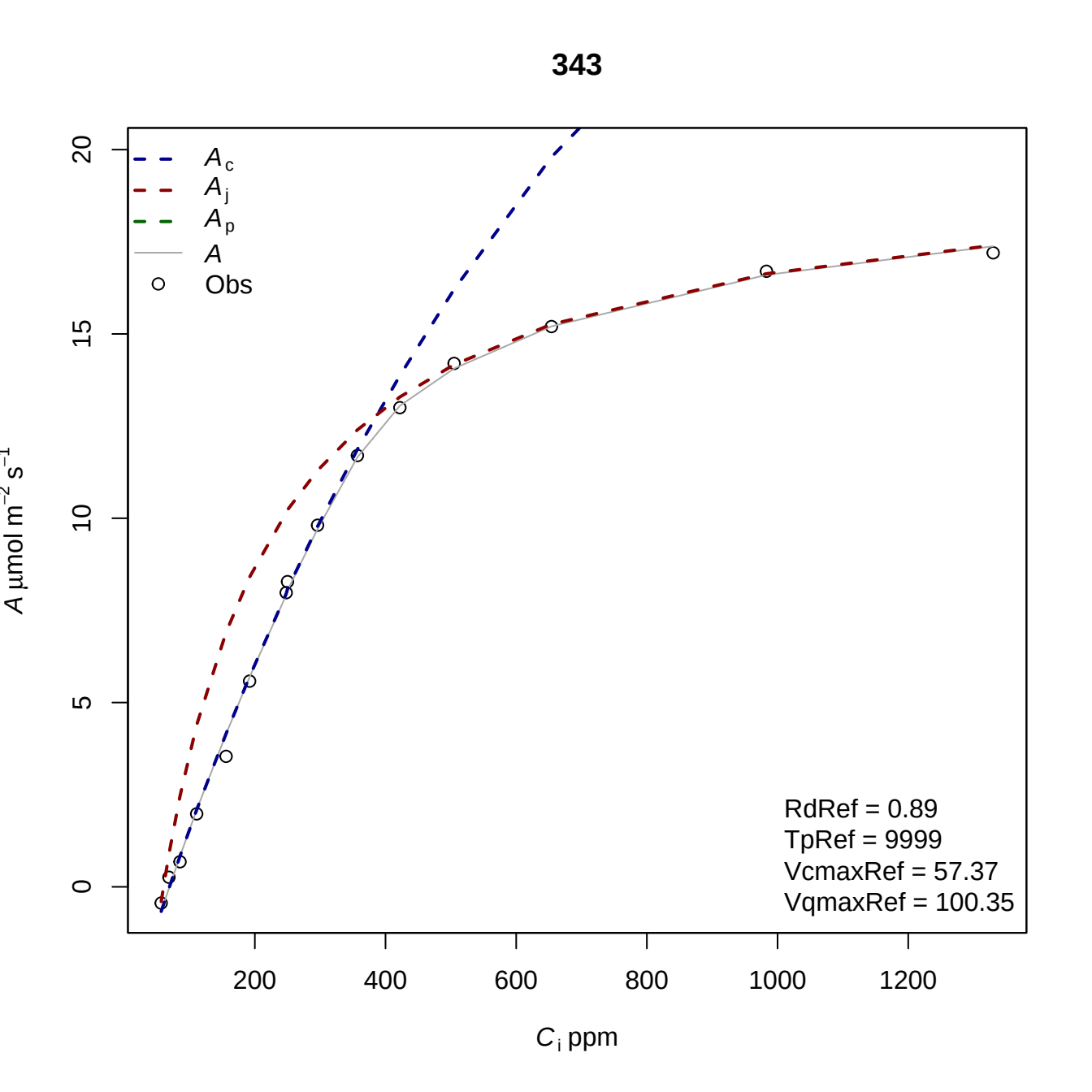

338
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.36
TpRef = 4.53
VcmaxRef = 42.68
VqmaxRef = 86.6
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

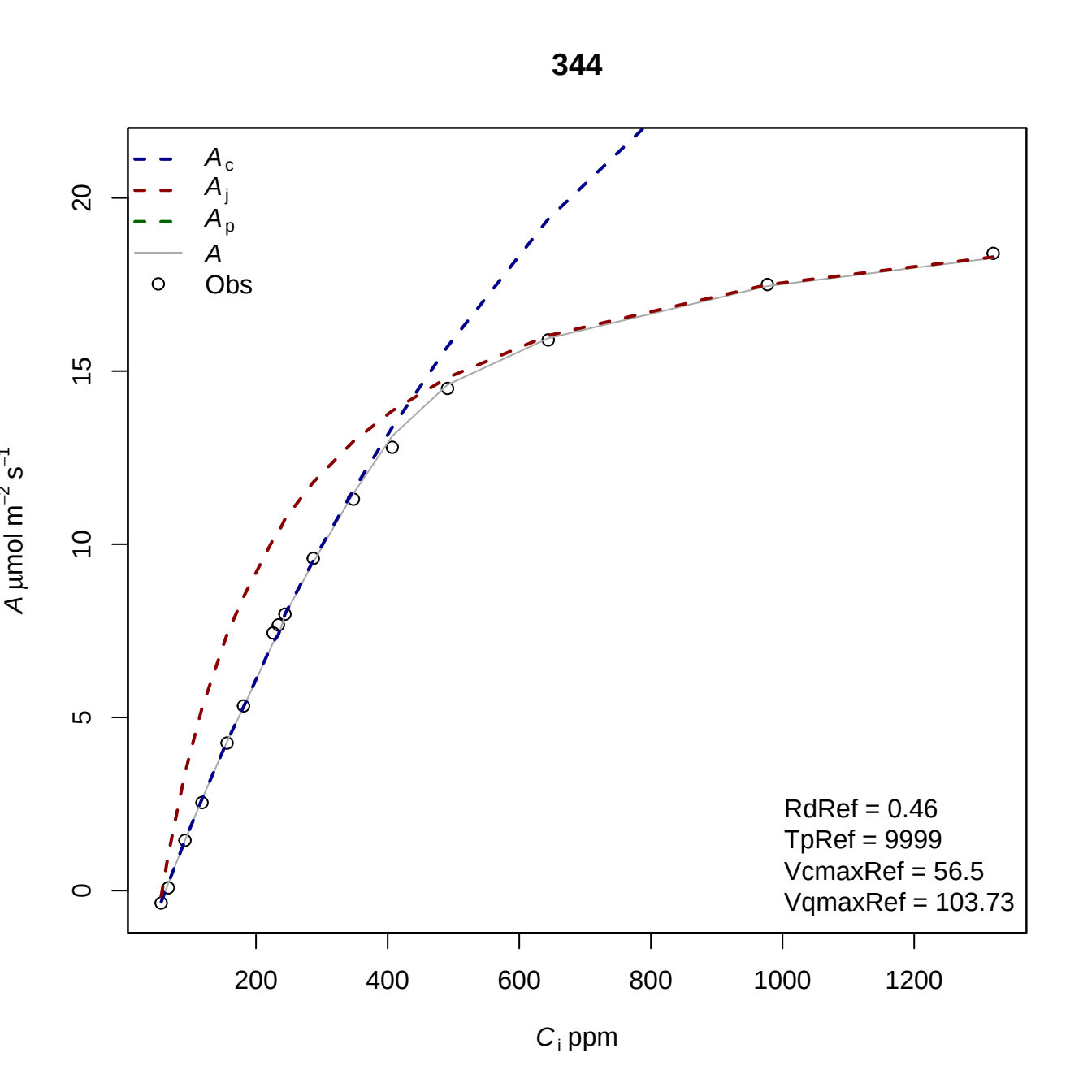

339
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.58
TpRef = 9999
VcmaxRef = 34
VqmaxRef = 74.62
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

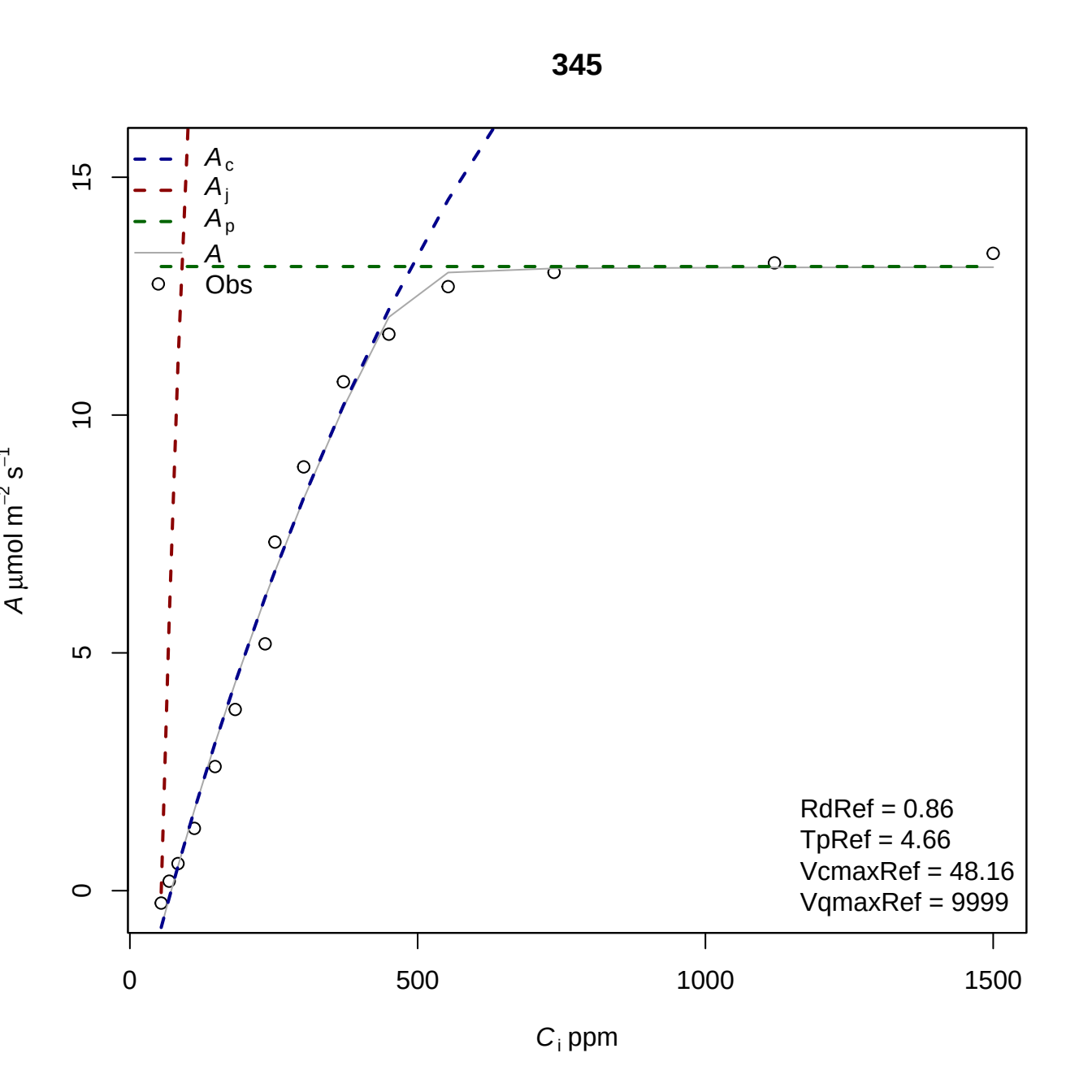

340
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.43
TpRef = 6.55
VcmaxRef = 61.5
VqmaxRef = 109.75
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

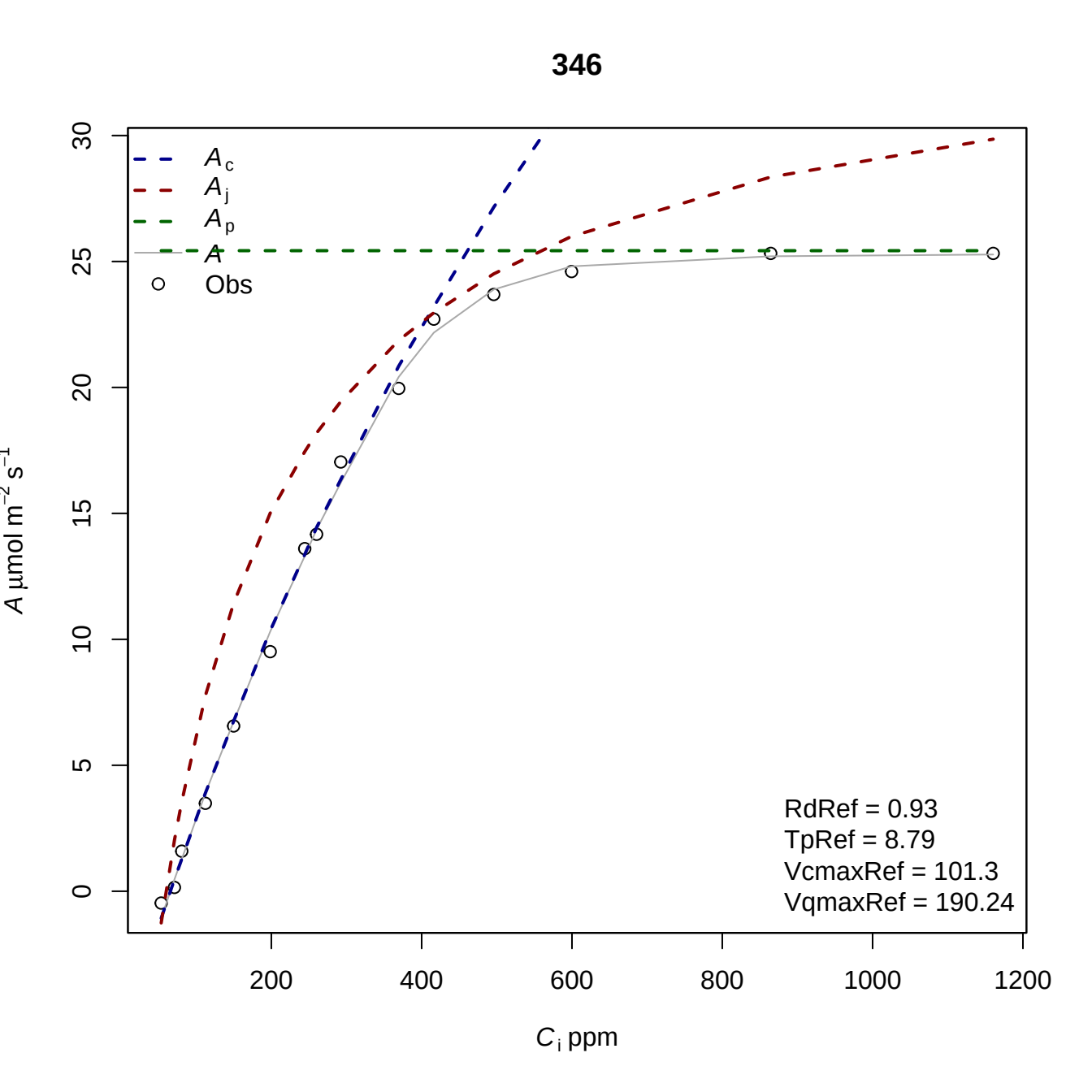

341
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.32
TpRef = 4.1
VcmaxRef = 43.31
VqmaxRef = 65.01
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
0
500
1000
1500

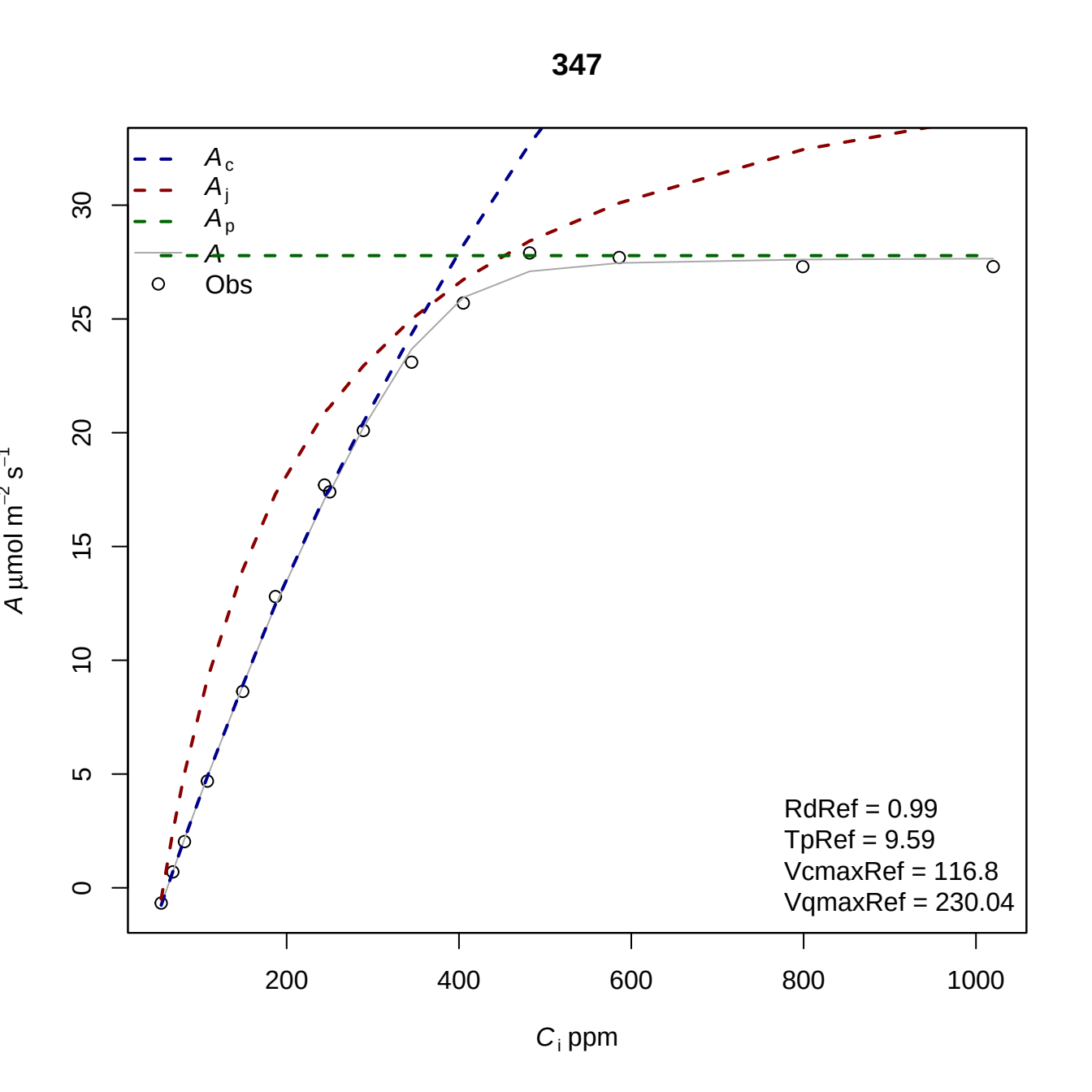

342
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.19
TpRef = 9999
VcmaxRef = 40.29
VqmaxRef = 62.37
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
1400

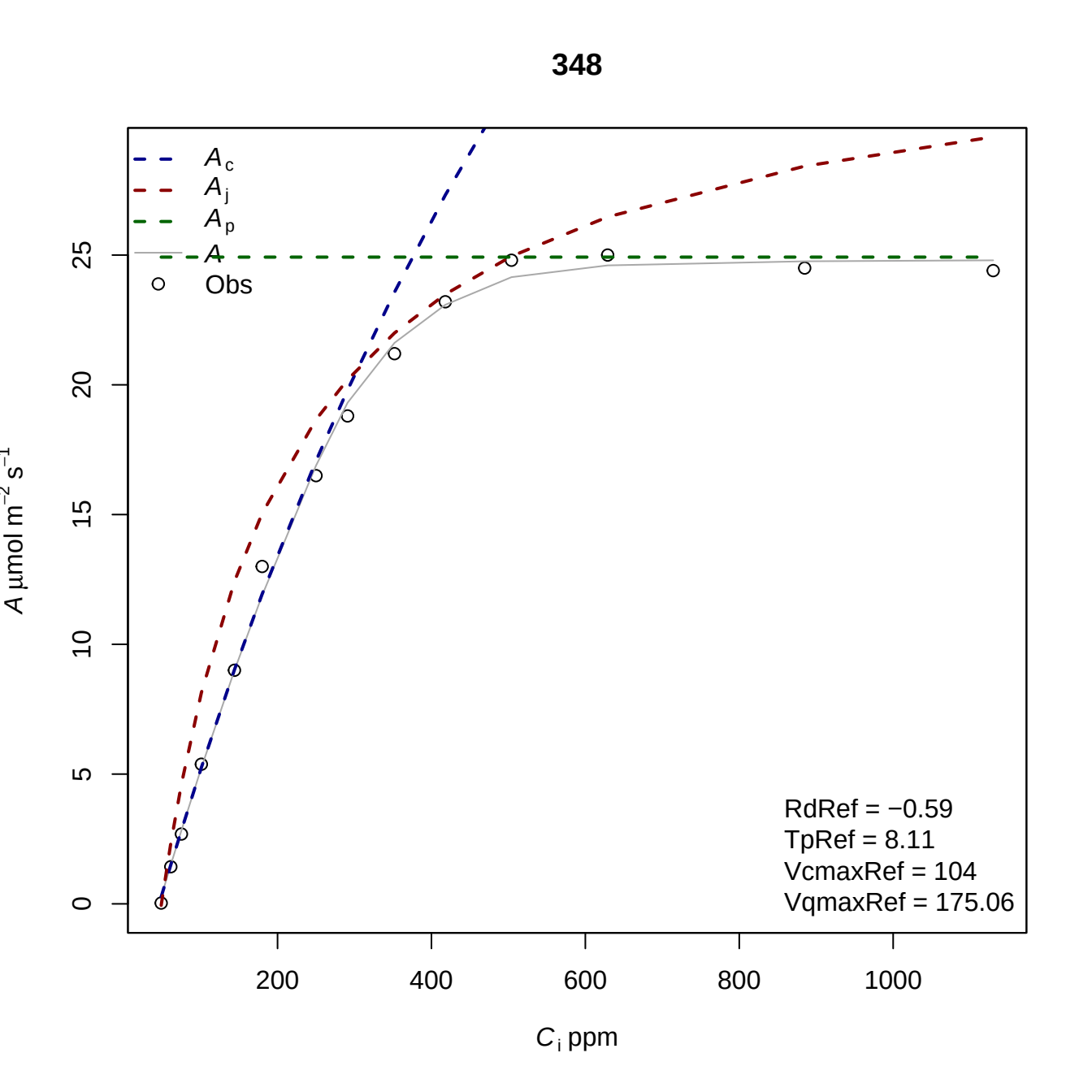

343
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.89
TpRef = 9999
VcmaxRef = 57.37
VqmaxRef = 100.35
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200

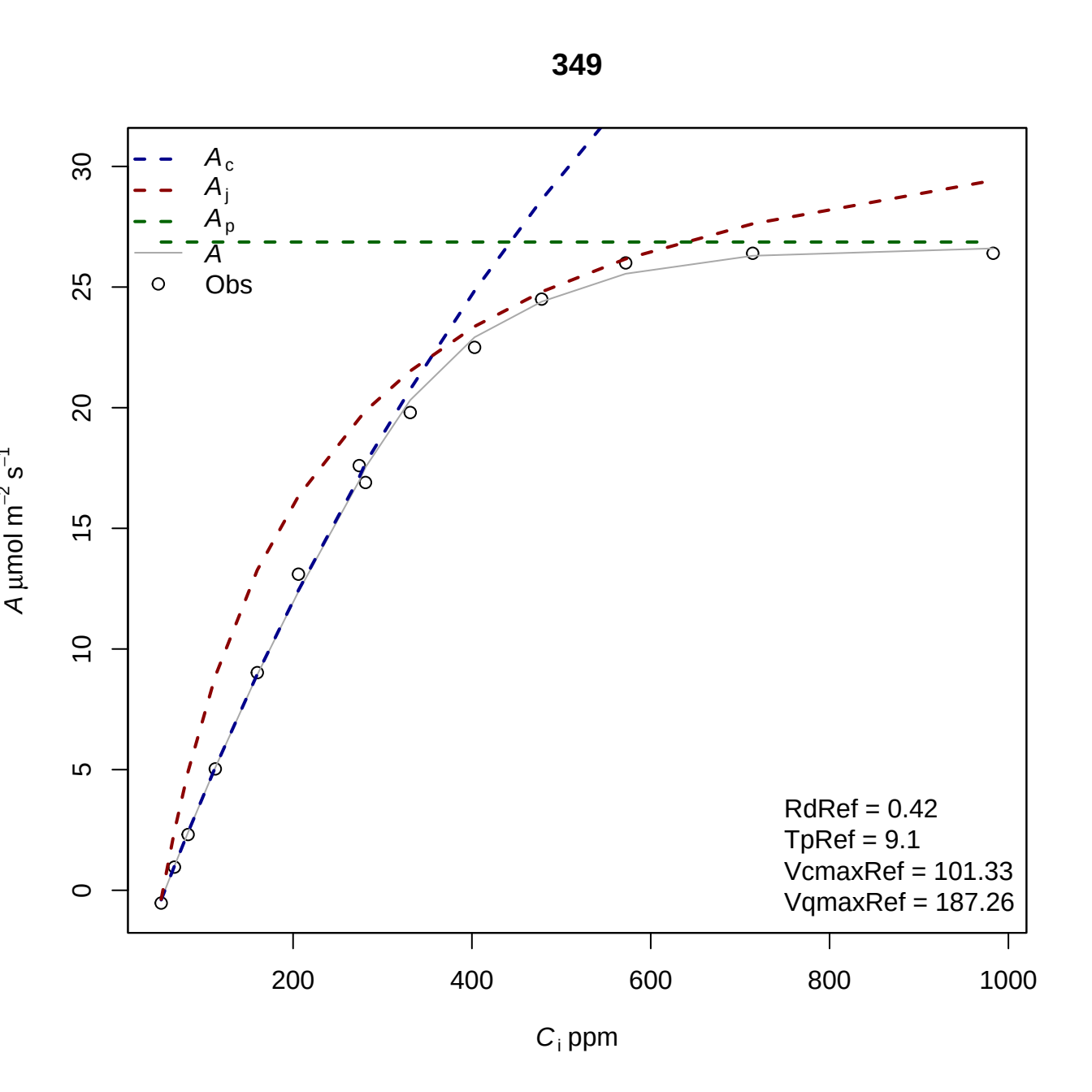

344
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.46
TpRef = 9999
VcmaxRef = 56.5
VqmaxRef = 103.73
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200

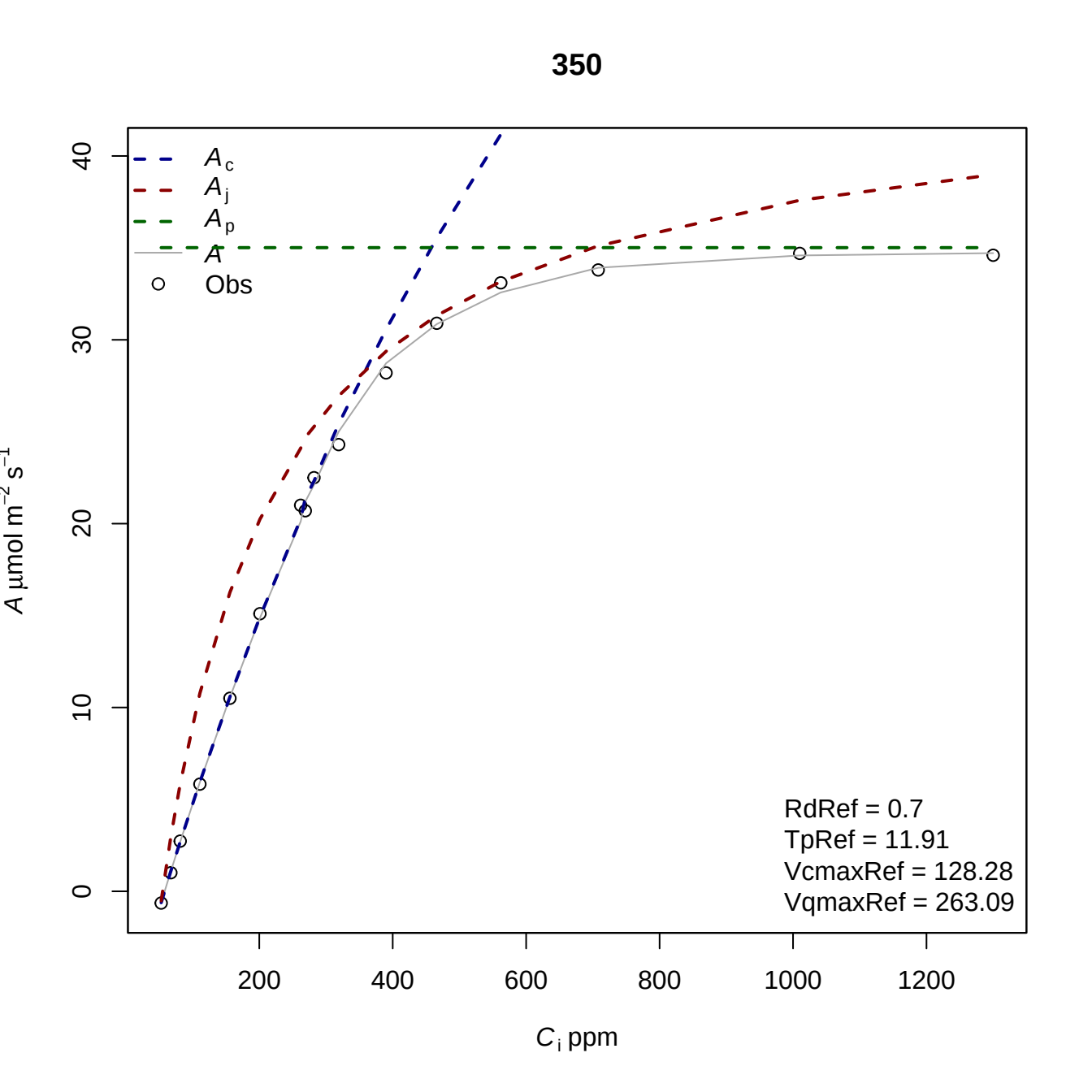

345
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.86
TpRef = 4.66
VcmaxRef = 48.16
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

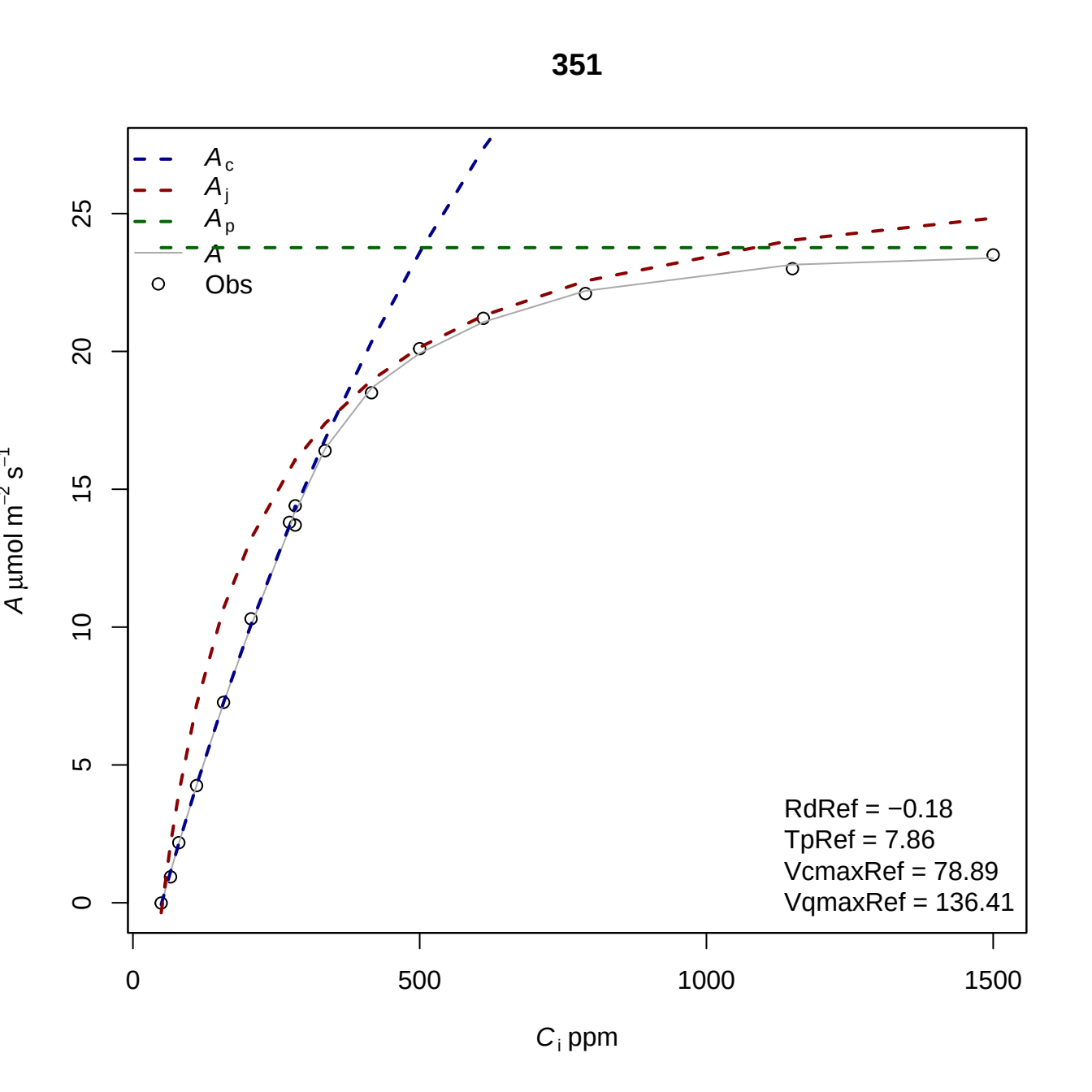

346
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.93
TpRef = 8.79
VcmaxRef = 101.3
VqmaxRef = 190.24
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

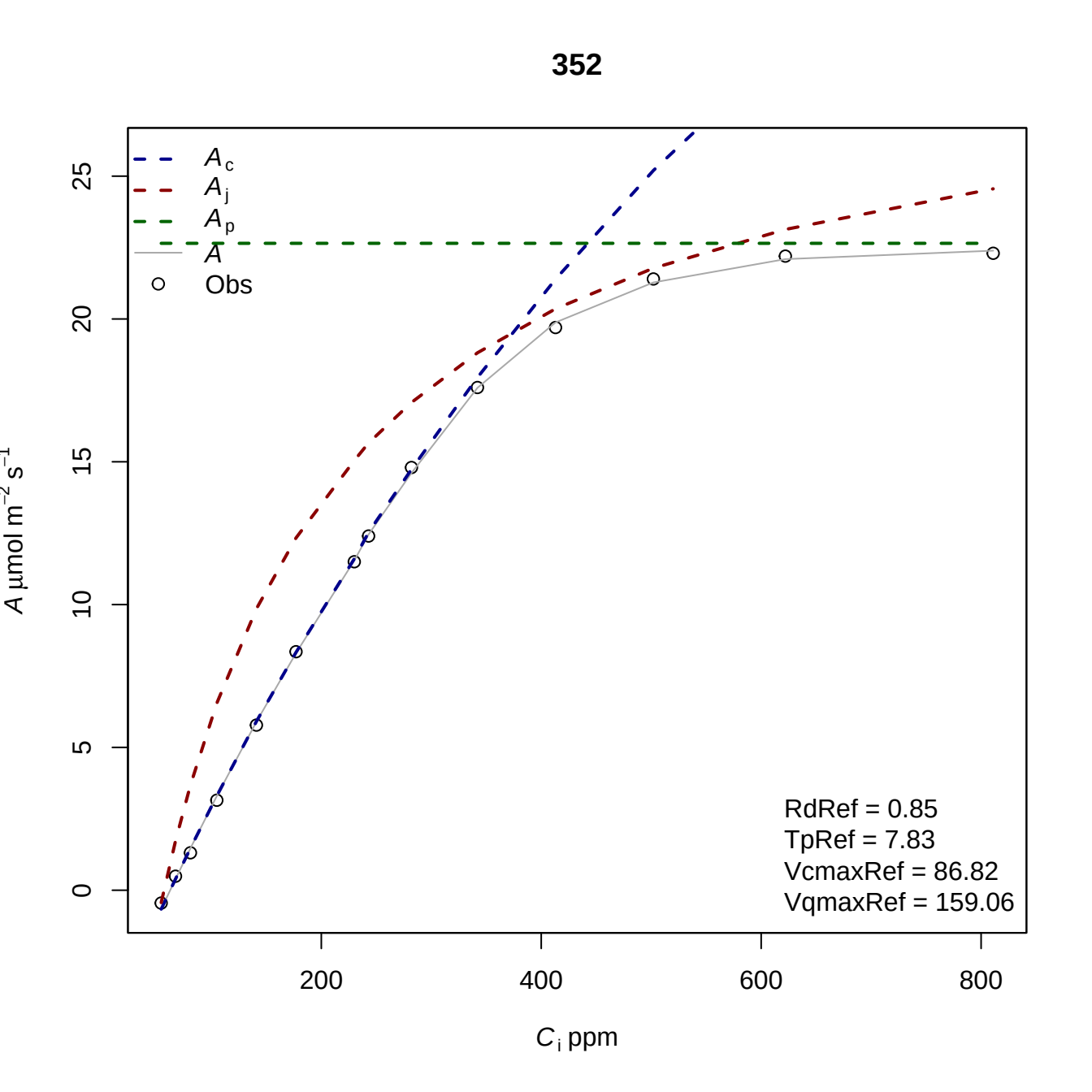

347
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.99
TpRef = 9.59
VcmaxRef = 116.8
VqmaxRef = 230.04
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000

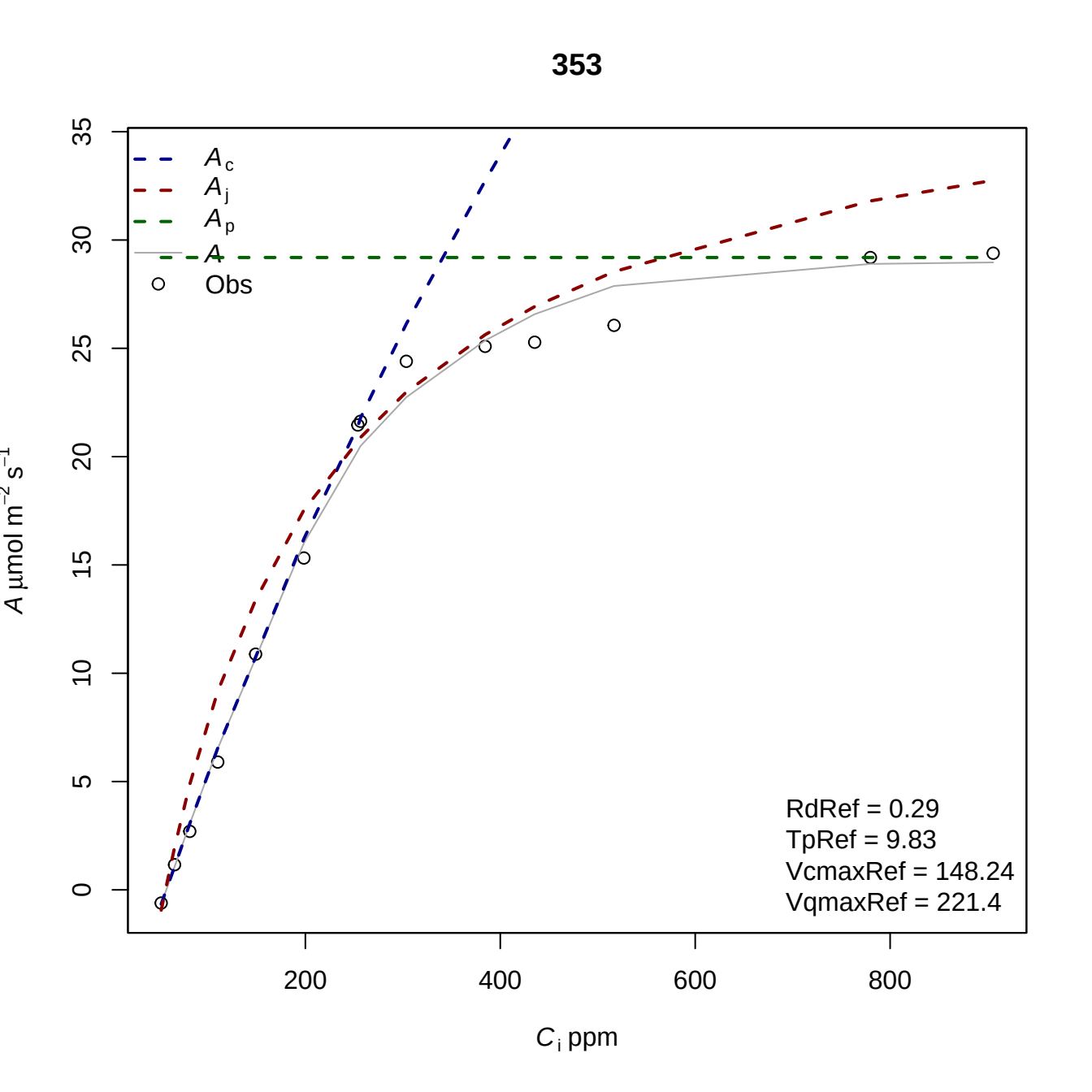

348
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.59
TpRef = 8.11
VcmaxRef = 104
VqmaxRef = 175.06
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000

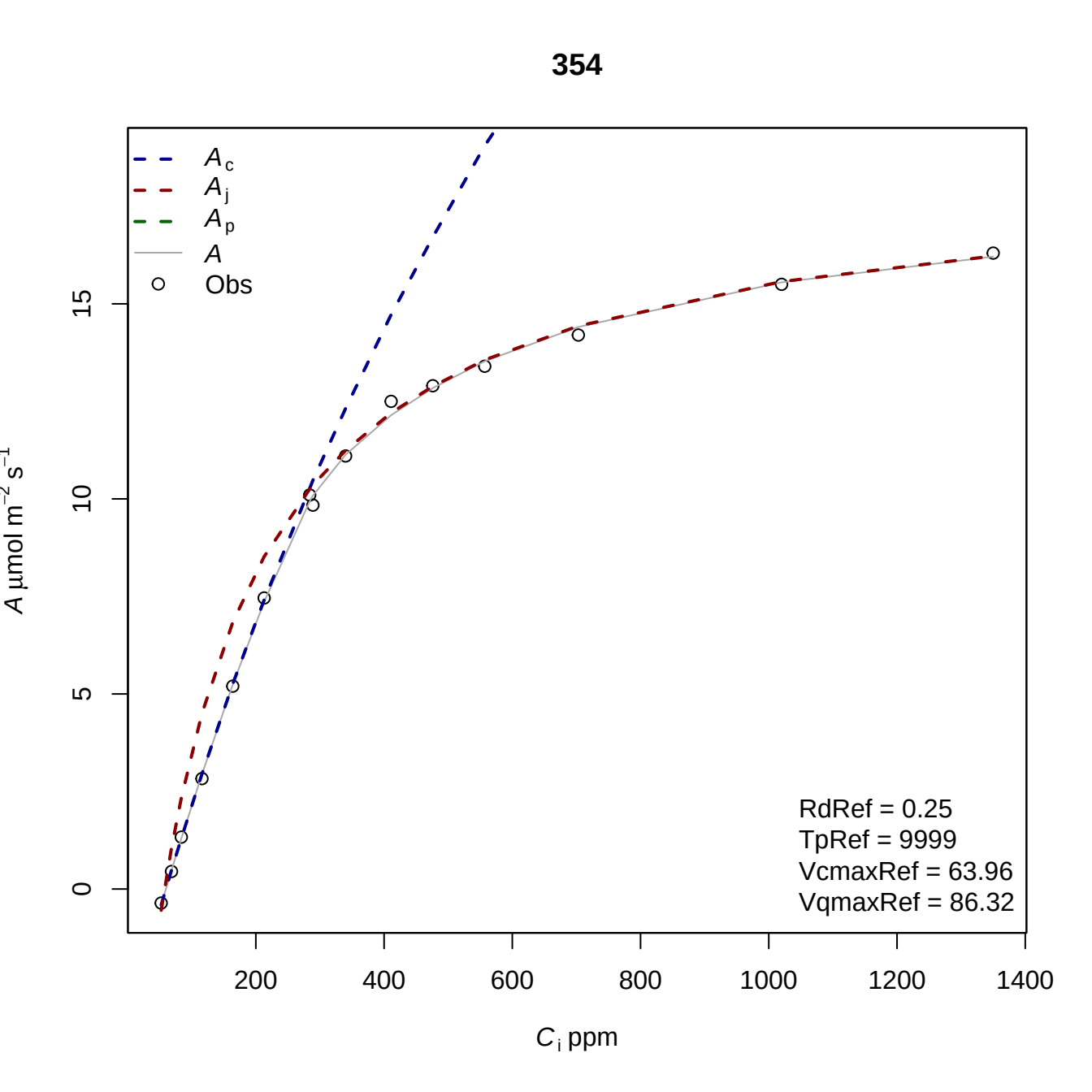

349
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.42
TpRef = 9.1
VcmaxRef = 101.33
VqmaxRef = 187.26
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000

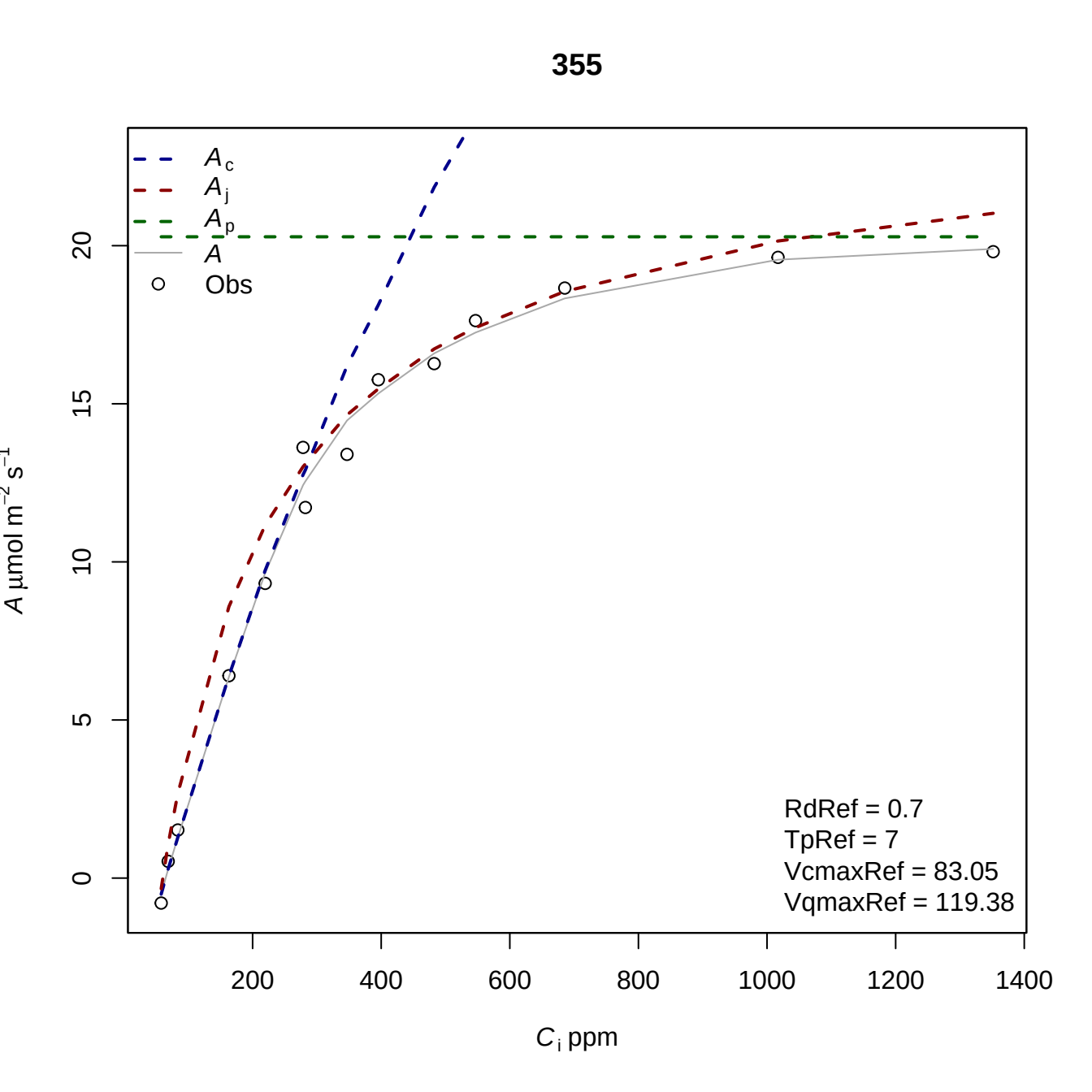

350
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.7
TpRef = 11.91
VcmaxRef = 128.28
VqmaxRef = 263.09
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

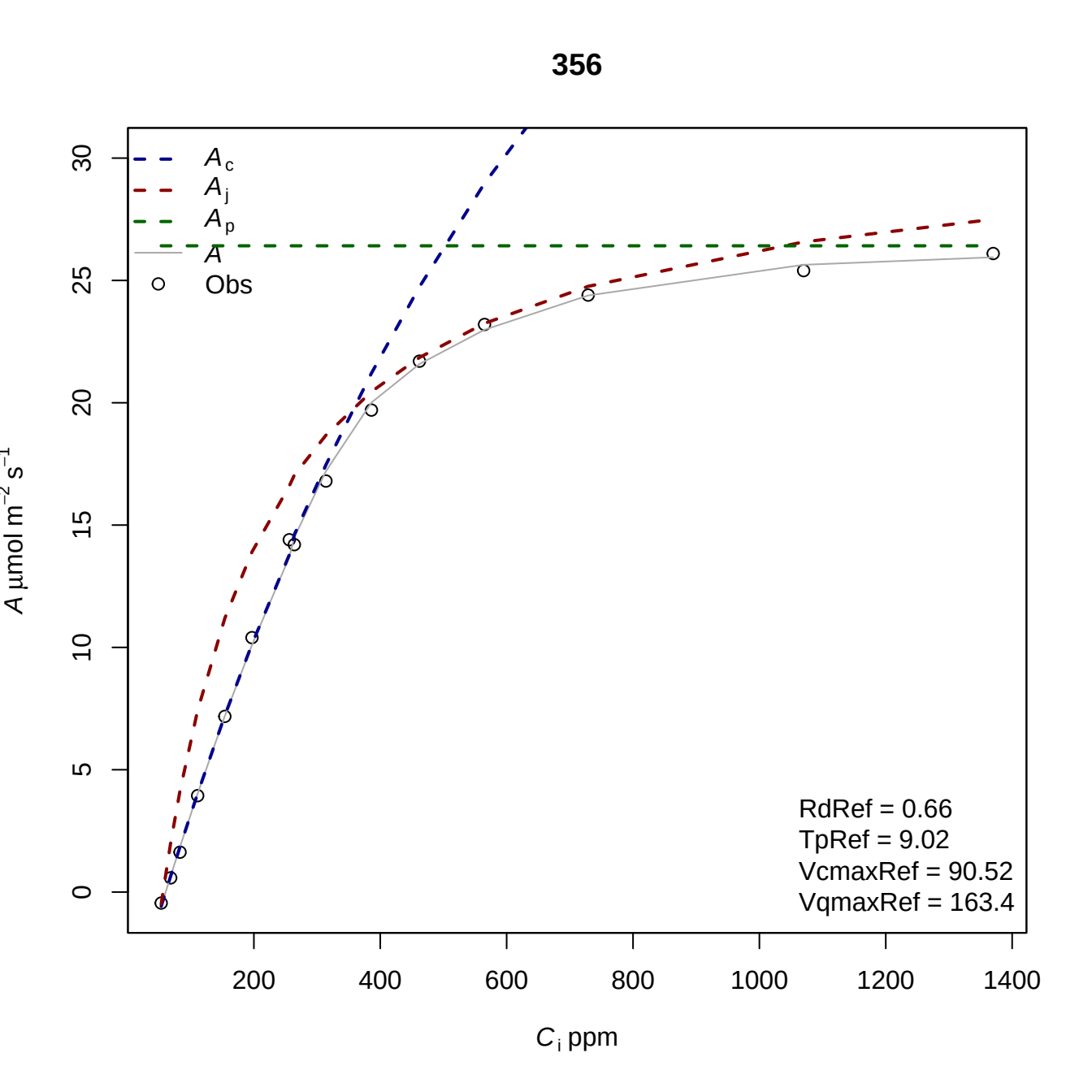

351
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.18
TpRef = 7.86
VcmaxRef = 78.89
VqmaxRef = 136.41
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

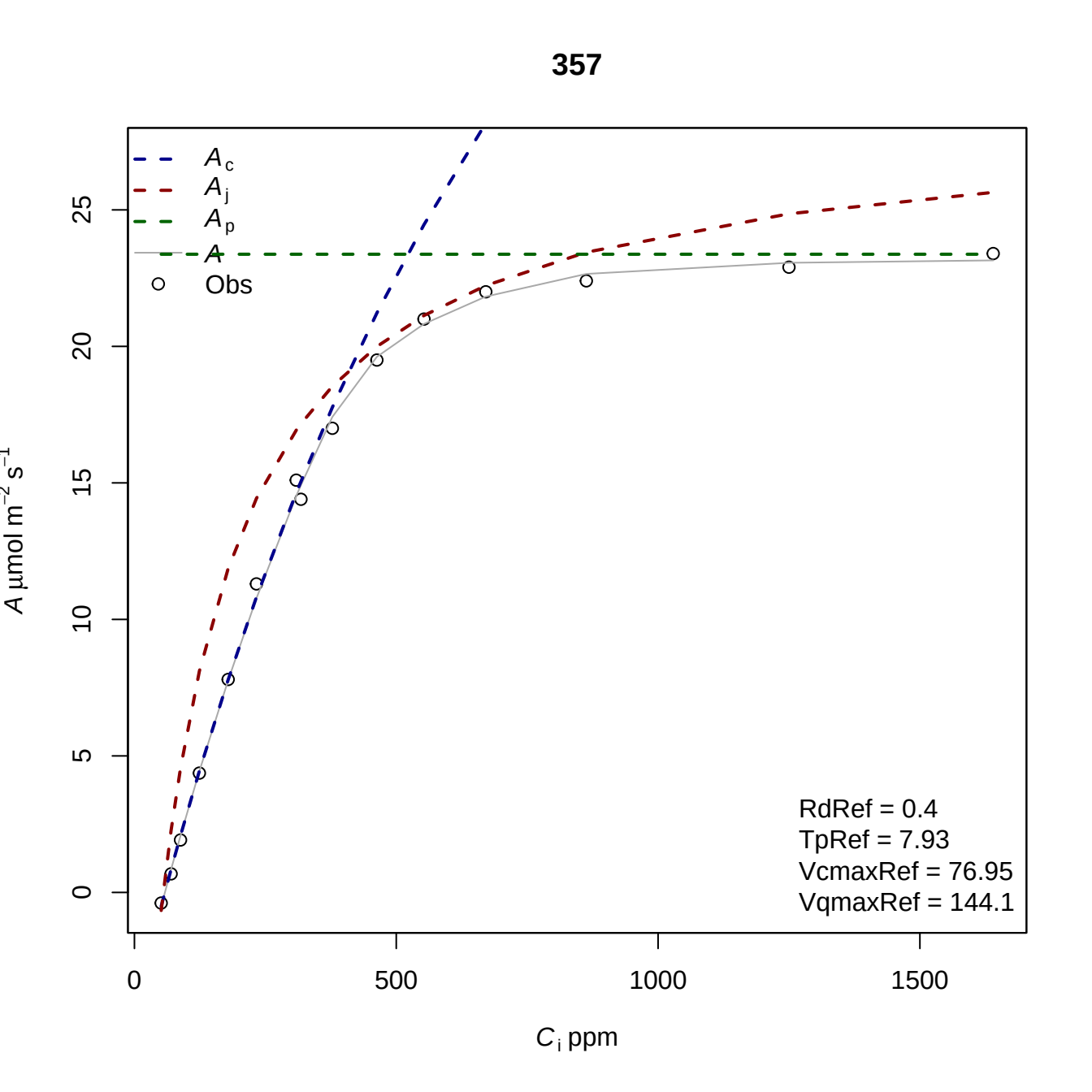

352
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.85
TpRef = 7.83
VcmaxRef = 86.82
VqmaxRef = 159.06
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800

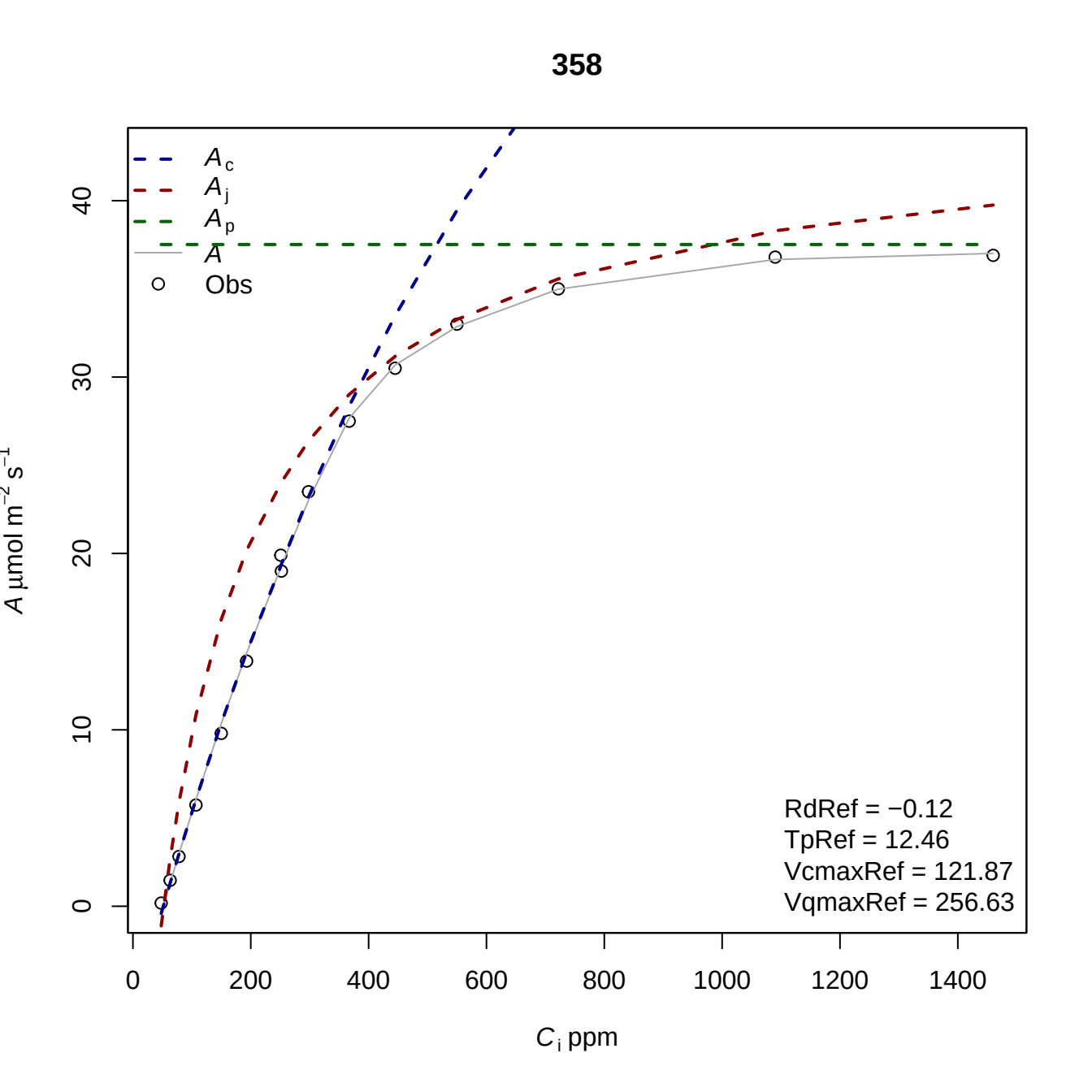

353
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.29
TpRef = 9.83
VcmaxRef = 148.24
VqmaxRef = 221.4
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
200
400
600
800

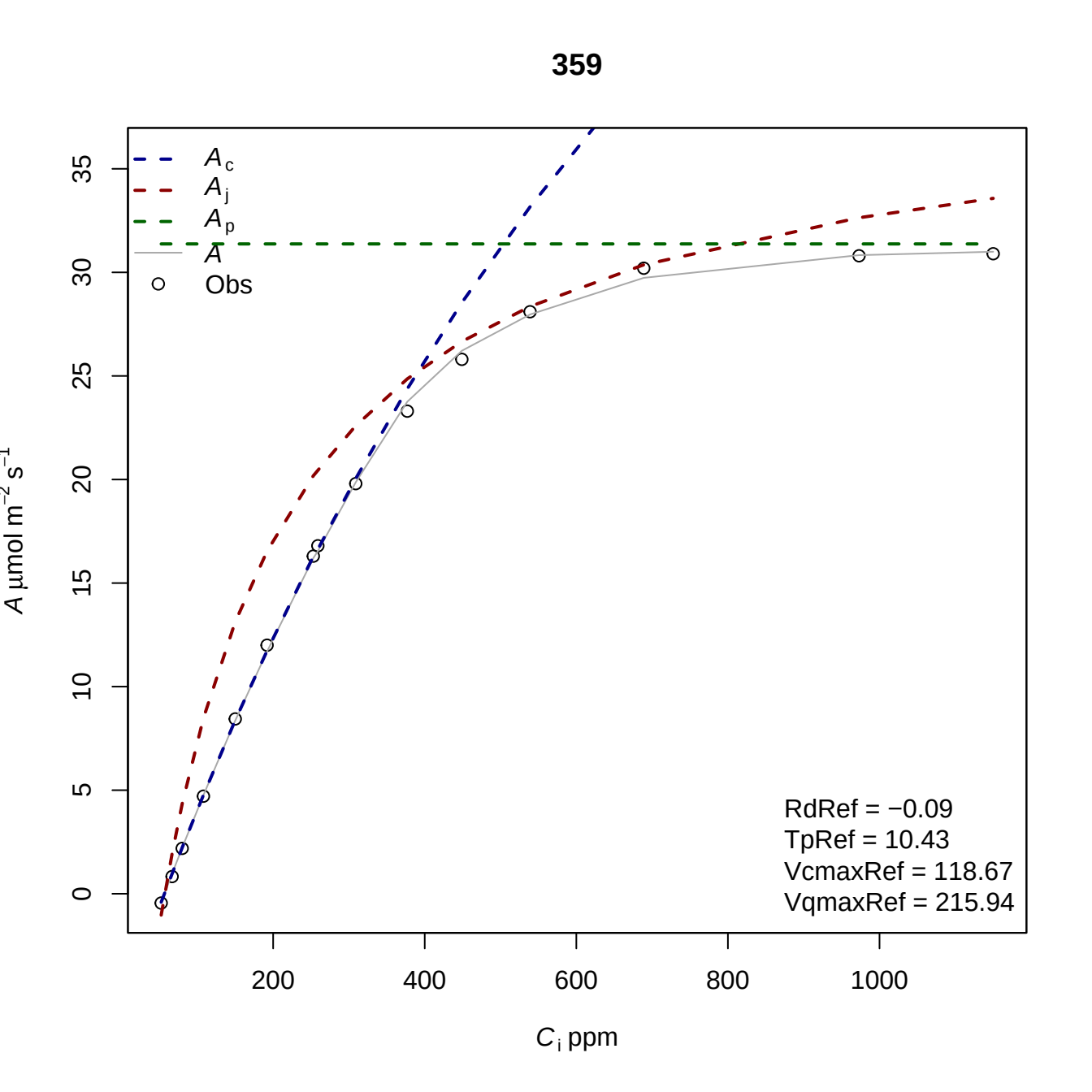

354
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.25
TpRef = 9999
VcmaxRef = 63.96
VqmaxRef = 86.32
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

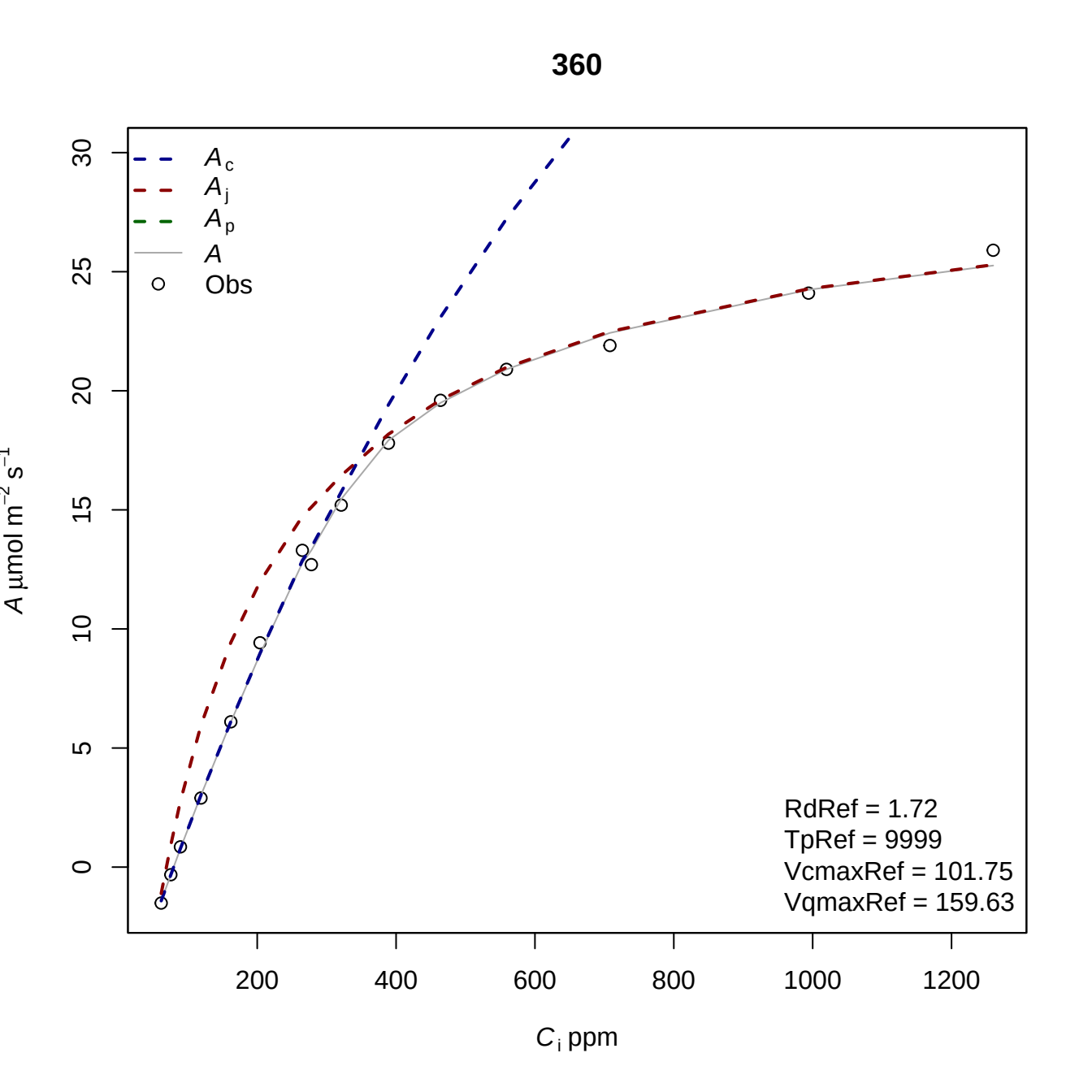

355
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.7
TpRef = 7
VcmaxRef = 83.05
VqmaxRef = 119.38
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

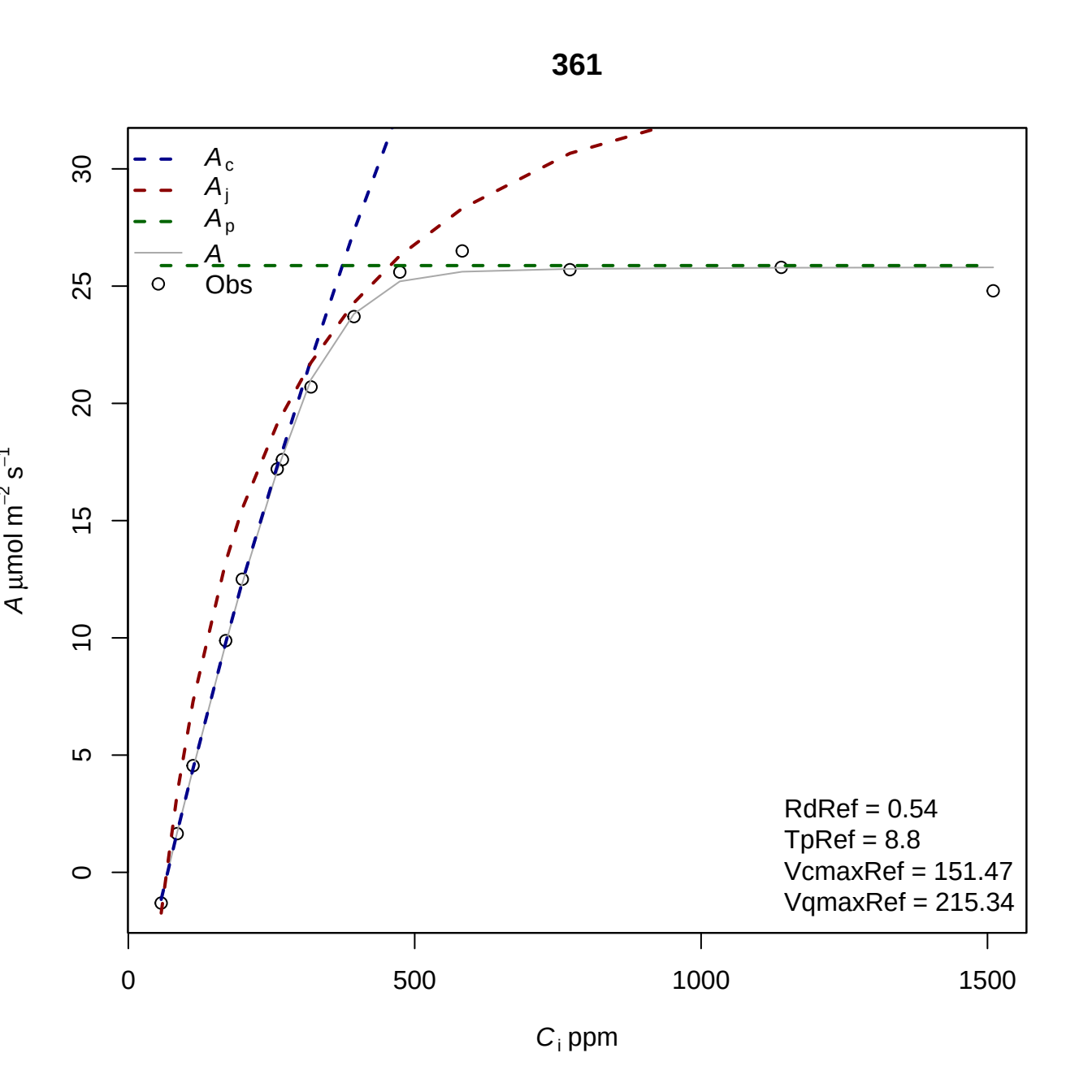

356
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.66
TpRef = 9.02
VcmaxRef = 90.52
VqmaxRef = 163.4
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200
1400

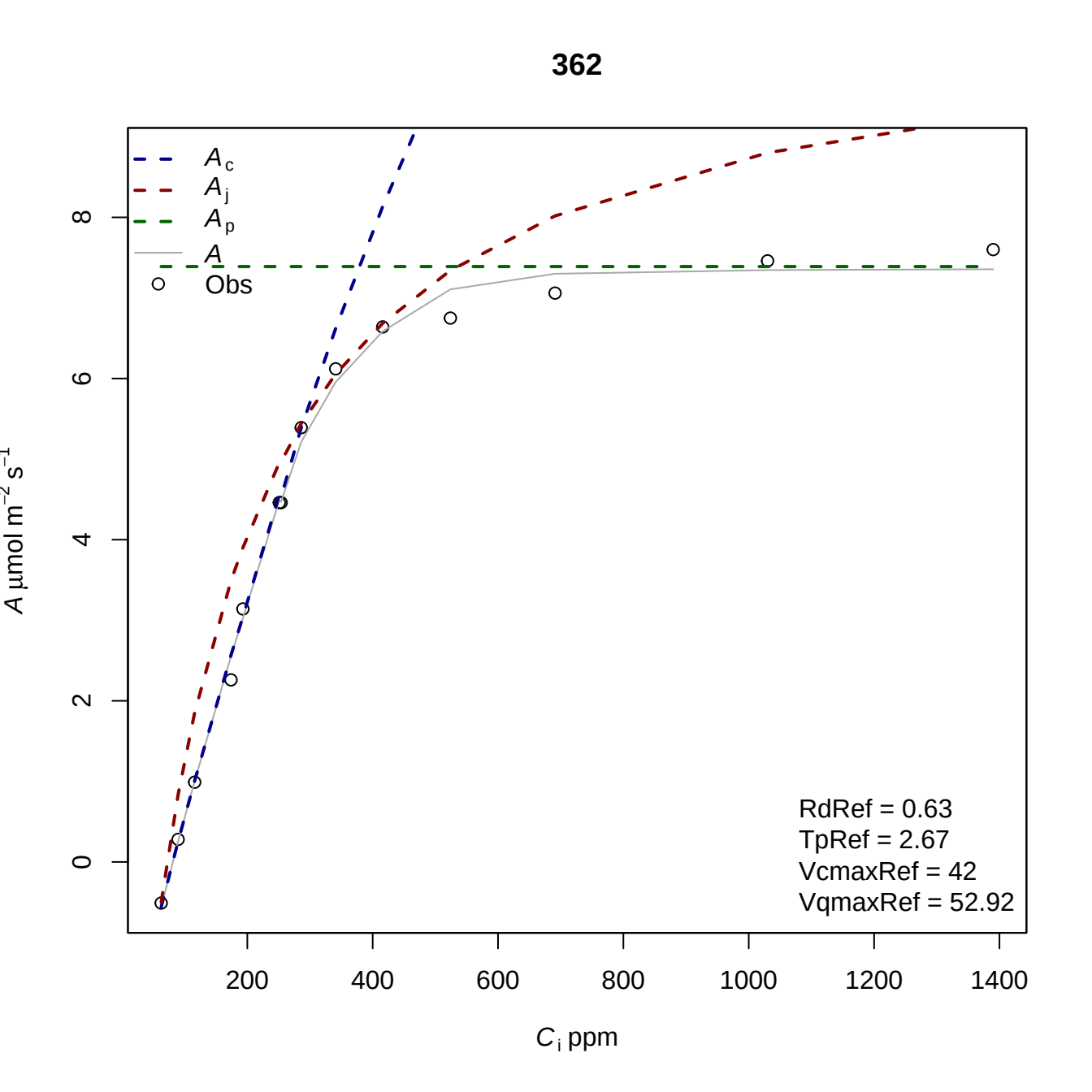

357
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.4
TpRef = 7.93
VcmaxRef = 76.95
VqmaxRef = 144.1
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

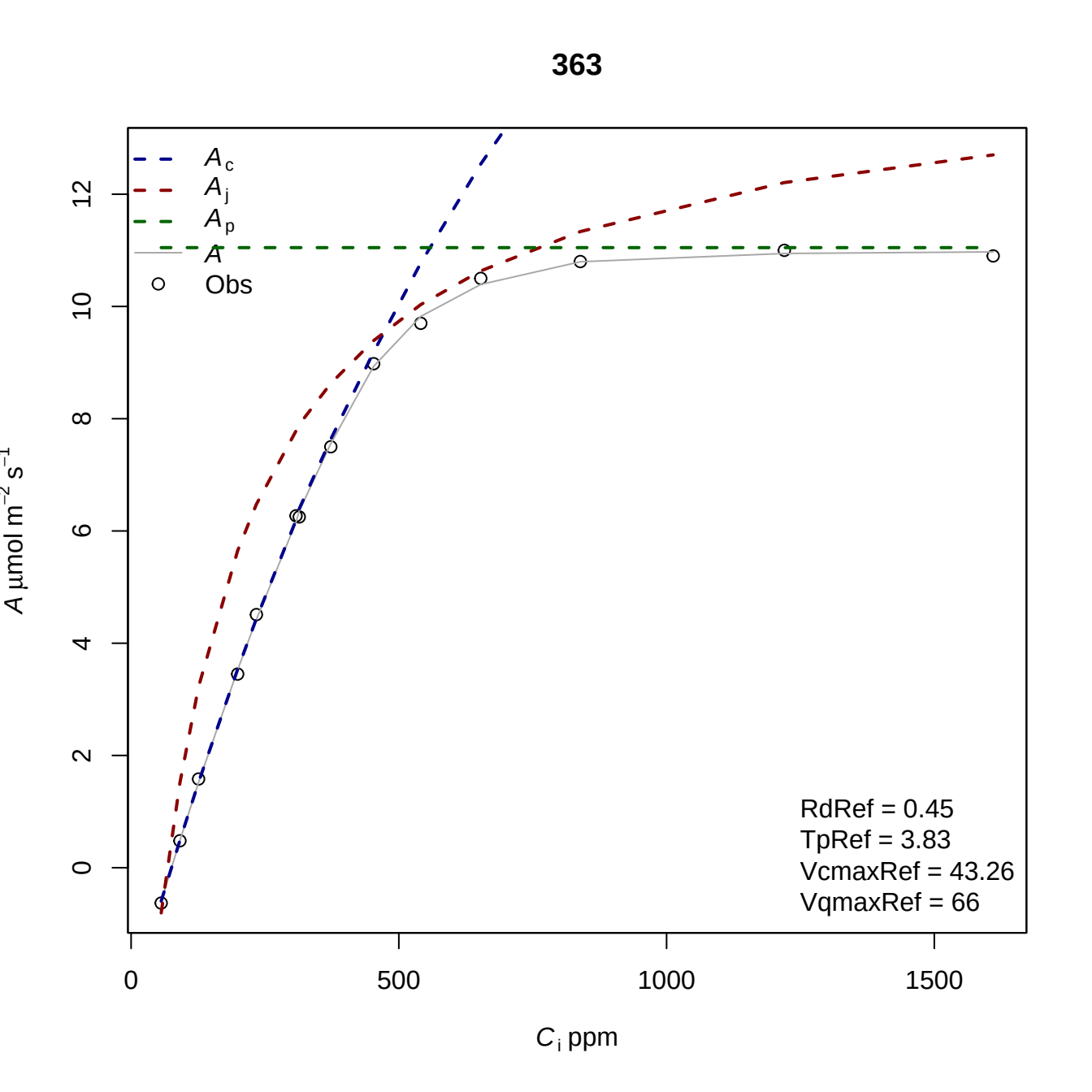

358
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.12
TpRef = 12.46
VcmaxRef = 121.87
VqmaxRef = 256.63
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200
1400

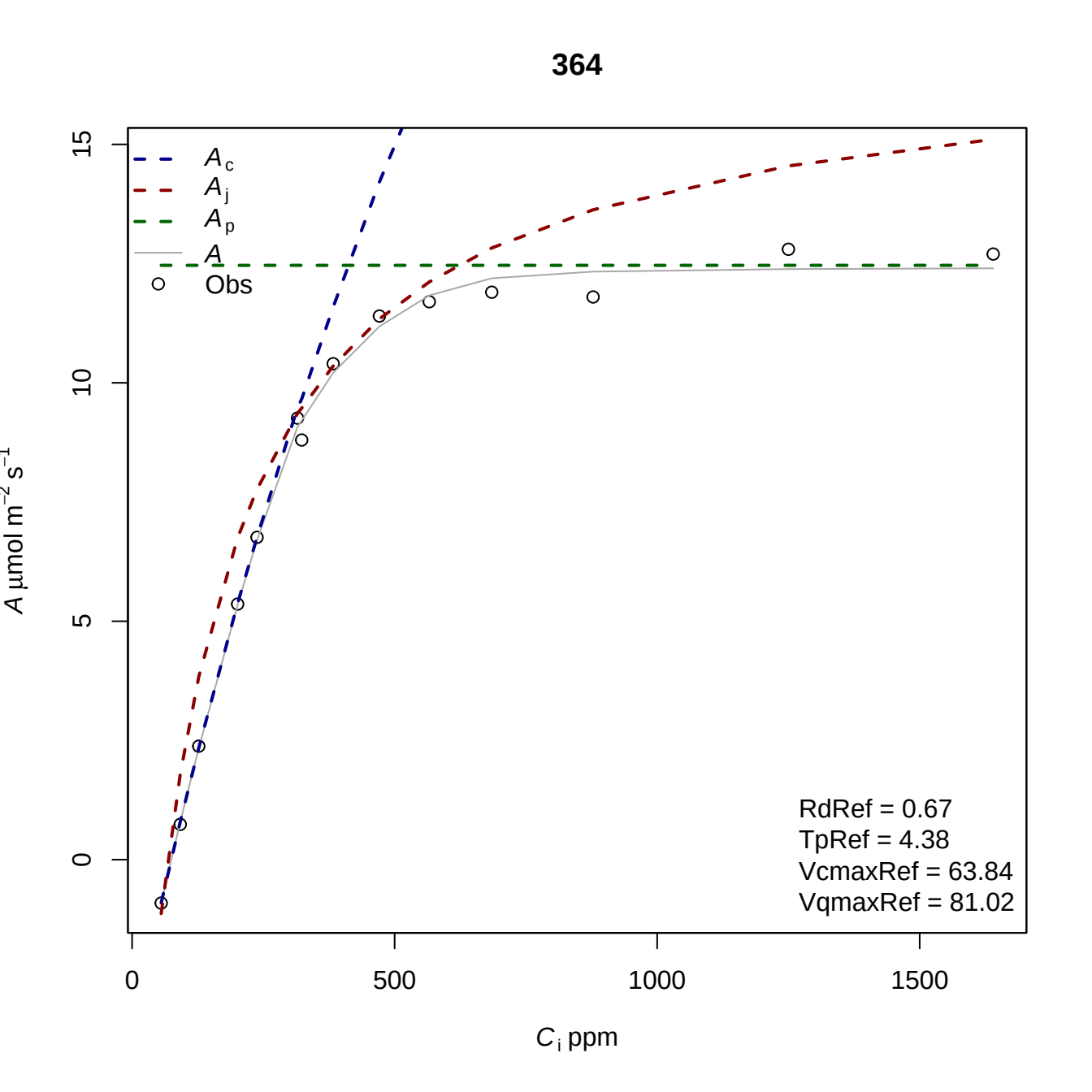

359
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.09
TpRef = 10.43
VcmaxRef = 118.67
VqmaxRef = 215.94
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
200
400
600
800
1000

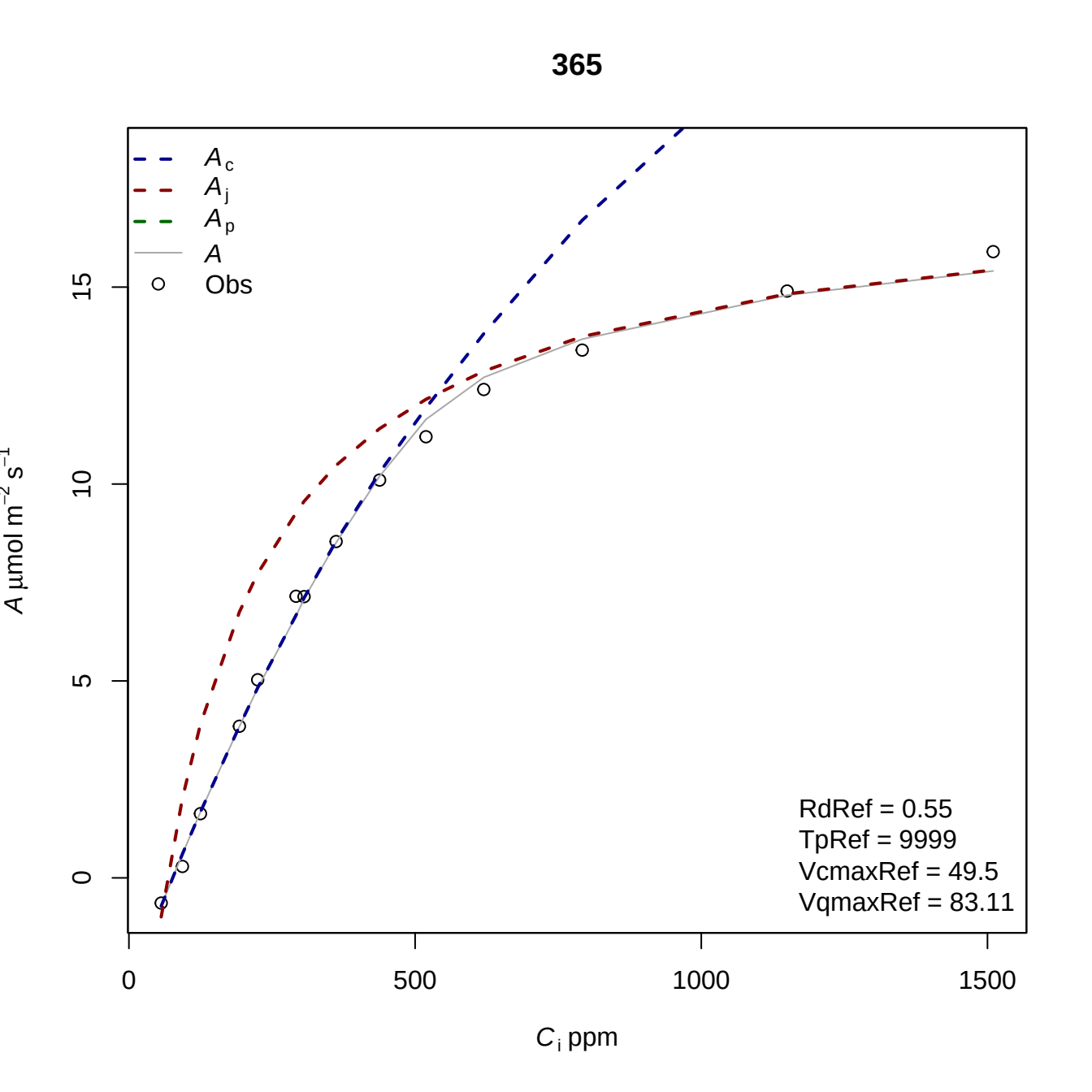

360
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.72
TpRef = 9999
VcmaxRef = 101.75
VqmaxRef = 159.63
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

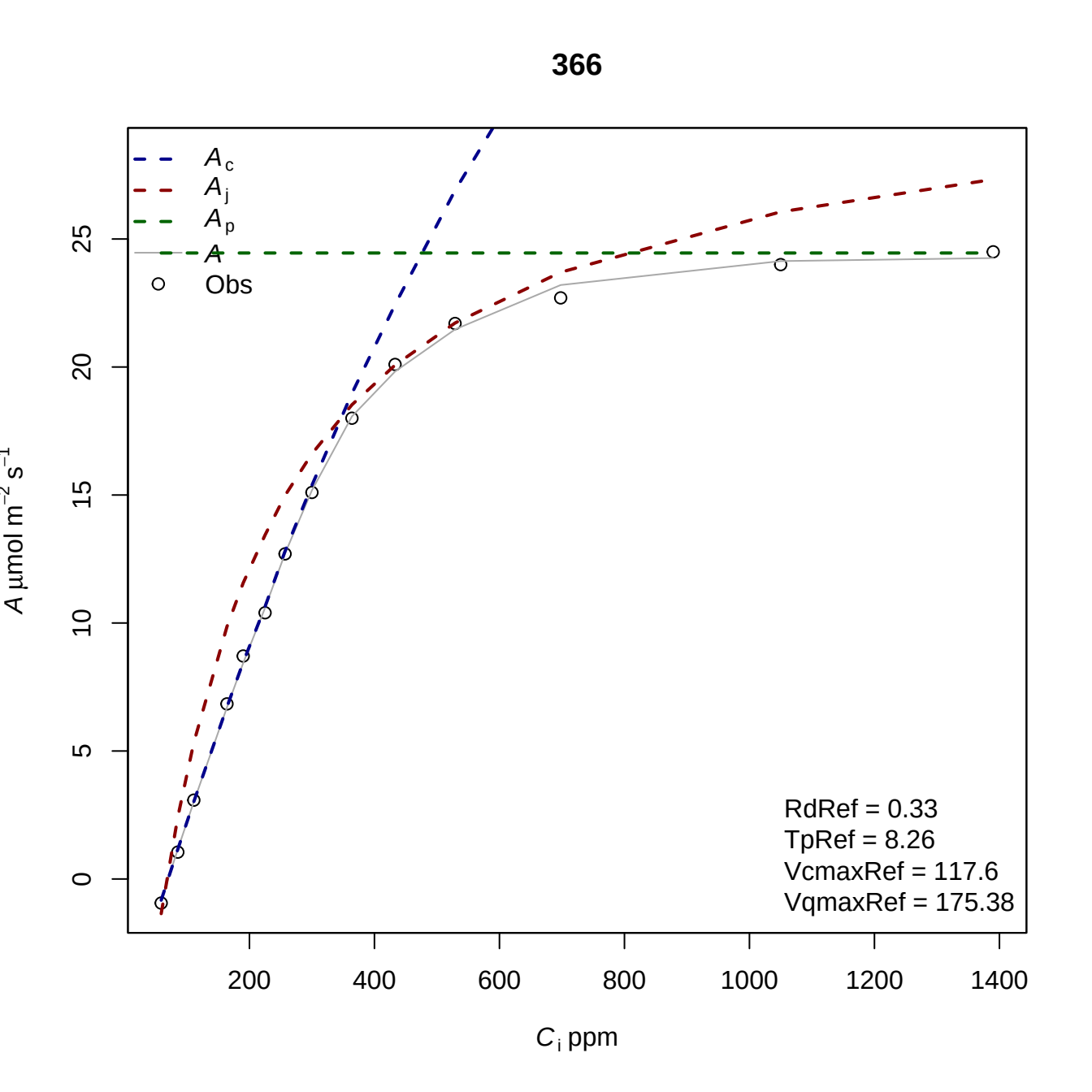

361
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.54
TpRef = 8.8
VcmaxRef = 151.47
VqmaxRef = 215.34
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

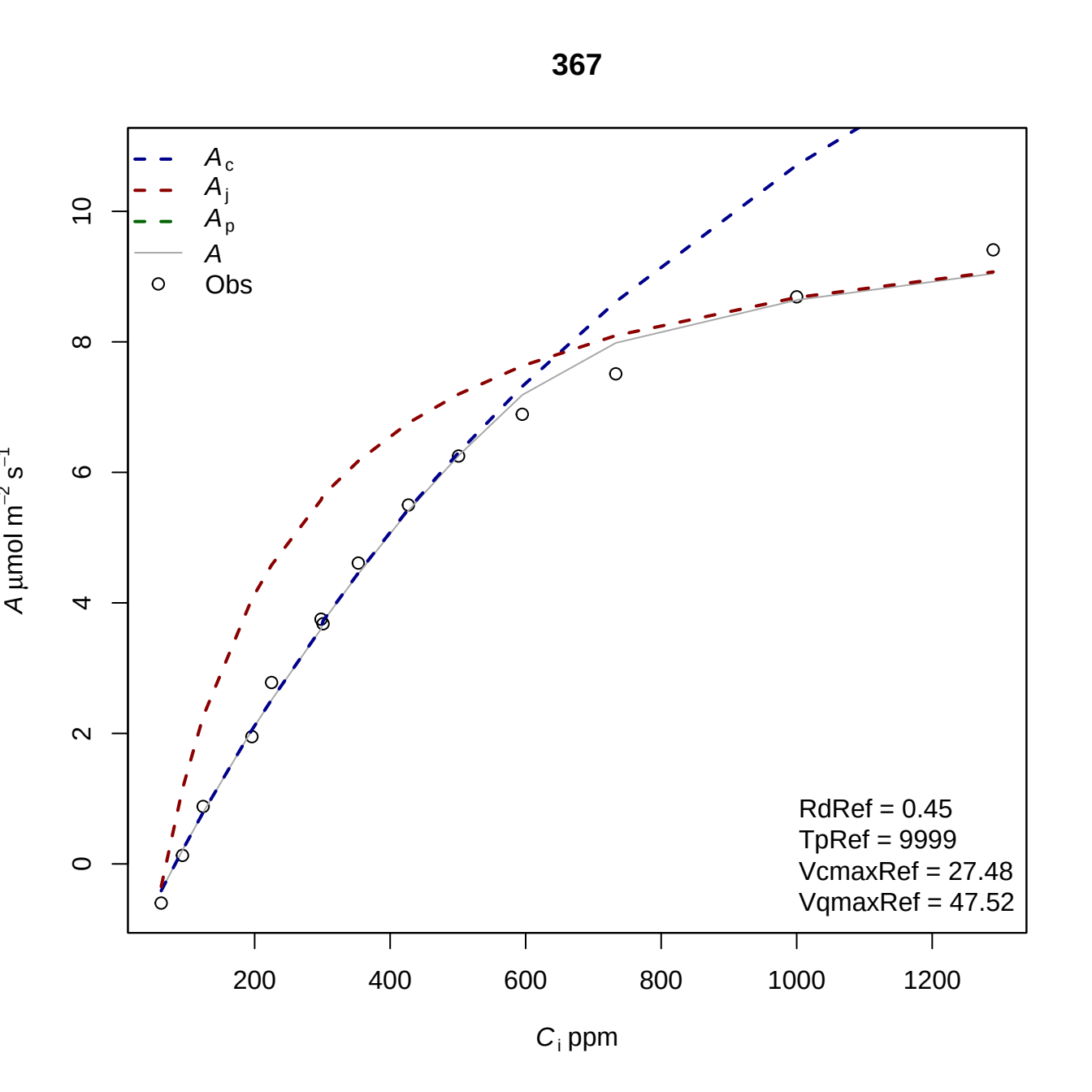

362
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.63
TpRef = 2.67
VcmaxRef = 42
VqmaxRef = 52.92
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
200
400
600
800
1000
1200
1400

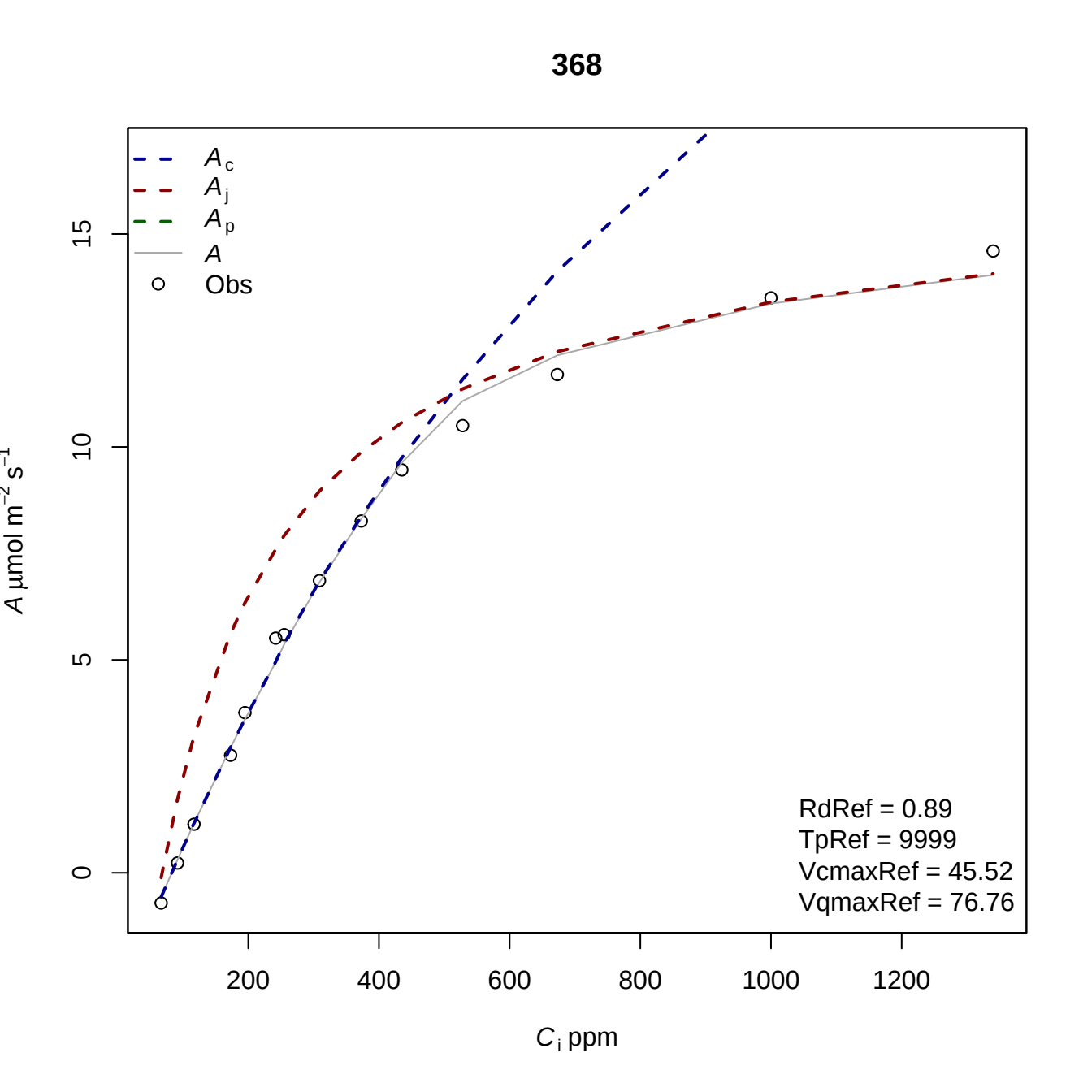

363
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.45
TpRef = 3.83
VcmaxRef = 43.26
VqmaxRef = 66
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

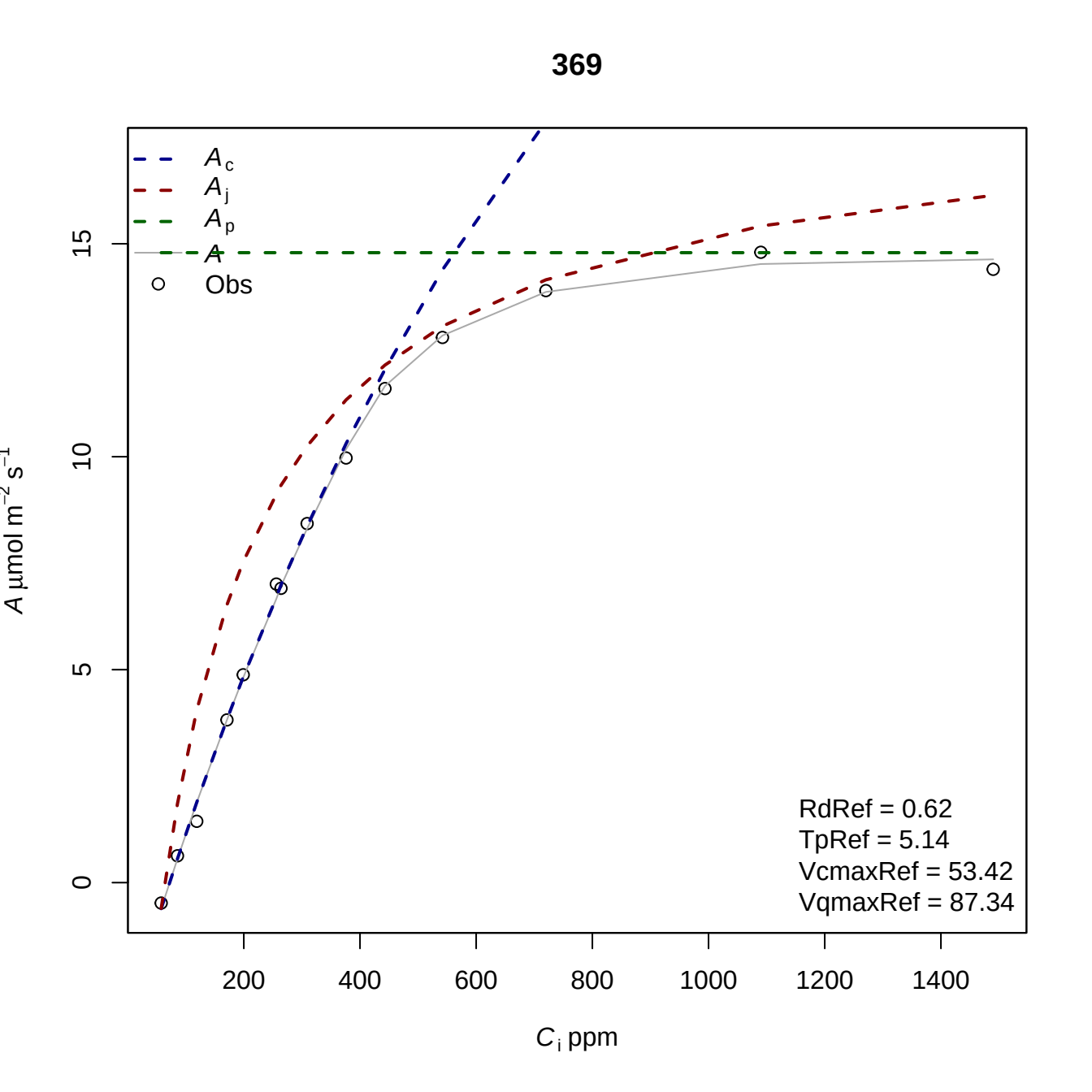

364
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.67
TpRef = 4.38
VcmaxRef = 63.84
VqmaxRef = 81.02
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

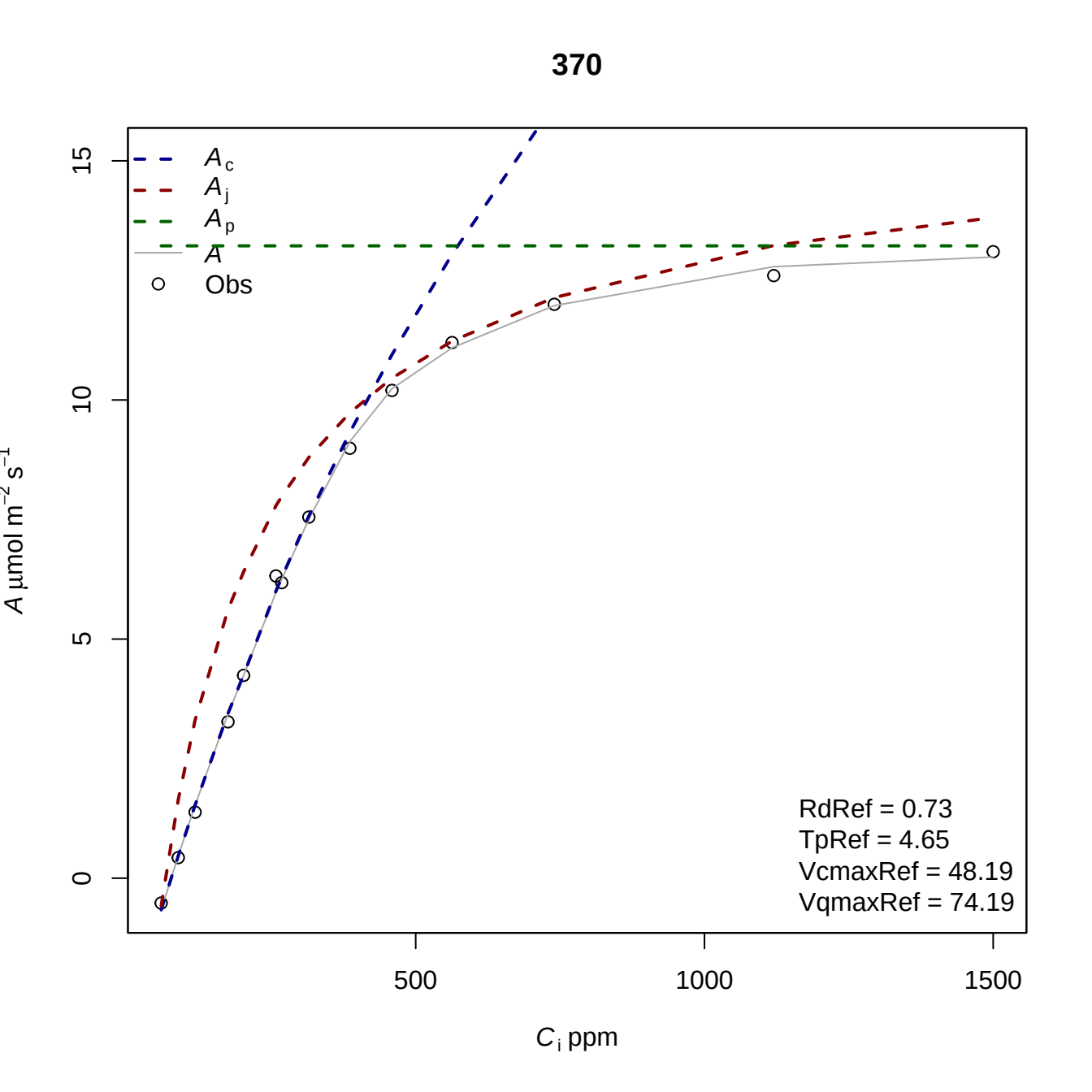

365
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.55
TpRef = 9999
VcmaxRef = 49.5
VqmaxRef = 83.11
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

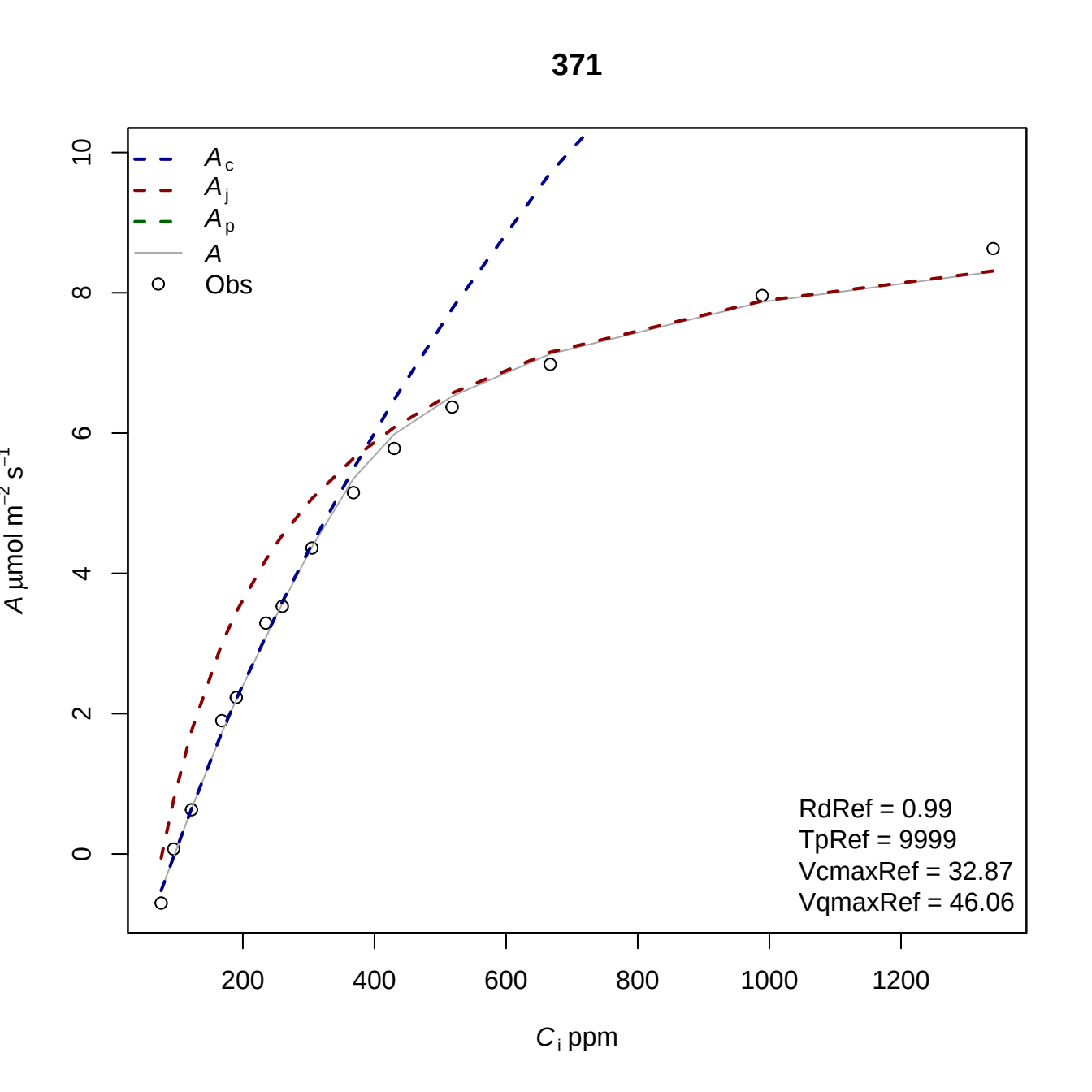

366
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.33
TpRef = 8.26
VcmaxRef = 117.6
VqmaxRef = 175.38
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200
1400

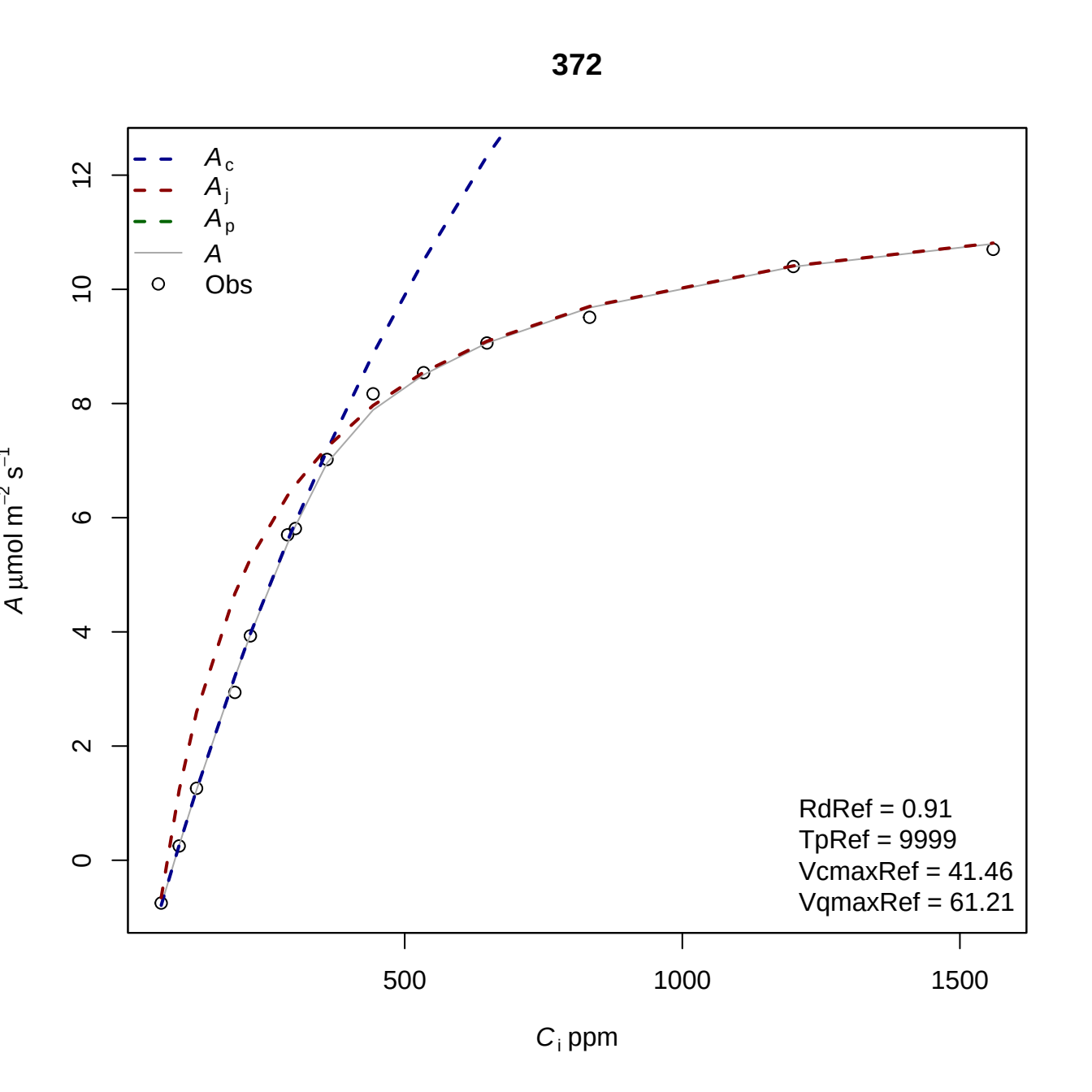

367
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.45
TpRef = 9999
VcmaxRef = 27.48
VqmaxRef = 47.52
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
200
400
600
800
1000
1200

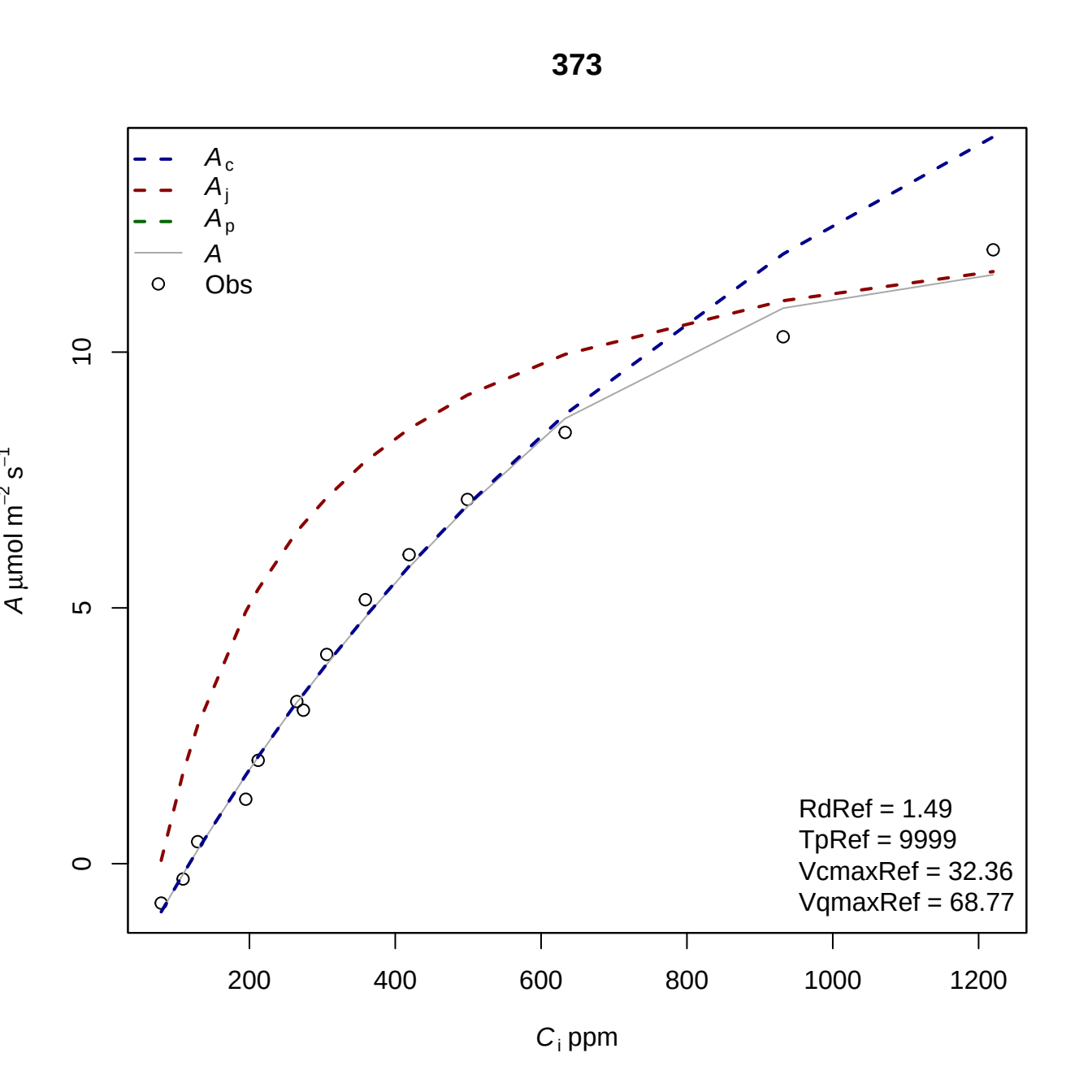

368
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.89
TpRef = 9999
VcmaxRef = 45.52
VqmaxRef = 76.76
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200

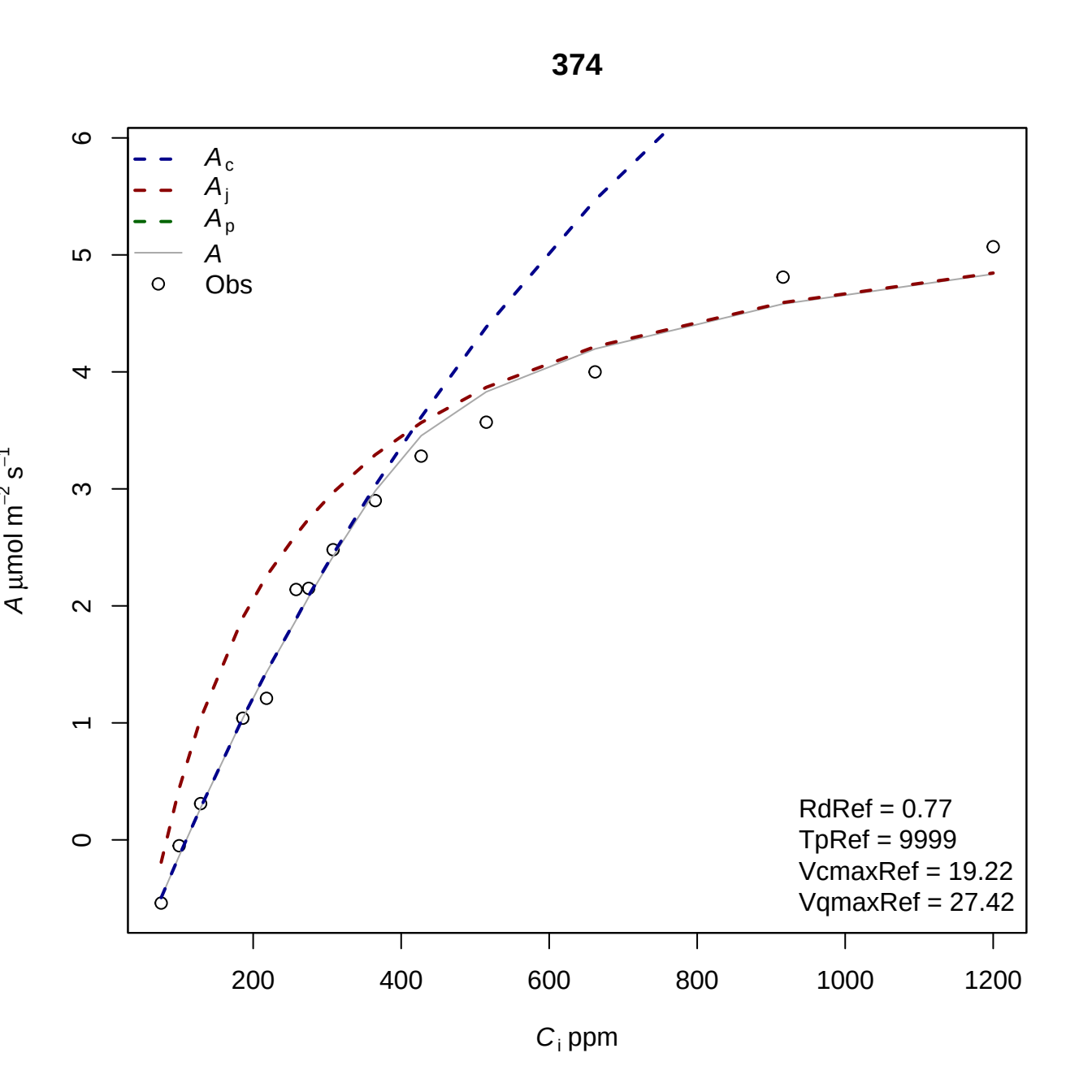

369
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.62
TpRef = 5.14
VcmaxRef = 53.42
VqmaxRef = 87.34
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

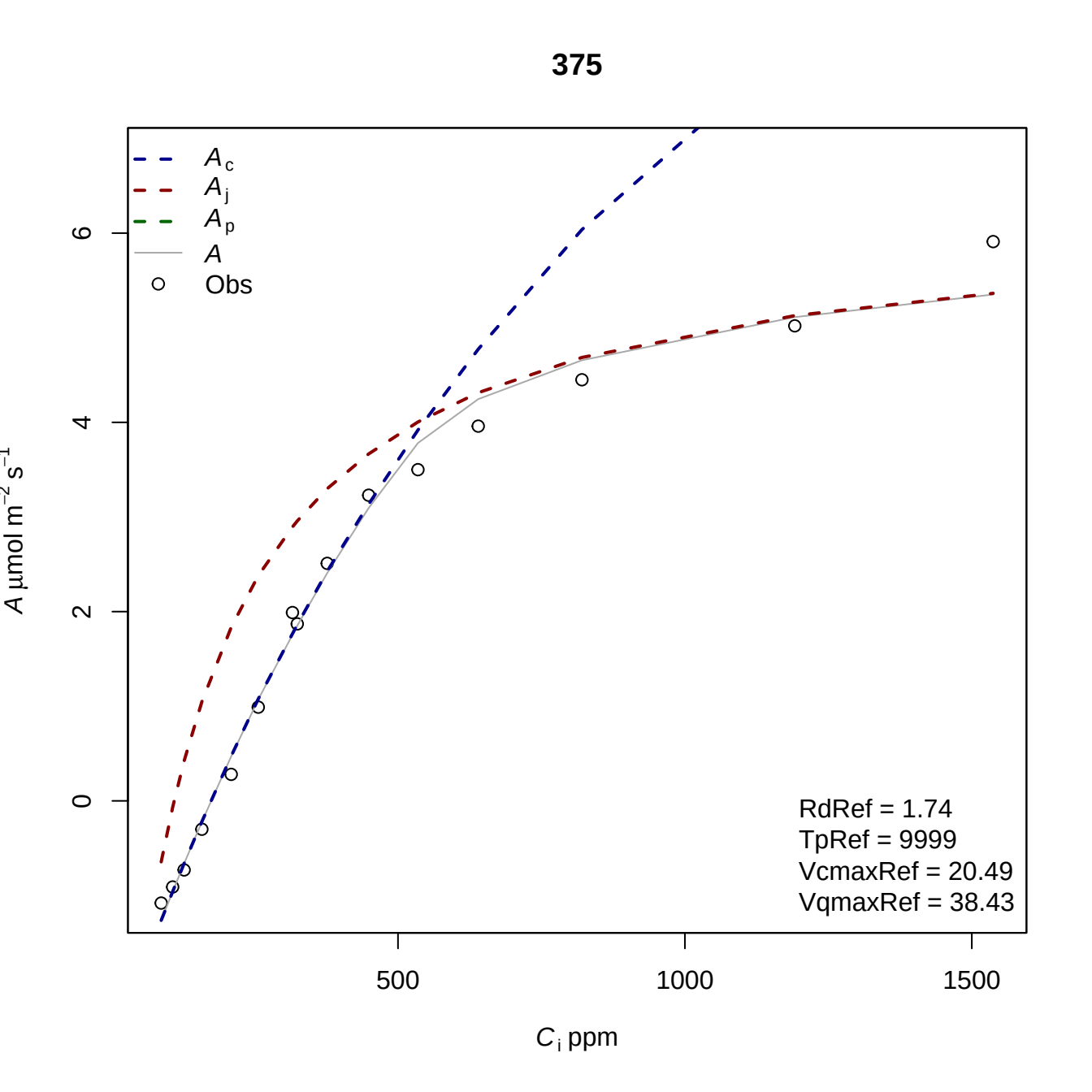

370
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.73
TpRef = 4.65
VcmaxRef = 48.19
VqmaxRef = 74.19
0
5
10
15
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

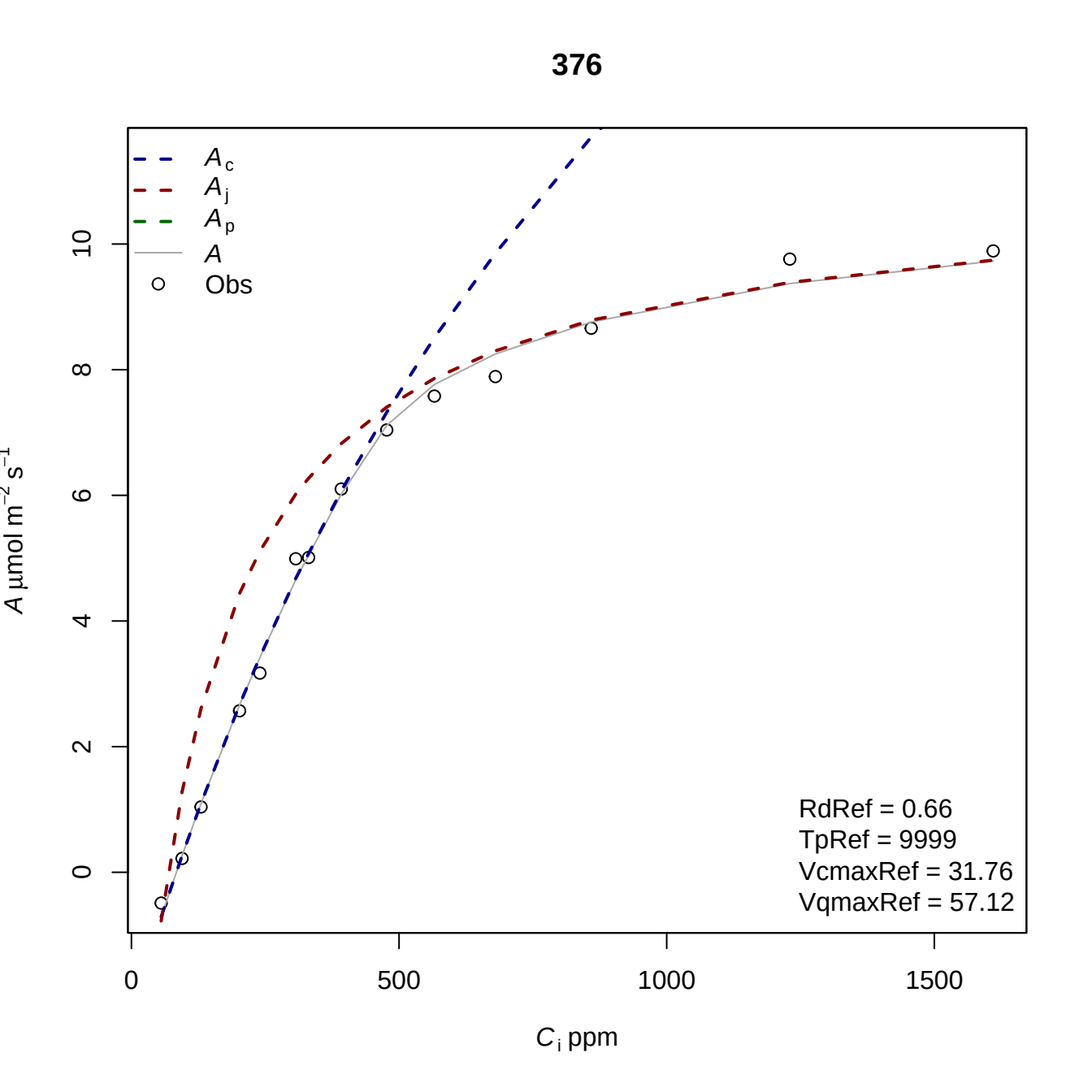

371
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.99
TpRef = 9999
VcmaxRef = 32.87
VqmaxRef = 46.06
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
200
400
600
800
1000
1200

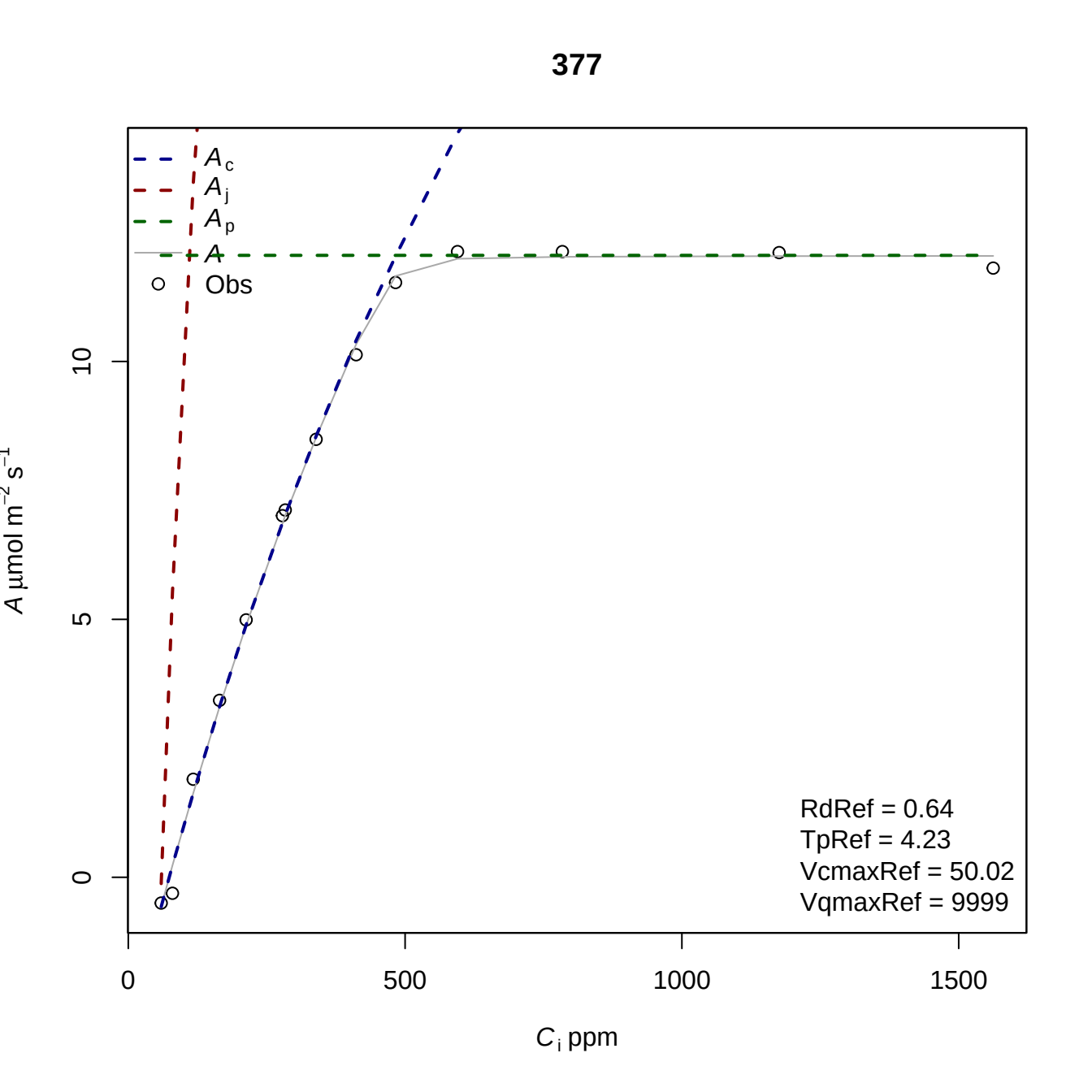

372
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.91
TpRef = 9999
VcmaxRef = 41.46
VqmaxRef = 61.21
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
500
1000
1500

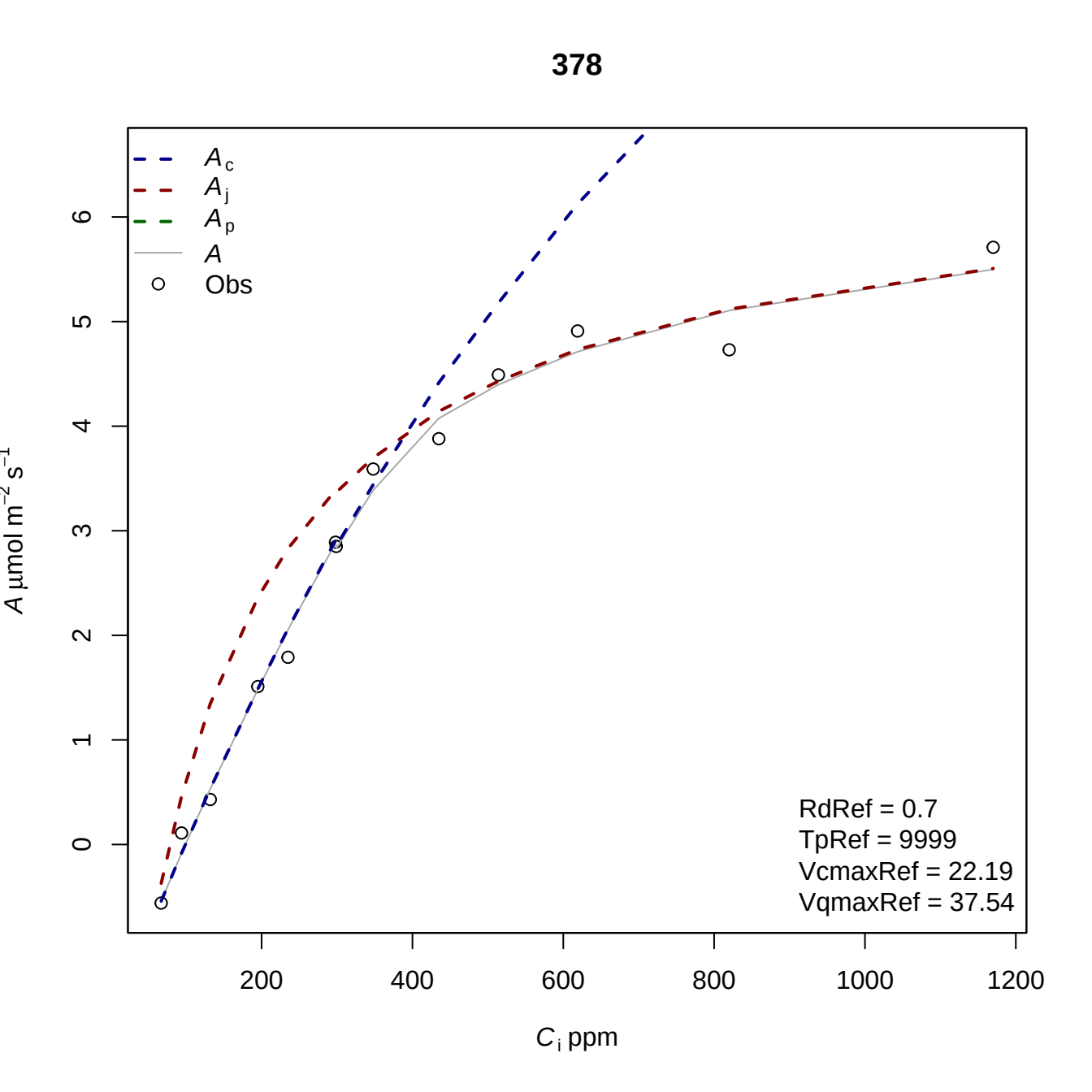

373
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.49
TpRef = 9999
VcmaxRef = 32.36
VqmaxRef = 68.77
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
200
400
600
800
1000
1200

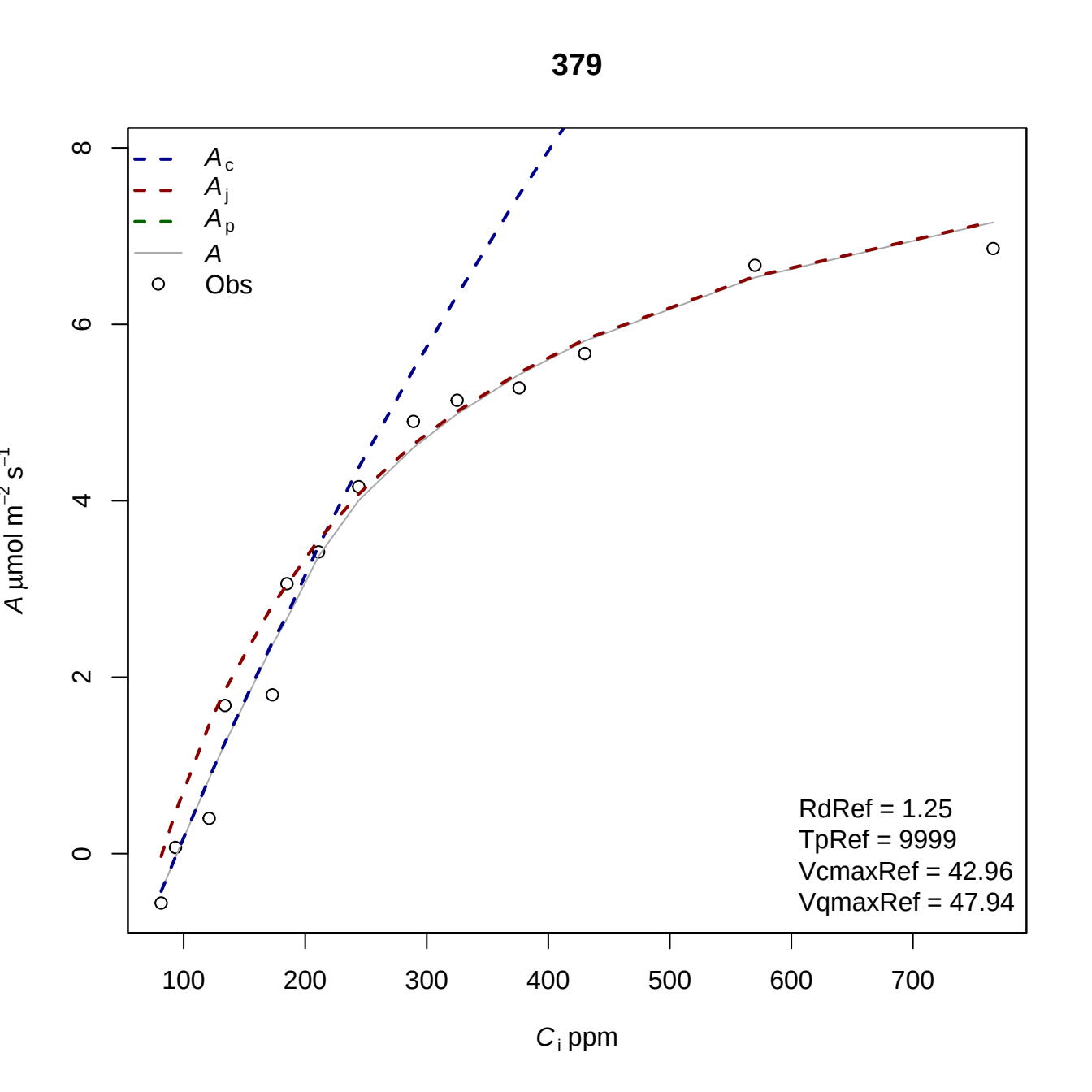

374
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.77
TpRef = 9999
VcmaxRef = 19.22
VqmaxRef = 27.42
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
200
400
600
800
1000
1200

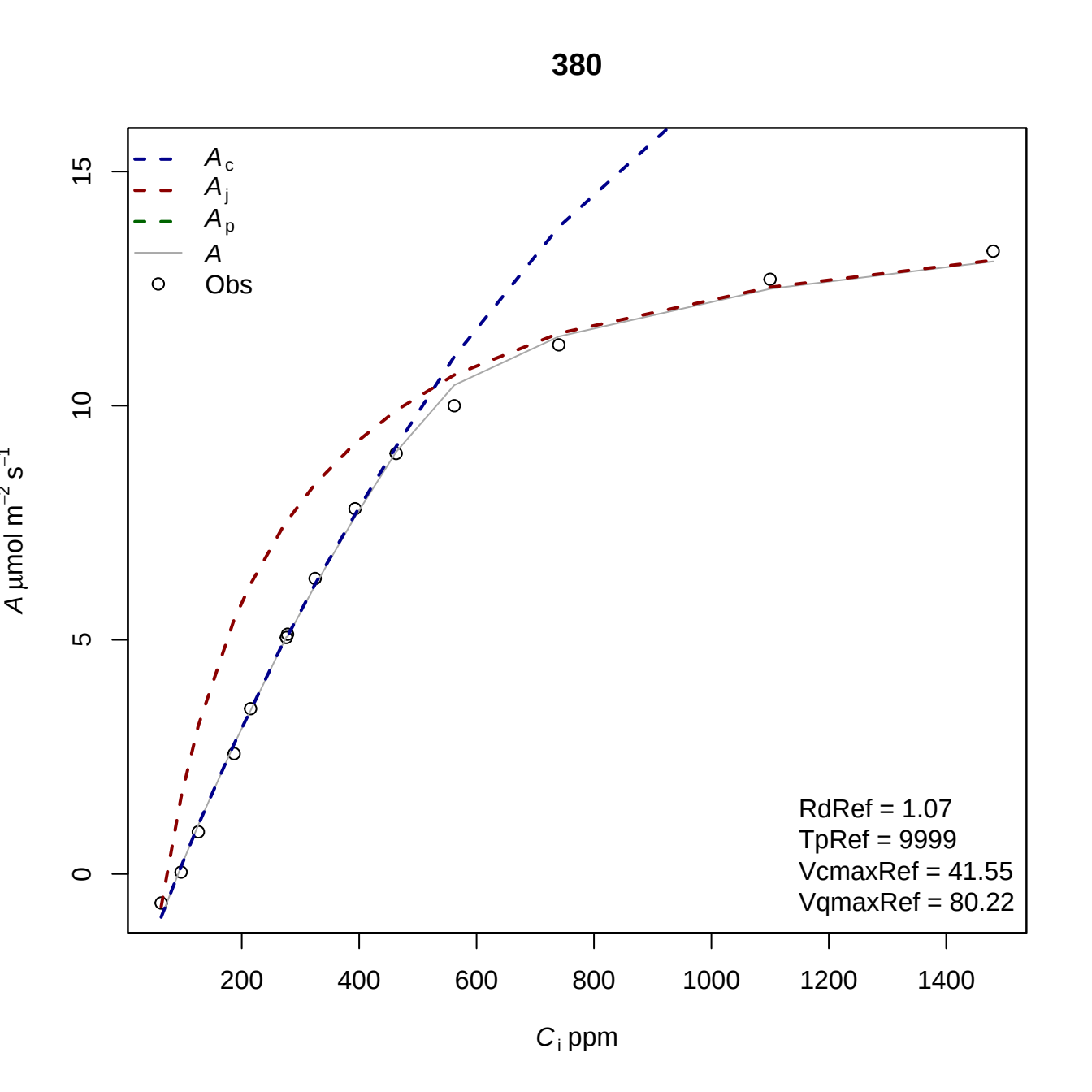

375
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.74
TpRef = 9999
VcmaxRef = 20.49
VqmaxRef = 38.43
0
2
4
6
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

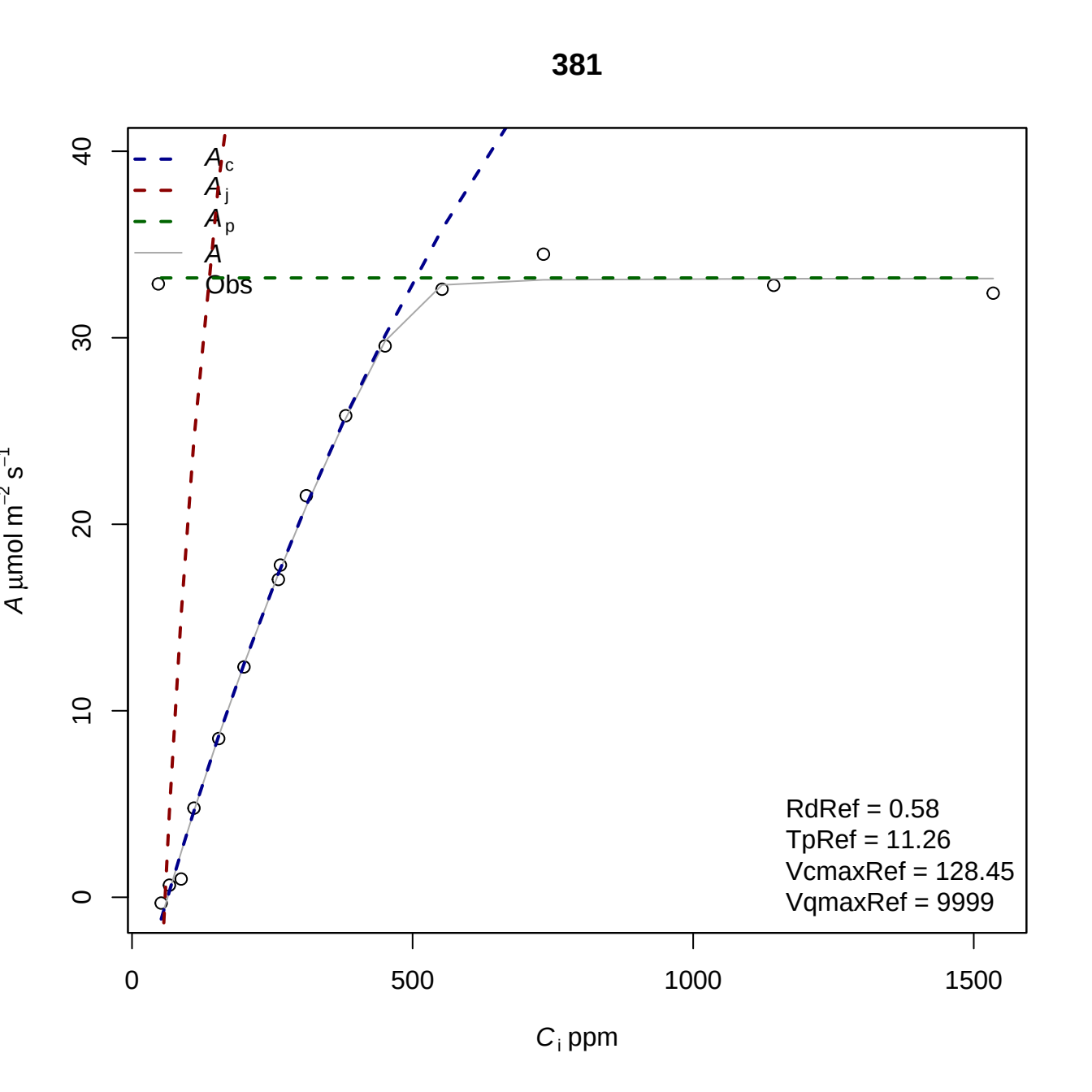

376
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.66
TpRef = 9999
VcmaxRef = 31.76
VqmaxRef = 57.12
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

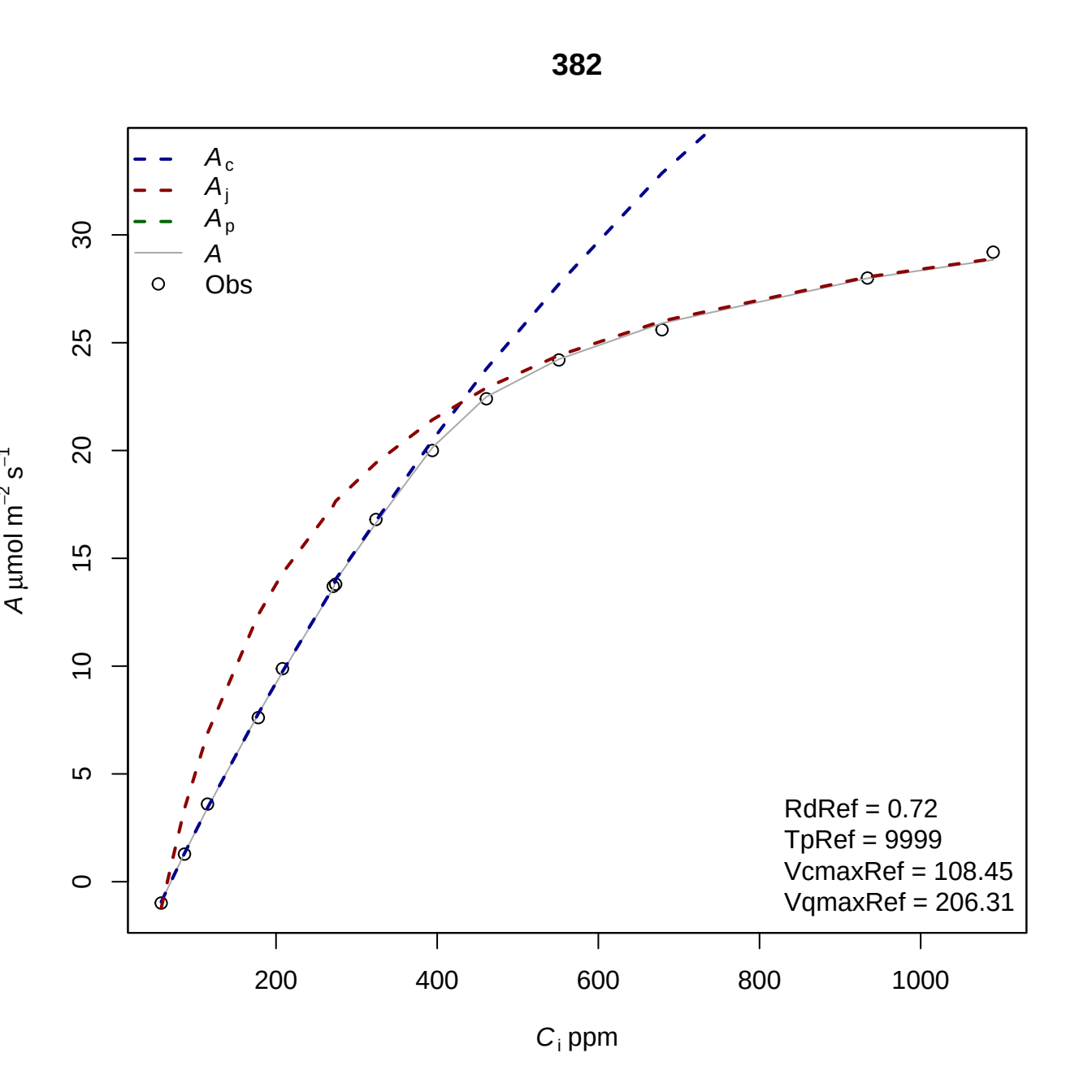

377
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.64
TpRef = 4.23
VcmaxRef = 50.02
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
0
500
1000
1500

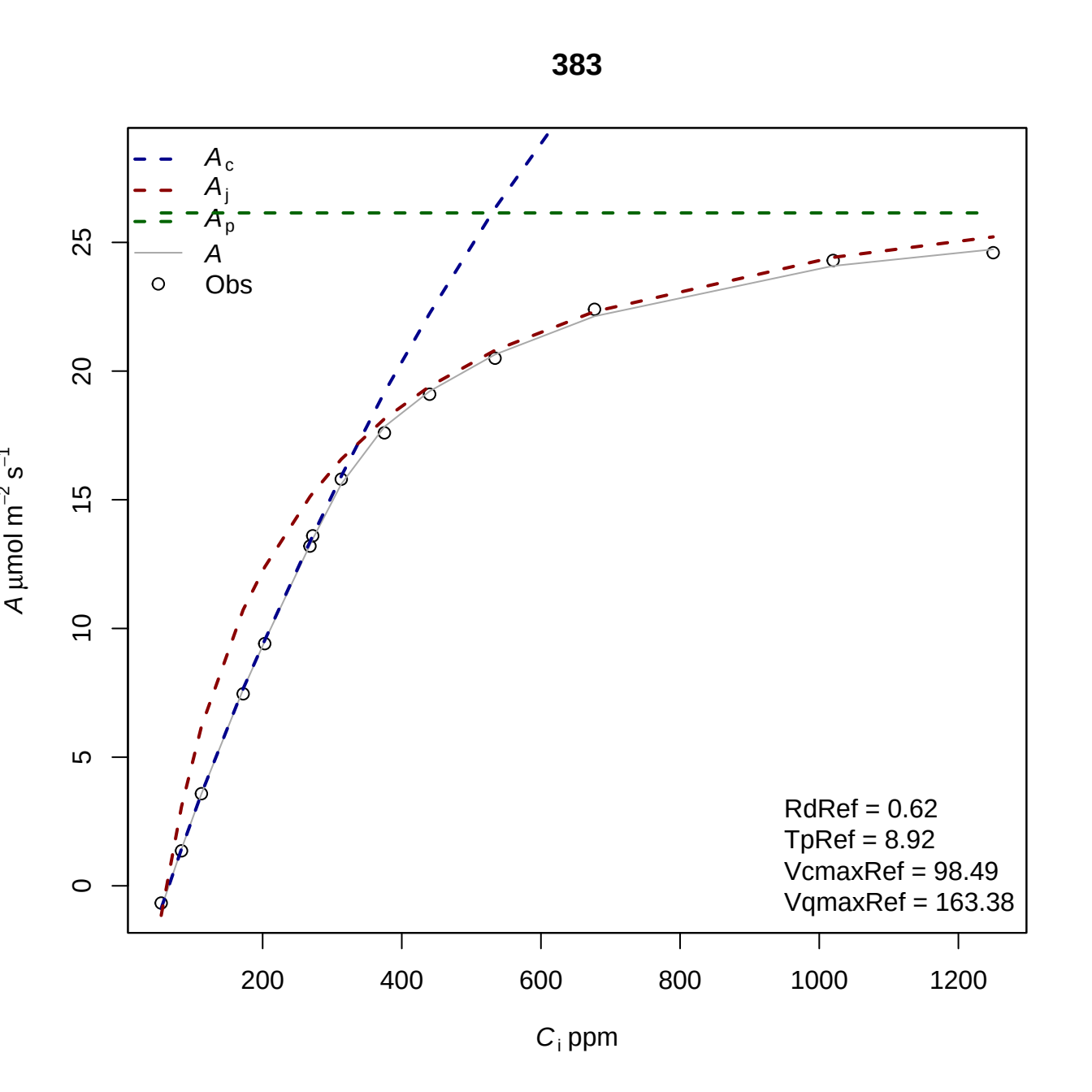

378
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.7
TpRef = 9999
VcmaxRef = 22.19
VqmaxRef = 37.54
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
200
400
600
800
1000
1200

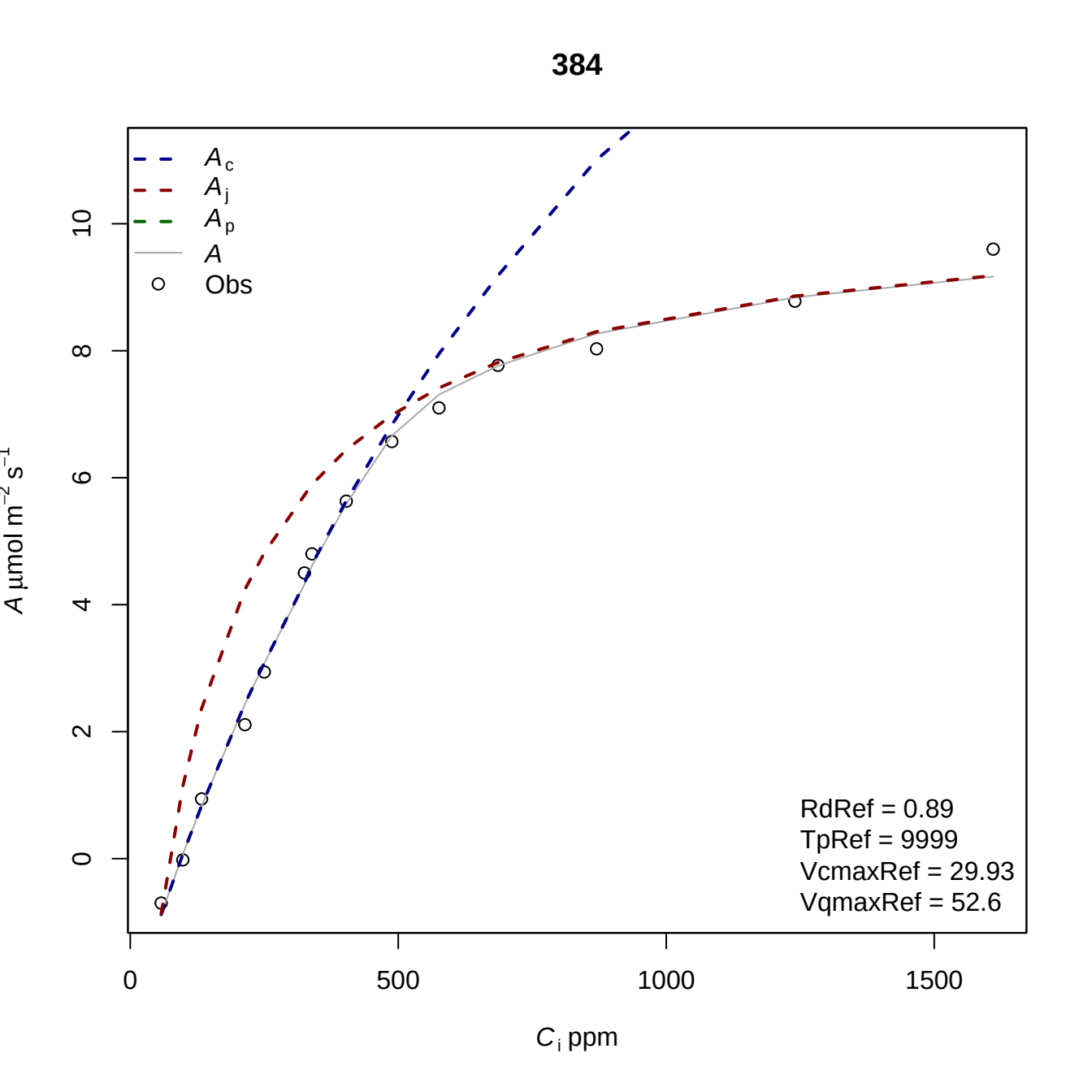

379
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.25
TpRef = 9999
VcmaxRef = 42.96
VqmaxRef = 47.94
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
100
200
300
400
500
600
700

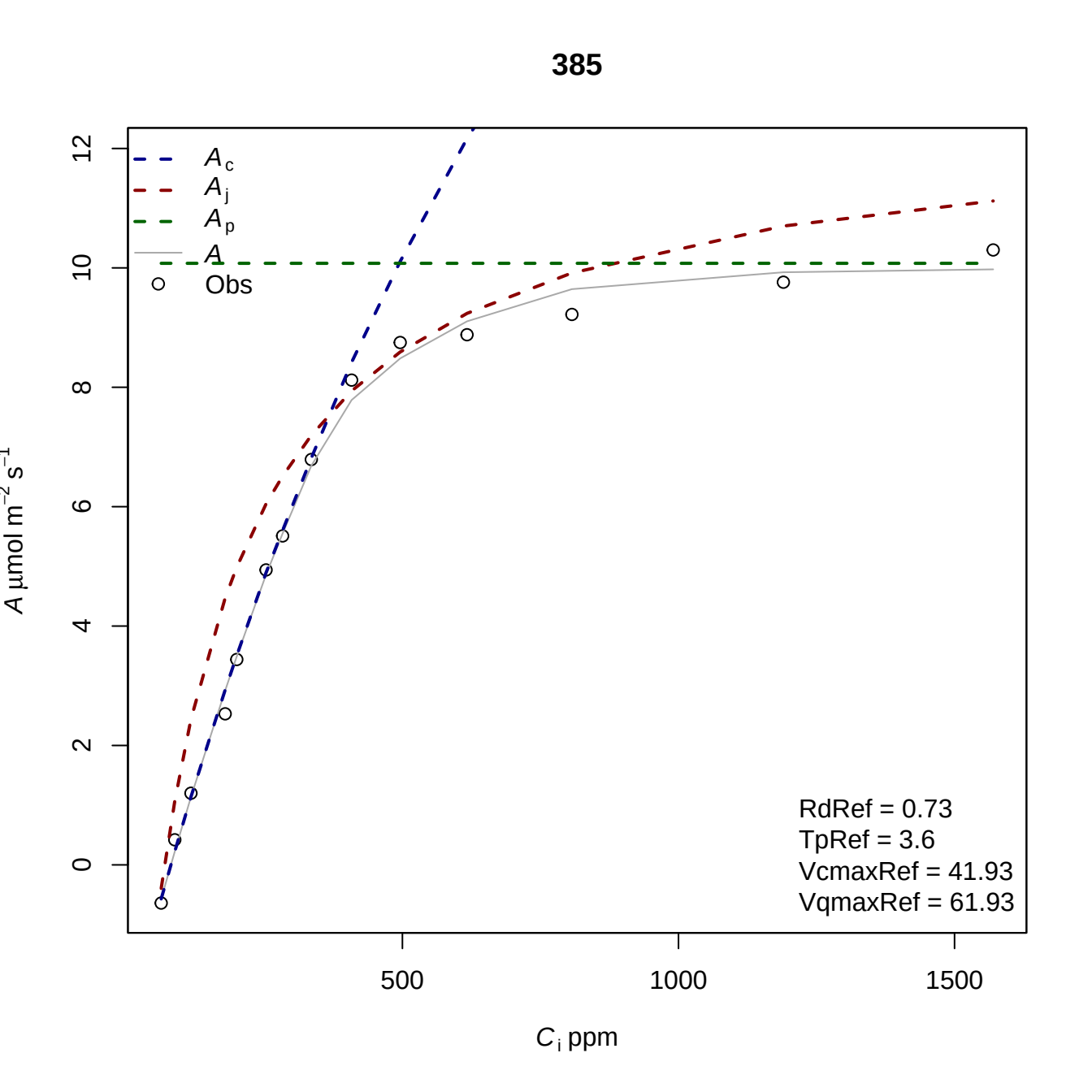

380
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.07
TpRef = 9999
VcmaxRef = 41.55
VqmaxRef = 80.22
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

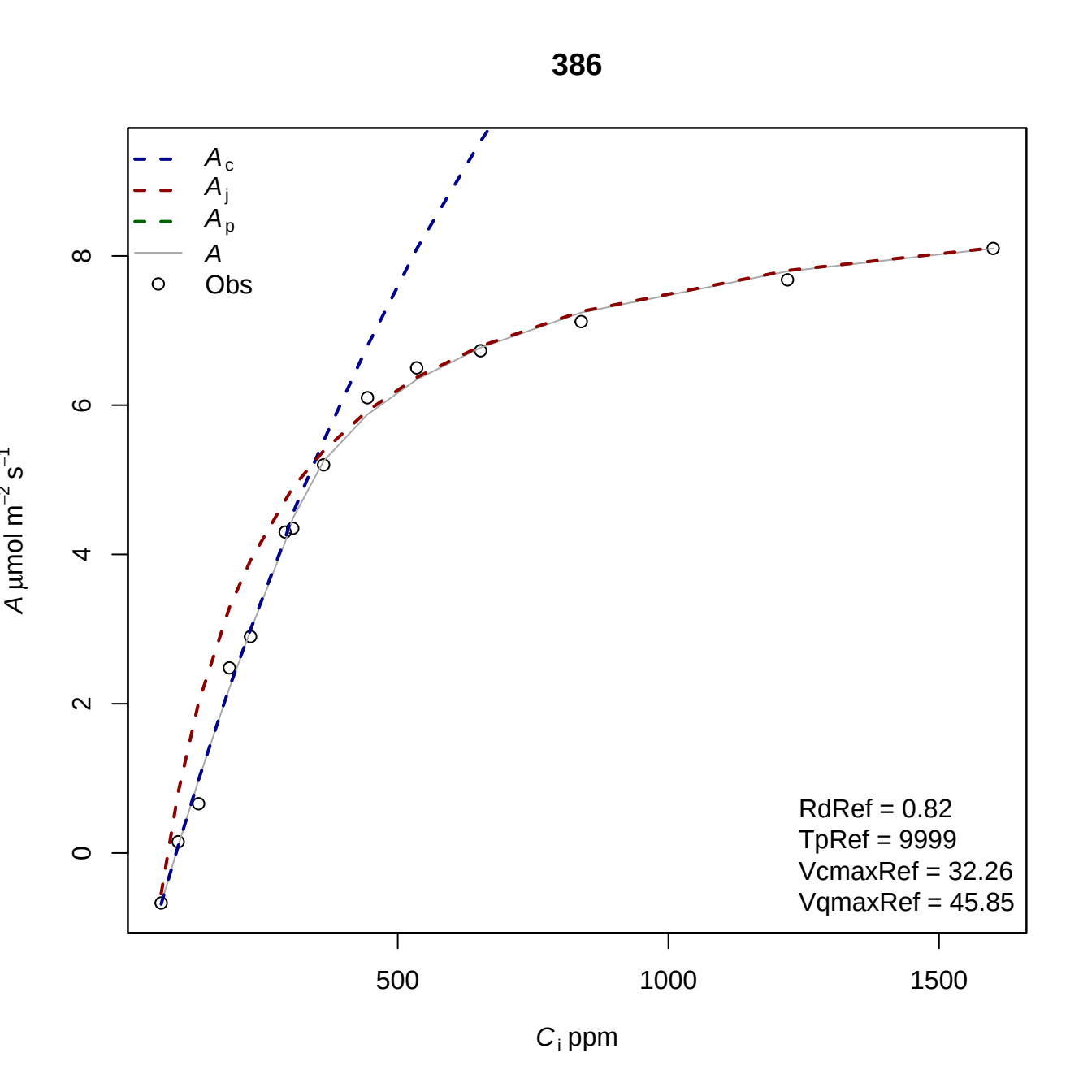

381
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.58
TpRef = 11.26
VcmaxRef = 128.45
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
500
1000
1500

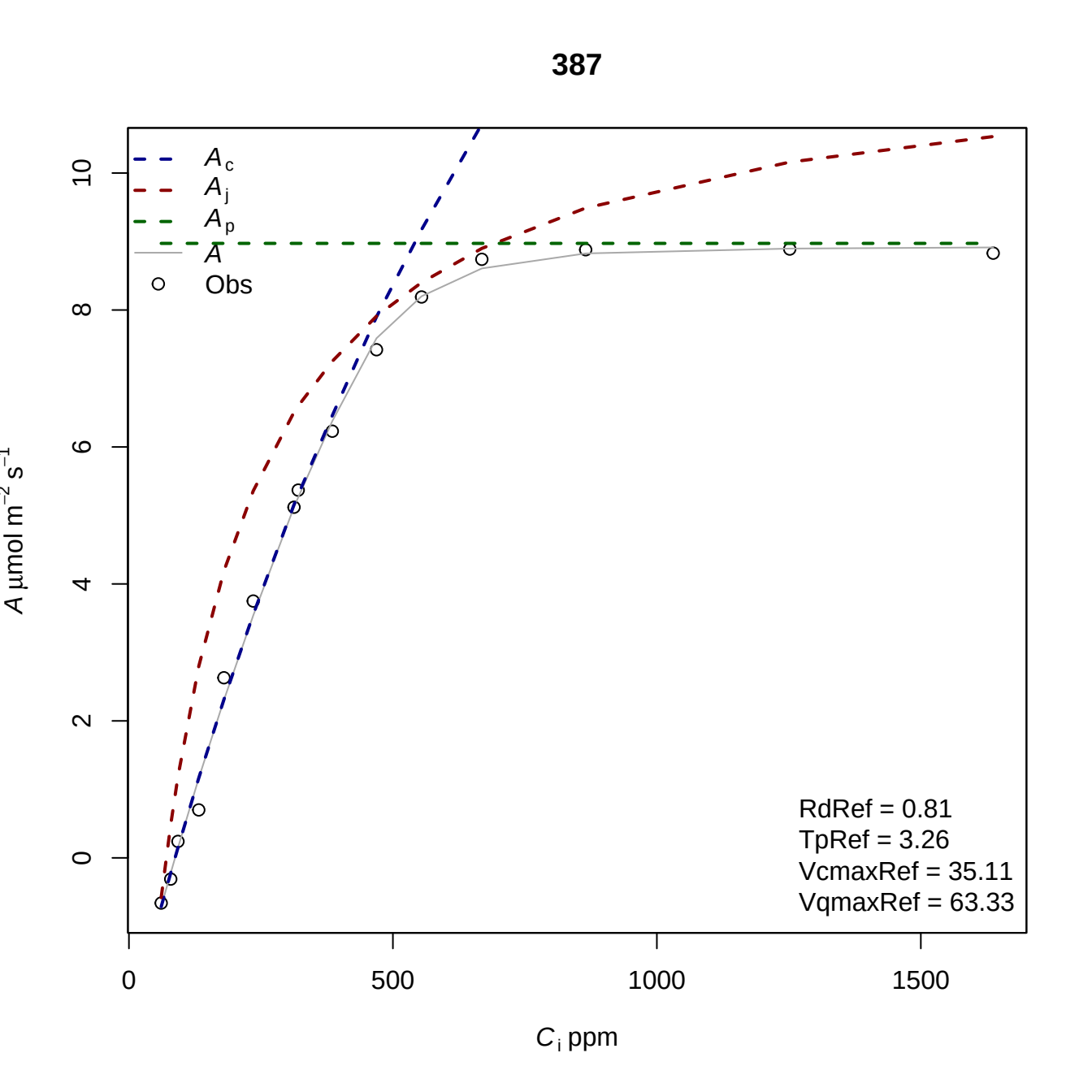

382
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.72
TpRef = 9999
VcmaxRef = 108.45
VqmaxRef = 206.31
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000

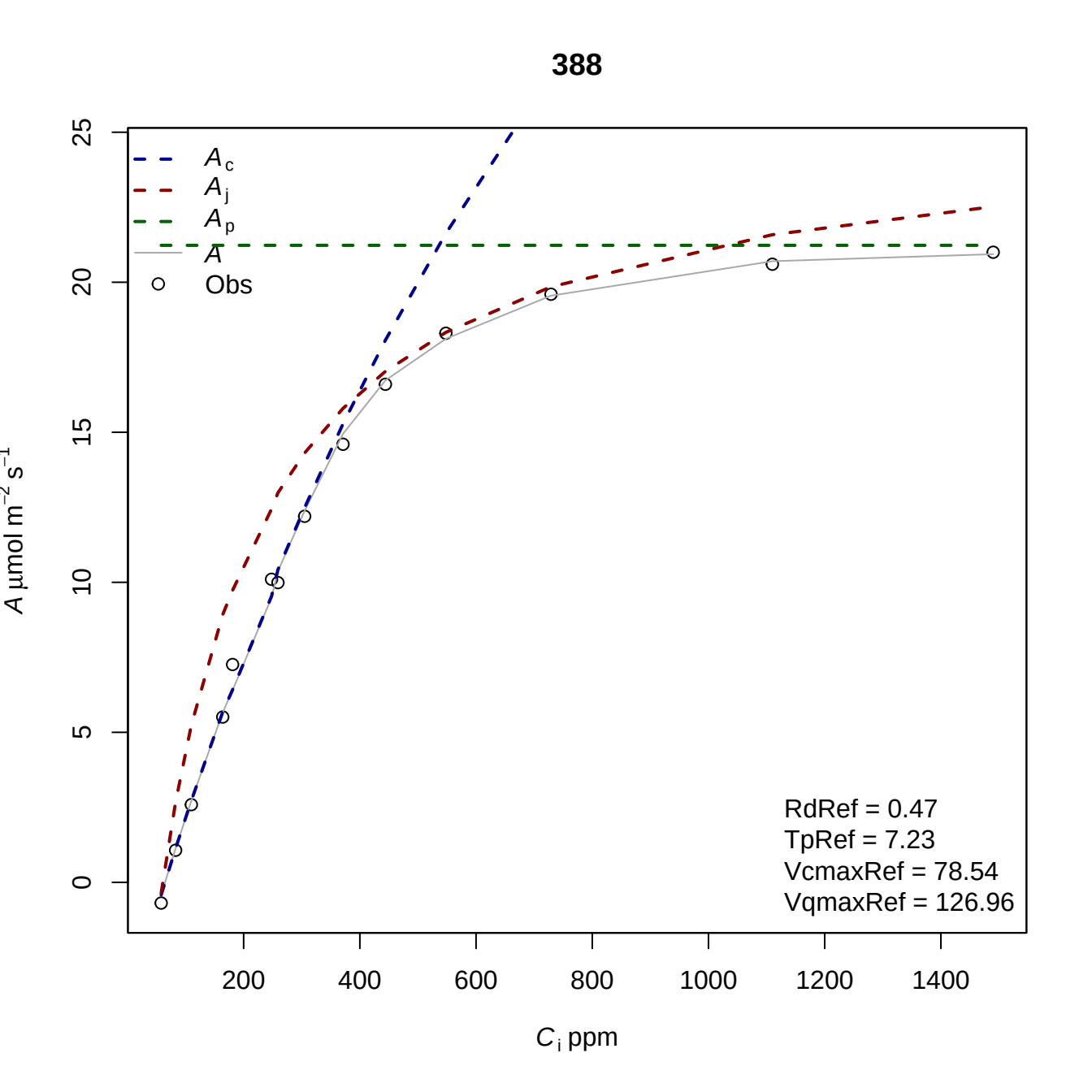

383
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.62
TpRef = 8.92
VcmaxRef = 98.49
VqmaxRef = 163.38
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

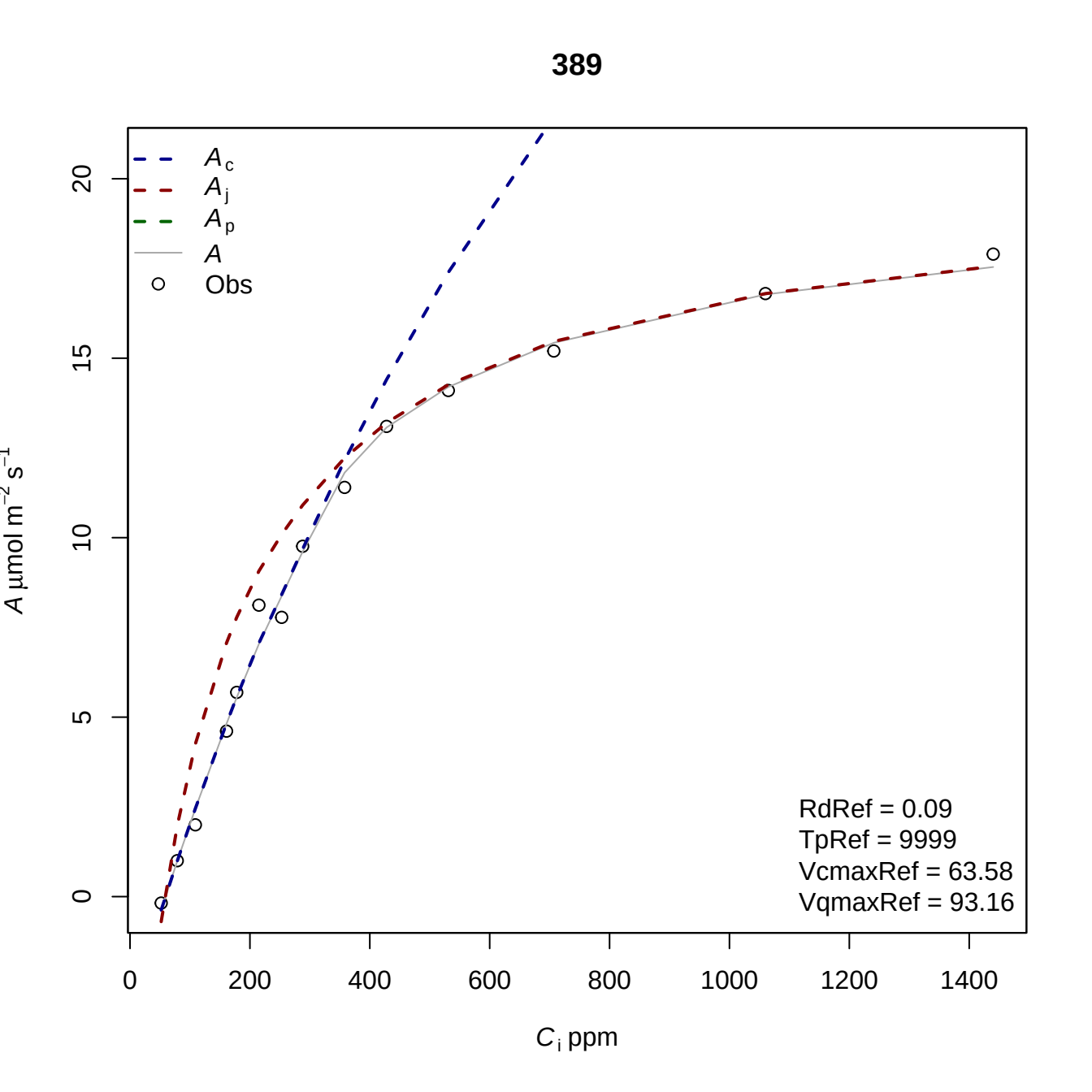

384
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.89
TpRef = 9999
VcmaxRef = 29.93
VqmaxRef = 52.6
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

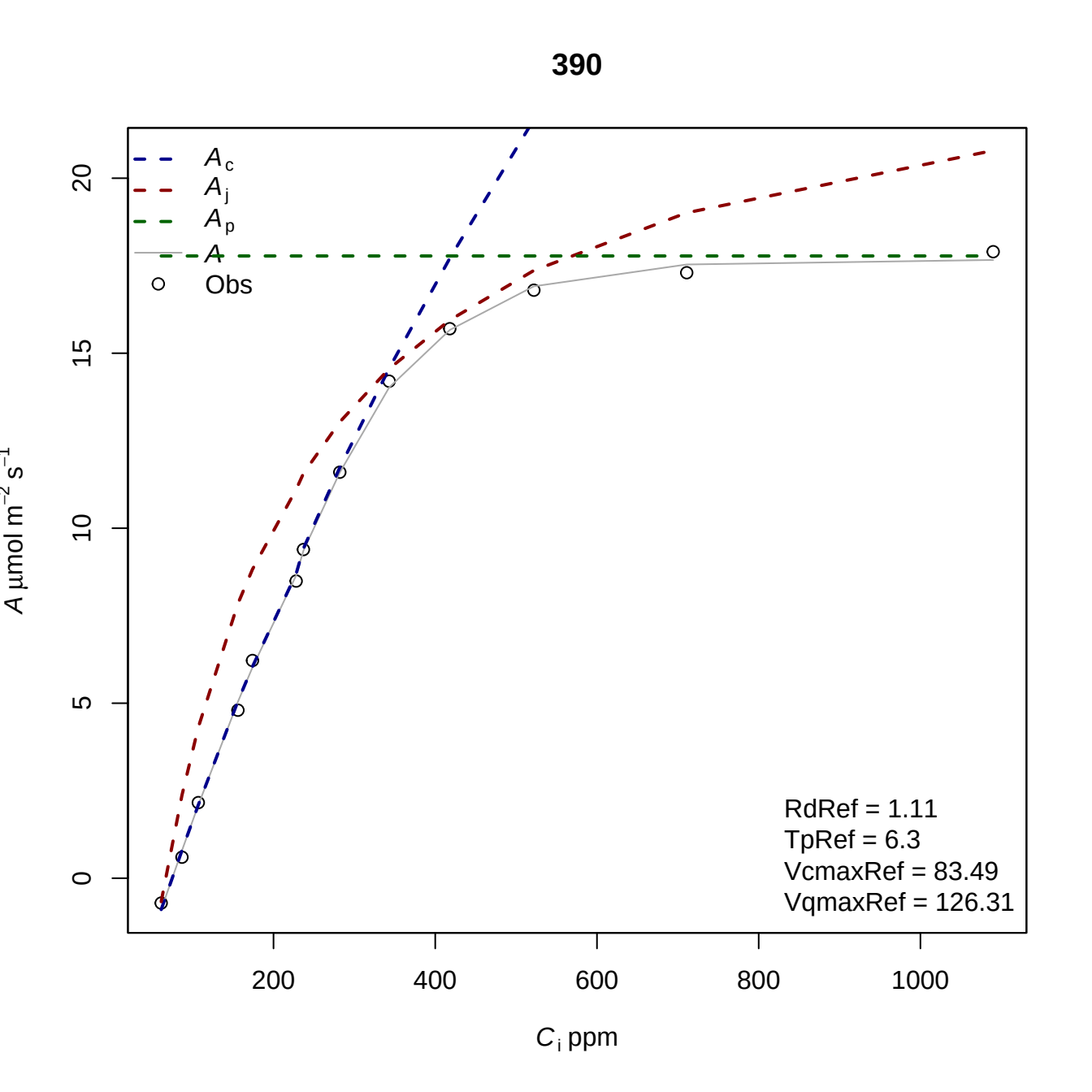

385
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.73
TpRef = 3.6
VcmaxRef = 41.93
VqmaxRef = 61.93
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
500
1000
1500

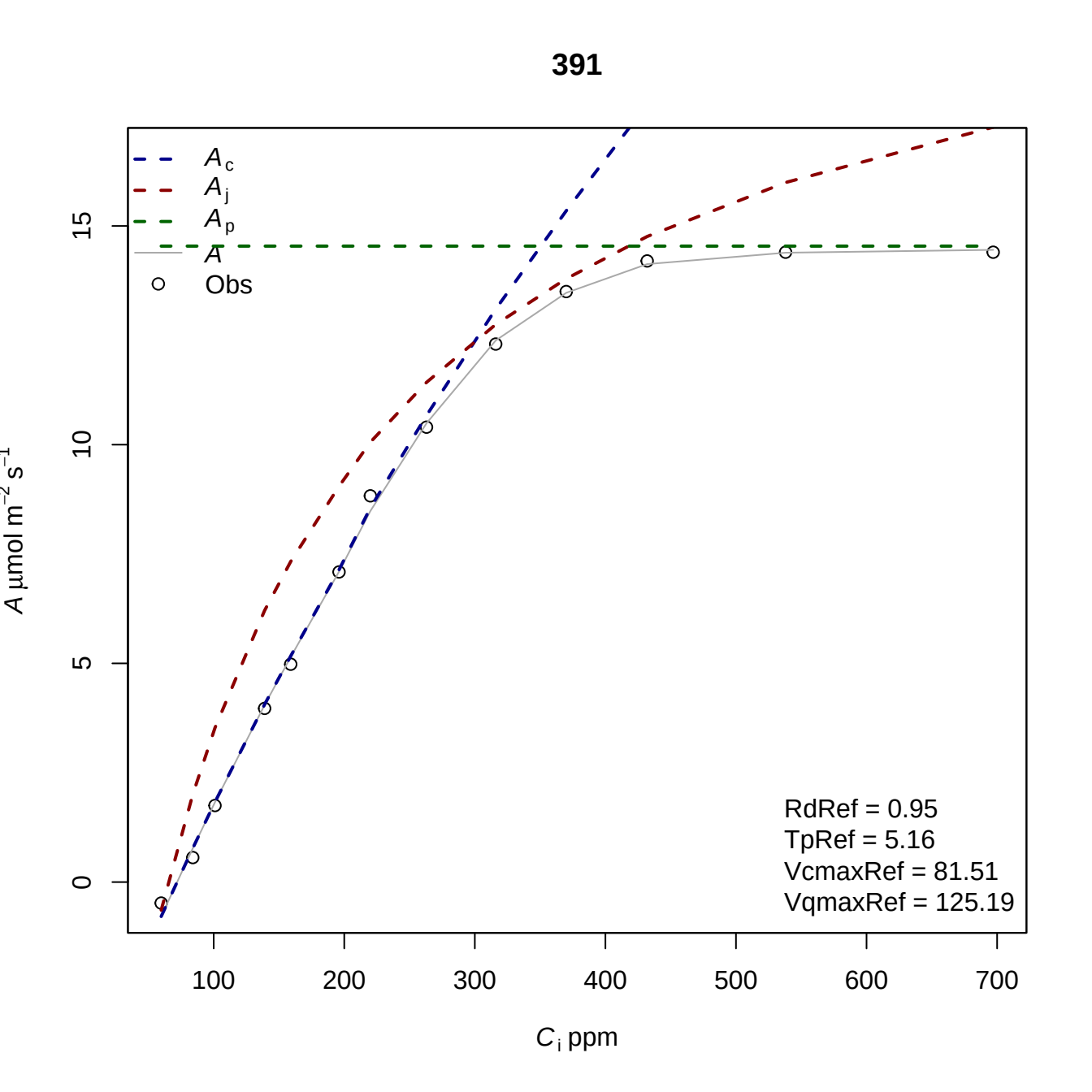

386
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.82
TpRef = 9999
VcmaxRef = 32.26
VqmaxRef = 45.85
0
2
4
6
8
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

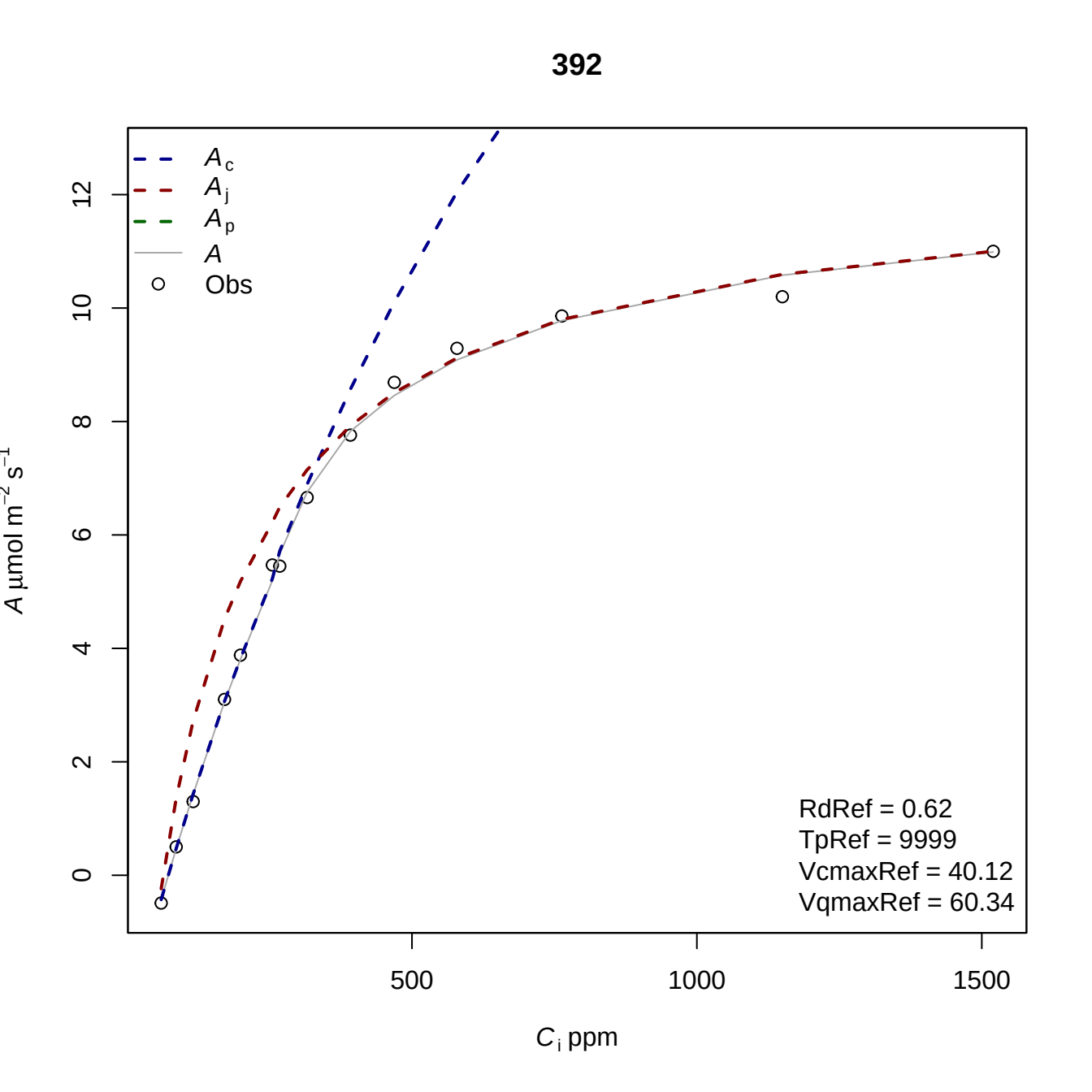

387
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.81
TpRef = 3.26
VcmaxRef = 35.11
VqmaxRef = 63.33
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

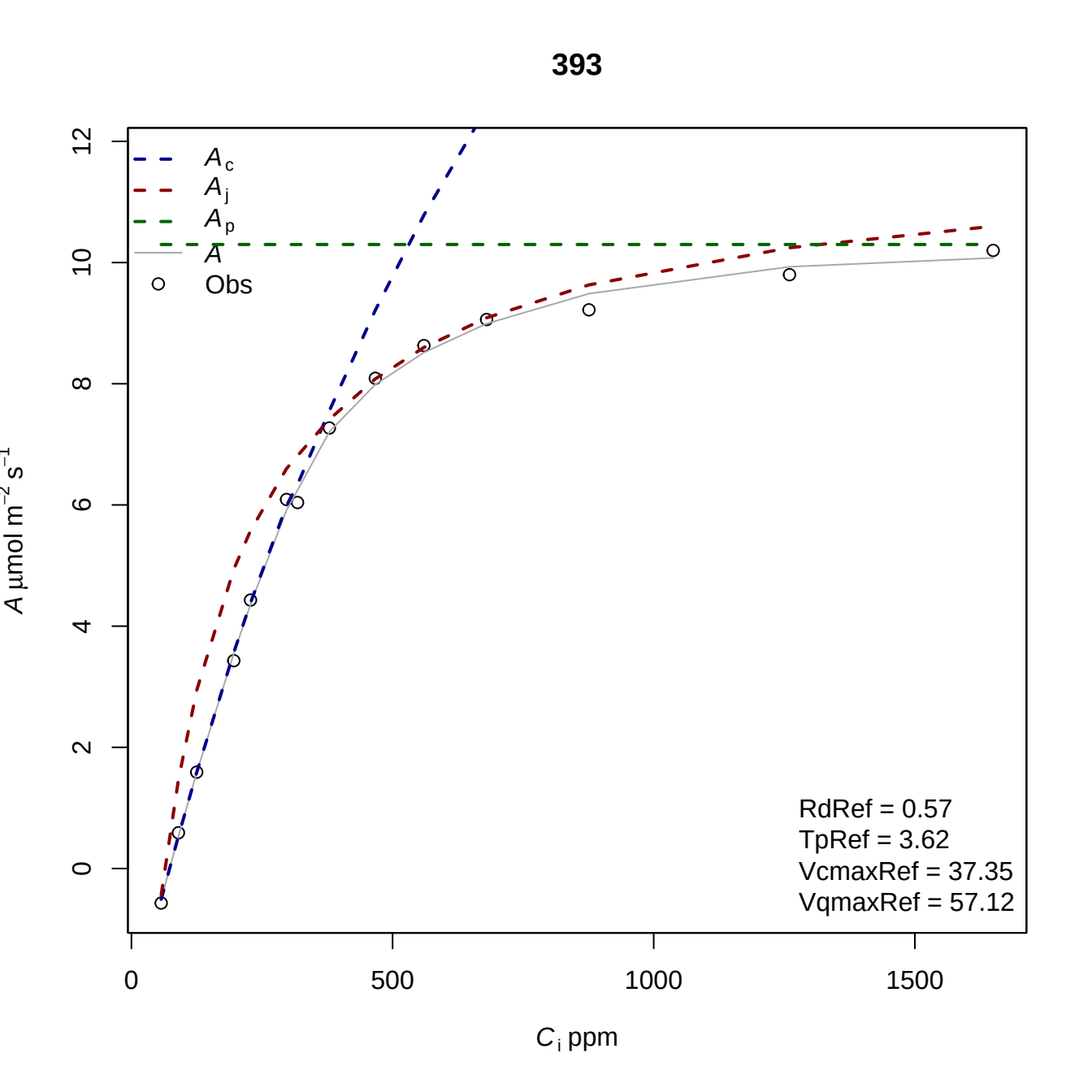

388
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.47
TpRef = 7.23
VcmaxRef = 78.54
VqmaxRef = 126.96
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200
1400

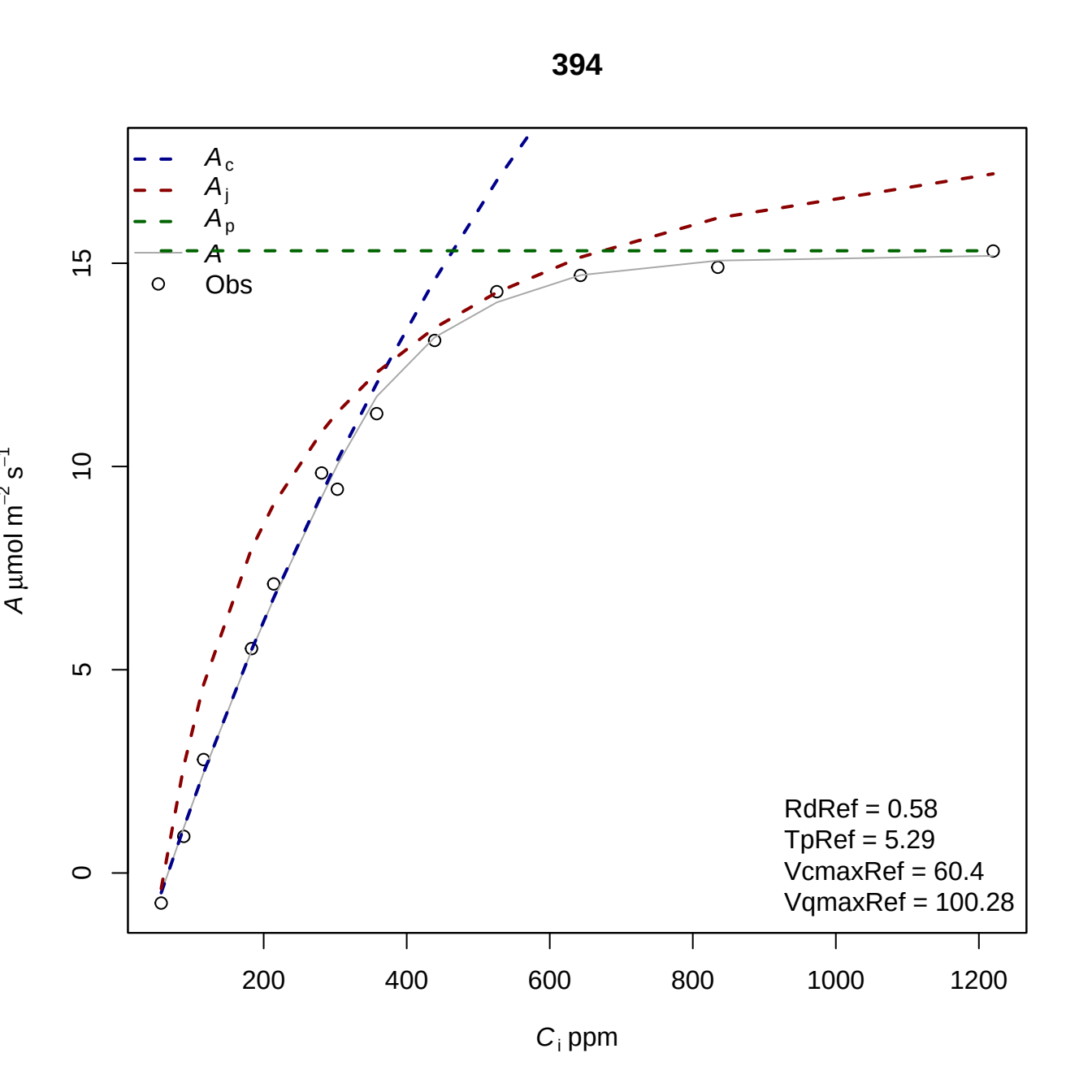

389
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.09
TpRef = 9999
VcmaxRef = 63.58
VqmaxRef = 93.16
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

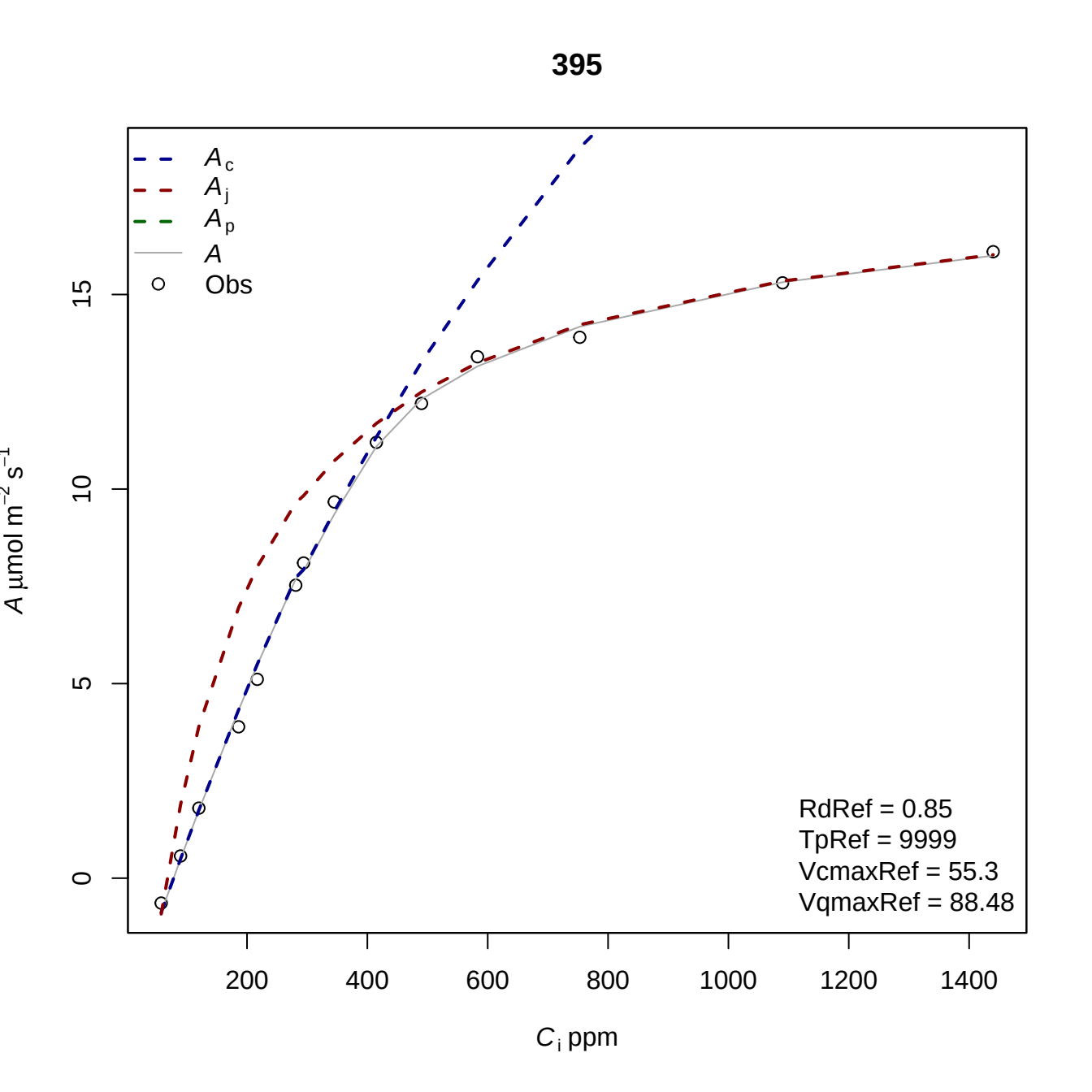

390
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.11
TpRef = 6.3
VcmaxRef = 83.49
VqmaxRef = 126.31
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000

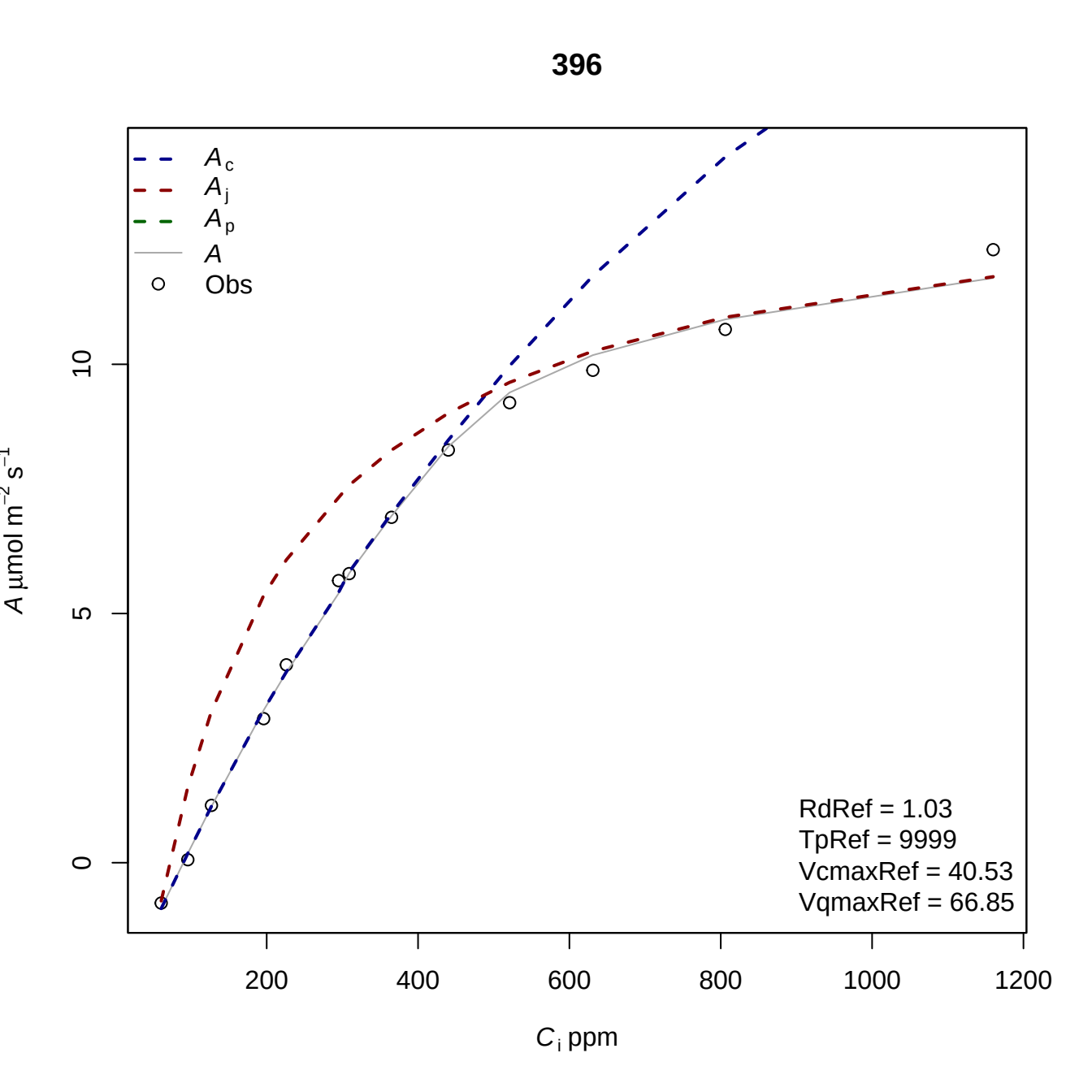

391
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.95
TpRef = 5.16
VcmaxRef = 81.51
VqmaxRef = 125.19
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500
600
700

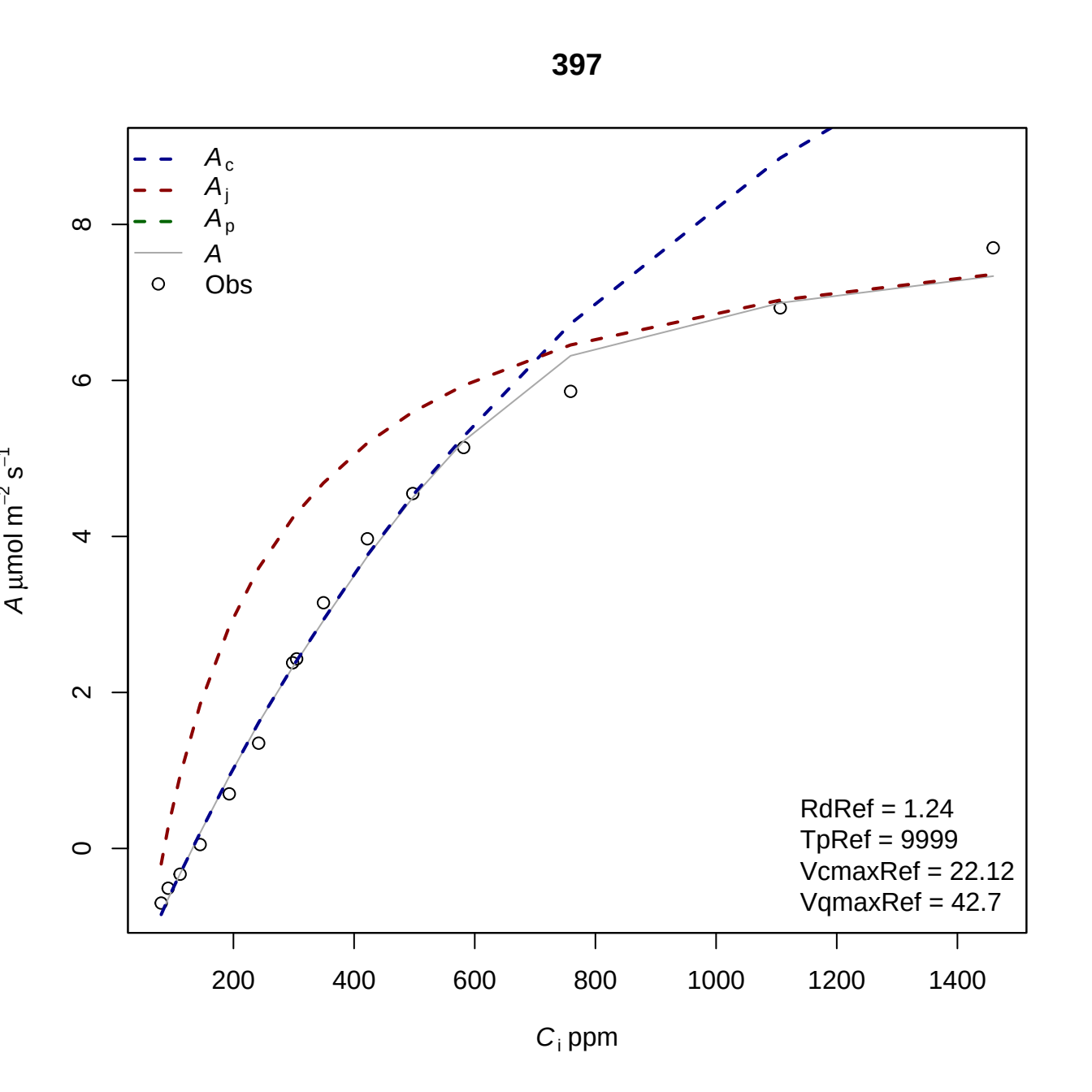

392
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.62
TpRef = 9999
VcmaxRef = 40.12
VqmaxRef = 60.34
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
500
1000
1500

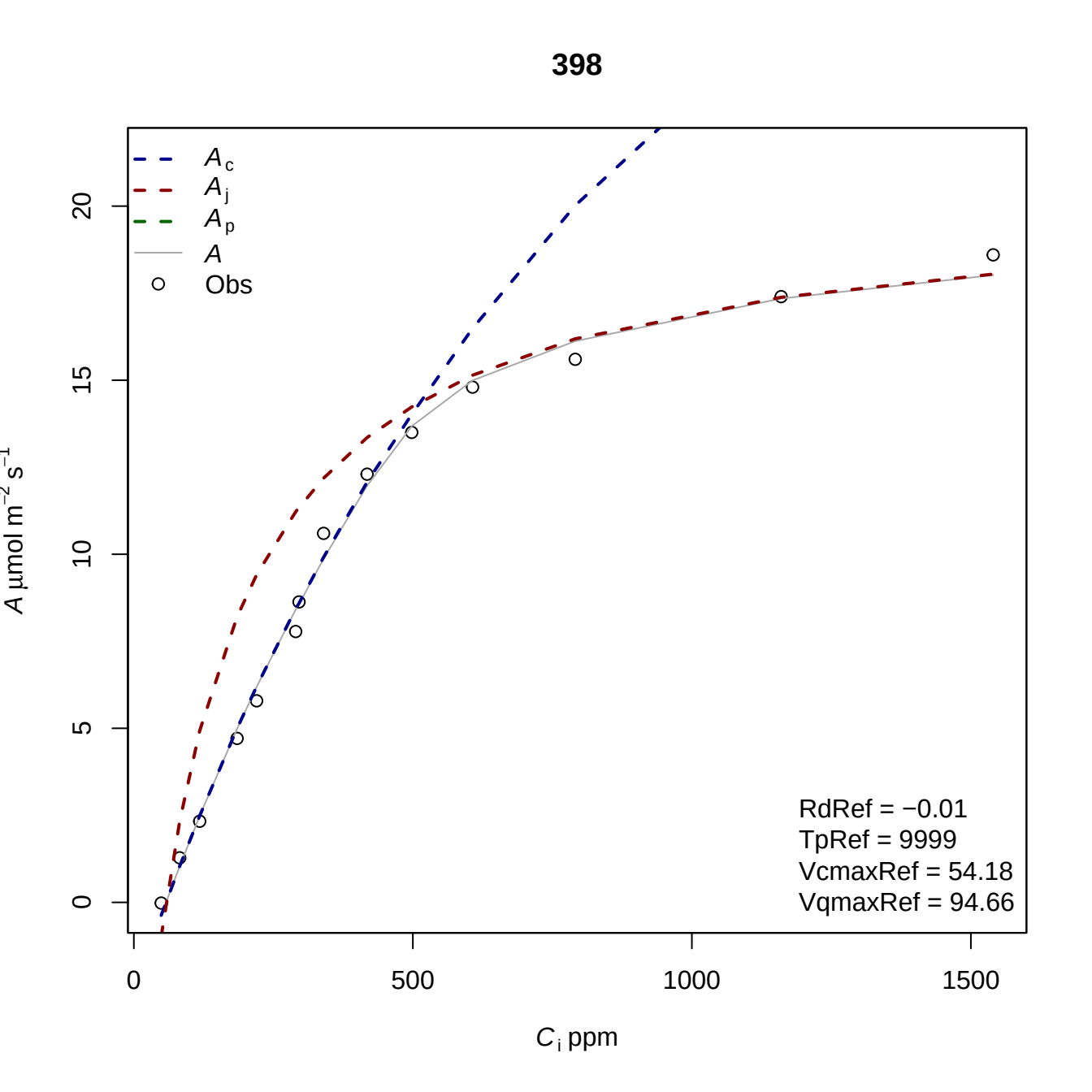

393
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.57
TpRef = 3.62
VcmaxRef = 37.35
VqmaxRef = 57.12
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

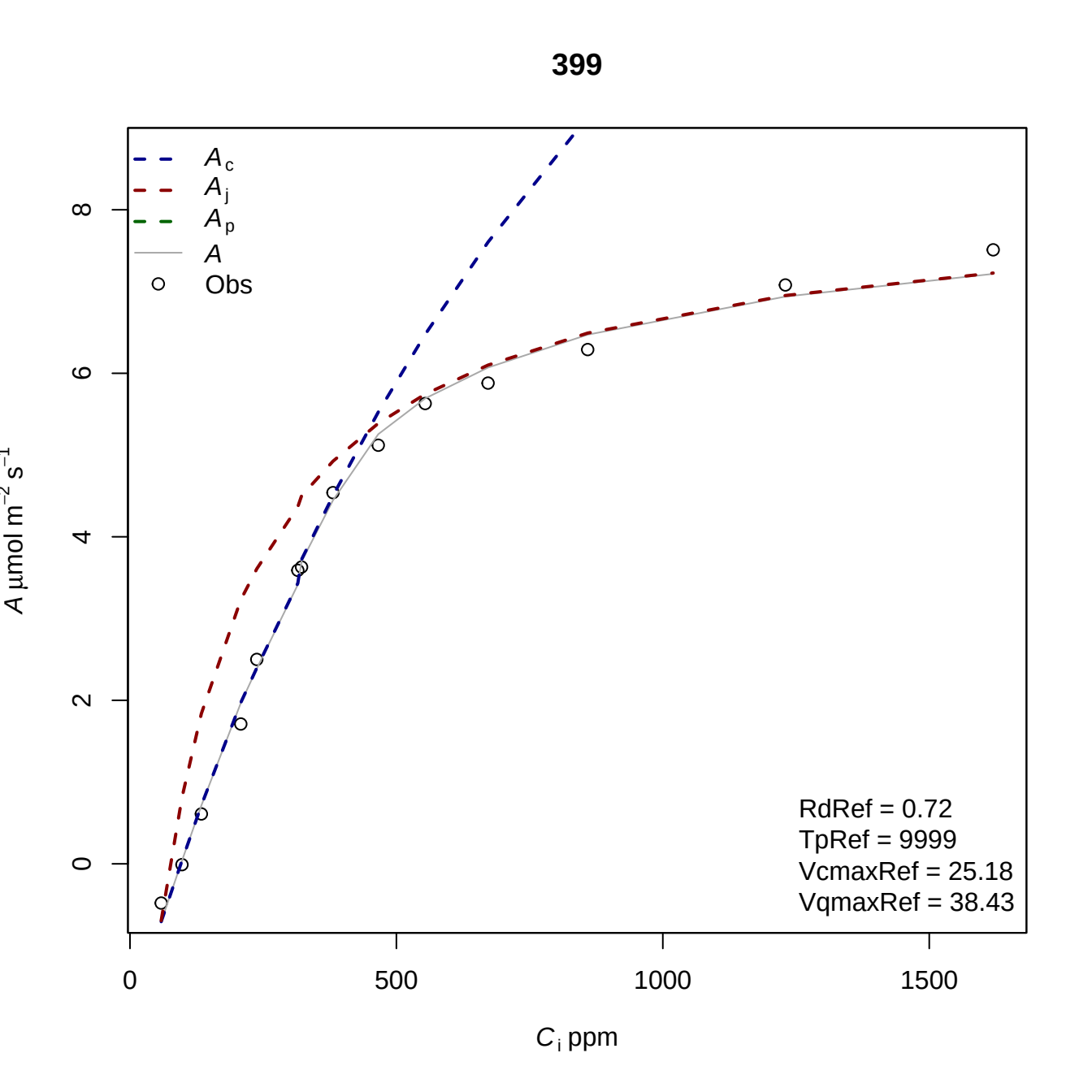

394
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.58
TpRef = 5.29
VcmaxRef = 60.4
VqmaxRef = 100.28
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200

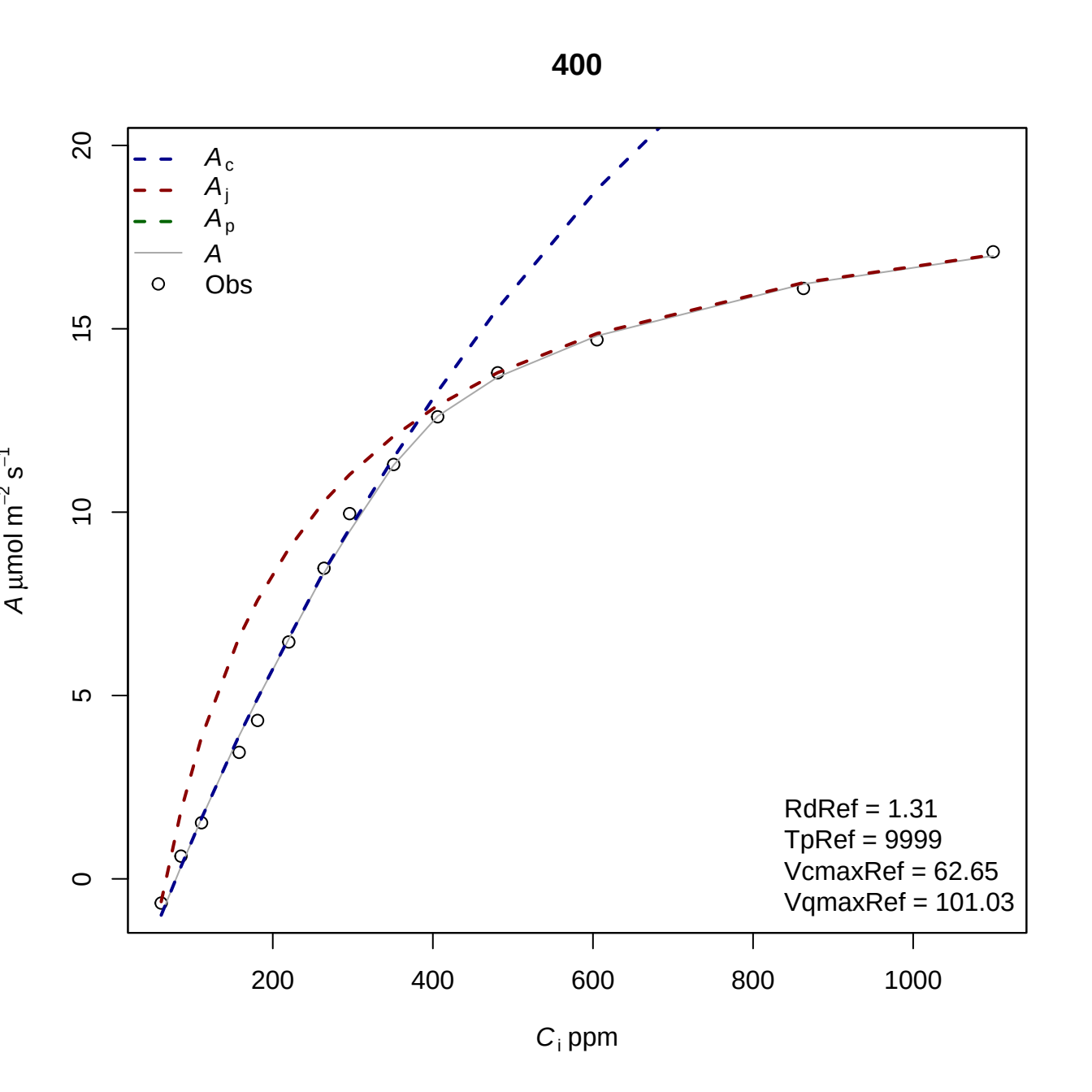

395
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.85
TpRef = 9999
VcmaxRef = 55.3
VqmaxRef = 88.48
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

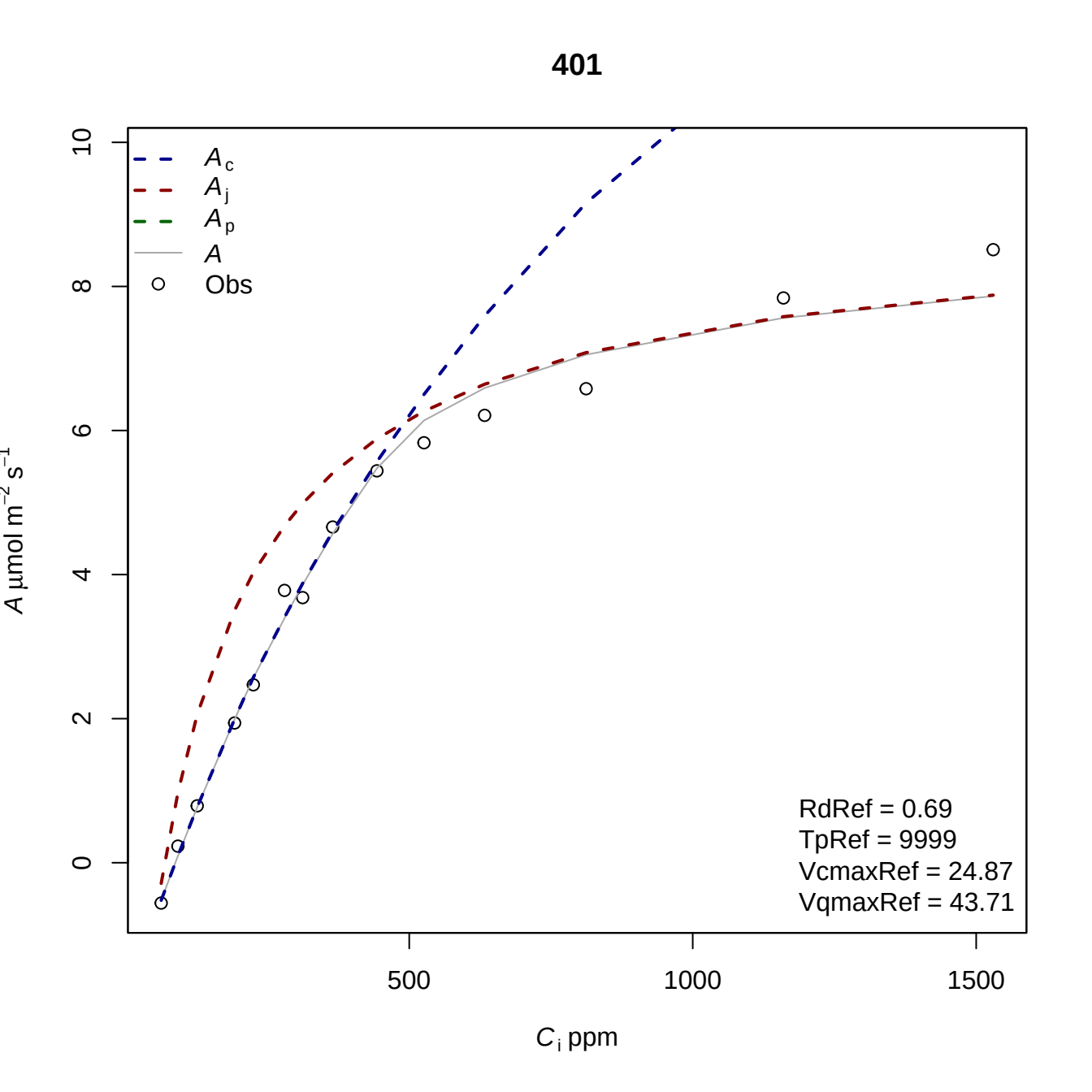

396
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.03
TpRef = 9999
VcmaxRef = 40.53
VqmaxRef = 66.85
0
5
10
200
400
600
800
1000
1200
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

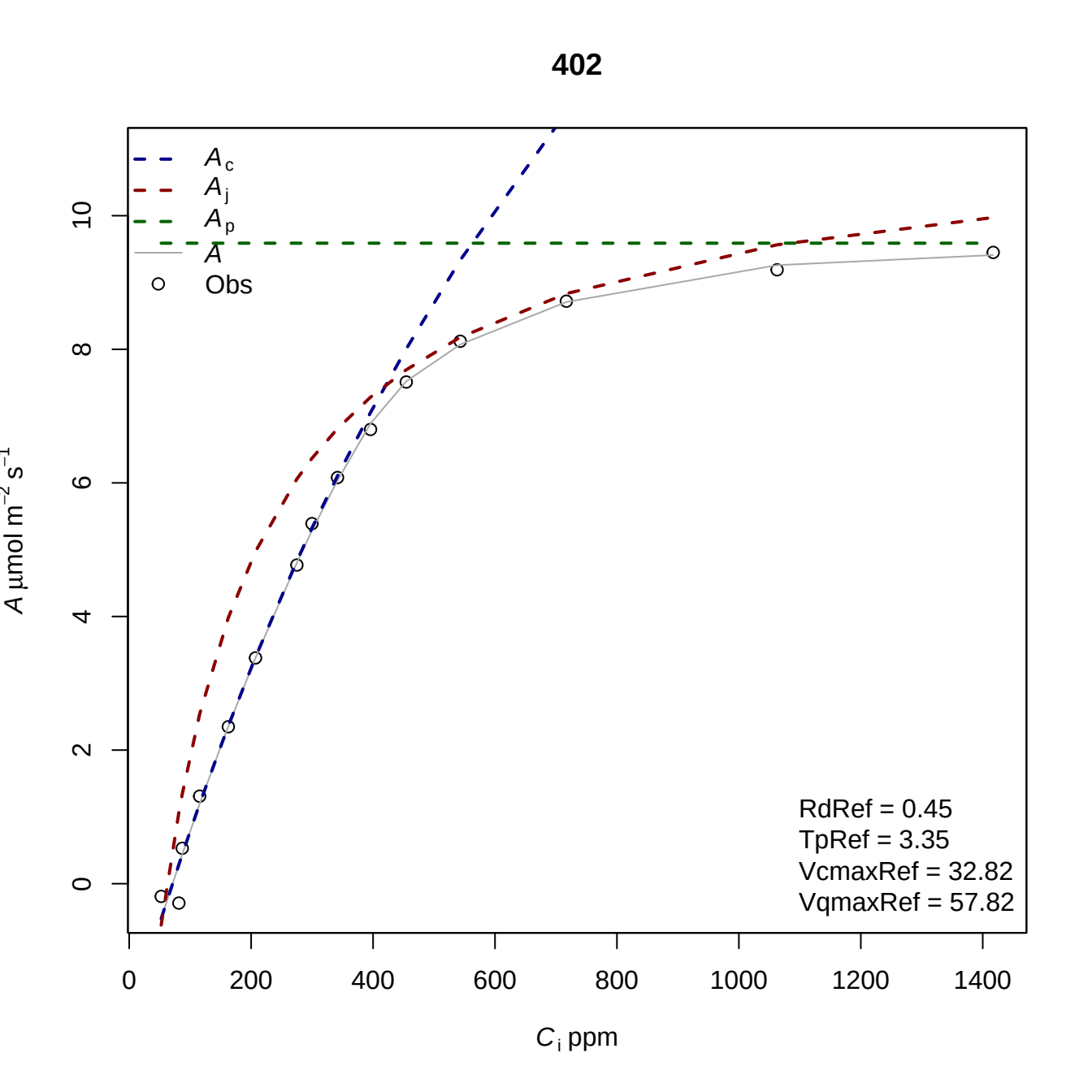

397
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.24
TpRef = 9999
VcmaxRef = 22.12
VqmaxRef = 42.7
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
200
400
600
800
1000
1200
1400

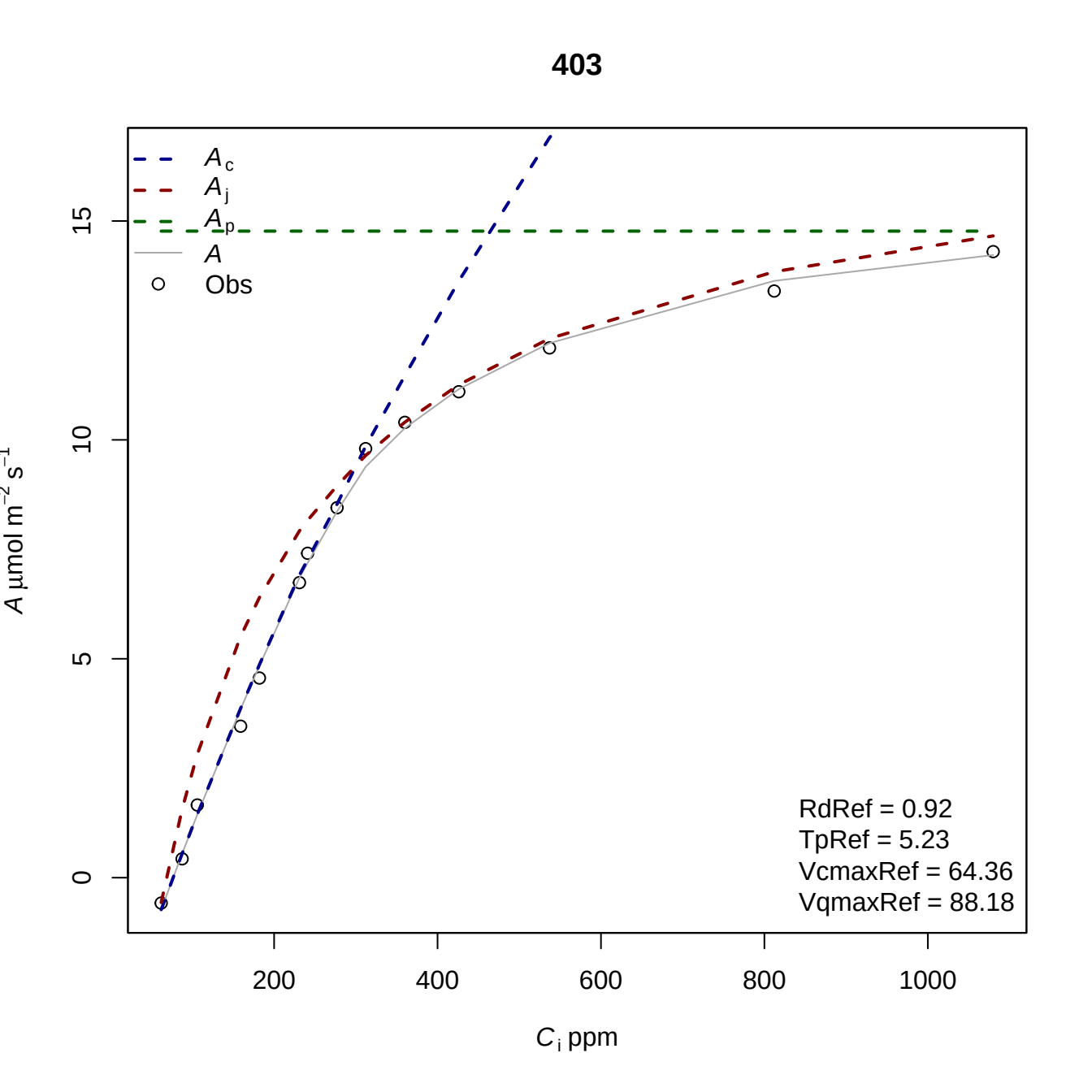

398
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.01
TpRef = 9999
VcmaxRef = 54.18
VqmaxRef = 94.66
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

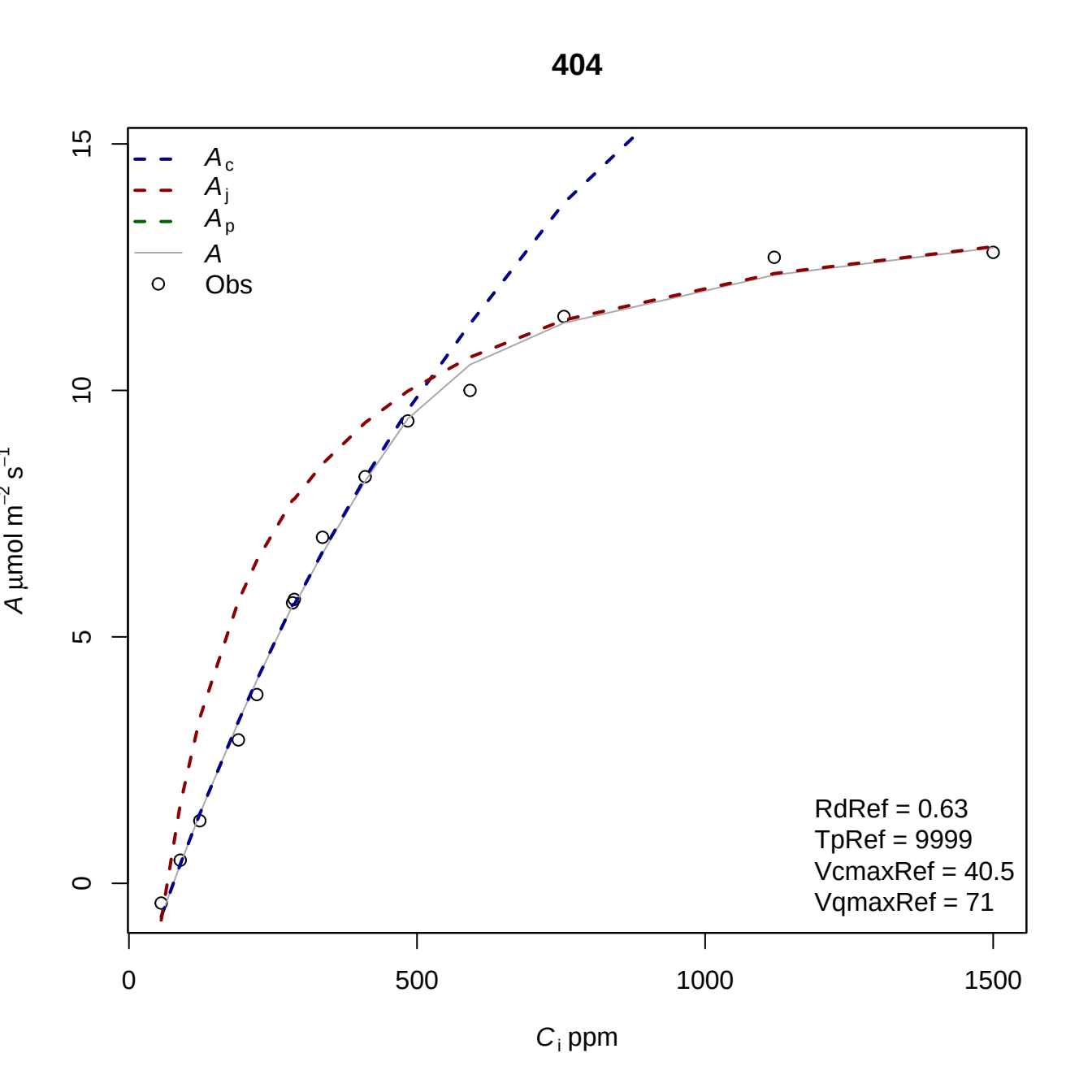

399
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.72
TpRef = 9999
VcmaxRef = 25.18
VqmaxRef = 38.43
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
0
500
1000
1500

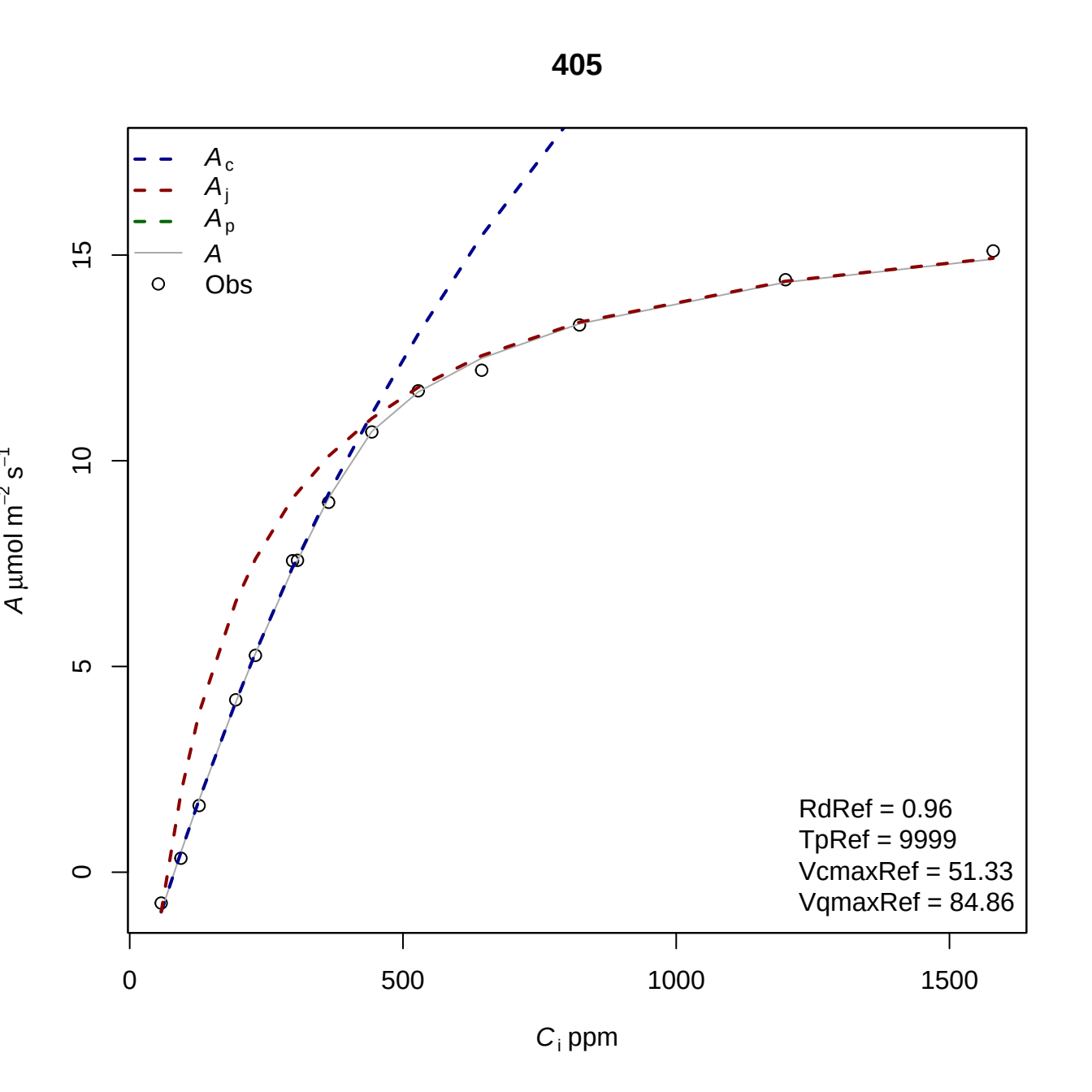

400
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.31
TpRef = 9999
VcmaxRef = 62.65
VqmaxRef = 101.03
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000

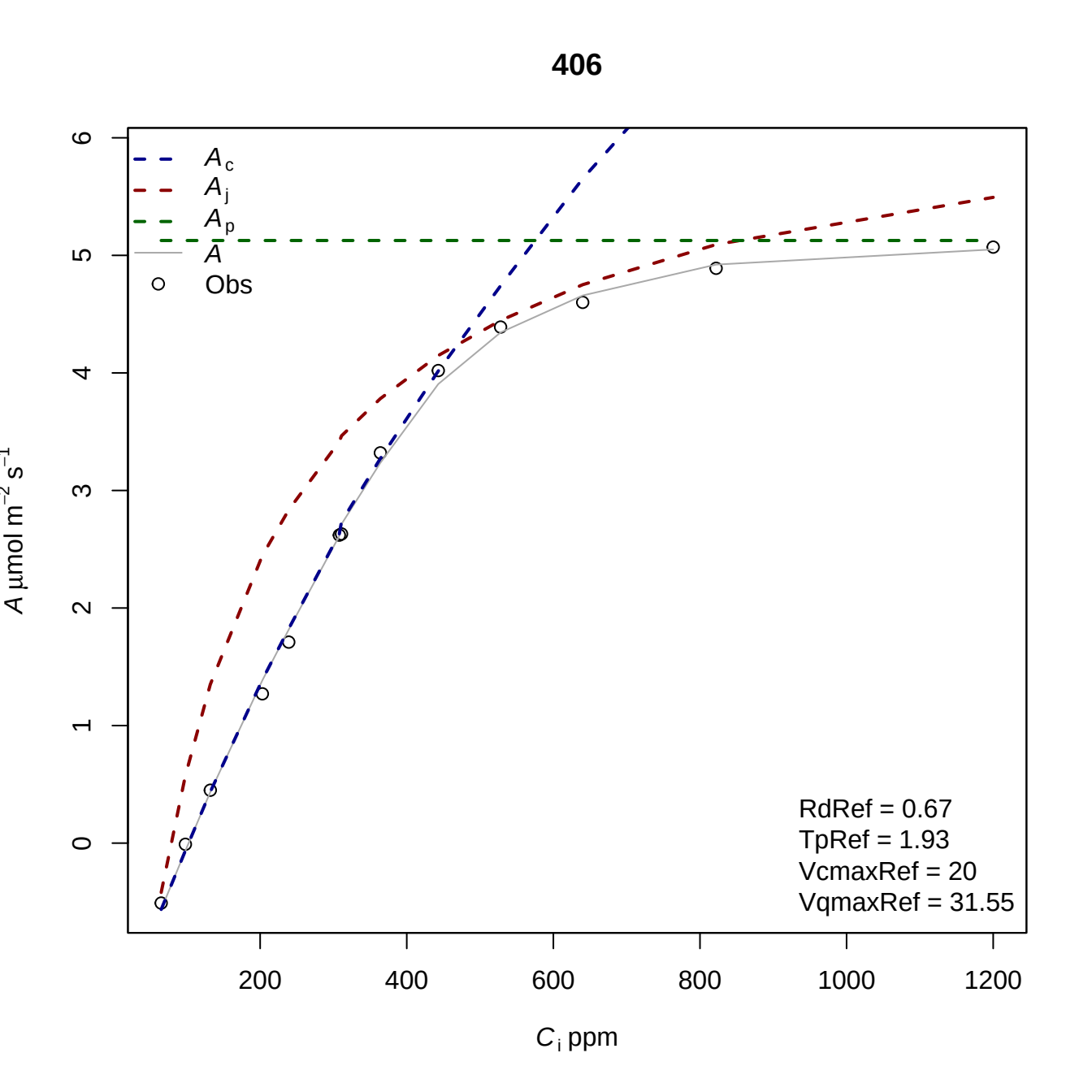

401
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.69
TpRef = 9999
VcmaxRef = 24.87
VqmaxRef = 43.71
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
500
1000
1500

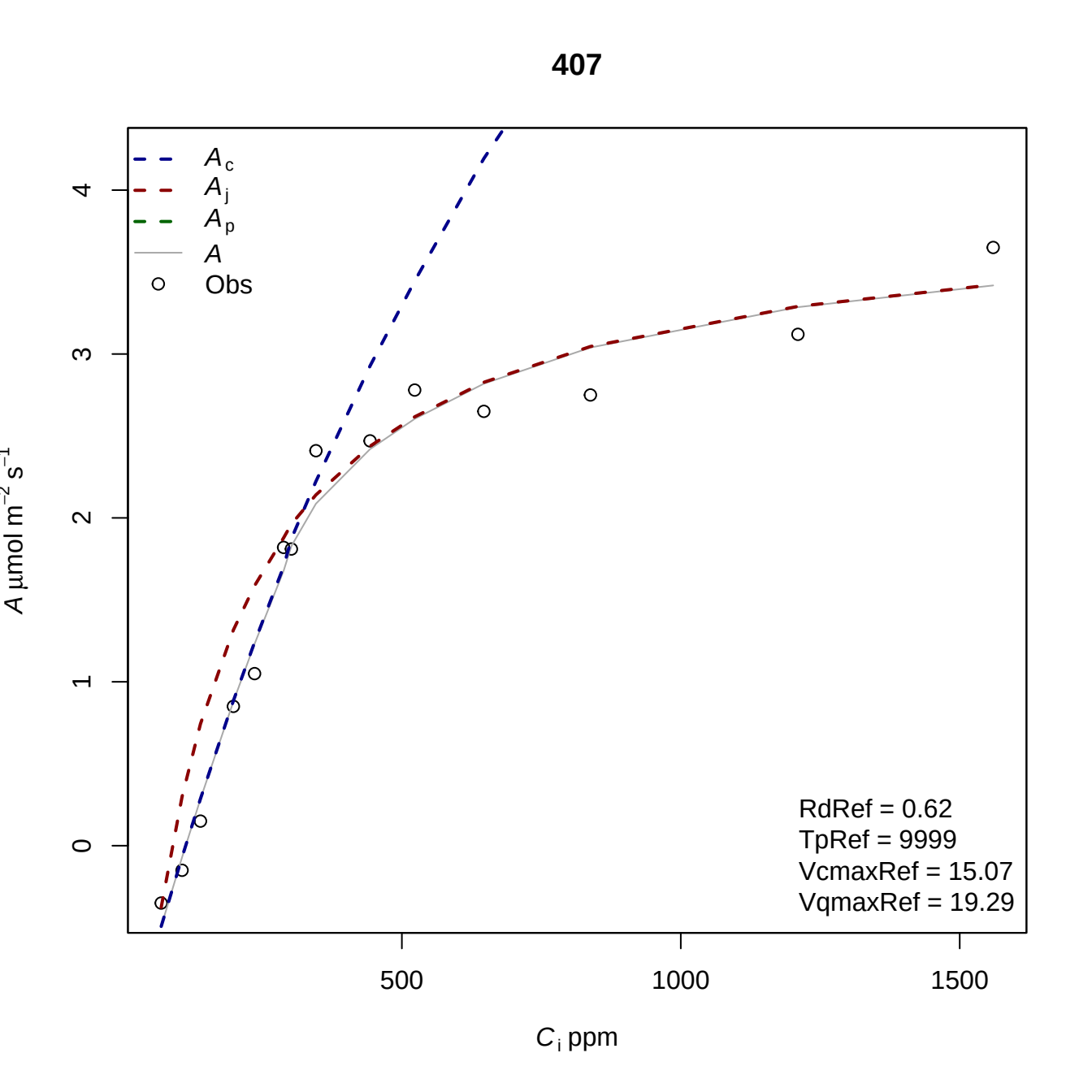

402
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.45
TpRef = 3.35
VcmaxRef = 32.82
VqmaxRef = 57.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

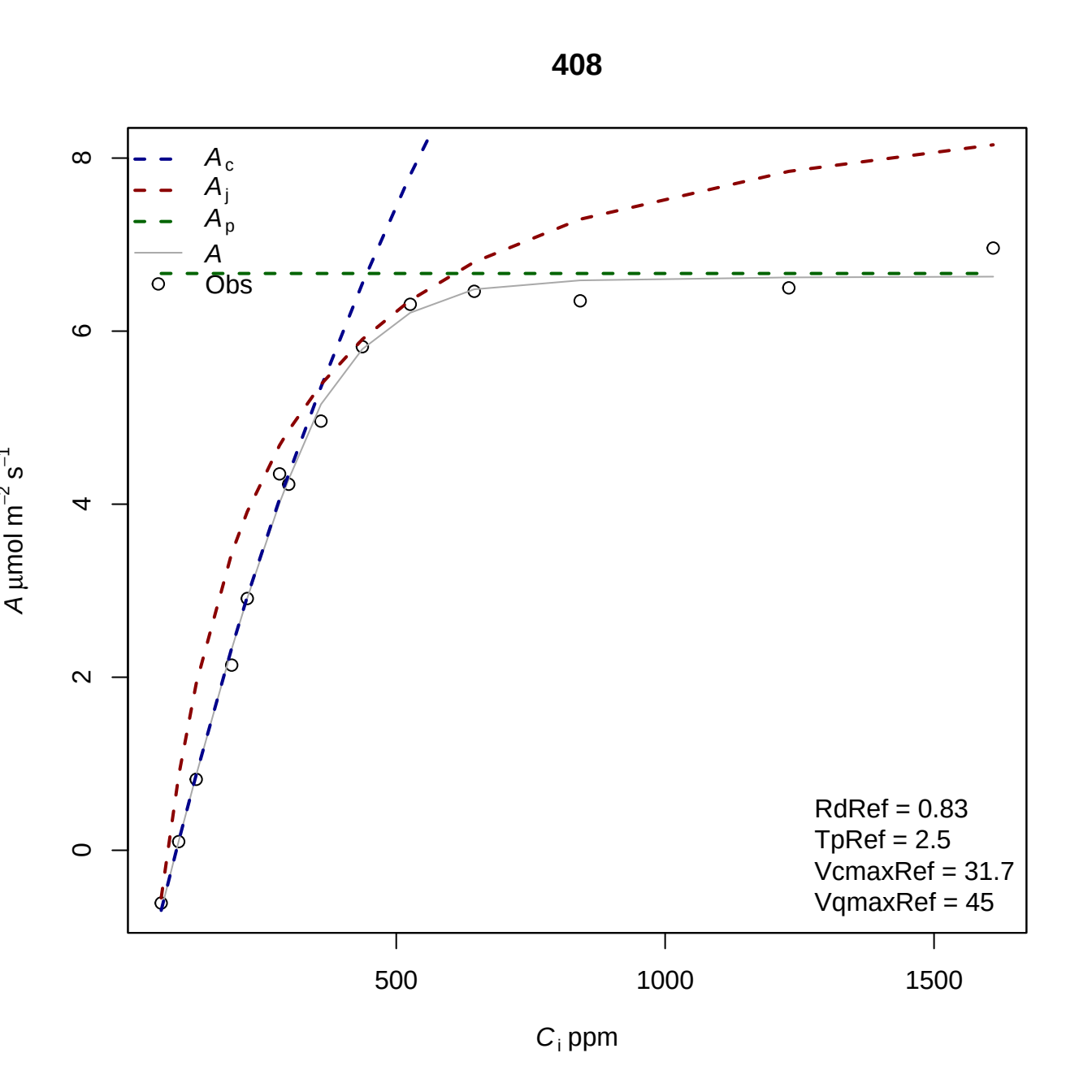

403
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.92
TpRef = 5.23
VcmaxRef = 64.36
VqmaxRef = 88.18
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

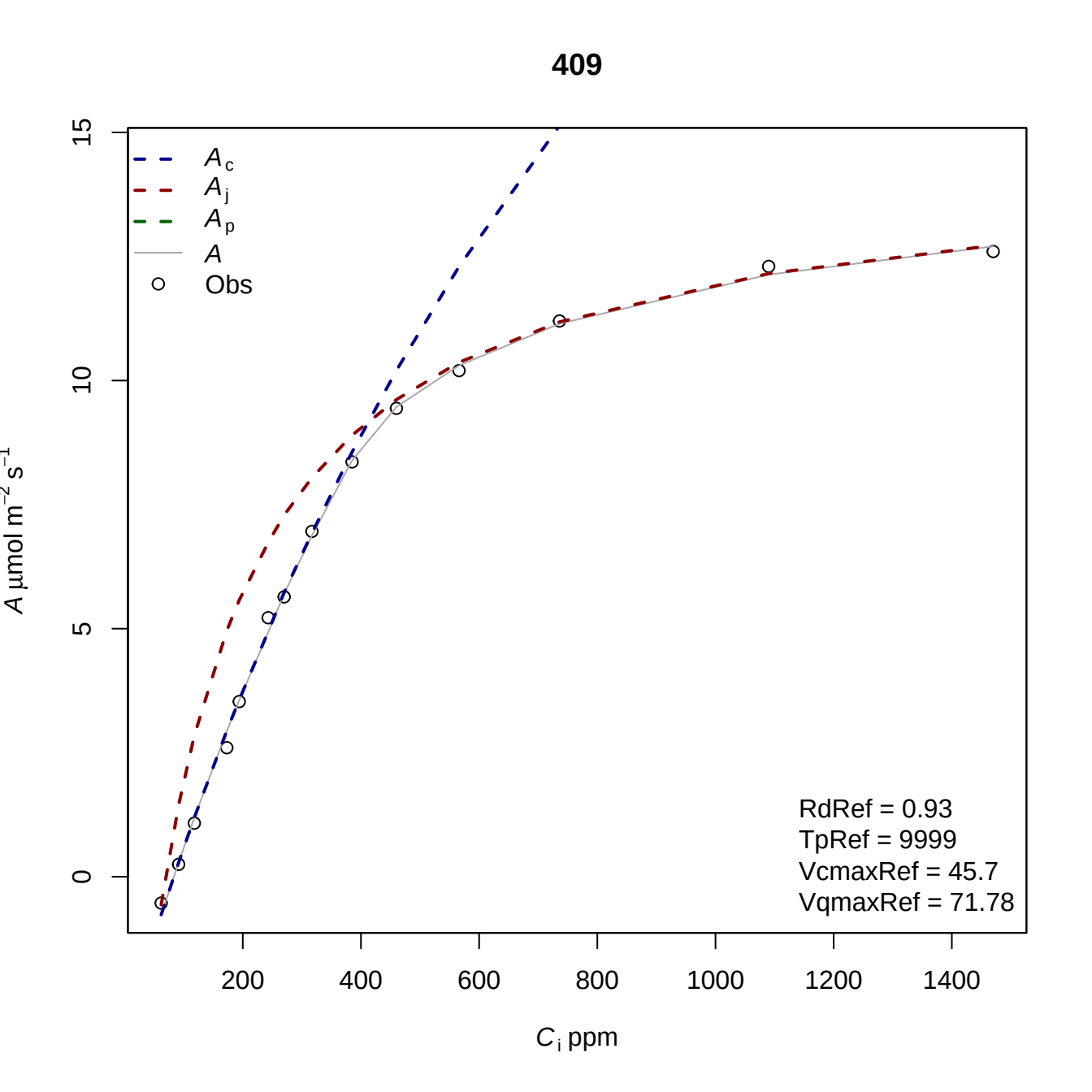

404
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.63
TpRef = 9999
VcmaxRef = 40.5
VqmaxRef = 71
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

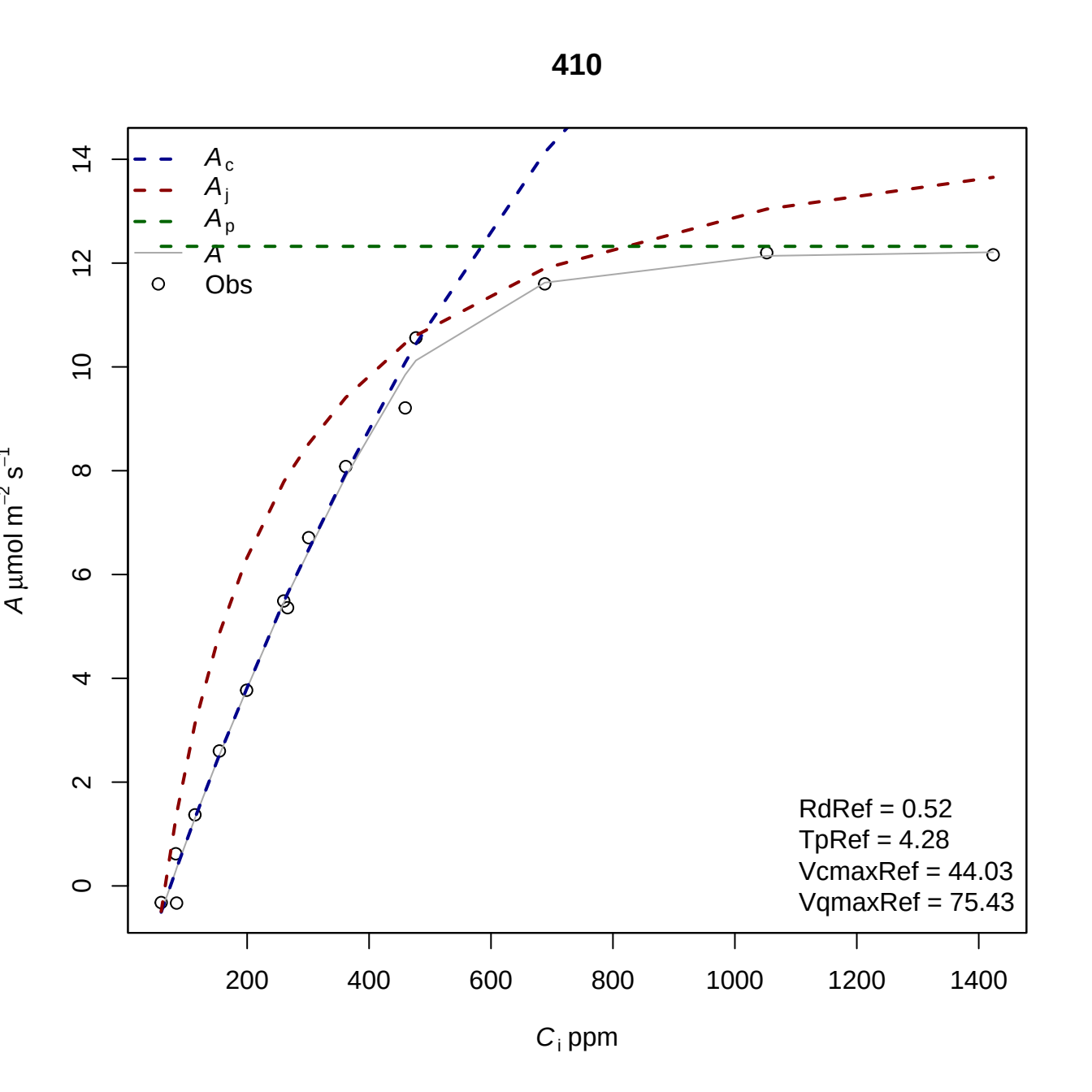

405
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.96
TpRef = 9999
VcmaxRef = 51.33
VqmaxRef = 84.86
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

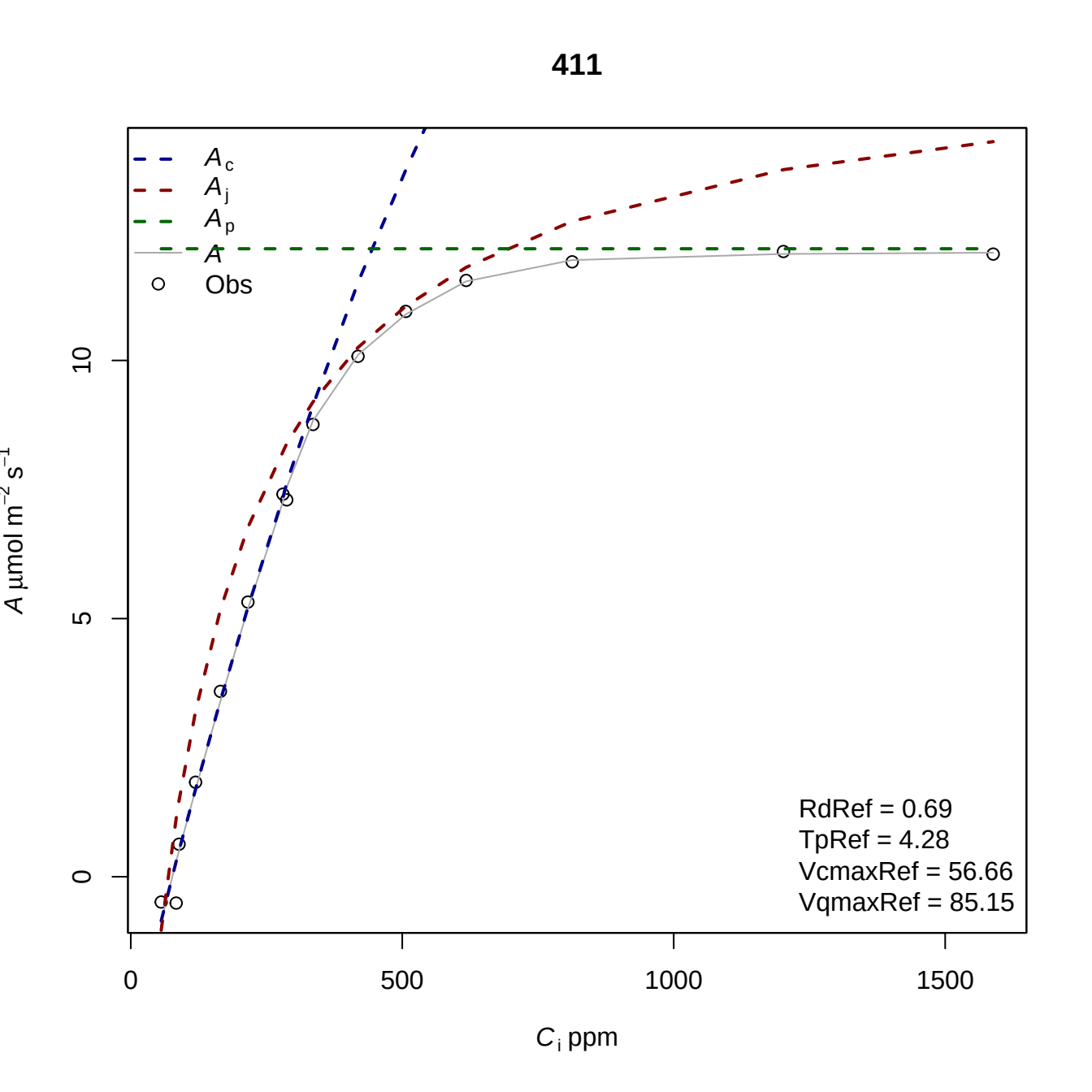

406
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.67
TpRef = 1.93
VcmaxRef = 20
VqmaxRef = 31.55
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
200
400
600
800
1000
1200

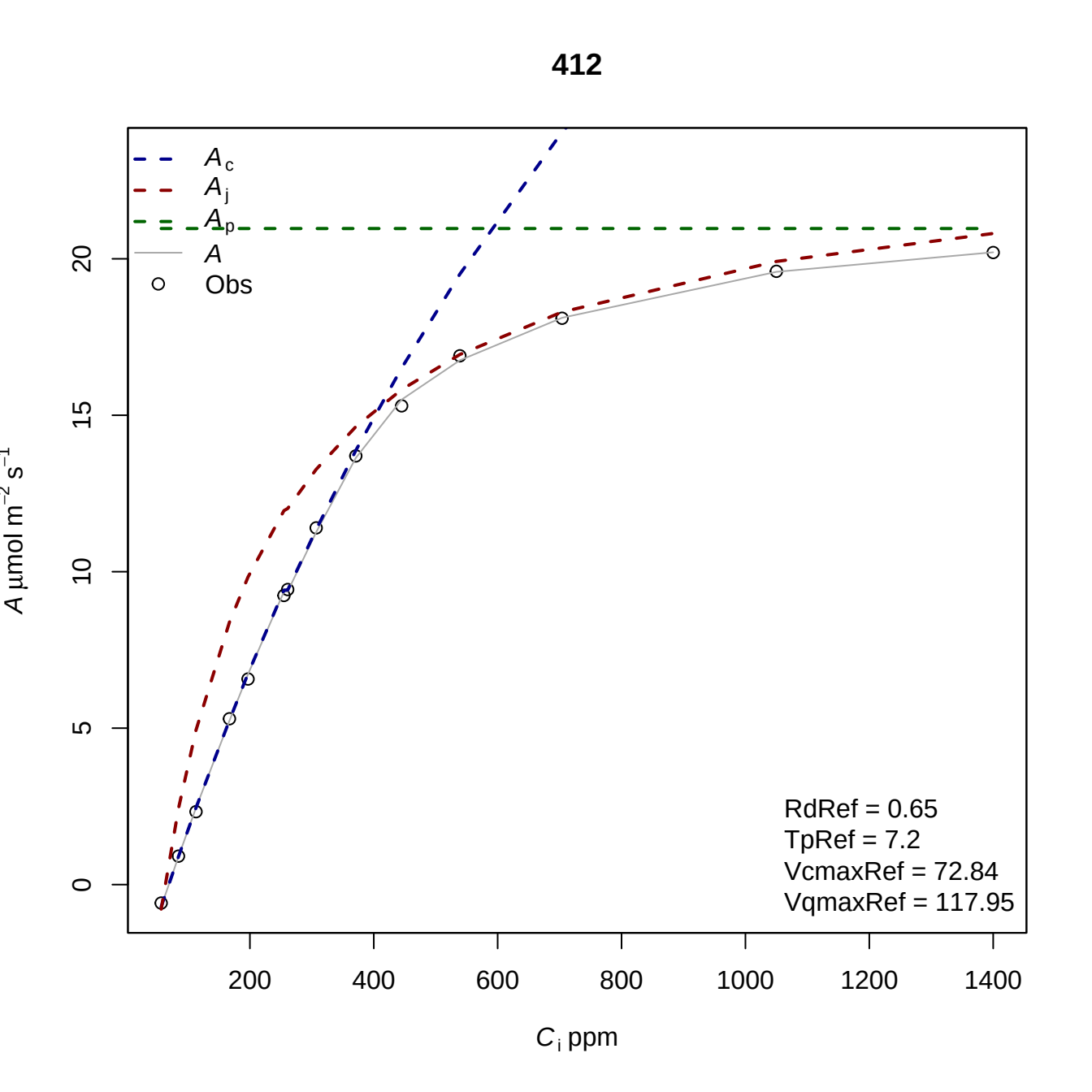

407
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.62
TpRef = 9999
VcmaxRef = 15.07
VqmaxRef = 19.29
0
1
2
3
4
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

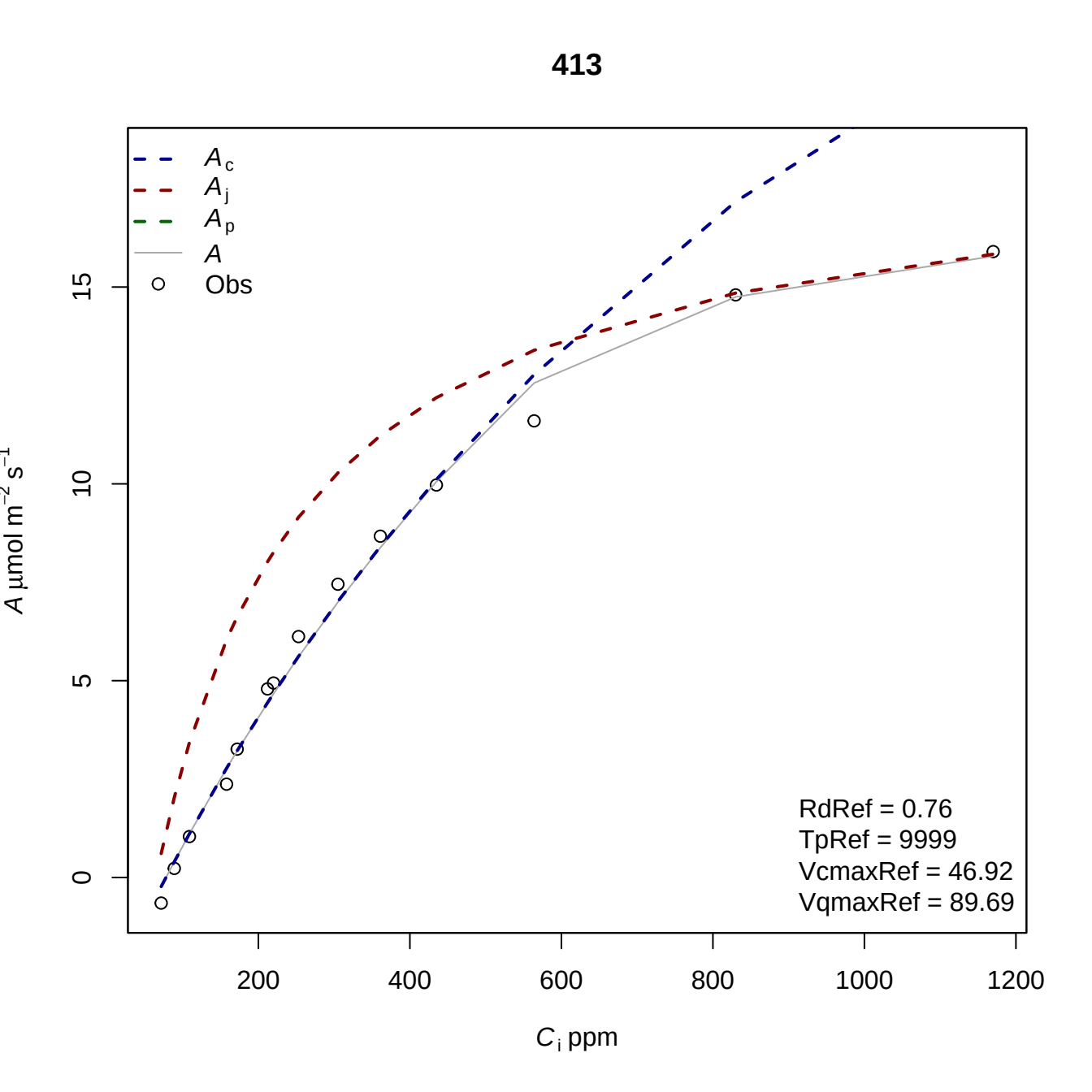

408
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.83
TpRef = 2.5
VcmaxRef = 31.7
VqmaxRef = 45
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
500
1000
1500

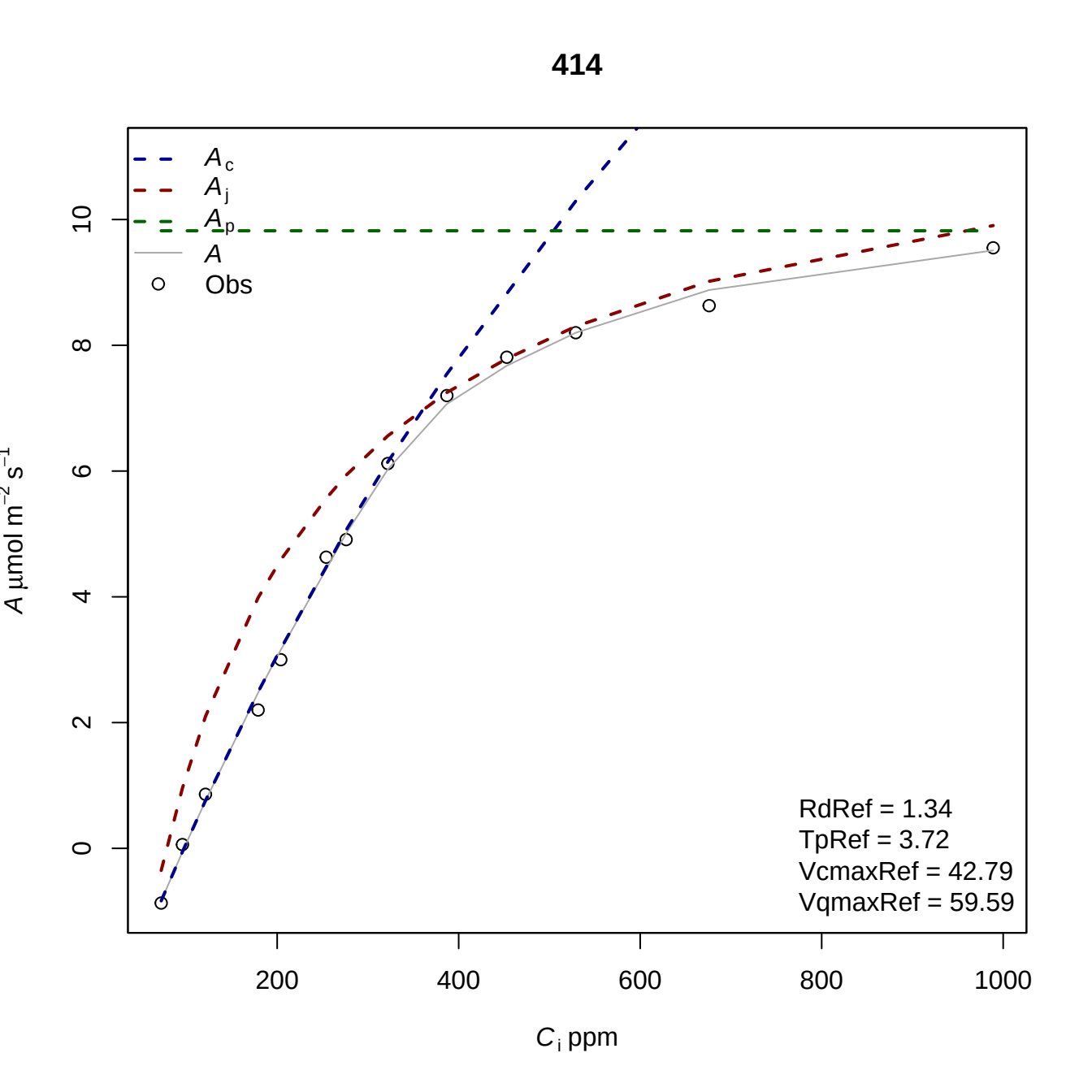

409
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.93
TpRef = 9999
VcmaxRef = 45.7
VqmaxRef = 71.78
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

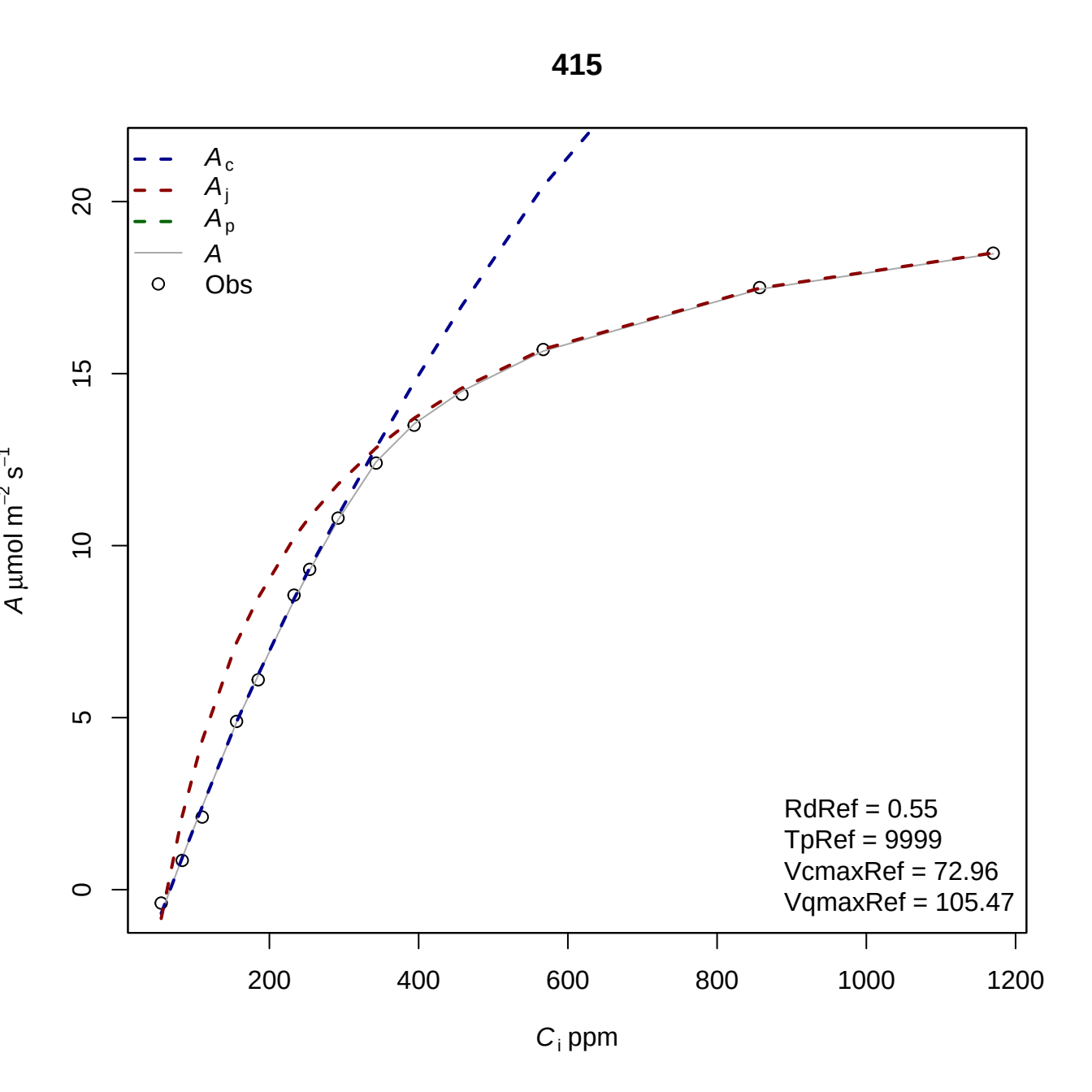

410
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.52
TpRef = 4.28
VcmaxRef = 44.03
VqmaxRef = 75.43
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
200
400
600
800
1000
1200
1400

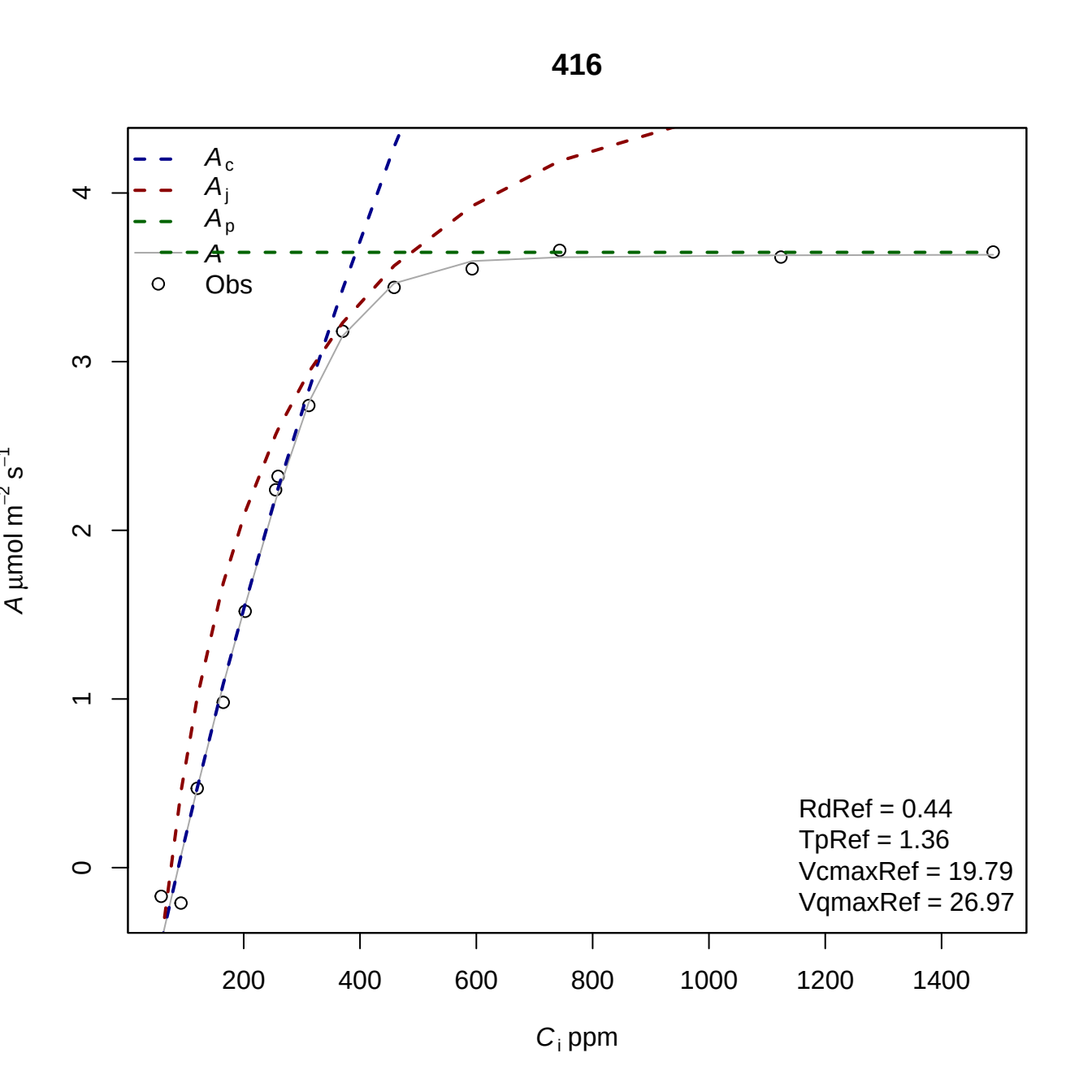

411
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.69
TpRef = 4.28
VcmaxRef = 56.66
VqmaxRef = 85.15
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$C_i$ ppm

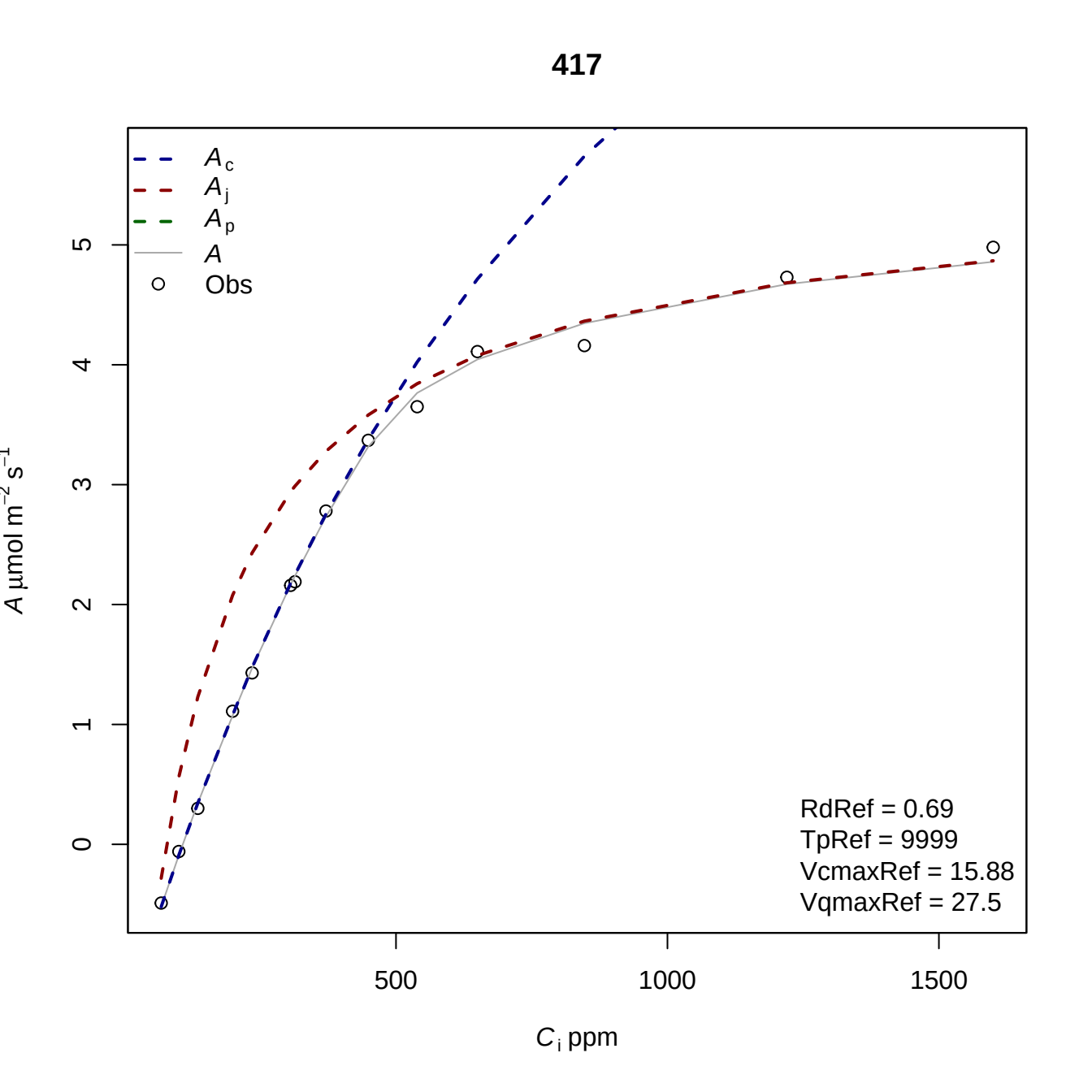

412
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.65
TpRef = 7.2
VcmaxRef = 72.84
VqmaxRef = 117.95
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

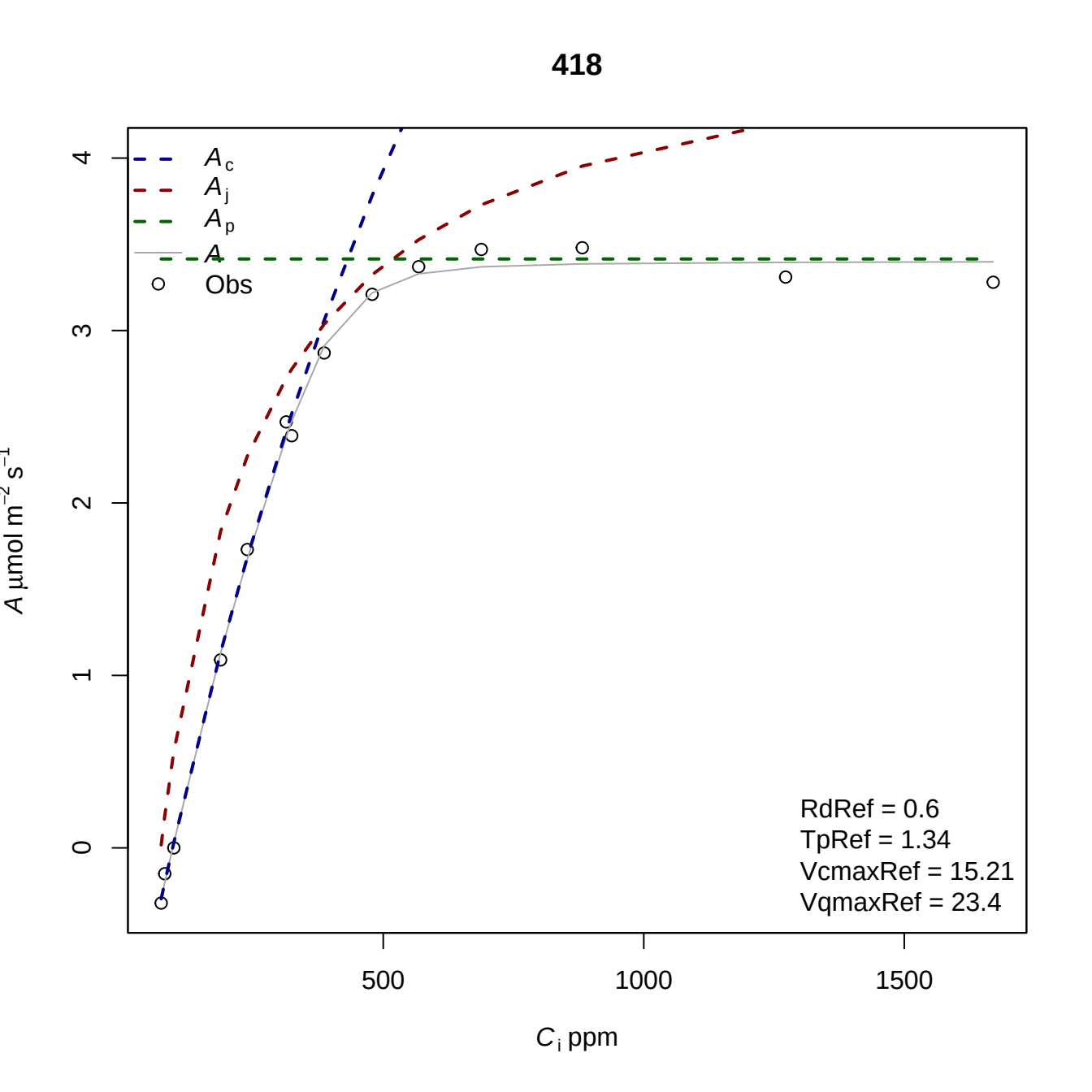

413
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.76
TpRef = 9999
VcmaxRef = 46.92
VqmaxRef = 89.69
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200

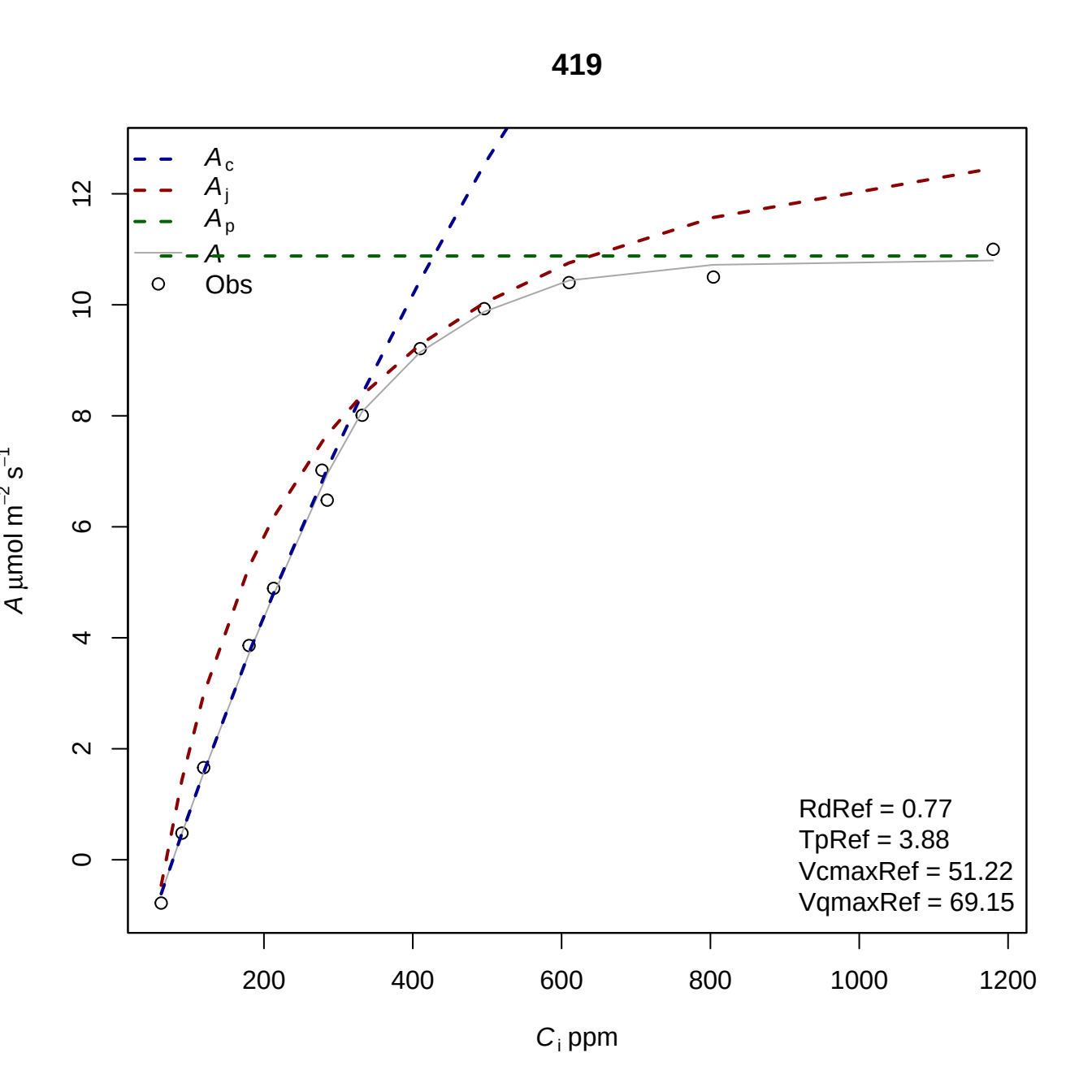

414
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.34
TpRef = 3.72
VcmaxRef = 42.79
VqmaxRef = 59.59
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
200
400
600
800
1000

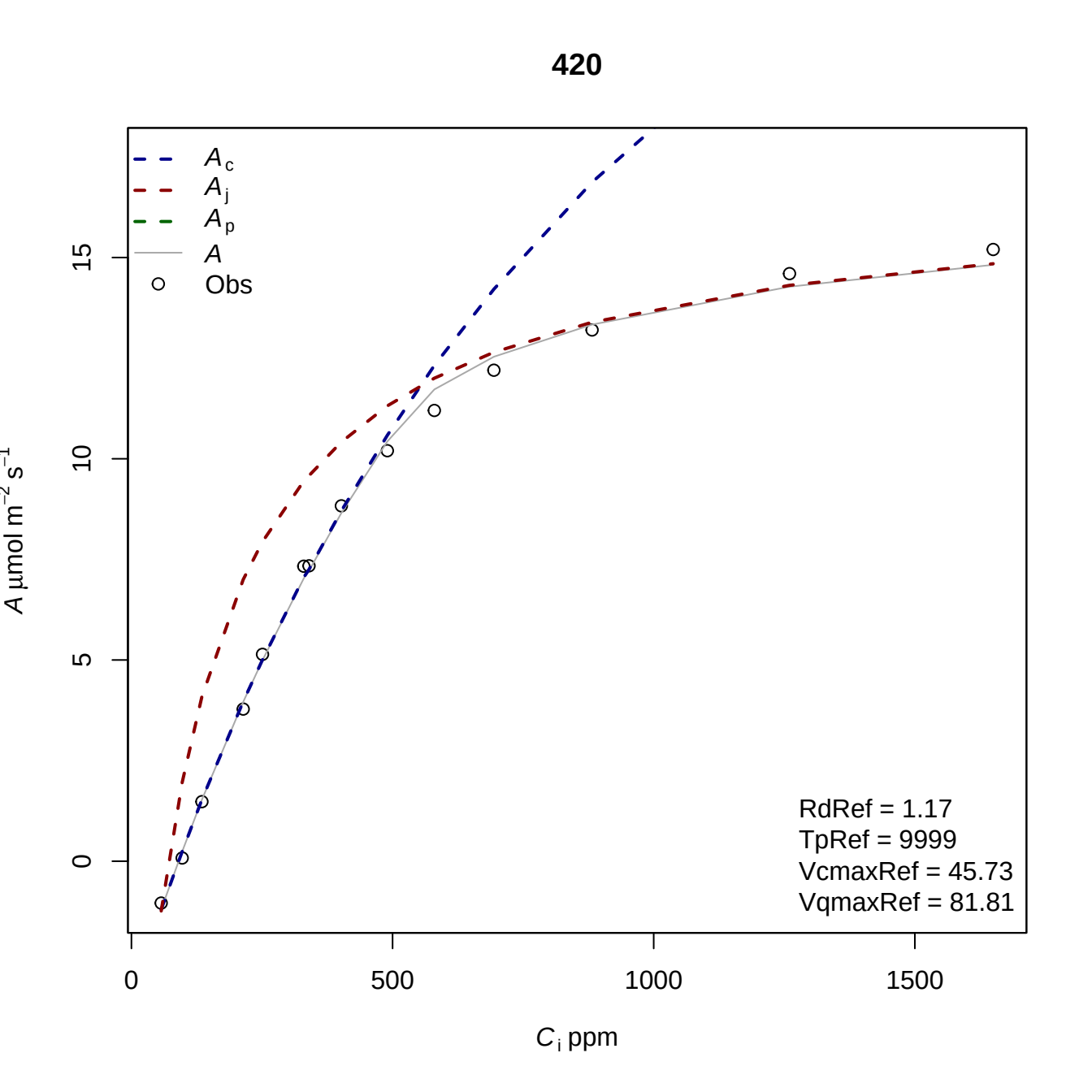

415
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.55
TpRef = 9999
VcmaxRef = 72.96
VqmaxRef = 105.47
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200

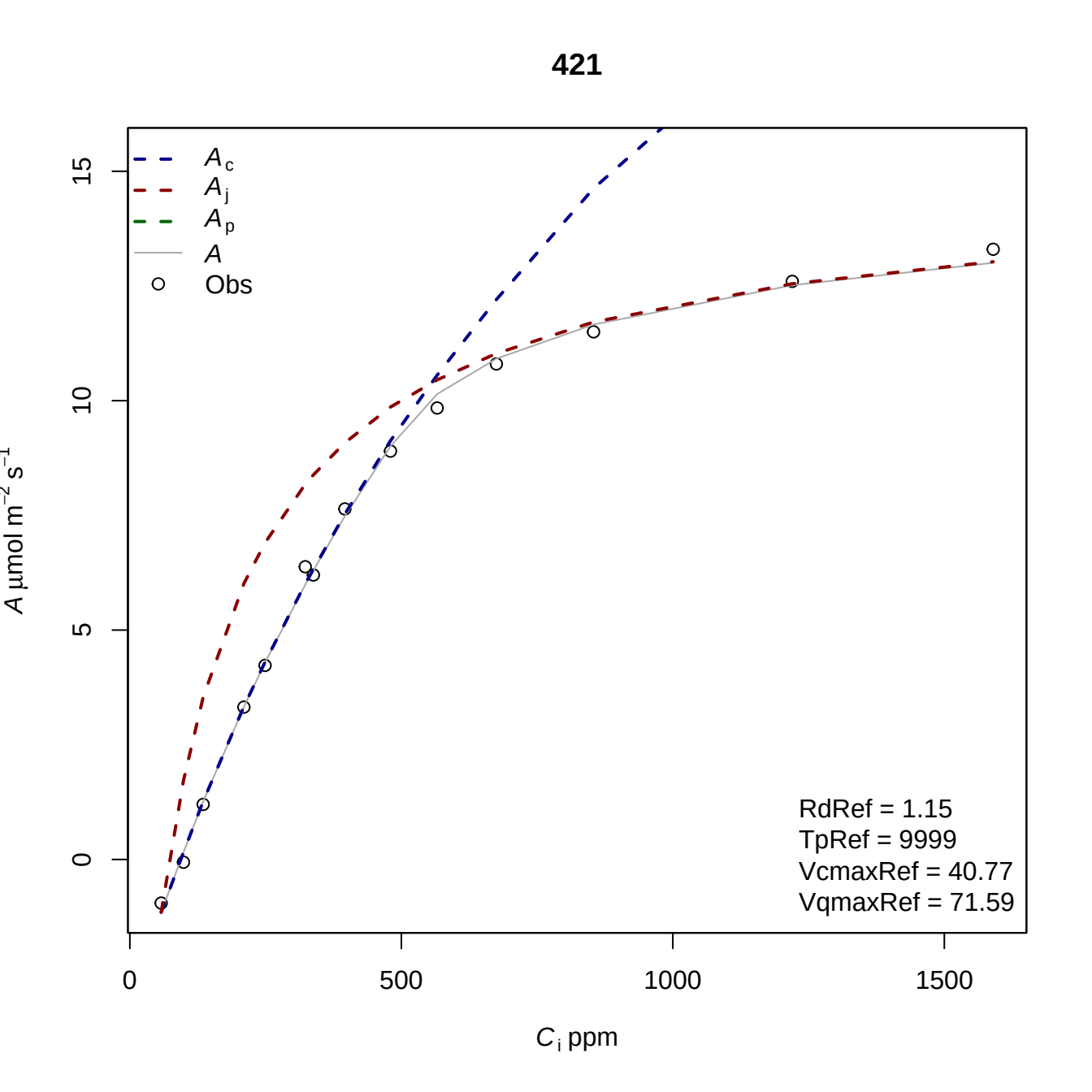

416
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.44
TpRef = 1.36
VcmaxRef = 19.79
VqmaxRef = 26.97
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
200
400
600
800
1000
1200
1400

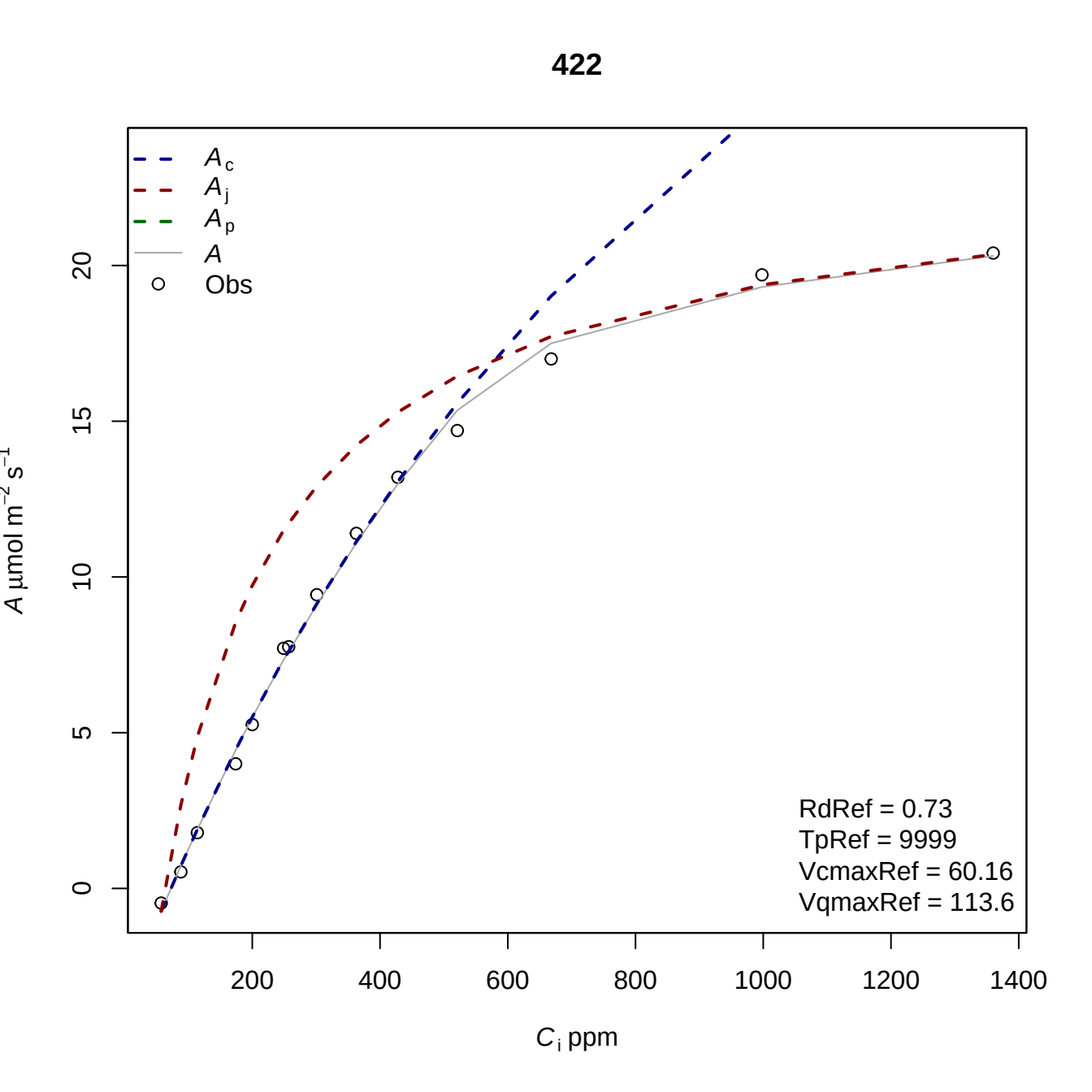

417
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.69
TpRef = 9999
VcmaxRef = 15.88
VqmaxRef = 27.5
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
500
1000
1500

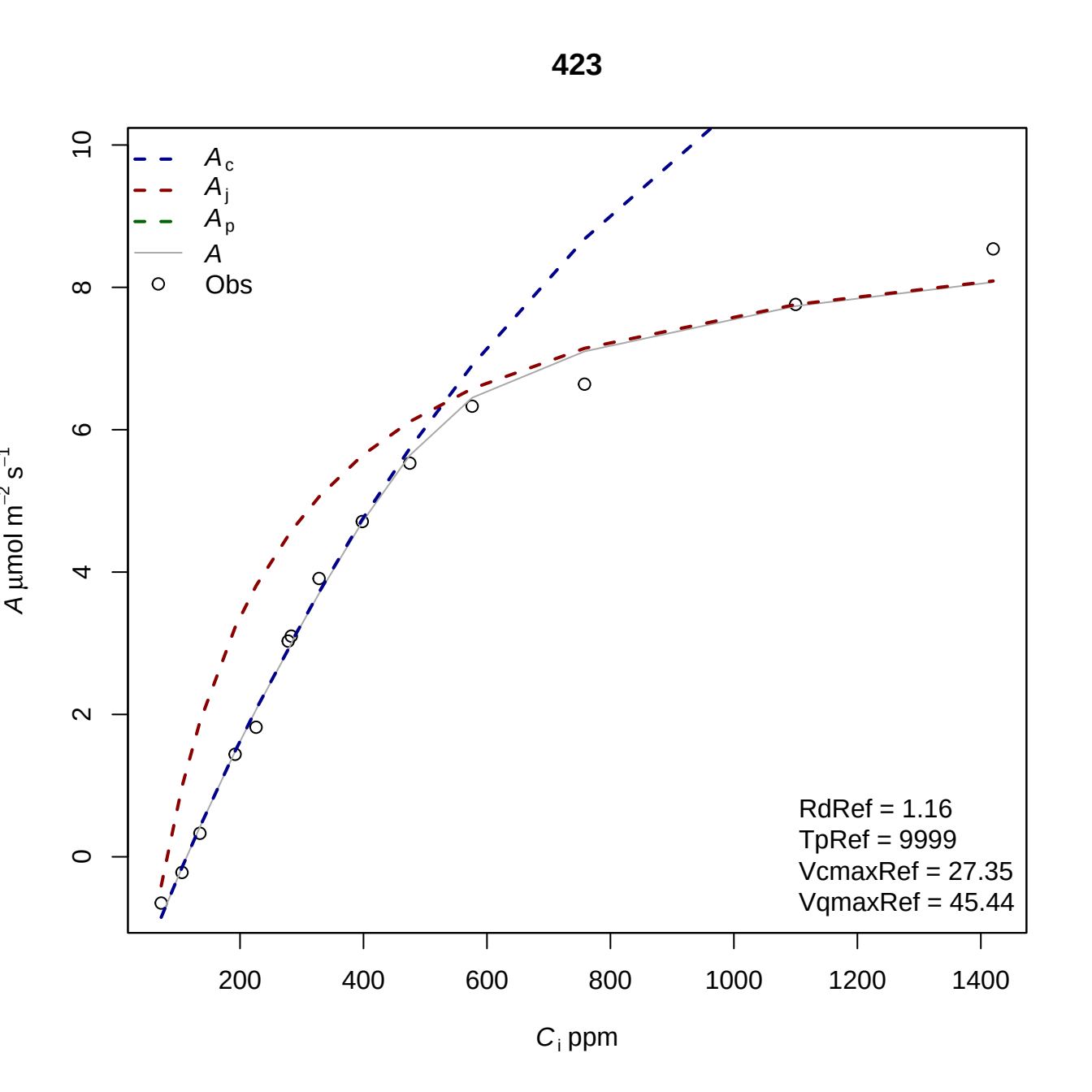

418
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.6
TpRef = 1.34
VcmaxRef = 15.21
VqmaxRef = 23.4
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
500
1000
1500

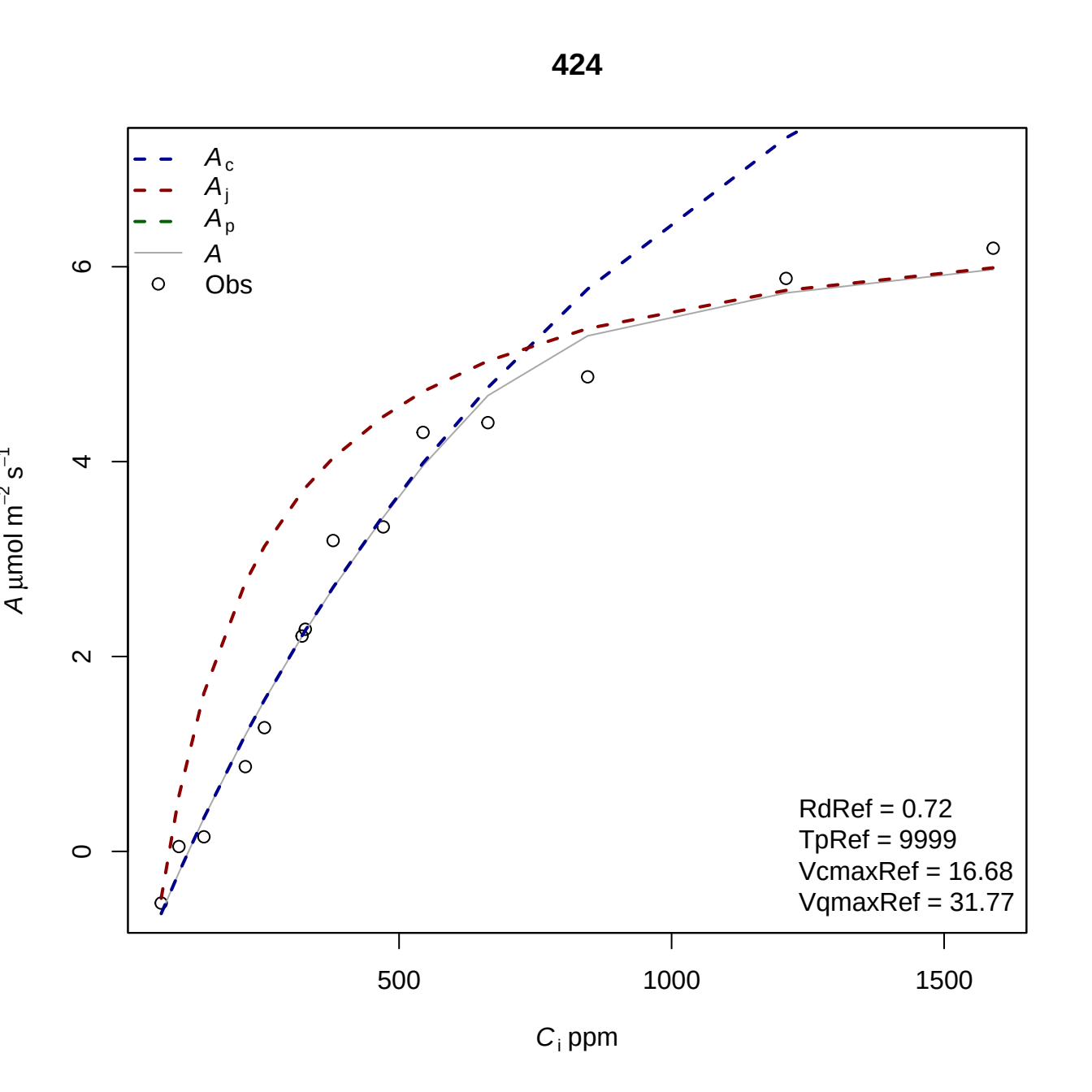

419
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.77
TpRef = 3.88
VcmaxRef = 51.22
VqmaxRef = 69.15
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200

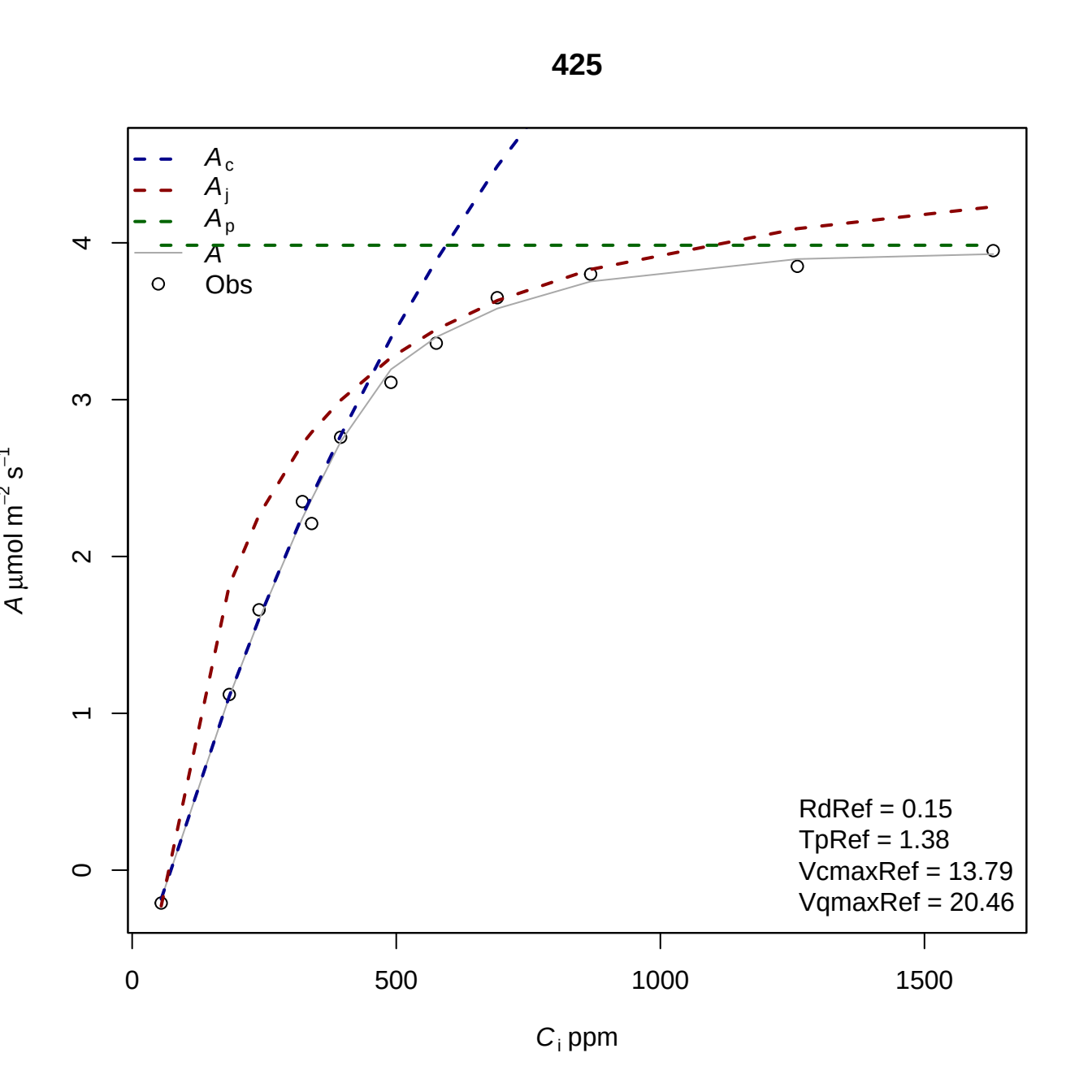

420
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.17
TpRef = 9999
VcmaxRef = 45.73
VqmaxRef = 81.81
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

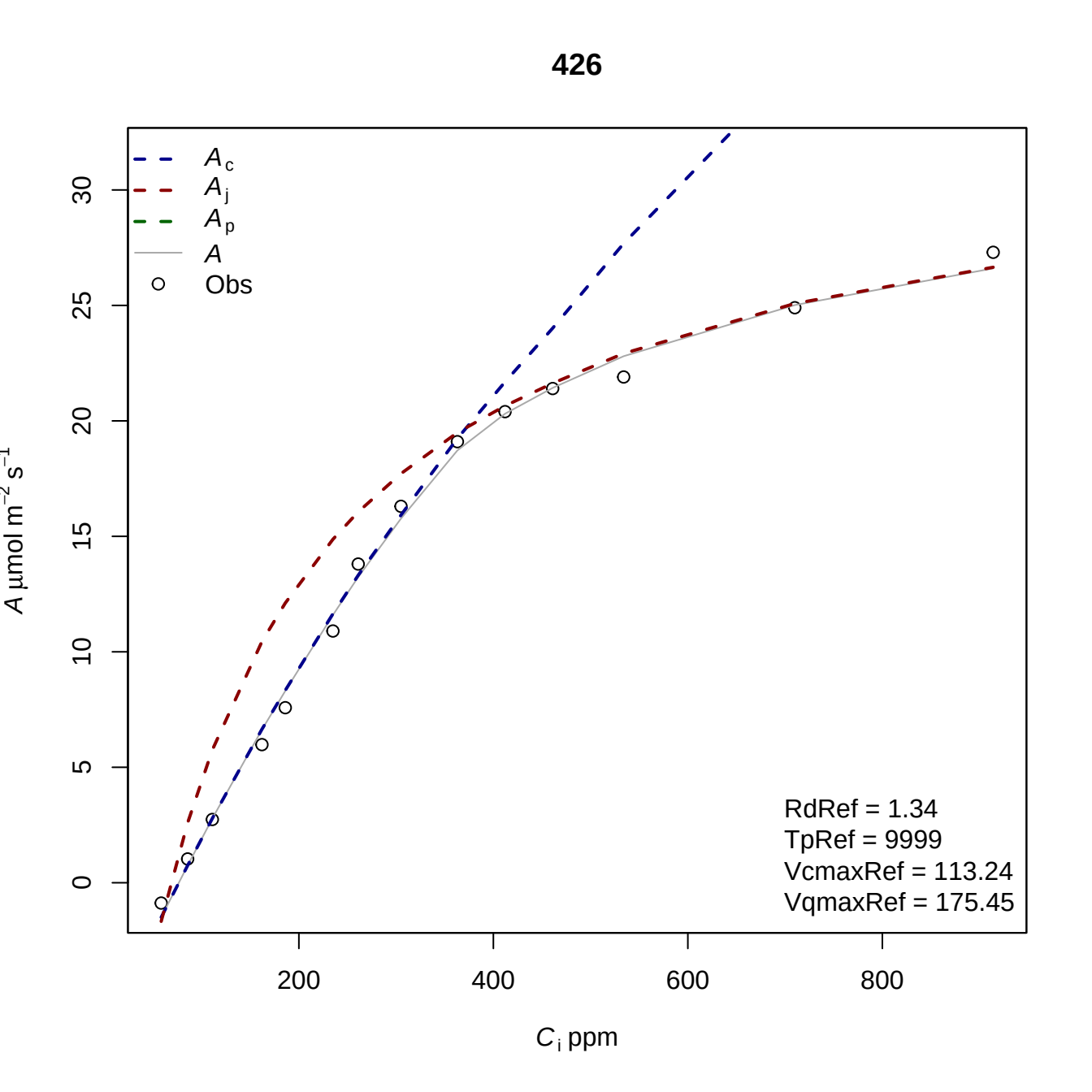

421
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.15
TpRef = 9999
VcmaxRef = 40.77
VqmaxRef = 71.59
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

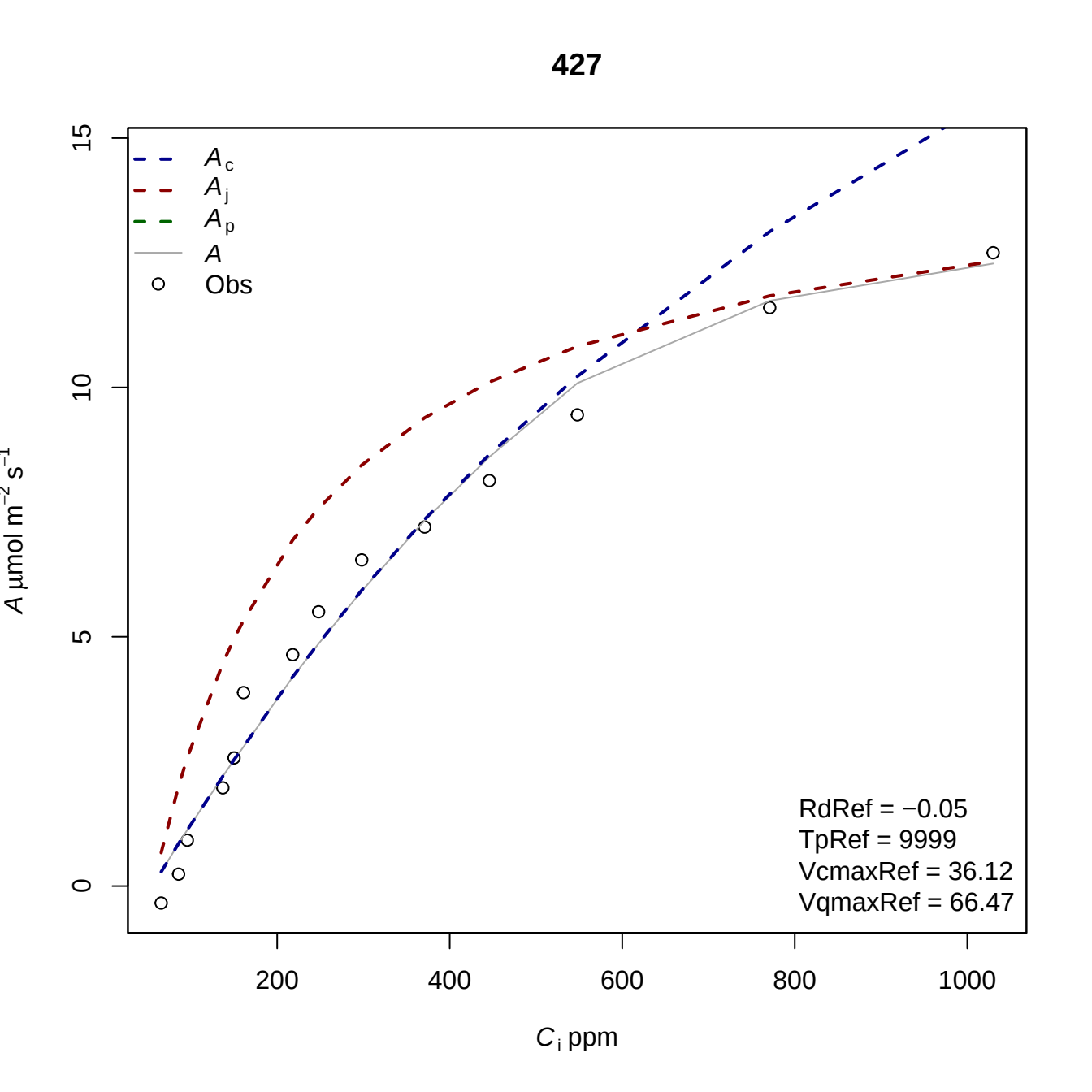

422
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.73
TpRef = 9999
VcmaxRef = 60.16
VqmaxRef = 113.6
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

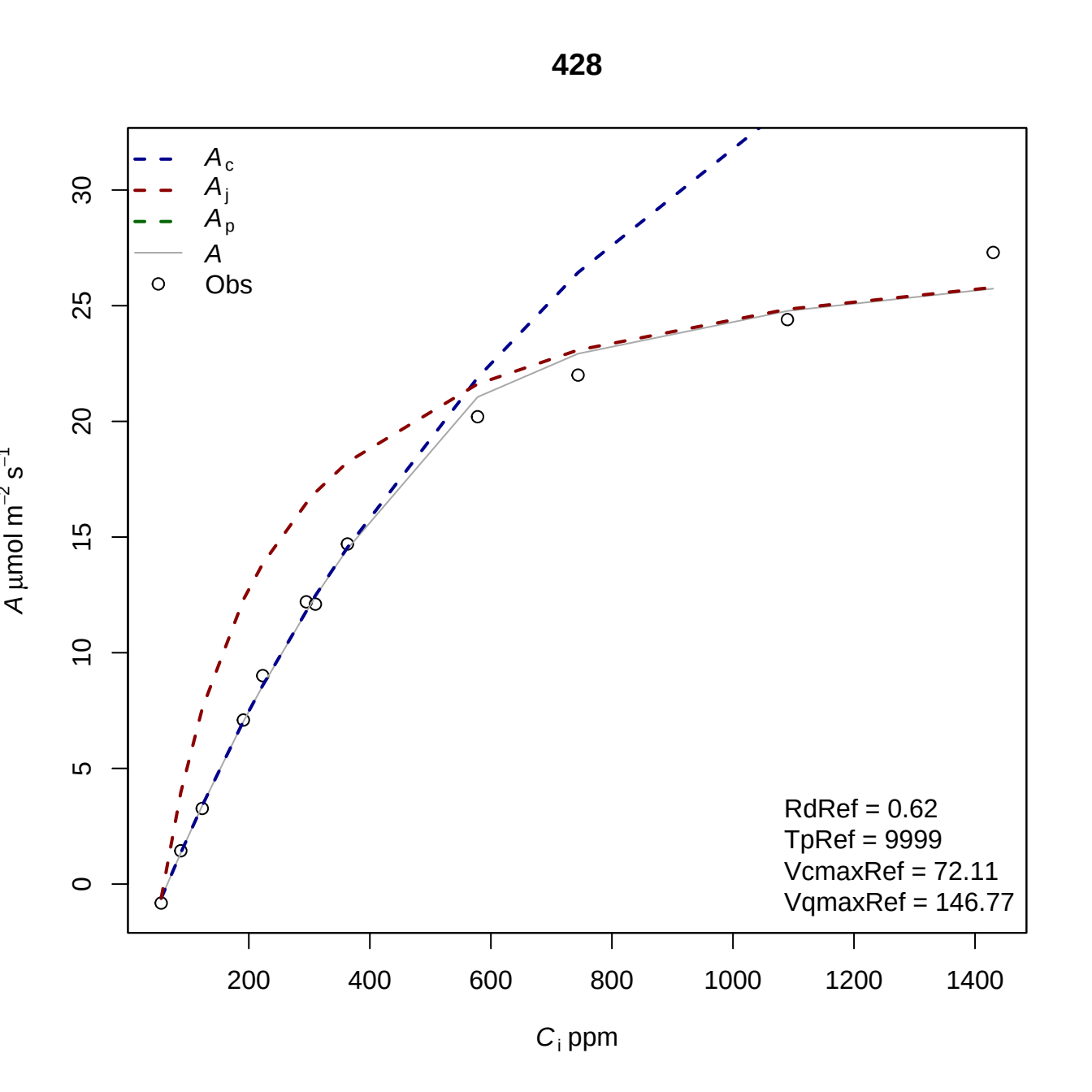

423
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.16
TpRef = 9999
VcmaxRef = 27.35
VqmaxRef = 45.44
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

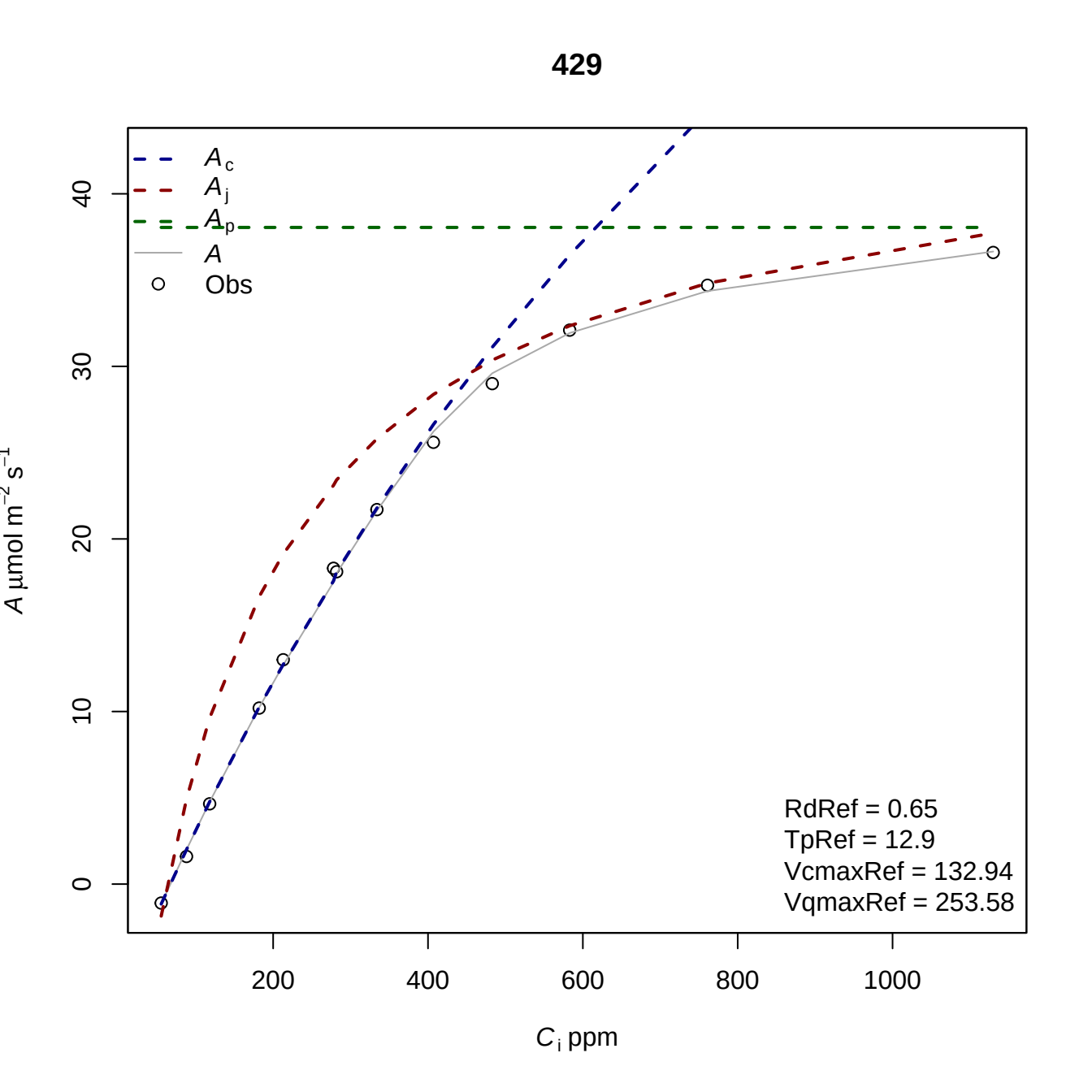

424
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.72
TpRef = 9999
VcmaxRef = 16.68
VqmaxRef = 31.77
0
2
4
6
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

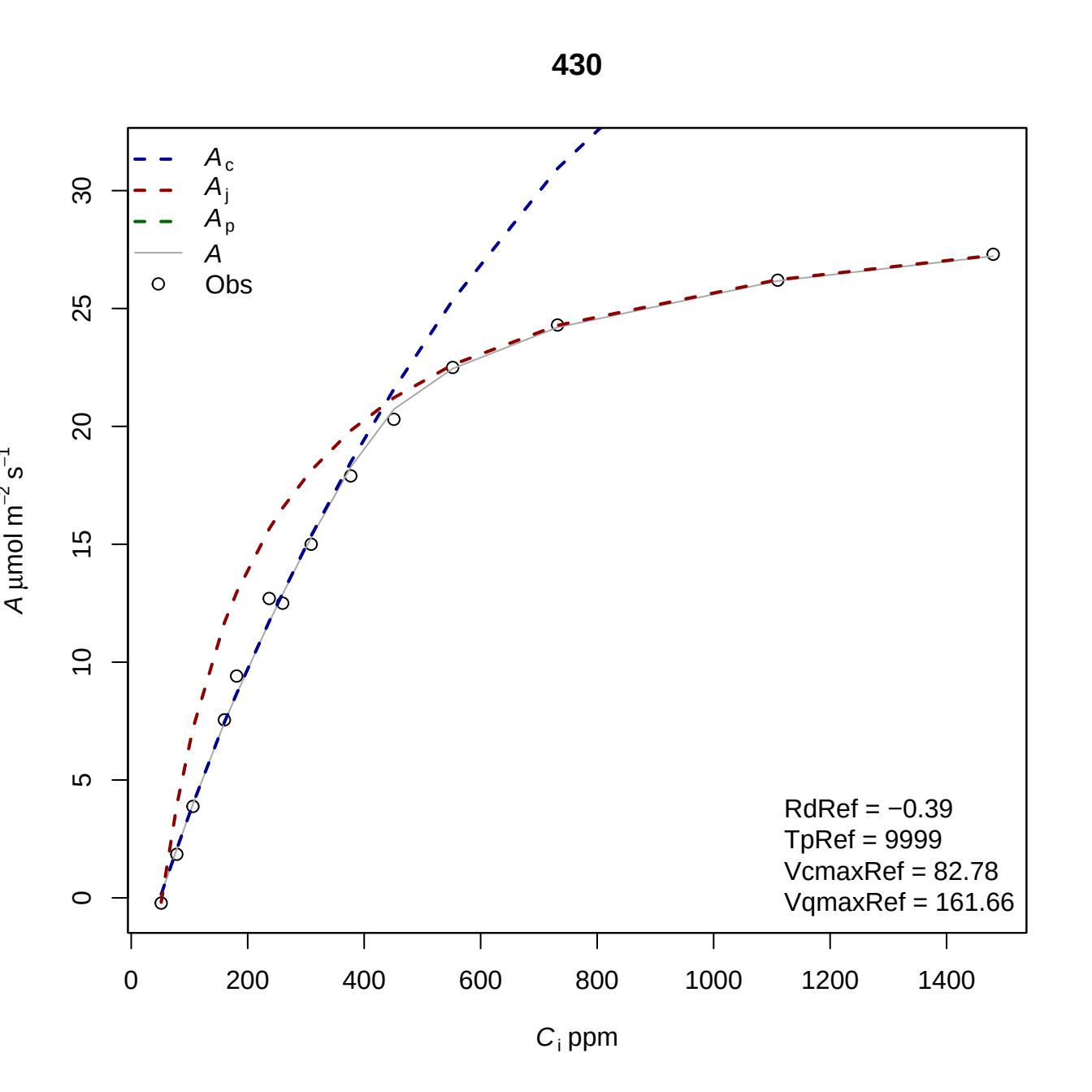

425
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.15
TpRef = 1.38
VcmaxRef = 13.79
VqmaxRef = 20.46
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
0
500
1000
1500

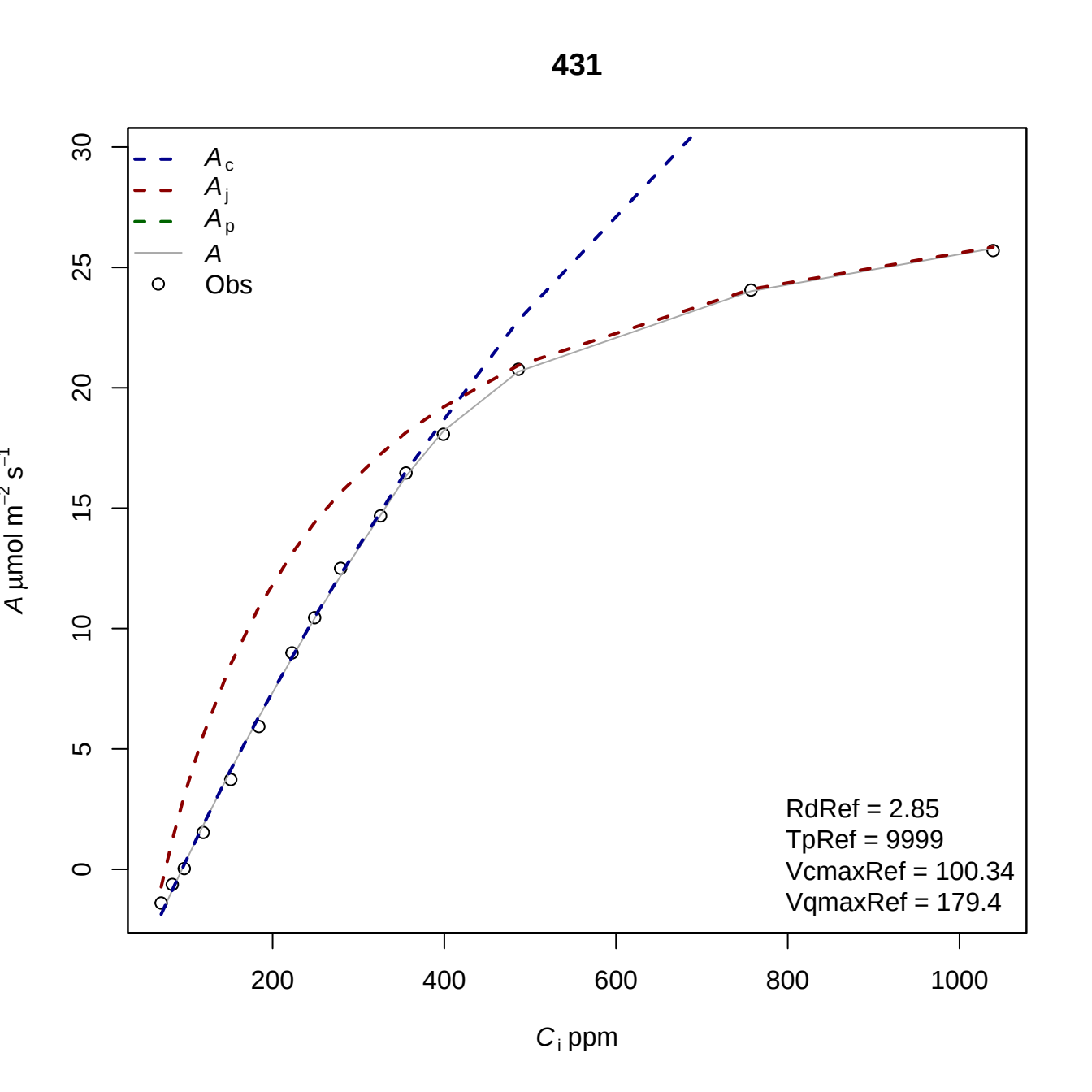

426
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.34
TpRef = 9999
VcmaxRef = 113.24
VqmaxRef = 175.45
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800

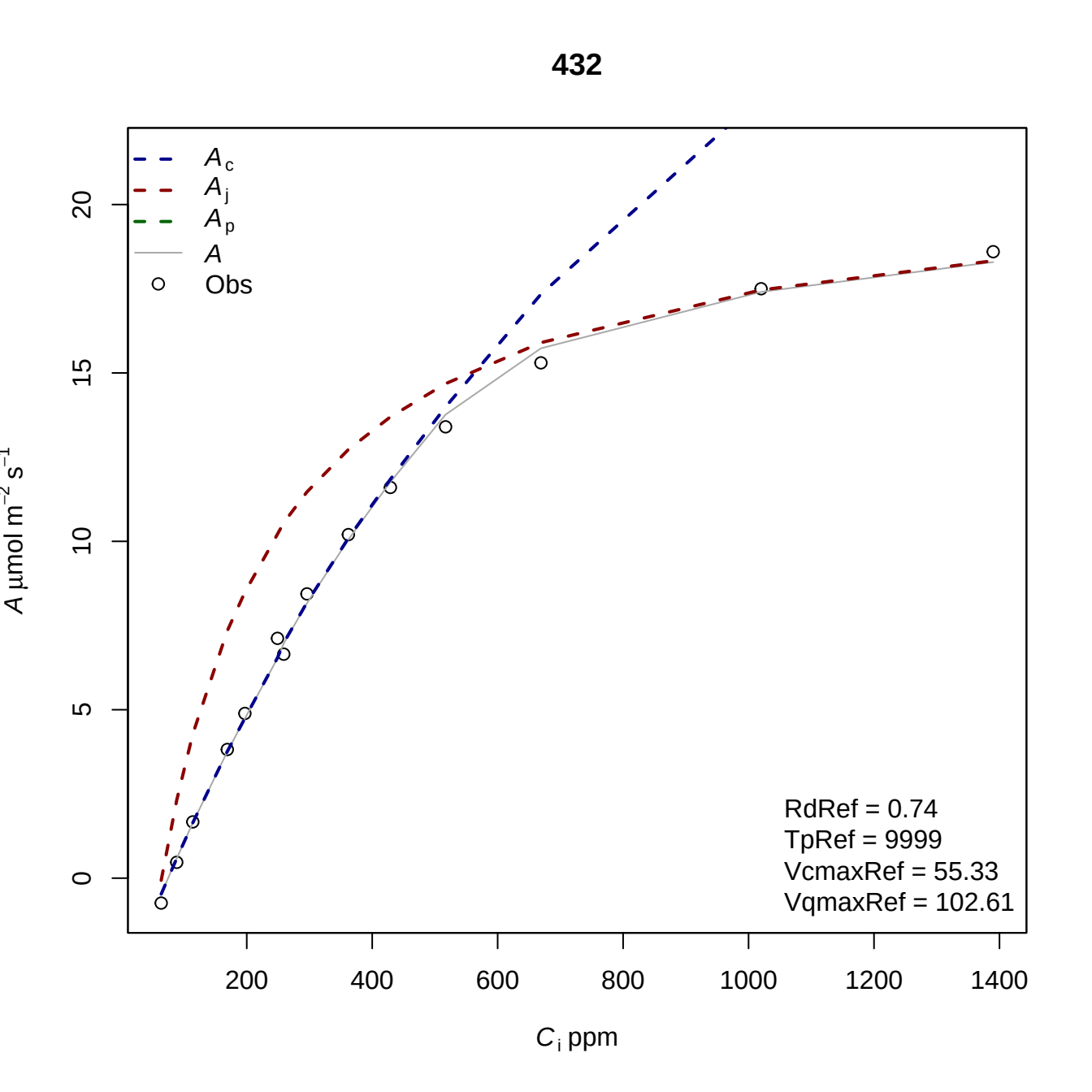

427
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.05
TpRef = 9999
VcmaxRef = 36.12
VqmaxRef = 66.47
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

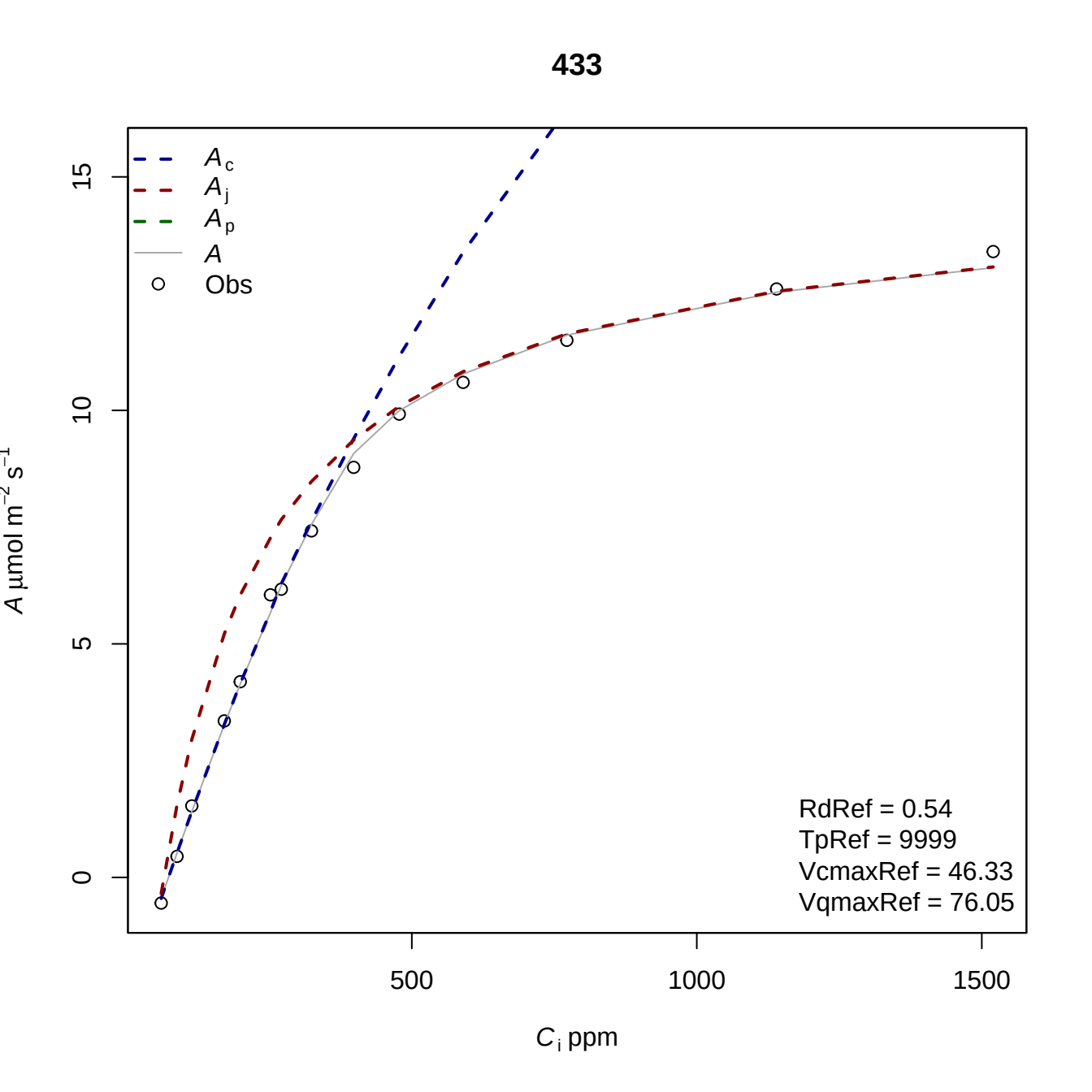

428
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.62
TpRef = 9999
VcmaxRef = 72.11
VqmaxRef = 146.77
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200
1400

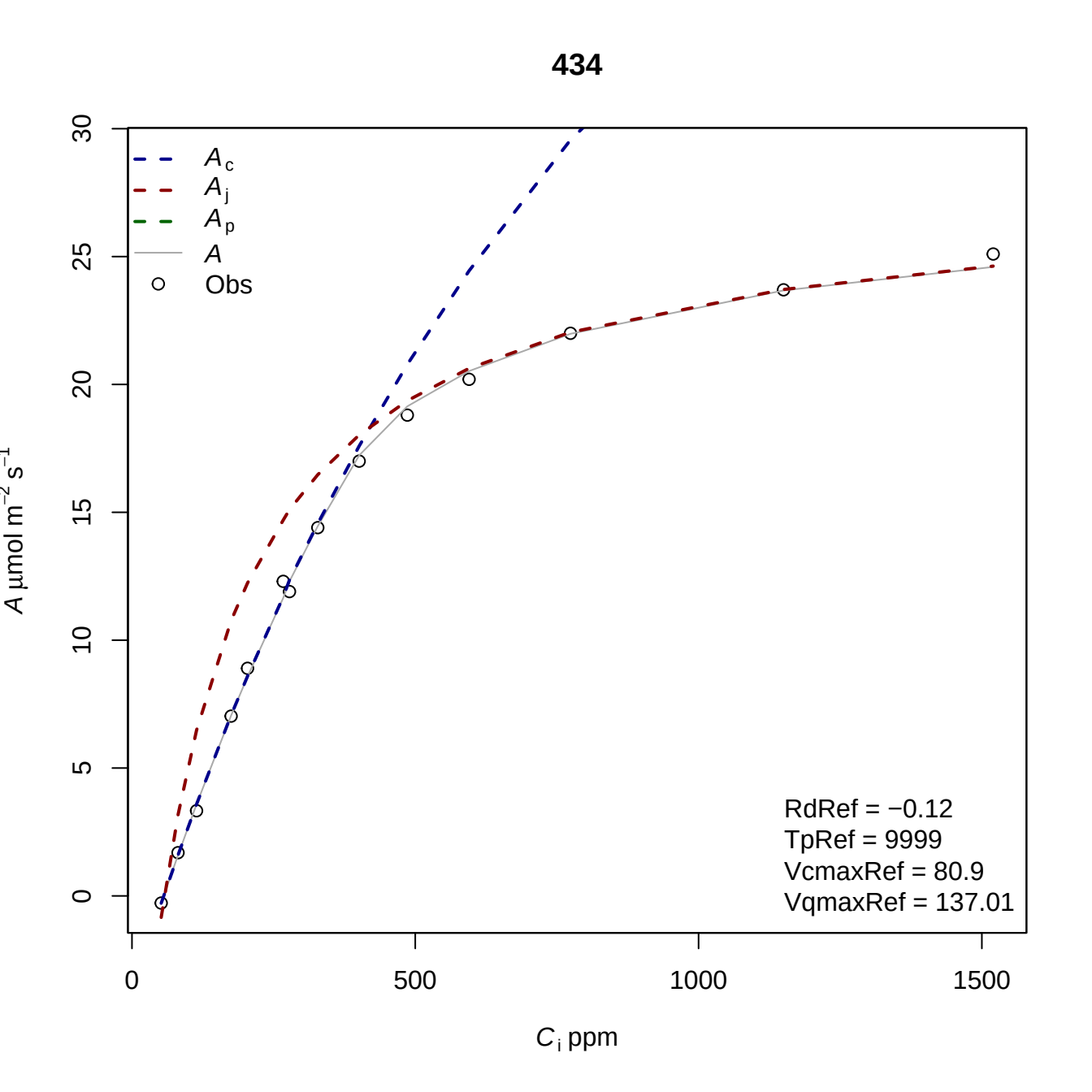

429
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.65
TpRef = 12.9
VcmaxRef = 132.94
VqmaxRef = 253.58
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000

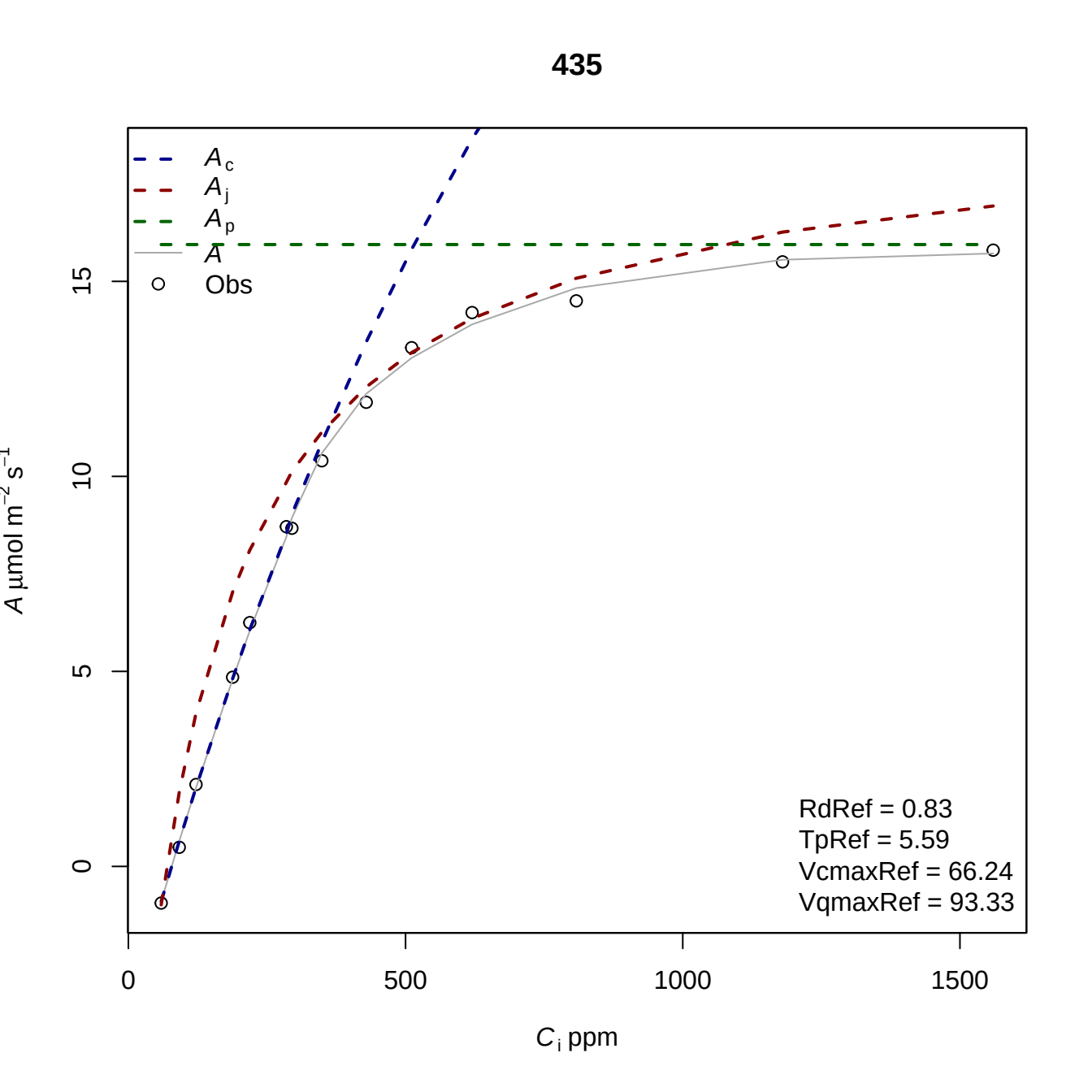

430
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.39
TpRef = 9999
VcmaxRef = 82.78
VqmaxRef = 161.66
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
1400

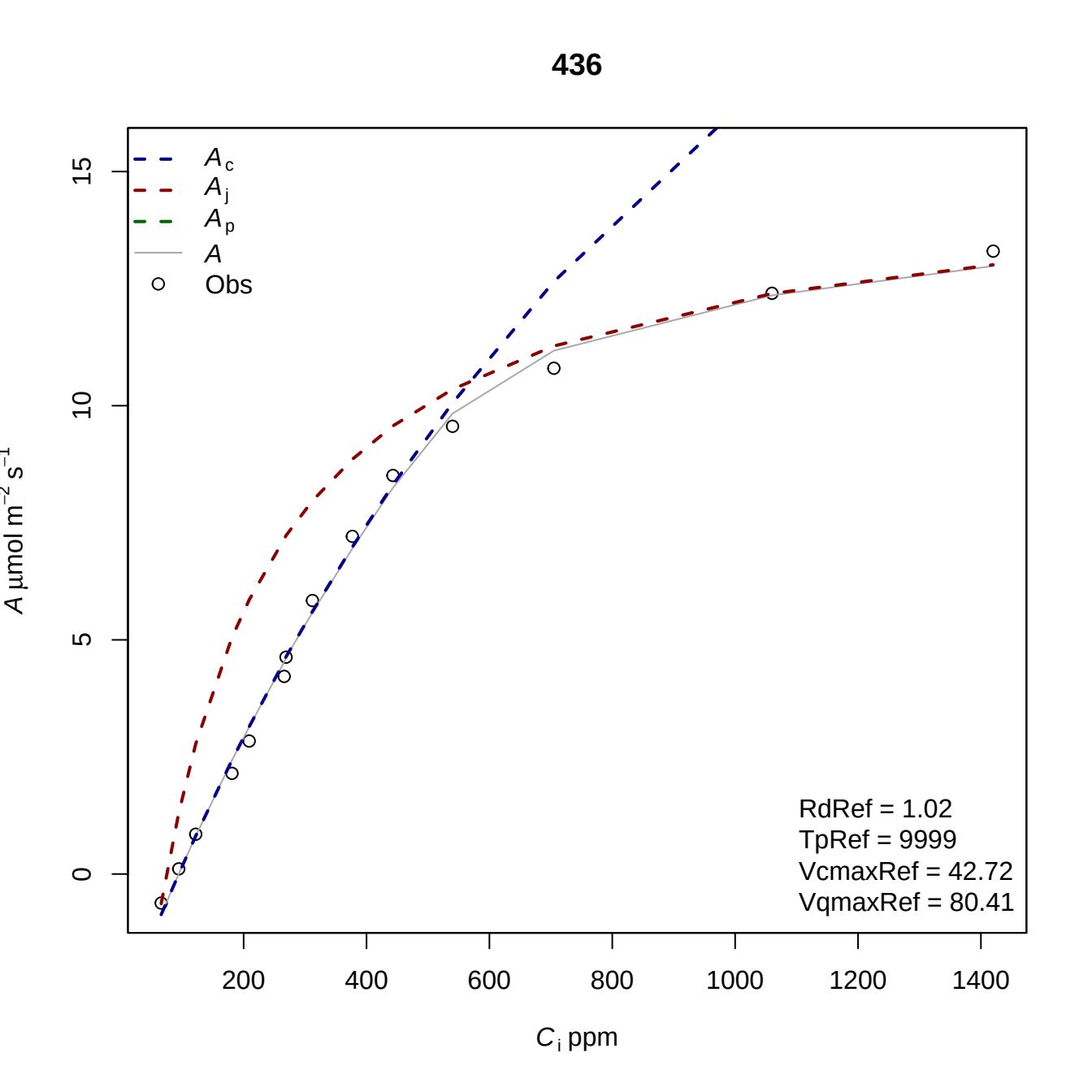

431
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.85
TpRef = 9999
VcmaxRef = 100.34
VqmaxRef = 179.4
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000

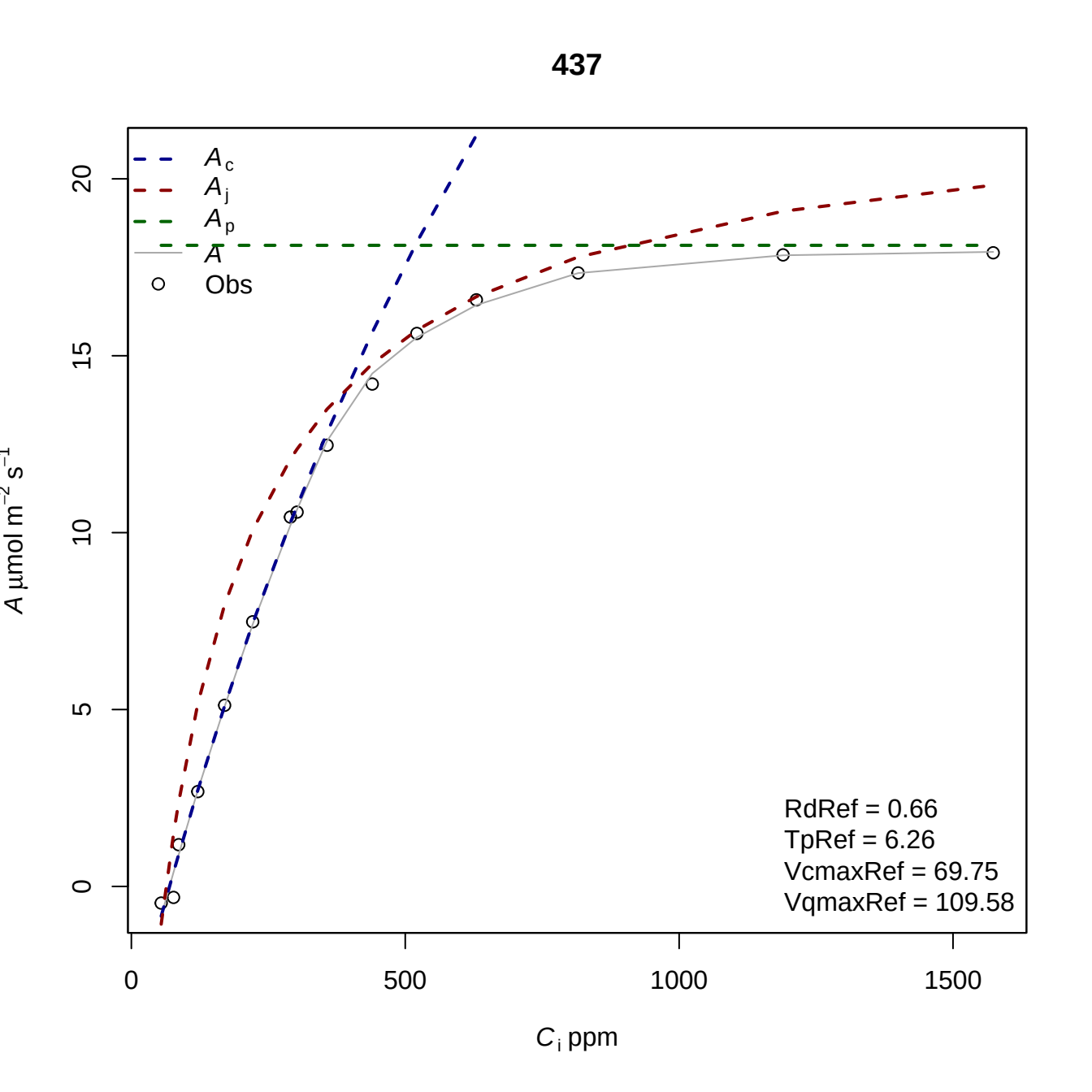

432
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.74
TpRef = 9999
VcmaxRef = 55.33
VqmaxRef = 102.61
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

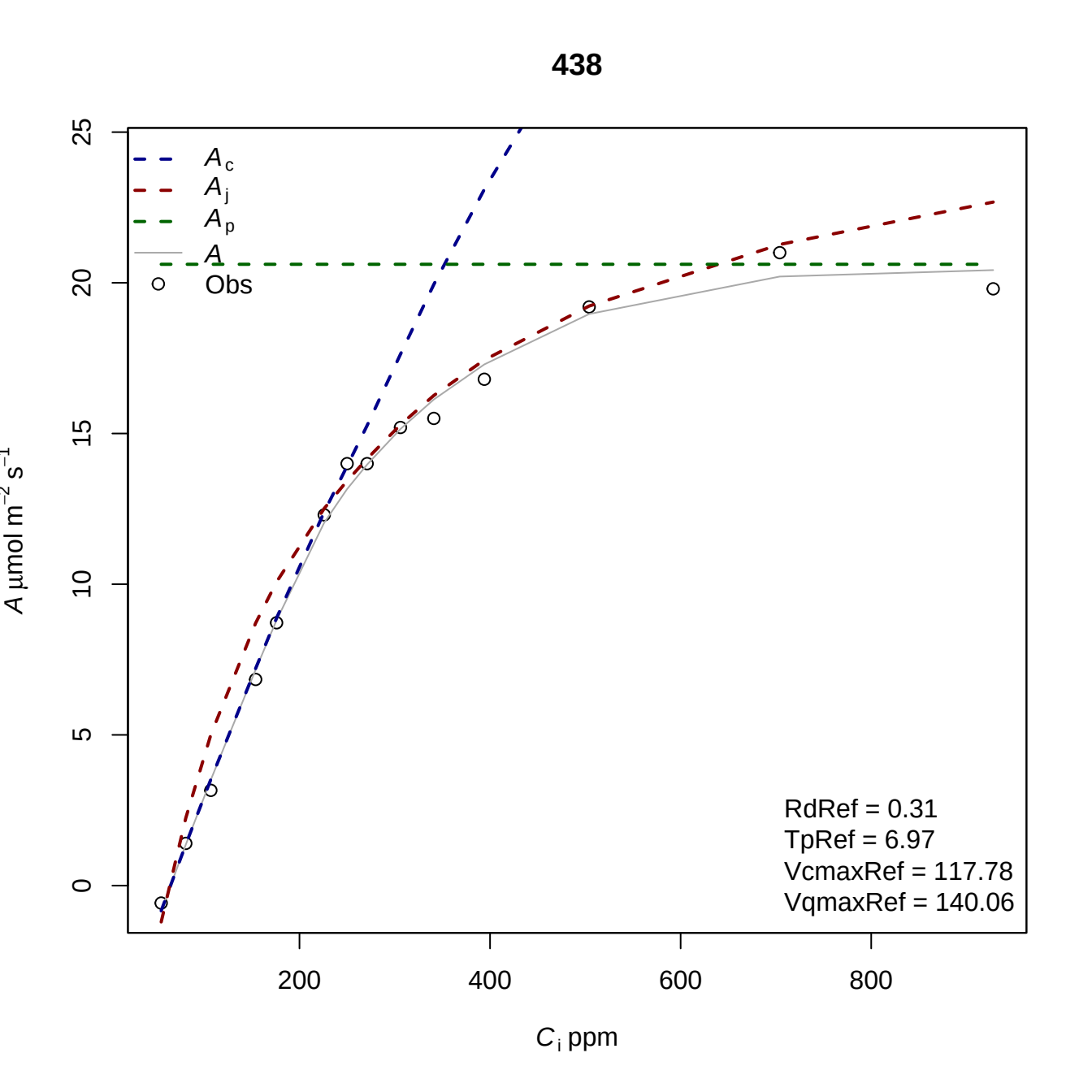

433
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.54
TpRef = 9999
VcmaxRef = 46.33
VqmaxRef = 76.05
0
5
10
15
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

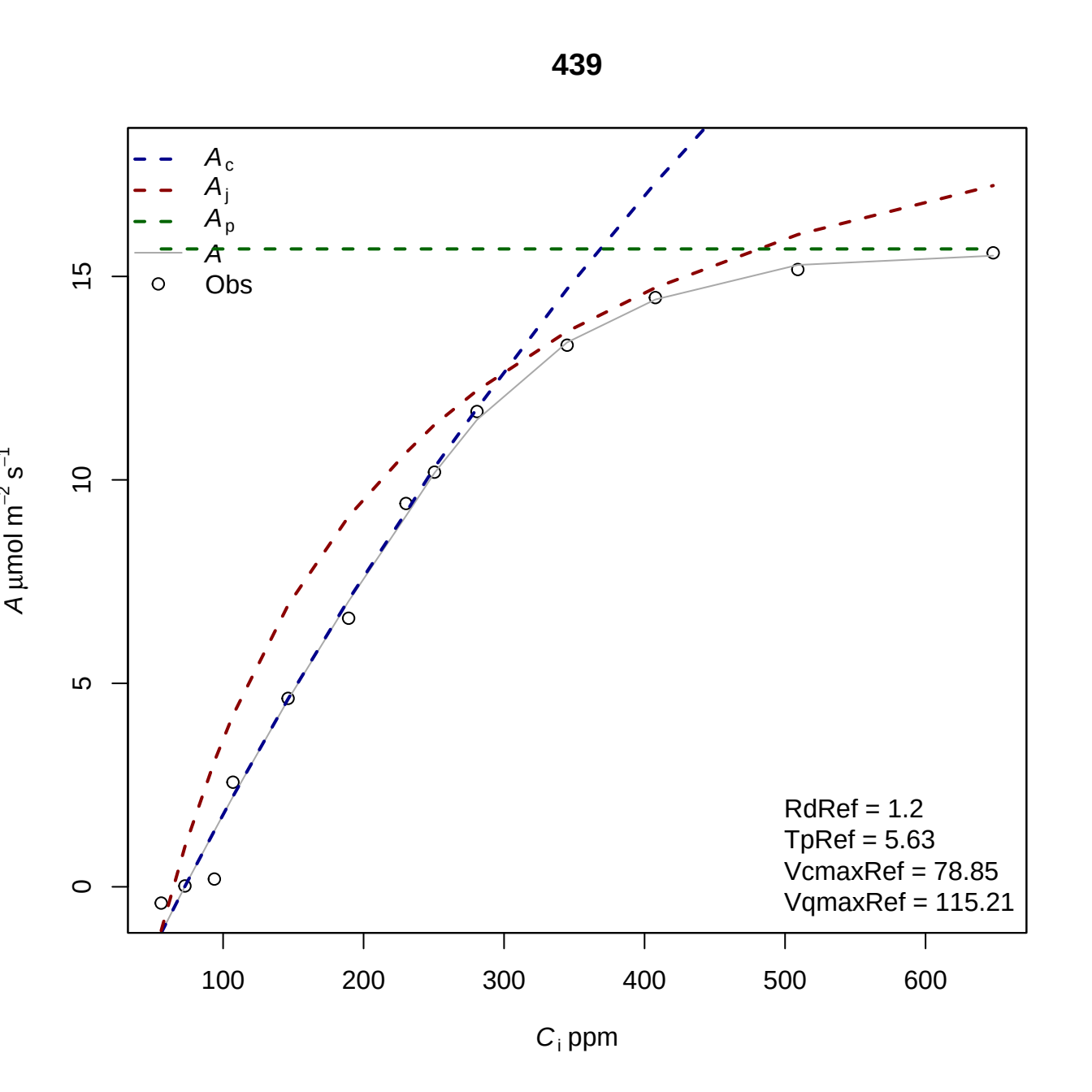

434
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = −0.12
TpRef = 9999
VcmaxRef = 80.9
VqmaxRef = 137.01
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

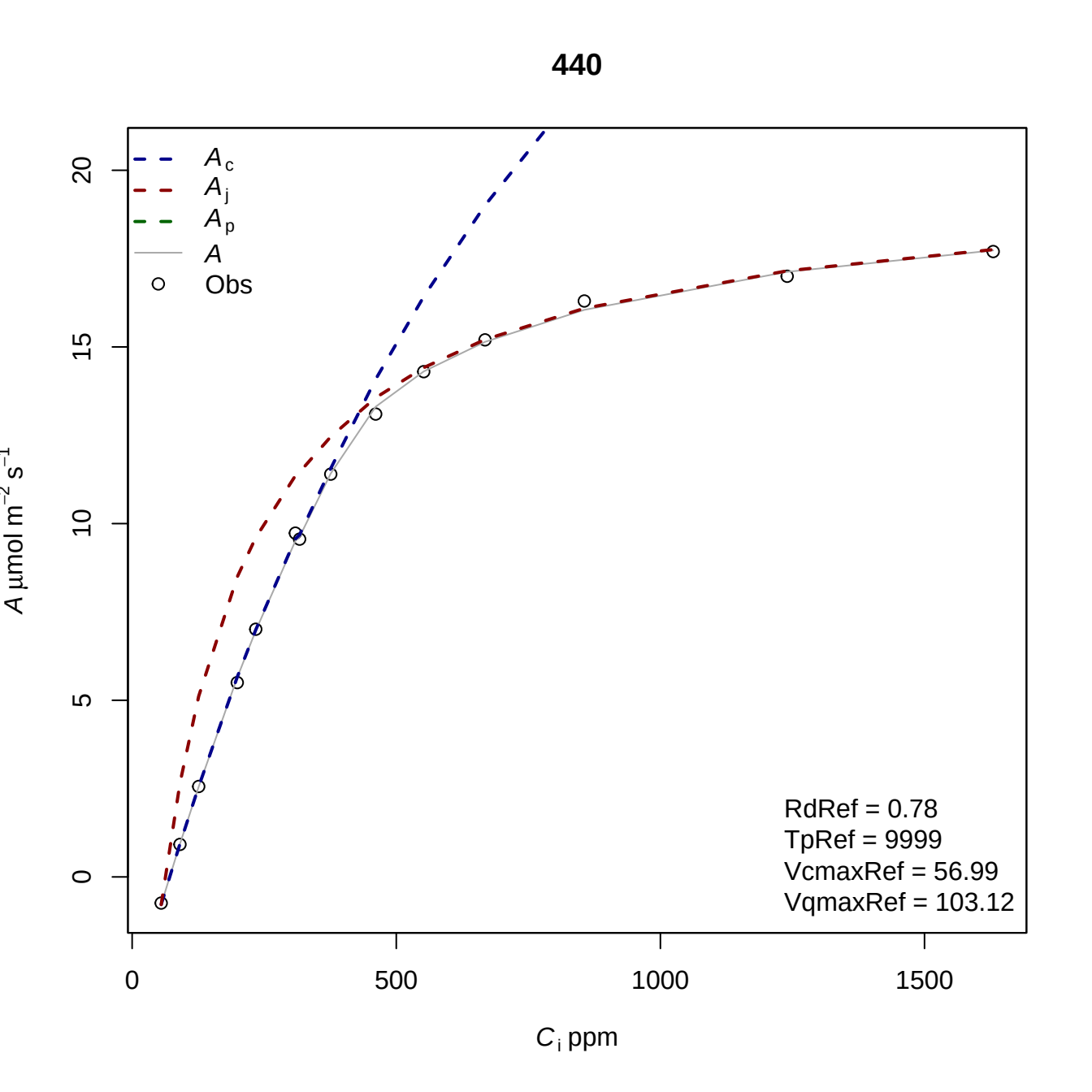

435
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.83
TpRef = 5.59
VcmaxRef = 66.24
VqmaxRef = 93.33
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

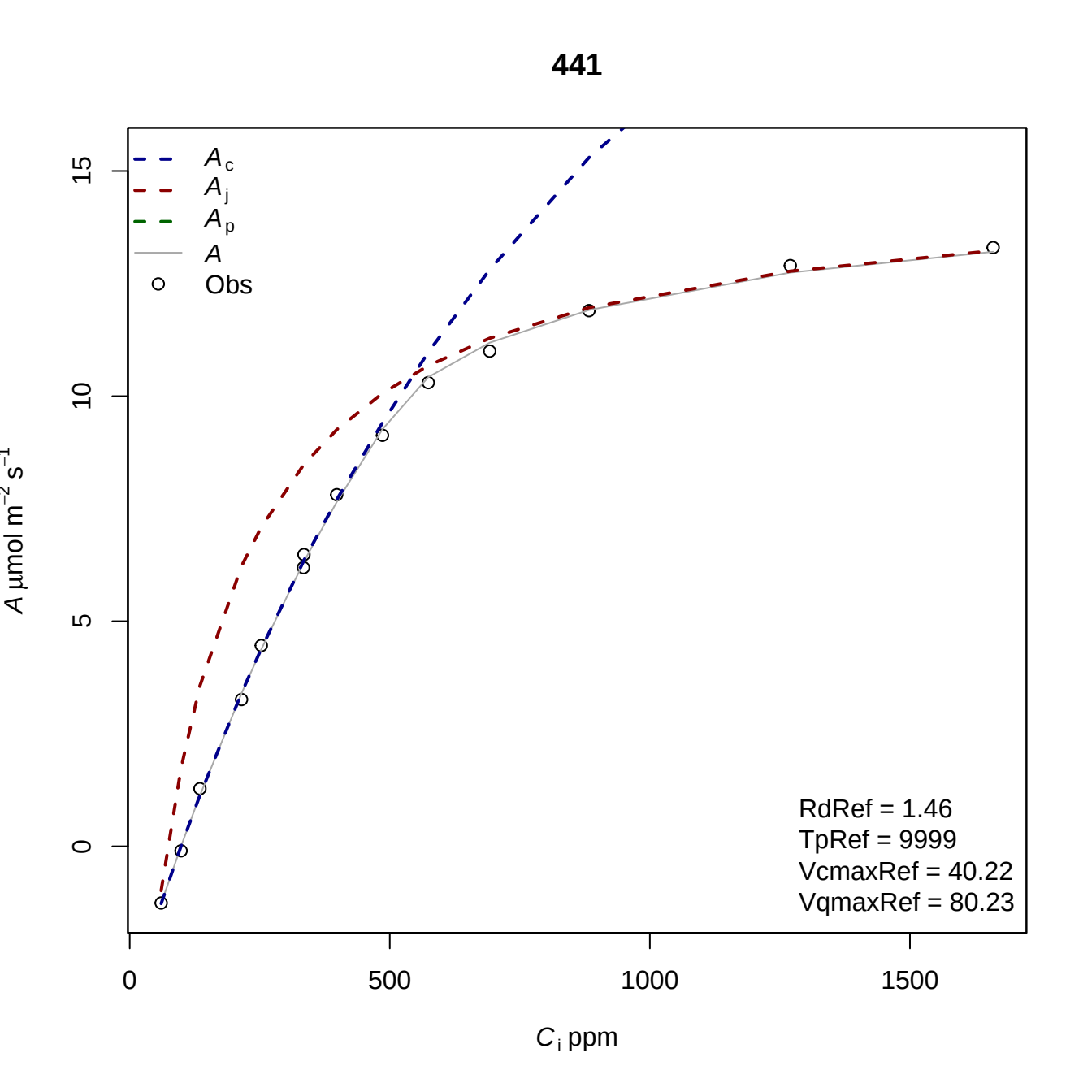

436
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.02
TpRef = 9999
VcmaxRef = 42.72
VqmaxRef = 80.41
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

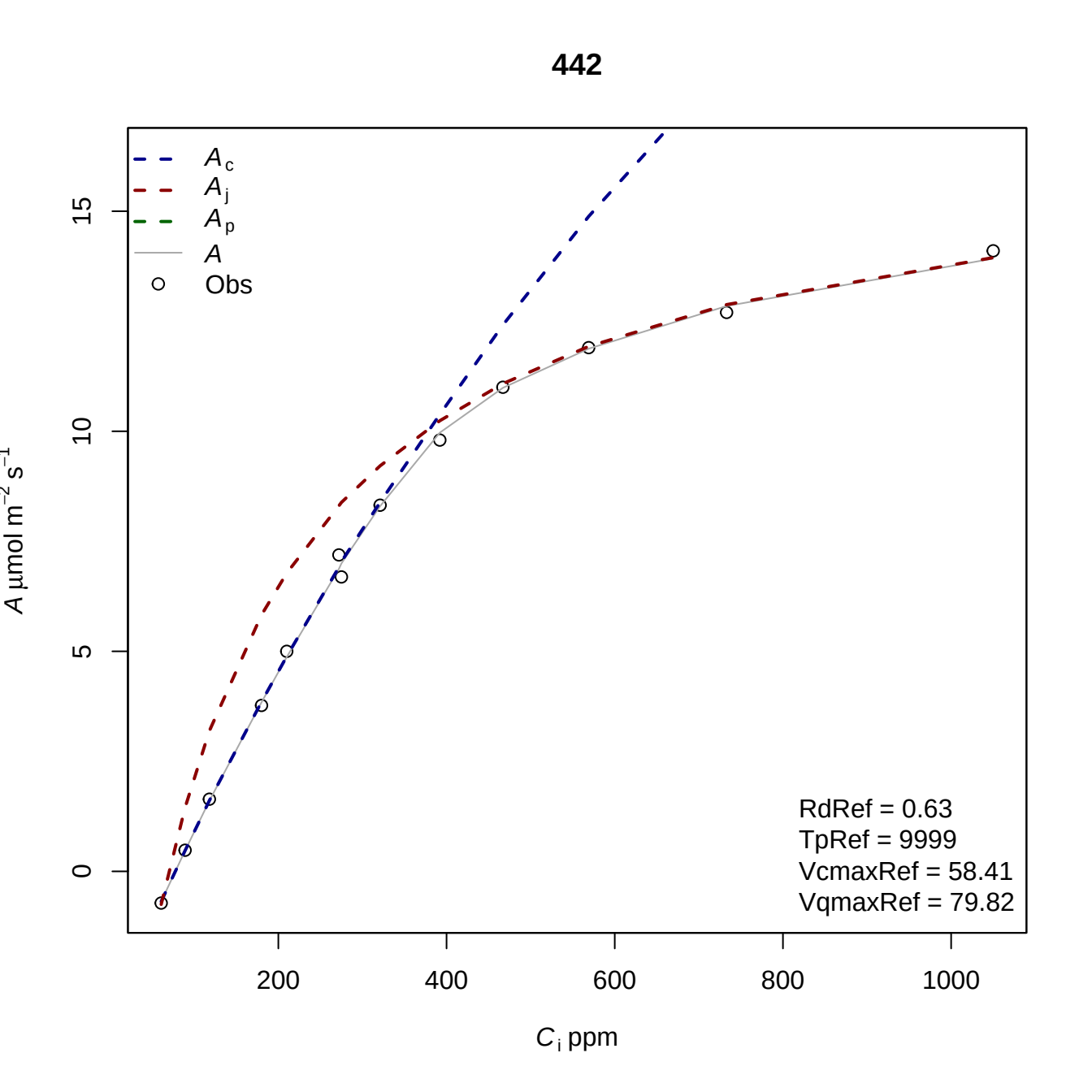

437
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.66
TpRef = 6.26
VcmaxRef = 69.75
VqmaxRef = 109.58
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

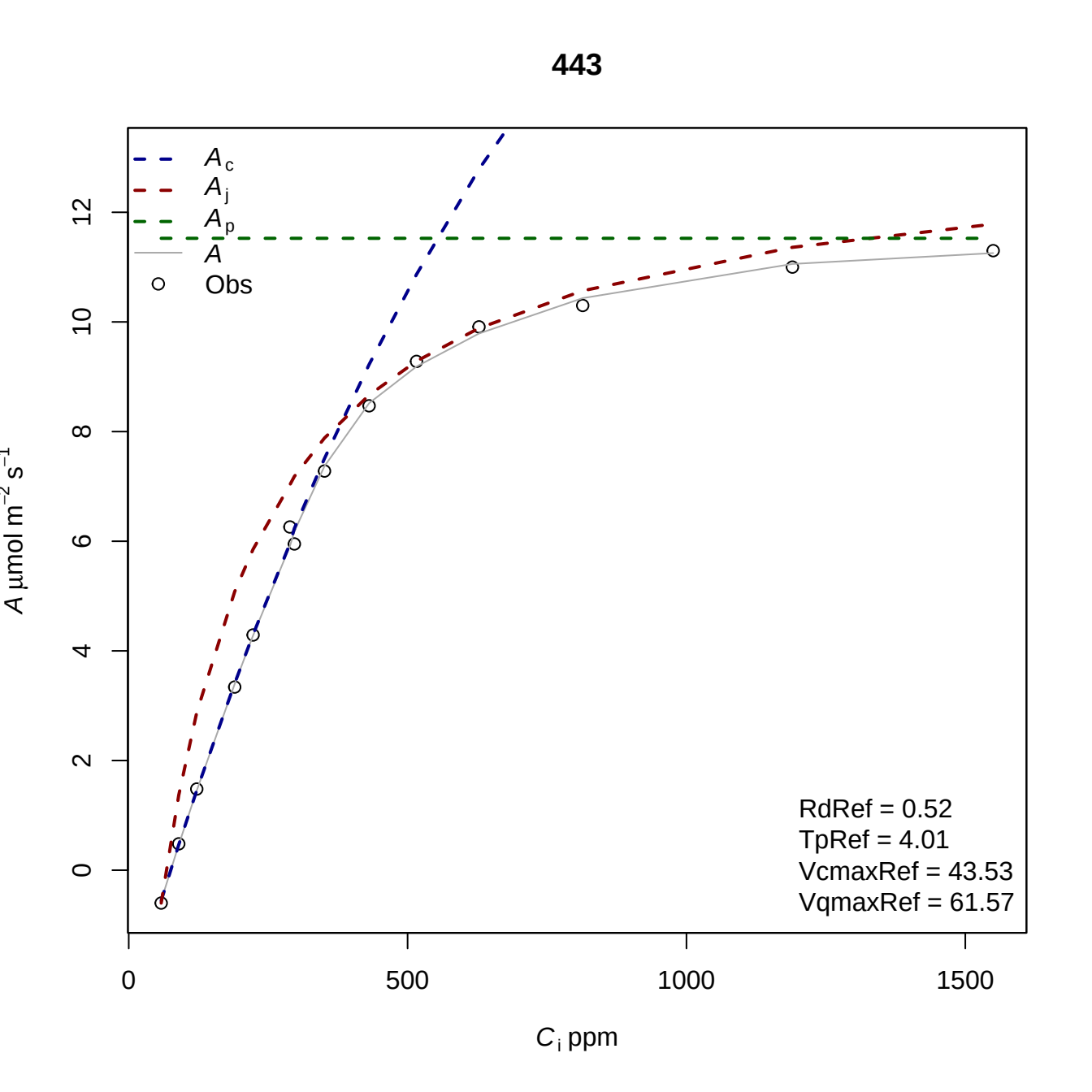

438
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.31
TpRef = 6.97
VcmaxRef = 117.78
VqmaxRef = 140.06
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800

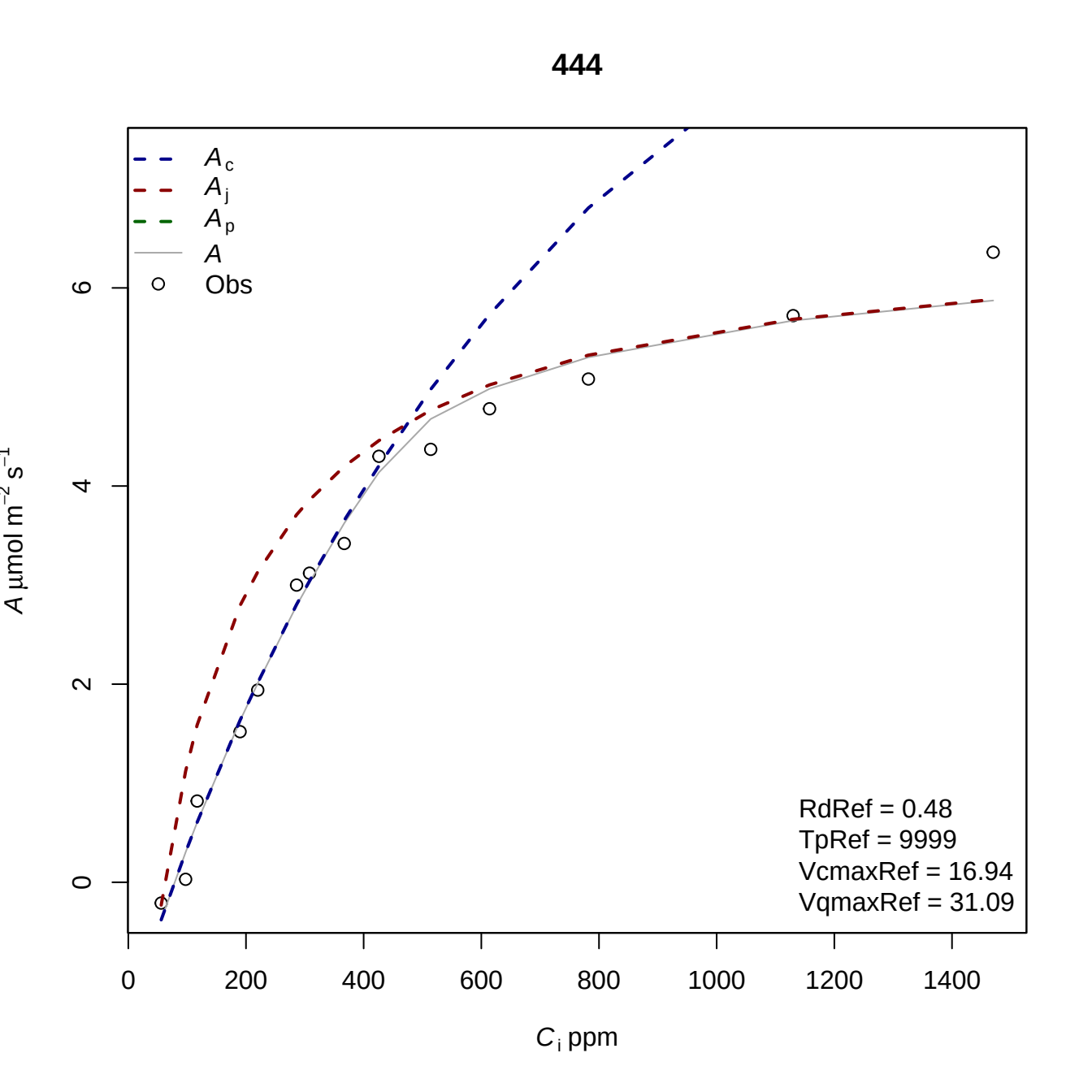

439
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.2
TpRef = 5.63
VcmaxRef = 78.85
VqmaxRef = 115.21
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500
600

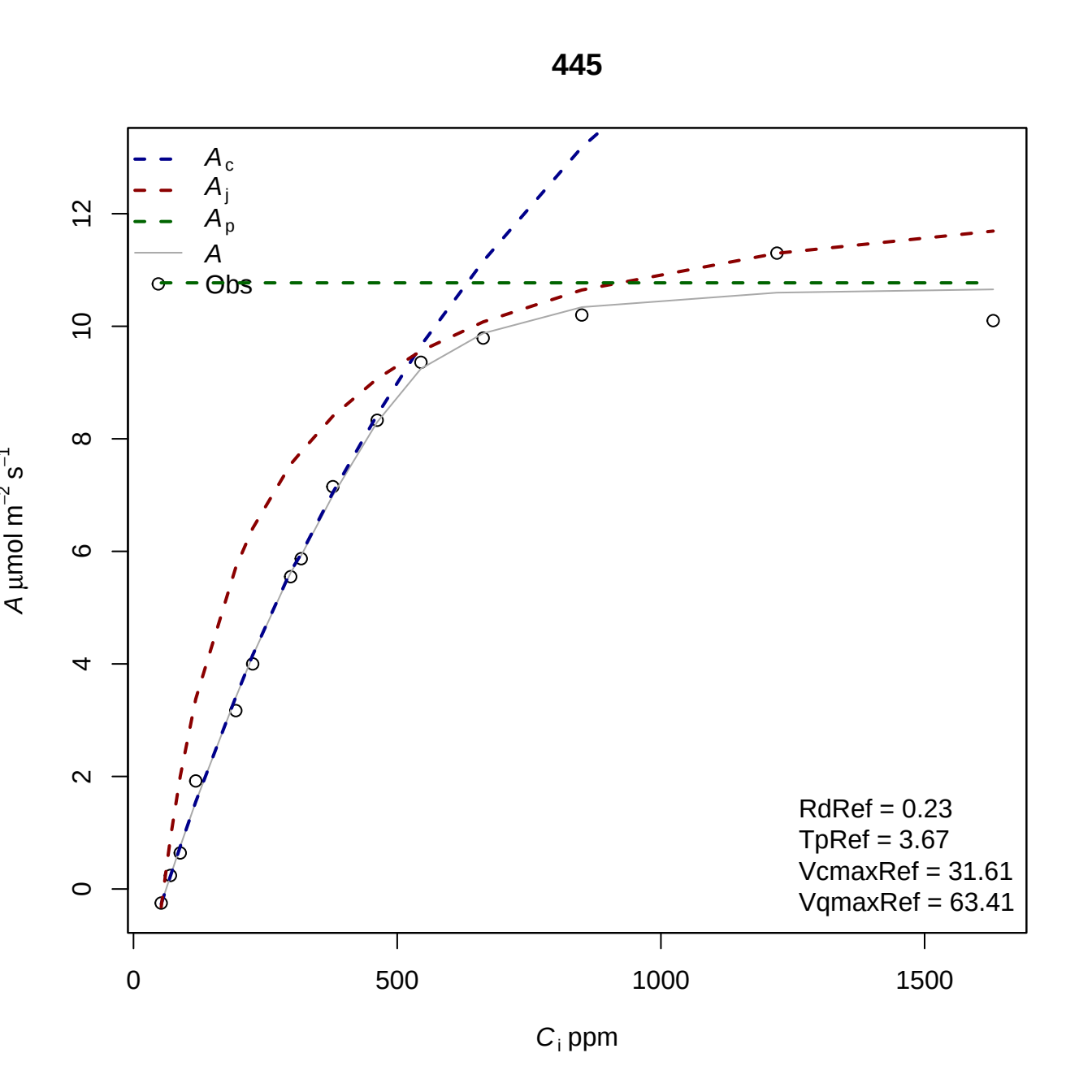

440
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.78
TpRef = 9999
VcmaxRef = 56.99
VqmaxRef = 103.12
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

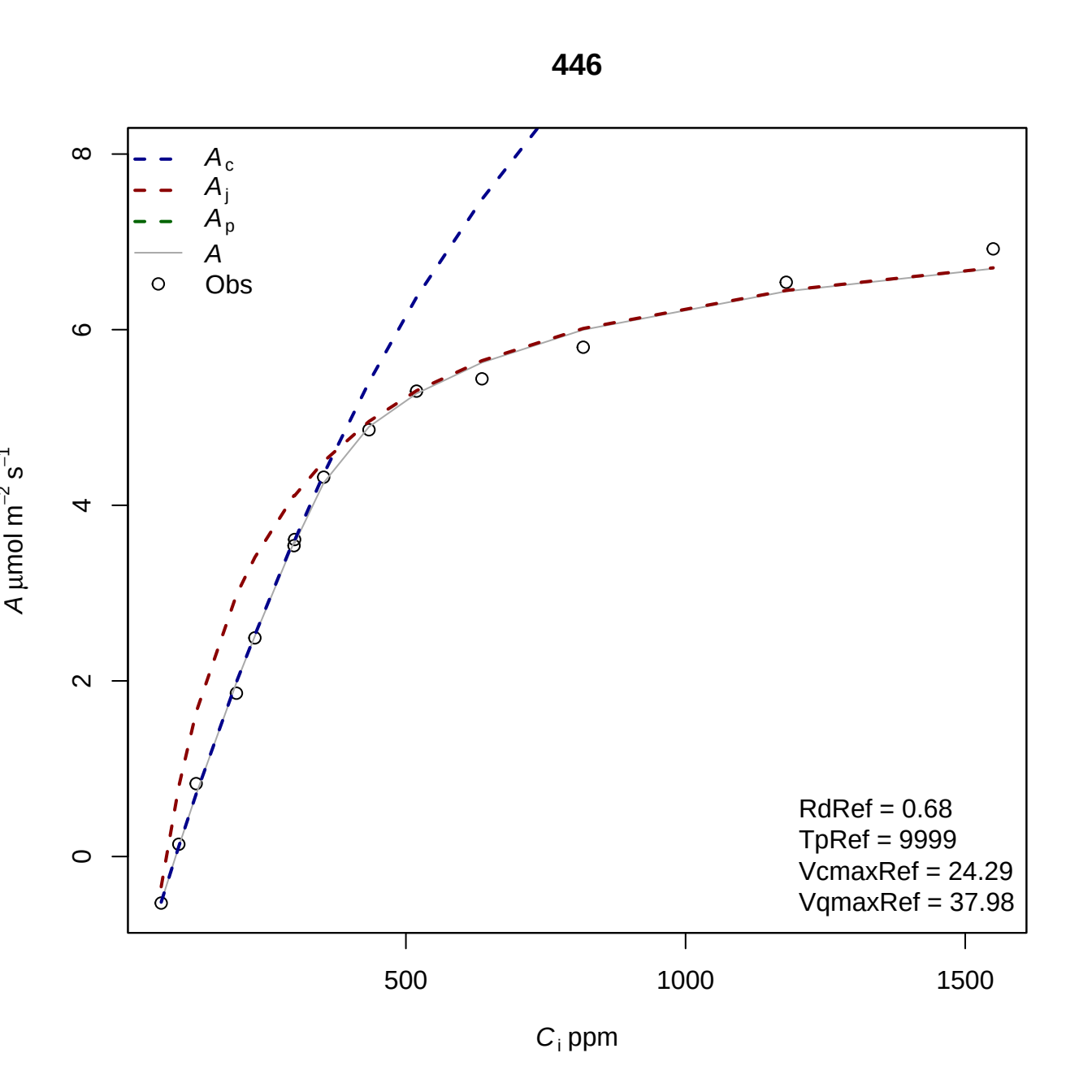

441
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.46
TpRef = 9999
VcmaxRef = 40.22
VqmaxRef = 80.23
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

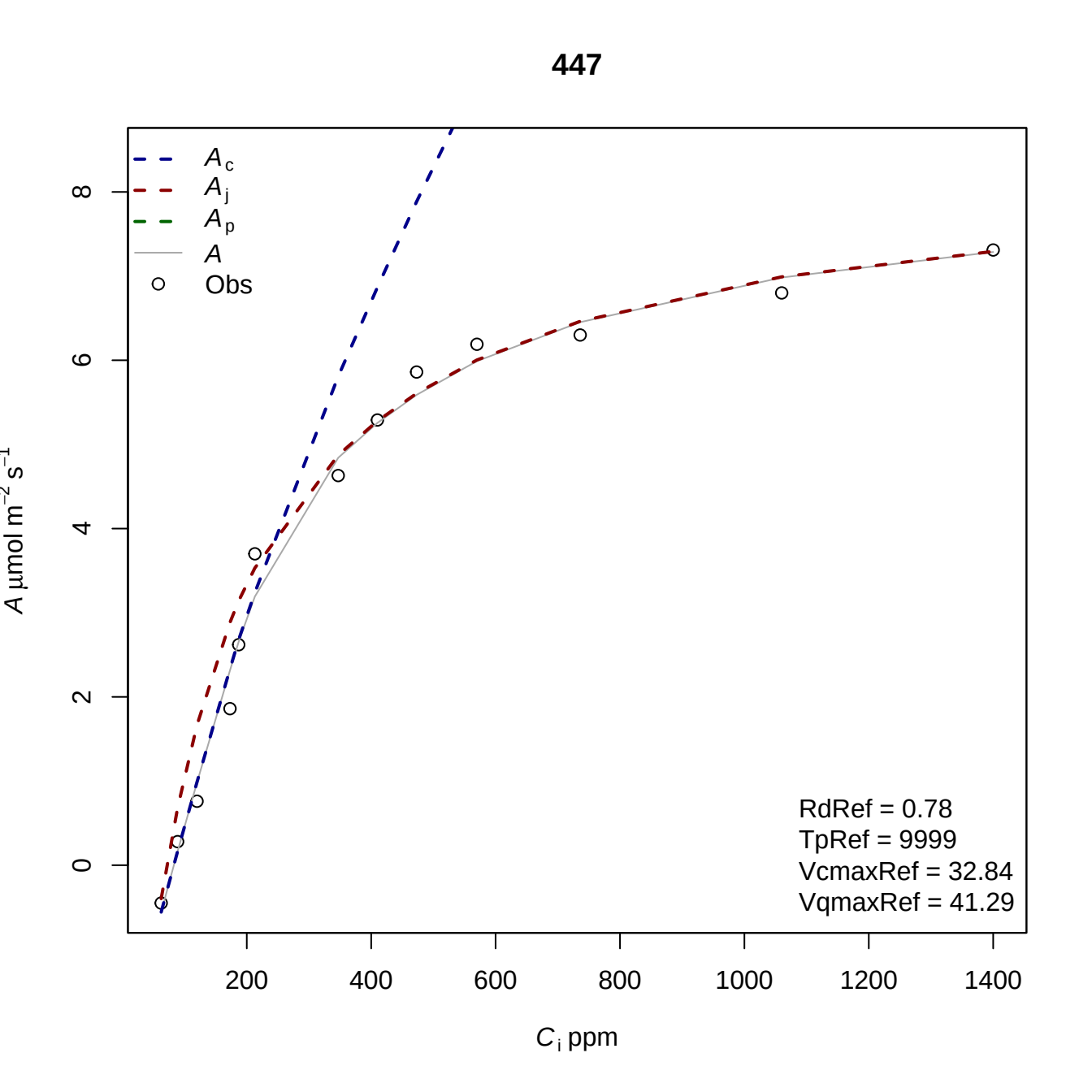

442
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.63
TpRef = 9999
VcmaxRef = 58.41
VqmaxRef = 79.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

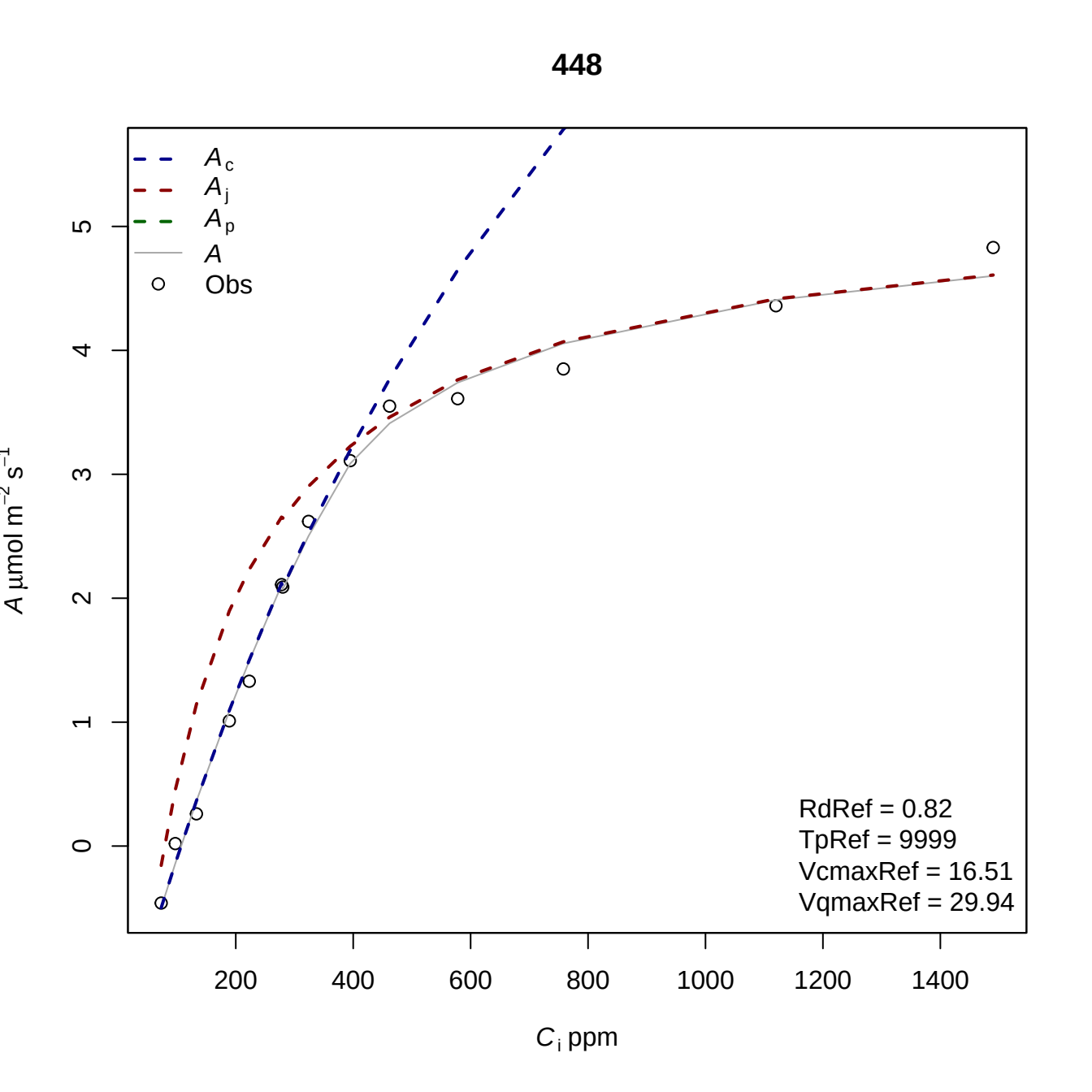

443
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.52
TpRef = 4.01
VcmaxRef = 43.53
VqmaxRef = 61.57
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

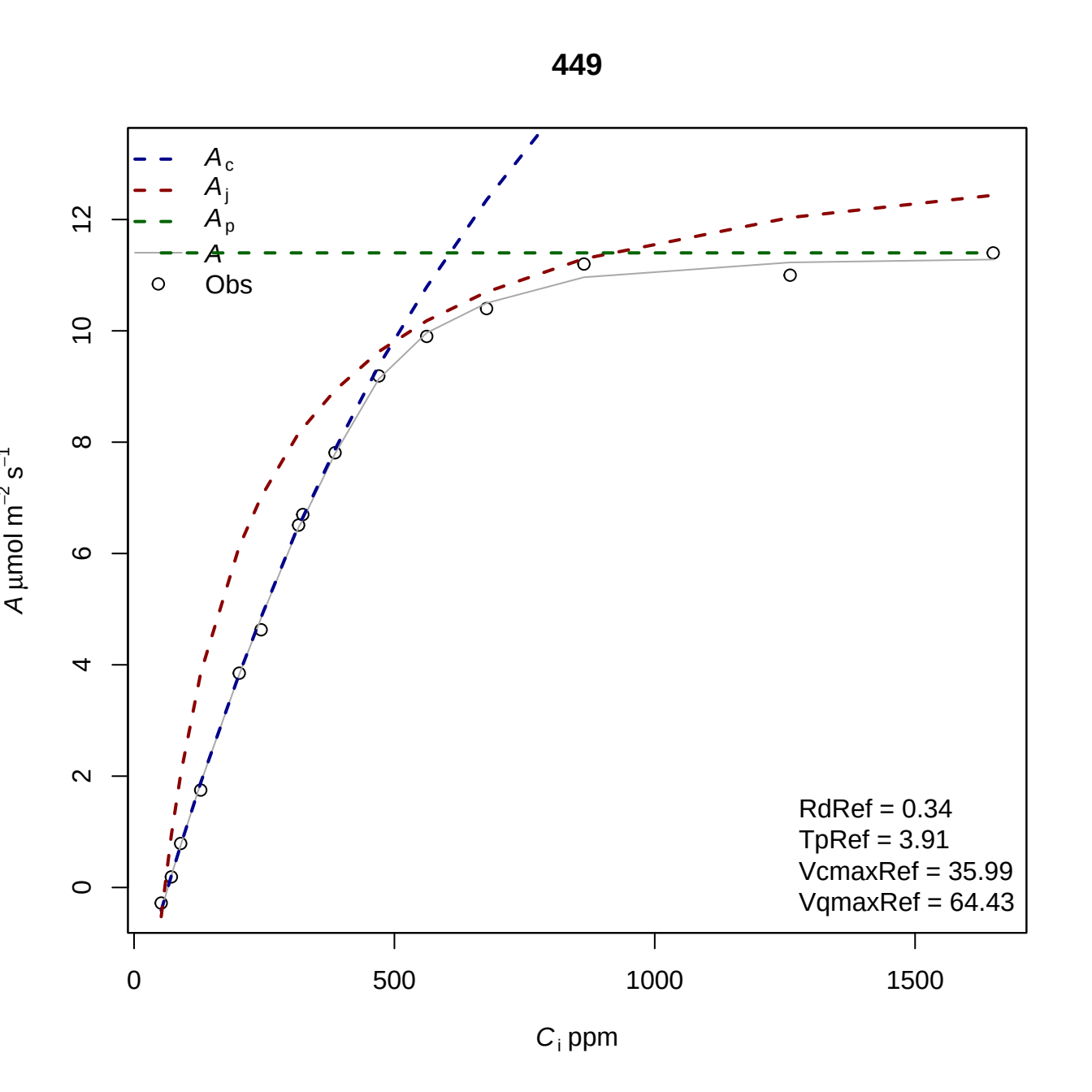

444
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.48
TpRef = 9999
VcmaxRef = 16.94
VqmaxRef = 31.09
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
0
200
400
600
800
1000
1200
1400

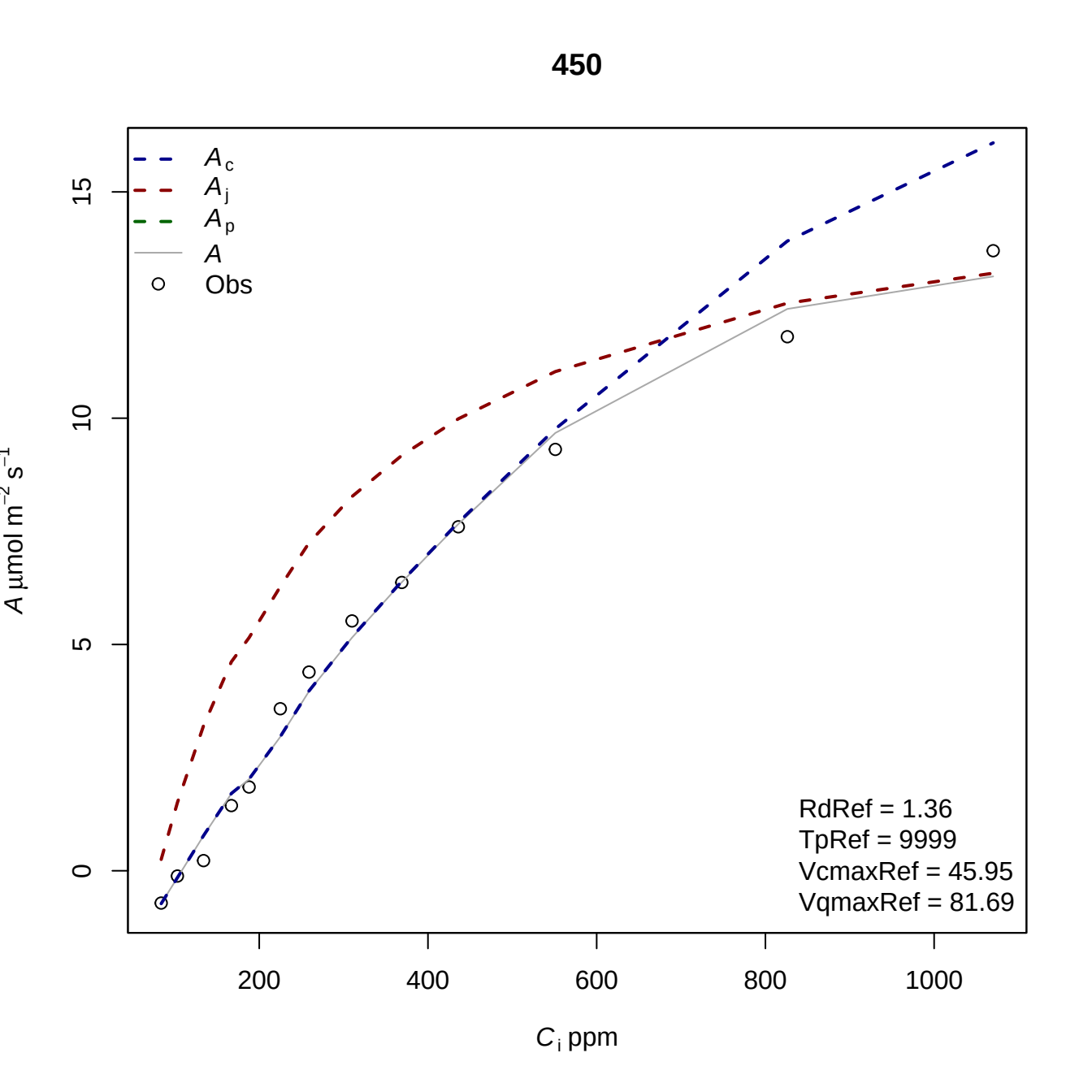

445
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.23
TpRef = 3.67
VcmaxRef = 31.61
VqmaxRef = 63.41
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

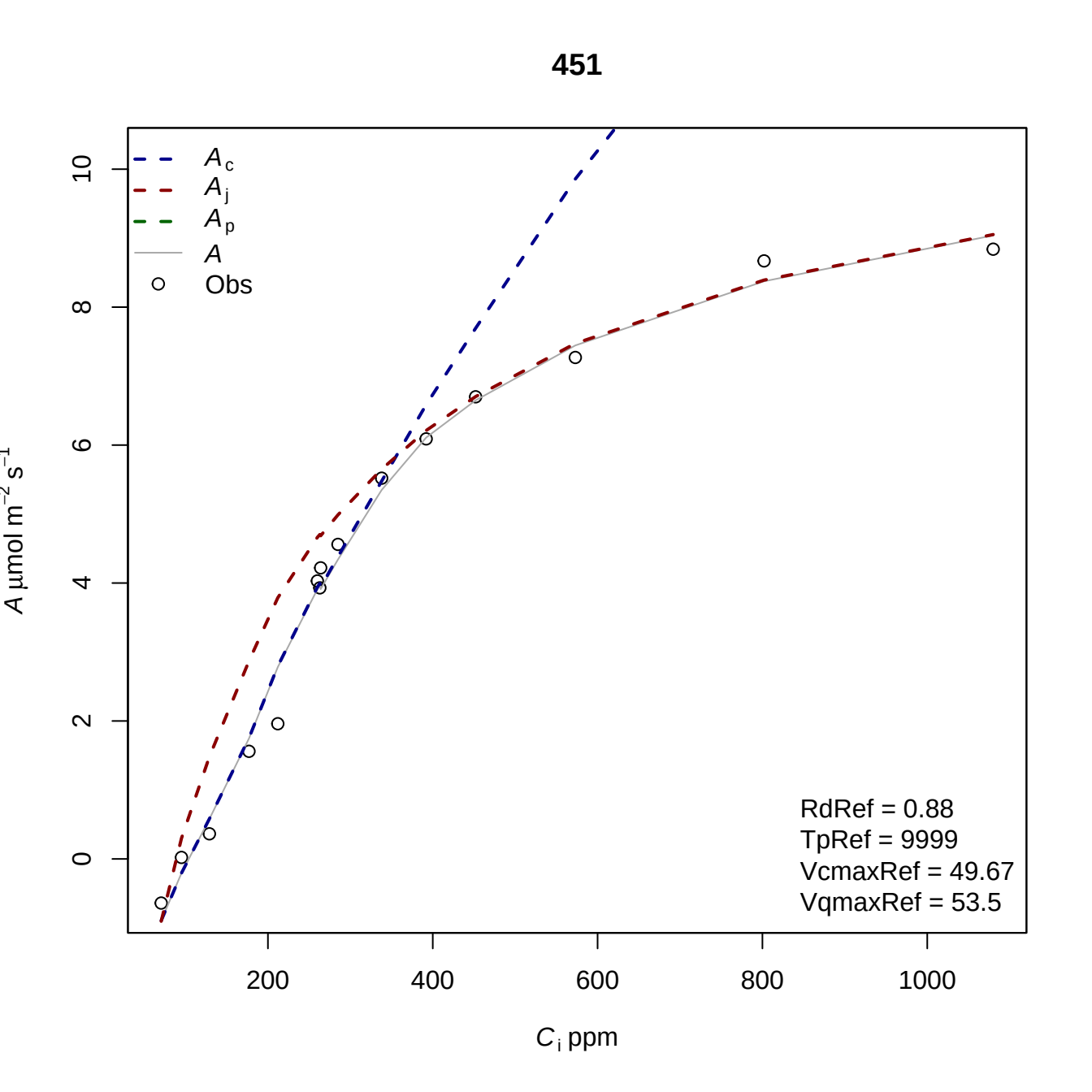

446
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.68
TpRef = 9999
VcmaxRef = 24.29
VqmaxRef = 37.98
0
2
4
6
8
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

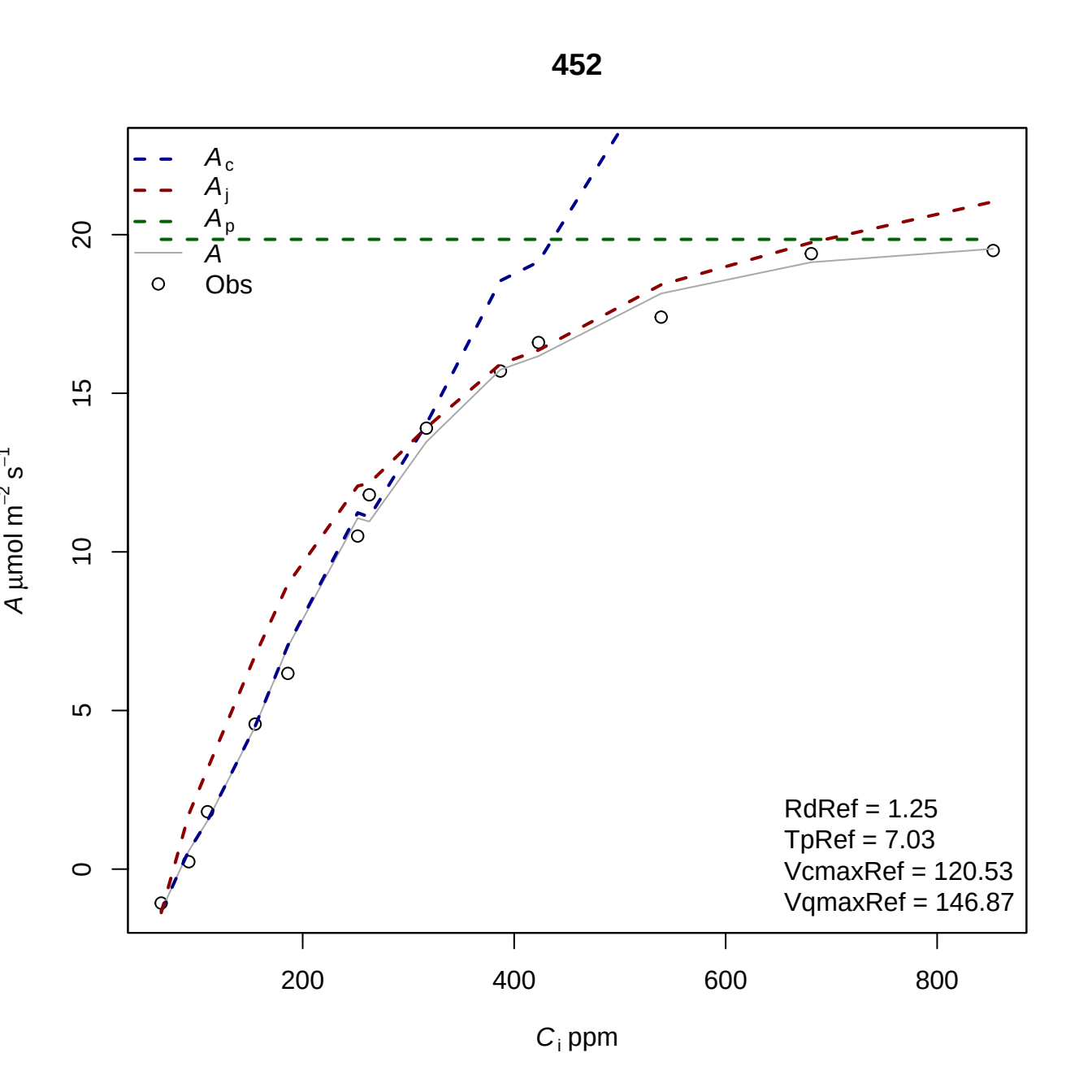

447
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.78
TpRef = 9999
VcmaxRef = 32.84
VqmaxRef = 41.29
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
200
400
600
800
1000
1200
1400

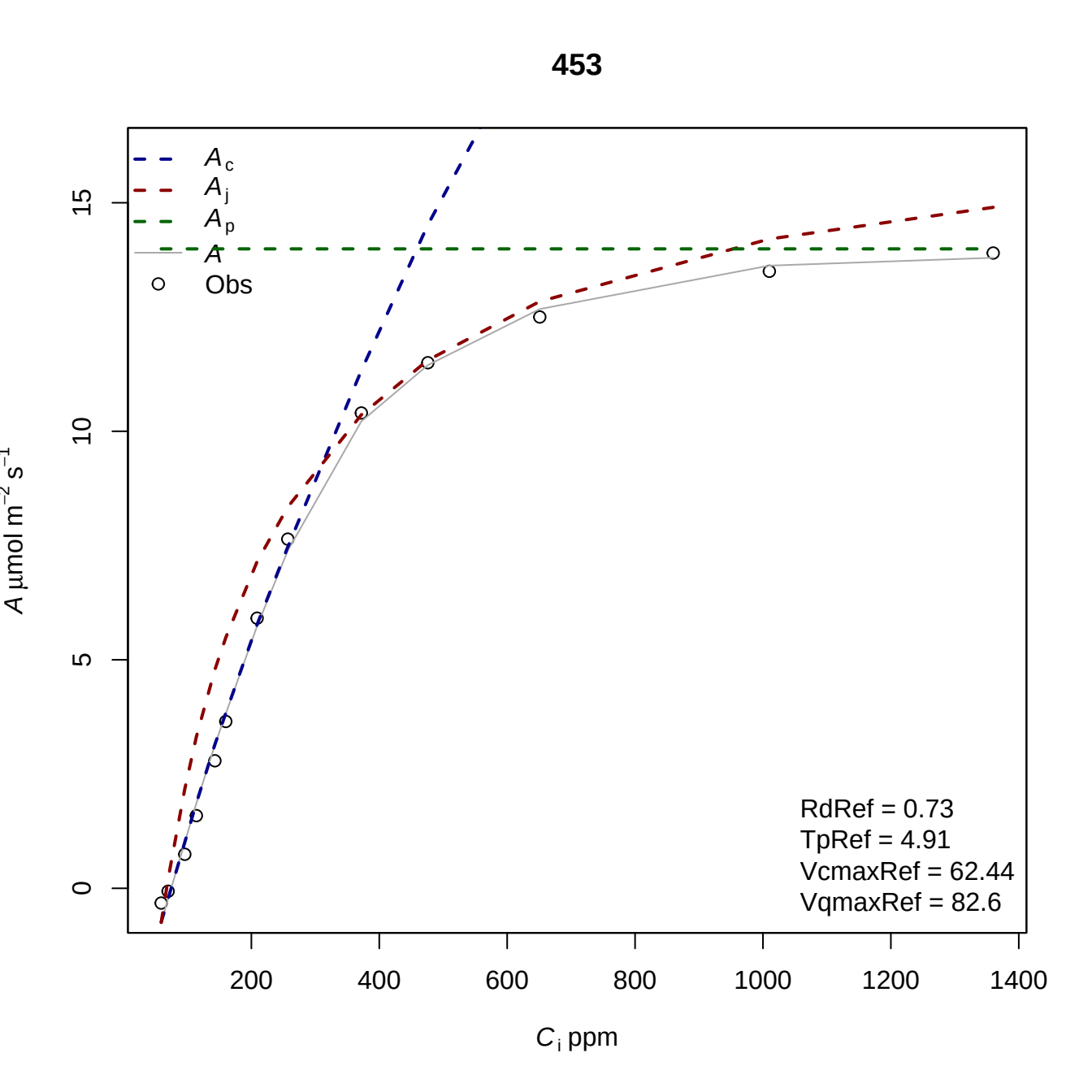

448
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.82
TpRef = 9999
VcmaxRef = 16.51
VqmaxRef = 29.94
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
200
400
600
800
1000
1200
1400

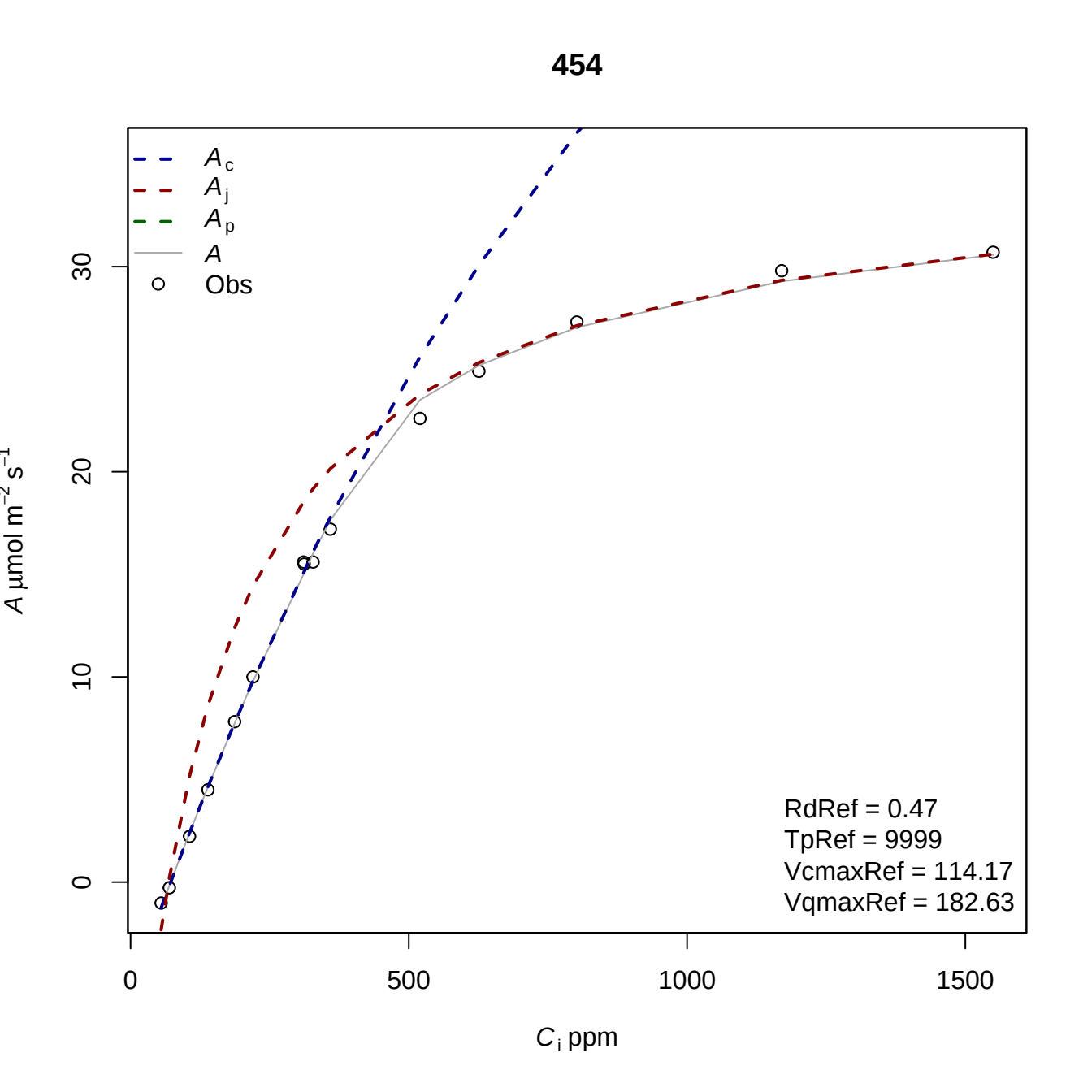

449
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.34
TpRef = 3.91
VcmaxRef = 35.99
VqmaxRef = 64.43
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

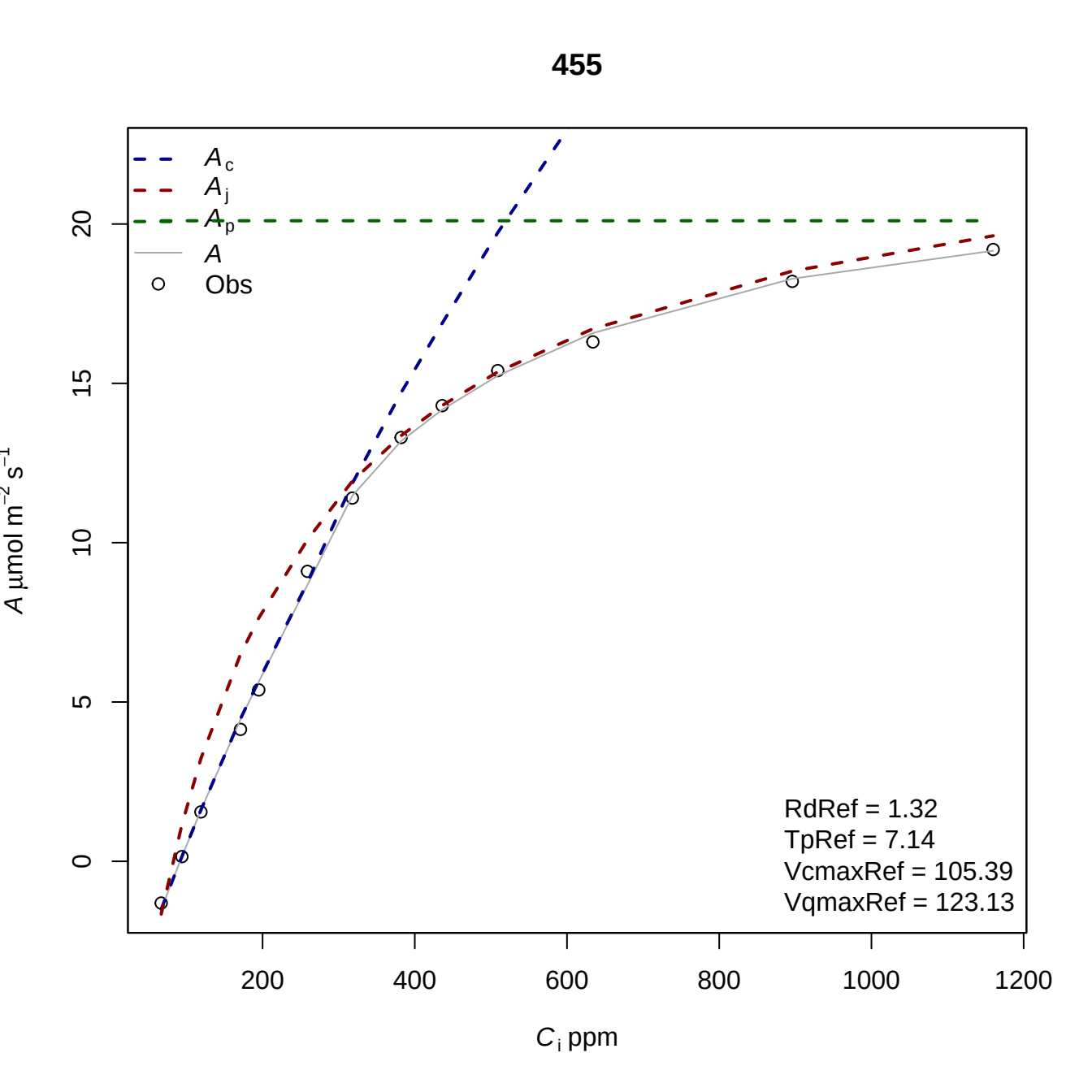

450
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.36
TpRef = 9999
VcmaxRef = 45.95
VqmaxRef = 81.69
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

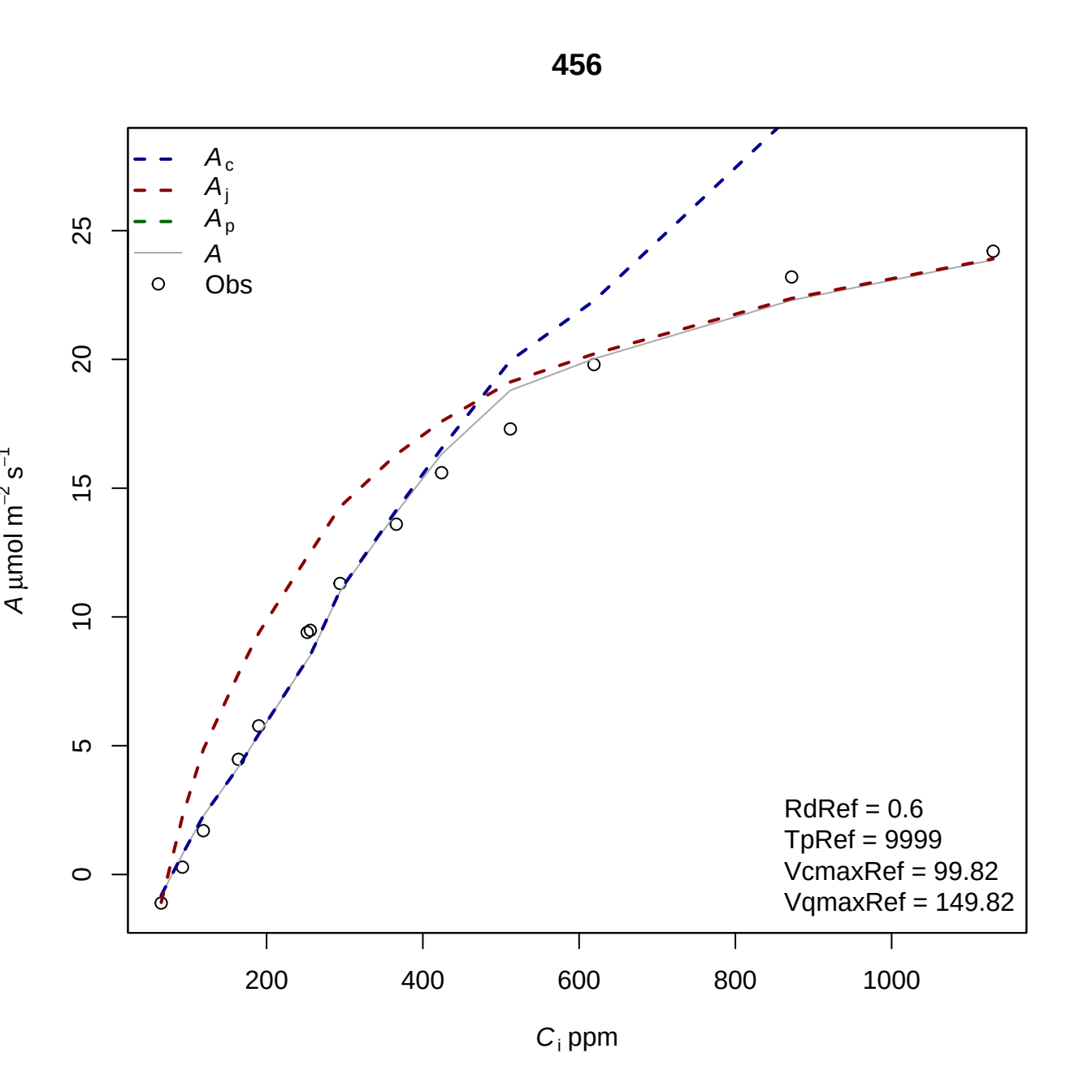

451
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.88
TpRef = 9999
VcmaxRef = 49.67
VqmaxRef = 53.5
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
200
400
600
800
1000

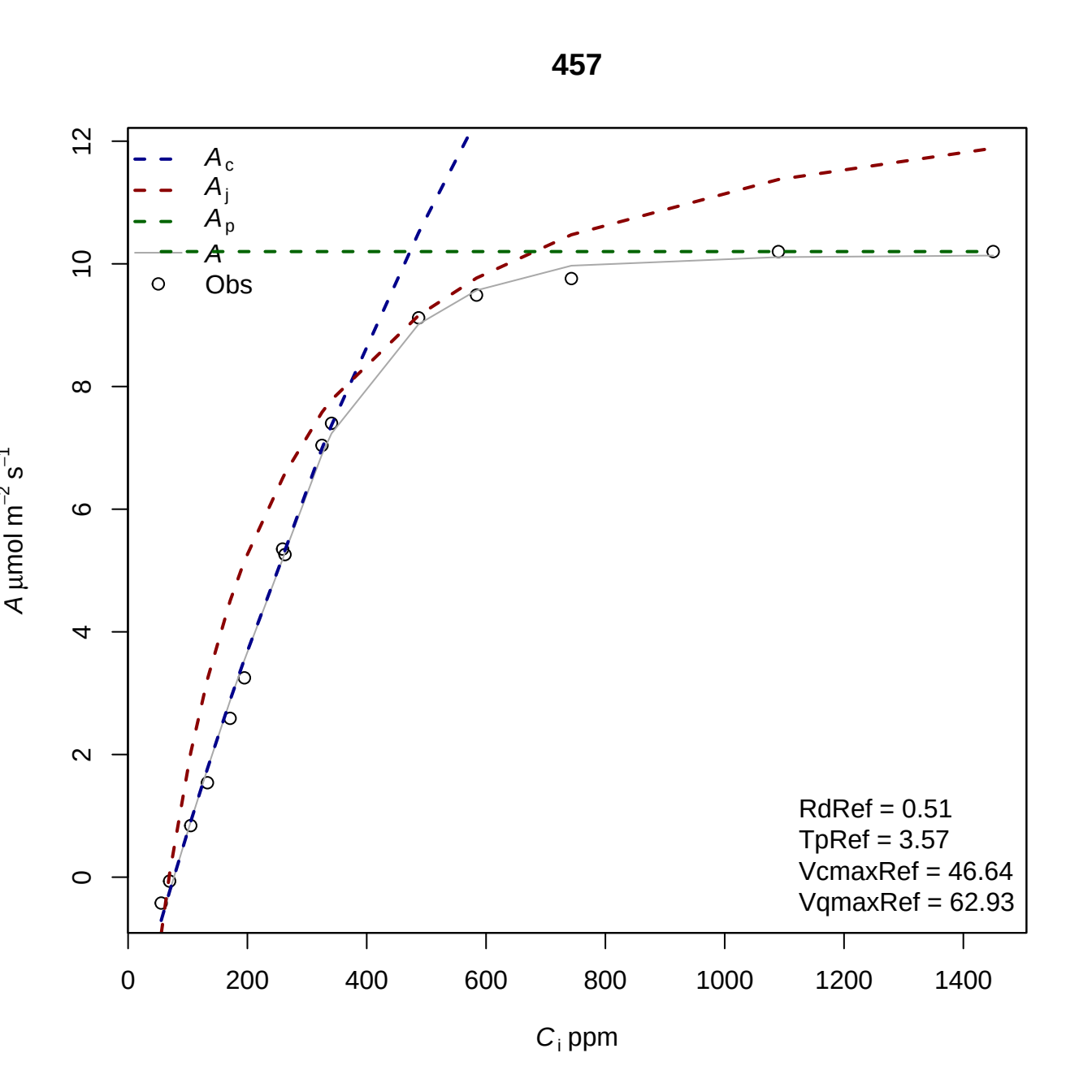

452
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.25
TpRef = 7.03
VcmaxRef = 120.53
VqmaxRef = 146.87
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800

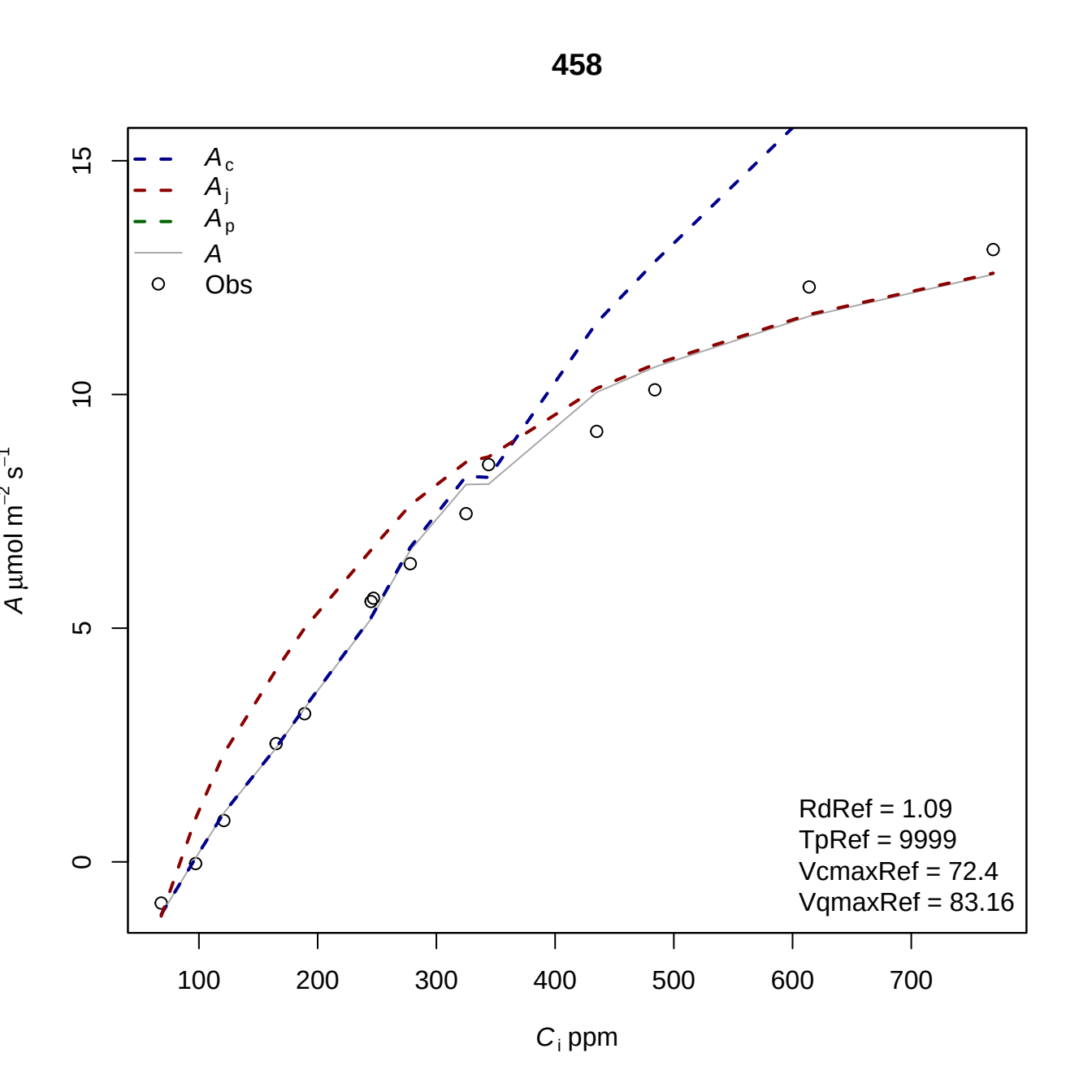

453
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.73
TpRef = 4.91
VcmaxRef = 62.44
VqmaxRef = 82.6
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

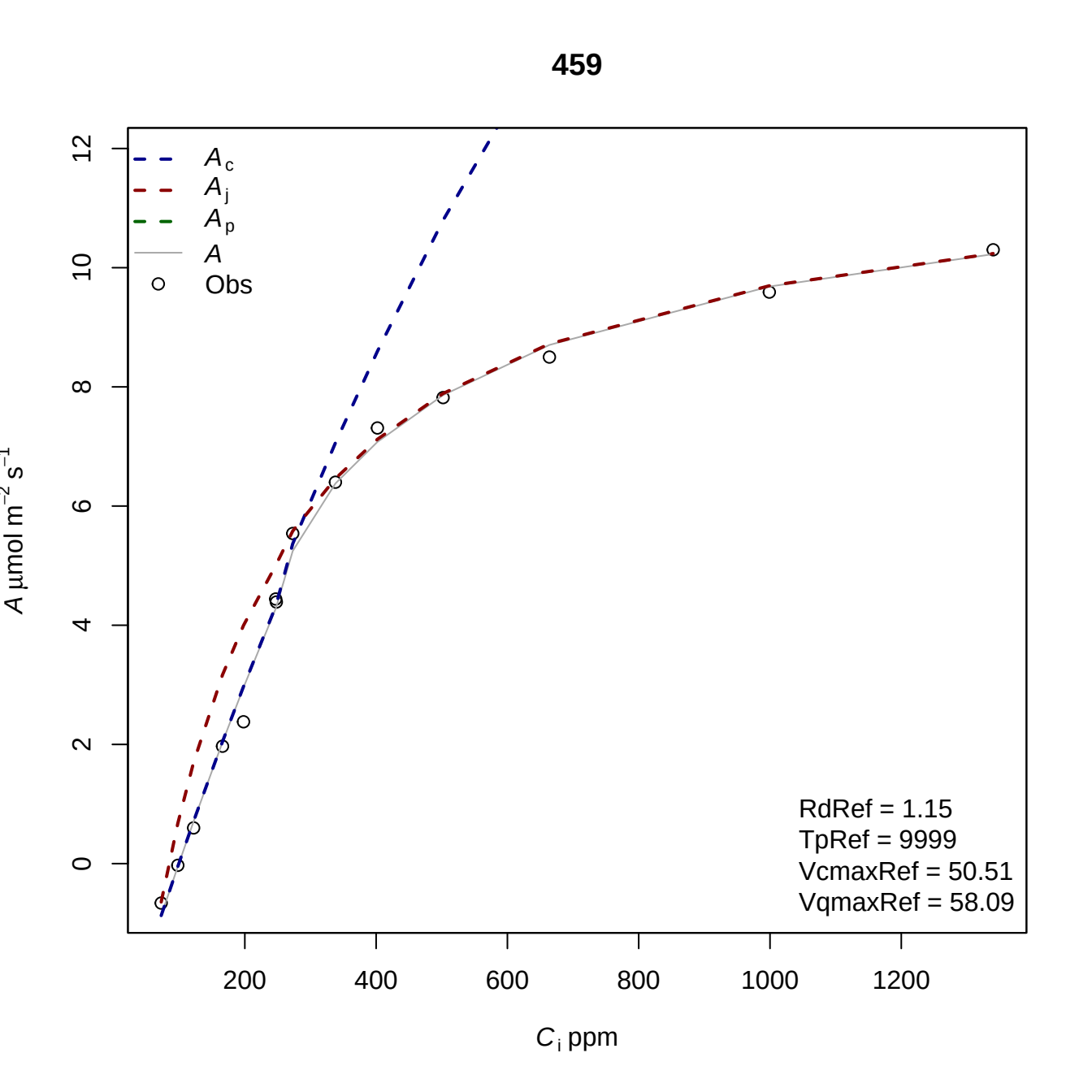

454
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.47
TpRef = 9999
VcmaxRef = 114.17
VqmaxRef = 182.63
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
0
500
1000
1500

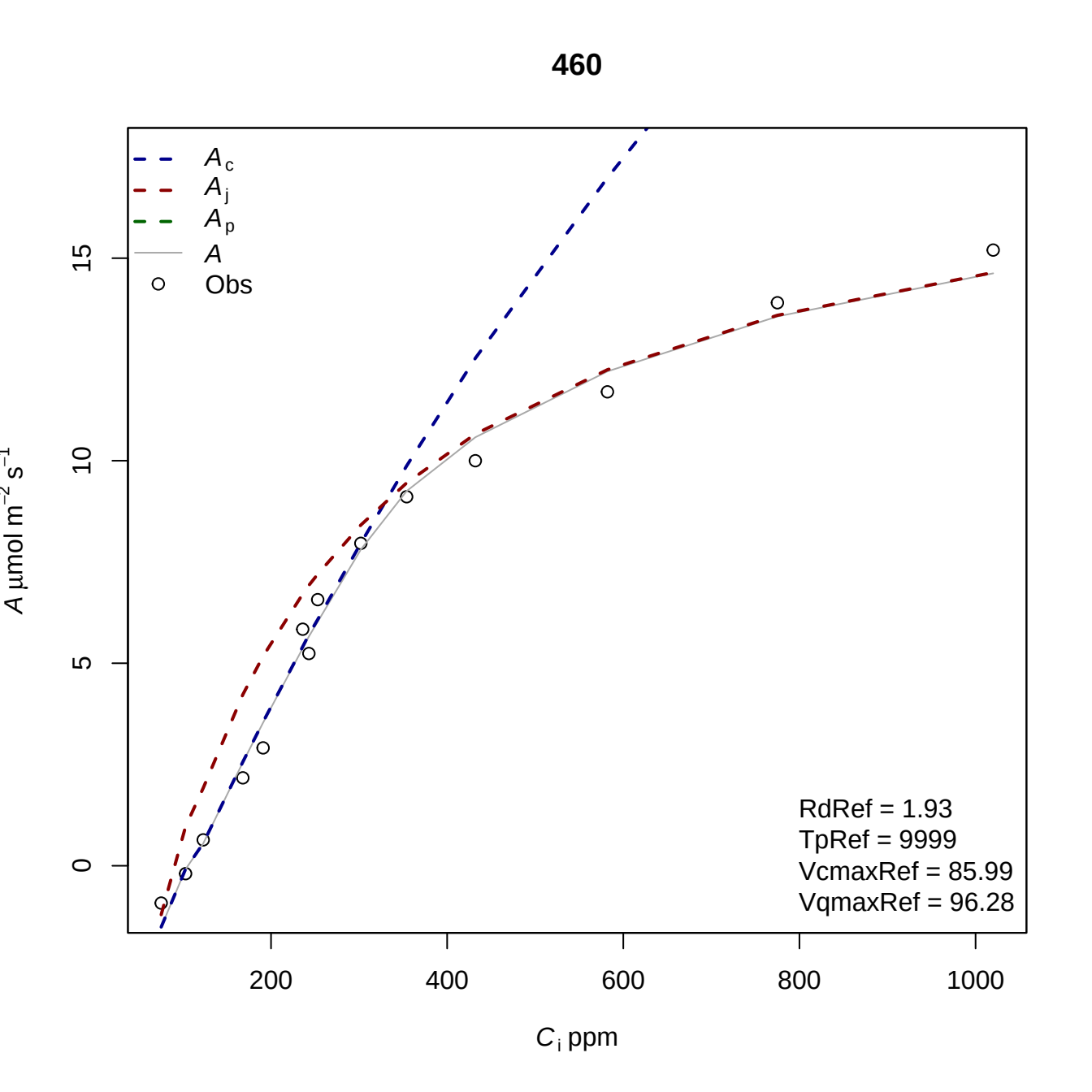

455
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.32
TpRef = 7.14
VcmaxRef = 105.39
VqmaxRef = 123.13
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200

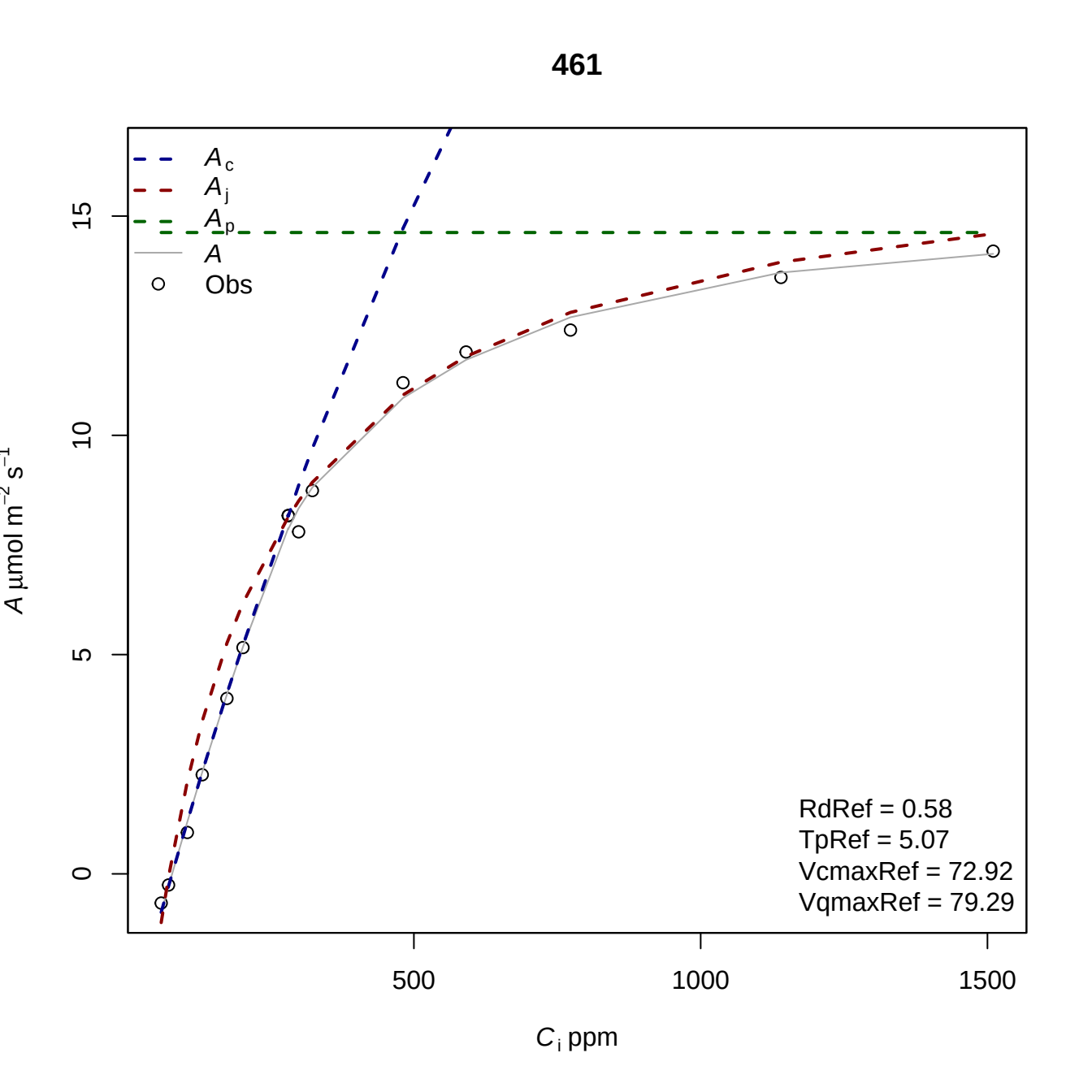

456
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.6
TpRef = 9999
VcmaxRef = 99.82
VqmaxRef = 149.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000

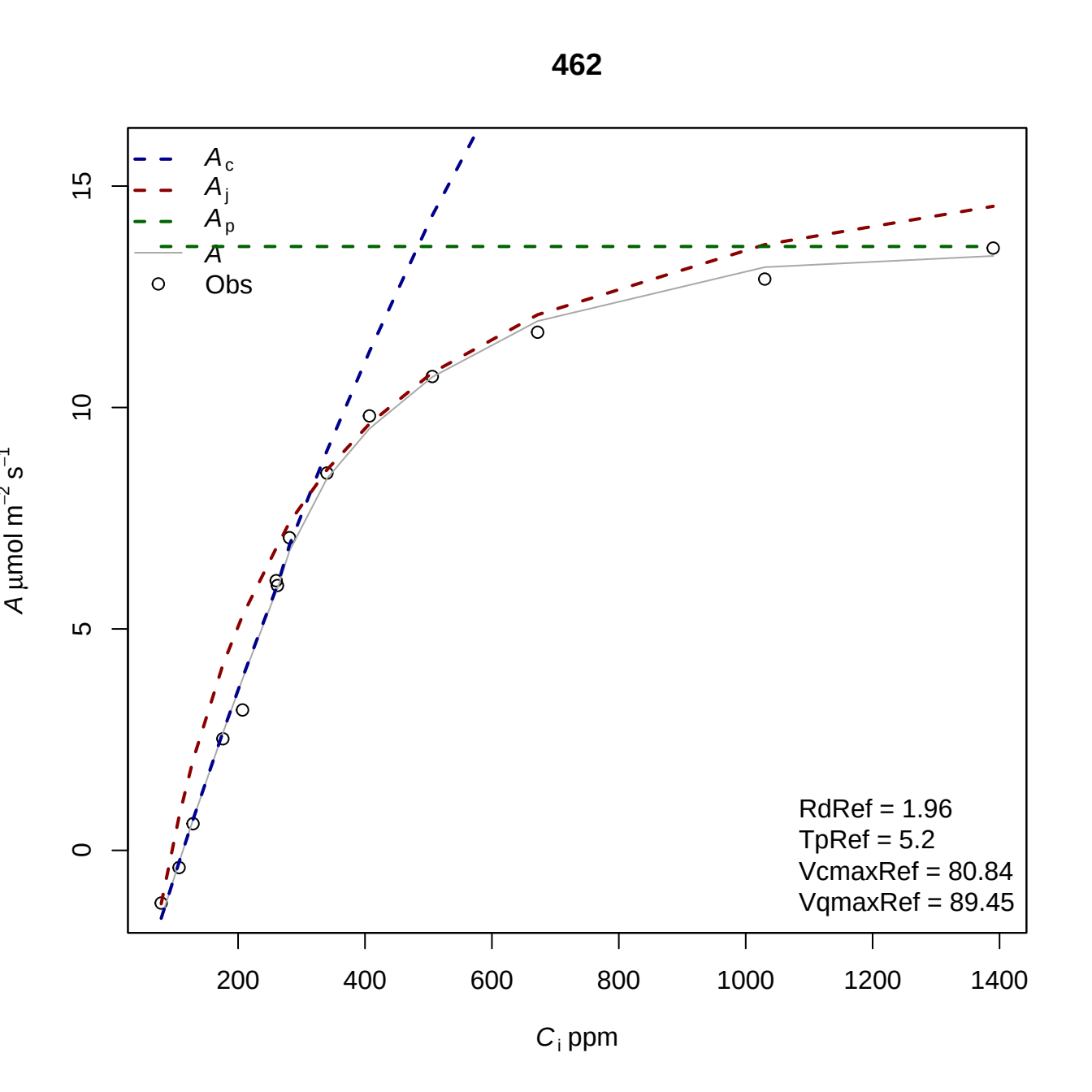

457
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.51
TpRef = 3.57
VcmaxRef = 46.64
VqmaxRef = 62.93
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

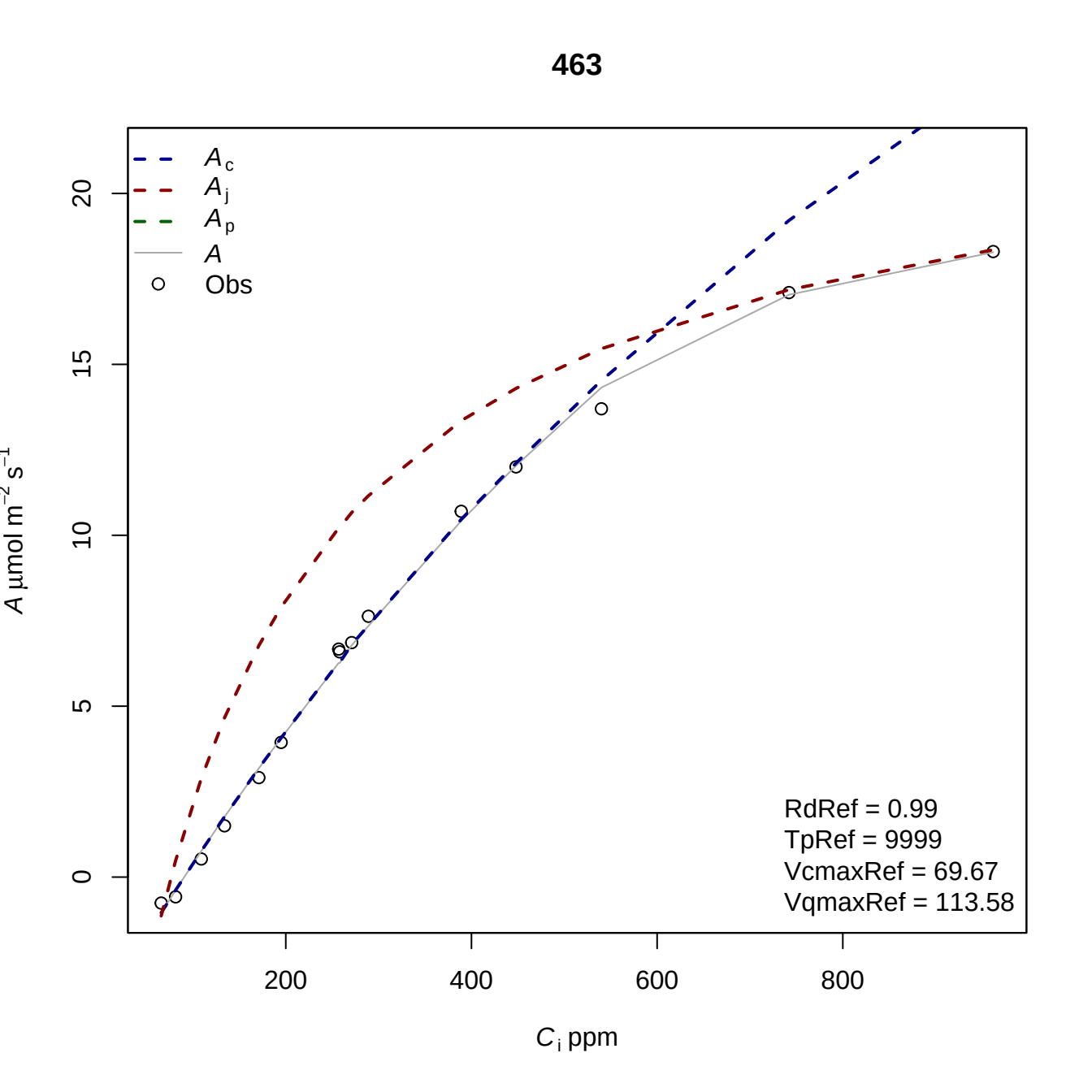

458
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.09
TpRef = 9999
VcmaxRef = 72.4
VqmaxRef = 83.16
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
100
200
300
400
500
600
700

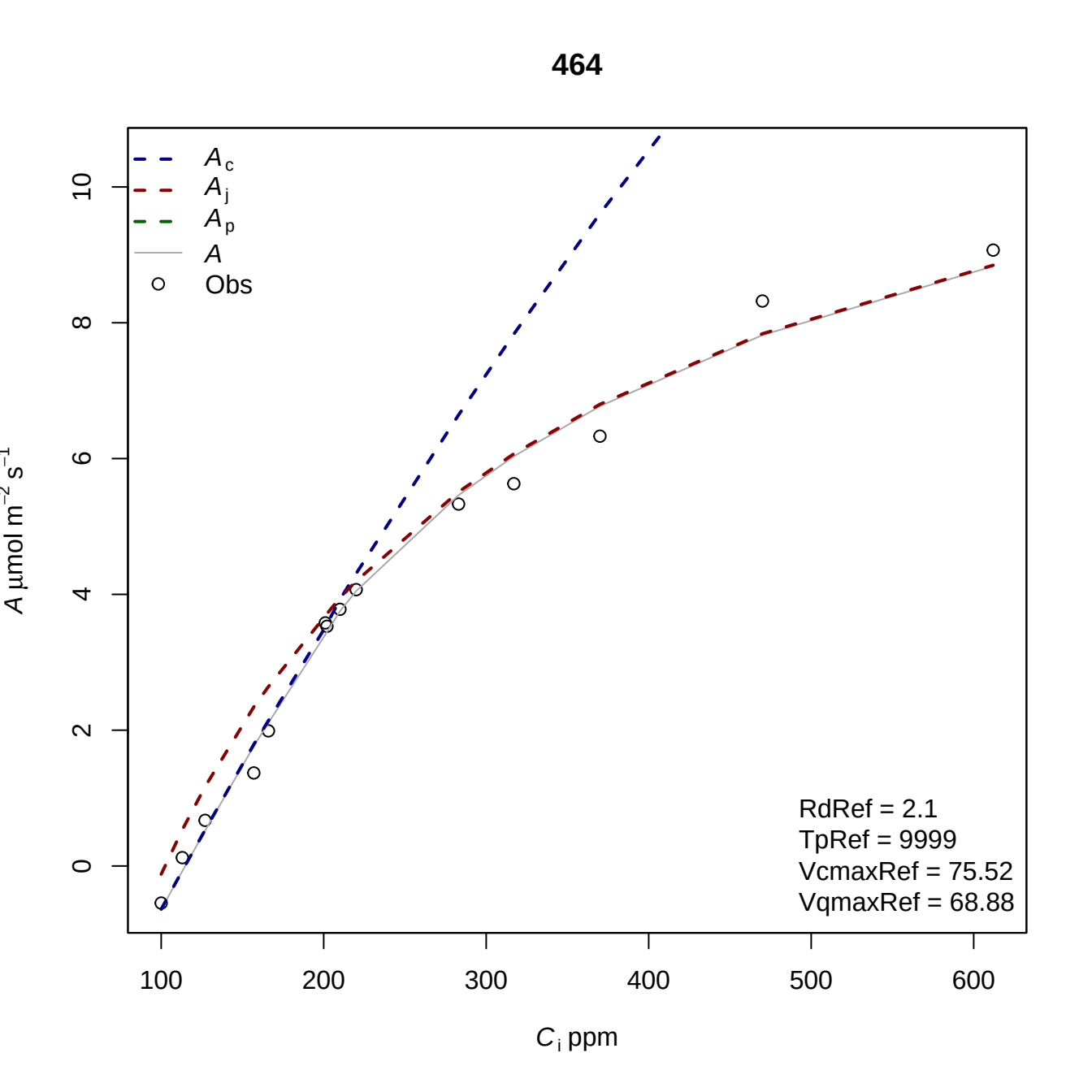

459
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.15
TpRef = 9999
VcmaxRef = 50.51
VqmaxRef = 58.09
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200

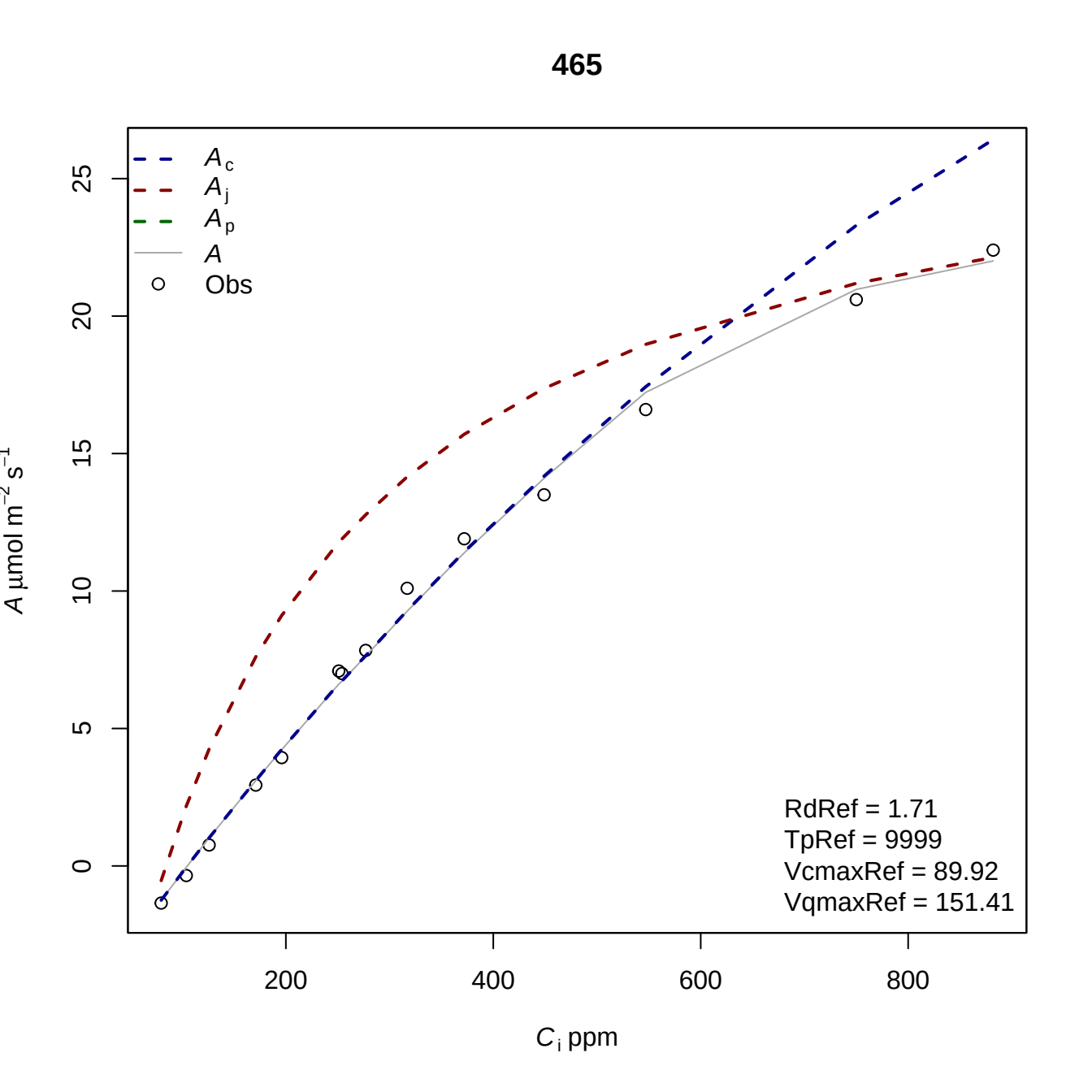

460
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.93
TpRef = 9999
VcmaxRef = 85.99
VqmaxRef = 96.28
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

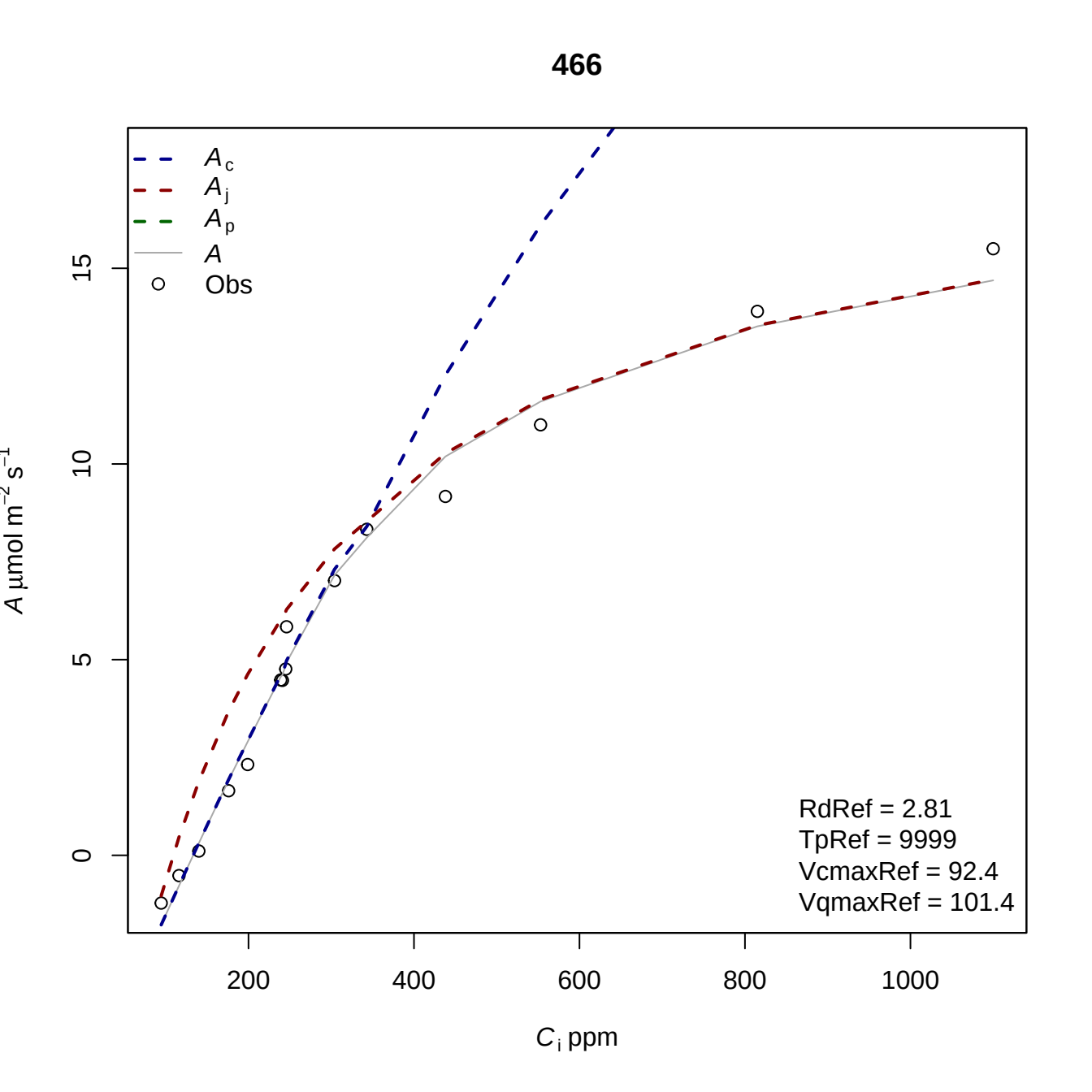

461
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.58
TpRef = 5.07
VcmaxRef = 72.92
VqmaxRef = 79.29
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
500
1000
1500

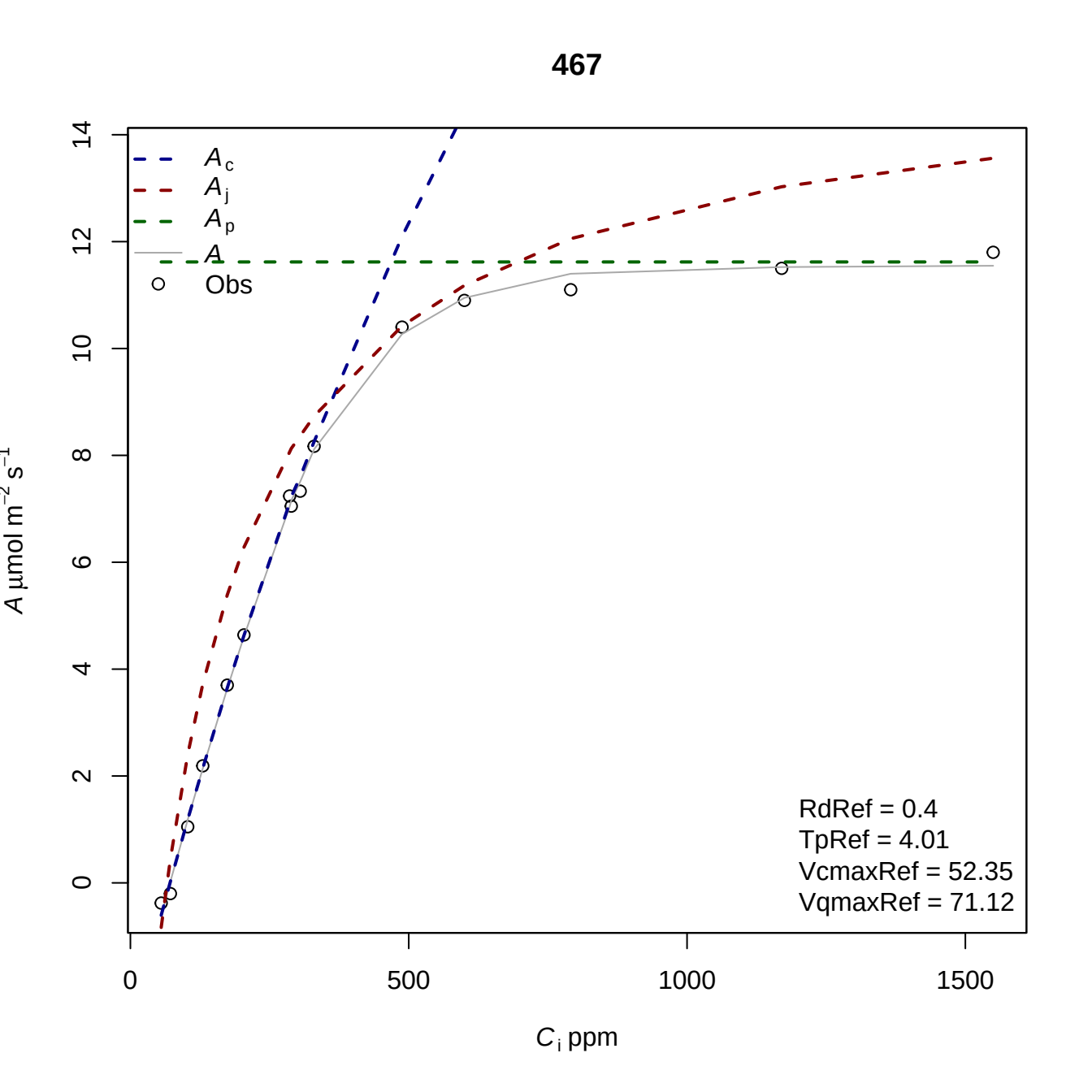

462
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.96
TpRef = 5.2
VcmaxRef = 80.84
VqmaxRef = 89.45
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

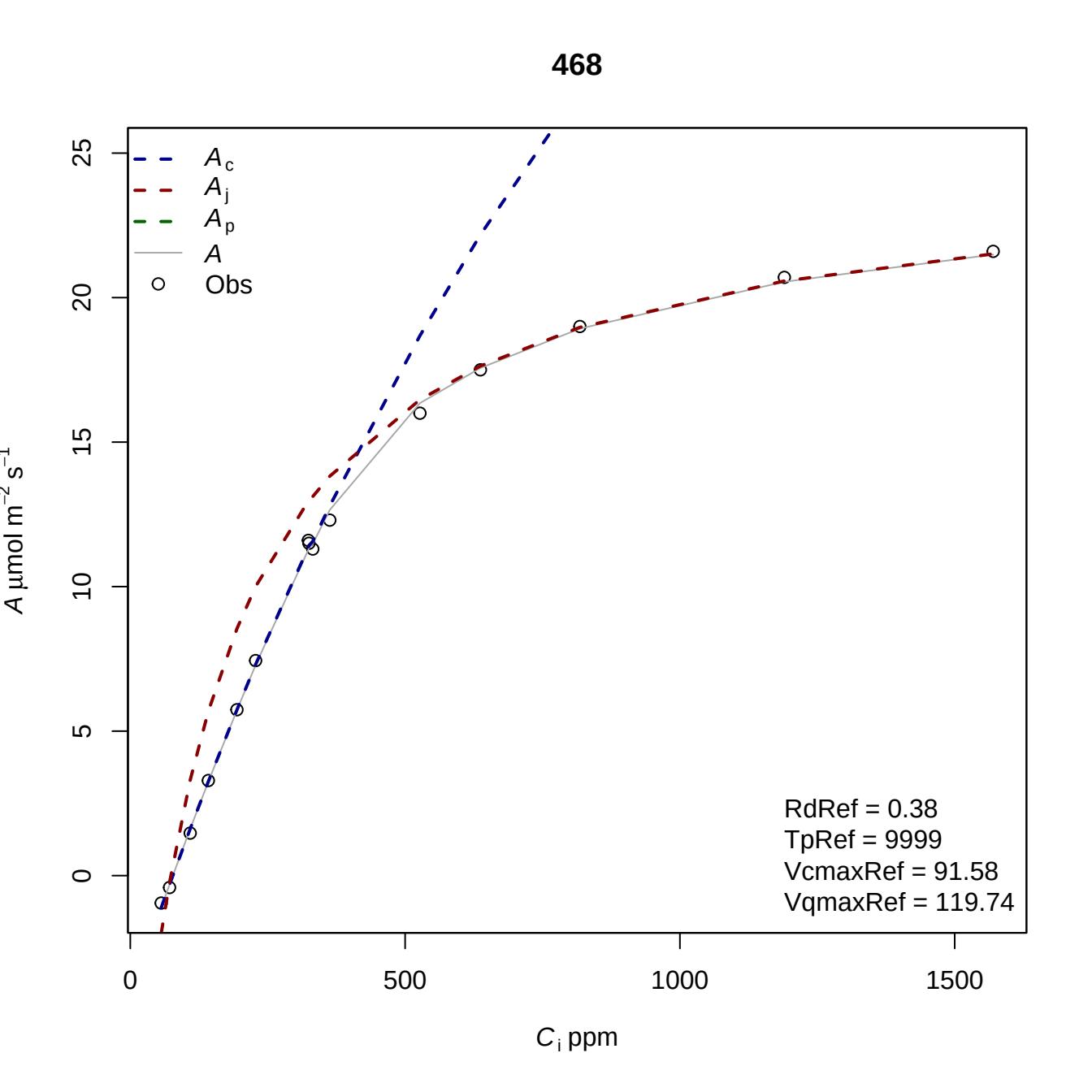

463
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.99
TpRef = 9999
VcmaxRef = 69.67
VqmaxRef = 113.58
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800

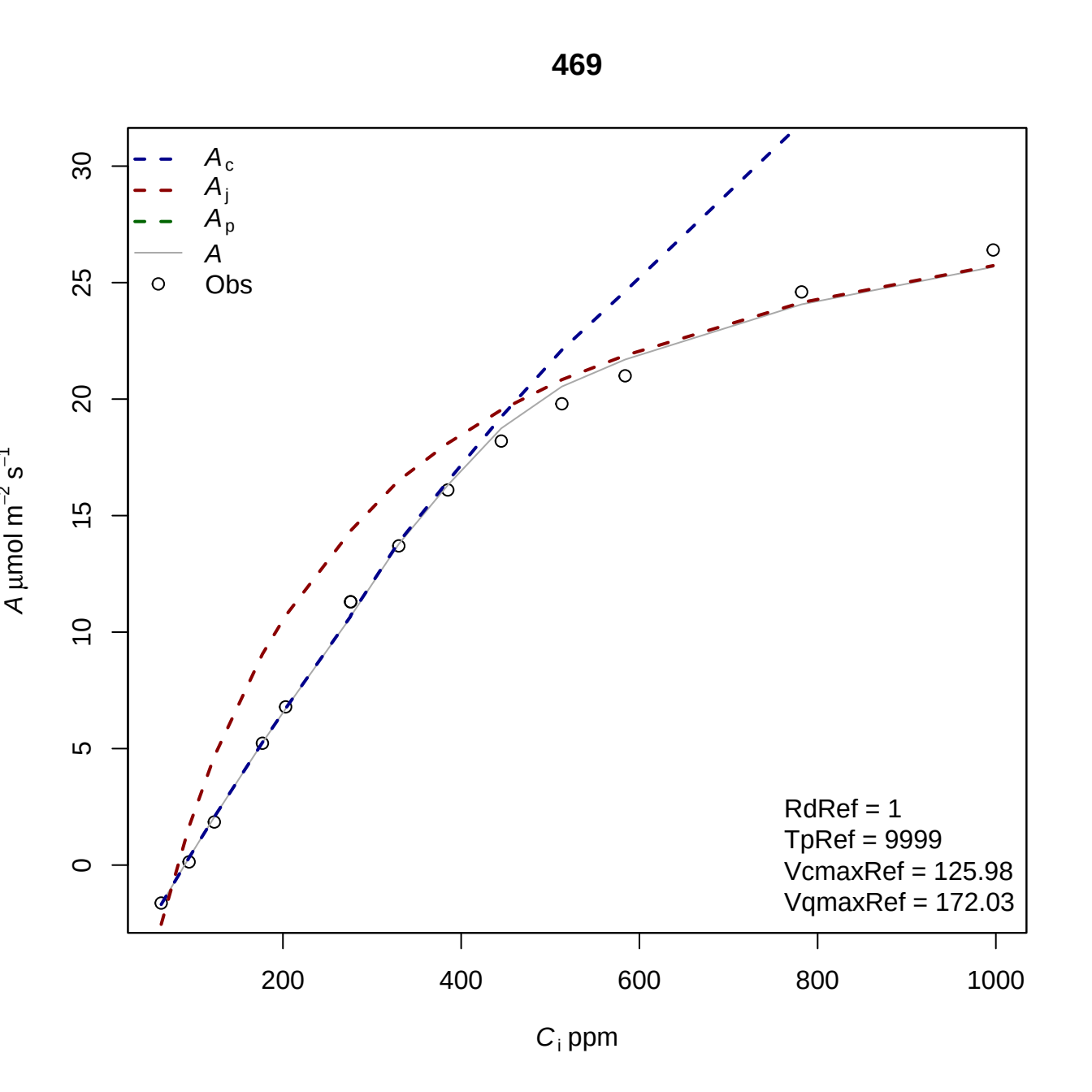

464
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.1
TpRef = 9999
VcmaxRef = 75.52
VqmaxRef = 68.88
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
100
200
300
400
500
600

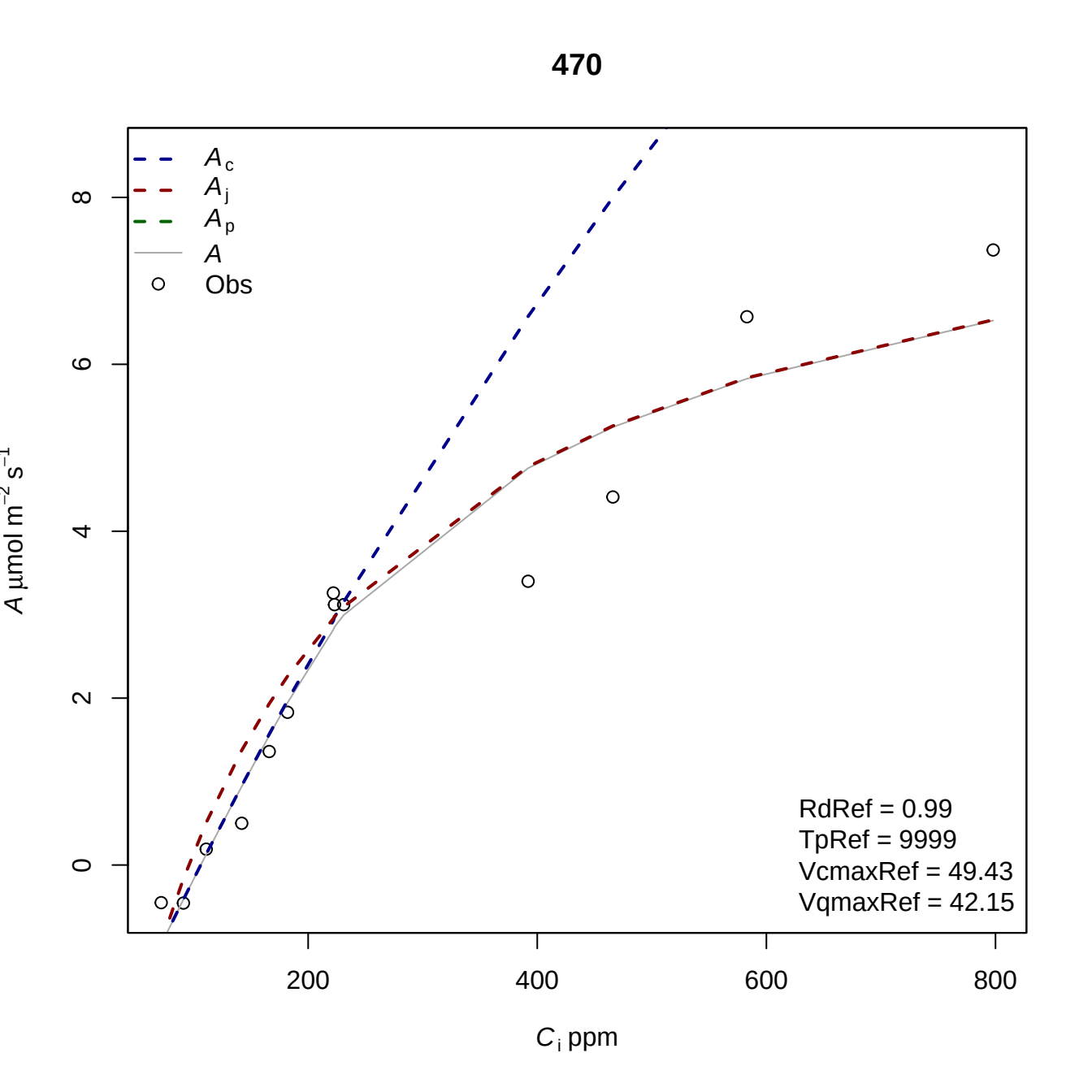

465
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.71
TpRef = 9999
VcmaxRef = 89.92
VqmaxRef = 151.41
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800

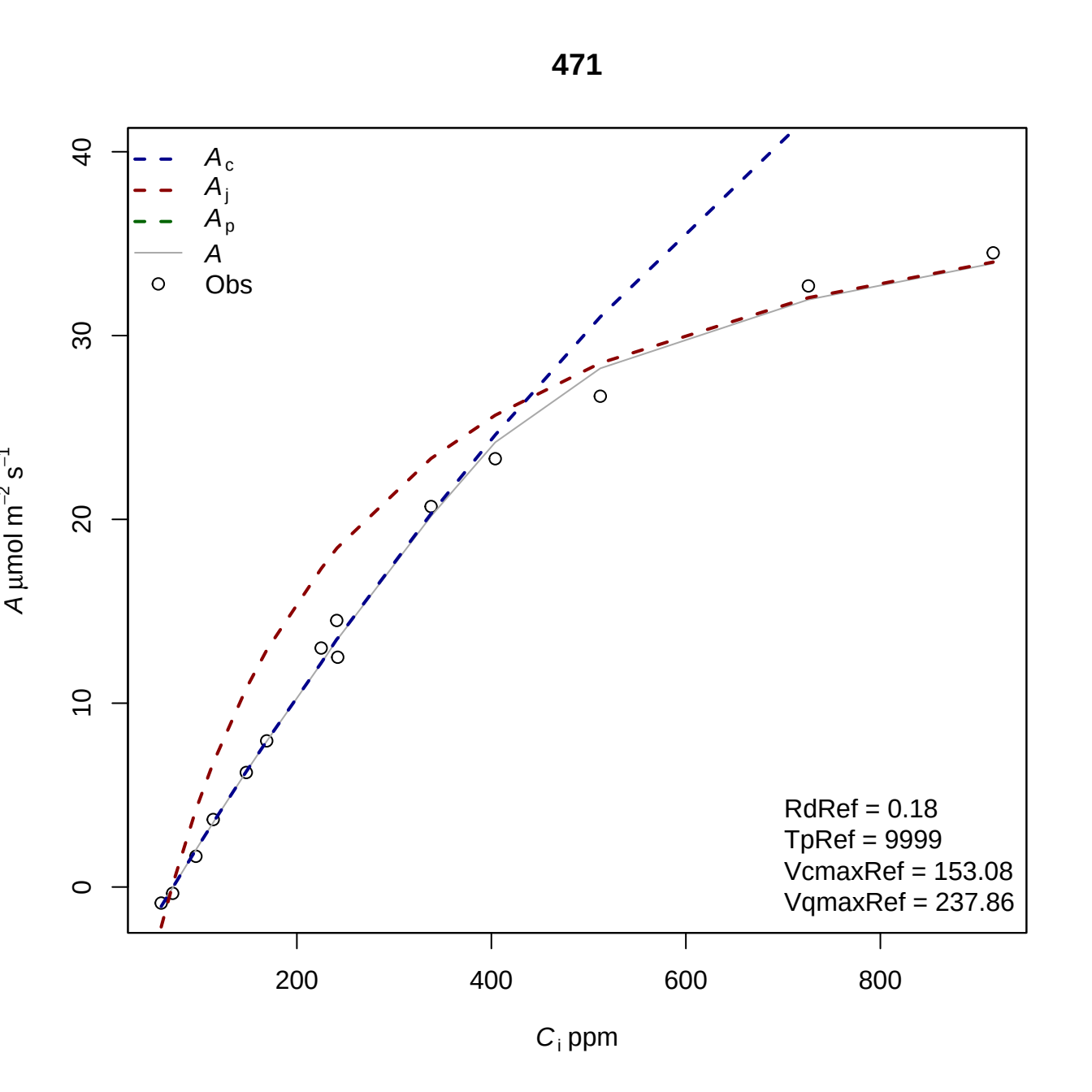

466
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.81
TpRef = 9999
VcmaxRef = 92.4
VqmaxRef = 101.4
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

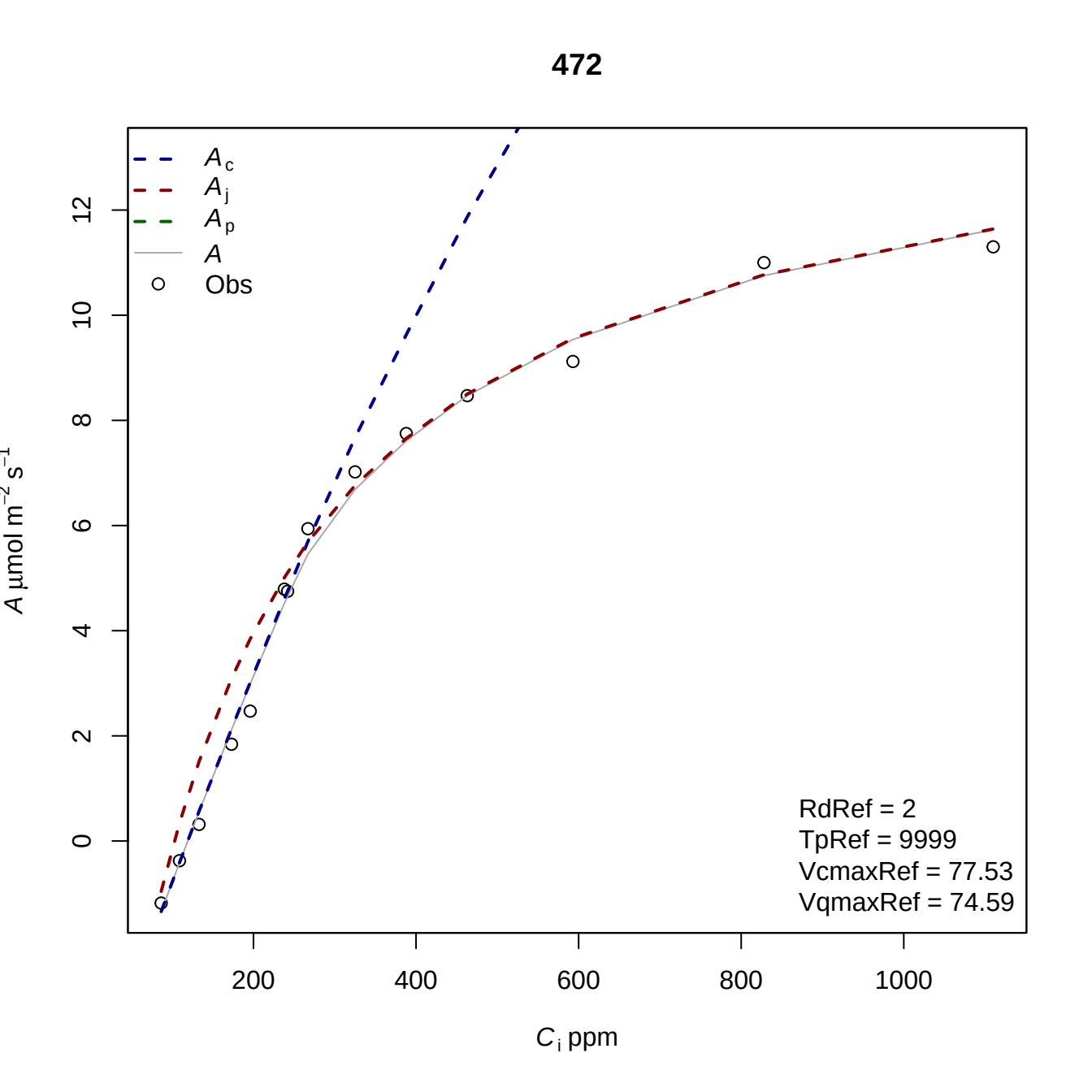

467
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.4
TpRef = 4.01
VcmaxRef = 52.35
VqmaxRef = 71.12
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
0
500
1000
1500

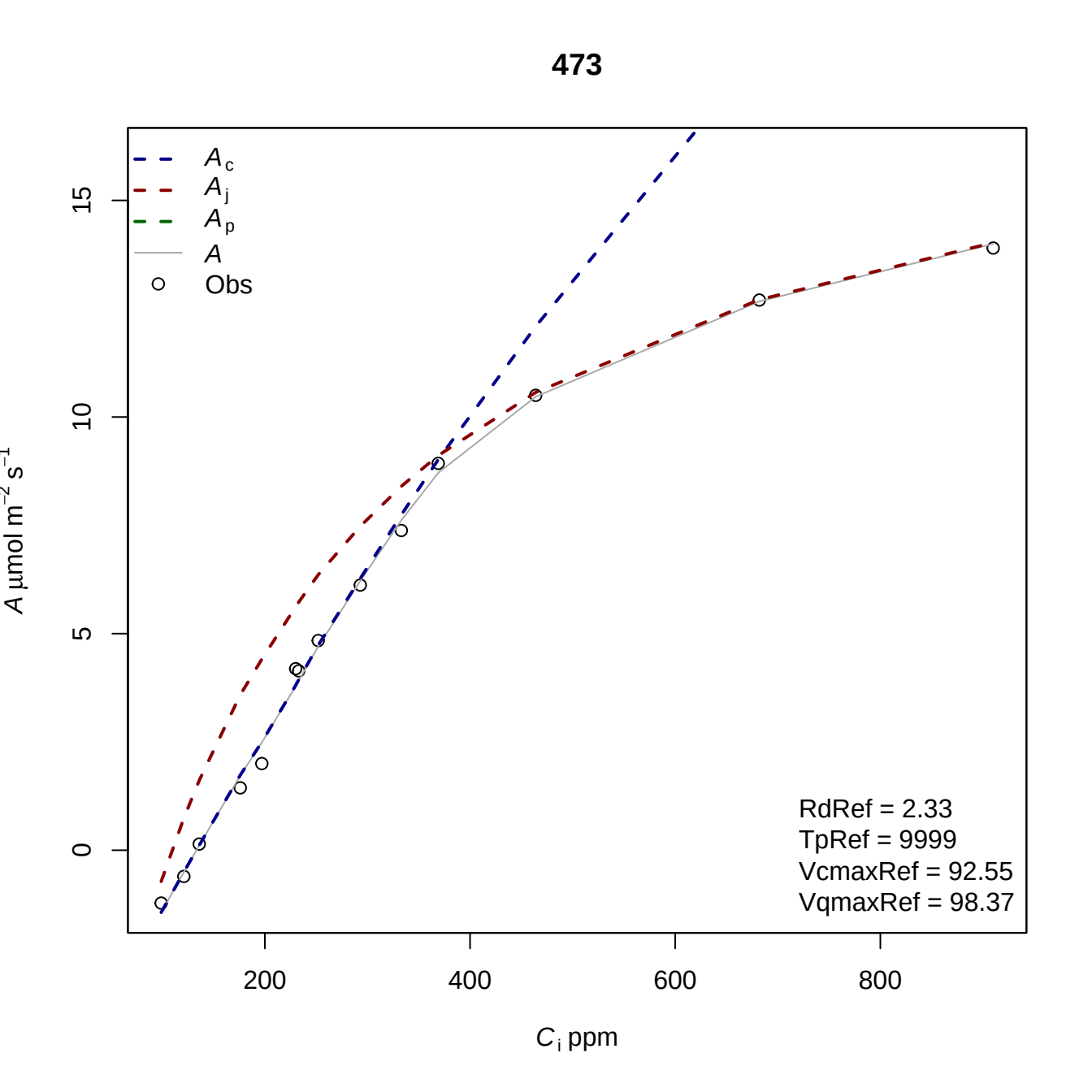

468
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.38
TpRef = 9999
VcmaxRef = 91.58
VqmaxRef = 119.74
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

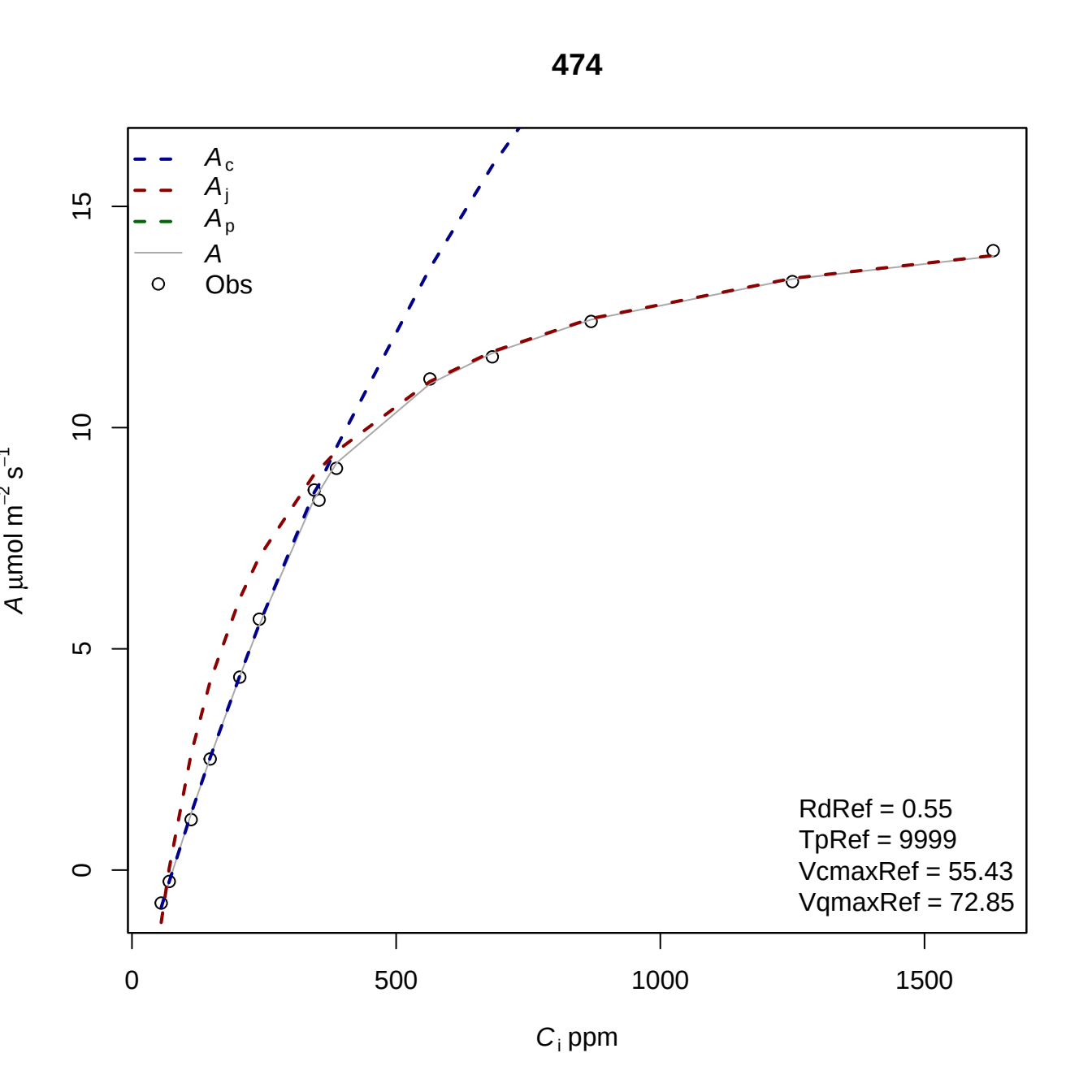

469
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1
TpRef = 9999
VcmaxRef = 125.98
VqmaxRef = 172.03
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000

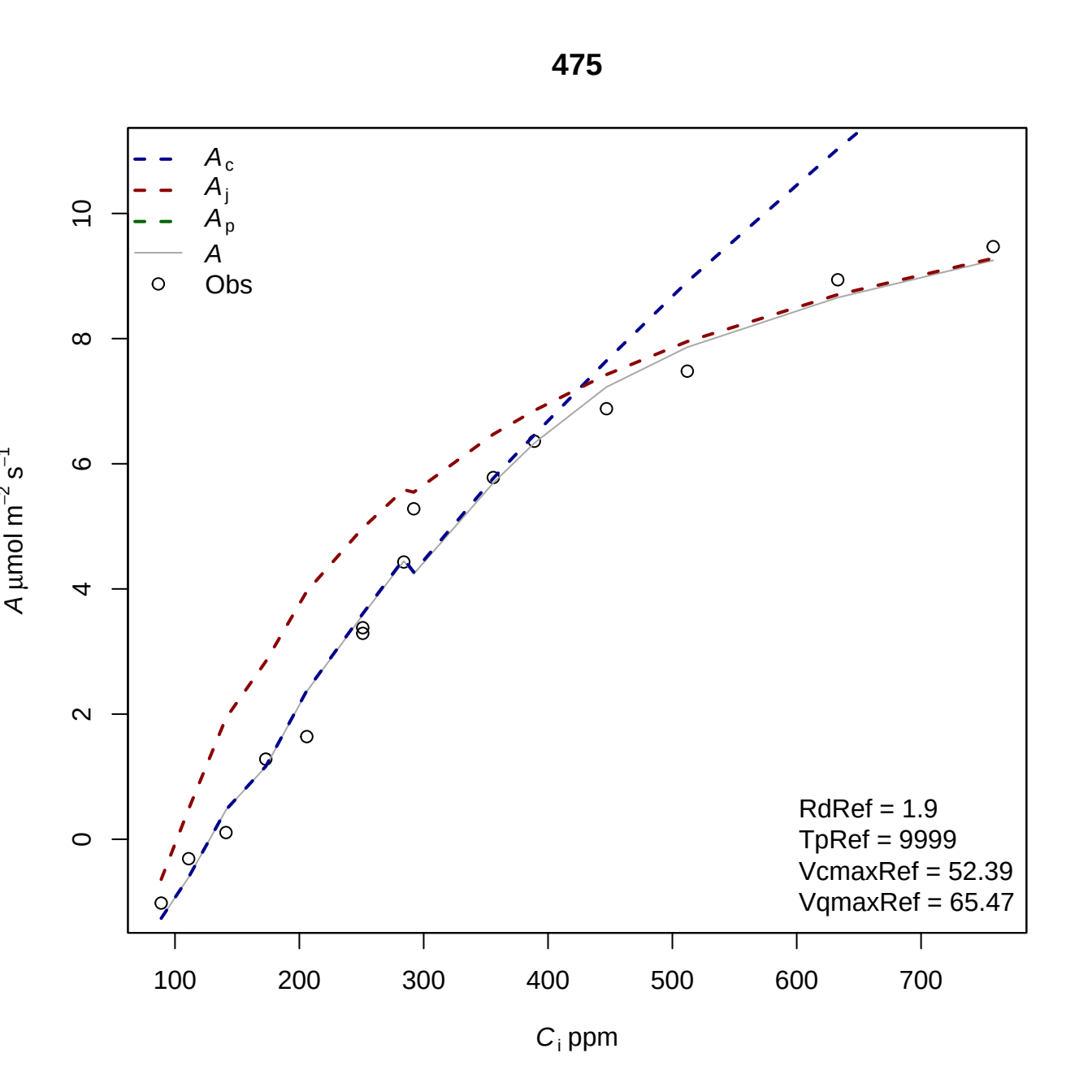

470
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.99
TpRef = 9999
VcmaxRef = 49.43
VqmaxRef = 42.15
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
200
400
600
800

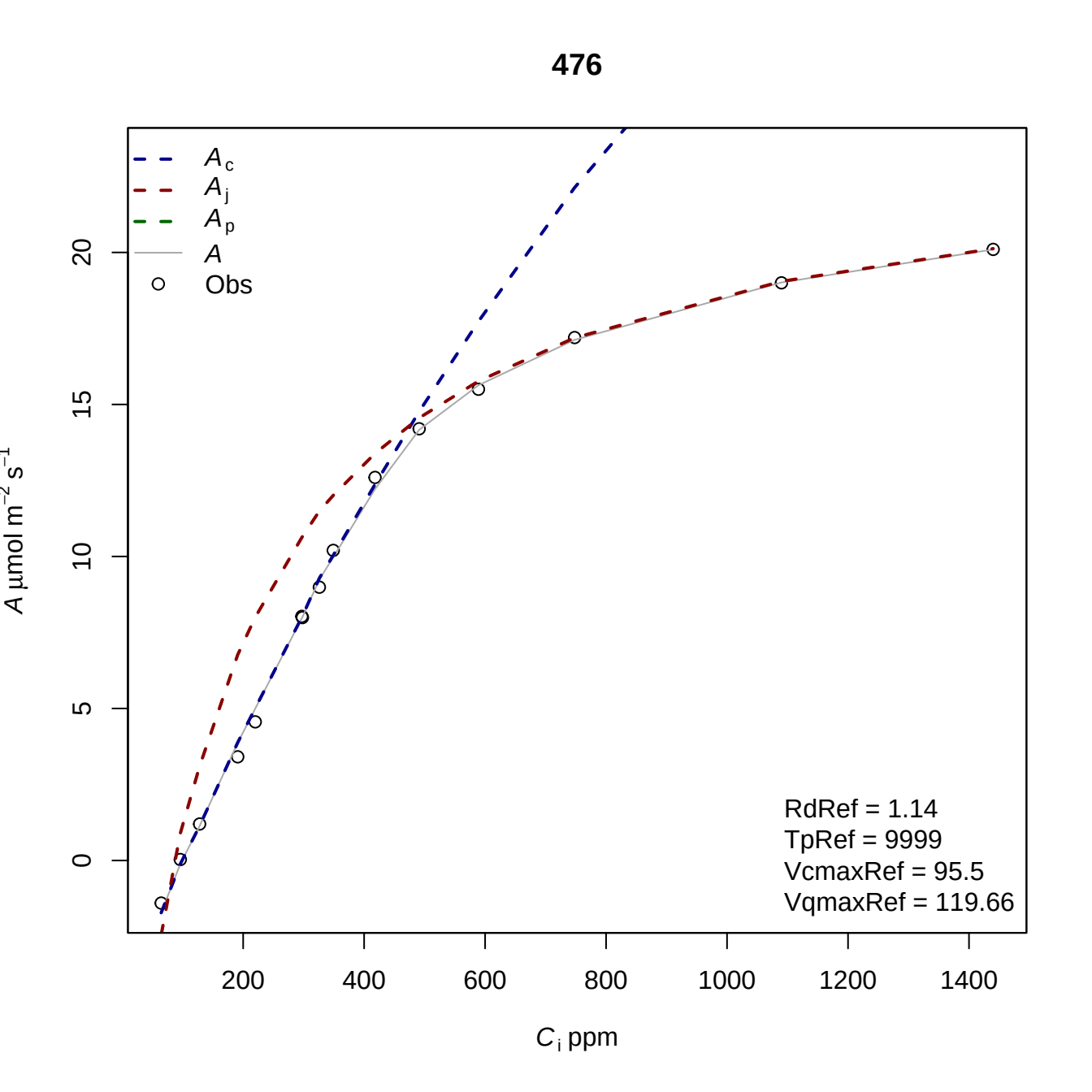

471
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.18
TpRef = 9999
VcmaxRef = 153.08
VqmaxRef = 237.86
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800

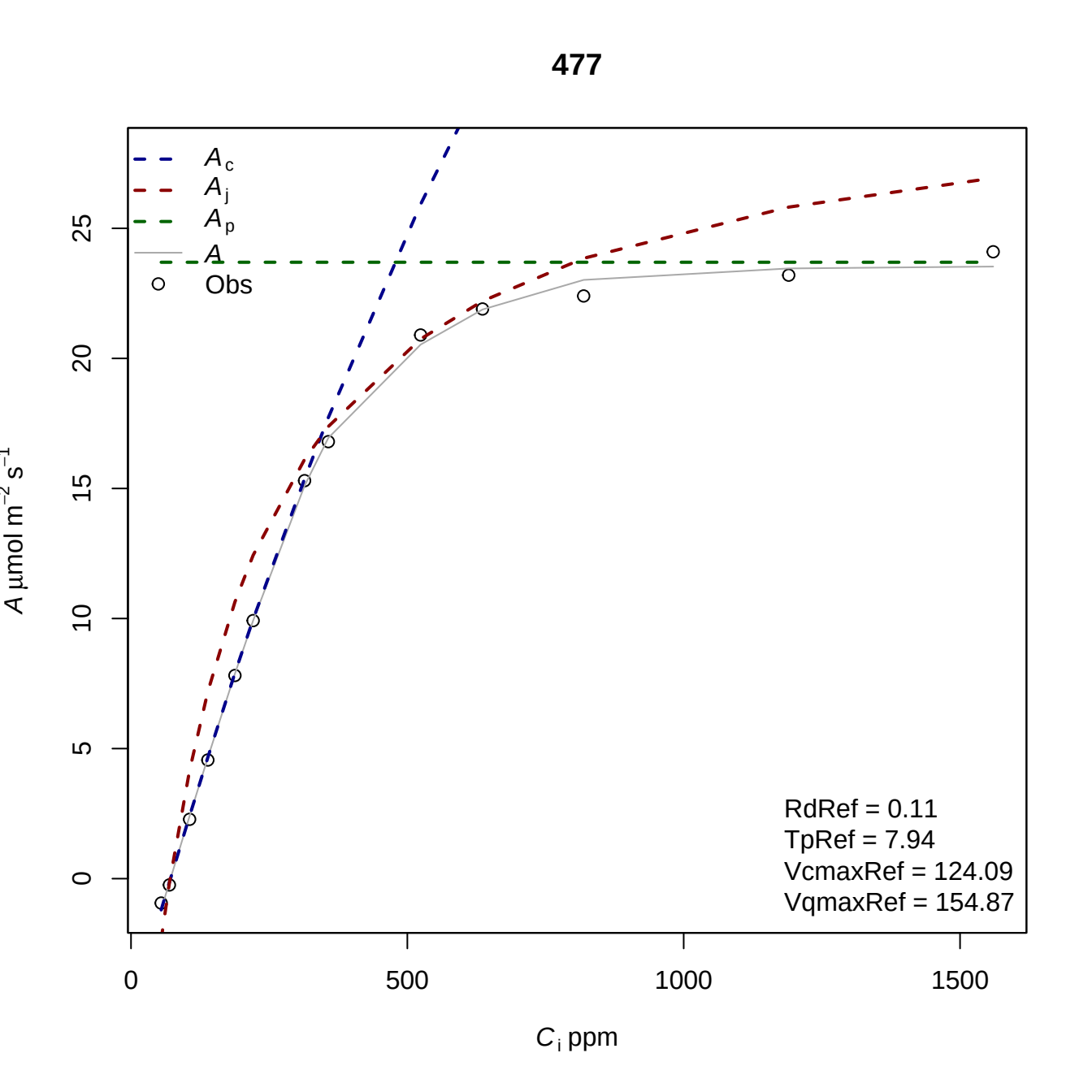

472
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2
TpRef = 9999
VcmaxRef = 77.53
VqmaxRef = 74.59
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800
1000

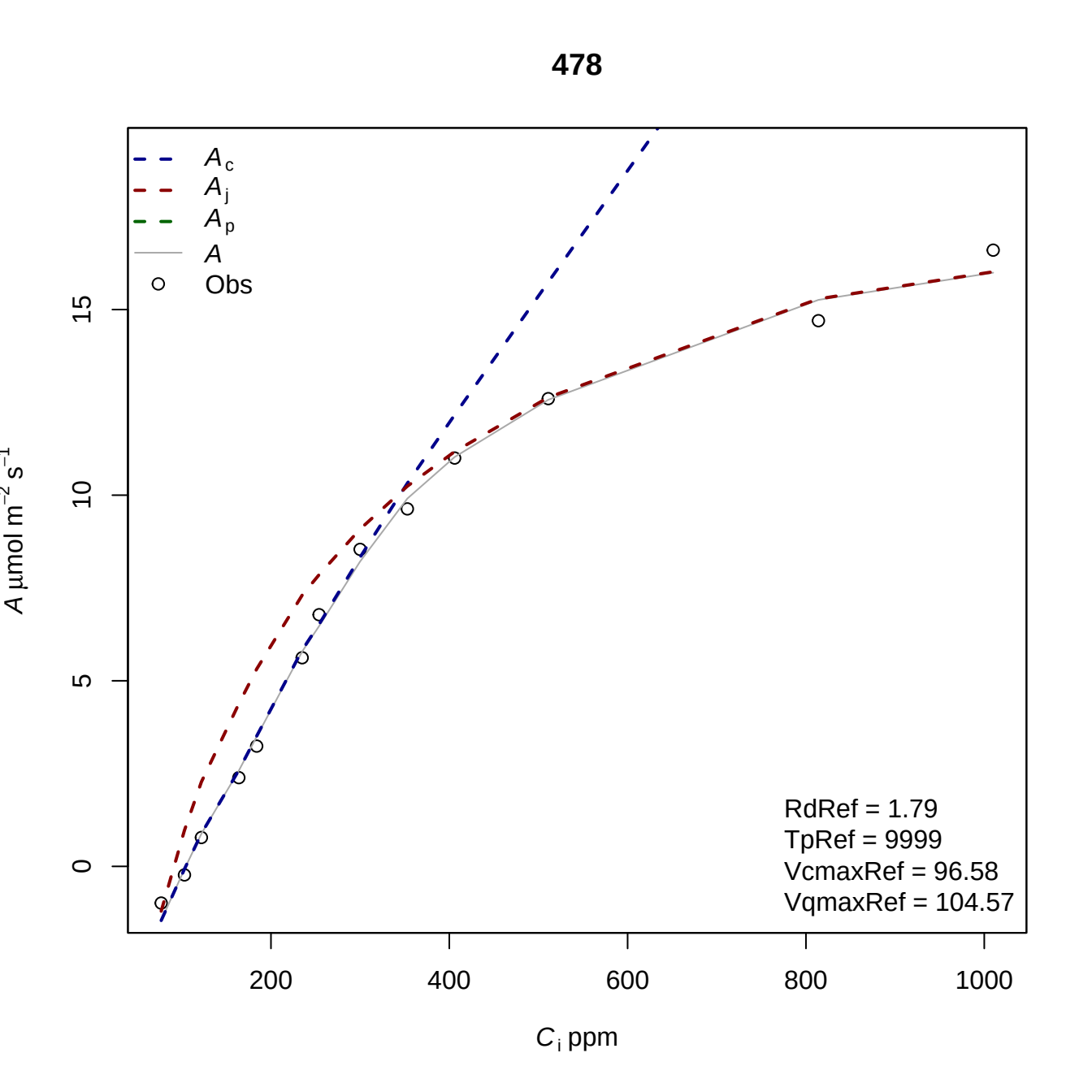

473
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.33
TpRef = 9999
VcmaxRef = 92.55
VqmaxRef = 98.37
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800

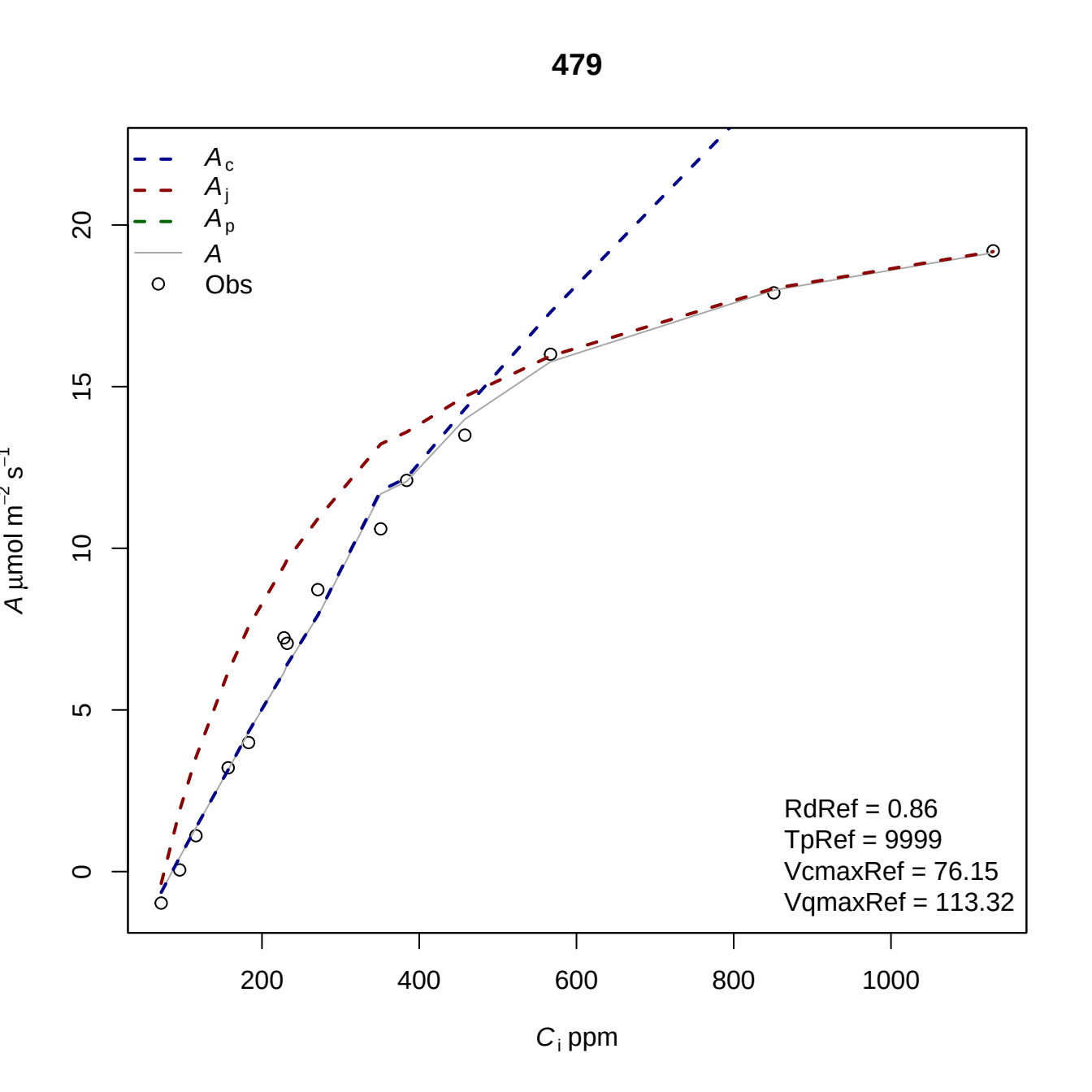

474
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.55
TpRef = 9999
VcmaxRef = 55.43
VqmaxRef = 72.85
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

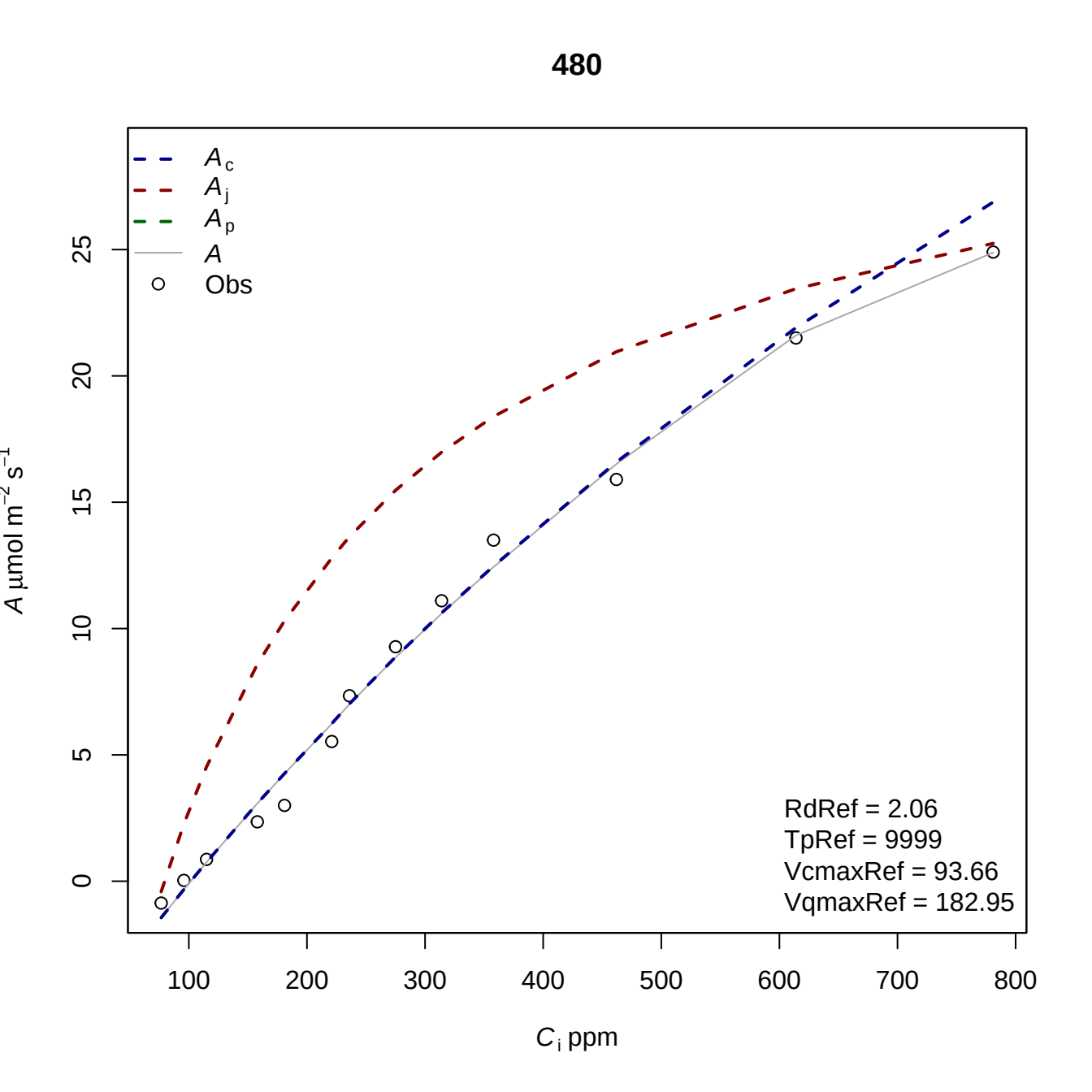

475
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.9
TpRef = 9999
VcmaxRef = 52.39
VqmaxRef = 65.47
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
100
200
300
400
500
600
700

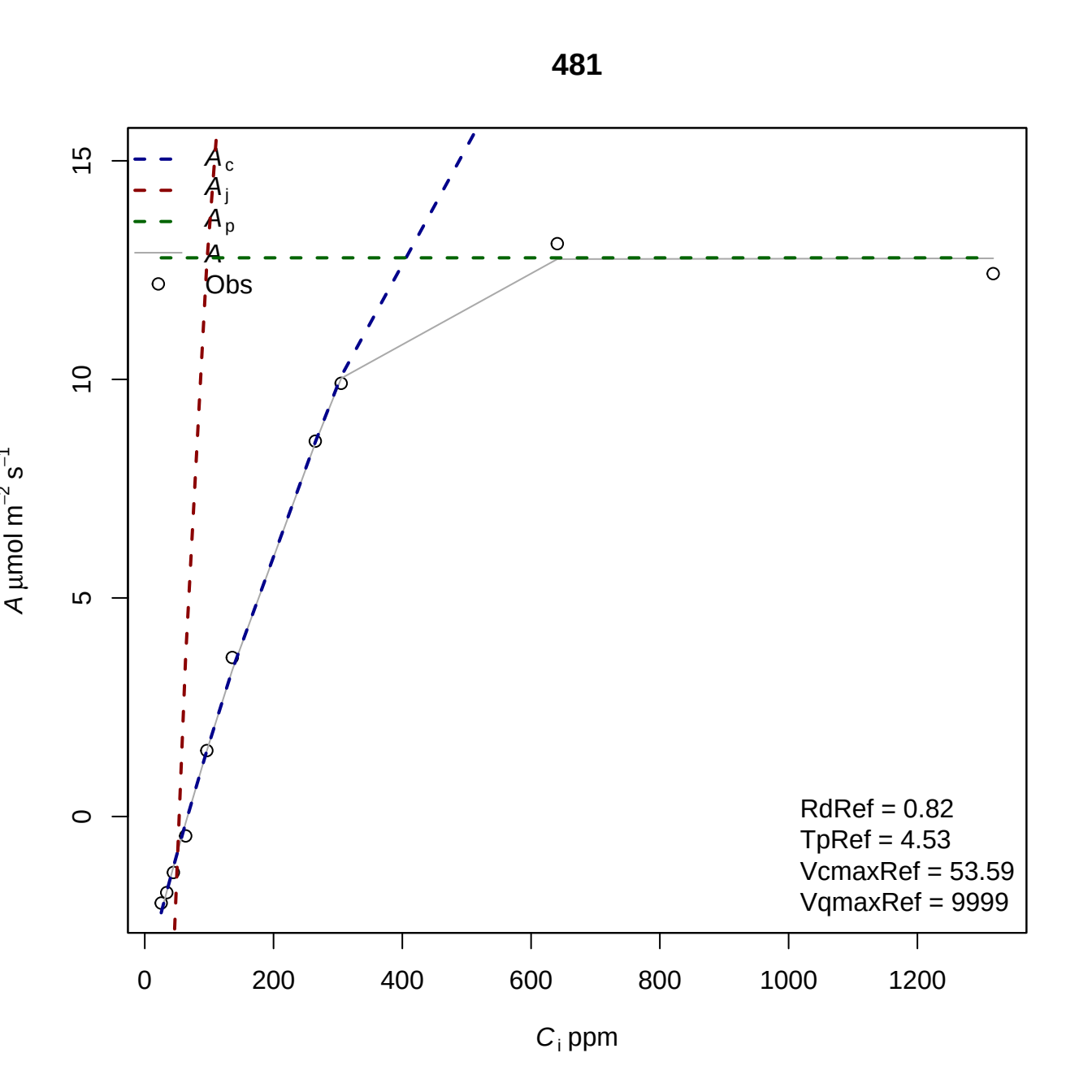

476
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.14
TpRef = 9999
VcmaxRef = 95.5
VqmaxRef = 119.66
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

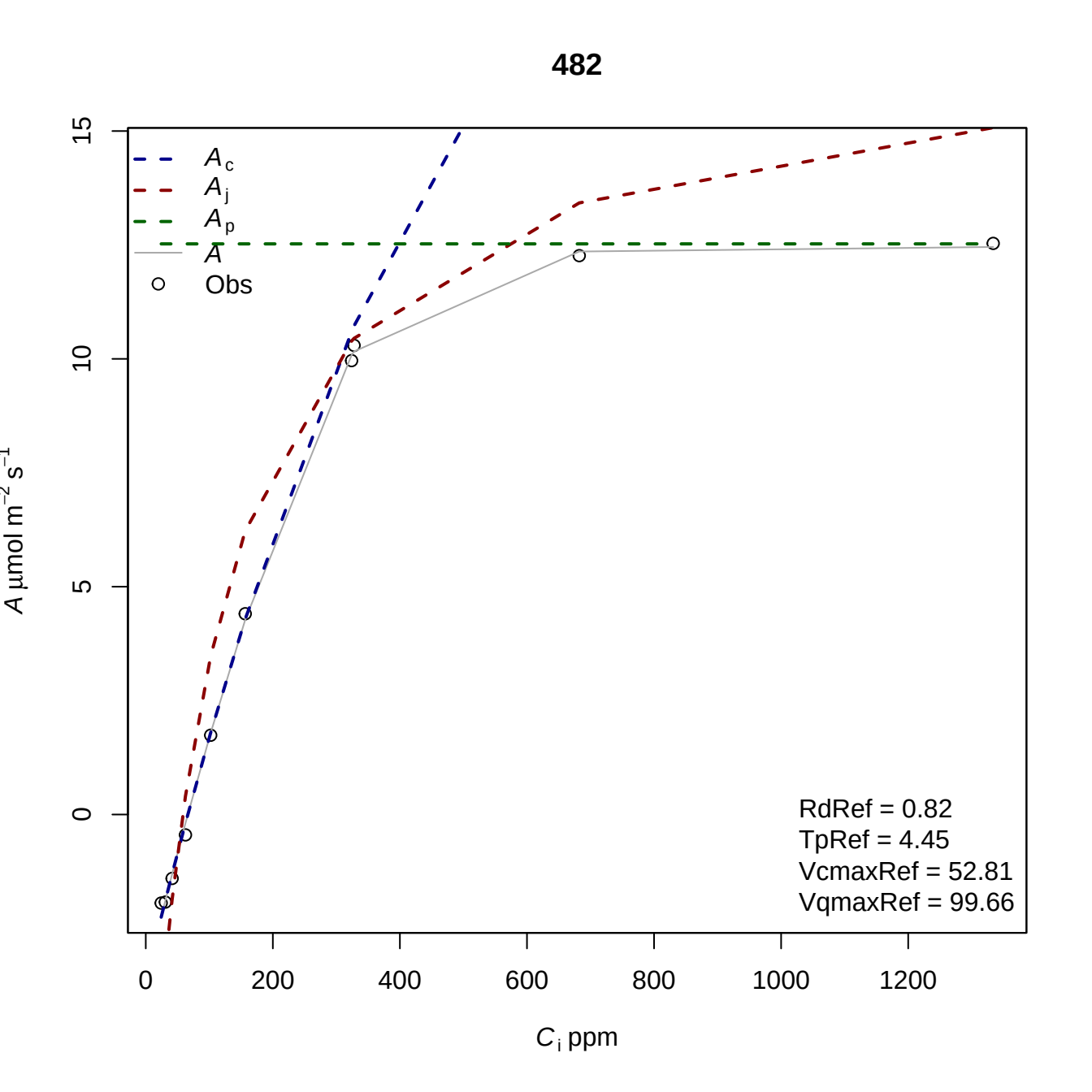

477
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.11
TpRef = 7.94
VcmaxRef = 124.09
VqmaxRef = 154.87
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

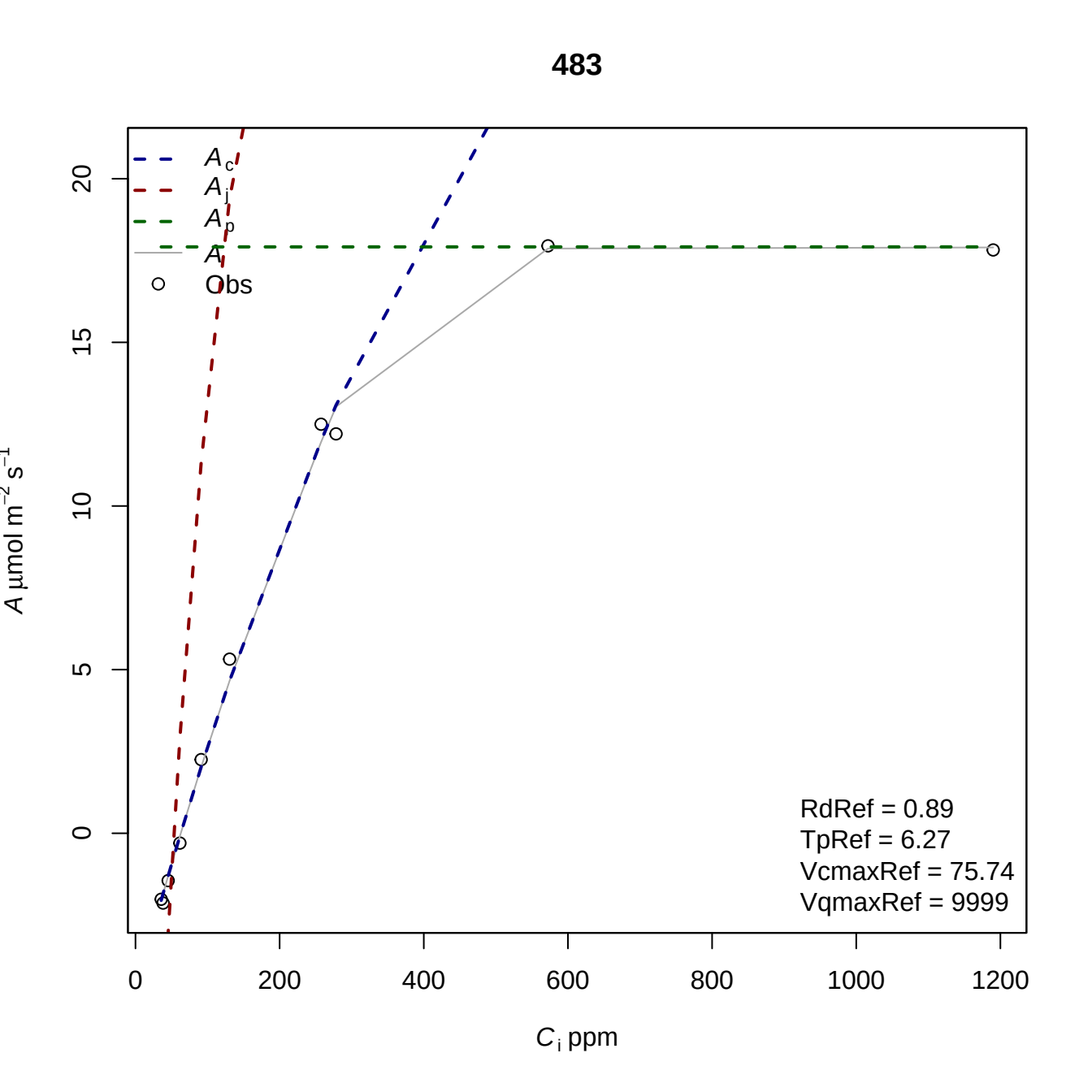

478
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.79
TpRef = 9999
VcmaxRef = 96.58
VqmaxRef = 104.57
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

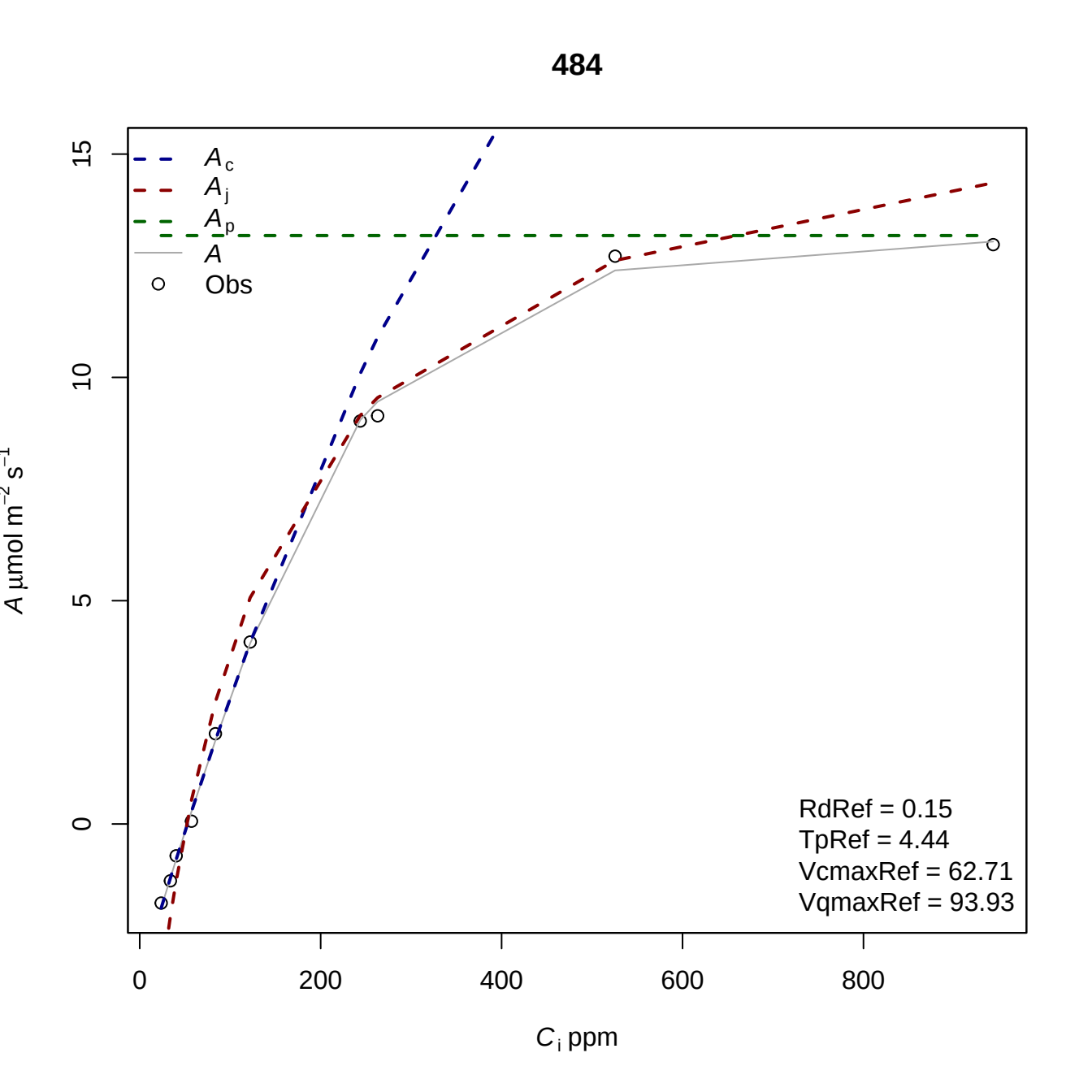

479
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.86
TpRef = 9999
VcmaxRef = 76.15
VqmaxRef = 113.32
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000

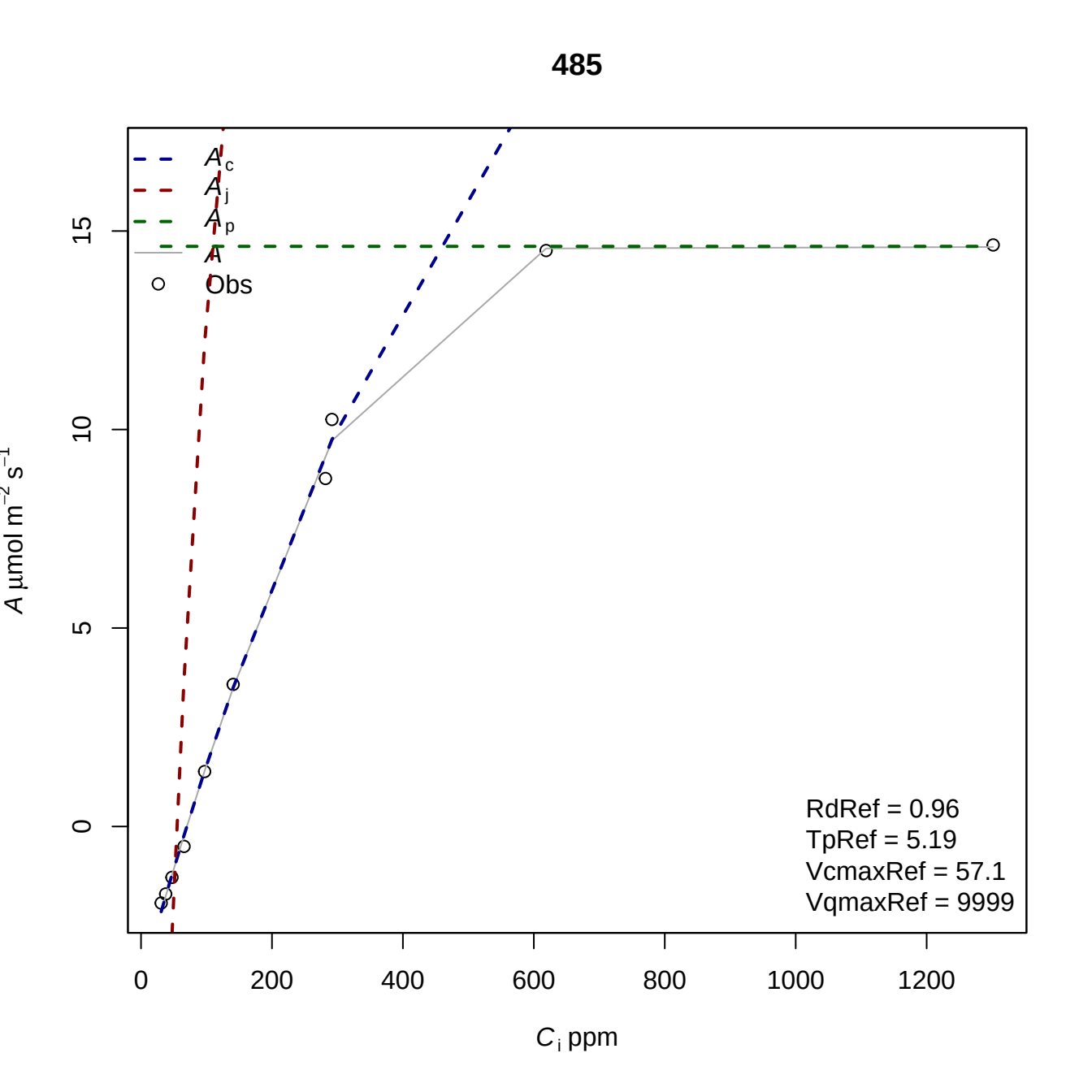

480
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.06
TpRef = 9999
VcmaxRef = 93.66
VqmaxRef = 182.95
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
100
200
300
400
500
600
700
800

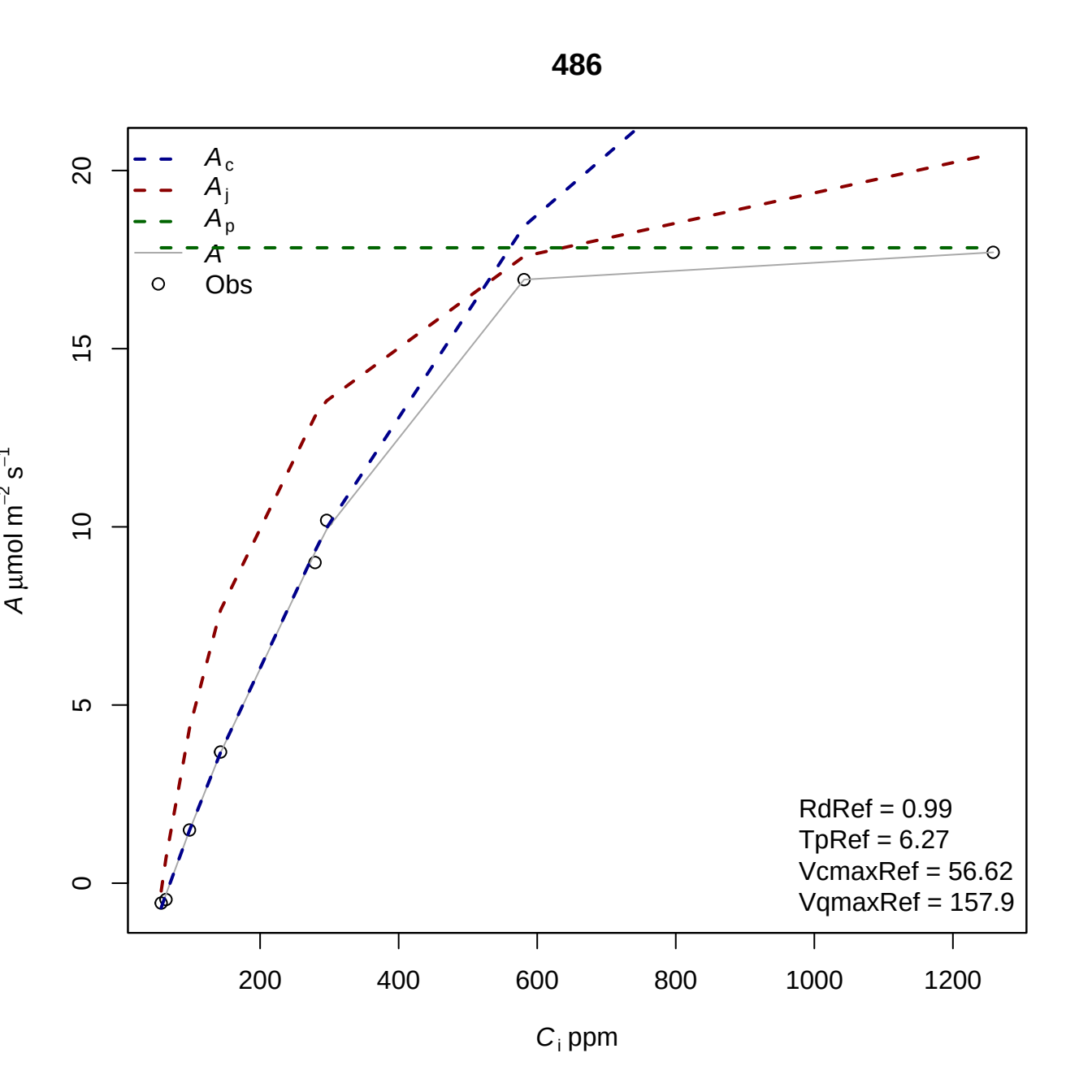

481
$A_c$
$A_j$
$A_p$
A
Obs
RdRef = 0.82
TpRef = 4.53
VcmaxRef = 53.59
VqmaxRef = 9999
A µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

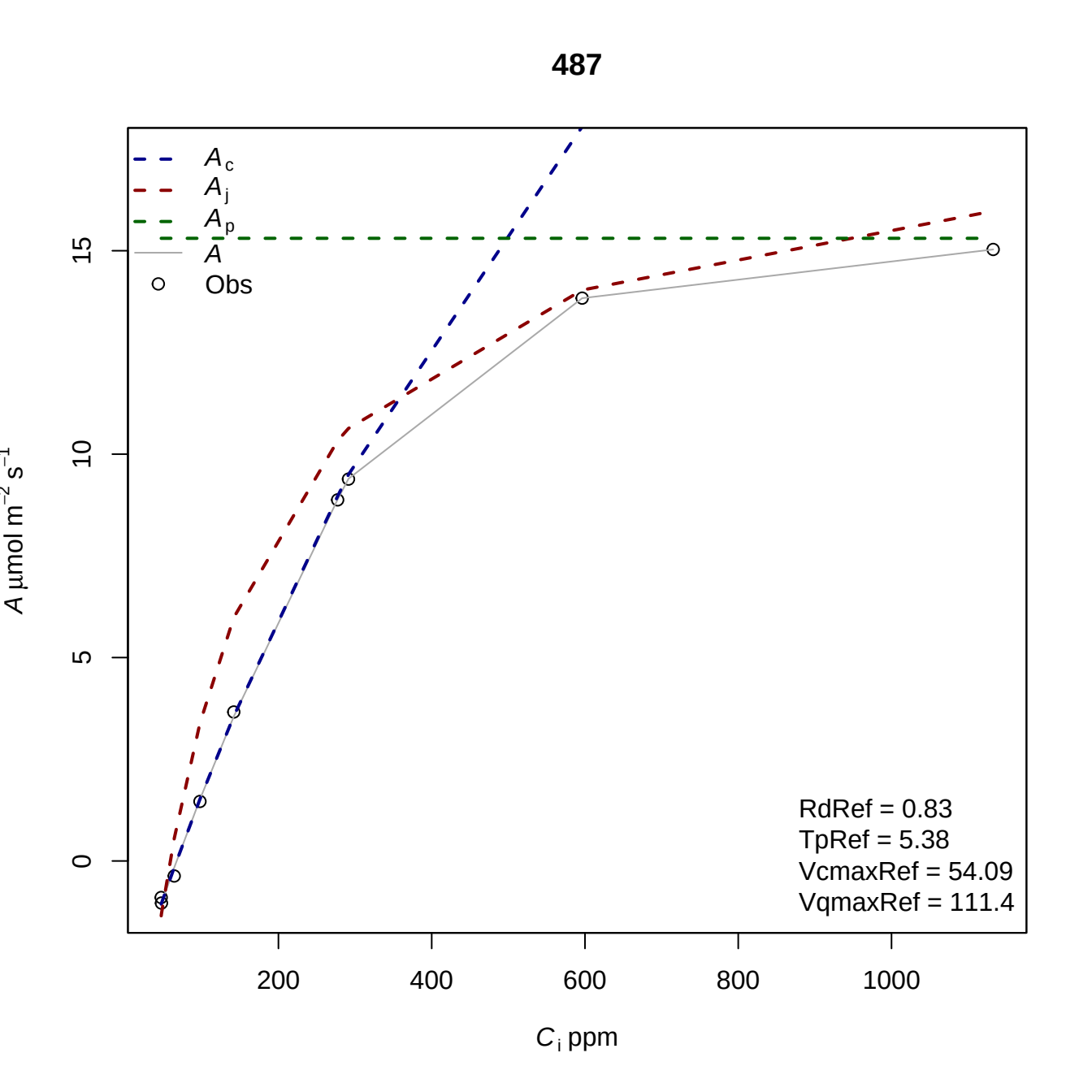

482
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.82
TpRef = 4.45
VcmaxRef = 52.81
VqmaxRef = 99.66
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

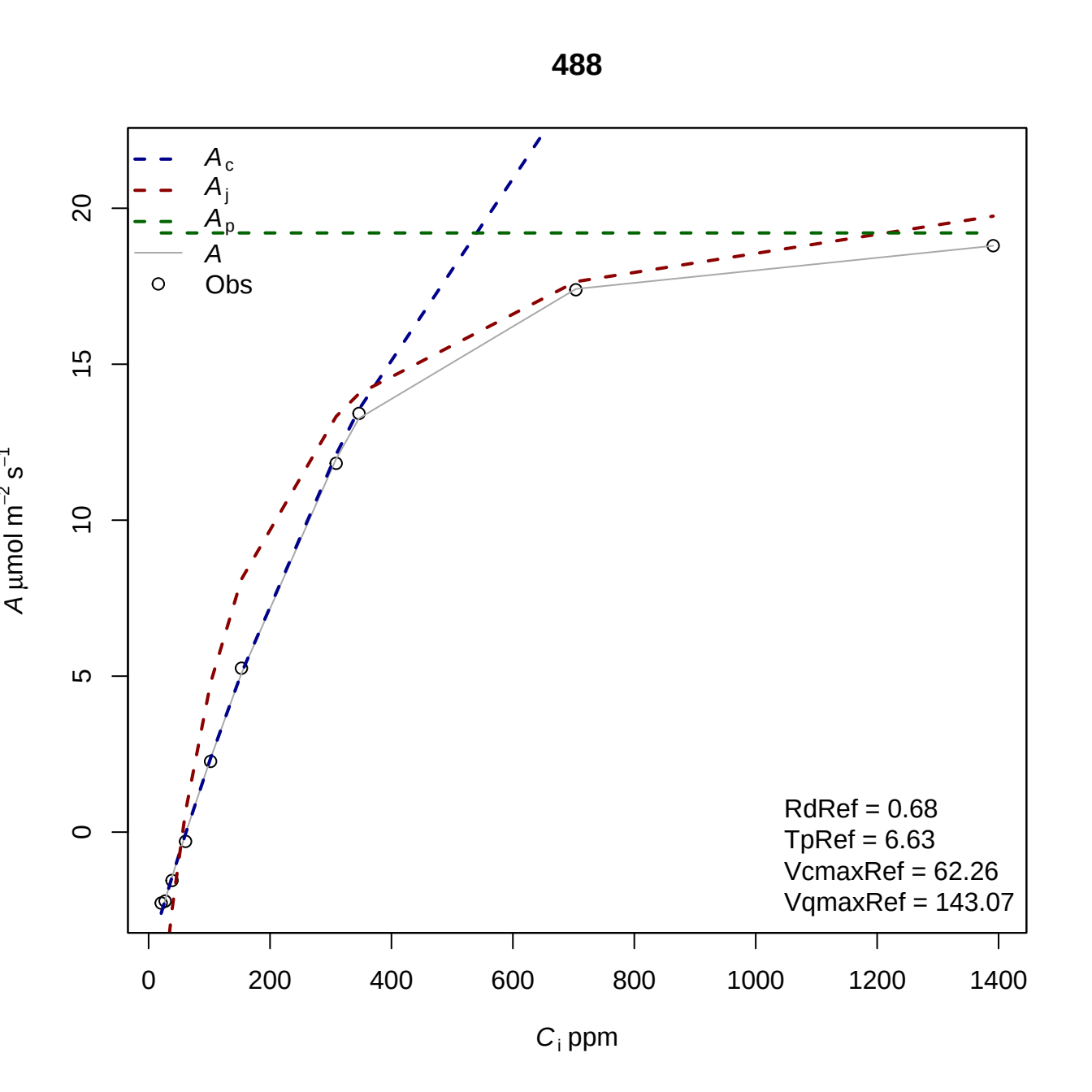

483
$A_c$
$A_j$
$A_p$
$A_n$
Obs
RdRef = 0.89
TpRef = 6.27
VcmaxRef = 75.74
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

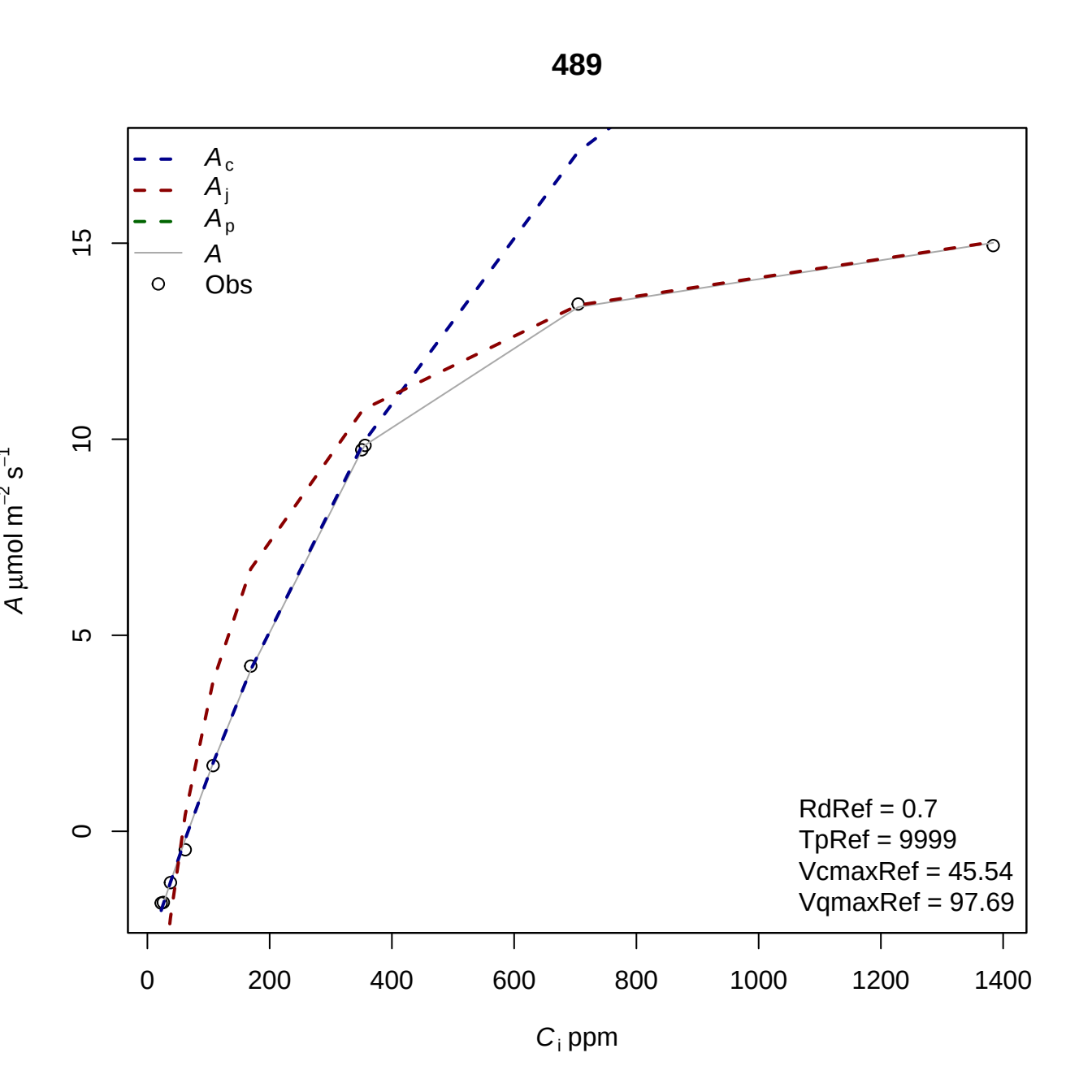

484
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.15
TpRef = 4.44
VcmaxRef = 62.71
VqmaxRef = 93.93
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800

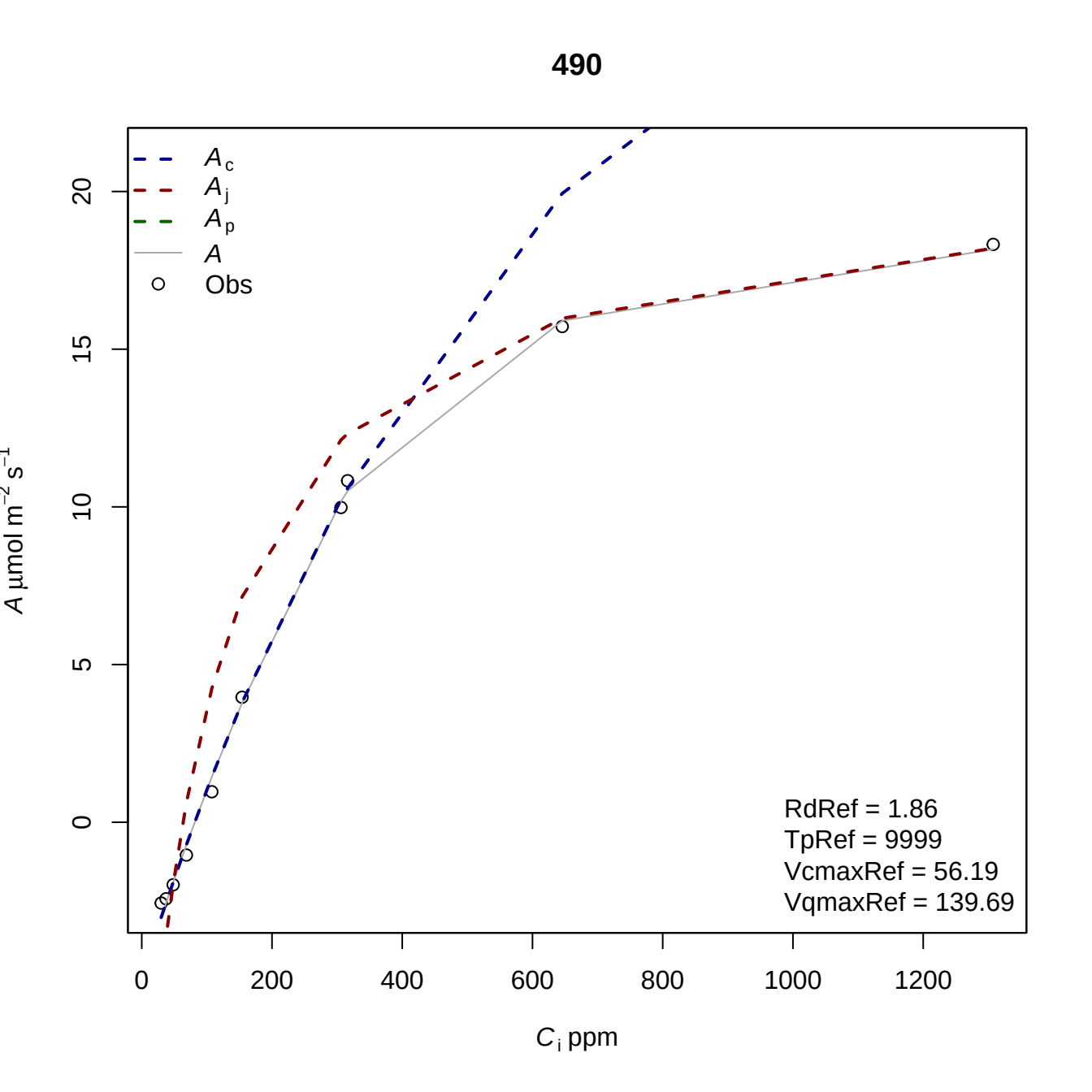

485
$A_c$
$A_j$
$A_p$
A
Obs
RdRef = 0.96
TpRef = 5.19
VcmaxRef = 57.1
VqmaxRef = 9999
A µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

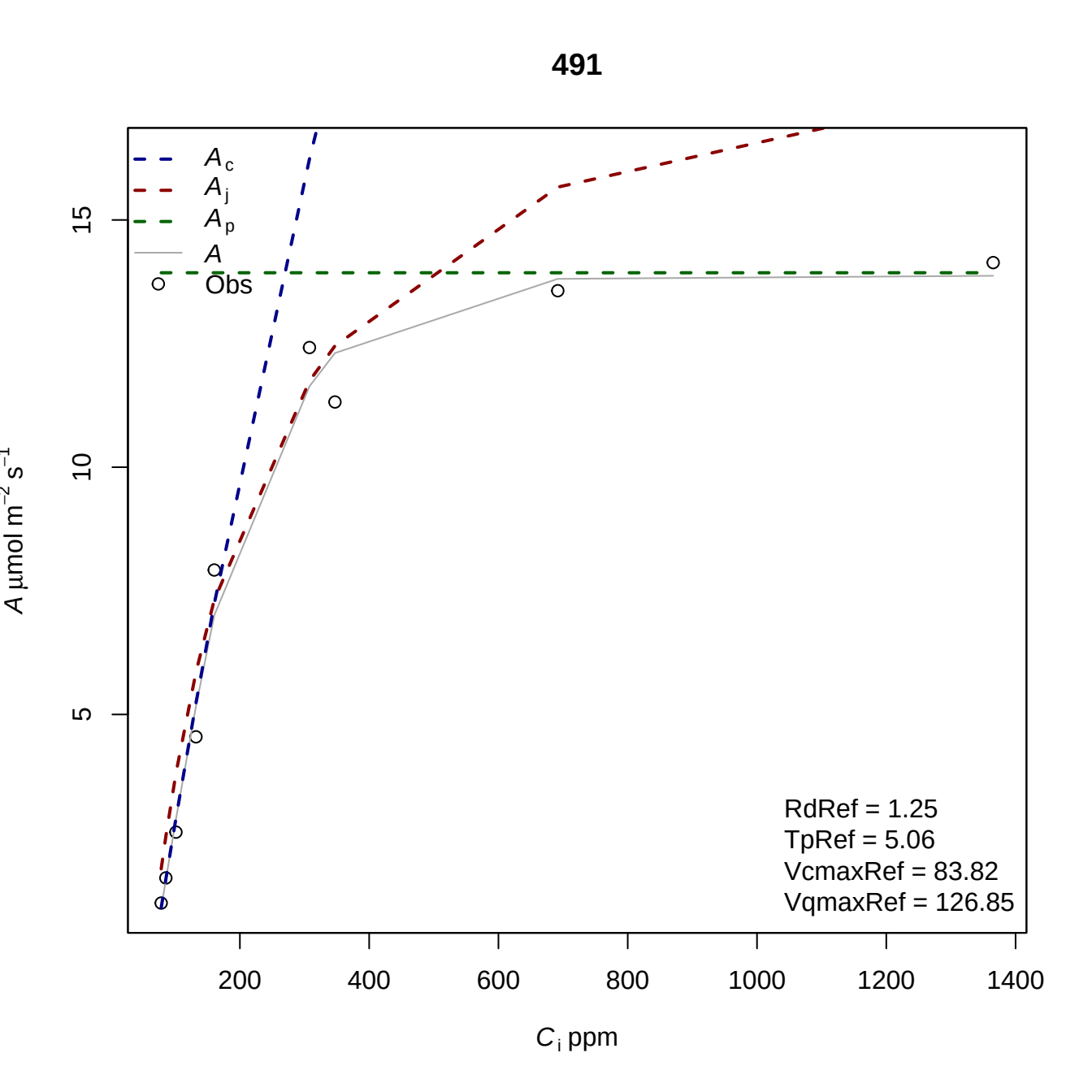

486
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.99
TpRef = 6.27
VcmaxRef = 56.62
VqmaxRef = 157.9
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200

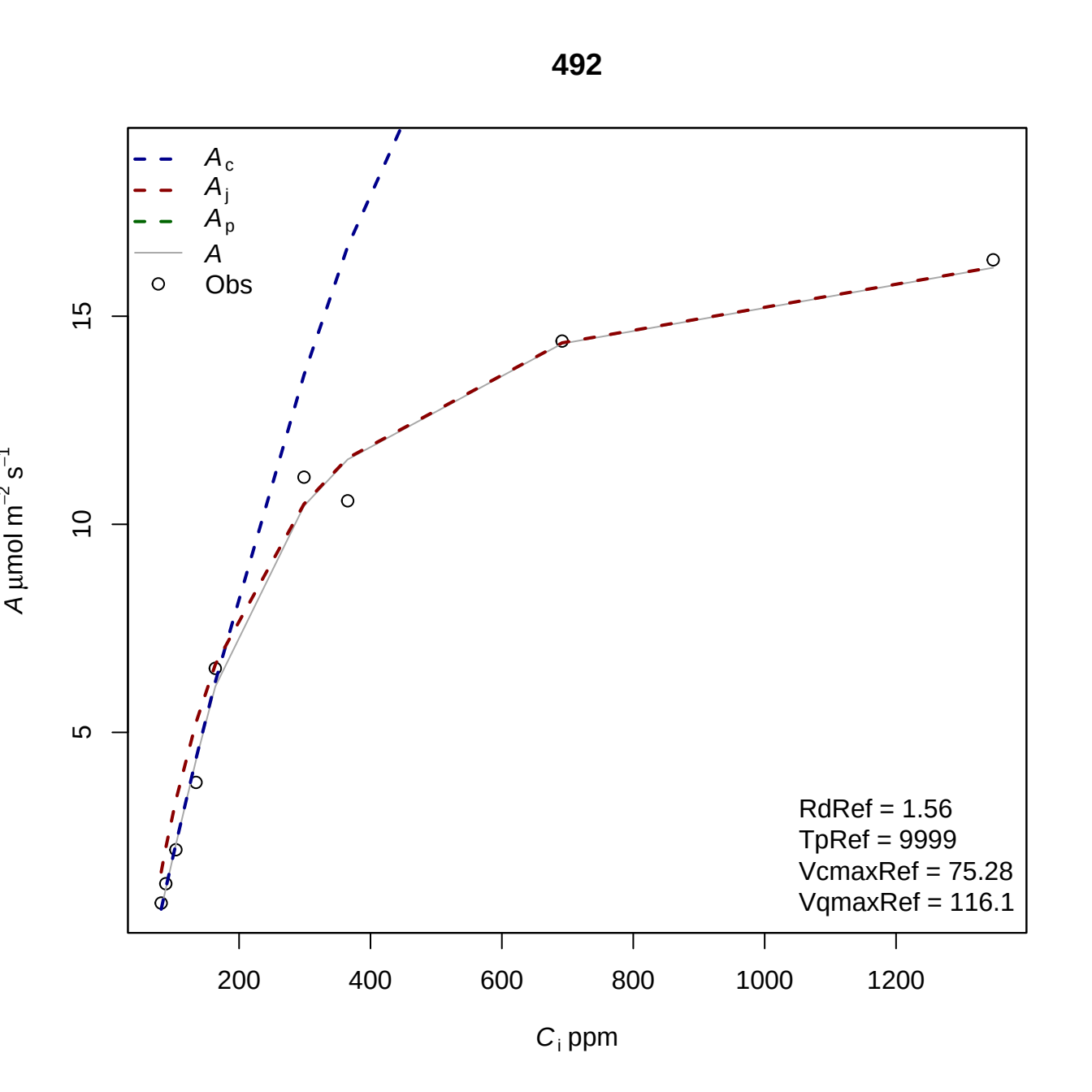

487
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.83
TpRef = 5.38
VcmaxRef = 54.09
VqmaxRef = 111.4
0
5
10
15
200
400
600
800
1000
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm

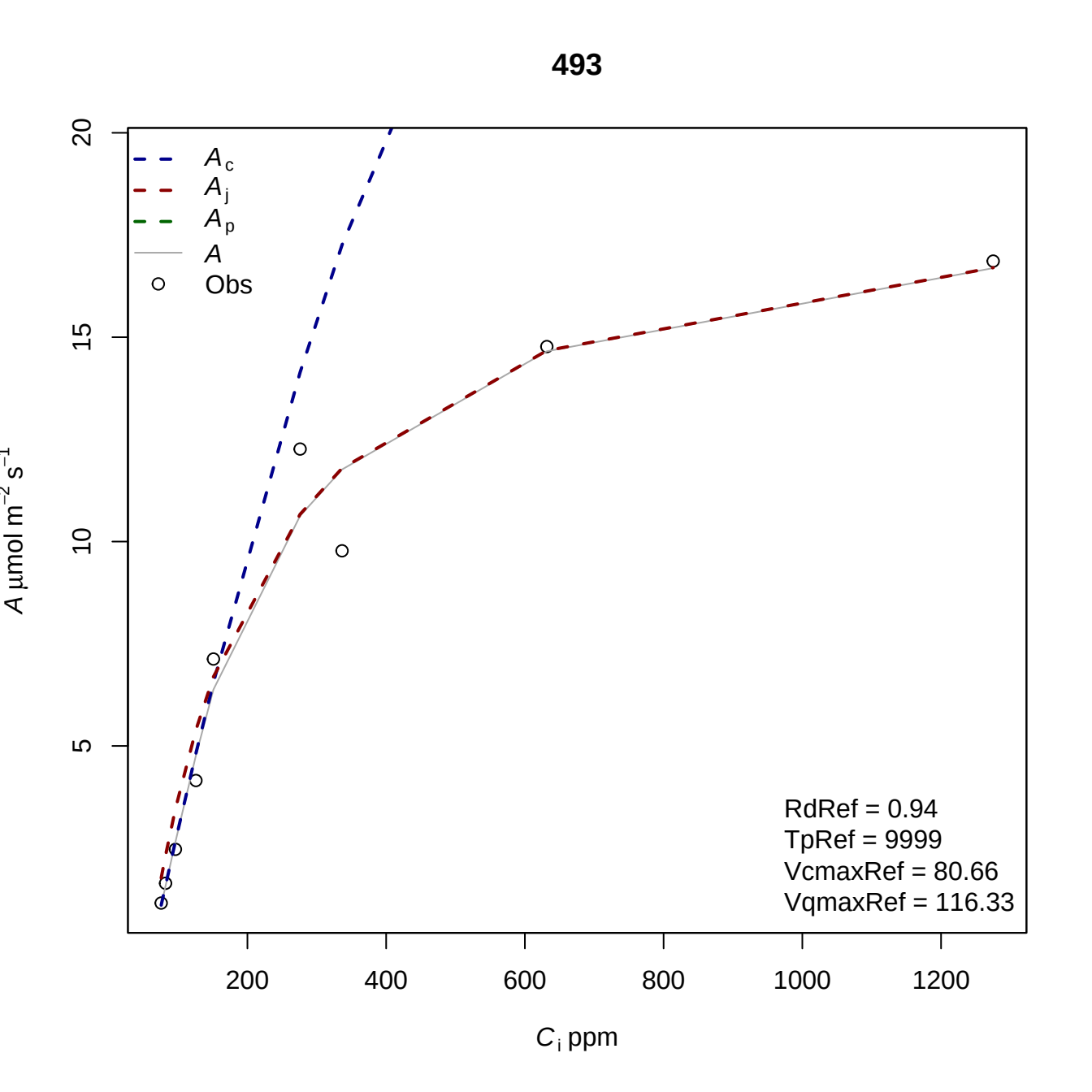

488
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.68
TpRef = 6.63
VcmaxRef = 62.26
VqmaxRef = 143.07
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

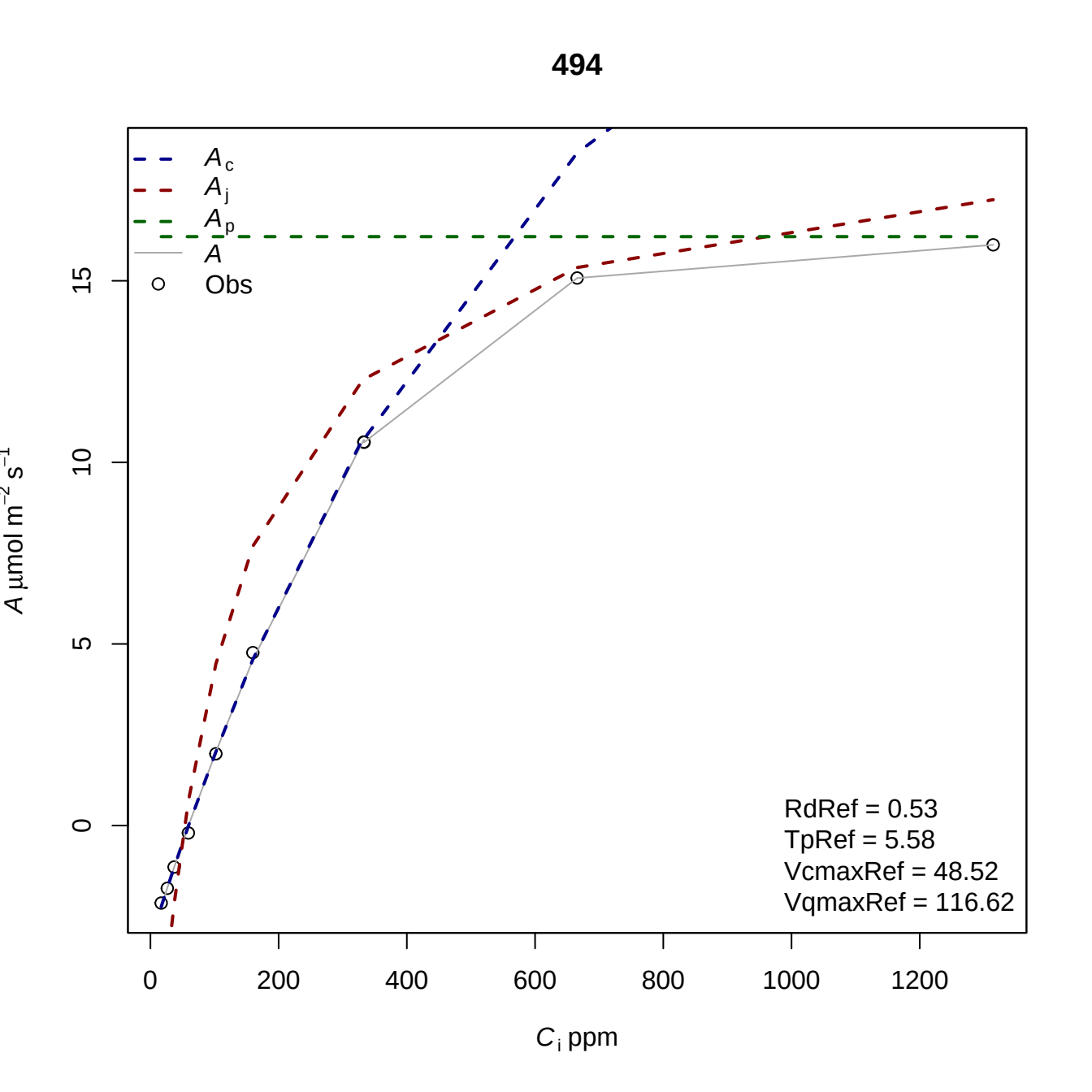

489
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.7
TpRef = 9999
VcmaxRef = 45.54
VqmaxRef = 97.69
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

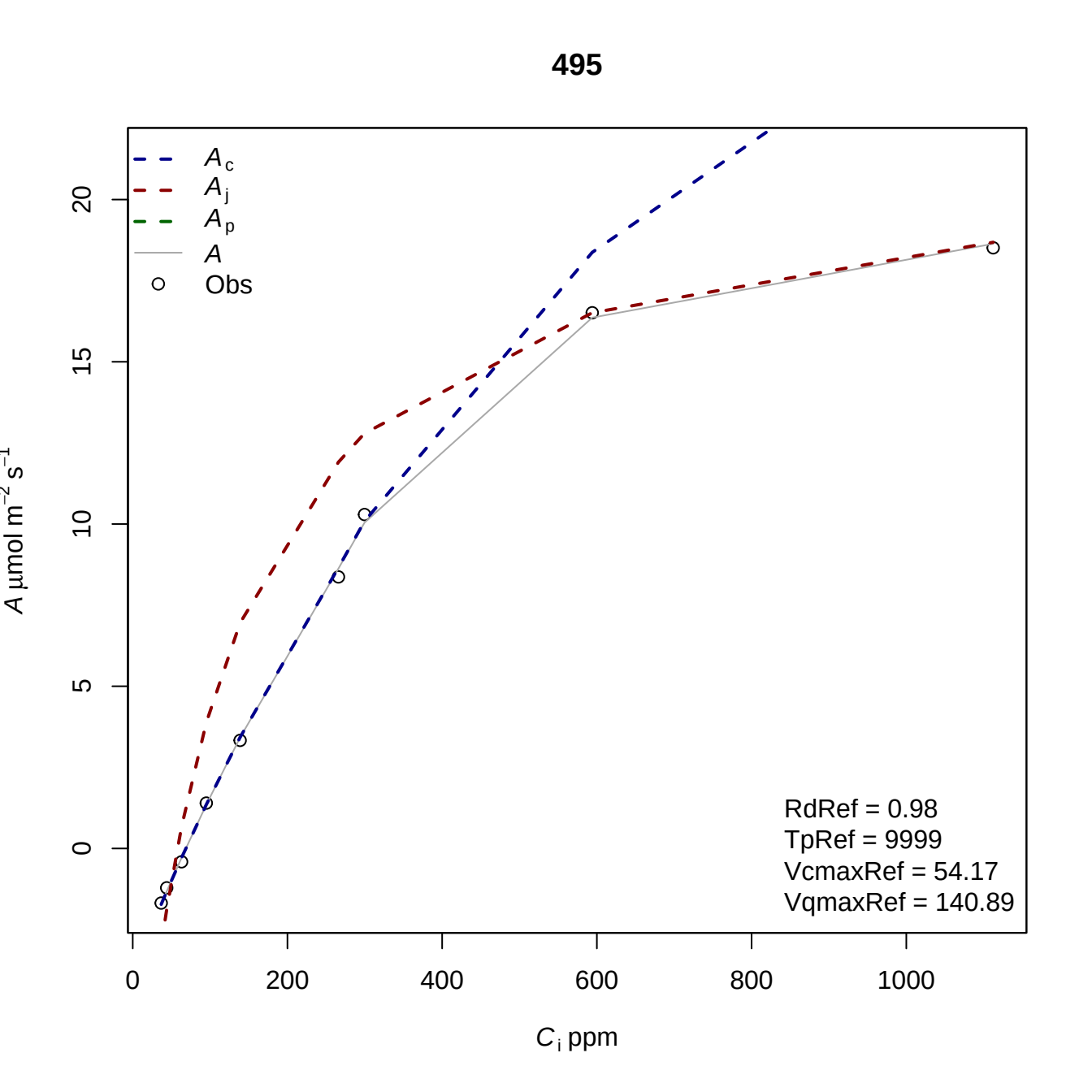

490
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.86
TpRef = 9999
VcmaxRef = 56.19
VqmaxRef = 139.69
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

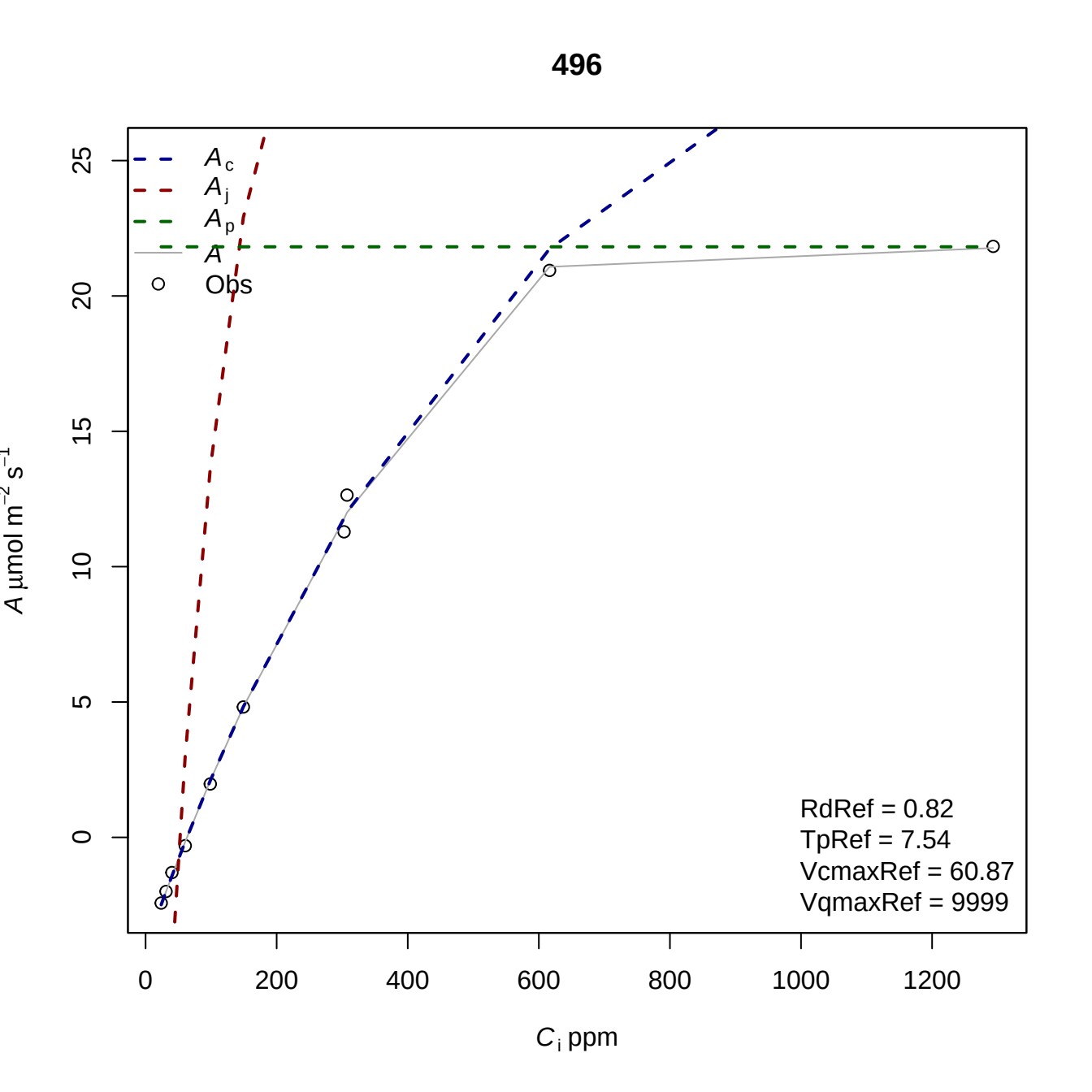

491
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.25
TpRef = 5.06
VcmaxRef = 83.82
VqmaxRef = 126.85
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
200
400
600
800
1000
1200
1400

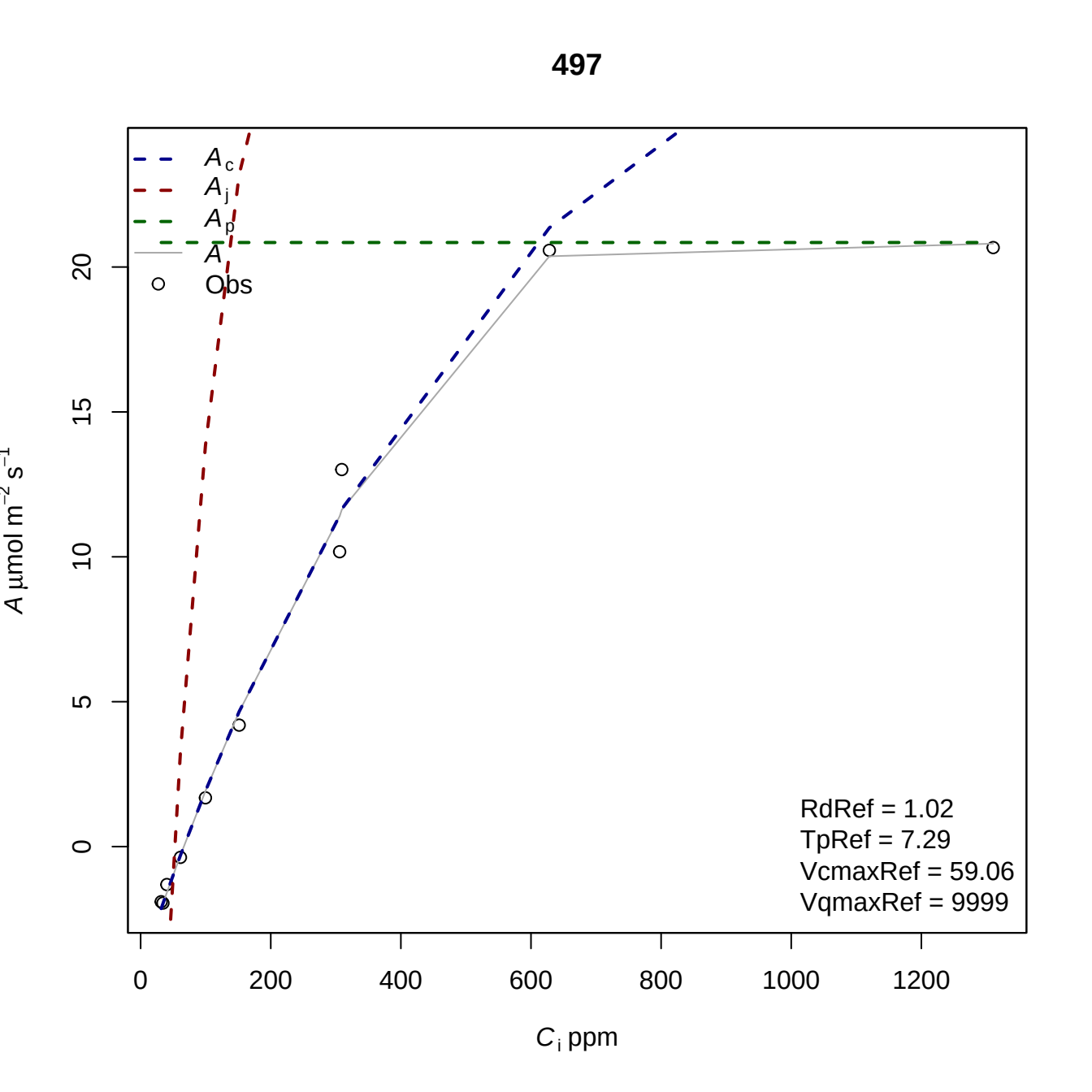

492
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.56
TpRef = 9999
VcmaxRef = 75.28
VqmaxRef = 116.1
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
200
400
600
800
1000
1200

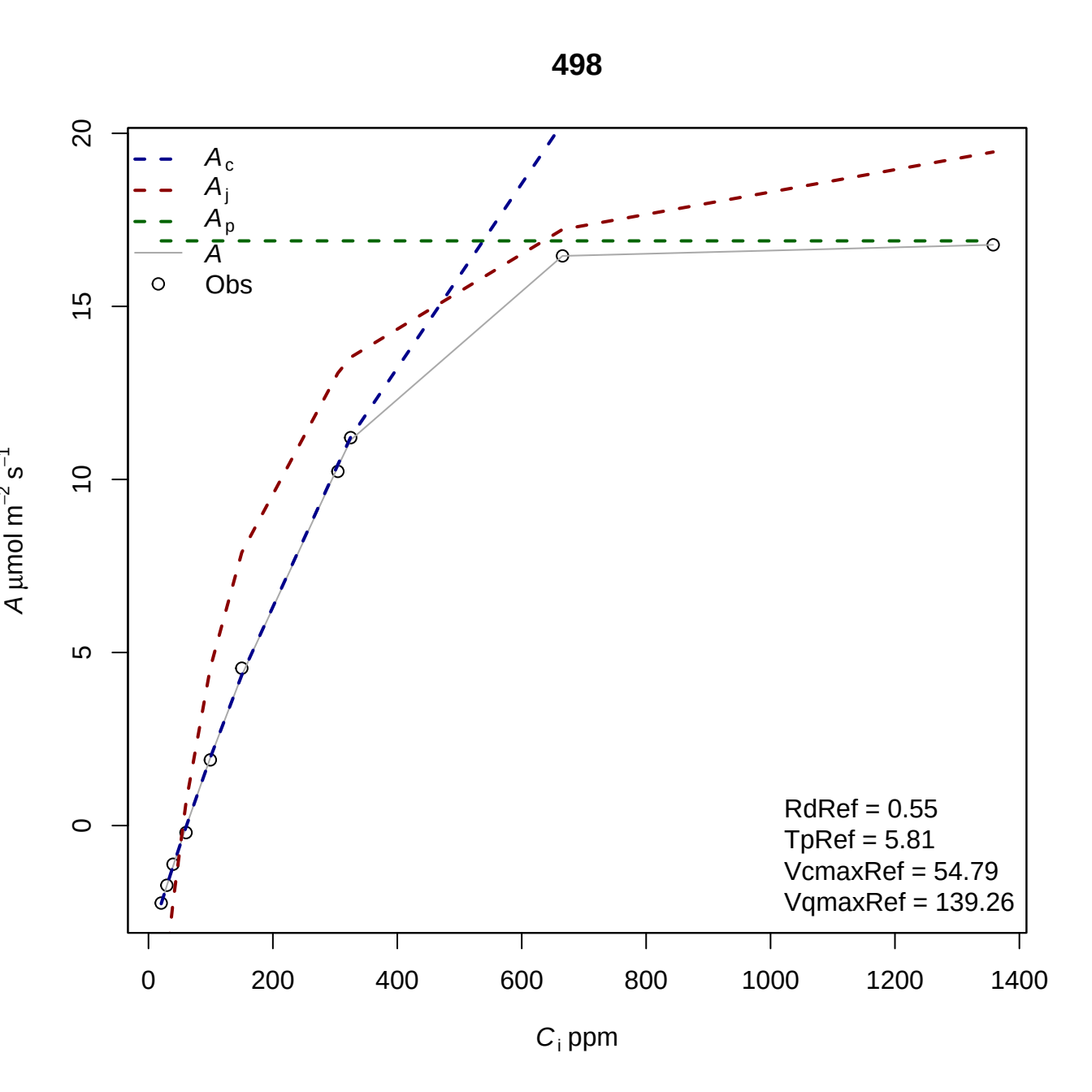

493
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.94
TpRef = 9999
VcmaxRef = 80.66
VqmaxRef = 116.33
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
20
200
400
600
800
1000
1200

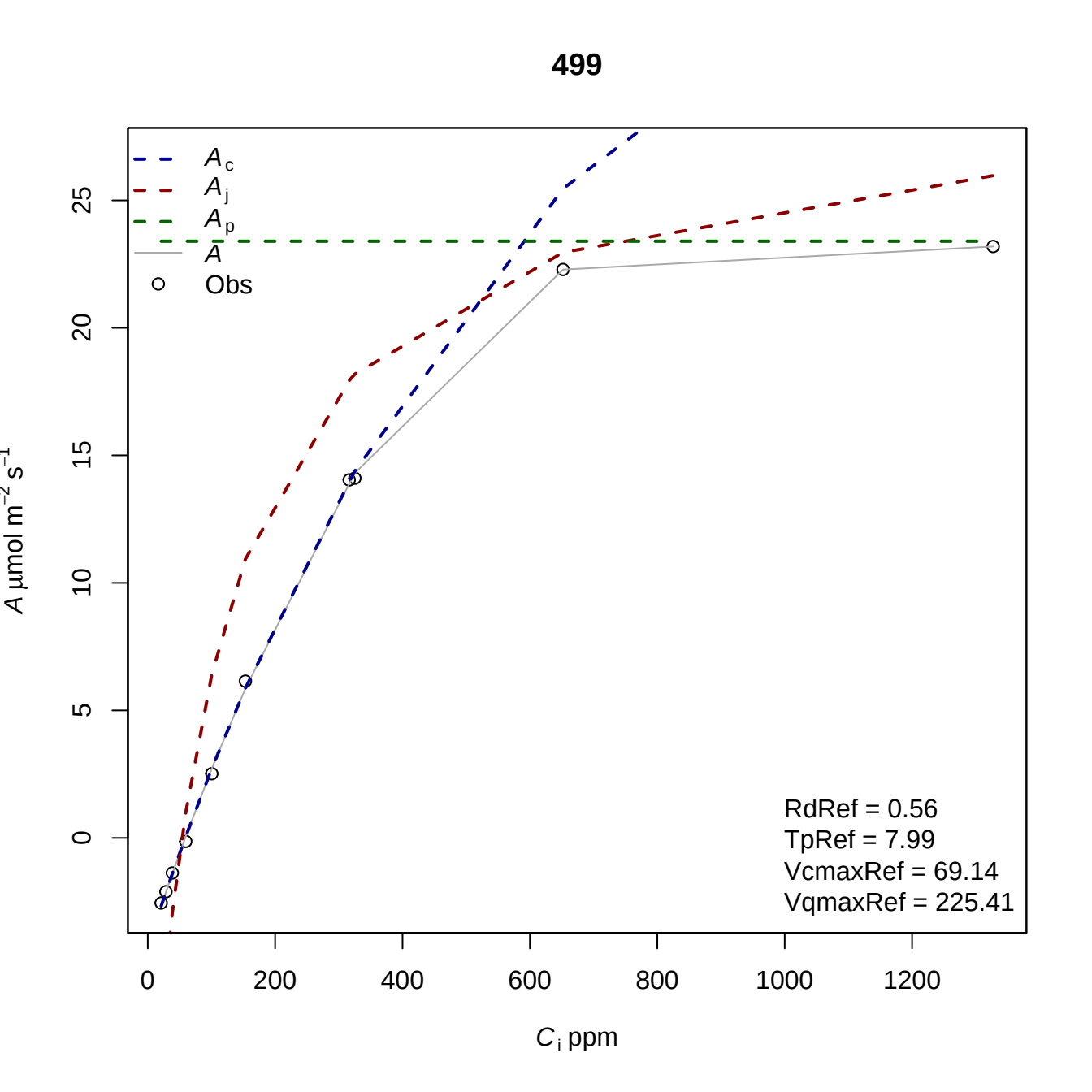

494
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.53
TpRef = 5.58
VcmaxRef = 48.52
VqmaxRef = 116.62
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

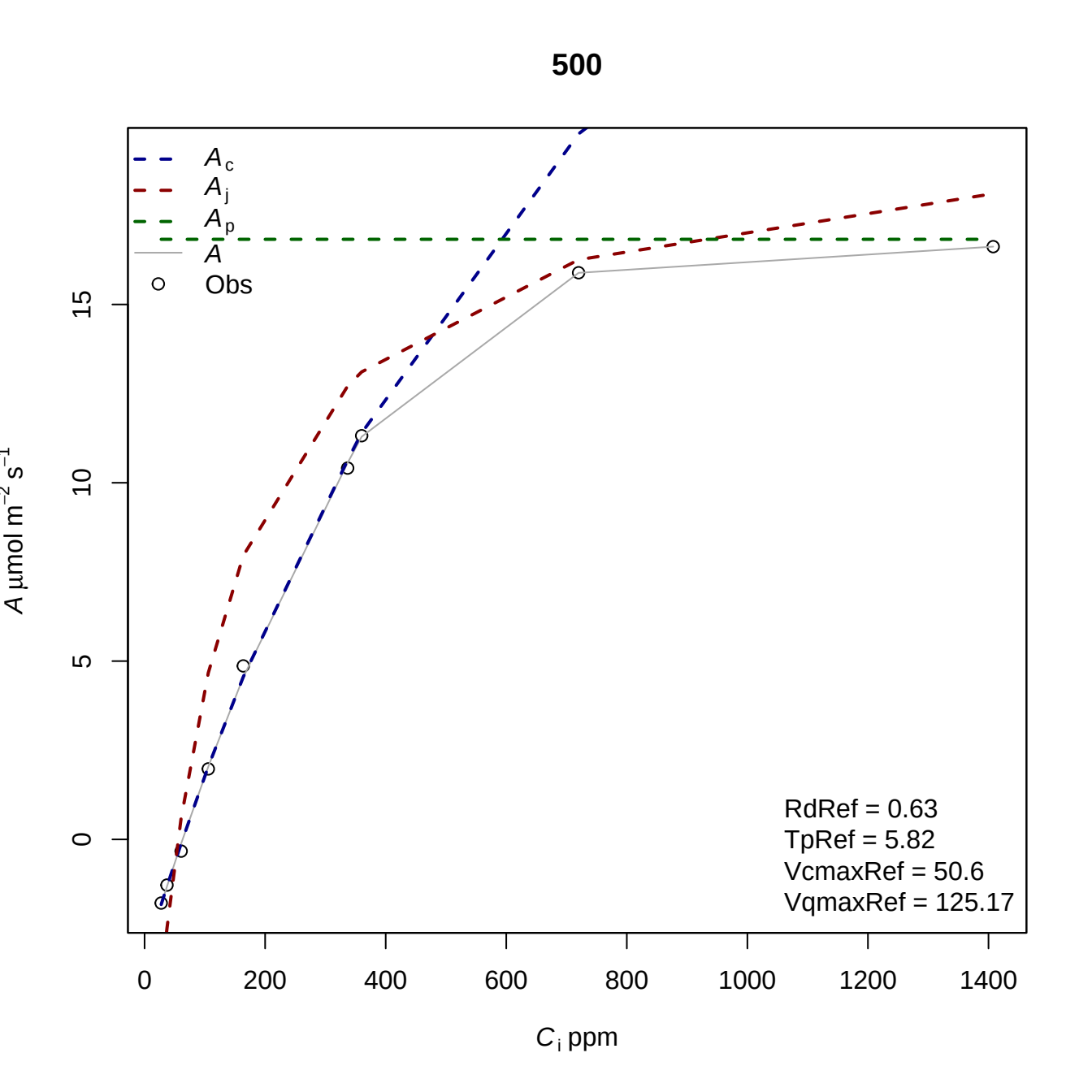

495
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.98
TpRef = 9999
VcmaxRef = 54.17
VqmaxRef = 140.89
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000

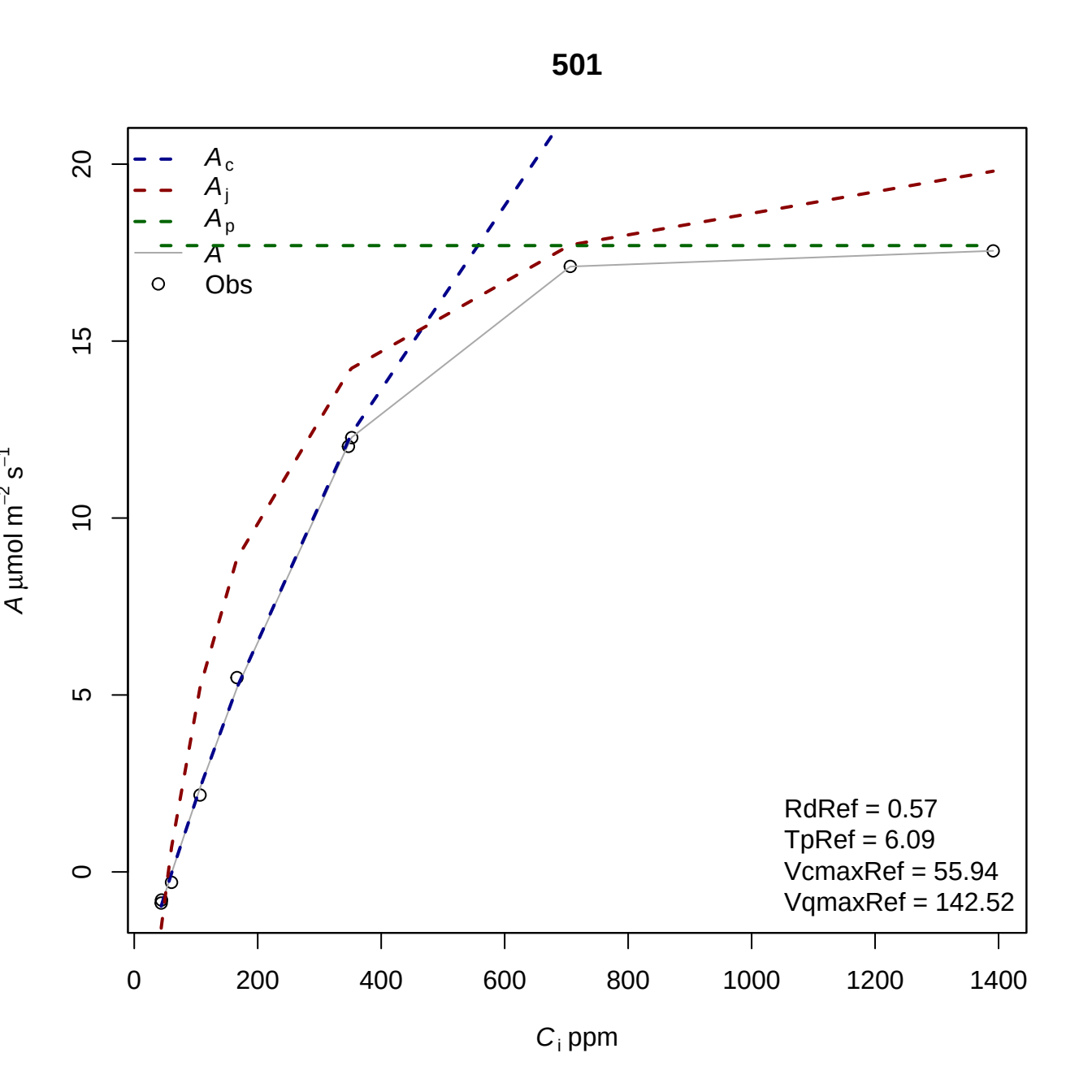

496
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.82
TpRef = 7.54
VcmaxRef = 60.87
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

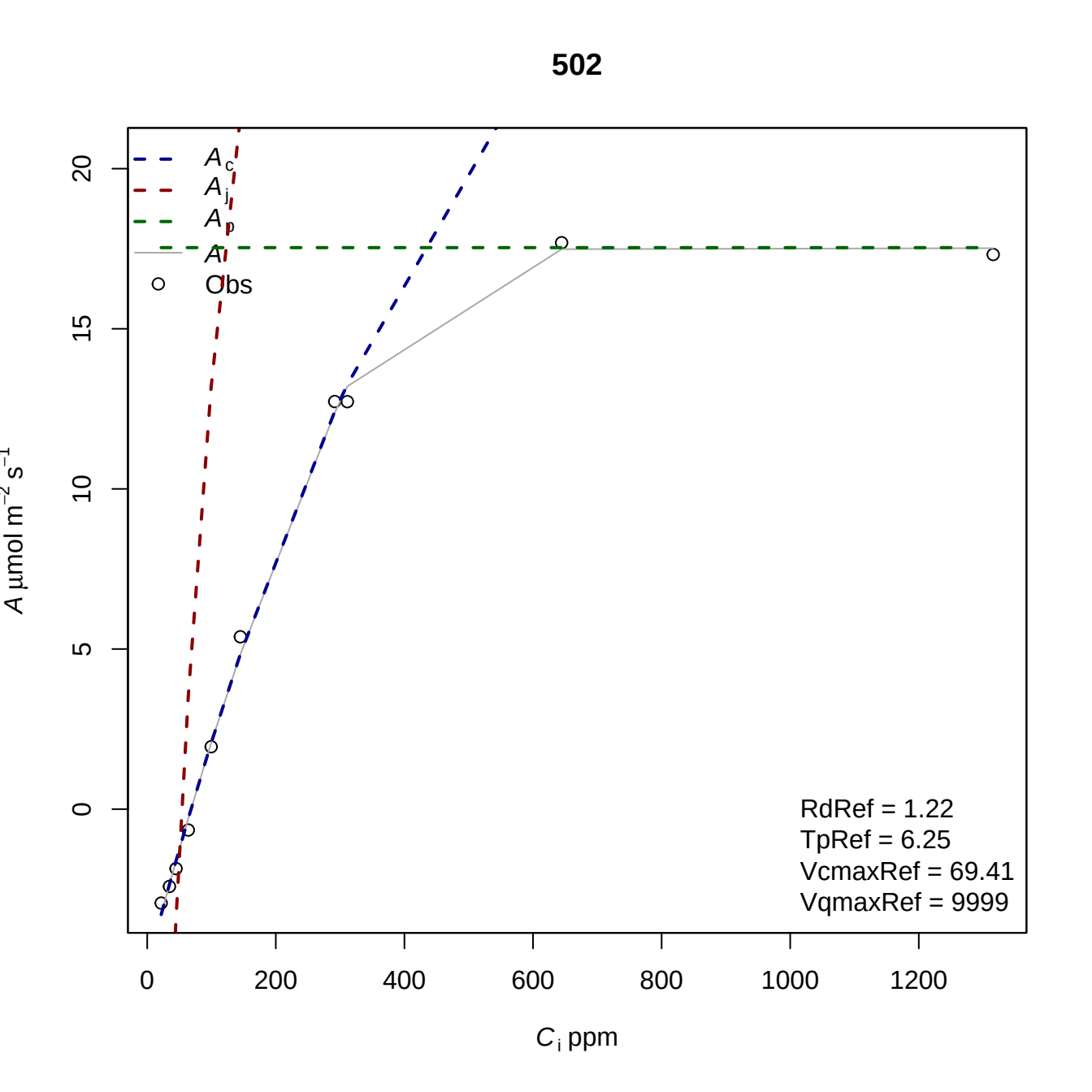

497
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.02
TpRef = 7.29
VcmaxRef = 59.06
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

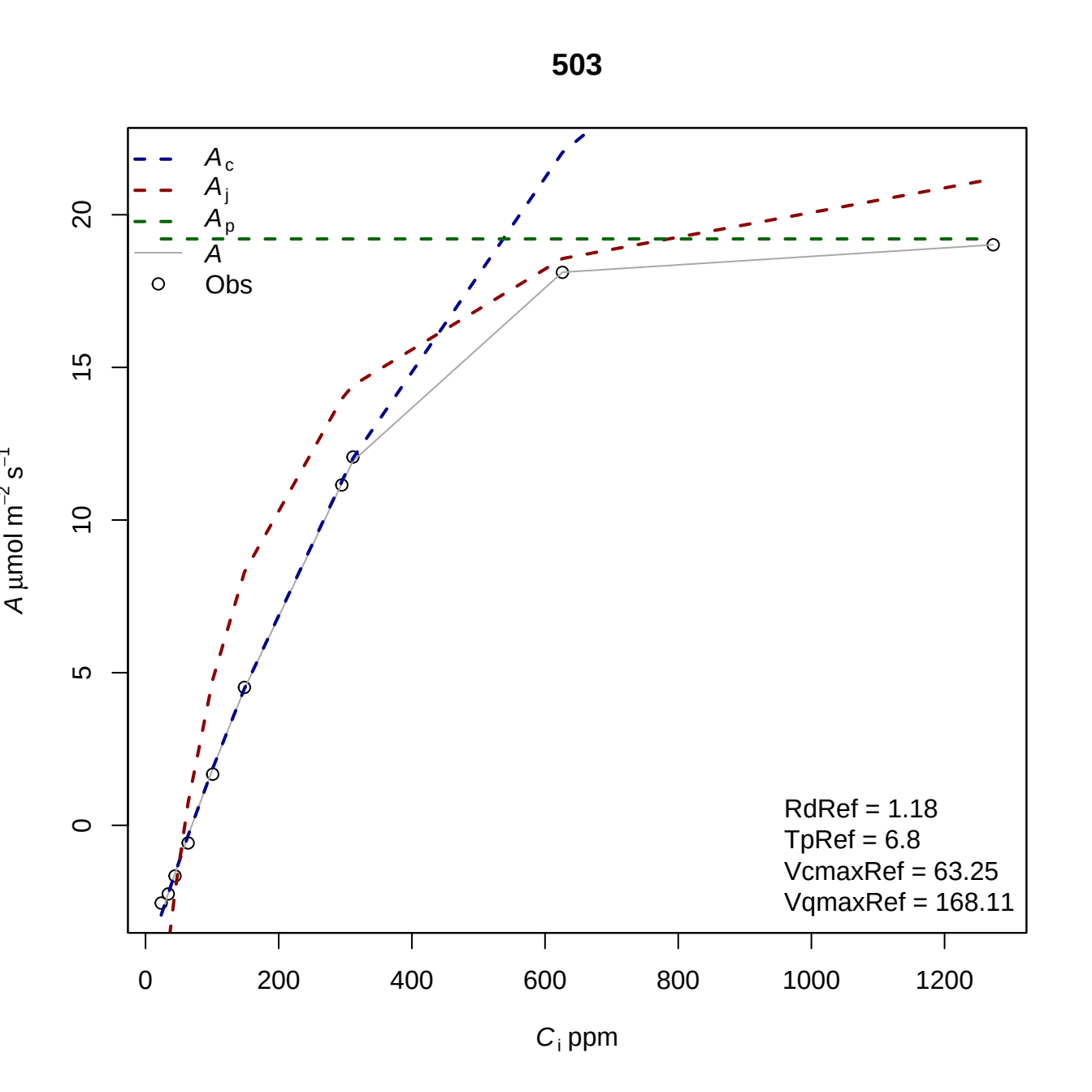

498
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.55
TpRef = 5.81
VcmaxRef = 54.79
VqmaxRef = 139.26
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

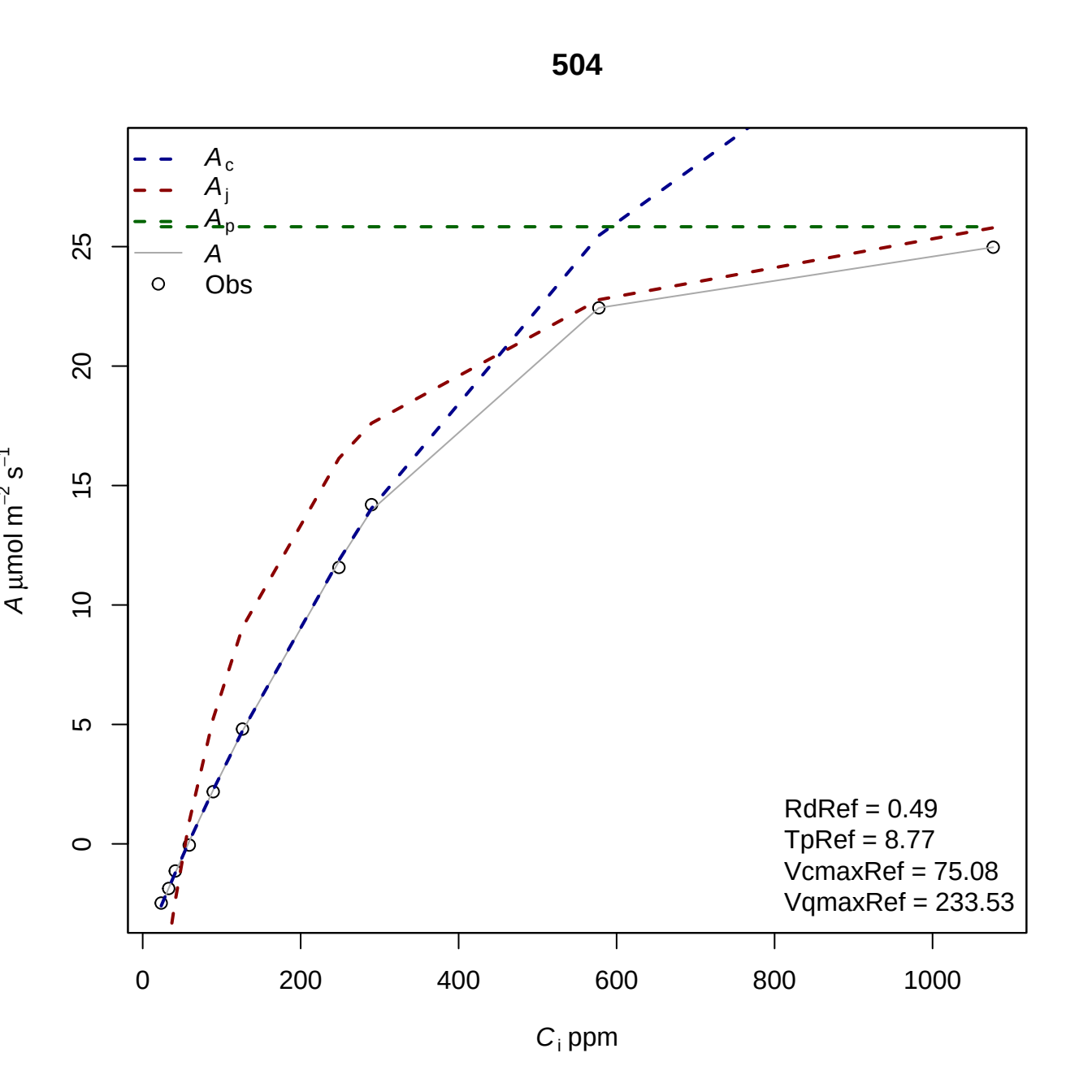

499
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.56
TpRef = 7.99
VcmaxRef = 69.14
VqmaxRef = 225.41
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

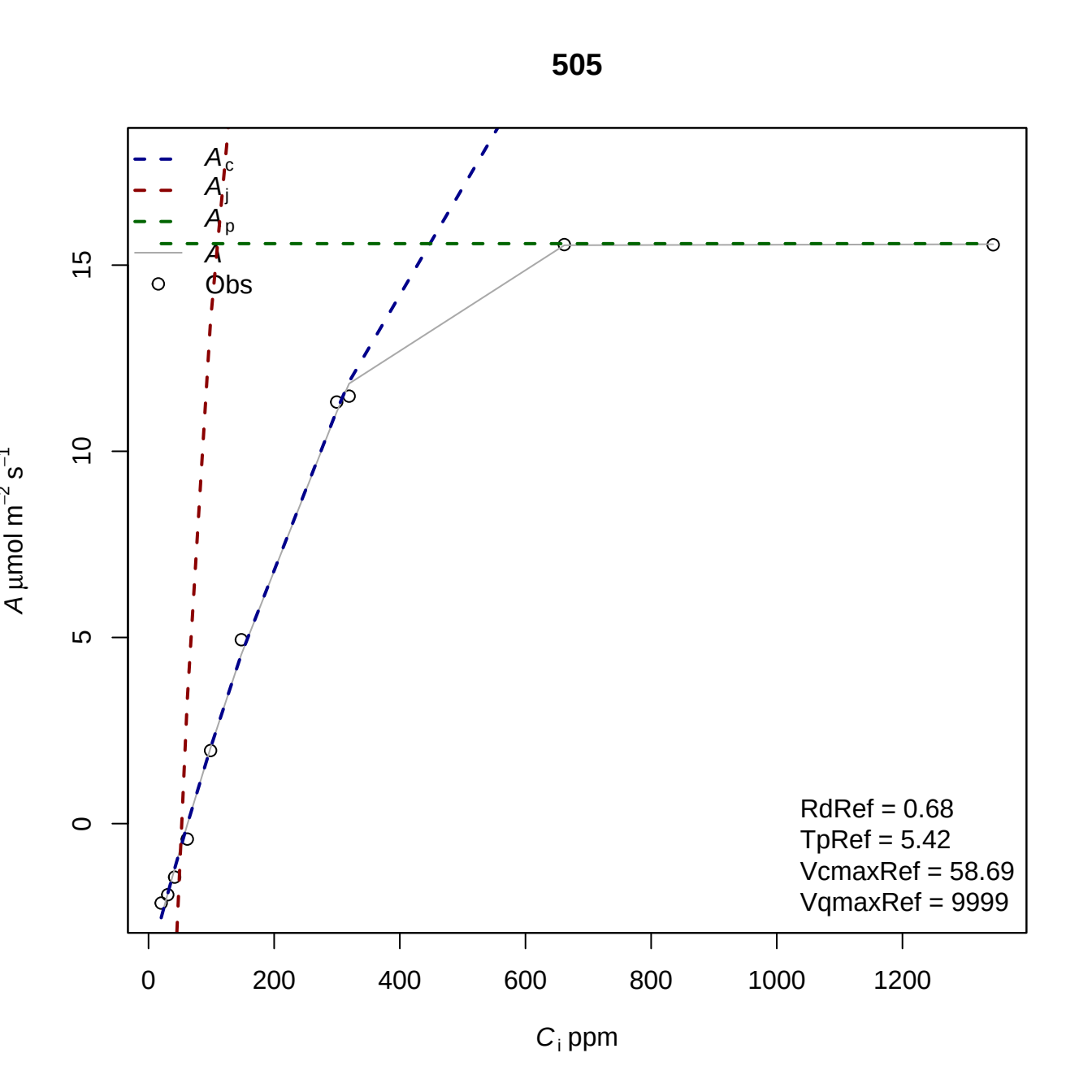

500
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.63
TpRef = 5.82
VcmaxRef = 50.6
VqmaxRef = 125.17
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

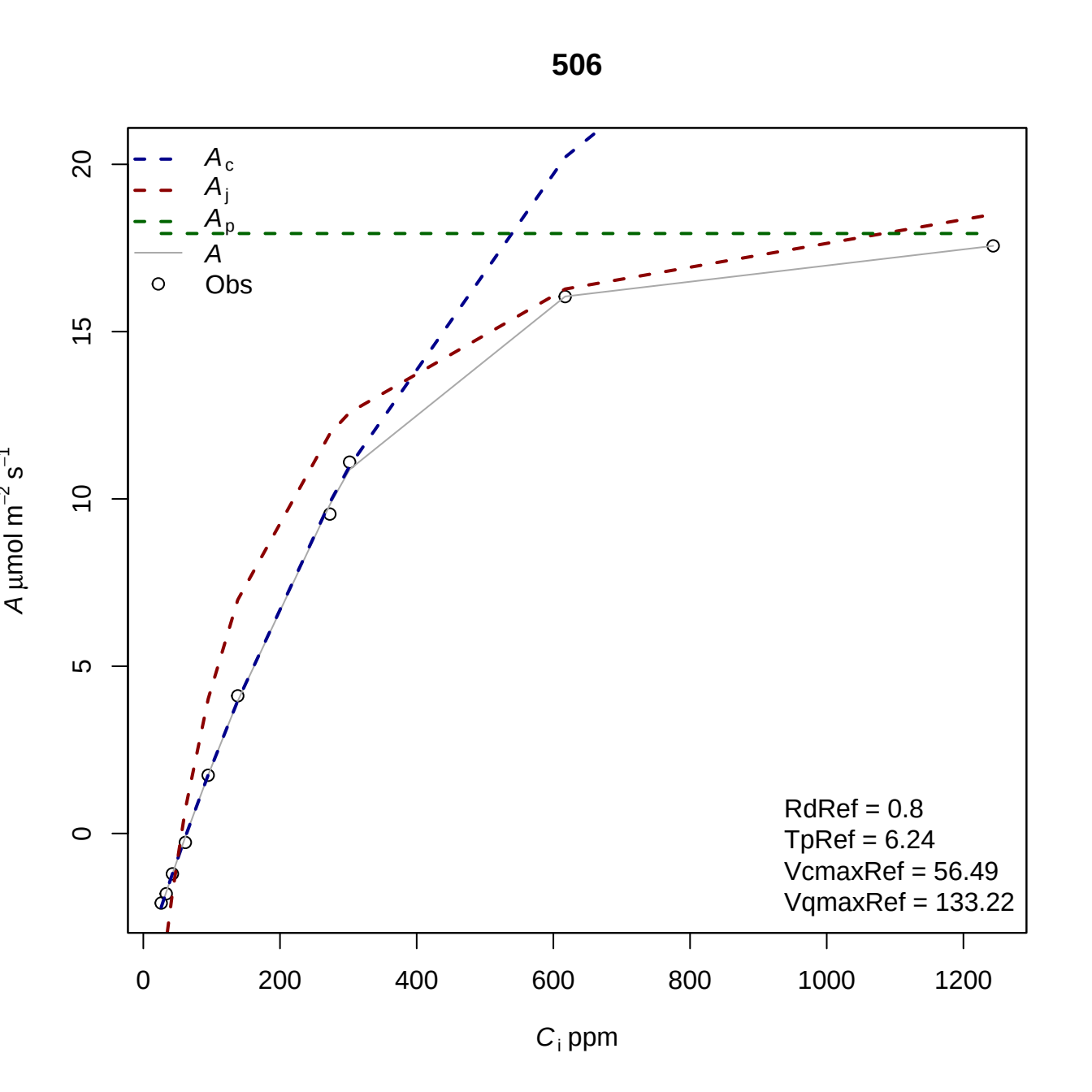

501
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.57
TpRef = 6.09
VcmaxRef = 55.94
VqmaxRef = 142.52
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

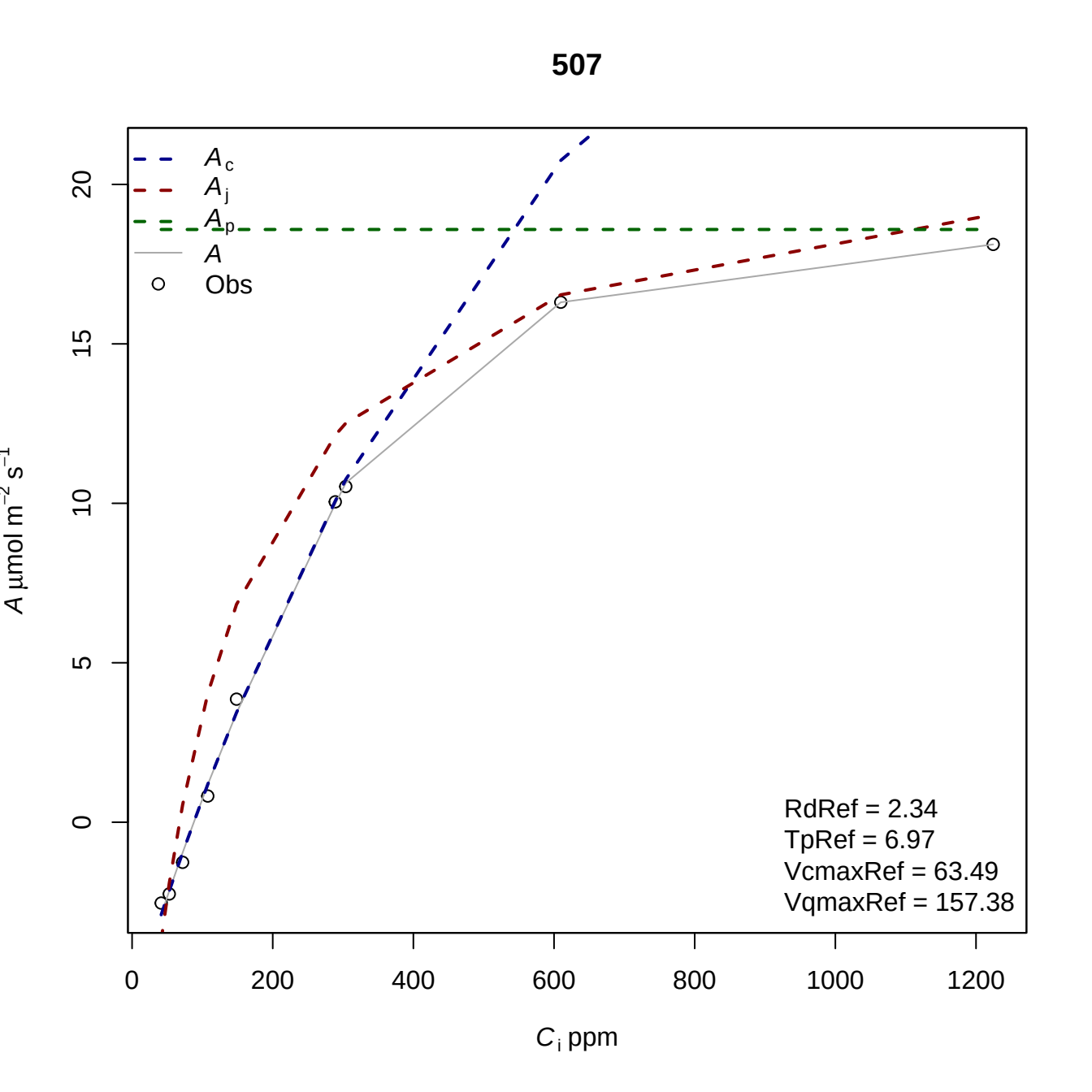

502
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.22
TpRef = 6.25
VcmaxRef = 69.41
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

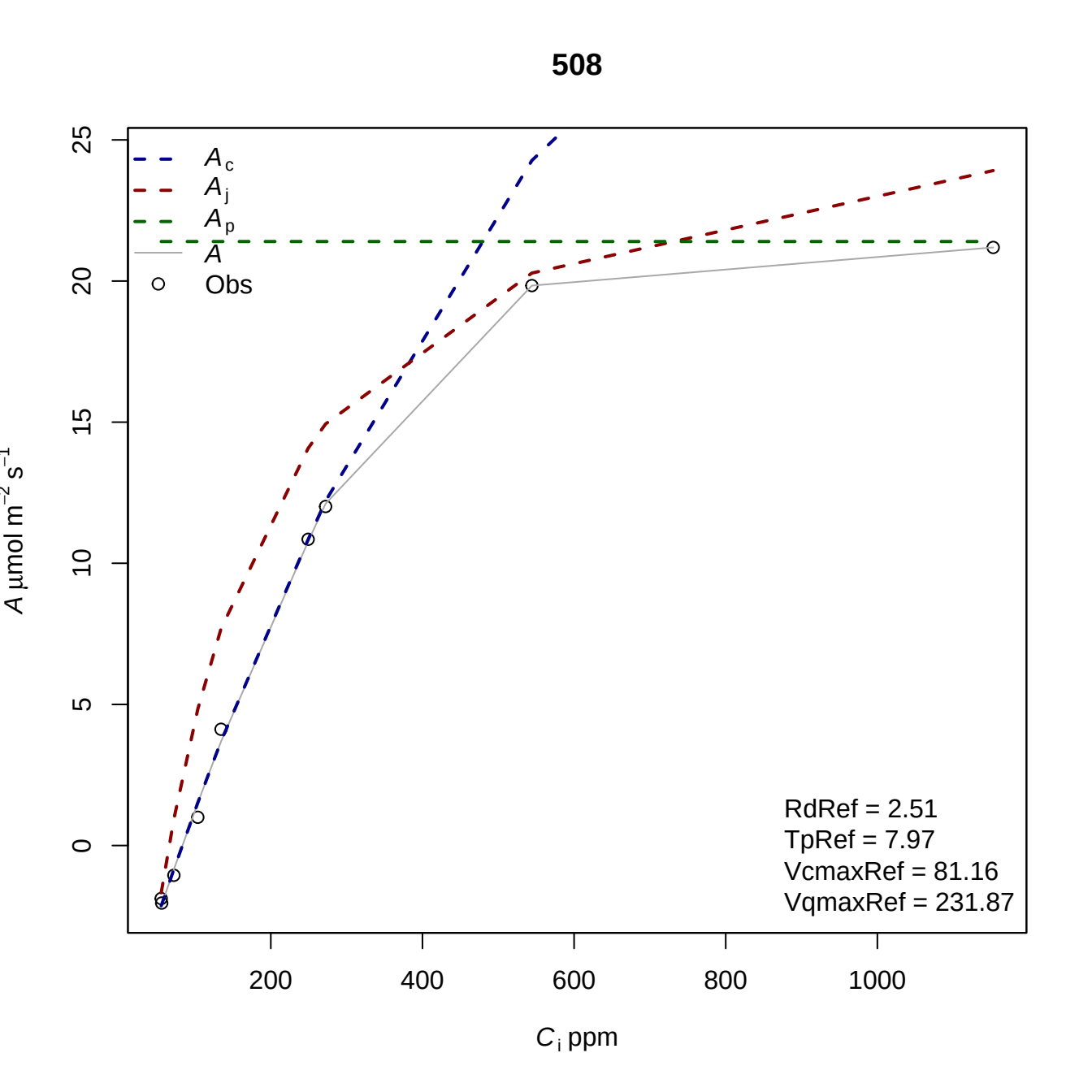

503
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.18
TpRef = 6.8
VcmaxRef = 63.25
VqmaxRef = 168.11
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

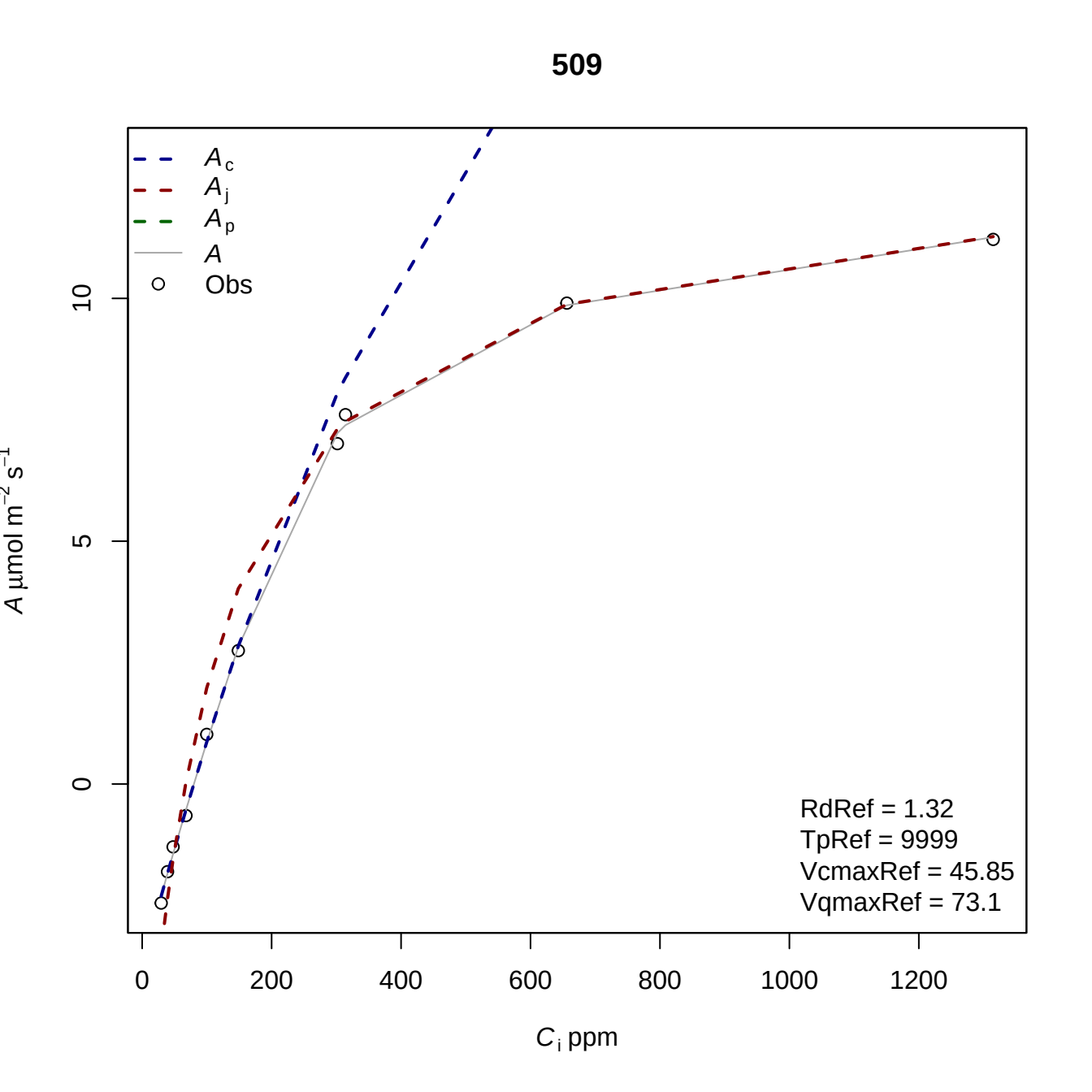

504
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.49
TpRef = 8.77
VcmaxRef = 75.08
VqmaxRef = 233.53
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000

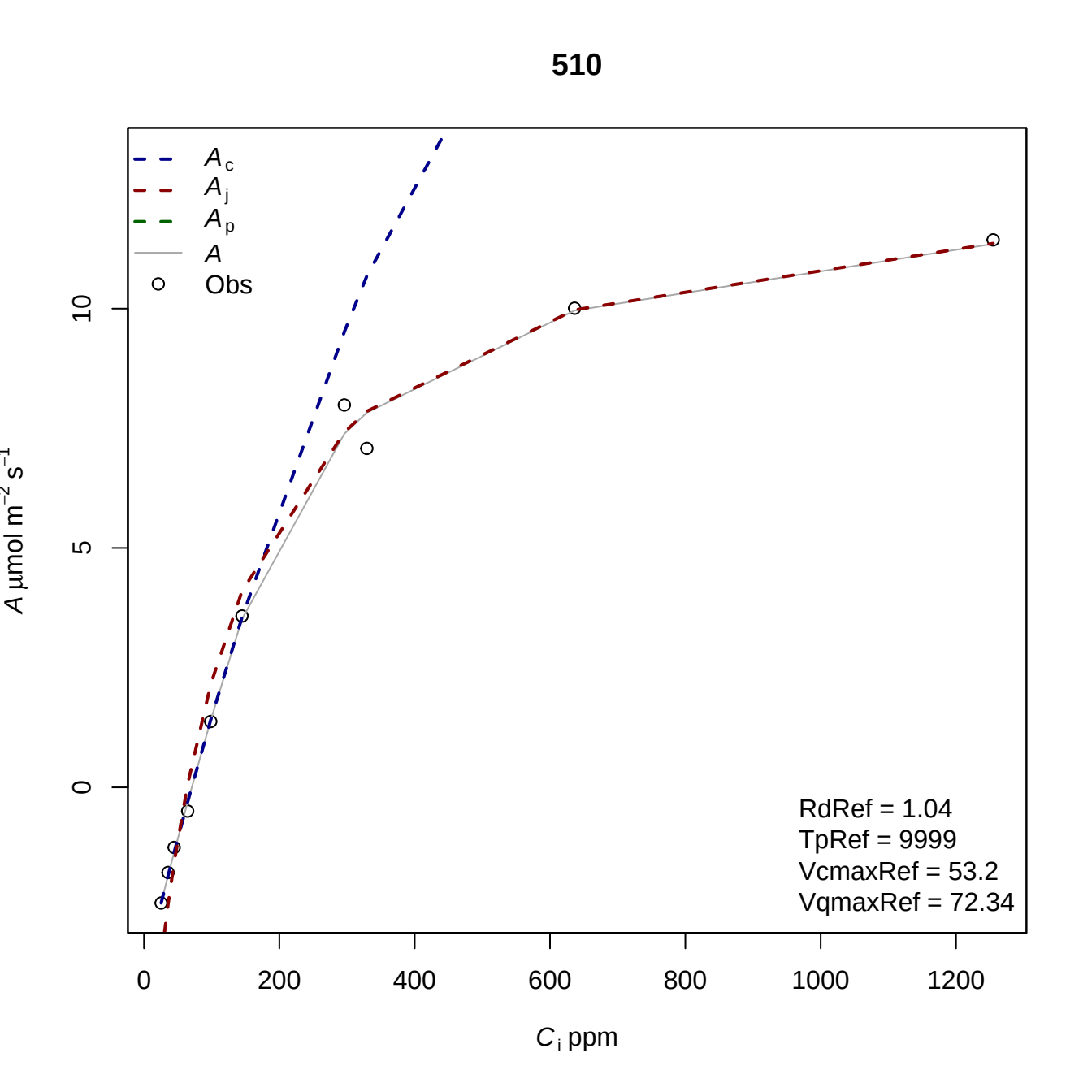

505
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.68
TpRef = 5.42
VcmaxRef = 58.69
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

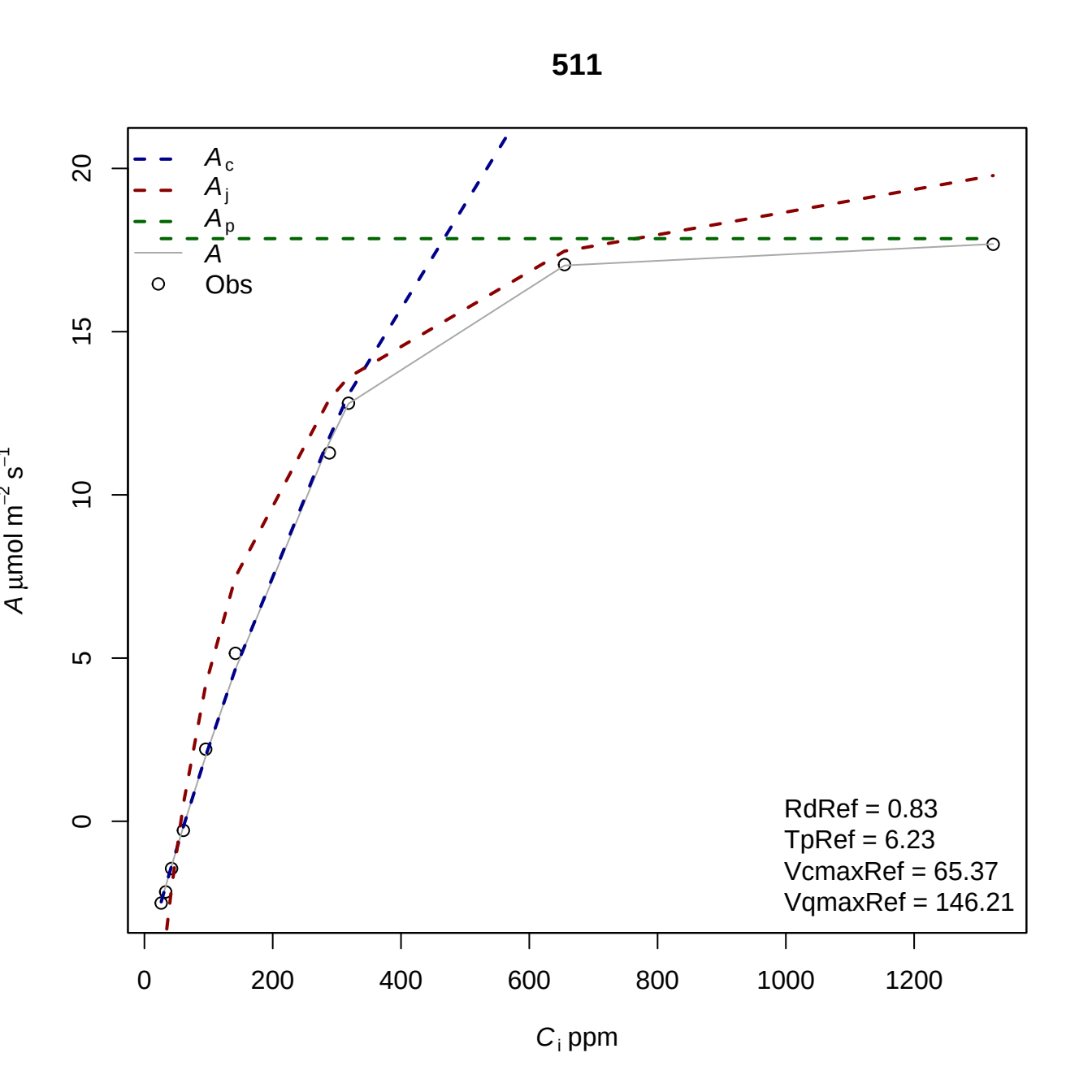

506
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.8
TpRef = 6.24
VcmaxRef = 56.49
VqmaxRef = 133.22
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

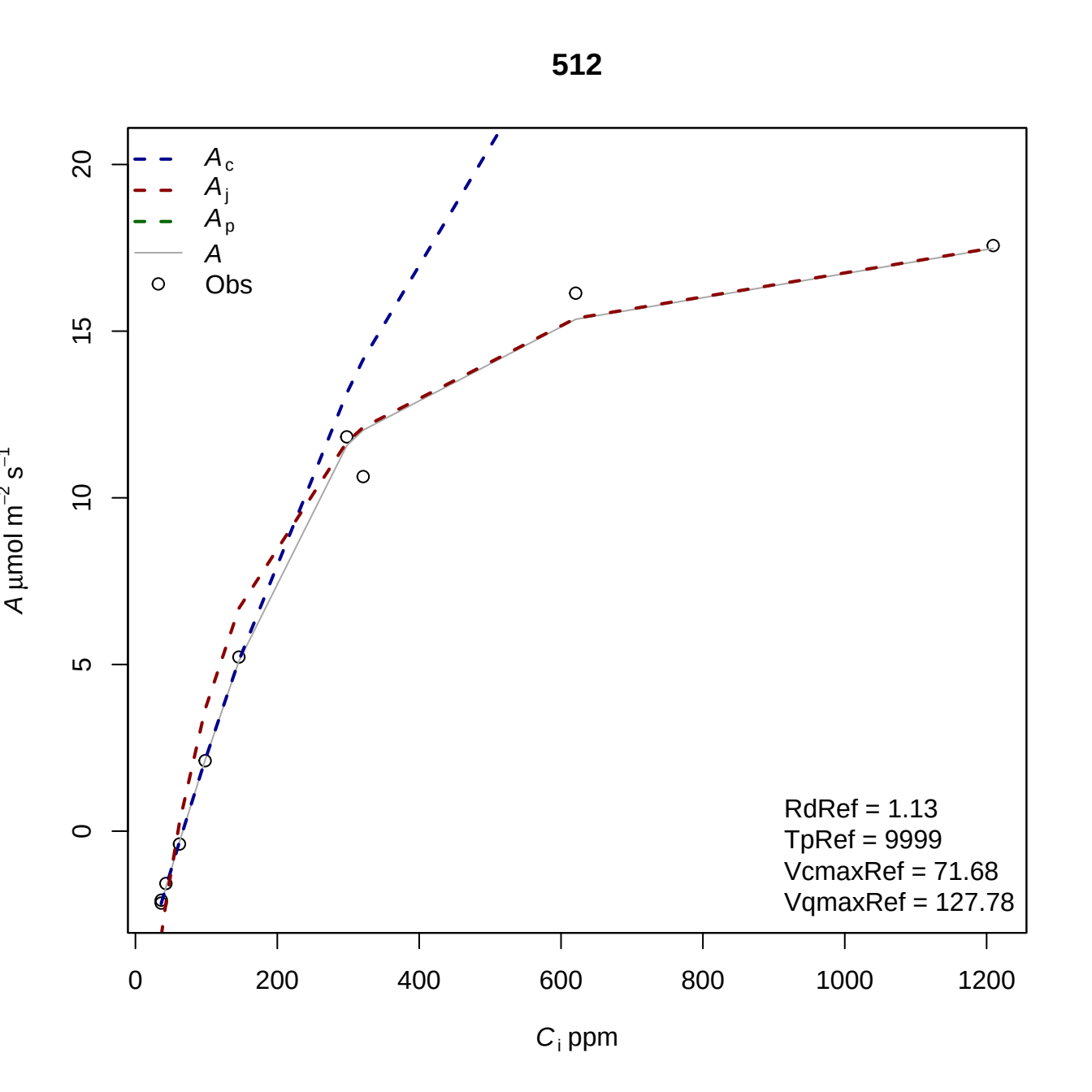

507
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.34
TpRef = 6.97
VcmaxRef = 63.49
VqmaxRef = 157.38
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

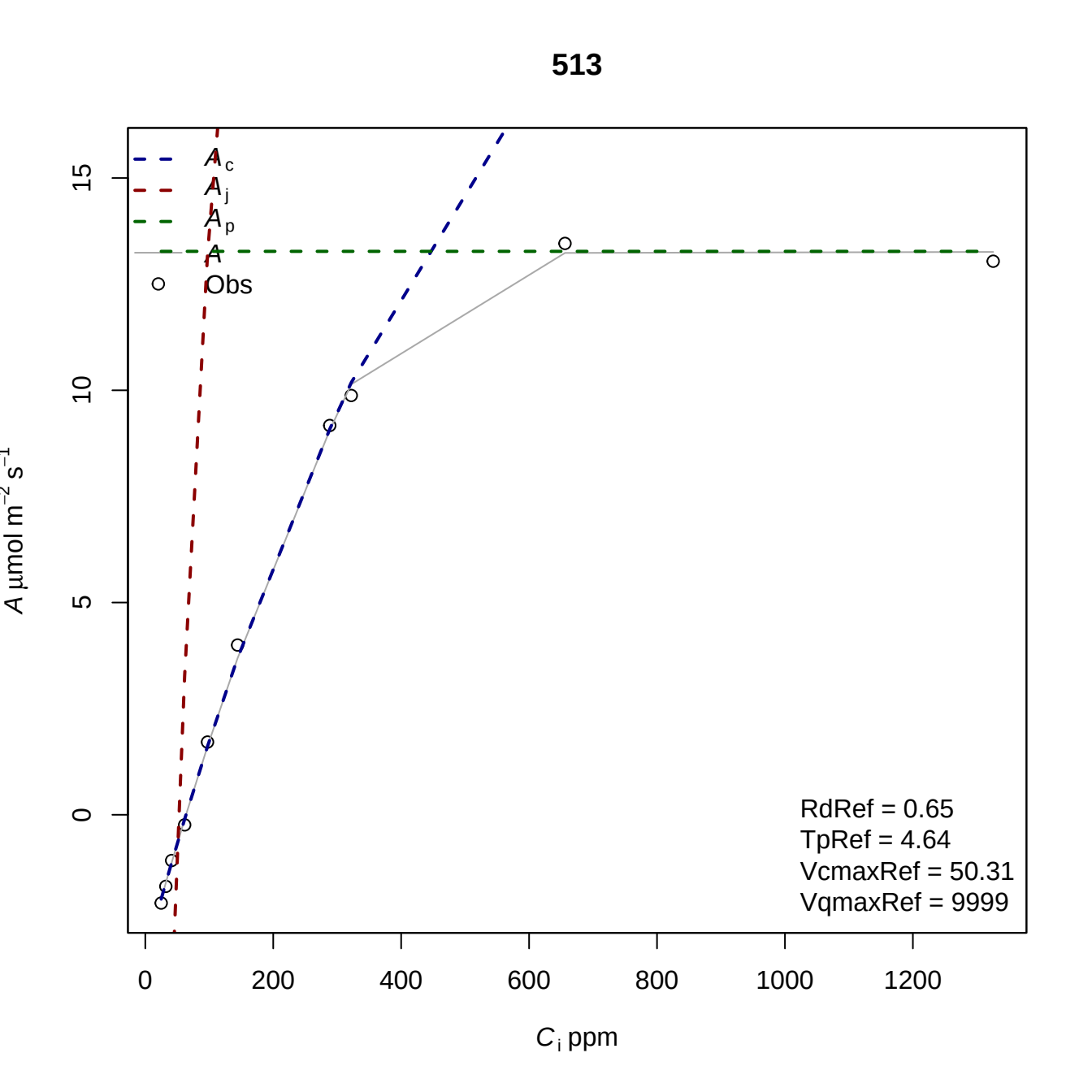

508
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.51
TpRef = 7.97
VcmaxRef = 81.16
VqmaxRef = 231.87
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000

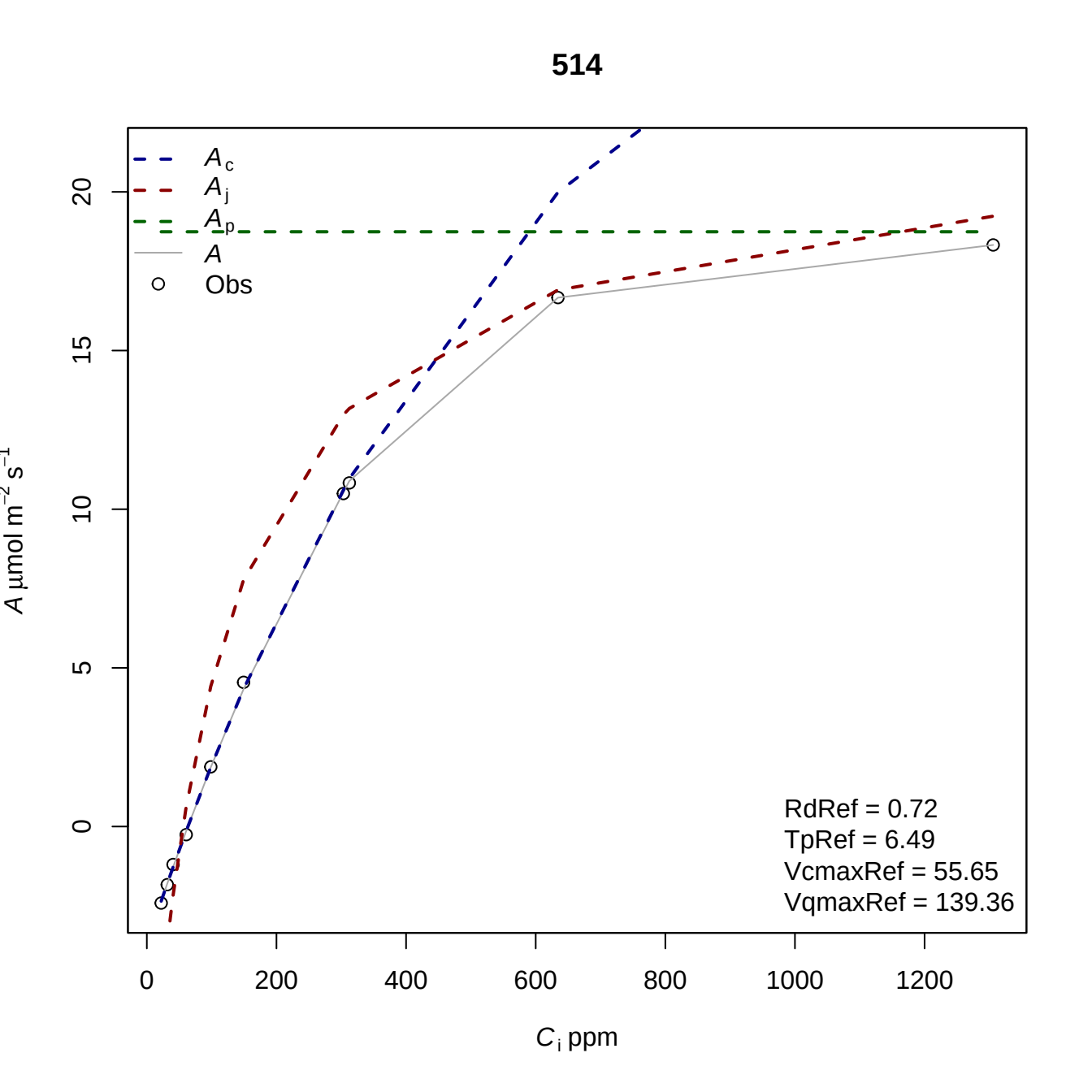

509
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.32
TpRef = 9999
VcmaxRef = 45.85
VqmaxRef = 73.1
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
0
200
400
600
800
1000
1200

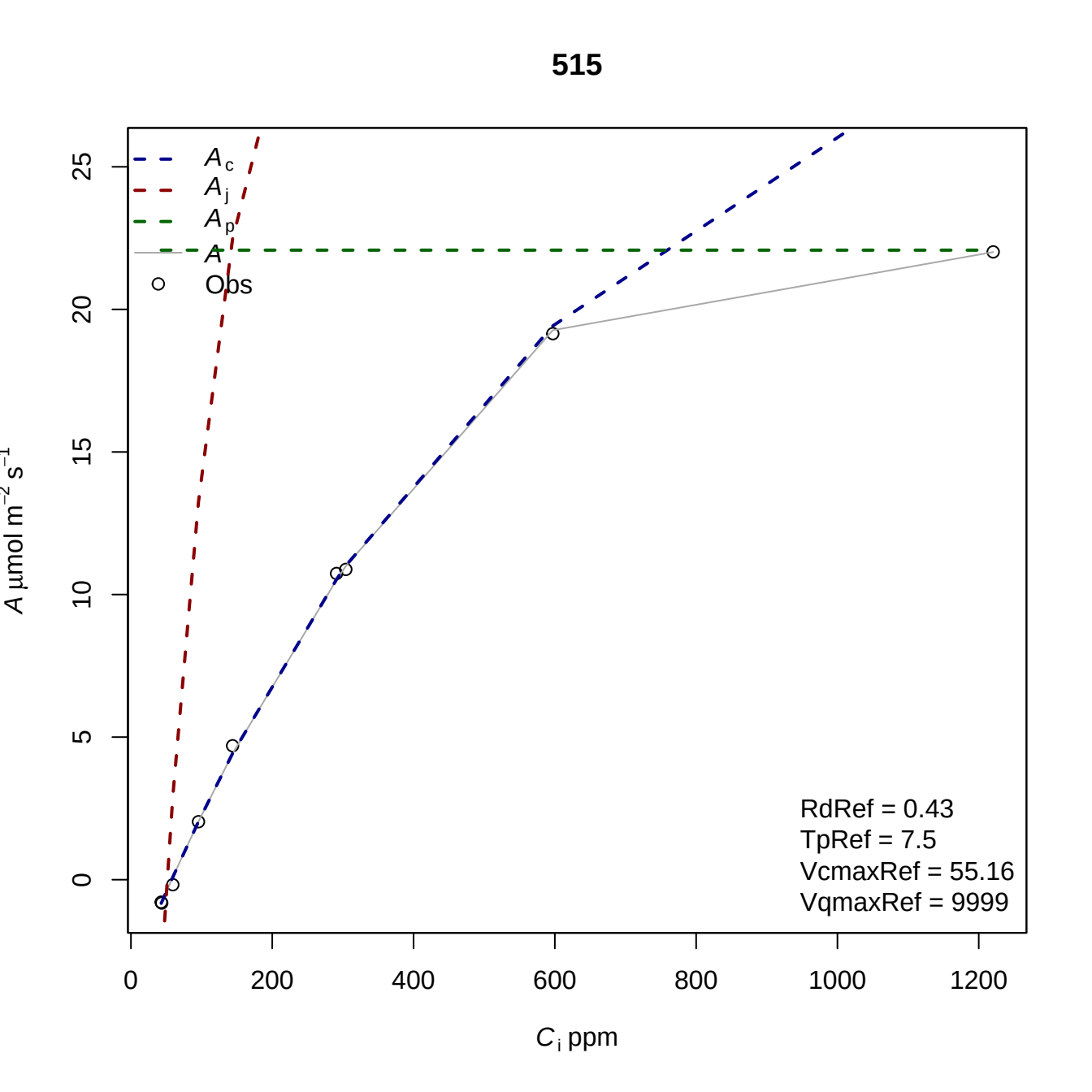

510
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.04
TpRef = 9999
VcmaxRef = 53.2
VqmaxRef = 72.34
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
0
200
400
600
800
1000
1200

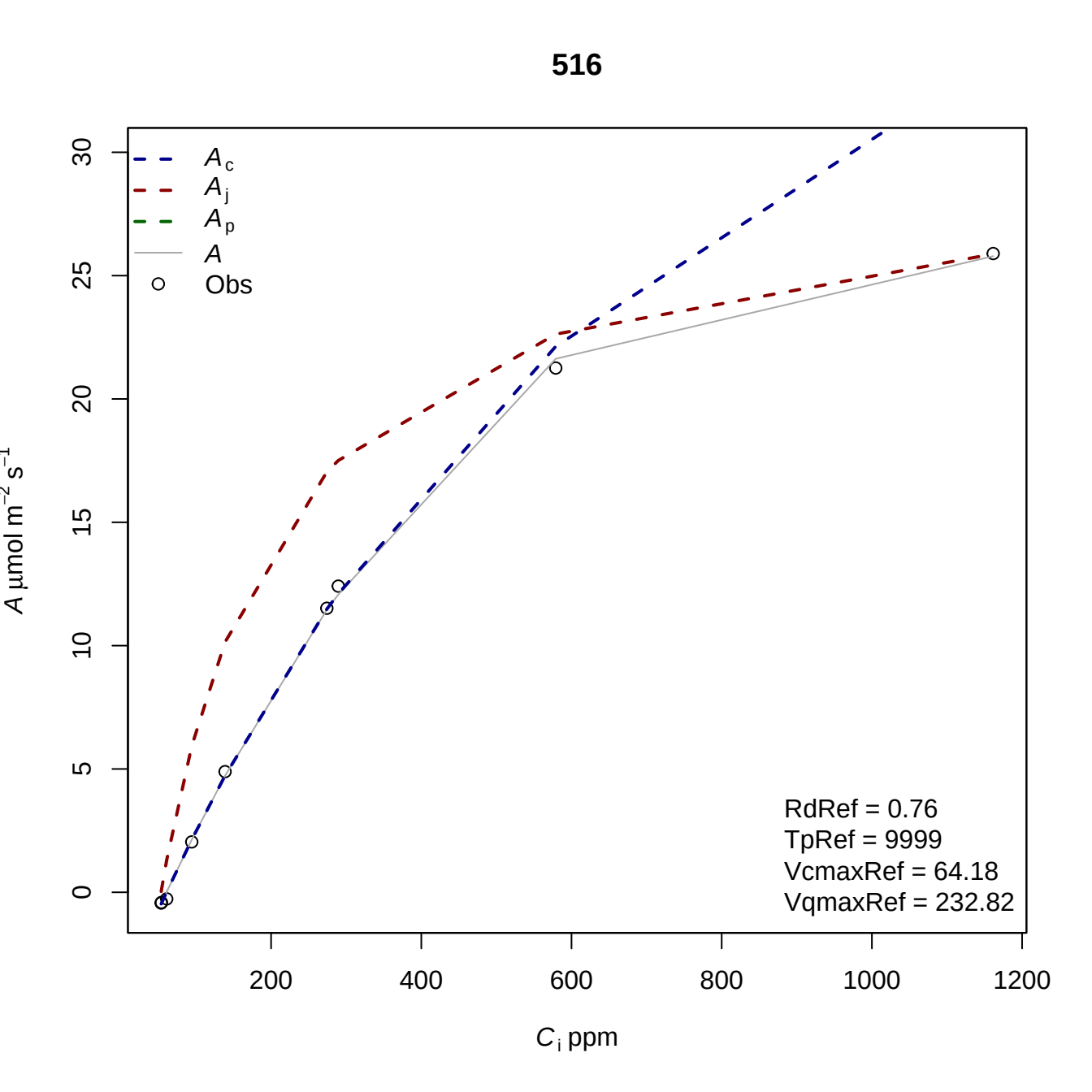

511
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.83
TpRef = 6.23
VcmaxRef = 65.37
VqmaxRef = 146.21
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

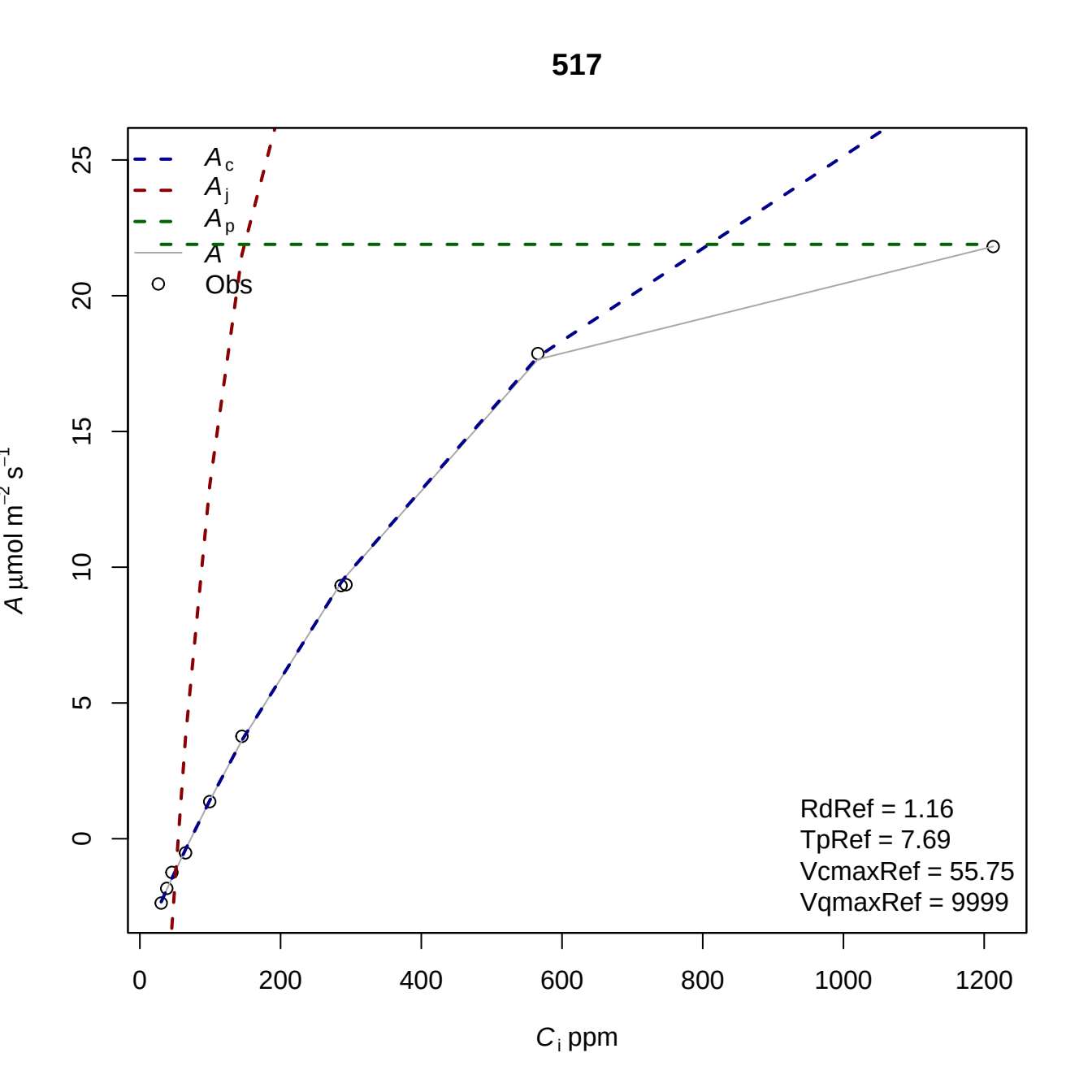

512
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.13
TpRef = 9999
VcmaxRef = 71.68
VqmaxRef = 127.78
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

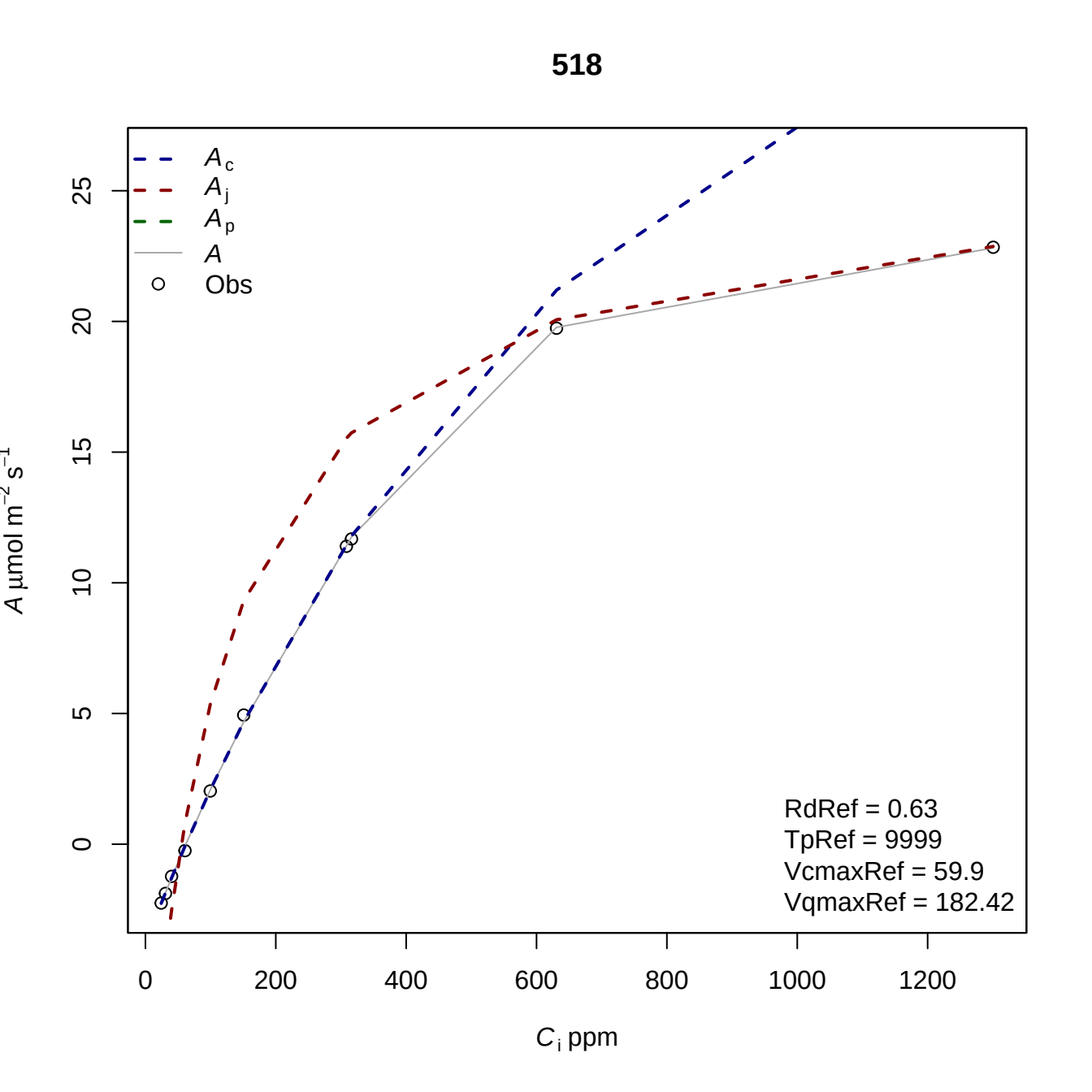

513
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.65
TpRef = 4.64
VcmaxRef = 50.31
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

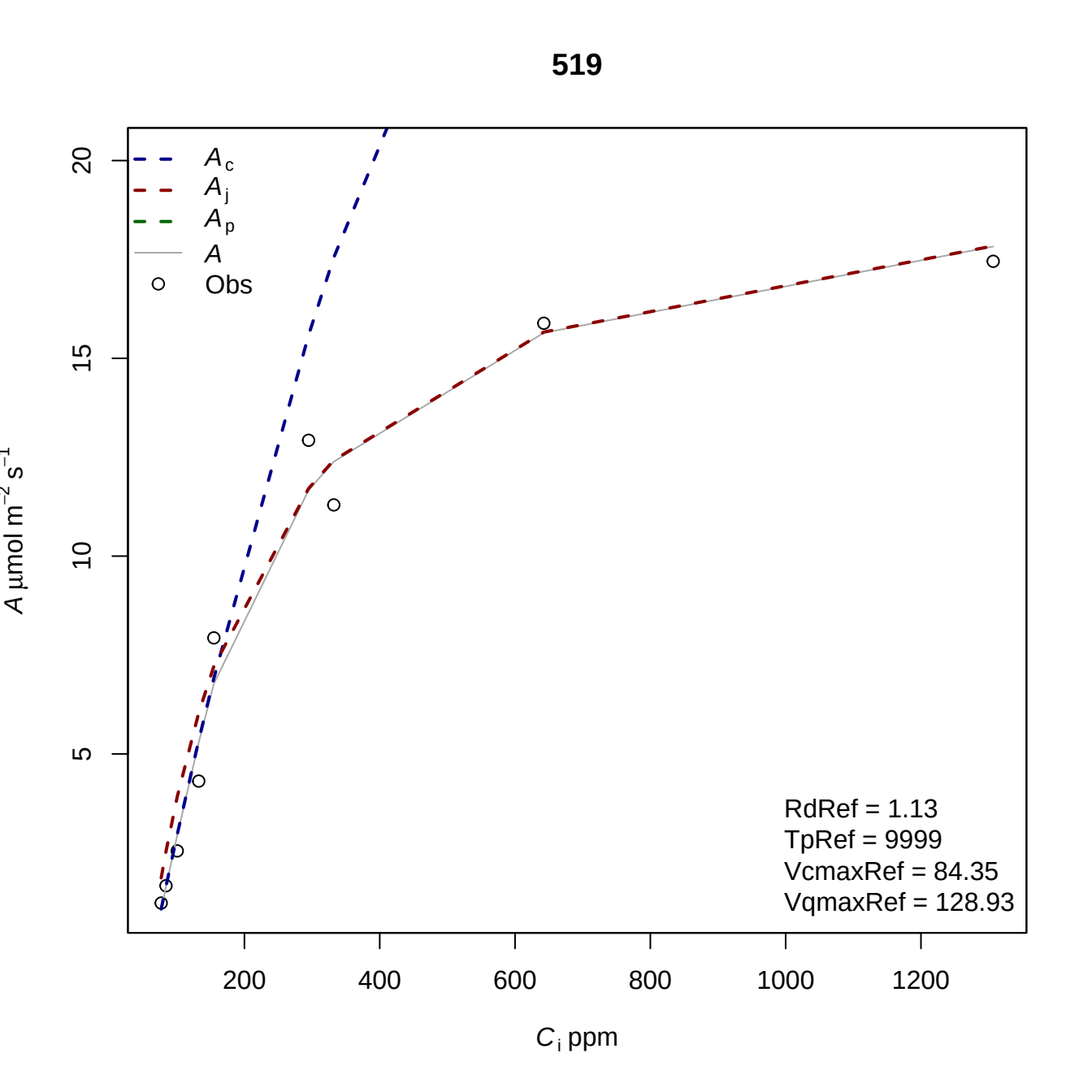

514
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.72
TpRef = 6.49
VcmaxRef = 55.65
VqmaxRef = 139.36
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

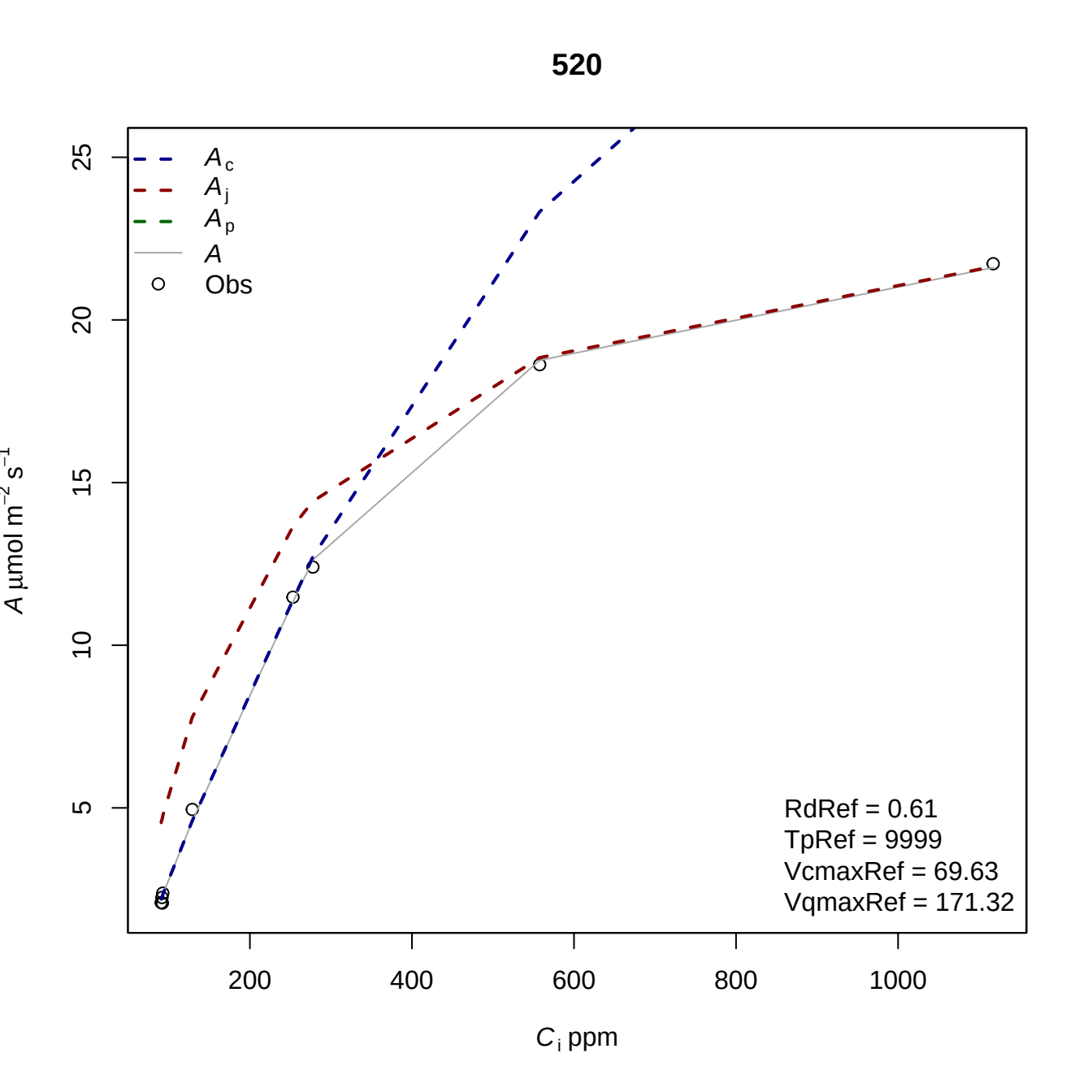

515
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.43
TpRef = 7.5
VcmaxRef = 55.16
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

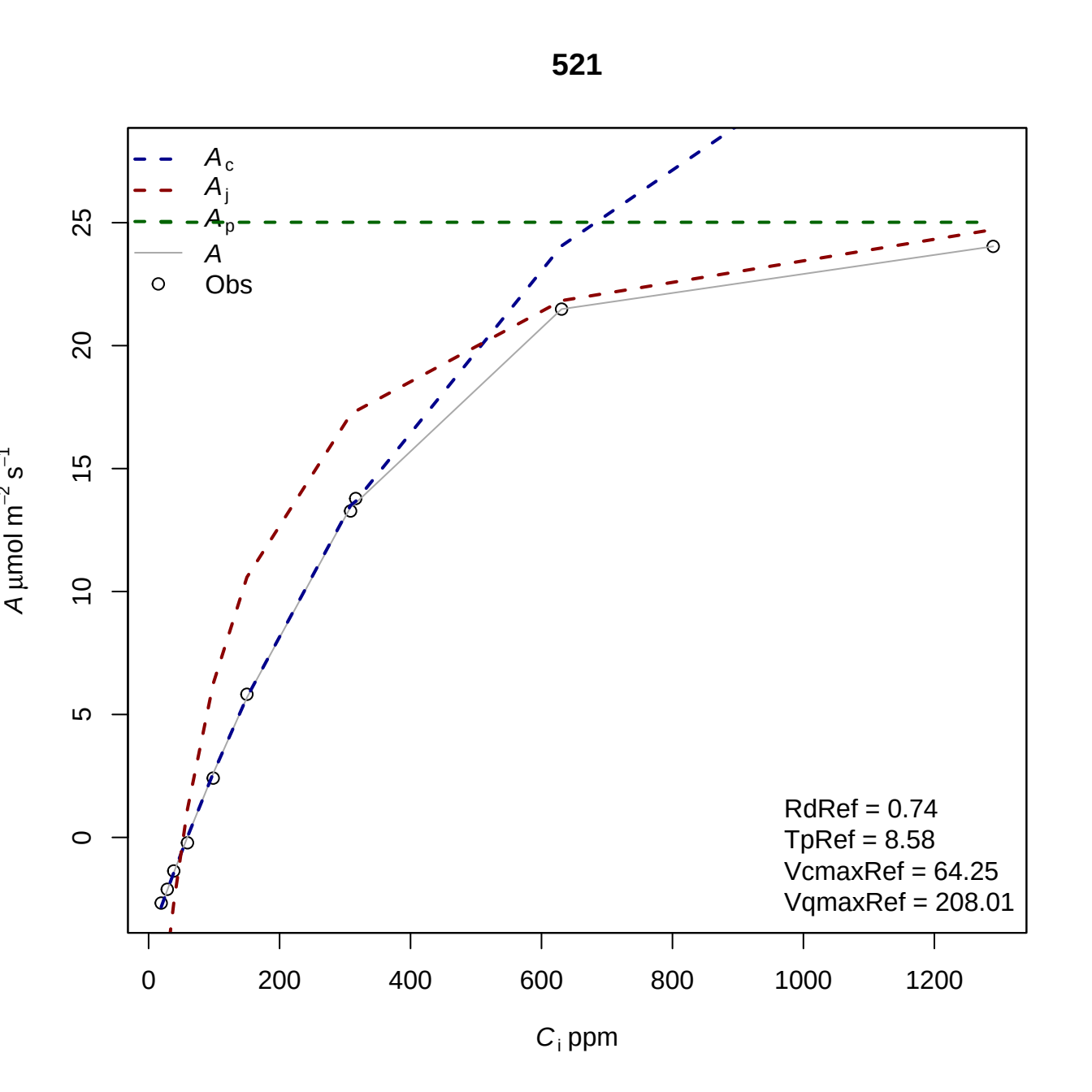

516
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.76
TpRef = 9999
VcmaxRef = 64.18
VqmaxRef = 232.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

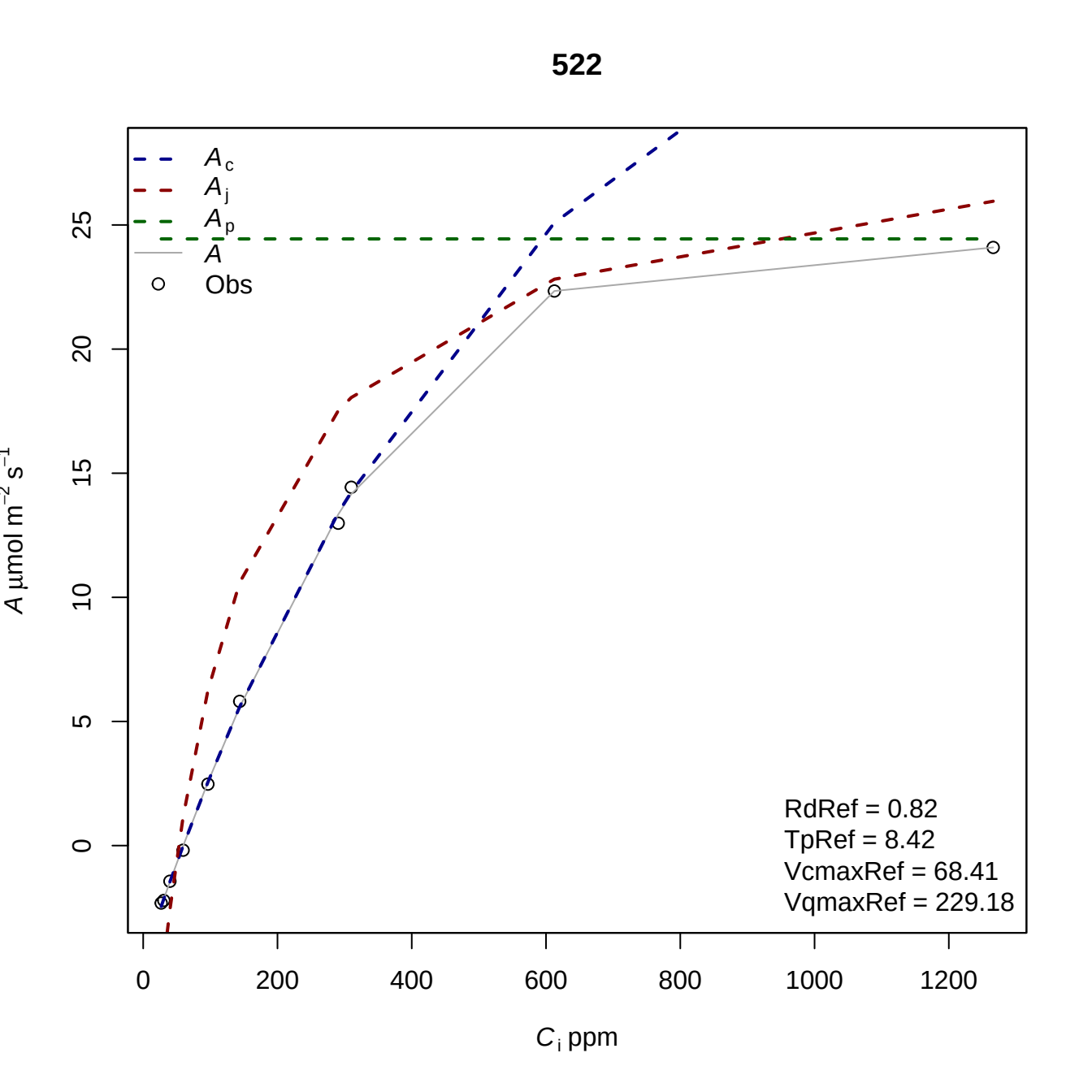

517
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.16
TpRef = 7.69
VcmaxRef = 55.75
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

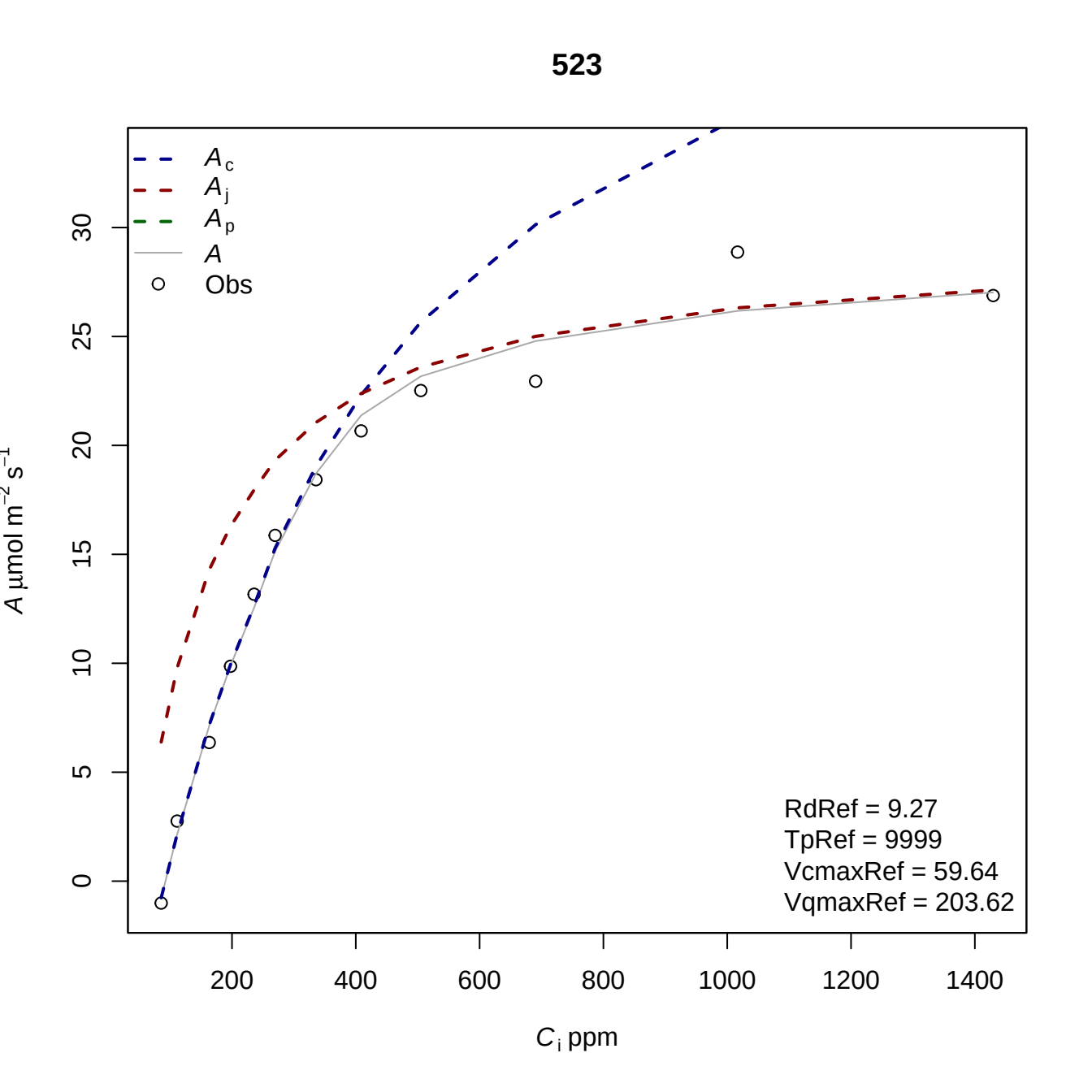

518
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.63
TpRef = 9999
VcmaxRef = 59.9
VqmaxRef = 182.42
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

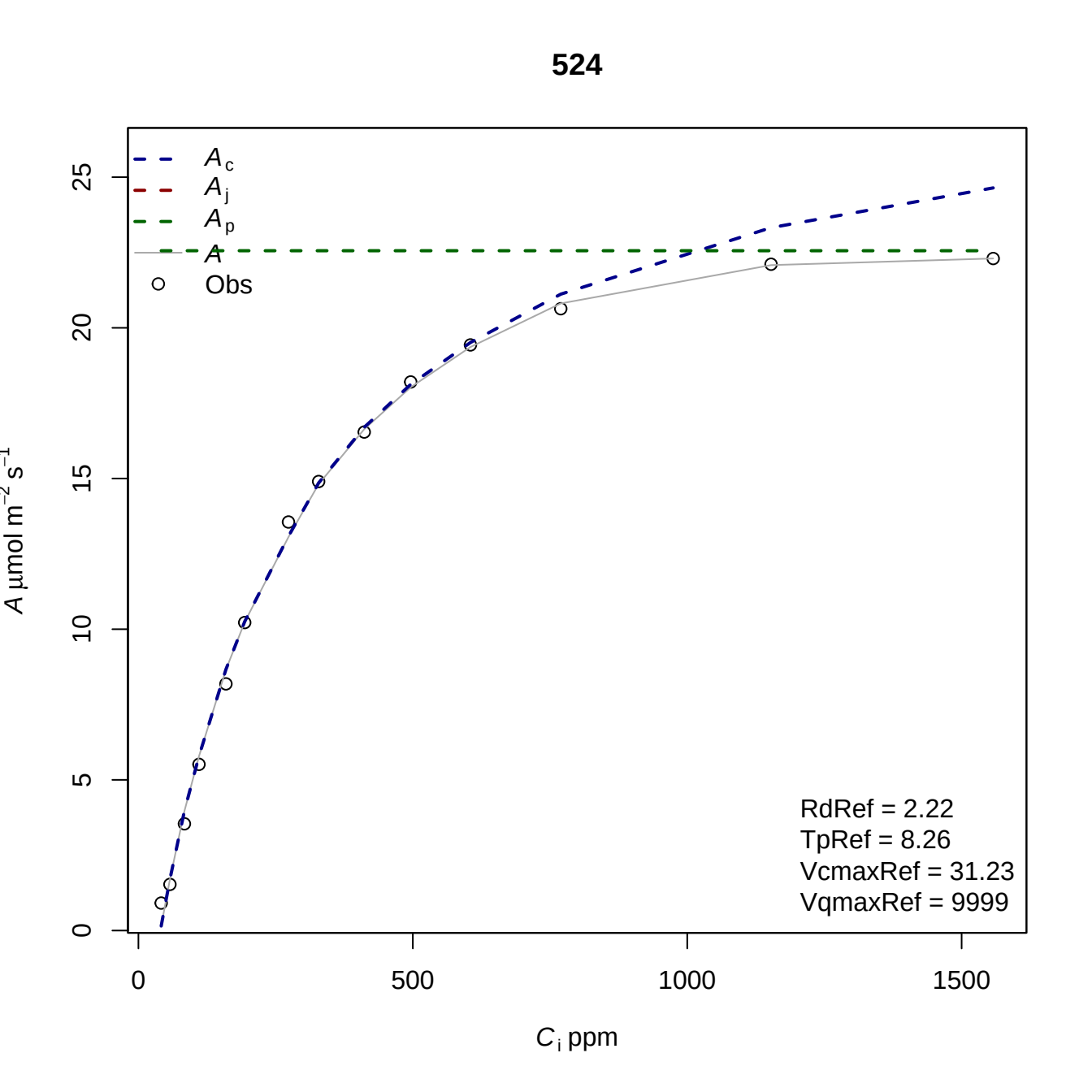

519
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.13
TpRef = 9999
VcmaxRef = 84.35
VqmaxRef = 128.93
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
20
200
400
600
800
1000
1200

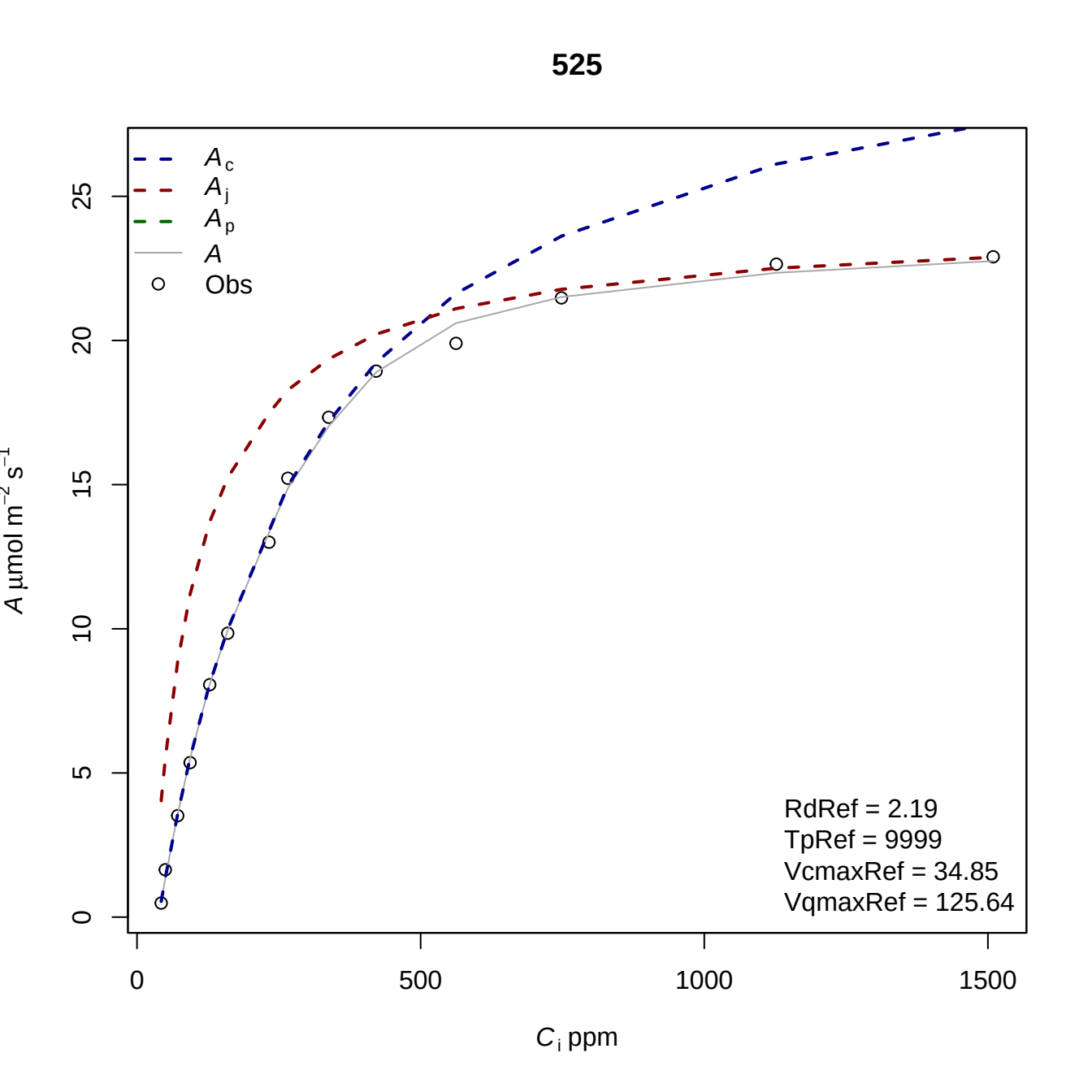

520
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.61
TpRef = 9999
VcmaxRef = 69.63
VqmaxRef = 171.32
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
5
10
15
20
25
200
400
600
800
1000

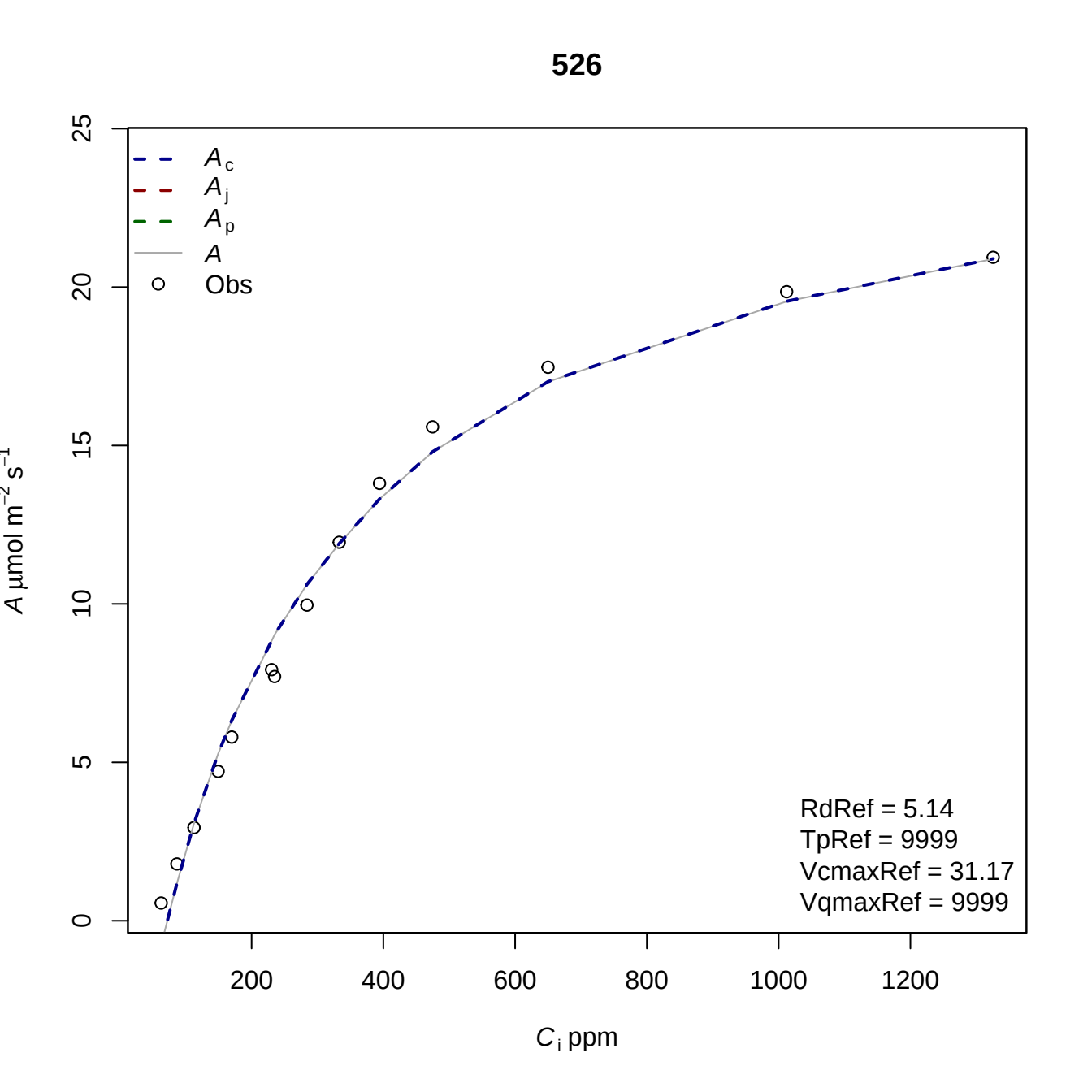

521
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.74
TpRef = 8.58
VcmaxRef = 64.25
VqmaxRef = 208.01
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

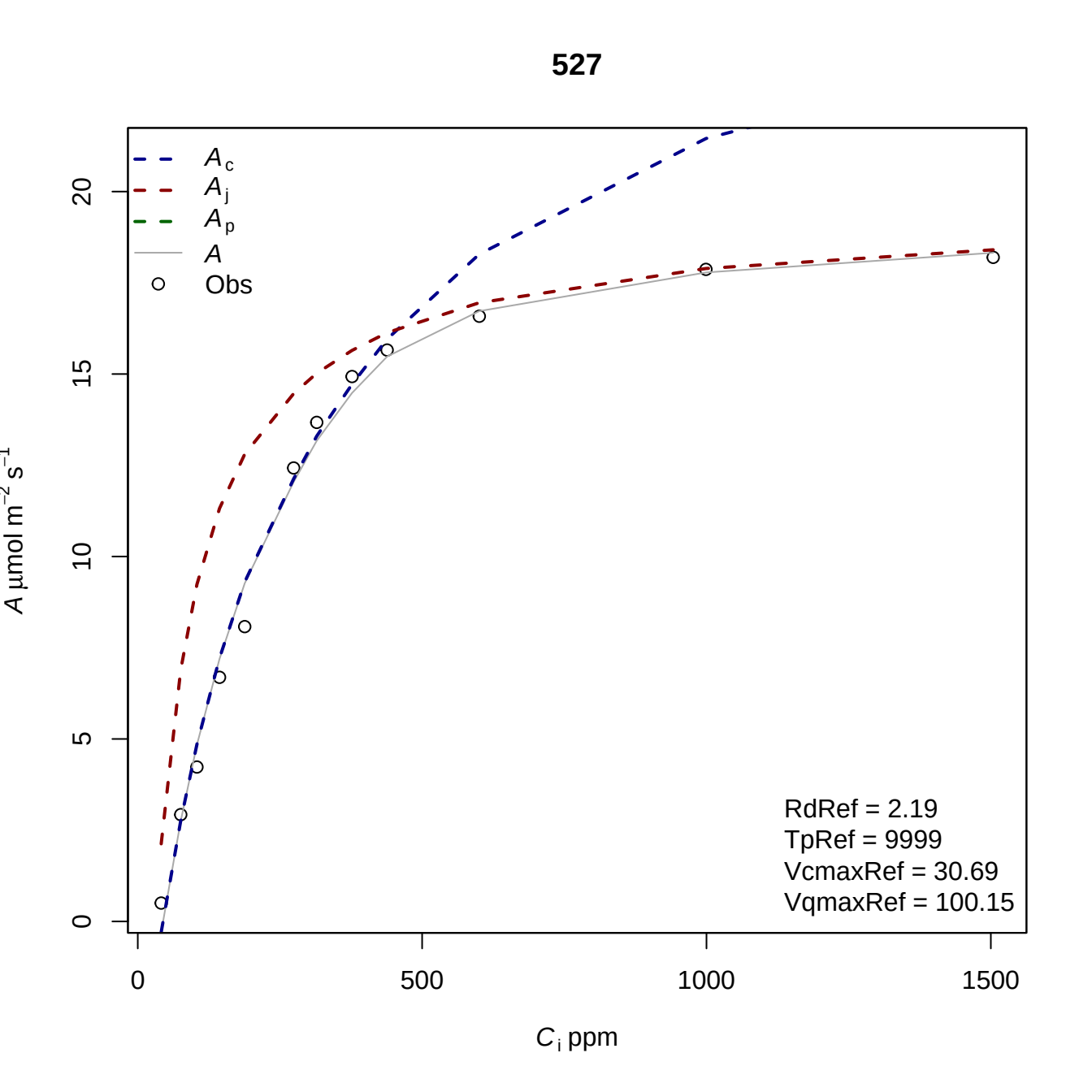

522
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.82
TpRef = 8.42
VcmaxRef = 68.41
VqmaxRef = 229.18
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

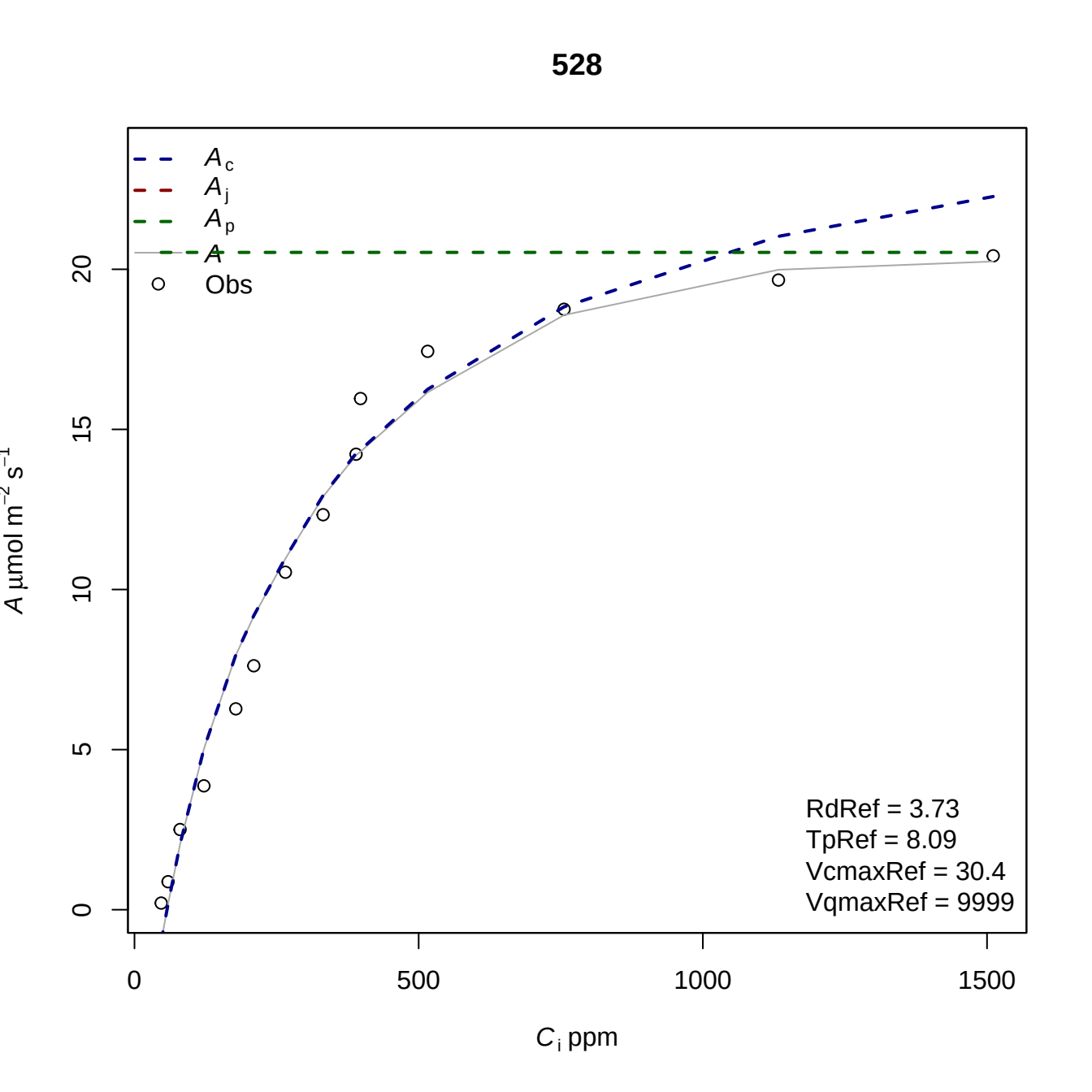

523
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 9.27
TpRef = 9999
VcmaxRef = 59.64
VqmaxRef = 203.62
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200
1400

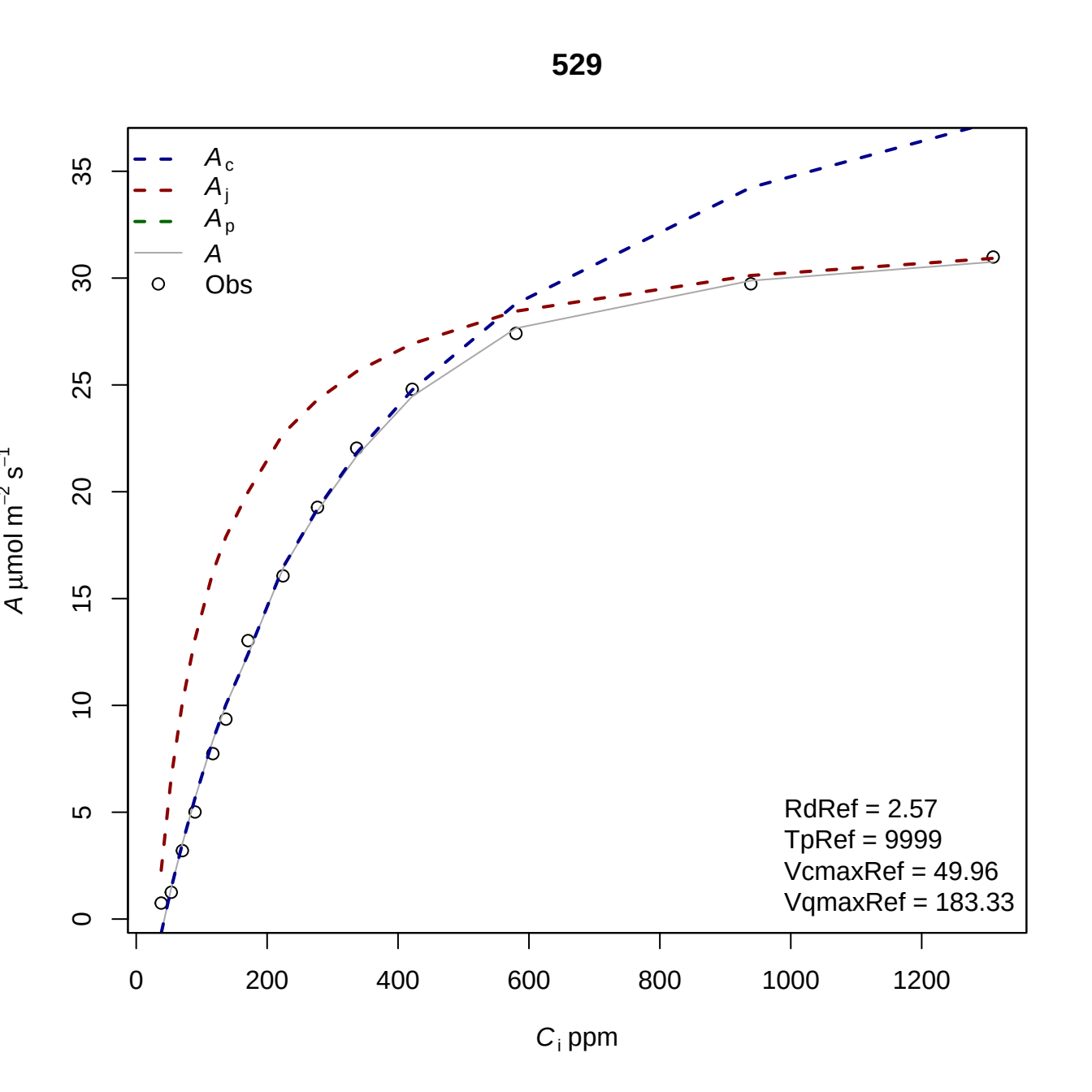

524
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.22
TpRef = 8.26
VcmaxRef = 31.23
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

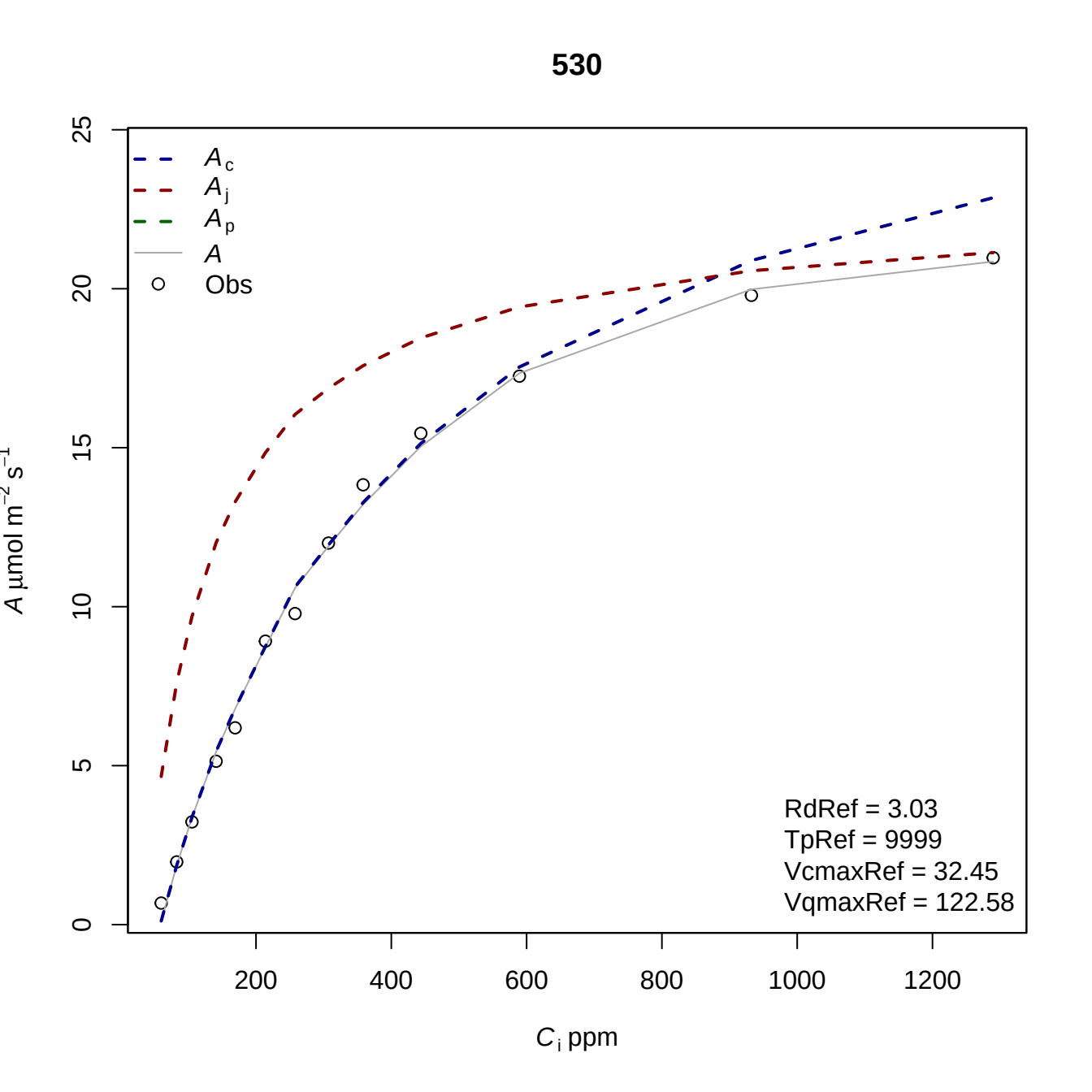

525
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.19
TpRef = 9999
VcmaxRef = 34.85
VqmaxRef = 125.64
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

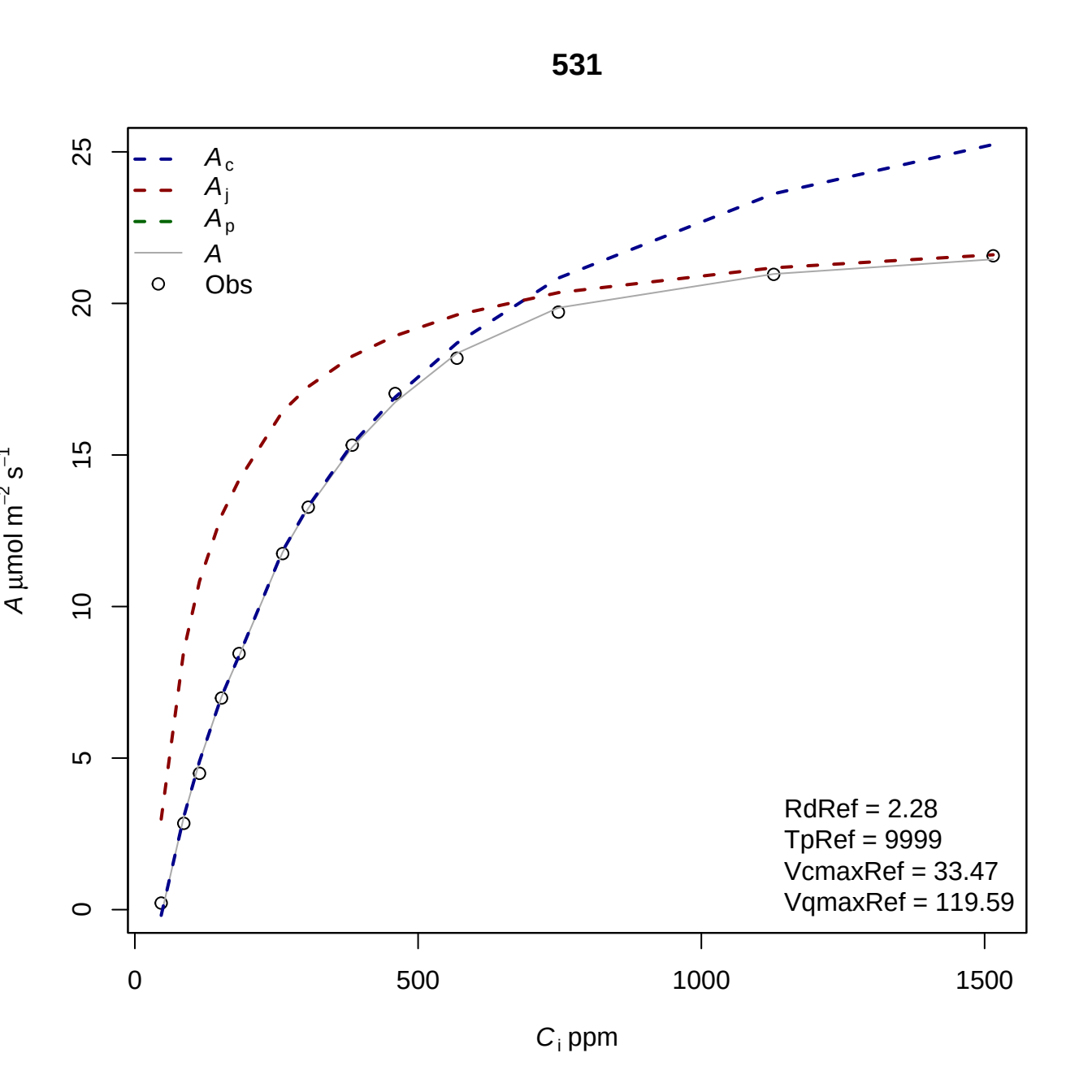

526
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 5.14
TpRef = 9999
VcmaxRef = 31.17
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

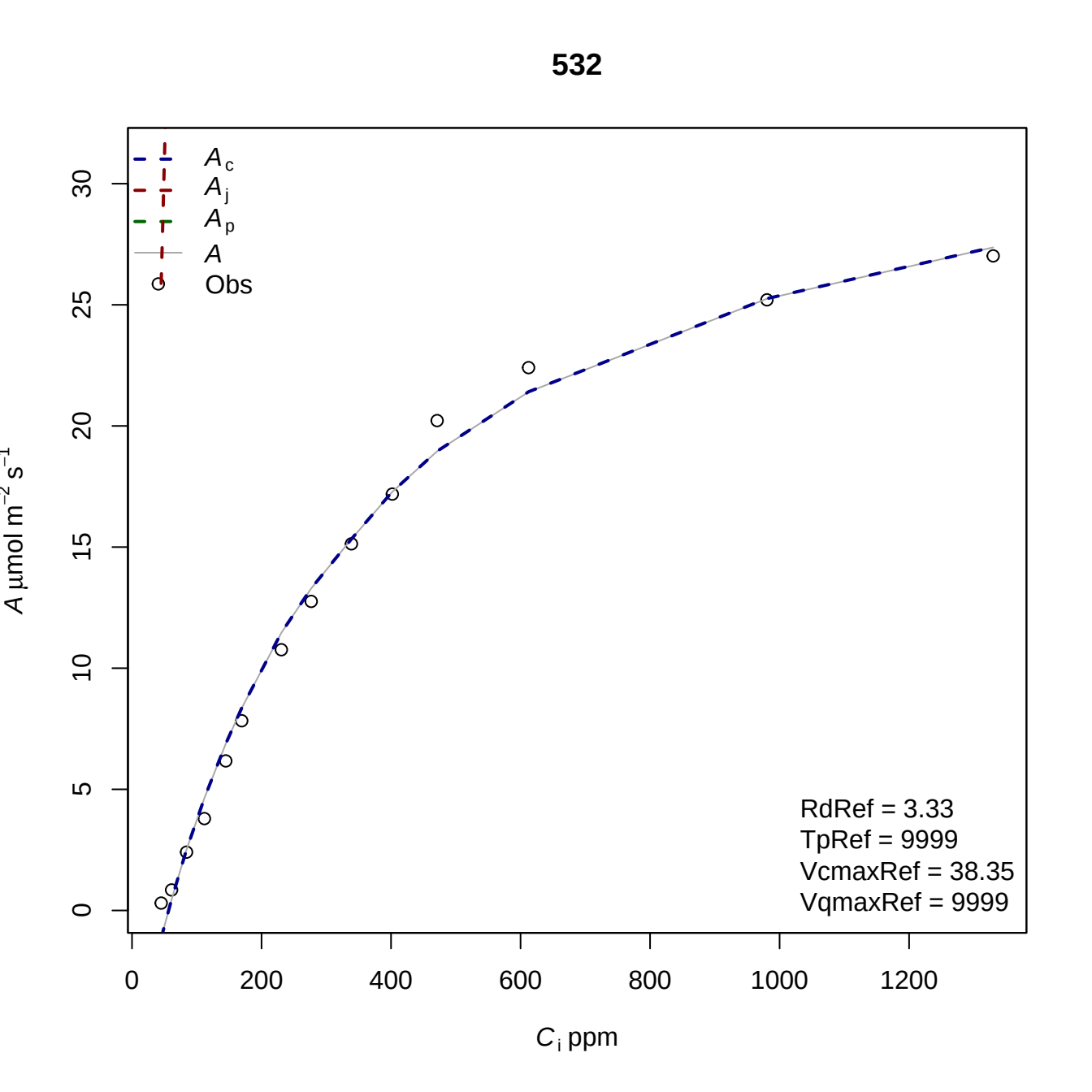

527
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.19
TpRef = 9999
VcmaxRef = 30.69
VqmaxRef = 100.15
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

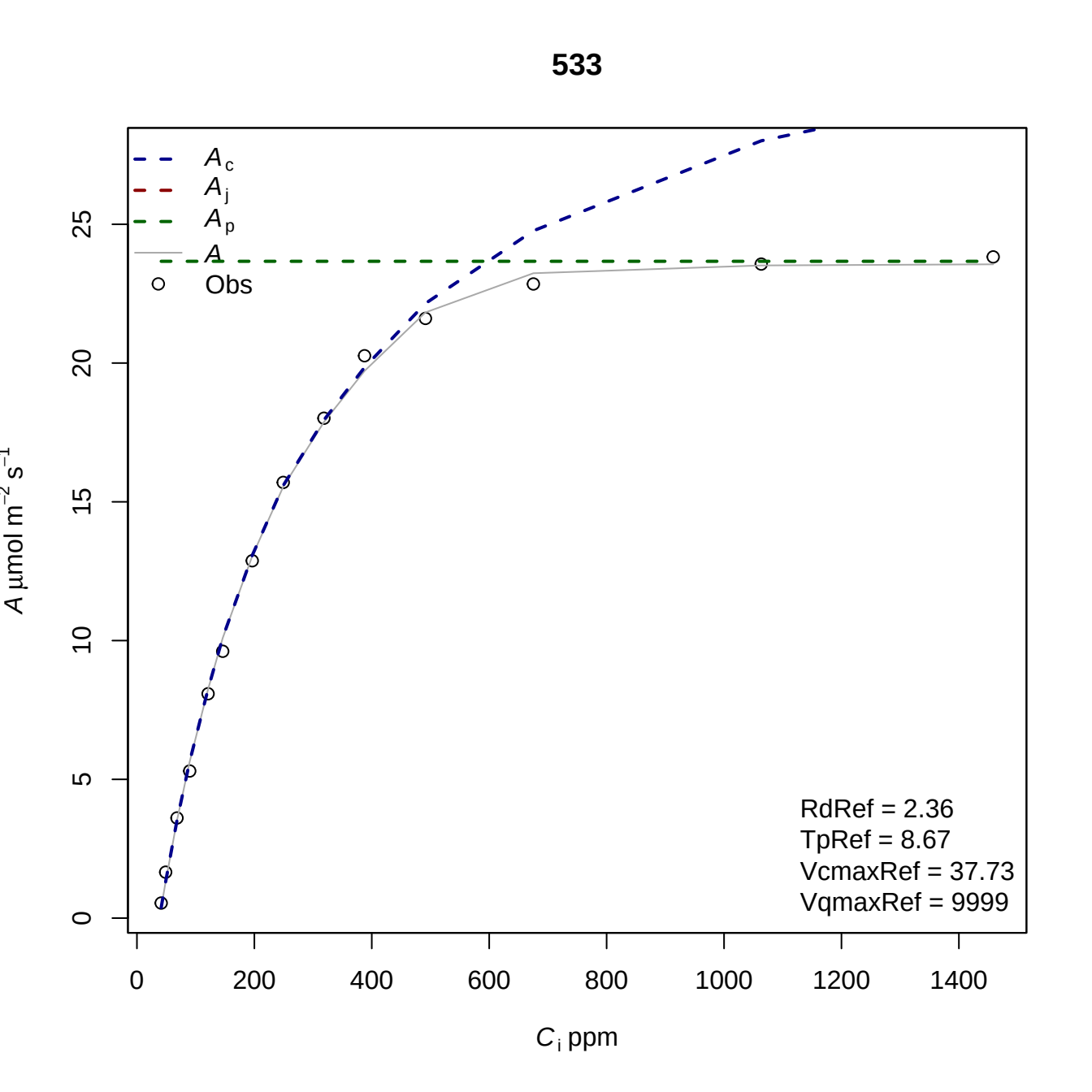

528
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.73
TpRef = 8.09
VcmaxRef = 30.4
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

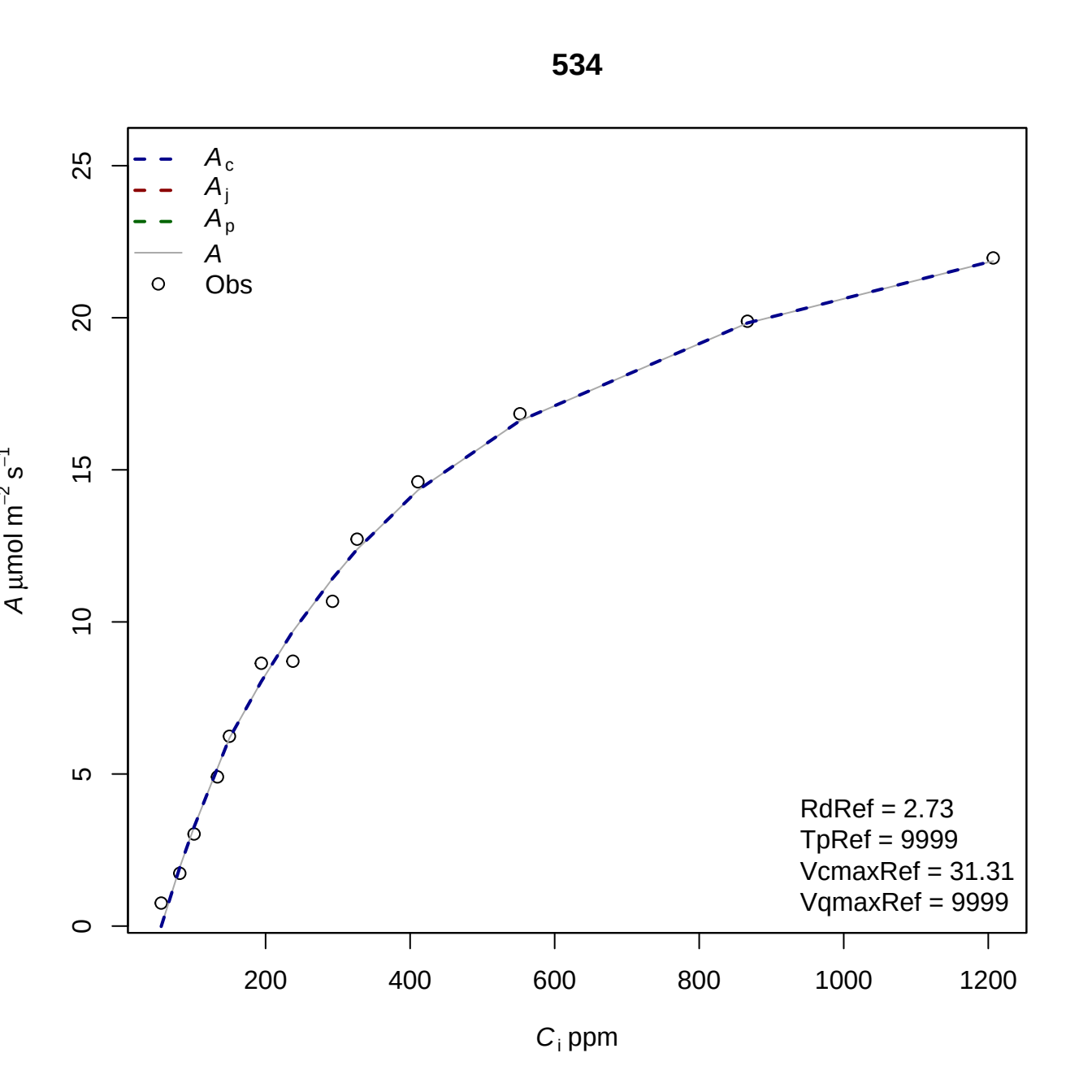

529
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.57
TpRef = 9999
VcmaxRef = 49.96
VqmaxRef = 183.33
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

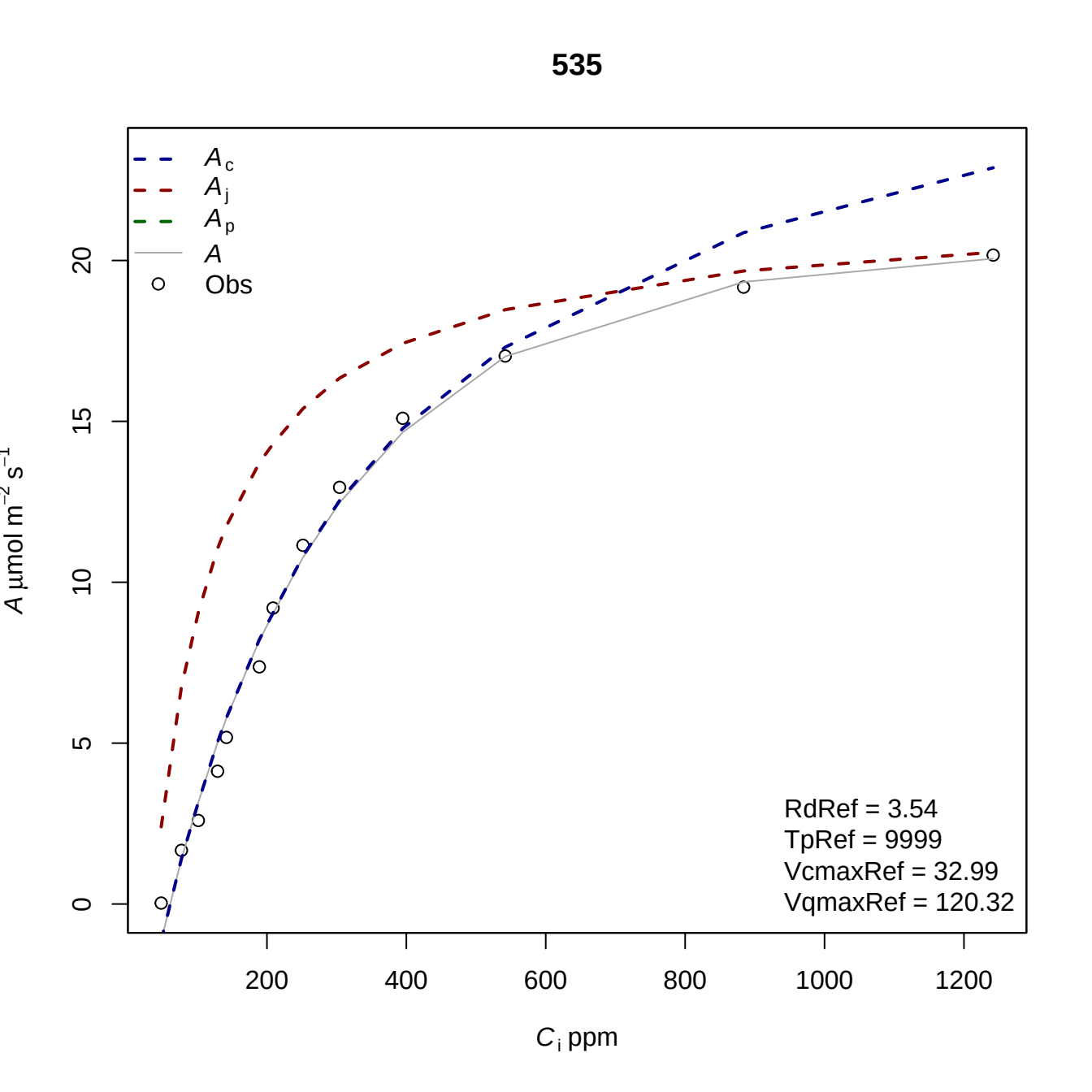

530
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.03
TpRef = 9999
VcmaxRef = 32.45
VqmaxRef = 122.58
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

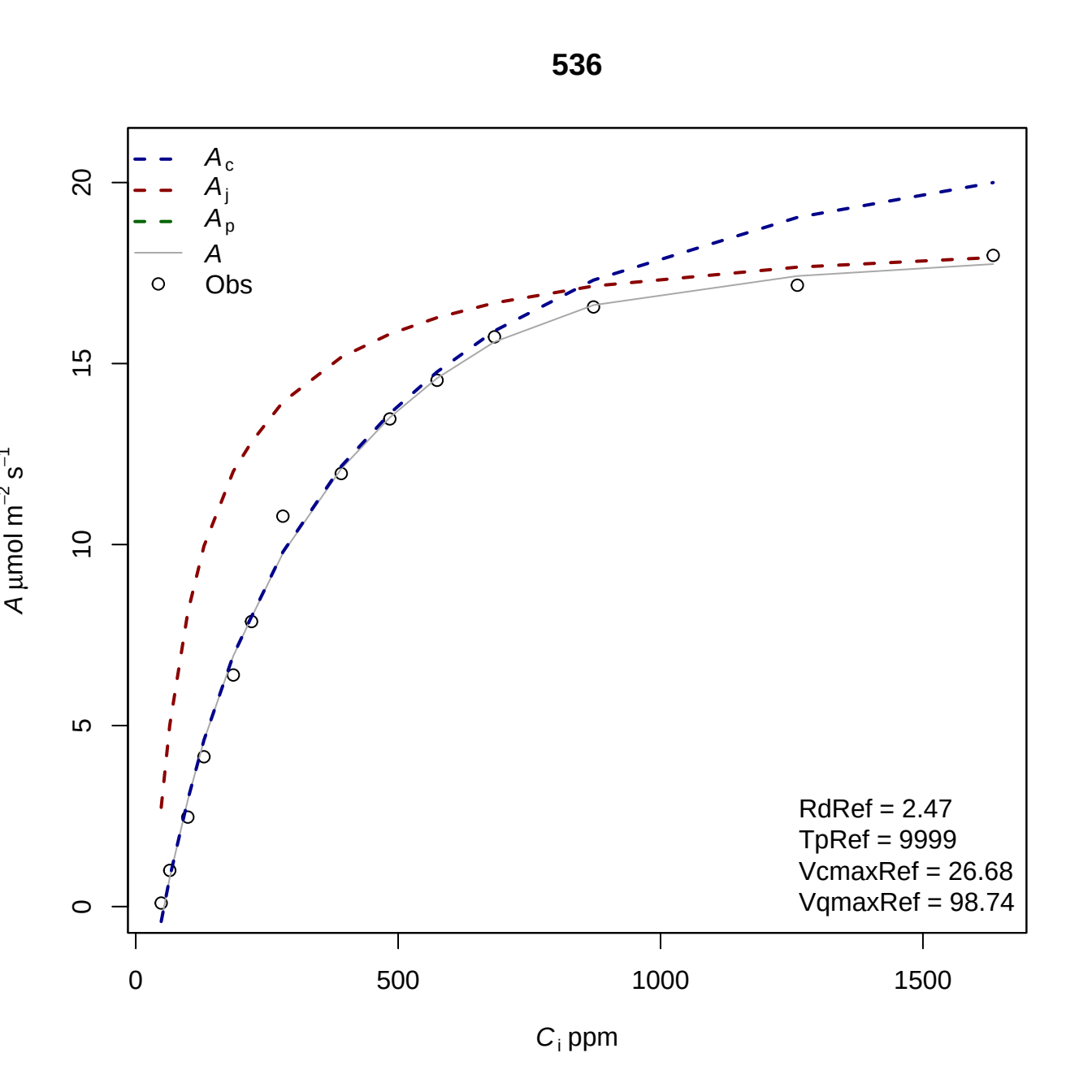

531
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.28
TpRef = 9999
VcmaxRef = 33.47
VqmaxRef = 119.59
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

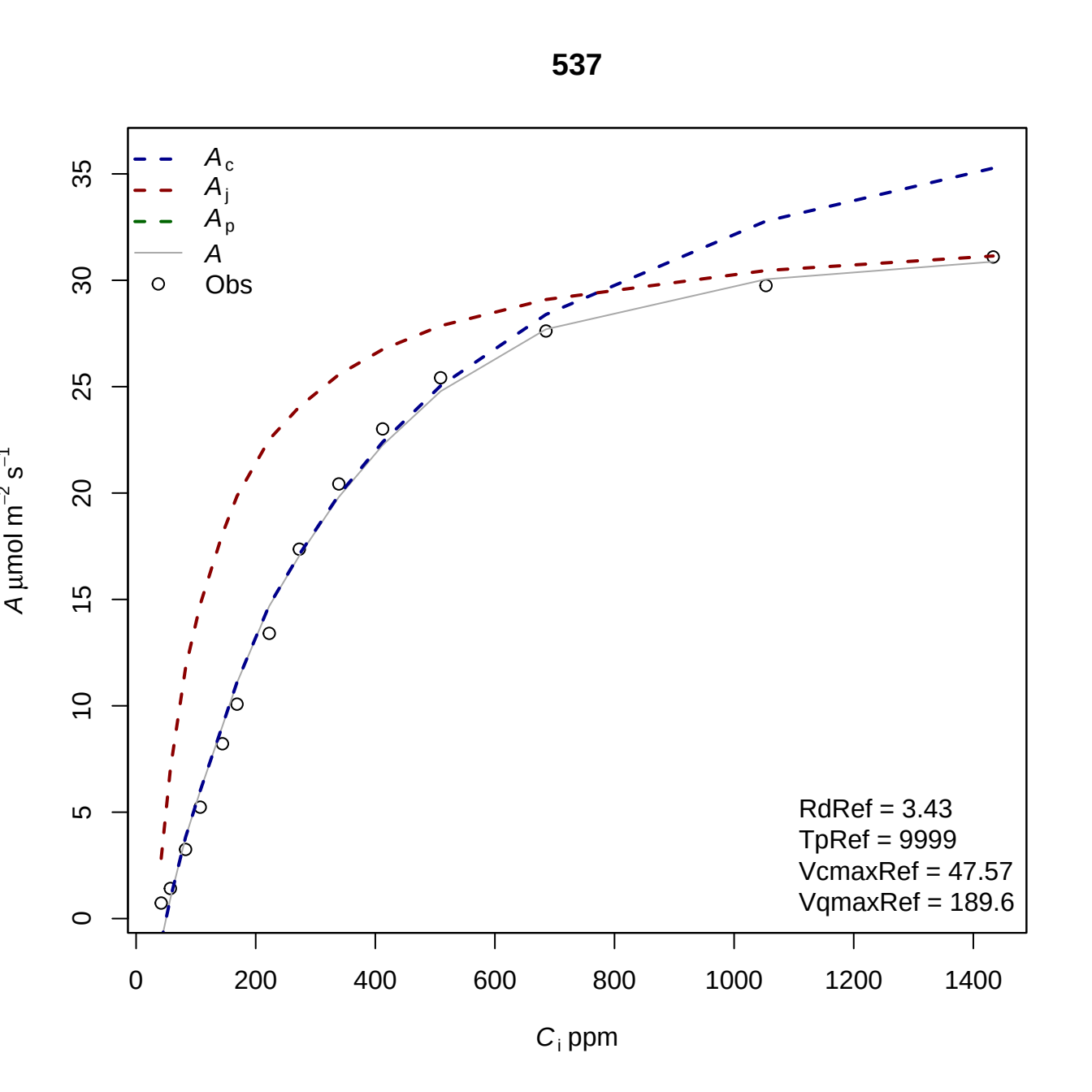

532
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.33
TpRef = 9999
VcmaxRef = 38.35
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

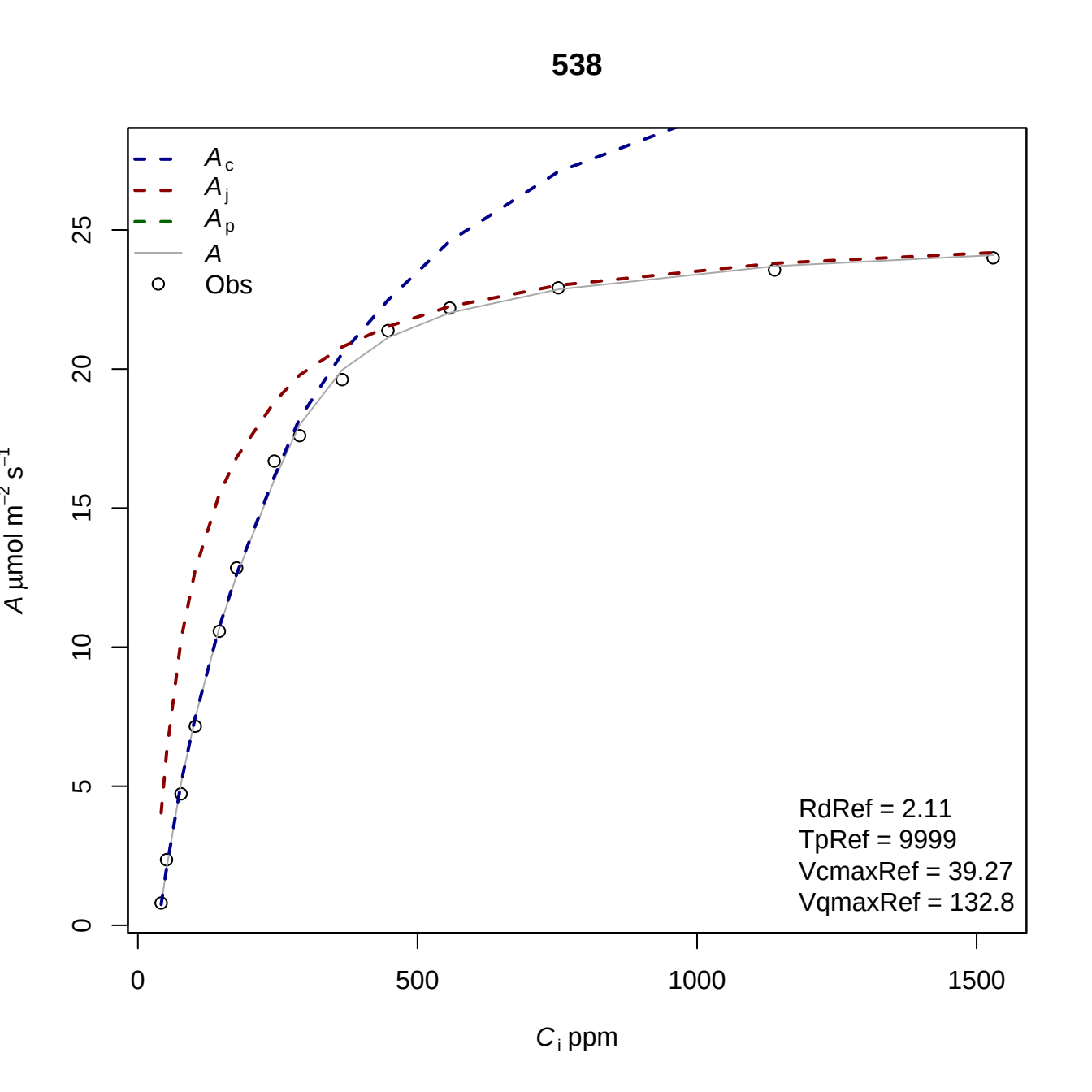

533
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.36
TpRef = 8.67
VcmaxRef = 37.73
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400

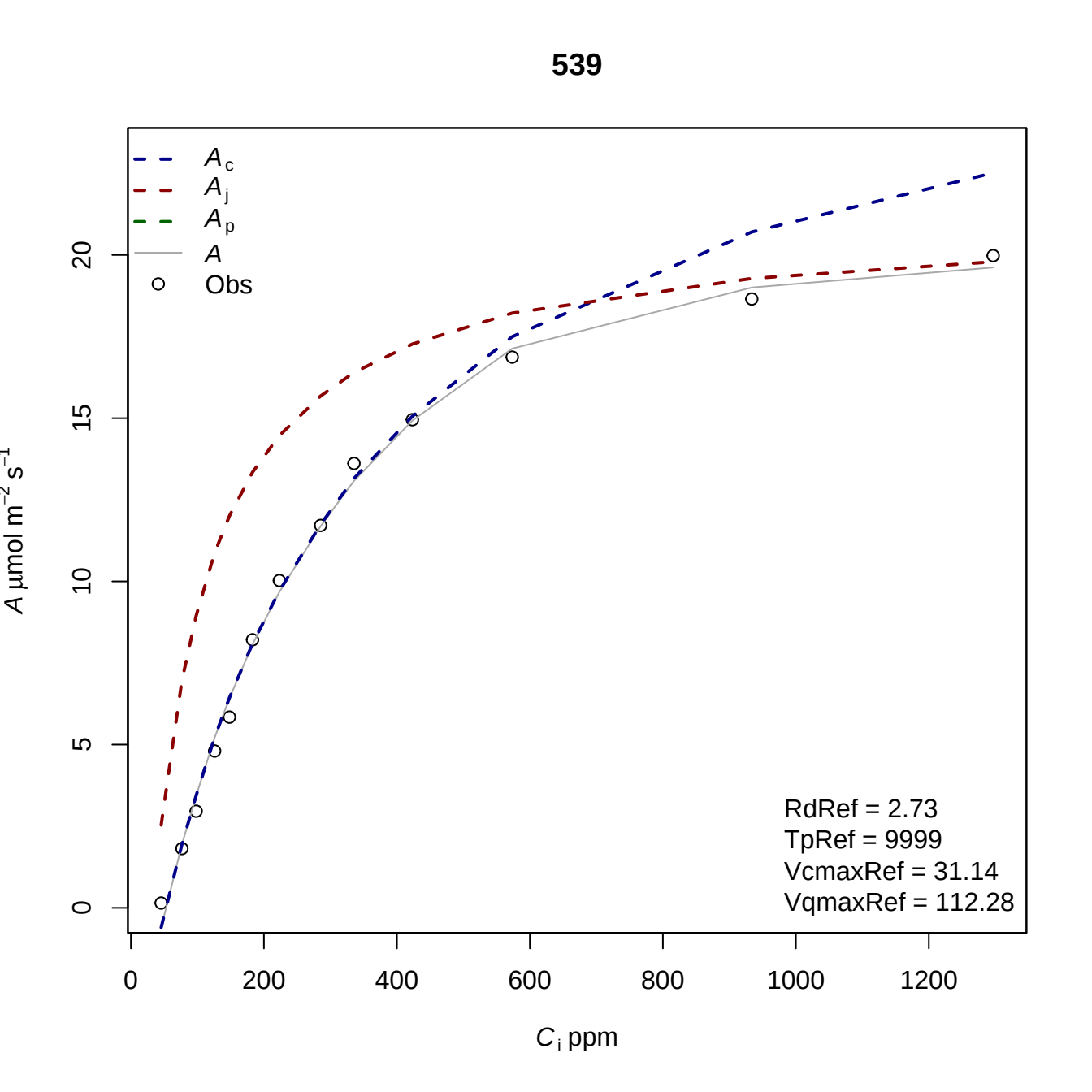

534
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.73
TpRef = 9999
VcmaxRef = 31.31
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

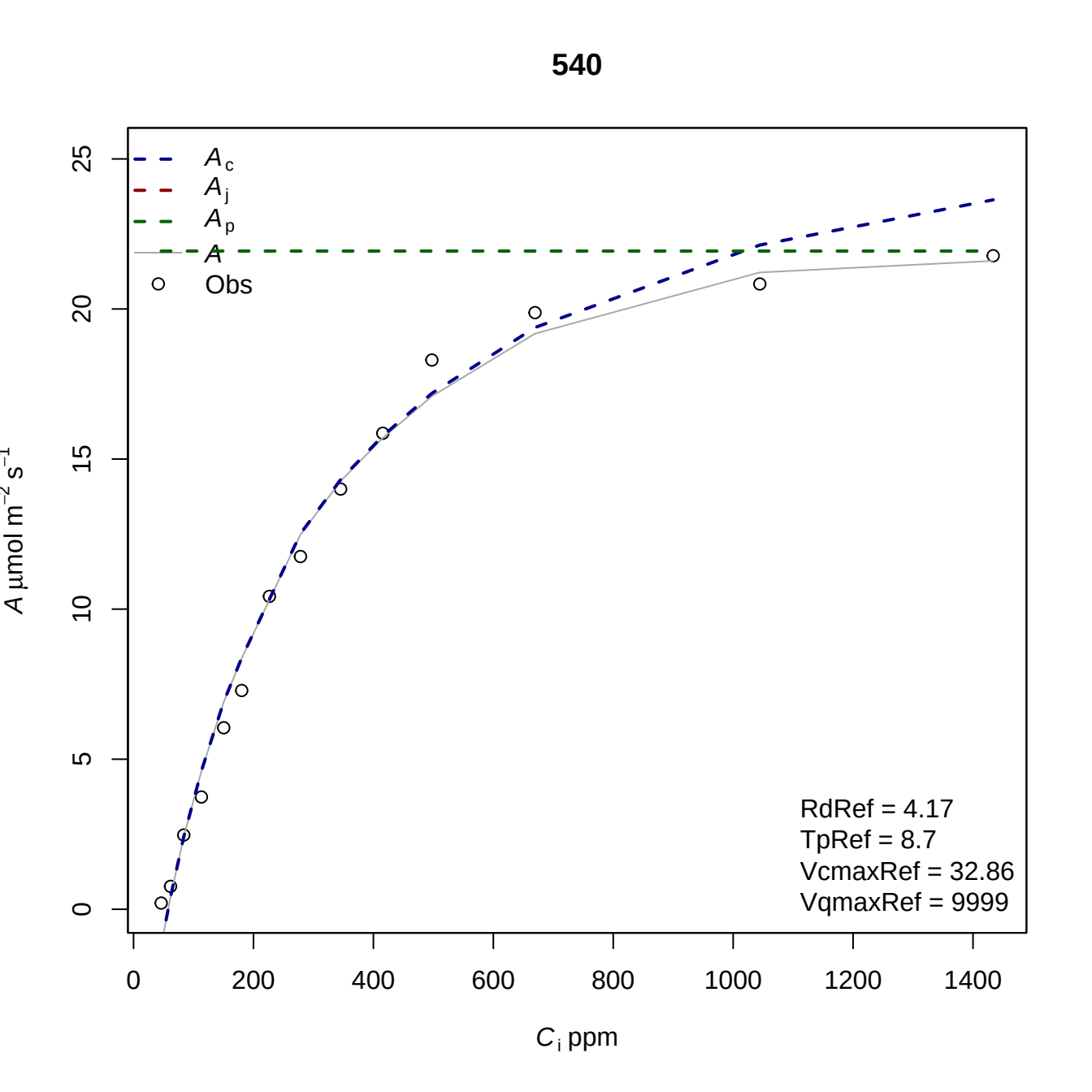

535
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.54
TpRef = 9999
VcmaxRef = 32.99
VqmaxRef = 120.32
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200

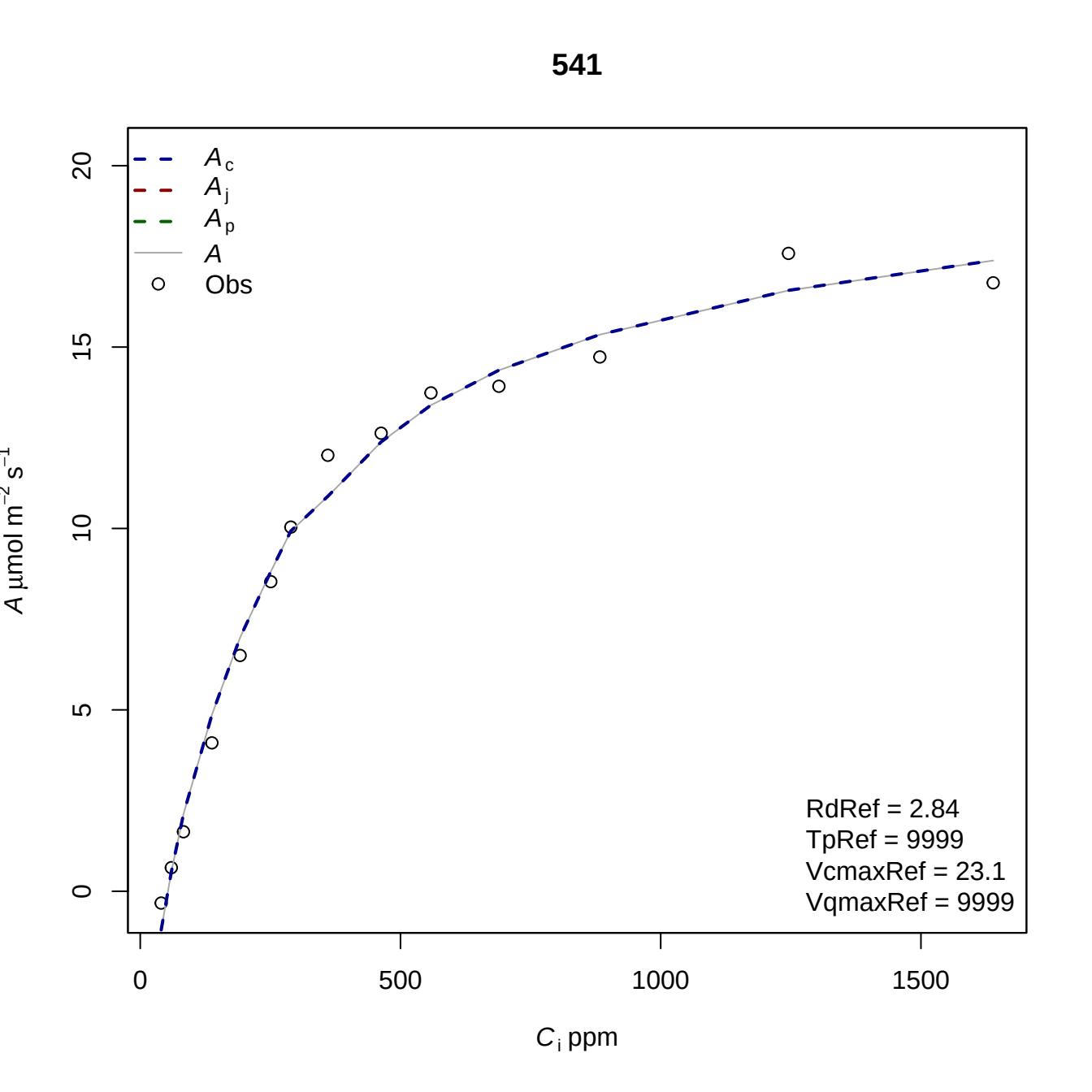

536
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.47
TpRef = 9999
VcmaxRef = 26.68
VqmaxRef = 98.74
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

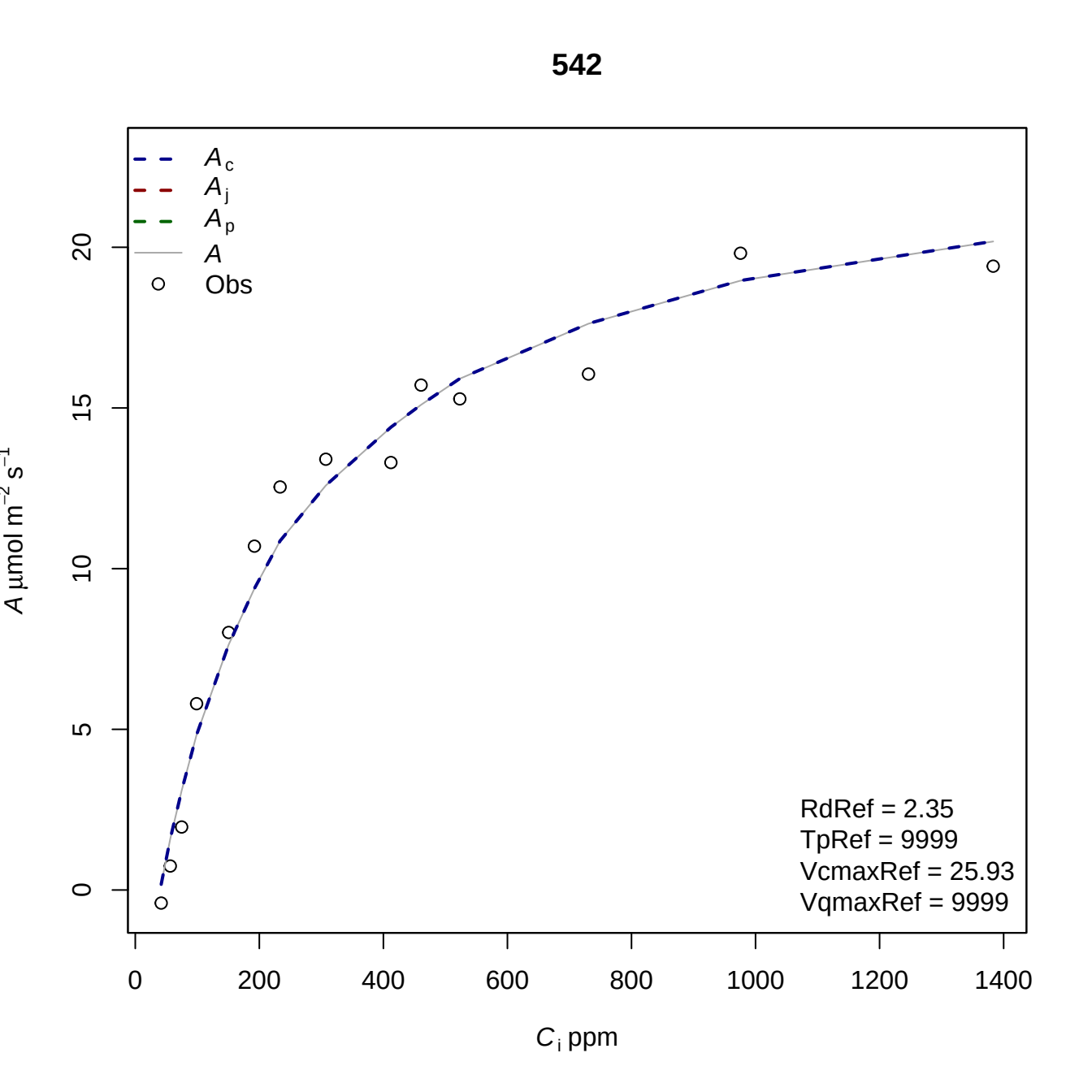

537
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.43
TpRef = 9999
VcmaxRef = 47.57
VqmaxRef = 189.6
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
1400

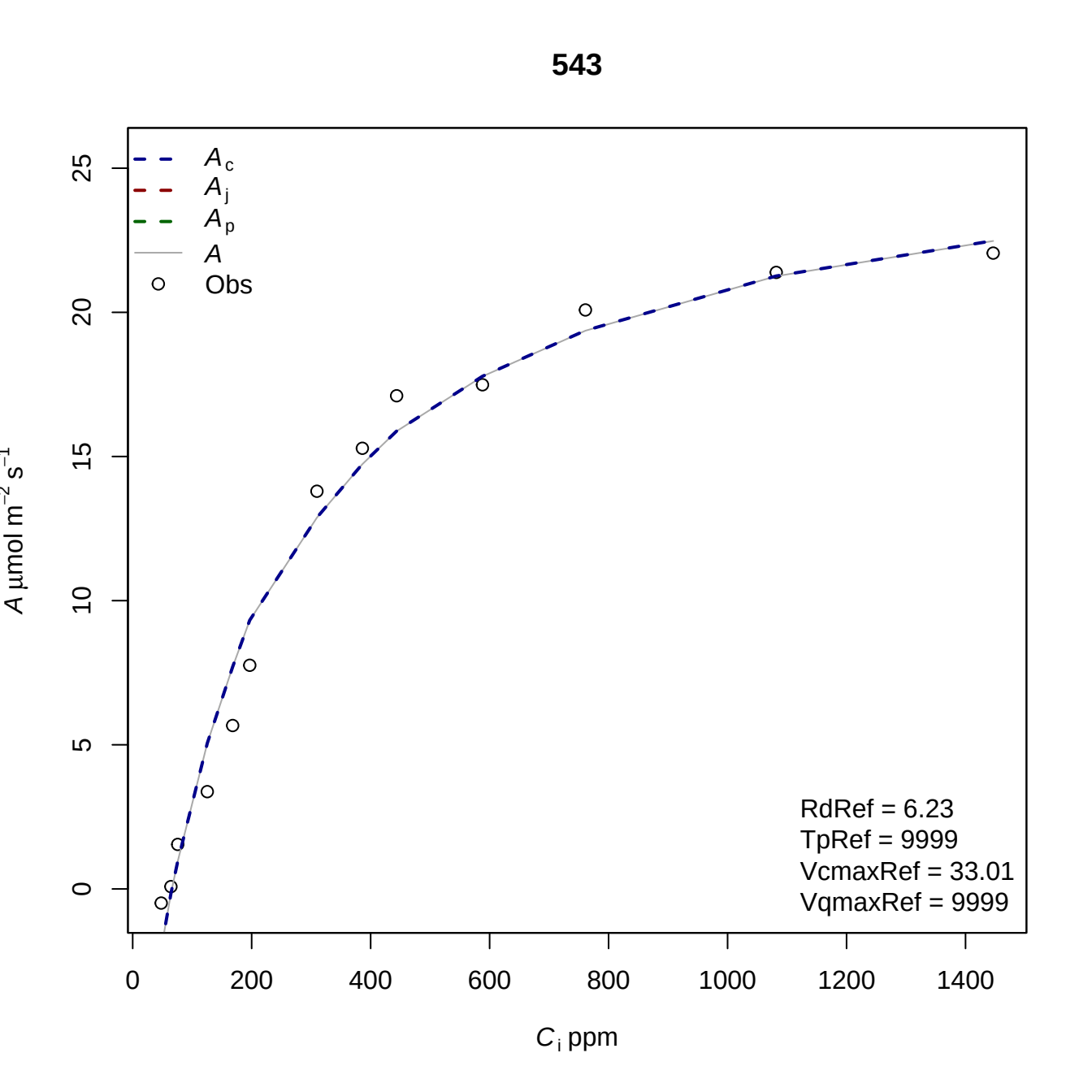

538
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.11
TpRef = 9999
VcmaxRef = 39.27
VqmaxRef = 132.8
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

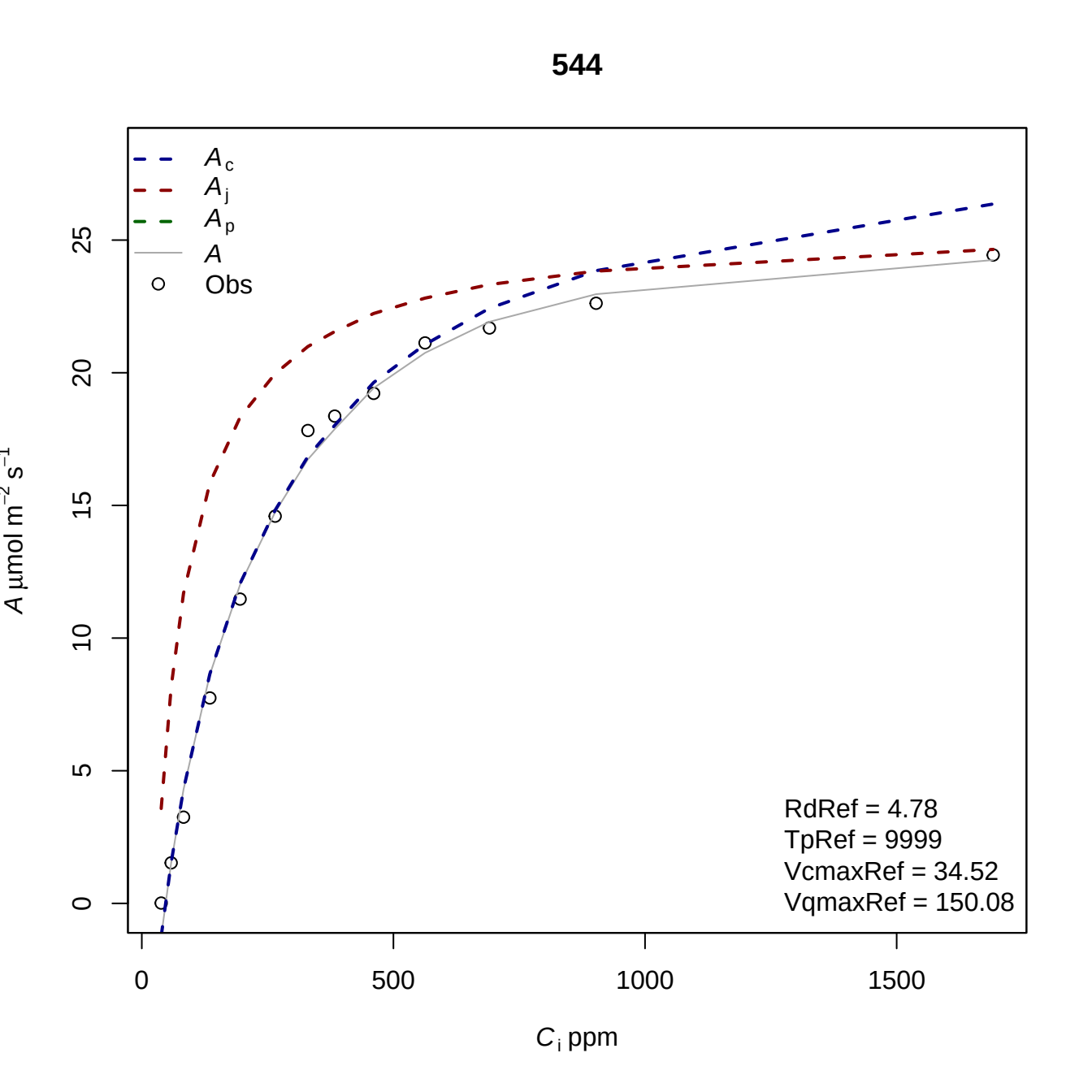

539
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.73
TpRef = 9999
VcmaxRef = 31.14
VqmaxRef = 112.28
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

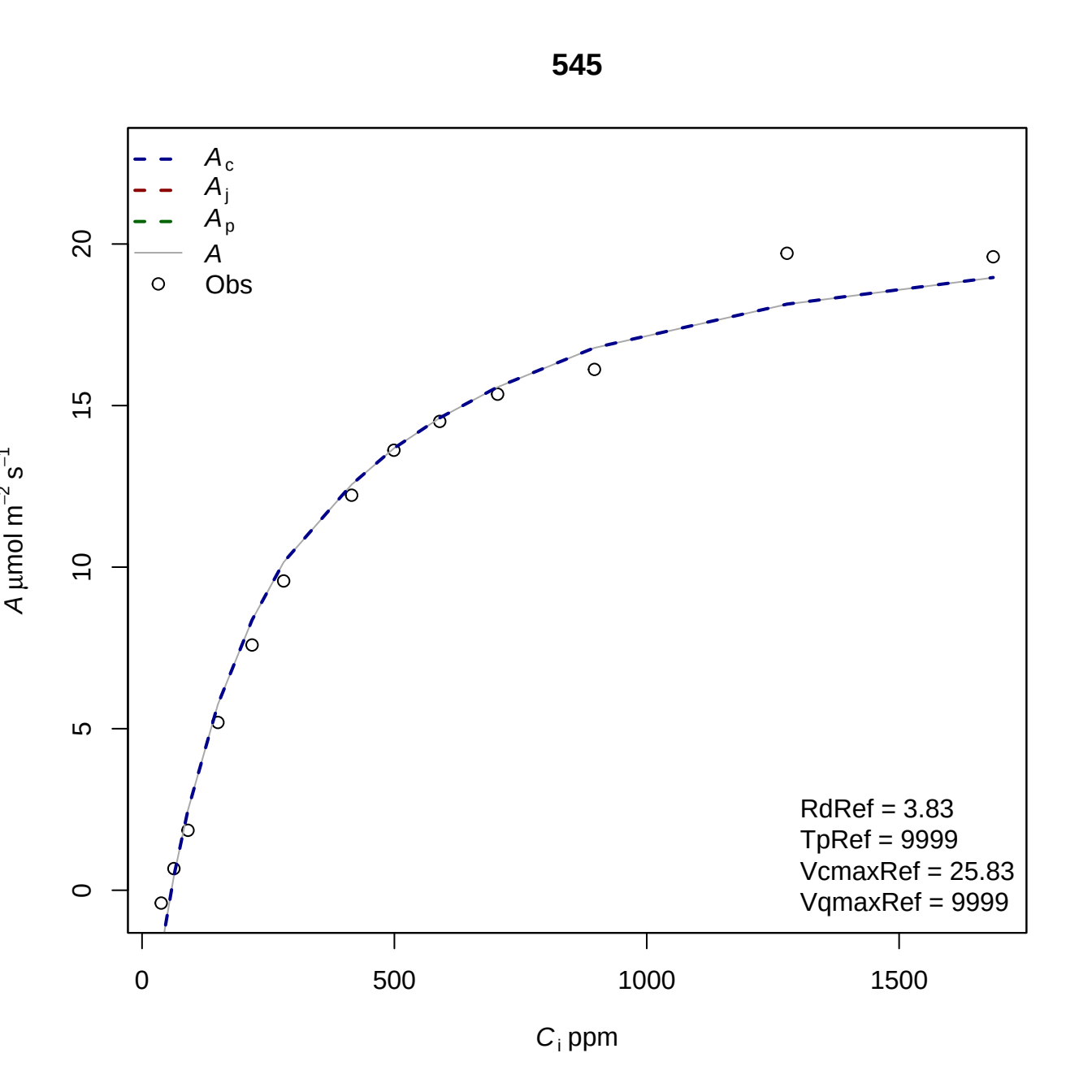

540
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.17
TpRef = 8.7
VcmaxRef = 32.86
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400

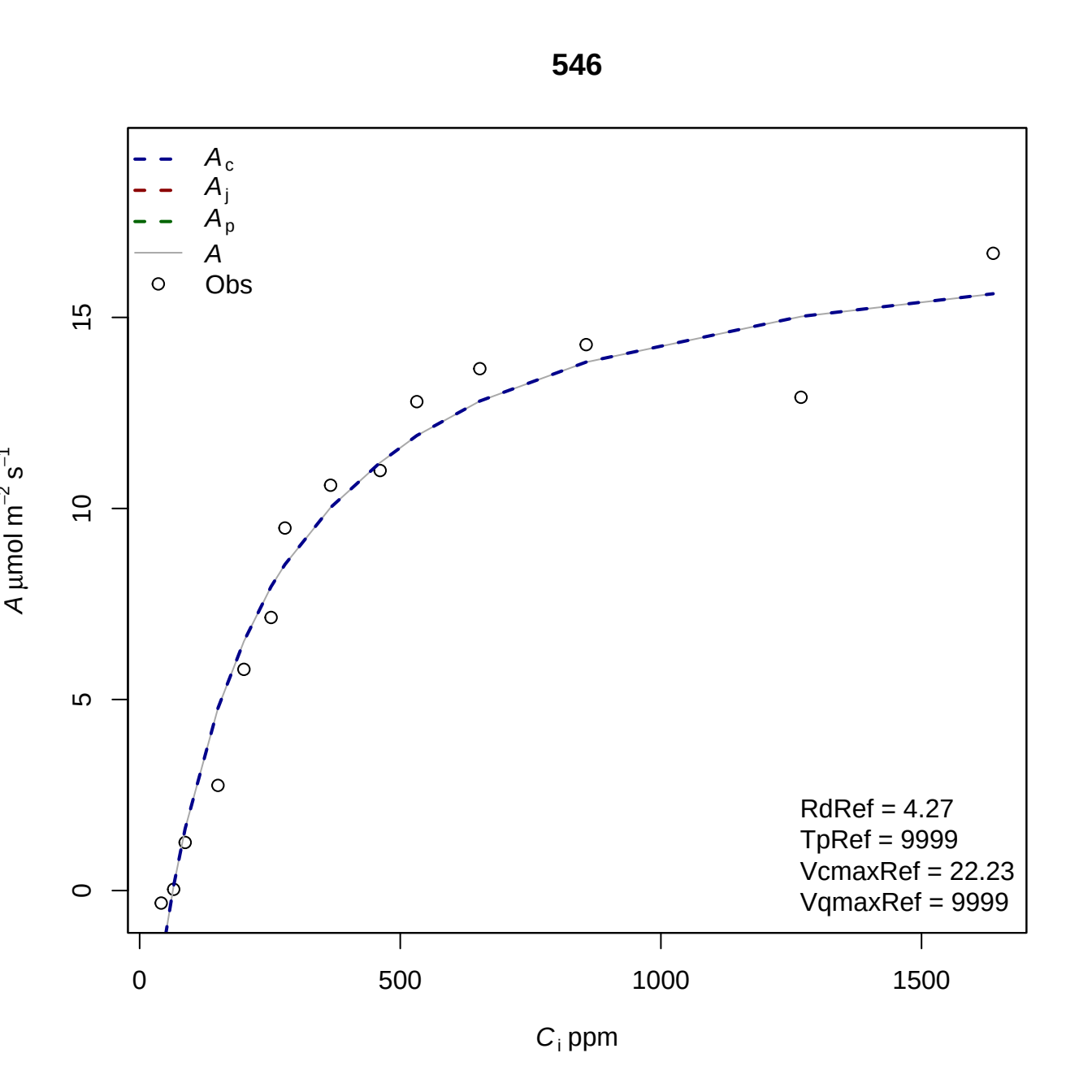

541
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.84
TpRef = 9999
VcmaxRef = 23.1
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

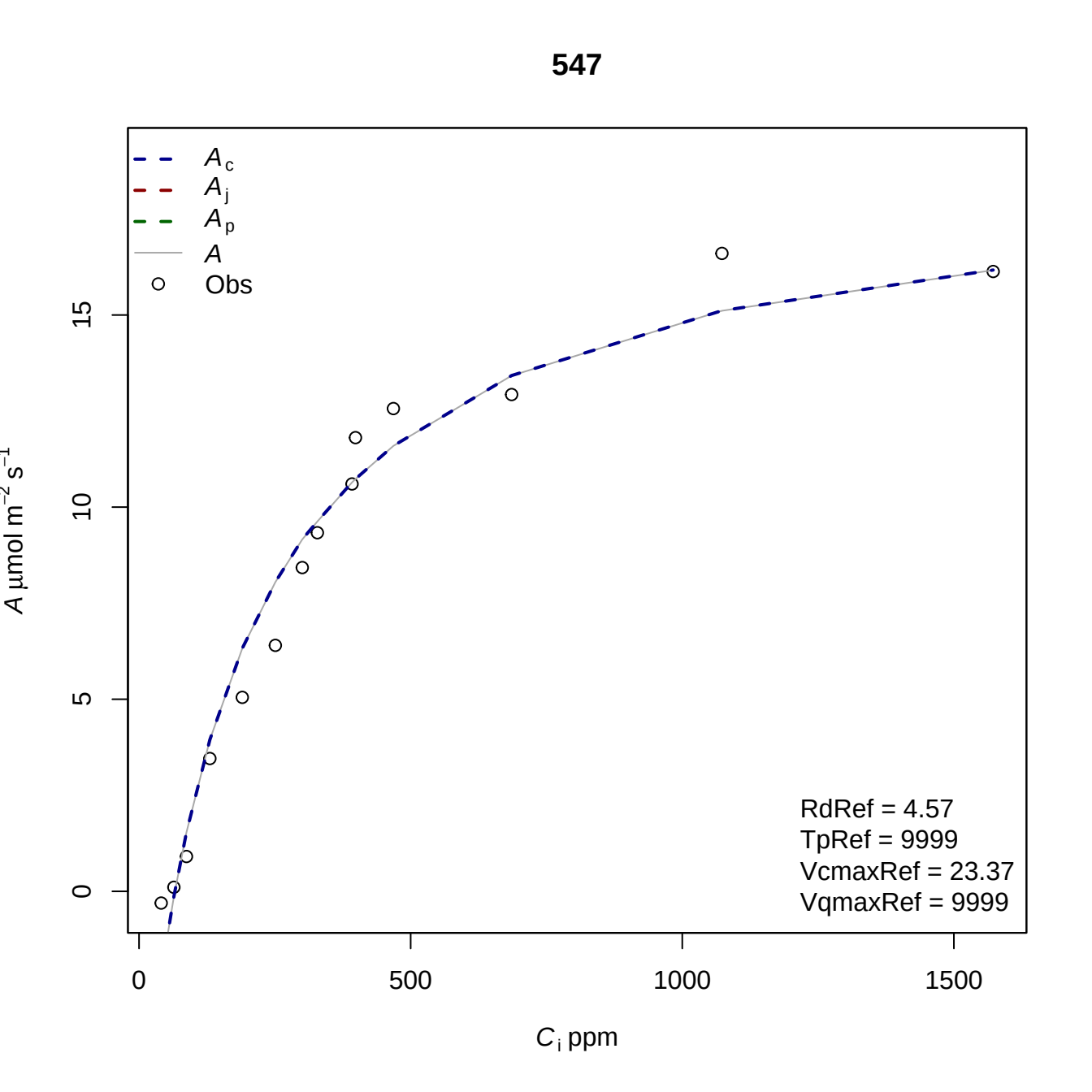

542
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.35
TpRef = 9999
VcmaxRef = 25.93
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

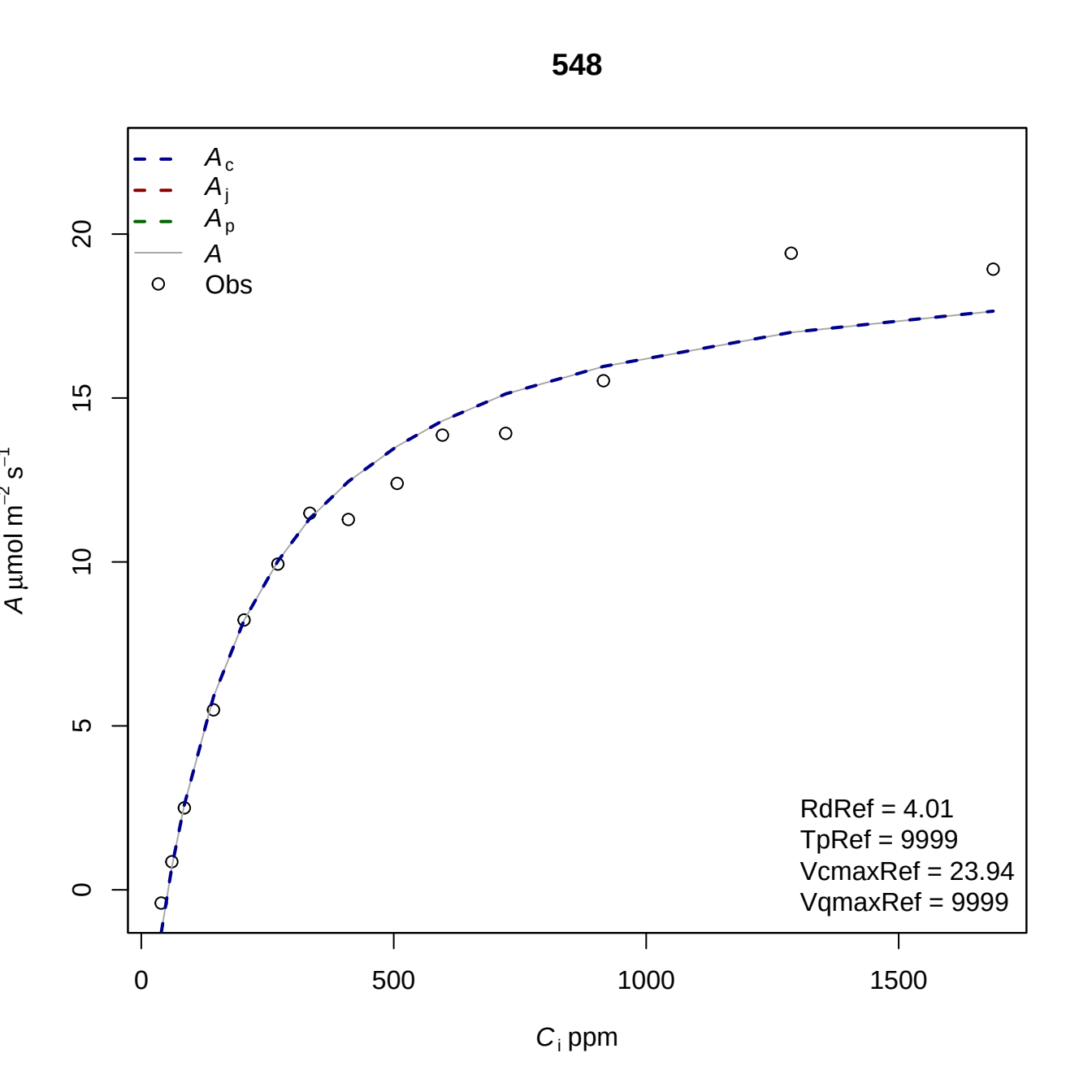

543
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 6.23
TpRef = 9999
VcmaxRef = 33.01
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400

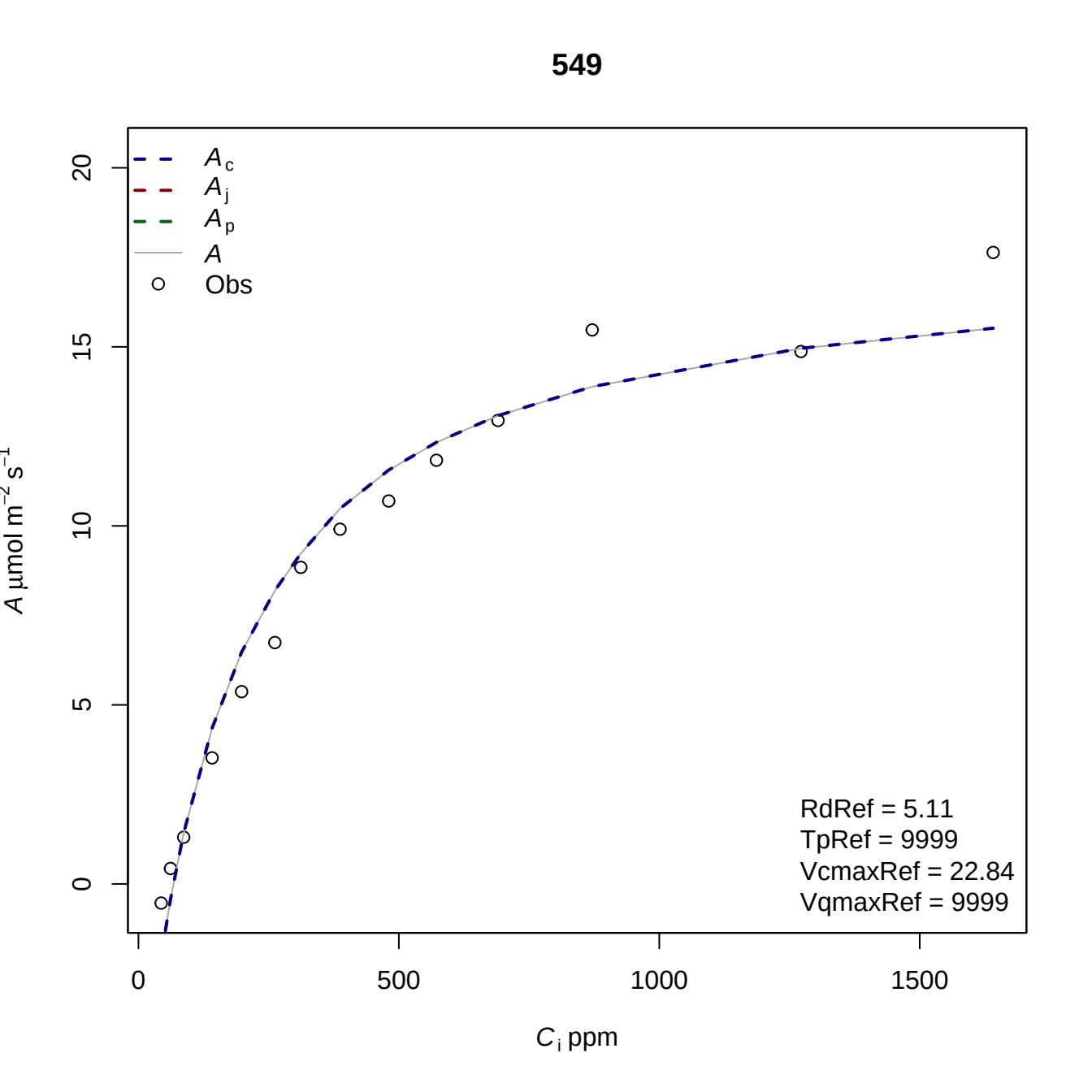

544
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.78
TpRef = 9999
VcmaxRef = 34.52
VqmaxRef = 150.08
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

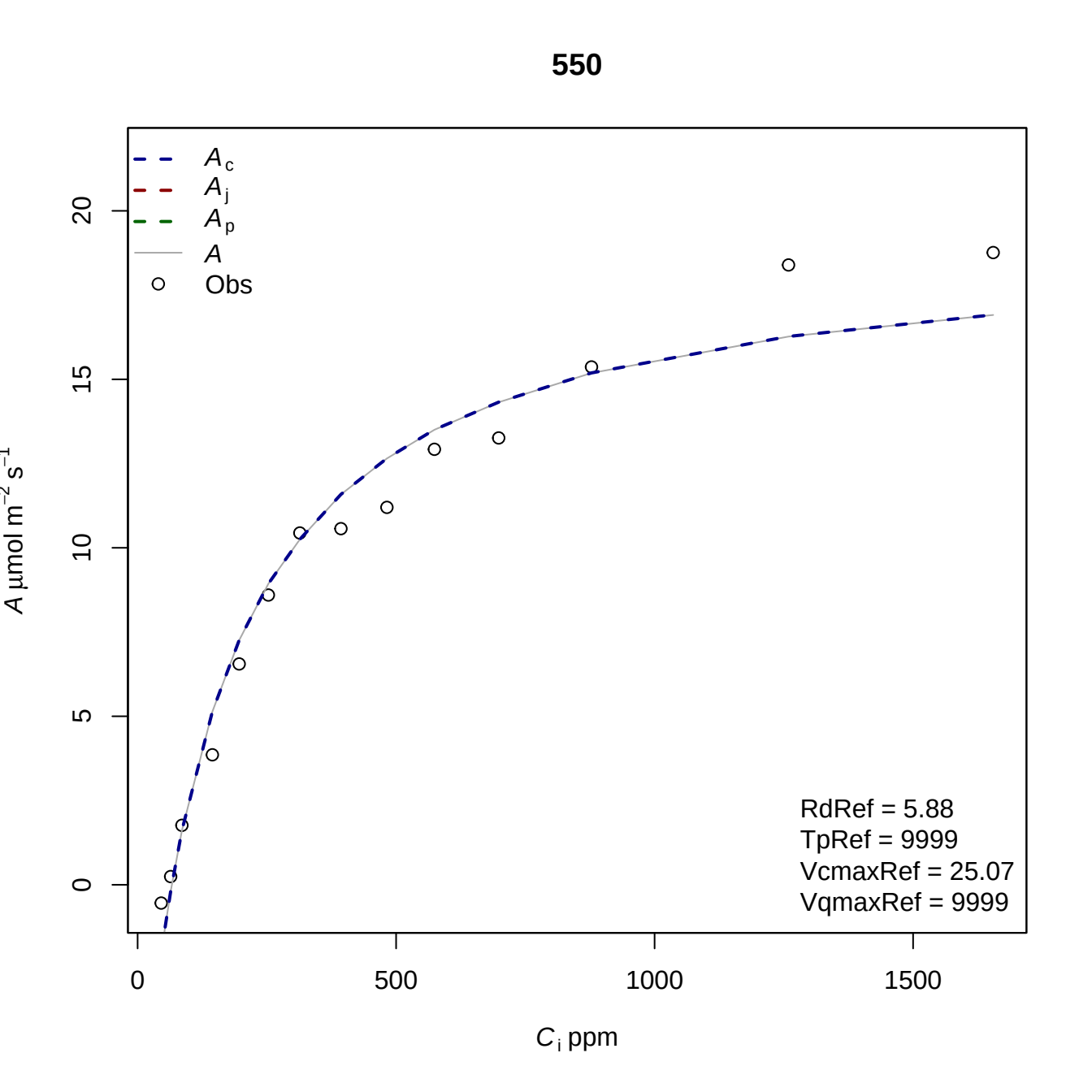

545
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.83
TpRef = 9999
VcmaxRef = 25.83
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

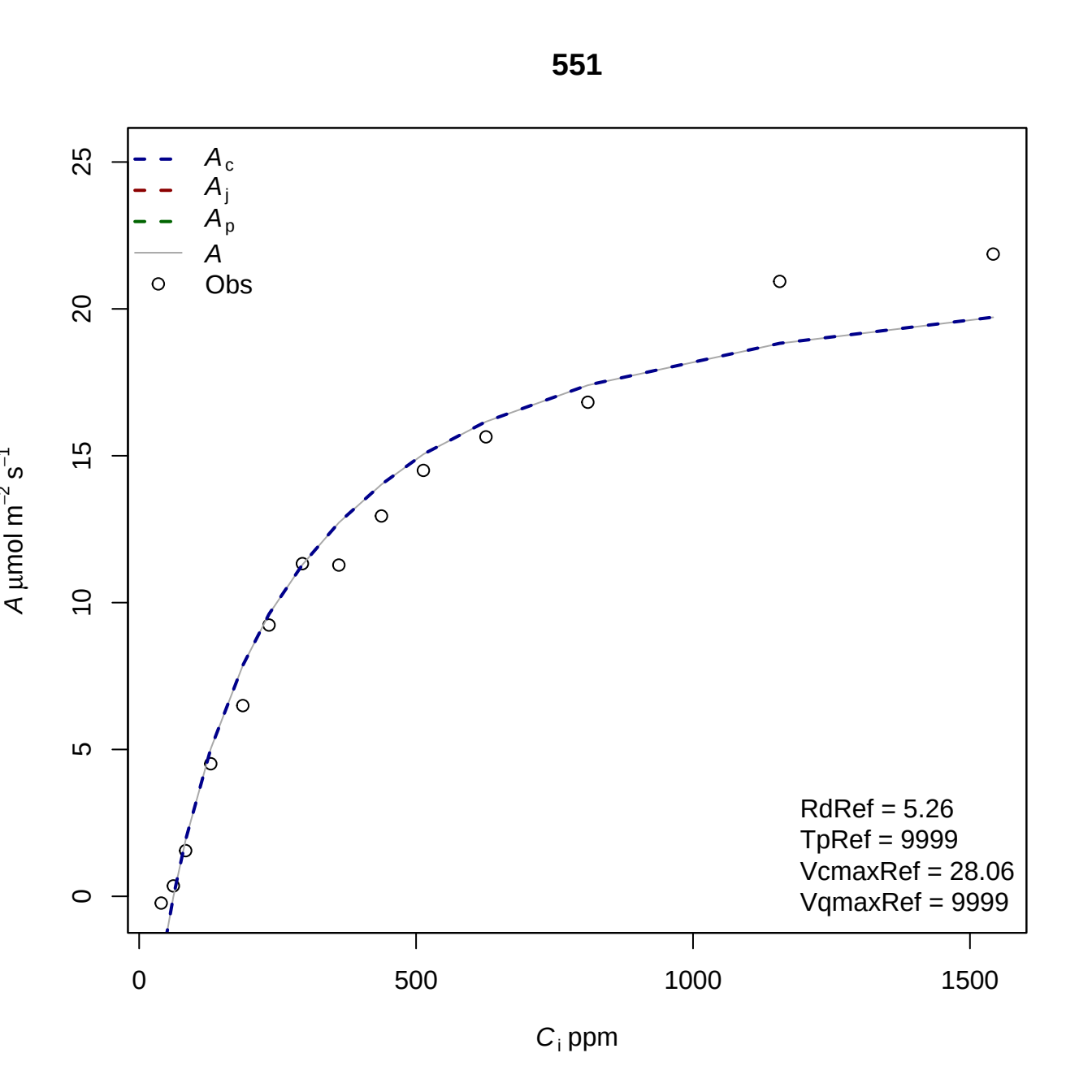

546
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.27
TpRef = 9999
VcmaxRef = 22.23
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

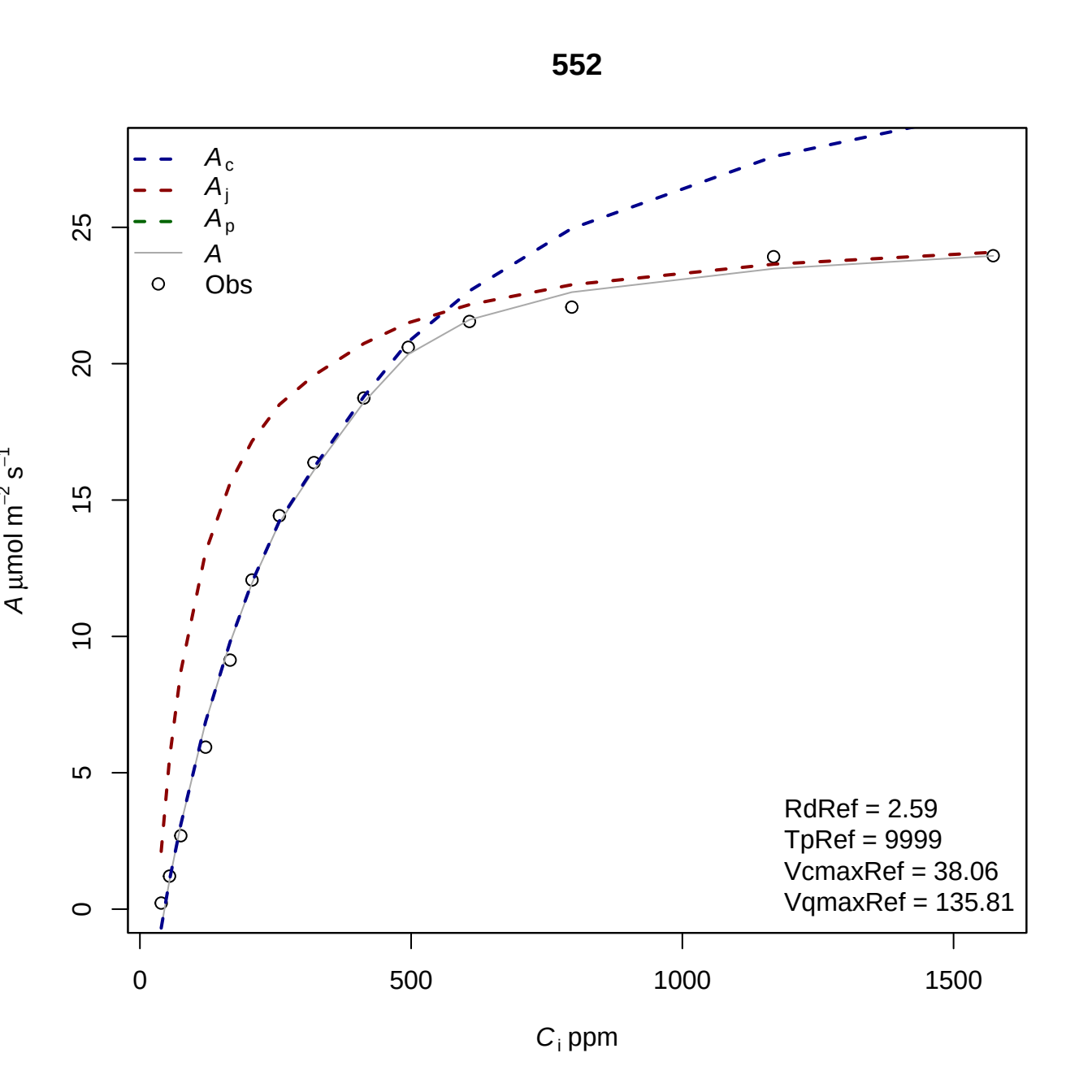

547
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.57
TpRef = 9999
VcmaxRef = 23.37
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

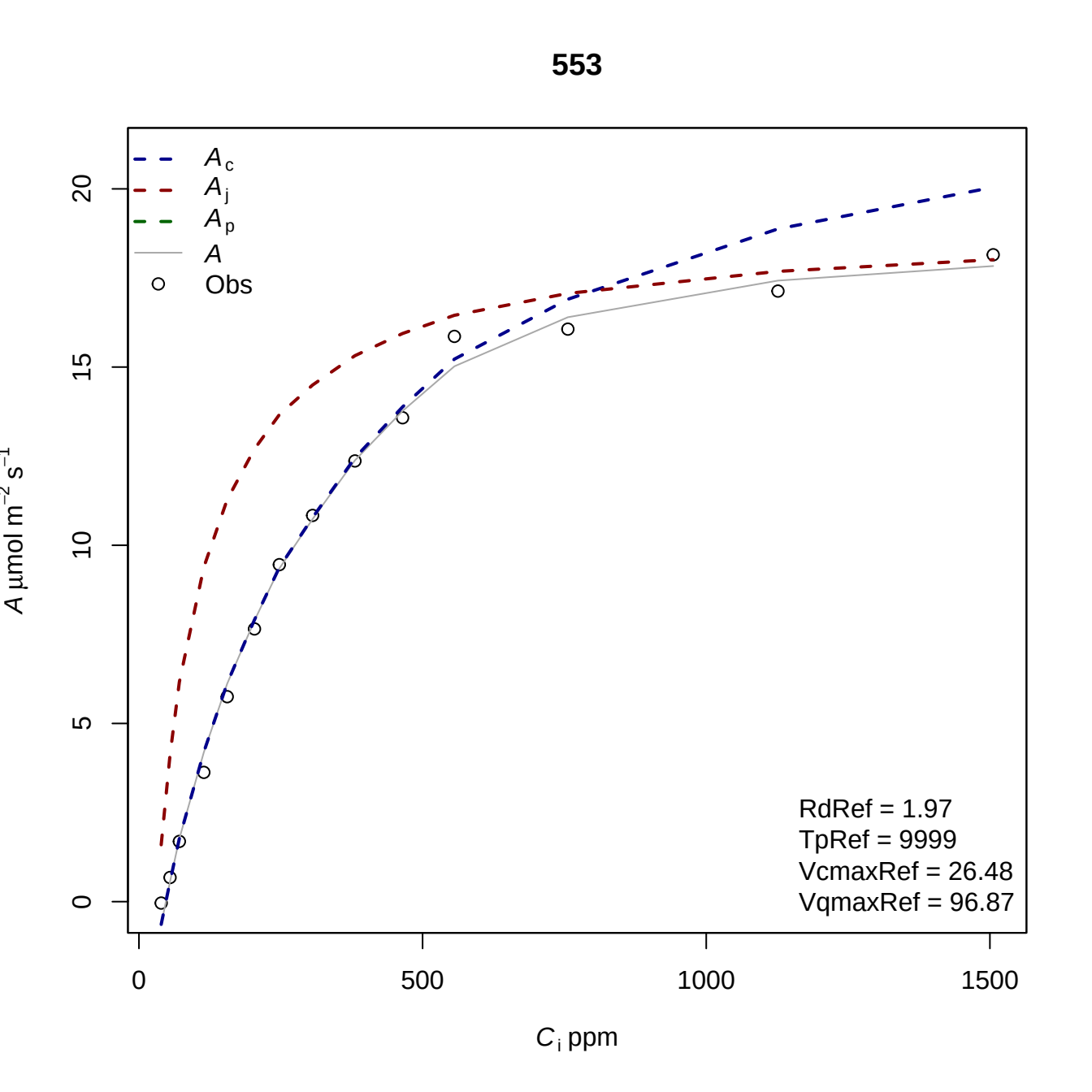

548
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.01
TpRef = 9999
VcmaxRef = 23.94
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

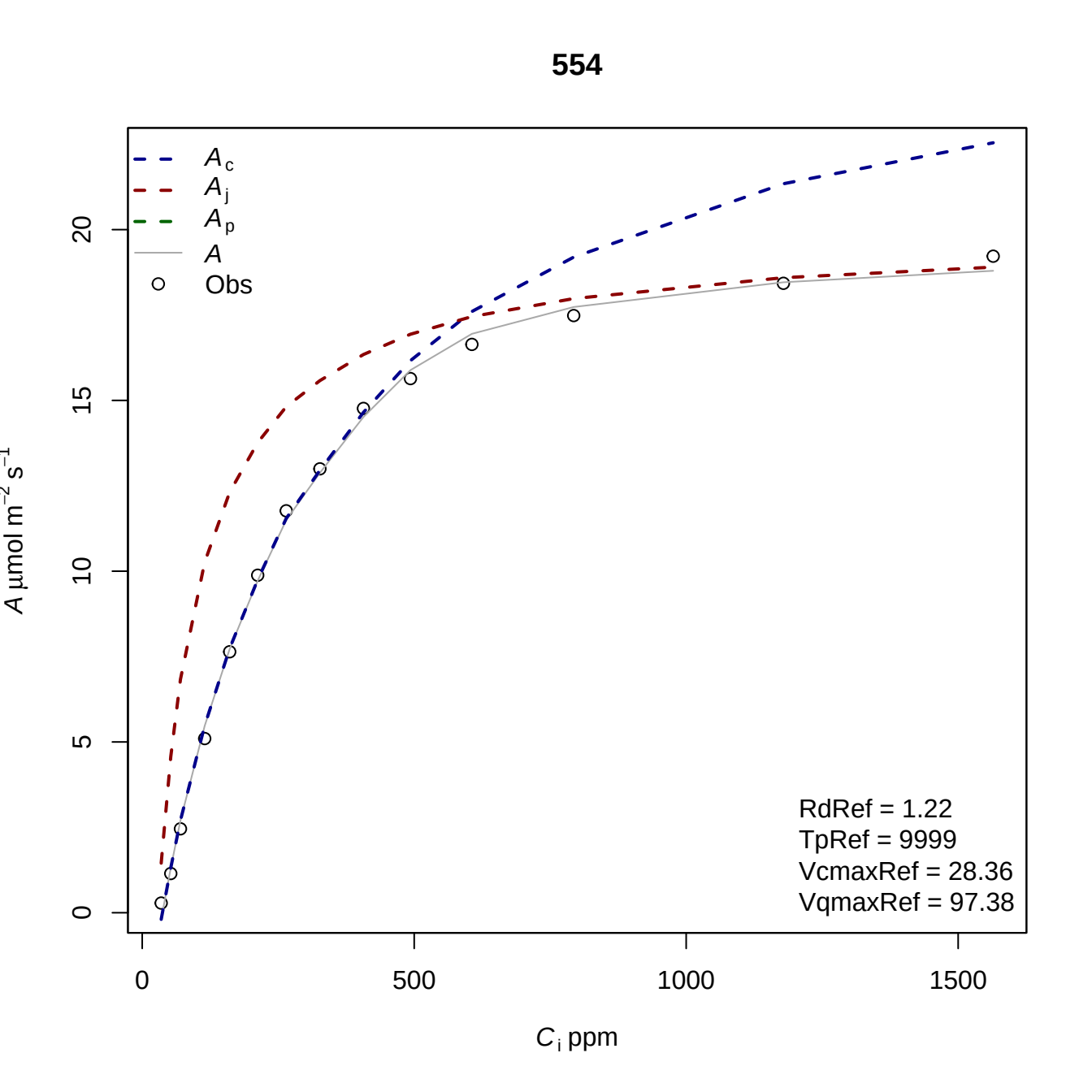

549
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 5.11
TpRef = 9999
VcmaxRef = 22.84
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

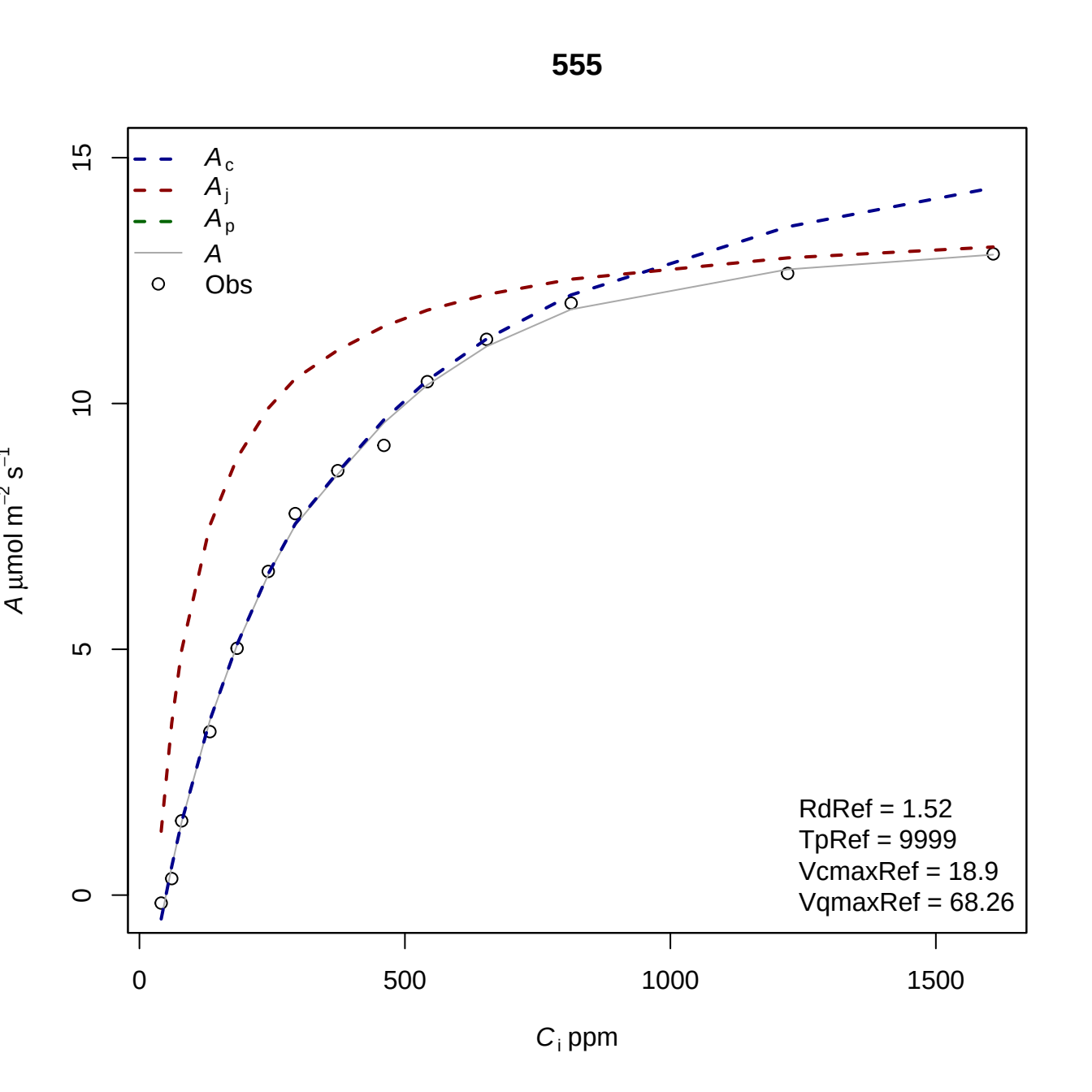

550
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 5.88
TpRef = 9999
VcmaxRef = 25.07
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

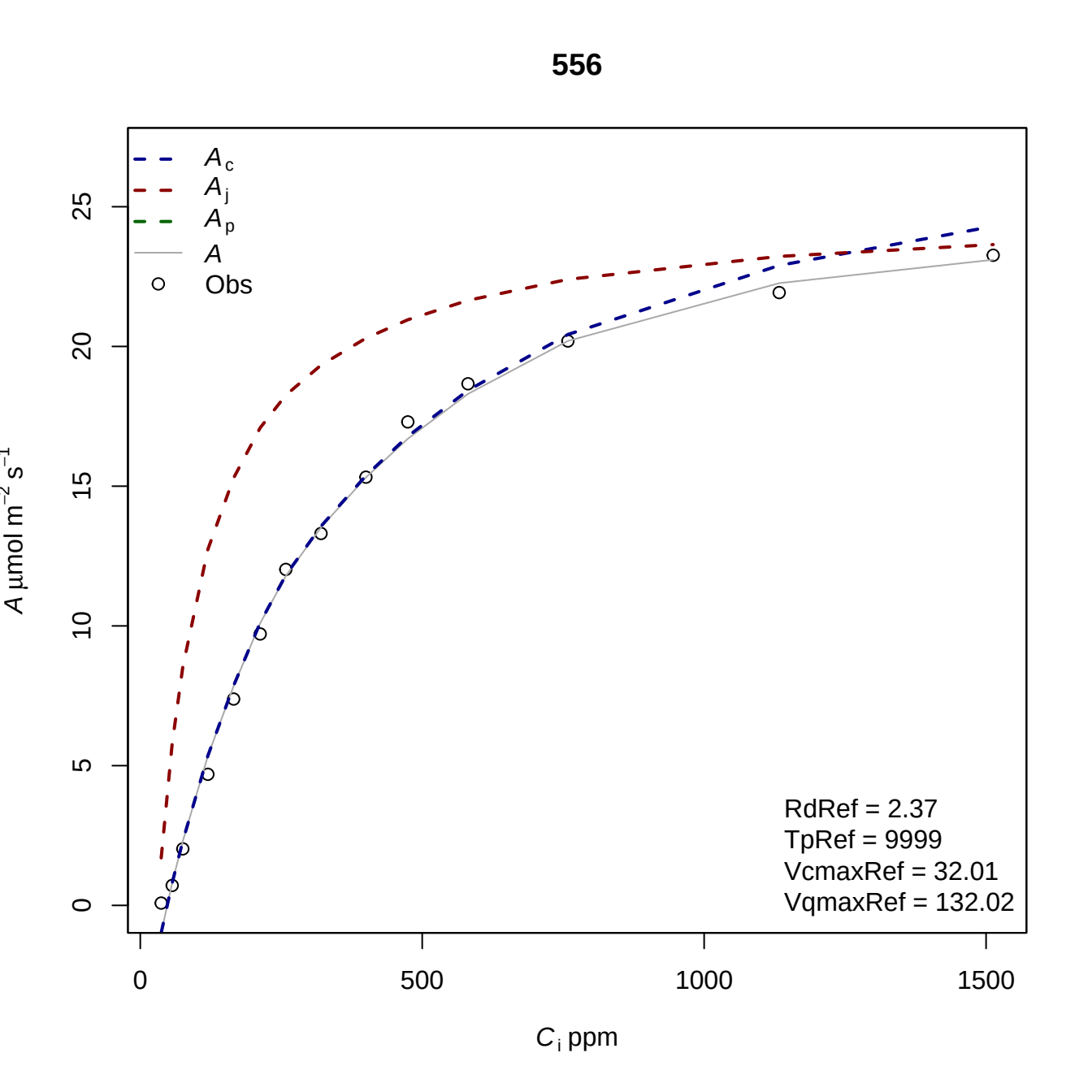

551
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 5.26
TpRef = 9999
VcmaxRef = 28.06
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

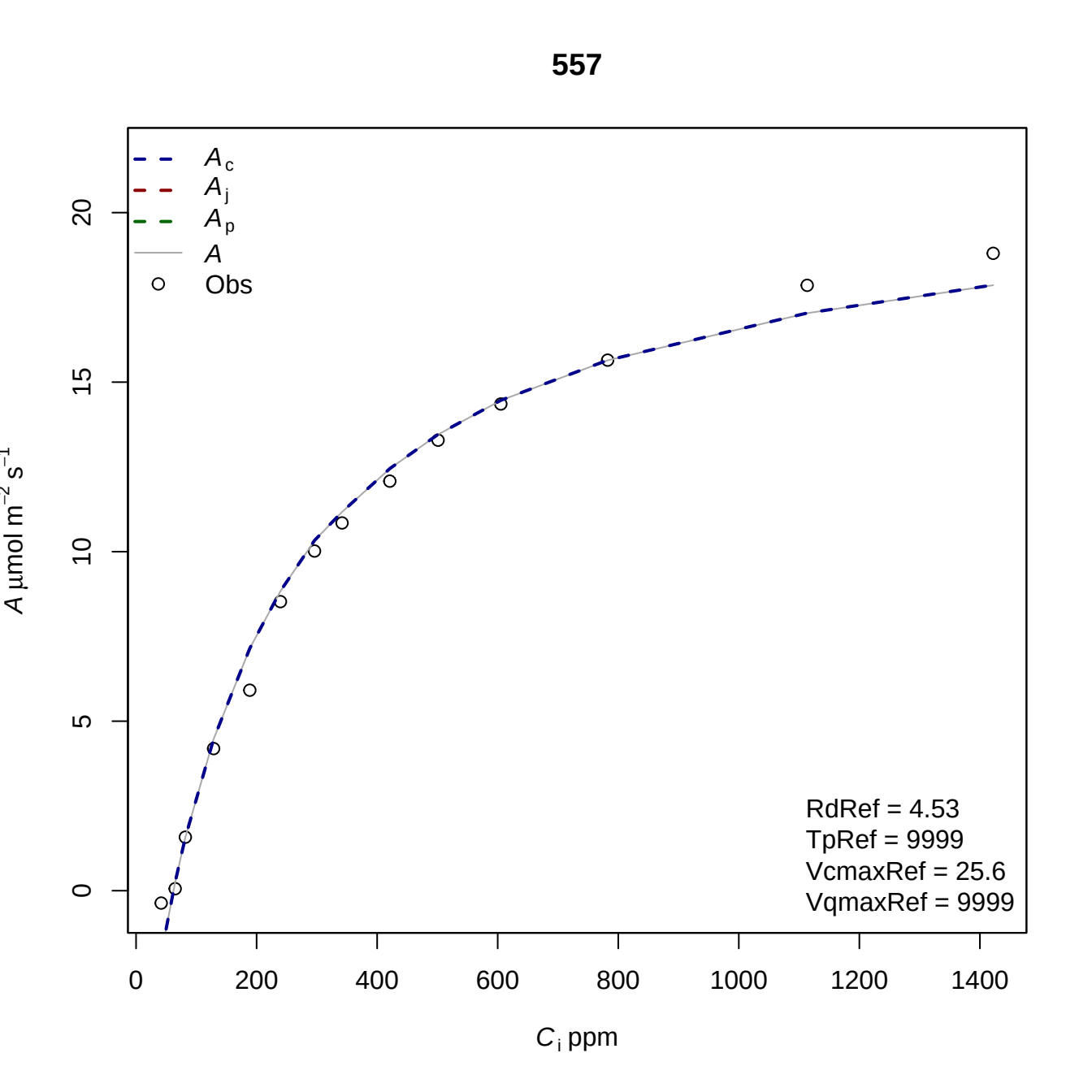

552
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.59
TpRef = 9999
VcmaxRef = 38.06
VqmaxRef = 135.81
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

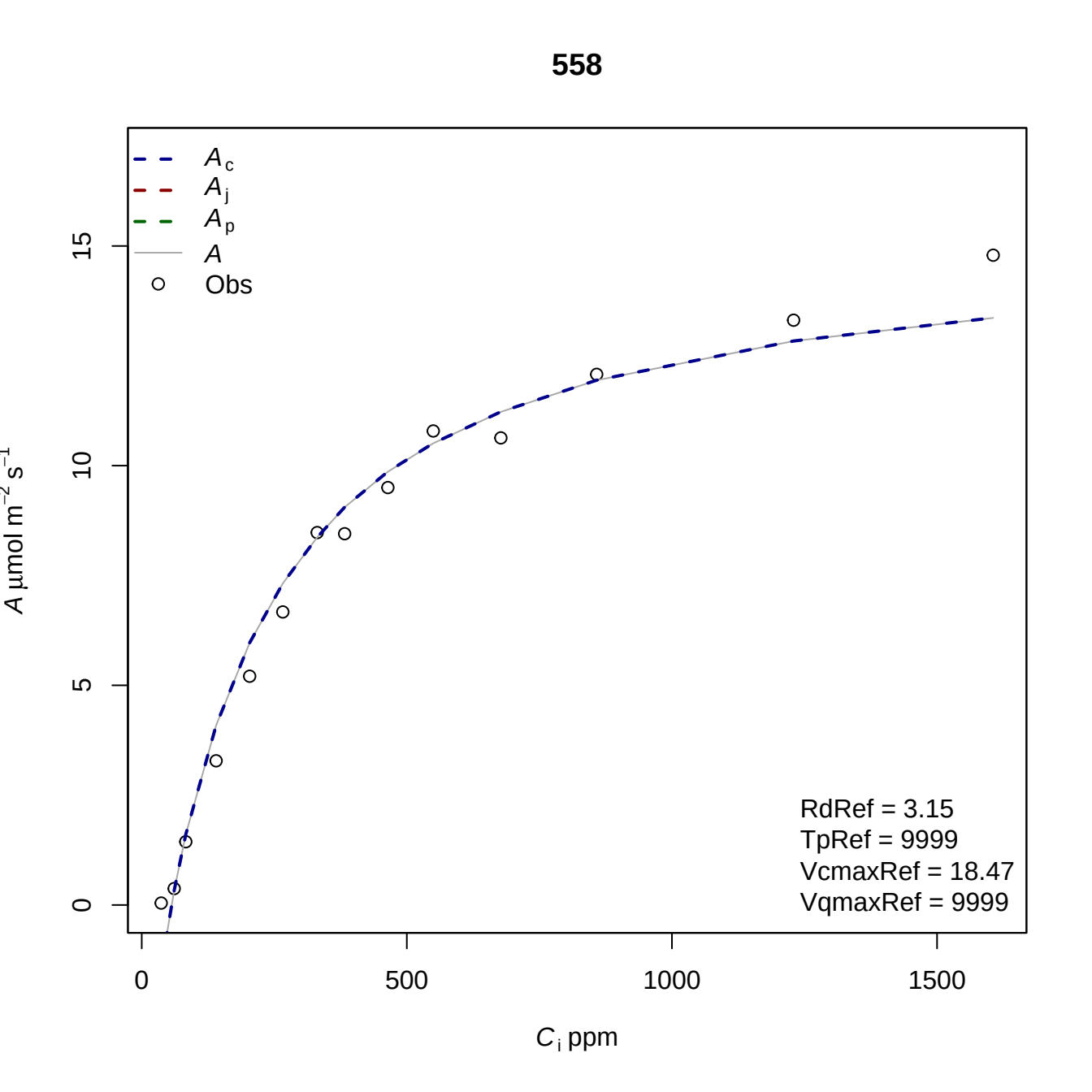

553
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.97
TpRef = 9999
VcmaxRef = 26.48
VqmaxRef = 96.87
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

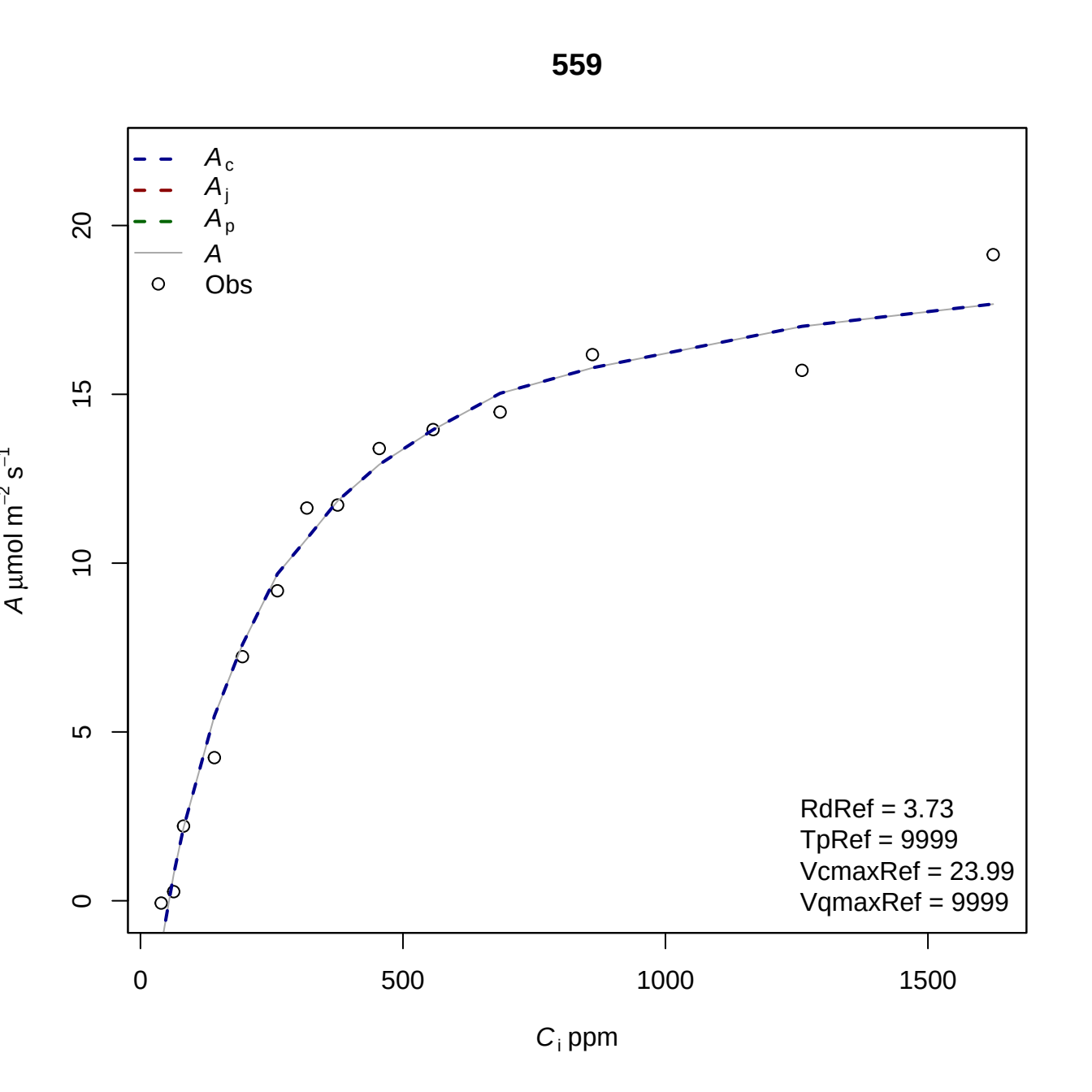

554
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.22
TpRef = 9999
VcmaxRef = 28.36
VqmaxRef = 97.38
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

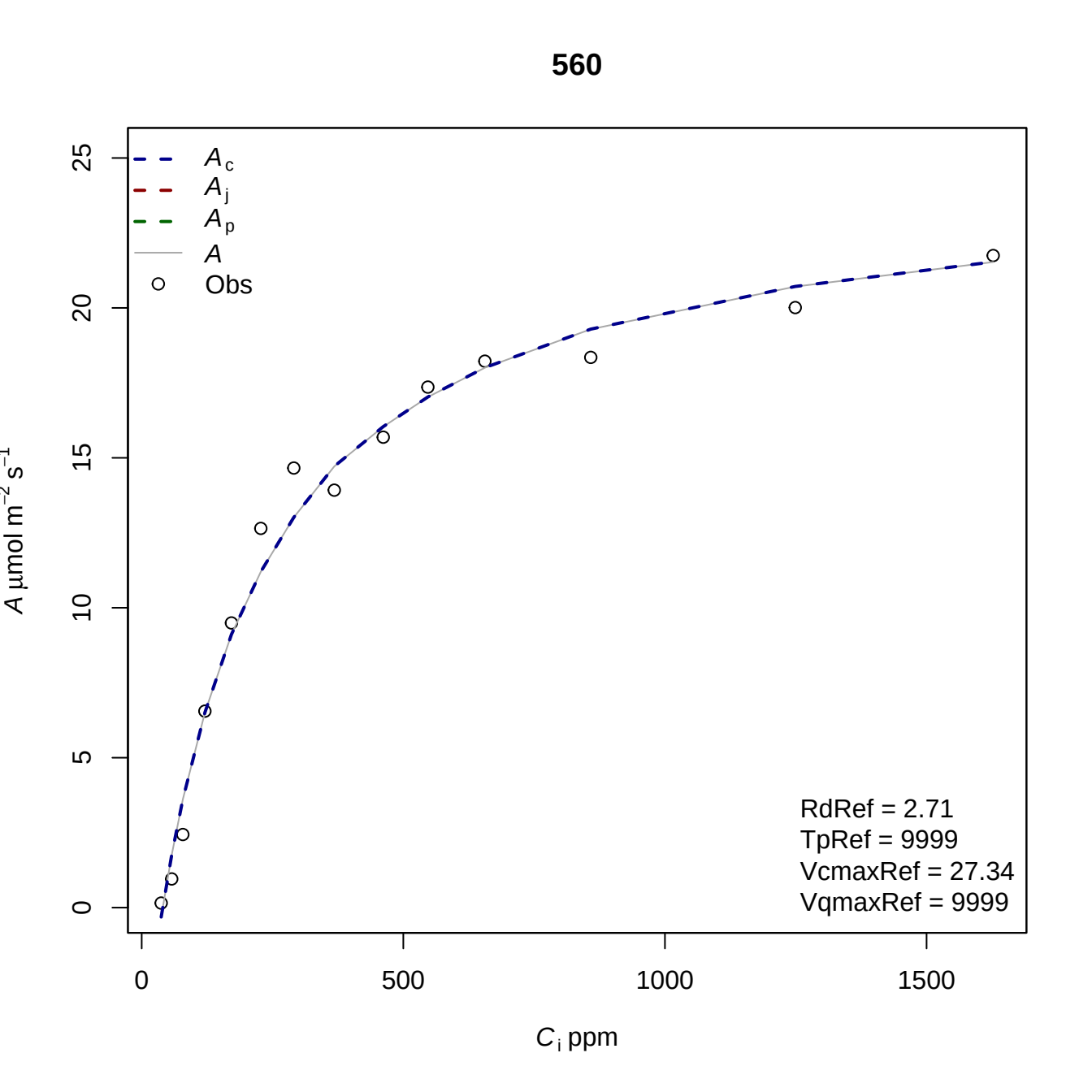

555
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.52
TpRef = 9999
VcmaxRef = 18.9
VqmaxRef = 68.26
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

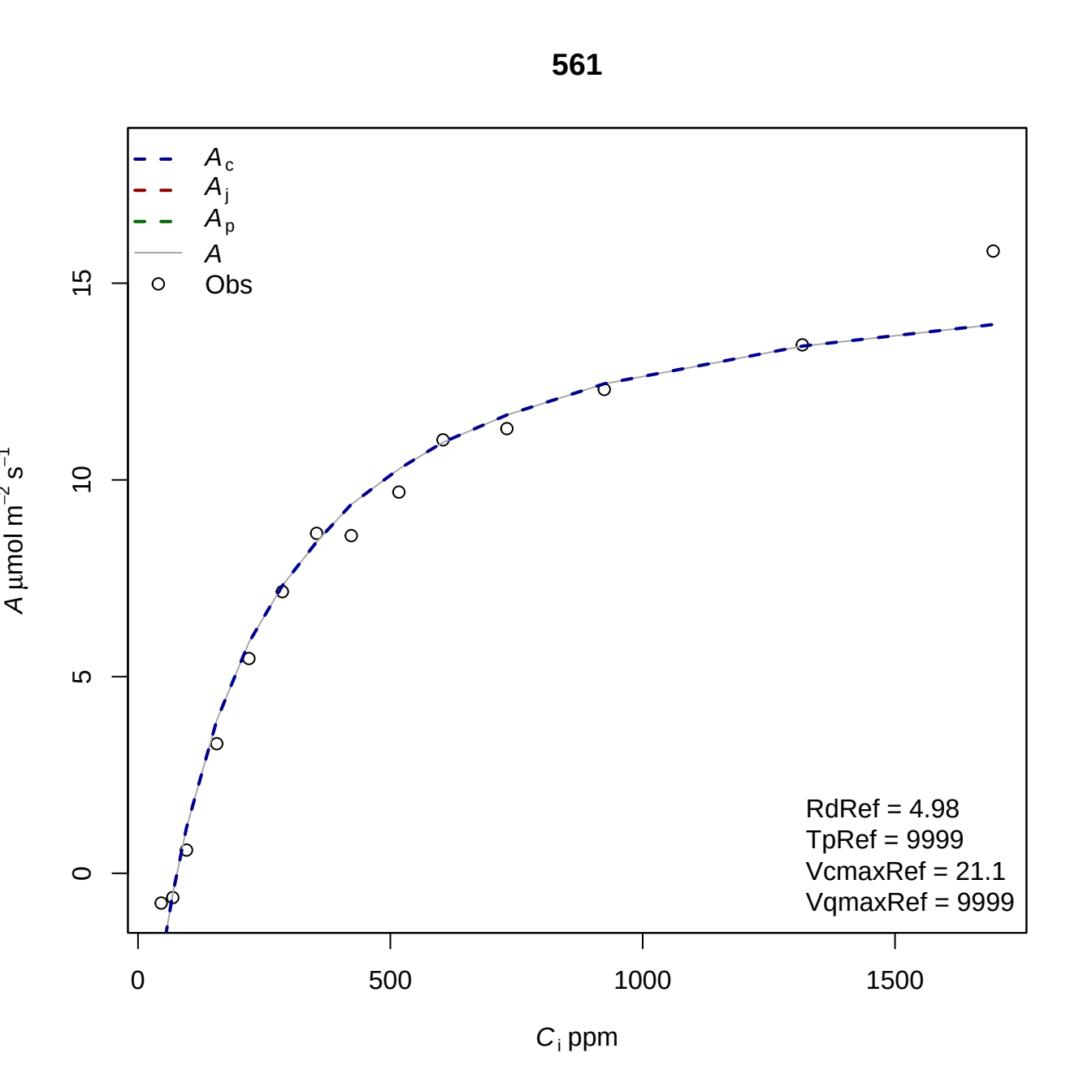

556
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.37
TpRef = 9999
VcmaxRef = 32.01
VqmaxRef = 132.02
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

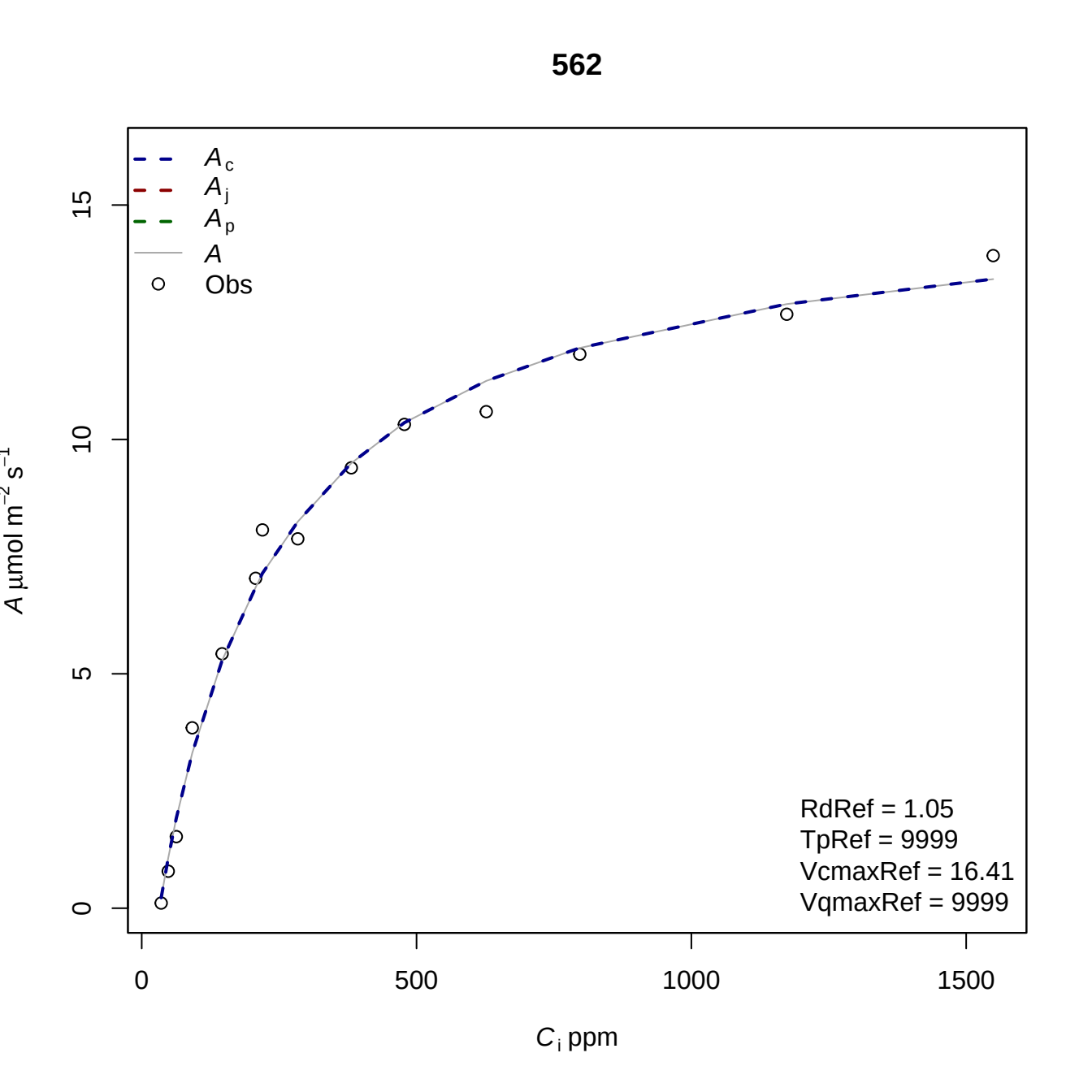

557
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.53
TpRef = 9999
VcmaxRef = 25.6
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

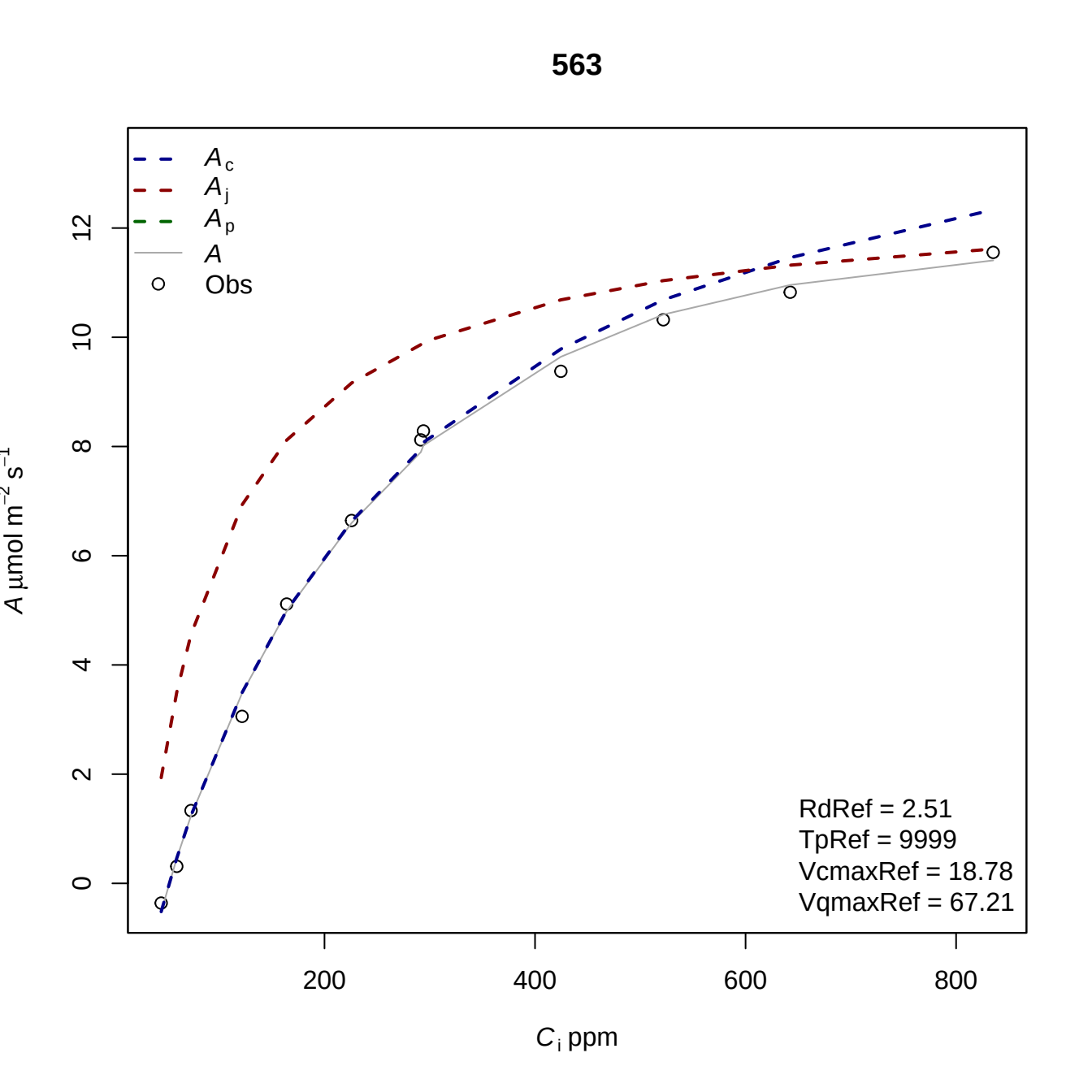

558
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.15
TpRef = 9999
VcmaxRef = 18.47
VqmaxRef = 9999
0
5
10
15
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$C_i$ ppm

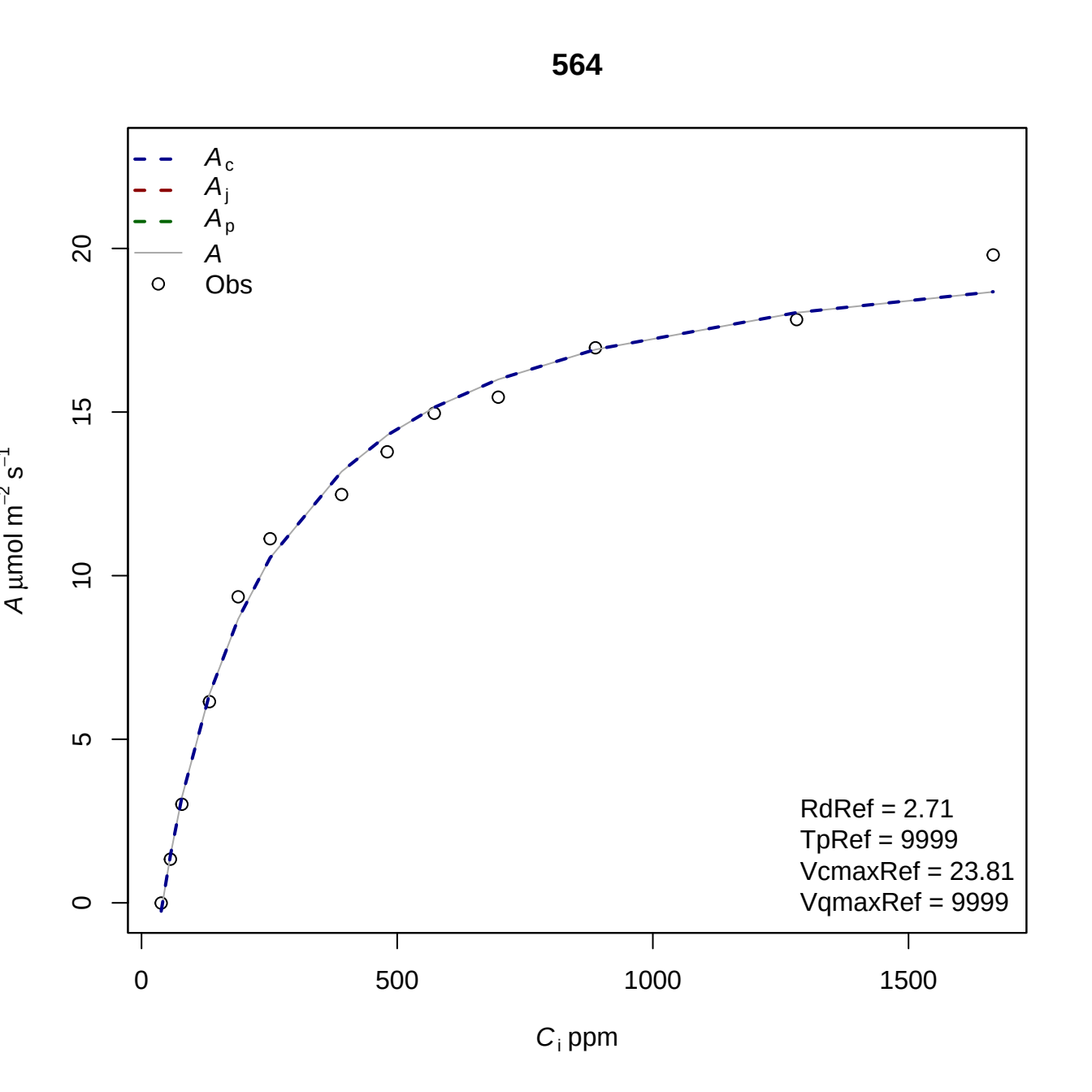

559
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.73
TpRef = 9999
VcmaxRef = 23.99
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

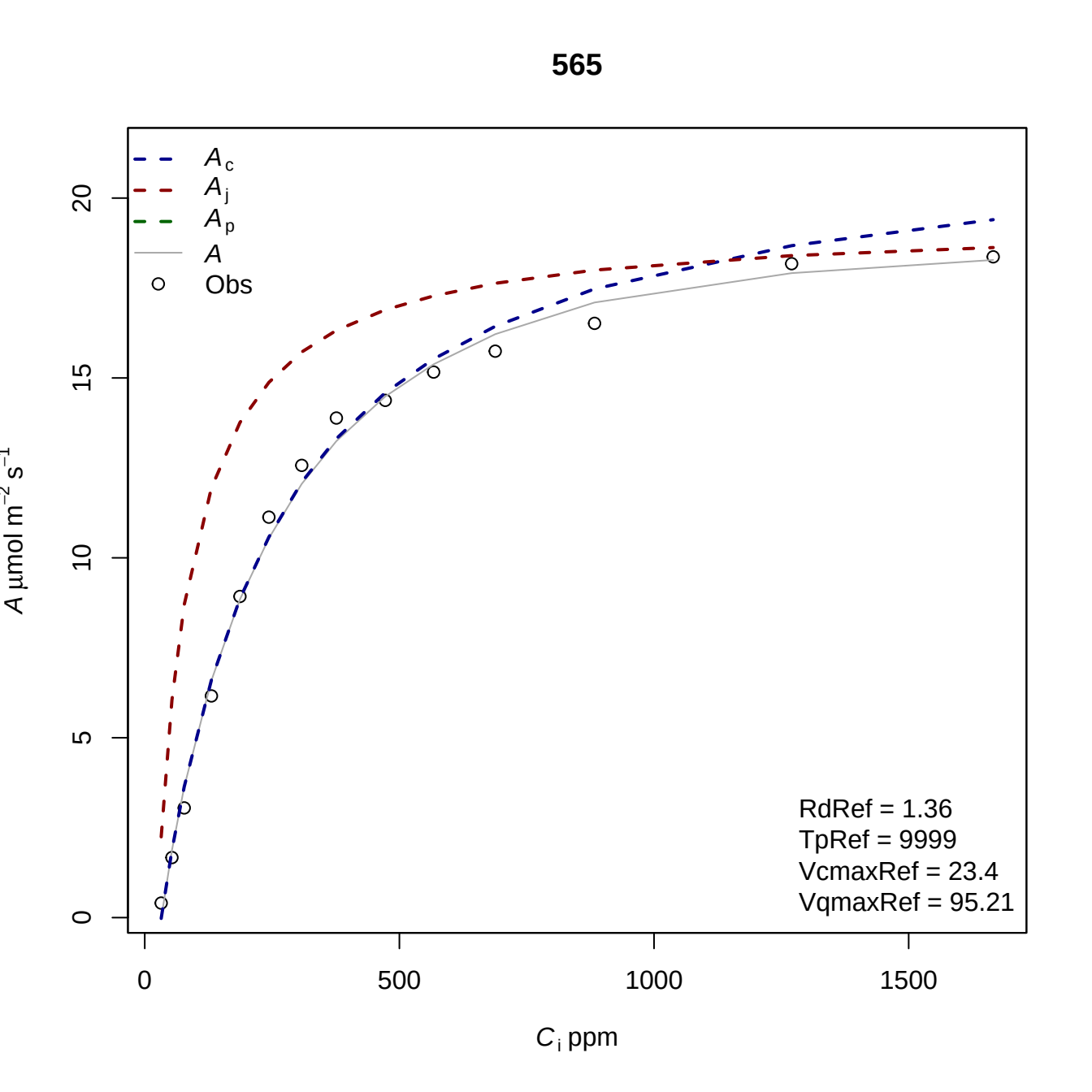

560
$A_c$
$A_j$
$A_p$
$A$
Obs
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
$C_i$ ppm
RdRef = 2.71
TpRef = 9999
VcmaxRef = 27.34
VqmaxRef = 9999

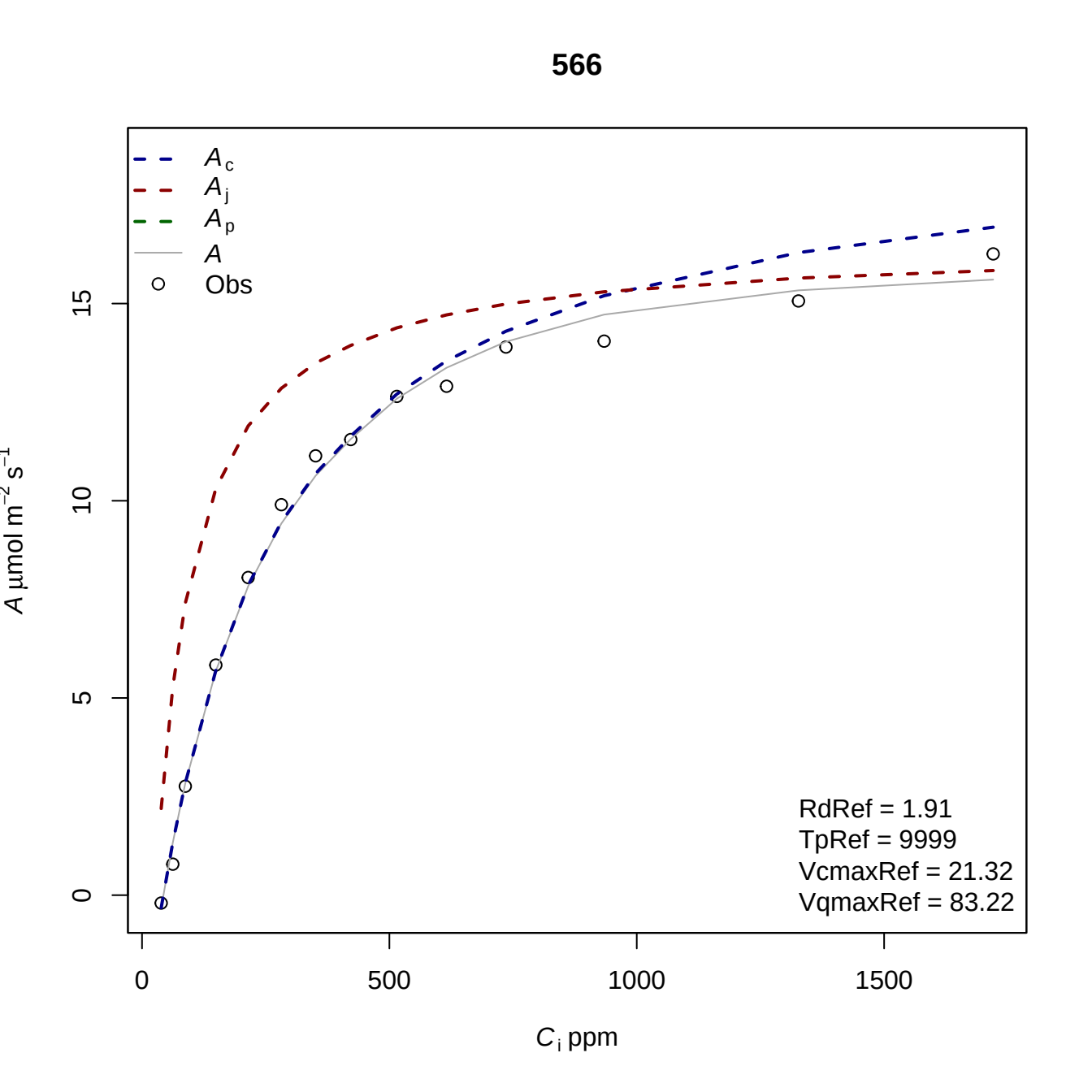

561
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.98
TpRef = 9999
VcmaxRef = 21.1
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

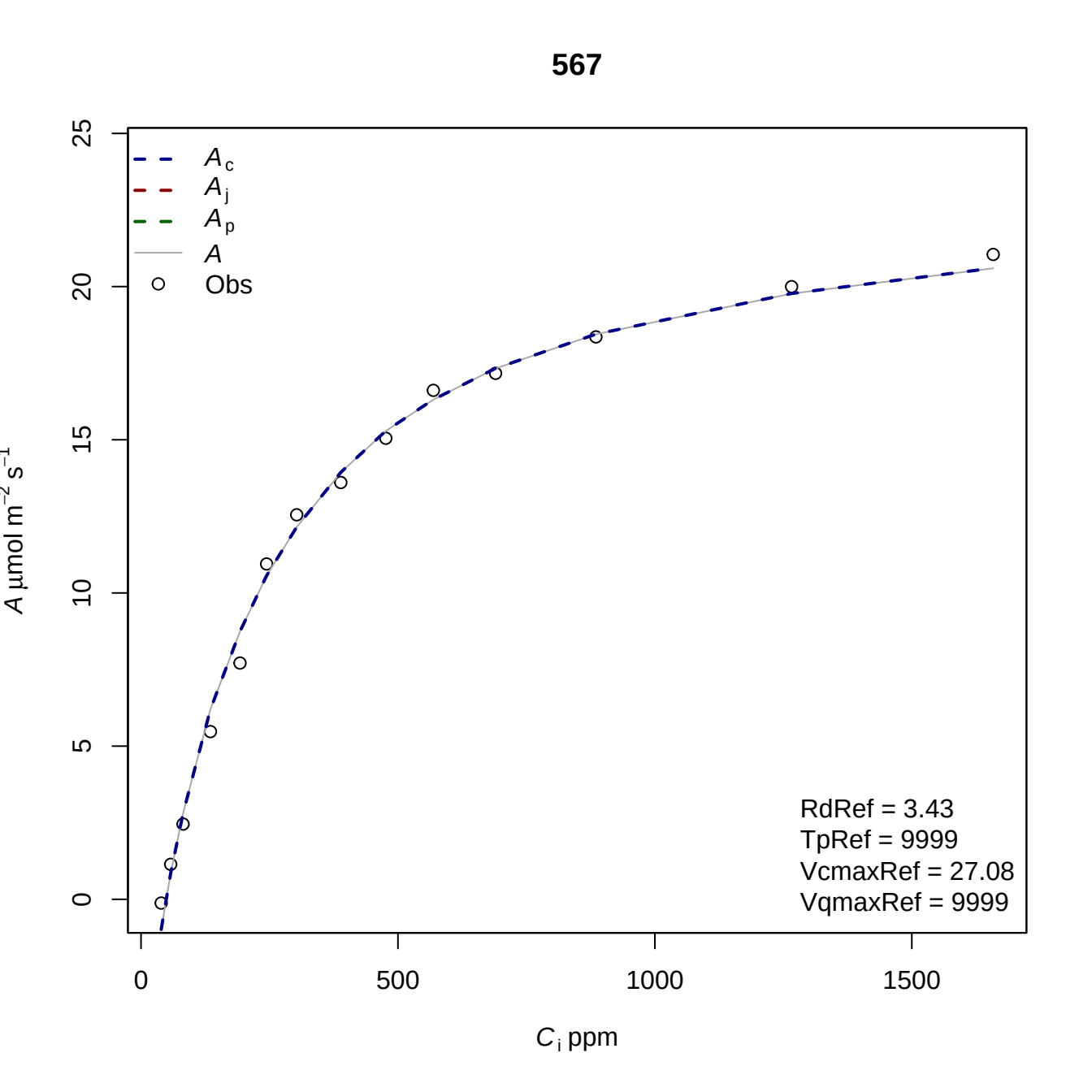

562
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.05
TpRef = 9999
VcmaxRef = 16.41
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

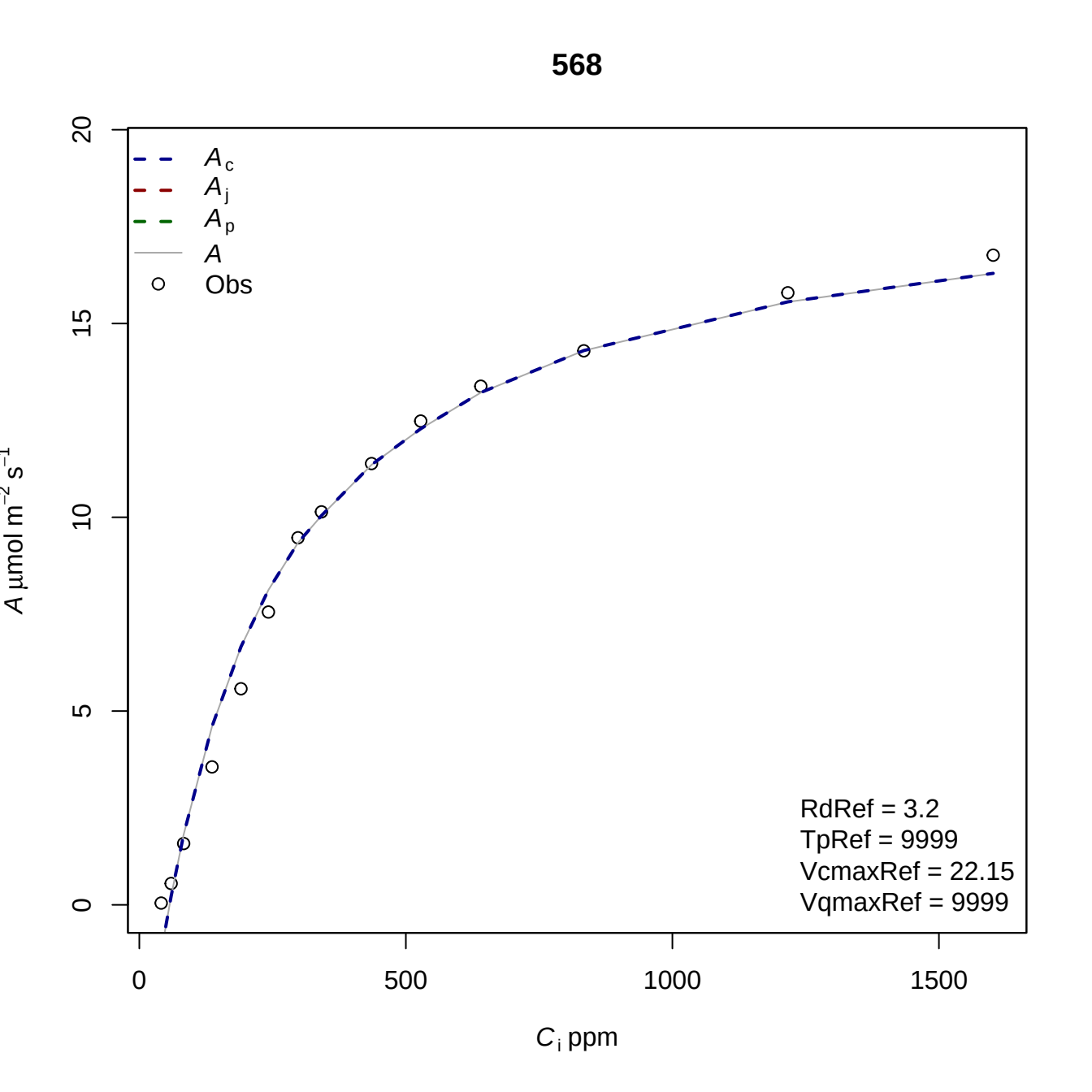

563
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.51
TpRef = 9999
VcmaxRef = 18.78
VqmaxRef = 67.21
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800

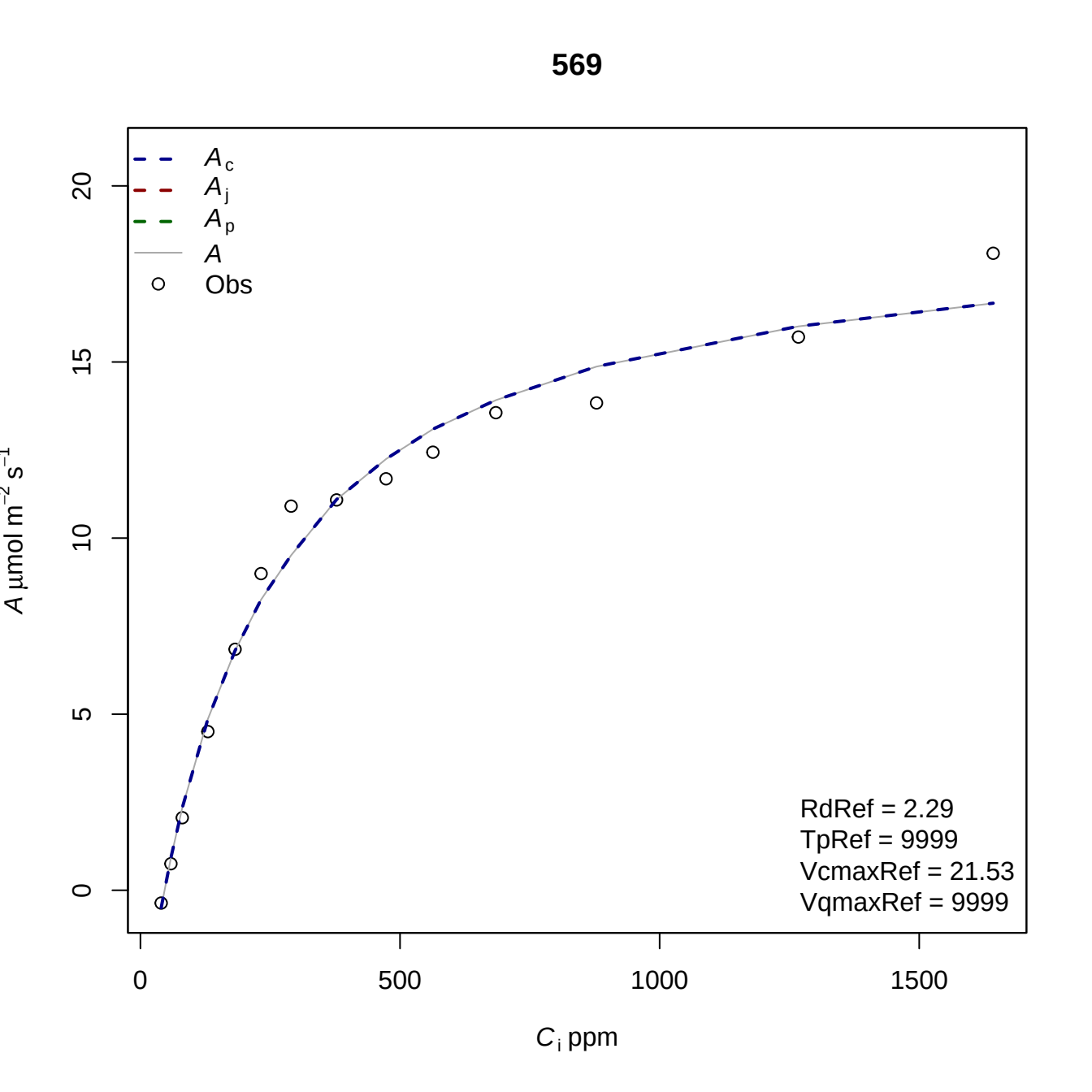

564
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.71
TpRef = 9999
VcmaxRef = 23.81
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

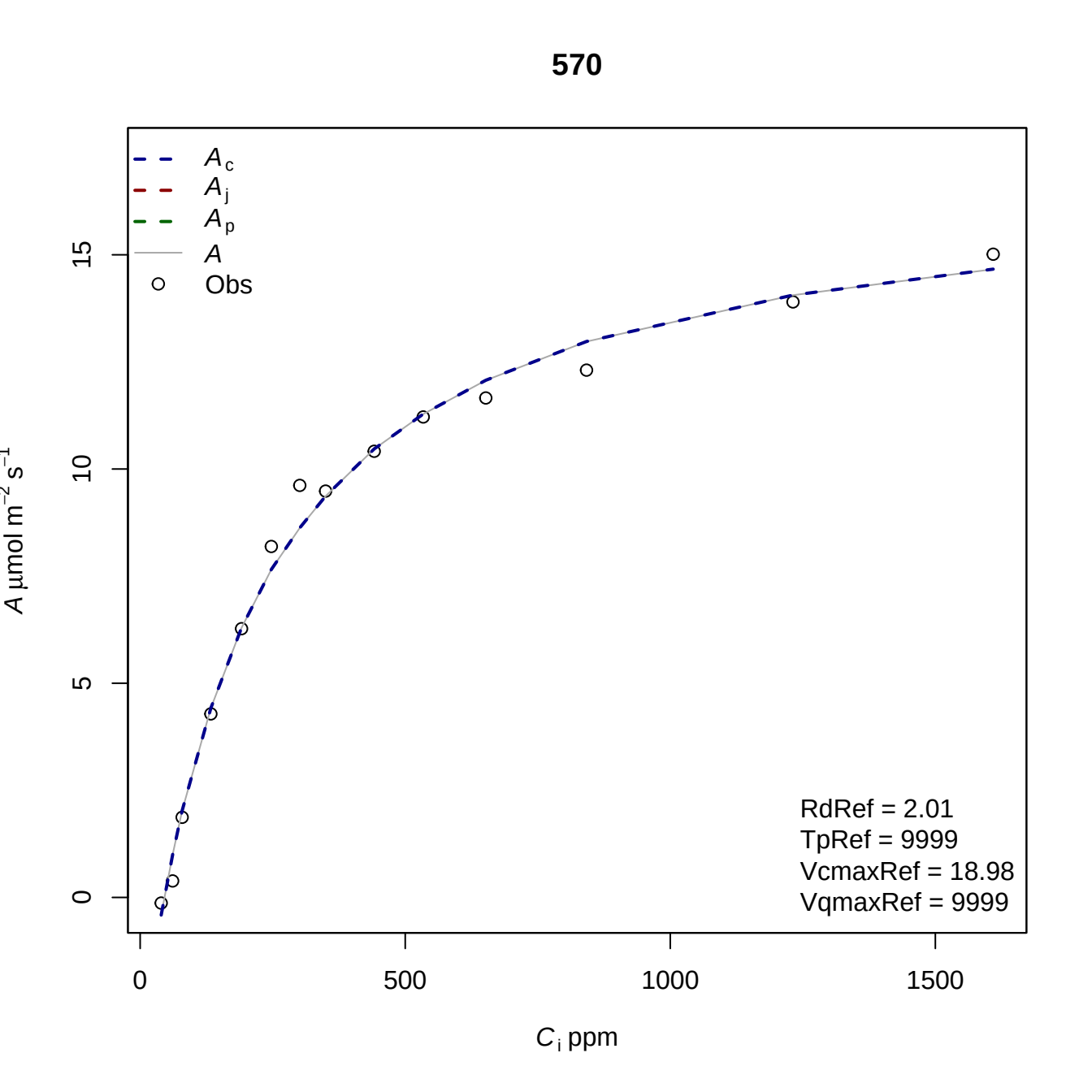

565
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.36
TpRef = 9999
VcmaxRef = 23.4
VqmaxRef = 95.21
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

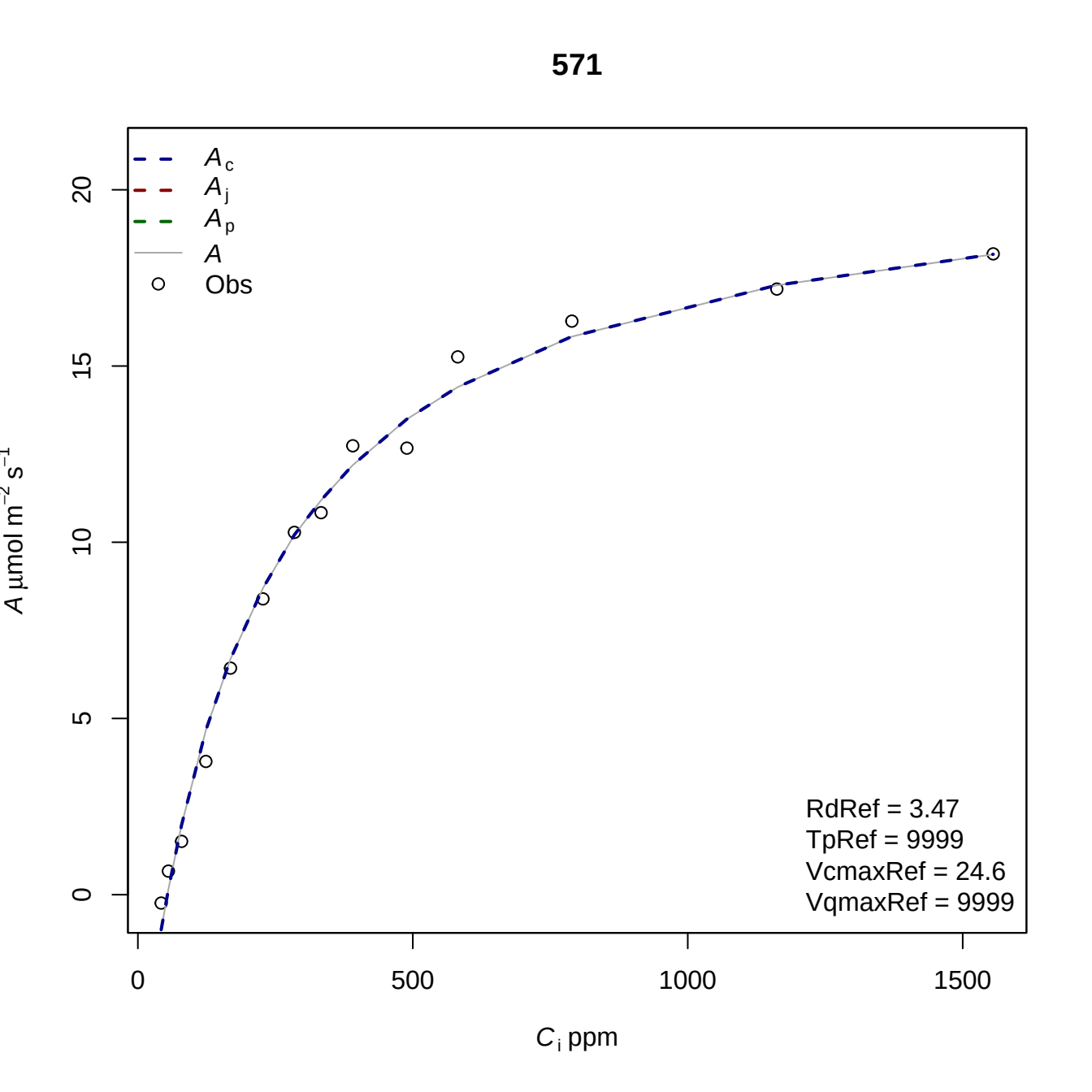

566
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.91
TpRef = 9999
VcmaxRef = 21.32
VqmaxRef = 83.22
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

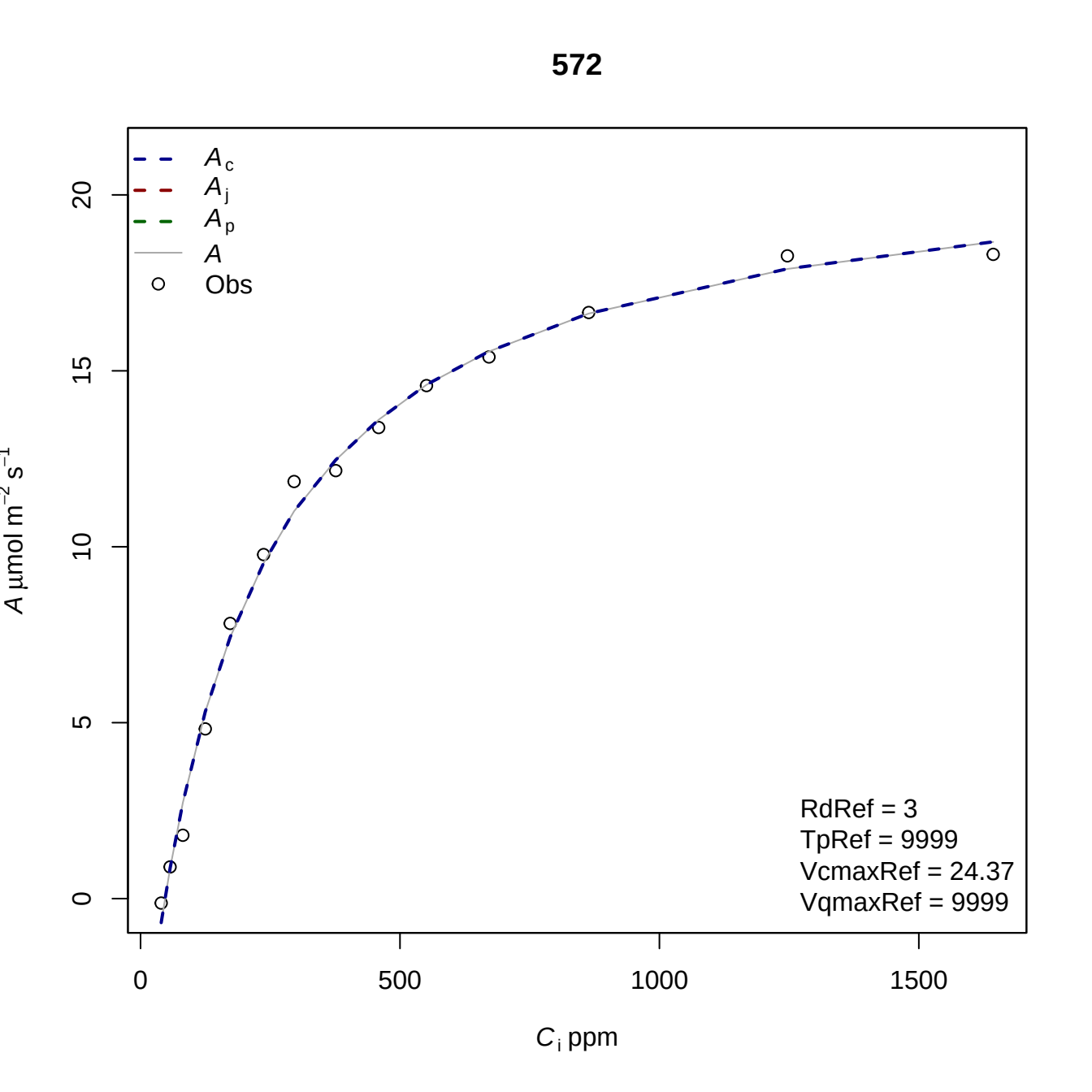

567
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.43
TpRef = 9999
VcmaxRef = 27.08
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

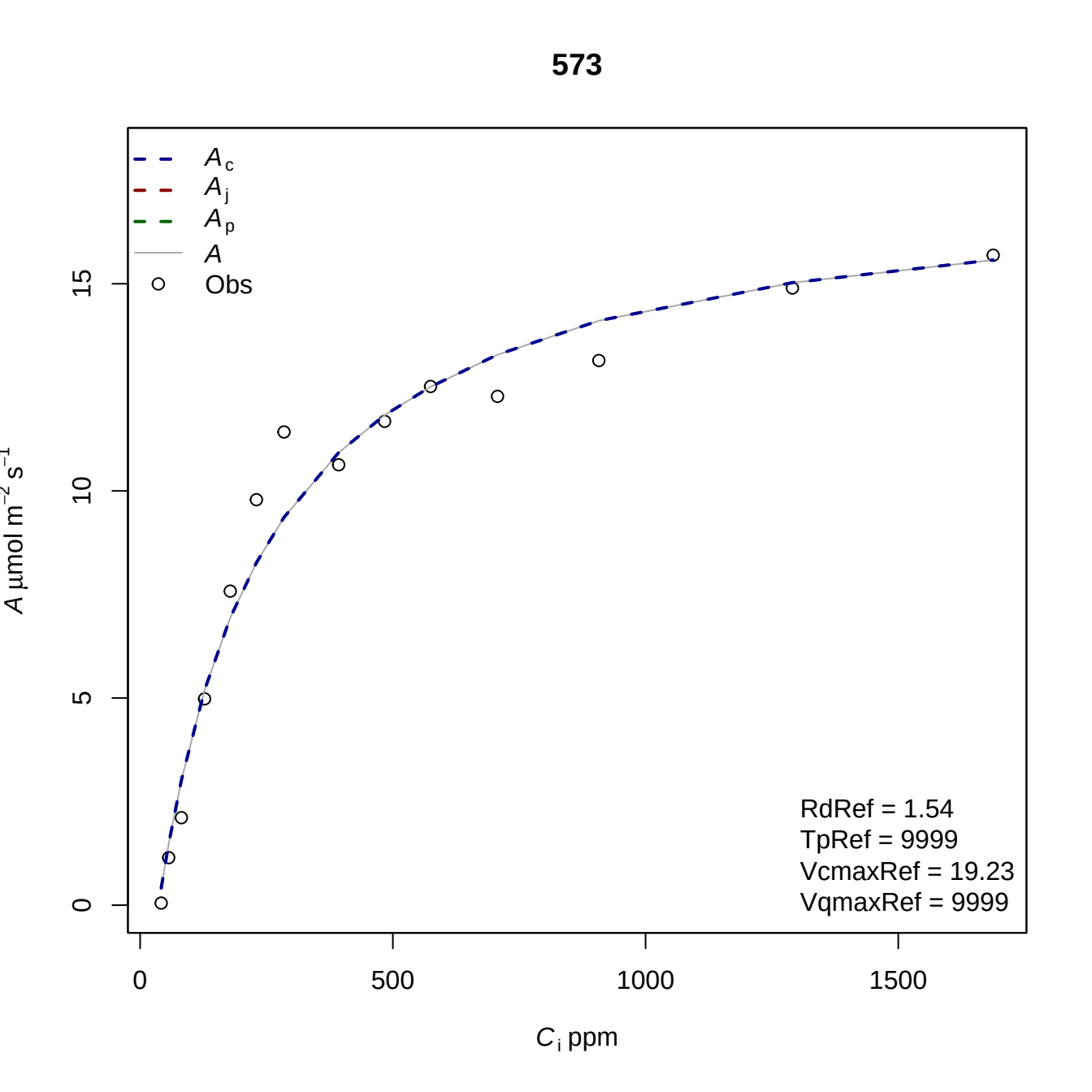

568
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.2
TpRef = 9999
VcmaxRef = 22.15
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

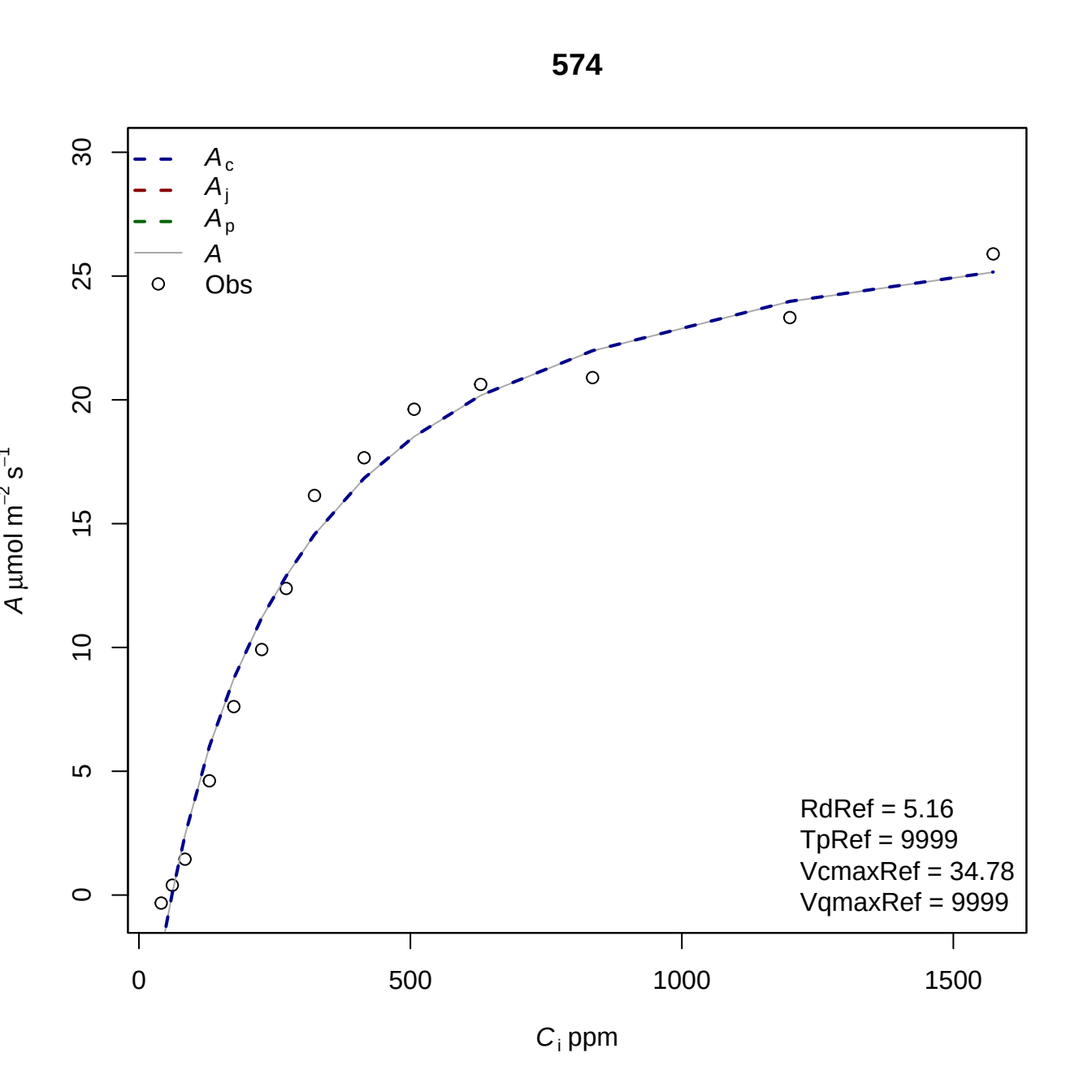

569
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.29
TpRef = 9999
VcmaxRef = 21.53
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

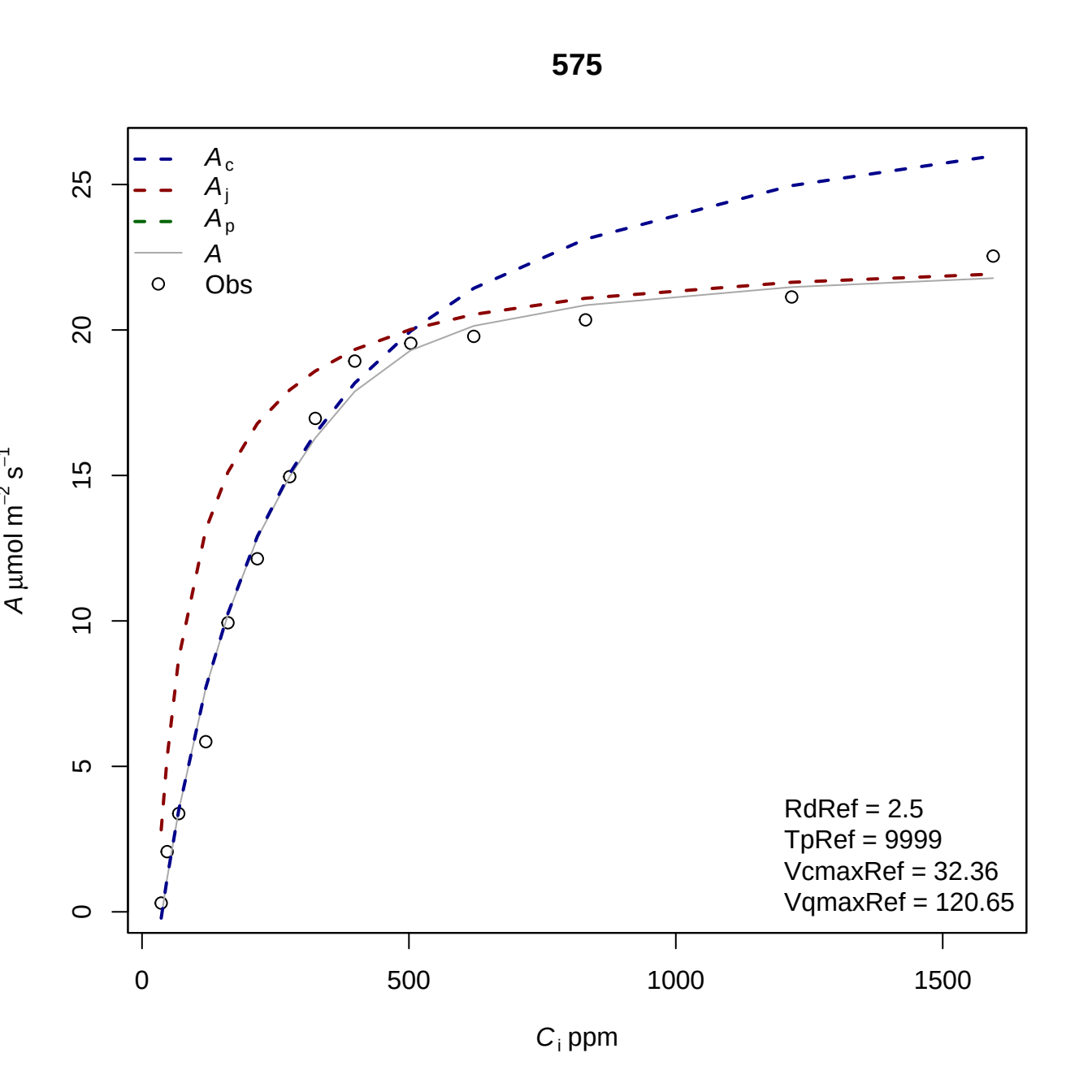

570
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.01
TpRef = 9999
VcmaxRef = 18.98
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

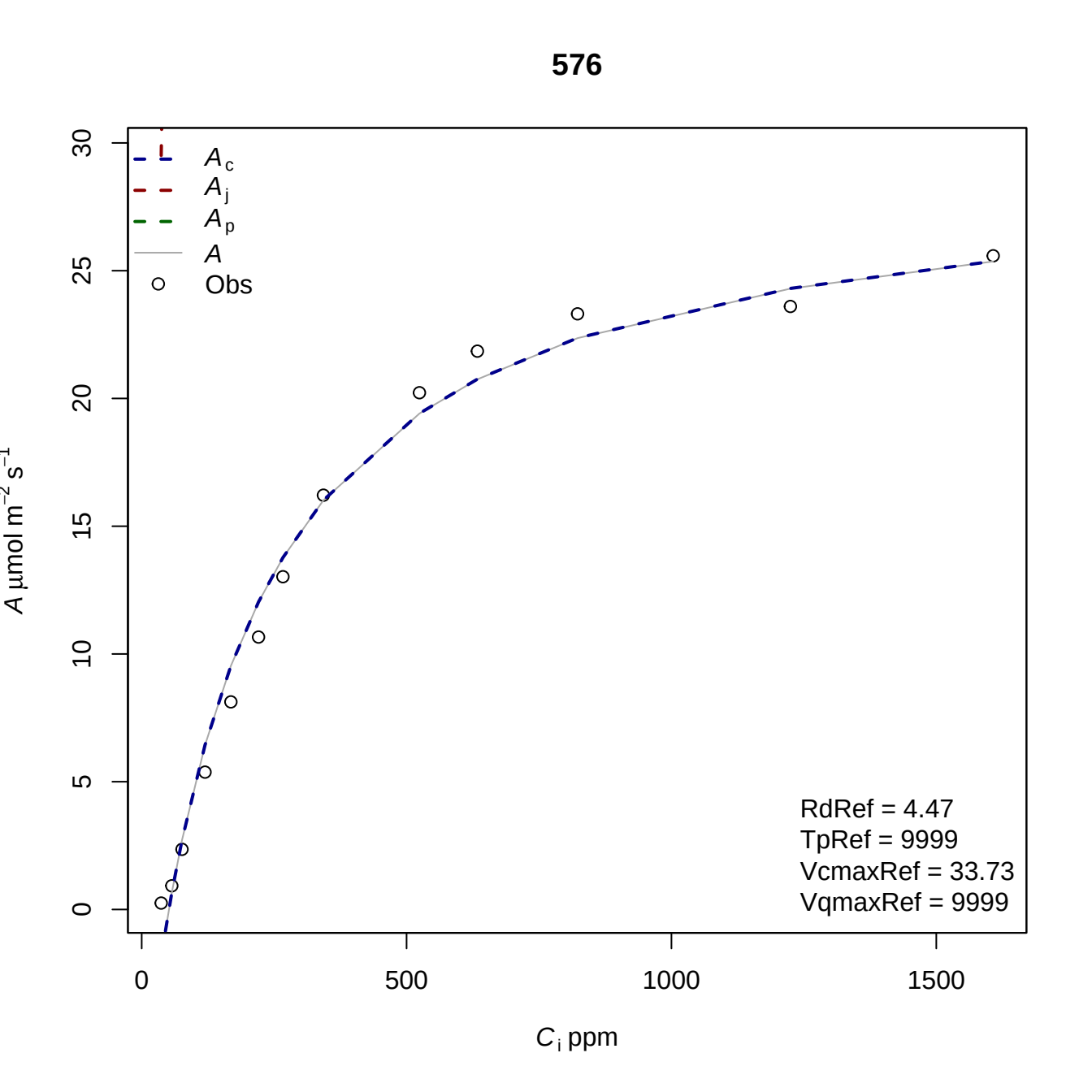

571
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.47
TpRef = 9999
VcmaxRef = 24.6
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

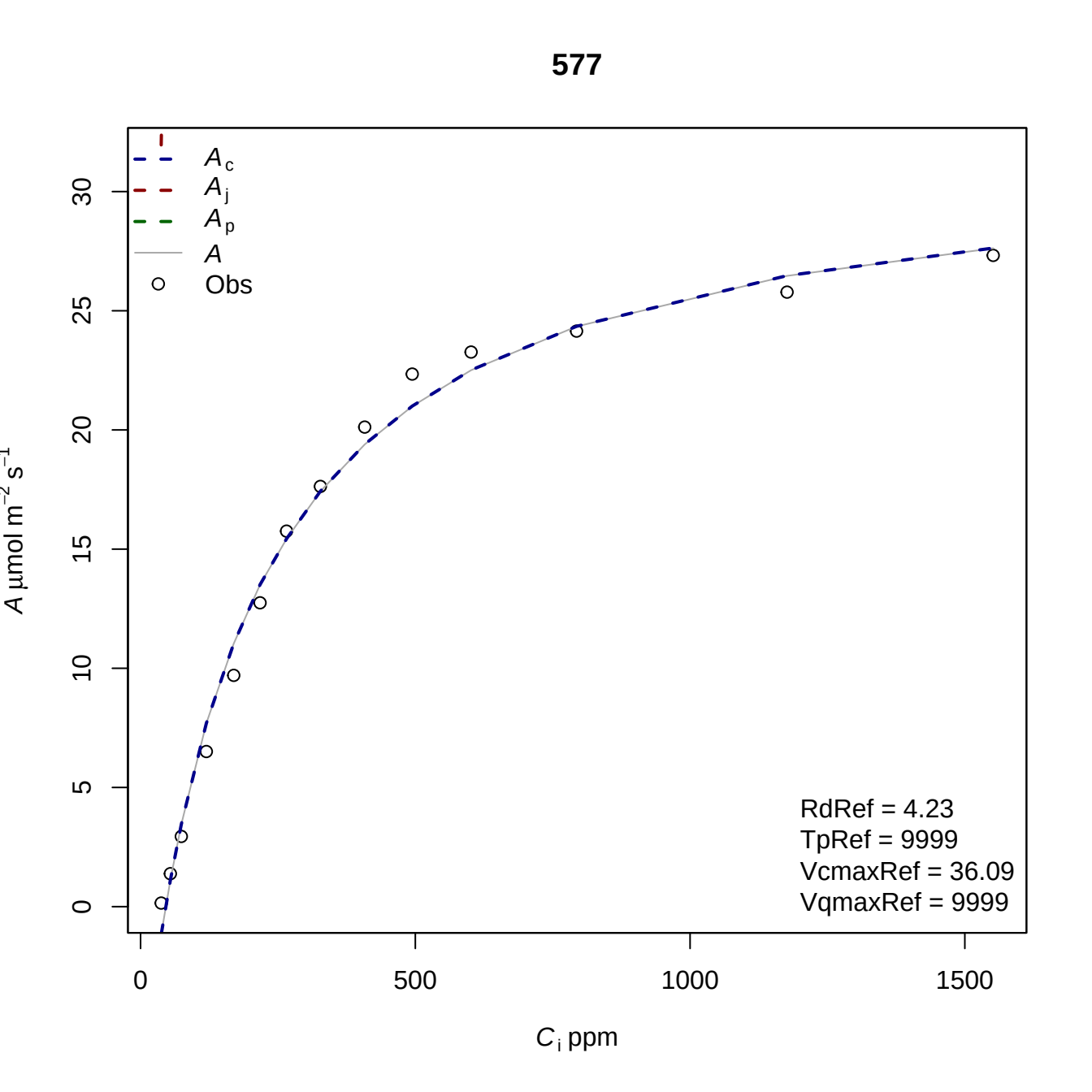

572
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3
TpRef = 9999
VcmaxRef = 24.37
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

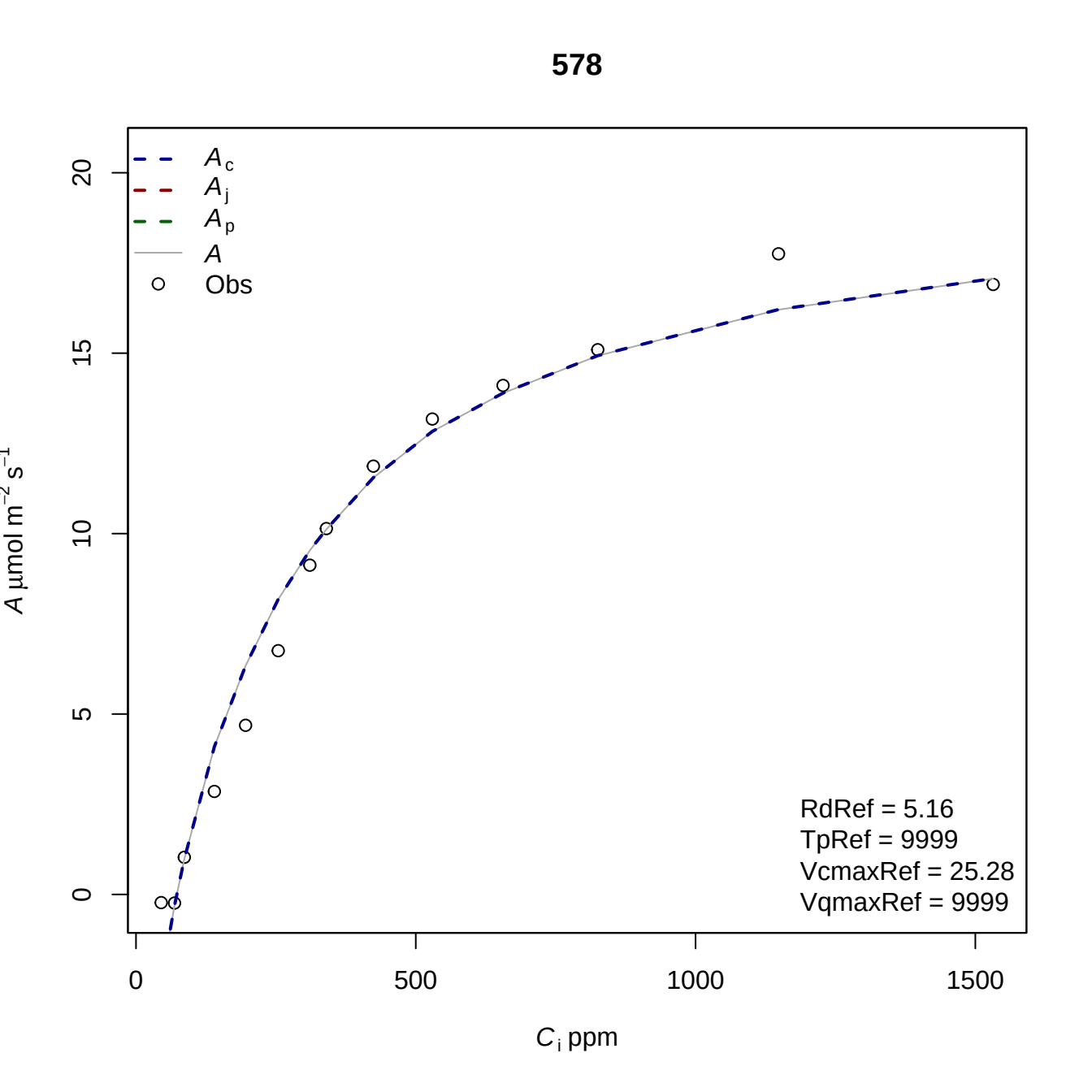

573
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.54
TpRef = 9999
VcmaxRef = 19.23
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

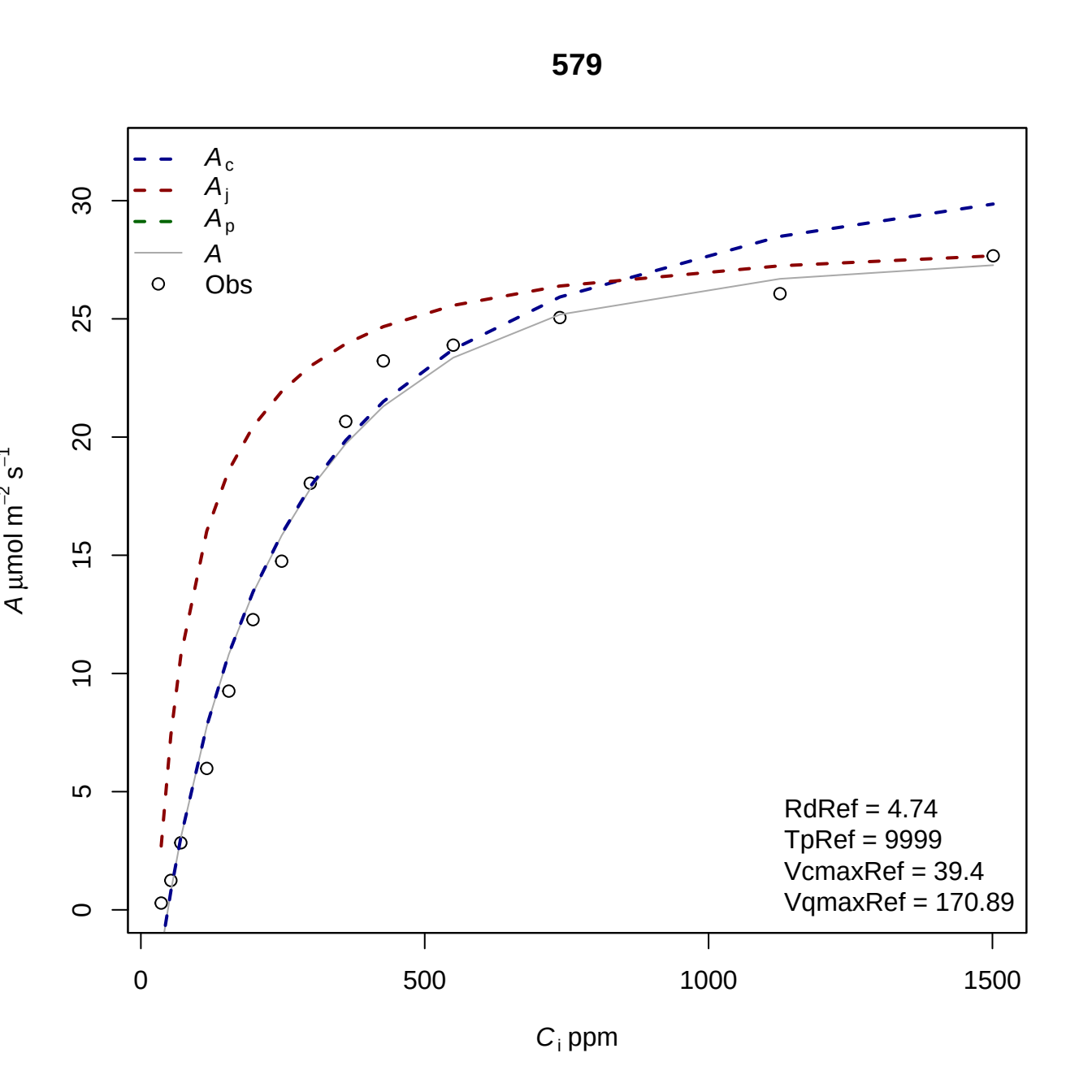

574
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 5.16
TpRef = 9999
VcmaxRef = 34.78
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

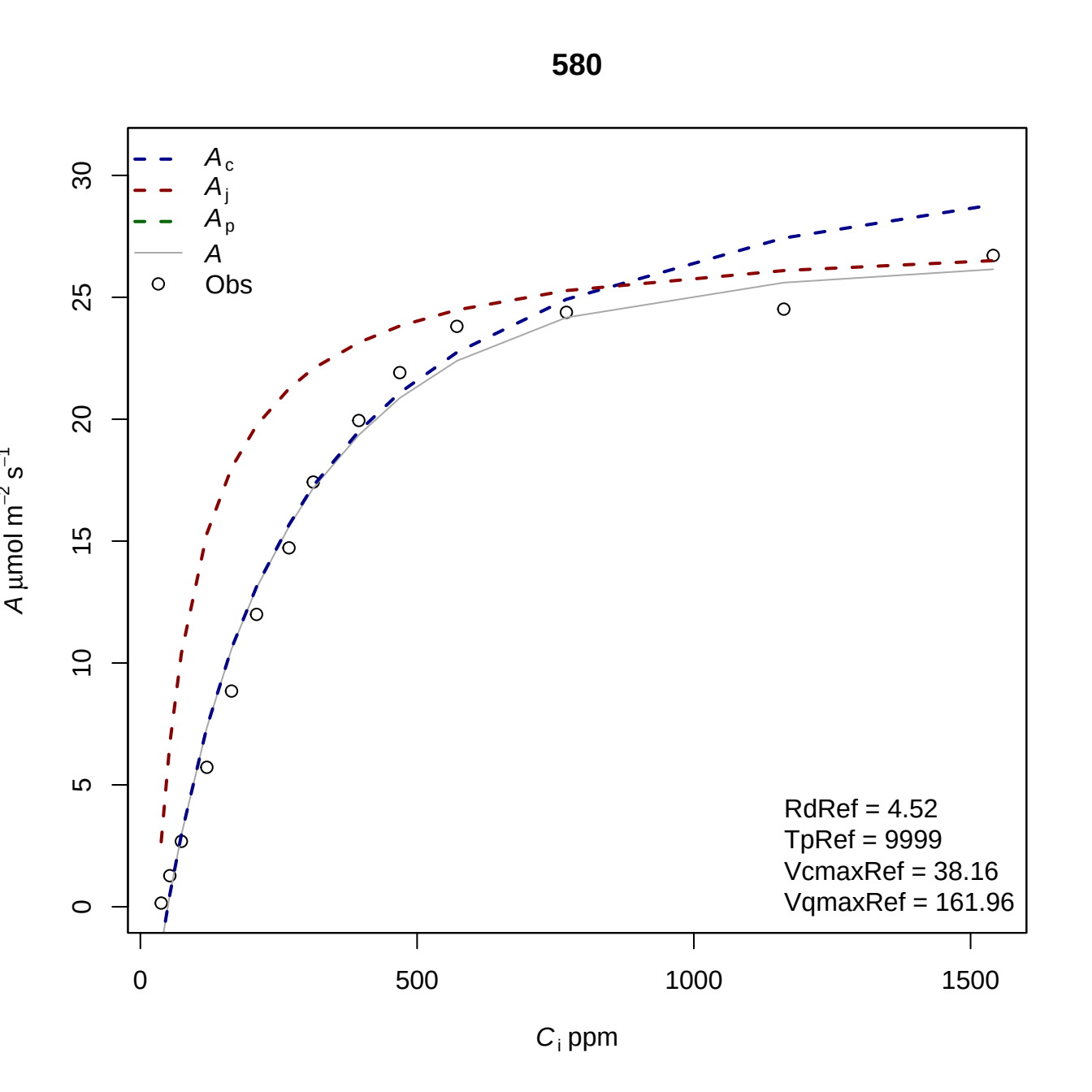

575
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.5
TpRef = 9999
VcmaxRef = 32.36
VqmaxRef = 120.65
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

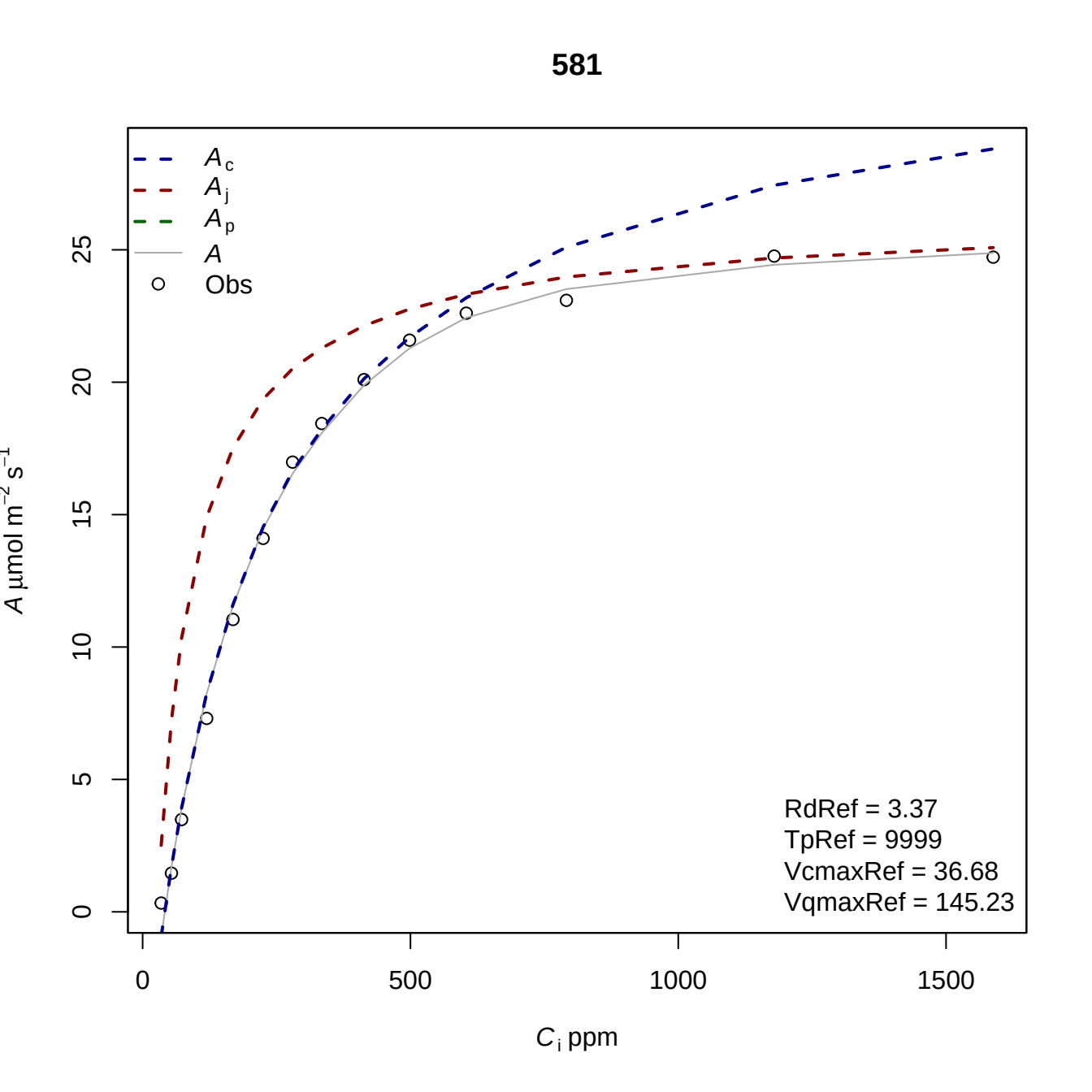

576
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.47
TpRef = 9999
VcmaxRef = 33.73
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

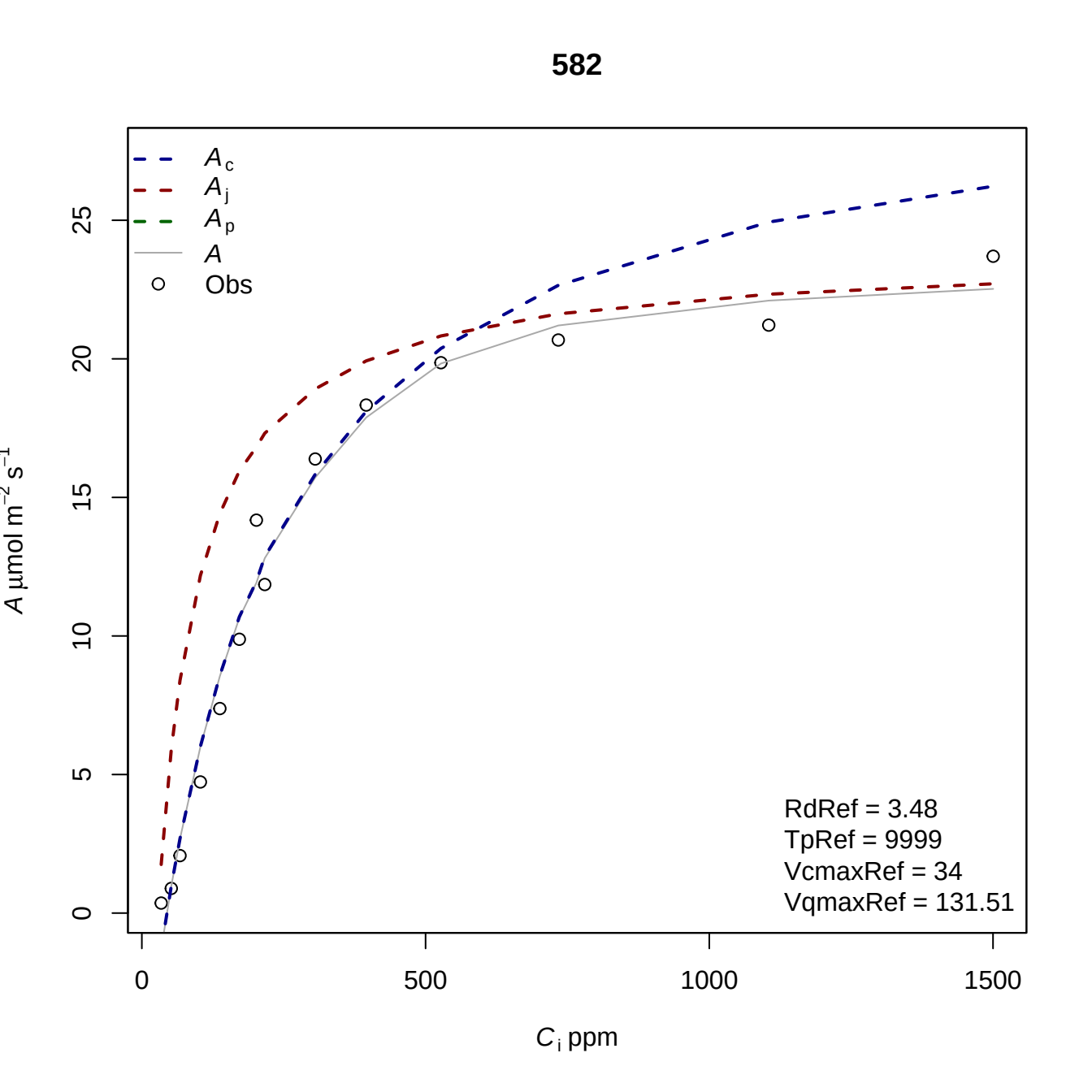

577
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.23
TpRef = 9999
VcmaxRef = 36.09
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

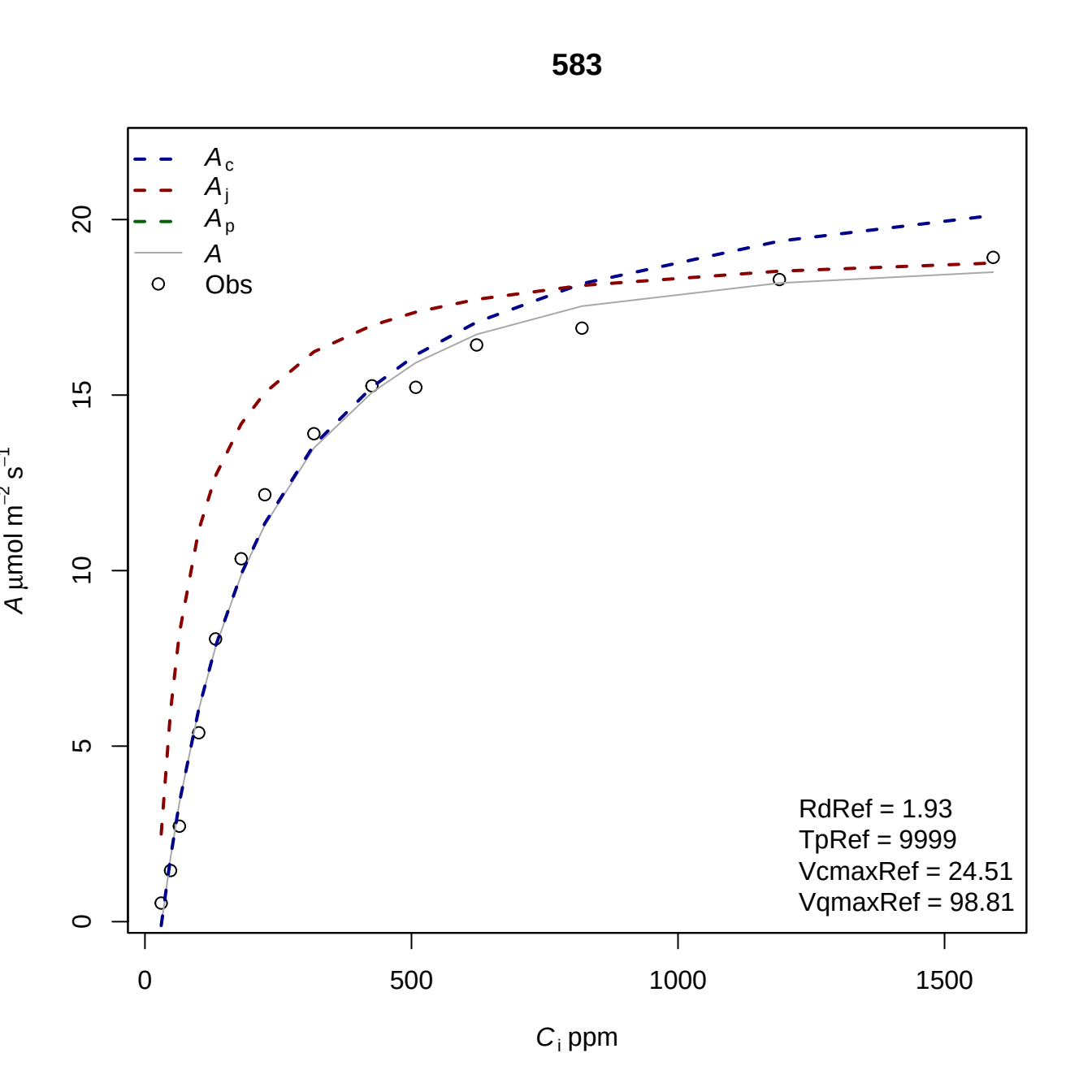

578
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 5.16
TpRef = 9999
VcmaxRef = 25.28
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

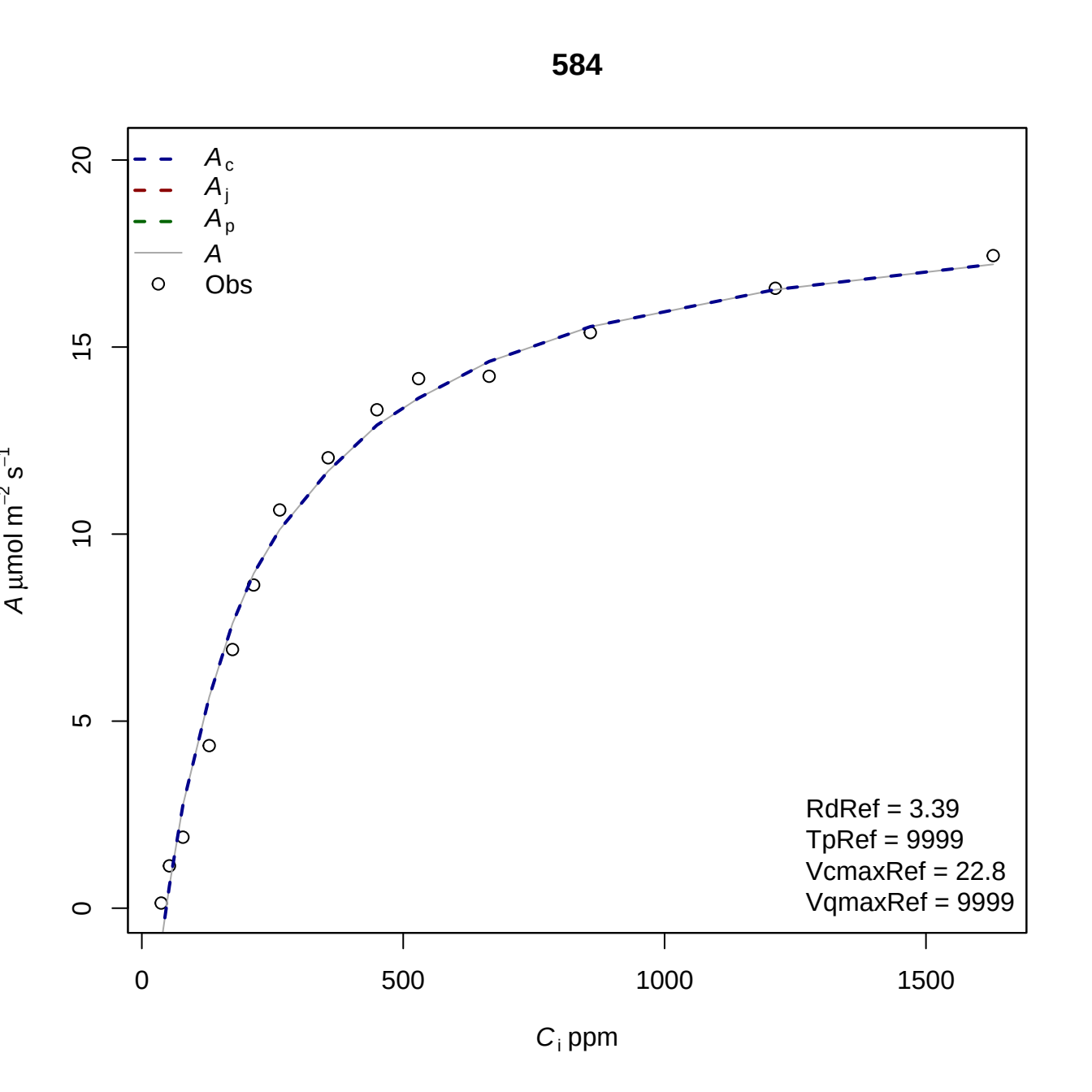

579
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.74
TpRef = 9999
VcmaxRef = 39.4
VqmaxRef = 170.89
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

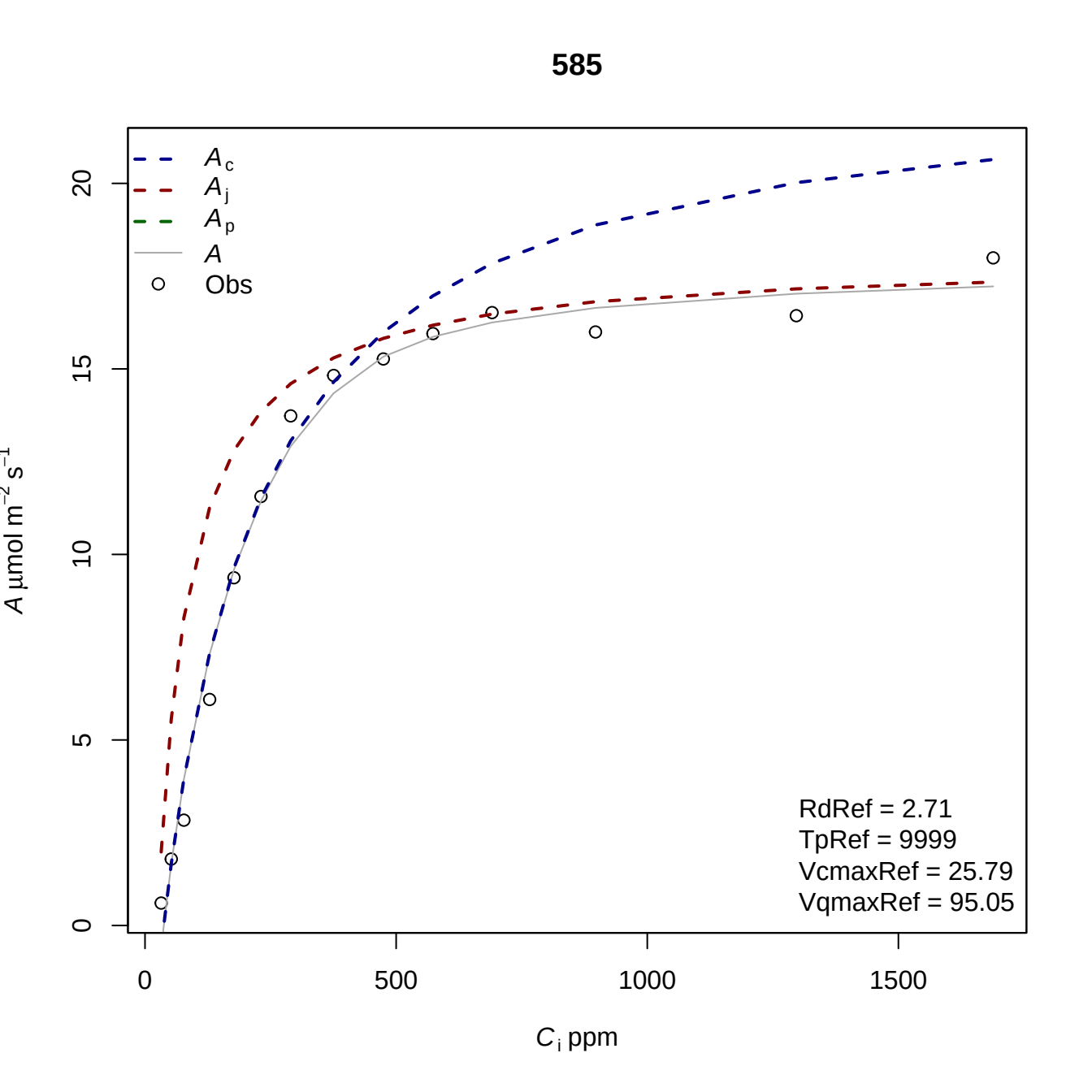

580
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 4.52
TpRef = 9999
VcmaxRef = 38.16
VqmaxRef = 161.96
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
500
1000
1500

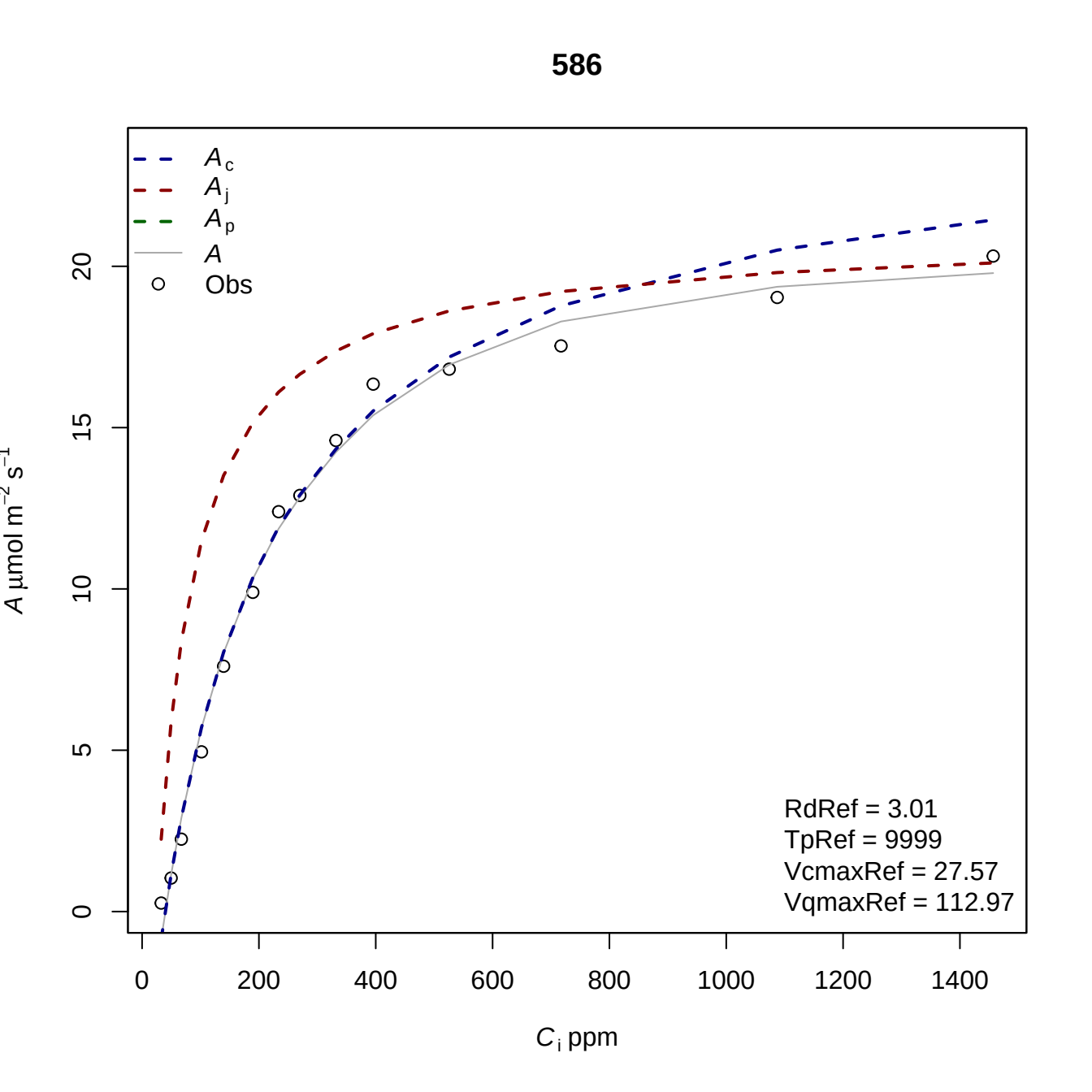

581
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.37
TpRef = 9999
VcmaxRef = 36.68
VqmaxRef = 145.23
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

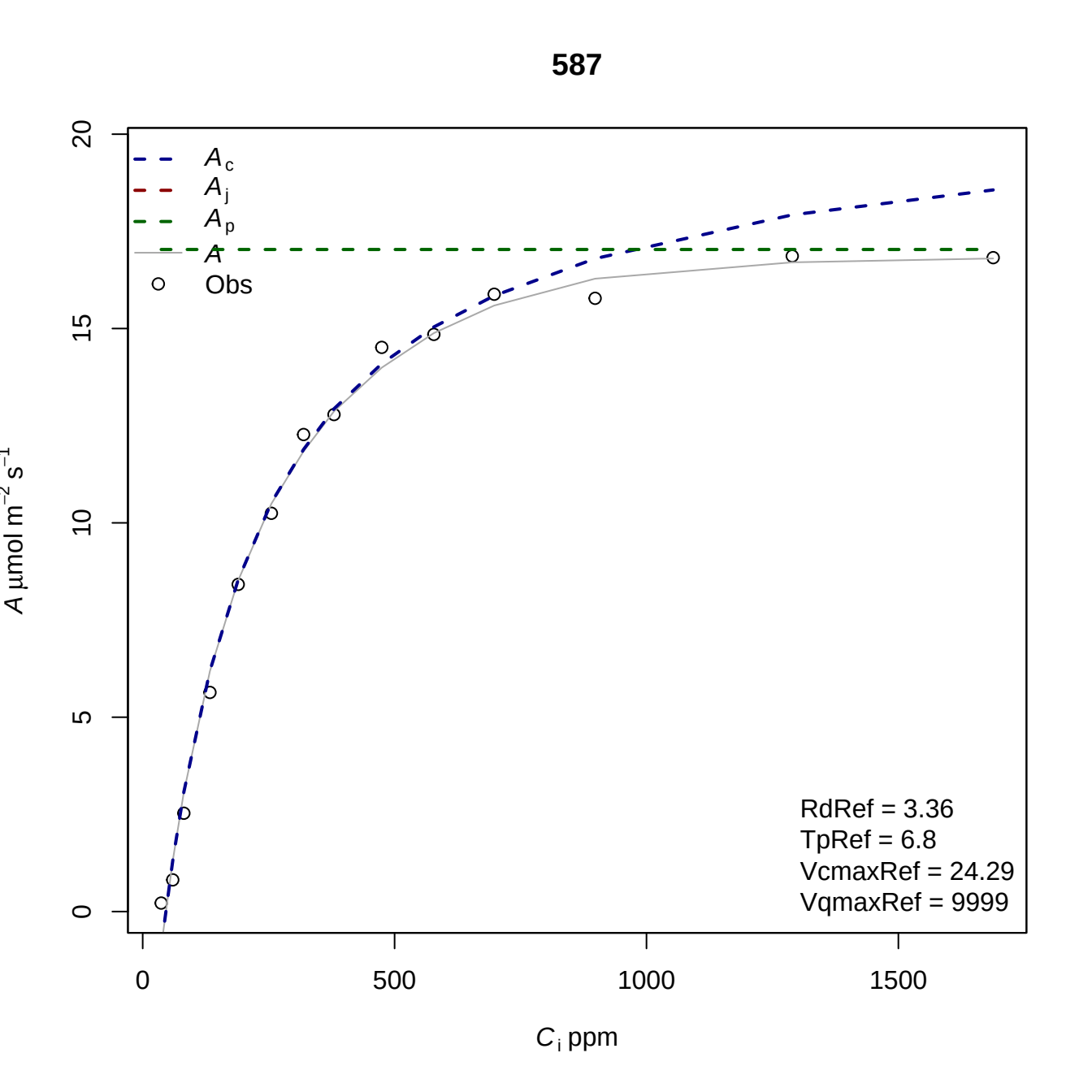

582
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.48
TpRef = 9999
VcmaxRef = 34
VqmaxRef = 131.51
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

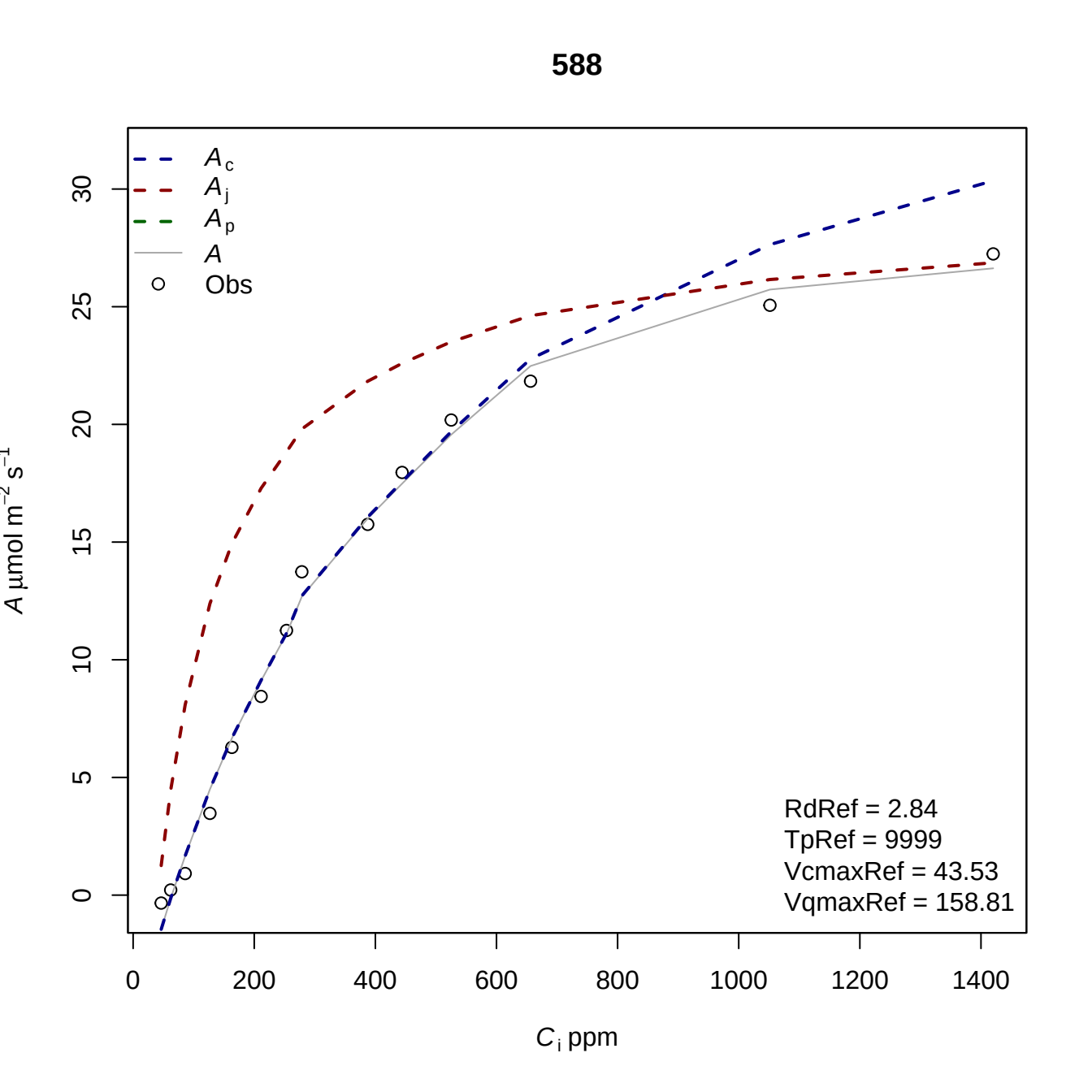

583
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.93
TpRef = 9999
VcmaxRef = 24.51
VqmaxRef = 98.81
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

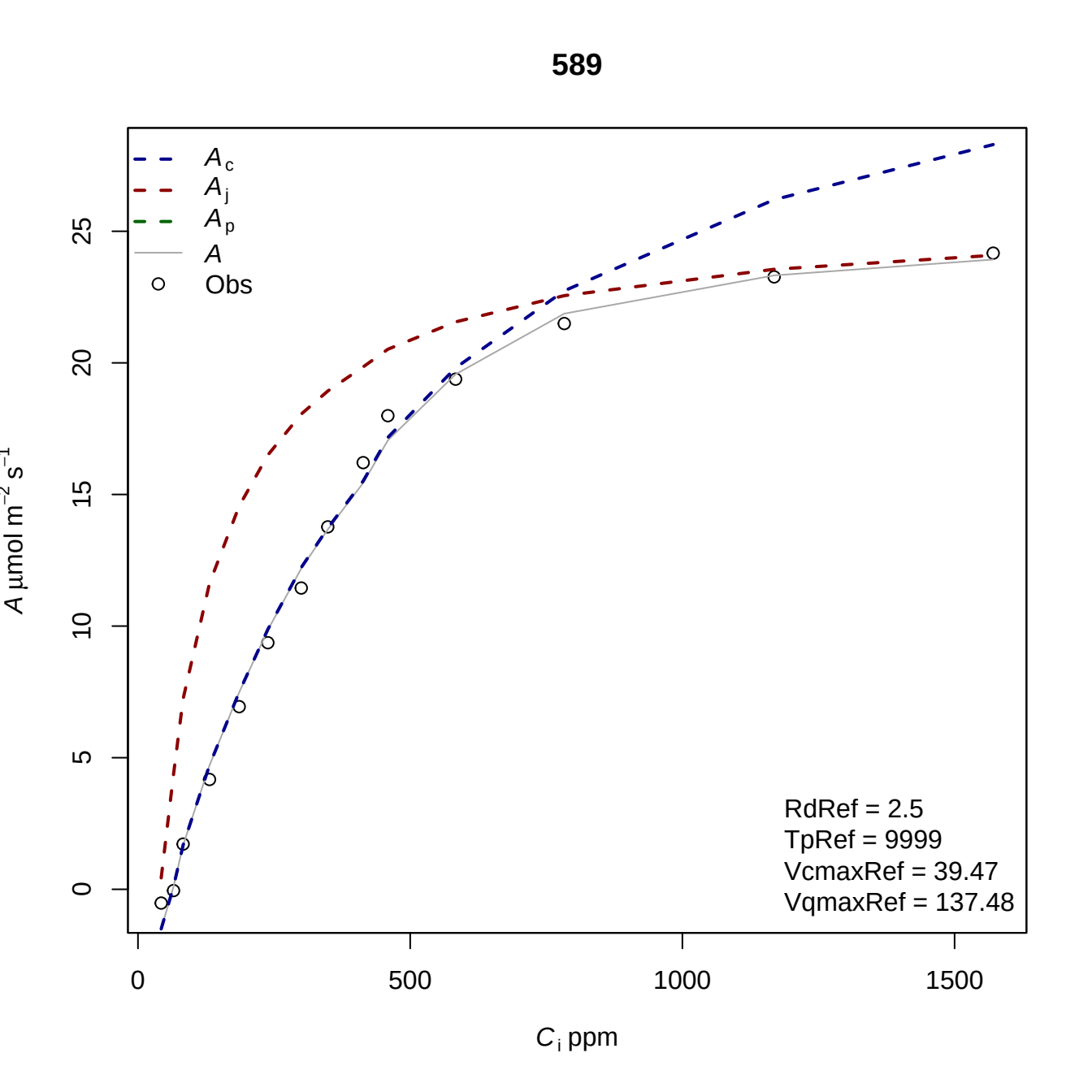

584
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.39
TpRef = 9999
VcmaxRef = 22.8
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

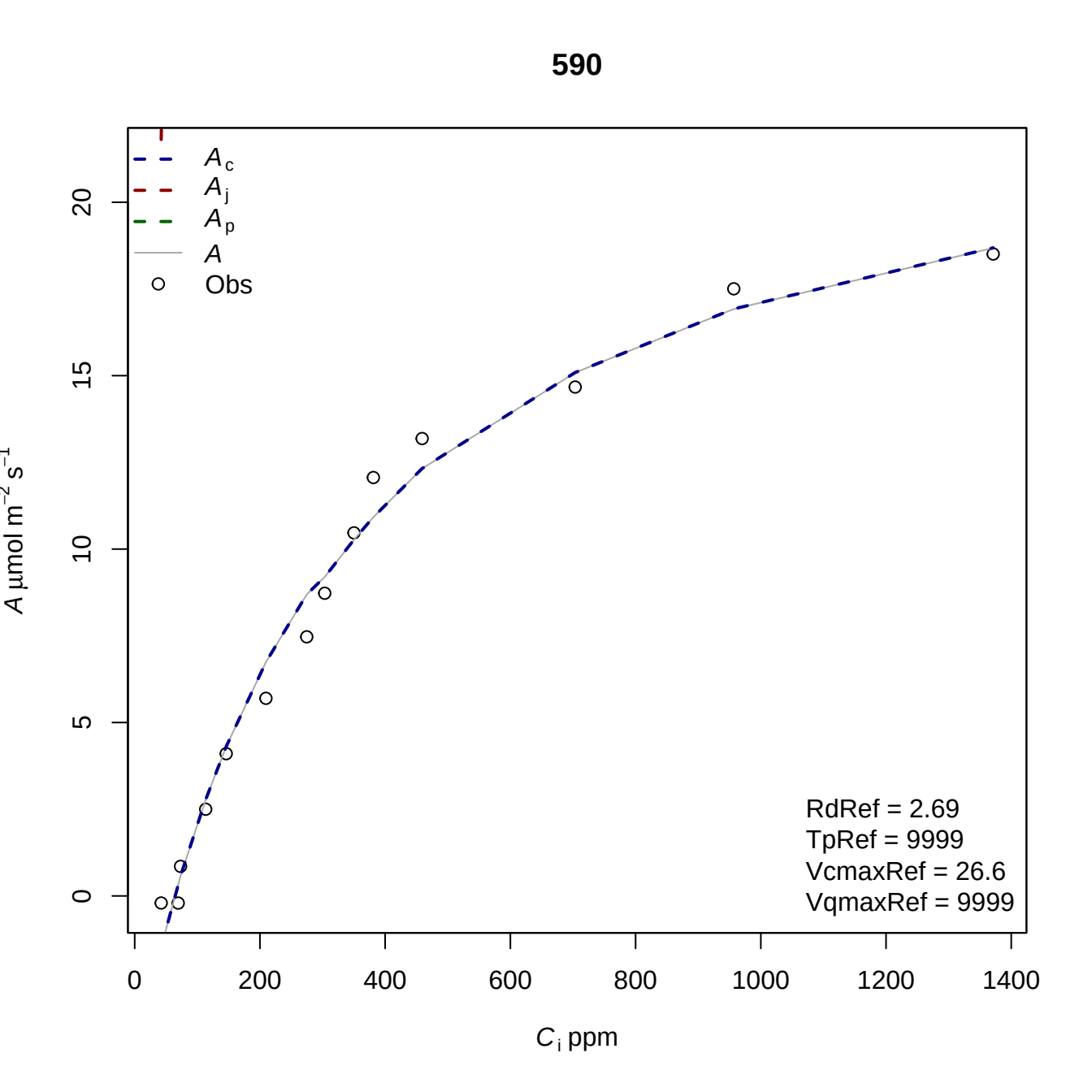

585
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.71
TpRef = 9999
VcmaxRef = 25.79
VqmaxRef = 95.05
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

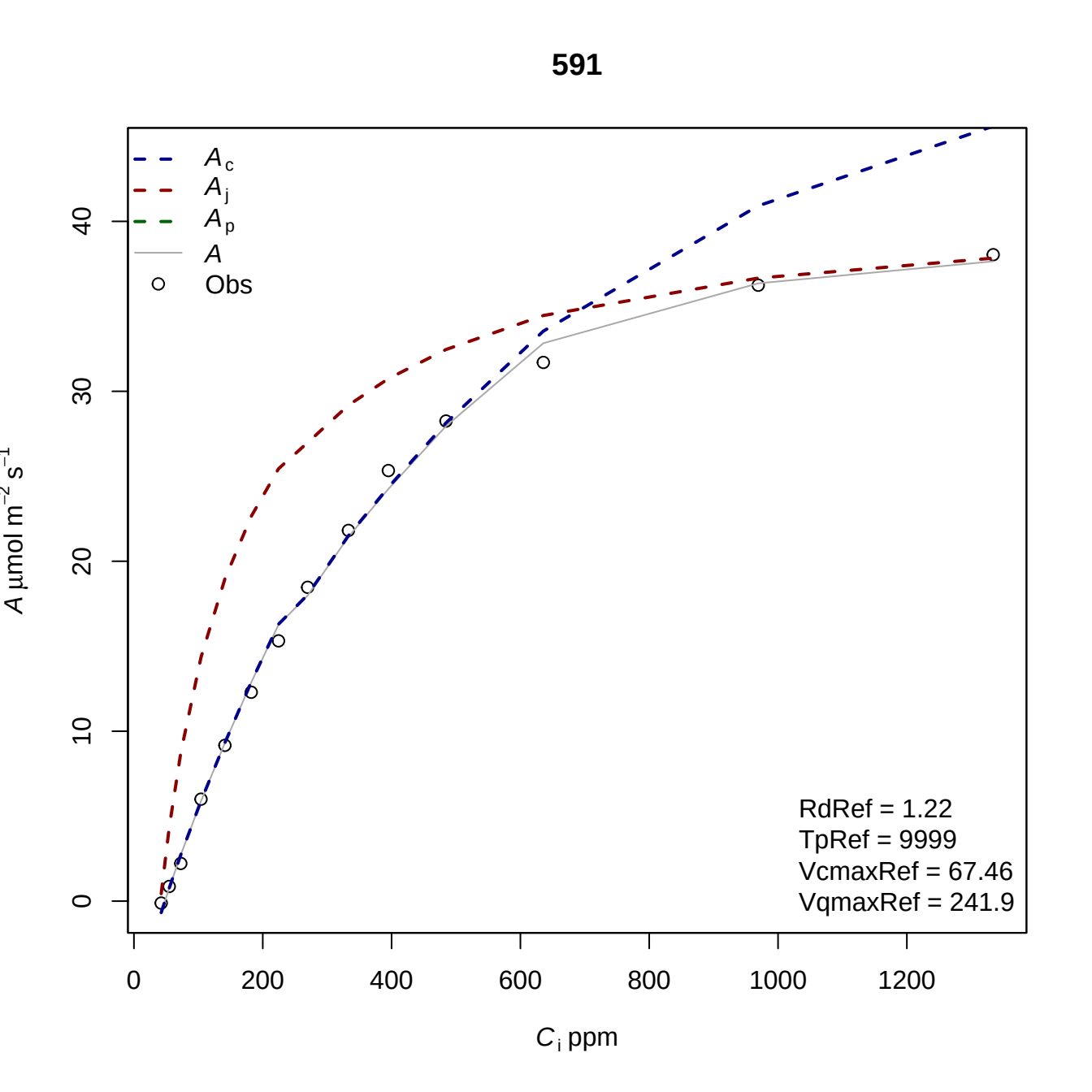

586
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.01
TpRef = 9999
VcmaxRef = 27.57
VqmaxRef = 112.97
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

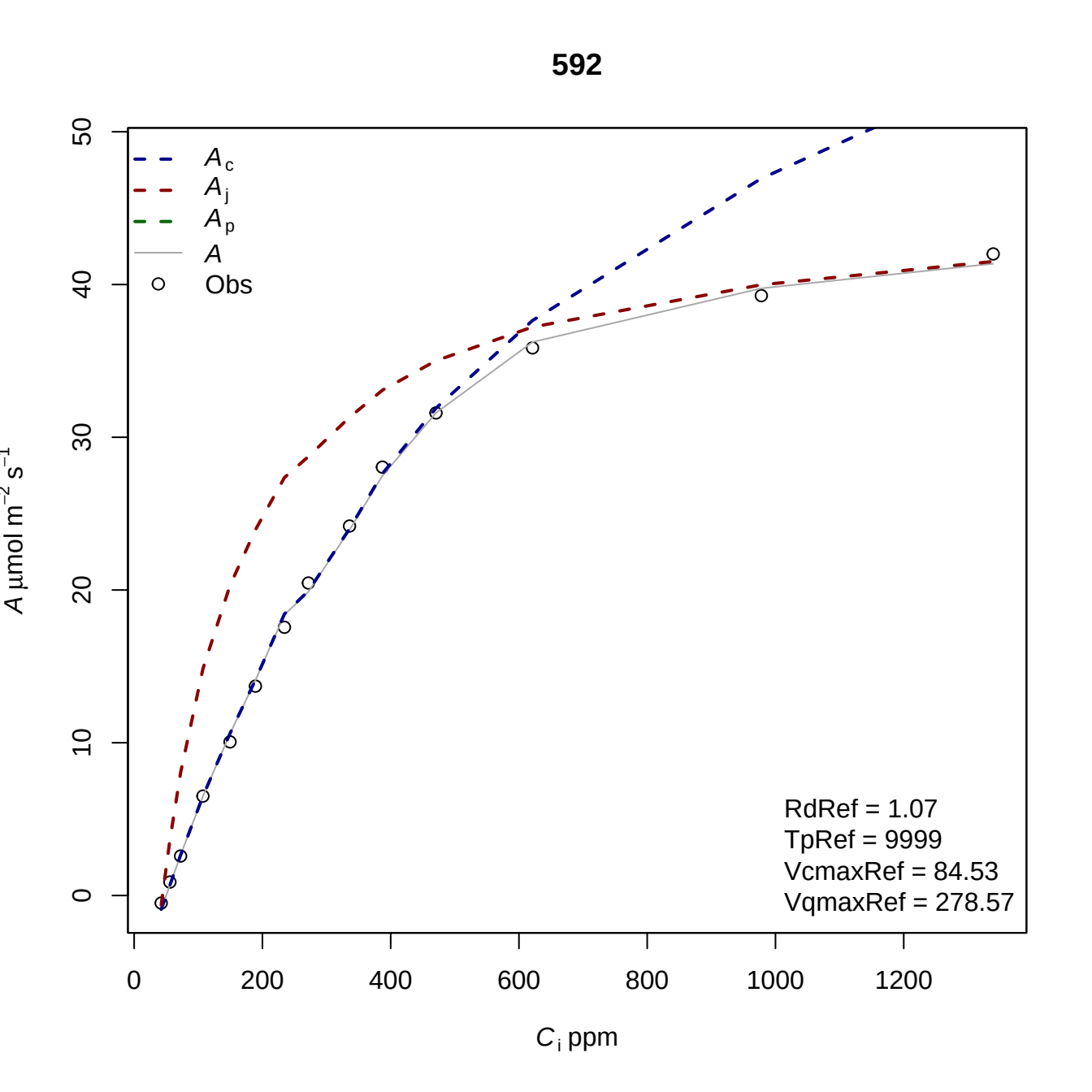

587
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 3.36
TpRef = 6.8
VcmaxRef = 24.29
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

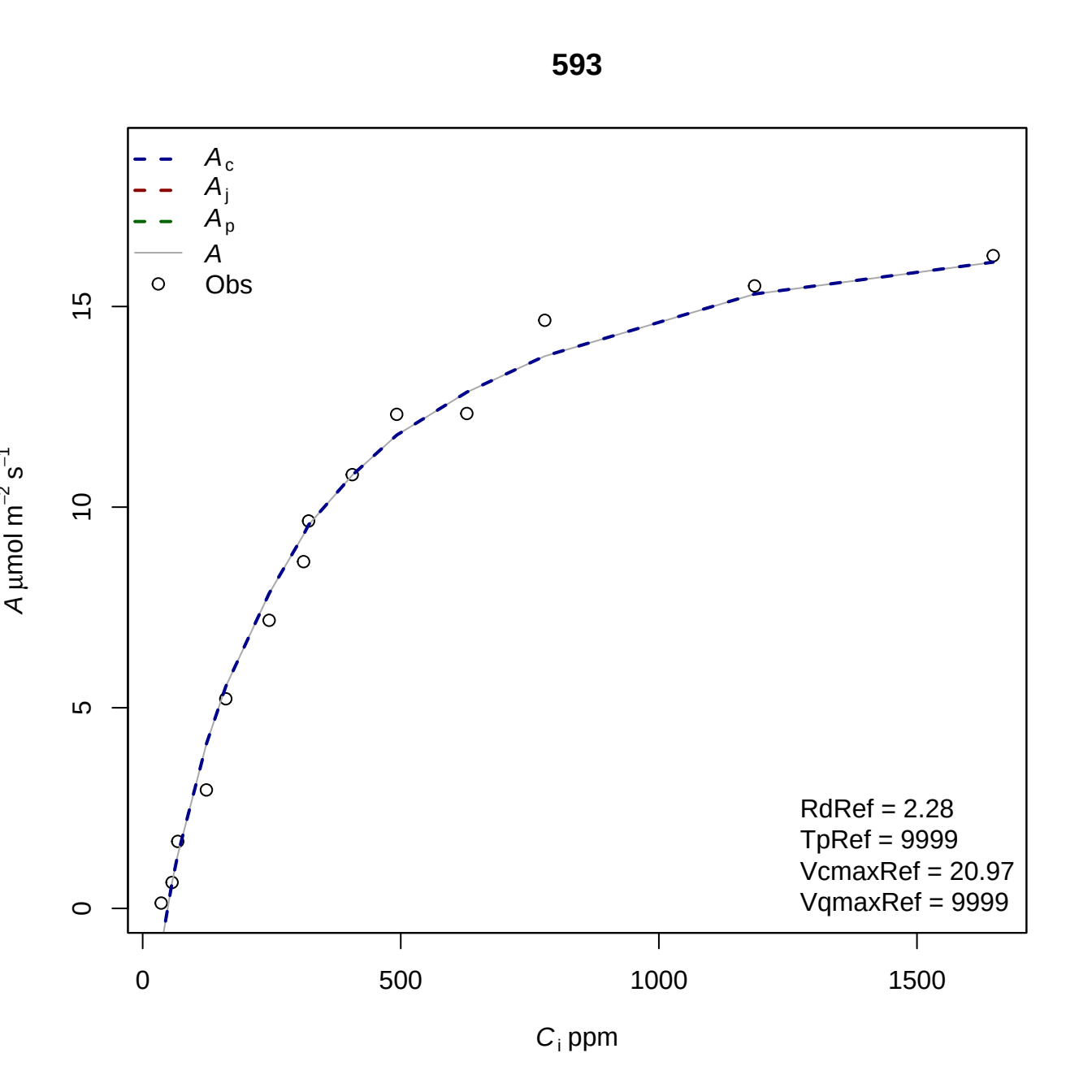

588
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.84
TpRef = 9999
VcmaxRef = 43.53
VqmaxRef = 158.81
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
1400

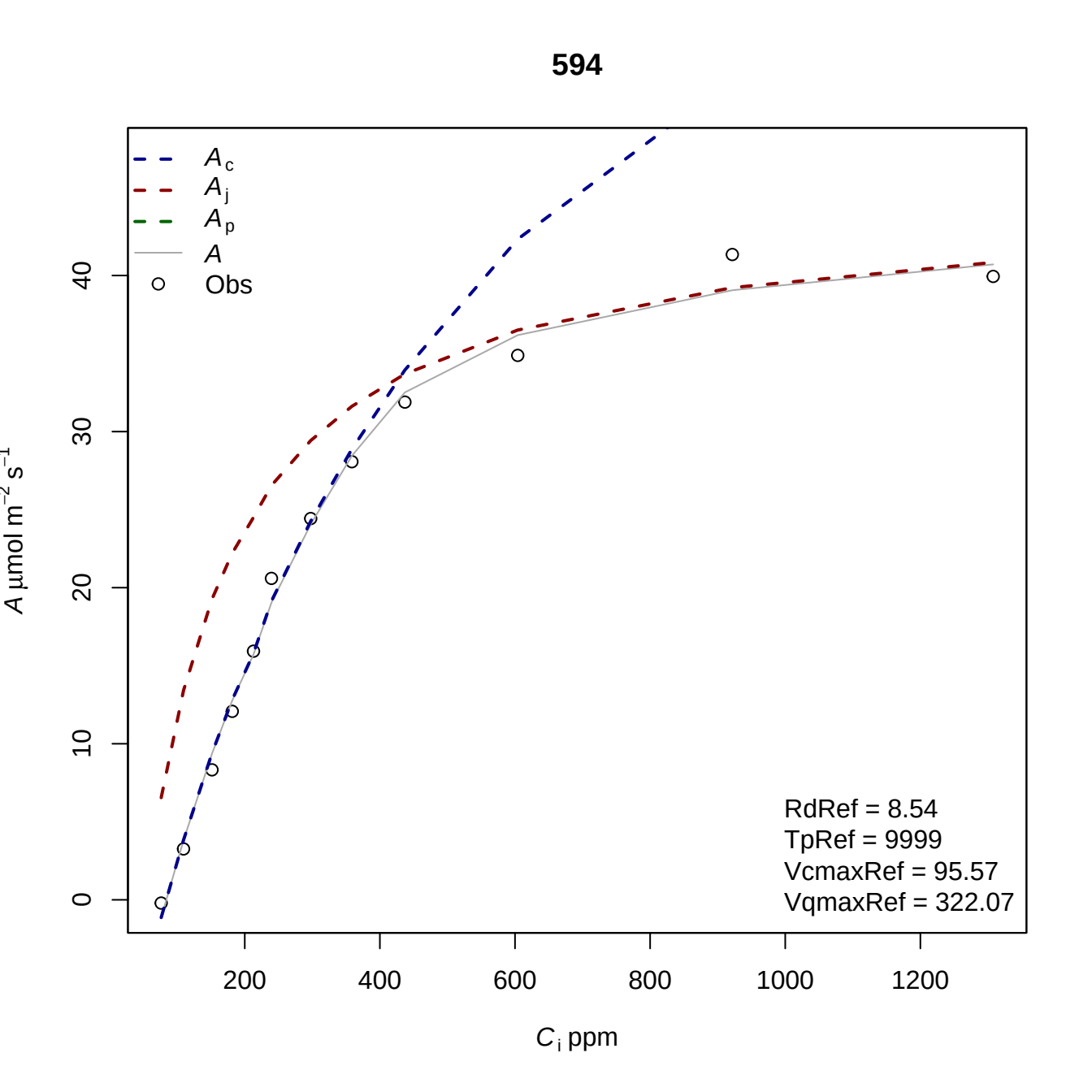

589
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.5
TpRef = 9999
VcmaxRef = 39.47
VqmaxRef = 137.48
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
500
1000
1500

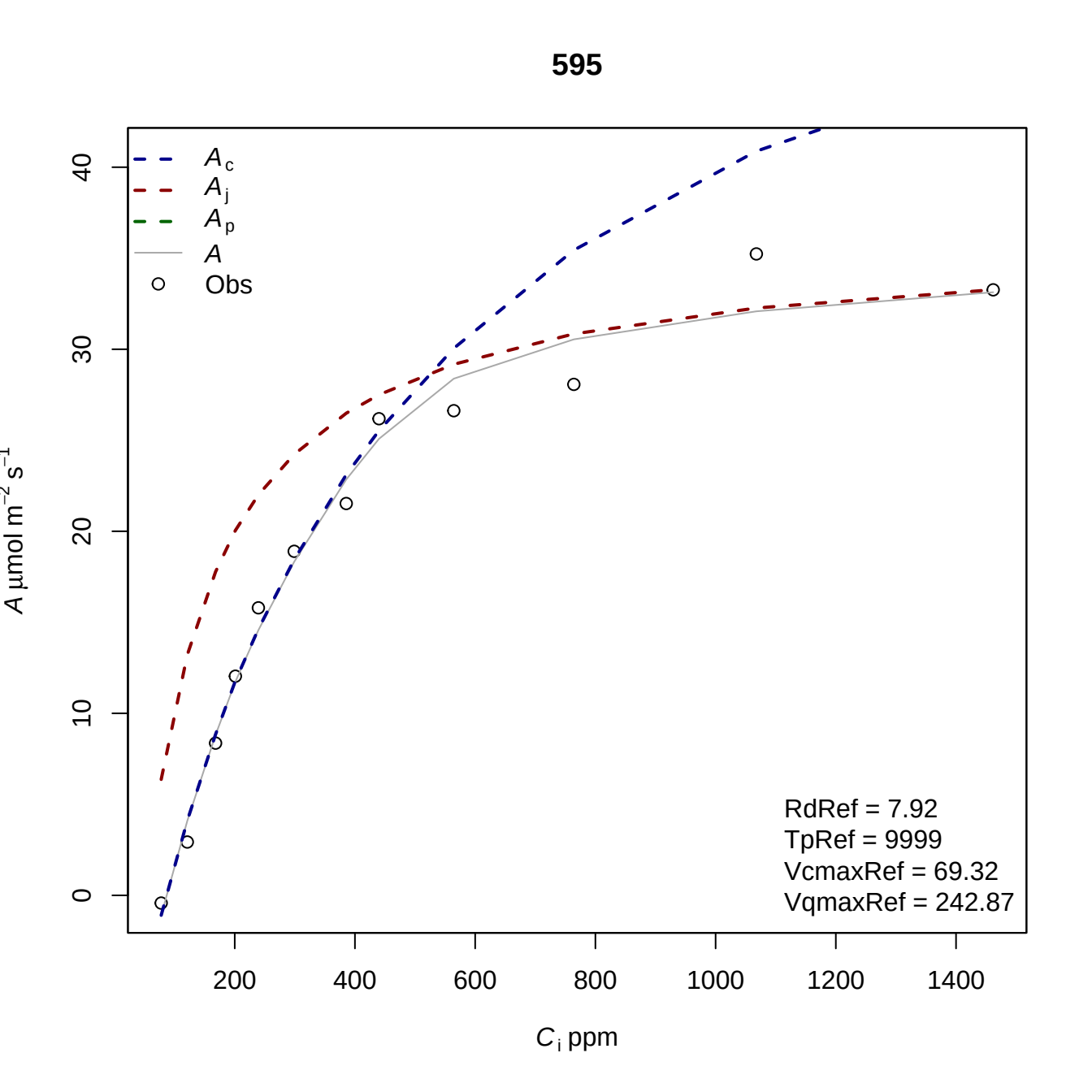

590
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.69
TpRef = 9999
VcmaxRef = 26.6
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

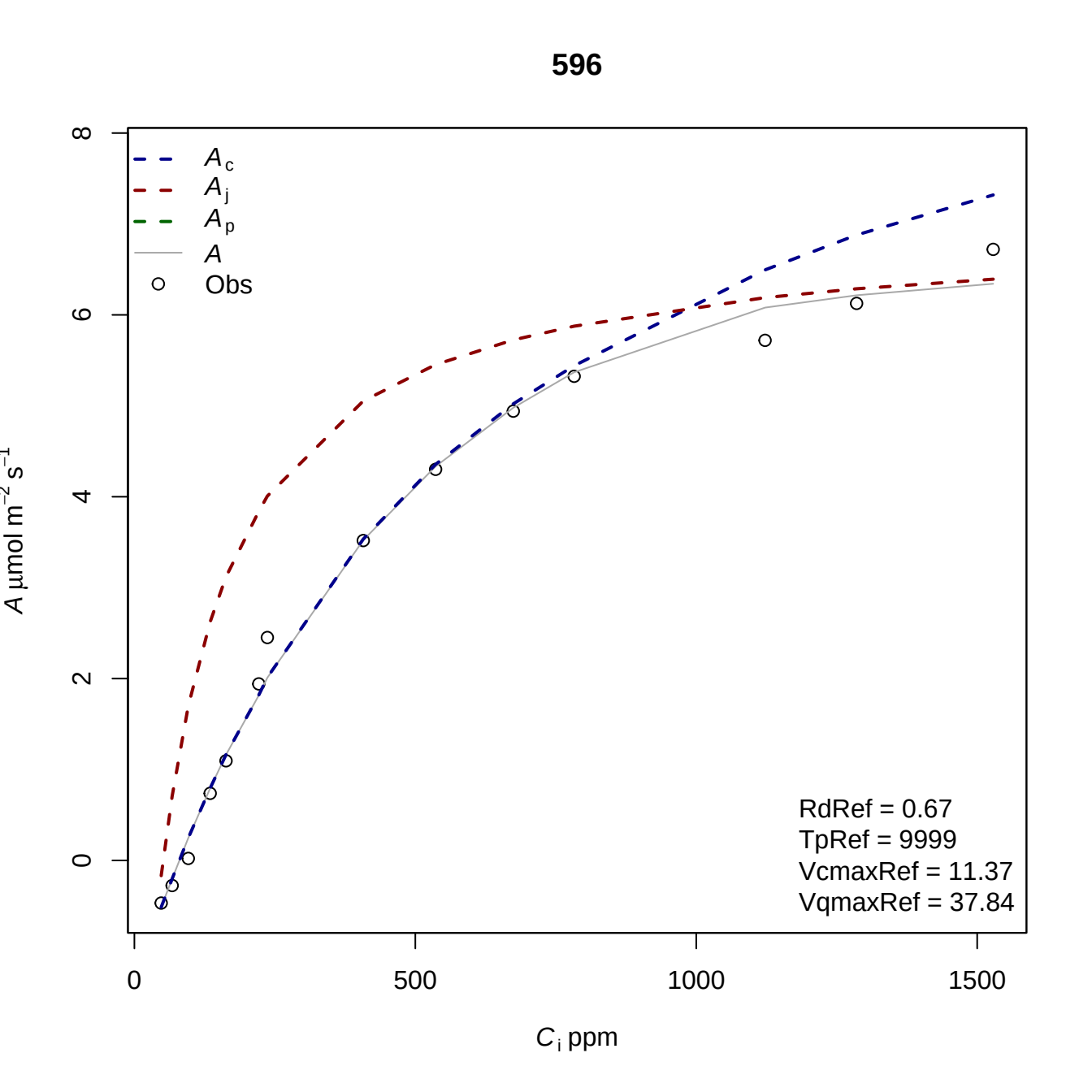

591
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.22
TpRef = 9999
VcmaxRef = 67.46
VqmaxRef = 241.9
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

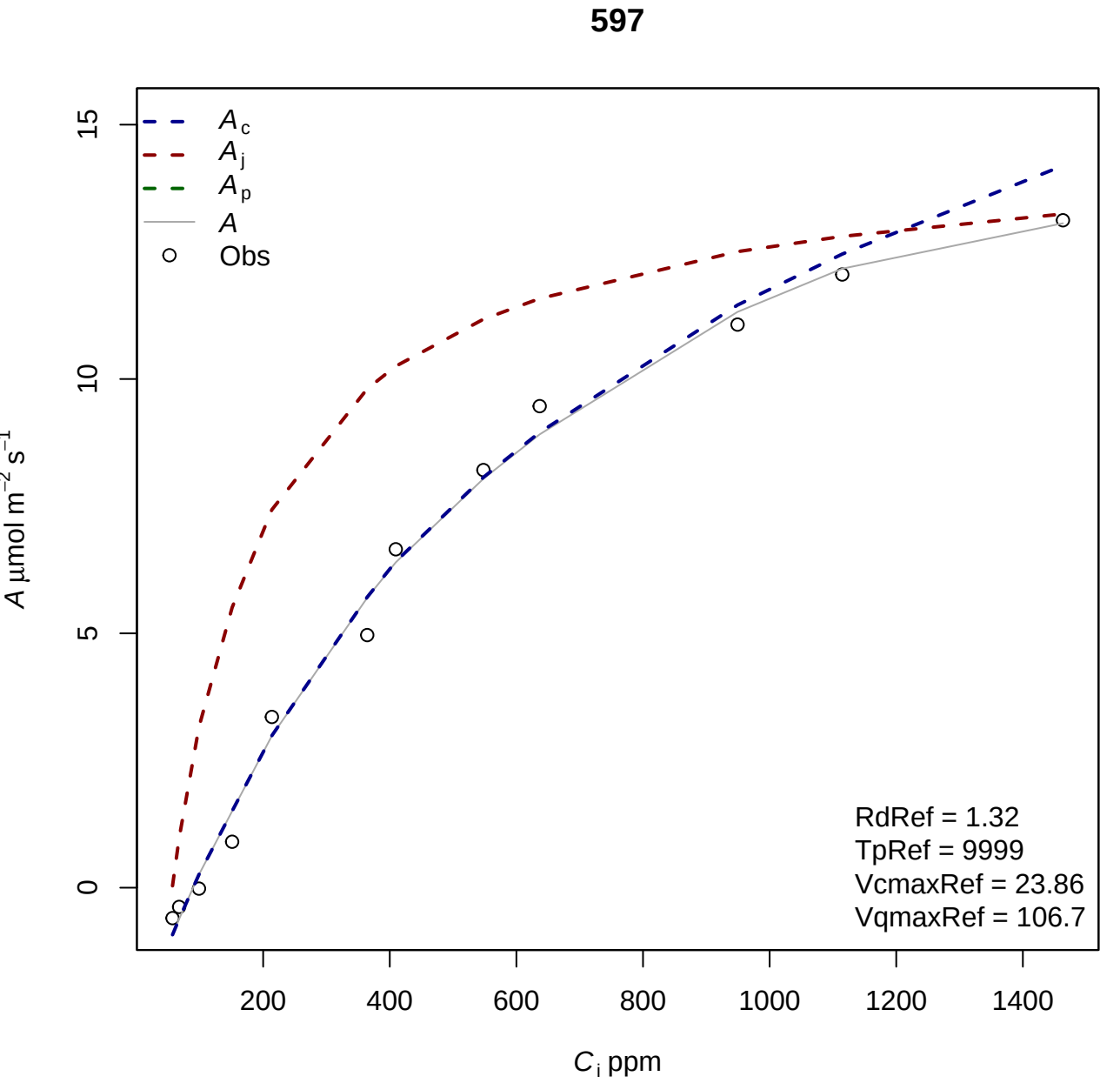

592
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.07
TpRef = 9999
VcmaxRef = 84.53
VqmaxRef = 278.57
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
50
0
200
400
600
800
1000
1200

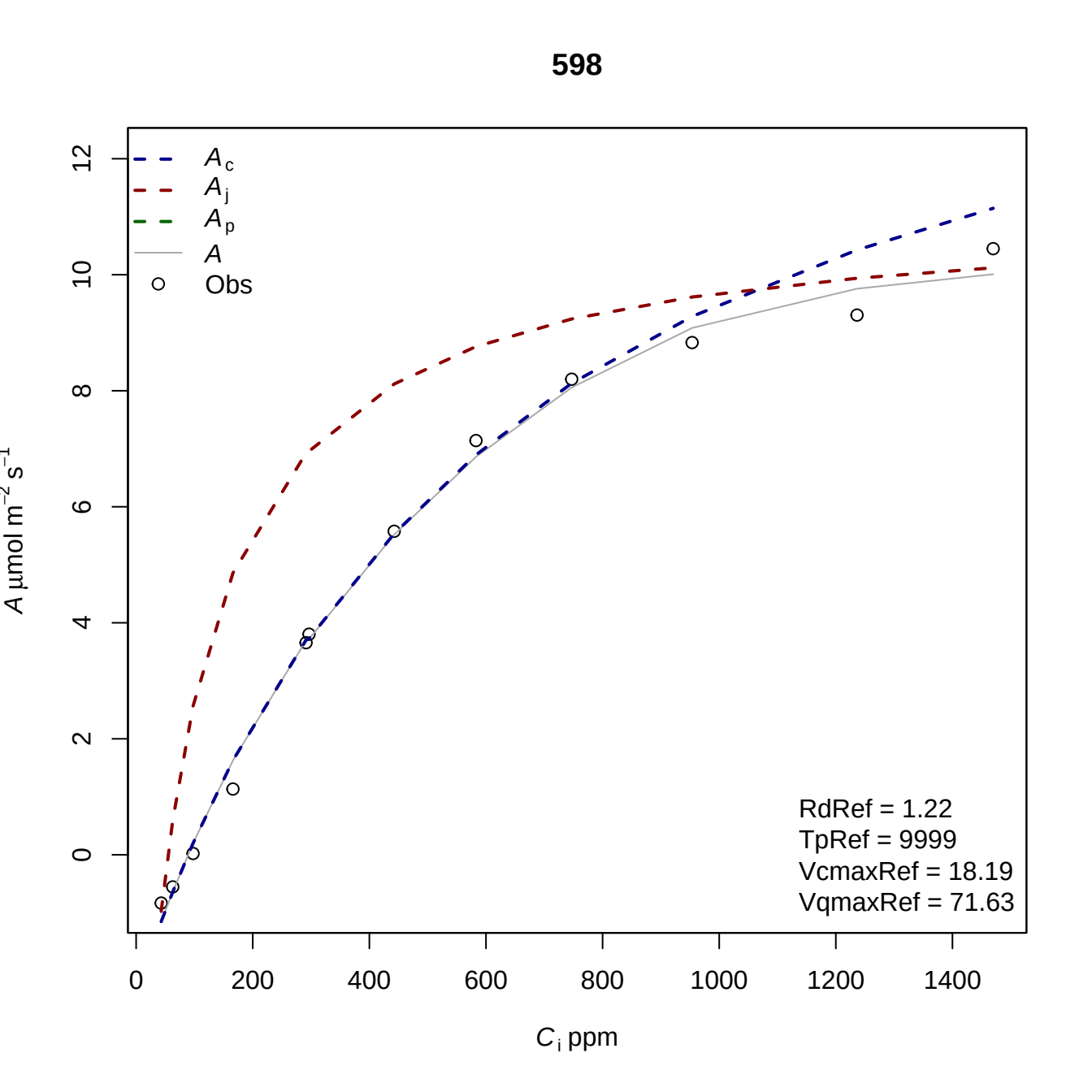

593
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.28
TpRef = 9999
VcmaxRef = 20.97
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

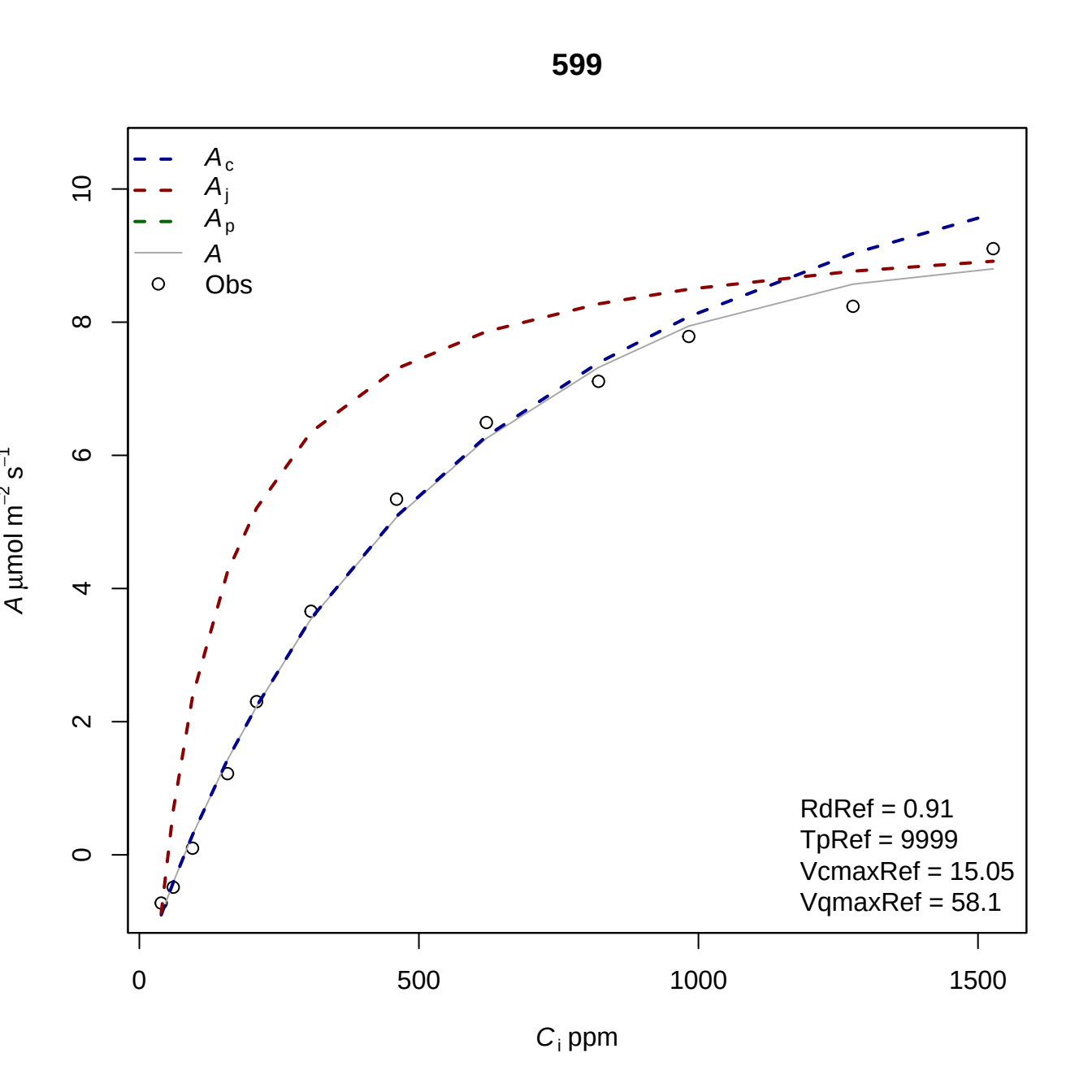

594
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 8.54
TpRef = 9999
VcmaxRef = 95.57
VqmaxRef = 322.07
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200

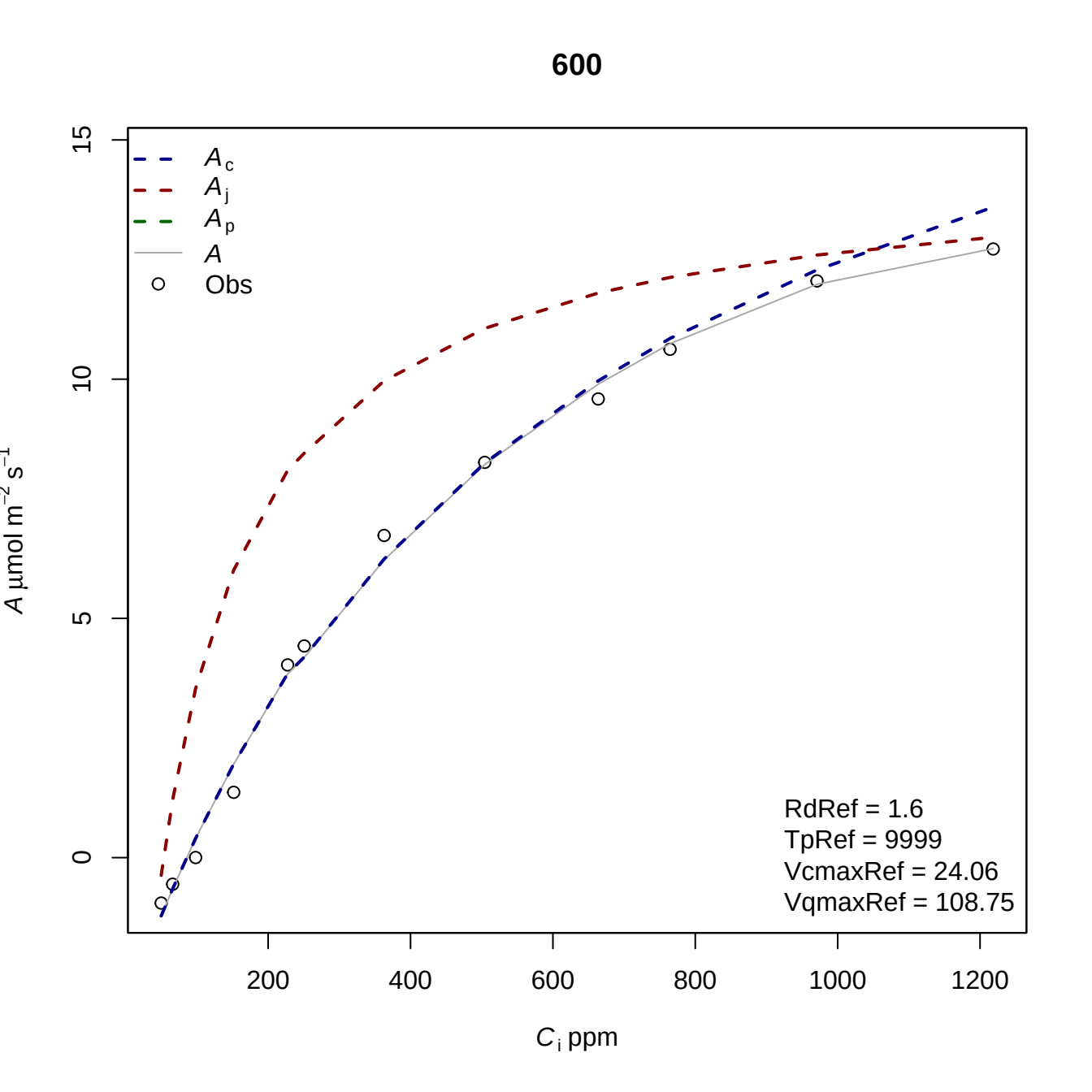

595
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 7.92
TpRef = 9999
VcmaxRef = 69.32
VqmaxRef = 242.87
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
10
20
30
40
200
400
600
800
1000
1200
1400

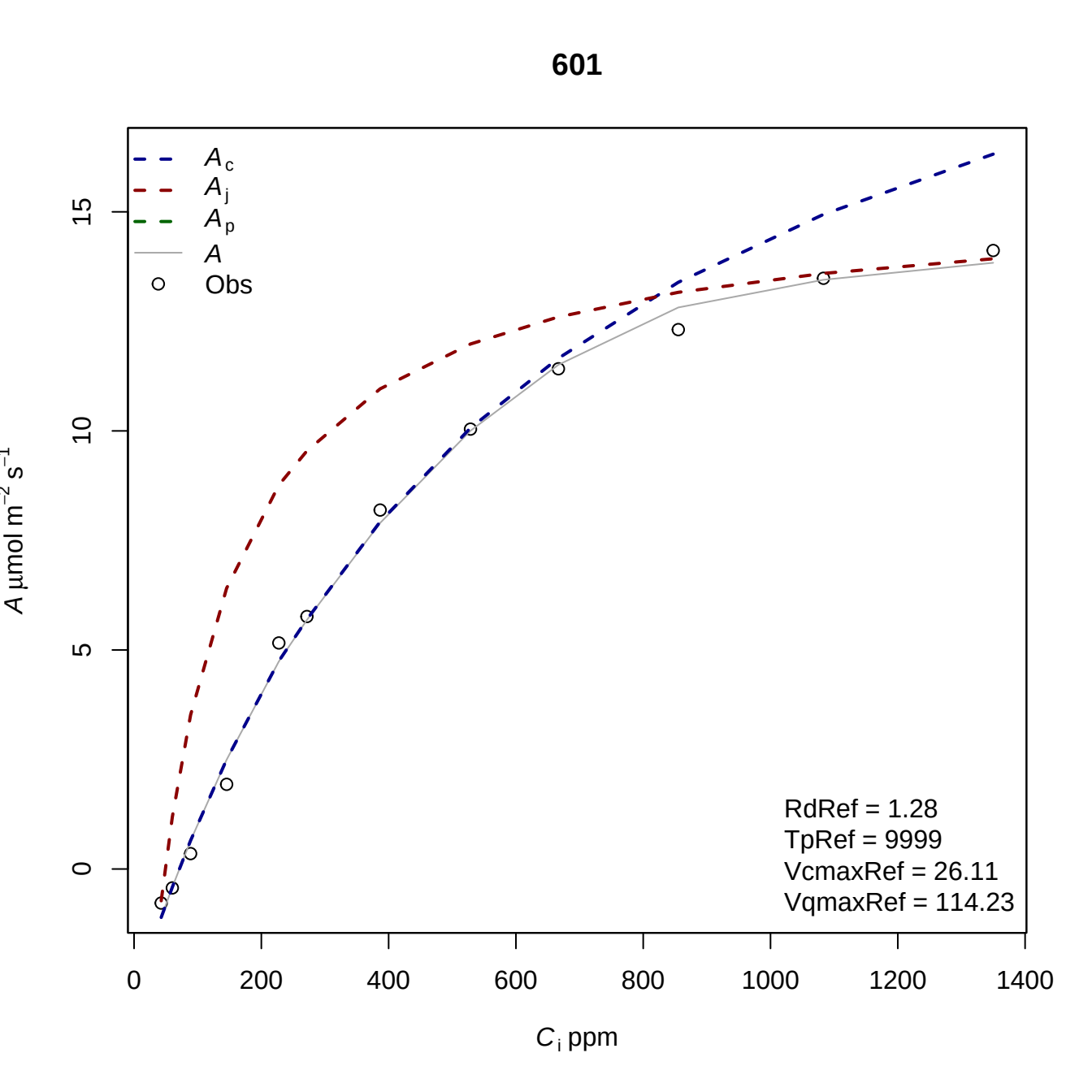

596
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.67
TpRef = 9999
VcmaxRef = 11.37
VqmaxRef = 37.84
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
0
500
1000
1500

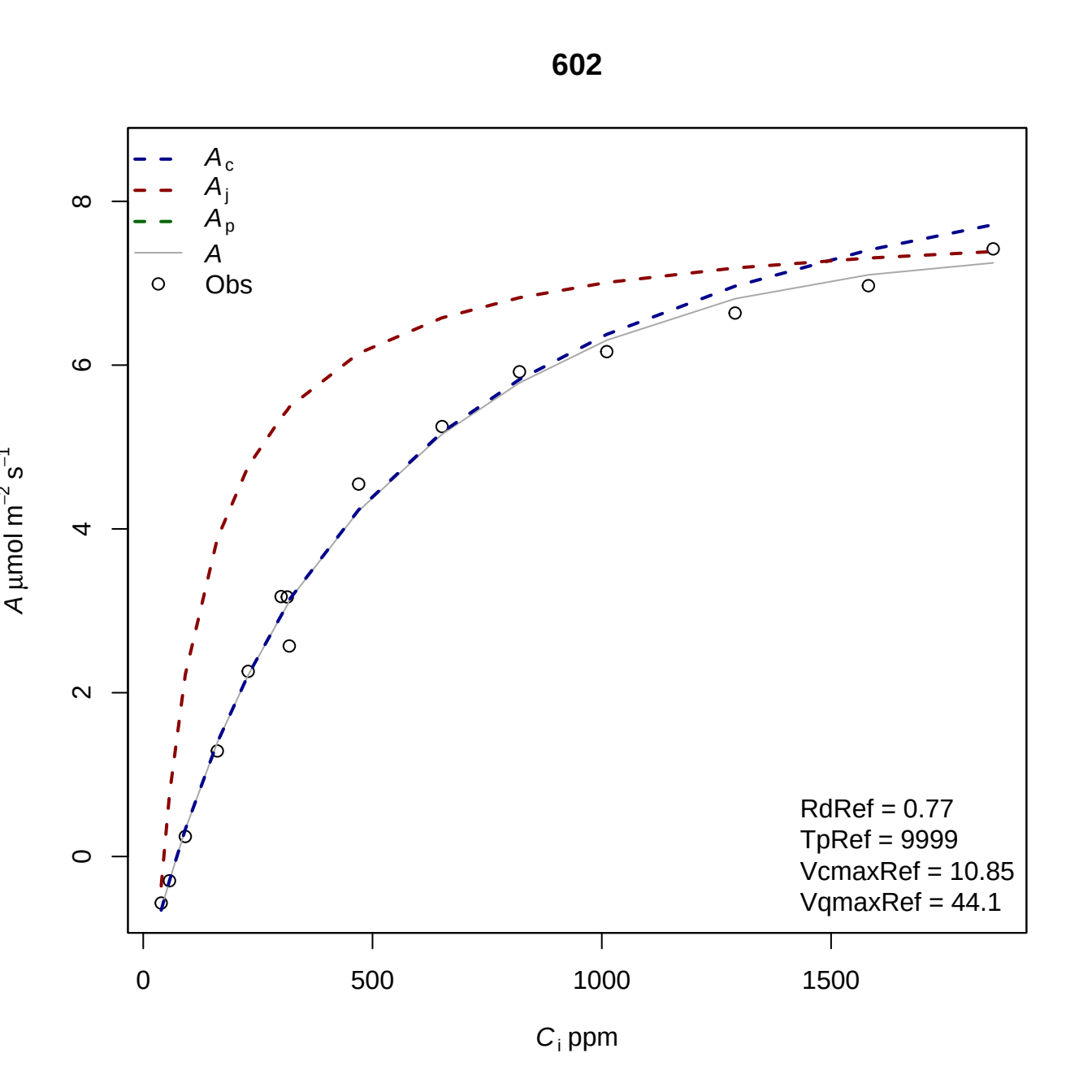

597
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.32
TpRef = 9999
VcmaxRef = 23.86
VqmaxRef = 106.7
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200
1400

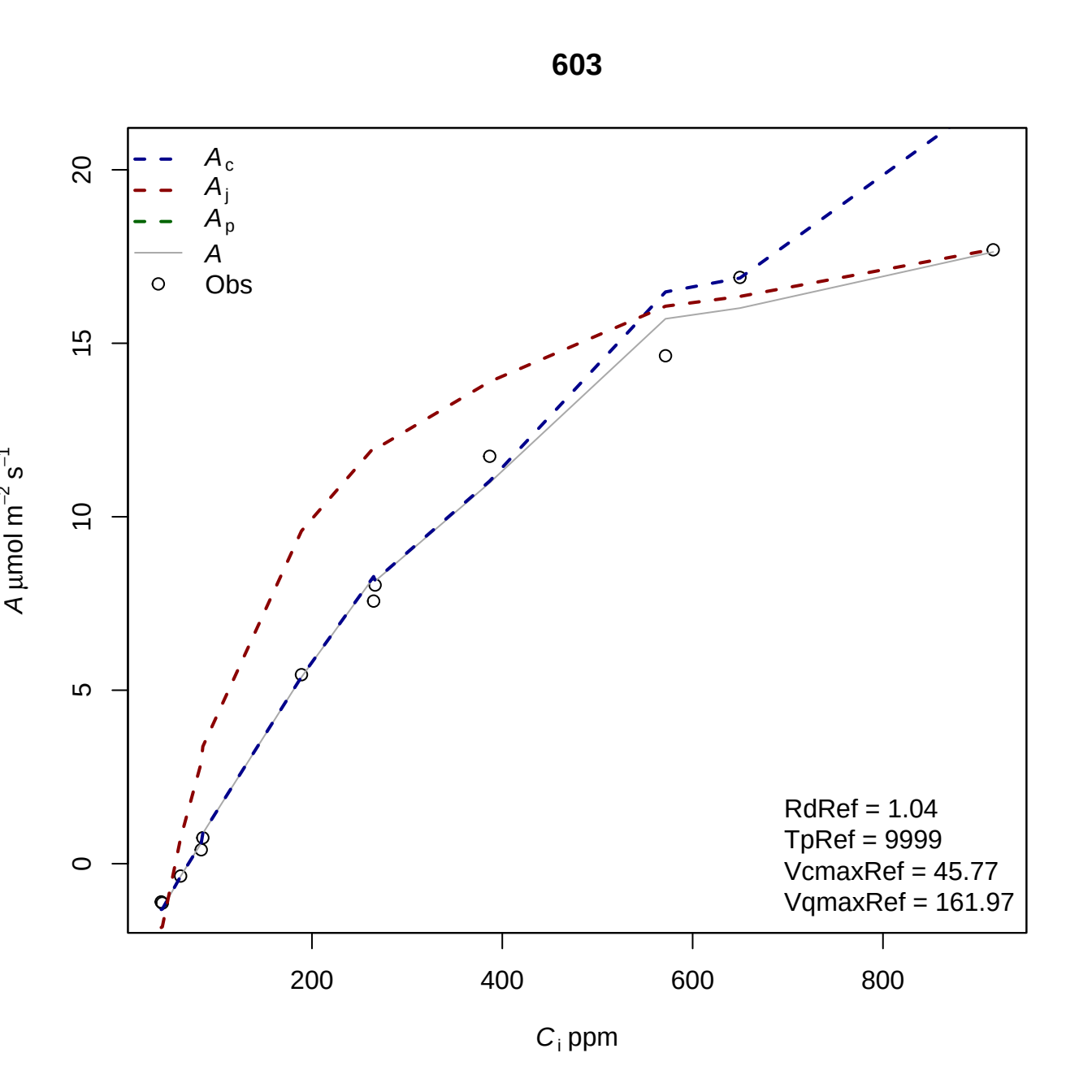

598
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.22
TpRef = 9999
VcmaxRef = 18.19
VqmaxRef = 71.63
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

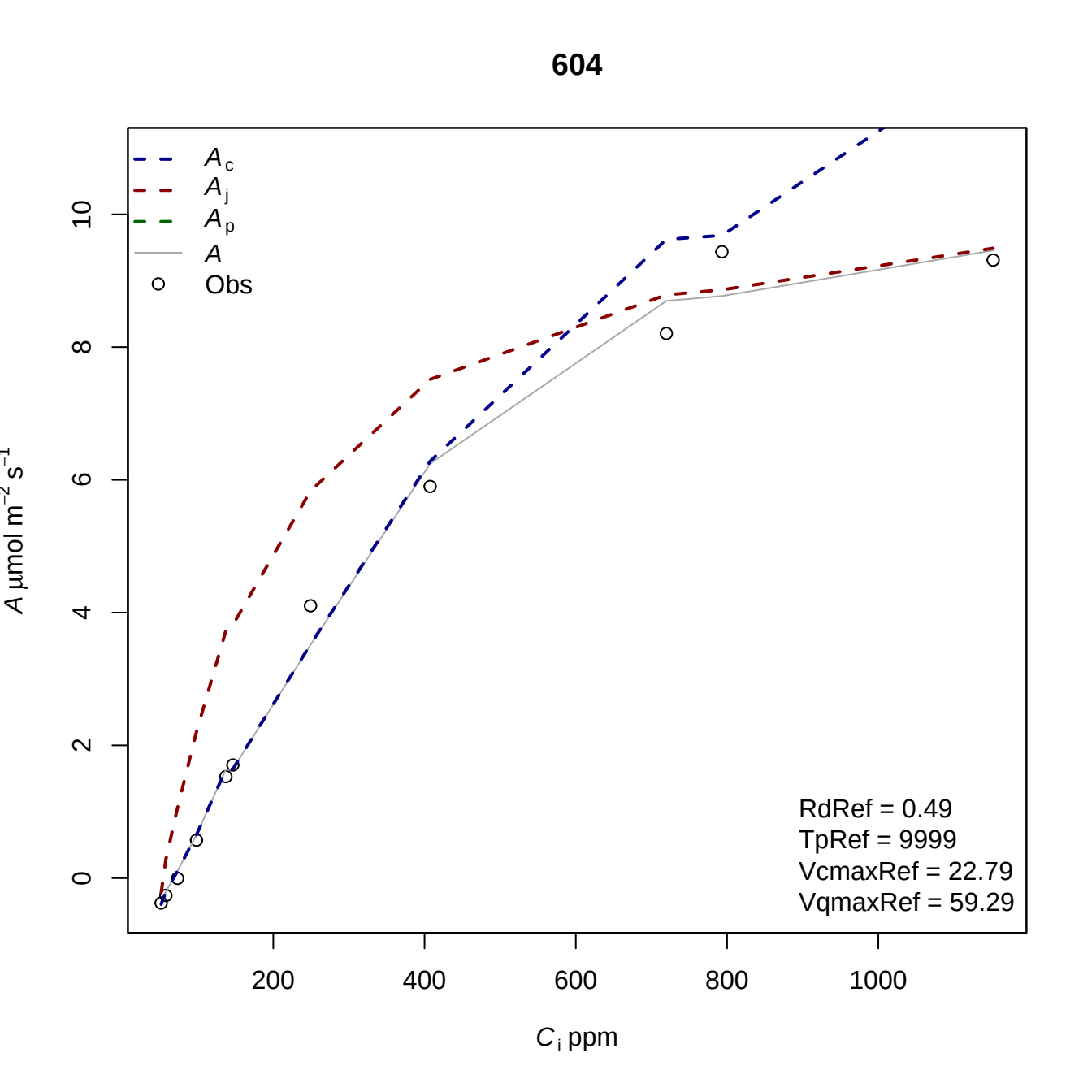

599
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.91
TpRef = 9999
VcmaxRef = 15.05
VqmaxRef = 58.1
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

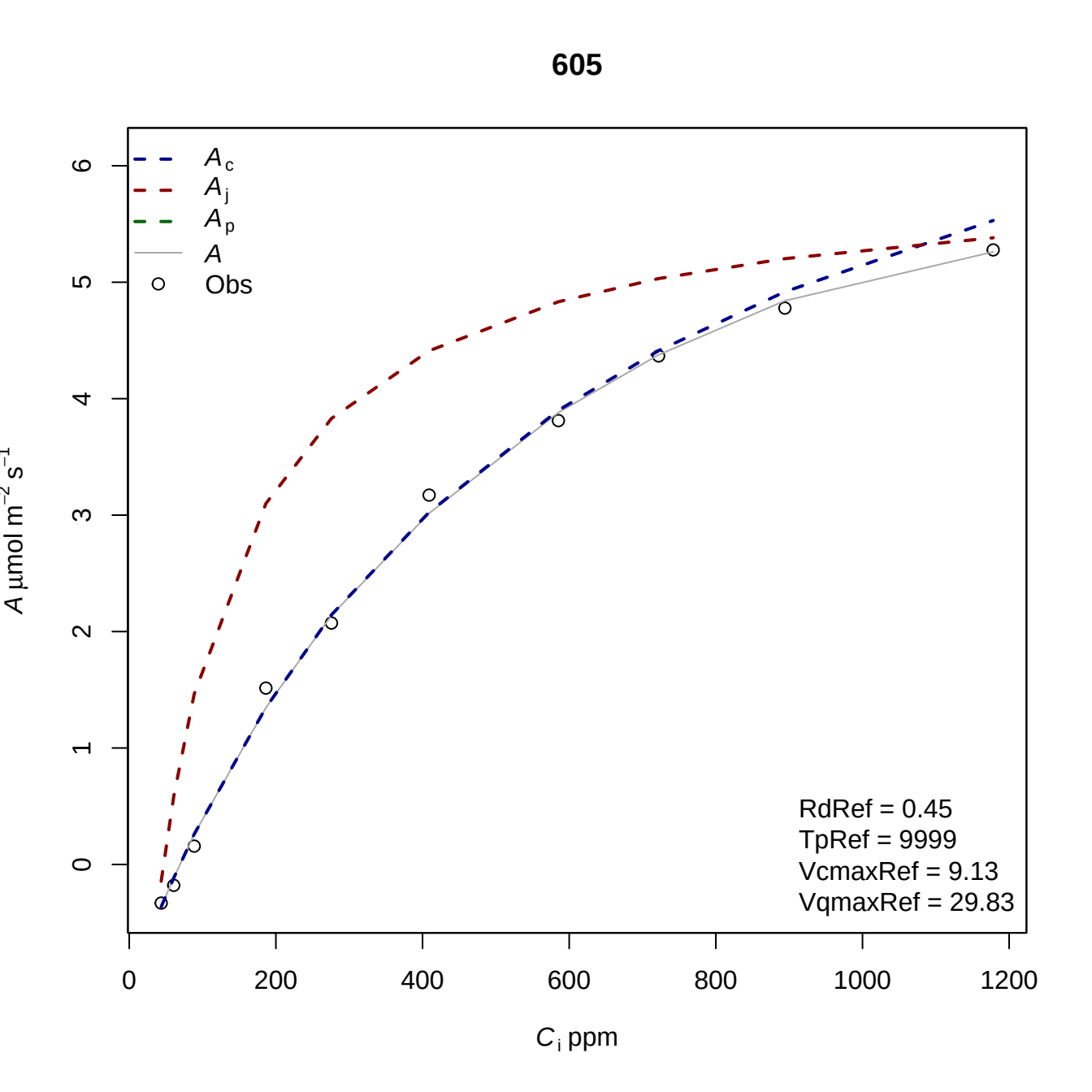

600
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.6
TpRef = 9999
VcmaxRef = 24.06
VqmaxRef = 108.75
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000
1200

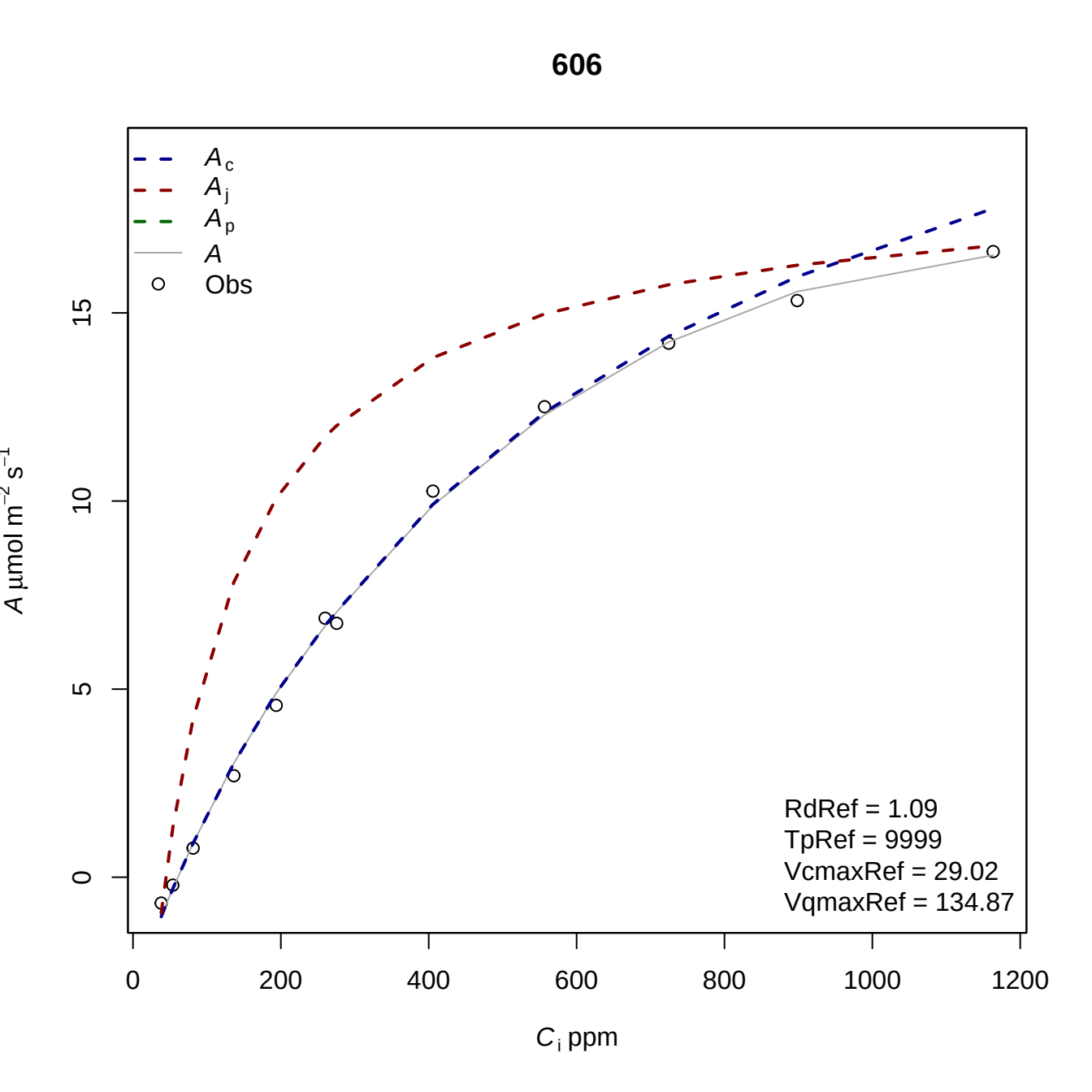

601
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.28
TpRef = 9999
VcmaxRef = 26.11
VqmaxRef = 114.23
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

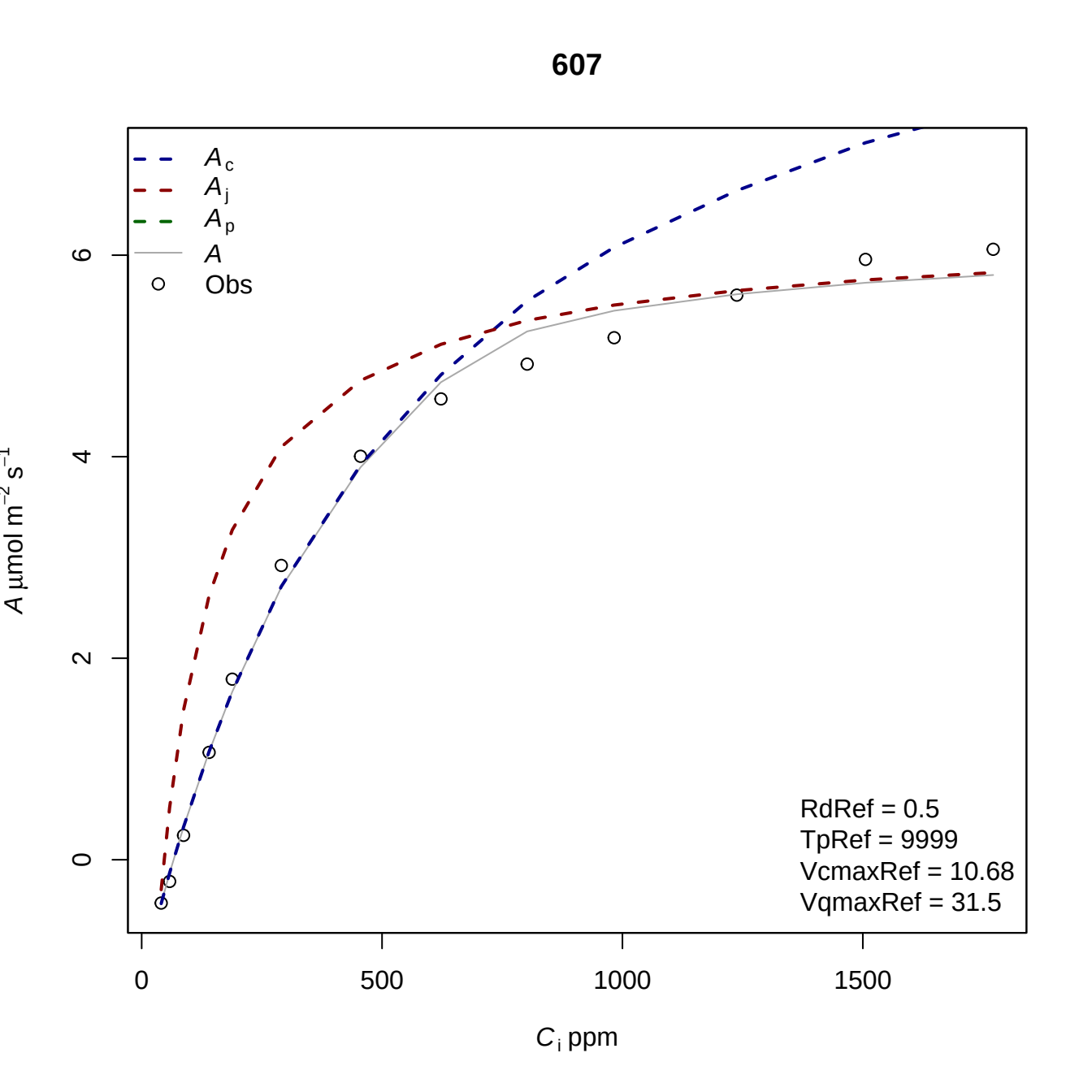

602
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.77
TpRef = 9999
VcmaxRef = 10.85
VqmaxRef = 44.1
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
0
500
1000
1500

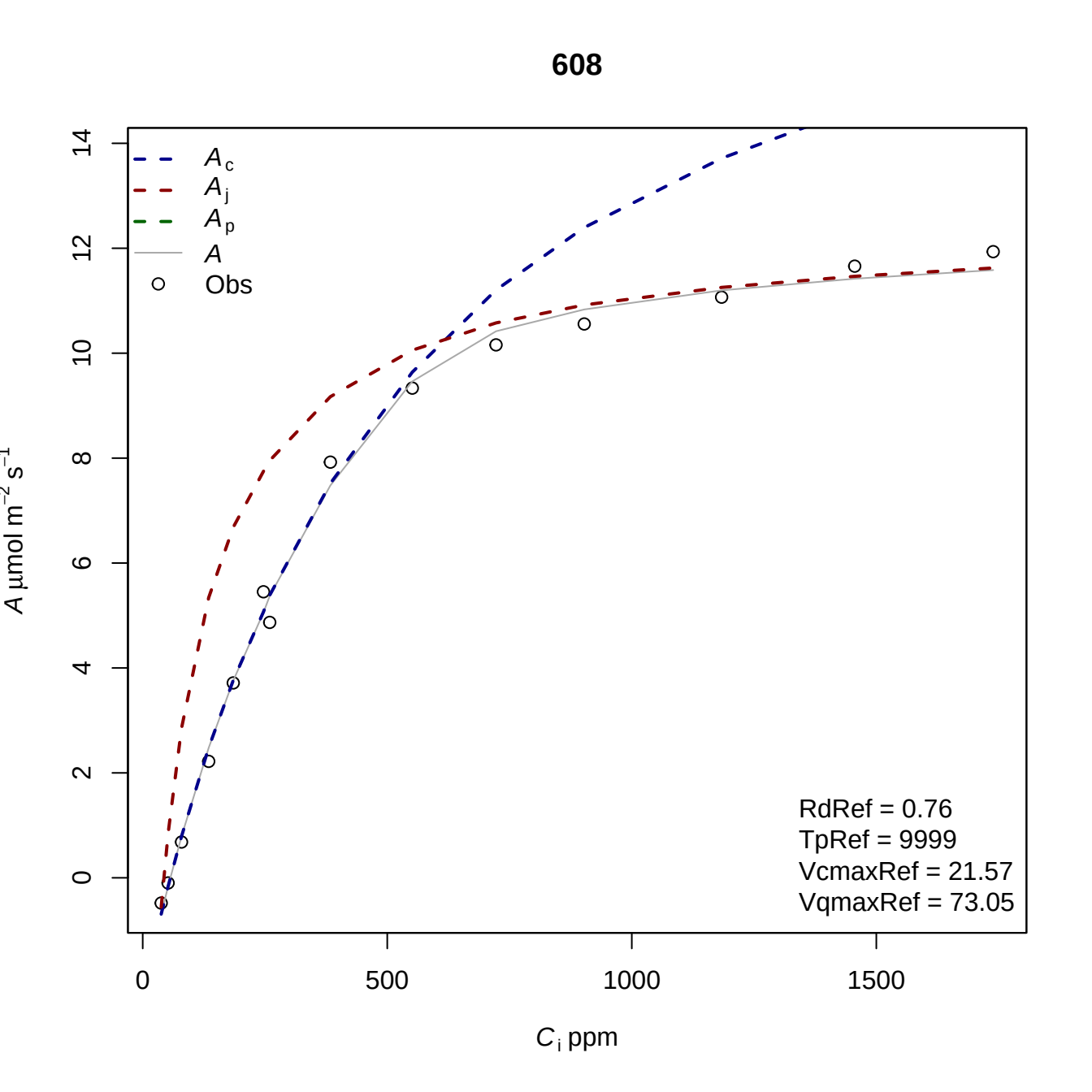

603
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.04
TpRef = 9999
VcmaxRef = 45.77
VqmaxRef = 161.97
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800

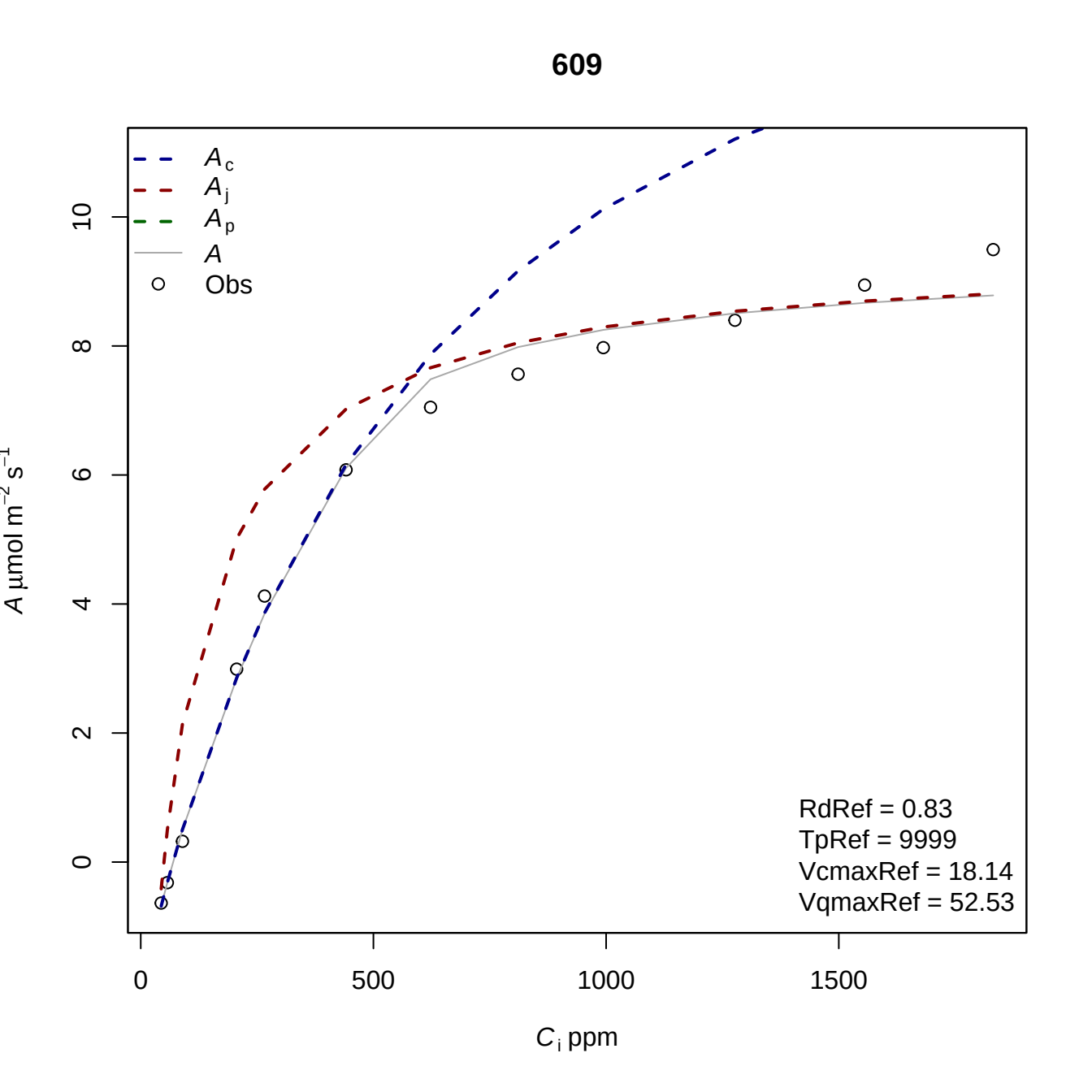

604
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.49
TpRef = 9999
VcmaxRef = 22.79
VqmaxRef = 59.29
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
200
400
600
800
1000

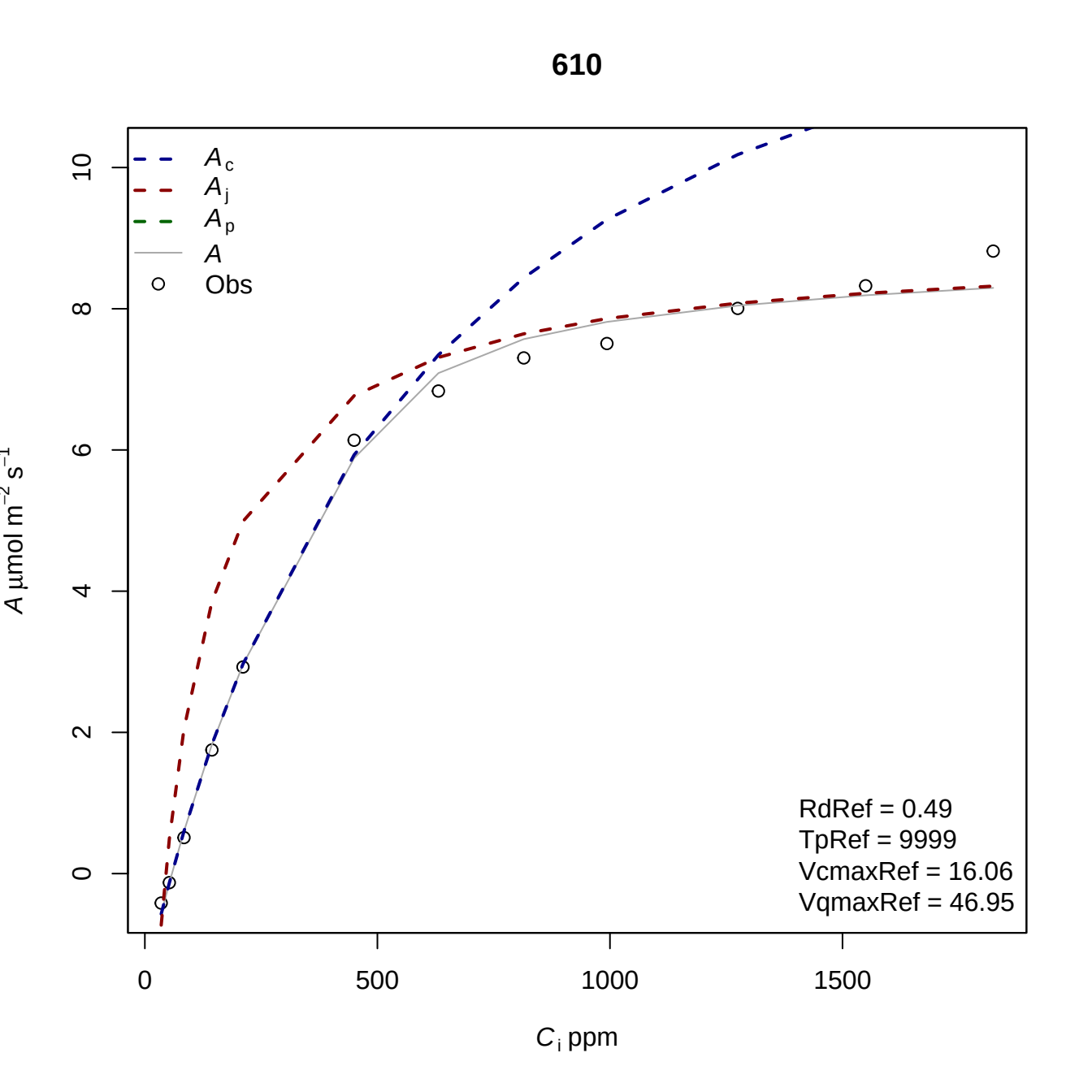

605
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.45
TpRef = 9999
VcmaxRef = 9.13
VqmaxRef = 29.83
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200

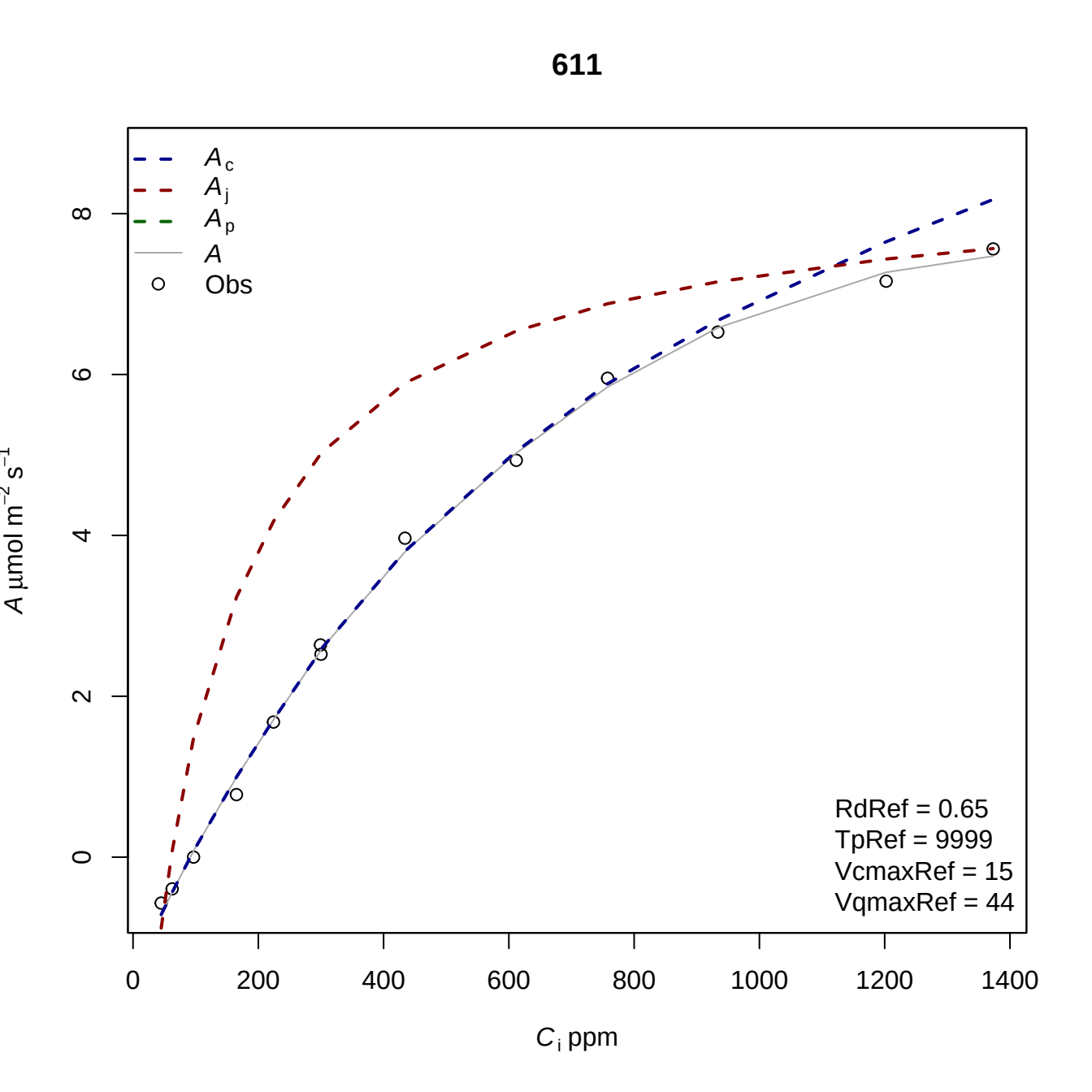

606
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.09
TpRef = 9999
VcmaxRef = 29.02
VqmaxRef = 134.87
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

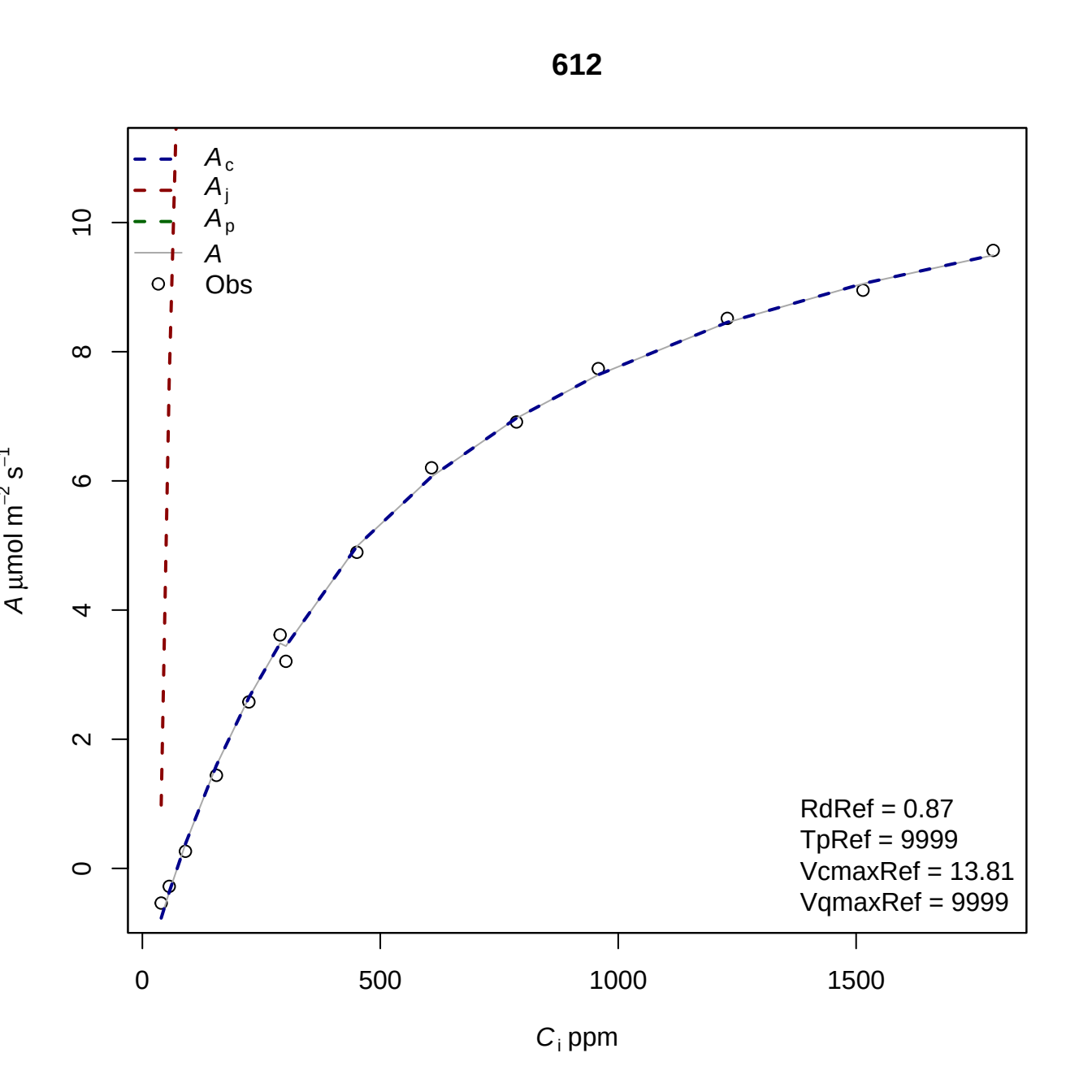

607
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.5
TpRef = 9999
VcmaxRef = 10.68
VqmaxRef = 31.5
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
0
500
1000
1500

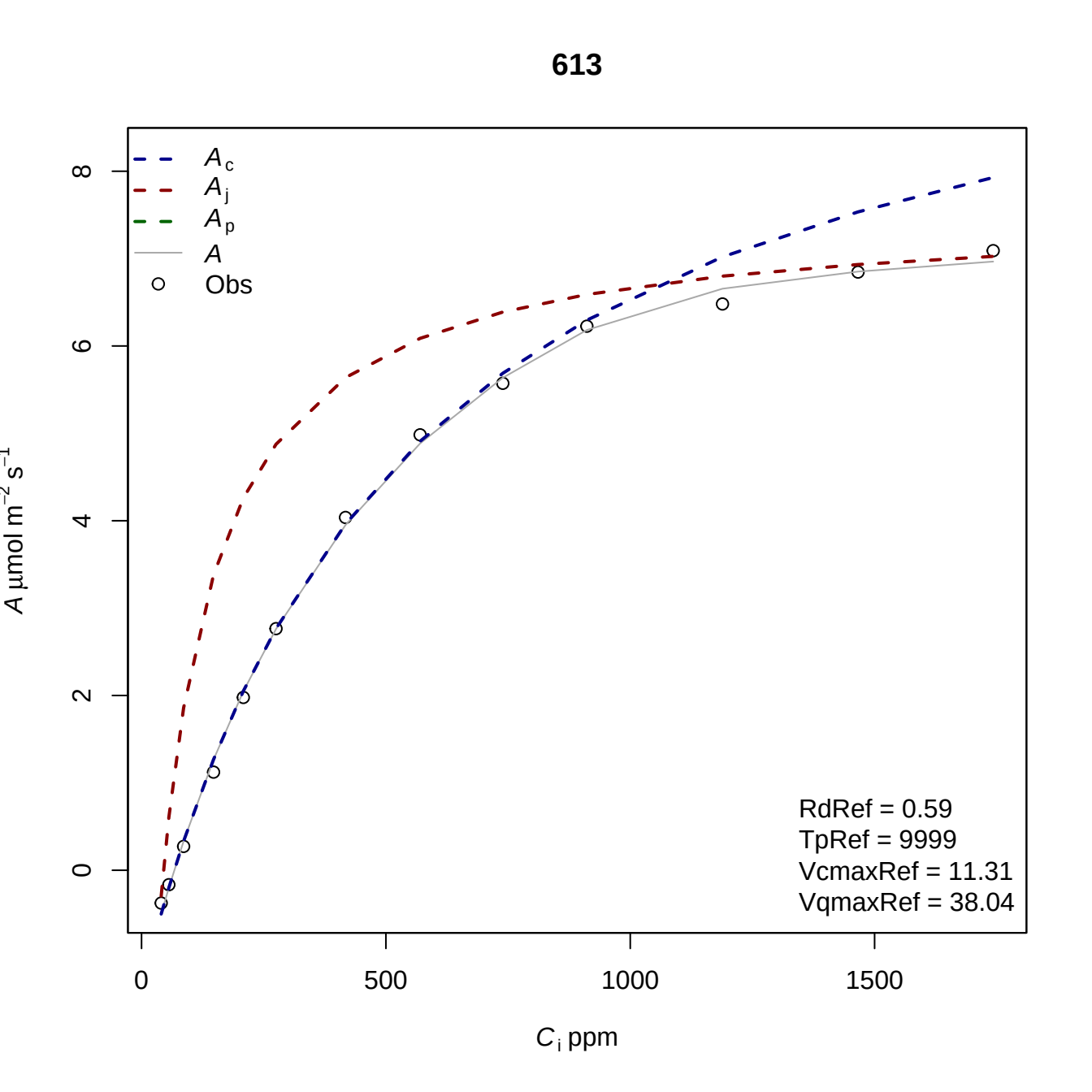

608
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.76
TpRef = 9999
VcmaxRef = 21.57
VqmaxRef = 73.05
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
0
500
1000
1500

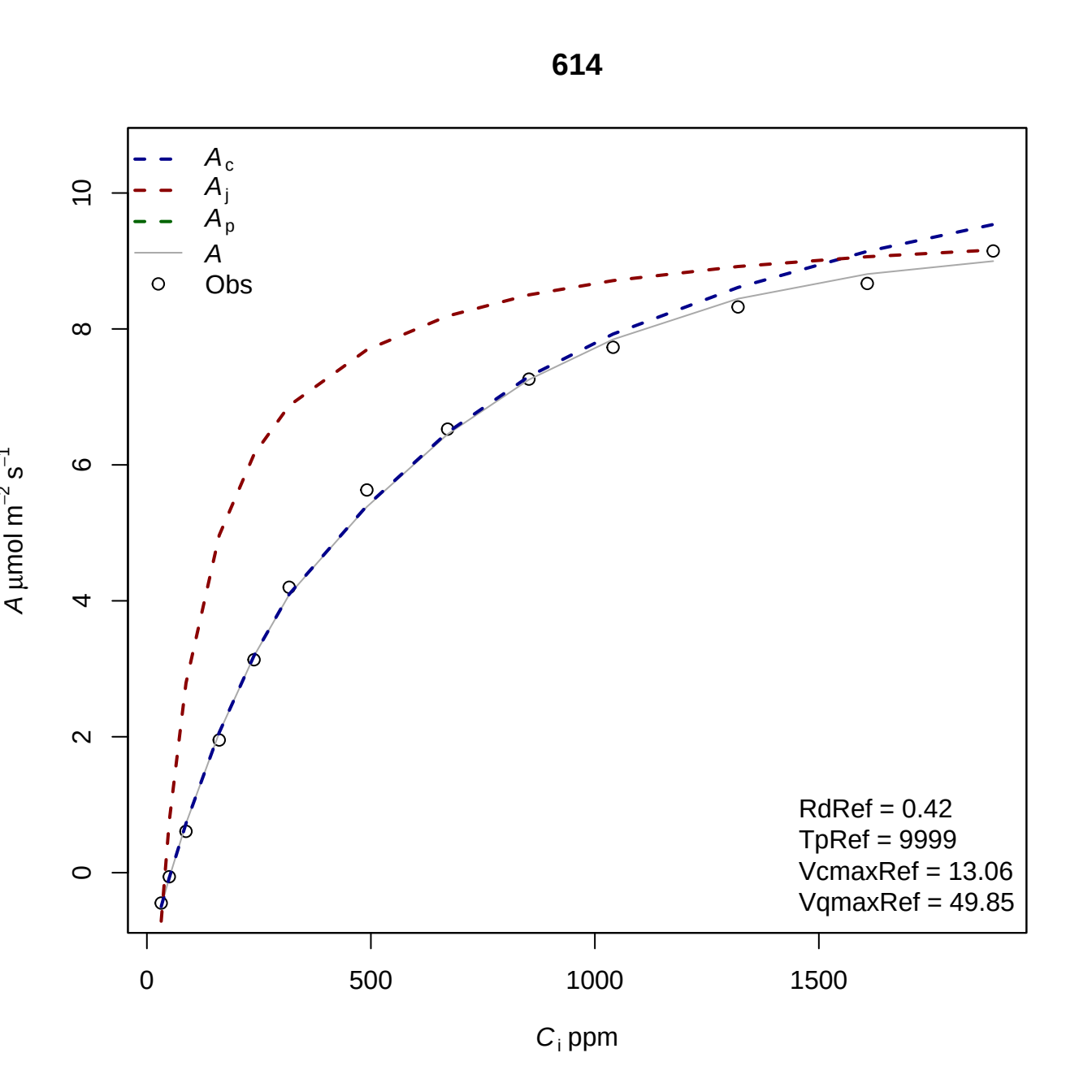

609
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.83
TpRef = 9999
VcmaxRef = 18.14
VqmaxRef = 52.53
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

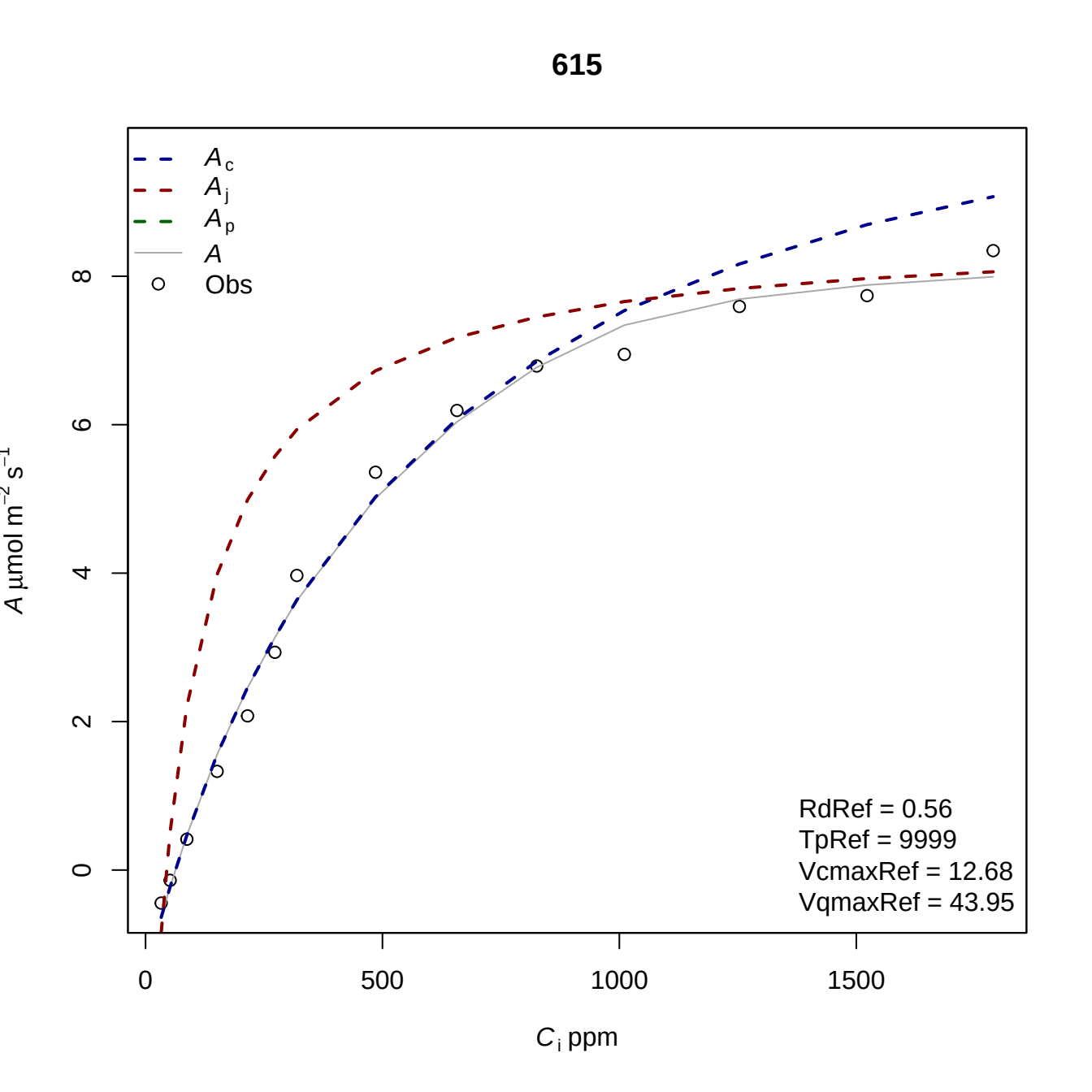

610
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.49
TpRef = 9999
VcmaxRef = 16.06
VqmaxRef = 46.95
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

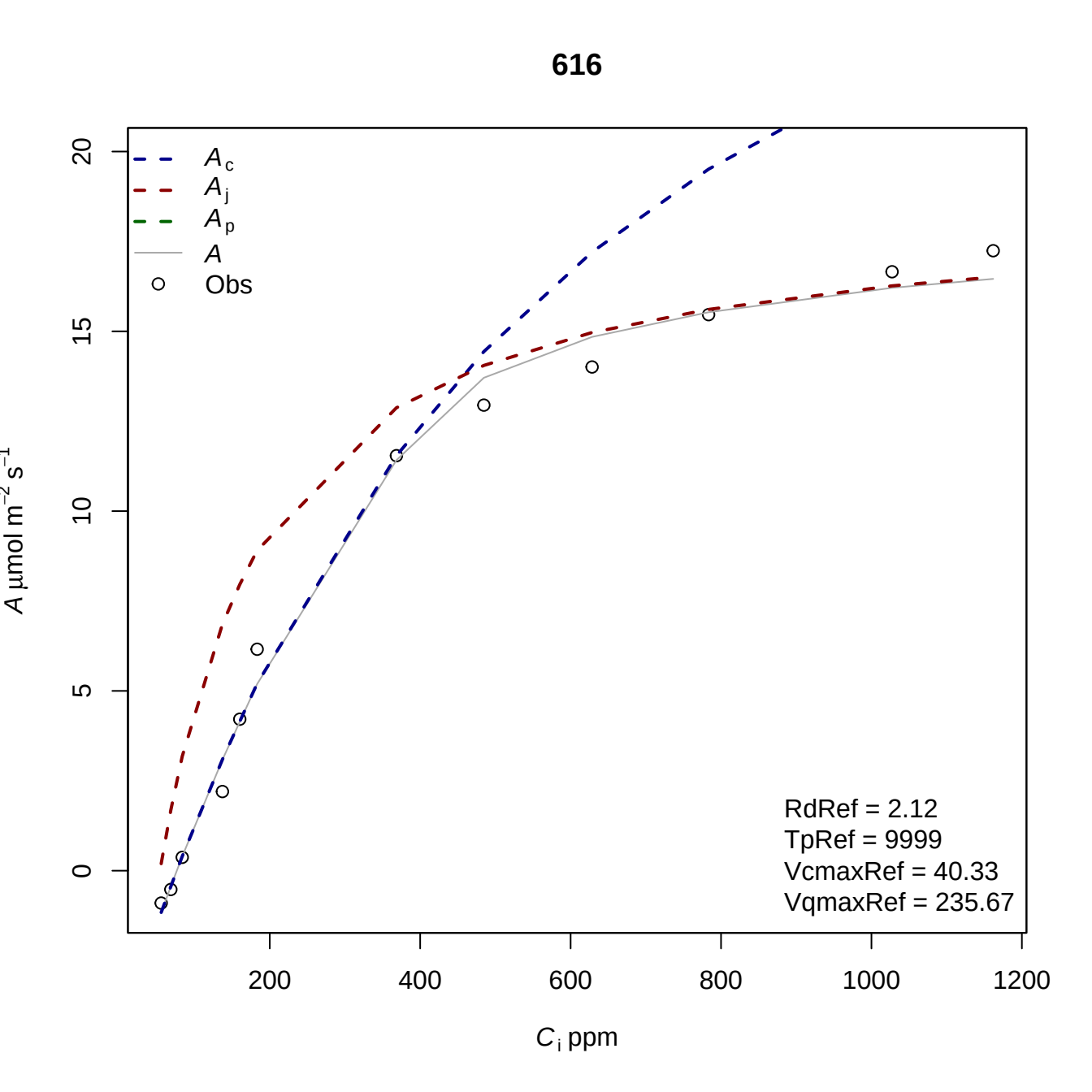

611
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.65
TpRef = 9999
VcmaxRef = 15
VqmaxRef = 44
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

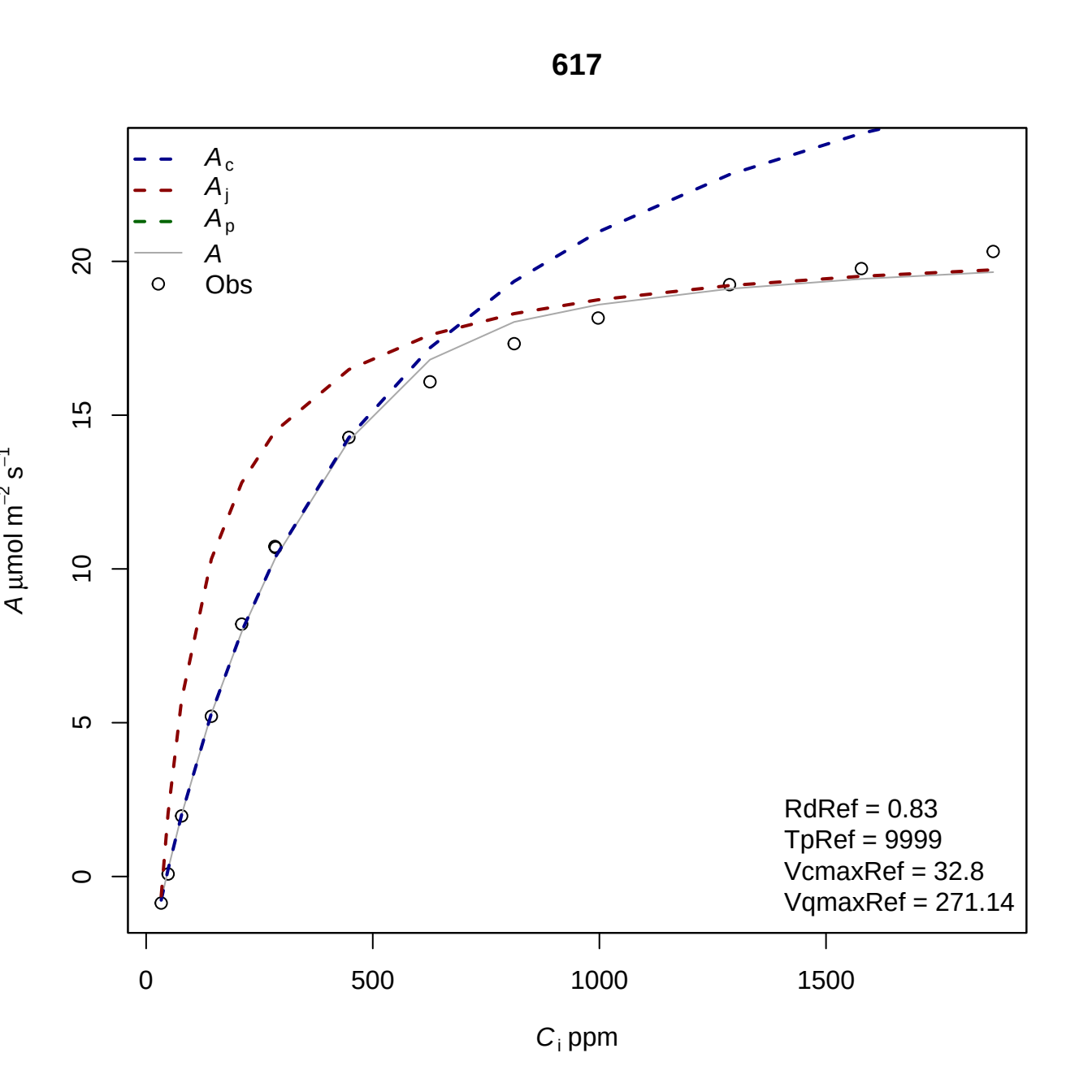

612
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.87
TpRef = 9999
VcmaxRef = 13.81
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

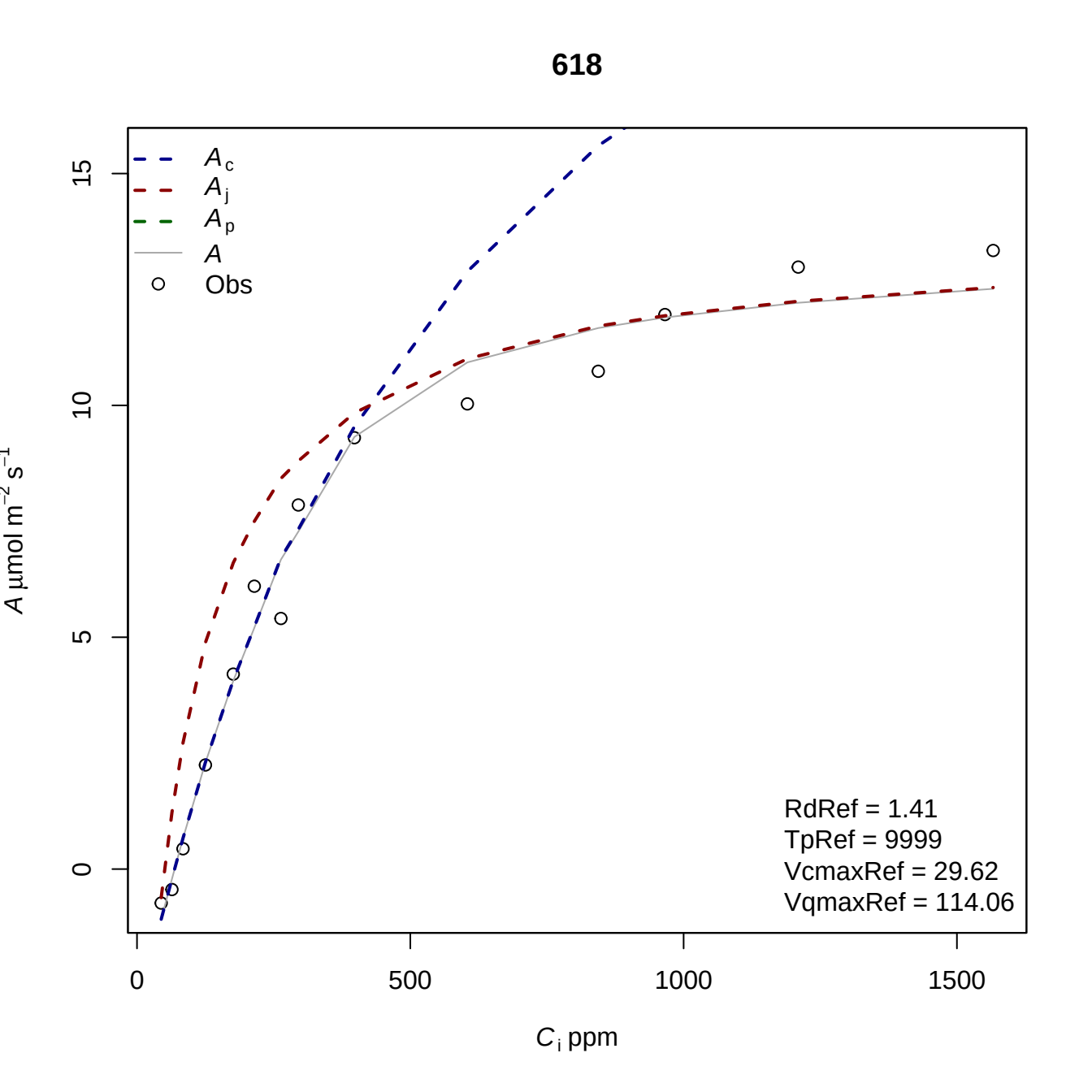

613
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.59
TpRef = 9999
VcmaxRef = 11.31
VqmaxRef = 38.04
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
0
500
1000
1500

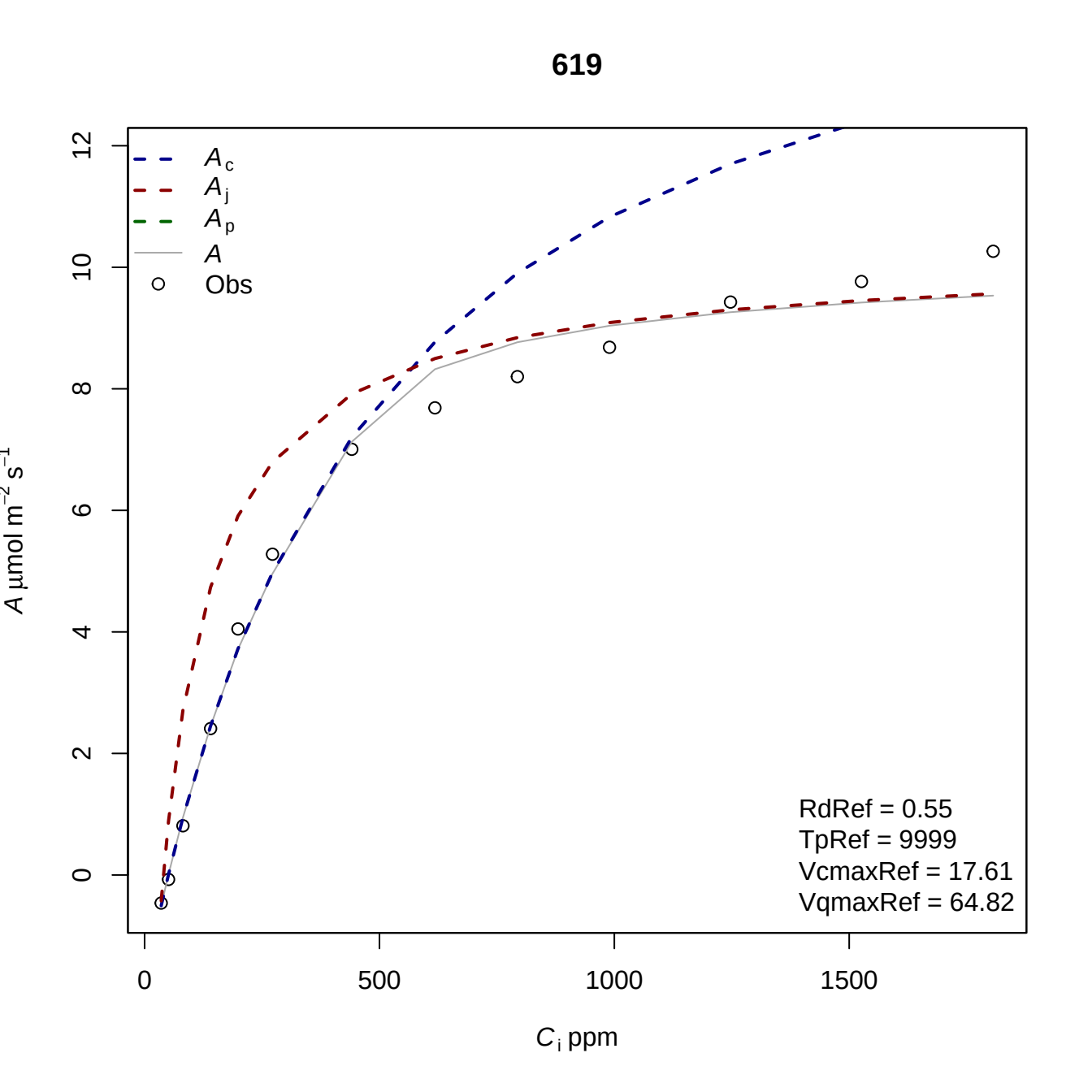

614
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.42
TpRef = 9999
VcmaxRef = 13.06
VqmaxRef = 49.85
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
500
1000
1500

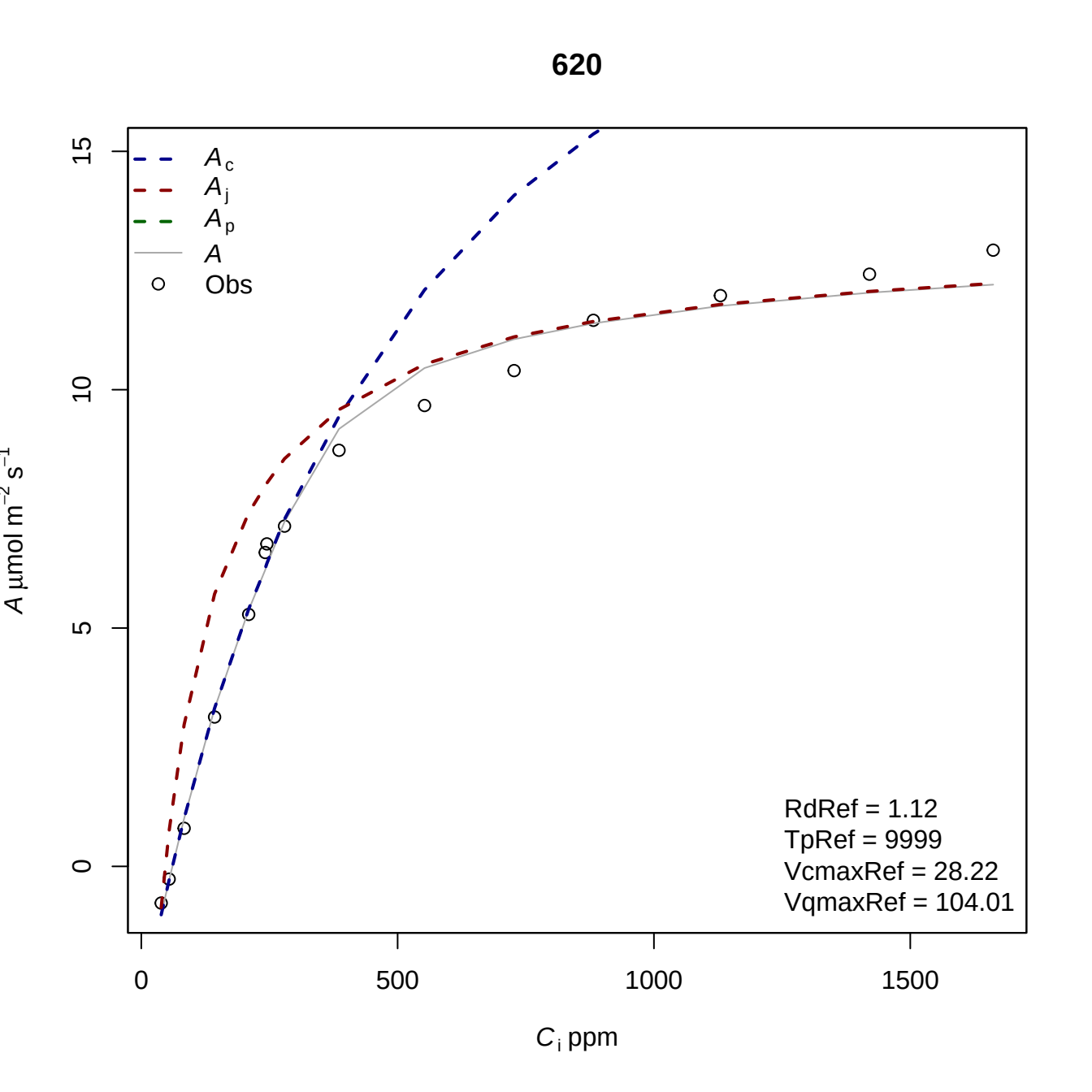

615
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.56
TpRef = 9999
VcmaxRef = 12.68
VqmaxRef = 43.95
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
0
500
1000
1500

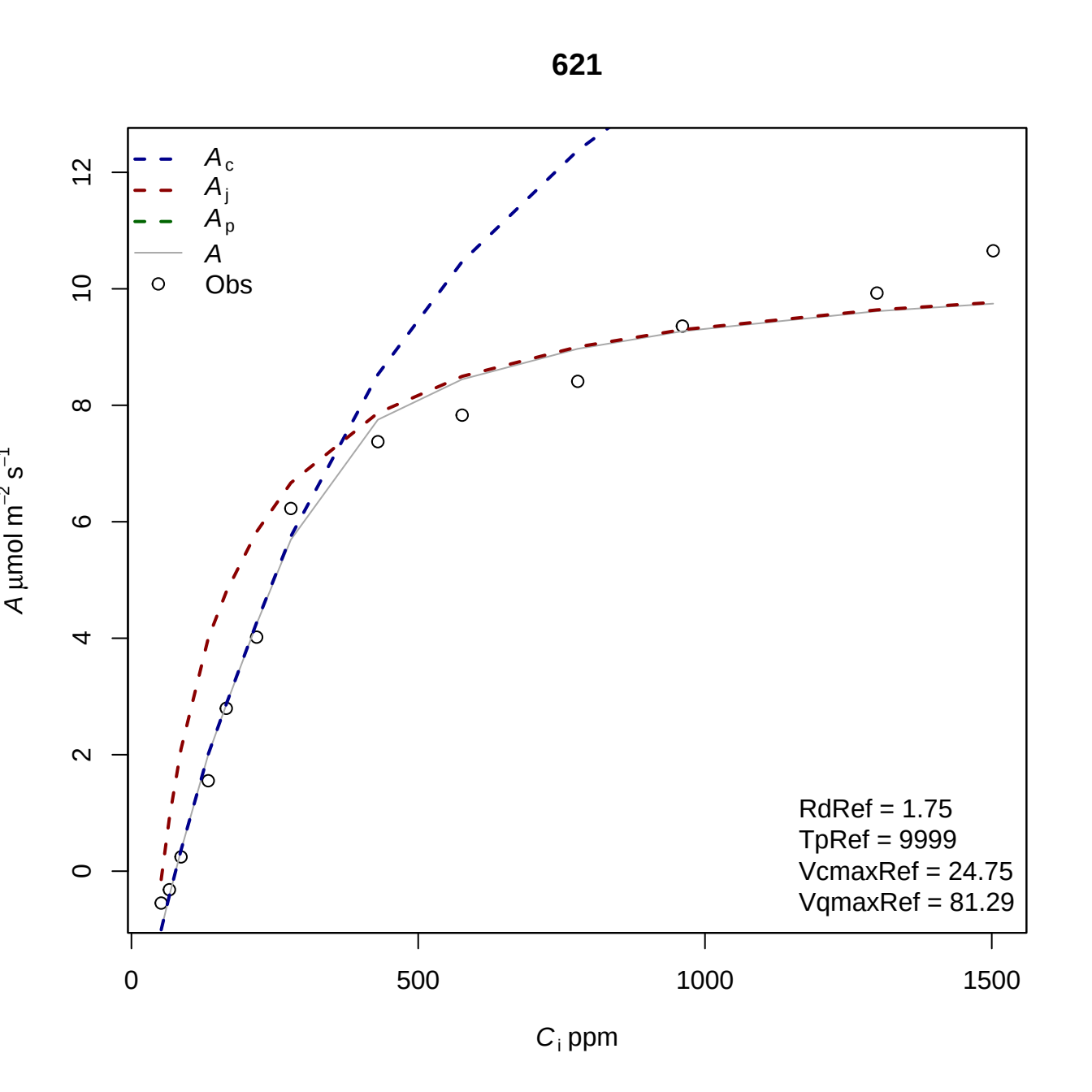

616
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 2.12
TpRef = 9999
VcmaxRef = 40.33
VqmaxRef = 235.67
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
200
400
600
800
1000
1200

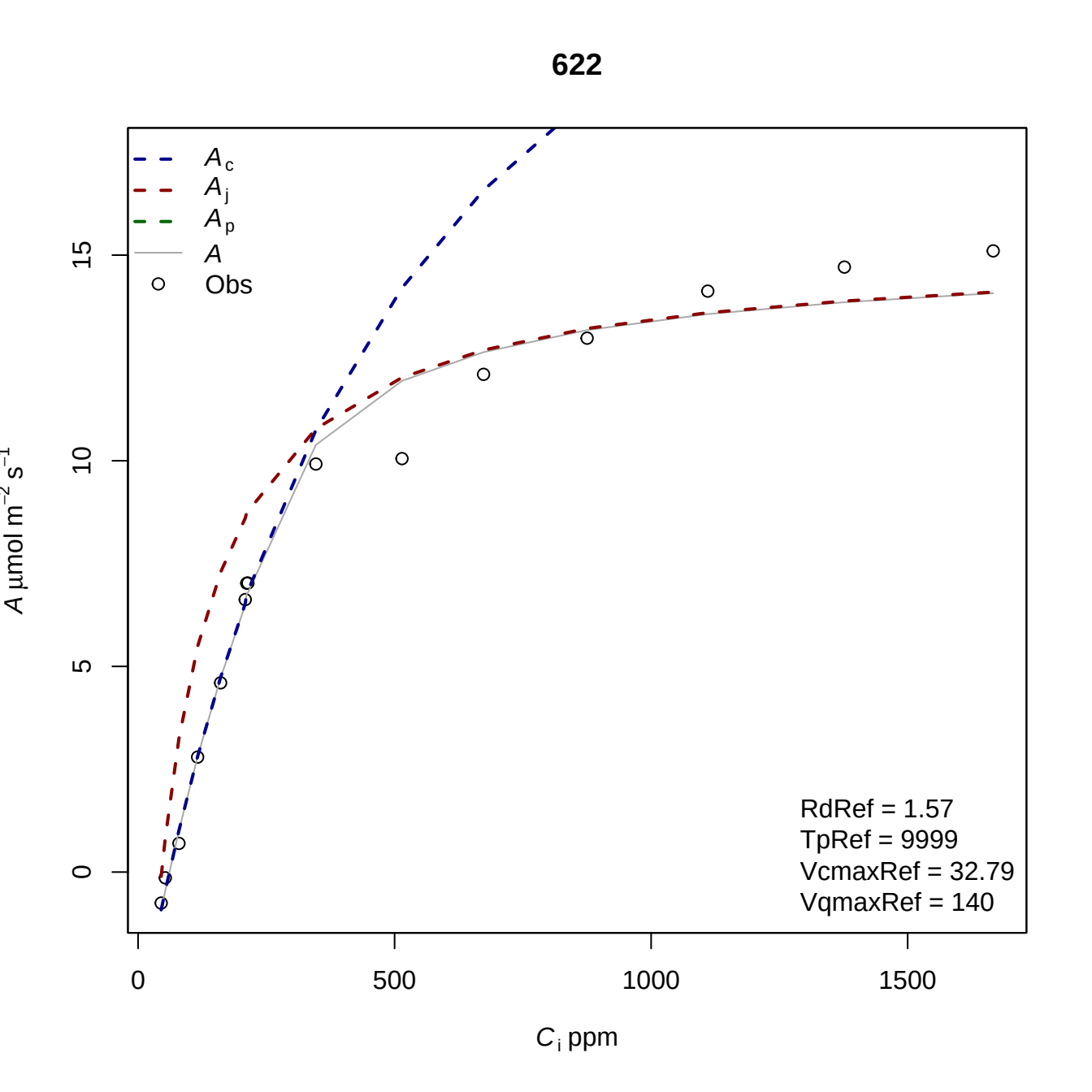

617
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.83
TpRef = 9999
VcmaxRef = 32.8
VqmaxRef = 271.14
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
500
1000
1500

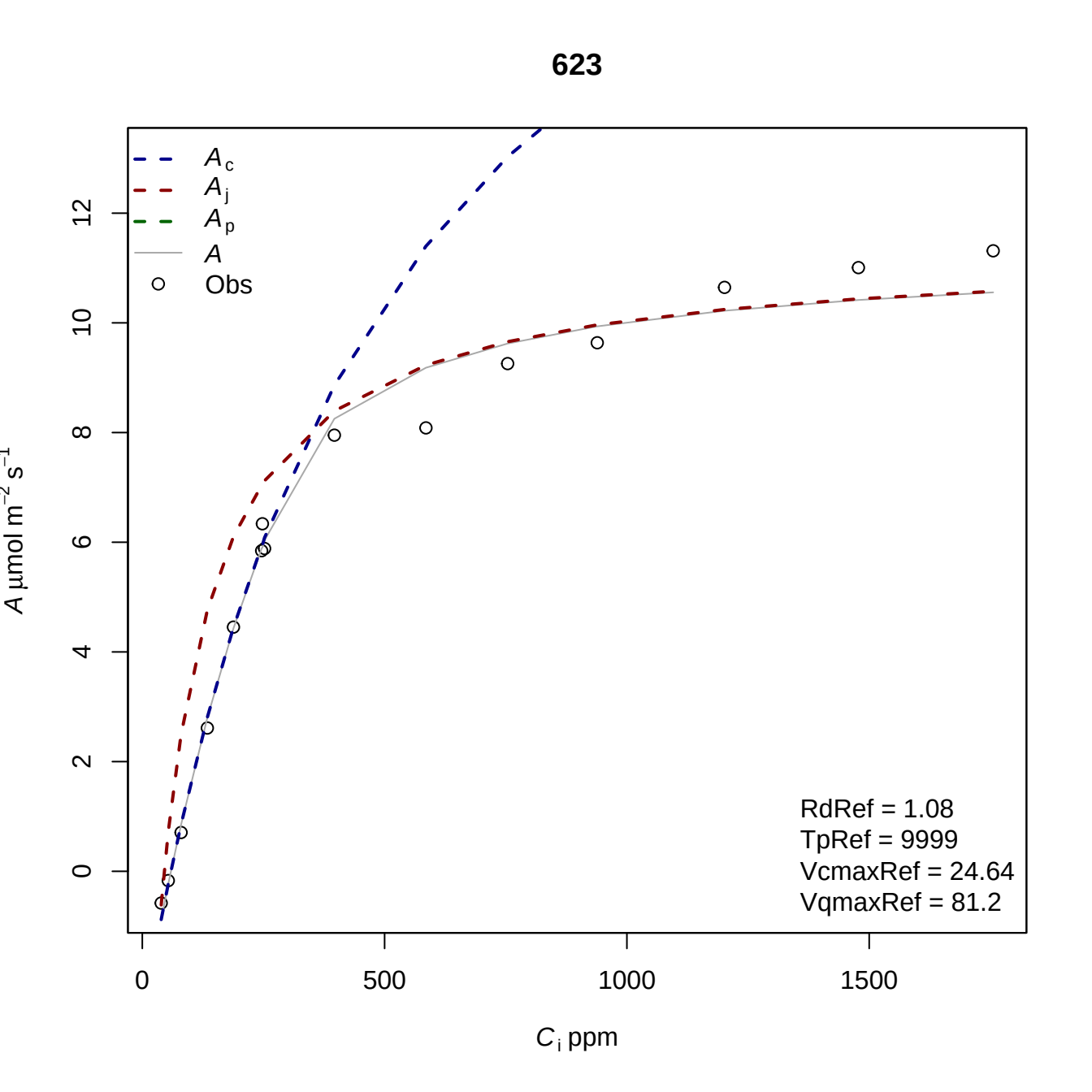

618
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.41
TpRef = 9999
VcmaxRef = 29.62
VqmaxRef = 114.06
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

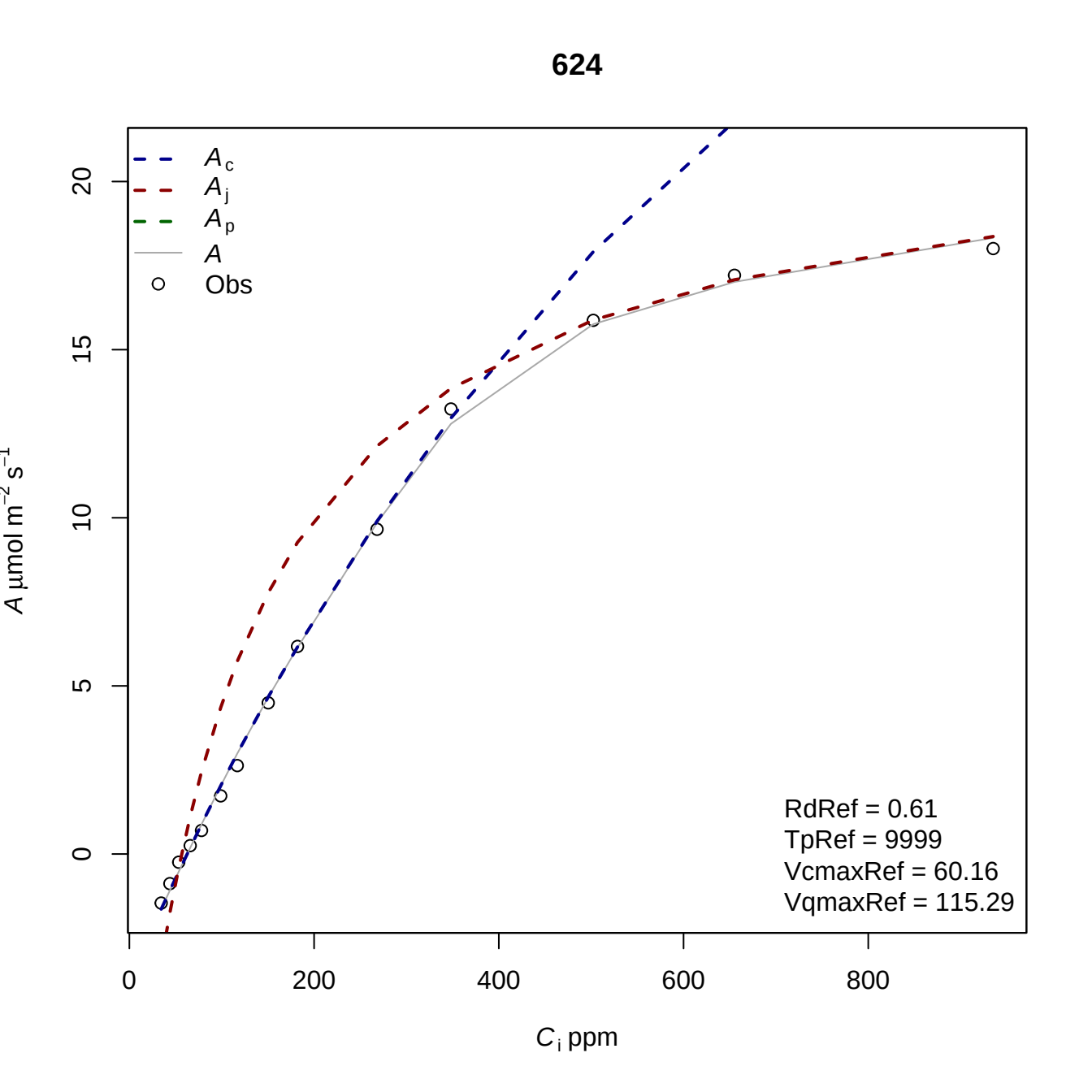

619
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.55
TpRef = 9999
VcmaxRef = 17.61
VqmaxRef = 64.82
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

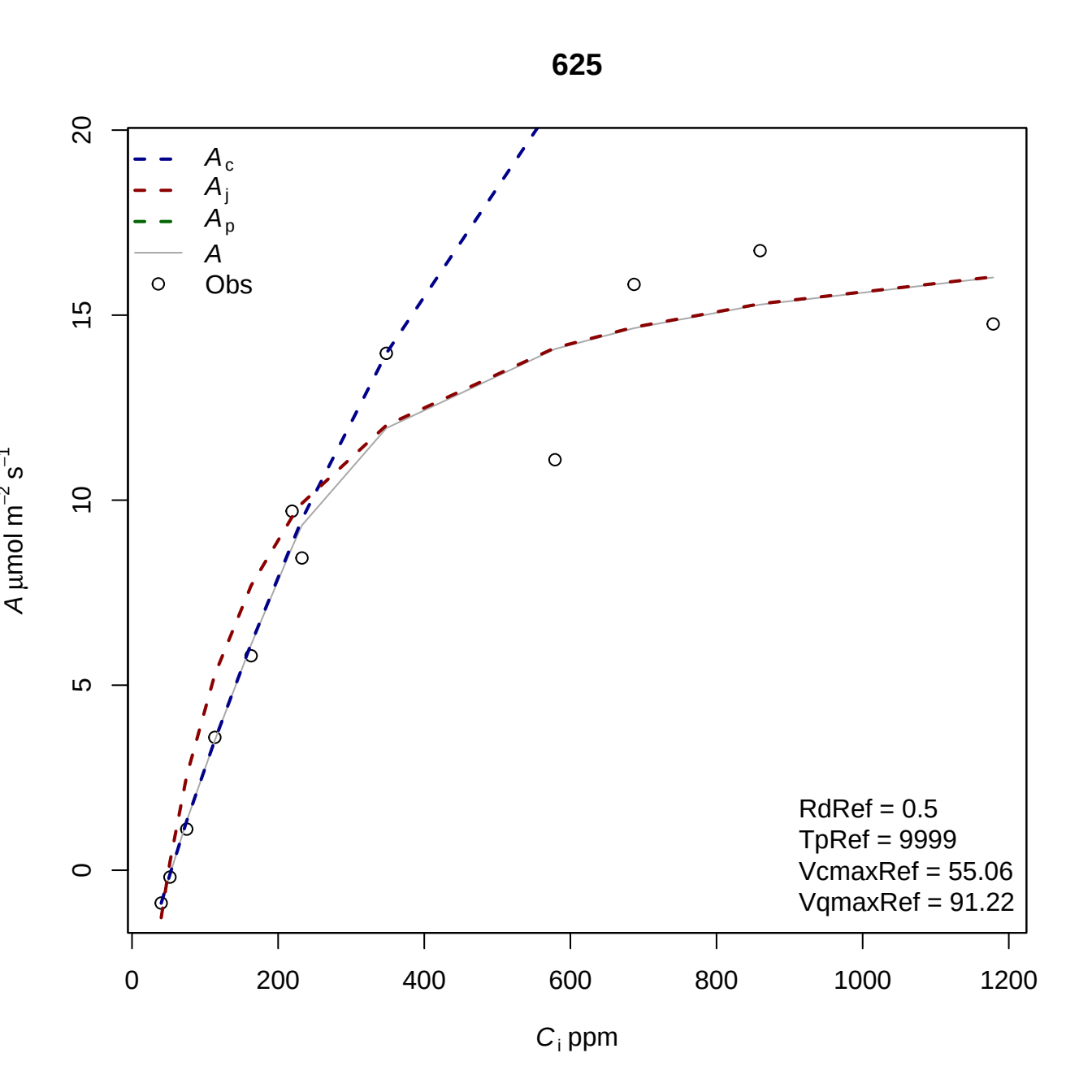

620
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.12
TpRef = 9999
VcmaxRef = 28.22
VqmaxRef = 104.01
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

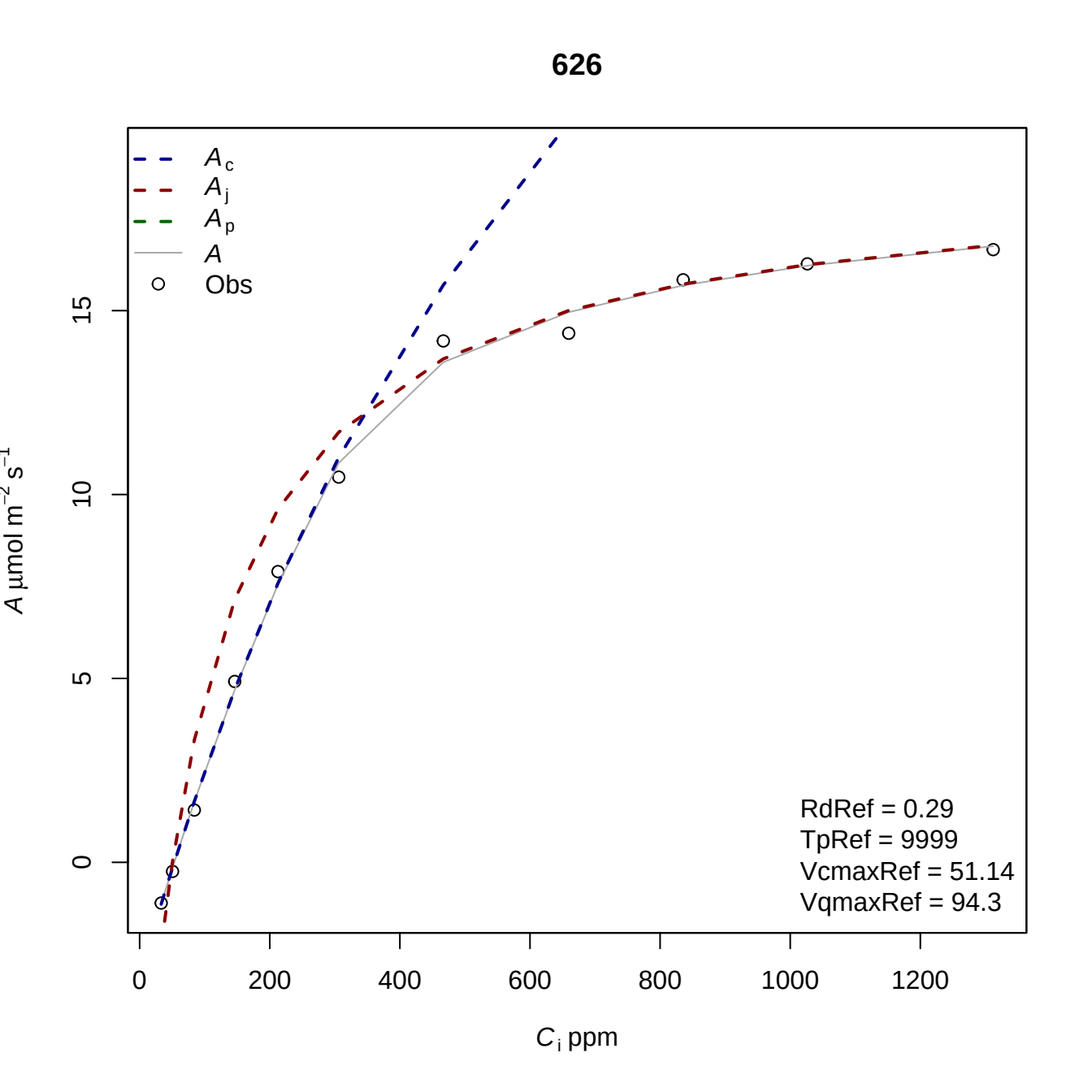

621
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.75
TpRef = 9999
VcmaxRef = 24.75
VqmaxRef = 81.29
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

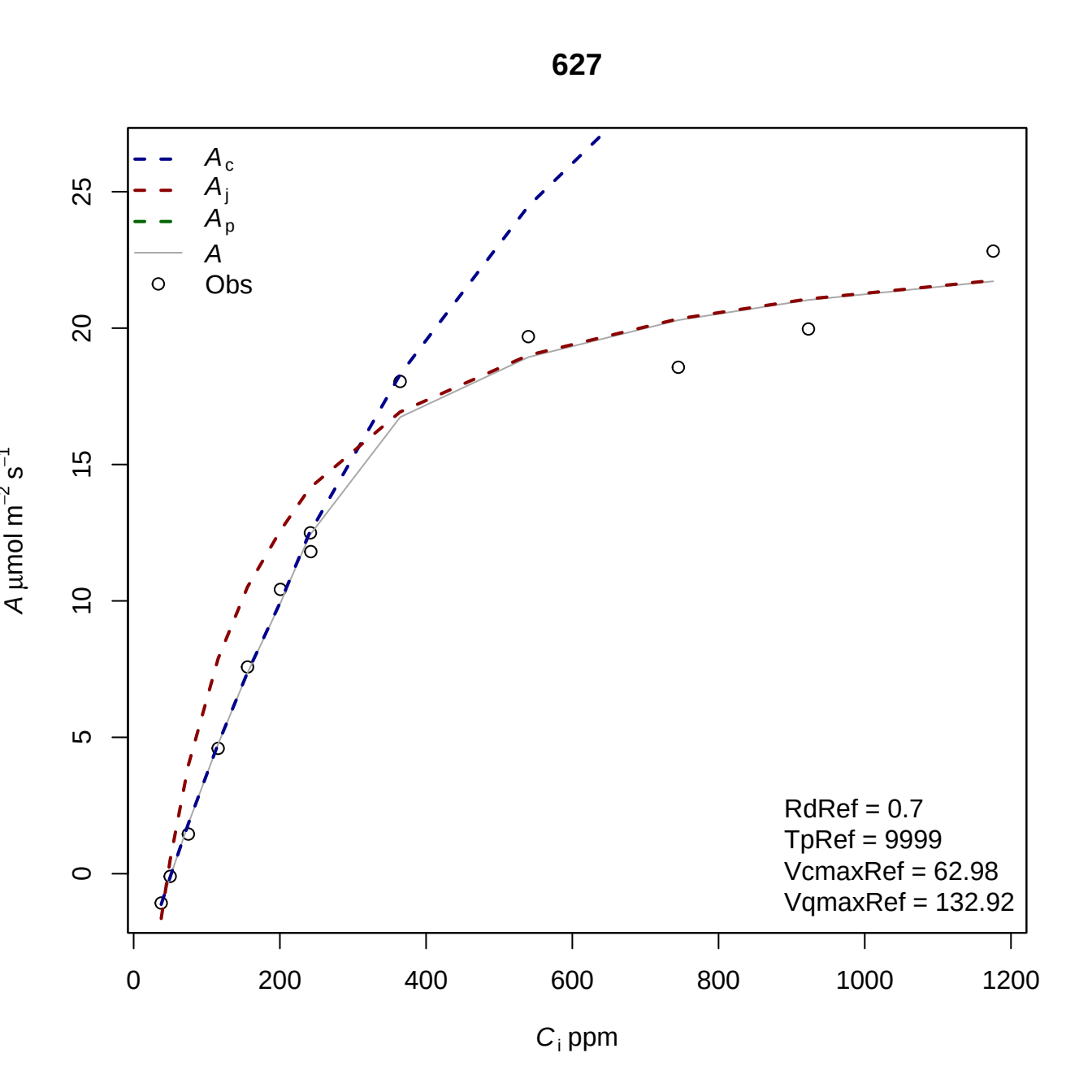

622
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.57
TpRef = 9999
VcmaxRef = 32.79
VqmaxRef = 140
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
500
1000
1500

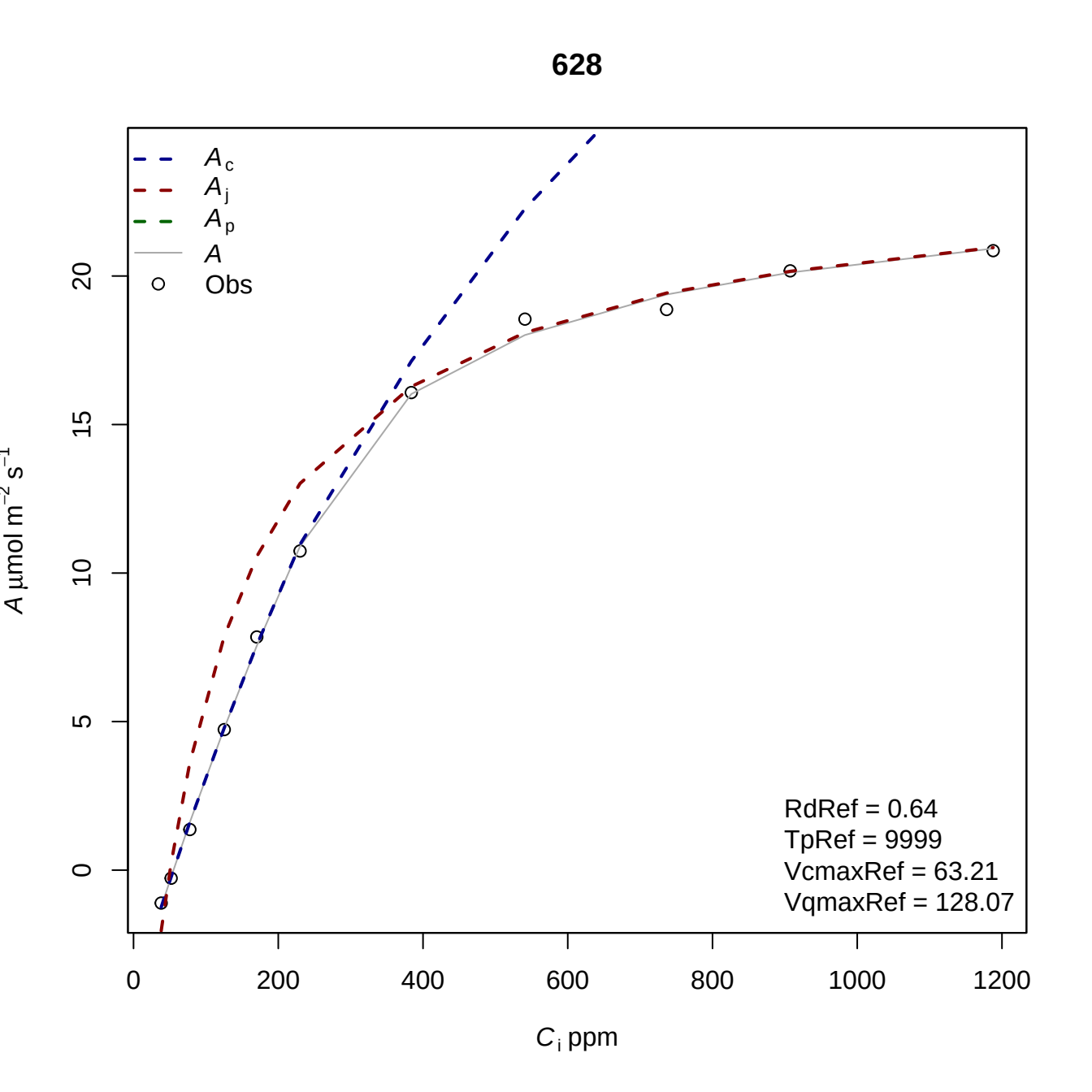

623
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.08
TpRef = 9999
VcmaxRef = 24.64
VqmaxRef = 81.2
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500

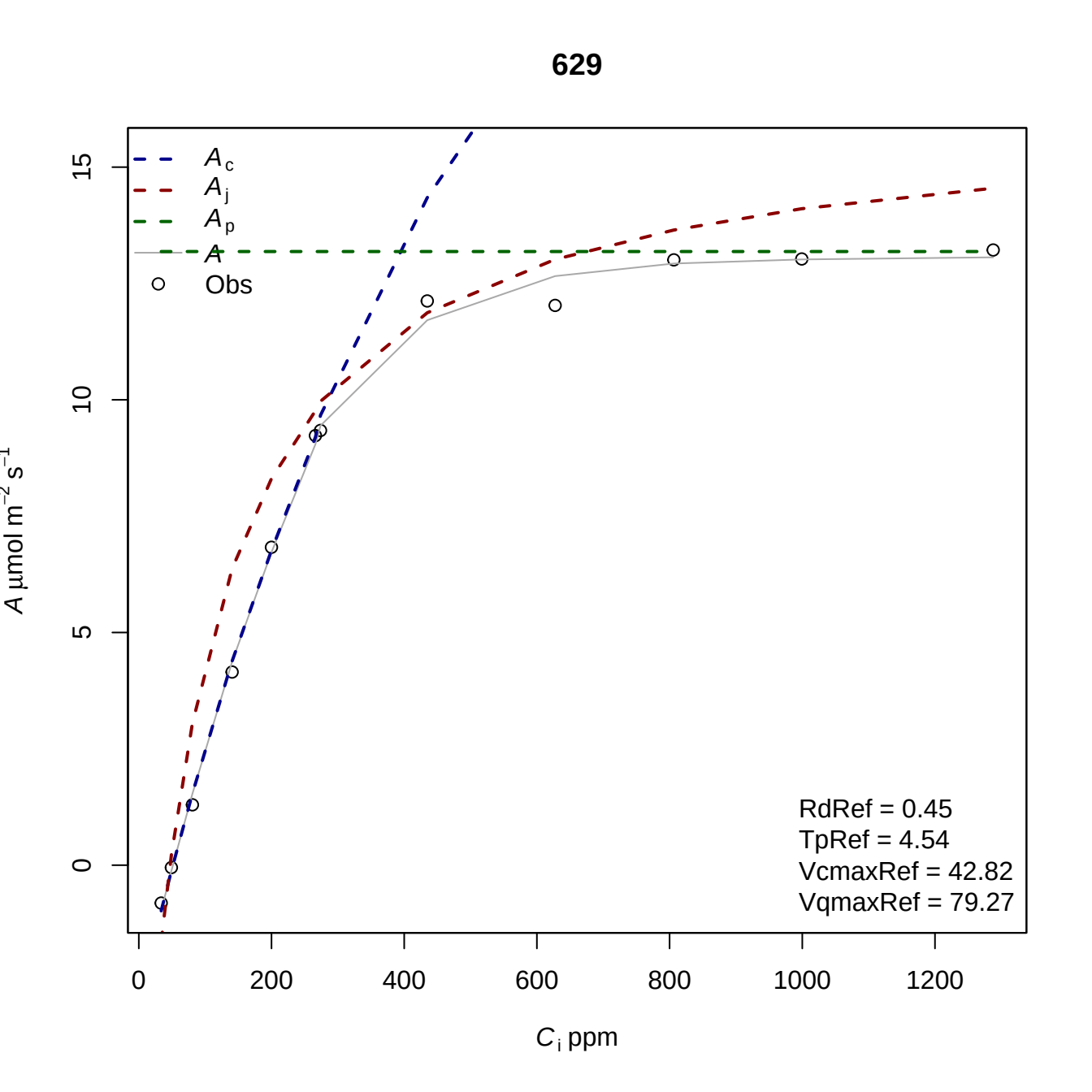

624
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.61
TpRef = 9999
VcmaxRef = 60.16
VqmaxRef = 115.29
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800

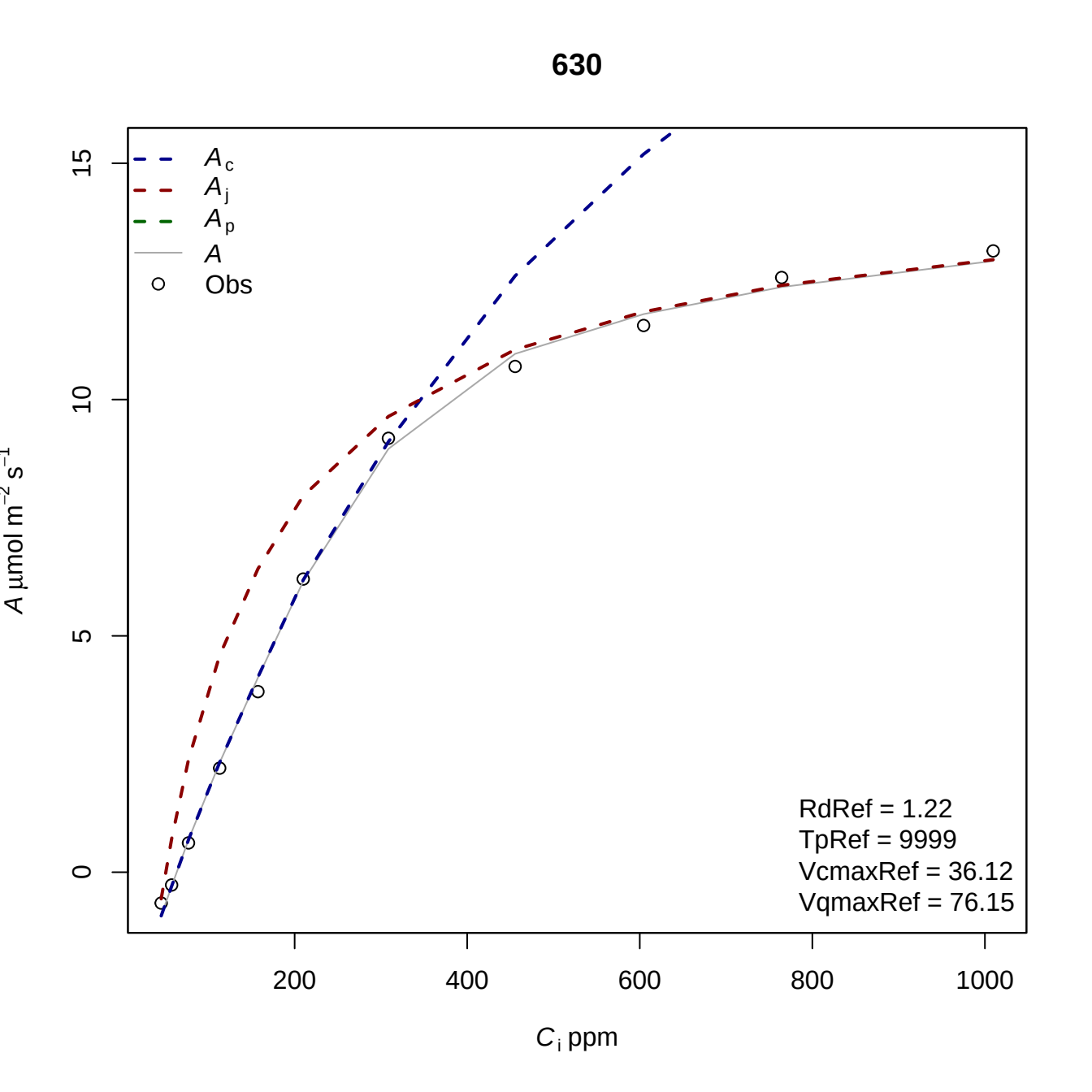

625
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.5
TpRef = 9999
VcmaxRef = 55.06
VqmaxRef = 91.22
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

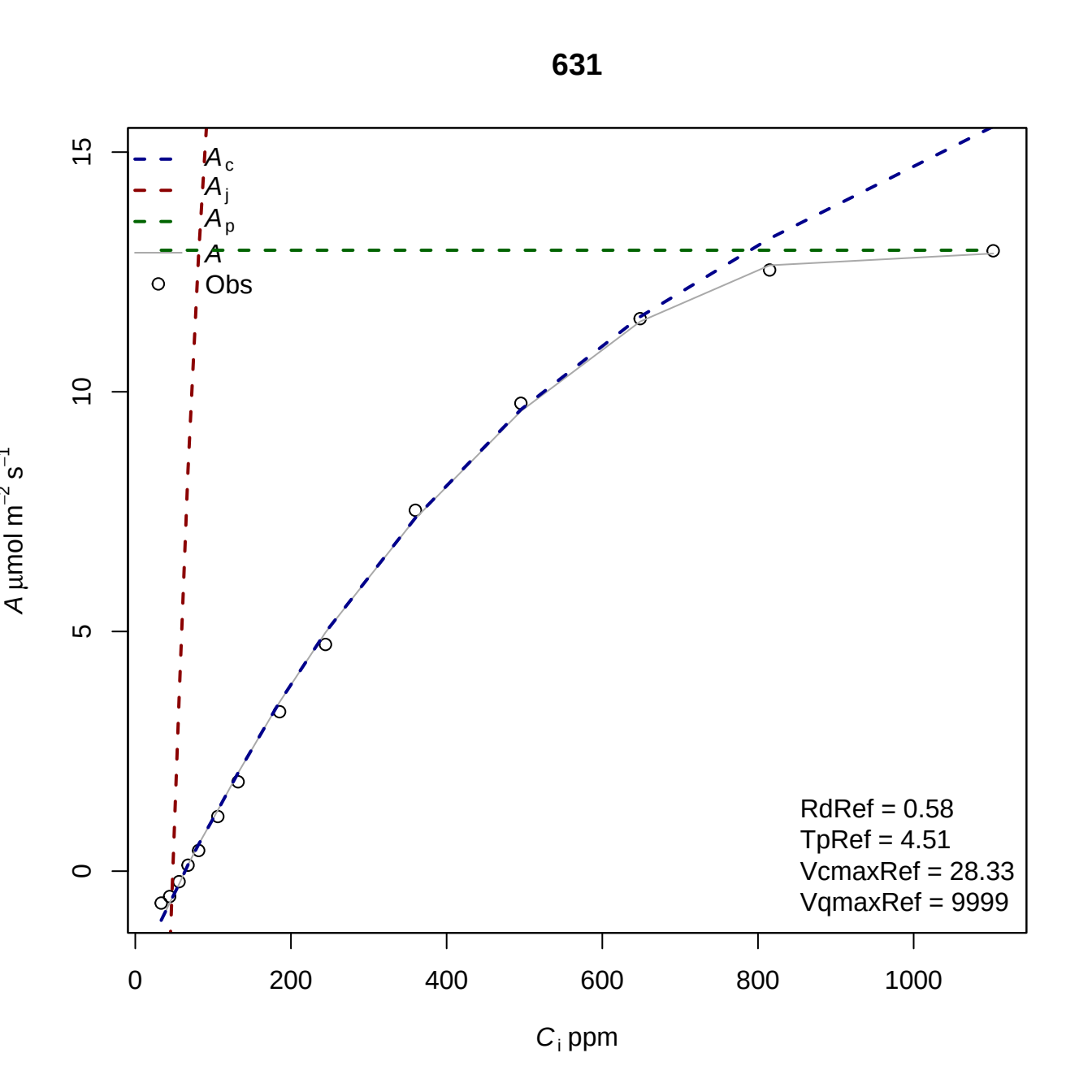

626
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.29
TpRef = 9999
VcmaxRef = 51.14
VqmaxRef = 94.3
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

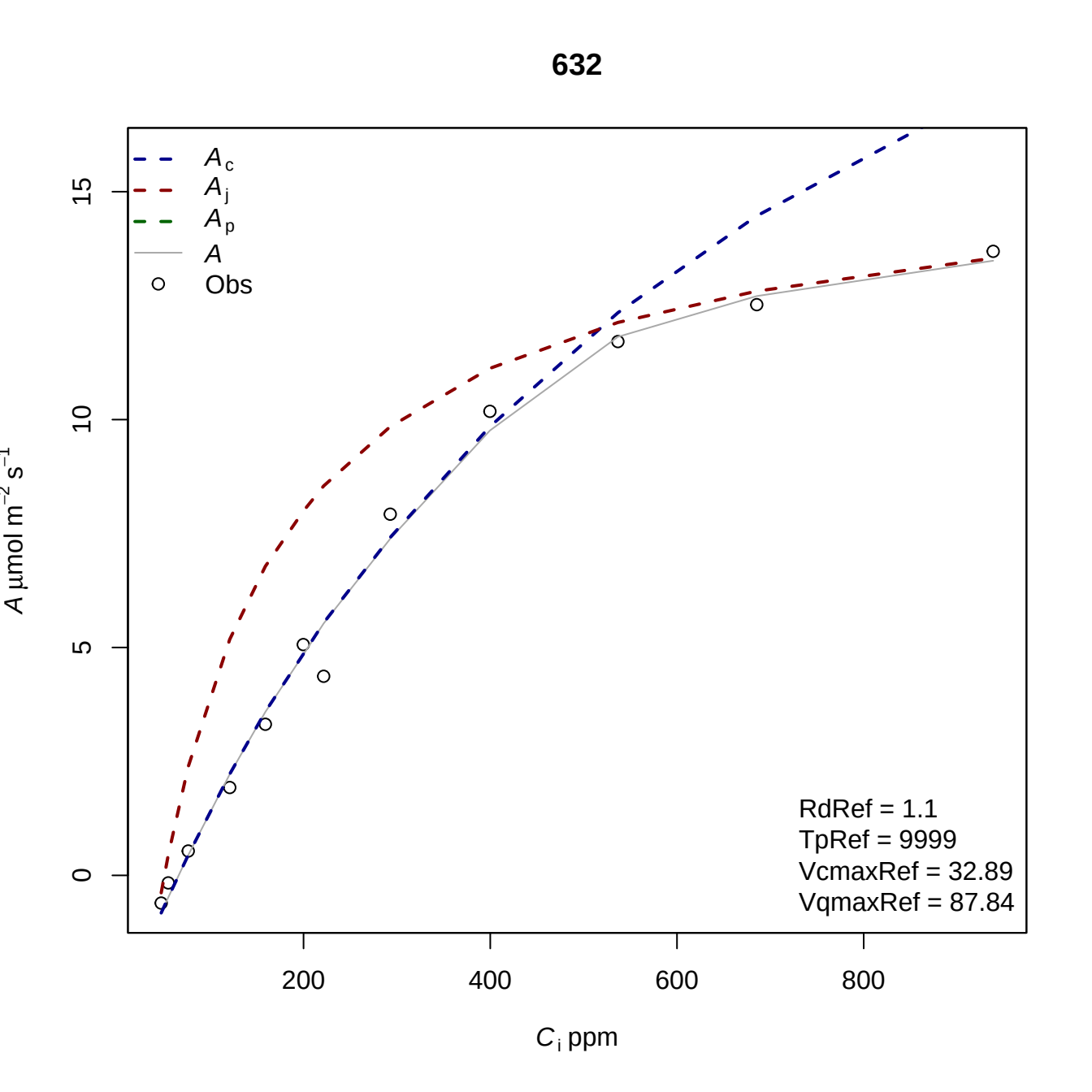

627
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.7
TpRef = 9999
VcmaxRef = 62.98
VqmaxRef = 132.92
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

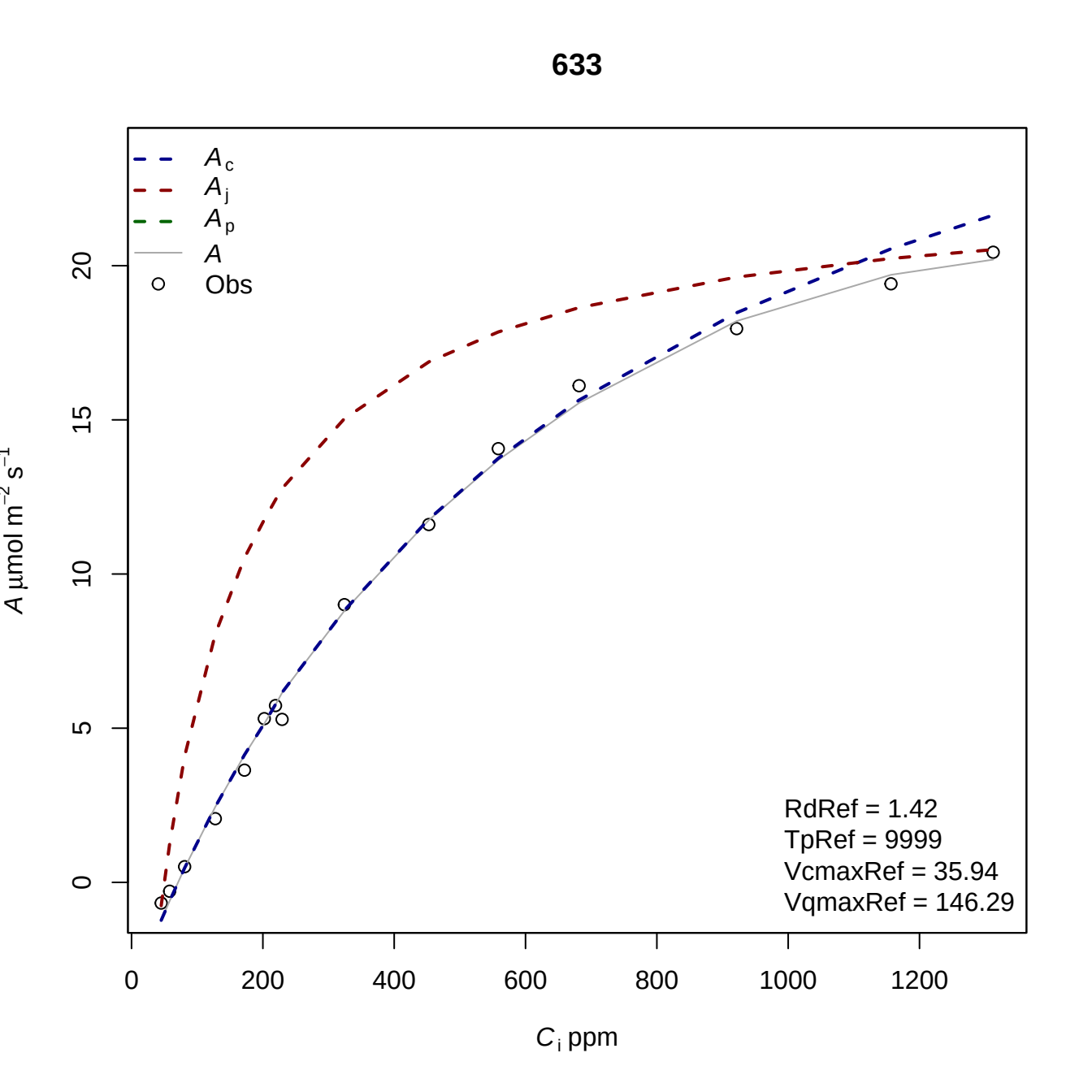

628
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.64
TpRef = 9999
VcmaxRef = 63.21
VqmaxRef = 128.07
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

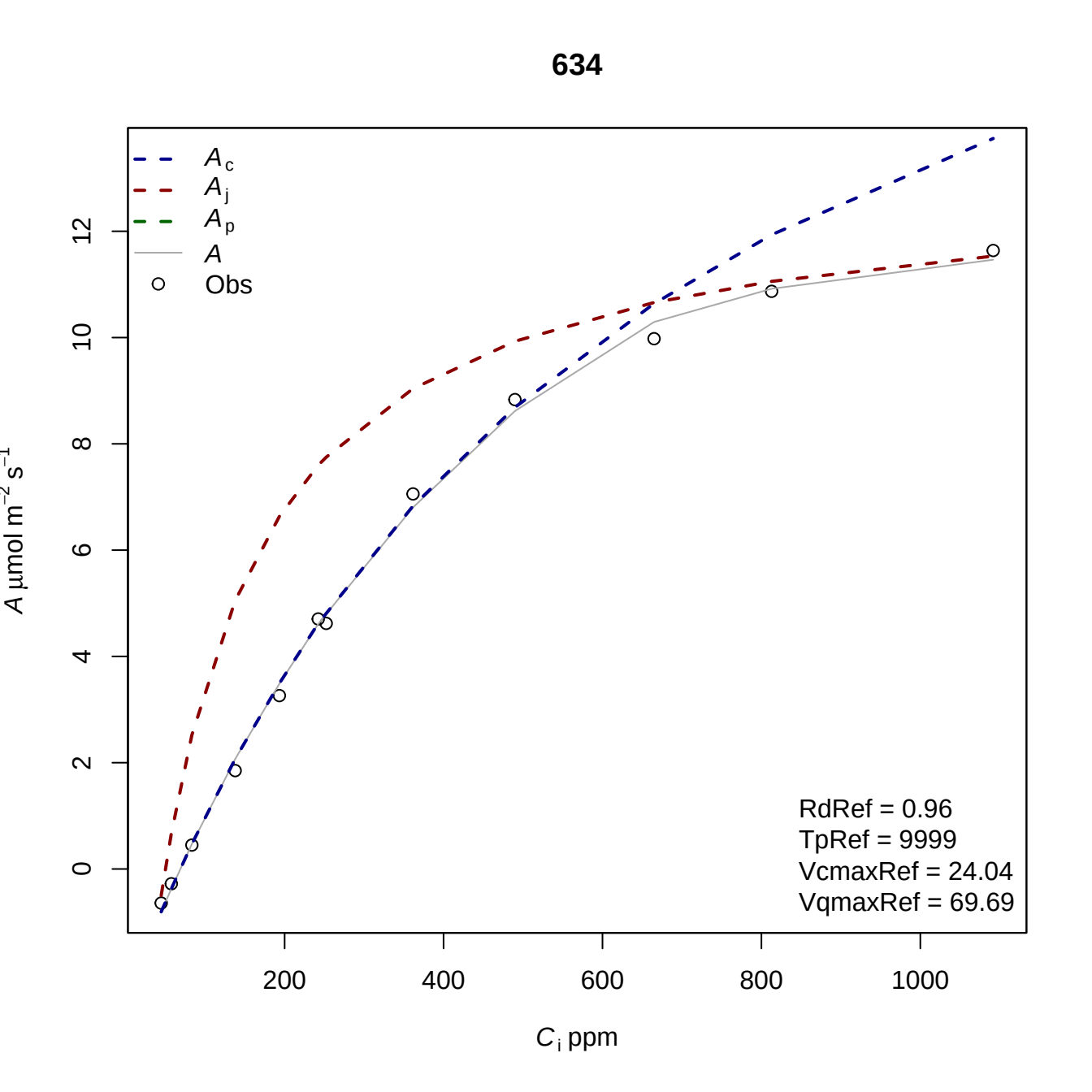

629
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.45
TpRef = 4.54
VcmaxRef = 42.82
VqmaxRef = 79.27
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000
1200

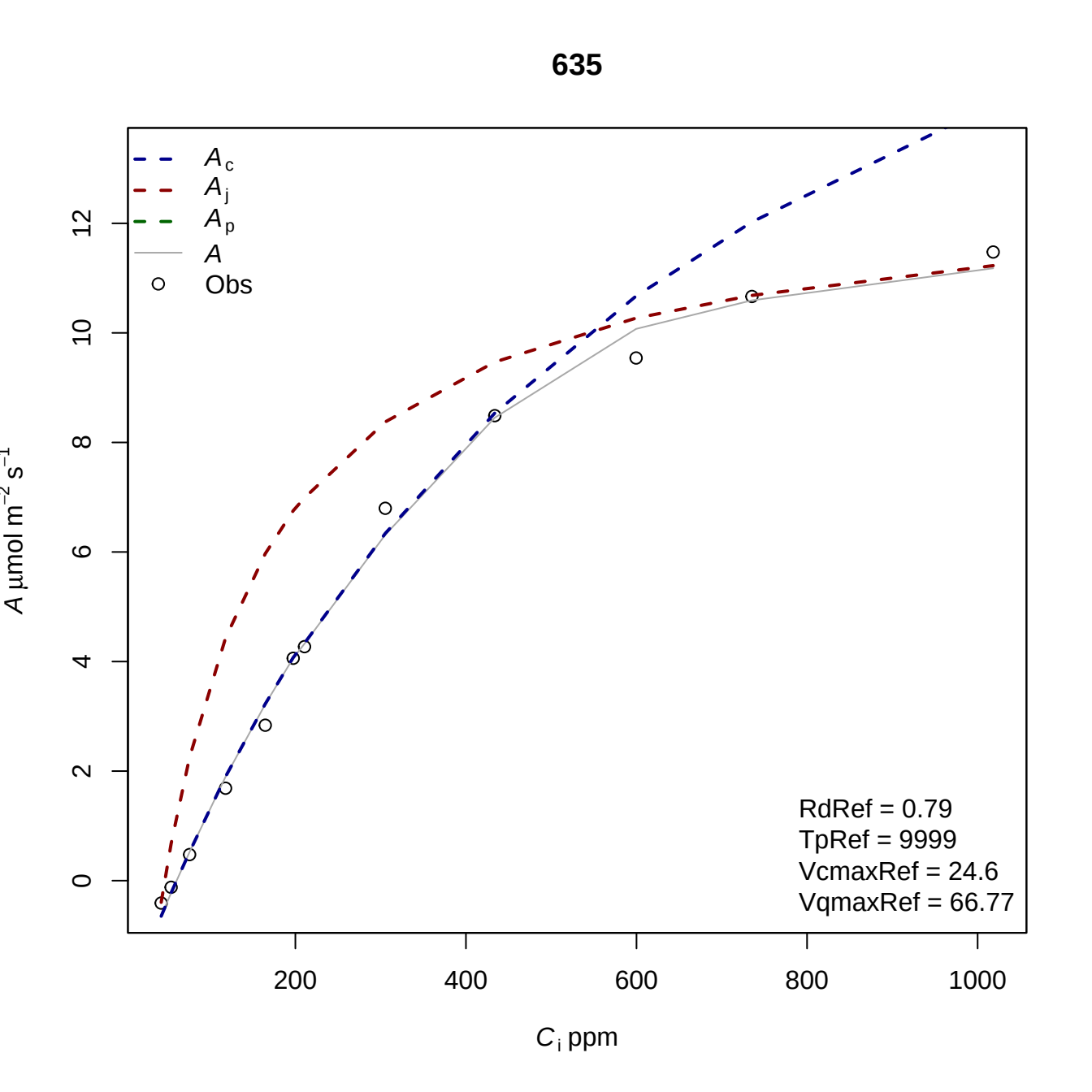

630
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.22
TpRef = 9999
VcmaxRef = 36.12
VqmaxRef = 76.15
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000

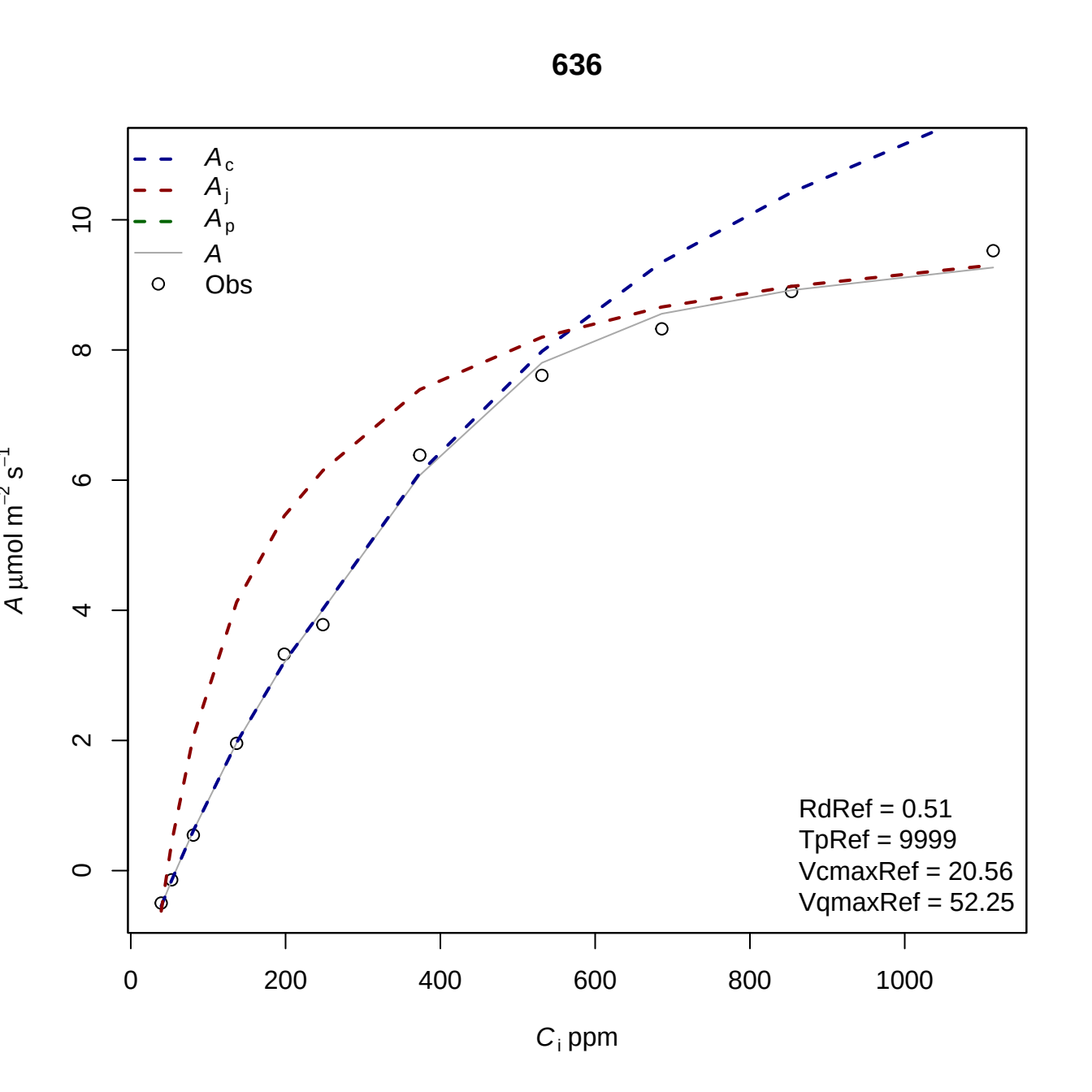

631
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.58
TpRef = 4.51
VcmaxRef = 28.33
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
0
200
400
600
800
1000

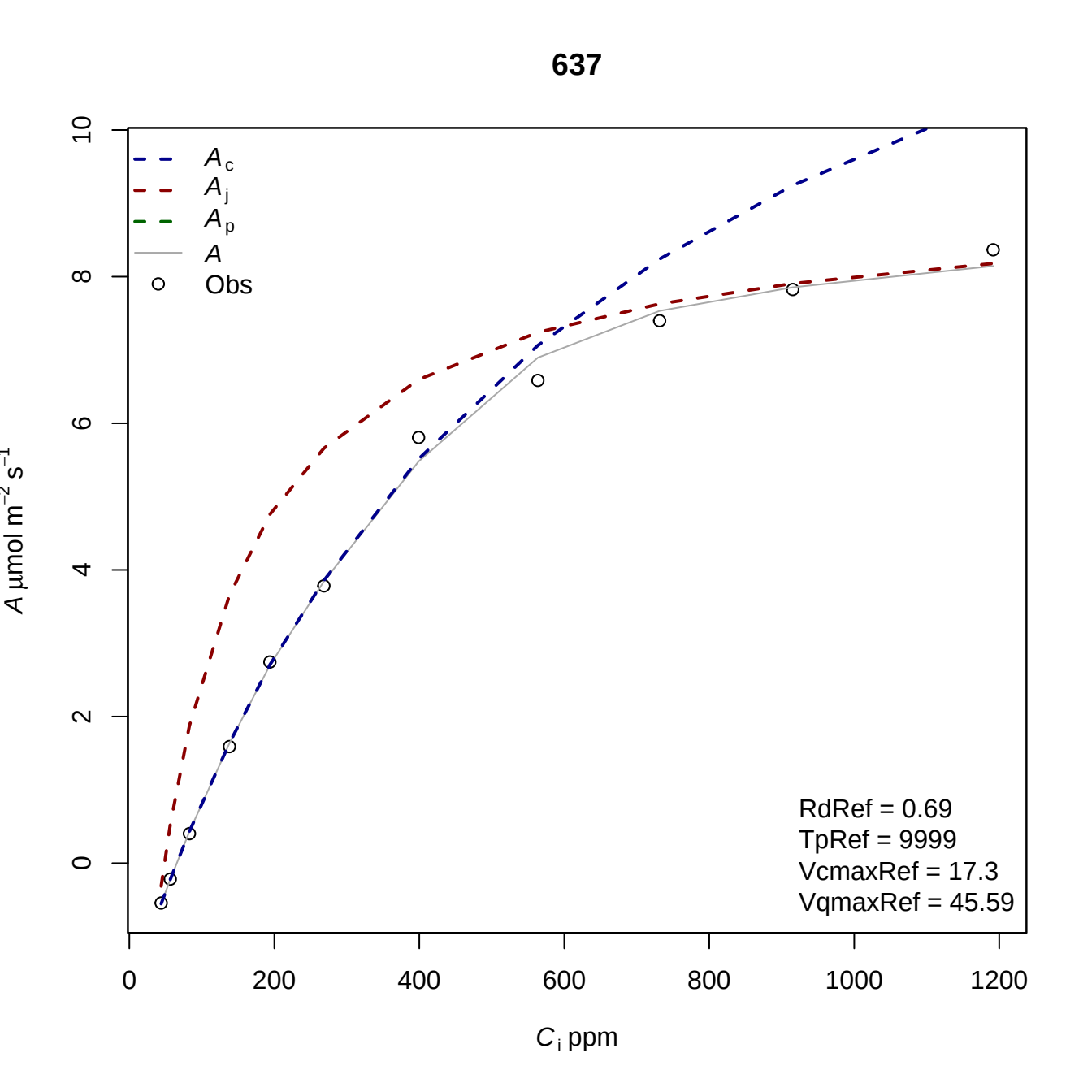

632
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.1
TpRef = 9999
VcmaxRef = 32.89
VqmaxRef = 87.84
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800

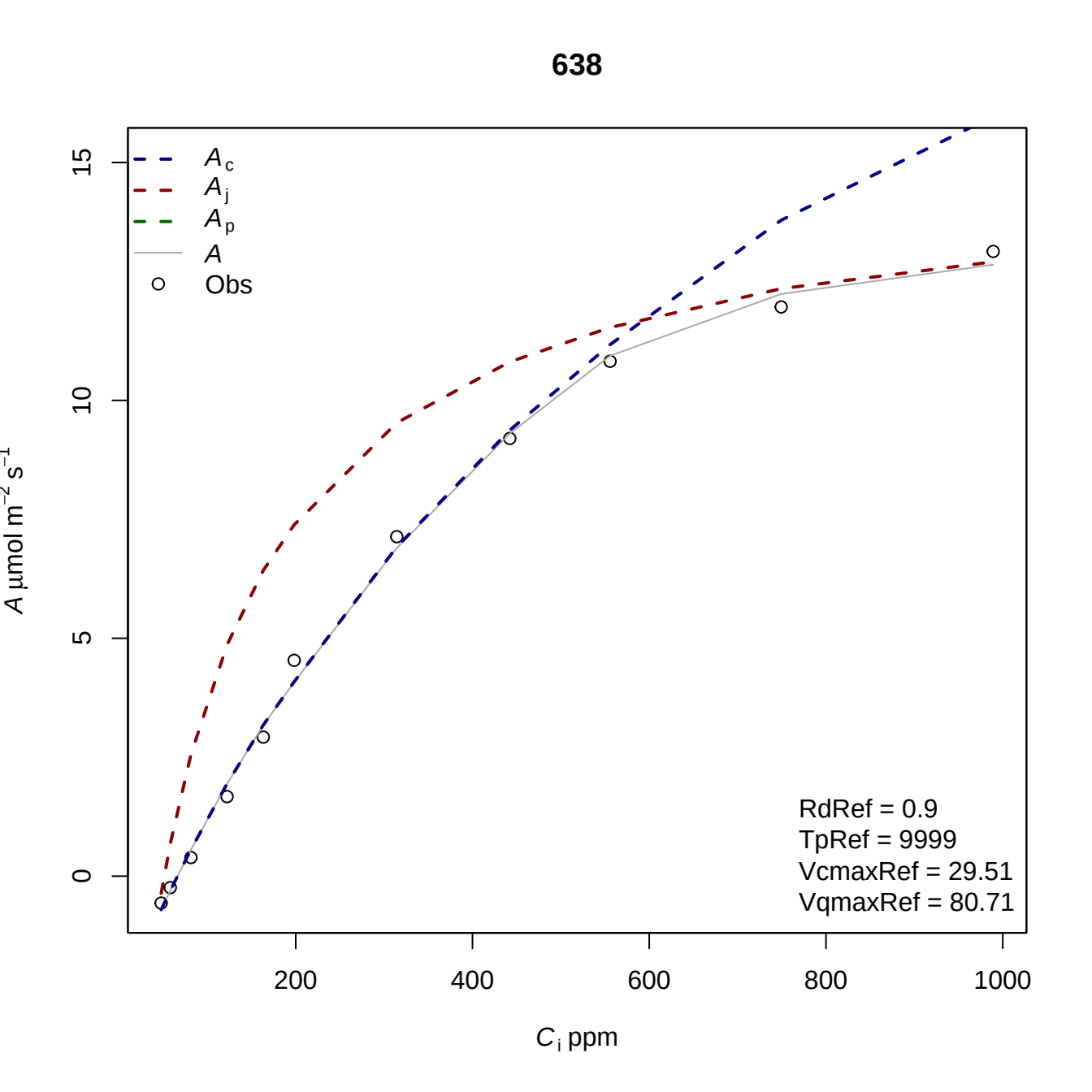

633
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.42
TpRef = 9999
VcmaxRef = 35.94
VqmaxRef = 146.29
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

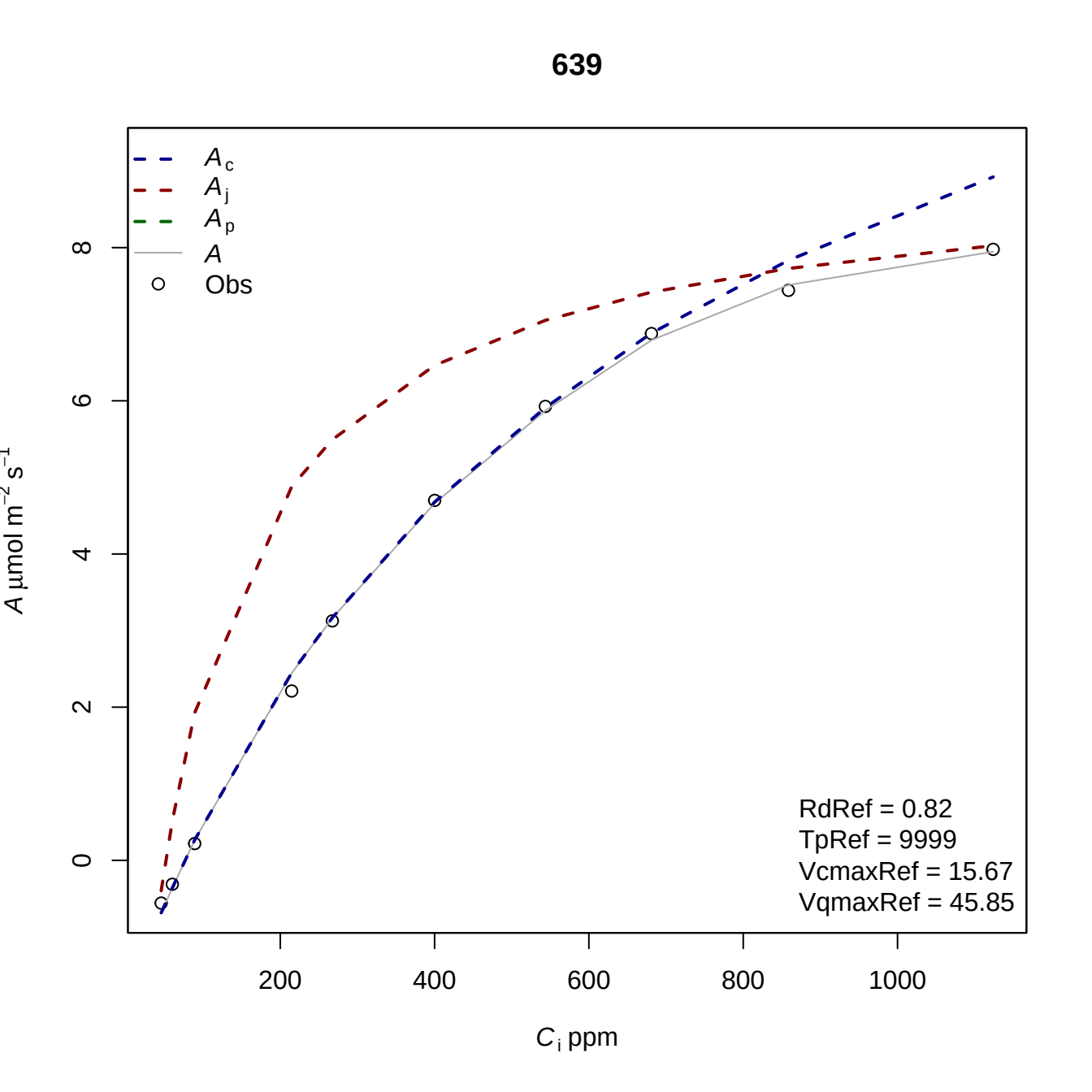

634
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.96
TpRef = 9999
VcmaxRef = 24.04
VqmaxRef = 69.69
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800
1000

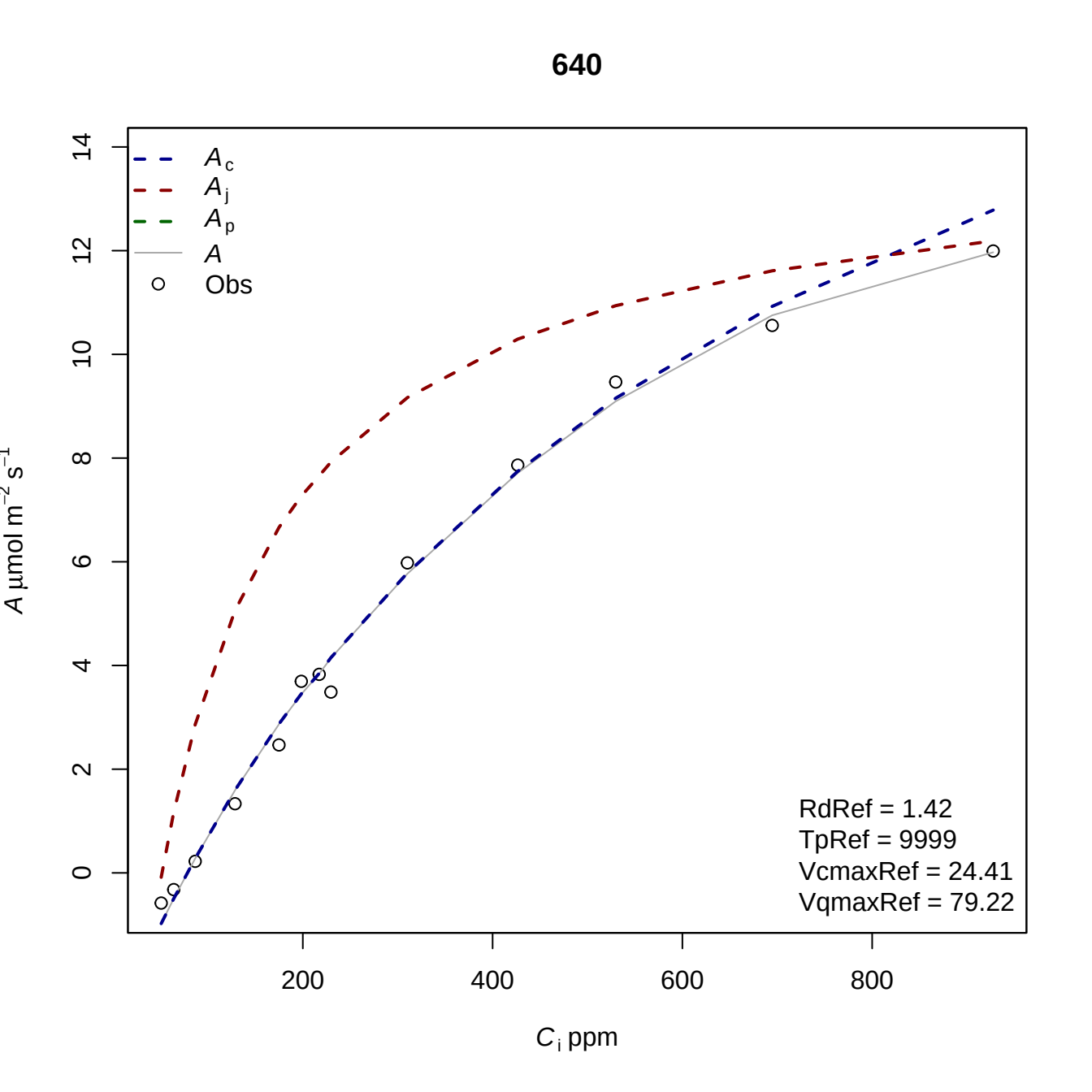

635
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.79
TpRef = 9999
VcmaxRef = 24.6
VqmaxRef = 66.77
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
200
400
600
800
1000

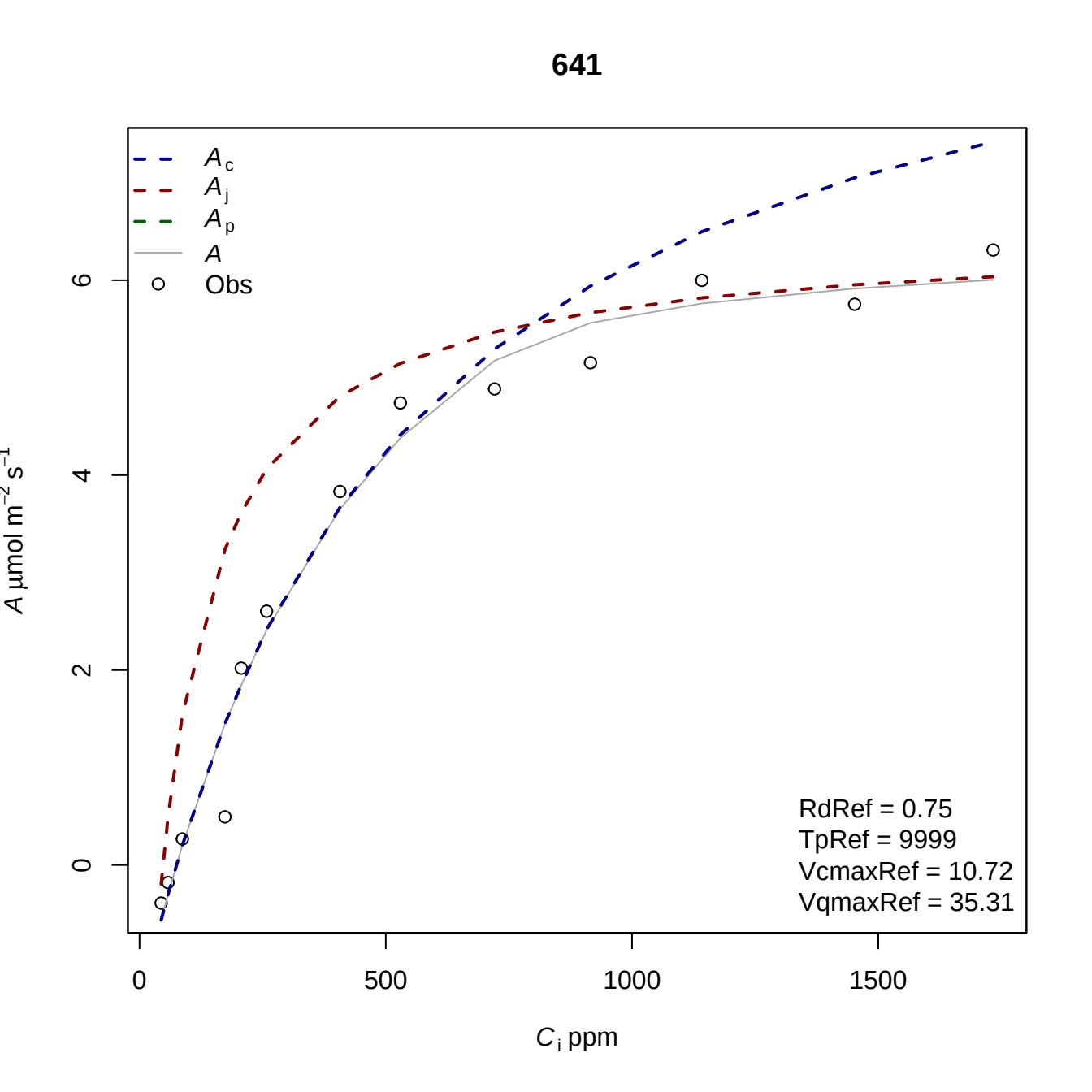

636
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.51
TpRef = 9999
VcmaxRef = 20.56
VqmaxRef = 52.25
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
200
400
600
800
1000

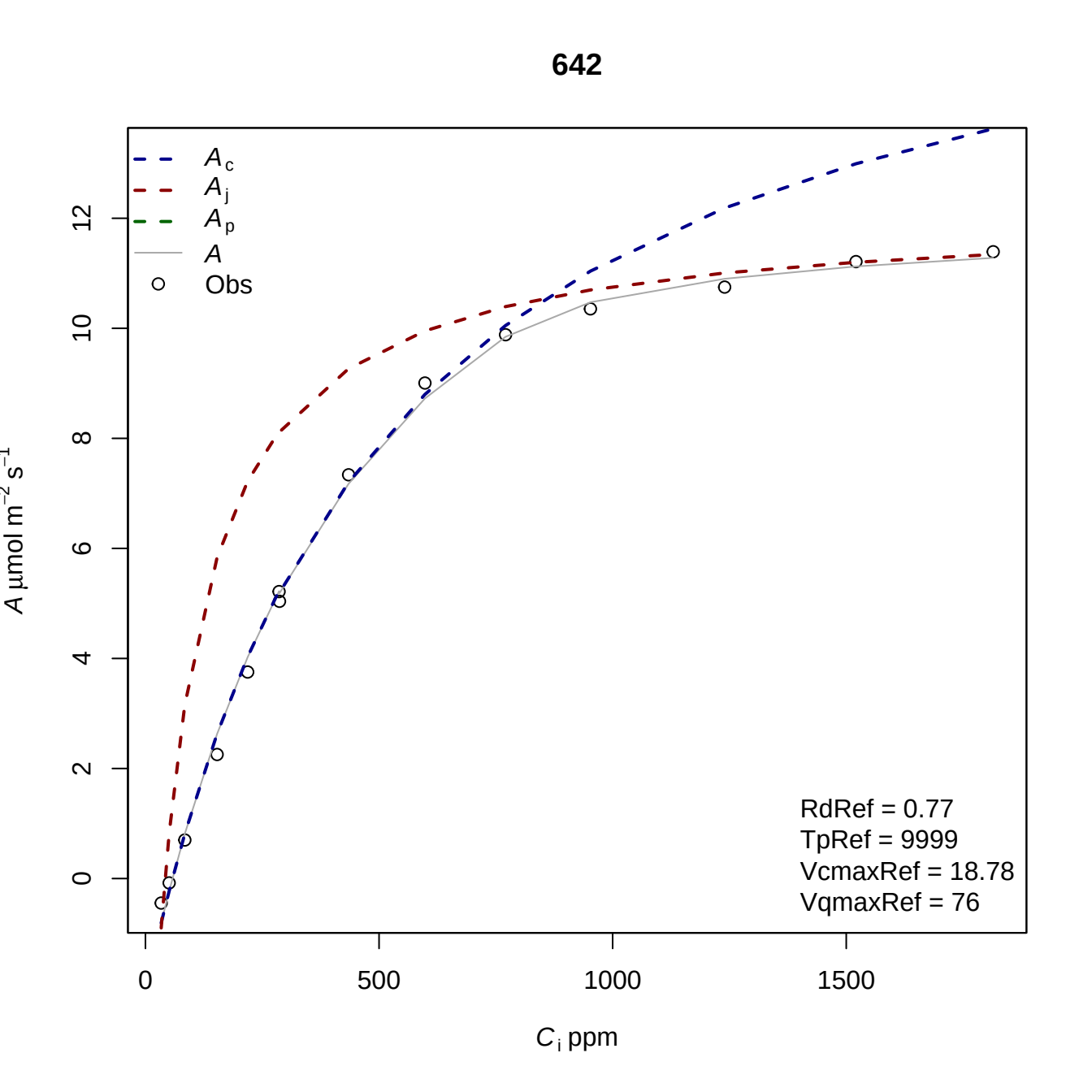

637
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.69
TpRef = 9999
VcmaxRef = 17.3
VqmaxRef = 45.59
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
0
200
400
600
800
1000
1200

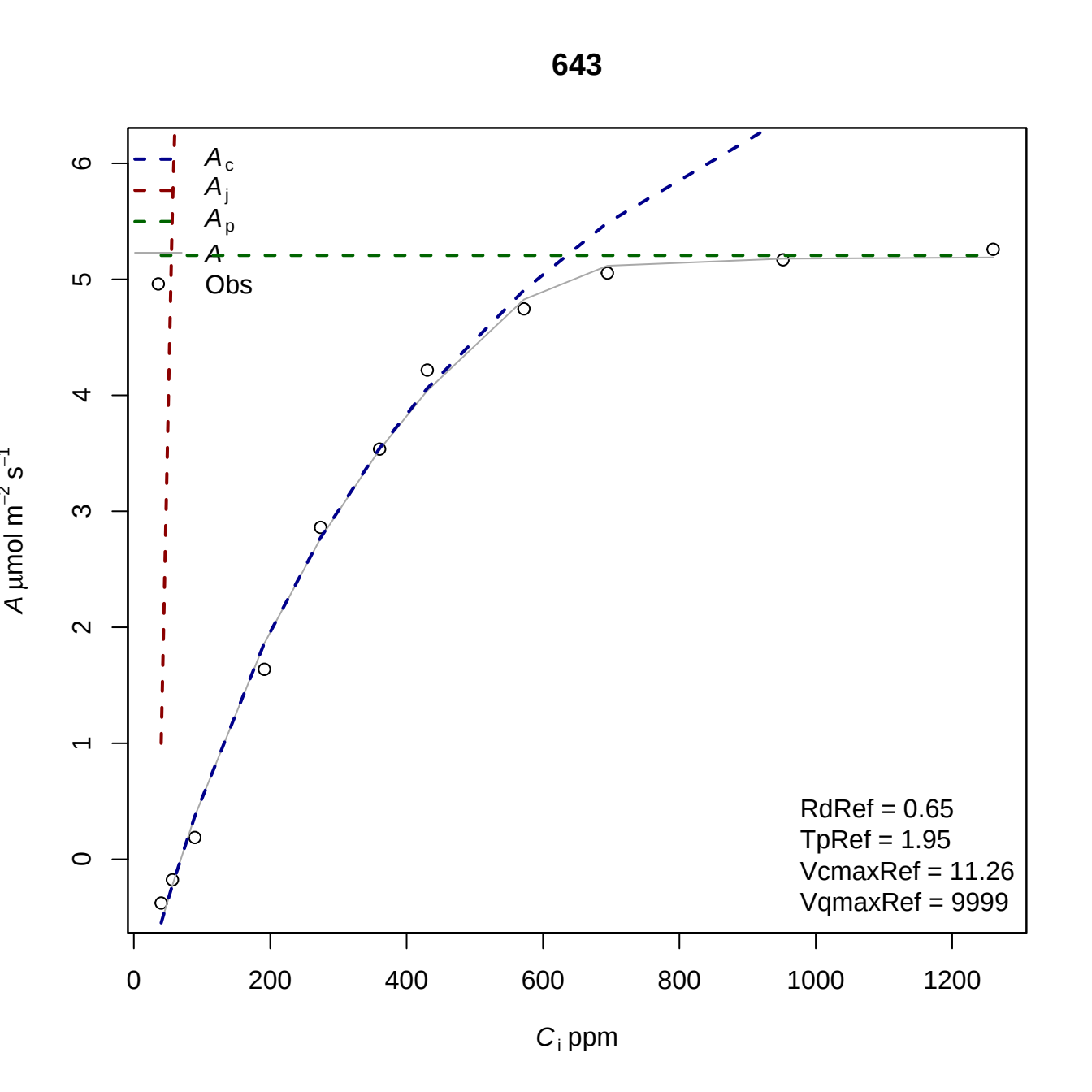

638
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.9
TpRef = 9999
VcmaxRef = 29.51
VqmaxRef = 80.71
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
5
10
15
200
400
600
800
1000


639
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.82
TpRef = 9999
VcmaxRef = 15.67
VqmaxRef = 45.85
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
200
400
600
800
1000


640
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 1.42
TpRef = 9999
VcmaxRef = 24.41
VqmaxRef = 79.22
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
14
200
400
600
800


641
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.75
TpRef = 9999
VcmaxRef = 10.72
VqmaxRef = 35.31
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
0
500
1000
1500


642
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.77
TpRef = 9999
VcmaxRef = 18.78
VqmaxRef = 76
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
2
4
6
8
10
12
0
500
1000
1500


643
$A_c$
$A_j$
$A_p$
$A$
Obs
RdRef = 0.65
TpRef = 1.95
VcmaxRef = 11.26
VqmaxRef = 9999
$A$ µmol m$^{-2}$ s$^{-1}$
$C_i$ ppm
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200